GMA eBook

Front-Office Sales & Trading
Interview Preparation

Contents

Chapter 1

Financial Markets Foundations

1.1 Purpose of financial markets

sales · trading · structuring core cross-asset

Every instrument in this book, from a share to a barrier option, exists for one reason: two people wanted opposite things and needed somewhere to meet. A company wants money now and will pay later; a saver has money now and wants more later. A farmer wants certainty about next year's price; a trader will take that uncertainty for a fee. Financial markets are the place those pairs find each other, and every product in the following chapters is a more precise way of doing it.

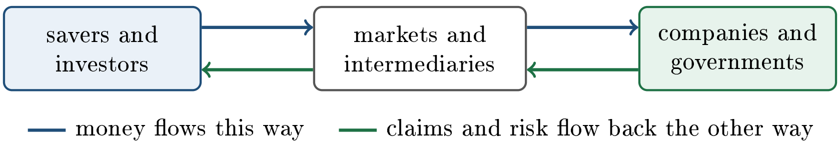
That sounds obvious, and it is the single most useful thing to hold onto while learning the rest, because it turns memorisation into reasoning. When you meet a structure you have never seen, the productive question is not "what is this called" but "who is on each side, and what is each of them trying to get rid of".

Four things markets do

1. **Move capital.** Savings sit with people who cannot use them productively; projects sit with people who have no savings. A market lets the first fund the second and be paid for it. This is what a share issue or a bond issue is, and it happens in the *primary* market, covered in primary versus secondary markets.
2. **Transfer risk.** Somebody exposed to a price they cannot control can hand that exposure to somebody willing to hold it. The airline that fixes its fuel cost has not made the risk disappear; it has paid someone to take it. Almost everything in the derivatives half of this book is this function, made precise.
3. **Provide liquidity.** A claim you can sell at any time is worth more than an identical claim you cannot, so a market that lets you leave makes people willing to enter in the first place. See liquidity.
4. **Discover prices.** The price that emerges from many people trading is a piece of public information: what the world collectively thinks something is worth right now. Companies, governments and central banks all use it. See price discovery.

INTUITION

Think of a village with a wheat farmer and a baker. The farmer's income depends on next autumn's wheat price and the baker's costs depend on the same number, so both of them lose sleep over it, in opposite directions. If they agree today on a price for next autumn's wheat, both stop worrying and can plan: the risk has not been destroyed, it has been *swapped* between two people who wanted opposite sides of it. That agreement is a forward contract, and it is the whole of finance in one sentence. Everything else is this idea with more precision, more counterparties, and someone in the middle making a market.



Every trade has two sides, and they want different things

The habit worth building from the first page is to read any instrument from both ends. A bond looks like one object, but the borrower and the lender are doing entirely different things with it, and understanding both is what lets you price it, sell it and know who will buy it.

REAL INTERVIEW QUESTION (JP Morgan · Sales)

What are bonds for, for the issuer and for the investor?

WHAT THEY TEST / ANSWER

Answer both halves separately, because the question is checking that you think about a security as a meeting point rather than as a product. **For the issuer**: a bond is borrowing. It raises money for a fixed term at a known cost, without giving away ownership, which is the key contrast with issuing shares, and the interest is typically tax-deductible where a dividend is not. It also lets the issuer choose the maturity to match what the money is for, so a thirty-year asset can be funded with thirty-year debt. **For the investor**: a bond is a contractual claim, so the cash flows are known in advance rather than discretionary; it ranks ahead of equity if the issuer fails; and it provides income and a way to match future liabilities, which is why insurers and pension funds are the largest holders. Then give the sentence that shows you see the trade rather than the two descriptions: the issuer is buying certainty about its funding cost and the investor is buying certainty about its income, and what changes hands in between is the risk that the issuer cannot pay, which is the credit spread. See fixed income for the instrument itself.

REAL INTERVIEW QUESTION (BNP · FX)

What is the FX market mainly used for?

WHAT THEY TEST / ANSWER

Split it into the real-economy purpose and the financial one, and give the relative sizes, because the surprise is part of the answer. **The original purpose** is settlement and hedging: cross-border trade and investment require converting currencies, and a company with revenue in one currency and costs in another needs to fix the rate, which is what forwards and options are for. **The larger reality** is that only a small fraction of daily FX turnover corresponds to trade in goods; most of it is financial, from investors hedging the currency exposure of foreign assets, from banks managing their own positions, and from speculation and market making that provide the liquidity everyone else uses. Say that liquidity provision explicitly rather than treating speculation as a separate and slightly disreputable activity, since it is the reason a corporate can hedge a large amount at a tight price at almost any hour. Then add the structural facts that mark out someone who follows the market: FX is over-the-counter and decentralised rather than exchange-traded, it runs continuously through the week rather than in sessions, and the dollar is on one side of the large majority of transactions, which is what makes it the funding and pricing currency for most of the world.

Is trading a zero-sum game?

This comes up constantly, in interviews and in arguments with people who dislike finance, and the honest answer has two parts.

For a single derivative contract, in pure cash terms, yes: one party's gain is the other's loss, because the contract creates no new wealth on its own. The same is roughly true of a day's trading in the secondary market.

But that is an incomplete accounting, because the two parties did not enter the contract to gamble; they entered it to change the shape of a risk they already had. The airline that fixed its fuel cost and lost money on the hedge because prices fell has not lost in any meaningful sense: it bought a year of being able to plan, and it paid for it. Value is created by the *transfer*, not by the cash flow, and that value does not show up in either party's profit and loss on the trade. The primary market is more obviously positive-sum still: the company that raised money built something.

The useful formulation to have ready is that **the trading is close to zero-sum and the market is not**, and that a desk earns its living by standing between two people who both benefit, which is what market making is.

KEY IDEA

When you meet any new instrument, ask three questions in order: who is on each side, what is each of them trying to get rid of, and who is in the middle. That is enough to reconstruct most of what a product does before anyone explains it, and it is the reasoning the rest of this book formalises.

INTERVIEW TRAP

Three beginner errors worth avoiding early. (1) Treating markets as a casino with extra steps: the gambling analogy fails because the participants mostly arrive with a risk they already own rather than one they created for entertainment. (2) Treating them as an unambiguous good: markets allocate capital well most of the time and badly some of the time, and the history chapter exists because those failures are large and instructive. (3) Learning instruments as names rather than as answers to a need, which works until the first product you have not seen before.

VISUAL / CODE TO BUILD: Figure

A single diagram of the system, built up in four steps the reader clicks through rather than shown at once. Step one: savers and users of capital, with money flowing one way and claims the other. Step two: add the intermediaries, showing where a bank, an exchange and a broker each sit. Step three: overlay a risk transfer on top of the capital flow, so the reader sees that the two are separate functions travelling through the same pipes. Step four: place four example instruments on the diagram, a share, a bond, a forward and an option, each drawn as an arrow between two specific participants, so the reader can see that the instruments are labels for paths rather than objects in their own right.

Where this goes next

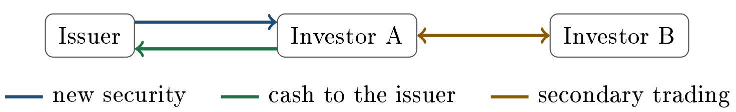
Where new securities are created and where existing ones change hands are separated in primary versus secondary markets, and the participants on each side are mapped in buy-side versus sell-side. The two functions that make the whole thing work, being able to leave a position and knowing what things are worth, are liquidity and price discovery. The instruments themselves start in equities and fixed income, and the pure risk-transfer instruments begin with forwards, which is the village agreement of this section written down precisely.

1.2 Primary vs secondary markets

sales · trading · structuring | core | equity · credit · cross-asset

One distinction organises the whole industry, including which desk you will end up sitting on: whether the money you are moving reaches the company that issued the security, or only another investor.

- **Primary market**: securities are *created* and sold for the first time. The issuer, a company or a government, receives the cash. This is a company raising capital.
- **Secondary market**: those same securities change hands between investors. The issuer receives nothing and is not a party to the trade. This is investors changing their minds.



only the first two arrows are the primary market; the issuer is not a party to the third

INTUITION

It is the difference between buying a new car from the manufacturer and buying a second-hand one from a neighbour. Only the first sends money to the factory. But nobody would buy the new car at a good price if there were no second-hand market at all, because they would be stuck with it forever. The two markets are separate and completely dependent on each other.

KEY IDEA

The secondary market sets the price of the primary market. An investor buying a new issue is asking one question before anything else: if I change my mind, can I sell this, and at what cost? The harder that is, the more yield or the bigger discount they demand at issue. So liquidity in the secondary market is not a nice-to-have for the issuer, it is a direct reduction in their cost of funding.

REAL INTERVIEW QUESTION (I-Kapital · Structured Products)

What is the difference between the primary market and the secondary market?

WHAT THEY TEST / ANSWER

Give the one-line distinction, then show you know why it is asked. Primary is issuance: the security is created and the proceeds go to the issuer. Secondary is trading between investors, with no cash reaching the issuer.

Then the three consequences that make the answer worth listening to. **Different jobs**: primary is origination and syndicate, secondary is sales and trading, and the front-office seat you are interviewing for is almost certainly on the second. The bank acts as an agent or underwriter on the primary side and as a principal or market maker on the secondary. **Different prices for the same thing**: a new issue usually comes with a concession to the issuer's existing curve, precisely to attract enough demand on the day. **One feeds the other**: the secondary price of an issuer's outstanding bonds is the reference against which their next issue is priced, so a deteriorating secondary market closes the primary market, which is exactly what happens in a crisis. In structured products specifically, the primary market is the note sold at 100 and the secondary market is the issuer buying it back later, which is a different market with one participant.

What actually happens on issue day

REAL INTERVIEW QUESTION (Natixis · Credit)

What happens on the day a bond is issued in the primary market?

WHAT THEY TEST / ANSWER

Walk the day in order, because the sequence is the answer.

Announcement. The issuer and its lead banks announce the deal in the morning: issuer, maturity, currency, format, and **initial price thoughts**, a deliberately conservative spread over a reference, typically the government curve or mid-swaps.

Books open. Investors place orders through the syndicate desk, each with a size and often a spread limit. The book builds through the morning and the leads publish updates on its size.

Guidance and revision. If the book is several times covered, guidance is tightened, the spread comes in, and some investors drop out or cut their orders. This is the price discovery, and it is a negotiation conducted through a book rather than on a screen.

Pricing and allocation. The final spread is set, the book is allocated, which is discretionary rather than pro rata, and the deal is priced. Allocation is where the issuer and the leads reward investors they want on the register.

Free to trade. The bonds start trading in the secondary market, usually the same day, and the immediate move is watched closely by everyone. Tightening means the deal was priced generously and the issuer left money on the table; widening, so the bonds trade below their issue price, means the concession was too small and the investors who bought are down on day one.

The concept to name explicitly is the **new-issue concession**: the extra spread over where the issuer's existing bonds trade, paid to compensate investors for taking down size in one go. Say that it is a genuine cost to the issuer and a genuine return to the investor, that it widens when markets are volatile and can go to zero or below when they are hot, and that measuring it against the issuer's secondary curve is exactly why the two markets cannot be discussed separately. Bond pricing itself is in bond pricing.

The equity version has the same skeleton with different names: an IPO is announced with a price range, built through a bookbuild, priced and allocated, and then trades. Two features are specific to it. The **greenshoe** lets the banks sell more shares than the deal size and buy them back in the market, which stabilises the price in the first days. The **lock-up** stops existing holders from selling for a set period, which limits the supply that can hit the new secondary market before it has settled.

When the secondary market has one participant

Most of this section assumes a secondary market with many buyers and sellers. Large parts of the structured-products world do not have one.

REAL INTERVIEW QUESTION (BNP · Exo)

On the secondary market, what is the point of trading with an external issuer, and who carries the default risk?

WHAT THEY TEST / ANSWER

Two separate questions, and the second is the one with a clean answer.

Who carries the default risk: the investor holding the note. A structured note is unsecured debt of its issuer, so if the issuer defaults the holder is a creditor alongside the bondholders, whatever the equity underlying did. This is true in the primary market and it does not change in the secondary market: buying a note from someone else transfers the issuer's credit risk to you along with the payoff.

Why trade with an external issuer: because it is the only way to get a real price. The issuer of a note is usually its only market maker, so a client wanting to sell is negotiating with the one counterparty that has no competitive pressure. Taking the risk with, or asking a price from, a different issuer gives you an independent mark, diversifies the credit away from a single name, and lets a distributor pick the best terms in the primary market rather than accepting one house's. The trade-off to name is that an external issuer's paper may be even less liquid and its credit weaker, so you are exchanging one problem for another unless you are paid for it.

Close on the honest point for a client: the secondary market for a structured note is thin by construction, and the bid on day two reflects the fees and the bid-offer that were embedded on day one. If a note is issued at 100 with 2% of distribution fees and a 1% bid-offer, the bid the next morning is nearer 97 than 100, with nothing having moved. That is a cost of the primary market showing up in the secondary market, and it is the number a client should be shown before they buy. See retail versus institutional structuring.

INTERVIEW TRAP

Three confusions. (1) Thinking a company makes money when its share price rises in the secondary market: it does not directly, though it makes the next capital raise cheaper. (2) Assuming the new-issue price is the fair price: it usually includes a concession, which is why day-one performance is scrutinised. (3) Assuming any security can be sold back at its quoted value; for anything issued in small size or by one house, the exit price is set by whoever is willing to make you a price, which may be nobody.

Where this goes next

Who is on each side of these trades is buy-side versus sell-side, the participants who make the secondary market work are market makers, and what "can I sell this" really means is liquidity and the bid-ask spread. The venues themselves are exchanges, and the note that lives this whole life cycle is in structured notes.

1.3 Buy-side vs sell-side

sales · trading · structuring | core | cross-asset

Almost everyone in markets works on one of two sides, and the words for them are unhelpfully named: the labels come from the flow of *services*, not of securities. The sell side sells execution, research, products and liquidity; the buy side buys them. Both of them buy and sell securities all day.

The distinction that actually matters is simpler and worth stating in one line: **who holds the position, and why**.

The two sides

	Sell side	Buy side
Who	investment banks, brokers, market makers	asset managers, funds, pensions, insurers
Role	intermediates and manufactures	owns the risk
Positions	a by-product of servicing flow	the point of the job
Wants to be	flat, or close to it	positioned, deliberately
Horizon	intraday to weeks	months to years
Paid by	bid-offer, fees, margin	management and performance fees
Client	the buy side	the end investor

The third and fourth rows carry the distinction. A bank's derivatives trader ends up long a position because a client sold it to them, not because they wanted it, and their job is to manage and reduce that risk while being paid for having provided the service. A hedge fund's portfolio manager put on the position because they believe it will make money, and their job is to be right. Both are called traders, and they are doing different things.

> **KEY IDEA**
>
> The sell side is paid for *providing a service* and would rather not hold risk; the buy side is paid for *taking a view* and holds risk on purpose. Every difference below, in horizon, in compensation, in what "a good day" means, follows from that one distinction.

Who is on the buy side

"Buy side" is not one thing, and knowing the types matters because they want completely different products.

- **Long-only asset managers**: run money against a benchmark, so their risk is measured relative to an index and their constraint is a mandate that often forbids leverage and short selling.
- **Hedge funds**: absolute return, wide latitude, leverage permitted, and typically the most sophisticated users of derivatives.
- **Pension funds and insurers**: have long-dated *liabilities* and buy assets and hedges to match them. They are the natural buyers of long-dated protection and the largest holders of bonds.
- **Corporates**: not investors at all, but hedgers, using markets to fix a cost or a revenue in their own business.
- **Sovereign wealth funds and central banks**: very large, very long horizons, often constrained by policy rather than by returns.

For anyone selling a structure, this list is the first question to ask about a client, because the same product is right for an insurer matching a liability and wrong for a fund seeking absolute return.

REAL INTERVIEW QUESTION (BNP · Fixed Income)

What is the difference between the buy side and the sell side?

WHAT THEY TEST / ANSWER

Lead with the functional distinction rather than the list of firms, since the list is easy and the distinction is the point. The **sell side** intermediates: banks and brokers execute, make markets, manufacture products and provide research and financing, and they are paid for the service through bid-offer, fees and structuring margin. Any position they hold is a by-product of having serviced a client, and their instinct is to hedge it down. The **buy side** owns the risk: asset managers, hedge funds, pension funds and insurers invest capital, either their own or their clients', and they are paid on the assets they manage and the returns they generate. Their positions are the job rather than a residue of it. Then add two things that mark out a candidate. First, the naming is about services, not securities: both sides buy and sell all day, and the sell side is so called because it sells execution and research to the buy side, which is worth saying because the terms confuse everyone at first. Second, the consequence for how each behaves: the sell side manages risk it did not choose over a short horizon and is judged on flow and on not losing money on inventory, while the buy side chooses its risk deliberately over a long horizon and is judged on performance, usually against a benchmark or a hurdle. If they ask which is harder, the honest answer is that they are different, and that the sell side is where you learn the products fastest, which is why the traffic between them runs mostly in one direction.

REAL INTERVIEW QUESTION (JP Morgan · EQD)

What is a hedge fund?

WHAT THEY TEST / ANSWER

Define it by what distinguishes it rather than by the word "hedge", which is historical and now misleading. A hedge fund is a pooled investment vehicle for professional or wealthy investors that pursues an **absolute return**, meaning it aims to make money in any market rather than to beat an index, and that operates under far fewer constraints than a regulated long-only fund: it can use leverage, sell short, concentrate, and hold illiquid or derivative positions. The compensation structure follows from that: a management fee on assets plus a performance fee on returns, historically quoted as two and twenty, often subject to a high-water mark so the manager is not paid twice for recovering the same losses. Then give the strategy taxonomy, briefly, because it is the natural follow-up: long-short equity, global macro, relative value and arbitrage, event-driven, credit, and quantitative or systematic. Close with why an equity derivatives desk cares, which is the reason this question is asked in an EQD interview: hedge funds are the most active and most demanding derivatives clients, they trade both directions rather than one, and unlike retail flow they can be on either side of a position, which is what makes them valuable to a desk that would otherwise warehouse the same risk from everyone. See retail versus institutional structuring for how differently the two client types are served.

Grey areas worth knowing

The clean split is a simplification, and interviewers sometimes probe the edges.

Proprietary trading firms and electronic market makers sit awkwardly. They take positions with their own capital, which sounds buy side, but their business is providing liquidity and earning bid-offer, which is the sell-side function. They are usually described as sell side by function and neither by structure.

A bank's own treasury manages the bank's balance sheet and funding, so it is an internal buy side of a kind, and the reason a structured note has a funding leg at all, as structured notes sets out.

Multi-strategy funds that internalise their own execution do part of what a broker would otherwise do for them, which is why the boundary is best treated as a description of function rather than of institution.

What it means for your career

The two sides are different jobs with different daily rhythms, and knowing which you are interviewing for is worth saying out loud when asked. Sell side is faster, more social, more product-driven, and you learn a very large number of instruments quickly because clients ask about all of them. Buy side is slower, more analytical, more solitary, and success is measured over quarters rather than days.

Movement between them is real but asymmetric: people move from sell side to buy side far more often than the reverse, because the sell side is where product knowledge and client access are built and because a buy-side seat is scarcer. That is also why an interviewer will not be offended if you say you want to end up on the buy side eventually; they will be unimpressed if you say it without being able to explain what the sell-side job involves first.

INTERVIEW TRAP

Three errors. (1) Thinking the names describe who buys and who sells securities: they describe who buys and who sells *services*. (2) Treating the buy side as one client type, when an insurer matching liabilities and a macro fund taking a view want opposite products. (3) Saying in an interview that you want the buy side without being able to describe the sell-side role you are applying for, which reads as using the desk as a waiting room.

VISUAL / CODE TO BUILD: Figure

A single map of the ecosystem with the participants placed by function rather than by name. The reader hovers over any participant and the diagram highlights who they trade with, what they pay for and what they are paid for, so the sell side lights up as a hub and the buy side as spokes. A toggle overlays a single trade, for instance a pension fund buying a hedge, and animates it through the chain, showing the broker, the market maker, the bank's trading book and the hedge in the listed market, with the fee or spread taken at each step labelled. The intended realisation for a beginner is that a single client trade touches four institutions, each paid for a different thing.

Where this goes next

The specific sell-side functions get their own sections: market makers and brokers, with the desk-level version in market making and flow trading. What the different roles inside a bank actually do day to day is different roles within the bank, and why the two sides exist at all is the purpose of financial markets.

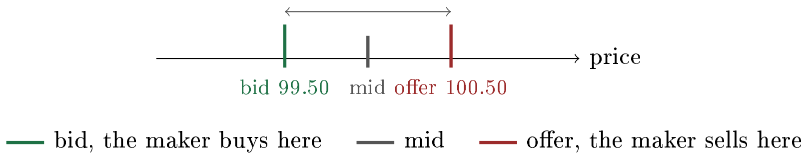
1.4 Market makers

sales · trading · structuring core cross-asset

When you decide to sell a share, somebody has to buy it at that moment. Not eventually, not when a matching buyer turns up, but now. The participant who stands ready to be that somebody, in both directions, all day, is the market maker, and understanding what they are compensated for is the foundation of most trading interviews.

What the job is

A market maker **quotes a two-way price**: a level at which they will buy, the **bid**, and a higher level at which they will sell, the **offer** or **ask**, each for a stated size. They do this continuously, and on many venues under a formal obligation to keep quoting even when they would rather not.



the double arrow is the spread, and everyone else trades *across* it

> **INTUITION**
>
> A market maker is a bureau de change. They will always change your euros for dollars and your dollars for euros, and the two rates are not the same. The gap is not a fee for a service: it is what they need in order to survive the fact that you get to choose which side to take, and that you may be changing money because you know something about the currency that they do not.

Why the two prices are different

The natural beginner's question is why they cannot just quote one price and earn a commission. The answer is the whole of this section.

> **KEY IDEA**
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> By quoting two-way, the market maker gives the entire market a **free option**: everyone else chooses whether to trade and on which side, and they will choose the side that suits them. Whoever holds an option must be paid for it, and the spread is that payment. It compensates for three things at once: **inventory risk**, since after trading the maker holds a position they did not want and the price can move before they can offload it; **adverse selection**, since some of the people trading against them know something; and the **operating cost** of being there at all. The anatomy of the spread itself is developed in the bid-ask spread.

WORKED EXAMPLE

A market maker quotes 99.50 bid, 100.50 offer, in size 100, on an instrument they believe is worth 100.

The good case. A seller hits the bid: the maker buys 100 units at 99.50. Minutes later a buyer lifts the offer: they sell 100 units at 100.50. They have bought and sold the same thing, hold no position, and have made $100 \times (100.50 - 99.50) = 100$. That is the business, repeated thousands of times.

The bad case. The seller hits the bid, and no buyer arrives. The maker is long 100 units at 99.50 and the market drifts to 98. The position is down $100 \times 1.50 = 150$, which wipes out the spread earned on the previous one and a half round trips.

Both cases come from the same quote. The market maker's skill is not in predicting direction; it is in setting a spread wide enough that the good case outnumbers the bad one, and in managing the inventory in between.

The three things a maker actually controls

A quote has more dimensions than beginners assume, and a trading interview will test all three.

- **The width.** How far apart the two sides are. Wider means safer per trade and less flow, because someone else's tighter quote gets hit first.
- **The level, or skew.** Where the pair sits. Moving both sides up makes your bid more attractive and your offer less attractive, so you buy more and sell less. This is how a maker steers what they end up holding without ever refusing a trade. When to skew rather than widen is worked out in the bid-ask spread.
- **The size.** How much you are good for. Showing less is the cheapest way to reduce risk, and it is the first lever a maker pulls in a name they do not understand.

REAL INTERVIEW QUESTION (Société Générale · EQD)

You have a bid-ask at 900, 700. If you get a buyer, how do you change your bid-ask?

WHAT THEY TEST / ANSWER

Fix the convention first, out loud: the bid is the lower of the two, 700, and the offer is the higher, 900, so the quote is 700 bid at 900 offered. Getting the convention right in the first sentence is part of the mark.

A buyer means they lifted your offer, so you have *sold* and you are now short. You want to buy back, and you want to discourage more buyers, so you **move the whole quote up**: raise the bid to attract a seller, and raise the offer so that a second buyer pays more. Say the numbers rather than the direction alone: something like 750 bid at 950, and keep the width roughly the same unless you have a reason to widen.

Two refinements earn the follow-up. Skewing is not a directional view, it is inventory management: you are not predicting the price will rise, you are pricing your own position. And if the same buyer comes back a second and a third time, the correct response is no longer just a skew but a widening as well, because repeated one-way flow is information rather than coincidence. That case is worked through in case studies, and the desk version in market making.

REAL INTERVIEW QUESTION (Société Générale · Rates)

A client asks you for a price on 100 calls and you quote a volatility of 20. Ten minutes later they ask for a price on 200 calls. Do you quote more than 20 or less than 20?

WHAT THEY TEST / ANSWER

More, if you are selling. Give the reason in one line before the nuances: a bigger trade costs more to hedge and carries more information, so it should be priced worse for the client, and on an offer that means a higher volatility.

Unpack it into the two effects. **Hedging cost**: twice the size is more than twice the difficulty, because you have to buy your vega back in a market that will move against you as you do it, and the larger your footprint the more it moves. **Information**: a client doubling up ten minutes later is showing you a direction, and repeated same-way interest is the classic sign that you are on the wrong side of someone with a reason.

The refinement that gets the nod is the symmetry: "more than 20" is only right on the side you are selling. If the second request were to *sell* you 200 calls, you would quote a *lower* volatility, for exactly the same two reasons. The general statement, which is the one to say, is that size widens the quote around your mid on both sides, and one-way repetition moves the mid itself.

Then add the sales point, because on a client-facing desk it matters: you say why. "The first hundred was on the screen price; the second two hundred moves my delta and my vega, so it comes at a different level" is a conversation that keeps the client. Silently repricing without explanation is how a franchise is lost, and repeating the quote unchanged to seem accommodating is the single most expensive habit a junior market maker can have. The fuller version of this question, including what to do when the follow-up size is fifteen times the first, is worked through in the bid-ask spread.

Who does this

Two kinds of firm. **Banks** make markets in the products their clients need, including many that never trade on an exchange, and treat it as part of a franchise: the spread is one revenue line among several, and being reliably there in a bad market is what earns the rest of the relationship. **Specialist trading firms** make markets on exchanges in listed products, compete on speed and on the width of the quote, and hold positions for a very short time.

The distinction that matters for an interview is against the broker, who never takes the other side and is paid a commission for finding a counterparty; that comparison is in brokers.

INTERVIEW TRAP

Three misunderstandings. (1) "The market maker wants the price to go up": they want *flow*, in both directions, and a strong directional view is a different business. (2) "The spread is their profit": it is their gross margin per round trip, before inventory losses, hedging costs and the trades where they were picked off, and in a bad week the net is negative. (3) "A wider spread means they are greedy": width is priced off risk, so it widens in volatile markets, in illiquid names and in large size, which is precisely when the service is most valuable.

Where this goes next

The intermediary who does not take the other side is in brokers, the spread's anatomy in the bid-ask spread, and what "can I trade this" means in liquidity. The mechanics of how quotes meet orders are market microstructure, and the professional version of this job, including how a real book is managed, is market making and inventory management.

1.5 Brokers

sales · trading · structuring core cross-asset

A broker arranges a trade between two other parties and does not take the other side of it. That one sentence separates a broker from a market maker, and almost everything else in this section is a consequence of it.

> KEY IDEA
>
> The distinction that matters is **agency** against **principal**. An agent finds you a counterparty and is paid a commission; its profit does not depend on where the market goes. A principal *is* your counterparty and takes the risk onto its own book; its profit depends entirely on what happens next. Everything else, the labels and the business models, follows from which of those two a firm is doing on a given trade.

Some firms do both, on different trades or on the same one, and the middle case has a name worth knowing. A **riskless principal** trade is one where the firm interposes itself legally, buying from one side and selling to the other, but does so simultaneously so it holds no market risk. It looks like principal in the documentation and like agency in the economics, and it exists because many clients want a single credit-worthy counterparty rather than exposure to whoever was on the other side.

The kinds you will actually hear named

- **Inter-dealer brokers.** The most important kind for anyone joining a trading floor. They sit between banks in over-the-counter markets, rates, credit and FX above all, and their real product is **pre-trade anonymity**: a dealer can show interest in size without the rest of the street learning who wants what. Voice broking survives in exactly the products where that matters most and where liquidity is too thin for a screen.
- **Agency and execution brokers.** They work orders in listed markets for institutional clients, charging commission, and compete on execution quality rather than on price.
- **Prime brokers.** A bundled service to hedge funds: financing, custody, clearing, and stock borrow from securities lending. A prime broker is emphatically *not* a pure agent, because it lends money and securities against collateral, and so it carries counterparty risk on every client.
- **Retail brokers.** Order routing for individuals, where the commission has largely gone to zero and the revenue has moved elsewhere, which is the subject of the conflicts below.

Why anyone pays for this

A broker is being paid to solve a search problem. In a market with a central limit order book the search problem is small, which is why commissions there have collapsed. In an over-the-counter market it is large: you do not know who wants the other side, and asking around reveals your intention to the people you are about to trade with. The broker knows who has been asking, and can put two sides together while telling neither who the other is until the trade is agreed. Information leakage, not matching, is the thing being priced.

> **INTUITION**
>
> Compare how the two business models earn. A market maker's revenue is the spread and its risk is inventory, so it makes money on volume and loses it on being caught the wrong way round. A broker's revenue is commission and its risk is having no clients, so it makes money on volume and does not care about direction. That is why brokerage revenue is stable and thin while market making is volatile and fat, and why the two businesses behave completely differently in a crisis: volumes spike, so brokers do well, while market makers can do either extremely well or extremely badly.

Conflicts, which is where the exam questions would be

Because a broker acts for a client, its obligations run to the client rather than to its own book, and the interesting cases are where that breaks down.

Best execution is the regulatory requirement to take sufficient steps to get the best result for the client across price, cost, speed and likelihood of settlement, not merely the best headline price. It is a process obligation, so it is satisfied by having a policy and evidence rather than by winning on every trade.

Payment for order flow is a retail broker being paid by a wholesale market maker for the right to execute its clients' orders. It is what replaced commission when commission went to zero, it is banned in several jurisdictions including the European Union, and the argument against it is exactly the agency conflict above: the routing decision is being made by whoever pays the broker rather than by what the client's order needs.

Research unbundling separated payment for research from payment for execution, so that commission could no longer quietly fund analyst coverage. It changed how the sell side is paid for research, and it is the reason research budgets are now negotiated explicitly.

REAL INTERVIEW QUESTION (Cargill · Commodity)

What strategies did you cover as a broker assistant?

WHAT THEY TEST / ANSWER

An experience question, and the trap is answering it as a list of product names. They want to know what you *saw*, and specifically whether you understood the business you were sitting in rather than the tasks you were given.

Structure it in three parts. **What the desk brokered**: the products and the client types, in one sentence each. **What you actually did**, honestly, since a broker assistant is usually confirming trades, maintaining the order log and running the screen, and saying so plainly is better than inflating it. **What you learned that you could not have learned from a book**, which is where the answer is won.

For a broking seat that last part is nearly always the same thing and it is worth saying: you saw that the firm makes money on flow and not on direction, that its inventory is relationships rather than risk, and that the price a client gets depends on information as much as on the level. If you saw a trade go wrong, describe it, because failures are where the mechanics become visible.

Then, since this is a commodity desk, make it concrete to the asset: whether the flow was physical or financial, whether you saw the basis between them, and whether the clients were producers hedging, consumers hedging, or funds taking a view. Naming which side of the market your clients were on tells the interviewer you understood who you were serving.

INTERVIEW TRAP

1. **Using "broker" and "dealer" interchangeably.** A dealer takes the other side; a broker does not. In the United States the combined firm is literally called a broker-dealer, which is a description of two activities rather than one.
2. **Assuming a prime broker is an agent.** It lends cash and securities against collateral and is exposed to its clients, as Archegos demonstrated expensively for several of them.
3. **Thinking voice broking is obsolete.** It persists precisely where anonymity and size matter and where the screen is too thin, which is most of the over-the-counter rates and credit market.
4. **Confusing best execution with best price.** It is a process obligation across several dimensions, not a guarantee on any single trade.
5. **Assuming zero commission means free.** The revenue moved to order flow payments, financing and spreads. Someone is still paying, and in the retail case it is the client, less visibly.

VISUAL / CODE TO BUILD: Interactive

A trade-routing sandbox. The reader is given an order, an instrument and a size, and chooses a route: an exchange order book, an agency broker working it over a day, an inter-dealer broker, or a request for quote to two dealers. The page simulates each and shows the total cost decomposed into commission, spread paid and market impact, so the reader can see the crossover directly: small listed orders are cheapest on the book, large or illiquid ones are cheapest through a broker despite the commission, and the difference is information leakage rather than fees. A toggle reveals what each counterparty learned about the reader's intention on each route, which is the part the cost numbers do not show.

1.6 Exchanges

sales · trading · structuring | core | equity · commodity · cross-asset

An exchange is a place with rules where standardised things are traded against a published price. That sentence contains the four features that make it different from picking up a phone, and each one is worth a line.

- **Standardisation.** You cannot trade whatever you like: the contracts are defined by the exchange, in fixed sizes, with fixed maturities and fixed delivery terms. That is what allows one buyer's contract to be identical to every other buyer's, which is what allows a market to form at all.
- **Centralisation.** All the interest meets in one order book instead of being spread across bilateral conversations, so the best bid and the best offer are visible.
- **Transparency.** Prices and volumes are published, which is where the "price of the market" that newspapers quote comes from.
- **Clearing.** Once a trade is matched, a central counterparty steps in between the two sides, so neither of them carries the other's credit risk. This is the least visible feature and the most important; it is developed in clearing houses.

The order book

Trading on an exchange means posting into or taking from a **central limit order book**. Two order types do most of the work. A **limit order** says "buy 500 at 99.50 or better" and sits in the book until it is filled or cancelled. A **market order** says "buy 500 now" and takes whatever the book is showing.

WORKED EXAMPLE

A book on a share:

Bids		Offers	
Size	Price	Price	Size
800	99.48	99.51	400
1200	99.49	99.52	600
300	99.50	99.53	500

The best bid is 99.50 for 300 and the best offer is 99.51 for 400, so the spread is one cent and nothing trades until somebody crosses it. A market order to buy 500 takes all 400 at 99.51 and then 100 at 99.52, so the average fill is $(400 \times 99.51 + 100 \times 99.52)/500 = 99.512$. The extra two tenths of a cent over the best offer is **slippage**, and it is why size and liquidity are the same conversation.

VISUAL / CODE TO BUILD: Interactive

A depth chart of the ladder above: cumulative bid size stepping down and to the left from the touch, cumulative offer size stepping up and to the right, with the spread as the flat gap between them. Let the reader set an order size and watch the fill price walk up the offer steps, so the average fill separates from the touch as size grows. The static table gives the three levels; what a reader needs to see is the *shape* of the liquidity behind them, which is what decides whether a given size is cheap or expensive to execute.

Orders are matched by **price-time priority**: better prices first, and among equal prices, whoever posted first. That simple rule is the reason speed matters on electronic venues, and it is developed in market microstructure.

Listed versus over-the-counter

The alternative to an exchange is to agree a trade directly with a counterparty, which is called **over-the-counter**. Most of the notional traded in the world is OTC, and most of what a beginner has heard of is listed.

REAL INTERVIEW QUESTION (Morgan Stanley · EQD)

What is the difference between the organised (listed) market and the OTC market?

WHAT THEY TEST / ANSWER

Give the four differences in order, then say why anyone would choose the second.

Standardisation: listed contracts are defined by the exchange, OTC contracts are negotiated line by line. **Counterparty risk**: a listed trade is novated to a clearing house that both sides face, with margin posted daily, while an OTC trade leaves you facing the other party, mitigated by collateral agreements rather than removed. **Transparency**: listed prices and volumes are public, OTC prices are known to the two parties and to whoever they ask. **Liquidity and exit**: a listed position can usually be closed by trading the same contract with anybody, whereas an OTC position is usually unwound with the same counterparty who quoted it, which is a weaker position to be in.

Then the point of the question: given that list, why does OTC exist at all? Because standardisation is exactly what a real hedger often cannot use. A company with a 137 million dollar receivable due on an awkward date cannot hedge it with a standard contract for a round amount on a quarterly date, and the residual mismatch may cost more than the OTC spread. So the trade-off is **tailoring against liquidity and credit**: OTC gives you the exact exposure at the cost of a wider price, a counterparty you must monitor, and a harder exit. Add the regulatory postscript, since it is the part that dates candidates: since the post-2008 reforms, standardised OTC derivatives such as vanilla interest rate swaps are required to be centrally cleared and largely traded on electronic platforms, so the two worlds have moved towards each other and the clean dichotomy is now a spectrum. The forward-versus-future case is worked in futures.

Knowing the venues

Interviewers use exchange names as a cheap check on whether you follow markets or only studied them. The list is short and worth simply knowing.

- **Equities**: Euronext (Paris, Amsterdam, Brussels, Lisbon, Dublin, Oslo, Milan), the London Stock Exchange, Deutsche Börse with its Xetra platform, and in the United States the NYSE, Nasdaq and Cboe. In Asia, the Tokyo exchange under JPX, Hong Kong's HKEX, and the Shanghai and Shenzhen exchanges.
- **Listed derivatives**: Eurex for European equity and rates futures and options, CME Group in the United States, and ICE, which also owns the NYSE.
- **Commodities**: NYMEX and COMEX within CME Group for energy and metals, CBOT for grains, ICE for Brent crude and soft commodities, and the London Metal Exchange for base metals.

REAL INTERVIEW QUESTION (BNP · Commodity)

What are the main exchanges? Can you name a contract on one of them?

WHAT THEY TEST / ANSWER

Organise by asset class rather than listing names at random, then be specific, because the second half of the question is the real one and a candidate who names an exchange but no contract has revealed that the knowledge is second-hand.

Energy: **Brent crude futures on ICE** and **WTI crude futures on NYMEX**, the two global oil benchmarks, plus **Henry Hub natural gas on NYMEX** and **TTF gas** for Europe. Metals: **copper and aluminium on the LME**, and **gold and silver on COMEX**. Agriculture: **corn, wheat and soybeans on CBOT**.

Then add a detail about one of them, which is what turns a list into a conversation. The LME is the interesting choice because it is genuinely different from every other exchange: it trades daily prompt dates rather than only monthly ones, reflecting its origins in physical metal delivery, and it operates warehouses, so a position there can end in real metal in a real shed. That structure is also why LME episodes tend to involve squeezes on physical inventory rather than pure price moves. Knowing one story like that is worth more than reciting ten exchange names.

REAL INTERVIEW QUESTION (Société Générale · Delta One)

What is the stock exchange of the Netherlands?

WHAT THEY TEST / ANSWER

Euronext Amsterdam, whose benchmark index is the **AEX**. Answer in three words and then add the two facts that make it worth having been asked. It is part of **Euronext**, the pan-European group that also runs Paris, Brussels, Lisbon, Dublin, Oslo and Milan on a single trading platform, which is why a European cash equities desk deals with one venue rather than six. And Amsterdam is generally considered the **oldest stock exchange in the world**, dating from the early seventeenth century and the shares of the Dutch East India Company, which is the origin of most of what primary versus secondary markets describes.

The reason this question exists is that it is unfailable if you read about markets and unanswerable if you have only read textbooks. Treat every question of this type the same way: give the name immediately, then one structural fact and one piece of context, and move on without padding.

INTERVIEW TRAP

Three slips. (1) Saying an exchange "sets the price": it publishes the price that buyers and sellers produce, and it sets the *rules*, not the level. (2) Believing listed always means liquid: an exchange lists many contracts that barely trade, and a listed far-dated option can be far less liquid than an OTC swap. (3) Describing OTC as unregulated: it is bilaterally negotiated, which is not the same thing, and large parts of it are now cleared, reported and margined by rule.

Where this goes next

Who quotes the prices sitting in the book is market makers, who routes your order to it is brokers, and what happens after the match is clearing houses. The cost of crossing the book is the bid-ask spread, the mechanics of the book itself are market microstructure, and the clearest listed-versus-OTC pair of instruments is in futures.

1.7 Clearing houses

sales · trading · structuring core cross-asset

Agreeing a price is the easy part of a trade. The hard part is that a derivative is a *promise about the future*, so between the handshake and the settlement there is a period, sometimes years, in which the other side might fail. A clearing house is the institution that removes that worry, and understanding how it does it explains a great deal about why listed markets and over-the-counter markets behave differently.

The problem: everyone owes everyone

Without a clearing house, each trade creates a bilateral link. Trade with ten counterparties and you have ten separate exposures, each of which you must assess, monitor and collateralise, and each of which could fail independently. Worse, the exposures do not cancel: if you are owed 100 by one bank and owe 100 to another, you still carry the full risk of the first failing while remaining obliged to pay the second.

bilateral: ten links cleared: five links

With five participants there are ten bilateral links; with fifty there are 1,225. The number grows with the square of the participants, which is why a market of any size cannot be run this way safely.

What a clearing house does

A **central counterparty**, or CCP, steps into the middle of every trade by **novation**: the original contract between A and B is legally replaced by two contracts, A with the CCP and the CCP with B. From that moment neither A nor B has any exposure to the other. Each faces only the clearing house, which is deliberately built to be the safest participant in the market.

Three things follow immediately.

Anonymity. Since you face the CCP, you no longer care who took the other side, which is what makes anonymous exchange trading possible at all. See exchanges.

Multilateral netting. All your positions face the same counterparty, so offsetting trades genuinely cancel. Buying 100 futures from one member and selling 100 to another leaves you flat with the CCP rather than carrying two live exposures.

Fungibility. Because every contract is with the same entity, contracts become interchangeable, which is what lets you close a position by trading with anyone rather than by finding your original counterparty.

> KEY IDEA
>
> A clearing house does not make risk disappear. It *concentrates* the credit risk in one institution and then makes that institution extremely hard to break, using margin, a default fund and strict membership rules. The price of doing so is that the risk you no longer carry as credit risk comes back to you as a *funding* obligation: you must post collateral, in cash, every day, including on the days when you can least afford it.

The machinery that makes it safe

Initial margin is collateral posted up front, sized to cover the loss the CCP would suffer while closing out your position if you defaulted, over a horizon of a few days and to a high confidence level. It scales with the volatility of what you hold, so it *rises* in a stressed market.

Variation margin is the daily, sometimes intraday, settlement of mark-to-market moves, paid in cash. This is the mechanism that stops exposure accumulating: a cleared position is brought back to zero value every day, so the CCP is never owed more than one day's move plus the initial margin buffer. It is also the reason a future and a forward differ economically, as futures explains.

The default waterfall is the ordered list of resources used if a member fails, and being able to recite it is a standard risk-interview answer:

1. the defaulting member's own margin;
2. the defaulting member's contribution to the default fund;
3. a tranche of the CCP's own capital, its "skin in the game";
4. the surviving members' default fund contributions, which is where losses become mutualised;
5. further assessments on surviving members, up to a cap.

Membership standards sit underneath all of it: clearing members must meet capital and operational requirements, and smaller firms access the CCP indirectly through a clearing member, which reintroduces a bilateral exposure to that member.

The trade-off nobody mentions until it matters

Clearing converts a credit problem into a liquidity problem, and that is not a free swap.

Under a bilateral arrangement with no collateral, a losing position costs you nothing until settlement. Under clearing, the same losing position demands cash *today*, and if volatility rises the initial margin requirement rises with it. So a market participant can be perfectly solvent, correctly positioned and still fail, simply because they cannot find the cash for a margin call on the morning it is due. That is not a hypothetical: margin calls are one of the standard transmission channels by which a price move becomes forced selling, which liquidity risk develops.

The second consequence is concentration. Making the CCP the counterparty to everything makes it systemically critical, so the institution designed to remove counterparty risk has become one of the largest single points of failure in the financial system. Regulators supervise them accordingly, and "who guarantees the guarantor" is a legitimate question rather than a clever one.

REAL INTERVIEW QUESTION (BNP · Market Risk)

You have a payer swap with a counterparty: explain how it works, who is exposed to rate moves, and does it imply a movement of counterparty risk?

WHAT THEY TEST / ANSWER

Take it in three parts, and note that the third is what the question is actually for. **Mechanics**: in a payer swap you pay fixed and receive floating over a schedule of dates on an agreed notional, and only the net difference changes hands on each payment date, with the notional itself never exchanged. **Rate exposure**: you gain when rates rise, since the floating leg you receive resets higher while the fixed leg you pay is unchanged, so a payer swap is short duration and your counterparty is on the other side of exactly that. **Counterparty risk, and here is the point**: yes, and it *moves with the market*. At inception the swap is worth roughly zero to both sides, so neither has meaningful exposure. As rates move the swap acquires value for one side, and that side is now owed money and therefore exposed to the other failing. So counterparty exposure on a swap is not a constant, it is a function of the market, and it switches sides when the swap crosses through zero. That is why it is measured as a *potential future exposure* over the life of the trade rather than as a current number, why it is charged for through CVA, and why the standard mitigants are a collateral agreement that resets the exposure to near zero daily, or clearing, which replaces the counterparty with a CCP that margins you every day. See counterparty risk and collateral and haircuts.

INTERVIEW TRAP

Four errors. (1) Saying a clearing house eliminates risk: it concentrates and mutualises credit risk, and converts what remains into a daily funding obligation. (2) Confusing initial with variation margin; the first is a buffer against a future close-out and the second is the settlement of today's move. (3) Treating counterparty exposure on a swap as fixed, when it starts near zero, grows with the market and can change sides. (4) Equating listed with cleared: since 2008 a large volume of OTC derivatives is centrally cleared, so the meaningful distinction is cleared versus uncleared.

VISUAL / CODE TO BUILD: Interactive

A clearing simulator. The reader sets a number of participants and watches the count of bilateral links grow with the square, then flips a switch to interpose a CCP and sees it collapse to a linear count, with the total exposure recomputed both ways so the netting benefit is a number rather than a picture. A second panel runs a member default through the waterfall step by step, letting the reader size the loss and watch it consume the defaulter's margin, then the default fund, then the CCP's own capital, then the surviving members' contributions, so mutualisation is something they see happen. A third panel makes the liquidity point: the reader holds a position, the market moves and volatility rises, and the tool shows the variation margin due today next to the increase in initial margin, which is the combination that turns a bad day into a forced unwind.

Where this goes next

Where trades are matched in the first place is exchanges, and the participants who quote the prices are market makers. The instrument whose economics are defined by daily margining is futures. The collateral machinery itself, including haircuts and eligibility, is collateral and haircuts, and the risk-management view of what is left after all of it is counterparty risk.

1.8 Liquidity

sales · trading · structuring | core | cross-asset

Liquidity is the most overloaded word in finance. Three different things are called by that name, they behave differently, and a candidate who does not separate them will answer the wrong question at some point in every interview. Separating them is most of this section.

Managing liquidity as a risk is liquidity risk; the spread itself is the bid-ask spread; who supplies liquidity and why is market makers. What is owned here is what the word means, how it is measured, and why it disappears.

> KEY IDEA
>
> **Market liquidity** is whether you can trade size quickly without moving the price. **Funding liquidity** is whether *you* can finance the positions you hold. **Corporate liquidity** is whether a company can meet its obligations as they fall due. The first is a property of a market, the second of a balance sheet, the third of an issuer. When someone asks "how liquid is it?", establish which one they mean before answering, because the evidence you would cite is different in each case.

Market liquidity has four dimensions

Quoting a single number for market liquidity is the commonest mistake, because it has at least four independent dimensions and an asset can be strong in some and weak in others.

- **Tightness**: how far apart the bid and the offer are. The cheapest to observe and the least informative, since a tight quote in tiny size tells you almost nothing.
- **Depth**: how much can trade at or near the touch. This is the dimension that matters for institutional size and the one screens understate, because a large part of the real depth is never displayed.
- **Immediacy**: how quickly a given size can be done. A bond that trades once a week is not illiquid in the sense of a wide spread, it is illiquid in the sense of time.
- **Resilience**: how fast the price recovers after a large trade. A market that gaps and stays gapped was never as liquid as the depth suggested.

Ordinary measures each capture one dimension and miss the others: the quoted spread measures tightness, displayed size at the touch measures a fraction of depth, average daily volume and turnover proxy for immediacy, and the Amihud measure, the average absolute return per unit of volume traded, is a usable proxy for price impact and therefore for depth. Resilience is the one with no cheap proxy at all: it has to be measured after the fact, from how quickly the price reverts once a large trade is done. A serious answer names the measure *and* the dimension it stands for, and says which dimension it has no measure for.

> **INTUITION**
>
> The trap in all of this is that liquidity is **conditional**. It is a description of what the market will do for you if nothing is wrong, and almost every measure of it is computed in exactly those conditions. It therefore has the unhelpful property of being highest when you do not need it and lowest when you do, which is not a coincidence but a mechanism: liquidity is provided by people with the balance sheet and the appetite to take the other side, and stress removes both at once. Anything you plan to sell in a crisis should be assessed on how it traded in the last one, not on how it trades today.

The two market senses reinforce each other

Market and funding liquidity are not independent, and the loop between them is the mechanism behind most liquidity crises. A price fall raises measured volatility, which raises haircuts and margin, which forces leveraged holders to sell, which pushes prices down further. That spiral is developed with numbers in collateral and haircuts; what matters here is the direction of causation, which runs both ways, and the consequence, which is that funding-driven selling is indistinguishable from information-driven selling while it is happening.

Liquidity is priced

Investors demand compensation for holding assets they may not be able to sell, and that compensation is the **liquidity premium**. It is why an off-the-run government bond yields more than the on-the-run bond of nearly identical maturity and identical credit: the difference is not credit, it is tradability, and it also shows up in the repo market, where the on-the-run issue goes special and its holder earns cheap funding, as repo markets sets out.

The same premium explains why private assets, small caps and long-dated structured products all offer more yield than their risk alone would justify, and why that extra yield is not free money: you are being paid to accept that you cannot leave.

REAL INTERVIEW QUESTION (Moody's · Credit)

How do you assess liquidity?

WHAT THEY TEST / ANSWER

Ask which liquidity, and then say why you asked, because on a credit desk the question is almost certainly about the *issuer* rather than about the market, and demonstrating that you know there are several is the first mark.

For a company, assess whether it can meet obligations as they fall due, and do it in three layers. *Sources*: cash on hand, undrawn committed credit facilities, and operating cash flow, with the emphasis on committed, since an uncommitted line is not liquidity. *Uses*: debt maturing in the next twelve to twenty-four months, capital expenditure that cannot be deferred, working capital swings and dividends. *Cushion*: the ratio between them, with the maturity wall as the specific thing to look for, and covenant headroom, because a covenant breach converts long-dated debt into immediate debt and turns a solvency question into a liquidity one overnight. Add the classic distinction: a company can be solvent and still fail, and most failures are liquidity events rather than solvency events.

For a security, assess tradability along the four dimensions above, and say which measures you would use for each.

For your own book, assess whether you can finance what you hold: the tenor of your funding against the tenor of your assets, how much of your funding is secured and against what collateral, and what happens to your margin requirements if prices fall.

Close on the point that ties them together and is the one a rating agency cares about: these three fail together. A stressed issuer's bonds become harder to trade at the same moment its banks reduce its lines, so an assessment that treats them as independent is understating the risk in all three.

REAL INTERVIEW QUESTION (BNP · FX)

What is FX? What is the daily volume?

WHAT THEY TEST / ANSWER

Two parts, and the second is the reason the first was asked.

What it is, in one sentence: the market in which one currency is exchanged for another, quoted as a pair, overwhelmingly over the counter rather than on exchange, traded continuously from the Asian open to the New York close, with spot, forwards and swaps as its core instruments and FX swaps rather than spot accounting for the largest share of turnover. That last clause is worth including because most candidates assume spot dominates and it does not; the instrument is developed in FX forwards and swaps.

The volume is measured every three years by the Bank for International Settlements in its triennial survey, and it runs in the trillions of dollars a day, making FX by a wide margin the largest and most liquid financial market there is. Give the current figure if you know it, and give the survey by name if you do not, because naming the source is worth more than a number you half-remember. Then note the caveats that make the number honest. The headline is reported net-net, so inter-dealer double-counting is stripped out, and an FX swap is counted once, on its long leg, so its two legs are not double-counted either. What does flatter the figure against intuition is tenor: a large part of the swap turnover is overnight or tom-next and is rolled again the following day, so the same underlying exposure is counted afresh each day. And any currency breakdown of the survey sums to 200%, because every trade has two currencies in it.

Then earn the follow-up by saying what the size implies rather than stopping at it: enormous turnover concentrated in a handful of pairs, so EUR/USD is genuinely liquid in a way that an emerging-market cross is not, and the depth in the majors is what makes FX the market where large hedges can actually be executed. Liquidity that is concentrated is still concentrated.

REAL INTERVIEW QUESTION (Barclays · Rates)

What is the point of cross-currency swaps? Are they a liquidity hedge?

WHAT THEY TEST / ANSWER

The purpose first. A cross-currency swap exchanges principal and interest in one currency for principal and interest in another. Its use is to fund in the currency you can borrow cheaply and end up with the currency you actually need, without running the FX risk, which is why an issuer sells a bond in the market with demand for its paper and swaps the proceeds back into its home currency. The pricing of the residual, the basis, is cross-currency basis.

Is it a liquidity hedge? The honest answer is: it hedges one liquidity and creates another, so say both.

It genuinely hedges *funding* liquidity in the sense the question is reaching for. A bank with dollar assets and euro deposits has a currency funding mismatch, and a cross-currency swap converts the euro funding into dollar funding for the life of the swap, removing the need to roll dollar borrowing in a market that may not be there. Term funding is the thing being bought.

But it is not a liquidity hedge without qualification, and this is where the answer is won. The swap is collateralised, so it generates margin calls, and those calls arrive in cash precisely when the currency moves against you. So a position taken to remove funding risk has introduced a new demand for liquid cash, correlated with market stress. The cross-currency basis itself widens in exactly those episodes, which means the hedge is most expensive to put on or roll at the moment you most want it. Both of those were visible in 2008 and again in March 2020.

So: a cross-currency swap hedges the currency mismatch in your funding, and converts a rollover risk into a margin risk. That is a good trade, and it is not the same thing as removing liquidity risk.

INTERVIEW TRAP

1. **Answering the wrong sense of the word.** Market, funding and corporate liquidity are three different questions. Establish which one before answering.
2. **Quoting the spread as the measure.** It captures tightness only, and a tight quote in tiny size is not liquidity.
3. **Trusting displayed depth.** Much of the real size never appears on a screen, and much of what appears will disappear before you can trade with it.
4. **Measuring liquidity in calm conditions.** Every measure is computed when nothing is wrong, which is the only time the answer does not matter.
5. **Treating the liquidity premium as free yield.** The extra return is payment for not being able to leave, and it is collected right up until the moment you need to.

VISUAL / CODE TO BUILD: Interactive

A liquidation simulator. The reader is given a position and a deadline and chooses how fast to sell; the page prices the trade-off between market impact, which grows with speed, and price risk, which grows with delay, and plots the cost of every schedule so the optimal one is visible rather than asserted. A regime switch then halves the displayed depth and doubles the spread without changing anything else, and the same schedule is re-run, so the reader sees directly that a position which was liquid on Monday's measures is not liquid on Friday's. A second panel shows the same asset's four dimensions as separate bars rather than one score, and lets the reader find assets that are tight but shallow, or deep but slow, which is the distinction a single number destroys.

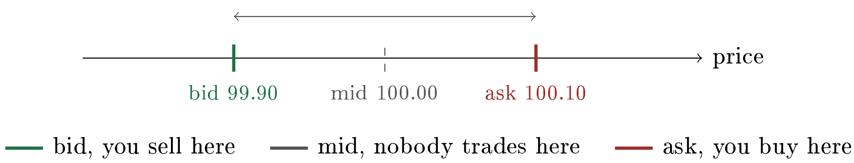
1.9 Bid-ask spread

sales · trading · structuring | core | cross-asset

There is no such thing as "the price". There are two prices: the one at which someone will buy from you and the one at which someone will sell to you, and the gap between them is the bid-ask spread. Every cost in trading, every market maker's revenue, and most of what separates a liquid market from an illiquid one lives in that gap.

Two prices, and which one is yours

The **bid** is the price at which the market maker *buys*, so it is the price at which *you sell*. The **ask**, also called the offer, is the price at which they sell, so it is the price at which you buy. The **mid** is the average of the two and is the closest thing to a fair value, but it is a number nobody trades at.



the double arrow is the spread: 0.20, or 20 basis points

The number that matters for a client is not the spread but the **half-spread**, because that is what a single trade costs you against fair value. Buying at 100.10 when the mid is 100 costs 0.10, or 10 basis points. Doing a round trip, buying and immediately selling, costs the full spread: buy 10,000 shares at 100.10 and sell them straight back at 99.90 and you are down 2,000 on a position worth about a million, having taken no view at all.

KEY IDEA

The spread is not a fee somebody chose to charge. It is the price of *immediacy*: the market maker is being paid to take the other side right now and hold the risk until they can lay it off. That is why the width tracks how risky and how slow that lay-off is likely to be, and why it widens in exactly the markets where you most want to trade.

What sets the width

Five things, and a candidate who can list them is describing the market maker's problem rather than reciting a definition.

- **Volatility.** The maker holds the position for some period before offloading it, so the wider the price could move in that window, the more they need to be paid.
- **Size.** A larger clip takes longer to unwind and moves the market while doing it, so the quote widens with size. There is a price for 100 and a different one for 100,000.
- **Adverse selection.** This is the deepest reason and the one candidates miss. If someone is willing to trade at your price, there is a chance they know something you do not, and you will lose to the informed and win from the uninformed. The spread is what makes that average out in your favour, so it widens when the proportion of informed flow rises, which is what happens around news and events.
- **Inventory.** A maker already long will show a keener bid on the side that flattens them, which shifts the whole quote rather than widening it. See below.
- **Competition and structure.** More makers quoting the same instrument compresses the spread; a minimum tick size puts a floor under it.

Skewing rather than widening

A market maker has two separate levers, and confusing them is a common interview error. **Widening** the spread charges more for immediacy on both sides, and is the response to uncertainty. **Skewing**, that is moving the mid up or down while keeping the width, is the response to inventory or to information: it makes one side of your quote more attractive to clients and the other less so, which lets you attract the flow that flattens you. How a desk sizes and automates that choice belongs to market making, which owns quote skewing in practice; what follows here is only the lever and when to reach for it.

REAL INTERVIEW QUESTION (Société Générale · EQD)

A client asks me for a price on 1,000 calls on an underlying. I quote a 20 to 22 market. Then he asks me again for the same price on another 1,000 calls. How do I react? And what if it is 15,000 calls?

WHAT THEY TEST / ANSWER

The second request is not a repeat, it is *information*, and reacting to it is the whole question. First establish the fact the question leaves implicit: ask whether he dealt, and on which side. If he lifted your offer at 22 and has come back for more, that tells you he is a buyer and that your 22 was probably too cheap. So **skew the market up**, to something like 21 to 23: you keep a similar width, you show a better bid because you are now short and would like to buy some back, and you make him pay more if he wants to keep going. Requoting 20 to 22 unchanged is the answer that fails, because it says you will let the same client run you over repeatedly without learning anything from the flow. **For 15,000** two things change at once. First, size: that is fifteen times the clip, it will take real time and market impact to hedge, so the quote widens as well as moving, and you would show something materially worse than 21 to 23. Second, and more important, ask why. A client who wanted 1,000 and now wants 15,000 may have information, or may be executing a large programme you are only seeing part of, and either way you should find out before you fill it rather than after. Say explicitly that you would check your existing position and your capacity to hedge before quoting, and that on a clip that size you might decline, or quote a price at which you are happy to be wrong, which is what a real desk does. The point being tested is whether you treat a quote as arithmetic or as a negotiation with someone who knows more than you.

The spread is what your hedge actually costs

For anyone running an option book the bid-ask stops being an abstraction, because every re-hedge crosses it. The theoretical gamma profit of theta as the cost of convexity assumes you re-hedge at the mid, and you do not.

WORKED EXAMPLE

A book with 1,000,000 of cash gamma per 1% move. The underlying moves 1%, so you re-hedge once at the end of the move, trading 1,000,000 of cash delta.

- Gamma profit: $\Gamma_{1\%}x^2/200 = 1{,}000{,}000 \times 1/200 = 5{,}000$.
- Hedging cost at a 20 basis point spread, so 10 basis points against mid: $1{,}000{,}000 \times 0.0010 = 1{,}000$.

Twenty percent of the theoretical profit went to the spread, on a single re-hedge in a liquid name. Re-hedge more often to capture the path more finely and you pay it more often; re-hedge less often and you capture less of the move. That trade-off is the real content of intraday hedging, and it is why transaction costs are not a detail in a gamma book, they are one of its main expenses.

REAL INTERVIEW QUESTION (RBC · Exo)

How do you hedge your vega on illiquid stocks?

WHAT THEY TEST / ANSWER

Start by naming the problem, which is that the hedge you want does not trade at a price worth paying. On an illiquid single name the listed option market is thin or absent, the bid-ask on what does trade is wide enough to consume the edge in the position, and the borrow needed for the delta leg may be expensive or unavailable, so hedging vega directly is often not economically possible. Then give the ladder of alternatives in order. **Proxy hedge**: trade vega in a liquid correlated instrument, the sector index or a liquid peer, and accept that you are now running basis risk between the two, which you should size and monitor rather than ignore. **Internalise**: look for offsetting risk elsewhere in the book, since the cheapest hedge is another client's opposite trade. **Restructure at source**: if this is a structured product, change the underlying, use a basket or an index rather than the illiquid name, or add a cap that reduces the vega you are taking on, which is a structuring solution rather than a trading one. **Price for it**: if you must warehouse the risk, charge for it up front, since an unhedgeable exposure is a cost the client should pay for rather than one the desk absorbs. Close on the general principle, which is what makes this a good answer: the bid-ask is not a nuisance around the theoretical price, it is a real constraint that determines what you can hedge and therefore what you should be willing to sell.

Quoted, effective and realised

Three measures are used and they differ, which is worth knowing because a market can look tighter than it is.

The **quoted spread** is the visible gap, and it applies only up to the displayed size. The **effective spread** is what you actually paid against the mid at the time you traded, which is worse than quoted if your order was larger than the top of the book and better if you were filled inside it. The **realised spread** is what the market maker actually earned once the price moved after the trade, which is smaller than the quoted spread precisely because of adverse selection: the trades that fill are disproportionately the ones that turn out to be right.

That last distinction is the quantitative version of the adverse-selection point, and it explains why market making is a genuine business rather than a toll booth: the maker collects the quoted spread on paper and keeps only the realised one.

INTERVIEW TRAP

Four errors. (1) Treating the mid as the price you will get: nobody trades at the mid, and a single trade costs the half-spread. (2) Quoting a spread without reference to size, when the whole point is that the price depends on the clip. (3) Widening when you should be skewing: uncertainty widens, inventory and information move the mid. (4) Ignoring the spread when computing the profit of a hedged position, when on a gamma book it routinely consumes a fifth of the theoretical gain.

VISUAL / CODE TO BUILD: Interactive

A quoting game with a visible cost meter. The reader is the market maker: they set a mid and a width, clients arrive with random sizes and directions, some of them informed and some not, and the reader must fill or decline. The scoreboard separates the two things that actually matter, the spread captured and the money lost to informed flow, so the reader can watch a too-tight market fill constantly and lose, and a too-wide one trade nothing at all. A second mode makes the inventory point: the reader accumulates a position and the tool shows what skewing the mid does to the arrival rate on each side, so keeping the width and moving the middle becomes a tool they use rather than a rule they read. A third panel takes a delta-hedged option position and plots realised profit against re-hedging frequency at several spread levels, showing the optimum move as the spread widens.

Where this goes next

The people quoting these prices are market makers, and the desk version of the activity is market making. The property the spread measures is liquidity, the process the trades feed is price discovery, and the machinery of order books and matching is market microstructure. What the spread does to a hedging programme is intraday hedging, and what it does to a structured book is hedging structured products.

1.10 Price discovery

sales · trading · structuring | core | cross-asset

Nobody sets the price of a share. Thousands of people with different information, different horizons and different needs each act on what they know, and out of that comes one number that everyone then treats as the truth. That process has a name, price discovery, and it is the single most useful idea in this chapter, because almost every market question a candidate is asked is really a question about it.

How a price actually moves

The mechanism is concrete, not philosophical. Someone with a reason to trade sends an order. It either sits in the book or crosses the spread and trades. The market maker on the other side, having been hit, now holds a position they did not choose and knows one new thing: somebody wanted to buy. They move their quote up, both because they are short and because that buyer might have known something. The next participant sees a higher price and updates their own view. Repeat a few thousand times a day.

KEY IDEA

A price does not move because the world changed. It moves because someone *traded* on the fact that the world changed. Price discovery is the conversion of private information into a public number, and it happens one order at a time, through the market makers who have to guess whether the flow hitting them is informed. That is why the same event can move a liquid market in seconds and an illiquid one not at all until somebody finally trades.

This also explains the beginner's most common confusion: a company reports excellent results and the share price falls. The price already contained the market's expectation. What moves a price is not the news, it is the **difference between the news and what was expected**, which is why traders talk about a number being "in the price" and why an economic release comes with a consensus forecast attached.

Prices as forecasts you can read backwards

Because a price aggregates expectations, you can run the process in reverse and extract what the market believes. This is one of the most practical skills in front office work, and every asset class has its version.

- An option's **implied volatility** is the volatility that makes the model reproduce the traded price, so it is the market's price for future movement, developed in implied versus realized volatility.
- A **forward rate** extracted from the curve is the market's expected future short rate plus a term premium, in forward rates.
- A **credit spread** is compensation for expected default losses plus a risk and liquidity premium, in credit spreads.

REAL INTERVIEW QUESTION (Société Générale · Credit)

How would you assess implied counterparty risk using only the information in the bond, with no CDS and no risk profile?

WHAT THEY TEST / ANSWER

Work from the price to the risk in four steps, and say what you are assuming at each one.

One, isolate the spread. Take the bond's yield and subtract the risk-free yield for the same maturity and currency, using the swap or government curve. What is left is the spread, and it is not pure credit: it also contains liquidity and, for some issuers, tax and regulatory effects. Say so, because pretending otherwise is the usual error.

Two, apply the credit triangle. To a good approximation the annual spread s compensates for the annual default probability p times the loss given default $(1 - R)$, so

$$s \approx p \times (1 - R), \qquad \text{hence} \qquad p \approx \frac{s}{1 - R}.$$

With a spread of 300 basis points and an assumed recovery of 40%, the implied annual default probability is $0.03/0.6 = 5\%$, and over five years that compounds to a cumulative probability of roughly $1 - e^{-0.05\times 5} = 22\%$. Flag that this is a *risk-neutral* probability, so it is larger than any statistical estimate because it includes the premium investors demand for bearing the risk, and that the recovery assumption is the weakest link since the answer scales directly with it.

Three, read the shape. One bond gives a point; several bonds from the same issuer give a curve, and its slope is informative. An upward-sloping credit curve is normal, and an *inverted* one, where short-dated paper trades at a wider spread than long-dated, is the market saying it fears a near-term event: that inversion is often the earliest signal available.

Four, read the rest of the term sheet. Seniority and security, covenants, whether the bond trades far below par, the size and currency of the issue, and whether recent trades were small and infrequent, which tells you how much of the spread is liquidity rather than credit. Close by naming the check you would do if you could: the CDS level, which is the cleaner read on the same risk, covered in credit default swaps.

Where discovery happens first

Not all venues discover equally. The price forms first where informed traders find it cheapest and quickest to act, which is usually the most liquid and most leveraged instrument on the risk, not necessarily the underlying itself. Index futures typically move before the basket of cash shares. Options can move before the underlying when the information concerns an event rather than a level, which shows up as the implied volatility and the skew moving while the spot does not. And in credit, the CDS market has often reacted before the bonds, because it is easier to sell protection than to short a bond.

The practical consequence for a junior on a desk: when two markets on the same risk disagree, the question is never which is right, it is which one the informed flow is going through.

What the price of a real thing is made of

REAL INTERVIEW QUESTION (CACIB · Commodity)

What mainly determines the price of petrol at the pump?

WHAT THEY TEST / ANSWER

Decompose it rather than answering "the oil price", because the decomposition is the answer and the first component is not the largest.

Crude oil is the raw input and the most visible driver, but it is only one part of the pump price. **Refining margin**, the crack spread, is a separate market with its own supply and demand: refinery outages, maintenance seasons and the gasoline-versus-diesel product mix move it independently of crude. **Distribution and retail margin** adds logistics and the station's own margin. **Taxes** are in most European countries the single largest component and often more than half the price at the pump, and their structure matters as much as their size: the excise duty is a fixed amount per litre, while value-added tax is proportional and is charged on the pre-tax price *plus* the excise, so a large block of the pump price does not move with crude at all.

That last point is the one to make explicitly, because it answers a question the interviewer has not asked yet: it is why the pump price moves far less than crude in percentage terms. If taxes and margins are a large fixed block, a 20% move in crude is a much smaller percentage move at the pump. Add the asymmetry that consumers notice, that pump prices rise quickly with crude and fall slowly, which is usually attributed to inventory cycles and to competitive dynamics at the retail level rather than to any single decision. Finish with the currency point: crude is priced in dollars, so for a European consumer the euro-dollar rate is a genuine second driver and one candidates almost always forget.

What information a price is discovering

REAL INTERVIEW QUESTION (Société Générale · EQD)

What information matters when pricing a company?

WHAT THEY TEST / ANSWER

Split it by what the information is used for, because a scattergun list is what most candidates give.

For the level: the cash flows the business generates and their growth, the capital structure, so debt as well as equity, the balance sheet, and the discount rate appropriate to the risk. That is valuation, and it sets a view rather than a price.

For the derivatives, which is what an equity derivatives desk actually needs: four inputs that have nothing to do with whether the company is good. The **dividend** stream, its size and its dates, because dividends drive the forward and are one of the largest hidden risks on a structured product. The **repo** or borrow cost, which sets the forward alongside dividends and can be enormous on a hard- to-borrow name. The **implied volatility surface**, by strike and maturity, since one number is not enough. And the **corporate event calendar**: earnings dates, capital increases, mergers, index inclusions, because an option spanning an earnings date is a different instrument from one that does not.

Then the sentence that shows which seat you are interviewing for: a fundamental analyst prices the company, a derivatives desk prices the *forward and the distribution around it*, and only the second is used to make a price on an option. Both are called "pricing a company" in casual conversation and they use almost disjoint sets of information.

When discovery breaks

Price discovery is a mechanism, and mechanisms fail. Three failures worth naming. When liquidity disappears there is no flow to aggregate, so the last traded price becomes stale and the instrument is marked to model rather than to market. When an instrument can only be sold and physical constraints bind, prices can reach levels that make no sense as a valuation, which is what produced a negative crude oil futures price in April 2020: holders of an expiring contract had nowhere to store the barrels and paid to get out. And when the same risk trades in two places with different participants, the two prices can disagree for longer than any arbitrage argument suggests, because arbitraging them requires capital that is itself scarce exactly when the gap opens.

INTERVIEW TRAP

Three misreadings. (1) Treating the last traded price as the value: in an illiquid instrument it may be hours old and unrepeatable, which is why marks are taken from quotes rather than from prints. (2) Expecting good news to move a price up, without asking what was expected. (3) Reading a risk-neutral probability out of a price and calling it a forecast: it includes a risk premium, so market-implied default and crash probabilities are systematically higher than statistical ones, and saying so is worth a mark on its own.

Where this goes next

The participants who do the aggregating are market makers, the conditions that make it possible are liquidity, and its cost is the bid-ask spread. The order-level mechanics are market microstructure. The three worked examples of reading a price backwards are implied versus realized volatility, forward rates and credit spreads.

1.11 Market Microstructure

sales · trading · structuring | core | equity · fx · cross-asset

Microstructure is the study of what the plumbing does to the price. Every other part of this book asks what something is worth; this section asks what you actually get when you try to trade it, and why those two numbers differ.

The boundaries are worth stating because the neighbouring sections are close. Who supplies liquidity and how they quote is market makers; what the spread is made of is the bid-ask spread; how information gets into prices at all is price discovery; whether you can trade size is liquidity. What is owned here is the *mechanism*: the book, the order types, the auctions, the venues, and what your own order reveals.

The limit order book

A limit order book is a queue of resting orders on each side. A **limit order** names a price and waits; a **market order** names a size and takes whatever the book offers. The best bid and best offer are the **touch**, and the gap between them is the spread.

> KEY IDEA
>
> Posting a limit order and hitting with a market order are not two ways of doing the same thing. A limit order **supplies** liquidity: you are paid the spread, and you are exposed to being filled only when someone wanted to trade against you, which is precisely when you were wrong. A market order **demands** liquidity: you pay the spread and get certainty. Almost every question in this section is about which side of that trade you are on.

Matching is normally by **price-time priority**: better prices execute first, and among orders at the same price, the one that arrived first does. That second half is why queue position has value and why speed is worth paying for, which is most of what latency competition is actually about.

Order types beyond the basic two exist to manage what you reveal. **Iceberg** orders display a fraction of their size and refresh, so that a large order does not announce itself. **Immediate-or-cancel** and **fill-or-kill** refuse to leave a resting remainder behind. **Pegged** orders track the touch or the mid rather than a fixed price. **Stop** orders are not orders at all until triggered, which is why they cluster and why the market moves quickly through the levels where they sit.

Auctions and fixings

Continuous trading is not the only mechanism, and it is not where most volume prints. Exchanges open and close with a **call auction**: orders accumulate without executing, and at the end a single price is chosen to maximise the volume that can trade. Auctions concentrate liquidity in one moment, which is why index funds, which must own the closing price, trade there, and why the close is by far the largest single liquidity event of the day.

That matters for anyone selling a product whose payoff references a level, because a level has to be observed somehow, and how it is observed is a design decision with real consequences.

REAL INTERVIEW QUESTION (Union Capital · FX)

How is the fixing of a structured product taken?

WHAT THEY TEST / ANSWER

Start by saying what a fixing is for, because that determines the answer: it is the observation that converts a market level into a contractual number, and the whole design problem is that whoever is hedging it has to be able to trade at, or close to, the level that gets used.

The methods, in order of how they trade off precision against manipulability.

- **The closing auction price** on the primary exchange. The single most common choice for equity underlyings, because it is the deepest moment of the day and the hedge can genuinely be executed at it.
- **An official published fix**, such as a benchmark FX rate computed over a window around a set time, or a central bank reference rate. Used where a recognised third-party number is wanted.
- **An average over a window**, whether a few minutes around the close or daily observations over weeks. Averaging is deliberately harder to move than a point observation, which is why it is used on illiquid underlyings and on the final fixing of large notes.

Then the reason it is a real question and not a definition. A point fixing on a single instant is the easiest thing in finance to manipulate, and it has been manipulated, which is why the FX benchmark window was lengthened after the fixing scandals. Averaging reduces that risk and introduces a different one: the desk hedging the note cannot trade an average, so it must execute across the whole window and accepts tracking error against the level it will be paid on.

Close on the hedging consequence, which is the part that makes it a desk answer. On a large note the hedge is a substantial order that must be executed into the same fixing that determines the payout, so the desk's own trading moves the number it is about to be measured on. That is why fixings on structured products are usually averaged rather than instantaneous, why big maturities are worked over several days, and why the fixing convention belongs in the term sheet discussion rather than in the small print. The broader hedging problem is hedging structured products.

Fragmentation and darkness

The same instrument usually trades in several places at once: the primary exchange, competing lit venues, systematic internalisers, and dark pools that publish nothing before the trade. Fragmentation makes the consolidated picture harder to see and is the reason smart order routing exists at all.

Dark venues exist for one reason: a large order that is visible moves the price against itself. Matching at the midpoint without displaying anything removes that impact. The cost is on the other side of the same coin. You give up certainty of execution, and you accept that whoever is willing to trade with you in the dark may be there because they know something, which is the adverse selection problem in its purest form.

What your order reveals

INTUITION

Every order carries information, and the person on the other side is pricing that information whether or not you meant to send it. If someone lifts your offer once, it is a trade. If they lift it three times in a row, it is a message: they know something you do not, or they have a large position to build, and either way your quote was wrong. Reading your own fills is the microstructure skill that transfers directly to a trading seat.

REAL INTERVIEW QUESTION (Société Générale · EQD)

You are asked for a bid/ask on the volatility of a tech name. It does not trade, and historical vol is 30. Your ask is hit; what do you do with your spread? The same happens a second time. And a third. So you have only sold, at an average vol of 34. Two days later you learn the accounts were falsified and the stock drops 25%: have you made or lost money? Why? What could you have done so that this was not the case?

WHAT THEY TEST / ANSWER

Take it in the three parts it is actually asking.

Did you lose money? Yes. You are short volatility at an average of 34 and the stock has just fallen 25%, so realised volatility will vastly exceed 34 and implied will gap far above it. Both the mark-to-market and the delta-hedging P&L go against you, and because you are short gamma the rehedging on the way down compounds it. Being short volatility into an accounting fraud is close to the worst version of this trade.

Why? Not because the price was wrong on the first trade, but because you did not update on the information the flow was giving you. One client lifting your offer is a trade. The same client lifting it three times, in a name with no listed volatility to check yourself against, is information: they have a view you do not have. The historical volatility of 30 was an anchor that described the past, and the buyer was pricing the future. Each fill should have made you more suspicious, not more comfortable.

What could you have done? Several things, and the good answer names more than one.

- **Raise the quote after every fill**, and by more each time. Going 32, 34, 36 is not enough when the flow is one-directional; a market maker who is being repeatedly hit on one side should move until the flow stops or reverses.
- **Skew rather than widen.** Move the offer up while leaving the bid, so you stay in the market for two-way business but stop supplying the side you are being picked off on.
- **Size discipline.** In an unlisted name with no reference volatility your maximum size should be small, and it should shrink as your position grows, not stay constant.
- **Hedge the wings.** Buy some downside from someone else, even at a poor price, so the tail you have sold is not open-ended.
- **Ask why.** On an over-the-counter trade you often know the counterparty. A client repeatedly buying volatility in a single mid-cap name is worth a phone call to the sales coverage.

The general lesson, and the one to state explicitly, is that in a market with no observable price your only information is your own flow. Quoting off historical volatility and refusing to update is not making a market, it is writing a free option to whoever is better informed.

Impact, and the algorithms built to manage it

Trading moves prices, and the move splits in two. **Temporary impact** is the concession you pay for demanding immediacy, and it decays once you stop. **Permanent impact** is the part that stays, because the market has inferred something from your trading. Empirically the cost of executing a given size grows roughly with the square root of that size relative to daily volume, which is why doubling an order does not double its cost but does not leave it unchanged either.

That trade-off is the whole reason execution algorithms exist. Trading fast pays impact; trading slowly pays price risk, since the market may move away while you wait. **VWAP** and **TWAP** schedules spread an order across a day or a window, **percentage-of-volume** adapts to what is actually trading, and implementation shortfall algorithms optimise the trade-off explicitly against a benchmark of the price when the decision was made. None of them removes the cost; they choose which form of it to pay.

INTERVIEW TRAP

1. **Treating a limit order as free.** It is a written option: you are filled exactly when the market has moved against your price.
2. **Reading the screen as the market.** Displayed depth is a fraction of real depth on some venues and a mirage on others, and fragmentation means no single screen shows the whole book.
3. **Ignoring your own fills.** One-directional flow is information, and the counterparty is not obliged to tell you twice.
4. **Assuming dark means safe.** No pre-trade impact, but the counterparty who meets you there selected themselves.
5. **Quoting a point fixing on an illiquid underlying.** It is the easiest number in finance to move, and the desk hedging it will be moving it too.
6. **Believing impact is linear.** Roughly square-root in size, so intuitions built on proportionality are wrong in both directions.

VISUAL / CODE TO BUILD: Interactive

A live order book the reader can trade against. One panel shows resting depth on both sides with the queue at each price level drawn as stacked blocks, so placing a limit order visibly joins the back of a queue and the reader watches their position advance or get overtaken as others cancel and join. Sending a market order walks the book and prints the average fill price against the touch, making slippage a thing that happens rather than a formula. A second mode runs a parent order of a stated size through VWAP, TWAP and percentage-of-volume schedules over a simulated day and reports each one's impact, timing risk and total shortfall, so the reader can see that no schedule is best in every path. A third mode reruns the same order with an informed counterparty present, so the reader experiences adverse selection as a fill pattern rather than as a definition.

1.12 Different role within the bank

sales · trading · structuring core cross-asset

"I want to work in markets" is not a job description. A trading floor contains half a dozen distinct front-office roles, each with a different skill, a different measure of success and a different day, and every interview will assume you know which one you are applying for. This section is the map, and it is worth reading before the technical chapters, because the same product looks different from each seat.

The front office

Role	Owns	Measured on
Sales	the client relationship	revenue generated from clients
Trading	the book and its risk	profit and loss, and risk discipline
Structuring	the product and its price	deals done, and their quality
Research	the view and its publication	influence and client demand
Quant, strats	the models and the tools	what the desk can price and see

Sales talk to clients, understand what they need, bring the request to the desk and negotiate the price on the client's behalf and the bank's at the same time. The skill is listening, translating and being trusted. **Trading** price the risk, take it on the book and manage it, which on most desks means market making rather than speculating, as market making sets out. **Structuring** sits between the two on anything non-standard: they design the payoff, price it, write the term sheet and make sure the thing can actually be hedged, which is payoff engineering. **Research** produces the house view. **Quants and strats** build the pricing models, the risk tools and the automation everyone else uses, and on a modern floor the boundary between a strat and a trader is thinner every year.

> **KEY IDEA**
>
> The three client-facing roles are distinguished by *what they are optimising*. Sales optimises the relationship, and is judged over years. Trading optimises the book, and is judged every evening. Structuring optimises the deal, and is judged on whether it printed and whether it can be hedged. Most floor arguments are two of those three objectives disagreeing, and understanding that is more useful than any org chart.

The other axis: which desk

Roles are one axis; the desk is the other. A floor is cut by asset class and then by product, and the same role means different work on each.

- **Cash equities**: shares, execution, agency business.
- **Delta One**: linear equity exposure without options, so futures, swaps, certificates, exchange-traded funds and the financing and borrow that go with them.
- **Flow derivatives**: listed and vanilla options in size, largely market making.
- **Exotics**: structured and path-dependent risk, the smallest and most technical seats.
- **Rates, credit, FX, commodities**: the same three roles again, with different instruments and a different client base.

REAL INTERVIEW QUESTION (Société Générale · Delta One)

What are the activities of a Delta One desk?

WHAT THEY TEST / ANSWER

Define the name first, because it explains the whole desk: Delta One means a delta of one, that is products whose value moves one for one with the underlying and which carry no optionality. So the desk covers **futures and index arbitrage**, **total return swaps** and other synthetic exposure, **exchange-traded funds**, including creating and redeeming units, **certificates and trackers**, and the **stock borrow and financing** that all of it depends on. Then say what unites them, which is the part that shows understanding rather than recall: none of these has meaningful gamma or vega, so the desk's risk is not about volatility, it is about *basis, financing and dividends*, that is the small differences between two instruments that should be equivalent and the cost of carrying one against the other. Add that this makes Delta One a balance-sheet and financing business as much as a trading one, which is why it sits close to the equity finance and securities-lending functions, and why its clients are heavily institutional. See ETFs and index products.

REAL INTERVIEW QUESTION (Société Générale · Delta One)

How does a Delta One desk make money?

WHAT THEY TEST / ANSWER

List the revenue lines, because there are several and they are not obvious from the products. **Bid-offer on flow**: clients pay a spread to get synthetic exposure they cannot or do not want to hold physically. **Financing spread**: the desk funds a physical position and charges the client a rate on the synthetic, and the gap is revenue, which is why this is a balance-sheet business. **Stock lending**: holding the physical hedge lets the desk lend the shares out and keep part of the borrow fee, which on hard-to-borrow names is material. **Dividend and basis**: the desk takes a view on, or is simply better informed about, expected dividends and the basis between a future and its underlying, and both feed the price it quotes. **Index events**: rebalances and corporate actions create predictable flow the desk is positioned to service. Then give the risk side, since a good answer never lists revenue alone: the desk is short the risk that its financing assumptions are wrong, that a dividend is cut, that a borrow becomes expensive or is recalled, and that a basis it treated as stable moves. Those are small numbers most of the time and occasionally very large ones, which is the characteristic shape of a Delta One profit and loss.

Beyond the front office

The floor is supported by functions a junior will interact with constantly and should be able to name.

- **Market risk** sets and monitors limits and independently measures the book's exposure. See market risk.
- **Product control** independently values the book and explains the daily profit and loss, which is the function that will challenge a trader's marks.
- **Credit risk** approves counterparties and sets exposure limits.
- **Operations** confirms, settles and reconciles every trade.

- **Compliance and legal** own the documentation, the conduct rules and the suitability framework.
- **Technology** runs the systems the whole chain sits on.

Calling these “back office” dismissively is a small but real interview error: a trader who cannot work with product control and operations is a liability, and juniors are assessed on exactly that cooperation.

The life of a trade

The single most useful thing to be able to describe, because it shows you understand how the roles connect rather than what they are called.

1. A client calls **sales** with a need, not a product.
2. Sales brings it to **structuring**, who turn the need into a payoff and ask trading what it costs.
3. **Trading** prices the risk, checks whether it can be hedged and at what cost, and returns an internal price.
4. Structuring adds the margin and sales quotes the client, then negotiates.
5. If it trades, **operations** confirm and settle it, **product control** values it independently, and **market risk** sees it appear in the limits.
6. **Trading** lives with the risk until maturity, and **sales** lives with the client for longer than that.

The last line is the one worth internalising: the trade ends and the relationship does not, which is why sales and trading sometimes want different prices on the same deal and why both positions are legitimate.

REAL INTERVIEW QUESTION (BNP · Fixed Income)

Tell us about your previous internship and a typical day.

WHAT THEY TEST / ANSWER

Structure the day around the desk’s rhythm rather than around your tasks, because that is what demonstrates you understood where you were sitting. **Before the open**: what you checked, the overnight moves, the economic calendar, the positions and reports that had to be right before anyone traded. **The open and the morning**: the flow, the requests you helped with, what the desk was watching. **The afternoon and the close**: marks, the profit and loss explanation, the work that only happens once the market is quiet. **The recurring ownership**: the one or two things that were genuinely yours, which is the part interviewers listen for, since a junior who owned a daily report end to end is more credible than one who helped with everything. Two things distinguish a strong answer. Be specific about products and numbers rather than saying “I supported the desk”, since specifics are the only evidence that you were there. And say what you learned about the *job* rather than about the market, for instance that the hardest part was not the pricing but getting a clean answer to a client quickly, which tells the interviewer you were paying attention to the work rather than to the theory. What the answer is really being scored on is covered in behavioral signals desks look for.

Choosing

There is no correct answer, but there is a wrong way to choose, which is by prestige. The honest questions are about the daily work: do you want to spend the day talking to people or looking at a screen; do you want to be measured tonight or over a year; are you happier owning a number or owning a relationship; do you enjoy building things that others use.

Two practical notes. Titles differ between banks, so "structurer" at one firm is a "sales-structurer" or a "strat" at another, and it is worth asking what the role actually does rather than assuming. And the boundaries are porous early in a career: juniors frequently move between structuring and trading, or between strats and trading, so the first seat is a starting point rather than a life sentence.

INTERVIEW TRAP

Four errors. (1) Saying "sales and trading" as though it were one job: they are different jobs with different skills, and an interviewer will ask which. (2) Describing trading as taking directional views, when most bank trading is market making and risk management. (3) Dismissing the control functions, which juniors are explicitly assessed on working with. (4) Choosing a desk without knowing what it trades, which is exposed instantly by one question about its products.

VISUAL / CODE TO BUILD: Figure

An interactive floor plan. The reader sees a trading floor laid out by desk, clicks any seat and gets what that role owns, what it is measured on, who it talks to and a sample hour of its day. A second mode animates the life of a trade: the reader picks a client request and watches it travel through sales, structuring, trading, operations, product control and risk, with each stop showing what was added and what could go wrong there. A filter by track, sales, trading or structuring, greys out the seats that are not on that path, so a reader can see their own possible careers rather than the whole floor at once.

Where this goes next

Which side of the market all of these sit on is buy-side versus sell-side. What the trading role actually involves day to day is market making and flow trading; the structuring role is payoff engineering; and the risk functions are market risk. How to talk about the role you are applying for is behavioral signals desks look for.

Chapter 2

Useful Market Culture

2.1 Geopolitics for Markets

sales · trading · structuring core cross-asset

Candidates lose this topic by knowing more about it than the interviewer and still failing the question. Being able to narrate a conflict is not the skill; the skill is saying what it does to a price and through which mechanism. This section is a translation layer, and it is deliberately short on history and long on transmission.

KEY IDEA

A geopolitical event reaches a price through exactly three channels, and almost every good answer names which one is doing the work.

1. **Supply.** Something physical becomes scarcer or dearer: energy, food, semiconductors, shipping lanes. This is the channel that moves commodities and, through them, inflation.
2. **Policy.** The response of central banks and governments, which is frequently larger than the event. A shock matters to rates through what it makes the central bank do, not directly.
3. **Risk premium.** Uncertainty rises, so investors demand more to hold risk. Equities de-rate, credit spreads widen, volatility rises, and capital moves to whatever is currently considered safe.

An event with no route through any of the three is news, not a market event, and saying so is a better answer than inventing a link.

The transmission map

Event	First order	Second order
Conflict in an energy region	oil and gas up	inflation up, policy tighter
Trade war, tariffs	import costs up	margins down, growth down
Sanctions on a large economy	that currency down	counterparty and settlement risk
Election with policy shift	that country's rates	currency, then sectors
Sovereign stress	that credit spread	contagion to peers
Shipping route disrupted	freight up	regional price spreads widen

Two habits make this usable. Always say which *leg* of the map you are on, because most people stop at the first column and the interesting answer is in the second. And always ask whether the shock is inflationary or deflationary, because that single question determines whether the central bank cushions it or compounds it, and therefore whether bonds rally.

Flight to quality, and the condition under which it works

The reflex answer to any geopolitical shock is that money moves to safety: the dollar, the yen, the Swiss franc, gold and government bonds. That is usually right and it is right for a reason worth stating, which is also the reason it sometimes fails.

INTUITION

Government bonds are a haven *conditional on the shock being disinflationary.* A shock that destroys demand pushes the central bank toward cutting, so bonds rally and the reflex holds. A shock that destroys *supply* pushes inflation up, so the central bank tightens into the shock and bonds sell off at the same time as equities. In that world there is no haven in duration at all, which is why 2022 surprised so many portfolios built on the assumption that bonds hedge equities.

The general lesson is the one to carry into any answer: the equity-bond correlation is not a constant, it is a function of whether the dominant shock is to demand or to supply.

Currencies follow their own logic. The dollar strengthens in most global risk events regardless of the source, because it is the funding and reserve currency and a stress event is a scramble for dollars. The franc and yen strengthen partly for haven reasons and partly because they fund carry trades that get unwound. A currency with a peg does something else entirely, which is why the peg gets tested rather than the rate moving.

REAL INTERVIEW QUESTION (Société Générale · FX)

Geopolitical news and its influence on FX.

WHAT THEY TEST / ANSWER

Give a framework, then one live example, then the caveat. The framework is what is being marked.

The framework: three channels into a currency.

Terms of trade. A commodity shock helps the currency of an exporter and hurts an importer, mechanically, through the trade balance. This is the cleanest channel and the one to lead with: an oil shock is bullish NOK and CAD and bearish JPY and INR, and you can say that without knowing anything about the politics.

Policy divergence. What the shock does to the two central banks involved. FX is a relative price, so what matters is not whether a shock is inflationary but whether it is *more* inflationary for one of the two, because the rate differential is what moves the pair.

Risk premium and safe-haven flow. Dollar strength in almost any global stress event, because a dollar shortage is what a stress event produces; franc and yen strength partly as havens and partly through carry unwinds.

Then be honest about the hierarchy, which is what distinguishes a real answer. Over days, risk premium and positioning dominate and the moves are large and noisy. Over months, policy divergence dominates. Over years, terms of trade and relative inflation dominate. A geopolitical headline usually moves a currency for a week and is then overtaken by whichever central bank reacts, so if you are asked what a specific event will do, ask what horizon they mean.

Close on positioning, since it is the part that explains the moves that look wrong. A well-anticipated event moves the price on the announcement of the *outcome*, not on the event, and if positioning is already one-sided the move on confirmation can be the opposite of what the news implies. That is why FX often sells the fact, and it is worth saying because it shows you understand that price reflects expectations rather than facts.

Tariffs: the case where the arithmetic is unintuitive

REAL INTERVIEW QUESTION (Bank of America · Rates)

What is the impact of customs tariffs on the equity market?

WHAT THEY TEST / ANSWER

The instinct is to reason proportionally, that a 25% tariff on part of the cost base is a moderate problem. Operating leverage says otherwise, and showing that is the answer.

Work an example. A company has revenue 100, cost of goods 60 of which 40 is imported, and other operating costs 25, so operating profit is **15**. A 25% tariff on the imported input adds $0.25 \times 40 = \mathbf{10}$ of cost.

Pass-through to prices	Operating profit	Change
none	5	−66.7%
half	10	−33.3%
full	15	0%

So a 25% tariff on 40% of revenue is a two-thirds cut in earnings if none of it can be passed on. That is the point: equities are claims on profit, not on revenue, so a cost shock is amplified by the ratio of revenue to profit. Here that ratio is nearly seven, which is why equity reactions to tariff announcements look wildly out of proportion to the tariff rate and are not.

Then say what determines where on that table a company sits, which is pricing power. A company with a differentiated product passes the cost on and the row is closer to the bottom; a commoditised producer facing untariffed competitors cannot pass on anything and sits at the top. So the correct market response to a tariff is not to sell the index, it is to separate the two, and the dispersion between them is usually the better trade than the direction.

Finish with the second-order effects, briefly, because they are what a rates desk asked this question actually cares about: tariffs raise consumer prices, which is inflationary and pushes the central bank toward tightening; they invite retaliation, which hits exporters; and the uncertainty itself defers corporate investment, which is a growth drag independent of the tariff. The net effect on a bond market therefore depends on whether the inflation impulse or the growth drag dominates, and that is genuinely uncertain rather than a thing to assert.

REAL INTERVIEW QUESTION (HSBC · desk not recorded)

What are the effects of a trade war on the economy?

WHAT THEY TEST / ANSWER

Structure it as who loses and how, rather than as a list, and be honest that the aggregate effect is negative while the distribution is what matters.

Direct. Tariffs raise input costs for importers and reduce demand for exporters. Both economies lose output relative to free trade; this is the least interesting part and the part everyone says.

Through prices. It is an **inflationary supply shock**, which is the awkward kind for a central bank: prices up and growth down at the same time, so cutting rates to support growth worsens inflation and raising them to fight inflation worsens growth. Naming that as a supply shock rather than a demand shock is the single most valuable sentence in this answer, because it explains why the policy response is contested rather than automatic.

Through uncertainty. This is frequently larger than the tariffs themselves. A firm that does not know what its supply chain will cost in two years defers the investment, and deferred investment is a real reduction in growth caused by the *announcement* rather than by the measure.

Through supply chains. Reconfiguration is slow, expensive and mostly permanent, and it does not reverse when the tariff does, which is why the effects outlive the policy.

Then the distribution, which is where the trades are. Domestically focused producers with pricing power gain relative to import-dependent ones; the currency of a country running a trade surplus tends to weaken as its export demand falls; and commodity exporters are affected mainly through global growth rather than through the tariffs directly. An aggregate answer of "everyone loses" is true and useless; naming the relative winners is the answer a desk wants.

REAL INTERVIEW QUESTION (CACIB · desk not recorded)

Give me three events in 2016 that impacted our business and their impacts on FX and rates (for example Brexit, Trump, the second Fed hike).

WHAT THEY TEST / ANSWER

The examples are supplied, so the marks are entirely in the transmission and in the pattern you draw from them. Take them one at a time, then say what they have in common.

Brexit, June 2016. Sterling fell sharply and did not recover, which is the terms-of-trade and growth-expectation channel repricing a country's future. Gilts rallied and the Bank of England cut and restarted asset purchases, so this was a *deflationary* shock for the UK in market terms even though the currency fall was inflationary for import prices, and the Bank chose to look through that. Volatility spiked before the vote and collapsed after it, which is the cleanest event-volatility pattern in recent memory.

The US election, November 2016. The instructive part is that the market moved *opposite* to the pre-positioning. The consensus was that the outcome would be risk-negative; equities rallied, the dollar rallied and Treasury yields rose sharply on expectations of fiscal expansion and deregulation. The lesson is the one worth stating: markets price policy expectations, not political preferences, and an event whose direction everyone agrees on is usually already in the price.

The December Fed hike. The second in a decade, and it mattered less for the 25 basis points than for confirming that the tightening cycle was real. Front-end yields rose, the dollar strengthened, and emerging-market currencies with dollar debt suffered, which is the standard transmission from US policy into the rest of the world.

Then the pattern, which is what turns three anecdotes into an answer. In all three, the durable move came from the **policy expectation** the event created rather than from the event itself, and in two of the three the direction surprised a consensus that had positioned for the opposite. That is the general shape of a geopolitical event in markets, and it is why the useful preparation is a transmission map rather than a view.

INTERVIEW TRAP

1. **Narrating the event instead of the transmission.** Knowing the history is not the question. Which channel, which asset, which direction, and over what horizon.
2. **Assuming bonds always rally.** They rally on demand shocks and sell off on inflationary supply shocks, and the equity-bond correlation flips with the type of shock.
3. **Forgetting that expectations are already priced.** An event everyone anticipates moves the market on the resolution, and often in the direction opposite to the news if positioning is one-sided.
4. **Reasoning proportionally about tariffs.** Operating leverage turns a cost shock into a much larger earnings shock, and the arithmetic above is worth being able to produce.
5. **Applying the last playbook.** Each shock differs in whether it hits supply or demand, and the previous crisis's correlations are the least reliable guide precisely when they are most confidently cited.
6. **Having a political opinion in an interview.** You are being asked what a market does, not what should happen. Answer in transmission and stop.

VISUAL / CODE TO BUILD: Interactive

A transmission-map explorer. The reader picks a shock from a list, an energy supply disruption, a tariff round, sanctions, a sovereign stress event, and the panel animates it flowing along arrows through the three channels into a set of asset boxes, each shaded by expected direction and thickness of arrow by expected magnitude. The value is that the second-order boxes light up a beat after the first-order ones, which is the habit the section is trying to build.

A single switch labelled "supply or demand" flips the shock's character and redraws the bond box from green to red, which makes the flight-to-quality condition visible in one gesture rather than as a caveat.

A second panel is the tariff calculator: sliders for import share of costs, tariff rate and pass-through, with a waterfall from revenue to operating profit before and after. Setting import share to 40%, tariff to 25% and pass-through to zero reproduces the two-thirds earnings fall in the section, and dragging pass-through to one shows it vanish, which is pricing power made quantitative.

Where this goes next

The supply channel in detail, meaning which conflicts move which barrel, is energy and global conflicts, with the market mechanics in energy markets. The policy channel is monetary policy, and the supply-versus-demand distinction that decides whether the central bank cushions or compounds a shock is inflation. When the currency in question cannot move, the shock goes into the reserves instead, in pegged currencies and FX crises. The scheduled events that carry most of this risk are macro calendars, the discipline of asking what an unscheduled one would do to a book is stress testing, and the interview form these questions actually take is market questions.

2.2 Energy and global conflicts

sales · trading · structuring core commodities · cross-asset

Energy is the channel through which geopolitics reaches a trading floor. A coup in a country that exports nothing moves nothing; a coup in a country that exports oil, or that sits beside a shipping lane, moves inflation expectations, then rates, then every asset class. That is why "what is going on in the world" is a real interview question rather than small talk, and why the good answer is a causal chain rather than an opinion. The general framework for reading geopolitics is in geopolitics for markets; the market mechanics of the products themselves are in energy markets. This section is the link between the two.

Why energy shocks are violent: inelasticity

INTUITION

If the price of restaurant meals doubles, you cook at home. If the price of petrol doubles you still drive to work, because the alternatives (move house, change job, buy a different car) take years. Energy demand is **inelastic in the short run**, and that single property is the reason a small physical shortage produces an enormous price move: since almost nobody can stop consuming, the price has to rise until someone is financially forced to.

Put a number on it, because this is what makes the answer rigorous. Estimates vary, but the short-run price elasticity of oil demand is consistently found to be small, on the order of 0.05 in absolute value. Take that as a working figure:

$$\frac{\Delta q}{q} = -0.05 \times \frac{\Delta p}{p} \quad\Longrightarrow\quad \frac{\Delta p}{p} = \frac{1}{0.05} \times \left|\frac{\Delta q}{q}\right| = 20 \times \left|\frac{\Delta q}{q}\right|.$$

So losing 1% of global supply requires roughly a 20% price rise to clear the market. That arithmetic explains everything that follows: why a single pipeline, refinery or strait can move the oil price by double digits, why OPEC's marginal barrels have such disproportionate power, and why energy is the asset class where geopolitical risk is priced most violently.

The other half of the same coin is that supply is inelastic too. A new field takes years, a refinery takes longer, and the only genuinely fast supply lever is **spare capacity**, which is concentrated in a handful of Gulf producers. When spare capacity is high the market absorbs disruptions with a shrug; when it is thin the same headline moves the price ten dollars. Knowing whether spare capacity is currently ample or thin is more useful than knowing the price.

Chokepoints: geography as a risk factor

Energy is produced where the geology is and consumed where the people are, so it has to move, and it moves through a small number of narrow places. Learn these, because they come up constantly.

- **Strait of Hormuz.** The exit from the Gulf, carrying roughly a fifth of the world's oil consumption plus a large share of seaborne LNG. There is no adequate bypass, so it is the single most consequential piece of water in commodity markets.
- **Bab el-Mandeb and Suez.** The Red Sea route between Asia and Europe. When it is unsafe, ships reroute around the Cape of Good Hope, which adds roughly ten days and a corresponding amount of freight cost and tied-up inventory, as happened from late 2023.
- **Strait of Malacca.** The artery for Middle East crude going to East Asia, and the reason Asian energy security is a naval question.

- **Turkish Straits** for Black Sea crude and grain, and the **Panama Canal**, whose transit capacity depends on rainfall, which is a useful reminder that climate is also a chokepoint risk.
- **Pipelines**, which are chokepoints that cannot be rerouted at all. Nord Stream is the definitive example: when flows stopped in 2022 there was no alternative delivery of that gas to Europe by any means other than buying LNG on the world market at whatever it cost.

> KEY IDEA
>
> A disruption's market impact is not proportional to the volume lost, it is proportional to the volume lost *divided by the ease of replacing it.* A million barrels a day removed from a well-supplied market with spare capacity and open sea lanes is noise. The same million barrels removed when inventories are low, spare capacity is thin and the alternative route is closed is a crisis. This is why traders watch inventories, spare capacity and freight rates rather than headlines.

The transmission chain to a trading floor

This is the answer to almost every energy-geopolitics question, and it is worth memorising as a sequence.

1. **Energy prices rise**, in the front of the curve first, so the curve backwardates, as in contango and backwardation.
2. **Headline inflation rises** directly through fuel and utility bills, then more slowly through everything that is transported or manufactured, see inflation.
3. **Central banks face a supply shock**, which is the hardest case for them: output falls and prices rise at the same time, so there is no interest rate that fixes both. Whether they tighten depends on whether inflation expectations are moving, see monetary policy.
4. **Rates rise, growth expectations fall**, so equities fall while the equity-rates correlation turns *positive*, which is the inflation-driven regime discussed in market color.
5. **Cross-sectional effects** dominate the index move: energy producers and energy-exporting currencies gain (NOK, CAD, and historically RUB), energy importers lose (JPY, EUR, INR), energy-intensive industrials and airlines suffer, and credit spreads widen for companies whose input costs cannot be passed on.
6. **Second-round effects**, which are where the real damage often is: natural gas is the feedstock for ammonia and therefore the dominant cost in nitrogen fertiliser, so a gas shock becomes a food shock a season later, and food inflation is historically the most politically destabilising kind in low-income countries. Gas also sets the marginal cost of electricity in many markets, so a gas shock is a power shock immediately, see spark spreads.

The episodes worth knowing, and what each one teaches

Do not memorise dates. Memorise the mechanism each episode demonstrates.

- **1973, the OAPEC embargo.** Prices roughly quadrupled. Lesson: a coordinated supply cut against inelastic demand, and the birth of energy security as a policy field, including the creation of strategic petroleum reserves and the IEA.
- **1979, the Iranian revolution.** Prices roughly doubled on a loss of supply that was, in volume terms, modest. Lesson: it is inventory behaviour and precautionary buying, not the barrels lost, that sets the price.
- **1990 to 1991, Iraq's invasion of Kuwait.** Two producers removed at once, prices roughly doubled, then fell sharply once the military outcome was clear and reserves were released. Kuwaiti wells were set alight in retreat and burned for months, and Iraqi exports stayed under

sanction for years afterwards. Lesson: war premia are priced on expectations and can deflate faster than the physical damage is repaired.

- **2014 to 2016, the shale price war.** OPEC declined to cut in the face of rising US shale output, and prices fell from around \$110 to under \$30. Lesson: supply shocks are not always upward, and a cartel's power depends on whether a higher-cost competitor can grow.
- **2020, Covid plus the OPEC+ dispute.** Demand collapsed while supply briefly rose, storage filled, and the expiring WTI contract settled below zero, told in full in physical vs derivative commodities. Lesson: the binding constraint can be physical storage rather than production.
- **2022, Russia's invasion of Ukraine.** European gas at TTF peaked above €300 per MWh, more than ten times its pre-2021 norm, as pipeline flows stopped and Europe had to buy LNG in competition with Asia. The response reshaped the market: LNG import terminals, demand destruction in energy-intensive industry, a G7 price cap on Russian crude, and a durable rerouting of trade flows. Lesson: when a pipeline is the chokepoint, the substitute is a globally-priced cargo, so a regional political event becomes a global price event.
- **From late 2023, the Red Sea.** Attacks on shipping pushed Asia-Europe traffic around the Cape. Lesson: a disruption can raise *freight and time* without removing a single barrel, and that is still a real cost, see shipping and freight.

REAL INTERVIEW QUESTION (CACIB · Commodity)

Is the price of oil elastic?

WHAT THEY TEST / ANSWER

Distinguish the two elasticities and give numbers, because the question is imprecise on purpose. **Demand** is highly *inelastic* in the short run, with estimates of the price elasticity typically well below 0.1 in absolute value, because consumption is tied to a stock of vehicles, aircraft, heating systems and industrial plant that cannot change quickly. It is considerably more elastic over a horizon of years, through efficiency, substitution and electrification. **Supply** is also inelastic in the short run, since new production takes years, with one exception that matters enormously, spare capacity, which is concentrated in a few Gulf producers and can be brought on in weeks. Then deliver the consequence, which is the actual point of the question: with a demand elasticity of about 0.05, clearing a 1% loss of supply requires roughly a 20% price move, so oil prices are structurally volatile and geopolitical risk is priced sharply. Finish with the two refinements that show real understanding: shale shortened the supply response from years to months and therefore damped the upper tail, and elasticity is asymmetric, since demand destruction at high prices is real while low prices do not produce a symmetric surge in consumption.

REAL INTERVIEW QUESTION (CACIB · Commodity)

What is the impact of the oil price on the economy?

WHAT THEY TEST / ANSWER

Answer with the transmission chain, and separate importers from exporters, since the question has no single answer. For an oil-*importing* economy a price rise is a **negative supply shock**: it is a transfer of income abroad, it raises headline inflation directly and core inflation with a lag through transport and manufacturing costs, it reduces real disposable income and therefore consumption, it worsens the current account, and it confronts the central bank with the one situation monetary policy cannot fix, since output is falling while prices rise. For an oil-*exporting* economy the same move is a terms-of-trade gain, strengthening the currency and the fiscal position, and creating the familiar Dutch-disease and sovereign-wealth-fund questions. Then add the three things that distinguish a good answer. The **cause** matters: a price rise driven by strong global demand is a very different signal from one driven by a supply disruption, and only the second is stagflationary. The **magnitude** is smaller than it used to be, because developed economies use meaningfully less oil per unit of GDP than in the 1970s. And the **second-round effects**, gas into fertiliser into food, and gas into power prices, are often where the political and credit stress actually shows up.

REAL INTERVIEW QUESTION (BP · Commodity)

What was the problem encountered by Middle East oil in the 1990s?

WHAT THEY TEST / ANSWER

They are checking whether you know the industry's own history. The defining event is **Iraq's invasion of Kuwait in August 1990** and the war that followed. Give the chain: two significant producers were removed from the market at once, prices roughly doubled on the invasion, and then fell back sharply once the coalition's military success became clear and consuming countries released strategic reserves, so the war premium deflated well before the physical situation was repaired. The physical damage was severe and lasting, with Kuwaiti wellheads set alight during the retreat and burning for months, requiring an unprecedented firefighting effort, and Iraqi exports were then constrained for years by United Nations sanctions and later channelled through the oil-for-food arrangement. Then draw out the lessons an oil company interviewer wants: geopolitical risk in this region is a permanent part of the cost of capital rather than an event, war premia are priced on expected outcomes and can disappear faster than production returns, strategic reserves exist precisely for this and were used, and the episode entrenched the security-of-supply logic that still drives diversification of both sources and routes. Noting that the 1990s ended with the opposite problem, a price collapse around 1998 on weak Asian demand and oversupply, shows you understand the decade rather than one headline.

REAL INTERVIEW QUESTION (SCB Group · Commodity)

Is the United States a net exporter or a net importer of oil?

WHAT THEY TEST / ANSWER

The answer needs one distinction, and giving it is the whole test. On **total petroleum**, meaning crude oil plus refined products and other liquids, the United States has been a **net exporter** since around 2020, a reversal produced by the shale revolution. On **crude oil alone** it remains a net *importer*, and that is not a contradiction: US refineries are configured for heavier and sourer grades than domestic light sweet shale crude, so the country imports heavy crude, largely from Canada, and exports both light crude and large volumes of refined products. Say that clearly, then add the market consequences, which is why a commodity desk asks. The export ban was lifted in 2015, which connected US crude to seaborne markets and structurally changed the WTI-Brent relationship. Being a net exporter changes the country's macro sensitivity to oil, since a price rise is no longer straightforwardly negative for the US as it is for Europe or Japan. And the grade mismatch means US refining margins depend on the spread between crude qualities rather than only on the flat price. Finish by flagging that this is a live statistic whose sign has changed within recent memory, so you would quote it with an as-of date.

REAL INTERVIEW QUESTION (HSBC · desk not recorded)

What is the impact of a low oil price?

WHAT THEY TEST / ANSWER

Resist the reflex that cheap energy is simply good, because the interesting part is the asymmetry. The **positives** are real: for importing economies it is a transfer of income from producers to consumers, raising real incomes and margins, and lowering headline inflation, which gives central banks room to be easier. The **negatives** are the part candidates miss. Energy capital expenditure collapses, and it is capital-intensive and regionally concentrated, so the investment and employment loss is immediate and geographically painful while the consumer gain is diffuse and slow to be spent. High-yield energy credit deteriorates sharply, and in 2015 and 2016 that was large enough to move whole credit indices. Oil-exporting sovereigns come under fiscal and currency stress, which becomes an emerging-market and geopolitical problem. And persistently low headline inflation can un-anchor inflation expectations, which central banks fear more than a one-off price fall. Then close with the two framing points that make it a strong answer: the **cause** matters, since a fall driven by weak demand is a bad signal about growth while one driven by extra supply is genuinely stimulative, and low prices sow the next upcycle by cutting the investment that future supply depends on, which is the mechanism behind the commodity cycle.

INTERVIEW TRAP

Four traps. First, **answering with opinions rather than mechanisms**: an interviewer wants the chain from event to price to rates to your book, not your view on a war, and political advocacy on a trading floor is a liability rather than a personality. Second, **scaling impact by volume lost** and ignoring inventories, spare capacity and routes, which is what actually determines whether a disruption matters. Third, **treating cheap energy as unambiguously good**, which misses the capex, credit and sovereign channels. Fourth, **quoting a stale statistic**, in a domain where the US net trade position, OPEC's spare capacity and the direction of flows have all reversed within a decade: always give an as-of date.

VISUAL / CODE TO BUILD: Interactive

A transmission-chain explorer. A world map shows producing regions and chokepoints with current flow volumes; the reader clicks a chokepoint or a producer and sets a disruption size and duration. The tool then walks the chain visibly, one panel at a time: barrels or cubic metres removed, replacement capacity available from spare capacity and inventories, the implied price move using an editable demand-elasticity slider so the reader can see the 20× multiplier appear from a 0.05 elasticity, then headline inflation, then a policy-rate and equity response, then a table of winners and losers by sector and currency. A second tab is a historical replay of the six episodes above, each showing the actual price path against what the elasticity arithmetic alone would have predicted, so the reader sees how much of a move is physics and how much is expectations.

Where this goes next

The general framework for turning political events into market views is geopolitics for markets, and the calendar of events and data that carry these shocks into prices is macro calendars. The macro machinery invoked above is inflation and monetary policy. The market mechanics of the products themselves are energy markets, physical vs derivative commodities and shipping and freight, and how to talk about all of it to a client is market color.

2.3 Major central banks

sales · trading · structuring | core | rates · cross-asset

Every price in this book discounts at a rate, and a committee sets the anchor for that rate. That is reason enough to know how these institutions work, but there is a more practical one: no topic is asked more reliably in a markets interview, on every desk and not only on rates, and it is the cheapest possible question to fail.

Who they are and what they are told to do

REAL INTERVIEW QUESTION (CACIB · Rates)

What is the ECB's mandate? The Bank of Japan's? The Fed's?

WHAT THEY TEST / ANSWER

The difference between them is the point, so lead with it: the Fed has two objectives and the others have one with something attached.

The Federal Reserve has a **dual mandate** written into its statute: maximum employment and stable prices, alongside moderate long-term interest rates. It targets 2% inflation on the personal consumption expenditures measure. Be careful with the framework, because it has changed and a dated answer is noticed: the 2020 statement adopted flexible *average* inflation targeting, under which the Fed would tolerate an overshoot after a period of undershooting, and the 2025 framework review removed that makeup strategy and returned to plain flexible inflation targeting. The 2% goal itself was never in question. The dual mandate is why the Fed reacts to the labour market in a way the ECB formally does not.

The ECB has a **primary objective of price stability**, defined as 2% inflation over the medium term on the harmonised index, symmetric so that overshoots and undershoots are equally undesirable. Only *without prejudice* to that may it support the general economic policies of the Union, which is a genuine hierarchy rather than a formality and reflects the institution's founding design.

The Bank of Japan targets 2% price stability, and the relevant context is that it spent decades fighting deflation rather than inflation, which is why its toolkit went furthest: negative rates and yield curve control, from which it has more recently been normalising. Say that you would check the current setting, because this is the one that has changed most.

Add **the Bank of England** unprompted, since it is usually the next question: a 2% CPI target, with a secondary objective to support growth and employment, and a formal requirement to write a public letter to the Chancellor when inflation misses by more than one percentage point in either direction, which is a transparency mechanism the others do not have.

Close on the practical consequence, which is what makes it a markets answer rather than a civics one: because the mandates differ, **the same data releases matter differently**. A strong US payrolls number moves Fed expectations directly; a strong euro area employment number moves ECB expectations only through its effect on the inflation outlook. That asymmetry is tradable and it is why the two curves do not move together.

The rate they actually set

A central bank does not set the rate on your mortgage, it sets a very short-term rate and lets the market do the rest.

- **The Fed** sets a **target range** for the federal funds rate and steers the effective rate inside it with administered rates: interest on reserve balances at the top and the overnight reverse repo facility at the bottom.
- **The ECB** sets **three** rates: the deposit facility rate, the main refinancing operations rate, and the marginal lending facility rate. The three form a corridor, and the one that matters is the **deposit rate**, since with abundant excess liquidity that is where money market rates settle. This is the detail that separates candidates, because "the ECB rate" is ambiguous and

the answer to any question about negative rates is the deposit rate.

- **The Bank of England** sets **Bank Rate**, a single number, which is the cleanest of the three.

Whatever the policy rate is, what everything else discounts off is the *overnight index* built on it, in OIS, LIBOR, SOFR and ESTR.

REAL INTERVIEW QUESTION (ECB · Market Risk)

What are the ECB's main lending facilities: the refinancing rate, overnight lending, TLTRO?

WHAT THEY TEST / ANSWER

Group them into standing facilities, open market operations and targeted programmes, and say what each is for.

Standing facilities, available on demand and forming the corridor. The **marginal lending facility** lends overnight against collateral at the top of the corridor, and the **deposit facility** takes overnight deposits at the bottom. A bank never needs to borrow in the market above the top or lend below the bottom, which is what makes it a corridor.

Open market operations, the regular refinancing. **MRO**, the main refinancing operation, is a weekly collateralised loan at the main refinancing rate, and **LTRO** is the same thing at longer maturity. These are the ordinary way the banking system obtains liquidity.

Targeted programmes. **TLTRO**, targeted longer-term refinancing operations, lends to banks for years at a rate that depends on how much they lend on to the real economy, which is the innovation: it is monetary policy with a condition attached, aimed at transmission rather than at the level of rates.

Then name the two things beyond lending, since the question is about the toolkit. **Asset purchases**, which act on the long end and the term premium rather than on the overnight rate, and their reversal. And **collateral policy**, the eligibility criteria and haircuts, which is an underrated instrument: widening what counts as collateral is itself an easing, and it is done without changing any rate.

Close on the purpose that unifies them: all of these exist to make sure the policy rate actually reaches the economy. A policy rate the banking system cannot transmit is a number in a press release, which is why the facilities, the collateral rules and the targeted operations receive as much attention from a rates desk as the rate itself.

How a decision reaches the world

REAL INTERVIEW QUESTION (Bank of America · Credit)

How do the Fed's rate decisions transmit to the yield curve and to the economy?

WHAT THEY TEST / ANSWER

Follow the chain, and be explicit that most of it happens before the decision.

To the curve. The policy rate sets the overnight rate directly. Everything longer is an *expectation* of the average future overnight rate plus a term premium, so a change in the policy rate moves the long end only to the extent that it changes expectations about the path. This is why a hike that was fully anticipated can leave ten year yields unchanged or even lower them, if the accompanying message suggests the cycle is ending. The front end tracks policy closely, the belly reflects the expected path, and the long end is dominated by the term premium and by inflation expectations, which is the subject of yield curves.

To the economy. Through four channels worth naming separately. **Borrowing costs**: bank lending, mortgages and corporate funding reprice off the curve. **Asset prices**: a higher discount rate lowers equity and property valuations and widens credit spreads, which affects wealth and collateral. **The exchange rate**: higher relative rates attract capital and strengthen the currency, which imports disinflation, covered in central banks and FX policy. **Expectations**: if households and firms believe inflation will return to target, they set prices and wages accordingly, and that channel works without any transaction at all.

Then the two points that make this a markets answer. The lags are **long and variable**, usually quoted as several quarters to two years, which is why policy is set on forecasts rather than on current data and why the debate is always about being early or late. And markets price the **expected path**, not the decision, so what moves a curve on the day is the surprise relative to what was already priced, which is the general principle of price discovery.

Reading them

The vocabulary is small and you are expected to use it correctly. **Hawkish** means leaning towards tighter policy, higher rates, more concern about inflation; **dovish** is the opposite, leaning towards support for growth and employment. A meeting can be a hawkish cut or a dovish hike, when the decision goes one way and the message the other, and being able to say that is a sign you follow the market rather than the headline.

What actually moves prices, in rough order: the **statement**'s changes from the previous one, the **projections** including the Fed's dot plot of individual rate expectations, the **press conference**, and the **dissents**. The decision itself is usually the least informative part, because it was priced beforehand.

REAL INTERVIEW QUESTION (Commerzbank · Rates)

If US demand rises and inflation rises, what will the Fed do and why?

WHAT THEY TEST / ANSWER

Tighten, and the value of the answer is in the reasoning and the caveats rather than in the direction.

The mechanism. Demand above the economy's capacity to supply produces inflation, and the Fed's dual mandate makes price stability an explicit objective, so it raises rates to slow demand: borrowing becomes more expensive, asset prices fall, the currency strengthens, and demand cools with a lag. The intellectual shorthand is a Taylor-type reaction function, where the policy rate rises more than one for one with inflation so that the *real* rate rises, which is the condition for policy to actually be restrictive.

The three qualifications that turn a textbook answer into a good one. First, the **source of the inflation** matters: this question stipulates demand, which monetary policy can act on. Supply-driven inflation, an oil shock or a supply chain disruption, is much harder, because raising rates does not create supply and only reduces inflation by destroying demand, which is why central banks often look through it. Second, the **other half of the mandate**: if the labour market is deteriorating at the same time, the Fed faces a genuine trade-off, and that trade-off is what makes meetings interesting. Third, **expectations**: the reaction is much stronger if long-run inflation expectations start to move, because an unanchoring is the failure the institution is most concerned about.

Close on the market answer rather than the economics one: what a desk trades is not whether the Fed hikes, it is **whether it hikes more than is already priced**. If the curve already prices four hikes and the data supports three, the correct trade is to receive rates even though the Fed is tightening. Saying that is the difference between an economics answer and a markets answer.

INTERVIEW TRAP

Four errors. (1) Saying "the ECB rate" without specifying which of the three, when the deposit rate is the one that matters. (2) Assuming a hike raises long yields, when it moves the long end only through expectations and can lower them. (3) Quoting stale policy settings, particularly for the Bank of Japan, which has changed most: know the current level or say you would check. (4) Answering with the decision rather than with the surprise relative to pricing, which is what markets actually trade.

Where this goes next

The curve that policy acts on is yield curves and the overnight benchmarks built on the policy rate are OIS, LIBOR, SOFR and ESTR. Each channel is worked through properly in monetary transmission mechanisms, with policy as a subject in monetary policy. The currency channel is central banks and FX policy, the calendar of the meetings and releases is macro calendars, and the interview versions of these questions are market questions.

2.4 International Financial Institutions

sales · trading · structuring | core | cross-asset · rates

These institutions look like general knowledge and are not. They matter to a desk in three concrete ways: one of them writes the capital rules that decide which businesses your employer is allowed to run, one of them is the reason a distressed sovereign's bonds move fifteen points in a morning, and several of them are large, frequent AAA issuers you will quote prices on. This section is about those three, and skips the organisational history.

The four that matter, and what each actually does

The IMF is a balance-of-payments lender of last resort to *countries*. A member in a currency crisis borrows against a **quota**, denominated in Special Drawing Rights, and the loan carries **conditionality**: fiscal and structural commitments the country must meet to keep drawing. It does not fund development and it does not lend to companies.

The World Bank does the opposite job on a different horizon: long-dated lending for development projects, with no stabilisation mandate. Where a markets desk meets it is not policy but paper, since it is a large and permanent AAA bond issuer.

The BIS is the central banks' bank, and it is the one with the most direct effect on a trading floor, because it hosts the **Basel Committee on Banking Supervision**. The Committee writes the capital and liquidity standards, Basel I through III and the subsequent revisions, including the FRTB market-risk framework and the LCR and NSFR. Those standards are the reason certain trades became uneconomic without ever becoming illegal.

The FSB, the Financial Stability Board, coordinates the national regulators and designates the globally systemically important banks that carry an extra capital surcharge.

Regionally: the **ESM** is the euro area's own bailout facility, built after the sovereign debt crisis, and the **EIB**, **EBRD** and **ADB** are development banks that are also, from a desk's point of view, bond issuers.

Why an IMF programme moves a bond

> **KEY IDEA**
>
> An IMF programme is the largest single event in a distressed sovereign's credit, and the mechanism is worth stating precisely: the programme replaces near-term *liquidity* risk with a medium-term *solvency and politics* question. The country gets hard currency it did not have, so the next coupon is paid, and spreads at the front of the curve compress hard. The long end moves much less, because the conditionality has to be delivered by a government that may not survive delivering it.
>
> The tradeable version of that sentence is that an IMF programme steepens a distressed sovereign's credit curve. That is a better answer than "spreads tighten", and it is the one that shows you know where the risk went rather than that it disappeared.

The second-order point is the one candidates miss. The IMF is a **preferred creditor**: it is repaid before private bondholders. So a large programme improves the near-term picture and simultaneously *subordinates* the existing bonds, which is why very large packages can be received badly by the long end of the curve.

Basel, and what capital costs a desk

The Basel framework is not background. It is the constraint that decides what a desk can hold, and it is worth being able to convert it into a price.

Capital requirements are expressed against **risk-weighted assets**: an exposure's notional multiplied by a risk weight that reflects its quality. A government bond in its own currency may carry a zero weight; a corporate loan may carry 100%. Basel III sets a minimum common equity ratio of 4.5% plus a 2.5% conservation buffer, and countercyclical and systemic surcharges take large banks to something in the region of 11%.

> WORKED EXAMPLE
>
> **Turning a risk weight into basis points.** A desk holds €1 billion of an asset with a 20% risk weight.
>
> $$\text{RWA} = 1{,}000 \times 0.20 = €200 \text{ million}, \qquad \text{equity required} = 200 \times 0.11 = €22 \text{ million}.$$
>
> If the bank targets a 12% return on equity, that position must earn $22 \times 0.12 = €2.64$ million a year purely to pay for the capital it consumes, which on €1 billion of notional is
>
> $$\mathbf{26.4} \text{ basis points a year.}$$
>
> Run the same calculation across risk weights and the framework's effect on behaviour becomes obvious:
>
> | Risk weight | Capital cost, bp per year |
> |---|---|
> | 0% | 0.0 |
> | 20% | 26.4 |
> | 50% | 66.0 |
> | 100% | 132.0 |
>
> A trade that earns 20 basis points is profitable at a zero risk weight and destroys value at a 20% weight, before any consideration of whether it is a good trade. That is why capital rules changed what banks do far more effectively than any prohibition would have, and it is the concrete answer to why an institution in Basel matters to a desk in Paris.

The SSA market, where you actually meet them

Sovereigns, supranationals and agencies form their own segment of the bond market, universally called **SSA**. The World Bank, EIB, ESM, KfW and their peers issue in benchmark size across currencies, are almost all rated AAA, and typically trade at a small spread over the relevant government curve.

Three things make the segment distinctive and are worth knowing if you are asked. The credit is effectively that of the shareholder governments rather than of any project, so the analysis is about who stands behind the paper. The issuance is heavily driven by currency swap arbitrage, since an issuer raises where it is cheapest and swaps back, so a supranational appearing in an unusual currency is usually a cross-currency basis story rather than a funding-need story. And the paper is high-quality liquid asset eligible, which means demand is partly regulatory rather than economic, and that demand does not behave like a normal investor's.

INTERVIEW TRAP

1. **Confusing the IMF with the World Bank.** Crisis lending to countries against conditionality, versus long-horizon development lending. Getting this backwards is the fastest way to look unprepared on a macro desk.
2. **Saying an IMF programme tightens spreads and stopping there.** It compresses the front and much less at the back, and the preferred-creditor status subordinates existing bondholders.
3. **Treating Basel as compliance trivia.** It prices trades. The 26.4 basis points above is a real cost that decides whether a business exists.
4. **Assuming the BIS regulates banks.** It does not. It hosts the committee that writes the standards; national regulators implement them, and they implement them differently and on different timetables, which is why the same trade has a different capital cost in two jurisdictions.
5. **Treating supranational paper as a single credit.** It is a claim on the shareholder governments, so a supranational is only as strong as the callable capital behind it.

VISUAL / CODE TO BUILD: Interactive

A capital-cost calculator. Sliders set notional, risk weight, the required common equity ratio and the bank's target return on equity, and the output is the annual capital charge expressed in basis points of notional, printed next to a second input for the trade's actual expected margin. The panel colours the trade green or red depending on which is larger.

The intended moment is fixing a plausible margin, say 30 basis points, and dragging the risk weight from 0 to 100%: the trade goes from comfortably profitable to value-destroying without anything about the trade changing. That single drag is the whole argument for why Basel belongs in a markets book rather than in a compliance one.

A second panel draws a distressed sovereign's credit curve and lets the reader apply an IMF programme of a chosen size. The front of the curve compresses sharply while the long end moves little, and at large programme sizes the long end widens as the preferred-creditor subordination outweighs the liquidity relief, which is the section's least intuitive claim shown rather than asserted.

2.5 Macro Calendars

sales · trading · structuring | core | cross-asset · fx · rates

The most useful single fact about market risk is that a large part of it is *scheduled*. Central bank decisions, inflation prints and employment reports arrive at published times on published dates, which means a desk knows in advance when it will be exposed and can choose whether to be. A calendar is therefore not an administrative document, it is a map of when risk happens.

KEY IDEA

Split every risk you carry into **scheduled** and **unscheduled**. Scheduled risk can be sized, hedged or avoided in advance, and it is priced in the options market before it arrives, so you can read what the market expects. Unscheduled risk, which is most geopolitics, cannot. Most of the value of knowing a calendar is that it converts the first category out of the second, and a trader surprised by a scheduled event has made an avoidable mistake rather than an unlucky one.

What is actually on it

Releases are tiered by how much they move markets, and the tiering is stable enough to memorise.

Tier one, which moves everything: central bank decisions and their press conferences, US CPI, and US non-farm payrolls. A desk plans around these.

Tier two, which moves the relevant asset class: PMIs and other survey data, retail sales, PPI, weekly jobless claims, GDP releases, and the central bank minutes published some weeks after a meeting.

Tier three: almost everything else, which matters only when it contradicts a tier-one reading or when a market is unusually focused on one variable.

The tiering is not fixed, and noticing when it shifts is a genuine skill. In an inflation-fighting regime CPI is the most important number in the world and payrolls is secondary; in a growth scare the ranking reverses. If asked which release matters most, the correct answer names the regime first.

Markets trade the surprise

KEY IDEA

A price does not move because a number is good. It moves because the number differs from what was expected, and the size of the move scales with how *surprising* the difference is, which means the deviation measured against the dispersion of forecasts rather than in its own units.

WORKED EXAMPLE

The same number, two different market reactions. Consensus for payrolls is 180,000 and the print is 200,000, a beat of 20,000.

If the standard deviation across forecasters is 40,000, the surprise is $20/40 = \mathbf{0.5}$ standard deviations, which is inside the noise, and the market barely moves.

If forecasters are tightly clustered with a standard deviation of 10,000, the same beat is $20/10 = \mathbf{2.0}$ standard deviations, and the market moves hard.

Identical headline, identical direction, completely different reaction. This is why an experienced answer to "what will the market do if payrolls beat" is a question back: by how much, and how wide is the distribution of forecasts.

Reading the expected move out of the options market

The options market prices scheduled events explicitly, and extracting that price is the most useful calendar skill.

> WORKED EXAMPLE
>
> **Backing out what the market expects from payrolls.** EURUSD overnight implied volatility is 8% on an ordinary day. Going into the payroll report the overnight implied is 16%.
>
> Convert both to one-day variance, with $T = 1/252$:
>
> $$\sigma^2 T\Big|_{16\%} = 1.0159 \times 10^{-4}, \qquad \sigma^2 T\Big|_{8\%} = 2.5397 \times 10^{-5}.$$
>
> The difference is the variance the market attributes to the event itself, and its square root is the expected one-off move:
>
> $$\sqrt{1.0159 \times 10^{-4} - 2.5397 \times 10^{-5}} \;=\; \mathbf{0.87}\%.$$
>
> So the market is pricing roughly an 87 basis point jump on the release, over and above a normal day's diffusion. That number is directly tradeable against your own view: if you think the release is a non-event, the overnight straddle is expensive; if you think it is decisive, it is cheap. Either way you now have the market's estimate rather than an impression.
>
> The related quantity is the straddle premium itself, which at 16% overnight implied is about $0.8\,\sigma\sqrt{T} = 0.81\%$ of spot, and that is the move the position needs just to break even.

Two consequences follow and both are worth stating.

Volatility crush. The event premium exists only while the event is in the future. The moment the number prints, overnight implied collapses from 16% back toward 8%, so a long straddle held through the release can be right about direction and still lose. Buying volatility into an event is a bet that the realised move exceeds the priced move, never a bet on direction, which is developed in strategy selection.

Positioning. If everyone expects a hawkish outcome and is positioned for it, a hawkish outcome can sell off, because the news that confirms a consensus removes the reason to add. That is the mechanism behind buying the rumour and selling the fact, and it is why a move in the wrong direction after a release is usually a positioning story rather than a mispricing.

REAL INTERVIEW QUESTION (CACIB · FX)

What is the big news on the US market (NFP), and what is its impact on the USD?

WHAT THEY TEST / ANSWER

Answer in three parts: what the release is, the transmission into the dollar, and the qualifications that show you have watched one.

What it is. Non-farm payrolls, published by the Bureau of Labor Statistics on the first Friday of most months at 8:30 New York time. It is the monthly change in employment excluding farms, and it comes with the unemployment rate and average hourly earnings in the same release, which matters because those two frequently drive the market more than the headline does.

The transmission. A strong print means a stronger economy, which means the Federal Reserve can stay tighter for longer, which raises US yields, which supports the dollar. State it as that chain rather than as "good data, strong dollar", because every link is a place where it can break.

The qualifications, which are the answer.

Surprise, not level. The market trades the deviation from consensus scaled by forecast dispersion, so a beat inside the forecast range does nothing.

Which component leads depends on the regime. When the Fed is worried about inflation, average hourly earnings can matter more than the headline count, and a strong jobs number with weak wages can be read as disinflationary and therefore dollar-negative. Saying this is what separates a candidate who has followed a payroll from one who has read about one.

Revisions. The prior two months are revised in the same release, and a large downward revision routinely outweighs a headline beat. A market that sells the dollar on a strong number has usually seen the revision.

Duration of the move. The immediate reaction is seconds and the follow-through is minutes to hours; by the following week the dollar is being driven by whatever the Fed has said since. Getting the horizon right is more important than getting the direction right.

Close by noting the options context if you have room: the overnight straddle prices the expected move in advance, so the interesting question is not which way the dollar goes but whether the realised move exceeds the priced one.

REAL INTERVIEW QUESTION (UBS · desk not recorded)

Did you hear about the ECB's meeting yesterday, and do you know what the decision was?

WHAT THEY TEST / ANSWER

This is a preparation check with a very low pass mark and a lot of failures, and the failure mode is bluffing. If you do not know, say so immediately and say what you do know, because a candidate who invents a decision has told the interviewer something far worse than that they missed a meeting.

If you do know, the good answer has three parts and takes twenty seconds.

The decision, precisely: the rate level and the direction of the move, if any.

The market reaction, which is what shows you were watching rather than reading a summary afterwards. Did the euro rally, did the front end reprice, did the reaction match the decision or contradict it?

Why it went that way, which is almost always about the guidance rather than the decision. A hold with a hawkish press conference moves markets more than a cut that everyone expected, because the decision was priced and the guidance was not. Naming that distinction between the decision and the guidance is the whole content of a good answer.

Then, whatever you knew, add the habit rather than the fact: say when the next meeting is and what the market currently prices for it, because forward-looking is the posture the question is really testing, and it also proves the first part was not luck.

INTERVIEW TRAP

1. **Reasoning from the level instead of the surprise.** A good number that everyone expected is not a market event.
2. **Ignoring forecast dispersion.** The same deviation is a non-event or a large move depending on how tightly forecasters agree.
3. **Buying an event straddle for direction.** You are buying realised against implied. Volatility crush means a correct directional call can still lose.
4. **Forgetting revisions.** Payrolls revises the prior two months in the same release, and the revision often outweighs the headline.
5. **Assuming the tiering is permanent.** Which release matters most is a function of the regime, and the answer to "which number matters" starts with what the central bank is currently worried about.
6. **Treating time zones casually.** A release is at a fixed local time in its own country, which is not a fixed time in yours, and being on the wrong side of a daylight-saving change is an embarrassing way to be short gamma into a print.

VISUAL / CODE TO BUILD: Interactive

An event-pricing panel. Inputs are the ordinary overnight implied volatility and the elevated implied going into the event; the output is the implied one-off move, computed as the square root of the variance difference, drawn as a shaded band around the current spot alongside the straddle break-even. Setting 8% and 16% reproduces the section's 0.87%, and dragging the elevated volatility shows how quickly the implied event move grows.

Beneath it, a surprise-to-move panel: the reader sets consensus, forecast dispersion and the actual print, and the panel reports the surprise in standard deviations and maps it onto an expected market move through a simple response function. Holding the headline beat fixed at 20,000 and dragging dispersion from 40,000 down to 10,000 turns a non-event into a large move without changing the number at all, which is the section's central claim performed rather than asserted.

A third element replays a release on a timeline: implied volatility elevated before, spot jumping at the print, implied collapsing immediately after, and a running P&L for a long straddle held through it. Watching that P&L end negative on a correct directional call is the clearest possible statement of what volatility crush is.

Where this goes next

The institutions whose meetings anchor the calendar are the major central banks, and what they are reacting to is monetary policy. The two tier-one prints get their own treatment: the inflation release in inflation and the payroll report in employment. The realised-against-priced framing behind the volatility crush is implied versus realised volatility, and choosing a structure once you have a view on it is strategy selection by market regime. The risk that is not on any calendar is geopolitics, and the interview form these questions take is market questions.

2.6 Trading floor terminology

sales · trading · structuring core cross-asset

In your first week on a desk you will understand perhaps half of what is said around you, and that is normal. What is not survivable is misunderstanding the words that commit money. Trading language is terse because speed matters and because most of it was designed for a telephone: it is compressed, unambiguous to insiders, and completely opaque to everyone else. This section teaches the spoken language of dealing. The alphabet soup of abbreviations lives next door in market jargon and abbreviations.

Price language

Term	Meaning
Bid	The price at which the quoting party will *buy*
Offer, or ask	The price at which the quoting party will *sell*
Mid	Halfway between bid and offer; a reference, not a tradeable price
Spread	The distance between bid and offer, the maker's compensation
Tick	The smallest price increment the instrument trades in
Pip	In FX, conventionally the fourth decimal, or the second for JPY pairs
Handle, big figure	The leading digits everyone omits when speaking
Choice	Bid equals offer, no spread
Level	A price for information; "level check" means no intention to trade
Indicative	Not tradeable. **Firm** or **live** means tradeable now

The convention that beginners get backwards: the quote is always from the *maker's* perspective, so **you buy on the offer and sell on the bid**, and you therefore lose the spread on a round trip. What determines the width of that spread is developed in the bid-ask spread.

Because the leading digits rarely change during a conversation, they get dropped. A euro-dollar market quoted as "fifty at fifty-three" on a handle of 1.08 means 1.0850 bid and 1.0853 offered. If you are not certain of the handle, ask, because assuming it is the single easiest way to trade at a price you did not intend.

Dealing language: the words that commit money

Word	What it commits you to
Mine	"I buy" at your offer
Yours	"I sell" to you at your bid
Done	Agreed, we have a trade
Nothing done	No trade, the price is not taken
Off	The quote is withdrawn, or you lost the competition
Your risk	My quote has expired, the market has moved, ask again
Subject	Not firm, subject to confirmation
I'll work it	I will try to execute over time rather than immediately
Leave it with me	I will come back; you are not filled
Away	It traded elsewhere, or your interest is not with me

KEY IDEA

Mine and **yours** are the two most dangerous words on a floor. They are said quickly, on a recorded line, and they create a binding trade. "Mine" means the speaker takes the offer and buys; "yours" means the speaker gives at the bid and sells. If there is any doubt at all, restate the whole trade in full before moving on: instrument, direction, size, price. "So I sell you fifty million at 1.0850, value Wednesday" takes four seconds and prevents the class of error that generates a mistrade, an apology, and a P&L adjustment someone has to explain.

Position and direction language

Long is owning it, **short** is having sold what you do not own, and **flat** or **square** is having no position. A **better buyer** means the flow you are seeing is skewed to the buy side. An **axe** is a position the desk wants to do, so the price improves on that side, which is the cheapest edge available to a client and is covered in axes and trade ideas.

Two asset-class conventions cause genuine confusion and are worth learning explicitly.

- **In swaps, you pay or receive, you do not buy or sell.** "Paying" means paying the fixed rate and receiving floating, so you gain when rates rise; "receiving" is the reverse. "Buying a swap" is ambiguous and a careful person avoids it. See interest rate swaps.
- **In bonds, price and yield move opposite ways**, so "better bid" means prices higher and therefore yields lower, and "the market is heavy" means prices are falling and yields rising. A statement about "rates going up" almost always means yields, and therefore bond prices going down.

Size, and how it is spoken

Sizes are always spoken in the market's own unit and almost never in full. A **yard** is one billion, **mio** is one million, and a **clip** is one tradeable parcel of a larger order. "Five by five" means five million on each side of a two-way quote. "In comp" means you are in competition with other dealers. "Full amount" means the quote is good for the whole size, as opposed to a price for a smaller clip.

Two things to insist on, both of which prevent expensive misunderstandings. State the **unit** the first time ("fifty million euros"), because "fifty" means different things on different desks. And state whether a size is **all or nothing**, since a partial fill on a hedge leaves you with a position you did not want.

Time and settlement language

- **Value date** or **settlement date**: when cash and assets actually move. **Spot** is the market's standard value date, conventionally two business days ahead in FX for most pairs.
- **T+1**, **T+2**: settlement measured in business days after the trade date.
- **Cut-off**: the deadline after which an instruction or an option exercise is too late.
- **Fix** or **fixing**: an officially published reference level at a stated time, used to settle contracts. On the floor the word is used for the moment as much as for the number, as in "do it on the fix", because that print is when a great deal of hedging flow is forced to trade. How a fixing is actually specified and observed is in market microstructure.
- **Roll**: moving a position from the expiring contract to the next one.
- **IMM dates**: the third Wednesday of March, June, September and December, on which a great many futures and swaps are standardised.
- **The close**: the reference price for marks, which is why the closing auction is the most important minute of the equity day.

Risk, P&L and error language

Greeks are used as adjectives about a person: "I am short gamma" means the book is, and "bleeding theta" means it is losing value with time. To be **picked off** is to be traded against by someone better informed; to be **gapped** is to have the market jump through your hedge; to be **stopped out** is to have a loss limit force you out. A **mark** is the price a position is valued at, **marking** is setting it, and a **stale mark** is one the market has left behind, which is a control problem as much as a P&L one.

When something goes wrong the vocabulary is specific, and using it correctly is how you get helped quickly: a **mistrade** is a trade agreed in error, **off-market** means at a price away from fair value, a **break** is a mismatch between two records, to **affirm** and then **confirm** is the post-trade agreement chain, and a **reprice** is an agreed correction. The rule that protects you: escalate immediately and in plain words. Every serious desk would rather hear "I think I have made an error" in the first minute than discover it in the reconciliation the next morning.

INTERVIEW TRAP

Four habits that mark someone out as unsafe rather than merely new. First, **guessing the handle** instead of asking, which trades you at a price you did not mean. Second, **saying "mine" or "yours" without restating the trade**, which is how mistrades happen. Third, **using "buy" and "sell" for swaps** instead of pay and receive, or being vague about whether "up" refers to a yield or a price. Fourth, **pretending to understand**. Asking "sorry, what does that mean" costs you nothing in the first month and saves a great deal later; every experienced person on the floor was told the same thing on their first day.

REAL INTERVIEW QUESTION (CACIB · desk not recorded)

What is the Bid/Ask? At what price do you buy and at what price do you sell?

WHAT THEY TEST / ANSWER

Answer the mechanics without hesitating, since hesitation is the only way to fail this. The bid is the price at which the quoting party buys and the offer, or ask, is the price at which they sell, and the quote is always from the market maker's perspective. So as the client, **you buy on the offer and you sell on the bid**, which means you cross the spread twice on a round trip and start every position with a small loss equal to the spread. Then add the three things that turn a definition into a good answer. The spread is the maker's compensation for providing immediacy and for bearing inventory and adverse-selection risk, developed in market making. Mid is the average of the two and is a valuation reference rather than a price you can trade, which matters because marking a book at mid overstates what it could actually be liquidated for. And the spread is not a constant: it widens with volatility, with size, with illiquidity and around events, see the bid-ask spread. If they push for the mnemonic, the quoting party always gets the better side of both, which is why "buy low, sell high" is not available to you at someone else's quote.

REAL INTERVIEW QUESTION (Société Générale · Fixed Income)

What does being long gamma mean in trading?

WHAT THEY TEST / ANSWER

Translate the jargon into behaviour, because the question is about what the phrase means in practice on a desk rather than about the definition of a second derivative. Being long gamma means your delta moves in your favour as the market moves: you get longer as it rallies and shorter as it falls, so rehedging means selling higher and buying lower, and you make money when the market actually moves, in either direction. You are typically long gamma because you own options, and you pay for the privilege in **theta**. Stop there unless they ask for more, because knowing where a terminology answer ends is itself a good signal, and the follow-ups are separate topics with their own answers: what the position is worth as the spot moves is gamma and convexity, how often to rehedge it is intraday hedging, and which structures are bought to carry it is volatility strategies. On a fixed income desk, add that the same language applies to convexity in bonds and swaptions, where a long-convexity position behaves in exactly this way with respect to yields.

REAL INTERVIEW QUESTION (HSBC · Rates)

What is DV01?

WHAT THEY TEST / ANSWER

A one-line definition, then the two things people actually use it for. DV01, the dollar value of a basis point, is the change in the present value of a position for a one basis point move in rates, quoted in currency per basis point, and it is the rates desk's universal risk unit because it converts every instrument, cash bonds, futures, swaps and options, into a single comparable number. Its two uses: **aggregation**, since summing DV01 across a book gives the book's rate exposure and it can be reported by curve bucket to show where the risk sits on the curve rather than only in total, and **hedging**, since matching DV01 between two instruments is how a hedge ratio is computed. Then the order of magnitude, which is what distinguishes someone who uses it from someone who has read the definition: a par swap or bond has a DV01 of roughly notional times duration times one basis point, so a ten-year position on €100m with a duration near 8.5 has a DV01 of about €85,000 per basis point. Finish with the two caveats: DV01 is a first-order measure, so it misses convexity, which matters for large moves and for long-dated positions, and a single number hides curve risk, which is why desks look at bucketed key-rate DV01s. The underlying mathematics is in duration and convexity.

VISUAL / CODE TO BUILD: Interactive

A floor-language trainer, since this material is learned by ear rather than by reading. Audio or text snippets of realistic dealing conversations play ("fifty at fifty-three in ten", "yours in five", "your risk", "nothing done") and the reader must answer what just happened: who bought, at what price, in what size, or that no trade occurred. Scoring emphasises the money-critical cases, so **mine** and **yours** appear most often and a wrong answer shows the resulting position and the P&L consequence of the misunderstanding. A second panel is a two-way glossary keyed by situation rather than alphabet (pricing, dealing, sizing, settlement, error handling) with a search box, and a third generates a full trade restatement from a snippet so the reader practises the confirm-back habit that prevents mistrades.

Where this goes next

The abbreviations and acronyms this section deliberately leaves out are in market jargon and abbreviations. The economics behind the quote itself is the bid-ask spread and market making, and the conversation this language is used in is RFQs. The rates conventions appear again in interest rate swaps and duration and convexity, and the desk roles who speak to each other in it are in the different roles within the bank.

2.7 Market jargon and abbreviations

sales · trading · structuring | core | cross-asset

Jargon is a compression scheme. A trader who says "I paid ten on the five year, up half a big figure, and my vega is in points" has said a great deal in one breath, and the only reason it sounds like noise is that the words have not been unpacked. This section unpacks them. It is deliberately a reference rather than an argument, because the failure mode it addresses is not a gap in reasoning: it is a candidate who understands the finance perfectly and answers a question about a different quantity from the one that was asked.

Two warnings before the lists. First, the sibling section trading floor terminology covers what people *say* to each other on a floor, the verbs and the etiquette. This section covers what they *write*: units, acronyms and screen mnemonics. Second, an abbreviation is not a substitute for the thing. Knowing that IRS expands to interest rate swap is worth exactly one sentence in an interview, and the sentence after it is the one that counts.

Units: the expensive kind of confusion

Almost every unit error in markets is one of four confusions, and all four are asked about.

- **Basis point (bp, bps).** One hundredth of a percent: $1\,\text{bp} = 0.01\% = 0.0001$. It exists because "rates rose by 1%" is ambiguous between an absolute and a relative move, and "rates rose by 100 bp" is not.
- **Percentage point versus percent.** A yield going from 4% to 5% has risen by one percentage point, or 100 bp, or 25% in relative terms. All three are correct and they are different numbers.
- **Vol point.** Implied volatility is quoted in percent, so "the vol went up one point" or "up one vol" means from 16% to 17%, an absolute move of 100 bp and a relative move of 6.25%. Vega is normally quoted per vol point, so a vega of 50,000 means a gain of 50,000 for that one point move.

- **Pip and big figure.** In FX, a pip is the last decimal of the standard quote: 0.0001 for most pairs, but 0.01 for yen pairs, which is the exception people forget. The big figure is the part of the quote everyone already knows and therefore stops saying, so "ninety bid" on a EURUSD trading at 1.0890 means 1.0890, not 90.

WORKED EXAMPLE

EURUSD moves from 1.0850 to 1.0900. In pips that is $(1.0900 - 1.0850)/0.0001 = 50$ pips, or half a big figure. In percentage terms it is $0.0050/1.0850 = 0.461\%$, which is 46.1 bp. So **50 pips is not 50 basis points**, and the gap between the two is a function of where spot happens to be. Quote FX moves in pips and rate moves in basis points, and never convert casually between them.

The same trade in USDJPY: a pair quoted at 150.25 moving up 50 pips is at $150.25 + 50 \times 0.01 = 150.75$, because the yen pip is 0.01. A candidate who applies 0.0001 here lands on 150.255 and has moved the market by a third of a basis point (0.005/150.25) instead of thirty-three (0.50/150.25).

The fourth confusion is about size rather than about moves.

- **Notional** is the face amount a derivative's payments are calculated on. A €100m interest rate swap exchanges interest on €100m; nobody pays €100m.
- **Market value** is what the position is worth today, which for a swap struck at the market is close to zero on day one and drifts away from zero afterwards.
- **Nominal** is used loosely, sometimes for notional and sometimes for the face value of a bond. If it matters, ask.

KEY IDEA

Notional measures how much risk you have, market value measures what you own. A desk with €5bn of notional and €2m of market value is not a large position wrongly reported, it is a normal swap book. Confusing the two makes every risk number you quote wrong by orders of magnitude, which is why the distinction is tested early.

REAL INTERVIEW QUESTION (Union Capital · FX)

If you pay out euros, are you long or short?

WHAT THEY TEST / ANSWER

Short. The rule is that you are long what you end up holding and short what you have parted with, so paying euros away leaves you short euros and long whatever you received.

Say the rule in general terms, because that is what is being tested: **long means you benefit when the price rises, short means you benefit when it falls**. In cash that reduces to what you hold. In FX it needs one extra step, because every trade is two currencies at once: selling euros for dollars makes you short EUR and long USD simultaneously, and there is no such thing as being short in isolation. When a trader says "I am short euro" without naming the other leg, the other leg is the dollar by convention.

Then extend it to the two cases that follow. In derivatives, long and short refer to the *contract*, not to the underlying: long a put means you own the put, which gives you a position that gains when the underlying falls, so you are long the option and short the underlying's direction. And in funding, paying out a currency you do not have means borrowing it, so a short position in a currency is a liability in it, which is what makes the carry trade work and what makes it dangerous.

Product acronyms

The following are the abbreviations you are expected to expand without hesitating. The right-hand column is the one line you owe before moving to the mechanics, which live in the sections that own them.

Short	Expansion	One line
IRS	Interest rate swap	Fixed against floating, one currency
OIS	Overnight index swap	Fixed against compounded overnight
FRA	Forward rate agreement	One future period's rate, fixed today
CCS	Cross-currency swap	Two currencies, notionals exchanged
TRS	Total return swap	Rent the whole return of an asset
CDS	Credit default swap	Insurance against a credit event
ETF	Exchange-traded fund	A listed wrapper around a portfolio
RC	Reverse convertible	Yield paid for selling a put
WOF	Worst-of	Payoff reads the weakest underlying
OTC	Over the counter	Bilateral, not on an exchange
CCP	Central counterparty	Sits between the two sides, clears
ISDA	Int. Swaps and Derivatives Assoc.	Its Master Agreement is what trades hang off
CSA	Credit support annex	The collateral rules attached to it
RFQ	Request for quote	Client asks several banks for a price
DV01	Dollar value of a basis point	Cash change per 1 bp of rates
MTM	Mark to market	Today's value, not the entry price

REAL INTERVIEW QUESTION (Société Générale · Rates)

Give the definition and examples of an IRS, a cross-currency swap, an FX swap and a cap, and draw them.

WHAT THEY TEST / ANSWER

Four instruments that all sound like swaps and are not the same thing. The discriminator is **what actually gets exchanged**, so answer with that.

IRS. One currency. Two parties exchange interest on a notional, one leg fixed and one leg floating, and the notional itself is never exchanged because it would cancel. The payer of fixed gains when rates rise. This is the instrument the whole rates market is quoted in, and its mechanics belong to interest rate swaps.

Cross-currency swap. Two currencies. Interest is exchanged in each currency, and unlike the IRS the **notionals are exchanged**, at the start and again at maturity, because they are in different currencies and no longer cancel. The standard interbank form is floating against floating with a spread on one leg, and that spread is the cross-currency basis.

FX swap. Two currencies, but no interest legs at all. It is a spot transaction and an offsetting forward transaction agreed together, so you exchange currencies now and reverse the exchange at a fixed rate later. Economically it is a collateralised loan in one currency against another, which is why it is a funding instrument and why it dominates the short end, covered in FX forwards and swaps.

Cap. Not a swap. It is a strip of options, one per period, each paying the excess of the reference rate over a strike, so it is a series of calls on a rate. The buyer of a cap is protected against rates rising and pays a premium for it, which is the difference that matters: a swap fixes the rate and costs nothing up front, a cap sets a ceiling and costs a premium, so the swap removes the upside as well and the cap does not.

Draw them as flows on a timeline, one arrow per exchange, and the picture makes the answer by itself: the IRS has two arrows per period in one currency, the cross-currency swap has arrows in two currencies plus a fat arrow at each end for the notionals, the FX swap has only the two fat arrows, and the cap has one arrow per period going in one direction only. That last detail, **arrows in one direction only**, is what makes it an option.

The screen and its mnemonics

Most market abbreviations reach you through a terminal, so the terminal's own vocabulary is part of the jargon.

REAL INTERVIEW QUESTION (Moody's · desk not recorded)

What can you do on Bloomberg?

WHAT THEY TEST / ANSWER

Do not list functions at random. Group them by what the job needs, then name examples.

The grammar first, because it shows you have actually used it: you type a **security**, press a **yellow key** that says what kind of thing it is (`Equity`, `Govt`, `Corp`, `Curncy`, `Comdty`, `Index`), then a **function mnemonic**, then `<GO>`. So `AAPL US Equity DES <GO>` is the description page for one stock. Every function is three or four letters and the whole system is that one pattern repeated.

Find and describe an instrument: `DES` for the description, `ALLQ` for who is quoting it and where. **Look at history**: `GP` for a price graph, `HP` for the historical price table, `GIP` for intraday. **Monitor a market**: `WEI` for world equity indices, `ECO` for the economic calendar and consensus forecasts, `TOP` and `NI` for news. **Analyse derivatives**: `OMON` for the listed option chain, `SWPM` for building and valuing a swap, `CDSW` for credit default swaps. **Communicate**: `MSG` for mail and instant Bloomberg for chat, which is where a large share of client dialogue actually happens. **Get the data out**: the Excel add-in, which is the part that matters most for the work you will be given in the first month.

Close honestly and usefully: say that the terminal is a tool you learn on the desk in weeks, that what transfers is knowing *which* question to ask of it, and that the skill worth claiming is pulling data into a spreadsheet or a script and checking it, not memorising mnemonics. Anyone can look up a function name.

REAL INTERVIEW QUESTION (BRED · desk not recorded)

Do you know BDH and BDP, and what is the difference between them?

WHAT THEY TEST / ANSWER

Both are Bloomberg's Excel formulas, and the difference is **one value against a series**.

BDP, Bloomberg Data Point, returns a single current value for one field on one security: the last price, the coupon, the maturity. It is a point in the present and it refreshes.

BDH, Bloomberg Data History, returns a **time series** for a field between two dates, so it spills down a column and you choose the frequency: daily, weekly, monthly. It is what you use to build a return series or a chart.

Add **BDS**, Bloomberg Data Set, unprompted if you know it: it returns bulk data that is a list rather than a number, such as the members of an index or a bond's cash flow schedule.

The practical point that shows you have used them rather than read about them: BDH returns an array, so it needs room beneath it and it breaks the moment something else is in the way, and it is slow enough that a sheet full of live BDH calls over long histories will crawl. The normal discipline is to pull history once, paste it as values, and keep the live calls for the small number of fields that genuinely need to move. Spreadsheet practice proper is in advanced Excel.

INTERVIEW TRAP

Four traps, all of them unit traps. (1) Saying a rate "rose 1%" when you mean one percentage point. (2) Treating pips as basis points, which is wrong by a factor that depends on the spot level. (3) Quoting vega without saying per what, since per vol point and per unit of volatility differ by a hundred. (4) Quoting a notional as though it were an exposure or a market value. None of these are knowledge failures, which is exactly why they are unforgiving: they read as carelessness with money.

Where this goes next

The spoken half of this vocabulary, the verbs a floor uses and the etiquette around them, is trading floor terminology, and the quoting convention underneath most of it is the bid-ask spread. The instruments abbreviated here are explained where they live, starting with interest rate swaps and FX forwards and swaps. The tooling side is advanced Excel, and the interview versions of these questions are collected in market questions.

Chapter 3

Macro & Economics

3.1 Economic growth

sales · trading · structuring core rates · cross-asset

Growth is the variable every other macro variable is measured against. Inflation is judged relative to how fast the economy can grow, policy is set relative to whether growth is above or below potential, and a bond's return depends on which of those two is true. This section is about what the number is, how it is reported, and the two or three conventions that cause candidates to give an answer that is out by a factor of four.

What GDP is

Gross domestic product is the market value of all final goods and services produced inside a country in a period. Final matters: counting the steel and the car it becomes would count the same output twice, which is the same double-counting problem the import line below solves. It is measured three ways that must agree by construction, since one person's spending is another's income and both equal the value of what was made. The expenditure version is the one worth memorising:

$$Y = C + I + G + (X - M),$$

consumption, investment, government spending, and net exports. Two things follow that people routinely get wrong.

- **Imports are subtracted because they were already counted**, not because imports are bad for an economy. They appear inside C, I and G, and the $-M$ removes what was not produced domestically. A rising import bill in a strong economy is arithmetic, not weakness.
- **Investment means physical capital formation**, plant, equipment, housing, inventories. Buying a share is not investment in this sense. Inventories in particular are small, volatile, and the reason a quarterly print can miss badly for reasons nobody cares about.

KEY IDEA

Real growth is what matters and nominal is what is measured. Output is observed in money, so the price effect must be stripped out using the GDP deflator. Do not subtract: the relationship is multiplicative. Nominal growth of 5% with a deflator of 3% gives real growth of $1.05/1.03 - 1 = \mathbf{1.94}\%$, not 2%.

The gap is small at these levels and stops being small when either number is large, which is exactly the situation, high inflation, where getting it right matters. The same approximation-versus-identity distinction runs through inflation.

The convention that produces wrong answers

WORKED EXAMPLE

The single most common error in a macro interview is comparing a US GDP print with a European one without adjusting for how they are quoted.

The United States reports quarterly GDP at a **seasonally adjusted annual rate**: the quarter's growth is compounded up as though it continued for a year. **The euro area and the UK** report the plain **quarter-on-quarter** change.

So a euro area print of 0.5% and a US print of 2.0% describe **almost exactly the same pace of growth**, because

$$1.005^4 - 1 = 2.015\%.$$

Quoted as they are published, one looks four times the other.

Say which convention you are using whenever you quote a growth number, and if you are given one without a convention, ask. An interviewer who hears "is that annualised or quarter-on-quarter" has learned more about you than any answer you could give afterwards.

A third convention, year-on-year, compares the quarter with the same quarter a year earlier. It is smoother and therefore better for seeing a trend, and it is slow, because it still contains three quarters of old news. A turning point shows up in the quarterly series and is invisible in the annual one for months.

Potential, and why the gap matters more than the level

INTUITION

Growth on its own says nothing. Two percent is strong for an economy that can only sustain one and weak for one that can sustain four. The reference point is **potential output**, what the economy can produce with its labour force and capital stock at normal utilisation, and potential growth is roughly the sum of labour force growth and productivity growth. That decomposition explains most cross-country differences in trend growth in one line: an ageing population with flat productivity has low potential growth and there is no policy that changes this quickly.

The **output gap** is actual output less potential. A positive gap means the economy is running hot, which generates inflation; a negative gap means slack. **This is why a central bank can tighten into what looks like decent growth**: the question is never the growth rate, it is the growth rate relative to potential. And potential is unobservable, estimated with wide error bars and revised afterwards, which is why reasonable people disagree about policy using the same data.

Reading it before it is published

GDP arrives late, quarterly, and is revised. Markets therefore trade the surveys that arrive monthly and predict it.

- **PMI**, the purchasing managers' index, is the flagship. It is a **diffusion index**: firms are asked whether activity is better, worse or the same as last month, and the index is the percentage answering better plus half the percentage answering the same. **50 means no change**, above 50 expansion, below 50 contraction.
- **Other surveys**, ifo in Germany, ISM in the US, consumer confidence, tell the same kind of story from different samples.
- **Hard data**, industrial production, retail sales, arrives later and is what the surveys were guessing at.

KEY IDEA

The property of a diffusion index that candidates miss: it measures **breadth, not magnitude**. A reading of 55 says more firms are improving than deteriorating; it does not say by how much. So a PMI can fall from 58 to 53 while output is still growing, because the number of firms improving fell while the improvement continued. **The direction of the index and the direction of the economy are not the same thing**, and confusing them is how a slowdown in the rate of improvement gets reported as a downturn.

REAL INTERVIEW QUESTION (Deutsche Bank · desk not recorded)

What are the main economic and monetary trends in the US, Europe and Asia: policy rates, GDP growth, PMI, unemployment, central banks, quantitative easing?

WHAT THEY TEST / ANSWER

This is a structure question disguised as a knowledge question. Nobody expects every figure; they expect you to have a framework, to know roughly where things stand, and to be honest about the difference. Give the framework, then the regions, then say what you would check.

The framework, in one sentence per region: where growth is relative to potential, what that implies for inflation, and therefore what the central bank is doing and what the market has already priced. Everything else hangs off that.

The United States. The developed economy with the highest potential growth, driven by population growth and productivity, and much the most domestically driven of the three. Do not extend that to the region comparison: emerging Asia has far higher potential growth than either the US or Europe, so the ranking only holds among developed economies. The Fed has a dual mandate, so it responds to the labour market directly as well as to prices, which is covered in major central banks. The variables to quote are the fed funds target range, payrolls and unemployment, and the ISM.

Europe. Lower potential growth, an older population, more open and therefore more exposed to global trade and to energy prices. The ECB has a primary price-stability objective rather than a dual mandate, so a given growth number implies a different policy response than the same number in the US. Watch the deposit rate, the composite PMI and the divergence between member states, since the euro area average often describes no member well.

Asia. Do not treat it as one thing. Japan is the special case, decades of deflation followed by a genuine normalisation, so it is the one where quoting a stale policy setting is most likely. China is a policy-driven cycle with property and credit at its centre rather than a consumer cycle, so the indicators that matter differ. The rest of the region is largely a function of those two plus the global trade cycle.

On quantitative easing, say what it is for rather than whether it is on: buying assets to act on the long end and the term premium when the policy rate can go no lower, and its reversal, balance sheet reduction, as the mirror image. Which phase each central bank is in has changed repeatedly and is exactly the sort of detail to check.

Then close the way a market participant would. What a desk trades is not the level of growth but the surprise relative to what is priced, so the useful question about any of these regions is where consensus sits and what would move it. That principle is price discovery. And say plainly that you follow these but would check the current readings before quoting them, naming the two or three you watch most. Admitting a gap costs nothing; quoting a stale number confidently costs a great deal.

INTERVIEW TRAP

Three errors. (1) Comparing a US annualised print with a European quarter-on-quarter one. A euro area 0.5% and a US 2.0% are the same pace. (2) Subtracting inflation from nominal growth instead of dividing. It is close at low levels and wrong when it matters. (3) Reading a falling PMI as a contracting economy. Anything above 50 is growth, and a diffusion index measures how many firms are improving, not by how much.

Where this goes next

The other half of every nominal number is inflation, and the input that most directly determines potential is employment. What a central bank does with an output gap is monetary policy, and the fluctuation of growth around its trend is economic cycles. The institutions reacting to these numbers are major central banks, the schedule on which the numbers arrive is macro calendars, and the reason only the surprise moves a price is price discovery.

3.2 Inflation

trading · structuring · sales | core | rates · cross-asset

Inflation is the macroeconomic variable a market professional is asked about most often, and the one where confident answers are most often wrong in the same three ways. It is worth getting precise, because every rates desk in the building is positioned against it and because the errors below are the ones interviewers use to sort candidates quickly.

> INTUITION
>
> Inflation is not the price of things. It is the *rate of change* of a weighted average of the prices of things, measured against the same month a year earlier. Almost every confusion in this section comes from forgetting one of those three qualifiers: rate of change, weighted average, against a year earlier.

The level and the rate are different things

When inflation falls from 8% to 3%, nothing has become cheaper. Prices are still rising, just more slowly, and the price level is higher than it has ever been. This is **disinflation** and it is what a successful central bank produces.

Deflation is a negative rate: the price level actually falls. It is a different animal and a far worse one, because it raises the real value of debt, encourages households to defer purchases, and cannot be fought by cutting a policy rate below zero by very much.

The gap between the two matters commercially. A household experiencing disinflation still feels that everything is expensive, because it is comparing the level to two years ago rather than the rate to last year. A politician talking about "bringing prices down" is either promising deflation or being imprecise, and it is almost always the second.

How the number is actually made

A statistical agency defines a basket of goods and services meant to represent what a representative household buys, prices it every month, and weights each item by its share of total spending. The published rate is the change in that weighted average.

Four construction details do real work and are worth knowing:

- **The weights are re-estimated** each year from household expenditure surveys, so the basket drifts. A component whose price is rising fast usually loses weight as people buy less of it, which mechanically damps the index.
- **Quality adjustment.** If this year's laptop costs the same as last year's but is twice as fast, the agency records a price *fall*. This is defensible and it is also a large part of why measured inflation feels lower than lived inflation.

- **Housing is handled differently everywhere.** The euro area HICP largely excludes owner-occupied housing costs, while the US CPI includes a large imputed rent component. Comparing the two headline numbers directly is comparing different baskets.
- **Chain-linking.** Indices are spliced year to year rather than held to a fixed basket for decades, which avoids the substitution bias a fixed basket accumulates.

The practical consequence: **CPI, HICP and PCE are not the same measure and do not print the same number.** HICP is the harmonised euro-area measure, and euro inflation swaps and euro-area linkers settle on the *ex-tobacco* variant of it rather than on the headline index, which is a detail worth having right because it is the index a trade actually pays on. US CPI is the measure US linkers settle on. PCE has different weights and a formula that allows more substitution, so it typically runs a few tenths below CPI, and it is the measure the Federal Reserve targets. Knowing which measure a question is about is half of answering it.

Headline and core, and why the gap is arithmetic

Headline inflation is the full basket. **Core** strips the volatile components, conventionally energy and food, and sometimes tobacco and administered prices. Core is not a sanitised number invented to make things look better; it is an attempt to see the persistent component that policy can actually influence, since a central bank cannot do anything about a gas pipeline or a drought.

The size of the gap is pure weighting arithmetic. Take a basket with roughly euro-area shares:

Component	Weight	Rate	Contribution
Energy	10%	+30.0%	+3.00 pp
Food, alcohol, tobacco	20%	+8.0%	+1.60 pp
Non-energy industrial goods	26%	+2.0%	+0.52 pp
Services	44%	+3.0%	+1.32 pp
Headline	100%		**+6.44%**
Core (last two rows)	70%	**+2.63%**	+1.84 pp

Energy is a tenth of the basket and supplies almost half of a 6.44% headline print. Core, the same two bottom rows expressed as a rate on their own 70% sub-basket, is $1.84/0.70 = 2.63\%$, close to target.

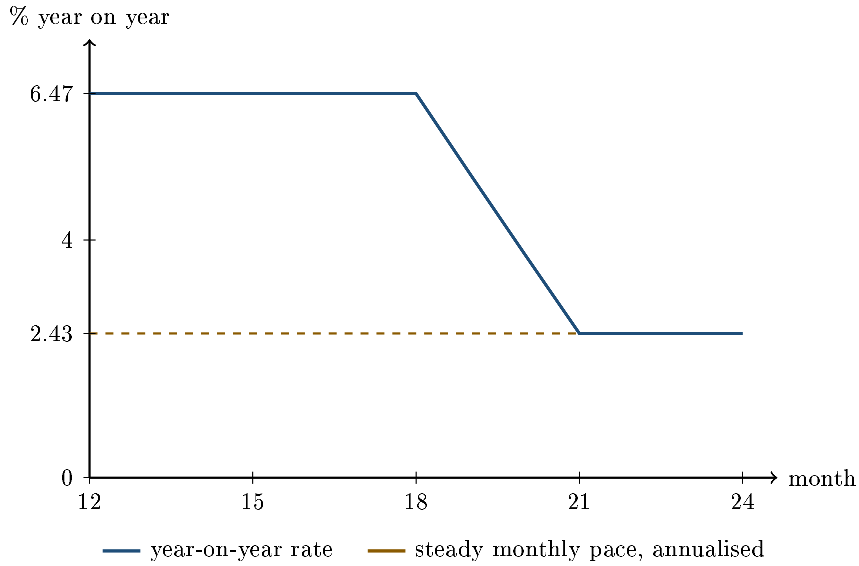
Now leave every other row untouched and let energy go from +30% to −10% the following year. Headline becomes $-1.00+1.60+0.52+1.32 = \mathbf{2.44}\%$ and core does not move at all. Headline has fallen four full points, ended up *below* core, and nothing about the underlying inflation problem has changed.

KEY IDEA

Headline is what households experience and what inflation products settle on. Core is what policy responds to. A trader can be right about the policy path, which tracks core, and still lose on a position that pays on headline. Which of the two a question is really about is usually the whole question.

Base effects: the year-on-year rate has a memory

A year-on-year rate compares this month to the same month a year ago, so it carries whatever happened twelve months ago for exactly twelve months and then drops it. That mechanical property is called a **base effect** and it is responsible for a large share of the moves that get explained on television as something else.



the entire fall over months 19 to 21 is three old months leaving the twelve-month window

WORKED EXAMPLE

Take a price index that rises **0.2% every month**, except for months 7, 8 and 9, where a one-off supply shock pushes it up **1.5% each month**. Nothing else ever happens.

A steady 0.2% a month annualises to $1.002^{12} - 1 = \mathbf{2.43}\%$. The three shock months add $1.015^3 - 1 = 4.57\%$ to the level, of which $1.002^3 - 1 = 0.60\%$ would have happened anyway, leaving $1.015^3/1.002^3 - 1 = \mathbf{3.94}\%$ of genuine excess.

Now read the year-on-year rate month by month:

Month	12 to 18	19	20	21	22	23 to 24
Month-on-month	0.2%	0.2%	0.2%	0.2%	0.2%	0.2%
Year-on-year	6.47%	5.10%	3.76%	2.43%	2.43%	2.43%

The year-on-year rate sits at 6.47% for seven straight months, then falls by more than four points in three months, and **not one monthly print changes anywhere in the table**. Prices are rising at exactly the same 0.2% a month throughout. The whole move is months 7, 8 and 9 leaving the comparison window.

Check the arithmetic at the top and bottom: $1.0647/1.0243 = 1.0394$, which is precisely the 3.94% of excess computed above. The peak and the floor differ by exactly the shock, and by nothing else.

KEY IDEA

When a year-on-year rate moves, the first question is whether anything happened this month or whether something merely stopped being counted. Central banks and forecasters know the base effects a year in advance, which is why a large expected fall in headline inflation can arrive with no policy reaction whatsoever.

Where inflation comes from

Three mechanisms, and a good answer names which one is operating rather than reciting all three.

Demand-pull. Spending runs ahead of what the economy can produce, so prices clear the gap. This is the kind monetary policy is built to handle: raise rates, cool demand, and the pressure subsides. It usually shows up in services and in wages rather than in goods.

Cost-push. An input gets more expensive, typically energy, food or shipping, and the cost passes through. Rates do nothing to a pipeline. A central bank facing pure cost-push inflation is choosing between tolerating a temporary overshoot and inducing a recession to offset a shock it did not cause, which is why these episodes produce the loudest arguments.

Second-round effects. This is where the distinction stops being academic. A cost-push shock becomes persistent if workers bargain for higher wages to restore real income and firms raise prices to restore margins. The initial shock is temporary; the wage-price dynamic it sets off is not. Central banks tolerate the first and fight the second, which is why they watch services inflation and wage growth so closely and why the labour market (employment) is read as an inflation indicator.

Expectations, and what anchoring means

Inflation is partly self-fulfilling. If everyone expects 2%, wage demands and price-setting cluster around 2% and the expectation delivers itself. If everyone expects 6%, the same mechanism delivers 6%. Expectations are therefore not a forecast of inflation so much as an input to it.

Anchored expectations mean long-horizon expectations barely move when a shock hits current inflation: the market believes the shock will be dealt with. That belief is the single most valuable asset a central bank owns, because it lets policy respond gently to a supply shock instead of aggressively. Losing it is expensive to reverse, which is the honest justification for the aggressive tightening cycles that follow large overshoots.

Expectations are read from surveys of households, firms and professional forecasters, and from market prices. The market-implied measure is breakeven inflation, which lives with the instruments in inflation-linked products together with the reasons a breakeven is not a clean expectation: it also contains an inflation risk premium and a liquidity premium.

Interview angle

REAL INTERVIEW QUESTION (JP Morgan · Sales)

Why is inflation high at the moment? (2023)

WHAT THEY TEST / ANSWER

The question is dated by design and the interviewer knows it. What is being tested is whether you can decompose an inflation episode rather than narrate one, so answer with the structure and let the specifics follow.

Decompose it. The 2021 to 2023 episode had all three mechanisms running at once, and the useful answer separates them. Demand: pandemic support left households with unusual savings and demand rotated violently from services into goods. Supply: shipping, semiconductors and then the energy shock after the invasion of Ukraine. Second round: by 2023 the energy shock was reversing, headline was falling fast on base effects, and yet core services inflation and wage growth were still elevated, which is exactly the signature of a shock that has propagated into the wage-price mechanism.

The 2023 detail that shows you were paying attention is that headline and core had separated, with headline falling sharply and core sticky. That gap is the base effect from the previous year's energy prices leaving the window, and it explains why central banks kept tightening while the headline number was collapsing. A candidate who says only "energy prices" has explained the 2022 headline and none of the 2023 problem.

Then be honest about the disagreement, because it was genuine and unresolved at the time: one camp read it as a supply shock that would unwind on its own, the other as demand inflation that would persist without policy. Say which way you lean and what evidence would change your mind. The answer to look for is something observable, such as whether wage growth converges to productivity growth plus target.

REAL INTERVIEW QUESTION (Natixis · Fixed Income)

What is the current inflation rate in France and in the euro area?

WHAT THEY TEST / ANSWER

This is a diligence check, not an economics question, and there is no clever way around it: if you want a rates job you are expected to know the last print. Look it up before every interview.

Answer it in the shape a desk would use rather than as a single number.

Name the measure and the month. Euro-area inflation means **HICP**, published by Eurostat, with a flash estimate at the end of the reference month and a final release a couple of weeks later. For France, distinguish the national CPI from the French HICP, which are different series with different baskets, and note that the harmonised one is what euro aggregates and euro inflation products use.

Give headline and core, since the gap between them is the interesting part and the whole point of the previous question.

Give the direction and the driver, not just the level: whether the last print was above or below expectations, and what moved it. A number with no direction attached signals that you memorised it that morning.

Then the honest caveat that improves the answer: France has run below the euro-area average through much of the recent episode, largely because of the electricity price shield and regulated tariffs, so the national number understates the underlying pressure relative to neighbours. That kind of observation is what the question is fishing for.

REAL INTERVIEW QUESTION (Natixis · Fixed Income)

Why does the Livret A rate not always follow the level of inflation?

WHAT THEY TEST / ANSWER

Because the Livret A is not indexed to inflation. It is set by a *formula* that references inflation, and every feature of that formula puts distance between the two.

The structure of the rule, which is what to describe rather than the exact coefficients, since those have been revised more than once. The rate is computed from an average of short-term euro money-market rates and an average of euro-area consumer price inflation excluding tobacco, taken over the preceding months, then rounded, subject to a floor, and reset on a fixed calendar rather than continuously.

Four sources of divergence follow from that, and they all push the same way:

- **Averaging** over past months means a spike in inflation enters the formula diluted by the quiet months around it.
- **Two fixed reset dates** a year mean the rate can be stale by up to six months while inflation moves.
- **The money-market leg** pulls the result toward the policy rate, which is deliberately below inflation early in a tightening cycle.
- **The floor and the rounding** truncate the result, and the floor binds precisely when inflation is very low.

Then the part that is not arithmetic at all, and the reason this is a good question: the government retains discretion to deviate from the formula, and has used it in both directions. It has capped the rate below the formula to protect the cost of the social housing that Livret A deposits finance, and it has held the rate above the formula when the formula would have cut it. The Livret A is an instrument of housing policy that happens to look like a savings account, so a purely mechanical answer misses what actually sets the rate.

Close on what it means for a saver: through a high-inflation period the real return is negative, which is the standard argument for inflation-linked products as the instrument that actually delivers indexation rather than merely referencing it.

INTERVIEW TRAP

Five inflation errors. (1) Saying prices will fall when you mean inflation will fall, which confuses disinflation with deflation. (2) Explaining a year-on-year move without checking whether it is a base effect, which is the single most common commentary mistake. (3) Comparing US CPI with euro-area HICP as though they measured the same basket; housing alone makes them incomparable. (4) Treating core as the number that matters to a position, when linkers and inflation swaps settle on headline. (5) Quoting a breakeven as the market's inflation forecast, when it also contains a risk premium and a liquidity premium.

Where this goes next

What a central bank does with this number is monetary policy, and who does it with what mandate is major central banks. How it is traded, hedged and priced is inflation-linked products, which owns the real versus nominal decomposition, breakevens and the whole instrument set. The labour market that turns a shock into a persistent problem is employment, the release calendar you watch it on is macro calendars, and the currency channel that imports it is central banks and FX policy.

3.3 Employment

sales · trading · structuring core rates · fx · cross-asset

Employment matters to a market for one reason above all others: it is half of the Federal Reserve's mandate, so it is one of the two variables that determine policy, and policy is what moves the rates curve. It matters for a second reason too, which is that the labour market is where an inflation shock either becomes entrenched or does not.

This section is about how to read the numbers rather than about labour economics. The release itself, as an event a desk trades, belongs to macro calendars.

The measures, and what each one hides

The unemployment rate is the share of the *labour force* without a job and looking for one. The definition carries its own trap: a person who stops looking leaves the labour force entirely, so the unemployment rate falls. It can therefore improve for the worst possible reason, and the fix is to read it alongside the participation rate.

The participation rate is the share of the working-age population in the labour force. It is the denominator behind the headline and it moves slowly for demographic reasons and quickly in a shock.

Payrolls count jobs rather than people, from a survey of employers. The unemployment rate comes from a separate survey of households. The two surveys measure different things by different methods and routinely disagree for months at a time, which is not a scandal but is a reason never to build a story on one month of either.

Wages are the bridge to inflation and are covered below.

KEY IDEA

Break-even payrolls is the number to know, because it converts a headline into a signal. The economy needs enough new jobs each month to absorb labour force growth; below that the unemployment rate rises, above it the rate falls.

With a labour force of about 168 million growing at 0.5% a year,

$$\frac{168{,}000{,}000 \times 0.005}{12} \approx \mathbf{70{,}000} \text{ jobs a month.}$$

So a print of 100,000 is a tightening labour market and a print of 50,000 is a loosening one, even though both are positive. Anyone reading payrolls as "jobs were added, therefore good" has skipped the only step that matters, and the break-even itself moves with demographics and immigration, which is why the same headline meant something different a decade ago.

Two relationships worth knowing, one of them shaky

Okun's law links unemployment to growth: unemployment falls when the economy grows faster than potential, with a coefficient of roughly a half in the United States,

$$\Delta u \approx -0.5\,(g - g^*).$$

So with potential growth at 2% and actual growth at 4%, unemployment falls by about 1 percentage point over the year. It is an empirical regularity rather than a law, and its usefulness is as a consistency check: a forecast with strong growth and rising unemployment needs an explanation.

The Phillips curve links unemployment to inflation, and it is where candidates get into trouble by stating it too confidently. The original version, that low unemployment produces high inflation, broke down badly in the 1970s, when high inflation and high unemployment coexisted. The modern version is *expectations-augmented*: inflation depends on expected inflation plus a term in the gap between unemployment and its non-accelerating rate. That formulation survives, but the slope is flat and unstable, so the honest position is that the relationship exists, is weak in normal times, and steepens when the labour market is very tight.

The good interview answer is not to pick a side but to say what the disagreement is about, since a central bank that believes the curve is steep will tighten pre-emptively on a strong labour market and one that believes it is flat will wait for inflation to appear.

Wages, and the arithmetic a central bank actually uses

The link from employment to inflation runs through **unit labour costs**, which is wage growth net of productivity growth:

$$\text{unit labour cost growth} \;=\; \text{wage growth} - \text{productivity growth.}$$

If wages grow 4% and productivity grows 1%, unit labour costs grow 3%, which is consistent with roughly 3% inflation at a constant profit share. That single subtraction is why a central bank is relaxed about 4% wage growth in a high-productivity economy and alarmed by it in a low-productivity one, and it is a far better answer than "wage growth is inflationary".

It is also the reason wage data can outweigh the headline jobs number in a release. A strong count with soft average hourly earnings tells a central bank that supply is expanding rather than that demand is overheating, and that reading is disinflationary.

Employment lags, and that changes how you trade it

INTUITION

Employment is a **lagging** indicator, and the lag is structural rather than statistical. Firms hoard labour into a downturn because hiring and training are expensive and they expect the weakness to be temporary, so they cut late. On the way out they add hours and temporary staff before they add permanent employees, so they hire late as well. The consequence is that the unemployment rate typically peaks *after* a recession has ended.

The trading consequence follows directly: a strong labour market is not evidence that a slowdown is not coming, and the market knows this, which is why bond yields can fall on strong employment data when the rest of the picture is deteriorating. Anyone who reasons from employment to the cycle without noting the lag will be confidently wrong at exactly the turning points.

The best-known attempt to extract a timely signal from a lagging series is the **Sahm rule**: when the three-month average unemployment rate rises half a percentage point above its lowest three-month average of the previous year, a recession has typically already begun. It is worth knowing because it is quoted, and worth qualifying for the same reason: it is a historical regularity with a small sample, and it identifies rather than forecasts.

INTERVIEW TRAP

1. **Reading a positive payroll number as a strong labour market.** Compare it against break-even, which is around 70,000 a month in the United States and moves with demographics.
2. **Celebrating a falling unemployment rate without checking participation.** The rate falls when people give up looking, which is the opposite of good news.
3. **Ignoring revisions.** The prior two months are revised in the same release and the revision frequently outweighs the headline.
4. **Stating the Phillips curve as though it were stable.** The modern expectations-augmented version is flat and unstable, and the interesting content is what the disagreement about its slope implies for policy.
5. **Treating wage growth as inflationary on its own.** Subtract productivity. Unit labour costs are what matters.
6. **Using employment to time the cycle.** It lags by construction, and it is strongest just before it is not.

VISUAL / CODE TO BUILD: Interactive

A break-even panel. Sliders set the labour force size and its growth rate, and the panel draws the monthly break-even payroll number as a horizontal line with a series of hypothetical prints as bars above and below it, colouring each bar by whether it tightens or loosens the labour market. Dragging labour force growth from 0.3% to 0.8% moves the line from 42,000 to 112,000, which shows that the same print can be either signal depending on demographics that nobody mentions on the day.

A second panel plots unemployment against the output gap with the Okun line through it, and lets the reader move growth and potential growth independently to see the implied path of unemployment, with a warning flag when the chosen combination is Okun-inconsistent.

A third overlays wage growth and productivity growth as two lines with the gap shaded as unit labour cost growth, alongside an inflation line. Raising wage growth while raising productivity by the same amount leaves the shaded band unchanged and the inflation line flat, which is the section's subtraction made visual and is the point most readers will not otherwise absorb.

3.4 Monetary policy

sales · trading · structuring | core | rates · cross-asset

Every asset is priced off a discount rate, and the short end of that discount rate is set by a committee. That is why monetary policy is not a macroeconomics topic that traders tolerate, it is the single most important input to every market, and why a front-office interview will always find its way to central banks. The good news is that the material is finite: a mandate, a corridor, a toolkit, and a reaction function. This section covers those four things and how markets read them. The channels through which policy reaches the real economy are developed in monetary transmission mechanisms, the institutions themselves in major central banks, and communication in forward guidance.

The mandate, and why the target is 2%

Most central banks have price stability as their primary objective, and most define it as inflation of about 2% over the medium term. The Federal Reserve additionally has a statutory employment objective, which is why it is called a dual mandate, and since 2008 every major central bank has acquired financial stability as a de facto third objective.

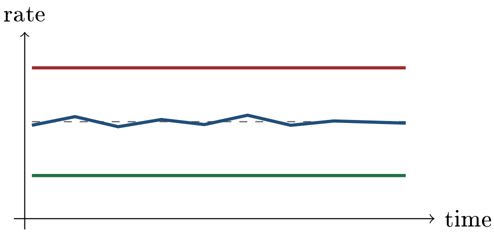
Why 2% rather than zero, which is the question interviewers use to separate people who have thought about it from people who have memorised it? Four reasons, and they compound.

- **Room to cut.** A positive inflation target means a positive nominal rate in normal times, which leaves space to cut in a recession before hitting the effective lower bound.
- **Deflation is worse than mild inflation.** Falling prices raise the real value of debt, encourage consumers to postpone purchases, and are hard to escape once expectations adjust.
- **Downward nominal wage rigidity.** People resist nominal pay cuts, so a little inflation lets relative wages adjust across sectors without anyone's pay packet visibly falling.
- **Measurement.** Price indices plausibly overstate true inflation slightly, so a measured 2% is closer to genuine stability than a measured zero.

How a policy rate is actually implemented

A committee does not set the rate at which banks lend to each other, it creates conditions that force that rate to a chosen level. The mechanism is a **corridor**.

- A **deposit facility** pays interest on reserves left at the central bank. No bank lends below this, so it is the **floor**.
- A **lending facility** lends against collateral at a penalty rate. No bank borrows above this, so it is the **ceiling**.
- **Open market operations** adjust the quantity of reserves so that the market rate settles where the committee wants it inside the corridor.



Two implementation regimes matter, and confusing them dates a candidate badly. In a **scarce reserve** or corridor system, the central bank supplies just enough reserves that the market rate sits near the middle of the corridor, so the quantity of reserves is the active lever. In a **floor** system, which is where the major central banks have operated since their balance sheets grew, reserves are abundant, so the market rate is pinned just above the deposit rate and the *deposit rate itself* becomes the policy instrument. This is why the ECB's deposit facility rate, not its main refinancing rate, is the number that steers euro money markets today, and why the Federal Reserve steers with interest on reserve balances and its overnight reverse repo facility inside a target range for the federal funds rate.

The toolkit

Conventional. The policy rate, open market operations and standing facilities as above, plus reserve requirements, which are largely dormant in developed markets but still an active tool in some emerging ones.

Unconventional, in the rough order they were adopted after 2008:

- **Liquidity provision**: longer-term refinancing against a wider collateral pool, which addresses a funding problem rather than a rate problem.
- **Forward guidance**: shaping expectations of the future path, which is powerful precisely because asset prices depend on the path and not the level, see forward guidance.
- **Quantitative easing**: large-scale asset purchases, treated in detail below.
- **Negative policy rates**: charging on reserves, used by the ECB from 2014 to 2022 and by others, on the argument that the effective lower bound is somewhat below zero because holding physical cash has its own costs.
- **Targeted lending**: cheap funding conditional on banks actually lending it on.
- **Yield curve control**: committing to buy whatever is needed to hold a chosen maturity at a chosen yield, the Bank of Japan's approach, which is a price target rather than a quantity target.

Quantitative easing, precisely

> **KEY IDEA**
>
> QE is an **asset swap**, not a transfer of cash to the public. The central bank creates reserves, which are a liability only usable between banks and the central bank, and uses them to buy bonds. Its balance sheet grows on both sides: bonds on the asset side, reserves on the liability side. The private sector ends up holding fewer long-dated bonds and more reserves or deposits. Nobody's cash machine dispenses new money as a direct consequence, which is the single most common misunderstanding of the whole policy.

Why do yields fall? Four channels, and a strong answer names more than one.

1. **Duration or portfolio-balance channel.** By removing long-dated bonds from private hands, the central bank reduces the interest rate risk that investors must be paid to bear, so the *term premium* compresses. This is the main mechanical effect on the long end.
2. **Signalling channel.** Committing to a large purchase programme signals that policy rates will stay low for a long time, which pulls down the expectations component of long yields.
3. **Portfolio rebalancing.** Investors who sold bonds must hold something, so demand spills into credit, equities and other risk assets, compressing spreads and raising valuations. This is why QE is felt as much in risk assets as in government bonds.
4. **Market functioning.** In a crisis the announcement of a buyer of last resort restores liquidity, which alone can collapse a yield spike, as in March 2020.

Why did it not produce the inflation many predicted? Because reserves are not credit. QE raises the monetary base, but broad money and inflation depend on banks and borrowers choosing to expand lending, and after a balance-sheet recession they did not. The 2021 to 2023 inflation episode is not a counterexample that vindicates the simple money-printing story either: it followed a supply shock combined with unprecedented *fiscal* transfers that went directly to households, which is a different mechanism from a reserve swap. Being able to make that distinction cleanly is a strong signal in an interview.

QT is the reverse. Passive QT lets maturing bonds roll off without reinvestment; active QT sells them. It drains reserves, and the risk is overshooting the point where reserves stop being abundant: in September 2019 US repo rates spiked sharply when reserves had fallen further than anyone realised, and the Fed had to inject liquidity immediately, see repo markets. QT is therefore run slowly, predictably, and with a willingness to stop.

The reaction function, with numbers

Markets do not trade the current policy rate, they trade the *path*. To forecast a path you need a view of the committee's reaction function, and the standard benchmark is the Taylor rule:

$$i \;=\; r^* + \pi + 0.5\,(\pi - \pi^*) + 0.5\,y,$$

where i is the nominal policy rate, r^* the neutral real rate, π current inflation, π^* the target, and y the output gap in percent.

WORKED EXAMPLE

Take $r^* = 0.5\%$, a target $\pi^* = 2\%$, current inflation $\pi = 3\%$ and an output gap of -1% (the economy running below potential):

$$i = 0.5 + 3 + 0.5(3 - 2) + 0.5(-1) = 0.5 + 3 + 0.5 - 0.5 = 3.5\%.$$

Now let inflation rise to 4% with the same gap: $i = 0.5 + 4 + 0.5(2) - 0.5 = 5.0\%$. So one extra point of inflation calls for **one and a half** points of policy rate.

That is not an accident, it is the **Taylor principle**: the coefficient on inflation exceeds one, so the *real* policy rate rises when inflation rises. If it did not, higher inflation would loosen policy in real terms and the process would be unstable. Two caveats worth stating out loud: r^* is unobservable and estimates of it move, which is where much of the genuine disagreement about policy lives, and no committee mechanically follows the rule, so it is a benchmark for judging whether policy is loose or tight rather than a forecast.

How markets read it

KEY IDEA

Hawkish and dovish are relative to what was already priced. A rate rise that the market had fully expected does nothing on the day; a rise of the expected size accompanied by an unexpected signal about the path moves everything. The tradeable object is the forward curve of expected policy rates, read off OIS, see OIS, LIBOR, SOFR and ESTR, and what moves is the **terminal rate** and the timing of the turn. This is why a candidate who says "rates up, equities down" without asking what was priced has not answered the question.

The first-order asset reactions to a genuine hawkish surprise: front-end yields rise and the curve bear-flattens, since the front end is anchored to policy while the long end also reflects growth; equity valuations fall through the discount rate, with long-duration growth names hit hardest; the currency strengthens on the rate differential; gold falls as real rates rise; and credit spreads widen if the market reads the tightening as a threat to growth rather than a response to strength. The mechanics of each are in yield curves and monetary transmission mechanisms.

REAL INTERVIEW QUESTION (Deutsche Bank · Rates)

How do central banks act on the economy?

WHAT THEY TEST / ANSWER

Answer in two layers: what they control directly, and how that reaches the economy. Directly they control only a very short-term interest rate and the size and composition of their own balance sheet. They implement the rate through a corridor, paying interest on reserves as a floor and lending at a penalty rate as a ceiling, and adjusting reserves through open market operations. From there, the transmission runs through several channels, and naming them is the substance of the answer: the **interest rate channel** into borrowing costs for households and firms; the **credit channel** through banks' willingness and capacity to lend; the **asset price and wealth channel**, since discount rates move equities, credit and housing; the **exchange rate channel**, where rate differentials move the currency and therefore import prices and competitiveness; and the **expectations channel**, which is the most important of all because what a committee is expected to do next matters more for asset prices than what it does today. Then add the two qualifications that make the answer sound professional: transmission works with long and variable lags, commonly a year or more, so policy is set on forecasts rather than on current data, and the tools are strong against a demand shock and weak against a supply shock, since no interest rate simultaneously fixes falling output and rising prices. The channels are developed in monetary transmission mechanisms.

REAL INTERVIEW QUESTION (Morgan Stanley · QIS)

What are the financial instruments of central banks to control the economy?

WHAT THEY TEST / ANSWER

Give a structured inventory rather than a list, splitting conventional from unconventional. **Conventional**: the policy rate itself, standing facilities (a deposit facility that floors the market rate and a lending facility that caps it), open market operations, usually short-dated repo against eligible collateral, and reserve requirements, now mostly dormant in developed markets. **Unconventional**: large-scale asset purchases, which is QE, buying government and sometimes corporate bonds to compress term premia and spreads; targeted long-term lending to banks conditional on onward lending; negative policy rates; forward guidance, which is a tool even though it costs nothing; yield curve control, which targets a yield rather than a quantity; FX intervention and swap lines with other central banks, which matter enormously in a dollar-funding squeeze; and emergency backstop facilities that make the central bank a buyer or lender of last resort in a specific market. Since this was asked on a quantitative desk, add the observation that ties it together: the instruments are really just two levers, the *price* of central bank money and the *quantity and composition* of the central bank's balance sheet, and every tool above is one of those two applied at a different point of the curve or to a different asset. Mentioning that the tradeable expression of all of this is the OIS curve, which is where the market's expected policy path is directly observable, shows how the macro connects to a book.

REAL INTERVIEW QUESTION (HSBC · Rates)

How does Quantitative Easing work?

WHAT THEY TEST / ANSWER

Be mechanical first, then give the channels, then the honest caveats. Mechanically, the central bank creates reserves and buys assets, usually long-dated government bonds, from banks and investors in the secondary market. Its balance sheet expands on both sides, bonds on the asset side and reserves on the liability side, and the private sector ends up holding fewer long bonds and more reserves and deposits: it is an **asset swap**, not a distribution of cash to households. Yields fall through four channels: compression of the **term premium**, because duration risk has been removed from private portfolios; a **signalling** effect, because a large programme implies policy rates stay low for longer; **portfolio rebalancing**, since sellers must hold something and spill into credit and equities; and **market functioning**, because a committed buyer restores liquidity in a dislocation, which is what made the March 2020 announcements so immediately effective. Then the caveats that distinguish the answer. QE did not produce the inflation many expected in the 2010s because reserves are not credit and broad money depends on banks lending, so the transmission was weak in a balance-sheet recession. It has distributional and financial-stability side effects, since it lifts asset prices, and it becomes entangled with fiscal policy when the central bank is the dominant buyer of its government's debt. And unwinding it, QT, drains reserves and must be done slowly, since the September 2019 repo spike showed how abruptly money markets tighten when reserves stop being abundant.

REAL INTERVIEW QUESTION (Natixis · Credit)

Which ECB rate are we talking about when we refer to negative yields (the deposit rate)?

WHAT THEY TEST / ANSWER

The **deposit facility rate**, and the question is really testing whether you know why that is the relevant one. The ECB has three policy rates: the deposit facility rate, the rate on the main refinancing operations, and the marginal lending facility rate. In a system with abundant excess liquidity, which is where the euro area has been since the asset purchase programmes, banks collectively have more reserves than they need, so the marginal use of a euro is to leave it at the deposit facility. That makes the deposit facility rate the effective floor *and* the binding policy rate, with short-term market rates such as ESTR trading a few basis points *below* it rather than near the refinancing rate. Below, not around: ESTR measures unsecured borrowing from lenders such as money market funds who have no access to the deposit facility themselves, so they accept slightly less than a bank can earn by simply leaving the cash at the central bank. The ECB took it negative in June 2014 and to -0.50% before exiting positive territory in July 2022, and that negative deposit rate is what dragged short-dated euro yields, and eventually a large stock of euro government bonds, below zero. Two additions that show depth: the ECB reviewed its operational framework in 2024 and narrowed the spread between the refinancing and deposit rates, which matters for how tightly market rates track the floor as excess liquidity declines; and on a credit desk the practical consequence of a negative floor is that it compresses the entire discount curve, pushes investors along the credit and duration spectrum in search of yield, and forces documentation questions about zero floors in floating-rate notes and loans.

REAL INTERVIEW QUESTION (ECB · Market Risk)

Which assets are bought under QE: government, corporate, ABS, covered bonds?

WHAT THEY TEST / ANSWER

All four, and the interesting half of the answer is why the list grew rather than what is on it. The Eurosystem's programmes were added in that order as each previous one ran into its own limit: covered bonds first, then asset-backed securities, then government and agency and supranational paper, which is by far the largest, then corporate bonds, and finally a pandemic programme that covered the same universe with more flexibility on how much of each was bought and when.

Then give the three constraints that shape it, because they are what a risk seat cares about. **Eligibility**: an asset must meet the same kind of credit-quality, currency and issuance criteria used elsewhere in the framework, which is what excluded the debt of a member state below the quality threshold. **Allocation**: sovereign purchases are made broadly in proportion to each country's share in the central bank's capital, so the programme is a monetary operation rather than a targeted transfer to one government, and issue and issuer limits cap how much of any one bond can be held. **Risk sharing**: only part of the sovereign portfolio sits on the joint balance sheet, with the rest held by national central banks at their own risk, which is the compromise that made a monetary union's QE possible at all.

Close on the principle that ties the list together, since that is what is really being tested. Moving from government paper to covered bonds to ABS to corporates moves the central bank progressively further out the credit spectrum, so each step buys more transmission and takes more credit risk onto a public balance sheet. That is why the corporate and ABS programmes are small relative to the sovereign one, and why the boundary between monetary and fiscal policy is argued about at exactly this point.

INTERVIEW TRAP

Four traps. First, **"QE prints money so it must be inflationary"**: it swaps assets for reserves, and reserves become inflationary only if they turn into credit and spending, which is why the 2010s did not look like the 2021 to 2023 episode. Second, **confusing the policy rate with market rates**: the committee sets an overnight rate, while a mortgage or a corporate bond yield depends on the whole curve, term premia and credit spreads. Third, **ignoring what was priced**: a hike is not automatically bad for equities, since only the surprise relative to the forward curve moves markets. Fourth, **treating balance sheet size as the stance**: composition and the expected *path* matter as much as the level, which is why an announced slowdown in purchases can tighten conditions before a single bond is sold.

VISUAL / CODE TO BUILD: Interactive

A policy cockpit in three panels. First, a **corridor simulator**: the reader sets the deposit and lending rates and the quantity of reserves, and watches the simulated overnight rate settle inside the corridor, then switches from a scarce-reserve to an abundant-reserve regime and sees the market rate pin itself to the floor, which makes the modern framework obvious rather than asserted. Second, a **Taylor rule calculator** with sliders for the neutral real rate, inflation, the target and the output gap, plotting the prescribed rate against the actual policy rate through history so the reader can see when policy was loose or tight, with the Taylor principle visible as the slope of prescribed rate against inflation exceeding one. Third, an **expectations panel**: the current OIS-implied policy path against a user-drawn path, with a readout of what a hawkish or dovish surprise of a given size implies for two-year yields, the equity discount rate and the currency, so the reader internalises that the surprise, not the level, is what trades.

Where this goes next

How these decisions reach the real economy is monetary transmission mechanisms, and how they are communicated is forward guidance. The institutions and their differing mandates are in major central banks, the variable being targeted is inflation, and the cycle policy is responding to is economic cycles. The tradeable expression is the curve, in yield curves and OIS, LIBOR, SOFR and ESTR, with the money-market plumbing in repo markets.

3.5 Monetary Transmission Mechanisms

sales · trading · structuring | core | rates · fx · cross-asset

A central bank controls exactly one thing: the rate at which banks lend to each other overnight. Everything a monetary policy decision is supposed to achieve, and everything it does to asset prices, happens through the chain that connects that one rate to spending decisions made by people who have never heard of it. That chain is the transmission mechanism, and understanding it is what separates a candidate who can say the ECB cut from one who can say what the cut will do.

Monetary policy covers the tools and what a central bank is trying to achieve. This section is about how the tools reach the economy, where the chain is fast, and where it breaks.

The six channels

KEY IDEA

1. **Interest rate.** The policy rate moves the curve, which moves the cost of borrowing for firms and households, which moves investment and consumption. The textbook channel, and the slowest.

2. **Credit and bank lending.** Rates change bank funding costs and the value of bank capital, which changes lending standards and volumes. This is the channel that fails in a banking crisis, which is why a rate cut can do nothing at all.

3. **Balance sheet.** Higher rates reduce the value of collateral, which tightens the borrowing constraints of the firms and households that were already constrained. This amplifies the first two channels and is why policy bites hardest on the most leveraged.

4. **Asset prices and wealth.** A lower discount rate raises equity and property values, and people who feel wealthier spend more.

5. **Exchange rate.** The rate differential moves the currency, which moves net exports and, faster than anything else, import prices. This is the quickest channel of the six and the dominant one in a small open economy.

6. **Expectations.** What the central bank says about the future path moves long rates today, at no cost and with no delay, which is the entire premise of forward guidance. It works until credibility is spent, and then nothing works.

Two channels worth quantifying

The reason to put numbers on transmission is that it explains why the same rate move is trivial in one economy and violent in another.

WORKED EXAMPLE

The mortgage channel, which is arithmetic rather than economics. Take a thirty-year mortgage. The monthly payment per 100,000 of principal is

$$P\,\frac{m}{1-(1+m)^{-360}}, \qquad m=\frac{r}{12}.$$

Mortgage rate	Monthly payment per 100,000
3%	421.60
4.5%	506.69
6%	599.55

Going from 3% to 6% raises the payment by **42%**. Read it the other way, which is how a buyer experiences it: at a fixed affordable monthly payment, the principal you can borrow falls by

$$1-\frac{421.60}{599.55} \;=\; \mathbf{29.7}\%.$$

A doubling of the mortgage rate removes almost a third of every buyer's purchasing power, with no change in their income and no change in the housing stock. That single number is most of what a central bank means by "policy is transmitting", and it is why housing is the most rate-sensitive sector in every economy.

WORKED EXAMPLE

The interest rate channel, amplified by leverage. A company with EBITDA of 100, depreciation of 30 and floating-rate net debt at 5%. Raise the rate by 100 basis points:

Net debt	Interest, before and after	Pre-tax profit	Change
2× EBITDA	10 → 12	60 → 58	−3.3%
4× EBITDA	20 → 24	50 → 46	−8.0%
6× EBITDA	30 → 36	40 → 34	−15.0%

The same 100 basis points is a rounding error at two turns of leverage and a fifth of the profit at six. This is the balance sheet channel made arithmetic, and it is why tightening shows up first in high-yield credit and in private equity rather than in the index: policy does not hit the average company, it hits the marginal borrower.

How QE transmits, which is a different question from what it is

REAL INTERVIEW QUESTION (HSBC · desk not recorded)

How does the mechanism of quantitative easing work?

WHAT THEY TEST / ANSWER

Say what it is in one line and then spend the answer on the channels, because the question asks about the mechanism.

One line. The central bank creates reserves and buys assets, mostly government bonds, from the private sector. It is not printing money in the sense people mean: it swaps one asset, a bond, for another, a reserve balance.

Then the four channels, in order of how much work each does.

Portfolio balance, or duration extraction. This is the main one. The central bank removes duration from a market that has to hold a fixed amount of it, so the remaining holders are pushed out of government bonds into corporate bonds, equities and everything else. The term premium falls, long yields fall, and risk assets rise. Note that this works on the *term premium* rather than on rate expectations, which is why QE flattens the curve rather than shifting it.

Signalling. Buying long bonds is an expensive commitment, so it credibly says policy rates will stay low for a long time, which lowers the expectations component of long yields as well.

Market functioning. In a stress event the central bank is the buyer of last resort, and simply being present restores liquidity. This is the channel that matters in March 2020 and it is largely absent in a calm QE programme.

Bank lending. Weakest in practice. Reserves pile up at the central bank rather than being lent, which is why the money multiplier story that predicted hyperinflation from QE did not happen.

Close with the limits, since that is the follow-up. QE is subject to diminishing returns because each purchase extracts less scarcity than the last; it has distributional consequences, since it works by raising asset prices and assets are unequally held; and it is very much easier to start than to stop.

REAL INTERVIEW QUESTION (HSBC · desk not recorded)

What would be the impact on FX, rates and equity markets if the ECB stopped its QE?

WHAT THEY TEST / ANSWER

Take the three markets in turn, and say the thing that governs all of them first: what matters is the change relative to what is *already priced*, so the honest answer distinguishes a well-telegraphed taper from a surprise.

Rates. Long yields rise, and specifically the *term premium* component rises, because the buyer that had been absorbing duration steps away and the private sector has to hold it again. So the curve steepens. Peripheral spreads widen more than core, since the programme had been compressing them disproportionately.

FX. The euro strengthens, through the rate differential and through the signal about the direction of policy. This is the fastest of the three reactions and it typically front-runs the others.

Equities. Two effects pointing in opposite directions, and naming both is the answer. The discount rate rises, which is negative and hits long-duration growth stocks hardest. But a central bank stops QE because the economy no longer needs it, so growth expectations are improving, which is positive. Which dominates depends on why the programme is ending: tapering into a recovery is broadly fine for equities, tapering into an inflation problem is not.

Then the sentence that gets you the follow-up. The 2013 taper tantrum is the reference point, and its lesson is that the damage came from the *surprise* rather than from the policy. A taper that is communicated well in advance is largely in the price by the time it happens, which is why central banks now spend months preparing markets for a decision that then moves nothing on the day.

Lags, and why a central bank looks wrong at the turn

Transmission is slow and the delay is uneven, which is the origin of the phrase *long and variable lags*. The exchange rate moves within minutes; asset prices within days; bank lending standards within a quarter or two; investment and hiring within a year; inflation last of all, with the peak effect commonly put at six to eight quarters after the move.

INTUITION

The consequence is that a central bank must act on a forecast rather than on the data in front of it, and will therefore always appear to be tightening into an economy that still looks fine and easing into one that still looks bad. That is not incompetence, it is the only thing the lag structure permits. It is also why the market pays far more attention to the projected path than to the current decision: the decision affects an economy two years away, and the path is what tells you about the two years.

REAL INTERVIEW QUESTION (CACIB · Fixed Income)

What can happen, and what is the risk, if policy rates are too high?

WHAT THEY TEST / ANSWER

Answer in ascending order of severity, because the question is really asking whether you understand that overtightening has its own failure modes rather than merely being cautious.

Demand. Rate-sensitive spending contracts first, housing and capital expenditure before consumption, through the arithmetic above.

Credit. The marginal borrower cannot refinance. Defaults rise first in high yield and in floating-rate borrowers, and the leverage table above shows why: the same move is trivial at two turns and severe at six.

Financial stability, which is the risk the question is pointing at. High rates mark down the value of every long-dated asset held at amortised cost, and an institution that funded long assets with short liabilities discovers the loss only when it is forced to sell. That is the 2023 regional bank episode, and it is the mechanism to name, because it is how a rate level that is merely restrictive becomes a solvency event.

Sovereigns and the fiscal channel. Higher rates raise debt service, which is a growing constraint for high-debt sovereigns and can force fiscal tightening into a slowdown.

Then the asymmetry, which is the real answer. The costs of overtightening are visible, immediate and reversible, since the central bank can cut. The costs of undertightening are slow and much harder to reverse, because once inflation expectations de-anchor it takes a far deeper recession to bring them back. That asymmetry is why central banks deliberately accept the risk of going slightly too far, and saying so is what turns this from a list of bad outcomes into an explanation of policy.

Where the chain breaks

- **The effective lower bound.** Below roughly zero the rate channel stops working and the central bank must use the balance sheet or its words instead.
- **Impaired banks.** If banks are undercapitalised they will not lend regardless of the rate, and the credit channel is severed. Japan in the 1990s and the euro area after 2011 are the standard examples.
- **Fixed-rate mortgage stock.** An economy where households hold thirty-year fixed mortgages transmits a rate rise very slowly, since only new buyers are affected, while an economy of floating or short-fixed mortgages transmits it within a year. The same central bank action has completely different force in the United States and in the United Kingdom, and it is the single most useful cross-country distinction to know.
- **A currency peg.** A country that pegs imports the anchor country's policy and has no independent transmission at all.

INTERVIEW TRAP

1. **Assuming a rate cut stimulates.** It stimulates through channels, and if the banks are impaired or borrowers are already at their constraint, it does very little.
2. **Ignoring the lag.** Policy set today is aimed at an economy eighteen months away, so judging a decision against this month's data is judging it against the wrong thing.
3. **Treating all economies as equally rate-sensitive.** The mortgage structure alone changes the speed of transmission by years.
4. **Confusing QE with printing money.** It swaps a bond for a reserve. The channel is the term premium, not the money supply, which is why the predicted inflation did not arrive.
5. **Reasoning about the average borrower.** Policy hits the marginal one, which is why the effects appear in high yield and in housing long before they appear in aggregate data.

VISUAL / CODE TO BUILD: Interactive

A transmission timeline. The reader applies a rate move and the panel animates it propagating through the six channels along a horizontal time axis, each channel drawn as a bar that fills at its own speed: the exchange rate within minutes, asset prices within days, lending standards within a quarter, investment within a year, inflation at six to eight quarters. Seeing the bars fill at wildly different rates is the whole of "long and variable lags" in one animation.

A second panel is the affordability calculator: a mortgage rate slider against monthly payment and maximum principal at a fixed budget, so the reader can drag 3% to 6% and watch borrowing capacity fall by 29.7% without anything else changing.

A third is the leverage panel: net debt as a multiple of EBITDA on one axis and the profit impact of a 100 basis point move on the other, drawn as a curve with the reader's chosen leverage marked. Dragging leverage from two to six turns a 3% profit hit into a 15% one, which is the balance sheet channel made visible and explains why credit spreads react to policy long before equity indices do.

3.6 Fiscal Policy

sales · trading · structuring | core | rates · fx · cross-asset

Fiscal policy is the government's two levers, what it spends and what it taxes. For a markets desk it matters in two distinct ways that are worth keeping separate: as a driver of growth and inflation, which is economics, and as the supply of government bonds and the credibility behind them, which is a rates trade. The second is the one that moves prices in a morning.

Monetary policy is set by an independent technocracy on a schedule; fiscal policy is set by a legislature on a political timetable, which is why it is slower to arrive, harder to reverse and far more prone to being priced as a risk premium rather than as a forecast.

The levers, and the ones that work without anyone deciding

Discretionary policy is a decision: a stimulus package, a tax cut, an infrastructure programme. It is slow, because it must be legislated and then actually spent, and the lag between the decision and the concrete being poured is routinely a year or more.

Automatic stabilisers are the parts of the budget that respond to the cycle with no decision at all. In a downturn tax receipts fall because incomes fall, and unemployment benefits rise because unemployment rises, so the deficit widens by itself and cushions the fall. They are fast, and their existence is why the measured deficit is a bad guide to the fiscal *stance*: a deficit that widens in a recession may represent no policy change whatsoever. The right measure is the **cyclically adjusted** primary balance, and saying that distinguishes an answer that has thought about it.

The multiplier, and why it is usually smaller than you expect

KEY IDEA

The fiscal multiplier is the change in output per unit of fiscal impulse. The textbook closed-economy version,

$$k = \frac{1}{1-c},$$

with a marginal propensity to consume c, gives 2.5 at $c = 0.6$ and is essentially never observed, because it ignores every way money leaks out of the circular flow.

WORKED EXAMPLE

Where the multiplier goes. Add the two largest leakages: income is taxed at rate t before it can be spent, and a share m of spending goes on imports rather than domestic output:

$$k = \frac{1}{1 - c(1-t) + m}.$$

At $c = 0.6$, $t = 0.30$ and $m = 0.25$,

$$k = \frac{1}{1 - 0.42 + 0.25} = \frac{1}{0.83} = \mathbf{1.20}.$$

The multiplier has fallen from 2.5 to 1.2 purely from taxation and openness, with no economics beyond arithmetic. That is the single most useful fact about fiscal stimulus and it explains a robust empirical pattern: small open economies have small multipliers, because the stimulus is spent abroad, and large closed ones have larger ones.

INTUITION

The modern result, and the one that separates a good answer, is that the multiplier depends on *what monetary policy is doing*. If the central bank is at the effective lower bound and cannot offset the stimulus, the multiplier is large, plausibly well above one. If the central bank is actively tightening to control inflation, it will simply offset the fiscal impulse with higher rates and the multiplier collapses toward zero.

So "what is the multiplier" has no context-free answer, and the correct response to the question is another question: what is the central bank doing at the time. That is also why fiscal stimulus into an inflationary economy is a poor trade for the economy and an excellent one for a rates short.

Debt dynamics: the one equation to know

Whether debt is sustainable is not a matter of the level. It is a matter of the direction, and the direction is governed by one relation. Writing d for debt as a share of GDP, r for the effective nominal interest rate on that debt, g for nominal GDP growth and pb for the primary balance as a share of GDP,

$$\Delta d \;=\; \frac{r-g}{1+g}\,d \;-\; pb.$$

Everything about sovereign solvency is in the sign of $r - g$.

WORKED EXAMPLE

The same debt level, two different worlds. Take debt at 100% of GDP.

If $r = 3\%$ **and** $g = 4\%$, then $\frac{r-g}{1+g}d = \frac{-0.01}{1.04} = \mathbf{-0.96}\%$ of GDP a year. Debt falls by about a percentage point of GDP every year even if the government runs a primary balance of exactly zero. Growth is doing the work, and this is how most post-war debt reductions actually happened.

If $r = 5\%$ **and** $g = 2\%$, then $\frac{0.03}{1.02} = \mathbf{+2.94}\%$ of GDP a year. The government must run a primary *surplus* of 2.94% of GDP, permanently, just to stop the ratio rising. For most political systems that is at or beyond the limit of what is deliverable, which is exactly why a rise in yields on a high-debt sovereign is treated as a regime change rather than as a price move.

Note also that the same arithmetic at 60% debt requires only a 1.76% surplus, so the level matters through its multiplication of the $r - g$ gap rather than in itself. A country with low debt can survive $r > g$; one with high debt cannot survive it for long.

The practical reading for a rates desk is that a sovereign's debt trajectory is a leveraged bet on the spread between its borrowing cost and its nominal growth, and that nominal growth includes inflation, which is why moderate inflation erodes debt ratios and why a disinflation is fiscally expensive.

Crowding out, and the interaction that decides everything

A fiscal expansion has to be financed, which means issuing bonds, which means the market must be persuaded to hold more duration. That pushes yields up, and higher yields discourage private investment. The effect is called **crowding out**, and how large it is depends on the state of the economy: with idle capacity it is small, and with the economy at full employment it can approach one for one, in which case the stimulus buys nothing.

The open-economy version is the Mundell-Fleming result and is worth being able to state. Under a floating exchange rate with mobile capital, fiscal expansion raises rates, which attracts capital, which strengthens the currency, which reduces net exports and offsets much of the stimulus. So *fiscal policy is weak and monetary policy strong under a float.* Under a fixed exchange rate the central bank must intervene to hold the peg, which prevents the currency appreciating, so fiscal policy is strong and monetary policy is unavailable. That reversal is the cleanest way to remember which regime does what, and it connects directly to pegged currencies.

When the market disciplines the government

For a markets desk this is the part that matters, because it happens in hours rather than in years.

A sovereign's yield contains a compensation for expected policy rates, a term premium for holding duration, and, when credibility is in question, a **fiscal risk premium**. The third component is normally invisible and can appear very suddenly.

The mechanism to be able to describe: an unfunded fiscal announcement raises expected issuance, which raises the term premium; if the announcement also suggests the fiscal framework is not binding, it adds a credibility premium on top; and if domestic institutions hold leveraged positions in the sovereign's own bonds, the yield rise forces them to sell into the move, which is a feedback loop rather than a repricing. The United Kingdom in September 2022 is the reference example, and its lesson is specific: the market repriced not the arithmetic of the package but the *institutional* question of whether anything constrained it, and the currency and the long end moved together, which is the signature of a credibility event rather than a rate event.

For a euro-area sovereign the same pressure shows up in the spread to Germany rather than in the outright yield, since the currency and the policy rate are shared, which is why the sovereign debt crisis was a spread crisis.

Interview angle

REAL INTERVIEW QUESTION (Moody's · Credit)

Is rising debt an issue? When is it more of an issue and when less? Give some examples.

WHAT THEY TEST / ANSWER

What is being tested is whether you judge debt by its level or by its trajectory. The answer that fails is "yes, past 100% of GDP it becomes dangerous", because it invents a threshold. The answer that works is that the level is close to uninformative on its own and the sign of $r - g$ decides.

Lead with the arithmetic. At debt of 100% of GDP, $r = 3\%$ against $g = 4\%$ retires about 0.96% of GDP a year even at a primary balance of exactly zero, while $r = 5\%$ against $g = 2\%$ demands a permanent primary surplus of 2.94% of GDP simply to hold the ratio flat. Identical debt, opposite trajectories, and nothing changed but two rates.

Then the qualifiers, which is the part a rating agency is listening for. Rising debt is *less* of a problem when it is denominated in a currency the sovereign itself issues, when its average maturity is long so a rate shock reaches the effective r only slowly, when it is held domestically, and when nominal growth sits above the borrowing cost. Japan is the standing example: a ratio far beyond any conventional threshold, carried for decades because r was held below g and the holders were domestic. It is *more* of a problem when the debt is in a foreign currency or under foreign law, when it is short-dated and therefore refinanced repeatedly at whatever the new rate is, when it is held by flight-prone foreign investors, and when the sovereign has given up its own monetary policy, which is the euro-area case where the strain surfaces in the spread to Germany rather than in the outright yield.

The follow-up trap is "so where is the threshold". There is not one, and saying so plainly is the answer: a sovereign at 60% with $r > g$ is on a worse path than one at 150% with $r < g$. What triggers a crisis is not a level being crossed but the market repricing whether the required primary surplus is politically deliverable, which is why these events arrive in a week rather than over a cycle.

INTERVIEW TRAP

1. **Reading the deficit as the stance.** Automatic stabilisers widen the deficit in a downturn with no policy change. Use the cyclically adjusted primary balance.
2. **Quoting a multiplier without the monetary context.** It is large at the lower bound and near zero when the central bank is offsetting, so the number is meaningless on its own.
3. **Judging debt by its level.** The level matters only as a multiplier of $r - g$. A country at 60% with $r > g$ and one at 150% with $r < g$ are on opposite trajectories.
4. **Forgetting that g is nominal.** Inflation erodes debt ratios and disinflation is fiscally expensive, which is an uncomfortable but real conflict between the two arms of policy.
5. **Assuming fiscal expansion is bullish for the currency.** Under a float the rate effect can strengthen it and the credibility effect can sink it, and which dominates is exactly what the market is judging.
6. **Treating the fiscal risk premium as always present.** It is zero almost all the time and then appears in a week, which is why positioning for it as a carry trade is expensive and reacting to it late is worse.

VISUAL / CODE TO BUILD: Interactive

A debt-trajectory panel. Sliders set the starting debt ratio, the effective interest rate, nominal growth and the primary balance, and the panel projects the debt ratio forward twenty years, drawing the path and printing the primary balance required to stabilise it.

The intended moment is setting r just below g, watching the path decline gently, and then nudging r above g and watching the same starting point turn into a curve that accelerates upward. That the sign of a single subtraction flips a benign path into an explosive one is the section's central claim, and it is far more convincing dragged than read. A second line showing the trajectory at 60% starting debt on the same parameters makes the point that the level acts as a multiplier.

Beneath it, a multiplier calculator with sliders for the propensity to consume, the tax rate and the import share, showing the multiplier collapse from 2.5 toward 1 as the leakages are dialled in, and a switch for the monetary regime that scales the result up at the lower bound and toward zero under an offsetting central bank.

3.7 Economic cycles

sales · trading · structuring | core | cross-asset

Nobody on a desk trades "the economy". They trade a view about which phase of the cycle the market is pricing against which phase they think is coming, and almost every macro conversation in an interview is really that question in disguise. The material is a vocabulary (the phases), a toolkit (the indicators), and one very useful mapping (phase to asset). What makes the difference is knowing that markets move ahead of the data, which is why the person who waits for confirmation is always late.

The phases, and what a recession actually is

The conventional four phases are **recovery**, **expansion**, **slowdown** and **recession**, and the useful way to hold them is as two dimensions rather than four boxes: is growth above or below trend, and is it accelerating or decelerating. The second derivative is what markets trade, which is why an economy can be growing strongly and equities can be falling.

> KEY IDEA
>
> **A recession is not "two consecutive quarters of negative GDP".** That shorthand is a media convention, and in the United States the body that actually dates cycles, the NBER's Business Cycle Dating Committee, uses a judgement across depth, diffusion and duration: a significant decline in activity, spread across the economy, lasting more than a few months, and visible in production, employment, real income and sales rather than in GDP alone. That is why a recession can be declared without two negative quarters, and why the dating is published long after the fact. Saying this correctly is a cheap and reliable way to sound like you have read past the headlines.

Cycles are also not periodic, and treating them as a clock is the classic amateur error. Economists distinguish an **inventory** cycle of roughly three to five years, driven by firms over- and under-stocking; a longer **capital expenditure** cycle, since capacity takes years to build and then arrives all at once; and a **credit** cycle, in which easy lending funds the leverage whose unwind causes the downturn. Overlaid on all of them is the **policy** cycle, because the central bank is reacting to the economy while the economy is reacting to the central bank, see monetary policy.

Indicators: what to watch, and in what order

Type	Examples	What it is good for
Leading	yield curve slope, PMI new orders, building permits, jobless claims, credit spreads, equity prices	anticipating the turn
Coincident	GDP, industrial production, payrolls, retail sales	confirming where you are
Lagging	unemployment rate, CPI, unit labour costs	confirming where you were

Two things follow that candidates routinely get wrong. The **unemployment rate is lagging**, so it is usually still falling when the expansion has already ended and still rising after the recovery has begun: it is the worst possible timing tool despite being the most politically salient number. And **surveys lead hard data**, so a purchasing managers' index, which asks firms what they are seeing now, turns before the production statistics that measure what they did last quarter. On a desk you also watch things that are not official statistics at all: freight rates, semiconductor orders, corporate guidance, bank lending standards, and the tone of primary issuance.

The curve is the market's cycle forecast

The single most useful cycle instrument is the shape of the yield curve, because it is the market's own opinion about the policy path, priced continuously and for free. The mechanics are in yield curves; what belongs here is the mapping from the four possible moves to the four phases.

Move	What is happening	Typical phase
Bull steepener	front end rallies more than the back	recession priced, cuts expected
Bear steepener	long end sells off more than the front	recovery, reflation
Bull flattener	long end rallies more than the front	late cycle, growth doubts
Bear flattener	front sells off more than the long end	tightening into a late expansion

KEY IDEA

An **inverted curve** says the market expects the policy rate to be lower in a few years than it is now, which is a statement that growth or inflation will weaken enough to force cuts. That is why inversion has preceded US recessions so reliably. Two qualifications keep the point honest: the lead time is long and variable, historically on the order of a year or more from inversion to recession onset, so it is useless as a timing signal, and the curve typically **re-steepens** before the recession arrives as the market moves from pricing tightening to pricing cuts, which means the disinversion is often the later and more urgent signal.

The rotation map

This is the payoff of the section, and it is what a sales or macro interview is fishing for. Treat it as a set of tendencies, not a law.

- **Recovery.** Policy still easy, growth turning up. Equities strong, led by cyclicals, small caps and high beta; credit spreads tighten hard from wide levels; the curve bull-steepens then bear-steepens; commodities turn up. Historically the best phase for risk assets, and it begins while the news is still bad.
- **Expansion.** Growth above trend, policy normalising. Equities grind higher with less drama, leadership rotates toward quality and earnings growth, the curve flattens as the central bank tightens, credit is quiet.
- **Slowdown, late cycle.** Capacity tight, inflation and wages rising, policy restrictive. Commodities and energy often outperform, the curve bear-flattens and may invert, equity leadership narrows and defensives begin to work, credit spreads stop tightening. This is the phase where being early is most expensive and being late is fatal.
- **Recession.** Government bonds rally as cuts are priced, the curve bull-steepens, defensives and quality outperform, credit spreads widen sharply, and equities bottom *before* the economy does.

KEY IDEA

Markets lead the economy, and the lead is not small. Equities have historically troughed while GDP was still contracting and while unemployment was still rising, because the price today discounts expectations of the future rather than the present. The practical consequence for an interview answer is that "wait for the data to confirm" is not a strategy, it is a description of arriving after the move. What you trade is the gap between the phase the market is pricing and the phase you think is coming.

REAL INTERVIEW QUESTION (Morgan Stanley · QIS)

Which market is the most representative of recession, in your view?

WHAT THEY TEST / ANSWER

They want a defended choice, not a survey, so pick one, justify it, and then show you know its weaknesses. The strongest answer is the **government bond curve**, specifically the slope between a short and a long maturity. Justify it on three grounds: it is a pure expectations instrument, since the slope is the market's forecast of the policy path and an inversion is literally the statement that rates must be cut; it is the deepest and most liquid market in the world, so it is the least distorted by flows and the hardest to push around; and its record of leading US recessions is the best of any single indicator. Then be honest about its limits, because that is what earns the mark: the lead time is long and variable, so it is a regime signal rather than a timing signal, and it is contaminated when the term premium is being moved by central bank purchases rather than by expectations, which is a genuine argument that recent inversions were less informative than earlier ones. Then name your runners-up and why they are runners-up: **credit spreads**, especially high yield, which price default risk directly and are excellent but move late and are pushed around by fund flows; **cyclical versus defensive equity** pairs or copper against gold, which are clean cross-sectional reads and avoid the index's composition problems; and **the front end of the curve** alone, which is the purest read on what the market thinks the central bank will be forced to do. Since this is a quantitative desk, close by saying you would not rely on one signal at all, you would combine them, because the individual indicators have low hit rates and their disagreements are themselves informative.

REAL INTERVIEW QUESTION (HSBC · Rates)

How has the US yield curve behaved recently?

WHAT THEY TEST / ANSWER

A live-market question, so the structure is what is being graded and inventing a level is the only way to fail badly. Answer in four parts and give your as-of date. First, the **level**: roughly where the two-year and ten-year sit and where the policy rate is relative to them. Second, the **shape and its direction of travel**: is the two-to-ten slope positive or negative, and has it steepened or flattened over the last quarter. Third, and this is the part that separates a rates candidate, **decompose the move** rather than describing it: a steepening is a bull steepener if the front rallied more, which means the market brought rate cuts forward, and a bear steepener if the long end sold off more, which means growth, inflation or supply expectations rose at the back. Say which one it was and what drove it, whether that was a data surprise, a policy communication, a fiscal or issuance story, or a flight to quality. Fourth, the **implication**: what the current shape implies about the number of cuts or hikes priced and about where the market thinks the terminal rate is. Then two things worth volunteering: an inverted curve is a growth warning with a long and unreliable lead, and the re-steepening out of inversion has historically been the more urgent signal, so I would watch the disinversion rather than the inversion. Finish by stating explicitly when you last looked, since being precise about the as-of date reads as discipline rather than ignorance.

The broad regional question, what the economic and monetary trends are across the US, Europe and Asia, is asked often enough to have its own treatment, and it is answered in economic growth where the growth measurement it leans on is defined. What this section adds to it is the phase: once you have said where each region is, the next sentence is which part of the cycle that puts it in and what the curve there is pricing.

REAL INTERVIEW QUESTION (Société Générale · desk not recorded)

Define a growth company versus a value company, and give examples of growth and value industry sectors.

WHAT THEY TEST / ANSWER

Define, exemplify, then connect it to the cycle, because that last step is what makes it a macro answer rather than a definition. A **growth** company is expected to grow earnings and revenue faster than the market, so most of its value sits in cash flows far in the future, it usually reinvests rather than paying dividends, and it trades on high multiples of current earnings. A **value** company trades cheaply relative to current fundamentals, on low price to earnings or price to book, typically with slower growth, more mature markets and higher distributions. Sectors: growth skews to technology, software, semiconductors, biotech and parts of consumer discretionary; value skews to banks, energy, utilities, telecoms, materials and industrials. Then the cycle and rates connection, which is the real content. Growth is a **long-duration** asset because its cash flows are far out, so it is far more sensitive to the discount rate: rising real yields hurt growth disproportionately, which is why value tends to outperform in a rising-rate, reflationary phase and growth in a falling-rate, low-growth one. Value is also more **cyclical** in the sectors listed, so it needs the economy to cooperate, and banks in particular benefit from a steeper curve. Two nuances that show you have thought about it: the style labels are constructed from ratios, so a stock can migrate between them and value screens can trap structurally declining businesses, which is the value trap; and a large part of what looks like a style call is really a sector and duration call in disguise.

INTERVIEW TRAP

Four traps. First, the **two-quarters definition** of recession, which is a media shorthand rather than how cycles are dated. Second, **treating the cycle as a clock**: expansions have varied from a couple of years to more than a decade, and “it is due” is not analysis. Third, **using lagging indicators to time**, above all the unemployment rate, which is usually at its best just as the expansion ends. Fourth, **applying the demand-shock playbook to a supply shock**: in a supply shock growth falls while inflation rises, so bonds do not hedge equities, the equity-rates correlation flips positive, and the rotation map above needs rewriting, as in energy and global conflicts.

VISUAL / CODE TO BUILD: Interactive

A cycle dashboard with a rotation backtest. The top panel plots a chosen set of leading, coincident and lagging indicators on one normalised axis with recession bands shaded, so the reader sees the ordering directly: surveys turning first, hard data next, unemployment last. A draggable marker on the timeline shows, for any historical date, what each indicator was saying *at that moment* rather than with hindsight, which is the single most instructive thing this tool can do. The second panel implements the rotation map: the reader assigns assets to phases, then the tool runs the allocation through history and reports the result against a static benchmark, with the phase labels applied both with hindsight and with a real-time rule, so the gap between the two shows how much of the strategy's apparent value came from knowing the phase in advance. A third panel plots the two-to-ten slope with recession onsets marked, and reports the distribution of lead times from inversion and from disinversion, making the "regime signal, not timing signal" point quantitative.

Where this goes next

The measurement of activity itself is economic growth and the price side is inflation, with the policy reaction in monetary policy and its channels in monetary transmission mechanisms. When a downturn becomes disorderly it belongs to financial crises. The instrument this section leans on most is developed in yield curves, the release schedule that moves it is macro calendars, and how to talk about any of it to a client is market color.

3.8 Balance of Payments

sales · trading · structuring | core | fx · rates · cross-asset

The balance of payments is the record of every transaction between a country and the rest of the world. It sounds like accounting and it is, but it is the accounting that determines whether a currency is under structural pressure, and every emerging-market currency crisis is legible in it in advance.

KEY IDEA

The balance of payments **always balances**. By construction,

$$\text{current account} + \text{capital account} + \text{financial account} + \text{errors} \;=\; 0.$$

So the interesting question is never whether it balances but *how*. A country running a current account deficit is, as an identity, importing capital to pay for it, and the only questions that matter are who is supplying that capital, on what terms, and what happens if they stop.

The three accounts

The current account is trade in goods and services, plus income on foreign investments, plus transfers such as remittances. It is the one people mean when they say a country runs a deficit.

The capital account is small and mostly technical: debt forgiveness and transfers of non-produced assets.

The financial account is the one that funds everything, and it is where the analysis lives. It records direct investment, portfolio investment, other investment including bank lending, and the change in official reserves.

The reframing that kills the reflex

National accounting gives a second identity that is worth more than the first:

$$\text{current account} = S - I,$$

national saving minus domestic investment. A current account deficit is therefore not a symptom of uncompetitiveness by definition; it is an excess of investment over saving, financed from abroad.

> INTUITION
>
> That reframing decides whether a deficit is benign. A country running a deficit because it is investing heavily in productive capacity is borrowing to build something that will service the debt, and the deficit is a sign of opportunity. A country running the same deficit because saving has collapsed and consumption is being funded by foreigners is borrowing to consume, and the deficit is a warning. Identical numbers, opposite meanings, and the distinction is invisible in the headline.
>
> So the correct first question about any current account deficit is not how large it is but what is on the other side of it: investment or consumption.

Financing it is a daily activity

The reason this matters to an FX desk is that a deficit has to be funded continuously, not annually.

> WORKED EXAMPLE
>
> **What a 5% deficit costs in flow.** A \$500 billion economy running a current account deficit of 5% of GDP needs
>
> $$0.05 \times 500 = \$25 \text{ billion a year}$$
>
> of net capital inflow, which is about **\$99 million every business day**, arriving reliably, forever, simply to keep the exchange rate where it is.
>
> Nothing guarantees that it arrives. If foreign investors decide one week that they would rather not, the adjustment has to happen through some combination of a weaker currency, higher domestic interest rates to attract the flow back, running down reserves, or a compression of imports through a recession. Usually all four, in that order, and that sequence is what a currency crisis *is*.

The quality of the financing matters more than the size

Two countries with identical deficits can be in completely different positions depending on what is funding them.

- **Foreign direct investment** is the best kind. It is illiquid by nature, a factory cannot be sold on a Tuesday, so it does not reverse in a panic.

- **Portfolio equity** is next. It reprices sharply but the capital has already arrived, and a falling equity market does not by itself force money out of the country.
- **Portfolio debt in local currency** shifts the currency risk onto the foreign investor, which is stabilising for the borrower and is why building a local-currency bond market is a standard emerging market policy objective.
- **Short-term external debt in foreign currency** is the dangerous one. It must be rolled continuously, the currency risk sits with the domestic borrower, and a depreciation raises the local-value of the debt at exactly the moment it becomes hard to refinance.

That last combination, a deficit funded by short-term foreign-currency borrowing, is the standard setup for the crises in financial crises, and the phrase for what happens is a **sudden stop**: the inflow does not merely slow, it reverses, and the adjustment that follows is violent because it has to happen in months rather than years.

Reserve adequacy, and two rules of thumb

Reserves are what a central bank can spend to smooth an adjustment, and there are two standard tests. Both are crude and both are used.

WORKED EXAMPLE

Testing a country's buffer. Reserves of \$60 billion, annual imports of \$120 billion, short-term external debt of \$70 billion.

Import cover. $60/(120/12) = \mathbf{6.0}$ months of imports, comfortably above the conventional three-month floor. On this test the country looks fine.

The Guidotti-Greenspan rule, that reserves should cover 100% of external debt falling due within a year: $60/70 = \mathbf{0.86}$. It fails, and would need another \$10 billion of reserves to pass.

The two tests disagree, and when they disagree the second one is the one to believe. Import cover was designed for a world where the risk was paying for goods; the risk that actually produces crises is refinancing debt, and that is what the second test measures. A country that looks safe on months of import cover and unsafe on short-term debt coverage is the classic profile going into a currency crisis.

What this means on a desk

Persistent current account *surpluses* tend to belong to currencies that behave as havens, since the country is a net exporter of capital and that capital comes home under stress: the yen and the franc are the standard examples, and their strength in a crisis is partly a repatriation flow rather than a judgement about the country.

Persistent *deficits* in emerging markets are the standard short candidate under global tightening, because the financing they depend on is precisely what a rise in developed-market rates withdraws. The trade is not about the deficit's level, it is about the sensitivity of the financing to external conditions, which is why the same deficit is untradeable in one year and the whole story in another.

Commodity exporters deserve a note of their own: their current account is largely a function of one price, so their currency is a leveraged position on that commodity, and the balance of payments is the mechanism through which FX transmits a terms-of-trade shock.

INTERVIEW TRAP

1. **Treating a current account deficit as bad in itself.** It is $S - I$. Ask whether the counterpart is investment or consumption before judging it.
2. **Looking only at the size of the deficit.** The composition of the financing decides the vulnerability, and short-term foreign-currency debt is the item that matters.
3. **Trusting import cover.** It is the friendlier of the two reserve tests and it measures the wrong risk. Check short-term external debt coverage as well.
4. **Forgetting the identity is an identity.** The balance of payments always balances; if the private inflow disappears, the adjustment happens through reserves, the currency or a recession, because arithmetic requires it.
5. **Assuming surpluses are always a sign of strength.** A surplus driven by collapsing imports is a recession, not an achievement, and that is how a crisis country's deficit closes.

VISUAL / CODE TO BUILD: Interactive

A funding panel. The reader sets GDP, the current account balance as a share of it, and the composition of the financing across FDI, portfolio equity, local-currency debt and short-term foreign-currency debt. The output is the required daily inflow in dollars and a vulnerability bar weighted by the flightiness of each financing type.

The teaching moment is holding the deficit constant at 5% and moving the composition from all FDI to all short-term foreign-currency debt: the daily funding requirement never changes and the vulnerability bar goes from green to red, which is the section's central claim performed rather than asserted.

A second panel is a sudden-stop simulator. The reader switches off the foreign inflow and chooses how the adjustment is absorbed across four sliders, currency depreciation, policy rate rise, reserve drawdown and import compression, with the constraint that they must sum to the missing flow. Discovering that the sliders are coupled, that refusing to let the currency move forces the others up, is the balance-of-payments identity felt rather than read, and it is also why a pegged country in this position ends up in a peg crisis.

3.9 Exchange rate regimes

trading · structuring · sales | core | fx · rates

A country has to decide what its currency is allowed to do. That decision is not a technical detail delegated to a back office; it determines what the central bank can target, how the economy absorbs a shock, and whether a currency can be attacked at all. Every dramatic event in pegged currencies and FX crises is downstream of a regime choice made years earlier.

INTUITION

An exchange rate can be a *price* or a *promise*, and it cannot be both. If it is a price it moves, and it absorbs shocks so the domestic economy does not have to. If it is a promise it does not move, and the domestic economy absorbs the shocks instead. Everything in this section is a consequence of that one sentence.

The spectrum

Regimes are usually presented as fixed versus floating, which is too coarse to be useful. The real object is a spectrum, and what changes along it is how much the central bank has committed away.

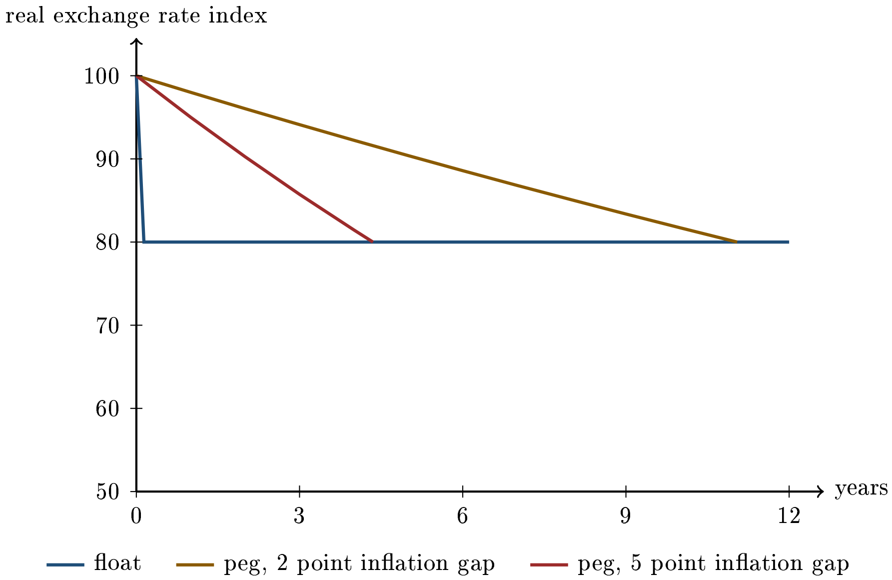
- **Free float.** No target and no intervention. The central bank sets the policy rate for domestic conditions and the currency goes where it goes. The dollar, the euro, sterling and the yen sit here.
- **Managed float.** No announced target, but the central bank intervenes to smooth moves it dislikes. Most large emerging economies are here, and most of them describe themselves as floating.
- **Crawling peg or band.** A target that moves on a pre-announced schedule, or a range inside which the rate is free. Used to disinflate gradually without an abrupt real appreciation.
- **Hard peg.** A fixed rate defended with reserves and with the policy rate. Domestic monetary policy is now in service of the peg.
- **Currency board.** A hard peg with a legal rule: domestic money is issued only against foreign reserves. Monetary policy has been abolished rather than reassigned. The mechanics are in pegged currencies and FX crises.
- **Dollarisation.** The foreign currency is simply adopted. There is no domestic money to defend, and no lender of last resort either.
- **Currency union.** A shared currency and a shared central bank. It is the hardest commitment of all, because there is no exchange rate left to abandon and no obvious procedure for leaving.

Moving down the list buys credibility and gives up flexibility, one unit at a time. The constraint that makes this a genuine trade-off rather than a menu is the impossible trinity, which is worked in central banks and FX policy: a country can have a fixed rate, free capital flows and an independent monetary policy, but only two of the three.

What you are actually choosing between

A fixed rate is imported credibility. A country with a history of high inflation can borrow the anchor of a country that does not, and the discipline is external and visible. That is a real benefit and it is why disinflation programmes so often start with a peg.

What it costs is the **adjustment mechanism**. When a country loses competitiveness, its prices and wages are too high relative to its trading partners and something must give. Under a float the nominal exchange rate does the work, overnight, at no political cost. Under a peg the nominal rate cannot move, so the *real* exchange rate has to be brought down the only other way available: domestic prices and wages must rise more slowly than everyone else's, for as long as it takes. This is **internal devaluation**, and its defining feature is how long it takes.



the same 20% real depreciation, reached overnight or over eleven years

WORKED EXAMPLE

A country needs a **20% real depreciation** to restore competitiveness. Domestic prices are sticky, so a nominal move passes through to the real rate roughly one for one in the short run.

Under a float, the nominal rate falls 20% and the job is done in an afternoon. The real exchange rate index goes from 100 to 80 and the adjustment cost is borne by importers and by anyone holding foreign-currency debt.

Under a peg, the nominal rate cannot move, so the index has to be walked down by running lower inflation than trading partners. If the gap is g per year, the index follows $100 \times (1-g)^t$ and reaches 80 after $\ln(0.8)/\ln(1-g)$ years:

Annual inflation gap	Years to a 20% real depreciation	Domestic inflation if partners run 2%
1 point	22.2	1%
2 points	11.0	0%
3 points	7.3	−1%
5 points	4.3	−3%

Read the third column, because it is where the argument actually lives. A two point gap already requires *zero* domestic inflation and still takes eleven years. Getting the adjustment done in four years requires **three percent outright deflation, sustained**.

And deflation is not a neutral speed setting. It raises the real value of every debt in the economy, so the fast path to competitiveness makes the debt burden worse at exactly the moment the country is least able to carry it. That is the trap, and no amount of resolve dissolves it: the arithmetic offers a decade of stagnation or a debt spiral.

KEY IDEA

A fixed exchange rate does not remove the need to adjust. It removes the fast, anonymous instrument for doing it and substitutes a slow one that runs through wages, employment and the real value of debt. Ask of any peg not whether it can be defended today, but what the country does when it needs a real depreciation.

What countries say and what they do

Regimes are classified two ways and the two disagree constantly. The **de jure** regime is what the country declares. The **de facto** regime is what the exchange rate data show it actually doing.

The systematic gap between them has a name: **fear of floating**. A large number of countries describe themselves as floating while intervening heavily and keeping the rate inside a narrow range, because they want the credibility of a float and the stability of a peg. The reasons are real. Foreign-currency debt makes a depreciation a balance-sheet event rather than a price change, import-heavy consumption baskets make it an inflation event, and neither is politically survivable.

The practical consequence for a trader is that the declared regime tells you little. What tells you something is realised volatility: a currency that has not moved more than a percent in a year is being managed, whatever the central bank's website says. That observation is the starting point for pricing options on it, which is covered in pegged currencies and FX crises.

When does sharing a currency make sense

The theory of **optimum currency areas** asks when a group of economies should give up their exchange rates against each other. Since the exchange rate is an adjustment mechanism, giving it up is only safe if something else adjusts instead. The criteria are all versions of that question:

- **Synchronised cycles.** If shocks hit all members at the same time and in the same direction, one policy rate suits everybody and the exchange rate was never needed.
- **Labour mobility.** If workers move from the depressed region to the booming one, unemployment adjusts instead of the currency.
- **Wage and price flexibility.** If wages fall where demand is weak, the real exchange rate adjusts internally without needing years of deflation.
- **Fiscal transfers.** If a shared budget moves money to the region in trouble, the adjustment is shared rather than borne locally.

Against these costs sits a genuine benefit that is easy to undersell: eliminating exchange rate risk and conversion costs inside the area deepens trade and financial integration, and that integration then tends to synchronise the cycles, which makes the union work better than it did at the start.

The euro area is the natural test case. It scores well on integration and poorly on labour mobility and fiscal transfers, and it has no wage flexibility to speak of. That combination is exactly the one for which the arithmetic above is unforgiving.

Interview angle

REAL INTERVIEW QUESTION (Barclays · Credit)

Explain the Greek crisis.

WHAT THEY TEST / ANSWER

The narration to avoid is the sequence of bailouts. What makes this a good interview question is that Greece is the cleanest demonstration in modern history of what a currency union costs, so answer it as a regime problem with a fiscal trigger rather than a fiscal problem with a regime footnote.

The setup. Joining the euro removed the currency risk premium from Greek borrowing, so Greek and German yields converged to within a few tens of basis points despite very different fiscal positions. Cheap money financed a decade of demand, unit labour costs rose far faster than Germany's, and Greece lost competitiveness steadily while the exchange rate that would normally have signalled it no longer existed. The revelation in 2009 that the deficit had been misreported was the trigger, not the cause.

Then the part that is this section. Greece needed a large real depreciation and had exactly one way to get it. Outside the euro the drachma falls twenty or thirty percent in a week. Inside it, the only route is running inflation below the rest of the euro area, and the table above says what that means: a two point gap takes over a decade, and compressing it into a few years requires outright deflation, which raises the real value of a debt stock that was already the problem. The adjustment did eventually happen, through roughly a quarter of lost output and unemployment above 25%, which is the price of doing with wages what a currency does overnight.

The two things worth volunteering. First, redenomination risk became a traded object: because leaving a currency union has no procedure, the market priced the possibility that a euro in an Athens bank was not the same asset as a euro in a Frankfurt bank, and that is a risk with no analogue under a floating regime. Second, the crisis showed that a monetary union without a fiscal union has no mechanism for regional shocks, which is the optimum currency area criterion the euro area most obviously fails.

The sovereign debt mechanics, the restructuring and the wider contagion belong with the European sovereign debt crisis; the point to make here is that the regime chose the adjustment path before the crisis started.

REAL INTERVIEW QUESTION (Barclays · Rates)

How would you go about trading bonds in local currency, RUB or TRY for example?

WHAT THEY TEST / ANSWER

The question is really about whether you understand that a local-currency bond is two positions in one, and that the second one is a bet on the exchange rate regime.

Split it first. A local-currency bond gives you local duration and currency exposure together. A foreign-currency bond from the same issuer gives you credit risk in a currency you can fund. These are different trades with different worst cases: the local bond can lose on a currency move while the credit is fine, and it is the currency leg that dominates the risk in practice.

Then ask what the regime lets you do, because it determines the instruments:

- **Is the currency deliverable?** If capital controls or convertibility restrictions bite, you hedge with non-deliverable forwards that cash-settle offshore, and you have introduced a basis between the onshore and offshore rate that can move violently on exactly the day you need the hedge.
- **Is the hedge worth doing?** Hedging a high-yield currency costs roughly the interest rate differential, which is most of the yield you came for. Hedged, you are left with the local credit and a thin spread; unhedged, you own the currency. Say which one you are being paid for.
- **Can you settle and hold it?** Local custody, withholding tax, settlement conventions and the possibility of controls being imposed while you are long are operational risks, and they are the ones that actually stop trades.

The regime point to close on. A managed currency looks stable until it does not, so realised volatility understates the risk and any position sizing based on it will be too large. The distribution is not the one the recent history suggests: long stretches of very little movement and then a single large move, which is the shape described in pegged currencies and FX crises. Size the position for the jump rather than for the calm.

INTERVIEW TRAP

Four regime errors. (1) Treating fixed and floating as the only two options, when almost every interesting case is in between. (2) Believing a country's declared regime rather than its realised volatility. (3) Assuming a peg removes the need for adjustment, when it only removes the fast instrument for it. (4) Reading a stable managed currency as a low-risk one, when the stability is a policy choice that can be withdrawn in a morning.

Where this goes next

How a peg is defended, why it breaks, and what the option market does about it are in pegged currencies and FX crises. The intervention toolkit and the impossible trinity are in central banks and FX policy. The system that tried a global fixed-rate regime and abandoned it is Bretton Woods, the episodes where regimes failed are major FX crises, and the inflation differential that drives every real exchange rate calculation here is inflation.

3.10 Sovereign Debt

sales · trading · structuring core rates · credit · cross-asset

Government bonds are the largest and most important asset class in the world, and their importance is structural rather than a matter of size: the sovereign curve defines what "risk-free" means in its own currency, everything else is priced as a spread to it, and government paper is the collateral that the entire financial system runs on. When a sovereign curve moves, nothing else is unaffected.

Fiscal policy covers whether the debt is sustainable. This section is about the instruments, what their prices contain, and the one distinction that decides whether a sovereign can be forced to default.

The instruments, by name

Every country's paper has a local name and interview questions use them without explanation.

- **France**: OAT, *Obligation Assimilable du Trésor*, plus BTF bills, and BTAN notes, which stopped being issued in 2013 with short-dated OATs taking their place; the inflation-linked versions are OATi, indexed to French inflation, and OAT€i, indexed to euro area inflation.
- **Germany**: Bund at ten and thirty years, Bobl at five, Schatz at two. The Bund is the euro area's benchmark and the reference against which every other euro sovereign is quoted.
- **Italy**: BTP. **Spain**: Bonos. **United Kingdom**: Gilts, with index-linked gilts a uniquely large share of the market. **United States**: Bills, Notes, Bonds and TIPS. **Japan**: JGBs.

REAL INTERVIEW QUESTION (Deutsche Bank · Rates)

What is an OAT?

WHAT THEY TEST / ANSWER

Give the expansion, the mechanics, and then the thing that makes it interesting, because the first two take ten seconds and the interviewer is waiting for the third.

The expansion. *Obligation Assimilable du Trésor*, the French government's medium and long-term bond, issued by the Agence France Trésor, typically fixed coupon with maturities from two years out to fifty, and the benchmark of the French curve.

What *assimilable* means, which is the part of the name that carries information and which most candidates skip. New issues are made fungible with an existing line rather than creating a new one: the Treasury taps the same bond repeatedly, so a single line grows to a very large size and stays liquid, instead of the market fragmenting into many small illiquid bonds. That is a deliberate liquidity policy embedded in the instrument's name.

Then the market context. OATs trade as a spread to the Bund, which is the euro benchmark, and that spread is the market's price for French credit and French political risk rather than for interest rates, since the policy rate is shared. Mention the inflation-linked variants, OATi indexed to French inflation and OAT€i to euro area inflation, because the existence of two different indices on the same issuer is a real detail. And if you have room, note that OATs are the collateral of the French repo market and are held in size by domestic insurers, which is why a dislocation in them is never only a rates event.

The distinction that decides everything

KEY IDEA

A sovereign borrowing in **its own currency**, which it controls, cannot be forced to default: it can always create the currency to pay. If it defaults it is choosing to, and the meaningful risk is not default but *inflation*, which is a default in real terms without being one in legal terms.

A sovereign borrowing in a **foreign currency**, or in a shared currency it does not control, *can* be forced. It must earn or borrow the currency to pay, and if the market stops lending it there is no printing press to fall back on.

That single distinction explains why Japan can carry debt above 200% of GDP without a crisis while Greece could not carry half of it, and it is the answer to almost any question about sovereign risk.

Euro area members are in the second category by construction, which is what made the sovereign debt crisis possible: member states issue in a currency their own central bank does not control, so they carry genuine default risk in a way the United States, the United Kingdom and Japan do not.

What a sovereign spread contains

REAL INTERVIEW QUESTION (BMCE · Credit)

If the spread between a German bond and a Spanish bond widens, what does that mean: has Spanish debt become more or less risky?

WHAT THEY TEST / ANSWER

More risky, and say so immediately, then explain what "risky" is doing in that sentence, because the mechanics are the answer.

The mechanics. Both bonds are in euros with the same policy rate, so the spread cannot be a rates differential. It is the extra yield investors demand to hold Spain rather than Germany, and yield rises when price falls, so a widening spread means Spanish bonds have underperformed and the compensation demanded has risen.

Then decompose it, because the spread is three things and knowing which one moved is the actual skill.

Credit risk: the probability that Spain does not pay, times the loss if it does not. Real for a euro member precisely because Spain does not control the currency it borrows in.

Liquidity: Bunds are the deepest market in the euro area and are the flight-to-quality destination, so in a stress the spread widens partly because Bunds richen rather than because Spain cheapens. Distinguishing those two is worth saying out loud, since they call for different trades.

Redenomination risk: the possibility that a bond is repaid in a currency other than the euro. This is unique to a currency union, it is usually zero, and it is what turns an ordinary widening into a discontinuous one.

Close with the caution. A wider spread means the *market prices* more risk, which is not the same as more risk existing; spreads overshoot in both directions, and the widening can be a Bund squeeze rather than a Spanish problem. Checking whether the Spanish yield actually rose, or only the German one fell, is the first thing to do and separates a market-aware answer from a textbook one.

WORKED EXAMPLE

Turning a spread into a default probability. The standard credit triangle says a spread compensates for expected loss:

$$\text{spread} \approx PD \times (1 - R),$$

so with a spread of 200 basis points and an assumed recovery of 40%,

$$PD \approx \frac{0.02}{0.60} = \mathbf{3.33}\% \text{ a year,}$$

and the cumulative probability over five years is $1 - (1 - 0.0333)^5 = \mathbf{15.6}\%$.

Two things to say about that number. It is a *risk-neutral* probability, so it embeds a risk premium and is systematically higher than any historical default frequency, which is why quoting it as "the market thinks there is a 16% chance" is loose. And the recovery assumption does enormous work: at 20% recovery the same spread implies 2.5% a year instead, so the sensitivity of the answer to an unobservable input is the honest caveat. CDS develops the relation properly.

Inflation-linked bonds, and the question that catches people

An inflation-linked sovereign bond indexes its principal to a price index, so the coupon and redemption grow with inflation and the quoted yield is a **real** yield. The difference between the nominal yield and the real yield of the same maturity is the **breakeven inflation rate**, which is what the market expects inflation to average, plus a risk premium.

REAL INTERVIEW QUESTION (Natixis · Fixed Income)

Why do inflation-linked OATs still lose value in a period of very high inflation?

WHAT THEY TEST / ANSWER

The question sounds like a contradiction and is not, and resolving it is the whole answer.

The resolution in one sentence. A linker is priced off the *real* yield, and real yields rise when a central bank tightens to fight inflation. The inflation accrual adds to the principal while the rise in real yields subtracts from the price, and over a sharp tightening the second effect is larger.

Put numbers on it. Take a linker with a real duration of about 9. Inflation runs at 8% over the year, so the accrual is +8%. But real yields rise by 2 percentage points, costing $9 \times 2 = 18\%$. The total return is

$$+8\% - 18\% \;=\; \mathbf{-10}\%,$$

a real loss of value in a high-inflation year, exactly as the question says.

Then the correction that makes it an answer. A linker never promised absolute protection. It promises protection *relative to a nominal bond of the same maturity.* Over the same period nominal yields rose by 3 percentage points on a duration of about 8, so the nominal bond returned -24%. The linker lost 10% and the nominal bond lost 24%, so the hedge worked: it outperformed by **14 percentage points** while still losing money.

Add the two refinements if you have room. The indexation is lagged, usually by around three months, so a linker under-protects at the very start of an inflation shock. And what a linker actually hedges is the difference between realised inflation and the *breakeven already priced in*: if the market has already priced 8% and inflation comes in at 8%, the linker delivers nothing extra, because you paid for that expectation on the way in.

When a sovereign does not pay

Sovereign default is a negotiation rather than a bankruptcy, because there is no court that can seize a country's assets and no process that can liquidate it. What happens instead is a restructuring: maturities are extended, coupons cut, principal written down, and the terms are agreed with enough creditors to bind the rest.

Two mechanisms are worth knowing. **Collective action clauses** let a supermajority of bondholders impose a restructuring on the minority, which exists precisely to stop **holdouts** refusing and suing for full payment, as happened for over a decade after Argentina's 2001 default. Since 2013 euro area sovereign bonds have carried standardised CACs, which is a direct institutional response to the Greek restructuring.

The market consequence is that sovereign recovery rates are far more variable than corporate ones, because they are political outcomes rather than the result of an asset sale, and that is the deepest reason the recovery assumption in the spread calculation above is so uncertain.

INTERVIEW TRAP

1. **Treating all government bonds as risk-free.** Only debt in a currency the issuer controls is free of forced default, and even that is not free of inflation risk.
2. **Reading a spread as pure credit.** It is credit plus liquidity plus, in a currency union, redenomination. Check whether the benchmark richened before concluding the periphery cheapened.
3. **Quoting a spread-implied default probability as a real one.** It is risk-neutral and includes a risk premium, and it is extremely sensitive to the recovery assumption.
4. **Expecting a linker to protect in absolute terms.** It hedges relative to a nominal bond of the same maturity, and it can lose a great deal while doing its job.
5. **Forgetting the indexation lag and the breakeven.** A linker pays off only on inflation above what was already priced, and it under-protects at the start of a shock.
6. **Assuming a high debt ratio means crisis.** Japan says otherwise. What matters is the currency of issuance and the $r - g$ arithmetic in fiscal policy.

VISUAL / CODE TO BUILD: Interactive

A spread decomposition panel. Two sovereign curves are drawn, a benchmark and a peripheral issuer, with the spread between them shaded and split into three stacked components: credit, liquidity and redenomination. Sliders move each one, and a separate control moves the *benchmark* yield alone, so the reader can produce a widening spread without the periphery moving at all and see the two cases side by side. That is the distinction the BMCE answer turns on, and it is much clearer as a picture.

A second panel converts spread to default probability, with sliders for the spread and the recovery rate and a curve of cumulative default probability against horizon. Holding the spread fixed and dragging recovery from 40% to 20% moves the implied annual probability from 3.3% to 2.5%, and pushing it the other way to 80% triples it to 10%, which makes the sensitivity to an unobservable input impossible to overlook.

A third is the linker calculator: inputs for realised inflation, the change in real yields and the change in nominal yields, with two bars showing the linker's and the nominal bond's total return and the difference labelled as the hedge's payoff. Setting the section's numbers produces −10% against −24%, and the reader can find the combinations where the linker still loses relative to the nominal bond, which is the case the question is really about.

3.11 Financial crises

sales · trading · structuring | core | cross-asset

Individual crises are narrated elsewhere in this book: 1929 in the 1929 crash, LTCM in Long-Term Capital Management, 2008 in the 2008 financial crisis, the euro area in the European sovereign debt crisis and 2020 in the Covid market crash. This section is about what they have in *common*, because the mechanism repeats even though the asset never does. Learn the anatomy and you can reason about a crisis you have never seen, which is exactly what an interviewer is testing when they ask about one you were too young to remember.

The taxonomy

Three basic kinds, which frequently arrive together.

- **Banking crisis.** Depositors or short-term creditors withdraw, and the institution cannot liquidate long assets fast enough to pay them.
- **Currency crisis.** A peg or a heavily managed exchange rate breaks under pressure, usually where reserves are finite and the domestic economy has borrowed in foreign currency.
- **Sovereign debt crisis.** A government's borrowing cost rises to the point where rolling its debt becomes impossible, which is often self-fulfilling.

The dangerous configurations are the combinations. A **twin crisis** pairs a banking and a currency crisis: the currency falls, foreign-currency debts held by domestic banks and firms explode in local terms, the banks fail, and the failure worsens the outflow. Add a sovereign whose banks hold its bonds and whose guarantee is what makes those banks solvent, and you have the **sovereign-bank loop** that defined the euro area episode.

The anatomy, in six phases

> INTUITION
>
> A crisis is not an explosion, it is a **drought followed by a fire**. The drought is the good years: nothing bad happens for long enough that everyone concludes nothing bad will happen, so they carry less water, build closer together and stop clearing the brush. None of that is irrational on its own, and each individual decision was rewarded. The fire itself can start anywhere and its cause is almost irrelevant. What determines whether you lose a field or a county is the dryness accumulated beforehand, and how connected everything has become.

1. **Benign conditions and easy credit.** Volatility is low, risk premia compress, lending standards loosen, and leverage rises because it is profitable and because everyone else is doing it. Minsky's formulation is the best one-liner in the field: **stability breeds instability**, since a long calm period makes risk-taking look safe and rewards those who take the most of it.
2. **Mismatch accumulates.** Somebody ends up funding long-dated, illiquid assets with short-dated liabilities. This is what a bank does by design, and what shadow banks, funds and vehicles do by accident or by choice. It is the structural precondition for every crisis, because it means solvency depends on the willingness of creditors to keep rolling.
3. **A trigger reveals losses.** The trigger is usually mundane and often not the largest exposure. What matters is that it makes losses *uncertain* rather than merely large: once nobody knows which counterparty is holding them, lenders stop lending to everyone, which is adverse selection closing a market rather than panic.
4. **Forced deleveraging.** Falling prices reduce equity, leverage rises mechanically, and margin calls and risk limits force sales into a market that already has no buyers. This is the self-reinforcing core, and it is worth doing with numbers.
5. **Contagion.** The problem spreads through channels that looked unrelated: common exposures, funding withdrawal, counterparty chains, and the informational channel where trouble at one institution is evidence about others that look like it.
6. **Policy response.** Liquidity, guarantees, recapitalisation, resolution, and rate cuts, in roughly that order of speed.

WORKED EXAMPLE

Why a small price move forces enormous selling. A leveraged investor holds €1,000m of assets against €100m of equity and €900m of debt, so leverage is 10×. Assets fall just 5%.

- Assets: €950m. Debt is unchanged at €900m, so equity is €50m.
- Leverage has jumped from 10× to $950/50 = 19\times$ with no action taken.
- To restore 10×, assets must come down to $10 \times 50 =$ €500m, so the investor must sell €450m of assets, which is **45% of the original book** in response to a 5% price move.

The general result is worth carrying, because it is one line and it explains the violence of every deleveraging episode. For assets A, leverage L and a price fall x, the sales required to hold leverage constant are

$$\text{forced sales} \;=\; A\,(L-1)\,x,$$

which here is $1{,}000 \times 9 \times 0.05 =$ €450m. Two readings follow immediately. The multiplier on the shock is $(L-1)$, so at 10× leverage every euro of price decline forces nine euros of selling. And equity is wiped out entirely when $x = 1/L$, that is a 10% fall here, so the distance between "uncomfortable" and "insolvent" is a single bad week.

Now close the loop: those €450m of sales push prices down further, which reduces equity again, which forces more sales. That is the **liquidity spiral**, and it is why a crisis looks nonlinear from the outside while every participant is behaving prudently by their own rules.

KEY IDEA

The spiral is a **fire-sale externality**: each seller is acting rationally, and the act of selling imposes a loss on everyone else holding the same asset, including on the seller's own remaining position. Nobody's individual restraint would fix it, which is why the resolution has to come from outside the private sector. This is the analytical case for a lender of last resort, and it is a much better answer than "governments bail out banks because they are politically connected".

Contagion channels, and why correlations go to one

- **Common exposure.** Different institutions hold the same asset, so one seller's price move marks down everyone else.
- **Funding.** Repo haircuts rise and lines are pulled, so an institution with no credit problem of its own loses its funding, see repo markets.
- **Counterparty.** Losses travel along derivative and lending chains, which is why the exposure profiles of counterparty risk matter most in precisely the states where they are hardest to measure.
- **Information.** A failure is evidence about anything that resembles the failed entity, so a run can spread to institutions that are genuinely solvent. Diamond and Dybvig's model of this is worth naming: a bank with a maturity mismatch has *two* equilibria, one where nobody runs and one where everybody does, and which one you get is a coordination outcome rather than a statement about the assets. Deposit insurance works by removing the bad equilibrium rather than by paying for it.
- **Forced selling of the unrelated.** When an investor must raise cash, they sell what is liquid, not what is troubled. This is why a crisis in one asset produces losses in everything, and it is

the mechanism behind the observation that **correlations go to one in a crisis**: it is not that the fundamentals suddenly align, it is that the marginal seller is the same person everywhere.

The policy playbook, and its price

Bagehot's rule still frames it: lend freely, to solvent institutions, against good collateral, at a penalty rate. Modern practice extends it to market-making of last resort, buying assets directly to restore functioning, and to guarantees and recapitalisation when the problem is solvency rather than liquidity, plus resolution regimes that let a failing bank be wound down without taking the payment system with it. The order matters: liquidity is fast and reversible, capital is slow and political, and confusing an insolvency problem for a liquidity one wastes both.

The price is **moral hazard**: an intervention that saves the system also rewards the risk-taking that created it, which is why the post-2008 settlement paired backstops with higher capital requirements, liquidity coverage ratios, resolution planning and bail-in debt. Being able to state the trade-off in one sentence, rather than picking a side, is what a good answer sounds like.

Early warning signs

Not a forecast, a checklist, and it is what a risk interview wants when it asks what would worry you.

- **Credit growing much faster than nominal GDP** for several years, which is the single most robust predictor in the literature.
- **Compressed risk premia**: spreads and implied volatility at multi-year lows, which means you are being paid little to bear risk precisely when leverage is highest.
- **Rising leverage and shortening funding**, especially where the leverage sits outside the regulated banking system and is therefore harder to see.
- **Asset prices detached from cash flows**, whether that is price to rent, price to earnings, or a valuation that only works if a growth rate persists for a decade.
- **External vulnerability**: a current account deficit financed by short-term foreign currency borrowing, which is the classic emerging-market setup.
- **A narrative that explains why the old rules no longer apply.** This one is not quantitative and it is the most reliable of the lot.

REAL INTERVIEW QUESTION (Société Générale · Credit)

Describe the subprime crisis to me as if you were talking to a child.

WHAT THEY TEST / ANSWER

This is a communication test, not a knowledge test, and the trap is to reach for jargon. Tell it as a chain of simple sentences, each following from the last. Banks lent money to people to buy houses, including people who probably could not repay, because house prices had been rising for years and everyone assumed that if a borrower failed you could just sell the house for more than the loan. The banks did not keep those loans, they bundled thousands of them together and sold slices of the bundle to investors all over the world, and because the bundles were large and complicated, almost nobody looked inside them. Then house prices stopped rising. Borrowers defaulted, so the slices were worth less than everyone had assumed, and the crucial part is that nobody knew *who* was holding the bad slices, so banks stopped lending to each other in case the other one was the sick one. Because banks borrow short and lend long, a bank that cannot borrow for a few days can fail even if it is basically sound, which is why the problem spread from American mortgages to the whole world in a matter of months. Governments and central banks eventually stepped in to lend the money nobody else would. Then, still in plain words, land the lesson: the mistake was not one bad loan, it was that everybody had quietly assumed the same thing, that house prices always go up, so when that turned out to be wrong everybody was wrong at the same time. Keep it under ninety seconds, use no acronym you have not just explained, and be ready to go straight into the technical version if they ask, which is developed in the 2008 financial crisis.

REAL INTERVIEW QUESTION (Natixis · desk not recorded)

What are the assumptions of the Black-Scholes model? What do you think of these mathematical models in periods of crisis?

WHAT THEY TEST / ANSWER

Two parts, and the second is where the marks are. Give the assumptions briskly, since they are developed in the Black-Scholes model: a geometric Brownian motion for the underlying with constant volatility, no jumps, continuous and costless hedging, constant known rates, no arbitrage, and frictionless markets with unlimited borrowing and lending. Then answer the real question with specifics rather than with "models fail in crises", naming which assumption breaks and what it costs. **Continuity** breaks first: crises are gaps, and a delta hedge set for a small move is useless through a jump, which is precisely how short-gamma books are destroyed. **Constant volatility** breaks: volatility clusters and spikes, so vega and the skew move violently, which is why the smile exists at all. **Costless hedging** breaks: spreads widen, liquidity disappears, and rehedging becomes expensive exactly when it is most needed. **Constant correlation**, in any multi-asset extension, breaks in the one direction that hurts, since diversification benefits evaporate as correlations move towards one because the marginal seller is the same everywhere. Then take a position, because they asked what you think: the models are not wrong so much as *misused*, since a model is a consistent language for interpolating between observed prices rather than a forecast of the world, and the failure is always in the layer above, where a calibrated number is treated as a fact and where risk limits are set from a normal distribution. Close with what a desk actually does about it: stress and scenario analysis alongside the model, gap and jump risk priced and limited explicitly, and a healthy suspicion of any position whose profitability depends on hedging continuously.

INTERVIEW TRAP

Four traps. First, **blaming the trigger**: the trigger is usually small and almost interchangeable, and the causes are the leverage and the mismatch accumulated in the calm before it. Second, **confusing illiquidity with insolvency**, which are different problems needing different tools, and mistaking one for the other is how policy responses fail. Third, **assuming diversification holds**, when the whole point of a crisis is that the marginal seller is the same person in every market, so correlations converge exactly when you were relying on them not to. Fourth, **treating each crisis as unprecedented**: the asset changes, the anatomy does not, and the candidate who can map a new episode onto the six phases above will always sound better than the one reciting a narrative.

VISUAL / CODE TO BUILD: Interactive

A crisis mechanics simulator in two parts. First, a **leverage spiral** panel: the reader sets initial assets, leverage and a price shock, and the tool runs the loop round by round, showing equity, leverage and forced sales at each step, with a market-impact slider that determines how much each round of selling moves the price. Setting impact to zero shows a single round; raising it shows the spiral diverging, which makes visible that the instability comes from the interaction of leverage with market depth rather than from either alone, and the closed form $A(L-1)x$ is displayed alongside so the first round can be checked by hand. Second, a **contagion network**: nodes as institutions with editable leverage, funding maturity and overlapping asset holdings, and the reader triggers a failure and watches it propagate through common exposures and funding links, with a toggle for a lender of last resort so the intervention's effect on the cascade is visible rather than asserted.

Where this goes next

The individual episodes are in the 1929 crash, Long-Term Capital Management, the 2008 financial crisis, the European sovereign debt crisis, the Covid market crash and major FX crises. The cycle that produces the preconditions is economic cycles and the policy response runs through monetary policy. On the risk side, the funding dimension is liquidity risk, the severe-scenario discipline is stress testing, and the regulatory settlement that followed 2008 is Basel III and IV.

3.12 Forward Guidance

sales · trading · structuring | core | rates · fx · cross-asset

Forward guidance is a central bank using words as an instrument: telling the market what it intends to do with rates in the future, in order to move prices today. It sounds like public relations and it is not. It is arithmetically the most powerful tool a central bank has, because almost every rate that matters to the economy is a function of the expected *path* rather than of today's level.

KEY IDEA

A long rate is, to a first approximation, the average of expected future short rates plus a term premium. So a central bank that changes what the market expects about the next two years has moved the two-year rate, without changing anything today and without spending anything. That is the whole mechanism, and it is why guidance became the central instrument once policy rates reached the effective lower bound: the current rate could not fall further, but the *expected path* still could.

WORKED EXAMPLE

What words are worth, in basis points. The policy rate is 4% and the market prices a cut of 25 basis points at each of the next three quarterly meetings, so the expected path over the coming year is

$$4.00,\ 3.75,\ 3.50,\ 3.25 \quad \text{averaging } 3.625\%.$$

The one-year rate sits near that average.

The central bank now says, credibly, that it does not expect to cut this year. The expected path becomes flat at 4.00%, the average moves to 4.000%, and the one-year rate reprices by

$$4.000 - 3.625 \;=\; \mathbf{37.5} \text{ basis points.}$$

Nothing happened. No rate changed, no asset was bought, no reserve was created. A sentence moved the front of the curve by 37.5 basis points, which on a one-year instrument is about 0.375% of notional, and moved every mortgage and corporate borrowing rate priced off it. That is why the market spends far more time parsing the statement than the decision.

The four kinds

- **Open-ended.** "Rates will remain low for an extended period." Vague, weak, and easy to abandon, which is also its attraction to the committee that issues it.
- **Time-contingent.** "At least through mid-2015." Concrete and therefore stronger, but it ties the bank to a calendar rather than to the economy, so it becomes wrong when the economy surprises.
- **State-contingent.** "Until unemployment falls below 6.5%." The strongest form, because it is a *rule* the market can price against incoming data rather than a promise it must take on trust. Its cost is that the chosen threshold can turn out to be the wrong variable, which is what happened when unemployment fell for the wrong reasons.
- **Qualitative and conditional.** "We will not raise rates until we see inflation sustainably at target." The modern compromise: a condition rather than a date, without a mechanical trigger.

INTUITION

The distinction that gets you credit in an interview is **Delphic** against **Odyssean** guidance. Delphic guidance is a *forecast*: the bank is telling you what it expects to do given its outlook, and if the outlook changes so does the policy. Odyssean guidance is a *commitment*: the bank binds itself to a course it will later want to abandon, and the binding is the point.

Odyssean guidance is far more powerful and far more dangerous, because it is deliberately **time-inconsistent**. Promising to hold rates low even after inflation returns only works if the market believes you will do the thing you will, at that moment, not want to do. That is also why guidance is a consumable resource: a bank that abandons a commitment has not merely changed policy, it has reduced the value of every future sentence it will say.

The reaction function is the thing, not the forecast

REAL INTERVIEW QUESTION (BMCE · Rates)

Is the Fed's policy hawkish or dovish if it expects high inflation, and why?

WHAT THEY TEST / ANSWER

The direct answer is **hawkish**, and then the whole value of the answer is in why that is not automatic.

The direct chain. A central bank with a price stability mandate that expects inflation above target should tighten, and because policy acts with long lags it must tighten on the forecast rather than waiting for the realised number. Expecting high inflation and doing nothing would be an admission that the mandate is not binding.

Then the correction that makes it a real answer. Hawkish and dovish describe the *reaction function*, not the forecast. Two banks with an identical inflation forecast can be one hawkish and one dovish, if one intends to respond and the other intends to look through it. So the question to ask back is: does the bank think the inflation is transitory, does it think the cost of tightening is higher than the cost of the overshoot, and has it said what it will do?

Which gives the sharper formulation. A bank that forecasts high inflation and guides to no change is *dovish*, and the market will read it that way regardless of the forecast, because the forecast is a description of the world and the guidance is a description of the bank. That distinction is what the question is testing.

Two refinements if you have time. First, what matters for the market is not the level of hawkishness but the *change* relative to what was priced: a hawkish bank that is less hawkish than expected produces a rally. Second, real rates are the actual stance: a nominal rate of 5% against expected inflation of 7% is a negative real rate and is stimulative however hawkish the language sounds. A candidate who says "hawkish, but the stance is what the real rate says" has answered better than the question asked.

Where to read it

Guidance arrives through four channels and they are not equally informative.

The statement is the binding text, negotiated word by word, and the market reads it as a *diff*: what changed since last time is the signal, and an unchanged paragraph is itself information.

The press conference is where the tone lives and where the statement gets qualified, which is why it routinely moves markets more than the decision did.

The projections, including the Federal Reserve's dot plot, show the committee's expected path. The correct way to read a dot plot is as a distribution rather than as a promise: the median is what gets quoted, the dispersion is what tells you how contested the path is, and neither is a commitment.

The minutes, published weeks later, reveal the balance of the debate. They matter most when the decision was close.

How it fails

Three failure modes are worth being able to name, because each is a recurring market event rather than a theoretical concern.

Withdrawing it faster than expected. The 2013 taper tantrum is the canonical case, and its lesson is that the damage came from the change in the expected path rather than from the policy itself.

Being forced to abandon it. Guidance issued in one inflation regime and repudiated in the next costs credibility precisely when credibility is the instrument, and the front end reprices violently because it had been anchored on a promise rather than on data.

Being too effective. Guidance that convinces the market rates will not move suppresses volatility and encourages leverage, so when the guidance does change the unwind is larger than the news warrants. A central bank that has successfully anchored expectations has also, quietly, built the positioning that will amplify its next surprise.

INTERVIEW TRAP

1. **Treating guidance as talk rather than as an instrument.** The worked example moves the one-year rate 37.5 basis points with no action at all.
2. **Confusing the forecast with the reaction function.** Hawkish and dovish describe what the bank will *do*, not what it expects.
3. **Reading the dot plot as a commitment.** It is a distribution of individual expectations, it is not voted on, and the dispersion carries as much information as the median.
4. **Ignoring what was already priced.** The market trades the change against expectations, so a hawkish central bank can produce a rally by being less hawkish than the market had assumed.
5. **Quoting the nominal stance.** A high nominal rate against higher expected inflation is a negative real rate and is stimulative, and the real rate is the stance.
6. **Assuming credibility is free.** Every abandoned commitment reduces the power of the next one, which is why banks are reluctant to make the strong state-contingent promises that would work best.

VISUAL / CODE TO BUILD: Interactive

A path-to-curve panel. The reader sets the expected policy rate at each of the next eight quarterly meetings by dragging a set of points, and the panel draws the implied curve out to two years alongside the resulting average, so the connection between the expected path and the traded rate is direct rather than asserted. A preset button labelled "guidance: no cuts this year" flattens the front of the path and prints the repricing in basis points, reproducing the section's 37.5.

A second panel is a credibility slider. The same guidance is issued with a credibility weight between zero and one, and the curve moves only partly toward the guided path, with the gap labelled as the market's discount for disbelief. Setting credibility low and watching the guidance achieve almost nothing is the clearest possible statement of why guidance is a consumable resource, and dragging it back up after a simulated abandonment, which resets it lower, makes the cost of breaking a promise a mechanic rather than a warning.

Chapter 4

History of Finance & Markets

4.1 The 1929 crash

trading · structuring · sales | core | equity · cross-asset

Nineteen twenty-nine is the event every market participant has heard of and very few can describe accurately. It is worth getting right for a reason that has nothing to do with history: the mechanism that turned a correction into a collapse is leverage meeting a margin call, and that mechanism has not changed in a hundred years. Only the instruments have.

Two different events, usually confused

The crash was a few days in October 1929. **The depression** was the three years that followed. Treating them as one event is the most common error, and it hides the fact that the crash by itself was survivable.

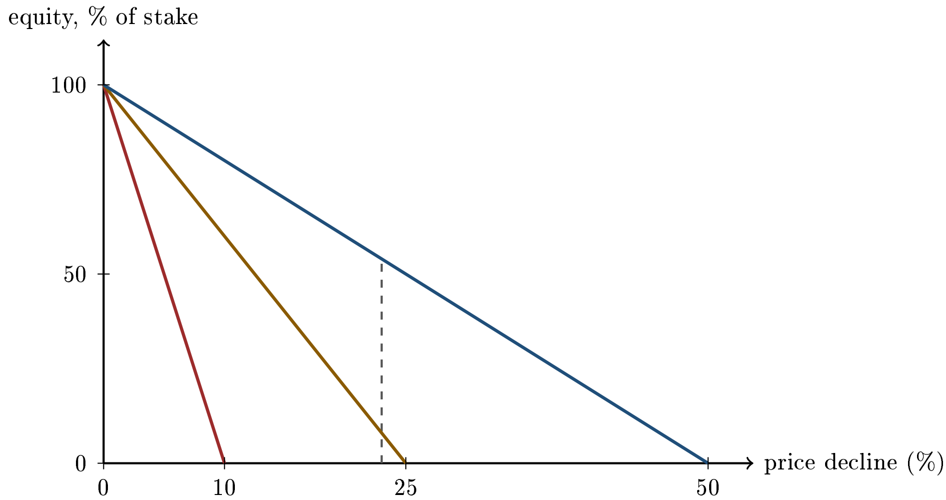
The Dow Jones Industrial Average peaked at **381.17** on 3 September 1929. It fell 12.8% on Monday 28 October and a further 11.7% on Tuesday 29 October, which compounds to $1 - 0.872 \times 0.883 =$ **23.0%** over two sessions. That is the crash, and it is roughly a bad fortnight by the standards of 2008 or March 2020.

What followed was not. The index kept falling for nearly three more years, bottoming at **41.22** on 8 July 1932, a decline of **89.2%** from the peak. The crash was an event; the depression was a process, driven by bank failures, monetary contraction and deflation, and it did far more damage.

Buying on margin, which is the part that matters

Through the 1920s, retail investors could buy stock with as little as **10% down**, the rest borrowed from the broker against the shares themselves. This is not the initial and variation margin of a modern cleared derivative, which is covered in collateral and haircuts; it is a loan secured on an asset whose price is the only thing supporting the loan.

The arithmetic of that arrangement is unforgiving and entirely mechanical.



10% margin   25% margin   50% margin
dashed guide at the 23% two-day fall of 28 and 29 October 1929

Put down a fraction m of the price and borrow the rest. After a decline d, your equity as a share of what you originally put up is $1 - d/m$: a straight line to zero at $d = m$. Leverage does not merely amplify the loss, it fixes the exact price at which you cease to exist.

WORKED EXAMPLE

Ten thousand dollars at 10% margin. You put up \$10,000, borrow \$90,000 and hold \$100,000 of stock.

Monday 28 October, the market falls 12.8%. Your position is worth $100{,}000 \times 0.872 =$ \$87,200 against a \$90,000 loan. Your equity is −**\$2,800**: the stake is gone and you owe the broker the difference. There is no version of waiting this out, because the broker sells you out on Monday to protect its own loan.

Tuesday adds 11.7%. Over the two days the position is worth $100{,}000 \times 0.872 \times 0.883 =$ \$76,998 against the same \$90,000, so equity is −**\$13,002**. A \$10,000 stake has become a \$13,000 debt in two sessions.

The same two days at today's 50% requirement. You put up \$10,000, borrow \$10,000 and hold \$20,000. After the same 23.0% fall the position is worth \$15,400 against a \$10,000 loan, so equity is **\$5,400**. You have lost 46% of your money, which is painful, and you still have a position and a choice.

Same market, same two days, same investor. The only difference is the leverage, and it is the difference between a bad week and insolvency.

KEY IDEA

Forced selling is what turns a decline into a crash. A margin call is not an opinion about value; it is an instruction to sell at whatever the market will pay, and it arrives at the worst moment by construction. The sales push prices lower, which triggers the next tier of margin calls, and the spiral runs until the leverage is gone. Every subsequent crash in this chapter contains the same loop with different collateral: portfolio insurance in 1987, repo haircuts on mortgage paper in 2008.

Why a crash became a depression

The 23% of late October did not cause what followed. Three policy failures did, and they are worth naming because each one is a live debate today.

Monetary contraction. The money supply fell by roughly a third between 1929 and 1933 as banks failed and deposits were destroyed. The Federal Reserve did not offset it. The modern consensus that a central bank should flood a banking panic with liquidity is a direct lesson from this, and it is why 2008 and 2020 looked the way they did.

The gold standard. A fixed exchange rate regime removed the option of easing, because loosening policy would have drained gold reserves. Countries that left gold earlier recovered earlier, which is the cleanest natural experiment in the history of exchange rate regimes. The full mechanism is in the end of the gold standard.

Trade contraction. The Smoot-Hawley tariff of 1930 raised US duties sharply, trading partners retaliated, and world trade collapsed. It is the standard exhibit for why a tariff war has no winner.

Underneath all three sits deflation, which raised the real burden of every debt in the economy exactly as described in inflation: falling prices meant falling nominal incomes against fixed nominal debts, so borrowers cut spending to service loans, which lowered prices further. The debt got heavier the harder anyone tried to pay it down.

The arithmetic of getting back

A drawdown and its recovery are not symmetric, and 1929 is the extreme illustration. Losing a fraction L requires a subsequent gain of $L/(1-L)$ merely to break even.

Drawdown	Gain needed to recover	Years at 7% a year
−20%	+25%	3.3
−50%	+100%	10.2
−80%	+400%	23.8
−89.2%	+824.7%	32.9

From 41.22 the index needed to multiply by $381.17/41.22 = 9.25$, a gain of **824.7%**, to reclaim its 1929 high. It did so in November 1954, **twenty-five years** after the peak. The right-hand column is what a steady 7% compounding would have taken, and the fact that the real recovery beat it says something about how far below fair value the trough was, not that any of it was pleasant to hold through.

KEY IDEA

The asymmetry is why risk management is about avoiding the deep drawdown rather than maximising the good years. A strategy that compounds well and occasionally loses 80% is not a good strategy with a flaw; it is a bad strategy, because the recovery arithmetic will not let the good years pay for it.

What survived, and why you work inside it

The market you work in is largely built from the response to this episode. The Securities Act of 1933 and the Securities Exchange Act of 1934 imposed disclosure and created the SEC. Glass-Steagall separated commercial from investment banking, a separation that lasted until 1999 and echoes in the Volcker Rule. Federal deposit insurance ended the retail bank run as a routine event.

And the Federal Reserve was given authority over margin requirements, which is why the 10% of 1929 is a legal impossibility today.

That last one is the point. Almost every rule that looks like bureaucratic friction on a modern desk exists because something in this section happened to somebody.

Interview angle

No question in the database asks about 1929 directly, which is itself informative: this is market culture rather than a technical screen. It surfaces obliquely, and when it does, three things separate a good answer.

Separate the crash from the depression. A candidate who says "the market fell 89% in 1929" has the number right and the event wrong, and it is an easy thing to check. The crash was 23% over two days; the 89% took until July 1932.

Go to the mechanism, not the narrative. The transferable content is margin and forced liquidation, and the way to show you understand it is the arithmetic above: at 10% margin a 10% fall is a total loss, so the wipeout price is known in advance. Any question about leverage, prime brokerage financing, or why a sell-off accelerates is the same question.

Have a view on what is different now, and be honest that some of it is not. Margin requirements, circuit breakers, deposit insurance and a central bank that acts as a lender of last resort are genuine structural changes. But leverage migrated rather than disappeared: it now sits in repo, in total return swaps, in options positioning and in the balance sheets of funds rather than in retail broker loans. The forced-seller loop still runs, which is exactly what 1987, 2008 and March 2020 demonstrated.

> **INTERVIEW TRAP**
>
> Four errors. (1) Quoting the 89% decline as though it happened in October 1929. (2) Treating margin as merely amplifying returns, when its distinctive feature is a known wipeout price and an involuntary seller. (3) Assuming the recovery arithmetic is symmetric, so that a 50% loss needs a 50% gain. (4) Believing leverage was regulated out of the system, rather than moved into instruments that are harder to see.

Where this goes next

The regime constraint that stopped policy from responding is the end of the gold standard and, in general terms, exchange rate regimes. The same forced-selling loop with different collateral runs in 1987 and in 2008. The deflation mechanism that made the debt heavier is in inflation, and the modern discipline built to stop a liquidation spiral before it starts is liquidity risk.

4.2 Bretton Woods system

sales · trading · structuring | core | fx · rates · commodities

For twenty-seven years the world's exchange rates were a policy decision rather than a market price. That sentence is worth sitting with, because everything a modern FX desk does exists only because the arrangement described here stopped working. Bretton Woods is not studied for nostalgia: it is studied because the dollar's position at the centre of the financial system, the existence of the IMF, and the reflex that a currency can be defended are all artefacts of a system designed in a New Hampshire hotel in July 1944.

This section covers the system as it was built and as it ran. Its collapse, and the floating world that replaced it, are the end of the gold standard.

What was agreed

Delegates from 44 nations met at Bretton Woods with the interwar period fresh in memory: competitive devaluation, capital flight and the trade collapse that followed the 1929 crash. The diagnosis was that a currency system with no rules invites every country to devalue its way out of a slump, and that all of them devaluing at once helps nobody. The design followed from the diagnosis.

- **A gold anchor, held by one country.** The dollar was fixed to gold at \$35 an ounce, and every other currency was fixed to the dollar. Only the dollar was convertible into gold, and only for foreign central banks, not for the public. So gold anchored the dollar and the dollar anchored everything else.
- **An adjustable peg, not a fixed one.** Currencies traded within a narrow band, conventionally one percent either side of parity. A country in what the Articles called *fundamental disequilibrium* could change its parity, but a change of more than ten percent needed the Fund's agreement. Devaluation was permitted and supervised rather than forbidden.
- **An institution to lend during the adjustment.** The IMF was created to provide short-term financing so that a country under pressure could defend its parity instead of devaluing at the first strain, with the World Bank alongside it for reconstruction. Both are described in international financial institutions.
- **Capital controls, assumed throughout.** Cross-border capital movement was restricted by design. This is the part most often left out, and it is the part that makes the rest coherent.

> KEY IDEA
>
> The reason capital controls were not an afterthought is the **impossible trinity**: a country can have a fixed exchange rate, free capital movement and an independent monetary policy, but only two of the three. Bretton Woods chose fixed rates and independent domestic policy, so it had to give up free capital movement. Every later stress on the system, and every peg that has broken since, is a country discovering it was trying to hold all three. The regimes themselves are set out in exchange rate regimes.

Note what the design was for. It was not built to make currencies immovable. It was built to make devaluation *negotiated* rather than unilateral, so that the adjustment one country needed did not become an act of aggression against its trading partners.

> WORKED EXAMPLE
>
> Sterling under the system devalued twice, and the two cases show the rules working as written.
>
> In 1949 the pound went from \$4.03 to \$2.80, a fall of $2.80/4.03 - 1 = -30.5\%$. In 1967 it went from \$2.80 to \$2.40, a fall of $2.40/2.80 - 1 = -14.3\%$.
>
> Both exceeded the ten percent threshold, so both required the Fund's agreement rather than a unilateral announcement. That is the mechanism: a large devaluation was available, but it was a public negotiation with an institution, which is precisely the friction the designers wanted. A country that could devalue quietly would do it too often.

The flaw that was identified in advance

INTUITION

The system needed the world to hold dollars as reserves. The world could only obtain those dollars if the United States sent more of them abroad than it took back, which means running a deficit. But the more dollars abroad, the larger the claim on a fixed stock of American gold, and the less credible the promise to convert at $35. So the system required the United States to do the one thing that would eventually destroy confidence in it. Supplying the world's reserve asset and remaining credibly convertible were incompatible in the long run.

This is the **Triffin dilemma**, named after the economist who set it out in 1960, more than a decade before the system ended. It is worth knowing by name because it is the cleanest example in monetary history of a structural flaw being correctly diagnosed while the structure was still standing, and because the same logic applies to any reserve currency, including the dollar today.

WORKED EXAMPLE

The magnitude of what has happened to the dollar's gold value is easy to state and hard to believe, and it is one division. Under the system a dollar bought one thirty-fifth of an ounce. Divide 35 by the current gold price to get the fraction of that gold a dollar still buys: at \$3,500 an ounce the answer is $35/3{,}500 = 0.01$, so a dollar buys one percent of the gold it was worth in 1971, a fall of 99%.

Treat the \$3,500 as an assumed level for the arithmetic rather than a quote. The point survives whatever the real number is: measured in gold, the currency that anchored the system has lost almost all of its value, and measured in goods it has not, which is why the gold price is a poor inflation measure and a good confidence measure.

What the system left behind

Bretton Woods ended in 1971, but three of its features outlived it: the institutions, the habit of defending a peg, and above all the dollar's central position, which no longer rests on gold but on the fact that everyone else uses it.

REAL INTERVIEW QUESTION (Credit Suisse · FX)

What does the level of the dollar imply in macro terms?

WHAT THEY TEST / ANSWER

Answer in three layers and do not stop at the first, because the first is trivial.

For the United States. A stronger dollar makes imports cheaper and exports dearer, so it is disinflationary and a drag on the traded sector. Around forty percent of S&P 500 revenue is earned abroad, so a strong dollar is a headwind to reported US earnings even when nothing happens to the underlying business.

For everyone else, which is the layer that matters. The dollar is the currency the rest of the world borrows in and trades in, and that is the surviving legacy of this system. So a strong dollar tightens global financial conditions directly: emerging market borrowers with dollar debt and local revenue see their debt burden rise in real terms, commodities priced in dollars become more expensive for non-dollar buyers, and trade finance becomes scarcer. **The dollar is therefore a global risk variable, not just an exchange rate**, and a sharp appreciation is one of the most reliable stress signals there is.

For the United States again, structurally. Because the world holds dollars as reserves, the US can borrow in its own currency at a lower rate than it otherwise would, the so-called exorbitant privilege, and it runs the persistent current account deficit that supplies the world with those dollars. That is the Triffin dilemma still running, now without a gold constraint to make it acute.

Close by saying what you would actually watch: the dollar index for direction, the cross-currency basis for whether dollar funding is genuinely scarce rather than merely expensive, and the reaction of emerging market spreads. Level alone tells you less than level plus the funding market.

REAL INTERVIEW QUESTION (CACIB · FX)

What is the difference between CNH and CNY?

WHAT THEY TEST / ANSWER

Same currency, two prices, because of capital controls, which makes it the clearest living example of the world Bretton Woods assumed.

CNY is the onshore renminbi, traded in mainland China under the authority of the People's Bank of China, which sets a daily reference rate and allows trading only within a band around it. Access is restricted to approved participants and approved purposes.

CNH is the offshore renminbi, traded mainly in Hong Kong, freely deliverable and priced by supply and demand. There is no band and no fixing.

They are the same currency and their prices differ, which cannot happen when capital moves freely, because arbitrage would close the gap instantly. It happens here precisely because capital does not move freely: the pipe between onshore and offshore is narrow and administered. So the **CNH minus CNY spread is a real-time reading of the pressure on the controls**, widening when offshore holders want out faster than the official channel allows.

Then make the point the question is testing, which is that you understand what convertibility means. Under Bretton Woods most currencies looked like CNY: a managed rate, an official channel and controls around it, with unofficial rates appearing wherever the controls bound. China today is not an anomaly, it is the arrangement that was normal for most of the twentieth century, and the freely floating major currency is the recent innovation. The market plumbing itself is in FX spot markets, and pegs under stress are in pegged currencies and FX crises.

INTERVIEW TRAP

Three errors. (1) Saying Bretton Woods was a gold standard: it was a *dollar* standard with a gold anchor at one remove, and only central banks could convert. (2) Saying exchange rates were fixed: they were adjustable, and the adjustment procedure was the point of the design. (3) Treating the system's end as a surprise, when the Triffin dilemma had named the mechanism eleven years earlier. Getting the third one right signals that you read history for mechanisms rather than for dates.

Where this goes next

The system's collapse and the floating regime that followed are the end of the gold standard. The institutions it created are international financial institutions, the taxonomy of what replaced the adjustable peg is exchange rate regimes, and what happens when a modern peg meets the impossible trinity is pegged currencies and FX crises. The interwar experience that motivated the whole design is the 1929 crash.

4.3 End of the gold standard

sales · trading · structuring | core | fx · commodities · cross-asset

Almost everything that makes a modern trading floor possible dates from one weekend in August 1971. Floating exchange rates, the FX market as we know it, currency derivatives, central banks with genuine discretion over interest rates, and gold as a freely traded asset rather than an official price: all of it follows from the decision to stop exchanging dollars for gold. It is also the cleanest available illustration of a principle that recurs everywhere in markets, that a fixed price survives only as long as the reserves and the political will behind it.

What was being ended

The system is described in the Bretton Woods system, so only the two load-bearing facts are needed here. The United States undertook to exchange dollars for gold with foreign official holders at $35 an ounce, and every other member pegged its currency to the dollar within narrow bands. Gold anchored the dollar, and the dollar anchored everything else.

> KEY IDEA
>
> **The Triffin dilemma is why it could not last**, and it is the single most valuable thing to take from this section because the logic generalises to any reserve-currency arrangement. The world needed dollars for reserves and for trade, and the only way to supply them was for the United States to run persistent external deficits. But the larger the stock of foreign-held dollars grew relative to the American gold stock, the less credible the promise to convert them became. So the system required the issuer to do the very thing that would eventually destroy confidence in it. Success and failure were the same action, differing only in degree, which means there was no policy path that kept the arrangement intact indefinitely.

The road to the exit

The strain was visible for more than a decade before the break.

- **Late 1950s onward.** European currencies returned to convertibility and world trade grew, so foreign dollar holdings rose steadily. Meanwhile American spending on the Vietnam war and on domestic programmes widened the deficit and raised inflation.
- **The London Gold Pool, from 1961.** Central banks intervened jointly to hold the free market price of gold at the official level. It worked while the pressure was modest, then failed: in March 1968 the pool was abandoned and a **two-tier** arrangement replaced it, with the official $35 price for central banks and a free market price for everyone else. Once the two prices could differ, the official one was a convention rather than an equilibrium.
- **Sterling's devaluation in 1967** showed that a major peg could break, which moved attention to the dollar.
- **Special Drawing Rights, created in 1969**, were an attempt to supply reserves without American deficits, that is, to solve Triffin directly. They arrived too late and too small.
- **1971.** Convertibility demands accelerated as holders converted dollars into gold while they still could, which is a run in every meaningful sense, conducted by governments.

The Nixon shock and the two years that followed

On 15 August 1971 the United States **suspended the convertibility of the dollar into gold**, alongside a temporary import surcharge and domestic wage and price controls. The gold window was described as closed temporarily. It never reopened.

- **December 1971, the Smithsonian Agreement.** An attempt to rebuild fixed rates at new levels: the dollar was devalued against gold from \$35 to \$38 an ounce and the permitted fluctuation bands were widened. It was intended as a new system and lasted barely a year.
- **February 1973.** A second devaluation, to \$42.22 an ounce, which convinced the market that the official price was being followed rather than defended.
- **March 1973.** The major currencies floated, and this time nobody tried to put the system back together.
- **1976, the Jamaica Accords.** Floating was formally accepted, gold was removed from the centre of the international monetary system, and the official price was abolished.

INTUITION

Read the sequence as a **defence of a fixed price that ran out of ammunition**, because that is a pattern you will meet again in major FX crises. A peg is a promise to trade at a level, so it is economically a short option position: the authority collects stability while conditions are calm and pays out reserves when they are not. Each stage above is a step in the classic sequence, defend by intervening (the Gold Pool), then segment the market (two tiers), then reset the level and claim it is now credible (Smithsonian), then reset again, then stop. The lesson is not that the officials were foolish, it is that a fixed price without a mechanism to change the underlying imbalance can only postpone.

Why it matters on a desk today

1. **The FX market exists.** With rates floating, currency risk became a continuous price rather than an occasional policy event, which created both the need to hedge it and the market to do so. Currency futures were listed in Chicago in 1972, and the option pricing paper that made derivatives tractable appeared in 1973, so the modern derivatives industry and floating exchange rates are almost exactly contemporaries.
2. **Monetary policy gained discretion, and had to earn credibility.** Under a gold anchor, a central bank's room for manoeuvre was limited by convertibility. Without it, the only constraint is the bank's own mandate and reputation, which is why the 1970s produced the Great Inflation, why that produced the Volcker tightening, and why the entire modern architecture of independent central banks with numerical inflation targets exists, see monetary policy. Fiat money is not a technical detail, it is the reason inflation expectations are a policy variable at all.
3. **The trilemma became the organising choice.** With capital mobile, a country can have at most two of a fixed exchange rate, free capital movement and an independent monetary policy. Bretton Woods chose fixed rates and policy independence, which required capital controls; as those controls eroded, the arrangement had to give. Every subsequent regime choice is a position on the same triangle, developed in exchange rate regimes.
4. **Gold became an asset instead of a price.** With no official rate, gold trades on supply, demand and above all on the opportunity cost of holding something that yields nothing, which is the real interest rate. The market mechanics belong to metals.
5. **The dollar kept its role without the anchor.** It remained the currency of invoicing, reserves and funding on the strength of network effects and market depth rather than convertibility, which is the source of both the so-called exorbitant privilege and of the global economy's sensitivity to American monetary policy.

REAL INTERVIEW QUESTION (UBS · Commodity)

Would you invest in gold right now?

WHAT THEY TEST / ANSWER

This is a framework question wearing a market question's clothes, so answer with the framework, take a position, and state what would change your mind. The framework: gold produces no cash flow, so it cannot be valued by discounting anything, and its price is essentially the market's required compensation for holding a zero-yield, zero-credit-risk asset. That gives four drivers to check in order. **Real rates**, the dominant one, since they are gold's opportunity cost, so falling real yields are the strongest case for owning it and rising real yields the strongest case against. **The dollar**, for the mechanical and common-driver reasons above. **Official demand**, since sustained central bank buying is a genuine change in the demand curve rather than a flow. And **portfolio role**, which is often the real reason to hold it: an asset with no credit risk and low or negative correlation to equities in some regimes, sized as insurance rather than as a return generator. Then the honest position: I would size it as a hedge against a specific risk I can name, most obviously a policy-credibility or inflation-persistence risk, rather than as a directional bet, because the historical Sharpe ratio of gold alone is unimpressive while its contribution inside a portfolio can be worth paying for. State what would change your mind, which is the beat that makes it a professional answer: a sustained rise in real yields, a reversal of official buying, or evidence that the crisis hedge is not working when tested. And give your as-of date rather than inventing a level.

INTERVIEW TRAP

Four traps. First, **saying the gold standard ended in 1971 and stopping there**: the convertibility suspension was August 1971, but fixed rates were patched twice and generalised floating only arrived in March 1973, with formal acceptance in 1976. Getting the sequence right is the whole point of the question. Second, **treating it as a policy error rather than a structural inevitability**: the Triffin dilemma means the arrangement was self-undermining, so the timing was a choice while the outcome was not. Third, **claiming fiat money caused inflation by itself**: removing the anchor removed a constraint, and what followed depended on what replaced it, which is why the same fiat system produced the 1970s and then four decades of low inflation. Fourth, **valuing gold with a discount model**: it has no cash flow, so the analysis is about opportunity cost and portfolio role, not about intrinsic value.

VISUAL / CODE TO BUILD: Interactive

A two-panel history explorer. The first plots the gold price and the American gold stock against foreign-held dollar claims from the 1950s to 1973, so the Triffin dilemma appears as two diverging lines rather than as an argument, with the Gold Pool period, the 1968 two-tier split, the August 1971 suspension and the two devaluations marked on the axis. The second is a regime laboratory built on the trilemma: a draggable triangle where the reader picks two of fixed exchange rate, free capital movement and monetary independence, and the panel shows which historical regimes occupied that corner and what broke them, with Bretton Woods sitting on the fixed-plus-independence vertex and capital controls named as the price it paid. A third strip plots gold against the US real yield over the whole floating era with a rolling correlation, so the dominant driver identified in the interview answers is visible rather than asserted.

Where this goes next

The system this section dismantles is the Bretton Woods system, and the framework for what replaced it is exchange rate regimes, with the pattern of pegs breaking in major FX crises and official intervention in central banks and FX policy. The discretion that central banks acquired is the subject of monetary policy, the inflation that followed is inflation, and gold's behaviour as a traded metal is metals.

4.4 The 1987 Crash

trading · structuring · sales | core | equity · cross-asset

On Monday 19 October 1987 the Dow Jones Industrial Average fell **22.6%** in a single session, the largest one-day percentage decline in its history, with index futures falling further still. There was no policy announcement, no war, no default. The market fell almost twice as far in one session as it did on the worst day of the 1929 crash, and the immediate cause was not news at all.

This section is in the book for two reasons that have nothing to do with history. It is the cleanest example of a mechanical trading strategy destroying the market conditions it depended on, and it is the reason the equity volatility skew exists.

What portfolio insurance was, and why it failed

By 1987 a large amount of institutional equity, plausibly in the tens of billions of dollars, was managed under **portfolio insurance**. The idea was elegant and, in a Black-Scholes world, exactly correct: rather than buying a protective put, you replicate one. A put's delta becomes more negative as the market falls, so you sell index futures as prices decline and buy them back as prices rise, and the resulting payoff is dynamic replication of downside protection without paying an option premium.

KEY IDEA

The strategy has a property its users treated as an implementation detail and which was in fact the whole risk: **it sells when the market falls**. That is fine when you are a small participant trading against a deep market. It is self-defeating when enough capital follows the same rule, because then the market you assumed you could sell into *is* the other portfolio insurers, all selling for the same reason at the same moment.

The general statement, which is what to carry away: **dynamic replication assumes a liquidity that dynamic replication itself destroys.** The strategy is not wrong; it is wrong at scale, and it has no way of knowing what scale it is at.

That sentence is not a historical observation. It applies to every delta-hedged short-gamma book that must buy as the market rises and sell as it falls, to CPPI structures with their mechanical de-risking rules, to leveraged strategies facing margin calls, and to any risk framework that tells many institutions to cut the same exposure on the same trigger. Every one of them is a portfolio insurer under a different name, and liquidity risk describes the spiral it produces.

The number that killed a model

WORKED EXAMPLE

How impossible was it? Equity index volatility of 20% a year gives a daily standard deviation of

$$\frac{20\%}{\sqrt{252}} = 1.26\%.$$

So a fall of 22.6% is

$$\frac{22.6}{1.26} = \mathbf{17.9} \text{ standard deviations.}$$

Under the normal distribution that Black-Scholes assumes, an event that far into the tail has a probability of roughly 10^{-72}. There have been about 3.5×10^{12} trading days since the Big Bang. So the model says a day like 19 October 1987 should occur about once in 10^{59} *times the age of the universe.*

It happened. Which does not mean the market did something impossible; it means the model was wrong about the tails, and by an amount that no recalibration of volatility can repair. You cannot fix a 10^{-72} by raising the volatility input, because the Gaussian tail falls away faster than any scaling can compensate. The distribution itself has to change, which is the argument for fat tails, for jump processes and for stress testing that does not sample from a fitted distribution at all.

The skew is a memory of this day

This is the point that makes 1987 a live subject on an options desk rather than a story.

INTUITION

Before October 1987, the equity index volatility surface was roughly flat. Options across strikes traded at similar implied volatilities, broadly as Black-Scholes says they should. After October 1987 they did not, and they never have again: out-of-the-money index puts have traded at persistently higher implied volatilities than out-of-the-money calls ever since.

So the skew is not a mathematical property of markets discovered by better modelling. It is a price the market started charging on a specific date, for a risk it had just been shown was real, and it has never stopped charging it. When equity skew says the shape reflects crash risk and the structural demand for downside protection, this is the event that created both.

The practical consequence for a trader is that skew is a risk premium with a history, not a static feature. It steepens when the memory is refreshed and flattens slowly when it is not, which is why selling it is a strategy with long quiet periods and rare, large losses, and why the people who sell it usually describe the last such loss as unprecedented.

What changed afterwards

Circuit breakers. US exchanges introduced market-wide trading halts triggered by percentage declines in the S&P 500, currently at 7%, 13% and 20%, on the theory that a pause allows information to disseminate and reduces the mechanical feedback. Whether they help is genuinely contested: they interrupt the spiral, and they also create a **magnet effect**, where approaching a halt accelerates selling by traders who want to exit before liquidity disappears.

The central bank template. The Federal Reserve's response, an immediate public commitment to supply liquidity, is the ancestor of every crisis response since, and it is where the market's expectation of official support in a crash begins.

Market structure. The crash exposed how badly the systems of the day handled volume, with order routing overwhelmed and quotes stale for hours, and much of the automation in market microstructure traces to that failure.

INTERVIEW TRAP

1. **Treating 1987 as a history question.** It is the origin of the skew, and the skew is priced into every equity option a desk trades today.
2. **Assuming a strategy that works in backtest works at scale.** Portfolio insurance was correct for one participant and catastrophic for all of them. Nothing in the backtest can reveal that, because the backtest assumes the market absorbs your trades.
3. **Repairing a fat-tail problem by raising volatility.** A 10^{-72} event cannot be brought into range by rescaling a Gaussian. The distribution is the problem.
4. **Believing circuit breakers prevent crashes.** They interrupt the mechanism and can accelerate the approach to it. They buy time rather than removing risk.
5. **Thinking this cannot recur because of the safeguards.** The specific mechanism was addressed; the general one, many participants forced to sell simultaneously by the same rule, recurs regularly, and 1998, 2008, 2020 and every volatility-target unwind are versions of it.

VISUAL / CODE TO BUILD: Interactive

A feedback simulator. The reader sets what fraction of the market follows a mechanical de-risking rule and how aggressively that rule sells per percentage point of decline, then applies an initial shock. The panel runs the loop: the shock triggers selling, the selling moves the price through a market-impact function, the new price triggers more selling.

The intended discovery is a threshold. At low participation the path damps out and looks like an ordinary bad day; above a critical share of the market the same initial shock produces a path that does not stop until the rule runs out of assets. Finding that threshold by dragging one slider is the section's central claim, and it makes the point that the danger is not the strategy but its share of the market.

A second panel draws two implied volatility surfaces side by side, one flat and one skewed, with a slider labelled "years since the last crash" that flattens the skew as it advances and snaps it steep when a crash event is triggered. Watching a decade of quiet slowly cheapen the downside, then reprice in a day, is the risk premium's history made visible, and it is the best argument the book can offer for why selling skew is a strategy with a long, comfortable losing tail.

Where this goes next

The general anatomy this episode is an instance of, and the other crises that share it, is financial crises. The shape it left on the surface is volatility smile and skew and, for equity specifically, skew and smile. The hedging technique whose scale caused it is delta neutrality and dynamic hedging, its modern retail descendant is capital-protected products, and the two risk disciplines that exist because a fitted distribution was not enough are liquidity risk and stress testing.

4.5 Long-Term Capital Management

trading · structuring · sales | core | rates · credit · cross-asset

Long-Term Capital Management is the most instructive failure in modern finance, and the reason is not that clever people were wrong. It is that they were largely *right* and still lost everything, because being right about where a price should end up says nothing about whether you can survive the path it takes to get there. That distinction, between the thesis and the funding, is the whole section, and it is why LTCM comes up in interviews twenty-five years later.

What the fund was

LTCM was founded in 1994 by John Meriwether, who had run the celebrated bond arbitrage group at Salomon Brothers. Its partners included Myron Scholes and Robert Merton, who received the Nobel Memorial Prize in economics in 1997 while the fund was running. The intellectual firepower is part of the story only because it explains why the market extended the fund so much credit and so little scepticism.

The strategy was **convergence arbitrage**. Find two instruments that are close substitutes and are trading at a spread that should close, buy the cheap one, sell the dear one, and wait. The classic example is on-the-run versus off-the-run Treasuries: the most recently issued bond trades richer than an almost identical older bond, purely because it is the one everyone trades, and that premium decays as it ages. Other trades were swap spreads, convergence between European sovereign bonds ahead of the euro, and later equity volatility and merger arbitrage.

INTUITION

Notice the shape of the payoff. A convergence trade earns a small, steady amount most of the time and loses a large amount rarely, when the spread widens instead of closing. That is the payoff of a **sold option**: you are collecting a premium for absorbing a risk that mostly does not happen. Nothing in the trade is an option contract, but the return profile is short optionality, and any strategy with that shape looks brilliant for exactly as long as the rare event has not occurred.

The individual spreads were small, so the returns had to come from size. The fund reported net returns above 40% in 1995 and in 1996, and around 17% in 1997, and it did so on positions that individually offered a few basis points.

The arithmetic that killed it

WORKED EXAMPLE

At the start of 1998 LTCM had roughly \$4.7bn of equity supporting a balance sheet of about \$125bn.

Leverage $= 125/4.7 = 26.6$ times, before counting derivatives, whose notional was a further order of magnitude larger.

Now invert it. The loss on assets that wipes out the equity entirely is $4.7/125 = 3.8\%$. **A 3.8% move against a portfolio of near-identical bonds destroys the firm.** Not a crash, not a default, a move that a single volatile week produces routinely in credit.

That is the entire lesson in one division, and it is worth doing in an interview rather than asserting. Leverage does not change your view, it changes how much of the path you can survive, and 1/leverage is the honest statement of how much room you have.

In August 1998 Russia defaulted on its rouble debt and devalued. The reaction was a flight to quality: investors bought the most liquid instruments and sold everything else. Every convergence trade LTCM held was, by construction, long the illiquid side and short the liquid one, so **every position lost at once**. The diversification the fund believed it had across countries and asset classes was diversification across trades that shared a single factor, and that factor had just been switched on.

Two things then compounded it. The trades were **crowded**: other desks had put on the same relative-value positions, so when everyone reduced at once the spreads widened further rather than snapping back. And the losses triggered margin calls, which forced selling into the move, which produced more losses. Equity fell from \$4.7bn to about \$400m in under four months, a loss of roughly \$4.3bn, or about 91% of the fund's capital.

WORKED EXAMPLE

The rescue is arithmetically tidy and worth knowing. On 23 September 1998 a consortium of 14 banks, convened by the Federal Reserve Bank of New York, injected \$3.625bn in exchange for 90% of the fund.

Check the implied valuation: \$3.625bn buying 90% implies a post-money value of $3.625/0.90 = \$4.03$bn, so the 10% left to the original partners was worth about \$400m. That matches what the fund's equity had fallen to, which is the point. The partners were not bought out, they were diluted to almost nothing by the only capital available.

No public money was used. The Fed convened the room, and the banks recapitalised the fund because an uncontrolled liquidation of \$125bn of positions into a market already under stress would have hit their own books harder than the cheque did. That reasoning, private money mobilised by a public convenor to avoid a disorderly unwind, is the template people reached for again in the 2008 financial crisis.

REAL INTERVIEW QUESTION (Allianz Global Investors · Market Risk)

Tell me about hedging strategies using derivatives and about changing the duration of a portfolio: which instruments, and what is the impact of leverage?

WHAT THEY TEST / ANSWER

Two halves. The instrument half is mechanical; the leverage half is where the answer is either good or forgettable, so spend your time there.

Instruments. To change a portfolio's rate sensitivity without trading the underlying bonds, you use interest rate swaps as the default, receiving fixed to lengthen duration and paying fixed to shorten it; bond futures where you want listed liquidity and daily margin; and options, caps and floors or swaptions, when you want to change the sensitivity in one direction only. Equity exposure is adjusted with index futures or total return swaps rather than by selling the book. The duration mechanics themselves are in duration and convexity.

Leverage, which is the real question. Derivatives change duration without using cash, and that is exactly what makes them dangerous: the position is embedded but the capital consumed is only margin, so the portfolio's sensitivity can be several times its equity without appearing large anywhere. Say the three consequences explicitly.

One, leverage does not change your expected return per unit of risk, it scales both. It is not alpha, it is a multiplier, and it multiplies the drawdown identically.

Two, it converts a market risk into a funding risk. An unlevered investor who is wrong for six months waits. A levered one meets margin calls in the meantime, and if the collateral runs out the position is closed at the worst possible moment regardless of whether the thesis was correct. This is the mechanism described in liquidity risk.

Three, the survivable move is the reciprocal of the leverage. At ten times, a 10% adverse move is terminal; at twenty-five times, 4% is. That single division is the discipline, and the example that makes it memorable is this section.

Close on what a risk manager would actually monitor: not the notional, but the margin consumed as a share of available liquidity, and how that number behaves under a stress scenario rather than under normal conditions.

What it taught the industry

KEY IDEA

The lesson is not "do not use leverage" or "models are wrong". It is that **correlation is a property of a regime, not of a portfolio**. LTCM's positions were uncorrelated in the data because the sample contained no episode in which liquidity itself was the scarce thing. When that episode arrived, every position was revealed to be the same position. Diversification measured on a calm sample is a description of calm.

REAL INTERVIEW QUESTION (Société Générale · Market Risk)

What is a stress test compared to what VaR studies?

WHAT THEY TEST / ANSWER

Different questions, and the difference is best stated as a distinction of kind rather than of severity.

VaR asks a question about the distribution you have observed: given the recent history of returns, what loss is exceeded only 1% or 5% of the time? It is statistical, it is calibrated on a sample, and it inherits every limitation of that sample. It says nothing about how bad the loss is once you are past the threshold, which is why Expected Shortfall exists.

A stress test asks a question about a scenario you specify: if Russia defaults, or rates move 200 bp, or every credit spread doubles, what happens to this book? It is not a probability statement at all. It does not care how likely the scenario is, and that is a feature: the point is to know the consequence, and let the committee decide how much of it to tolerate. The methods behind both are in value at risk and stress testing.

Then make the historical point, because it is what turns a definitions answer into an argument. **LTCM is the reason the industry stopped treating VaR as sufficient.** The fund's positions had a benign VaR because the historical sample contained no flight to quality of that severity, and no statistical measure fitted to that sample could have flagged the exposure. What would have flagged it was the question a stress test asks: what happens to this book if liquidity disappears and every spread widens together? That question requires no distributional estimate, only the willingness to ask it.

Finish with the practical relationship rather than choosing a winner. VaR is the daily number that sizes normal risk and drives limits; the stress test is the one that tells you whether you survive. A desk needs both, and the useful stress scenarios are usually historical episodes rather than invented ones, precisely because a real crisis correlates things in ways nobody would have thought to specify.

INTERVIEW TRAP

Three errors. (1) Saying LTCM's trades were wrong: most of them converged eventually, which is the point, since the fund was gone before they did. (2) Saying it was bailed out by the government: the Fed convened a room, the money was private, and the distinction matters because it is the argument that was reused in 2008. (3) Blaming the models generically. The failure was specific and repeatable: risk estimated on a sample containing no liquidity crisis, and leverage sized against that estimate. Name that mechanism and the answer sounds like understanding rather than like a moral.

Where this goes next

The template of a convened rescue, and the same lessons at systemic scale, are in the 2008 financial crisis. The measures argued about here are value at risk and stress testing, the mechanism that turns a losing position into a forced sale is liquidity risk, the vehicle type is placed in buy-side vs sell-side, and the desk-level version of carrying a position you cannot exit is inventory management.

4.6 2008 financial crisis

sales · trading · structuring | core | credit · cross-asset

This is the single most asked historical question in finance interviews, and most answers are a list of names in roughly the right order. What distinguishes a good one is a *chain*: cheap money produced bad loans, securitisation turned bad loans into apparently safe securities, short funding financed them, and when the assumption underneath the whole structure failed, the funding disappeared from institutions that were not obviously connected to housing at all. The general anatomy is in financial crises; this section is the specific machine, because the instruments and the sequence are what get asked about.

The build-up: four links in one chain

1. **Cheap money and a housing boom.** Low policy rates after 2001, large capital inflows into dollar assets, and a widespread belief that house prices do not fall nationally, which had been true in the United States for the length of most careers.
2. **Origination degraded, because nobody kept the loan.** The **originate-to-distribute** model meant a lender's income came from writing and selling a mortgage rather than from being repaid, so underwriting standards became a cost rather than a protection. Adjustable-rate mortgages with low introductory payments made the loans affordable only until they reset, which embedded a scheduled wave of stress.
3. **Securitisation manufactured safety.** Mortgages were pooled into residential mortgage-backed securities, sliced into tranches so that losses hit the bottom first, and the lower slices were repackaged into collateralised debt obligations and sometimes into CDOs of CDOs. Tranche mechanics belong to tranches; the point here is the *effect*: the process converted a pool of individually risky loans into a large volume of paper rated AAA, which is what allowed it to be bought by investors mandated to hold only high-quality assets.
4. **Short funding carried it.** The paper was financed in repo, through asset-backed commercial paper and in off-balance-sheet vehicles, so long-dated illiquid assets sat against liabilities that had to be rolled continuously, in some cases overnight. Broker-dealer leverage in the mid-

2000s ran at multiples of thirty times equity or more, which by the arithmetic in financial crises means a fall of one thirtieth, about 3.3%, in asset values is enough to erase the equity.

> KEY IDEA
>
> **The whole structure was a bet that mortgage defaults would not happen together.** Tranching does not remove credit risk, it transforms it into **correlation risk**: a senior tranche only takes losses if a large fraction of the pool defaults at once, so its value depends on the assumed correlation of defaults rather than on their average rate. The models used low correlations estimated from a period in which regional housing markets had indeed behaved independently. When the downturn turned out to be national rather than regional, correlation went to one, the senior tranches that could only fail together failed together, and the safety was revealed as an assumption rather than a structure. The signs are developed in credit correlation: an equity tranche is long correlation and a senior tranche is short it, and 2008 is what being short correlation looks like when you did not know you were.

Two amplifiers sat alongside. The **ratings** were produced under an issuer-pays model on models fed by the same correlation assumptions, so the label that made the paper widely holdable was manufactured by the same logic that failed. And the **super-senior** risk that banks could not sell was insured, often by monoline insurers and by AIG, which wrote protection without holding capital against a scenario they treated as impossible. That is the textbook case of **wrong-way risk**: the protection was worth least exactly when it was needed, because the event that triggered it was the event that bankrupted the seller, see counterparty risk.

The sequence, and where the hinge actually was

- **2006.** US house prices peak and begin to fall.
- **Early 2007.** Subprime originators fail as delinquencies rise. Losses are still described as contained to a small corner of the market.
- **June to August 2007.** Two Bear Stearns credit funds collapse, and in August BNP Paribas suspends redemptions on three funds because it cannot value their holdings. That sentence is the real beginning: not a loss, an *admission that the assets could not be priced*, which is what closed the asset-backed commercial paper market and made banks stop lending to each other.
- **September 2007.** A depositor run on Northern Rock, the first on a British bank in well over a century, showing that the funding problem had reached retail confidence.
- **March 2008.** Bear Stearns cannot roll its repo funding and is sold to JPMorgan with official support. The market reads this as evidence that a large dealer will not be allowed to fail outright.
- **7 September 2008.** Fannie Mae and Freddie Mac are placed into conservatorship.
- **15 September 2008.** Lehman Brothers files for bankruptcy, the largest in United States history.
- **16 September 2008.** AIG is rescued, because its written protection made it a node the system could not lose. On the same day the Reserve Primary Fund **breaks the buck**, valuing its shares below one dollar after writing off Lehman paper.
- **October 2008.** Coordinated rate cuts, government guarantees of bank funding, recapitalisation programmes including TARP, and central bank swap lines to supply dollars abroad.
- **2009.** Zero rates, quantitative easing, bank stress tests, and an equity market low in March while the economy was still contracting.

KEY IDEA

Why Lehman was the hinge, in one sentence: it was not the size of the loss, it was that a disorderly failure turned a solvency question into a system-wide funding question. Three mechanisms did the work. There was no **resolution regime** for a large dealer, so bankruptcy meant client assets frozen and derivative positions abruptly uncertain rather than an orderly transfer. Money market funds held Lehman paper, so when one broke the buck the run spread to the part of the system investors treated as cash, which froze commercial paper and therefore the funding of ordinary non-financial companies. And after Bear Stearns the market had priced an implicit guarantee, so Lehman's failure repriced the credit of *every* counterparty at once. That is the difference between a large loss and a crisis: uncertainty about who is solvent stops lending to everyone.

The regulatory legacy, which is why this matters operationally

Nearly every constraint a desk works under today is an answer to something in the sequence above, and being able to make that mapping is what turns a history answer into a professional one.

- **More and better capital, plus a leverage ratio**, because risk weights had allowed thirty-times leverage to look adequately capitalised. See Basel III and IV and capital requirements.
- **Liquidity rules**, the coverage and stable-funding ratios, because the failures were funding failures rather than capital failures in the first instance.
- **Central clearing and margin requirements** for standardised derivatives, because bilateral counterparty webs made exposures unknowable, see mandatory clearing.
- **A capital charge for CVA**, because the crisis losses on counterparty credit were mark-to-market losses on deteriorating counterparties rather than actual defaults, see CVA, DVA and FVA.
- **Resolution regimes, bail-in debt and living wills**, so that the Lehman choice between a disorderly bankruptcy and a public rescue is not the only one available.
- **Stress testing as a supervisory tool**, because value at risk calibrated on calm data had said the positions were safe, see stress testing.
- **Restrictions on proprietary risk-taking** inside deposit-taking institutions, and reform of rating agency reliance and securitisation disclosure.

REAL INTERVIEW QUESTION (Barclays · Credit)

Explain the 2008 crisis.

WHAT THEY TEST / ANSWER

Do not produce a chronology. Produce a **chain**, in four links, and then say what the hinge was. **Link one**, cheap credit and a housing boom, on the belief that US house prices do not fall nationally. **Link two**, originate-to-distribute: lenders earned fees for writing mortgages they immediately sold, so underwriting quality stopped being their problem, and adjustable-rate loans built in a scheduled wave of payment shocks. **Link three**, securitisation: pools were tranched so that losses hit the bottom slices first, which converted risky loans into a large volume of AAA-rated paper that mandated investors could buy, and the critical point is that tranching turns credit risk into *correlation* risk, so the senior paper was safe only if defaults did not happen together, which was calibrated on a period when housing was regional rather than national. **Link four**, funding: that paper was financed overnight in repo, in asset-backed commercial paper and in off-balance-sheet vehicles, at leverage above thirty times, so a few percent of asset decline erased the equity.

Then the trigger and the hinge. House prices fell, the correlation assumption failed, and the decisive moment was not a loss but an inability to *value*: once nobody knew which counterparty held the bad paper, funding markets closed to everyone, which is adverse selection rather than panic. Lehman was the hinge because there was no resolution regime, so its failure froze client assets and derivative exposures, money market funds holding its paper broke the buck and the run reached the part of the system treated as cash, and because the Bear Stearns rescue had priced an implicit guarantee that Lehman's failure withdrew from every counterparty at once.

Close with two things that mark a serious candidate. Name the amplifiers rather than only the causes: issuer-paid ratings built on the same correlation models, and monolines and AIG selling protection against a scenario they held no capital for, which is wrong-way risk in its purest form. And connect it to your own job: the leverage ratio, the liquidity coverage ratio, central clearing, the CVA capital charge, resolution planning and supervisory stress testing all exist because of a specific failure in that chain, so if you know the chain you can derive the rulebook.

REAL INTERVIEW QUESTION (Moody's · Credit)

How does a CDO work?

WHAT THEY TEST / ANSWER

Asked at a rating agency, so give the structure correctly and then be honest about what the crisis taught about rating it. The mechanics, which are developed in tranches: a special purpose vehicle buys a portfolio of credit assets, loans, bonds or other structured paper, and funds itself by issuing notes in a **waterfall**. Interest and principal flow to the most senior notes first, and losses are absorbed from the bottom up, so an equity tranche takes the first few percent, mezzanine tranches sit above it, and the senior and super-senior notes are only touched once the tranches below are exhausted. That subordination, not the quality of any individual loan, is what allows senior notes to be rated far above the average rating of the pool. Cash structures own the assets; synthetic ones take the exposure by selling credit default swap protection, see credit default swaps.

Then the part that matters at a rating agency. The senior tranche's risk is **correlation** risk, not average default risk: it fails only if a large fraction of the pool defaults together, so its rating is a statement about the joint distribution rather than about individual credits. An equity tranche is long correlation and a senior tranche is short it, as in credit correlation. The 2008 experience is the case study: correlations were calibrated on data in which US regional housing markets had behaved independently, the downturn was national, and senior tranches that could only fail together did. Add the second-order lessons, since they are what the industry changed: repackaging tranches into CDOs of CDOs made the sensitivity to that one assumption much larger while appearing to diversify; the issuer-pays model created a conflict on the very parameter that mattered most; and a rating is a statement about a distribution, so it should be presented with its sensitivity to the correlation input rather than as a single letter.

INTERVIEW TRAP

Four traps. First, **answering with a chronology instead of a chain**: the dates are worth little without the causal mechanism, and the chain is what lets you reason about the next crisis. Second, **saying subprime losses caused the crisis**: total subprime losses were small relative to global wealth, and what turned them into a systemic event was leverage, maturity mismatch and uncertainty about where they sat. Third, **treating Lehman as simply too big**: the problem was disorderly failure without a resolution regime, plus the withdrawal of an implicit guarantee the market had priced after Bear Stearns. Fourth, **blaming the models alone**: the correlation assumption was wrong, but the incentives to prefer that assumption, in origination, in ratings and in insurance, are why it survived scrutiny for so long.

VISUAL / CODE TO BUILD: Interactive

A securitisation machine the reader can break. Panel one builds the chain: a pool of mortgages with adjustable sliders for default rate, loss given default and **default correlation**, feeding a tranche waterfall that shows each tranche's expected loss and implied rating live. The intended discovery is that raising the average default rate hurts the equity tranche while raising the correlation is what destroys the senior tranche, which makes the section's central point experimentally rather than rhetorically, and a CDO-of-CDOs toggle shows the correlation sensitivity amplifying while the apparent diversification improves. Panel two is a funding run: the reader sets leverage and the share of funding that is overnight, then applies an asset shock and watches haircuts rise, forced sales trigger and equity erode, with a resolution-regime switch that changes the outcome of a dealer failure from a system-wide freeze to an orderly transfer. Panel three is a timeline scrubber over 2006 to 2009 showing spreads, equity, house prices and the interbank basis together, with each event on the axis, so the reader sees how early the credit market moved relative to the headlines.

Where this goes next

The general anatomy this episode instantiates is financial crises, and the instruments at its centre are in tranches, credit correlation and credit default swaps, with the funding market in repo markets. The European sequel is the European sovereign debt crisis and the next liquidity event is the Covid market crash. The rulebook it produced runs through Basel III and IV, mandatory clearing and the Volcker rule, and the risk-management practices through counterparty risk and stress testing.

4.7 European sovereign debt crisis

trading · structuring · sales | core | rates · credit

Between 2010 and 2012 the euro area discovered that it had built a currency union in which a government bond could default, a banking system was not separable from its sovereign, and no institution had the job of stopping either. The crisis that followed is the most important recent event for anyone working in European rates or credit, and its resolution is the reason the market you trade today looks the way it does.

The convergence trade, and what it hid

From 1999 the euro removed currency risk from intra-European government bonds, and the market concluded that it had removed credit risk as well. Greek, Italian, Spanish and Portuguese ten-year yields converged to within a few tens of basis points of the Bund despite fiscal positions that had not converged at all.

That was not entirely irrational. Sovereign bonds carried a zero risk weight in bank capital rules, they were accepted as collateral by the ECB on near-identical terms regardless of issuer, and there was a widespread assumption that a euro member would not be allowed to fail. Each of those three was a policy choice, and each was later revealed to be the mechanism that transmitted the crisis rather than a reason there would not be one.

The revelation in October 2009 that Greece's deficit had been substantially misreported was the trigger. What made it a systemic event rather than a Greek problem was the arithmetic below.

The equation that governs a sovereign crisis

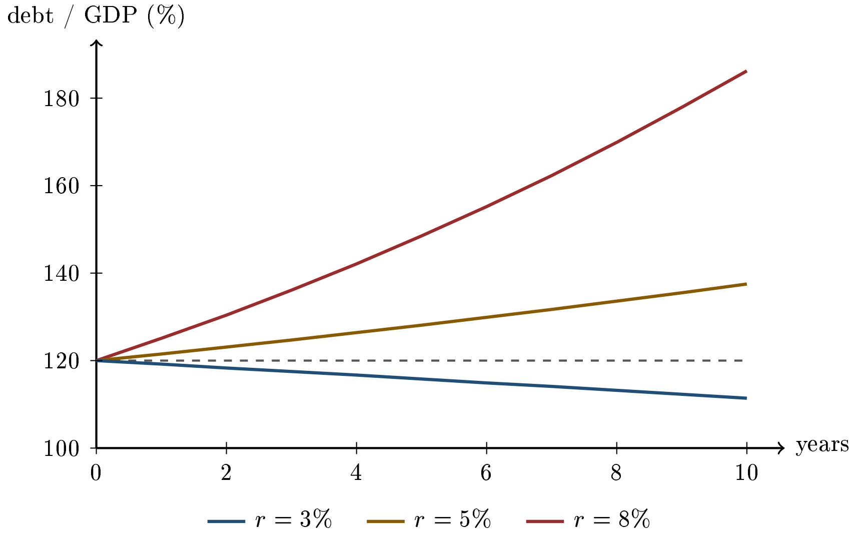
A government's debt to GDP ratio d evolves according to one relation, and it is worth being able to write it down:

$$d_{t+1} = d_t \cdot \frac{1+r}{1+g} - pb,$$

where r is the nominal interest rate on the debt, g is the nominal growth rate of GDP, and pb is the primary balance, the budget surplus *before* interest payments, as a share of GDP.

Everything follows from the sign of $r - g$. If growth exceeds the interest rate the debt ratio falls on its own and a government can run deficits indefinitely. If the interest rate exceeds growth, the ratio compounds upwards and only a primary surplus can hold it still. Setting $d_{t+1} = d_t$ gives the surplus required to stabilise:

$$pb^* = d \cdot \frac{r-g}{1+g}.$$



same country, same 2% primary surplus, same 2% growth: only the borrowing cost differs

WORKED EXAMPLE

A country starts at **120% debt to GDP**, grows at 2% in nominal terms, and runs a primary surplus of **2% of GDP**, which is a genuinely demanding fiscal stance sustained by few governments for long. Vary only the interest rate.

Borrowing cost	Debt/GDP after 5 years	after 10 years	Surplus needed to stabilise
$r = 3\%$	115.8%	111.4%	1.18% of GDP
$r = 5\%$	128.1%	137.5%	3.53% of GDP
$r = 8\%$	148.5%	186.3%	7.06% of GDP

Same country, same austerity, same growth. At 3% the debt is falling and the programme works. At 8% the debt reaches 186% in a decade despite the surplus, and holding the ratio still would have required a primary surplus of **7.06% of GDP, every year, indefinitely**.

That last figure is the crisis in one number. Sustained primary surpluses above about 5% of GDP are close to unknown in the historical record. So once yields reached 8%, the debt was no longer a liquidity problem being mistaken for a solvency problem; it had become a solvency problem *because* of the yield.

And that is what makes it self-fulfilling. A wider spread raises r, which raises the surplus required, which forces more austerity, which lowers g, which raises the required surplus again. Fear of default produces the conditions for default. Two identical countries can sit at either of two equilibria, solvent or not, decided by nothing more than which one the market believes.

KEY IDEA

A sovereign borrowing in a currency it does not issue is a credit, not a rate. The United Kingdom and Japan carried comparable or higher debt ratios through the same period without a crisis, because their central banks can always meet a maturing bond in domestic currency. Euro members could not, which is why a currency union turned government bonds into instruments with genuine default risk and why the yields diverged the way credit spreads do.

The doom loop

The second mechanism is why a sovereign problem became a banking problem and back again. Domestic banks are the largest holders of their own government's bonds, held at a zero risk weight and used as collateral for funding.

- The sovereign spread widens, so bank assets fall in value and bank capital falls with them.
- Weaker banks lend less, so growth falls, so g falls and the sovereign's arithmetic deteriorates.
- If the banks need rescuing, the cost lands on the sovereign, so the sovereign spread widens further.

Each step makes the next one worse, and the loop runs in both directions. Ireland and Spain entered from the banking side, with a property bust that a solvent state then absorbed onto its own balance sheet; Greece entered from the fiscal side and infected its banks. The destination was the same.

Redenomination risk, which had no precedent

By mid-2012 the market was pricing something with no analogue in credit: not that Italy would fail to pay, but that it might pay in a currency that no longer existed. Leaving a currency union has no legal procedure, so a euro in a Milan bank and a euro in a Frankfurt bank stopped being the same asset.

This showed up in places a pure credit story cannot explain: in the divergence between bonds governed by domestic law and by English law, in cross-border deposit flight, and in the TARGET2 balances of the national central banks. It is the risk that the regime itself might change, which is discussed as a general problem in exchange rate regimes.

What actually ended it

Three years of bailouts, austerity programmes and a Greek restructuring did not stop the crisis. The restructuring of March 2012 was the largest in sovereign history and imposed losses on private holders of roughly half the face value and considerably more in present value terms, and spreads elsewhere kept widening afterwards.

What ended it was Mario Draghi's statement in July 2012 that the ECB would do whatever it took to preserve the euro, followed in September by **Outright Monetary Transactions**: a commitment to buy the bonds of a member state under conditions, without a stated limit.

> **KEY IDEA**
>
> OMT has never been used, and that is the point. A self-fulfilling panic has two equilibria, and a credible unlimited backstop removes the bad one. Once the market accepted that the ECB would not permit a liquidity-driven default, the liquidity-driven default stopped being worth positioning for, so the promise never had to be honoured. This is the same logic as a deposit guarantee that ends a bank run without a single withdrawal.

The institutional legacy is what you work inside now: the ESM as a permanent rescue facility, banking union with supervision moved to the ECB, a resolution regime that bails in creditors rather than taxpayers, and the sovereign spread to Bund as the standing measure of European risk.

Interview angle

REAL INTERVIEW QUESTION (CACIB · Credit)

What do you think about the Greek crisis? What may happen and what impact on other European countries?

WHAT THEY TEST / ANSWER

An opinion question, so it wants a framework and a position rather than a summary. Give the mechanism first, because it makes the view checkable.

Diagnose it. Greece had a debt stock whose arithmetic only worked at low borrowing costs, and it borrowed in a currency it could not issue. Once the spread widened, the primary surplus required to stabilise the ratio moved beyond anything historically achieved, and the market's fear became the reason for the outcome it feared. Two identical countries can sit at either of two equilibria; the spread decides which.

Then the contagion channels, which is the part the question is actually asking about, and there are three:

- **Direct exposure.** Foreign banks holding Greek paper take losses, and this turned out to be the smallest channel because the exposures were manageable.
- **The doom loop.** Every euro sovereign is held mainly by its own banks, so a repricing of sovereign risk repriced every domestic banking system at once.
- **Information.** The largest channel, and the one people underrate. Greece did not transmit an exposure; it transmitted the fact that a euro sovereign could default at all. Once that was known, the market repriced Portugal, Ireland, Spain and Italy on their own arithmetic, and Italy is large enough that its arithmetic is a systemic question.

Now take a position. The defensible one is that Greece was solvable and Italy was not, not because Italy's finances were worse, but because Italy's debt stock was too large to be rescued by any facility that existed. That is why the crisis only ended when the backstop became unlimited rather than larger, and it is why the answer to "what may happen next" is a question about the ECB's willingness to act rather than about any national budget.

Finish with what would change your mind, since that is what distinguishes a view from a recital: a durable rise in nominal growth relative to yields fixes the arithmetic quietly, and a sustained period of r above g across the large sovereigns brings the whole problem back regardless of any fiscal rule.

REAL INTERVIEW QUESTION (CACIB · Fixed Income)

What is the level of the Italian ten-year rate? What does the curve look like? Compare it with the German curve.

WHAT THEY TEST / ANSWER

Know the current levels before the interview; there is no substitute. But the question is really testing whether you read a European sovereign curve as a rate or as a credit, so answer in three steps.

Quote the spread, not just the yield. The number a European rates desk actually watches is the BTP-Bund spread, the Italian ten-year yield less the German. Giving both levels and the difference in basis points, and saying whether the spread has widened or tightened recently, demonstrates immediately that you know which of the two numbers carries information. The Bund is the risk-free benchmark; the Italian yield is that plus a sovereign credit and redenomination premium.

Describe both curves and say why they differ. The general reading of curve shape belongs with yield curves. What is specific here is that the two curves share a policy rate and a currency, so the front ends are anchored together and essentially all of the divergence is in the long end. That is the signature of a credit spread rather than a different rate expectation: the market is not forecasting different ECB policy for Italy, it is pricing different repayment risk.

Then the point that makes it an informed answer. The spread is the direct descendant of the crisis and it remains the standing gauge of European sovereign risk. It moves on Italian politics, on budget submissions to the Commission, and on any question about ECB support, far more than on the growth and inflation news that moves the Bund. So the two curves have different drivers even though they share a central bank, and a trader long BTP against Bund is running a political and institutional position, not a rates view.

INTERVIEW TRAP

Four errors. (1) Calling it a debt crisis without saying why comparable debt ratios in the UK and Japan produced nothing similar; the answer is the currency the debt is issued in. (2) Treating the sovereign and its banks as separate credits, when the doom loop is the whole mechanism. (3) Believing austerity failed because it was insufficient, when the arithmetic shows a surplus above 7% of GDP was required and has essentially never been achieved. (4) Describing OMT as a bond-buying programme, when its defining feature is that announcing it was sufficient and it has never been used.

Where this goes next

Why a currency union removes the adjustment mechanism, and what internal devaluation costs, is exchange rate regimes. The banking crisis that preceded this one and left the sovereigns carrying the bill is the 2008 financial crisis. The institution that ended it and the tools it used are in major central banks, how to read the curves the spread is measured on is yield curves, and the spread-and-default vocabulary applied to companies rather than states is corporate bonds.

4.8 COVID Market Crash

trading · structuring · sales | core | cross-asset · equity

The February and March 2020 crash was the fastest bear market on record: the S&P 500 fell about **34% in 23 trading days**, and the VIX closed above 80, above its highest close in 2008. It is worth studying not for the speed but for what broke, because the things that broke were the ones every risk framework had assumed could not.

WORKED EXAMPLE

How fast is fast. The 2008 bear market fell roughly 57% over about seventeen months, some 355 trading days, an average of $57/355 = 0.16\%$ a day. The 2020 fall was $34/23 = 1.48\%$ a day, roughly **nine times faster**.

That ratio is the whole risk-management story. Every measure that assumes you can react, reduce, hedge or negotiate assumes a rate of change, and 2020 delivered a rate of change nine times what the previous benchmark crisis had produced. A stop-loss discipline, a limit framework or a margin process calibrated on 2008 was calibrated on the wrong speed.

The three phases

The repricing, late February into early March, was orderly in the sense that equities fell and government bonds rallied, exactly as a flight-to-quality should.

The dash for cash, from around 9 March, was not. Investors began selling *everything*, including US Treasuries, because what they needed was cash rather than safety. Treasury yields rose while equities fell, gold fell, and the dollar surged. This is the phase that matters, because correlations did not merely rise toward one, they *inverted* against the assumption underlying almost every hedge in existence.

The policy response, from 23 March, was unprecedented in speed and scope, including central bank purchases extended for the first time to corporate credit. The market bottomed the same week, which is the origin of the reflex that policy support arrives before fundamentals improve.

INTUITION

The single most useful lesson is that **the safe asset is only safe if you are not the one who needs to sell it**. In a leverage unwind the most liquid asset is sold first, precisely because it can be sold, so the deepest market in the world is where the selling appears. A hedge portfolio built on "Treasuries rally when equities fall" was correct in every historical sample and wrong in the week it was needed, and liquidity risk explains why: this is a funding event wearing a market event's clothes.

What actually broke in Treasuries

The mechanism is concrete and worth being able to describe, because it is the clearest recent example of a leveraged relative-value trade becoming a systemic event.

Hedge funds ran a large **basis trade**: long cash Treasuries, short Treasury futures, capturing the small spread between them, financed in repo at very high leverage because the position looks nearly riskless. When volatility spiked, futures margins rose and repo haircuts widened, so the position had to be cut. Cutting it meant selling cash Treasuries into a market where every other holder of the trade was doing the same, which widened the basis further, which forced more cutting.

The result was a week in which the most liquid market on earth had visibly impaired liquidity, with bid-ask spreads on off-the-run bonds multiples of normal, and it ended only when the central bank became the buyer. The general shape, a mechanically forced seller with no natural counterparty, is the same one as 1987 and as every crisis in between.

The problem the exotics desks had

Banks issue autocallables and similar structures in enormous size, and the investor in an autocall is structurally *short* dividends, because higher expected dividends lower the forward and hurt both the autocall trigger and the downside put. The dealer takes the other side, so the industry as a whole ends up **long dividends**, in size, across many years of a dividend strip. That position is not a view, it is a by-product of selling a product clients want.

What made 2020 different from an ordinary drawdown is that dividends did not fall because earnings fell. They were **cancelled**, quickly and in many cases by regulatory instruction, with European banks suspending distributions outright. The loss was therefore not a repricing along a distribution the desks had modelled but a discrete administrative event with no market mechanism behind it, and long-dated dividend futures, the natural hedge, fell violently at the same time.

The shape worth extracting is not a Greek anyone was measuring badly: it is a *crowded structural position, held in one direction by an entire industry, in a variable that could be set to zero by a decision rather than by a market.* What that did to the Greeks across the regime, and to the hedging flows the desks then had to execute, belongs to Greeks across market regimes and to hedging structured products.

Dispersion: the crisis that separated sectors

REAL INTERVIEW QUESTION (Bloomberg · desk not recorded)

Which sectors have benefited from the COVID crisis, and why?

WHAT THEY TEST / ANSWER

Answer by mechanism rather than by list, because the list is obvious and the mechanism is not.

Three mechanisms produced winners.

Forced substitution of the physical for the digital. Software, cloud infrastructure, video conferencing, e-commerce and payments gained because activity that had been happening in person had to happen somewhere, and the shift pulled years of adoption into months.

Direct demand for the response itself. Pharmaceuticals and medical equipment, and logistics and delivery as the fulfilment layer for everything above.

Balance sheet and duration. This one is financial rather than operational and is the answer that gets credit: rates collapsed, so long-duration growth equities were discounted at a much lower rate, and their valuations rose for reasons that had nothing to do with the virus. Distinguishing that from operational benefit is the point, because it predicts which of the winners would give it all back when rates rose.

Close with the durability question, which is what a Bloomberg interviewer is really asking. Separate a permanent change in behaviour from a pull-forward of demand. Remote collaboration is partly permanent; a surge in home fitness equipment or garden furniture was demand borrowed from the future, and borrowed demand reverses. The good answer names at least one winner it expects to be temporary and says why.

REAL INTERVIEW QUESTION (Bloomberg · desk not recorded)

Which sectors have not benefited from the COVID crisis, and why?

WHAT THEY TEST / ANSWER

The mirror, and it is a better question, because the losers separate into two groups with completely different outlooks and conflating them is the error.

Group one: demand deferred. Airlines, hotels, restaurants, live events, commercial aerospace. The activity was prevented rather than destroyed, so the revenue returns when the constraint lifts. The risk here is not the demand, it is **survival**: these are high fixed cost, heavily leveraged businesses, and the question is whether the balance sheet lasts long enough to see the recovery. That is a credit question dressed as an equity one.

Group two: demand structurally reduced. Office real estate, business travel, city-centre retail. Here the behaviour change may be permanent, so the recovery is partial by construction, and the correct analysis is about the new level rather than about the timing of the return.

Then two mechanisms worth adding. Energy was hit through a collapse in transport demand meeting a supply war, an unusually pure combination of both blades of the scissors. And banks were hit twice over, through expected credit losses and through a collapse in net interest margins as rates went to zero, which is why financials underperformed even though the crisis did not originate with them, in direct contrast to 2008.

Close on the general point. The dispersion between winners and losers was far larger than the index move, so the crisis was a much better environment for relative-value and dispersion trades than for direction, and saying that turns a sector list into a market observation.

Negative oil, in one paragraph

On 20 April 2020 the expiring WTI futures contract settled below zero. Storage at the delivery point was full, so a holder facing physical delivery had nowhere to put the barrels and paid to be relieved of the obligation. The mechanics belong to physical versus derivative commodities, which owns the question; what belongs here is that it is the clearest demonstration in modern markets that a futures price is a price for delivery at a place under a contract, and that “the price of oil” is a convenient fiction the rest of the time.

INTERVIEW TRAP

1. **Assuming government bonds always rally in a crisis.** In a funding event the most liquid asset is sold first because it can be.
2. **Calibrating processes on 2008's speed.** 2020 was about nine times faster per day, and every process that assumes you have time to react was built on the wrong number.
3. **Treating a dividend cut as a small repricing.** For desks structurally long dividends from structured issuance it was a discrete administrative event with no market mechanism behind it.
4. **Reading a sector as a winner without asking why.** Operational benefit and duration-driven valuation gains look identical in a return series and behave completely differently when rates rise.
5. **Conflating deferred demand with destroyed demand.** The first is a solvency question about surviving to the recovery; the second is a question about the new steady state.
6. **Concluding the safeguards worked.** The market functioned because the central bank became the buyer, which is a different statement, and the basis trade that broke it has since been rebuilt.

VISUAL / CODE TO BUILD: Interactive

A correlation-regime panel. Equity and Treasury returns are plotted against each other over a rolling window with a date slider running from January 2020 forward. The scatter starts in the familiar negative-correlation quadrant, and as the slider crosses into the second week of March the cloud rotates into positive correlation before rotating back after the policy response. Watching a hedge relationship invert and then repair, on the same axes, is the section's central lesson delivered in one drag.

A second panel is a drawdown-speed comparison: cumulative decline against trading days elapsed, with 1987, 2008 and 2020 overlaid from their respective peaks. 2008 goes deeper, 2020 goes steeper, and a readout of percent per day makes the nine-times figure concrete rather than rhetorical.

A third shows a dividend strip: expected dividends by year drawn as bars, with a control that cancels a chosen fraction of near-dated dividends and reprices the strip. Applying a full one-year cancellation and watching the whole curve reprice, not only the cancelled year, shows why the exotic desks' loss was so much larger than the dividends actually foregone.

4.9 Major FX crises

sales · trading · structuring | core | fx · rates · cross-asset

Currency crises are the most repetitive events in financial history. The countries change, the decade changes, and the mechanism does not: a government promises an exchange rate, the promise is cheap to keep for years, and then it is not. What makes them worth studying is that the shape of the loss is always the same, and it is a shape you already recognise from options.

This section is about the episodes. The mechanics of running, defending and breaking a peg are pegged currencies and FX crises, and the regime taxonomy is exchange rate regimes.

> KEY IDEA
>
> **A peg is a short volatility position taken by a central bank.** It collects a premium in the form of stability, credibility and cheap foreign borrowing, every day, for as long as it lasts. It pays out once, discontinuously, and the payout is unbounded in the sense that nobody knows in advance how far the rate goes when it is released. Every crisis below is that same short option being exercised, which is also the payoff shape that destroyed Long-Term Capital Management.

1992: sterling and the ERM

Britain joined the European Exchange Rate Mechanism in October 1990 at a central rate of DM 2.95 with a 6% band, which put the floor at DM 2.778, a shade under 5.9% below parity ($2.778/2.95 - 1 = -5.83\%$). The commitment was to keep sterling inside that band.

The problem was the impossible trinity in its purest form. Germany, reunifying, needed high rates. Britain, in recession, needed low ones. With capital free to move and the rate fixed to the mark, Britain had no independent monetary policy left, so it was running German policy in a British recession. Speculators did not need a view on sterling's fair value. They needed only to observe that the domestic cost of the defence was rising and that no democratic government pays it indefinitely.

> WORKED EXAMPLE
>
> On 16 September 1992 the Bank of England raised Bank Rate from 10% to 12%, and announced a further increase to 15% that was never implemented. That is a 5 percentage point rise promised in a single day, in a recession, with unemployment rising.
>
> Work out what that means for the other side of the trade. A speculator short sterling pays the interest differential to hold the position, so a 5 point increase in UK rates raises the cost of a one-month short by roughly $5\% \times 1/12 = 0.42\%$ of notional. If the devaluation you expect is 15%, you can afford that carry for about three years.
>
> The defence therefore failed on arithmetic, not on nerve. **Rate rises punish the speculator by a few tenths of a percent a month and punish the domestic economy permanently.** Britain left the mechanism that evening and sterling fell roughly 15% against the mark over the following weeks.

The uncomfortable epilogue is that leaving was good for Britain: with policy back under domestic control the recovery began shortly afterwards. That is worth saying in an interview, because it makes the general point that **a peg breaking is often the resolution of a problem rather than the problem itself**.

REAL INTERVIEW QUESTION (Moody's · desk not recorded)

How is the FTSE 100 doing these days? What about sterling? Is there a connection?

WHAT THEY TEST / ANSWER

There is a connection, it is usually **negative**, and the reason is structural rather than sentimental. Give the mechanism first, then offer the levels.

The mechanism. The FTSE 100 is a list of very large companies domiciled in London and earning most of their revenue elsewhere: energy, mining, pharmaceuticals, global consumer brands. A large majority of index revenue is non-sterling. So when sterling falls, those foreign earnings translate into more pounds, reported earnings rise in sterling terms and the index rises, without anything having improved. The FTSE 100 is closer to a portfolio of foreign-currency earnings streams than to a bet on the British economy.

The contrast that proves it. The FTSE 250 is far more domestic, so it behaves the opposite way: weak sterling usually goes with a weak 250 because it reflects weaker domestic prospects. If you are asked about the connection, naming the 100 versus 250 divergence is the answer that shows you understand it rather than having memorised the correlation sign.

The historical illustration. This relationship was at its most visible in the two episodes where sterling repriced sharply, the 1992 exit from the ERM and the 2016 referendum: in both, the domestic index and the international index moved in opposite directions on the same news. A single currency move produced two different equity answers because the two indices are not exposed to the same thing.

Then be honest about levels. Say that you would check, give an approximate range if you follow it, and never state a precise level you have not seen that morning. A confident stale number is worse than an admitted gap, and the interviewer is testing the mechanism, which does not go out of date.

The emerging market template

Mexico in 1994, Thailand and the wider Asian crisis in 1997, Russia in 1998, Argentina in 2001, Turkey and Argentina again more recently. Different countries, one recipe.

- **A pegged or heavily managed rate**, which suppresses measured volatility and makes the currency look safe.
- **Borrowing in foreign currency**, because the peg makes it look cheap and apparently riskless. This is the fatal ingredient: the debt is in dollars and the revenue is not.
- **A current account deficit** financed by short-term inflows that can leave faster than they arrived.
- **Reserves that look adequate** against a normal outflow and are trivial against a simultaneous one.

INTUITION

The trap is that devaluation, the normal remedy for an overvalued currency, is the thing that makes the debt unpayable. A country whose debt is in dollars cannot devalue its way out of trouble, because the devaluation multiplies the debt in local terms at the exact moment the economy contracts. That is why these crises are so violent: the adjustment mechanism and the solvency constraint point in opposite directions, so there is no gentle path, which is sometimes called original sin.

REAL INTERVIEW QUESTION (JP Morgan · desk not recorded)

How are emerging markets doing?

WHAT THEY TEST / ANSWER

Do not answer with a view on an index. Answer with the framework, then a view, then say what would change your mind.

The four things that actually drive EM as an asset class, in rough order of importance. *The dollar*, because most EM external debt is dollar denominated, so a strong dollar tightens EM financial conditions mechanically. *US real rates*, because they set the return available with no EM risk at all and therefore the hurdle capital must clear to travel. *Chinese growth*, because it drives commodity demand and half the complex is commodity exporters. *Idiosyncratic politics*, which decides dispersion within the complex once the first three have set the direction.

Then differentiate, because the phrase itself is a category error. Emerging markets contain commodity exporters and commodity importers, whose interests are opposite; countries with credible independent central banks and countries without; and local-currency debt, where you take the currency risk, against hard-currency debt, where you take default risk instead. Saying "EM is cheap" without saying which of these you mean is the mistake the question is looking for.

Give a view and the trigger. Something of the form: the constructive case rests on the dollar and US real rates having peaked, the risk case is a renewed dollar rally or a sharp Chinese slowdown, and the indicator I would watch is the trade-weighted dollar alongside local-currency yields relative to their own history. Then say plainly that you would check current levels before committing.

The historical anchor, if the conversation goes there, is that every major EM crisis since 1994 has been preceded by a strong dollar, a peg or heavy management, and foreign-currency borrowing. Those three together are the condition to look for, not any single valuation metric.

2015: the franc, and the crisis where the peg was too strong

Every episode above involves a currency that markets wanted to sell. The Swiss case is the mirror image and is more instructive for a trader.

The Swiss National Bank set a floor of 1.20 in EURCHF in September 2011, promising to buy euros without limit to stop the franc appreciating. That is a defence with no theoretical reserve constraint, since a central bank can create its own currency indefinitely. The constraint is instead the size of the balance sheet it accumulates, and the losses on it.

WORKED EXAMPLE

On 15 January 2015 the SNB abandoned the floor without warning and cut its policy rate to -0.75%. EURCHF traded as low as roughly 0.85 intraday, against a floor of 1.20.

In EURCHF terms that is $0.85/1.20 - 1 = -29\%$. Quoted the other way, the franc rose $1.20/0.85 - 1 = +41\%$ in minutes. The rate settled near 1.04 by the close, still $1.04/1.20 - 1 = -13\%$ below the floor.

The trading lesson is **gap risk**. There was no path from 1.20 to 0.85; the market did not trade through the intervening levels in any size, so a stop loss at 1.19 was not filled at 1.19. Positions sized on the assumption that you can exit at your stop were sized on an assumption that does not hold at exactly the moment it matters, which is liquidity risk in its purest form. Several retail brokers were made insolvent that morning by client accounts that went negative faster than they could be closed.

The standing policy that surrounded the floor and outlived it, the years of negative rates, the intervention programme and the balance sheet it built, is peg mechanics rather than a historical episode, and it is set out with what it means for Swiss corporates in pegged currencies. What this section keeps is the exit: the morning the floor went, and why a defence that faced no reserve limit still ended discontinuously.

INTERVIEW TRAP

Three errors. (1) Saying a central bank can always defend its currency: it can print unlimited amounts of its own currency to *weaken* it, but it can only sell the reserves it has to *strengthen* it, so the two directions are not symmetric and the defence against depreciation is the one with a hard limit. (2) Treating the speculator as the cause, when the position only exists because the policy was already inconsistent. (3) Assuming you can exit at your stop level. The 2015 franc is the standing counter-example, and citing it is a fast way to show you think about execution and not only about direction.

Where this goes next

The mechanics of pegs, defences and reserve adequacy are pegged currencies and FX crises, and the regime taxonomy behind them is exchange rate regimes. The system these episodes are the descendants of is the Bretton Woods system, the 1998 Russian devaluation as seen from a levered book is Long-Term Capital Management, the market that does the repricing is FX spot markets, and the policy side is central banks and FX policy.

4.10 Evolution of derivatives

sales · trading · structuring core cross-asset

Derivatives are usually taught as a set of instruments to memorise. They are much easier to hold in your head as a history, because every one of them was invented to solve a specific problem for somebody, and the conventions that look arbitrary are almost always fossils of that problem. The history also has a shape that repeats, and recognising the shape is what lets you say something intelligent about a product you have never traded.

Before the theory: three thousand years of forwards

The forward contract is not a financial innovation, it is a commercial one, and it is old. Clay tablets record agreements to deliver goods at a future date at an agreed price. Aristotle relates the story of Thales securing the right to use olive presses ahead of a harvest, which is a call option in everything but name. Options on tulip bulbs traded in the Netherlands in the 1630s.

The first recognisably *modern* market is the rice exchange at Dojima in Osaka in the eighteenth century, where contracts were standardised, netted and settled in cash rather than by delivery. In the West the decisive step was the Chicago Board of Trade, founded in 1848 to serve grain merchants who needed to fix prices between harvest and shipment. Over the following decades the essential apparatus appeared: standardised contract terms, a clearing house interposed between buyer and seller, and margin. Those three inventions are what make a futures market possible, because together they let a buyer stop caring who the seller is, see clearing houses.

> KEY IDEA
>
> **Every derivative exists because a real risk was sitting with someone who did not want it.** Grain futures exist because a farmer and a miller face opposite risks; currency forwards exist because an importer's costs and revenues are in different money; swaps exist because a borrower's liabilities and its income are in different rates or currencies. If you cannot name the natural hedger, you are probably looking at a product that exists for a reason other than risk transfer, and that is a useful thing to notice.

The 1970s: three things happen at once

The modern derivatives industry is not much older than the people running it, and it begins with a coincidence that is not really a coincidence.

1. **The risk appeared.** Fixed exchange rates ended and currencies floated, as told in the end of the gold standard, and inflation made interest rates volatile. Suddenly every international company and every bank had price risks that had simply not existed as continuous variables before.
2. **The venue appeared.** Chicago listed currency futures in 1972, and the Chicago Board Options Exchange opened in 1973 as the first exchange for listed options, bringing standardisation and a clearing house to a product that had previously traded bilaterally and thinly.
3. **The theory appeared.** The Black-Scholes and Merton papers were published in 1973, giving a defensible price and, more importantly, a *hedge*, which is what let a dealer quote a two-way market in something it did not want to hold outright. See the Black-Scholes model.

Interest rate futures followed within a few years, and in 1981 the currency swap between IBM and the World Bank is conventionally cited as the beginning of the swap market, which grew from that single bilateral trade into the largest derivatives market in the world. The International Swaps and Derivatives Association was founded in 1985, and the master agreement it produced is arguably as important as any pricing model, because standardised documentation with close-out netting is what made bilateral trading scalable, see counterparty risk.

The pattern that repeats

KEY IDEA

Every generation of derivatives has followed the same **lifecycle**, and knowing it is worth more than any individual date:

(1) A new risk becomes tradeable because a market floats, deregulates or grows. **(2) Bespoke OTC products appear**, priced with wide margins because few people can value or hedge them. **(3) Volume attracts competition**, terms converge, and documentation standardises. **(4) The product is cleared or listed**, which collapses the bid-offer and the capital it consumes. **(5) Margins compress to nothing**, and the desk that used to make money on it moves to the next bespoke thing.

So the same desk sold currency forwards at a wide spread in the 1970s, vanilla swaps in the 1980s, exotic equity options in the 1990s, credit derivatives in the 2000s and multi-asset structured notes after that, each time earning its margin during phase two and its volume during phase four. This is why "what is your edge" on a mature product is a hard question and why structuring exists at all.

The other regularity is that **crises accelerate the standardisation step**. Portfolio insurance was implicated in the 1987 crash and led to circuit breakers; LTCM in 1998 changed how dealers looked at leverage and at liquidity in a crowded trade; and the credit derivatives boom that ended in 2008, described in the 2008 financial crisis, produced the largest single push in the history of the industry, from bilateral OTC towards central clearing, initial margin and standardised terms, see mandatory clearing.

From the 1980s to now, briefly

- **1980s.** Swaps become an industry, documentation standardises, and the first wave of exotics and structured notes appears as clients ask for payoffs rather than instruments.
- **1990s.** Equity exotics and structured products mature, quantitative modelling becomes a competitive advantage rather than an academic exercise, credit default swaps are developed as a way to move loan risk off bank balance sheets, and two spectacular failures, in 1994 and 1998, teach the industry about leverage and about model risk.
- **2000s.** Credit derivatives and structured credit grow explosively, volatility becomes an asset class in its own right with variance swaps and listed volatility futures from 2004, and the notional outstanding in OTC derivatives reaches numbers that are large enough to mislead anyone who confuses notional with risk.
- **After 2008.** Central clearing and margining for standardised products, capital charges that make bespoke risk expensive to hold, mandatory reporting, electronification of execution, an industry devoted to compressing and optimising existing portfolios, and the replacement of the interbank benchmarks the whole swap market had been written on with overnight risk-free rates. The net effect is that the plain products became cheaper and more liquid while the complex ones became more expensive to carry, which pushed innovation towards payoffs that are cheap to hedge with cleared building blocks.

INTERVIEW TRAP

Three traps. First, **confusing notional with risk**: a hundred million of ten-year swap notional and a hundred million of equity are not remotely comparable exposures, and quoting gross notional outstanding as a measure of systemic danger is the mark of someone reading the wrong number. Second, **describing derivatives as inherently speculative**: the instrument is neutral, and the same swap is a hedge for one side and a position for the other, which is precisely why a market can exist. Third, **treating the history as decoration**: the reason a swap has an ISDA master behind it, why futures have margin, and why the market moved to clearing are all answers to specific past failures, so the history is the explanation of the rulebook.

REAL INTERVIEW QUESTION (Société Générale · desk not recorded)

Give the definition and purpose of derivatives.

WHAT THEY TEST / ANSWER

Definition first, in one sentence: a derivative is a contract whose value is *derived* from the level of something else, an underlying asset, rate, index or event, so it lets you take or transfer exposure to that underlying without necessarily owning it. Name the four building blocks briskly, since everything else is a combination of them: forwards and futures, options, and swaps, developed in forwards, options and swaps. Then the purposes, and there are four rather than one. **Hedging**, which is the original reason they exist and the answer to give first: a producer and a consumer face opposite risks, and a derivative lets each shed the one they do not want. **Price discovery**, since a liquid forward curve aggregates expectations better than any survey. **Efficiency and access**: derivatives let you take a precise exposure with a small outlay, gain exposure to something you cannot easily hold physically, and separate one risk from a bundle, for example taking credit risk without funding the bond. **Risk transformation**, which is what structuring does: reshaping a payoff into the profile a client actually wants. Then add the two-sided point that shows judgement rather than salesmanship: the same features that make derivatives efficient, embedded leverage and non-linear payoffs, are what make them dangerous when the user does not understand the exposure, which is why the industry's accidents have usually been mis-sold or misunderstood products rather than mispriced ones. Say the sentence that ties it together: a derivative does not create or destroy risk, it moves it, and the question is always to whom and whether they know it.

REAL INTERVIEW QUESTION (Société Générale · Delta One)

Why equity derivatives?

WHAT THEY TEST / ANSWER

A motivation question, so it needs to be specific to *equity* derivatives rather than a generic enthusiasm for markets, and specific to the desk in front of you. Three strands, and pick the ones you can defend with something you have actually done. **The intellectual content**: equity derivatives sit exactly where probability, calculus and real markets meet, so the job is quantitative without being detached, and a mispriced parameter is a position rather than an abstraction. **The breadth**: the asset class runs from a listed future to a bespoke multi-underlying note, so the same desk touches flow trading, volatility, dividends, repo, correlation and client structuring, and on a Delta One desk specifically the work sits on the financing, index and ETF machinery that everything else is built on, which is unusually concrete. **The feedback loop**: positions are marked daily and hypotheses are tested by the market rather than by an argument, which is a strong preference to state honestly if you hold it. Then anchor it in evidence: name a product you have looked at, a question that puzzled you and what you did about it, or a market episode you followed and what it taught you, because a claim about motivation is only credible if it comes with a fact. Close with why *this* desk, referring to what it actually does, and if you are asked what you like less, answer honestly with a real cost of the job rather than a disguised strength.

REAL INTERVIEW QUESTION (UBS · EQD)

How do derivatives work? What is your expectation when you buy a call or a put, and what are the key advantages of options?

WHAT THEY TEST / ANSWER

Three parts, so signpost them and keep the first two short. **How they work**: the contract's value depends on an underlying, and the dealer on the other side does not simply take the opposite view, it *hedges*, which is the point most candidates miss. A dealer that sells you a call buys delta in the underlying and rebalances as spot moves, so it is not betting against you, it is selling you a package it can manufacture, which is why a two-way market exists at all. **What you expect**: buying a call you expect the underlying to rise, or more precisely to rise by more than the premium implies, and buying a put the reverse; and the sharper version, which is worth saying, is that you are also taking a view on *volatility*, since you profit if realised movement exceeds what you paid for, so a directional view and a volatility view are bundled unless you hedge the delta. **Advantages**: capped and known downside for the buyer against unbounded upside on a call, which is the asymmetry you are paying for; leverage, since a small premium controls a large notional; precision, since you can choose a strike and a maturity to match a view or an exposure rather than accepting the linear payoff of the underlying; the ability to express non-directional views such as a range or a volatility view at all; and capital and access efficiency. Then give the costs, because an answer that lists only advantages sounds like it is selling: you pay premium that decays, you can be right about direction and still lose money on timing or on paying too much volatility, and the payoff structures are far easier to misunderstand than a share. Payoff mechanics are developed in options and the sensitivities in first and second order Greeks.

VISUAL / CODE TO BUILD: Interactive

A timeline the reader can interrogate rather than scroll. The horizontal axis runs from the nineteenth century to the present, with three stacked bands: **instruments** appearing as they were invented, **crises** as markers, and **rules** as they were introduced, so the causal arrows from a failure to a rule are visible as vertical alignment. Clicking any instrument opens its lifecycle position on the five-stage model (new risk, bespoke OTC, standardisation, clearing, compressed margins) with the approximate dates of each transition, which turns the pattern into something the reader can apply to a product not on the chart. A second panel plots gross notional outstanding against a risk measure such as gross market value over the same period, side by side, to make the notional-is-not-risk point unmissable, with a toggle showing how much of the total moved from bilateral to cleared after 2008.

Where this goes next

The instruments themselves are built up in forwards, futures, options and swaps, with the pricing framework in the Black-Scholes model. The macro break that created the modern industry is the end of the gold standard, the crisis that reshaped it is the 2008 financial crisis, and the resulting market structure is in mandatory clearing and clearing houses. Where the bespoke end of the business now lives is structured notes.

4.11 Key figures in financial history

sales · trading · structuring | core | cross-asset

On a trading floor, names are used as shorthand for ideas. Somebody says "that is a very Minsky setup" or "we are pricing it Black-Scholes flat" or "he is running a Kerviel", and everyone in earshot is expected to decode it instantly. In an interview, the same shorthand shows up as a test of whether you actually read around your subject or only crammed formulas the week before. This section is the who-is-who: for each name, the one idea they contributed, the year, and why a desk still cares.

Three families, and they map onto three different interview moods. The **theorists** gave us the equations we still price with, so their names come up in technical rounds. The **central bankers** set the price of money, so their names come up in market-culture rounds. The **practitioners**, including the ones who blew up, are where interviewers check that you know what the models leave out.

KEY IDEA

Never name-drop without the idea attached. "Markowitz" on its own is worth nothing; "Markowitz, 1952, showed that a portfolio's risk depends on correlations and not just on the individual variances, which is why diversification is the one free lunch in finance" is worth a lot. One name, one idea, one year, one consequence.

The theorists: where the pricing machinery came from

Louis Bachelier (1900). A French mathematician who, in a doctoral thesis called *Théorie de la spéculation*, supervised by Henri Poincaré, modelled the price of a French government bond as a random walk. He wrote down Brownian motion as a mathematical object five years before Einstein used it for physics, and he used it to value options on the Paris Bourse. His model let prices go negative, because he assumed the *price* moved as a Brownian motion rather than its logarithm, and that limitation is exactly why the modern convention is lognormal. Bachelier was then almost entirely forgotten for fifty years. Note the irony that desks quoted rate options in "normal" (that is, Bachelier) volatility when euro rates went negative, because the lognormal convention breaks there and his does not. The underlying process is developed in the section on Brownian motion.

Kiyosi Itô (1944 onwards). A Japanese mathematician who built the calculus that makes stochastic models tractable. Ordinary calculus assumes paths are smooth; a Brownian path is not, and its wiggle contributes a term that ordinary calculus misses. Itô's lemma is the correction, and it is the single tool that turns "the stock follows a random process" into a partial differential equation you can solve. Everything downstream, Black-Scholes included, runs through it. The statement and proof sketch live in Itô's lemma.

REAL INTERVIEW QUESTION (Natixis · EQD)

Itô's formula.

WHAT THEY TEST / ANSWER

What they test is whether you can produce the correction term without hesitating, because a candidate who has only memorised the Black-Scholes closed form will stall here. For a process $dX_t = \mu\, dt + \sigma\, dW_t$ and a smooth function $f(t, X_t)$,

$$df = \Big(\partial_t f + \mu\, \partial_x f + \tfrac{1}{2}\sigma^2\, \partial_{xx} f\Big) dt + \sigma\, \partial_x f\, dW_t.$$

The whole point is the $\frac{1}{2}\sigma^2 \partial_{xx} f$ term, which appears because $(dW_t)^2 = dt$ rather than being negligible. Say that sentence out loud: it is the difference between reciting and understanding. Then add the punchline that this second-derivative term is Gamma, so Itô is the formal reason a trader gets paid for convexity.

Harry Markowitz (1952). His paper *Portfolio Selection* did something that sounds obvious now and was not then: it treated risk as a number (the variance of returns) and showed that the risk of a portfolio depends on how its assets *co-move*, not just on how risky each one is alone. That single move created the efficient frontier, mean-variance optimisation, and the whole language of correlation in asset allocation. Nobel prize in 1990, shared with Merton Miller and William Sharpe.

WORKED EXAMPLE

Take two assets, each with expected return 8% and volatility $\sigma = 20\%$, and hold half of each. Portfolio variance is

$$\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho\,\sigma_1\sigma_2 = 0.25(0.04) + 0.25(0.04) + 2(0.25)\rho(0.04) = 0.02 + 0.02\,\rho.$$

Now read off three correlations:

- $\rho = 1$: $\sigma_p^2 = 0.04$, so $\sigma_p = 20\%$. No benefit at all.
- $\rho = 0$: $\sigma_p^2 = 0.02$, so $\sigma_p = \sqrt{0.02} \approx 14.1\%$.
- $\rho = -1$: $\sigma_p^2 = 0$, so $\sigma_p = 0\%$. Risk is fully cancelled.

The expected return is 8% in all three cases, because expectation is linear and does not care about correlation. Same return, less risk: that is what people mean by the free lunch. With a risk-free rate of 2%, the reward-per-unit-of-risk ratio goes from $(8-2)/20 = 0.30$ when $\rho = 1$ to $(8-2)/14.14 \approx 0.42$ when $\rho = 0$.

William Sharpe (1964). He asked what happens when *everyone* optimises the way Markowitz described, and got the Capital Asset Pricing Model: in equilibrium, only the risk you cannot diversify away should be rewarded, and that risk is measured by beta against the market. Two artefacts carrying his name are still on every desk: beta, from this paper, and the Sharpe ratio (excess return divided by volatility), which he introduced two years later in 1966 as the reward-to-variability ratio. Jack Treynor, John Lintner and Jan Mossin reached the same model independently at roughly the same time, which is a good detail to have ready.

REAL INTERVIEW QUESTION (Natixis · QIS)

Explain the CAPM and Markowitz portfolio theory.

WHAT THEY TEST / ANSWER

They are testing whether you can tell the two apart, because candidates routinely merge them. Markowitz is *normative*: given expected returns, volatilities and correlations, here is how you should build a portfolio, and the set of best risk-return trade-offs is the efficient frontier. CAPM is *positive*: if everybody does that, prices must settle so that expected excess return is proportional to beta,

$$\mathbb{E}[R_i] - r_f = \beta_i\big(\mathbb{E}[R_m] - r_f\big), \qquad \beta_i = \frac{\mathrm{Cov}(R_i, R_m)}{\mathrm{Var}(R_m)}.$$

Then volunteer the criticisms before they ask: variance is a poor risk measure when returns have fat tails and are skewed, correlations rise toward one exactly when you need diversification most, and a single market factor is empirically too thin, which is what motivated the later multi-factor work. That last sentence is what separates a memorised answer from a considered one.

Eugene Fama (1970). He formalised the efficient market hypothesis, the claim that prices already reflect available information, in weak, semi-strong and strong forms. Later, with Kenneth French, he showed that size and value factors explain returns the single-factor CAPM cannot, which is the ancestor of every factor and systematic strategy desk today. Nobel prize in 2013, shared with Lars Peter Hansen and, pointedly, Robert Shiller, who spent his career arguing that markets are *not* efficient and that bubbles are real. Being able to name both sides of that split is a good sign in an interview.

Fischer Black, Myron Scholes and Robert Merton (1973). The single most consequential result in derivatives. Black and Scholes published *The Pricing of Options and Corporate Liabilities* and Merton published *Theory of Rational Option Pricing* in the same year. The idea that changed everything is not the formula but the argument behind it: an option can be *replicated* by continuously trading the underlying and a bond, so by no-arbitrage its price must equal the cost of that replicating strategy, and the expected return of the stock drops out entirely. That is why the drift μ does not appear in the formula, and it is the question interviewers use to check you did not just memorise $N(d_1)$. The model, its assumptions and its derivation are owned by the Black-Scholes section.

Two facts about the prize that candidates get wrong. It was awarded in 1997 to Scholes and Merton only, because Fischer Black had died in 1995 and the prize is not awarded posthumously. And it is not, strictly, a Nobel prize: it is the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel, established in 1968 rather than by Nobel's will. Neither point wins you an offer, but getting Black's name into the sentence shows you know the history rather than the poster.

The generation that fixed Black-Scholes. The 1987 crash (see the 1987 crash) broke the model's central assumption of constant volatility, and the equity skew that appeared afterwards never went away. The response defines the modern vol desk:

- **Robert Engle** (ARCH, 1982) and **Tim Bollerslev** (GARCH, 1986) modelled volatility as clustering and time-varying instead of constant. Engle shared the 2003 Nobel with **Clive Granger**, whose cointegration work underpins every pairs and relative-value strategy. See ARCH and GARCH models.
- **Steven Heston** (1993) gave the tractable stochastic volatility model, in which volatility has its own random process correlated with the spot. See the Heston model.
- **Bruno Dupire**, and independently **Emanuel Derman** and **Iraj Kani** (1994), showed how to read a volatility surface off market prices as a deterministic function of spot and time, which is local volatility. See local volatility.
- **Patrick Hagan** and co-authors (2002) published SABR, still the market standard for interest-rate smiles. See the SABR model.

The equity-specific shape of the skew and why it persists are covered in equity skew and smile.

Year	Milestone
1900	Bachelier models prices as Brownian motion
1944	Itô builds stochastic calculus
1952	Markowitz: portfolio selection and diversification
1964	Sharpe: CAPM and beta (the Sharpe ratio follows in 1966)
1970	Fama: the efficient market hypothesis
1973	Black, Scholes and Merton: option replication
1982	Engle: ARCH, volatility clustering
1993	Heston: tractable stochastic volatility
1994	Dupire, Derman and Kani: local volatility
2002	Hagan and co-authors: SABR

The central bankers: the people who set the price of money

The institutions themselves are covered in major central banks and the mechanics in monetary policy. What belongs here is the handful of individuals whose names are used as shorthand for a regime.

Paul Volcker (Fed chair 1979 to 1987). He inherited US inflation running near 15% and killed it by pushing the policy rate to roughly 20% in 1981, accepting a recession in which US unemployment peaked at 10.8% in late 1982. Inflation was down to the low single digits by 1983. "Volcker shock" is the standing example of a central bank buying credibility with pain, and it is the reference every commentator reached for during the 2022 tightening cycle. He later gave his name to the Volcker rule restricting proprietary trading at deposit-taking banks.

Alan Greenspan (1987 to 2006). He took office about two months before Black Monday and responded by flooding the system with liquidity, which is the origin of the phrase "the Greenspan put": the belief that the Fed will always step in to catch a falling market. He also gave the market the phrase "irrational exuberance", in a speech in December 1996, roughly three years before the dot-com peak. His long tenure is the standard illustration of how an implicit central-bank backstop changes the price of risk, and by extension the demand for tail hedges.

Ben Bernanke (2006 to 2014). An academic scholar of the Great Depression who then had to run a central bank through the worst crisis since it, which is about as neat as history gets. He is the name attached to quantitative easing, zero interest-rate policy, and the emergency facilities of 2008, covered in the 2008 financial crisis. He received the 2022 economics prize, with Douglas Diamond and Philip Dybvig, for work on banks and financial crises.

Mario Draghi (ECB president 2011 to 2019). On 26 July 2012, in London, with Spanish and Italian yields at levels that implied a euro breakup, he said the ECB would do "whatever it takes to preserve the euro", and added, "believe me, it will be enough". Peripheral spreads collapsed on a sentence, before any programme existed; the Outright Monetary Transactions framework was announced weeks later and was never actually used. It is the cleanest example in modern markets that a credible central bank's *words* are a policy instrument, which is the whole subject of forward guidance. Background in the European sovereign debt crisis.

> **INTUITION**
>
> Think of a central banker as the referee in a contact sport. Volcker showed that a referee who is willing to send someone off early controls the whole match. Greenspan showed that a referee known to be lenient changes how hard everyone tackles. Draghi showed that a referee only has to *say* they will send someone off, provided nobody doubts they would.

The practitioners: what the models leave out

George Soros (1992). On 16 September 1992, later called Black Wednesday, his Quantum Fund was the most visible seller of sterling while the Bank of England defended its band inside the European exchange rate mechanism. The Bank raised its base rate from 10% to 12% and announced a further move to 15% in the same day, then abandoned the band and let sterling float. Reported profit for the fund was around one billion dollars. The lesson desks actually take from it is not about one investor: it is that a currency peg defended with finite reserves is a short option position, which is developed in pegged currencies and FX crises.

John Meriwether and LTCM (1994 to 1998). A fund whose partners included Myron Scholes and Robert Merton, running highly levered convergence trades that were individually sensible and jointly identical. When Russia defaulted in August 1998, every position moved the wrong way at once and the liquidation itself moved prices further against them. Fourteen banks recapitalised the fund for about \$3.6bn on 23 September 1998, in a meeting organised by the New York Fed. It is the canonical demonstration that leverage plus correlated positions plus illiquidity is a different animal from any one of those risks alone. Full account in Long-Term Capital Management.

Warren Buffett. Worth knowing for exactly one line in a derivatives interview: in the 2002 Berkshire Hathaway letter to shareholders he called derivatives "financial weapons of mass destruction". Candidates quote it as if it settles an argument. The considered response is that his objection was specifically to opaque, long-dated, hard-to-value contracts with concentrated counterparty exposure, that Berkshire itself sold multi-billion-dollar long-dated equity index puts, and that most of what a flow desk trades is cleared, margined and short dated. Knowing the nuance is more valuable than knowing the quote.

Jim Simons. A geometer who left academia and founded Renaissance Technologies in 1982, and whose Medallion fund became the standing proof that systematic, high-turnover, statistically driven trading can work at scale. He is the name you reach for if you are asked why quantitative strategies deserve a seat next to discretionary ones.

Hyman Minsky and Nassim Taleb. Two different attacks on the same blind spot. Minsky's financial instability hypothesis says stability itself breeds fragility: a long calm period encourages more leverage, which makes the eventual turn violent. The term "Minsky moment", the point where levered holders must sell, was coined by Paul McCulley in 1998 about the Russian crisis and was everywhere in 2008. Taleb, a former options trader, spent *Dynamic Hedging* (1997), *Fooled by Randomness* (2001) and *The Black Swan* (2007) arguing that the tails of financial distributions are far fatter than a normal model implies and that risk systems calibrated to the middle of the distribution are calibrated to the part that does not matter. That critique is why every desk supplements Value at Risk with stress scenarios.

The rogue traders. Three names, one structural lesson each. **Nick Leeson** destroyed Barings, a 233-year-old bank, in February 1995 with losses of about £827m: he ran unauthorised Nikkei futures and short straddle positions and hid them in error account 88888, and the fatal governance flaw was that he controlled both trading and settlement. **Jérôme Kerviel** cost Société Générale €4.9bn, announced in January 2008, running enormous unauthorised directional positions in European equity index futures masked by fictitious offsetting trades, exploiting control gaps he knew from a previous middle-office role. **Bruno Iksil**, the "London Whale", lost JPMorgan roughly \$6.2bn in 2012 on credit index tranche positions that had grown so large the desk was effectively the market and could not exit without moving prices against itself. Read them as three failure modes: segregation of duties, position verification, and concentration relative to market size. They are the reason for the controls described in operational risk, and they come up more often than candidates expect, because a desk hiring a junior cares a great deal about whether you understand why the rules exist.

INTERVIEW TRAP

Four traps. First, saying Fischer Black won the Nobel prize; he died in 1995 and the 1997 award went to Scholes and Merton. Second, saying Bachelier wrote down the Black-Scholes model; he used arithmetic Brownian motion, which allows negative prices, and there was no replication argument. Third, attributing the whole of LTCM to “the model was wrong”; the model was roughly right and the leverage, crowding and funding were what killed it. Fourth, and most common, quoting the Buffett “weapons of mass destruction” line to an interviewer who sells derivatives for a living without being able to say what he actually objected to.

REAL INTERVIEW QUESTION (UBS · EQD)

Do you think Black-Scholes works well?

WHAT THEY TEST / ANSWER

The trap is that both a flat yes and a flat no are wrong, and they are watching which way you fall. The answer with a spine: Black-Scholes is not a description of reality and was never meant to survive as one, since it assumes constant volatility, continuous frictionless hedging, lognormal returns and no jumps, and every one of those is false. It works because it is a *quoting convention*: an invertible, universally agreed map between a price and a single number, the implied volatility, which traders then risk-manage as the real underlying. The existence of the smile is the market openly admitting the model is wrong and correcting for it strike by strike. Close by naming what you would use instead and when, local volatility for a consistent surface, stochastic volatility when the dynamics of the smile matter for the payoff you hold, referenced in stochastic volatility models.

VISUAL / CODE TO BUILD: Interactive

A scrollable, filterable timeline of the people in this section on a single horizontal axis from 1900 to today, with three colour-coded lanes (theorists, central bankers, practitioners). Clicking a marker opens a card with the person, the year, the one-line contribution, and a live link to the section of this book where the idea is developed. A second toggle overlays crisis episodes (1929, 1987, 1992, 1998, 2008, 2020) so the reader can see how each wave of theory follows a wave of losses. The intended “aha” is that finance theory is reactive: the smile follows 1987, credit modelling follows 2008.

Where this goes next

The episodes these people lived through are the rest of this chapter, from 1929 to the COVID crash, and the instruments they invented are traced in the evolution of derivatives. The theory itself is built properly in the mathematics and derivatives chapters, starting with options and Black-Scholes. And the practical payoff is in market questions, where being able to attach a name, a year and a consequence to an idea is precisely what is being scored.

Chapter 5

Asset Classes Overview

5.1 Equities

sales · trading · structuring core equity

This is where the book starts from zero. If you do not yet know what a share is, this section is the one that has to work, because everything in the derivatives chapters is built on top of it.

What a share actually is

A company is financed by two kinds of claim. Lenders put in money and are owed a fixed amount back. Shareholders put in money and own *whatever is left* once everyone else has been paid. A **share**, or **stock**, is one unit of that leftover claim.

That single word, leftover, carries three entitlements and one consequence.

- **The residual cash flows.** After suppliers, employees, tax and interest, what remains belongs to shareholders, and part of it may be paid out as a **dividend**.
- **A vote** at the annual meeting, one per share for ordinary shares, which is how ownership becomes control.
- **A claim in liquidation**, ranked *last*, behind every creditor. In a bankruptcy shareholders usually receive nothing.

KEY IDEA

Being last in line is not a detail, it is the entire character of the asset class. Because debt is paid first and is roughly fixed, everything that happens to the value of the business, good or bad, lands on the equity amplified. That is why equity is the most volatile of the major asset classes, why a company's equity can lose 40% on news that moves its bonds by two points, and why an equity holder's position is economically a **call option on the firm's assets** struck at the value of its debt: limited downside at zero, unlimited upside, and worth more when the firm's assets are more volatile. That correspondence is the basis of the structural credit models in structural versus reduced-form models, and it is also the deepest reason the equity volatility smile slopes downwards, in the volatility smile and skew: as a firm's value falls, its leverage rises, and its equity gets riskier.

INTUITION

Think of a flat worth 300,000 with a 250,000 mortgage. The bank's claim is fixed; your equity is the 50,000 on top. If the flat gains 10%, worth 30,000, your stake rises by 60%. If it falls 10%, your stake more than halves. You did not take a 10% risk, you took a leveraged one, and you did it simply by being the residual claimant. A shareholder is in that position on a company.

How a shareholder gets paid

Two channels, and knowing that there are exactly two is worth saying in an interview. A **dividend** distributes cash directly. A **buyback** uses the same cash to repurchase shares, which leaves each remaining share owning a larger slice; it is economically similar to a dividend, and differs in tax treatment, in flexibility, and in the signal it sends.

Dividends matter far more to a derivatives desk than to an investor, because they are a scheduled, forecastable cash leak from the underlying, and they are a traded risk in their own right.

REAL INTERVIEW QUESTION (Société Générale · EQD)

Demonstrate the influence of dividends on the price of the share.

WHAT THEY TEST / ANSWER

The result to prove is that on the ex-dividend date the share price falls by approximately the amount of the dividend, and the proof is an arbitrage, not an observation.

Consider a share trading at 100 that will pay a dividend $D = 5$, with the ex-date tomorrow. Buy the share today at S_{cum}, receive the 5, and sell it tomorrow at S_{ex}. Ignoring tax, transaction costs and overnight interest, that strategy costs S_{cum} and returns $S_{\text{ex}} + 5$. If $S_{\text{ex}} > S_{\text{cum}} - 5$, the trade makes money with no risk, and everyone does it until the gap closes; if $S_{\text{ex}} < S_{\text{cum}} - 5$, the mirror trade does. So $S_{\text{ex}} = S_{\text{cum}} - D = 95$.

The intuition behind the algebra is the one to state first: paying a dividend does not create value, it moves 5 of cash out of the company and into the shareholder's pocket. The company is worth 5 less, the shareholder is not richer, and nothing has been earned.

Two refinements earn the follow-up. In practice the observed drop is often slightly *less* than the full dividend, which is usually attributed to differential taxation of dividends and capital gains across the shareholder base. And the announcement is a completely different event from the payment: an unexpected dividend cut moves the share because of what it says about the business, while the ex-date drop is pure mechanics and no information at all. Candidates who conflate the two are giving the wrong answer to whichever question was asked. What dividends do to *options* is a different question again, answered through the forward in forwards.

Size, float and indices

The **market capitalisation** of a company is its share price times the number of shares outstanding. It is the market's price for the whole equity claim, and it is not the value of the company: adding net debt gives the enterprise value, which is what you would actually have to pay to own the business outright.

The **free float** is the portion of shares genuinely available to trade, excluding stakes held by founders, families, the state or long-term strategic holders. Index providers weight by free float rather than by full market capitalisation, precisely because a fund tracking the index can only buy what is available.

An **index** is a rule for combining share prices into one number, and the rule matters. Almost all the ones you will hear quoted, the CAC 40, the Euro Stoxx 50, the S&P 500, the FTSE 100, weight by free-float capitalisation, which is why the free float just defined is the number index providers actually use; the two famous survivors of an older convention, the Dow Jones Industrial Average and the Nikkei 225, are price weighted instead. How those rules are built, what the divisor does and what happens at a rebalancing, is indices.

One distinction belongs here, because it decides whether the dividends discussed above show up at all: a **price index** excludes them, while a **total return index** reinvests them. The headline CAC 40 quoted in the news is a price index. At a 3% dividend yield, twenty years of reinvestment compounds to $1.03^{20} = 1.81$, so the total return version ends roughly 81% higher than the price version on the same underlying shares. Any comparison of long-run index performance that does not say which version is being used is meaningless. The full taxonomy, including the net return version that reinvests after withholding tax, is in indices.

REAL INTERVIEW QUESTION (Natixis · Credit)

Among these companies (Total, Renault, Vinci, Engie, Plastic Omnium, Bonduelle, Leclerc), which are in the CAC 40?

WHAT THEY TEST / ANSWER

Sort them into three groups rather than guessing name by name, because the sorting is what is being tested.

Large caps that belong in the index: TotalEnergies, Renault, Vinci and Engie are all large listed French companies of the size and liquidity the CAC 40 selects from. **Listed but too small**: Plastic Omnium and Bonduelle are listed French companies of mid-cap size, so they sit in the broader SBF 120 or below rather than in the 40. **Not listed at all**: Leclerc is the trap. It is a cooperative of independent retailers, not a listed company, so it cannot be in any index, and spotting that is worth more than the rest of the answer.

Then say the thing that protects you: index membership is reviewed periodically on free-float capitalisation and turnover, so names do enter and leave, and the right framing is the selection *criteria* plus a check of the current list rather than a memorised roster. Give the criteria out loud, free-float market capitalisation and traded volume among the largest French listings, and you have answered the question even for a name you were unsure about. The practical reason a desk cares is that index inclusion and exclusion force tracking funds to trade, which makes review dates real liquidity events.

What an equity desk actually trades

Very little of a bank's equity business is buying shares. The ladder runs from the cash share upwards, each rung adding leverage, tailoring or both: **cash equities**, then **futures** on indices and single names, then **ETFs**, covered in ETFs and index products, then **swaps**, then **listed and OTC options**, and finally the **structured products** built from them. Three risks are specific to this asset class and run through all of it: dividends, which are forecast and traded as their own market; **borrow**, the cost of locating shares to short, which can be enormous on a squeezed name and enters every forward; and **corporate actions**, since splits, rights issues, spin-offs and takeovers all require derivative contracts to be adjusted rather than simply repriced.

REAL INTERVIEW QUESTION (Société Générale · EQD)

What is the difference between a TRS, an equity swap and an index swap?

WHAT THEY TEST / ANSWER

Start by saying that these are not three species but one instrument described at three levels of generality, which is the answer most candidates miss.

A **total return swap** is the general contract: one leg pays the total return of a reference asset, price performance *plus* dividends and other distributions, and the other leg pays a funding rate, typically an overnight reference plus a spread. The reference asset can be anything, an equity, a bond, a loan portfolio, an index.

An **equity swap** is a total return swap whose reference is an equity, and in common usage it is the single-name version. An **index swap** is the same thing with an index as the reference. So index swap is a subset of equity swap, which is a subset of TRS.

Then give the two things that actually differ in practice, because that is where the mark is. **Dividend treatment**: a true total return leg passes dividends through, but the terms specify gross, net of withholding tax, or a fixed percentage, and that choice is a real economic negotiation rather than a formality. **Why the client uses it**: exposure without funding the position or appearing on the register, access to a market they cannot trade directly, or a tax or regulatory treatment they prefer, while the bank hedges by holding the underlying and earns the funding spread. Close on the risks: the client keeps market risk and adds counterparty risk to the bank, the bank carries the balance sheet, and both sides carry the dividend and borrow assumptions embedded in the spread.

INTERVIEW TRAP

Four confusions. (1) Market capitalisation is not the value of the company: it is the value of the equity, and enterprise value adds the debt. (2) A share price is not comparable across companies: a 5 euro share is not cheaper than a 500 euro one, because the number of shares differs; only the capitalisation and the multiples compare. (3) The ex-dividend drop is not a loss and carries no information; the dividend *announcement* is a different event entirely. (4) Comparing a price index to a total return index, which quietly omits several decades of dividends.

Where this goes next

The other asset classes are in fixed income, foreign exchange and commodities; the packaged versions of this one are ETFs and index products and structured products. The price of an equity in the future, including the dividends and borrow discussed here, is forwards, the right to buy it is options, and the shape of its volatility surface is the volatility smile and skew.

5.2 Fixed income

sales · trading · structuring | core | rates · credit

Equity is a claim on whatever is left. Fixed income is the other side of the same company: a claim on a *fixed* amount, paid first. That one difference produces an asset class with completely different mathematics, a different market structure and a different set of desks, and it is much the larger of the two by outstanding amount.

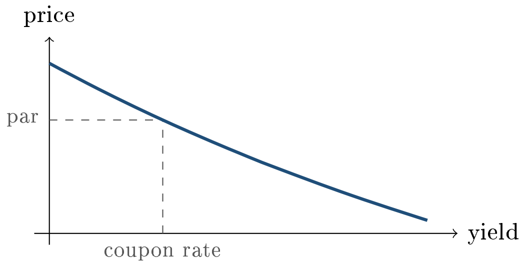
What a bond is

A **bond** is a tradable loan. The issuer, a government, a company or a bank, borrows a **principal** (also called the face value, nominal or par), promises to pay **coupons** at fixed dates, and to repay the principal at **maturity**. The lender's return is fixed at the moment they buy, provided two things: the issuer pays, and they hold to maturity.

> **INTUITION**
>
> A bond is a rental agreement on money. You hand over 100, the borrower pays you an agreed rent each year, and gives the 100 back at the end. Nothing about that promise changes if interest rates move. What changes is what your promise is *worth to somebody else*, because they can now rent their money out at a different rate.

That last sentence is the whole of fixed income mathematics in one line, and it is worth seeing why.

> **REAL INTERVIEW QUESTION** (JP Morgan · Sales)
>
> *What is a bond, and how does its price move with interest rates?*
>
> **WHAT THEY TEST / ANSWER**
>
> Define it, then prove the direction rather than asserting it, because "prices and yields move inversely" is a sentence anyone can memorise.
>
> A bond is a tradable debt instrument: fixed coupons on fixed dates and principal repaid at maturity, with the holder ranking ahead of shareholders and behind secured creditors.
>
> Its price moves **inversely** to interest rates, and here is the argument. You hold a bond paying a 3% coupon. Rates rise, and new bonds of the same issuer and maturity are issued paying 5%. Nobody will buy your 3% bond at 100 when they can have 5% for the same 100, so your bond has to fall in price until buying it at that lower price gives the same 5% return. The cash flows never changed; the discount rate applied to them did. Formally, price is the present value of fixed cash flows, so raising the discount rate lowers every term in the sum.
>
> Then add the two refinements that separate a prepared candidate. **Sensitivity grows with maturity**: a 30-year bond loses far more than a 2-year bond for the same move, because more of its value sits in distant cash flows, which is duration, in duration and convexity. And the relationship is **not a straight line**: the price falls by less than the linear estimate when rates rise and gains more than it when they fall, which is convexity, and it is why the curve below bends. Close with the practical version for a client: the loss on a rate rise is a mark-to-market loss, and an investor who holds to maturity and suffers no default still receives exactly the return they signed up for.



price of a five-year 3% annual-coupon bond against its yield
downward sloping, and curved: the curvature is convexity

KEY IDEA

Three facts fall straight out of that picture and are asked constantly. If the yield **equals** the coupon, the bond trades **at par**. If the yield is **above** the coupon, it trades at a **discount**, below 100. If the yield is **below** the coupon, it trades at a **premium**, above 100. You never need to compute anything to place a bond in one of those three states.

REAL INTERVIEW QUESTION (BRED · Rates)

For a bond trading at par with a 3% coupon, what is the yield to maturity?

WHAT THEY TEST / ANSWER

3%. Say the number immediately, then the one-line reason, because the question is testing whether you understand the definition or only memorised a formula.

The yield to maturity is the single discount rate that makes the present value of the bond's cash flows equal to its price. At par, the investor pays 100, receives 3 a year, and gets 100 back: there is no capital gain or loss to amortise, so the return is exactly the coupon rate, and discounting a 3% coupon stream at 3% gives precisely 100.

Two caveats worth volunteering. The yield to maturity assumes coupons are *reinvested* at the same yield, which is a convention rather than a fact, so the realised return differs if reinvestment rates differ. And it is a single number applied to every cash flow, whereas the correct valuation discounts each one at its own zero rate. The yield to maturity is only a summary of that curve, weighted by one bond's own cash flows, so two bonds of the same maturity but different coupons have different yields to maturity off the very same curve. That is why desks price off zero curves rather than off yields, in bond pricing.

The instrument families

"Fixed income" covers more than bonds, and an overview answer should name the families.

- **Government bonds**: the benchmark of each currency, from Treasuries in the United States to Bunds, OATs and BTPs in the euro area. Deepest liquidity, and the reference against which everything else is quoted.
- **Money market instruments**: Treasury bills, commercial paper, certificates of deposit, repo. Under one year; bills and commercial paper are usually issued at a discount with no coupon, while a certificate of deposit pays its interest at maturity and repo is a collateralised loan at a rate.

- **Corporate and financial bonds**, where the credit spread rather than the rate is the main risk, developed in credit.
- **Floating-rate notes**, whose coupon resets against a reference rate, so they carry almost no duration and are mostly a credit instrument.
- **Inflation-linked bonds**, whose principal is indexed to a price index, in inflation-linked products.
- **Derivatives**: interest rate swaps, futures, FRAs, caps, floors and swaptions, which is where most of the risk is actually traded and hedged, in interest rate swaps.

REAL INTERVIEW QUESTION (BMCE · Fixed Income)

What is the difference between the bond market and the money market?

WHAT THEY TEST / ANSWER

Maturity is the headline, and the consequences are the answer.

The **money market** is short term, conventionally under one year: Treasury bills, commercial paper, certificates of deposit and repo. Instruments are typically issued at a discount rather than paying coupons, duration is small so price sensitivity to rates is small, and the purpose is **liquidity and cash management**, both for corporates parking cash and for banks funding themselves overnight. The **bond market** is longer term, coupon bearing, with meaningful duration, and its purpose is **financing and investment**.

Then give the three consequences that make this more than a definition. **Risk**: money market instruments carry little rate risk and their main exposure is the issuer's short-term credit, whereas bonds carry substantial rate and spread risk. **Who uses them**: treasurers, money market funds and bank liquidity desks in the first, real-money investors, insurers and pension funds in the second, because the second is where liabilities of matching length can be hedged. **Where policy acts**: a central bank sets the overnight rate directly in the money market, and its effect on long bond yields is indirect, working through expectations of future short rates plus a term premium, which is exactly what the yield curve encodes, in yield curves. Add that the pivot between the two is the repo market, which is how bond positions are financed with money market cash, and you have connected the two rather than merely contrasted them.

How the market is organised

Two structural facts distinguish this asset class from equities and are worth knowing before an interview.

It is a dealer market, not an exchange. Most bonds trade over the counter between clients and dealers, quoted by request rather than continuously in a public book. One issuer has one share line and can have dozens of bonds outstanding, so liquidity is fragmented by maturity and by issue: the most recently issued bond of a given maturity, the **on-the-run**, trades far more tightly than the older **off-the-run** bonds around it, even though the credit is identical. That gap is a liquidity premium and it is directly tradable.

Quoting conventions differ by market. Government bonds are quoted in price, credit is often quoted in spread, swaps in rate, and futures in price with their own tick conventions. Prices are quoted **clean**, excluding accrued interest, while the buyer actually pays the **dirty** price including it, so that the quoted price does not jump on every coupon date. Asking which convention is in use before quoting a number is a habit interviewers notice.

INTERVIEW TRAP

Four errors. (1) Saying a bond is safe: it carries interest rate risk even when the issuer is riskless, and a long-dated government bond can lose more in a year than an equity index. (2) Confusing the coupon with the yield: the coupon is fixed at issue and never changes, the yield moves every second. (3) Comparing yields across bonds with different maturities or credit qualities as if they were the same number. (4) Forgetting accrued interest, and so quoting a clean price to somebody who is asking what they will pay.

Where this goes next

The mathematics of this asset class lives in the fixed income and rates chapter: bond pricing, yield curves, duration and convexity and interest rate swaps. What happens when the issuer might not pay is credit, the reference rates everything is quoted off are OIS, LIBOR, SOFR and ESTR, and the financing that makes the whole market work is repo markets.

5.3 Credit

sales · trading · structuring core credit

Credit is the asset class of *being repaid.* Lend money to a company and you are promised a fixed stream of payments, which is the fixed-income part; the credit part is the chance that the promise is not kept. Everything in this asset class is a way of taking, avoiding, pricing or repackaging that chance.

This section positions credit against the other asset classes and gives the one idea a reader should carry into the rest of the book. The instruments, the models and the market mechanics are the credit chapter's, and are referenced rather than repeated here.

Two risks in one instrument

A corporate bond's yield splits into two parts:

$$\text{yield} \;=\; \underbrace{\text{risk-free rate}}_{\text{fixed income}} \;+\; \underbrace{\text{credit spread}}_{\text{credit}}.$$

Buy the bond and you own both. That matters because the two move for completely different reasons and often in opposite directions: a recession pushes rates down and spreads up, so a corporate bond can be flat while both of its components are moving violently.

Almost every credit instrument exists to separate the two. An asset swap converts the fixed coupon into a floating one and leaves you with the spread alone. A credit default swap is the spread with no bond and, ignoring the small rate sensitivity of discounting the premium leg, no rate exposure, which is why it became the market's standard way of trading a credit view; see credit default swaps. Recognising which of the two risks a product actually gives you is the first question to ask about anything in this asset class.

The shape of credit returns

Here is the property that makes credit unlike equity, and the single most useful thing to know about it.

If everything goes well, you receive the spread. That is the whole upside: a bond cannot repay you more than it promised. If things go badly you can lose most of the principal. So credit returns are small and positive most of the time and occasionally very negative, which is the opposite shape to equity and the same shape as a **short option position**.

WORKED EXAMPLE

A portfolio of 100 corporate bonds, equally weighted, each paying a spread of 150 basis points over the risk-free rate, with a recovery rate of 40% if a name defaults.

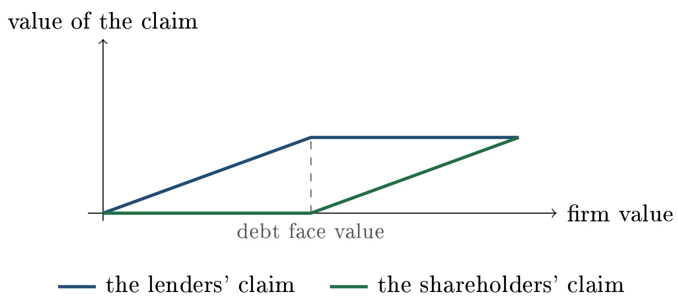
- Spread income: 1.50% of the portfolio per year.
- One default costs $(1 - 0.40) \times 1\% = 0.60\%$ of the portfolio.
- So $1.50/0.60 = 2.5$ defaults in a year wipes out the entire year's spread income.

Two and a half names out of a hundred. That is the arithmetic every credit investor lives with, and it explains the culture of the asset class: the job is mostly about avoiding losers rather than about picking winners, because there are no winners, only names that pay what they promised.

KEY IDEA

Owning a corporate bond is economically **lending at the risk-free rate and selling a put on the company**. The spread is the option premium, the strike is roughly the face value of the debt, and default is exercise. That is why credit returns are negatively skewed, why credit and equity volatility are linked, and why an equity derivatives trader and a credit trader are looking at two views of the same firm.

The same idea, drawn from the other end, is the capital structure. Split a firm's value between its two claims and each is an option on the firm's assets: the lenders own the firm subject to a cap at what they are owed, and the shareholders own everything above that.



The lenders' line is a bought bond plus a sold put; the shareholders' line is a call struck at the debt. Both are claims on the same assets, which is why credit and equity should not, and usually do not, disagree for long. The formal version of this is the structural model in structural versus reduced-form models.

What the spread is actually paying you

The spread is not the expected loss, it is more, and knowing the decomposition is a standard interview answer.

$$\text{spread} \;\approx\; \underbrace{\text{PD} \times (1 - \text{recovery})}_{\text{expected loss}} + \text{risk premium} \;+\; \text{liquidity premium}.$$

With a 1% annual default probability and a 40% recovery, the expected loss is $1\% \times 0.60 = 60$ basis points. A bond paying 150 basis points is therefore compensating you 90 basis points beyond what you expect to lose, for bearing the uncertainty around that number and for the fact that you may not be able to sell when you want to. The persistent gap between spreads and realised losses is a real feature of the market, and treating the whole spread as free money is the classic way to be wrong about credit for several years and then catastrophically right about it in one. Spreads themselves are credit spreads.

How the market is segmented

Credit is unusual among asset classes in that a formal label determines who is allowed to own what, and that label moves prices.

REAL INTERVIEW QUESTION (Natixis · Credit)

From which ratings does it become high yield?

WHAT THEY TEST / ANSWER

The boundary sits between **BBB-** and **BB+** on the Standard and Poor's and Fitch scales, and between **Baa3** and **Ba1** at Moody's. BBB- and Baa3 are the lowest investment-grade ratings; anything below is high yield, also called speculative grade or, less politely, junk. Give the scale around it so the boundary is anchored rather than memorised: AAA, AA, A, BBB are investment grade and BB, B, CCC and below are not, with the plus and minus or the 1, 2, 3 notches inside each band. Then say why the line matters, because that is the real question: it is not a smooth gradation but a cliff. Many investment mandates and insurance and bank capital rules are written in terms of investment grade, so a downgrade from BBB- to BB+ forces some holders to sell regardless of what they think of the company, which is why spreads widen far more than the change in default probability alone would justify. A bond crossing that line downwards is called a fallen angel and one crossing upwards a rising star, and both are traded as events in their own right. See rating agencies for the agencies and their methodologies.

REAL INTERVIEW QUESTION (Moody's · Credit)

Why are credit ratings important?

WHAT THEY TEST / ANSWER

Answer at three levels and do not stop at the first, which is where most candidates finish. **Information**: a rating is a standardised, comparable opinion on creditworthiness across thousands of issuers that most investors could not analyse individually, which lowers the cost of participating in the market at all. **Regulation and mandates**: this is the level that actually moves prices. Capital requirements for banks and insurers, central bank collateral eligibility, and the investment mandates of most funds are written using ratings, so a rating is not just an opinion, it is a permission. A downgrade can force a sale by holders who have no view on the company. **Pricing and issuance**: an issuer's rating sets the spread it pays and therefore its cost of capital, which is why companies manage to a target rating and treat losing one as a real financial event. Then give the honest critique, since answering this at Moody's without acknowledging it would be naive: ratings are through-the-cycle and slow-moving, they were badly wrong on structured credit before 2008, and the issuer-pays model creates a conflict of interest. The market's response is to use them as a segmentation and a constraint rather than as a live risk measure, and to look at spreads and credit default swaps for the current view, which is exactly why both exist alongside ratings.

The instruments, in one paragraph

Corporate bonds are the cash instrument and the largest part of the market, developed in corporate bonds. **Loans** are the bank equivalent, less liquid and usually floating rate and secured. **Credit default swaps** are the derivative, letting you buy or sell default protection without owning the bond. **Credit indices**, iTraxx in Europe and CDX in North America, package a basket of names into one liquid instrument and are how most macro credit views are expressed. **Structured credit**, from tranches to collateralised loan obligations, slices a pool of credit risk into pieces with different seniority, which is the same capital-structure idea applied a second time.

REAL INTERVIEW QUESTION (Morgan Stanley · Fixed Income)

A default?

WHAT THEY TEST / ANSWER

A short question that rewards a precise answer. A default is a failure to meet the terms of a debt obligation, and the three standard forms are worth naming: **failure to pay** a coupon or principal when due, usually after a short grace period; **bankruptcy** or an equivalent insolvency proceeding; and **restructuring**, where the terms are changed to the lenders' disadvantage, for example a maturity extension or a coupon reduction. Then make the distinction that shows you have worked with the instruments: in the derivatives market these are not adjectives, they are the defined *credit events* in the contract, and which ones are included varies by region and by contract type, with restructuring treated differently in Europe and North America. That definitional precision is the whole basis of a credit default swap paying out or not, and disputes are settled by a determinations committee rather than by argument. Finish with what happens next, because it is the part that determines the loss: after a credit event the recovery is established by an auction, and the holder's loss is one minus that recovery, which is why the pair of numbers that matters is never just the default probability. See recovery rates.

How it differs from the other asset classes

- **Versus equity**: same company, senior claim, capped upside, and a payoff that is short an option rather than long one.
- **Versus rates**: a corporate bond contains a government bond, so a credit investor is a rates investor whether they meant to be or not, until they strip it out.
- **Liquidity**: credit trades over the counter, in thousands of individual bonds from the same issuers, so it is far less liquid than equities or government bonds. This is not a footnote; it is why the indices exist and why liquidity premia are a real part of the spread.

INTERVIEW TRAP

Four errors. (1) Treating a corporate bond as a single risk, when it is a rate and a spread that routinely move in opposite directions. (2) Treating the spread as free income, when part of it is expected loss and part is compensation for illiquidity. (3) Forgetting the asymmetry: the upside is the spread and the downside is most of the principal, so a credit book's good years look nothing like its bad ones. (4) Assuming a rating is a live risk measure, when it is a slow-moving, through-the-cycle opinion whose main market effect is regulatory.

VISUAL / CODE TO BUILD: Interactive

A credit return simulator. The reader sets a spread, a default probability, a recovery rate and a portfolio size, and the tool simulates many years of outcomes and plots the distribution of annual returns. The intended realisation is visual and immediate: the distribution has a tall narrow peak just above the risk-free rate and a long thin tail to the left, and the median year looks nothing like the mean. A slider for portfolio size shows diversification narrowing the peak without removing the tail, and a correlation slider shows the tail returning as defaults cluster. A second panel draws the capital-structure diagram with a movable firm value, showing the debt and equity claims and the implied credit spread updating together, so the reader sees the structural model as one picture rather than two.

Where this goes next

Everything here is developed in the credit chapter: the cash instrument in corporate bonds, the price of risk in credit spreads, the labels in rating agencies, the two components of loss in default risk and recovery rates, and the derivative in credit default swaps. The interest-rate half of a corporate bond is fixed income, and the other claim on the same company is equities.

5.4 Foreign Exchange

sales · trading · structuring | core | fx

This is the orientation. Every mechanism named below has its own section in the FX chapter: spot, forwards and swaps, interest rate parity, options, policy and pegs. What is owned here is what makes FX a different *kind* of asset from the others in this chapter, because that difference explains most of what follows.

KEY IDEA

An exchange rate is a **relative price**, not a price. There is no such thing as being long FX: every position is long one currency and short another, so every view is a comparison. That single structural fact distinguishes FX from every other asset class. A share can rise while everything else does; a currency cannot, because it is measured in the things it would rise against.

Several familiar habits break as a consequence. There is no earnings stream, no dividend, no coupon and no book value, so the valuation anchors that work elsewhere have no analogue. The only carry is the interest rate differential between the two currencies. And "the market went up" is meaningless without saying against what, which is why FX people quote pairs and not levels.

The market, briefly

FX is the largest and most liquid market there is, and its structure follows from being a relative price: it is over the counter rather than exchange traded, decentralised across banks and electronic venues rather than centralised in one book, and open continuously from the Asian morning to the New York evening because somebody somewhere always needs to settle.

Conventions worth carrying: a pair is quoted as **base/quote**, and the number is how many units of the quote currency buy one unit of the base, so EUR/USD at 1.10 means 1.10 dollars per euro and "buying EUR/USD" means buying euros. Majors are the pairs involving the dollar; crosses are the rest. Spot normally settles two business days forward. The full treatment is FX spot markets.

The participants are more varied than in most asset classes, and knowing who they are explains the flow: corporates hedging real commercial exposure, central banks managing reserves and sometimes intervening, real-money investors hedging the currency risk in foreign assets, macro funds taking outright views, and banks intermediating all of it.

REAL INTERVIEW QUESTION (Société Générale · Fixed Income)

What do you know about FX?

WHAT THEY TEST / ANSWER

Wide open, so impose a structure rather than listing facts. Four headings takes ninety seconds and covers everything a follow-up could come from.

What it is: the market in relative prices between currencies, the largest and most liquid there is, over the counter and continuous. Say the relative-price point explicitly, because it is the thing that distinguishes the asset class and most candidates skip it.

The instruments: spot, outright forwards, FX swaps (which are the majority of turnover, not spot), non-deliverable forwards for restricted currencies, and options. Note that an FX swap is a funding instrument rather than a directional one, which is why the volumes are so large.

What drives it: interest rate differentials and the expected path of policy, growth and terms of trade, capital flows, and risk sentiment, with a handful of currencies behaving as havens in stress. Add the honest caveat that separates a real answer from a textbook one: FX is the hardest major asset class to forecast, and at horizons under a year a random walk beats most structural models, which is a well-documented result rather than a piece of modesty.

What is distinctive: no natural long, no yield except the differential, a peculiar two-sided relationship with central banks who are both participants and policymakers, and the fact that the largest players are not trying to make money on it at all, since a corporate hedging receivables is minimising variance rather than seeking return.

Then stop and let them choose the follow-up. If they push, the most likely directions are parity, the carry trade and how options differ, and this chapter's FX sections cover each.

Carry, which is the defining FX trade

If there is one trade that characterises the asset class, it is carry: fund in a low-yielding currency and hold a high-yielding one, and collect the difference. It is worth working through, because it exposes exactly what is strange about FX.

WORKED EXAMPLE

An AUD/JPY carry position, priced. Take spot at 100 yen per Australian dollar, a one-year Australian rate of 4% and a Japanese rate of 0.5%.

Borrow 100 yen, buy 1 Australian dollar, and deposit it. After a year you hold 1.04 Australian dollars and owe $100 \times 1.005 = 100.50$ yen.

- If spot is **unchanged** at 100, your Australian dollars are worth 104 yen and you repay 100.50, so you make **3.50 yen** on 100 borrowed, which is the rate differential.
- Your break-even spot is $100.50/1.04 = \mathbf{96.63}$. Anything above it and you profit; below it and you lose.
- At 90 you lose 6.90, and at 110 you make 13.90. Both sit 10.40 away from the 3.50 you collect if spot does not move, so a ten-yen move is worth three times the entire year's differential: it is a leveraged directional bet with a coupon attached.

Now the point. The one-year forward, by interest rate parity, is $100 \times 1.005/1.04 = 96.63$: exactly your break-even. So the market already prices the Australian dollar to fall by 3.4% over the year, and the carry trade is precisely a bet that it *will not*. Carry is not free money hiding in the rate differential; it is a position against the forward.

That the forward is a poor predictor, and that carry has historically earned positive returns, is one of the oldest puzzles in the subject. The usable explanation is that the return is compensation for crash risk: carry currencies fall slowly and violently rather than gradually, so the strategy earns a small premium most of the time and loses several years of it in a few days. Any answer that treats carry as an arbitrage has missed it.

REAL INTERVIEW QUESTION (Union Capital · FX)

Buying AUD/JPY for a carry trade: which currency do you borrow and which do you deposit, and if the pair rises do you gain or lose?

WHAT THEY TEST / ANSWER

Work it from the quoting convention rather than from memory, because that method survives any pair they name.

AUD/JPY means yen per Australian dollar, so the base is the Australian dollar. Buying the pair means buying the base, so you are **long Australian dollars and short yen**. Being short a currency is borrowing it, so you **borrow yen and deposit Australian dollars**. That is the right way round for a carry trade, since Japanese rates are the lower of the two, and it is why the yen has been the classic funding currency for decades.

If the pair rises, more yen per Australian dollar, then the currency you own has appreciated against the one you owe, so you are a **winner**, on top of the positive carry you were already earning. Both legs work in your favour, which is exactly what makes the trade popular and exactly what makes it crowded.

Then volunteer the risk, because it is what the question is set up for. The two legs also fail together. A carry trade is short the funding currency, and the funding currency is typically the safe haven, so when risk sentiment turns, everyone unwinds the same position at once, the yen rallies sharply, and the loss on the spot leg dwarfs years of accumulated differential. The usual description is that a carry trade goes up by the stairs and down by the lift, and saying so tells the interviewer you know the position is short a tail rather than long a yield.

Not every currency is freely tradeable

The last structural feature has no analogue in equities. Currencies differ in **convertibility**, and the regime determines what you can trade at all: freely floating, managed, pegged, or restricted, with capital controls limiting who may hold the currency. The macro treatment is exchange rate regimes and the way pegs break is pegged currencies and FX crises.

REAL INTERVIEW QUESTION (Union Capital · FX)

What is an NDF, and name some of the currencies concerned?

WHAT THEY TEST / ANSWER

What it is: a non-deliverable forward is an FX forward that settles in cash, in a convertible currency, rather than by exchanging the two currencies. You agree a rate today, and at maturity the difference between that rate and an official fixing is paid in dollars. No restricted currency ever moves.

Why it exists: because some currencies cannot be freely delivered offshore. Capital controls stop a foreign investor from holding or transferring the currency, so a deliverable forward is impossible, but the *risk* is still real and still needs hedging. The NDF separates the economic exposure from the physical settlement, which is the whole invention.

Which currencies: the classic non-deliverable markets are the Korean won, the Indian rupee, the Taiwan dollar, the Brazilian real and, historically and now only partly, the Chinese renminbi, where an onshore and an offshore market coexist. Name a few rather than reciting a list, and if you are unsure of one, say the category instead: emerging Asian and Latin American currencies with capital controls.

Then add the two things that make it a desk answer. NDFs price off a **fixing**, so fixing risk and the integrity of that fixing are real exposures, not technicalities. And NDF pricing does not obey covered interest parity cleanly, because the arbitrage that enforces parity requires being able to borrow and deliver the currency, which is exactly what is restricted. The gap between the NDF-implied rate and onshore rates is therefore a live measure of how tight the controls are, and it widens in stress.

REAL INTERVIEW QUESTION (Bank of America · Exo)

How does news impact your currency?

WHAT THEY TEST / ANSWER

Answer in terms of the channel, not the headline, because the same news moves a currency differently depending on which channel it runs through.

The rate channel, which dominates in normal conditions. Anything that changes the expected path of policy changes the rate differential and therefore the currency: a strong inflation print or a hawkish central bank speech makes the currency's rates higher relative to others, and it appreciates. This is why FX reacts to data releases in proportion to how much they move the front end of the curve rather than in proportion to how important they sound.

The growth and terms-of-trade channel. News about growth, exports or commodity prices moves currencies whose economies depend on them, which is why the Australian dollar tracks iron ore and the Norwegian krone tracks oil.

The risk channel, which can invert the first one. In a genuine risk-off event, capital moves to havens regardless of rate differentials, so the yen and the Swiss franc can rally on bad news about themselves. A candidate who notices that the sign of the rate channel can be overridden has said the most interesting thing available here.

Then two points of craft. What moves a currency is the *surprise* relative to what was priced, not the level of the number, so the reaction depends on positioning as much as on the data. And the largest reactions come from news that changes the policy *reaction function* rather than one data point, which is why a central bank changing its framing moves FX more than a single print ever does.

INTERVIEW TRAP

1. **Talking about a currency without naming the other one.** Every FX statement is a comparison; "the dollar is strong" is incomplete until you say against what.
2. **Getting the base and quote the wrong way round.** Buying EUR/USD is buying euros. Derive it from the convention rather than recalling it, and it will hold for any pair.
3. **Treating carry as free money.** The forward already prices the differential away; carry is a bet against the forward, paid for by taking crash risk.
4. **Assuming all currencies are deliverable.** A large part of the emerging market complex trades as non-deliverable forwards against a fixing.
5. **Forecasting FX with a model and no humility.** Under a year, the random walk is a hard benchmark to beat, and saying so is a strength rather than an admission.

VISUAL / CODE TO BUILD: Interactive

A carry trade laboratory. The reader picks a pair and two interest rates and the page draws the position's profit against terminal spot as a straight line, marking the break-even at the forward and shading the region where the differential alone is enough. A slider for holding period shows the carry accumulating linearly while the spot exposure stays the same size, which is the asymmetry the strategy lives on. A second panel replays historical episodes as a path, so the reader watches a position earn a small amount for many months and lose it all in a fortnight, and a positioning gauge shows how crowded the trade was going in. The intended lesson is the one the text states: the line through the forward is the break-even, and everything to the left of it is what you are being paid to accept.

5.5 Commodities

sales · trading · structuring | core | commodity

Every other asset class in this chapter is a claim on cash flows. A commodity is a physical thing that has to be dug up, grown, shipped, stored and eventually burned or eaten. That difference is not a curiosity, it is the source of every feature that makes this asset class behave unlike the others, and interviewers test whether you have understood it or merely learned the vocabulary.

Four things a barrel has that a share does not

- **No cash flow.** A commodity pays no dividend and no coupon, so it cannot be valued by discounting anything. Its price comes from supply and demand for the physical material, and the only way to hold it is to pay to keep it.
- **Storage costs.** Holding it is expensive, and for some commodities close to impossible: electricity cannot be stored at scale, gas needs caverns, cattle need feeding. This asymmetry is why the shape of a commodity curve carries information that an equity forward curve does not.
- **Quality and location.** There is no single "oil": there are grades, each with a sulphur content and a density, each delivered somewhere specific. Two prices for the same commodity in different places are different prices for genuinely different things.
- **It gets consumed.** Someone actually needs the material at a date, which produces seasonality, squeezes and inelastic short-run demand. A share is never urgently required by a refinery on Tuesday.

REAL INTERVIEW QUESTION (Barclays · Structured Products)

When you buy oil on the market, what are you actually buying?

WHAT THEY TEST / ANSWER

Almost never a barrel. You are buying a **futures contract**, and the correct answer is to state everything that contract specifies, because each item is a risk.

A standard contract fixes the **grade** (Brent, WTI, a defined API gravity and sulphur content), the **quantity** (1,000 barrels), the **delivery month**, the **delivery location** (Cushing, Oklahoma for WTI, a North Sea or cash-settled basis for Brent), and the **settlement mechanism**, physical delivery or cash. So what you own is a claim to a specific quality of oil, in a specific place, in a specific month.

Then give the two consequences that make this the interesting question it is. If you hold the contract into expiry you may be obliged to **take delivery**, which is why financial investors roll into the next month before then, and why a fund's return differs from the change in the spot price by the cost of rolling. And because delivery is at one location, the price can decouple from anywhere else if that location's logistics bind: the negative WTI settlement in April 2020 happened because holders of an expiring contract had no storage available at Cushing and paid to be relieved of the obligation. Close with the alternatives, since the question invites them: an ETF, which holds futures and inherits the same roll, a swap, or the physical itself, which requires a tank. Only the last one is really "buying oil". The physical-versus-derivative distinction is developed in physical versus derivative commodities.

One commodity, many prices

REAL INTERVIEW QUESTION (Huxley · Commodity)

What explains the Brent-WTI spread?

WHAT THEY TEST / ANSWER

Three things, and they are the same three that explain every benchmark spread in the asset class, so answer generally and then instantiate.

Quality. Both are light sweet crudes, and they differ slightly in density and sulphur, which changes what a refinery can make from them and therefore what it will pay. This is the smallest of the three effects between these two grades.

Location and logistics. This is the main driver. WTI is priced at Cushing, Oklahoma, an inland hub connected by pipeline, while Brent is a waterborne North Sea grade with direct access to seaborne markets. A landlocked benchmark can only clear locally, so when US production or pipeline and storage constraints bind at Cushing, WTI weakens against Brent regardless of global demand. The spread is, in effect, the cost and availability of moving barrels from one to the other, and it is bounded by that transport cost when logistics are working and unbounded when they are not.

Regional supply and demand, including policy. Anything that shifts the balance in one region and not the other moves the spread: North Sea field outages, US shale volumes, export restrictions, freight rates.

Finish with the general rule, which is what the interviewer is really checking: the price of a commodity is always *a grade, at a place, at a date*, and a spread between two benchmarks is the market pricing the difference in those three coordinates rather than a mispricing. The same logic explains regional gas spreads and metal premiums.

The curve is the asset class

Because you cannot easily hold the physical, a commodity investor holds futures, and their return depends on the *shape* of the futures curve as much as on the spot price. If the curve slopes upward, each roll sells a cheap expiring contract and buys a dearer one, which is a drag; if it slopes downward, the roll is a gain. Those two states have names, contango and backwardation, and they are developed in contango and backwardation. What matters at overview level is the mechanism that produces them: the curve is a statement about the cost and the benefit of holding the physical between now and then.

REAL INTERVIEW QUESTION (Louis Dreyfus · Commodity)

Given insurance and storage costs over three months, when is it worth buying the calendar spread and storing?

WHAT THEY TEST / ANSWER

Set it up as an arbitrage and then compare two numbers, because the answer is a threshold, not an opinion.

The trade is **cash and carry**: buy the physical today at S, pay to store and insure it, finance the purchase, and simultaneously sell the three-month future at F to lock the exit price. It is worth doing when

$$F - S \;>\; \underbrace{\text{storage} + \text{insurance}}_{\text{physical carry}} + \underbrace{r \times S \times T}_{\text{financing}}.$$

Put numbers on it. Spot 80, three-month future 82, storage 0.50 per month, insurance 0.10 per month, financing at 5% a year. Carry is $3 \times 0.60 = 1.80$ physical plus $80 \times 0.05 \times 0.25 = 1.00$ financing, so 2.80 in total against a spread of 2.00. Do not do it: the calendar spread does not pay for the carry, and you would lose 0.80. If the future were 83.50 the spread of 3.50 would cover the 2.80 and lock in 0.70 of profit per barrel.

Then say the three things that turn a formula into a trader's answer. This arbitrage is what **caps contango**: a curve cannot stay steeper than full carry for long, because everybody does this trade until it flattens, unless storage is physically full, which is exactly when the cap breaks. There is **no matching floor** on the other side, because you cannot short the physical you do not have, which is why backwardation can be arbitrarily deep and why the two states are not symmetric. And the practical requirements are real: you need available storage capacity, the quality must be deliverable against the contract, and you carry basis risk if your storage location differs from the delivery point. Storage economics in full are in storage costs and the benefit of holding the physical is convenience yield.

The sectors, and why they do not behave alike

"Commodities" is not one market. The families have almost nothing in common except that they are physical.

- **Energy**: crude and refined products, natural gas, power, emissions. The largest and most financialised, with refining and generation margins traded in their own right as crack and spark spreads.
- **Metals**: base metals such as copper and aluminium, traded on the London Metal Exchange with its physical warehouse system, and precious metals, where gold behaves less like a commodity than like a currency or a store of value, with a large above-ground stock relative to annual production.
- **Agriculture**: grains, oilseeds, softs and livestock, dominated by seasonality, weather and harvest cycles, with a fresh crop resetting supply every year.

REAL INTERVIEW QUESTION (UBS · Commodity)

What kind of shapes do commodity markets have, and what specificities have you learned about them?

WHAT THEY TEST / ANSWER

Answer in two layers: the shapes, and then what produces them, because anyone can name the shapes.

Shapes. A curve can slope up, slope down, or do neither monotonically. The third case is the one worth mentioning first, because it is specific to this asset class: gas and power curves are **seasonal**, with winter contracts above summer ones year after year, so the curve is a repeating wave rather than a slope, and agricultural curves reset at each harvest. Naming that shows you have looked at a real curve.

What produces them. Storability is the organising variable. A well-stored, easily financed commodity such as gold has a curve pinned close to full cost of carry, so it is almost always gently upward sloping and arbitrage keeps it there. A commodity that is hard or expensive to store, or where users need the physical now, can invert sharply, because there is no way to arbitrage a shortage by shorting inventory you do not have. And a non-storable such as power has effectively no link between one delivery period and the next, so its "curve" is a set of nearly independent markets and the usual carry arguments do not apply at all.

Then the specificities worth naming: prices are bounded below by production costs only loosely and in the long run, short-run demand is very inelastic so small supply shocks produce large price moves, the market is driven by inventories rather than by earnings, and the participants include producers and consumers who are hedging rather than speculating, which shapes the flow the desk sees. Close on the point that follows from all of it: in commodities the *spread* trades, calendar, location and quality, are often where the risk and the information are, rather than the outright price. The full treatment is in forward curves and energy markets.

INTERVIEW TRAP

Four errors. (1) Talking about "the price of oil" without a grade, a location and a month: those three make it a price. (2) Assuming a commodity fund tracks the spot: it holds futures, so its return includes the roll, which over a long contango period can dominate. (3) Applying equity forward logic: there is no dividend, the carry is storage net of the benefit of holding the physical rather than a borrow rate, and there is no reason for the curve to be an expectation of the future spot. (4) Treating gold as a typical commodity: its above-ground stock dwarfs annual production, so inventory arguments that work for copper do not apply to it.

Where this goes next

The physical side and its instruments are physical versus derivative commodities, the economics of holding the material are storage costs and convenience yield, and the curve states named here are contango and backwardation. The sectors are developed in energy markets, metals and agricultural commodities, with the processing margins in crack spreads.

5.6 Cryptocurrencies

sales · trading · structuring core crypto

Start with a finding, because it is the most useful thing this section can tell a candidate. The question database behind this book holds about 3,400 real interview questions from European front-office processes, and **not one of them is about cryptocurrency**. Not one mentions bitcoin, a blockchain, a stablecoin or a token.

That is not an accident of sampling and it is worth taking seriously: on the sales, trading and structuring desks these interviews come from, crypto is not part of the technical assessment. So the honest advice is to know this section and no more. Understand what the asset class is, why it does not fit the valuation frameworks in the rest of this book, and what is genuinely novel about its market structure. Do not prepare it at the expense of the Greeks.

> **KEY IDEA**
>
> Every other asset class in this chapter is a *claim*: equity is a claim on profits, a bond is a claim on contractual cash flows, a commodity is a claim on something with a use. A cryptocurrency is a **bearer digital asset with no issuer and no cash flows**, so none of the valuation machinery in this book applies to it. That single fact explains almost everything that is different about the asset class, including why its price behaves the way it does.

What it is, in market terms

A cryptocurrency is an entry on a distributed ledger maintained by a network rather than by an institution. Ownership is control of a private key, which makes it a bearer instrument: whoever holds the key holds the asset, and there is no registrar to appeal to if it is lost or stolen. There is no issuer, so there is no counterparty and no default, and equally no obligation on anyone to pay you anything.

Supply is set by protocol rather than by policy. Bitcoin's is capped at 21 million units on a published schedule, which is the source of the analogy with a commodity: it is scarce because its rules say so, in the way gold is scarce because the earth says so. The analogy is imperfect in an important way, since gold has industrial and ornamental uses that put a floor of sorts under it and a cryptocurrency has none.

Why valuation is a genuine problem, not a hard one

The methods in this book all start from cash flows. Discount a bond's coupons; discount a company's expected profits; for a commodity, model storage, convenience yield and the physical balance, as commodities sets out. A cryptocurrency has none of these inputs, so there is no fundamental anchor to discount toward and the price is whatever supply and demand say it is at that moment.

That is not a criticism, it is a description, and it has three consequences a markets professional should be able to state.

- **No anchor means no mean to revert to.** Trends can persist far longer and corrections go far deeper than in an asset with an earnings or coupon floor.
- **Volatility is several times higher** than an equity index, and it does not decay after a shock the way an index's does, because there is no valuation argument pulling it back.

- **Narrative is the fundamental.** Price is driven by adoption, regulation, liquidity and flow rather than by anything that can be forecast in a spreadsheet, which is why the asset class is analysed with flow and positioning tools rather than with discounted cash flow.

The market structure is genuinely different

This is the part worth knowing, because it is where a derivatives professional finds something new rather than something missing.

It never closes. Trading runs continuously, so there is no official close, no fixing and no overnight gap in the usual sense. Every convention in this book that depends on a daily close, from settlement prices to variation margin timing, has to be redefined.

It is fragmented and the venue is the risk. Liquidity is split across many exchanges with no consolidated tape, so the same asset trades at slightly different prices in different places. More importantly, most venues are not the neutral utilities described in clearing houses: they combine the roles of exchange, broker, clearing house and custodian in one company. When that company fails, as FTX did in 2022, clients lose the assets themselves rather than merely a counterparty on a contract. In traditional markets those functions are deliberately separated, and the reason why is exactly this.

Custody is a first-class risk. Holding a bearer asset means the operational question of where the keys live is a genuine market risk, not an operations detail, which is why institutional participation grew only once regulated custodians and, later, exchange-traded funds gave access without the key-management problem. Venue fragmentation, custody and the rest of the plumbing are the subject of market structure in crypto.

The one instrument worth learning: the perpetual future

Crypto's most interesting contribution to market structure is the **perpetual future**, a futures contract with *no expiry date*. Since it never converges to spot by maturing, it is pinned to spot by a different mechanism: a periodic **funding rate** paid directly between longs and shorts, typically every few hours.

If the perpetual trades above spot, longs pay shorts; if it trades below, shorts pay longs. The payment scales with the gap, so it makes holding the rich side expensive and the cheap side profitable, and arbitrage pushes the contract back toward spot. That is a real innovation: it replaces convergence-at-maturity with a continuous carry, and it means the funding rate is a directly observable price of leveraged positioning, which is a piece of information the traditional futures market does not publish.

Two things follow that are worth saying if the subject comes up. The funding rate is the crypto equivalent of the basis in futures, so a persistently positive funding rate is the market telling you that leveraged longs are crowded, which is exactly the positioning signal that precedes liquidation cascades. And because these contracts are heavily levered and margined in the volatile asset itself, forced liquidations are mechanical and fast, which is why crypto drawdowns are sharper than the underlying flow alone would suggest. The contract itself is developed in perpetual futures and the mechanism that pins it in funding rates; what is claimed here is only its place in the asset class.

How institutions actually access it

Regulated exposure has arrived through familiar wrappers rather than through the native market: listed futures and options on established derivatives exchanges, spot exchange-traded funds approved in the United States in early 2024, and structured notes referencing crypto underlyings issued in the same format as any other note, which is structured notes with an unusual underlying rather than a new product. For a bank desk the interesting problems are therefore the ordinary ones, hedging a very high implied volatility, funding, and whether the underlying can be borrowed, applied to an asset whose bid-ask and venue risk are much worse than anything in the bid-ask spread.

> **INTERVIEW TRAP**
>
> Three errors, and one of them is about the interview rather than the asset. (1) Applying a valuation framework that needs cash flows and then presenting the output as a target price: there is no anchor, and saying so is more credible than producing a number. (2) Treating a crypto exchange as equivalent to a regulated exchange, when most combine functions that are deliberately separated elsewhere and the failure mode is loss of the asset rather than of a contract. (3) Over-preparing this asset class for a front-office interview. The database says it does not come up; the Greeks come up on almost every page of it.

A note on sourcing

Every other section in this book carries at least one real, attributed interview question. This one carries none, and that is deliberate: there is no cryptocurrency question in the database to carry, and inventing one would be worse than leaving the section without. If the question set is extended in future and crypto questions appear, this is the section they belong in. Until then, the absence is itself the most accurate thing the book can report about how much this asset class matters in these interviews. Questions about market levels and current themes, which is where crypto would surface if it surfaced at all, are handled in market questions.

Where this goes next

The asset classes that do have valuation anchors are equities, fixed income and commodities, and the scarcity analogy is best understood against the last of those. The contract mechanics that the perpetual future modifies are in futures, and the market-structure assumptions it breaks are in clearing houses and the bid-ask spread. The asset class in depth, rather than as one entry in a survey of asset classes, is the subject of its own chapter, starting at blockchain fundamentals.

5.7 Structured Products

sales · structuring · trading | core | equity · cross-asset

Every other section in this chapter describes a market. This one does not, and saying so is the most useful thing it can do.

> **KEY IDEA**
>
> A structured product is **not an asset class**. It is a *wrapper*: a legal and cash-flow container that repackages instruments from the other asset classes into a single security with one price and one ISIN. There is no structured-product market in which prices are discovered. Its risk decomposes, without residue, into the risks of the things inside it plus the credit of whoever issued the wrapper. Anyone who cannot perform that decomposition is not in a position to say what they own.

That is why this section sits in an asset-class chapter without describing one, and why it stays at the level of what the wrapper is and what it contains. The products themselves are chapter 19: notes, autocallables, reverse convertibles, capital protection, and how a desk carries the risk in hedging structured products.

The two components

REAL INTERVIEW QUESTION (Société Générale · Structured Products)

What are the two main components of a structured product?

WHAT THEY TEST / ANSWER

A bond and a derivative. Almost every structured product in existence is a fixed-income instrument issued by a bank, which handles the capital, plus an option package, which handles the payoff. Say it in that order, because the bond is the part candidates forget and it is the part that determines how much can be spent on the interesting half.

The bond component is usually a zero-coupon note issued by the bank. It is what makes the product a security you can hold, and it is why the investor is exposed to the issuer's credit for the whole life of the trade, unsecured. It is also the source of the budget, as below.

The derivative component is whatever payoff was sold: a call for participation, a short put for yield, digitals for coupons, barriers for conditionality. This is where the client's view lives.

Then add the sentence that shows you understand rather than recite. The two components are not independent, they are joined by an **arithmetic constraint**: the product is issued at par, so the present value of the bond plus the price of the derivative plus the fees must equal 100. That equation is the whole of structuring. Everything a client can ask for, more participation, a higher coupon, more protection, is a request to move money between those three terms, and one of them always has to give.

If they push for a third component, the honest answer is that fees are the third, and they are not a rounding error.

WORKED EXAMPLE

Where the budget comes from, priced. A five-year capital-guaranteed note on an equity index at 100, with the issuer funding at 3% and the index at 20% volatility.

- **The bond.** To return 100 at maturity with certainty costs $100e^{-0.15} = 86.07$ today. That is the price of the guarantee.
- **The budget.** The remaining $100 - 86.07 = \mathbf{13.93}$ is everything available for the payoff and the fees.
- **The option.** A five-year at-the-money call on the index is worth 24.33 at those parameters. The budget does not cover one.
- **The participation.** It buys $13.93/24.33 = \mathbf{57}\%$ of one, so the note offers 57% of any index rise. Take 2% of upfront fees out first and the budget falls to 11.93 and the participation to **49%**.

Three things follow immediately, and they are the reason this arithmetic is worth doing once. **Capital protection is expensive**, and its cost is the entire gap between 86.07 and 100. **The participation is whatever is left**, not a number the issuer chose to be generous with. And **fees come out of the payoff, not out of the protection**, so two percent of the notional is closer to fifteen percent of the participation. The sizing method in general is payoff design, and the product is capital protection.

Why both sides do it

A structured product exists because it solves a problem for the buyer that the underlying instruments do not, and because it is profitable for the issuer. Both halves are worth being able to state, since interviewers ask for one and are impressed by the other.

For the buyer. A payoff shaped to a view, in one security they are allowed to hold. Many institutional buyers face mandates that permit bonds and forbid derivatives, and a note is a bond. Retail buyers get access to options they could not trade directly, in a form with a single price, a single tax treatment and no margin calls. And the payoff can be conditioned in ways no listed instrument offers.

For the issuer. Three things at once: funding, often cheaper than issuing a plain bond of the same maturity; a margin embedded at issue; and risk that partially offsets what the trading book already holds, which is why a bank would rather sell you a structure whose hedge it can use than one whose hedge it must buy.

INTUITION

The uncomfortable version, which is worth saying because a good interviewer is waiting for it: a structured product's fee is invisible. There is no commission line, only a price of 100 for something worth slightly less than 100, and the difference is disclosed as a percentage in a document nobody reads. That is not fraud, it is the business model, and it is why the regulatory attention has landed on disclosure and suitability rather than on the instruments themselves. A candidate who describes structured products without mentioning where the margin is has described half of it.

REAL INTERVIEW QUESTION (Société Générale · Structured Products)

Explain how a reverse convertible, an autocall and a capital-guaranteed product work (payoff, return, risk profile).

WHAT THEY TEST / ANSWER

Three products, three columns, and the fastest way to keep them straight is to say what option the investor is *selling* in each, because that is what determines the whole risk profile.

Capital-guaranteed: the investor *buys* an option. Bond plus a call, so the capital comes back whatever happens and the upside is a participation in the index rise. Return is low and asymmetric upward; risk is issuer credit and opportunity cost. This is the only one of the three where the investor is long optionality, which is exactly why it pays the least.

Reverse convertible: the investor *sells* a put. Bond plus a short put, so you receive a high fixed coupon and, if the underlying falls below the strike, you are repaid in shares rather than cash. Return is a known high coupon; risk is the full downside of the underlying below the strike. Developed in reverse convertibles.

Autocall: the investor sells a put *and* sells the right to end the trade early. Conditional coupons, early redemption at par if the underlying is above a trigger on an observation date, and capital at risk below a barrier at maturity. Return is a higher coupon than the reverse convertible for the same barrier, paid for with the uncertain life; risk is the same downside plus reinvestment risk when it calls early. Developed in autocallables.

Then the ordering that ties it together and is the answer to "which pays most": the yield ranks capital-guaranteed, then reverse convertible, then autocall, and it ranks in exactly the order of how much optionality the investor has sold. Nothing else is going on. Anyone offering a higher coupon is asking you to sell something more.

REAL INTERVIEW QUESTION (Natixis · Structured Products)

Which option strategies are used inside a structured product to leverage the payoff (a call spread, for example)?

WHAT THEY TEST / ANSWER

Start from the constraint, because that is what makes the question answerable: the budget is fixed, so leveraging the payoff always means selling something to buy more of something else. Then give the techniques in order of how commonly they appear.

- **Call spread**, the example given. Sell an upside call to fund more participation below it. The client gives up the tail they were least likely to reach and buys a higher participation over the range they expect, which is why it is the standard first move.
- **Selling a put**, whether vanilla or knock-in. By far the largest source of budget, and the reason yield products pay what they do. It converts capital protection into conditional protection.
- **Digitals** instead of continuous payoffs. A fixed coupon on a condition is cheaper than a proportional payoff, so the headline number can be larger.
- **Worst-of on a basket**. Referencing the worst performer of several underlyings makes the downside option the client sells far more valuable, because the worst of n names is much likelier to breach a barrier than any single one of them, and the extra premium goes straight into the budget. That is why worst-of coupons dwarf single-underlying ones, and why the client is short dispersion as well as short the market.
- **Barriers and knock-outs**, which remove states of the world from the payoff and are refunded in budget.
- **Lengthening the maturity**, which grows the bond discount and therefore the budget itself.

Then close on the principle rather than the list, because that is what is being tested. Every one of these raises the headline number by selling optionality, and the client is always the seller. So the right way to present any of them is to name what has been given up in the same breath as what has been gained. A structurer who can only describe the enhancement has not finished the sentence.

REAL INTERVIEW QUESTION (Société Générale · Structured Products)

What is the payoff and the pricing of a zero-coupon?

WHAT THEY TEST / ANSWER

Simple in itself, and asked here because it is the component candidates skip.

Payoff: a single payment of the face value at maturity, and nothing before. No coupons, which is the whole definition.

Pricing: the discounted face value, $P = N/(1+y)^T$ with annual compounding or Ne^{-yT} continuously, where the rate is the *issuer's* funding rate rather than the risk-free rate, since the investor is taking that issuer's credit. So 100 in five years at 3% is worth 86.07 today.

Then say why the question was asked on a structuring desk, which turns a recall answer into a good one. The zero-coupon is the capital component of every structured product, and its price *is* the constraint the whole structure is built against: the gap between it and par is the entire option budget. Two consequences follow that a structurer lives with. Higher rates mean a cheaper zero and therefore a bigger budget, which is why capital-protected products are easy to sell when rates are high and nearly impossible when they are near zero. And a wider issuer funding spread has the same effect, which is why a weaker issuer can always quote a better product, and why the coupon and the credit are the same conversation.

INTERVIEW TRAP

1. **Treating it as an asset class.** It is a wrapper. Decompose it or you do not know what you own.
2. **Forgetting the issuer.** The investor is an unsecured creditor of the bank for the full term, which is the risk that actually materialised in 2008.
3. **Reading the headline coupon as a return.** It is the price of the optionality the investor sold, and it is conditional.
4. **Assuming the fee is small because it is not itemised.** It comes out of the option budget, so a fee of two percent of notional can be a seventh of the participation.
5. **Ignoring the rate environment.** The budget is the gap between the zero-coupon price and par, so the same structure is generous at 4% rates and impossible at 0%.
6. **Comparing two products by their headline numbers.** Compare what optionality each one requires the investor to sell; the yields always rank in that order.

VISUAL / CODE TO BUILD: Interactive

A structuring bench with a visible budget. A bar at the top shows 100 split live into three blocks, the zero-coupon bond, the option package and the fees, and the reader drags the boundaries. Sliders for the funding rate, the maturity and the volatility resize the blocks automatically, so raising rates visibly enlarges the option budget and raising volatility visibly shrinks what that budget can buy. Below, the resulting payoff is drawn and relabelled as the reader switches the option package between a call, a call spread, a short put and a worst-of, with the participation or coupon recomputed each time. The intended moment is dragging the fee block from zero to two percent and watching the participation fall by a great deal more than two percent, which is the section's least comfortable point made arithmetically rather than rhetorically.

5.8 ETFs and index products

sales · trading · structuring | core | equity · cross-asset

An index is a number, and a number cannot be bought. Everything in this section exists to convert that number into something you can hold, and the interesting part is that there are several ways to do it, each with a different set of risks hiding inside what looks like the same exposure.

What an ETF is, and the one mechanism that makes it work

REAL INTERVIEW QUESTION (BNP · Delta One)

What is an ETF?

WHAT THEY TEST / ANSWER

A one-line definition then the mechanism, because the mechanism is what the question is really about. An **exchange-traded fund** is an investment fund, holding a portfolio designed to track an index, whose shares are listed and trade continuously on an exchange like any share.

The obvious follow-up is why it trades near the value of what it holds. A traditional closed-end fund can trade at a persistent discount to its net asset value, and an ETF does not, because of **creation and redemption**. A small group of institutions, the **authorised participants**, can at any time deliver a basket of the underlying securities to the fund and receive newly created ETF shares, or hand back ETF shares and receive the basket. That makes the ETF and its underlying basket convertible into each other, so if the ETF trades above the basket, an authorised participant buys the basket, creates shares and sells them, and if it trades below, they do the reverse. The arbitrage is what pins the price to the net asset value, not any promise from the issuer.

Two consequences worth adding. The ETF's real liquidity is the liquidity of its *underlying basket*, not its own screen volume, which is why a thinly traded ETF on a liquid index can still be dealt in size. And the mechanism can break: if the underlying market is halted, closed for a time zone or genuinely illiquid, creation stops working and the ETF price can detach from its stated net asset value, which is when an ETF becomes a price discovery instrument in its own right rather than a wrapper.

How it is built

REAL INTERVIEW QUESTION (Amundi · Delta One)

How is an ETF constructed, physical or synthetic?

WHAT THEY TEST / ANSWER

Two families, and the honest answer names the risk each one keeps.

Physical replication. The fund actually owns the securities. *Full replication* holds every constituent at index weight, which works for a liquid index of a few hundred names. *Sampling*, or optimised replication, holds a subset chosen to track the index closely, which is what you do for an index with thousands of illiquid constituents or a broad bond index. The risk kept is **tracking error** from the sampling, from rebalancing costs and from cash drag, plus operational complexity in markets that are hard to access.

Synthetic replication. The fund holds a *substitute basket* of liquid securities and enters a **total return swap** with a bank: the fund pays the return of its basket and receives the index return. The tracking is close to exact by construction, which is the appeal, and it is often the only practical route to an index that is expensive or restricted to access physically. The risk kept is **counterparty risk** on the swap provider, mitigated by collateral, by resetting the swap when the mark-to-market exceeds a threshold, and in Europe by regulatory exposure limits. A candidate should say plainly that this risk is reduced, not removed, and that the collateral basket is worth looking at because it need not resemble the index.

Then the two points that decide real cases. **Tax**: on some indices a synthetic structure receives the index return gross while a physical fund suffers withholding tax on dividends, which can be worth more than the fee difference. And **securities lending**: a physical fund can lend its holdings and earn revenue that offsets its fee, which a synthetic fund cannot do in the same way. So the choice is not "physical is safer", it is a trade of counterparty risk against tracking, tax and lending revenue, and it is made index by index. Lending mechanics sit alongside repo markets.

What it actually costs

The headline management fee is the number that is advertised and it is not the number that matters. Three costs stack up, and only their combination is visible in the result.

- The **management fee**, quoted annually. Broad developed-market equity index ETFs compete at single-digit basis points; specialist, emerging-market and thematic products are far higher, commonly several tens of basis points.
- The **trading cost**: the bid-offer you cross on the exchange plus any premium or discount to net asset value at the moment you trade. For a short-horizon holder this dominates the fee.
- The **tracking difference**, which nets everything: fee, taxes, rebalancing costs, cash drag, and securities lending revenue *received.*

WORKED EXAMPLE

A fund charges a 20 basis point fee and earns 8 basis points from lending its holdings. Before taxes and rebalancing costs its tracking difference is roughly $20 - 8 = 12$ basis points a year, not 20. On some indices, lending revenue and favourable tax treatment can exceed the fee, and the fund's tracking difference is close to zero or even slightly positive. Compare funds on tracking difference against the same index over the same period, never on the advertised fee.

What happens during its life

REAL INTERVIEW QUESTION (Amundi · Delta One)

Give examples of events that can occur during the life of an ETF.

WHAT THEY TEST / ANSWER

Group them, because the list is long and a grouped list shows you have run one rather than read about one.

Index events. Periodic rebalancing and constituent changes force the fund to trade on a known date, which is a real liquidity event and a well-known one, so the trades are anticipated by others. Index methodology changes, capping rules and free-float updates do the same on a smaller scale.

Corporate actions in the underlying. Dividends, which the fund accumulates or distributes and which create a temporary cash position, plus splits, rights issues, mergers, spin-offs and delistings, each requiring the holding to be adjusted rather than simply revalued.

Fund-level events. Distributions, share class launches, currency-hedged share classes with their own monthly hedge roll, changes of index provider, fee cuts in response to competition, changes or additions of swap counterparty for a synthetic fund, and share splits of the ETF itself.

Stress events. Suspension of trading in underlying names, a closed underlying market when the ETF's own market is open, which regularly makes an ETF on foreign shares trade away from a stale net asset value, creation or redemption being suspended, and in the extreme a flash crash where the ETF price dislocates for minutes. The most serious of these is the primary market failing while the secondary market keeps trading, because that is exactly when the arbitrage that defines the product stops working.

End of life. Funds that do not gather assets are closed and liquidated, or merged into another fund, and holders are cashed out at net asset value on a set date. Mentioning closure unprompted is a good signal, because it is the event that most surprises retail holders.

The wider delta one family

An ETF is one member of a family of instruments that all deliver the return of something with a delta of one, which is where the desk name comes from.

REAL INTERVIEW QUESTION (JP Morgan · Delta One)

What delta one products do you know?

WHAT THEY TEST / ANSWER

List them, then say what unites them, then say where the desk's money comes from, because that last part is what distinguishes someone who understands the seat.

The products: index and single-stock **futures**; **forwards**; **total return swaps** and equity swaps; **ETFs** and index certificates or trackers; **ADRs and GDRs**, which are a claim on a foreign share; and baskets or custom index products. Some desks also include certificate wrappers that carry no embedded optionality, such as participation certificates.

What unites them: a linear payoff, delta of one to the underlying and no optionality, so there is no gamma and no vega to speak of. The risk is not the direction, because that is hedged, it is everything embedded in the *difference* between the instrument and the underlying: financing, dividends, borrow, tax and index tracking.

Where the money comes from: financing spreads on swaps and certificates, securities lending, dividend and tax arbitrage, index arbitrage between a future or ETF and its basket, and the bid-offer on flow. Say it plainly: a delta one desk is a financing and inventory business wearing an equity coat, which is why it is judged on balance sheet and on the borrow book rather than on directional profit and loss. The listed cousin of these products is futures, and the non-linear family, where the delta stops being one, is structured products.

INTERVIEW TRAP

Four mistakes. (1) Judging an ETF's liquidity by its own traded volume rather than by its underlying basket, which is what an authorised participant can actually create against. (2) Comparing funds on the advertised fee rather than on tracking difference against the same index. (3) Calling synthetic replication risky and physical safe, when the real comparison is counterparty risk against tracking, tax and lending revenue. (4) Assuming an ETF price equals net asset value when the underlying market is shut: it is then the ETF that is discovering the price, and the published net asset value that is stale.

Where this goes next

What the index is made of is equities, the venue the shares trade on is exchanges, and the participants who keep the quote tight are market makers. The listed linear instrument on the same exposure is futures, the financing that makes the whole family work is repo markets, and the non-linear wrappers are structured products.

Chapter 6

Mathematics for Markets

6.1 Limits, Continuity, Derivatives, Integrals

trading · structuring core cross-asset

Calculus is in this book for a specific reason rather than for completeness. Every Greek is a derivative, every option price is an integral, and the places where a function fails to be smooth are exactly the places a desk loses money. The mathematics is not hard; knowing which object corresponds to which piece of a trading book is what an interview is checking.

Continuity, and why the exceptions are where the risk is

A function is **continuous** at a point if small changes in the input produce small changes in the output, and **differentiable** there if it also has a well-defined slope. Differentiability is the stronger condition: the absolute value function is continuous everywhere and has no derivative at zero.

KEY IDEA

That single example is the whole of option risk in miniature. A vanilla payoff $\max(S-K, 0)$ is continuous but not differentiable at the strike, so its delta jumps and its gamma is infinite there *at expiry*, which is why pinning at the strike on expiry day is difficult and why gamma is largest at the money.

A digital or barrier payoff is worse: it is *discontinuous*. As spot approaches the barrier the delta does not merely change quickly, it becomes unbounded, and no finite hedge exists. That is why barrier options are hedged with a spread of vanillas and a shifted barrier rather than with the underlying, and it is a statement about differentiability rather than about finance.

Derivatives are Greeks

The dictionary is worth stating explicitly once, because it converts a mathematics question into a trading one:

$$\Delta = \frac{\partial V}{\partial S}, \quad \Gamma = \frac{\partial^2 V}{\partial S^2}, \quad \mathcal{V} = \frac{\partial V}{\partial \sigma}, \quad \Theta = \frac{\partial V}{\partial t}, \quad \rho = \frac{\partial V}{\partial r}.$$

The Greeks are covered in their own chapter and the second-order expansion that turns them into a P&L identity belongs to Taylor expansions. What belongs here is the observation that every one of those is an ordinary partial derivative, so anyone who can differentiate can produce a Greek from a formula, and the chain rule is the only technique required.

The chain rule earns one worked remark, because it is where sign errors come from. Black-Scholes is written in terms of d_1 and d_2, which are themselves functions of S, σ and t, so differentiating the price means differentiating through them. The reason delta comes out as the clean $N(d_1)$, rather than as a mess of terms in $N'(d_1)$, is that those terms cancel exactly, and knowing that the cancellation happens is worth more in an interview than being able to grind it out.

An option price is an integral

This is the sentence that makes the second half of calculus matter. A derivative's price is a discounted expectation, and an expectation of a function of a continuous random variable is an integral:

$$V \;=\; e^{-rT}\,\mathbb{E}[\,\text{payoff}(S_T)\,] \;=\; e^{-rT}\int_{-\infty}^{\infty} \text{payoff}(x)\,f(x)\,dx.$$

Everything a pricing library does is either evaluate that integral in closed form, approximate it on a grid, or sample from it. Black-Scholes is the case where the integral has an answer; Monte Carlo is the case where it does not.

REAL INTERVIEW QUESTION (Natixis · EQD)

Compute the Gaussian integral.

WHAT THEY TEST / ANSWER

State the result, then give the trick, because the trick is the entire question.

$$I \;=\; \int_{-\infty}^{\infty} e^{-x^2}\,dx \;=\; \sqrt{\pi}.$$

Why it is not straightforward. The function e^{-x^2} has no elementary antiderivative, so no amount of integration by parts or substitution in one dimension will produce it. Saying that first shows you know why the standard toolkit fails here.

The trick: square it and go to polar coordinates.

$$I^2 \;=\; \int_{-\infty}^{\infty} e^{-x^2} dx \int_{-\infty}^{\infty} e^{-y^2} dy \;=\; \iint_{\mathbb{R}^2} e^{-(x^2+y^2)}\,dx\,dy.$$

Substituting $x = r\cos\theta$, $y = r\sin\theta$, so that $x^2 + y^2 = r^2$ and the area element becomes $r\,dr\,d\theta$,

$$I^2 \;=\; \int_0^{2\pi}\int_0^{\infty} e^{-r^2}\,r\,dr\,d\theta \;=\; 2\pi\left[-\tfrac{1}{2}e^{-r^2}\right]_0^{\infty} \;=\; 2\pi\cdot\tfrac{1}{2} \;=\; \pi,$$

so $I = \sqrt{\pi}$, taking the positive root since the integrand is positive. The reason it works is that the extra factor of r from the Jacobian is exactly what makes the radial integral elementary, and pointing that out is the difference between reciting the trick and understanding it.

Then connect it to finance, which is what an EQD desk is asking for. Substituting $x = u/\sqrt{2}$ gives

$$\int_{-\infty}^{\infty} e^{-u^2/2}\,du \;=\; \sqrt{2\pi},$$

which is precisely the normalisation constant in the standard normal density $\frac{1}{\sqrt{2\pi}}e^{-u^2/2}$. So this integral is the reason $\sqrt{2\pi}$ appears in every option pricing formula you will ever write, and $N(d_1)$ and $N(d_2)$ in Black-Scholes are integrals of exactly this object with finite limits, which is why they have no closed form and are evaluated numerically.

When there is no closed form

REAL INTERVIEW QUESTION (Société Générale · desk not recorded)

Write an algorithm that computes the integral of a function, with the bounds given as input.

WHAT THEY TEST / ANSWER

Give the simplest correct method, then the better one, then say when neither applies. The progression is the answer.

The structure of all of them is the same: partition $[a, b]$ into n intervals of width $h = (b - a)/n$ and sum an approximation of the area on each. The methods differ only in what shape they fit.

Rectangles fit a constant, using the value at one end. Error is $O(h)$, so halving the step halves the error. Simple, and rarely good enough.

Trapezoids fit a straight line, averaging the two endpoints. Error is $O(h^2)$. This is the one to write if asked for code without further qualification.

Simpson's rule fits a parabola through each pair of intervals, weighting the pattern $1, 4, 2, 4, \ldots, 4, 1$ and scaling by $h/3$. Error is $O(h^4)$, which is a very large improvement for the same number of function evaluations, and it requires n to be even.

Show the difference with a number, because the orders are abstract until you do. Take $\int_0^1 e^{-x^2} dx$, whose true value is 0.7468241, with only ten intervals:

Method	Estimate	Error
Rectangles	0.7778168	3.1×10^{-2}
Trapezoids	0.7462108	6.1×10^{-4}
Simpson	0.7468249	8.2×10^{-7}

Same ten function calls, and Simpson is four orders of magnitude more accurate than rectangles. That is the case for knowing the convergence order rather than just the method.

Then the practical points, which is where a desk-relevant answer separates itself. Handle the case $b < a$ by returning the negative. Singularities and infinite limits need a substitution rather than a finer grid, since no partition fixes an unbounded integrand. And crucially: all of this is one-dimensional. Quadrature cost grows exponentially with dimension, so above roughly three or four variables the answer is Monte Carlo, whose error is $O(1/\sqrt{N})$ regardless of dimension. That crossover is the real content of the question for anyone pricing a multi-asset product, and pricing libraries covers how a library routes between them.

Three techniques worth having

Integration by parts, $\int u\, dv = uv - \int v\, du$, which is how most expectations involving a density get turned into something tractable.

Differentiating under the integral sign. If $V = \int g(x, \theta)\, dx$ then $\partial V/\partial\theta = \int \partial g/\partial\theta\, dx$ under mild conditions. This is not an exotic trick; it is the justification for computing a Greek by differentiating the payoff inside the expectation rather than bumping the price, which is the *pathwise* method and the ancestor of adjoint differentiation.

Symmetry. The integral of an odd function over a symmetric interval is zero. A surprising number of terms in derivations vanish for this reason alone, and spotting it saves more time under pressure than any technique.

INTERVIEW TRAP

1. **Confusing continuity with differentiability.** A payoff can be continuous and have no delta at a point, which is exactly what happens at a strike at expiry.
2. **Trying to integrate e^{-x^2} directly.** It has no elementary antiderivative. Square it and go to polar coordinates, or recognise it as the normal density and use its known normalisation.
3. **Refining the grid at a singularity.** No partition fixes an unbounded integrand; change variables instead.
4. **Using quadrature in high dimensions.** Cost grows exponentially with the number of variables while Monte Carlo does not, and the crossover arrives at three or four.
5. **Forgetting Simpson needs an even number of intervals.** It is the most common bug in the code version of this question.
6. **Treating $\sqrt{2\pi}$ as a magic constant.** It is the Gaussian integral, and being able to say where it comes from is a cheap way to sound like you understand the formula rather than having memorised it.

VISUAL / CODE TO BUILD: Interactive

A quadrature panel. The reader picks a function and the number of intervals, and the panel draws the fitted shapes over the true curve, rectangles as steps, trapezoids as chords, Simpson as parabolic arcs, with the error printed and a log-log plot of error against n beneath. The three lines on that plot have visibly different slopes, one, two and four, which makes the convergence order something the reader reads off a graph rather than takes on trust.

A second panel makes the dimensionality argument concrete: cost to reach a fixed accuracy plotted against dimension, with a quadrature curve rising exponentially and a flat Monte Carlo line crossing it. Dragging the accuracy target moves the crossing point, and it usually sits lower than readers expect.

A third animates the polar-coordinate trick: the square of the Gaussian integral drawn as a surface over the plane, then re-sliced into annuli, with the factor of r from the Jacobian shown as the growing circumference of each ring. Seeing where the extra r comes from is what turns the trick from something remembered into something understood.

6.2 Taylor expansions

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Almost every risk number on a trading floor is a Taylor coefficient. Delta and gamma, duration and convexity, vega and volga are not separate inventions; they are the first and second terms of the same expansion applied to different functions. Understanding the expansion itself, and in particular understanding its *error*, tells you when those numbers can be trusted and when they quietly stop working.

INTUITION

A Taylor expansion replaces a complicated function near a point with the simplest function that agrees with it there: same value, same slope, same curvature, and so on. Each extra term buys accuracy over a wider neighbourhood. The whole practical question is how wide.

The statement, and the part people skip

For a function smooth enough near a,

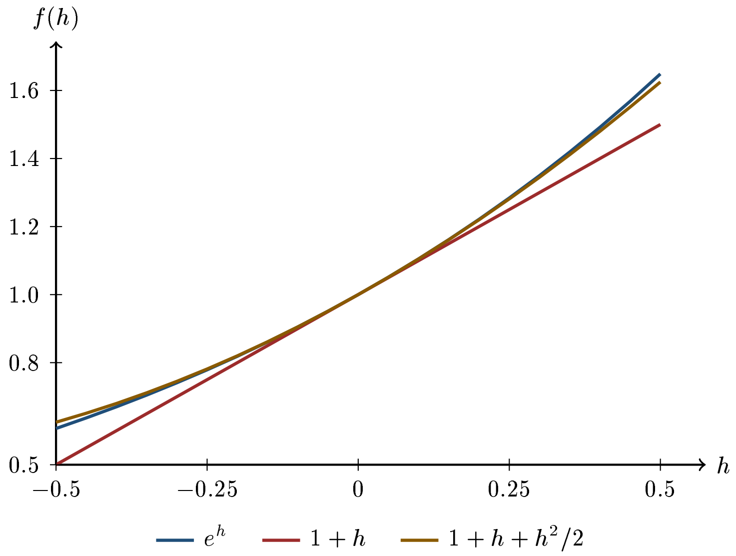
$$f(a+h) \;=\; f(a) + f'(a)\,h + \tfrac{1}{2}f''(a)\,h^2 + \tfrac{1}{6}f'''(a)\,h^3 + \cdots + R_n(h).$$

The part usually skipped is R_n, the remainder, and it is the only part that tells you anything about accuracy. In Lagrange form, truncating after the term in h^n leaves

$$R_n(h) = \frac{f^{(n+1)}(\xi)}{(n+1)!}\,h^{n+1}$$

for some ξ between a and $a+h$.

Two consequences do all the work in practice. First, **the error of an nth-order expansion scales as h^{n+1}**: halve the step and a first-order error falls by four, a second-order error by eight. Second, the size of the error depends on the next derivative, so a function with a large or badly behaved higher derivative is one where the expansion fails sooner than the step size alone suggests.



all three are indistinguishable near zero and separate steadily away from it

At $h = 0.1$ the linear approximation of e^h is wrong by 0.00517 and the quadratic by 0.00017. At $h = 0.5$ the same two are wrong by 0.14872 and 0.02372. The step grew by a factor of five; the linear error grew by 28.8 and the quadratic by 138.8, against the h^2 and h^3 predictions of 25 and 125. The scaling law is right, and here the errors run slightly *above* it, because every derivative of e^h is positive so the terms you dropped all push the same way. That is not a universal direction: where the derivatives alternate in sign, as they do for a discounted cash flow, the dropped terms partly cancel and the error comes in under the law, which is what the bond example below does.

Every risk number you will meet is a coefficient in this sum

Write the value of a position as a function of what it depends on and expand. For an option in spot alone,

$$\Delta V \;\approx\; \underbrace{\delta}_{f'}\,\Delta S \;+\; \tfrac{1}{2}\underbrace{\gamma}_{f''}\,(\Delta S)^2,$$

and for a bond in yield,

$$\frac{\Delta P}{P} \;\approx\; -D\,\Delta y \;+\; \tfrac{1}{2}C\,(\Delta y)^2.$$

These are the same equation. The Greeks as partial derivatives are set out in first and second order Greeks, duration and convexity and how to compute them in duration and convexity, and the decomposition of a day's profit into these terms in Greeks and P&L explain. What belongs here is the error.

Real positions depend on several variables at once, and the multivariate expansion has cross terms that the single-variable version hides:

$$\Delta V \approx \sum_i \frac{\partial V}{\partial x_i}\Delta x_i + \frac{1}{2}\sum_i\sum_j \frac{\partial^2 V}{\partial x_i \partial x_j}\Delta x_i \Delta x_j.$$

The diagonal terms are the familiar second-order Greeks. The off-diagonal ones, the cross sensitivity of spot to volatility for instance, are the reason a book can be flat on every individual sensitivity and still lose money when two things move together.

WORKED EXAMPLE

How fast the error actually grows. Take a ten-year bond with a 5% annual coupon priced at par, so its modified duration is 7.7217 and its convexity is 74.998 on the conventions in duration and convexity. Reprice it exactly and compare.

Yield move	Exact	First order	Error	Second order	Error
+25 bp	−1.907%	−1.930%	0.023	−1.907%	−0.0002
+100 bp	−7.360%	−7.722%	0.362	−7.347%	−0.0133
+300 bp	−20.130%	−23.165%	3.035	−19.790%	−0.3399

Now check the scaling rather than admiring the fit. From 25 to 100 basis points the step quadruples, and the first-order error grows by a factor of **15.57** against the h^2 prediction of 16, while the second-order error grows by **62.58** against the h^3 prediction of 64. From 100 to 300 basis points the step triples, and the two errors grow by **8.39** and **25.48** against predictions of 9 and 27.

The theory is confirmed to about two significant figures across a twelvefold range of step sizes, and it is confirmed in the safe direction: the observed errors are slightly smaller than the scaling law implies.

The practical reading. For a 25 basis point move, duration alone is wrong by 2 basis points of price and nobody cares. For a 300 basis point move, duration alone is wrong by *three percent of the position* and convexity is doing more work than most of the rest of the risk report. The rule that follows is not "use second order", it is that a second-order number has a range of validity which shrinks as the cube root of the accuracy you demand.

Where the expansion simply does not work

A Taylor expansion needs derivatives to exist. Several things that trade every day do not oblige.

- **A payoff with a kink**, such as a vanilla option at expiry, has no second derivative at the strike. Away from expiry the smoothing of the option price makes the expansion fine; at expiry it fails exactly where the money is.
- **A payoff with a jump**, such as a digital or a barrier, is worse: the first derivative blows up near the trigger, so delta is not merely inaccurate but unbounded. No number of extra terms rescues this, which is why those products are handled by replication rather than by expansion in barriers, digitals and Asians.
- **Large moves**, where the neighbourhood assumption is simply false. This is why a risk framework built on Greeks needs a separate scenario or stress process rather than a bigger ΔS.
- **Functions whose series does not converge to them.** Smoothness is not enough in general; the expansion can be perfectly well defined and still not equal the function. It is a mathematical curiosity rather than a desk problem, but it is why "infinitely differentiable" is not the same claim as "equal to its Taylor series".

The second-order term has a life of its own

There is a use of the expansion that has nothing to do with approximating a single move, and it is the deepest thing in this section. Expand f around the mean of a random variable X and take expectations:

$$\mathbb{E}[f(X)] \;\approx\; f(\mathbb{E}[X]) + \tfrac{1}{2}f''(\mathbb{E}[X])\,\mathrm{Var}(X).$$

The first-order term vanishes, because the expected deviation is zero. The second-order term does not, because the squared deviation is always positive. So **for a convex function the expected value exceeds the value at the expected point**, which is Jensen's inequality, and the gap is proportional to the variance.

KEY IDEA

This is where volatility enters a price at all. A linear instrument has $f'' = 0$ and its value depends only on where the underlying is expected to be. A convex one has $f'' > 0$ and its value depends on how much the underlying is expected to *move*, which is why gamma is worth paying for, why convexity is worth paying for, and why two positions with identical delta can have entirely different values. The correction $\frac{1}{2}f''\sigma^2$ is the same object in all of them.

The approximation is itself an approximation, and it is worth knowing which way it errs. Take $f = e^x$ and let X be $\pm s$ with equal probability, so the mean is zero and the variance is s^2:

s	True gap $\cosh(s) - 1$	$\frac{1}{2}f''\mathrm{Var} = s^2/2$	Shortfall
0.2	0.020067	0.020000	0.000067
0.5	0.127626	0.125000	0.002626
1.0	0.543081	0.500000	0.043081

The correction is excellent for small moves and *understates* the true gap as they grow, because the fourth-order term is also positive. Applied to a book, that means a gamma-based estimate of the value of convexity is conservative for a long position, which is the comfortable direction, and optimistic for a short one, which is not.

Interview angle

No question in the database says the words *Taylor expansion*, and that is not an oversight in the database. Interviewers test the applications: the Greeks, duration and convexity, the approximation of an at-the-money call, the decomposition of a day's P&L. Every one of those questions is this section wearing a costume, and most are asked and answered in the sections that own them. The one below is the expansion with the costume half off, because it names both orders and asks for the correction explicitly.

REAL INTERVIEW QUESTION (Citi · EQD)

First and second order approximation of a bond price, with convexity.

WHAT THEY TEST / ANSWER

What is being tested is whether the two orders are a formula you memorised or a truncated series you understand, and the tell is whether you volunteer the error without being asked.

Write it as an expansion, not as two rules. The bond price is a function of yield, so expand it around the current yield and divide by the price:

$$\frac{\Delta P}{P} = \underbrace{-D\,\Delta y}_{\text{first order}} + \underbrace{\tfrac{1}{2}C\,(\Delta y)^2}_{\text{second order}} + O\big((\Delta y)^3\big),$$

where D is modified duration and C is convexity. Say what each term is: the first is the slope, and it is symmetric, so on its own it claims a rise and a fall of the same size cost and earn the same amount. The second is the curvature, it is positive for an ordinary bond, and it therefore adds value on both sides, which is why the price falls by *less* than duration predicts when yields rise and rises by *more* when they fall. How D and C are computed is duration and convexity; what belongs to the expansion is the rest of this answer.

Put numbers on it, using the ten-year 5% par bond above, where $D = 7.72$ and $C = 75.0$. For a 100 basis point rise, first order gives -7.72% and the convexity term adds $\frac{1}{2}\times 75.0\times 0.01^2 = +0.37\%$, so second order gives -7.35% against an exact -7.36%. Duration alone was out by 36 basis points of price; adding one term cut that to about 1.

Then the part that separates candidates, which is the range of validity. The residual after the second-order term scales as the *cube* of the move, so the approximation does not degrade gently, it degrades eightfold every time the move doubles. At 25 basis points the second-order error is two hundredths of a basis point and irrelevant; at 300 basis points it is a third of a percent of the position and no longer is. State the number and then state where you would stop trusting it, because an approximation quoted without its range is the half of the answer that is not being marked.

If pushed further, two additions land well. Convexity is not free: it is priced, so a bond that offers more of it for the same duration yields less, which is the same trade-off as paying for gamma. And for a callable or mortgage-backed bond convexity turns *negative* in the region where the option matters, so the second-order term flips sign and the comfortable asymmetry above works against you.

INTERVIEW TRAP

Four errors. (1) Quoting a second-order approximation without any sense of its range of validity, which is the entire content of the remainder term. (2) Believing more terms always help, when for a non-smooth payoff no number of terms works. (3) Forgetting the cross terms in the multivariate expansion, which is how a book flat on every individual Greek still loses on a joint move. (4) Writing $\mathbb{E}[f(X)] = f(\mathbb{E}[X])$, which is false for anything convex and is the single most expensive line of reasoning in this book.

Where this goes next

The Greeks as the partial derivatives of an option value are first and second order Greeks, the bond version of the same expansion is duration and convexity, and what the second-order term is worth to a trading book is gamma and convexity. Using the expansion to explain a day's profit is Greeks and P&L explain, and the products where it breaks down entirely are barriers, digitals and Asians.

6.3 Linear algebra fundamentals

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Almost every quantity a desk reports is a linear algebra object wearing a different name. A risk report is a matrix of sensitivities. A portfolio's variance is a quadratic form. A correlation surface is a matrix that has to satisfy a condition most people never check. A curve bootstrap is a linear system. None of this requires advanced mathematics, but it does require the right mental picture, and the right picture is not the one taught first.

This section gives that picture and stops there. Bases and spans are vector spaces, determinants are matrices and determinants, and eigenvalues, which are what actually make a correlation matrix well behaved, are eigenvalues and eigenvectors.

A matrix is a map, not a table

The table is how a matrix is stored. What it *is* is a rule that takes a vector in and gives a vector out, and does so linearly: scaling the input scales the output, and adding two inputs adds their outputs.

> **KEY IDEA**
>
> Read Ax as "what A does to x", never as "a grid times a list". Every useful fact follows from that reading. Matrix multiplication AB is doing B first and then A, which is why the inner dimensions must agree and why $AB \neq BA$ in general: putting on your socks and then your shoes is not the same as the reverse. The identity matrix does nothing. An inverse undoes.

The reason this matters on a desk is that the objects really are maps.

- A **sensitivity matrix** maps market moves to profit and loss: give it a vector of risk factor changes and it returns the change in value of each position.
- A **factor model** maps a small vector of factor returns to a large vector of asset returns.
- A **bucketing or mapping matrix** takes cash flows at their true dates and assigns them to the standard tenors a risk system uses.
- A **covariance matrix** is the odd one out, and it is the subject of the rest of this section.

The one expression worth memorising

If a portfolio holds weights $w = (w_1, \ldots, w_n)$ in assets whose returns have covariance matrix Σ, then the portfolio variance is

$$\sigma_p^2 \;=\; w^\top \Sigma\, w \;=\; \sum_i \sum_j w_i\, \Sigma_{ij}\, w_j .$$

This is a **quadratic form**, and it is where linear algebra earns its place in finance. Unpacked, it says the portfolio's variance is the sum of every asset's own variance weighted by the square of its holding, plus a covariance term for every pair, counted twice.

WORKED EXAMPLE

Two assets: $\sigma_1 = 20\%$, $\sigma_2 = 30\%$, correlation 0.5. The covariance is $0.5 \times 0.20 \times 0.30 = 0.03$, so

$$\Sigma = \begin{pmatrix} 0.04 & 0.03 \\ 0.03 & 0.09 \end{pmatrix}.$$

For an equally weighted portfolio, $w = (0.5, 0.5)$:

$$w^\top \Sigma w = 0.25(0.04) + 2(0.25)(0.03) + 0.25(0.09) = 0.01 + 0.015 + 0.0225 = 0.0475,$$

so $\sigma_p = \sqrt{0.0475} = 21.8\%$.

Compare that with the weighted average of the two volatilities, $0.5(20\%) + 0.5(30\%) = 25\%$. The 3.2 points of difference is the diversification, and it comes entirely from the correlation being below one. **Volatilities do not add, and the quadratic form is the statement of how they do combine.**

The condition nobody checks

A covariance matrix is not an arbitrary symmetric grid of numbers. Because $w^\top \Sigma w$ *is* a variance, and a variance cannot be negative, Σ must satisfy

$$w^\top \Sigma\, w \;\geq\; 0 \qquad \text{for every } w.$$

A matrix with this property is called **positive semidefinite**. It is not a technical nicety: a matrix that fails it is claiming that some portfolio has negative variance, which is not a modelling approximation but an impossibility.

INTERVIEW TRAP

The usual way to produce an invalid matrix is to build it **pairwise**. Estimate each correlation separately, perhaps from different samples or by asking different traders for their view, assemble the grid, and it will very often not be positive semidefinite. Correlations are not free to be chosen independently: fixing some of them constrains the rest. A pricing library handed such a matrix will either fail loudly, which is the good case, or silently produce prices that admit arbitrage.

WORKED EXAMPLE

Take three assets and claim that asset 1 is 0.9 correlated with both 2 and 3, while 2 and 3 are -0.9 correlated with each other:

$$C = \begin{pmatrix} 1 & 0.9 & 0.9 \\ 0.9 & 1 & -0.9 \\ 0.9 & -0.9 & 1 \end{pmatrix}.$$

Every entry is individually a legal correlation. Now take $w = (1, -1, -1)$ and compute $w^\top C w$. There are three diagonal terms, each equal to 1, and six off-diagonal terms, each equal to -0.9 once the signs of w are applied:

$$w^\top C w = 3 - 6(0.9) = 3 - 5.4 = -2.4.$$

A negative variance. The matrix is therefore impossible, and one line of arithmetic found it without any machinery.

The underlying constraint is that with $\rho_{12} = \rho_{13} = 0.9$, the third correlation cannot fall below $\rho_{12}\rho_{13} - \sqrt{(1-\rho_{12}^2)(1-\rho_{13}^2)} = 0.81 - 0.19 = 0.62$. If two assets both track a third closely, they must track each other. The general test, and what to do about a matrix that fails it, is in eigenvalues and eigenvectors and matrix decompositions.

Solving a linear system

The other half of what a desk uses linear algebra for is $Ax = b$: given what a map does and the output you want, find the input. Bootstrapping a curve from observed instrument prices is this, and so is finding the hedge ratios that neutralise a set of exposures.

There are exactly three possible outcomes, and knowing which one you are in is more useful than any solution technique.

- **Exactly one solution.** The map is invertible: it loses no information, so the output determines the input. This is the case a well-posed calibration is designed to be in, with as many independent instruments as unknowns.
- **No solution.** The target b is not something A can produce. In practice this is over-determination: more market quotes than free parameters, and they are mutually inconsistent because of bid-offer and stale data. You do not solve it, you fit it, minimising the error.
- **Infinitely many solutions.** The map collapses distinct inputs to the same output, so the data does not pin down the answer. This is under-determination: more hedge instruments than constraints, or a model with more parameters than the market prices. You must add a criterion, typically smoothness or minimum cost, to choose among them.

INTUITION

The two failure cases are the normal ones in practice, and the mistake is treating them as errors. "No solution" means your data disagrees with itself and you should be fitting, not solving. "Infinitely many" means your problem is genuinely under-specified and you are about to make an arbitrary choice, so you should make it deliberately. A calibration that runs without complaint has usually had one of these decisions made for it silently.

Where this goes next

The structure underneath all of this is vector spaces, the invertibility test is matrices and determinants, and the tools that make a covariance matrix usable, testing it, repairing it and factoring it for simulation, are eigenvalues and eigenvectors and matrix decompositions. Measuring size and angle is norms and inner products, the desk application is aggregation of Greeks, and doing any of it in code is NumPy and pandas.

6.4 Vector spaces

sales · trading · structuring | core | cross-asset

Most people meet vectors as arrows and never find out why anyone cared. The reason to care, in finance, is that a **payoff is a vector** and a **portfolio is a recipe for adding vectors together**. Once you see that, a surprising amount of derivatives theory turns out to be linear algebra wearing a costume: replication is asking whether one vector lies in the span of some others, a redundant security is a linear dependence, market completeness is a statement about dimension, and an arbitrage is a linear dependence whose prices do not agree. This section builds the machinery from nothing and then hands you that dictionary.

From arrows to axioms

A **vector space** over the real numbers is a set V whose elements can be added together and multiplied by numbers (called **scalars**), such that the operations behave the way arithmetic leads you to expect. Precisely, addition is associative and commutative, there is a zero element with $v + 0 = v$, every v has a negative $-v$ with $v + (-v) = 0$, and scalar multiplication satisfies $1v = v$, $\alpha(\beta v) = (\alpha\beta)v$, $\alpha(u + v) = \alpha u + \alpha v$ and $(\alpha + \beta)v = \alpha v + \beta v$.

That list looks like bookkeeping, and the point of it is generality: *anything* satisfying it gets the whole theory for free. Examples that matter here:

- $\mathbb{R}^n$, lists of n numbers. A portfolio of n assets is a vector of holdings.
- **Payoff space.** If the world can end in n distinguishable states, a claim is described by what it pays in each, which is again a point of $\mathbb{R}^n$.
- **Functions.** The set of all functions from the real line to itself is a vector space, since you can add two functions and scale one. An option payoff seen as a function of the final spot lives here, and so does a whole yield curve seen as a function of maturity.

> INTUITION
>
> A vector space is a **world in which mixing is allowed**. The only two things you may do are combine ingredients and change the amount of each, so everything reachable in that world is some recipe of the form $\alpha_1 v_1 + \cdots + \alpha_k v_k$. Almost every question in this section is really the same question asked from different angles: given the ingredients you have, what can you make, what is the shortest ingredient list that still makes everything, and did you bring anything you did not need?

Span, independence, basis, dimension

Let $v_1, \ldots, v_k$ be vectors in V.

- A **linear combination** is any $\alpha_1 v_1 + \cdots + \alpha_k v_k$ with real coefficients. The set of all of them is the **span**, written $\mathrm{span}(v_1, \ldots, v_k)$, and it is always a **subspace**: a subset that is itself a vector space, meaning it contains zero and is closed under addition and scaling.
- The vectors are **linearly independent** if the only way to write zero as a combination of them is with every coefficient zero. Equivalently, and this is the useful phrasing, *no one of them can be built out of the others.* If they are dependent, at least one is redundant and can be deleted without shrinking the span.
- A **basis** is a spanning set that is also independent: enough ingredients to make everything, with nothing wasted.
- Every basis of a given space has the same number of elements, and that number is the **dimension**. This is the theorem that makes counting arguments valid.

Once a basis $e_1, \ldots, e_n$ is fixed, every vector has a *unique* list of **coordinates**, the coefficients in $v = \sum_i \alpha_i e_i$. Uniqueness is exactly what independence buys you, and it is why a replicating portfolio is unique when it exists.

KEY IDEA

The finance dictionary. Fix the states of the world, so payoffs live in $\mathbb{R}^n$.

Linear algebra	Finance
Vector	A claim, described by its payoff in each state
Linear combination	A portfolio
Span of the traded assets	Everything that can be replicated
Linear dependence	A redundant security
Dependence with mismatched prices	An arbitrage
Span is the whole space	A **complete** market
Coordinates in a basis	The replicating portfolio, and it is unique

So "can I hedge this" and "is this vector in that subspace" are the same question, and the law of one price is the statement that pricing is a *linear* function on the span.

Worked example: replication is solving for coordinates

WORKED EXAMPLE

Two states, two assets. Tomorrow the world is either up or down. A bond pays 1 in both states, so it is the vector $(1, 1)$. A stock pays 2 up and 0.5 down, so it is $(2, 0.5)$. Both trade at a price of 1 today.

Are they a basis? They are independent unless one is a multiple of the other, and $(2, 0.5)$ is not a multiple of $(1, 1)$, so the two vectors are independent. Two independent vectors in a two-dimensional space are a basis, therefore they span $\mathbb{R}^2$ and **the market is complete**: every payoff can be replicated.

Replicate a call struck at 1. Its payoff is $(\max(2-1, 0), \max(0.5-1, 0)) = (1, 0)$. We need coefficients a on the bond and b on the stock with

$$a\,(1, 1) + b\,(2,\ 0.5)\ =\ (1,\ 0),$$

which is the pair of equations $a + 2b = 1$ and $a + 0.5b = 0$. Subtracting gives $1.5b = 1$, so $b = 2/3$, and then $a = -0.5 \times 2/3 = -1/3$. Check both equations: $-1/3 + 4/3 = 1$ and $-1/3 + 1/3 = 0$. So the replicating portfolio is **two thirds of a share, funded by borrowing one third of a bond**.

Price it. Because pricing is linear on the span, the call must cost what the portfolio costs: $a \times 1 + b \times 1 = -1/3 + 2/3 = 1/3$.

Check by another route. The bond pays 1 and costs 1, so the interest rate is zero. The risk-neutral probability q of the up state solves $2q + 0.5(1-q) = 1$, giving $1.5q = 0.5$ and $q = 1/3$, so the call is worth $q \times 1 + (1-q) \times 0 = 1/3$. The two methods agree, as they must, because risk-neutral pricing is what linearity on a complete span looks like from the other side.

WORKED EXAMPLE

Three states, two assets: incompleteness. Now let the world have three states and keep two traded assets. Their span is at most a two-dimensional subspace of $\mathbb{R}^3$, which is a plane inside a room. Any payoff off that plane simply cannot be replicated, so the linear system has no solution and no-arbitrage alone does not pin down a price: it only produces a *range* of prices consistent with the traded assets. This is **market incompleteness**, and it is a dimension count rather than a subtlety. Adding a third independent asset, an option say, restores a basis and with it a unique price, which is exactly why liquid options are said to complete the market.

Linear maps and the rank-nullity theorem

A **linear map** $T : V \to W$ satisfies $T(\alpha u + \beta v) = \alpha T(u) + \beta T(v)$. Two subspaces come with it: the **kernel**, the vectors sent to zero, and the **image**, everything actually reached. The rank-nullity theorem says

$$\dim(V)\ =\ \underbrace{\dim(\text{image})}_{\text{rank}}\ +\ \underbrace{\dim(\text{kernel})}_{\text{nullity}}.$$

Read it on a hedging problem, where T maps a vector of positions to the vector of risk exposures they generate. The **rank** is how many independent risks you can actually control with the instruments you have, so if the rank is less than the number of risk factors, some combination of risks is unhedgeable no matter how you trade. The **kernel** is the set of position combinations that generate no exposure at all, which are precisely the trades that leave your risk unchanged, and a non-trivial kernel means your hedge is not unique. Both facts are dimension counts you can do before any pricing.

Change of basis, and why it is the whole point of factor models

The same vector has different coordinates in different bases, and choosing the right basis is usually the entire content of a piece of financial modelling. A yield curve observed at ten maturities is a vector in $\mathbb{R}^{10}$, and in the obvious basis its coordinates are the ten yields, which move together and are hard to think about. In a different basis, whose first three elements look like a parallel shift, a slope and a curvature, almost all of the observed variation sits in the first three coordinates and the rest is close to noise.

Nothing about the curve changed. The information is identical, and only the description became useful, which is what a change of basis does. Building that particular basis from the data is principal component analysis, and it needs eigenvectors and an inner product, developed in eigenvalues and eigenvectors and norms and inner products.

INTERVIEW TRAP

Four traps. First, **confusing the span with the whole space**: three vectors in $\mathbb{R}^3$ span it only if they are independent, and three dependent ones span a plane or a line, which in market terms is an incomplete market that superficially looks complete because you counted instruments rather than independent ones. Second, **assuming a spanning set is a basis**: extra vectors do not break the span, but they destroy the uniqueness of the coordinates, so the replicating portfolio stops being unique. Third, **counting instruments instead of independent risks** when sizing a hedge: two instruments that load on the same single factor hedge one risk, not two. Fourth, **forgetting that all of this is linear**: the moment a payoff depends on the path, or a constraint like a short-selling ban appears, the set of achievable outcomes stops being a subspace and these tools stop applying directly.

REAL INTERVIEW QUESTION (UBS · Structured Products)

If you cannot buy a stock, how do you replicate it?

WHAT THEY TEST / ANSWER

The question is a span question, so answer it as one and then give the practical routes. The principle: you need a set of tradeable instruments whose payoffs, combined linearly, reproduce the stock's payoff in every state, and any such combination will do because the replicating portfolio is unique only up to the independence of what you are using. Then the routes, from cleanest to most approximate. **Put-call parity**: long a call and short a put at the same strike and maturity gives you the stock's payoff plus a fixed cash offset, so combined with a bond it is a synthetic long, developed in replication strategies. **A forward or future** on the name, which is the same exposure with the financing separated out. **A total return swap**, where a bank holds the shares and passes you the performance, which is how this is usually done when the constraint is regulatory or operational rather than about price. **A basket or proxy**: if the name itself is untouchable, the closest achievable vector is a combination of correlated names or an index with the sector loadings adjusted, which is a projection rather than a replication and therefore leaves a residual. Then finish with the two questions a desk asks immediately, because they are what the interviewer is testing: *why* you cannot buy it, since a restricted list, a borrow constraint, a foreign-ownership limit and a liquidity problem each rule out different substitutes, and what **basis risk** remains, since every route above except the first two leaves a residual that has to be sized, monitored and paid for.

The synthetic forward, long a call and short a put at the same strike and maturity, is the cleanest example in finance of one vector written in another basis, and it is built in replication strategies with the practical caveats that go with it. The linear-algebra content of it belongs here and is worth stating on its own: the three instruments are **linearly dependent**, so given any two of a call, a put and a forward the third is redundant and must be priced by the other two. That is what put-call parity, owned by no-arbitrage pricing, says in the language of this section. It also holds state by state, so it needs no model and no volatility input, which is why a deviation from it is a trade rather than a view.

VISUAL / CODE TO BUILD: Interactive

A two-panel span explorer, because this material is visual and almost never taught visually. Panel one works in three states so the payoff space is a room the reader can rotate: traded assets appear as arrows, their span renders as a line or a plane, and a target payoff is shown as a point that is either inside the span (with its coordinates displayed, so the replicating portfolio appears) or outside it (with the shortest distance shown, which previews the projection idea from norms and inner products). Adding a third independent asset makes the plane fill the room, so completeness is witnessed rather than defined, and making the third asset dependent leaves the plane unchanged, which is the counting trap in the pitfall box made visible. Panel two takes real yield curve data and lets the reader switch between the raw-tenor basis and a level-slope-curvature basis, showing the coordinates in each and the variance captured by the first k of them, so the value of a change of basis is a number rather than a claim.

Where this goes next

The computational machinery for deciding independence, rank and solving these systems is in matrices and determinants, with the general setting in linear algebra fundamentals. Length, angle and projection, which is what you need when a payoff is *not* in the span and you want the closest thing that is, are in norms and inner products. Choosing a basis from data leads to eigenvalues and eigenvectors and diagonalization. On the finance side, the ideas here are the mathematical content of no-arbitrage pricing and replication strategies.

6.5 Matrices and Determinants

trading · structuring core cross-asset

A matrix is two things at once and confusing them is the source of most difficulty: it is a *table of numbers*, which is how a covariance matrix or a set of factor loadings is stored, and it is a *linear map*, which is what multiplying by it does. Almost every matrix question in finance is really a question about the second reading applied to an object introduced as the first.

The formula that makes this a finance topic

Portfolio variance is a matrix expression and nothing else:

$$\sigma_p^2 \;=\; w^\top \Sigma\, w,$$

with w the vector of weights and Σ the covariance matrix. Every risk number a desk quotes is built from that quadratic form, so the properties of Σ are not a mathematical aside, they decide whether the risk system produces a sensible answer.

Three consequences follow immediately and are worth stating.

A variance cannot be negative, so $w^\top \Sigma w \geq 0$ for every w. That is the definition of **positive semi-definiteness**, and it is a requirement on Σ rather than a property it happens to have.

Σ is symmetric, because the covariance of A with B equals that of B with A. Symmetry is what makes the whole eigenvalue machinery in eigenvalues well behaved.

And a correlation matrix is the same object rescaled to unit diagonal, which is why the two are used interchangeably in conversation and must not be in code.

What a determinant actually means

The determinant of a square matrix is the factor by which the linear map scales volume. That one sentence carries everything useful.

KEY IDEA

$\det(M) = 0$ means the map collapses space onto a lower dimension: it squashes a volume to zero. Equivalently the columns are linearly dependent, the matrix is **singular**, and it has no inverse.

In finance the reading is direct. A covariance matrix with zero determinant means one asset is an exact linear combination of the others, so it carries no independent risk: the portfolio has fewer genuine dimensions than holdings. A *near*-zero determinant means near-collinearity, which is much more common and much more dangerous, because the matrix is invertible in principle and the inverse is garbage in practice.

A **correlation matrix** is the covariance matrix rescaled to unit diagonal, $\rho_{ij} = \sigma_{ij}/(\sigma_i\sigma_j)$, and what it is and what it fails to capture belong to independence and correlation. What belongs here is the linear algebra: *which* symmetric matrices with unit diagonal are correlation matrices at all.

Not all of them. The matrix must be positive semi-definite, because otherwise some portfolio has negative variance, which is not a modelling inconvenience but an arbitrage in the risk system. The smallest example that bites is three assets all pairwise correlated at the same ρ:

$$\det \;=\; 1 - 3\rho^2 + 2\rho^3 \;=\; (1-\rho)^2(1+2\rho),$$

which is non-negative only for $\rho \geq -\frac{1}{2}$. So three assets **cannot** all be pairwise correlated at -0.6: the determinant is -0.512 and no such world exists. Correlations are not free parameters ranging independently over $[-1, 1]$; they live on a curved set, and the constraint binds at values people write down casually.

The practical consequence is that estimated correlation matrices are frequently *not* positive semi-definite, because different pairs are estimated over different windows or filled in by hand. The standard repair is to project onto the nearest valid matrix by flooring the negative eigenvalues at zero and rescaling the diagonal, which is an eigenvalue operation and belongs to eigenvalues. And estimating an $n \times n$ matrix requires $n(n-1)/2$ parameters from a sample that is usually far too short, which is why practitioners impose structure, a single factor, a block structure or shrinkage toward a constant correlation, rather than using the raw estimate.

Conditioning: why a valid matrix can still be useless

Singularity is a binary condition and is rarely the practical problem. The practical problem is being close to it.

WORKED EXAMPLE

Two assets at 99% correlation. The correlation matrix is

$$\begin{pmatrix} 1 & 0.99 \\ 0.99 & 1 \end{pmatrix}, \qquad \det = 1 - 0.99^2 = 0.0199.$$

It is positive definite, so it is a perfectly valid correlation matrix. Its eigenvalues are $1 + 0.99 = 1.99$ and $1 - 0.99 = 0.01$, so its **condition number** is

$$\frac{1.99}{0.01} \;=\; \mathbf{199}.$$

The inverse has entries of $1/0.0199 = 50.25$ and -49.75. So a mean variance optimiser, which multiplies by that inverse, will produce weights of the order of fifty times its inputs, in opposite directions on the two assets. A one percent error in an estimated correlation propagates into a weight error up to **199 times** larger, and correlations are estimated from short samples of noisy data.

That is the entire reason optimised portfolios come back with enormous offsetting long and short positions in similar assets, and why practitioners constrain, regularise or shrink rather than trusting the optimiser. The matrix is not wrong; it is ill-conditioned, and no amount of care in the arithmetic repairs an unstable input.

INTUITION

The operational rule that follows: **do not compute an inverse**. If you need $M^{-1}b$, solve the linear system $Mx = b$ instead, which is faster, more accurate and does not produce a large intermediate object whose entries are meaningless on their own. Forming an explicit inverse is almost always a sign that a numerical routine was written from the textbook definition rather than from the algorithm, and matrix decompositions covers how it is actually done.

The types worth recognising

- **Symmetric**, $M = M^\top$. Covariance and correlation matrices. Real eigenvalues and orthogonal eigenvectors, which is what makes them tractable.
- **Positive semi-definite**, $x^\top Mx \geq 0$ for all x. Required of any covariance matrix, and the precondition for a Cholesky factorisation, which is how correlated random numbers are generated for a simulation.
- **Orthogonal**, $M^\top M = I$. A rotation: it preserves lengths and angles and has determinant ± 1. Eigenvector matrices of symmetric matrices are orthogonal, which is why principal component analysis works.
- **Triangular**. The output of the decompositions, and the reason they are useful: a triangular system is solved by substitution in $O(n^2)$ rather than $O(n^3)$.
- **Sparse**. Most entries zero, common in large risk systems, and worth naming because the right algorithm for a sparse matrix is a different algorithm rather than the same one on more zeros.

INTERVIEW TRAP

1. **Assuming any symmetric matrix of correlations is valid.** It must be positive semi-definite, and the three-asset example shows the constraint is binding at values people write down casually.
2. **Using correlation and covariance interchangeably in code.** They differ by a diagonal rescaling, and the bug is silent because both are symmetric with plausible entries.
3. **Inverting a matrix.** Solve the system. An explicit inverse is slower, less accurate and rarely what you actually needed.
4. **Trusting an optimiser fed a near-singular matrix.** A condition number of 199 amplifies estimation error by up to 199 times, and that is what produces the enormous offsetting positions optimisers are famous for.
5. **Estimating a large matrix from a short sample.** An $n \times n$ correlation matrix has $n(n-1)/2$ free parameters (a covariance matrix has $n(n+1)/2$), and beyond a handful of assets the raw estimate is mostly noise. Impose structure.
6. **Reading a determinant as a size.** It is a volume scaling factor, and it is small when the map is nearly degenerate rather than when the entries are small.

VISUAL / CODE TO BUILD: Interactive

A determinant panel. A unit square is drawn alongside its image under a 2×2 matrix whose four entries the reader controls, with the area of the image printed and labelled as the determinant. Dragging the columns toward parallel collapses the parallelogram onto a line and the number to zero, which makes "singular" something seen rather than defined. A sign switch appears when the columns cross, which is the geometric meaning of a negative determinant.

A second panel is a correlation validity explorer for three assets. Three sliders set the pairwise correlations and the panel plots the point inside the region of valid combinations, shading it green when the determinant is non-negative and red outside. Setting all three equal and dragging them downward turns the point red exactly at $-\frac{1}{2}$, reproducing the section's bound; and moving them independently shows that the valid region is a curved body rather than a cube, which is the more general lesson.

A third takes the reader's correlation matrix, computes its condition number, and runs a mean variance optimisation twice, once on the estimate and once on the estimate perturbed by a small random error, showing the two sets of weights side by side. At low correlation the bars barely move; at 0.99 they invert. That is the argument for constraints, performed.

6.6 Eigenvalues and eigenvectors

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A matrix generally rotates and stretches whatever it acts on, which makes it hard to reason about. An eigenvector is a direction where it only stretches. Along that direction the map does nothing but multiply by a number, so a complicated object collapses into a scalar. The whole subject is the search for the directions in which a problem becomes one-dimensional.

Formally, $v \neq 0$ is an **eigenvector** of A with **eigenvalue** λ if

$$Av = \lambda v.$$

Finding them from the characteristic equation is matrices and determinants; the procedure for rewriting a whole matrix in terms of them is diagonalization. This section is about what they mean and what a desk does with them.

Why symmetric matrices are the good case

Every covariance and correlation matrix in finance is symmetric, and symmetric matrices have properties that no general matrix has.

- The eigenvalues are all **real**. No complex numbers appear, which is not guaranteed in general.
- The eigenvectors can be chosen **orthogonal**, so they form a set of mutually perpendicular directions. For a covariance matrix, those particular directions are **uncorrelated** portfolios, and that is the fact everything else rests on.
- The sum of the eigenvalues is the **trace**, the sum of the diagonal entries. For a covariance matrix the diagonal holds the variances, so the eigenvalues **redistribute the total variance without changing it**.

KEY IDEA

A covariance matrix's eigenvectors are the portfolios whose returns are uncorrelated with one another, and each eigenvalue is that portfolio's variance. The matrix has not been approximated, only re-expressed: n correlated assets have been rewritten as n independent ones. Because the trace is preserved, the eigenvalues also tell you *how the total risk is distributed* across those independent directions, and that distribution is what principal component analysis reports.

WORKED EXAMPLE

Take the two-asset covariance matrix from linear algebra fundamentals, with $\sigma_1 = 20\%$, $\sigma_2 = 30\%$ and correlation 0.5:

$$\Sigma = \begin{pmatrix} 0.04 & 0.03 \\ 0.03 & 0.09 \end{pmatrix}.$$

The trace is $0.04+0.09 = 0.13$ and the determinant is $0.04(0.09)-0.03^2 = 0.0036-0.0009 = 0.0027$. The two eigenvalues must therefore sum to 0.13 and multiply to 0.0027, which gives

$$\lambda = \frac{0.13 \pm \sqrt{0.13^2 - 4(0.0027)}}{2} = \frac{0.13 \pm \sqrt{0.0061}}{2},$$

so $\lambda_1 = 0.1041$ and $\lambda_2 = 0.0260$. Check: they sum to 0.13, the trace.

As shares of the total, $0.1041/0.13 = 80\%$ and $0.0260/0.13 = 20\%$. So four fifths of the risk in this pair lives in a single direction, roughly "both assets together", and one fifth in the direction that trades one against the other. That is the entire idea of principal components, done by hand on the smallest possible example.

The test that decides whether a matrix is legal

A symmetric matrix Σ is **positive semidefinite** if and only if all of its eigenvalues are non-negative. This is the practical form of the requirement introduced in linear algebra fundamentals, where the argument was that $w^\top \Sigma w$ is a variance and cannot be negative. The two statements are the same statement, and the eigenvalue version is the one you can actually check.

The connection is direct: if v is an eigenvector with eigenvalue λ, then $v^\top \Sigma v = \lambda\, v^\top v$. A negative eigenvalue therefore hands you, for free, the portfolio with negative variance. **The eigenvector of the most negative eigenvalue is the worst offending portfolio**, which is why the test is not just a yes or no: it tells you where the problem is.

WORKED EXAMPLE

Return to the impossible correlation matrix from the previous section, where asset 1 is 0.9 correlated with both 2 and 3 while 2 and 3 are −0.9 correlated with each other:

$$C = \begin{pmatrix} 1 & 0.9 & 0.9 \\ 0.9 & 1 & -0.9 \\ 0.9 & -0.9 & 1 \end{pmatrix}.$$

Its eigenvalues are 1.9, 1.9 and −0.8. Two checks confirm them: they sum to $1.9+1.9-0.8 = 3$, which is the trace, as it must be for any correlation matrix of three assets; and they multiply to $1.9 \times 1.9 \times (-0.8) = -2.888$, which is the determinant.

The eigenvector belonging to the negative eigenvalue is $(1, -1, -1)$, and it can be verified in one multiplication:

$$C\,(1,-1,-1)^\top = (-0.8,\ 0.8,\ 0.8)^\top = -0.8\,(1,-1,-1)^\top.$$

That vector is exactly the portfolio used in the previous section to expose the negative variance, and now it is no longer a lucky guess. Its variance is $v^\top Cv = \lambda\, v^\top v = -0.8 \times 3 = -2.4$, matching the earlier arithmetic. **The eigen-decomposition finds the counterexample rather than requiring you to think of one.**

Once a matrix has failed the test, the standard repair is to work in the eigen basis: replace every negative eigenvalue with zero, or with a small positive number, rebuild the matrix from the unchanged eigenvectors, and rescale the diagonal back to ones so that it is a correlation matrix again. It differs from the original only in the directions that were impossible. Be precise about how good it is, because this is a standard follow-up: the clipping step alone gives the *nearest positive semidefinite matrix* in the Frobenius sense, but the rescaling that restores a unit diagonal gives that property up, so the result is a legal correlation matrix and not in general the nearest one. Computing the genuinely nearest correlation matrix takes an iterative method, and that, along with the factorisation used afterwards to simulate correlated paths, is in matrix decompositions.

Principal components, and the three curve movements

Applied to a covariance matrix, this procedure has a name: **principal component analysis**. Sort the eigenvalues from largest to smallest, and each eigenvector is a "component", an uncorrelated direction of variation, with its eigenvalue's share of the trace giving the fraction of total variance it explains. When a few components explain almost everything, a high-dimensional problem has a low-dimensional description.

The classic result in markets is the yield curve. Run PCA on daily changes in yields across maturities and the first three components are always recognisable, and always in the same order.

- **Level**, an eigenvector that is roughly flat across maturities, so all yields move together. It typically explains the large majority of the variance, commonly reported around 85 to 90%.
- **Slope**, an eigenvector that is negative at the front and positive at the back, so the curve steepens or flattens. Usually the next largest, in the region of 5 to 10%.
- **Curvature**, a U-shaped eigenvector where the belly moves against the wings. Usually a couple of percent.

The exact shares vary by market and by period and should be quoted as approximations. The robust statement is the ordering and the fact that three numbers describe almost all of the movement of an object with a dozen or more maturities. The curve itself is yield curves.

REAL INTERVIEW QUESTION (UBS · Rates)

What is a steepener?

WHAT THEY TEST / ANSWER

A position that profits when the yield curve steepens, meaning long-maturity yields rise relative to short-maturity yields. Say that, then say the three things that make it a real answer.

How it is put on. Receive fixed at the short maturity and pay fixed at the long one, or equivalently buy the short bond and sell the long one, sized so the position has no directional exposure to the overall level of rates. That sizing matters: a steepener is a bet on the *relationship* between two points, so you weight the legs to be neutral to a parallel shift, otherwise you own a rates direction bet with a curve bet attached.

Why the trade exists as its own category. Because curve movements decompose into level, slope and curvature, and those three are close to uncorrelated. A steepener isolates the second one. That is not a convention, it is the eigenstructure of the curve: the principal components of yield changes come out as a flat direction, a monotone direction and a U-shaped one, so the market's vocabulary of outright, steepener and butterfly is the market naming its own eigenvectors. Saying that is what distinguishes the answer.

What you are actually betting on. Steepeners are typically expressions of a view that policy is about to ease, since the front end tracks expected policy and falls first, or that term premium is about to rise on supply or inflation risk. Bull steepener means the curve steepens with yields falling, driven by the front end; bear steepener means it steepens with yields rising, driven by the long end. The pair of terms exists because the same curve shape can arrive from opposite directions and they are different trades.

If you want the last point, note that the second component explains far less variance than the first, so a slope bet has a smaller natural size than an outright, which is why curve trades are typically run at larger notionals to reach comparable risk. That is the eigenvalue share translated into a position size.

When the first eigenvalue eats everything

INTUITION

"Correlations went to one" is a statement about eigenvalues. If every asset moves together, the first eigenvector is close to "everything, equally weighted" and its eigenvalue absorbs most of the trace, leaving almost nothing for the other directions. In calm markets the variance is spread across many components and there is genuine diversification. In a crisis the first component's share rises sharply, and the number of directions you were actually diversified across falls, sometimes to one. The first eigenvalue's share of the trace is therefore a usable measure of how concentrated the market's risk currently is, and it moves well before anyone announces a crisis.

The market question that puts this to work, how the correlation between the Nikkei and the dollar has evolved, is answered in ADRs and GDRs, where the unhedged-versus-hedged tracker variance makes the consequence concrete. The eigenvalue reading of it is the one above: a correlation is the visible trace of a shared driver, so when the driver changes the first eigenvalue's share of the trace moves, and the number of directions you were diversified across changes without either asset changing at all.

> **INTERVIEW TRAP**
>
> Three errors. (1) Treating a small negative eigenvalue as a rounding artefact. It usually is numerically small, and it still means some portfolio has negative variance, so it must be repaired rather than ignored. (2) Reading the first principal component as "the market" in every dataset. It is whatever the data's dominant shared direction happens to be, and on some datasets it is an artefact of scaling, which is why PCA is normally run on correlations or on standardised returns. (3) Quoting the yield curve's variance shares as fixed constants. The ordering is robust, the percentages are not.

Where this goes next

The characteristic equation that produces these numbers is matrices and determinants, and rewriting a matrix entirely in its eigen basis is diagonalization. The factorisations used to generate correlated random numbers are matrix decompositions, which feed Monte Carlo methods. The object the curve components describe is yield curves, and the desk-level version of collapsing many exposures into a few numbers is aggregation of Greeks.

6.7 Diagonalization

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You already know that a symmetric matrix has real eigenvalues and an orthogonal set of eigenvectors, and that the sign of those eigenvalues decides whether a covariance or correlation matrix is legitimate. All of that is established in eigenvalues and eigenvectors. This section is about the *procedure* that turns those facts into something you can compute with: rewriting a matrix in the coordinate system built from its own eigenvectors, so that an $n \times n$ tangle of interacting numbers becomes n independent dials.

> **INTUITION**
>
> A matrix is a machine that takes a vector in and pushes a vector out. In general the output points somewhere unrelated to the input, so the machine is hard to picture. But most machines have a few special directions where they do nothing but stretch. If you rebuild your coordinate system out of exactly those directions, the machine stops looking like a tangle of n^2 numbers and starts looking like n independent dials, one stretch factor per axis. Diagonalization is that change of coordinates. Nothing about the object changes; you just stop looking at it from an awkward angle.

The factorisation

An $n \times n$ matrix A is **diagonalizable** when there exists an invertible matrix P and a diagonal matrix D such that

$$A = PDP^{-1}, \qquad D = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}.$$

The columns of P are the eigenvectors of A and the diagonal entries of D are the matching eigenvalues, *in the same order*. Read the product right to left and it tells a story: P^{-1} translates a vector into eigenvector coordinates, D stretches each coordinate independently, and P translates back. Relating A and D through P in this way is called a **similarity transformation**, and similar matrices are the same linear map seen in two different bases.

The condition is exactly this: **A is diagonalizable if and only if it has n linearly independent eigenvectors.** That is what makes P invertible. A sufficient but not necessary quick test: if the n eigenvalues are all distinct, the matrix is diagonalizable.

For the matrices finance actually cares about the question never arises, because a real symmetric matrix is always orthogonally diagonalizable, with P chosen so that $P^{-1} = P^\top$. That result belongs to eigenvalues and eigenvectors; its practical payoff for this procedure is that you never invert anything, you transpose. Orthonormal bases are in norms and inner products.

When it fails. Not every matrix is diagonalizable, and the standard counter-example is the shear

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Its characteristic polynomial is $(1-\lambda)^2$, so $\lambda = 1$ appears twice: its **algebraic multiplicity** is 2. But solving $(A - I)v = 0$ gives $v_2 = 0$, so every eigenvector is a multiple of $(1, 0)^\top$ and the eigenspace is only one dimensional: its **geometric multiplicity** is 1. One independent eigenvector is not enough to build an invertible P, so no diagonalization exists. The general rule: diagonalizable means geometric multiplicity equals algebraic multiplicity for every eigenvalue.

> **KEY IDEA**
>
> Diagonalizable and invertible are different properties and neither implies the other. The identity matrix is both. The shear above is invertible and not diagonalizable. The zero matrix, and any matrix with a zero eigenvalue but a full set of eigenvectors, is diagonalizable and not invertible. Interviewers ask this precisely because candidates blur the two.

Carrying out the procedure

WORKED EXAMPLE

Diagonalize $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ and then use it.

Step 1, the eigenpairs. Try $v = (1,1)^\top$: $Av = (3,3)^\top = 3v$. Try $v = (1,-1)^\top$: $Av = (1,-1)^\top = 1 \cdot v$. So $\lambda_1 = 3$ and $\lambda_2 = 1$.

Step 2, normalise and assemble. The two eigenvectors are orthogonal, as the spectral facts promise for a symmetric matrix, so divide each by its length $\sqrt{2}$ and stack them as columns:

$$Q = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad \Lambda = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}.$$

Step 3, check it. Never trust an assembled factorisation without multiplying it back out:

$$Q\Lambda Q^\top = \tfrac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 3 & 3 \\ 1 & -1 \end{pmatrix} = \tfrac{1}{2}\begin{pmatrix} 4 & 2 \\ 2 & 4 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} = A.$$

Step 4, get something for free. Powers of A now cost almost nothing, because the inner $Q^\top Q$ pairs cancel:

$$A^5 = Q\Lambda^5 Q^\top, \qquad \Lambda^5 = \begin{pmatrix} 243 & 0 \\ 0 & 1 \end{pmatrix}, \qquad A^5 = \tfrac{1}{2}\begin{pmatrix} 244 & 242 \\ 242 & 244 \end{pmatrix} = \begin{pmatrix} 122 & 121 \\ 121 & 122 \end{pmatrix}.$$

Confirm it the laborious way to be sure: $A^2 = \left(\begin{smallmatrix} 5 & 4 \\ 4 & 5 \end{smallmatrix}\right)$, then $A^4 = \left(\begin{smallmatrix} 41 & 40 \\ 40 & 41 \end{smallmatrix}\right)$, and $A^4A = \left(\begin{smallmatrix} 122 & 121 \\ 121 & 122 \end{smallmatrix}\right)$. Same answer, far more arithmetic.

That last step generalises to the reason diagonalization earns its place. For any k,

$$A^k = PD^kP^{-1},$$

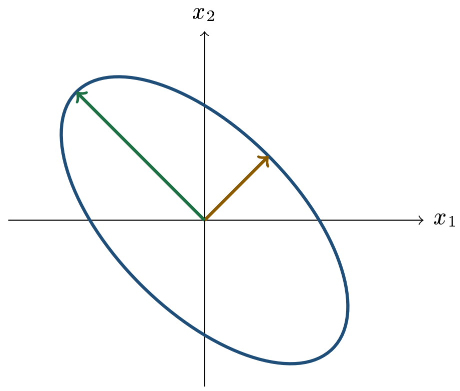
so a repeated matrix multiplication collapses into n scalar powers. It is also why the long-run behaviour of any linear recursion, a transition matrix, an AR process, a binomial lattice, is governed by its largest eigenvalue: raised to a high power, every other term shrinks to nothing beside it.

Diagonalizing a quadratic form

The second payoff is the one a desk meets daily. A **quadratic form** is the scalar $x^\top Ax$ with A symmetric. Portfolio variance $w^\top \Sigma w$ is the canonical example and is treated as an object in linear algebra fundamentals and matrices and determinants. What diagonalization adds is a change of variable. Substitute $A = Q\Lambda Q^\top$ and set $y = Q^\top x$, which is simply x read in the eigenvector coordinates:

$$x^\top Ax \;=\; x^\top Q\Lambda Q^\top x \;=\; y^\top \Lambda y \;=\; \sum_{i=1}^{n} \lambda_i\, y_i^2.$$

Every cross term has vanished. A tangle of n^2 interactions has become a weighted sum of squares, and because $y_i^2 \geq 0$ the sign of the whole expression is governed entirely by the signs of the λ_i. That is the mechanical reason the eigenvalue test for definiteness works; the test itself, and what to do when a correlation matrix fails it, belong to eigenvalues and eigenvectors.



— level set of $x^\top Ax$ — first eigenvector, semi-axis $\propto 1/\sqrt{\lambda_1}$ — second eigenvector, semi-axis $\propto 1/\sqrt{\lambda_2}$
the larger the eigenvalue, the faster $x^\top Ax$ grows, so the shorter the axis

The picture is the whole idea in one shape. A quadratic form's level set is a tilted ellipse, and the tilt is exactly the cross term you are trying to remove. The eigenvectors are the ellipse's own axes. Rotating your head until you are aligned with them is what $y = Q^\top x$ does, and in those coordinates the ellipse is axis-aligned and the cross term is gone.

REAL INTERVIEW QUESTION (Société Générale · EQD)

What is the correlation sensitivity of a call on a basket? Can you make money in Gamma if one of the elements of the Gamma matrix is negative?

WHAT THEY TEST / ANSWER

Two questions, and the second one is the real test. On the first, a basket call is long correlation: higher correlation means less diversification inside the basket, so the basket's volatility rises and the option is worth more. The pricing of correlation sits in equity correlation and dispersion.

On the second, the answer is yes, and the reason is the change of variable above. Second-order PnL on a multi-underlying book is a quadratic form,

$$\text{PnL} \;\approx\; \tfrac{1}{2}\,\Delta S^\top \Gamma\,\Delta S \;=\; \tfrac{1}{2}\sum_i \lambda_i y_i^2, \qquad y = Q^\top \Delta S.$$

What decides whether you are convex is the sign of the *eigenvalues* of Γ, not the sign of its individual entries. A negative off-diagonal cross-Gamma Γ_{ij} is perfectly compatible with a matrix whose eigenvalues are all positive, in which case every possible move makes money. Conversely, if the eigenvalues are of mixed sign you are long Gamma along the eigenvectors with positive eigenvalues and short Gamma along the ones with negative eigenvalues, so the honest answer is "it depends which way the market moves, and I can tell you exactly which directions pay". That is the sentence they are waiting for. Single-name Gamma dynamics are in Gamma and convexity.

Interviewers reach the same machinery from the risk side, by asking how you would build a variance-covariance matrix for a portfolio of options. That is a risk-factor mapping and estimation question rather than a diagonalization one, and it is treated in Value at Risk and descriptive statistics. The single point that belongs here is the reason such a matrix is usually near rank deficient: implied volatilities across strikes and tenors move together, so the surface is carried by its first few components, which is the eigen-decomposition owned by eigenvalues and eigenvectors.

INTERVIEW TRAP

Three recurring errors, all about the procedure rather than the arithmetic. First, assuming every matrix is diagonalizable; symmetric ones always are, general ones need not be, and the shear is the counter-example to keep in your pocket. Second, confusing diagonalizable with invertible. Third, assembling P and D with the eigenvalues in a different order from the eigenvector columns, which produces a factorisation that silently reconstructs the wrong matrix. That is why step 3 of the worked example exists: multiplying back out takes ten seconds.

VISUAL / CODE TO BUILD: Interactive

A change-of-basis animator. The reader edits a symmetric 2×2 matrix and the panel draws the level set of $x^\top Ax$ in the original axes, then smoothly rotates the frame through $y = Q^\top x$ until the ellipse is axis-aligned, with a live readout of the two representations showing the cross term shrinking to zero. A second control raises the matrix to a power k by both routes side by side, repeated multiplication against $Q\Lambda^k Q^\top$, displaying the operation count for each so the saving is measured rather than asserted, and letting k run large enough that the dominant eigenvalue visibly takes over. The intended "aha" is that diagonalization is not a new fact about the matrix, it is the same matrix written in the only coordinates where it is easy.

Where this goes next

The computational cousins of this factorisation, LU, Cholesky and the singular value decomposition, are in matrix decompositions; the SVD is the version that survives on non-square and non-symmetric matrices, which is what you reach for when the matrix is a data panel rather than a covariance. The statistical reading of the eigen-decomposition, and the repair of a correlation matrix that fails the eigenvalue test, are in eigenvalues and eigenvectors. The probabilistic meaning of the correlations that fill these matrices is in independence and correlation, the simulation that needs a matrix square root is in Monte Carlo methods, and the tooling for doing any of this on real data is in NumPy and pandas.

6.8 Matrix decompositions (LU, Cholesky, SVD)

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A decomposition rewrites a matrix as a product of simpler ones. That sounds like an exercise until you notice what it buys: solving a linear system becomes two easy substitutions instead of one hard inversion, generating correlated random numbers becomes a single matrix multiplication, and the question of whether a correlation matrix is even legitimate turns into a question about whether an algorithm completes. All three of the decompositions here earn their place on a desk, and one of them, Cholesky, is quietly running inside every Monte Carlo engine in the building.

INTUITION

Factorising a matrix is the same move as factorising a number. Knowing that $1{,}001 = 7 \times 11 \times 13$ tells you things that staring at 1,001 does not, and it makes some operations trivial that were awkward before. A matrix decomposition does exactly that: it finds the structure hiding inside a table of numbers, so that operations you care about become easy, and so that properties you care about become visible.

LU: solving systems without inverting anything

LU writes a square matrix as $A = LU$ with L lower triangular and U upper triangular (in practice with row swaps, $PA = LU$). It is just Gaussian elimination with the bookkeeping kept.

The point is efficiency in a specific and common situation. To solve $Ax = b$, substitute to get $L(Ux) = b$, then solve $Ly = b$ by **forward substitution** and $Ux = y$ by **back substitution**. Both are trivial because triangular systems unwind one variable at a time. And the factorisation depends only on A, so if you must solve $Ax = b$ for many different right-hand sides, which is exactly what happens when a curve is rebuilt every time a quote moves, you pay for the decomposition once and reuse it.

KEY IDEA

Never compute an inverse to solve a linear system. Forming A^{-1} and multiplying is slower, and numerically worse, than factorising once and substituting. In code this is the difference between `solve(A, b)` and `inv(A) @ b`, and the first is the correct habit. The only time you genuinely want an explicit inverse is when the entries of the inverse are themselves the answer.

Cholesky: the one you will actually use

If A is symmetric and **positive definite**, it factors as

$$A \;=\; LL^\top,$$

with L lower triangular and positive diagonal entries. This is Cholesky, and it is roughly twice as fast as LU because it exploits the symmetry.

Its importance in finance is that it **manufactures correlation**. Suppose you want a vector x of random variables with covariance matrix Σ. Start with independent standard normals z, which any generator produces, factor $\Sigma = LL^\top$, and set $x = Lz$. Then

$$\mathrm{Cov}(x) \;=\; \mathbb{E}[Lz\,z^\top L^\top] \;=\; L\,\mathbb{E}[zz^\top]\,L^\top \;=\; LIL^\top \;=\; LL^\top \;=\; \Sigma,$$

using only that the z are uncorrelated with unit variance. Three lines, and it is the whole basis of multi-asset Monte Carlo: correlated Brownian motions, basket and worst-of options, correlated default times, and multi-factor rate models all rest on it.

WORKED EXAMPLE

Two correlated variables, by hand. Take the correlation matrix

$$\Sigma = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}, \qquad L = \begin{pmatrix} 1 & 0 \\ \rho & \sqrt{1-\rho^2} \end{pmatrix}.$$

Verify by multiplying out: the top-left entry of $LL^\top$ is 1, the off-diagonal is ρ, and the bottom-right is $\rho^2 + (1-\rho^2) = 1$, so $LL^\top = \Sigma$ exactly.

With $\rho = 0.6$ we get $\sqrt{1-0.36} = \sqrt{0.64} = 0.8$, so

$$x_1 = z_1, \qquad x_2 = 0.6\, z_1 + 0.8\, z_2.$$

Check it directly: x_2 has variance $0.6^2 + 0.8^2 = 0.36 + 0.64 = 1$, and the covariance of x_1 and x_2 is 0.6, since only the z_1 term is shared. That two-line formula is what a pricing library is doing when it simulates a two-asset basket, and being able to write it from memory is a reasonable expectation on a quantitative desk.

KEY IDEA

Cholesky is also a test. The factorisation exists precisely when the matrix is positive definite, and the algorithm fails, by requiring the square root of a negative number, exactly when it is not. So "does Cholesky complete" is a practical check that a covariance or correlation matrix is legitimate, and the failure is informative rather than annoying: it tells you your matrix implies a portfolio with negative variance, which cannot exist.

WORKED EXAMPLE

A correlation matrix that cannot be real. Suppose someone hands you pairwise correlations of 0.9 between A and B, 0.9 between A and C, and -0.9 between B and C:

$$\Sigma = \begin{pmatrix} 1 & 0.9 & 0.9 \\ 0.9 & 1 & -0.9 \\ 0.9 & -0.9 & 1 \end{pmatrix}.$$

Expanding along the first row,

$$\det \Sigma = 1\,(1 - 0.81) - 0.9\,(0.9 + 0.81) + 0.9\,(-0.81 - 0.9) = 0.19 - 1.539 - 1.539 = -2.888.$$

A negative determinant means at least one negative eigenvalue, so the matrix is not positive semidefinite and Cholesky will fail on it. The intuition is worth more than the arithmetic: **if B and C are both nearly identical to A, they cannot be nearly opposite to each other.** Correlations are geometric constraints on angles between vectors, not free parameters, and this is why a matrix assembled from pairwise estimates over different sample windows, or adjusted by hand for a stress scenario, so often turns out to be inadmissible.

The standard repairs are to compute the eigen-decomposition, floor the negative eigenvalues at zero or at a small positive number and rebuild, or to solve the **nearest correlation matrix** problem properly, which finds the closest admissible matrix in a chosen norm. Either way, say out loud what you changed and by how much, because you have altered the risk you are reporting.

SVD: the decomposition that always exists

Every matrix, square or not, factors as

$$A = U\Sigma V^{\top},$$

with U and V orthogonal and Σ diagonal with non-negative entries called the **singular values**. Geometrically, any linear map is a rotation, then a stretch along perpendicular axes, then another rotation, which is a genuinely surprising fact and the reason the SVD is the most useful decomposition in applied mathematics.

Four things it gives you:

- **Rank and conditioning.** The number of non-zero singular values is the rank, and the ratio of the largest to the smallest is the **condition number**, which measures how much a small error in the inputs is amplified in the answer. A near-singular covariance matrix has a huge condition number, which is the formal version of "these two risk factors are nearly the same thing, so the optimiser will produce nonsense".
- **Low-rank approximation.** Keeping the largest k singular values gives the best rank-k approximation, in a precise sense. Applied to a covariance matrix this is principal component analysis, which is how a yield curve becomes level, slope and curvature, developed in eigenvalues and eigenvectors and diagonalization.
- **Least squares and the pseudo-inverse.** When a system has no exact solution or too many, the SVD produces the minimum-norm least-squares answer, which is the numerically robust way to fit a regression or to calibrate more parameters than the data supports, see linear and multiple regression.
- **Repairing matrices.** It is the same machinery as the eigenvalue flooring above, with one caveat worth saying out loud: for a symmetric matrix the singular values are the *absolute*

values of the eigenvalues, so the two views agree only where the matrix is already positive semidefinite. Do the flooring on the eigen-decomposition, because flooring a singular value would turn a negative eigenvalue into a positive one rather than into zero, which repairs the symptom and keeps the error.

Decomposition	Requires	Use it for
$PA = LU$	square, invertible	solving $Ax = b$, repeatedly, cheaply
$A = LL^\top$	symmetric positive definite	correlated simulation, and as a PSD test
$A = U\Sigma V^\top$	nothing	rank, conditioning, PCA, least squares
$A = Q\Lambda Q^\top$	symmetric	repairing a matrix, by flooring negative eigenvalues

REAL INTERVIEW QUESTION (Société Générale · EQD)

Construction of correlated Brownian motions.

WHAT THEY TEST / ANSWER

Give the construction, prove it in one line, and then raise the practical issues, because on an equity derivatives desk this is a working procedure rather than a theory question. The construction: to simulate n Brownian motions with correlation matrix Σ, draw a vector z of independent standard normals at each time step, Cholesky-factor $\Sigma = LL^\top$, and set the correlated increments to $\sqrt{\Delta t}\, Lz$. The proof is that $\mathrm{Cov}(Lz) = L\,\mathbb{E}[zz^\top]\,L^\top = LL^\top = \Sigma$, using only that the z are independent with unit variance. Give the two-asset case explicitly, since they often ask for it: $W_1 = z_1$ and $W_2 = \rho z_1 + \sqrt{1-\rho^2}\, z_2$, which is the 2×2 Cholesky factor written out. Then the practical points that separate a good answer. The correlation matrix must be positive semidefinite or the factorisation does not exist, and a matrix built from pairwise estimates over different windows frequently is not, so you repair it by flooring negative eigenvalues or solving the nearest-correlation-matrix problem and you disclose the change. The decomposition is computed *once* outside the simulation loop, not per path, which is the difference between a fast engine and a slow one. Correlation in the Gaussian world is only a linear dependence measure, so if the product's risk is in joint tail events you may need a copula rather than a correlation matrix. And for variance reduction you would use the same z across scenarios with antithetic or quasi-random sequences, which interacts with the factorisation because the dimension ordering matters for low-discrepancy sequences.

REAL INTERVIEW QUESTION (Tikehau Capital · Market Risk)

How do you build a variance-covariance matrix for a portfolio of options?

WHAT THEY TEST / ANSWER

Answer in three steps and be explicit that options make this harder than equities, because that is the point of the question. **Step one, choose risk factors, not instruments.** You do not build a covariance matrix over the options themselves, you build it over the underlying risk factors: spot or index returns, implied volatilities by strike and maturity (usually reduced to a few surface components rather than every point), interest rates by key tenor, dividends and, where relevant, credit and FX. **Step two, map each position to those factors** through its sensitivities, delta and vega at minimum, with gamma if you want the second-order term, so the option book becomes a vector of factor exposures, as in Value at Risk. **Step three, estimate the covariance matrix** of the factors, deciding the window and weighting scheme (equally weighted versus exponentially weighted, which trades responsiveness against noise), the return frequency, and how you handle non-synchronous data across time zones, which induces spurious low correlations. Then the issues specific to this problem, which is where the marks are. The matrix is often **near-singular** because implied volatility points are almost perfectly correlated with each other, so you reduce the surface to a few principal components before estimating rather than inverting a nearly rank-deficient matrix. Estimation with more factors than observations produces a matrix that is not full rank at all, which is why shrinkage towards a structured target is standard. You should verify positive semidefiniteness explicitly, with a Cholesky attempt or the eigenvalues, and repair it if needed. And you should say the honest limitation: a covariance matrix captures linear co-movement, so for an option book with material convexity it describes small moves well and large ones badly, which is why it is complemented by full-revaluation historical simulation and by stress testing.

REAL INTERVIEW QUESTION (Société Générale · EQD)

Tell me about the correlation matrix.

WHAT THEY TEST / ANSWER

An open prompt, so give the structure, the constraints and then the practical failure modes, which is the order of increasing sophistication. **Structure**: a square symmetric matrix with ones on the diagonal, entry (i, j) being the correlation between assets i and j, each in $[-1, 1]$, obtained from the covariance matrix by scaling out the volatilities. **The constraint people forget**: it must be positive semidefinite, meaning every eigenvalue is at least zero, because $w^\top \Sigma w$ is the variance of the portfolio with weights w and a variance cannot be negative. That is not a technicality, it is what makes the correlations mutually consistent, and the geometric reading is that correlations are cosines of angles between vectors, so if A is nearly parallel to B and to C then B and C cannot be nearly opposite. Give the concrete example if they want one: 0.9, 0.9 and -0.9 on three assets has a negative determinant, so it cannot describe any real portfolio. **Practical failure modes**: pairwise estimation over different sample windows breaks positive semidefiniteness, hand-adjusting a single entry for a stress scenario usually breaks it too, and correlations are unstable and tend towards one in a crisis precisely when diversification is being relied upon. **What you do about it**: check with Cholesky or the eigenvalues, repair by flooring negative eigenvalues or by solving for the nearest correlation matrix, reduce dimension with principal components when the matrix is large relative to the data, and shrink towards a structured target. On an equity derivatives desk, close with the point that connects it to the book: for a basket or worst-of product, correlation is not an input you estimate and forget, it is a traded risk with its own bid-offer, so the matrix used for pricing and the matrix used for risk are usually not the same object, see independence and correlation.

INTERVIEW TRAP

Four traps. First, **inverting a matrix to solve a system**, which is slower and less accurate than factorising and substituting. Second, **assuming any symmetric matrix of plausible correlations is usable**: it must be positive semidefinite, and matrices assembled pairwise or adjusted by hand often are not. Third, **re-factorising inside the Monte Carlo loop**, which can dominate the run time of an otherwise sensible engine. Fourth, **ignoring the condition number**: a nearly singular covariance matrix will still invert numerically, and it will hand you an optimiser output with enormous offsetting positions that is pure noise amplification.

VISUAL / CODE TO BUILD: Interactive

A three-tab numerical laboratory. **Cholesky**: the reader sets a correlation matrix with sliders, sees L computed live, and watches a scatter of simulated points change shape as ρ moves, with the algorithm turning red at the exact moment the matrix stops being positive semidefinite, so the failure is experienced rather than described. A three-asset version with the $0.9/0.9/-0.9$ preset shows the determinant going negative as the third correlation is dragged down, which makes the geometric constraint visible. **SVD**: an image or a covariance matrix reconstructed from its largest k singular values, with the reconstruction error and the condition number displayed, so low-rank approximation and conditioning become one idea. **Curve PCA**: real yield curve data decomposed into components, with the reader choosing how many to keep and seeing both the variance explained and the reconstructed curves against the originals, connecting this section directly to the change-of-basis argument in vector spaces.

Where this goes next

The matrix algebra underneath is in matrices and determinants, and the eigenvalue view that the SVD generalises is in eigenvalues and eigenvectors and diagonalization, with the change-of-basis idea they all serve in vector spaces. On the finance side, the correlation input is independence and correlation, the risk application is Value at Risk, the fitting application is linear and multiple regression, and the code these routines live in is Python for finance.

6.9 Norms and Inner Products

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A norm measures how big something is; an inner product measures how much two things point the same way. That is the whole subject, and the reason it is in a markets book is that both of those questions are asked constantly on a trading floor under other names. How risky is this position is a norm. How good is this hedge is an inner product.

Norms: three ways to be big, three risk philosophies

For a vector x, the family that matters is

$$\|x\|_1 = \sum_i |x_i|, \qquad \|x\|_2 = \sqrt{\textstyle\sum_i x_i^2}, \qquad \|x\|_\infty = \max_i |x_i|.$$

These are not three notations for the same idea; each corresponds to a different statement about what you are afraid of.

The L_1 norm adds up magnitudes. It is total turnover, total notional, total transaction cost. Used as a penalty it produces *sparse* answers, setting most weights to exactly zero, which is why it is the right penalty when you want a portfolio with few positions rather than a small position in everything.

The L_2 norm is Euclidean distance, and squaring it gives variance. Every mean-variance result, every least-squares regression and every tracking error is an L_2 statement. Used as a penalty it shrinks all weights smoothly toward zero without eliminating any.

The L_∞ norm is the worst single element. It is the natural language of limits and stress: a desk that cares about the largest single exposure rather than the aggregate is thinking in L_∞, and a stress limit is an L_∞ constraint while a VaR limit is closer to an L_2 one.

Choosing the norm is choosing the risk you are managing, and disagreements about risk measures are frequently disagreements about which norm, conducted without either party naming one.

Correlation is a cosine

KEY IDEA

The inner product of two vectors is $\langle x, y\rangle = x^\top y$, and the angle between them satisfies

$$\cos\theta \;=\; \frac{\langle x, y\rangle}{\|x\|\,\|y\|}.$$

Take x and y to be two series of demeaned returns. Then the numerator is the covariance, the denominators are the volatilities, and

$$\boxed{\rho \;=\; \cos\theta}$$

Correlation is literally the cosine of the angle between two return series. Uncorrelated means orthogonal, at right angles. Perfectly correlated means parallel. Perfectly anti-correlated means antiparallel.

Two consequences fall out of that identification and are worth more than the identification itself.

Cauchy-Schwarz is the correlation bound. The inequality $|\langle x, y\rangle| \le \|x\|\|y\|$ says exactly that $|\rho| \le 1$. There is no separate theorem about correlations being bounded; it is a statement about angles, and cosines cannot exceed one.

The correlation triangle inequality becomes obvious. If A and B are 60 degrees apart, which is $\rho = 0.5$, and B and C are also 60 degrees apart, then A and C cannot be more than 120 degrees apart, so

$$\rho_{AC} \;\ge\; \cos(120^\circ) \;=\; -0.5.$$

For three assets all at the same pairwise correlation the three angles must close, so $3\theta \le 360^\circ$, giving $\theta \le 120^\circ$ and $\rho \ge -\frac{1}{2}$. That is the identical bound the determinant produces, and here it takes one line of geometry instead of a cubic. When two derivations of a constraint agree from completely different directions, the constraint is real rather than an artefact of the method.

Projection is hedging

This is the idea that makes the section worth a trader's time.

Given an exposure y and a hedge instrument x, the best hedge is the **orthogonal projection** of y onto x: the multiple of x that leaves a residual perpendicular to x. The coefficient is

$$\beta \;=\; \frac{\langle y, x\rangle}{\langle x, x\rangle} \;=\; \frac{\mathrm{Cov}(y, x)}{\mathrm{Var}(x)} \;=\; \rho\,\frac{\sigma_y}{\sigma_x},$$

which is simultaneously the minimum-variance hedge ratio, the regression slope of y on x, and a projection. They are the same object.

INTUITION

The orthogonality of the residual is not a technicality, it is the definition of having finished. "The residual is perpendicular to the hedge instrument" means precisely "there is no remaining exposure that this instrument could have removed". If the residual still had a component along x, you would trade more x.

So a hedger and a statistician running a regression are performing the same operation and calling the answer different things, and the residual, the alpha and the basis risk are the same vector.

WORKED EXAMPLE

How much risk a good hedge actually removes. An exposure with 20% volatility, a hedge instrument with 15%, correlation 0.8.

The hedge ratio is $\beta = 0.8 \times 20/15 = \mathbf{1.0667}$ units of the instrument per unit of exposure. The residual volatility is

$$\sigma_y\sqrt{1-\rho^2} \;=\; 20\% \times \sqrt{1-0.64} \;=\; 20\% \times 0.6 \;=\; \mathbf{12}\%.$$

So a correlation of **0.8** removes only **40%** of the volatility. It removes 64% of the *variance*, which is ρ^2, and variance is the quantity people quote while volatility is the quantity that shows up in a P&L. The full picture:

Correlation	**Variance removed**	**Volatility removed**
0.50	25%	13.4%
0.80	64%	40.0%
0.90	81%	56.4%
0.95	90%	68.8%
0.99	98%	85.9%

Read the first row. A hedge with a correlation of 0.5, which sounds like a meaningful relationship, removes thirteen percent of the risk. This is the single most useful number in the section, because proxy hedges are routinely justified by a correlation that sounds high and delivers very little, and hedging that relies on a proxy should be sized against this table rather than against the correlation.

Orthogonal bases, in one paragraph

If a set of vectors is mutually orthogonal, each component can be computed independently of the others, which is why orthogonality is worth manufacturing. That is the entire motivation for principal components: rotate the correlated risk factors into an uncorrelated set so that each can be measured, limited and hedged separately, which eigenvalues develops. The price is that the new factors are combinations rather than tradeable things, so you gain independence and lose interpretability, and which of those you need depends on whether you are measuring risk or trading it.

INTERVIEW TRAP

1. **Quoting correlation as hedge effectiveness.** It is ρ^2 that measures variance removed and $1 - \sqrt{1-\rho^2}$ that measures volatility removed, and at $\rho = 0.5$ the latter is 13%.

2. **Using the wrong norm for the question.** A limit framework that constrains the aggregate says nothing about the largest single position, and vice versa. Name the norm you are managing.

3. **Forgetting that correlation is a cosine.** Once you see it, every bound on correlations is a statement about angles and most of them become obvious.

4. **Treating the residual as noise.** It is basis risk, and it is the entire risk that remains after a proxy hedge, which is exactly the risk that shows up in a single-name event.

5. **Assuming orthogonal factors are tradeable.** Principal components are combinations, and hedging one requires trading all the underlying instruments.

VISUAL / CODE TO BUILD: Interactive

A two-vector panel drawn in the plane. Two return series are shown as arrows with a draggable angle between them, and the panel prints the correlation as the cosine alongside the angle in degrees. Dragging to 90 degrees prints zero, and the identification stops being a formula.

The projection is drawn on the same picture: the component of one vector along the other, in one colour, and the residual perpendicular to it in another, with the hedge ratio and the residual volatility printed. Watching the residual arrow shorten as the angle closes, and reading off that it is still 60% as long at a correlation of 0.8, is the section's central number delivered as a picture.

A third element adds a third vector and shades the region of angles that the first two permit, reproducing the $\rho \geq -\frac{1}{2}$ bound as a geometric impossibility: the reader simply cannot drag the third arrow into the forbidden region, because the angles will not close.

6.10 Constrained optimization and Lagrange multipliers

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Nobody on a trading floor optimises anything freely. Every decision is made against a limit: a risk limit, a capital charge, a balance-sheet constraint, a client's budget. Lagrange multipliers are the mathematics of deciding under constraints, and the reason they matter here is not the technique but the by-product. **The multiplier is a price**, specifically the price of the constraint, and once you see that, a whole class of desk arguments becomes arithmetic.

INTUITION

At an unconstrained optimum the gradient is zero: no direction improves things. At a *constrained* optimum the gradient need not be zero at all. It only has to point straight into the constraint, so that every improving direction is one you are not allowed to take. The multiplier measures how hard the objective is pushing against the wall.

The method

To maximise $f(x)$ subject to $g(x) = c$, form the Lagrangian

$$\mathcal{L}(x, \lambda) \;=\; f(x) - \lambda\,\big(g(x) - c\big),$$

and set all its partial derivatives, including the one in λ, to zero. The condition in x says

$$\nabla f(x) \;=\; \lambda\,\nabla g(x),$$

that the gradients of objective and constraint are parallel, and the condition in λ just restores $g(x) = c$.

The geometry is the whole content. ∇f points in the direction of steepest improvement and ∇g points perpendicular to the constraint surface. If they are not parallel, part of ∇f lies *along* the surface, so you can slide in that direction and do better while staying feasible. Only when they are parallel is there nothing left to exploit. The multiplier λ is simply the ratio of their lengths.

The multiplier is a shadow price

Let $V(c)$ be the optimal value achieved when the constraint level is c. The result that makes this section useful is

$$\frac{dV}{dc} \;=\; \lambda.$$

The multiplier is exactly the rate at which the objective improves if the constraint is relaxed by one unit. It is a price with units: profit per unit of risk limit, revenue per unit of balance sheet, return per unit of capital.

KEY IDEA

This turns an argument into a calculation. “Should this desk get more risk limit?” is not a matter of seniority; it is a question of whether the desk’s multiplier is higher than the multiplier of whoever currently holds the limit. Every scarce resource on a trading floor has a shadow price, and the optimisation that allocates it computes that price whether or not anyone looks at it.

Allocating a risk budget

WORKED EXAMPLE

Three independent desks. Desk i has expected P&L μ_i per unit of exposure and volatility σ_i, and its **information ratio** is $IR_i = \mu_i/\sigma_i$. Take $IR = 2.0$, 1.5 and 1.2.

Maximise total expected P&L $\sum_i \mu_i w_i$ subject to a variance budget $\sum_i w_i^2\sigma_i^2 = V$, the desks being independent so the variances add. The Lagrangian gives

$$\mu_i - 2\lambda w_i \sigma_i^2 = 0 \quad\Longrightarrow\quad w_i = \frac{\mu_i}{2\lambda\,\sigma_i^2},$$

so **each desk's risk allocation $w_i\sigma_i$ is proportional to its information ratio**. Here that is 2.0 : 1.5 : 1.2, or **42.55%, 31.91% and 25.53%** of the risk budget. Not equal shares, and not everything to the best desk either.

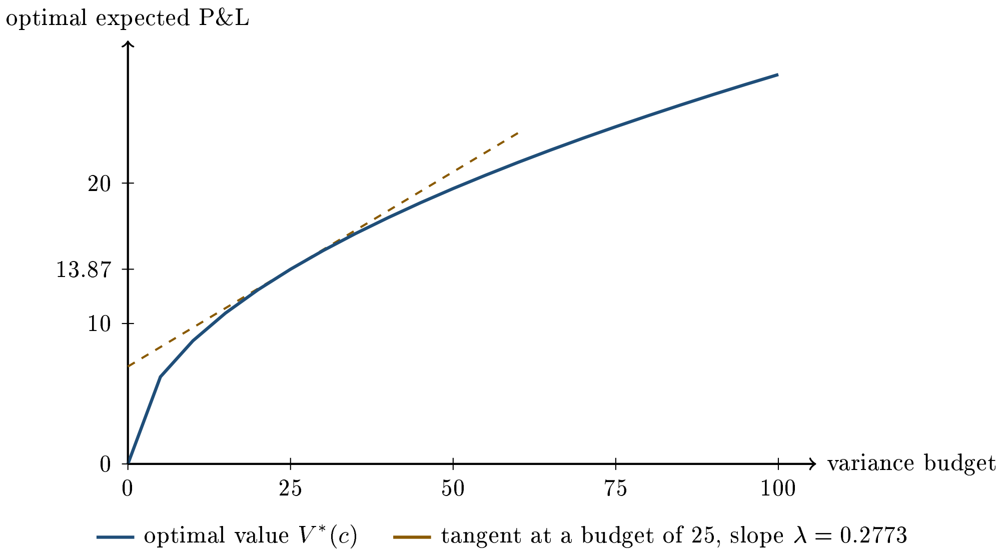
Substituting back, the optimal expected P&L is $K\sqrt{V}$ with

$$K \;=\; \sqrt{\textstyle\sum_i IR_i^2} \;=\; \sqrt{4+2.25+1.44} \;=\; \sqrt{7.69} \;=\; \mathbf{2.7731}.$$

That number is worth pausing on. The combined information ratio, 2.77, is larger than the best individual desk's 2.0. Information ratios add in quadrature across independent sources, which is the entire quantitative case for running several uncorrelated strategies rather than the single best one. It is also why correlation between desks is expensive: the moment they stop being independent, the quadrature no longer holds and the combined ratio falls.

And the multiplier. Differentiating $V^* = K\sqrt{V}$ gives $\lambda = K/(2\sqrt{V})$. At a variance budget of 25, so five units of volatility, the optimum is 13.87 and $\lambda = 0.2773$. Double the budget to 50 and the optimum rises to 19.61, a gain of exactly $\sqrt{2}$, while λ falls to 0.1961, a fall of exactly $1/\sqrt{2}$.

More risk buys more profit, and each additional unit buys less than the last.



the multiplier is the slope of this curve, and it falls as the budget grows

When the constraint is an inequality

Real limits are inequalities: risk must be *at most* the limit, not exactly it. The extension is the **Karush-Kuhn-Tucker** conditions, and one of them is the only piece of this machinery that traders quote by accident.

For $g(x) \leq c$, the optimum satisfies $\lambda \geq 0$ together with

$$\lambda\left(g(x) - c\right) \;=\; 0,$$

which is **complementary slackness**: either the constraint binds and its multiplier can be positive, or it does not bind and its multiplier is exactly zero. There is no third case.

> **KEY IDEA**
>
> This is precisely how a risk limit behaves. If a desk is nowhere near its limit, the limit is worth nothing to it and raising it changes nothing. If the desk is pinned against it, the limit has a positive price and every unit of relief is worth exactly λ. When somebody says a limit is not binding, they are stating that its multiplier is zero, and the correct response is that there is therefore no case for changing it.

The sign condition $\lambda \geq 0$ matters too. It says relaxing a constraint can never make you worse off, which is obvious in words and easy to get backwards in algebra.

Interview angle

REAL INTERVIEW QUESTION (Natixis · QIS)

What is the efficient frontier?

WHAT THEY TEST / ANSWER

Most candidates describe the picture. Describe the optimisation that produces it, because the picture is a consequence and the question is really testing whether you know that.

Define it as the solution of a constrained problem. For each target expected return m, solve: minimise portfolio variance $w^\top \Sigma w$ subject to $w^\top \mu = m$ and $w^\top \mathbf{1} = 1$. Two constraints, so two multipliers. The frontier is the set of (σ, m) pairs traced out as m varies, and a portfolio is efficient when no other portfolio offers more return for the same risk or less risk for the same return.

Then say what the shape means. It is a hyperbola in return-volatility space, opening to the right, with the minimum-variance portfolio at its leftmost point. Everything below that point is dominated: same volatility, worse return. The curvature comes entirely from correlation. If every asset were perfectly correlated the frontier would collapse to a straight line and there would be no diversification benefit; every unit of curvature is a unit of correlation below one.

Now the multiplier, which is the part that separates answers. The multiplier on the return constraint is the marginal variance cost of demanding one more unit of return, and its reciprocal is the slope of the frontier plotted in return-variance space. So the frontier is not merely a set of portfolios; it is the schedule of *prices* of return in units of risk, and it is steepest where risk is cheapest. Adding a risk-free asset turns the problem into finding the tangency point, which is where that price is best, and the tangent line from the risk-free rate is the capital market line.

Finish with the honest limitation, since a QIS interviewer is waiting for it. The frontier is computed from estimated means, variances and correlations, and it is far more sensitive to the estimated means than to anything else. Small errors in expected returns produce wildly different optimal weights, which is why practitioners constrain the weights, shrink the estimates, or fall back on minimum-variance and risk-parity constructions that do not require return forecasts at all. Quoting the frontier without that caveat suggests you have seen it in a textbook and not used it.

INTERVIEW TRAP

Four errors. (1) Treating λ as an algebraic nuisance to be eliminated, when it is the answer to the more interesting question. (2) Forgetting that a first-order condition finds a stationary point, not necessarily a maximum; the second-order behaviour still has to be checked, which is Taylor expansions again. (3) Getting the sign of λ wrong on an inequality constraint, which asserts that relaxing a limit hurts. (4) Pricing a non-binding constraint as though it were worth something, in defiance of complementary slackness.

Where this goes next

The second-order conditions that decide whether a stationary point is actually an optimum are Taylor expansions. The limits this machinery allocates in practice are risk limits frameworks, and the desk-level version of the same trade-off, deciding whether a position earns its capital, is book management. The aggregation problem that supplies the covariance matrix is aggregation of Greeks.

Chapter 7

Probability Theory

7.1 Discrete and continuous probability

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Every price in this book is an average over things that have not happened yet. Before any of that machinery can be built, one distinction has to be completely solid: whether the uncertain quantity takes *separate* values you can list, or values on a *continuum* you cannot. Almost every confusion beginners have about option pricing traces back to blurring those two cases, so this section builds both carefully and shows exactly where they meet.

The framework, in three objects

- The **sample space** Ω: the set of everything that could happen. For a single die it is $\{1, 2, 3, 4, 5, 6\}$, and for a share price in a year it is the positive half-line $(0, \infty)$.
- An **event**: any subset of Ω. "The die shows an even number" is $\{2, 4, 6\}$. "The share ends above 100" is $(100, \infty)$.
- A **probability** $\mathbb{P}$: a rule assigning a number to each event, obeying three conditions. Probabilities are never negative; $\mathbb{P}(\Omega) = 1$; and for events that cannot both happen, the probability of either is the sum of the two.

Two consequences are used constantly and are worth stating before anything else. The **complement rule**, $\mathbb{P}(\text{not } A) = 1 - \mathbb{P}(A)$, which turns many hard countings into easy ones. And **inclusion-exclusion** for events that can overlap,

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B),$$

where subtracting the overlap is what stops you double counting.

REAL INTERVIEW QUESTION (BNP · Market Risk)

From a 32-card deck, I draw one card. What is the probability of drawing neither an ace nor a spade?

WHAT THEY TEST / ANSWER

Use the complement and inclusion-exclusion together, and say the deck out loud first, because that is where the error usually is: a 32-card deck runs from seven to ace in four suits, so there are 8 spades and 4 aces.

The event "an ace or a spade" has probability

$$\mathbb{P}(A \cup S) = \frac{4}{32} + \frac{8}{32} - \frac{1}{32} = \frac{11}{32},$$

where the subtracted card is the ace of spades, counted in both sets. So "neither" is $1 - 11/32 = 21/32 \approx 0.656$.

Then give the check, because it is faster than the calculation and catches the classic slip. Count the cards that qualify directly: the three non-spade suits have 8 cards each, of which one is an ace, leaving 7 each, so $3 \times 7 = 21$. Both routes give 21, and an answer of $20/32$, obtained by forgetting that the ace of spades was counted twice, is the mistake the question exists to catch. The habit worth showing is doing it both ways in ten seconds rather than doing it once confidently.

Discrete: probability lives on the points

A random variable is **discrete** when its possible values can be listed. It is described by a **probability mass function**, $p(x) = \mathbb{P}(X = x)$, and the masses sum to one:

$$\sum_x p(x) = 1.$$

Every question about a discrete variable is answered by adding up the right masses. That is the whole method, and it means a discrete model can be priced with arithmetic alone, with no calculus anywhere.

REAL INTERVIEW QUESTION (BNP · EQD)

A stock takes discrete random values from 1 to 20, all equally likely. What is the price of an option struck at 12 on this stock?

WHAT THEY TEST / ANSWER

State the assumptions before computing, because the question is deliberately stripped down: European exercise, the given distribution taken as the risk-neutral one, and zero interest rates so no discounting. Under those assumptions the price is the expected payoff, and the expectation is a sum.

The payoff $\max(S - 12, 0)$ is zero for $S \leq 12$ and takes the values $1, 2, \ldots, 8$ for $S = 13, \ldots, 20$. Each has probability $1/20$, so

$$C = \frac{1}{20}\sum_{k=1}^{8} k = \frac{1}{20} \times \frac{8 \times 9}{2} = \frac{36}{20} = 1.8.$$

Then add the two remarks that lift it above arithmetic. Only 8 of the 20 outcomes pay anything, so the probability of finishing in the money is $8/20 = 40\%$, and the option is worth 1.8 while paying an average of 4.5 *when* it pays: price equals probability times conditional payoff, which is exactly the structure of the Black-Scholes formula, where $N(d_2)$ is the probability term. And check the consistency: the expected value of the stock is 10.5, so for this to be a risk-neutral distribution the forward must be 10.5, which with zero rates means a spot of 10.5. Saying that shows you know a price cannot be computed from a distribution without knowing which measure it is under, which is developed in no-arbitrage pricing.

Continuous: probability lives on intervals

Now let the variable take any value in a range. Something breaks immediately. If a share price in a year can be any positive real number, what is the probability it is exactly $100.0000\ldots$? It has to be **zero**: there are uncountably many candidate values, and if each had positive probability the total would be infinite.

KEY IDEA

For a continuous random variable, **single points have probability zero, and only intervals have positive probability**. Probability is therefore described not by a mass at each point but by a **density** $f(x)$, whose integral over an interval gives the probability of landing in it:

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f(x)\,dx, \qquad \int_{-\infty}^{\infty} f(x)\,dx = 1.$$

The density is not a probability and can exceed 1; only areas under it are probabilities. Sums over masses become integrals over densities, and that single substitution converts every discrete formula in this chapter into its continuous counterpart.

A useful consequence: for a continuous variable, $\mathbb{P}(X \leq a)$ and $\mathbb{P}(X < a)$ are the same number, because the single point contributes nothing. For a discrete variable they differ by exactly $p(a)$, and that difference is where careless answers go wrong.

REAL INTERVIEW QUESTION (Société Générale · Delta One)

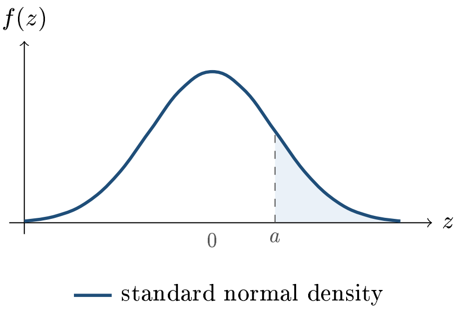
Draw the curve of a normal distribution and give the coordinates of each point.

WHAT THEY TEST / ANSWER

Draw the bell, then hang numbers on it, because "each point" means the landmarks, not a table of values. For the standard normal, mean 0 and standard deviation 1, with density $f(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$:

- **The peak**, at $z = 0$, of height $1/\sqrt{2\pi} = 0.3989$. That the maximum is about 0.4 rather than 1 is the detail that shows you know a density is not a probability.
- **The inflection points**, at $z = \pm 1$, exactly one standard deviation out, of height 0.2420. The curve is concave between them and convex outside, which is how the eye locates the standard deviation on a drawn bell.
- **The tails**, at $z = \pm 2$ the height is 0.0540, and the curve approaches zero without ever reaching it, so the support is the whole real line.
- **The areas**, which are what actually get used: about 68% within one standard deviation, 95% within two, 99.7% within three. Exactly, those are 0.6827, 0.9545 and 0.9973.

Finish with symmetry and scaling: the curve is symmetric about the mean, so $\mathbb{P}(Z \le -a) = \mathbb{P}(Z \ge a)$ and the median equals the mean, and a general normal is obtained by $X = \mu + \sigma Z$, which shifts the peak and stretches the width without changing the shape. If the interviewer is on an equity derivatives desk, add the sentence they are waiting for: share prices are not modelled as normal but as *lognormal*, so it is the log return that gets this bell, which keeps the price positive and produces a right-skewed distribution for the price itself, in probability distributions.



the shaded area, not any height on the curve, is $\mathbb{P}(Z > a)$

The object that covers both: the CDF

Densities and masses look different. The **cumulative distribution function**

$$F(x) = \mathbb{P}(X \le x)$$

is defined identically for both, is always non-decreasing, and always runs from 0 to 1. Its shape tells you which case you are in: a **staircase** with a jump at each possible value for a discrete variable, where the jump height *is* the probability mass; a **smooth increasing curve** for a continuous one, whose slope is the density.

That reading gives the general statement: **a jump in the CDF is an atom of probability, and a slope is a density.** A distribution is allowed to have both, and such mixed cases are not exotic in finance, they are the normal situation.

WORKED EXAMPLE

Take a call option struck at K on a continuously distributed share price. The share price S_T is continuous, with no atoms. But the *payoff* $\max(S_T - K, 0)$ is not: it equals exactly zero on the whole event $\{S_T \le K\}$, which has positive probability. So the payoff has an **atom at zero** of size $\mathbb{P}(S_T \le K)$ and a continuous density above it. A single transformation turned a purely continuous variable into a mixed one.

The same thing, more sharply, happens with a digital: a payoff of 1 above K and 0 below turns a continuous variable into a purely *discrete* one taking two values. This is why digitals and barriers are awkward to hedge. The instrument sits exactly on the boundary between the two worlds this section has been separating, and everything that is difficult about them, the exploding delta at the strike, the replication by a narrow spread, follows from that.

INTERVIEW TRAP

Four errors. (1) Treating a density value as a probability: $f(x)$ can be larger than 1, and only areas are probabilities. (2) Carrying the discrete habit into the continuous case and asking for $\mathbb{P}(S_T = 100)$, which is zero, when the question needed an interval. (3) Carrying the continuous habit into the discrete case and treating $\mathbb{P}(X \le a)$ and $\mathbb{P}(X < a)$ as equal, when they differ by the mass at a. (4) Adding probabilities of events that overlap, which is what inclusion-exclusion exists to prevent.

Where this goes next

The specific distributions that fill this framework are probability distributions, their summary numbers are expectation, variance and higher moments, and how information changes a probability is conditional probability and Bayes' theorem. The puzzles built on all of it live in brain teasers, and the reason any of this prices an option is no-arbitrage pricing.

7.2 Probability Distributions (Bernoulli, Binomial, Poisson, Normal, Lognormal)

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Five distributions do almost all the work in a markets interview. Knowing their shapes, their parameters and how they turn into one another is worth more than knowing twenty of them, because the questions that get asked are about which one applies and why.

Moments are expectation and variance, the limit theorems that connect these distributions are convergence, and the model that uses the last of them is Black-Scholes. What is owned here is the distributions themselves.

The discrete three

Bernoulli(p) is one trial with two outcomes. Mean p, variance $p(1-p)$, which is maximised at $p = \frac{1}{2}$: an even bet is the most uncertain one. It is the atom everything else is built from, and in finance it is a default, a barrier touch, or a digital payoff.

Binomial(n,p) counts successes in n independent Bernoulli trials. Mean np, variance $np(1-p)$. It is the distribution of the one-period-repeated tree in no-arbitrage pricing, and it is the right answer whenever a question involves a fixed number of independent yes-or-no events.

Poisson(λ) counts events in a fixed interval when they arrive independently at a constant average rate. Mean λ, variance λ: the equality is its signature, and noticing it is often the point of the question. It is the limit of a binomial when n is large and p small with $np = \lambda$ held fixed, which is why it models rare events: defaults in a large portfolio, and jumps in jump-diffusion models.

The continuous two that matter

Normal(μ, σ^2) is the limit of sums. The central limit theorem says that averages of many independent contributions converge to it regardless of what they individually looked like, which is why it appears everywhere something is an accumulation of small independent effects, and why it is a poor model when the effects are neither small nor independent.

Lognormal is what you get when the *logarithm* is normal, so $S = e^X$ with X normal. It is strictly positive, right-skewed, and it is the distribution Black-Scholes puts on the price of an asset.

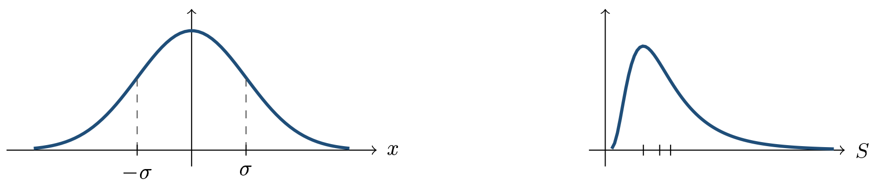
KEY IDEA

Prices are lognormal because **returns compound**. A price cannot go negative and it moves in percentages rather than in absolute amounts, so the thing that accumulates additively is the *log* return, not the price. Add many independent log returns, apply the central limit theorem, and the log price is normal, so the price is lognormal. That is the entire argument, and it is worth being able to give in two sentences.

The consequence people get wrong is that the lognormal's three centres are different, and they are ordered. With $\ln S \sim N(\mu, \sigma^2)$:

$$\text{mode} = e^{\mu-\sigma^2} \; < \; \text{median} = e^{\mu} \; < \; \text{mean} = e^{\mu+\sigma^2/2}.$$

So the average outcome is above the typical one, and the gap widens with volatility. That gap is the same $\sigma^2/2$ term that appears as volatility drag in implied vs realised volatility, and it is why a fund can have a positive average return and a median investor who lost money.



Left, —— the normal density: symmetric, with its inflection points at $\pm\sigma$ marked on the axis. Right, —— the lognormal: zero at the origin, strictly positive, right-skewed. Its three x-axis ticks are, left to right, the mode $e^{-\sigma^2} = 0.70$, the median 1 and the mean $e^{\sigma^2/2} = 1.20$, for $\sigma = 0.6$ and $\mu = 0$: mode below median below mean.

Drawing the bell and naming its landmarks, the peak at $1/\sqrt{2\pi}$, the inflection points at $\pm\sigma$ and the 68/95/99.7 areas, is discrete and continuous probability. One constant from it is worth carrying here, because it reappears where nobody expects it: the peak height $1/\sqrt{2\pi} = \mathbf{0.399}$ is the same 0.4 that shows up in the at-the-money option approximation, since an at-the-money option's value is proportional to the density at the money. A candidate who can say why those two numbers are the same number is doing better than one who has memorised either.

REAL INTERVIEW QUESTION (BNP · Trading)

Plot the distribution of a log-normal law.

WHAT THEY TEST / ANSWER

Draw it, then say the three things about it that a normal does not have.

The shape: zero at the origin and for all negative values, since the variable cannot be negative; rising to a single peak fairly early; then a long right tail decaying slowly. Do not draw it symmetric, which is the commonest error and the whole reason the question is asked.

The three centres are distinct and ordered: mode, then median, then mean, left to right. Give the formulas if you can, $e^{\mu-\sigma^2}$, e^{μ} and $e^{\mu+\sigma^2/2}$, and note that they separate further as volatility rises.

Why we use it: because the log of the variable is normal, and log returns are what add up. Say the two consequences that matter on a desk. A lognormal price can never be negative, which is the right property for a share price and the wrong one for an interest rate, which is why rates models moved away from lognormal after rates went negative. And the mean sitting above the median is exactly the $\sigma^2/2$ that separates the arithmetic and geometric average return.

If they ask you to relate the drawing to options, the honest link is that the Black-Scholes price is an expectation under a lognormal law, and that the right tail you have just drawn is what a call option is paid on, which is why the price rises with volatility: widening the distribution adds mass to the tail you own and none to the part you have already given away.

The relationships worth knowing

Interviewers like these because they test structure rather than recall.

- **Bernoulli** repeated n times gives **binomial**.
- **Binomial** with n large and p small, np fixed, tends to **Poisson**. Rare events in many trials.
- **Binomial** with n large and p fixed tends to **normal**. This is the central limit theorem, and it is also why a binomial tree converges to Black-Scholes.
- **Normal** exponentiated gives **lognormal**. Additive becomes multiplicative.
- **Exponential** is the waiting time between Poisson events, and it is memoryless, which is the property that makes it the right model for a hazard rate.

REAL INTERVIEW QUESTION (Société Générale · Market Risk)

What law could the return distribution follow, and do you know any other law?

WHAT THEY TEST / ANSWER

Give the standard answer, then dismantle it, because the second half is what a market risk desk is asking for.

The standard answer: log returns are usually modelled as normal, so prices are lognormal. It is the assumption behind Black-Scholes and behind the simplest value-at-risk calculations, and it is justified by the central limit theorem applied to many small independent price increments.

Then say why it is wrong, with the three specific failures, because naming them is what separates this answer.

- **Fat tails.** Real returns produce far more large moves than a normal allows. A five standard deviation daily move should be a once-in-several-thousand-year event and happens repeatedly per decade, so excess kurtosis is positive rather than zero.
- **Negative skew.** In equities, large moves are more often down than up, so the distribution is not symmetric. This is the same asymmetry that shows up as volatility skew in option prices.
- **Volatility clustering.** Returns are close to uncorrelated but their *squares* are strongly autocorrelated, so the independence assumption fails even where the marginal shape is tolerable. Calm periods and turbulent periods come in runs.

The alternatives, in the order a risk desk would reach for them: **Student-t**, the simplest fix, giving fatter tails with one extra parameter; **jump-diffusion**, a normal plus a Poisson jump component, which produces fat tails and skew with an economic story attached; **stochastic volatility** models such as Heston, which generate clustering and skew endogenously; **GARCH**, the econometric route to clustering; and **extreme value theory** for the tail alone, which is what a stress-testing framework actually uses.

Close on the judgement, since this is a market risk seat: the normal is kept not because anyone believes it but because it is tractable and aggregates cleanly across a book. The correct posture is to use it for the centre of the distribution and to size tail risk with something else entirely, which is why value-at-risk is reported alongside stress tests rather than trusted on its own.

INTERVIEW TRAP

1. **Saying returns are lognormal.** *Prices* are lognormal; log *returns* are normal. Getting this backwards is an instant tell.
2. **Drawing the lognormal symmetric.** It is skewed right and bounded at zero, and the drawing is the question.
3. **Confusing the mean and the median.** They differ by the $\sigma^2/2$ term, and the difference is the entire content of several interview questions.
4. **Reaching for Poisson without the rare-event condition.** It needs a constant arrival rate and independence; correlated defaults break it completely.
5. **Forgetting that Poisson's mean equals its variance.** It is the fastest way to identify or reject it in a puzzle.
6. **Assuming the central limit theorem applies.** It needs independence and finite variance. Markets routinely violate the first and occasionally the second.

VISUAL / CODE TO BUILD: Interactive

A distribution bench with two panels. On the left the reader picks a distribution and moves its parameters, watching the density redraw with the mean, median and mode marked as separate vertical lines, so the lognormal's three centres visibly separate as volatility rises while the normal's stay locked together. On the right, the relationships are shown as animations rather than claims: a binomial with a slider on n visibly converges to a normal as n grows with p fixed, and to a Poisson as p shrinks with np held constant. A third mode overlays an empirical histogram of real daily equity returns on a fitted normal, so the fat tails and the negative skew are seen rather than asserted, with a counter reporting how many observations fall beyond four standard deviations against how many the normal predicts.

7.3 Expectation, variance, higher moments

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A random variable is an object with infinitely many numbers hidden inside it. A desk cannot trade an object; it trades a handful of numbers extracted from it. Those numbers are the **moments**: the average (expectation), the spread (variance), the asymmetry (skewness), the tail weight (kurtosis). Almost every sentence spoken on a trading floor is a statement about one of them. "This option is worth 4.20" is an expectation. "The stock is a 25-vol name" is a standard deviation. "Equity index skew is steep" is a third moment. "Nothing is ever normal" is a fourth moment.

This section owns those four numbers for the whole book. It builds them from the definitions in discrete and continuous probability, then shows the three or four algebraic facts that do all the work in an interview.

Expectation: the probability-weighted average

INTUITION

Imagine playing a game a million times and writing down your result each round. At the end you add the million numbers and divide by a million. That single number is the expectation. You do not actually need the million rounds: each possible outcome shows up in proportion to its probability, so you can shortcut straight to "value times how often it happens, summed over everything that can happen".

For a discrete random variable X taking values $x_1, x_2, \dots$ with probabilities $p_i = \mathbb{P}(X = x_i)$,

$$\mathbb{E}[X] \;=\; \sum_i x_i\, p_i.$$

For a continuous random variable with density f, the sum becomes an integral,

$$\mathbb{E}[X] \;=\; \int_{-\infty}^{\infty} x\, f(x)\, dx.$$

Same idea in both cases: sweep over every possible value, weight it by how much probability sits there. More generally, for any function g,

$$\mathbb{E}[g(X)] \;=\; \sum_i g(x_i)\, p_i \qquad \text{or} \qquad \mathbb{E}[g(X)] \;=\; \int_{-\infty}^{\infty} g(x)\, f(x)\, dx.$$

Note what this does *not* say: you do not need the distribution of $g(X)$, only the distribution of X. That small remark is the whole of option pricing in embryo, since a European option price is $\mathbb{E}[g(S_T)]$ for a payoff function g, computed under the risk-neutral distribution of S_T (see the Black-Scholes model).

KEY IDEA

Linearity is the workhorse. For any random variables X, Y and constants a, b,

$$\mathbb{E}[aX + bY] \;=\; a\,\mathbb{E}[X] + b\,\mathbb{E}[Y].$$

This holds **always**, with no independence assumption whatsoever. Nine expectation brain teasers out of ten are solved by writing the quantity as a sum of simple pieces and taking the expectation of each piece separately, even when the pieces are horribly dependent.

The most useful special case is the **indicator trick**. Let $\mathbf{1}_A$ be the random variable worth 1 if event A happens and 0 otherwise. Then $\mathbb{E}[\mathbf{1}_A] = \mathbb{P}(A)$. So counting problems ("how many of these events happen on average?") collapse into a sum of probabilities.

WORKED EXAMPLE

Roulette, and why the house edge does not care how you bet. Take a European single-zero wheel: 37 pockets numbered 0 to 36, and a straight-up bet on one number pays 35 to 1 (you keep your stake and receive 35 times it).

All 100 on one number. You gain +3500 with probability 1/37 and lose your 100 with probability 36/37:

$$\mathbb{E}[\text{PnL}] = \tfrac{1}{37}(3500) + \tfrac{36}{37}(-100) = \frac{3500 - 3600}{37} = -\frac{100}{37} \approx -2.70.$$

50 on each of two numbers. With probability 2/37 one of them hits: that chip returns $50 \times 35 = 1750$ while the other chip loses 50, so the PnL is +1700. With probability 35/37 neither hits and you lose 100:

$$\mathbb{E}[\text{PnL}] = \tfrac{2}{37}(1700) + \tfrac{35}{37}(-100) = \frac{3400 - 3500}{37} = -\frac{100}{37} \approx -2.70.$$

Identical, and linearity explains why without any arithmetic. Every euro placed straight-up has expectation $\frac{1}{37}(35) + \frac{36}{37}(-1) = -\frac{1}{37}$. Total expectation is the sum over euros, so it is $-100/37$ however you scatter them. The house edge is $1/37 \approx 2.70\%$ of turnover, and no allocation of a fixed stake can move it.

REAL INTERVIEW QUESTION (Barclays · Credit)

At roulette: you put 100 euros on one number, what is your expected gain? You put 100 euros on two numbers, what is your expected gain?

WHAT THEY TEST / ANSWER

What they test is whether you reach for linearity or for a case tree. Give the number, $-100/37 \approx -2.70$ euros, then immediately say "and it is the same for two numbers, because expectation is linear: every euro staked is worth $-1/37$ of a euro, so spreading the stake cannot change the total." State your convention out loud (European wheel, 37 pockets, 35 to 1) since an American double-zero wheel has 38 pockets and an edge of $2/38 \approx 5.26\%$. The follow-up is usually "so what does change?": the answer is the *variance*, which collapses as you spread the stake, and that is exactly why a casino is happy to take many small bets and a bank runs position limits rather than expectation limits.

REAL INTERVIEW QUESTION (Morgan Stanley · QIS)

N coats, N people: what is the expected number of people who get the right coat?

WHAT THEY TEST / ANSWER

The answer is 1, for every N. Let $X_i = \mathbf{1}\{\text{person } i \text{ gets their own coat}\}$. By symmetry $\mathbb{P}(X_i = 1) = 1/N$, so $\mathbb{E}\big[\sum_i X_i\big] = \sum_i \mathbb{E}[X_i] = N \cdot \frac{1}{N} = 1$. What is really being tested is whether you notice that the X_i are strongly dependent (if $N-1$ people have their own coat, the last one certainly does) and whether that stops you. It does not: linearity never needed independence. The follow-up trap is the variance, where dependence *does* matter. For a uniform random permutation, $\mathbb{E}[X(X-1)] = N(N-1) \cdot \frac{1}{N(N-1)} = 1$, so $\mathbb{E}[X^2] = 2$ and $\mathrm{Var}(X) = 2 - 1^2 = 1$ as well, for every $N \geq 2$.

Variance: how far the outcomes scatter

Expectation alone is a poor description of a risk. A coin flip paying ± 1 and a coin flip paying $\pm 1{,}000{,}000$ both have expectation zero, and no desk would treat them alike. The second number measures spread.

INTUITION

Take every outcome, measure how far it lands from the average, square that distance so overshoots and undershoots cannot cancel, and average the squares. That is the variance. Squaring is what makes the algebra work, and it is also what makes the units wrong, so we take the square root at the end and call it the standard deviation, which lives in the same units as the thing itself.

$$\operatorname{Var}(X) \;=\; \mathbb{E}\big[(X-\mathbb{E}[X])^2\big] \;=\; \mathbb{E}[X^2]-\big(\mathbb{E}[X]\big)^2, \qquad \sigma(X) \;=\; \sqrt{\operatorname{Var}(X)}.$$

The second form is the one you compute with. Two consequences you should be able to quote instantly:

- $\operatorname{Var}(aX+b) = a^2\operatorname{Var}(X)$. Adding a constant shifts the distribution without stretching it, so it cannot change the spread; multiplying by a scales the standard deviation by $|a|$.
- $\mathbb{E}[X^2] = \operatorname{Var}(X)+\mathbb{E}[X]^2 \geq \mathbb{E}[X]^2$, so a squared quantity is always worth more than the square of the average. Variance is exactly the gap.

For a sum, the covariance term appears and does not go away:

$$\operatorname{Var}(aX+bY) \;=\; a^2\operatorname{Var}(X)+b^2\operatorname{Var}(Y)+2ab\operatorname{Cov}(X,Y), \qquad \operatorname{Cov}(X,Y)=\rho\,\sigma_X\sigma_Y.$$

Only when $\operatorname{Cov}(X,Y)=0$ do variances simply add. What ρ means, how it behaves, and why zero correlation is weaker than independence are developed in independence and correlation.

WORKED EXAMPLE

Two-asset portfolio. Half your money in A ($\sigma_A = 20\%$), half in B ($\sigma_B = 30\%$), correlation $\rho = 0.4$. The portfolio variance is

$$0.5^2(0.20)^2+0.5^2(0.30)^2+2(0.5)(0.5)(0.4)(0.20)(0.30) = 0.0100+0.0225+0.0120 = 0.0445,$$

so $\sigma_P = \sqrt{0.0445} = 0.2110$, that is 21.1%. The naive weighted average of the two volatilities would have been 25%. The 3.9 points of difference are the whole of diversification, and they come entirely from the fact that $\rho < 1$.

WORKED EXAMPLE

Why volatility scales with $\sqrt{T}$. Suppose daily returns are independent with the same daily standard deviation σ_d. Over n days the return is the sum of n daily returns, so the *variances* add: $\operatorname{Var} = n\,\sigma_d^2$, and therefore $\sigma_n = \sigma_d\sqrt{n}$. With $\sigma_d = 1\%$ and 252 trading days, the annualised figure is $1\% \times \sqrt{252} = 1\% \times 15.87 \approx 15.9\%$. This is the reason every quoted volatility is annualised and why the square root is everywhere in option pricing. Independence is doing real work here: with autocorrelated returns the $\sqrt{n}$ rule breaks, which is the practical content of implied versus realized volatility.

Which average? The moment-level statement

"The average" is not one object. Which of the three classical means is the right one depends on whether the quantities add, multiply, or are rates over a fixed denominator, and that choice, with the return-series version of it, is developed in descriptive statistics.

What belongs here is the part that follows from this section's own machinery. For positive numbers the three are always ordered,

$$\text{harmonic} \;\leq\; \text{geometric} \;\leq\; \text{arithmetic},$$

with equality only when every value is identical, and that ordering is not a coincidence: it is **Jensen's inequality** applied to log (concave) and to $1/x$ (convex), which is why the gaps widen with dispersion. The moment-level consequence is the one to carry into pricing: to first order the compounded growth rate is the arithmetic mean minus $\sigma^2/2$, so the more volatile the series, the further the compounded outcome falls below the naive average. That $-\sigma^2/2$ is the same term that appears in the drift of a lognormal stock price in Black-Scholes, and Jensen itself is developed below.

REAL INTERVIEW QUESTION (Société Générale · Rates)

A car does one lap of a circuit at 60 km/h. What constant speed must it do on the second lap so that its average speed over the two laps is 120 km/h?

WHAT THEY TEST / ANSWER

It is impossible. Average speed is total distance over total time, which is a harmonic average of the lap speeds, not an arithmetic one. Let the lap length be L. Two laps at an average of 120 km/h would take $2L/120 = L/60$ hours in total, and the first lap alone already consumed $L/60$ hours. The second lap would have to be completed in zero time. What they test is whether you reflexively answer "180" by averaging the two speeds. Say the words "average speed is a harmonic mean, so I cannot just average the numbers", then produce the budget-of-time argument. The general result is worth knowing: to double your average over two equal laps you would need infinite speed, and more generally the target average can never exceed twice the first-lap speed.

Higher moments: shape beyond spread

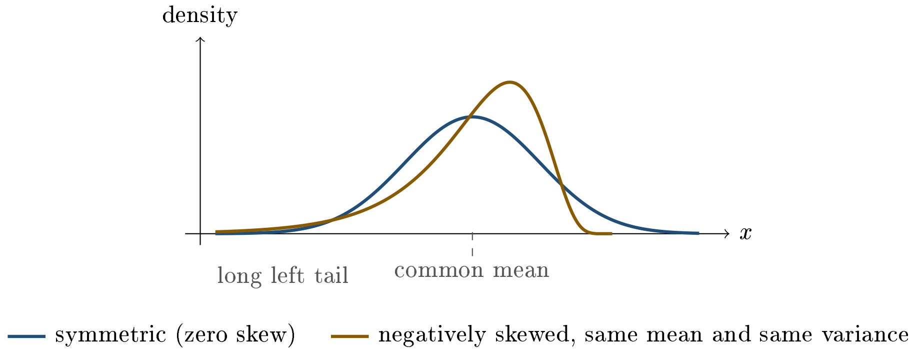
Two distributions can share a mean and a variance and still be completely different animals. The next two numbers describe how.

The k-th **central moment** is $\mu_k = \mathbb{E}[(X-\mu)^k]$. Dividing by σ^k removes the units and gives the standardised versions everyone quotes:

$$\text{skewness} \;=\; \frac{\mathbb{E}\big[(X-\mu)^3\big]}{\sigma^3}, \qquad \text{kurtosis} \;=\; \frac{\mathbb{E}\big[(X-\mu)^4\big]}{\sigma^4}, \qquad \text{excess kurtosis} \;=\; \text{kurtosis} - 3.$$

Skewness measures asymmetry. The odd power keeps the sign, so outcomes far below the mean contribute negatively and outcomes far above contribute positively. Negative skew means a long left tail: most days are quietly fine, and the rare bad day is very bad. Equity indices are the textbook case, and they are why index put options trade at a higher implied volatility than calls, the shape described in the volatility smile and skew and, for single stocks and indices specifically, in equity skew and smile.

Kurtosis measures tail weight. The even fourth power ignores direction and punishes distance, so it is dominated by the extremes. The Gaussian benchmark is exactly 3, which is why the market speaks in *excess* kurtosis: positive excess kurtosis means fatter tails and a more peaked centre than a normal distribution with the same variance. Real asset returns have positive excess kurtosis at every horizon that matters, which is the single most consequential failure of the Gaussian assumption inside Value at Risk and stress testing.



REAL INTERVIEW QUESTION (Société Générale · Market Risk)

What are skewness and kurtosis?

WHAT THEY TEST / ANSWER

Give the formulas, then say what each one *buys you*, because the second half is what separates a memoriser from someone who has looked at a return series. Skewness is the standardised third central moment and measures asymmetry: equity index returns are negatively skewed, so the left tail is the long one, and that asymmetry is priced as the equity skew. Kurtosis is the standardised fourth moment; the normal distribution sits at 3, so quote excess kurtosis, and observe that essentially every financial return series has positive excess kurtosis, so a Gaussian VaR understates the tail. The follow-up on a market risk desk is practical: "what do you do about it?" The honest answers are fat-tailed or mixture distributions, historical rather than parametric VaR, and moving from VaR to Expected Shortfall, which averages over the tail instead of merely locating its edge. Note also that empirical third and fourth moments are dominated by a handful of observations, so they are noisy and unstable in small samples, a point developed in descriptive statistics.

Jensen's inequality: why convexity has value

One inequality connects all of this to what an option desk actually sells.

KEY IDEA

If g is a **convex** function, then

$$\mathbb{E}\big[g(X)\big] \;\geq\; g\big(\mathbb{E}[X]\big),$$

with strict inequality whenever X is genuinely random and g is strictly convex. For a concave g the inequality flips. In words: for a convex payoff, the average of the outcomes beats the outcome of the average.

The special case $g(x) = x^2$ recovers $\mathbb{E}[X^2] \geq \mathbb{E}[X]^2$, whose gap is the variance. The special case that pays salaries is $g(x) = \max(x - K, 0)$, the call payoff, which is convex. So

$$\mathbb{E}\big[\max(S_T - K, 0)\big] \;\geq\; \max\big(\mathbb{E}[S_T] - K, 0\big).$$

An option on a random stock is worth strictly more than an option on a stock pinned at its forward, and the gap widens with the dispersion of S_T. This is the precise reason volatility has a price at all. The trader's version of the same inequality is that positive gamma is a convex position, so it profits from movement in either direction, developed in gamma and convexity, and the purest way to trade the second moment directly is the variance swap.

VISUAL / CODE TO BUILD: Interactive

A four-moment sandbox. The reader drags four sliders (mean, standard deviation, skewness, excess kurtosis) and the panel redraws a density matching those moments, side by side with a fixed normal reference of the same mean and variance. Three readouts update live: (i) the 1% quantile, so the reader watches VaR move when only the third and fourth moments change; (ii) the price of a 10% out-of-the-money put; (iii) the price of a 10% out-of-the-money call. The intended "aha" is that mean and variance can be held perfectly constant while the tail risk and the option prices move a long way. Implement with a Cornish-Fisher or Gram-Charlier expansion for the density, and expose the four moments plus the three readouts as a table for accessibility.

INTERVIEW TRAP

Five ways candidates lose points here. (1) Claiming linearity of expectation "requires independence". It never does. Independence is needed for $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$ and for variances to add, not for expectations to add. (2) Writing $\mathbb{E}[g(X)] = g(\mathbb{E}[X])$. This is false for every non-linear g, and the error has a name and a sign: Jensen. (3) Averaging rates arithmetically. Speeds over a fixed distance, and prices per unit for a fixed budget, are harmonic averages. (4) Quoting kurtosis where excess kurtosis is meant, or forgetting that the normal benchmark is 3. (5) Assuming every distribution has moments at all. A Cauchy distribution has no mean, and a Student-t with 3 degrees of freedom has no finite kurtosis. If the moment does not exist, no amount of sample data will make your estimate converge, which is the failure mode behind many blown-up tail models.

Where this goes next

The named distributions whose moments you should know cold are in probability distributions. Conditioning turns expectation into the single most powerful computational tool in the subject, via the tower property, in conditional probability. What happens to an average as the sample grows, and in what sense it converges, is convergence concepts, which is also what licenses Monte Carlo pricing. A process whose conditional expectation never drifts is a martingale, the formal backbone of risk-neutral pricing. On the estimation side, turning realised data into estimated moments is descriptive statistics. And the teasers that live on linearity and the indicator trick are collected in brain teasers.

7.4 Independence and correlation

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Correlation is the most used and most abused number in finance. It is the parameter that prices a basket, that makes diversification work, that decides whether a worst-of is cheap, and that stops behaving exactly when the answer matters. This section is the probability underneath it. What desks do with it lives in equity correlation and dispersion strategies.

Two different ideas

Independence means knowing X tells you nothing about Y: the joint distribution factorises, $\mathbb{P}(X \in A,\, Y \in B) = \mathbb{P}(X \in A)\,\mathbb{P}(Y \in B)$. It is a statement about the whole distribution.

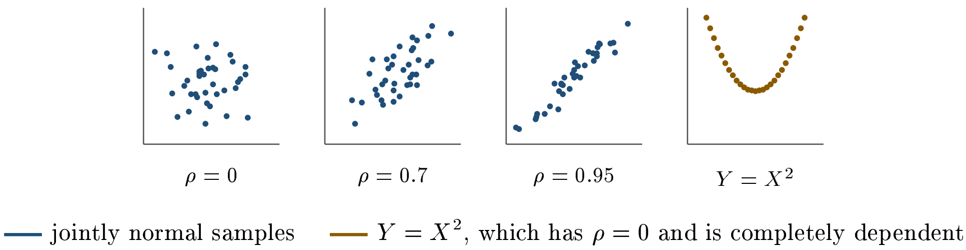
Correlation measures one specific thing: the strength of the *linear* relationship between them. Define covariance and then standardise it,

$$\mathrm{Cov}(X,Y) = \mathbb{E}\big[(X - \mathbb{E}X)(Y - \mathbb{E}Y)\big], \qquad \rho_{XY} = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y} \in [-1, 1],$$

where the bounds come from the Cauchy-Schwarz inequality, with $|\rho| = 1$ exactly when Y is an affine function of X.

The relationship between the two ideas runs one way only.

> KEY IDEA
>
> Independent implies uncorrelated. **Uncorrelated does not imply independent.** The standard counterexample takes one line: let X be standard normal and $Y = X^2$. Then $\mathrm{Cov}(X,Y) = \mathbb{E}[X^3] = 0$, so they are exactly uncorrelated, and yet Y is a deterministic function of X, which is as dependent as two variables can be. Correlation sees only the straight line through the data, and there is none.



The fourth panel is the one to remember. It has the same correlation as the first, zero, and none of the same behaviour.

The one formula that does the work

Almost every application of correlation in this book is a variance of a sum:

$$\mathrm{Var}(aX + bY) \;=\; a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\,\rho\,\sigma_X\sigma_Y,$$

and in general, for a weighted sum, $\mathrm{Var}\big(\sum_i w_i X_i\big) = \sum_{i,j} w_i w_j \rho_{ij} \sigma_i \sigma_j$. That is the basket variance formula, the portfolio variance formula and the source of the correlation exposure in every multi-asset product.

REAL INTERVIEW QUESTION (Natixis · EQD)

I have a call on a basket, 50% on underlying one and 50% on underlying two. What is the volatility of this basket? What is the impact of correlation on the price of this call?

WHAT THEY TEST / ANSWER

The volatility comes straight from the variance of a sum with equal weights:

$$\sigma_B^2 \;=\; \tfrac{1}{4}\sigma_1^2 + \tfrac{1}{4}\sigma_2^2 + \tfrac{1}{2}\rho\,\sigma_1\sigma_2.$$

Check the two boundary cases out loud, since that is what shows you understand rather than recall. At $\rho = 1$ this collapses to $\sigma_B = \frac{1}{2}(\sigma_1 + \sigma_2)$, the simple average, because the two assets are then the same asset. At $\rho = -1$ it collapses to $\frac{1}{2}|\sigma_1 - \sigma_2|$, which is zero if the two volatilities are equal, since the basket is then a constant. And with $\sigma_1 = \sigma_2 = 30\%$ and $\rho = 0$, the basket volatility is $30\%/\sqrt{2} = 21.2\%$, which is the diversification effect in one number. **The impact on the call** follows immediately: basket volatility is increasing in ρ, and a call is increasing in volatility, so a **long basket call is long correlation**. The option-pricing side of that, and why the basket is not exactly lognormal even when its components are, belongs to dispersion strategies; what this section gives you is the reason, which is the $2ab\rho\sigma_1\sigma_2$ term and nothing more.

Where diversification stops

Extend the same formula to n assets, each with volatility σ, each pair correlated ρ, equally weighted:

$$\sigma_B^2 \;=\; \sigma^2\left(\frac{1}{n} + \frac{n-1}{n}\rho\right) \xrightarrow[n\to\infty]{} \sigma^2\rho.$$

So the basket volatility does not go to zero as you add names; it goes to $\sigma\sqrt{\rho}$.

WORKED EXAMPLE

Thirty percent volatility on every name, pairwise correlation 0.3.

Number of names	1	5	10	50	limit
Basket volatility	30.0%	19.9%	18.2%	16.8%	16.4%

Five names capture most of the available diversification and fifty add almost nothing. The floor at $\sigma\sqrt{\rho} = 30\% \times \sqrt{0.3} = 16.4\%$ is the part that cannot be diversified away, and it is why a portfolio manager who owns two hundred stocks still has market risk, and why index volatility is far from zero.

Correlations cannot be chosen freely

A correlation matrix is not a collection of independent numbers. It must be **positive semi-definite**, because $\mathrm{Var}(\sum w_i X_i) = w^\top \Sigma w$ has to be non-negative for every portfolio w, and that constrains the entries against each other.

Estimating it, and what goes wrong

REAL INTERVIEW QUESTION (Société Générale · EQD)

How do you calculate the correlation between an asset A and an asset B?

WHAT THEY TEST / ANSWER

Give the recipe precisely, then the choices it hides, because the choices are the answer. **Recipe**: take a common history of prices, convert to *returns*, usually log returns, then compute $\hat{\rho} = \sum_t (r_t^A - \bar{r}^A)(r_t^B - \bar{r}^B) / \big((T-1)\hat{\sigma}_A \hat{\sigma}_B\big)$, which is the sample covariance over the product of the sample standard deviations. **The choices**: the observation frequency, since daily, weekly and monthly correlations of the same pair genuinely differ; the window length, which trades responsiveness against noise; whether returns are simple or log; and, for assets in different time zones or with different holidays, the alignment of the two series, since a stale price in one market mechanically depresses the measured correlation. Then give the two qualifications an interviewer is listening for. This is *realised* correlation, which is not the *implied* correlation that a basket option prices at, and the gap between them is a tradable spread. And the estimator assumes the relationship is linear and stable, when the empirical fact is that correlations rise together in stressed markets, so the number you estimate in calm conditions is systematically the wrong one for the scenario you most need it in. If you want a measure that is robust to the linearity assumption, mention rank correlation such as Spearman's, and copulas as the general framework for separating the marginal distributions from the dependence between them.

INTERVIEW TRAP

Four errors. (1) Reading zero correlation as independence: $Y = X^2$ is the counterexample and it takes one line. (2) Treating the entries of a correlation matrix as free parameters, when positive semi-definiteness ties them together and can rule a value out entirely. (3) Expecting diversification to remove risk: the basket volatility floor is $\sigma\sqrt{\rho}$, not zero. (4) Using a calm-market correlation estimate for a stress scenario, when the empirical regularity is that correlations converge toward one exactly when the diversification is being relied on.

VISUAL / CODE TO BUILD: Interactive

A correlation sandbox with three panels. Panel A is a scatter plot with a correlation slider, so the reader watches the cloud tighten into a line, plus a menu of dependence structures that all have $\rho = 0$ (the parabola, a circle, a checkerboard) to make the point that the coefficient does not see them. Panel B is the diversification curve: the reader sets σ and ρ and watches basket volatility fall as names are added and then flatten at $\sigma\sqrt{\rho}$, with the floor drawn as a horizontal asymptote. Panel C is a correlation matrix the reader edits by hand, with a live eigenvalue readout that turns red the moment the matrix stops being positive semi-definite, and a button that projects it back to the nearest valid matrix so the repair step is something they perform rather than read about.

Where this goes next

The moments this section builds on are in expectation, variance and higher moments. The market versions of correlation are equity correlation and credit correlation, the trade that expresses a view on it is dispersion strategies, and the risk application that depends most heavily on the correlation matrix behaving is value at risk.

7.5 Conditional probability

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Probability without information is a starting point. Every real question arrives with something already known, and the whole skill is updating cleanly on it. This section builds the three tools that do the updating: restricting the sample space, the multiplication rule, and the law of total probability. Inverting a conditional probability, which is one specific further step, is Bayes' theorem, and the collection of puzzles built on all of this is brain teasers. The questions used below are here as teaching vehicles for the three tools.

The definition, and the picture behind it

For events A and B with $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A \mid B) \;=\; \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

INTUITION

Learning that B happened does not change the world, it changes the *list of worlds you are still in*. Everything outside B is crossed off, and what remains has to add up to one again, so you divide by $\mathbb{P}(B)$ to renormalise. Conditioning is cropping the picture and rescaling it, nothing more.

That picture is also the most reliable way to answer a hard-looking question: rather than reaching for a formula, **enumerate the outcomes consistent with what you were told, and count**. The formula is the bookkeeping; the crossing-off is the idea.

REAL INTERVIEW QUESTION (BNP · Trading)

There are 100 coins: 50 have both faces black, and 50 have one white face and one black face. A coin lies on the table showing a black face. What is the probability that the other face is white?

WHAT THEY TEST / ANSWER

One third, and the reason most people say one half is that they condition on the *coin* when the information is about the *face*.

Count faces, not coins. The 50 double-black coins contribute $50 \times 2 = 100$ black faces. The 50 mixed coins contribute one black face each, so 50 more. Being shown a black face means you are equally likely to be looking at any one of those $100 + 50 = 150$ black faces. Of them, exactly 50 sit on a coin whose other side is white. So

$$\mathbb{P}(\text{other face white} \mid \text{shown black}) = \frac{50}{150} = \frac{1}{3}.$$

Say the diagnosis out loud, because it is the transferable part: the natural but wrong answer treats the two coin types as equally likely *given* the observation, when in fact a double-black coin is twice as likely to have produced a black face. Information about an outcome reweights the causes that could have produced it, in proportion to how readily each one produces it. That is the sentence that also solves the three-prisoners problem, the Monty Hall problem and every medical-testing question, and it is formalised in Bayes' theorem.

Tool two: the multiplication rule

Rearranging the definition gives the form that is actually used:

$$\mathbb{P}(A \cap B) = \mathbb{P}(B)\,\mathbb{P}(A \mid B),$$

and chaining it along a sequence,

$$\mathbb{P}(A_1 \cap A_2 \cap \cdots \cap A_n) = \mathbb{P}(A_1)\,\mathbb{P}(A_2 \mid A_1) \cdots \mathbb{P}(A_n \mid A_1 \cap \cdots \cap A_{n-1}).$$

This is how any sequential experiment is priced: walk the path, multiply the conditional probabilities as you go.

REAL INTERVIEW QUESTION (Société Générale · EQD)

An urn contains $N-1$ red balls and one blue ball. N candidates for an internship each draw one ball in turn, without replacement, and whoever draws the blue ball gets the internship. Which position, from 1 to N, has the best chance?

WHAT THEY TEST / ANSWER

They all have exactly the same chance, $1/N$, and the interviewer wants to see you prove it rather than assert it.

Take candidate k. For them to win, the first $k-1$ draws must all be red and their own draw must be blue. By the chain rule,

$$\mathbb{P}(k \text{ wins}) = \underbrace{\frac{N-1}{N} \cdot \frac{N-2}{N-1} \cdots \frac{N-k+1}{N-k+2}}_{\text{first } k-1 \text{ draw red}} \times \underbrace{\frac{1}{N-k+1}}_{k \text{ draws blue}}.$$

The product telescopes to $\frac{N-k+1}{N}$, and multiplying by the last factor leaves $\frac{1}{N}$, independent of k.

Then give the one-line argument, which is the better answer and the one that generalises: imagine the balls drawn out and laid in a row before anyone looks. The blue ball is equally likely to occupy any of the N positions, by symmetry, so every candidate has probability $1/N$. There is nothing to compute.

The intuition worth stating for the sceptic is that the later candidates face a trade-off that exactly cancels: they are less likely to get a turn at all, and more likely to win if they do. Note also what would break the symmetry, since interviewers probe it: if the draws were *with* replacement, or if the game stopped early, or if candidates saw the earlier results and could decline, the positions would no longer be equivalent.

Tool three: the law of total probability

If $B_1, \dots, B_n$ are mutually exclusive and cover everything, then for any event A

$$\mathbb{P}(A) = \sum_i \mathbb{P}(A \mid B_i)\, \mathbb{P}(B_i).$$

In words: split the world into cases, solve each case, and weight by how likely the cases are. Most difficult probability questions become easy the moment the right split is chosen, and choosing it is the skill.

REAL INTERVIEW QUESTION (JP Morgan · Rates)

A box contains 50 red balls and 50 black balls, so drawing a red one at random has probability one half. You are given a second box and may distribute the balls between the two boxes however you like. A box is then chosen at random and one ball drawn from it. How do you maximise the probability of drawing red?

WHAT THEY TEST / ANSWER

Set up the law of total probability first, because it makes the optimisation obvious. The box is chosen with probability one half each, so

$$\mathbb{P}(\text{red}) = \tfrac{1}{2}\,\mathbb{P}(\text{red} \mid \text{box } 1) + \tfrac{1}{2}\,\mathbb{P}(\text{red} \mid \text{box } 2),$$

which is the average of the two boxes' red fractions. So the problem is to split the balls to maximise that average.

The trick is to make one box *certain*. Put a single red ball alone in box 1, and the remaining 49 red and 50 black in box 2:

$$\mathbb{P}(\text{red}) = \tfrac{1}{2}(1) + \tfrac{1}{2}\left(\frac{49}{99}\right) = \tfrac{1}{2} + 0.2475 = 0.7475.$$

Just under 75%, against 50% before.

Two things finish the answer. **Why this is optimal**: box 1 cannot do better than 1, and given that, box 2 is best off holding as few black balls as possible relative to red, which means keeping every remaining ball, so any other split is worse. And **why it works at all**: the equal weighting of the boxes is what you are exploiting, since it lets one ball carry the same weight in the average as ninety-nine do. That is the general lesson, and it is the same effect that makes an equal-weighted average of unequal groups misleading, which is worth saying because it is the version of this that appears in real data rather than in a box.

Conditioning through time

When a process repeats, conditioning on "nobody has won yet" resets the problem to where it started, and the whole infinite sequence collapses into one round plus a geometric factor.

REAL INTERVIEW QUESTION (BNP · EQD)

Paul, Marc and John shoot at a basketball hoop in turn. Paul starts and scores with probability $1/2$; if he misses, Marc shoots and scores with probability $1/3$; if Marc misses, John shoots and scores with probability $1/4$. If John misses it is Paul's turn again, and so on until somebody scores, at which point the game stops. What is the probability that Marc wins?

WHAT THEY TEST / ANSWER

Two steps: one round, then the geometric sum.

One round. Marc wins in the first round if Paul misses and Marc scores: $\frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$. Nobody scores in a round with probability $\frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} = \frac{6}{24} = \frac{1}{4}$, a telescoping product that is worth pointing out because it makes the arithmetic trivial.

The sum. Conditional on a round ending with no score, the game restarts identically, so

$$\mathbb{P}(\text{Marc wins}) = \frac{1}{6} \sum_{k \geq 0} \left(\frac{1}{4}\right)^k = \frac{1}{6} \times \frac{1}{1 - 1/4} = \frac{1}{6} \times \frac{4}{3} = \frac{2}{9}.$$

Now do the check that turns a right answer into a convincing one: compute the other two and verify they sum to one. Paul wins with probability $\frac{1/2}{3/4} = \frac{2}{3}$, John with $\frac{1}{2} \cdot \frac{2}{3} \cdot \frac{1}{4} / \frac{3}{4} = \frac{1}{9}$, and $\frac{2}{3} + \frac{2}{9} + \frac{1}{9} = 1$. The game ends with probability one, so anything else means an error. Note the ordering, $\frac{2}{3} > \frac{2}{9} > \frac{1}{9}$: shooting first is worth far more than the individual accuracies suggest, because each player only shoots if everyone before them has missed.

Why any of this matters on a desk

Every price is a conditional expectation. The forward is today's expectation of a future price *given everything known today*, an implied volatility is the market's view of future movement conditional on the current state, and a survival probability read out of a credit spread is conditional on the issuer not having defaulted yet. When a desk says a number is "priced in", it means the number is already in the conditioning set.

Two habits follow. Always ask **what is being conditioned on**, because the same question has different answers under different information. And keep **conditioning separate from causation**: that defaults are more likely conditional on a wide spread does not mean the spread causes them, and the same trap in reverse is the one below.

INTERVIEW TRAP

Four errors. (1) Swapping the two sides: $\mathbb{P}(A \mid B)$ is not $\mathbb{P}(B \mid A)$, and treating them as equal is the prosecutor's fallacy, unpicked in Bayes' theorem. (2) Conditioning on the wrong object, such as the coin rather than the face, which is what makes the answer above one half instead of one third. (3) Forgetting to renormalise after restricting the sample space, so the surviving probabilities no longer sum to one. (4) Assuming independence in order to multiply, when a conditional probability was required; independence is a property to be justified, not a convenience, and it is treated in independence and correlation.

Where this goes next

Inverting a conditional probability is Bayes' theorem, the case where conditioning changes nothing is independence and correlation, and averaging under a conditional distribution is expectation, variance and higher moments. The framework underneath it all is discrete and continuous probability, and the puzzles are in brain teasers.

7.6 Bayes' Theorem

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Bayes' theorem is one line of algebra and a great deal of trouble. The algebra is the easy part and nobody gets it wrong; what interviewers are testing is whether you condition on the *right* event, and whether you remember the base rate. Both failures are common in people who can state the formula perfectly.

Conditional probability defines $P(A \mid B)$; this section is about inverting it, which is a different skill. Puzzle technique in general is brain teasers.

The statement, and the useful form

From the definition of conditional probability twice over,

$$P(A \mid B) = \frac{P(B \mid A)\,P(A)}{P(B)}, \qquad P(B) = P(B \mid A)P(A) + P(B \mid A^c)P(A^c).$$

The three pieces have names worth using: $P(A)$ is the **prior**, what you believed before; $P(B \mid A)$ is the **likelihood**, how well the evidence fits the hypothesis; and $P(A \mid B)$ is the **posterior**, what you believe after. The denominator is just a normalisation, and expanding it as above is the step that forces you to consider the alternative explanation, which is the step people skip.

KEY IDEA

The evidence tells you how *likely* the observation is under each hypothesis. It does not tell you how likely the hypothesis is. Converting one into the other requires the prior, and when the prior is small no amount of ordinary evidence rescues it. **A test being 99% accurate does not make a positive result 99% likely to be true** when the thing being tested for is rare. That single confusion is the content of most Bayes questions.

WORKED EXAMPLE

Why an accurate test is not a reliable one. A screening test detects a condition with 99% sensitivity, has a 99% specificity, and the condition affects 0.1% of the population. You test positive. What is the probability you have it?

$$P(D \mid +) = \frac{0.99 \times 0.001}{0.99 \times 0.001 + 0.01 \times 0.999} = \frac{0.00099}{0.01098} = \mathbf{9.0}\%.$$

Not 99%. The reason is visible in the denominator: among 100,000 people, 100 have the condition and 99 of them test positive, while 99,900 do not have it and 999 of them test positive anyway. There are ten times as many false positives as true ones, purely because there are a thousand times as many healthy people.

Change nothing except the prevalence, from 0.1% to 10%, and the same test gives $P(D \mid +) = \mathbf{91.7}\%$. The test did not improve. The base rate did.

That example is worth carrying into markets, because the structure recurs constantly. A signal that flags 99% of crashes and gives a 1% false alarm rate is nearly useless if crashes are rare, and a risk system tuned to catch every genuine breach will drown a desk in alerts. The same arithmetic explains why fraud detection, credit scoring and anomaly monitoring all struggle: they are inference problems with tiny priors.

The odds form, which is faster

Dividing Bayes' theorem by itself for the complementary hypothesis clears the denominator entirely:

$$\underbrace{\frac{P(A \mid B)}{P(A^c \mid B)}}_{\text{posterior odds}} = \underbrace{\frac{P(A)}{P(A^c)}}_{\text{prior odds}} \times \underbrace{\frac{P(B \mid A)}{P(B \mid A^c)}}_{\text{likelihood ratio}}.$$

Posterior odds are prior odds times the likelihood ratio, which is both easier to compute in your head and easier to reason with: evidence *multiplies* odds. In the example above the prior odds are 1:999 and the likelihood ratio is 99, giving posterior odds of 99:999, which is the 9.0% already computed. Doing it this way takes ten seconds and no paper, which is the point in an interview.

Conditioning on the right event

REAL INTERVIEW QUESTION (Société Générale · Fixed Income)

Brain teaser: a woman has two children, one of them is a girl. What is the probability that the other is a girl?

WHAT THEY TEST / ANSWER

One third, under the reading the question is normally given, and the reason it is asked is that almost everyone says one half.

Enumerate rather than reason verbally, because verbal reasoning is exactly what fails here. With two children the equally likely outcomes by birth order are $\{BB, BG, GB, GG\}$. The information "at least one is a girl" eliminates BB and leaves three equally likely cases, $\{BG, GB, GG\}$. In only one of them is the other child a girl, so the answer is $\frac{1}{3}$.

The reason the intuition fails is worth stating: the information given is about the *pair*, not about a particular child. It does not identify which child is a girl, so it removes one of four cases rather than fixing one child's sex and leaving the other free.

Then volunteer the ambiguity, because it is the strongest thing you can say and a good interviewer is waiting for it. If instead you *meet* one of the children and she is a girl, the answer is $\frac{1}{2}$: you have now conditioned on a specific child, which leaves the other child's sex untouched. Same words in English, two different conditioning events, two different answers. Naming that distinction is what the question is really testing, and it is the same distinction that makes people misread conditional statements about markets.

The assumptions are worth flagging in one clause too, since they are doing real work: boys and girls equally likely, and independent between children. Neither is exactly true and both are intended.

INTUITION

Bayes' theorem is not what is hard. Deciding precisely what you have been told is what is hard. The two-children question turns entirely on the difference between "a girl exists in this pair" and "this particular child is a girl", which are different sets and give different answers, and the habit that prevents the error is writing the event down formally before computing anything.

The complementary skill, recognising when a problem is invariant under relabelling and needs no computation at all, belongs to brain teasers, which owns puzzle technique and is where the symmetry arguments live. The two-children question stays here despite being posed as a brain teaser because it is a conditioning question rather than a puzzle-craft one.

INTERVIEW TRAP

1. **Ignoring the base rate.** The single most common error. A 99% accurate test for a one-in-a-thousand condition yields a positive result that is wrong ninety percent of the time.
2. **Confusing $P(B \mid A)$ with $P(A \mid B)$.** "Most crashes are preceded by an inverted curve" says nothing about how often an inverted curve is followed by a crash.
3. **Conditioning on the wrong event.** Write the event down formally. "At least one is a girl" and "this one is a girl" are different sets.
4. **Reaching for the theorem before checking the structure.** Some conditioning problems collapse to a one-line argument, and brain teasers covers the techniques for spotting them. Bayes is the tool when the structure genuinely requires it.
5. **Forgetting that evidence multiplies odds, not probabilities.** The odds form is faster and it is much harder to get wrong under pressure.
6. **Treating the prior as optional.** Without it there is no posterior, and choosing not to state one is choosing an implicit one.

VISUAL / CODE TO BUILD: Interactive

A base-rate machine. A population of ten thousand dots is drawn, split by prevalence into two coloured groups, and then split again by the test result, so the four cells of the confusion matrix are visible as areas rather than numbers. Sliders for prevalence, sensitivity and specificity resize the blocks live, and the posterior is reported beside them. The intended moment is holding a 99% accurate test fixed and dragging prevalence from 10% down to 0.1%, watching the true positives shrink to a sliver inside a sea of false alarms while the headline accuracy never moves. A second tab runs the same machine on a trading signal, with base rate as the frequency of the event and the two error types priced in money, so the reader can find the prevalence below which acting on the signal loses money however accurate it is.

7.7 Convergence concepts (almost sure, in distribution, L^2)

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A sequence of numbers either converges or it does not. A sequence of *random* variables can converge in several genuinely different senses, and the differences are not pedantry: each one answers a question a desk actually asks. Does my simulation get the right answer if I run it long enough? How wrong is it after ten thousand paths? Does my discretely rebalanced hedge reproduce the payoff? Three questions, three modes of convergence.

The three modes, and what each one is for

Let $X_1, X_2, \ldots$ be random variables and X a limit.

Almost sure convergence, written $X_n \to X$ a.s., means $\mathbb{P}\big(\lim_n X_n = X\big) = 1$: the sequence converges along essentially every possible path of the world. It is the most demanding of the three pathwise, and the least practically useful, because it says nothing about speed. It is what the **strong law of large numbers** delivers: the running average of independent, identically distributed draws converges almost surely to the mean, provided the mean exists.

Convergence in L^2, or in mean square, means $\mathbb{E}\big[(X_n - X)^2\big] \to 0$: the expected squared error vanishes. This is the engineer's mode, because the quantity it controls is an error you can measure and budget. It is how the Itô integral is constructed, and it is the natural way to state that a discretely rebalanced delta hedge reproduces a payoff: not that it works on every path, which is false, but that its mean squared replication error goes to zero as rebalancing becomes more frequent.

Convergence in distribution, written $X_n \Rightarrow X$, means the cumulative distribution functions converge, $F_n(x) \to F(x)$ at every point where F is continuous. It is the weakest mode, and note what it does *not* say: the variables need not get close to each other at all, only their laws do. This is what the **central limit theorem** delivers, and it is also the sense in which a binomial tree converges to Black-Scholes as the time step shrinks.

> KEY IDEA
>
> The modes are ordered. Almost sure convergence implies convergence in probability, and so does L^2 convergence; convergence in probability implies convergence in distribution. None of the reverse implications holds, and almost sure and L^2 convergence are not comparable: neither implies the other. Practically: **the law of large numbers tells you the simulation is right, the central limit theorem tells you how wrong it still is, and L^2 is the language for hedging error.**

The two theorems that do all the work

The **law of large numbers**: for independent draws $Y_1, \dots, Y_n$ from a distribution with mean μ,

$$\bar{Y}_n = \frac{1}{n}\sum_{i=1}^{n} Y_i \;\longrightarrow\; \mu.$$

The **central limit theorem**: if the variance σ^2 is finite, the error is asymptotically normal,

$$\sqrt{n}\,\big(\bar{Y}_n - \mu\big) \;\Longrightarrow\; \mathcal{N}\big(0, \sigma^2\big)\,, \qquad \text{so} \qquad \text{standard error} \;=\; \frac{\sigma}{\sqrt{n}}.$$

Everything below is a consequence of that $\sqrt{n}$.

Why the average is the price at all is the law of large numbers: the payoffs on independent paths are independent draws from the payoff distribution, so their average converges to the expectation that the price is defined as. Two conditions carry that argument, and both are where it fails rather than where it holds. The paths must be drawn from the correct, *risk neutral* distribution, and the payoff must have a finite mean, so a simulation of a distribution with heavy enough tails converges to nothing useful however long it runs. The accuracy is then the central limit theorem, which is where the $\sigma/\sqrt{n}$ above comes from. The method itself, its steps and the machinery for beating that constant, belongs to Monte Carlo methods.

WORKED EXAMPLE

Suppose the payoff has a standard deviation of 20 across paths. With $n = 10{,}000$ paths the standard error is $20/\sqrt{10{,}000} = 0.20$, so a 95% confidence interval is roughly $\pm 1.96 \times 0.20 = \pm 0.39$ around the estimate.

To reduce the error to 0.02, one thousandth of the payoff's dispersion, you need

$$n = \left(\frac{20}{0.02}\right)^2 = 1{,}000{,}000 \text{ paths},$$

a hundredfold increase in work for a tenfold improvement in accuracy. That is the practical meaning of $1/\sqrt{n}$: **four times the computation buys half the error**, and no amount of hardware changes the exponent.

REAL INTERVIEW QUESTION (Société Générale · desk not recorded)

Explain the mathematical concept behind a Monte Carlo simulation, using the unit circle to help you.

WHAT THEY TEST / ANSWER

They are asking you to build the method out of the two theorems above, using the oldest example there is, so do exactly that and keep the circle concrete.

The construction. Draw points uniformly at random in the unit square. A point (U_1, U_2) lands inside the quarter circle when $U_1^2 + U_2^2 \le 1$, and the chance of that is the quarter circle's area over the square's, which is $\pi/4$. Define $Y_i = \mathbf{1}\{U_{i,1}^2 + U_{i,2}^2 \le 1\}$, a Bernoulli draw with mean $p = \pi/4$. The estimator is $\hat{\pi}_n = 4\bar{Y}_n$. That is the whole idea, and it generalises without changing: replace the indicator by any payoff and the area by any expectation.

Why it works is the law of large numbers: $\bar{Y}_n \to p$ almost surely, so $\hat{\pi}_n \to \pi$. **How fast** is the central limit theorem: the standard error is $4\sqrt{p(1-p)/n} = 1.642/\sqrt{n}$. Put numbers on it, because that is what turns a recited definition into an answer. With $n = 10{,}000$ the standard error is 0.0164 and the 95% interval is about ± 0.032, so you have π to roughly one decimal place. Two decimal places, meaning a 95% half-width of 0.005, needs $n = (1.96 \times 1.642/0.005)^2 \approx 414{,}000$ draws, and three decimal places needs a hundred times more again, about 41 million.

Then say the honest thing, which is the mark: as a way of computing π this is terrible. Any deterministic quadrature rule beats $1/\sqrt{n}$ comfortably in two dimensions. What makes Monte Carlo indispensable is that its error rate does not depend on the dimension at all, while a grid's cost grows exponentially in it, so the method loses in two dimensions and wins in twenty. The circle is the illustration, not the use case. Which payoffs actually force you to reach for it is in path dependency and scenario analysis, and the machinery for shrinking that 1.642 is in Monte Carlo methods.

Beating the constant, and beating the rate

Since $1/\sqrt{n}$ cannot be improved by running longer, the effort goes into shrinking the σ in front of it.

Those techniques are a numerical-methods problem rather than a probability one, and the catalogue of them, antithetic variates, control variates, importance sampling, stratification, and quasi-Monte Carlo, which attacks the rate rather than the constant, is developed in Monte Carlo methods. What belongs here is the reason the effort has to go there at all. The exponent in $n^{-1/2}$ is a theorem about independent averages, not an implementation detail, so it cannot be improved by better code or more hardware; only the σ in front of it is available to work on, and only a different sampling scheme changes the exponent itself.

Where these modes show up elsewhere

- A **binomial tree** converges to the Black-Scholes price as the step shrinks. That is convergence in distribution: the discrete price law approaches the lognormal, in the Black-Scholes model.
- **Realized volatility** measured over more and more frequent observations converges to the true volatility of the process, which is why a variance swap can be replicated at all.
- **Discrete delta hedging** reproduces a payoff in L^2 as rebalancing becomes continuous, and the residual error at a finite frequency is the gamma term that a trader actually lives with.
- **Estimating a mean return** from history is the law of large numbers applied to a sample too short for it, which is why expected returns are estimated badly and volatility comparatively well.

INTERVIEW TRAP

Four errors. (1) Quoting a Monte Carlo price without a standard error, which is a number without an accuracy. (2) Expecting a tenfold accuracy gain from ten times the paths: it buys a factor of $\sqrt{10} \approx 3.2$. (3) Assuming the central limit theorem applies regardless: it needs a finite variance, and a payoff with heavy enough tails has a sample average that is not asymptotically normal. (4) Confusing the modes, in particular reading convergence in distribution as though the variables themselves were converging, when only their laws are.

Where this goes next

The distributions these theorems act on are probability distributions and the moments they converge to are expectation, variance and higher moments. The model whose price a tree converges to is the Black-Scholes model, and the products that make simulation unavoidable are path-dependent products and autocallables.

Chapter 8

Statistics & Time Series

8.1 Descriptive Statistics

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Nothing in this section is difficult and all of it is asked, because the questions are not really about the formulas. They are about whether you pick the right summary for the data in front of you, and finance supplies data that punishes the default choices.

There are three means, and the data decides

> **KEY IDEA**
>
> The **arithmetic** mean is right for things that *add*. The **geometric** mean is right for things that *multiply*, which includes every return series. The **harmonic** mean is right for *rates* averaged over a fixed quantity of the thing in the denominator, which is why it appears in speeds and in price-earnings ratios. They satisfy AM $\geq$ GM $\geq$ HM always, with equality only when every observation is identical, so choosing the wrong one is not a rounding error, it is a bias with a known sign.

REAL INTERVIEW QUESTION (Société Générale · EQD)

I run one lap of a track at 5 km/h. How fast must I run the second lap to average 10 km/h over the two laps?

WHAT THEY TEST / ANSWER

It is impossible, and recognising that is the entire question. Do not attempt a number.

Work in time rather than in speed, which is the move the puzzle is built to reward. Let the lap be L. To average 10 km/h over two laps, the total time must be $2L/10 = L/5$. But the first lap at 5 km/h already took $L/5$. The budget is exhausted before the second lap starts, so no finite speed works, and even infinite speed only reaches the average in the limit.

Why the intuition fails: average speed is total distance over total time, so it is a **harmonic** mean of the two lap speeds, not an arithmetic one. The arithmetic answer, 15 km/h, is wrong because it averages speeds while the physics averages times, and slow laps consume disproportionate time. The harmonic mean of 5 and v can never reach 10 precisely because the first term already caps it below.

Then generalise it, because that is what the interviewer is buying: to double an average by adding one more observation, you need the second observation to be infinite whenever the target is twice the first. The same structure appears in finance whenever a rate is averaged over a fixed denominator: the price-earnings ratio of an index is the harmonic mean of its constituents' ratios weighted by market value, and averaging them arithmetically overstates it, sometimes badly, because a single near-zero-earnings company produces an enormous ratio that an arithmetic average lets dominate.

REAL INTERVIEW QUESTION (Barclays · EQD)

Give an example where the mean of the returns cannot be neglected.

WHAT THEY TEST / ANSWER

Read the question as asking where the *choice* of mean matters, and give the cleanest possible example.

The example: a position gains 50% and then loses 50%. The arithmetic mean return is $(+50-50)/2 = \mathbf{0}$. The actual wealth is $1.5 \times 0.5 = 0.75$, so you have lost **25%**, and the geometric mean return is $\sqrt{0.75} - 1 = \mathbf{-13.4\%}$ per period. A summary statistic said zero and the account said minus twenty-five.

Why: returns compound, so what matters is the product of $(1 + r_i)$, and the mean of a product is governed by the geometric mean. The arithmetic mean is an unbiased estimator of the *expected single-period return*, which is a genuinely different quantity from the growth rate you will experience over many periods.

The size of the gap is worth quoting, because it makes the effect predictable rather than anecdotal:

$$\text{GM} \;\approx\; \text{AM} - \frac{\sigma^2}{2},$$

so the shortfall grows with the square of volatility. At 50% volatility the drag is 12.5 points, which is most of the 13.4 above. This is the same $\sigma^2/2$ that separates the mean and median of a lognormal in distributions and that appears as volatility drag in implied vs realised.

Where it bites in practice, which is the answer to "cannot be neglected": comparing a leveraged product to its underlying, since leverage multiplies volatility and therefore multiplies the drag quadratically; quoting a fund's average annual return when the investor experienced the compounded one; and any backtest that reports an arithmetic average of period returns as though it were a growth rate. In all three the arithmetic mean is not merely imprecise, it is systematically optimistic.

Dispersion, and one convention worth knowing

Variance is the mean squared deviation and standard deviation is its root, which restores the units of the data and is why volatility is quoted as a standard deviation rather than a variance.

The divisor is the detail interviewers probe. Dividing by $n - 1$ rather than n gives an unbiased estimator of the population variance, correcting for the degree of freedom used by estimating the mean. In finance the mean is usually *not* estimated but set to zero, because over daily horizons the drift is negligible relative to the noise and because a zero-mean estimator is exactly additive across periods, which matters for the variance swap convention in realised volatility. So the finance convention divides by n and subtracts no mean, and knowing that this is a deliberate departure rather than an error is the point.

Shape: the third and fourth moments

Skewness and kurtosis, the standardised third and fourth moments, are developed in higher moments. Two things about them belong here because they are conventions rather than mathematics. A normal distribution has kurtosis exactly 3, so the quantity usually quoted is **excess kurtosis**, kurtosis minus 3, and saying which one you mean is worth a clause since the two differ by exactly that. And the empirical facts for equity returns are negative skew and positive excess kurtosis:

the market falls faster than it rises and produces far more extreme days than a normal allows, both of which are visible in option prices as well as in history, through the volatility skew and the wings of the smile.

The consequence is the reason a descriptive-statistics section cares. Any summary built on the first two moments alone, which includes variance, a normal value-at-risk and a Sharpe ratio, is blind to exactly the two features that make financial returns dangerous. A strategy that sells options has an attractive mean and standard deviation and a terrible third and fourth moment, and reporting only the first two is how that strategy gets sold.

Robust alternatives, and when to reach for them

Every statistic above is sensitive to outliers, and financial data is mostly outliers. The robust versions are worth having: the **median** instead of the mean, the **interquartile range** or the **median absolute deviation** instead of the standard deviation. They have a higher breakdown point, meaning a larger fraction of the sample must be corrupted before the statistic is destroyed: the mean is destroyed by a single bad observation while the median survives until half the sample is.

The judgement is knowing when a robust statistic is the right answer and when it is a way of discarding the data that matters. For describing a typical day, the median is better. For sizing risk, the outliers *are* the risk, and a robust statistic that ignores them is answering a question nobody asked.

Two variables

Covariance is the mean product of deviations and depends on the units of both variables; correlation divides it by the two standard deviations to give a unitless number in $[-1, 1]$. The definitions belong to independence and correlation; the point to carry here is what correlation does *not* capture.

It measures **linear** dependence only. Two variables can be perfectly dependent and have zero correlation, and $Y = X^2$ with X symmetric about zero is the standard counterexample. In markets this matters because the dependence that hurts is usually not linear: assets that behave independently in normal conditions move together in a crash, so a correlation estimated over a calm sample describes precisely the regime in which the number does not matter.

REAL INTERVIEW QUESTION (Tikehau Capital · Market Risk)

How would you build a variance-covariance matrix for a portfolio of options?

WHAT THEY TEST / ANSWER

The trap is in the word *options*, and spotting it is the answer. A covariance matrix describes the joint behaviour of **risk factors**, not of instruments, and an option is a non-linear function of its factors. So you do not build the matrix on the options at all.

Step one, choose the risk factors. For an equity option book that is typically the underlying levels, points on the volatility surface, points on the rate curve and the dividend curve. The matrix is built on *those* and is the same matrix whatever instruments you hold, which is the point.

Step two, estimate it from the historical returns of those factors, choosing the window and the weighting deliberately: an equally weighted window responds slowly and is stable, an exponentially weighted one responds quickly and is noisy, and the choice is a statement about which regime you believe you are in.

Step three, map the portfolio onto the factors through its sensitivities, which is where the options enter: the delta and vega ladders, and for a portfolio with meaningful convexity the gamma terms as well. The portfolio's variance is then $w^\top \Sigma w$ with w the vector of sensitivities.

Then the two caveats that make it a market risk answer. The mapping is a **linear approximation**, so it is wrong by exactly the convexity of the book, which is the largest part of an option portfolio's risk in a large move; that is why this approach is a parametric value-at-risk and why desks with real gamma use historical simulation or Monte Carlo instead, treated in value at risk. And a covariance matrix estimated on many factors from limited history is **ill-conditioned**, so it must be regularised, shrunk toward a structured target or reduced by principal components, or it will produce confidently wrong risk numbers for portfolios that happen to sit in its noisiest directions.

INTERVIEW TRAP

1. **Averaging returns arithmetically.** Returns compound; the geometric mean is the growth rate and the arithmetic mean is systematically higher by about $\sigma^2/2$.
2. **Averaging rates arithmetically.** Speeds, yields and price-earnings ratios averaged over a fixed denominator are harmonic means, and the arithmetic version overstates them.
3. **Mixing kurtosis and excess kurtosis.** They differ by exactly 3, and a normal is 3 or 0 depending on which you meant.
4. **Quoting only mean and standard deviation.** That is blind to skew and tails, which is exactly what makes a short-option strategy look attractive on paper.
5. **Reading zero correlation as independence.** It rules out linear dependence only, and market dependence is not linear where it counts.
6. **Estimating correlation in calm markets.** It rises when everything falls, so a calm-sample estimate describes the regime in which you did not need it.

VISUAL / CODE TO BUILD: Interactive

A summary-statistics sandbox with a real return series. The reader sees a histogram with the mean, median and mode marked separately, and a panel reporting all three means, the two dispersion measures and the third and fourth moments. Dragging a single observation into the tail updates every statistic live, so the reader watches the mean and standard deviation move sharply while the median and interquartile range barely respond, which is what "robust" means made physical. A second mode fits a normal to the same data and overlays it, with a counter of how many observations fall beyond four standard deviations against how many the normal predicts. A third panel takes two series and plots them against each other with the correlation displayed, letting the reader construct a perfect parabola and watch the correlation read near zero, and then splits the sample into calm and stressed periods to show the correlation in each, which is usually the most persuasive chart in the section.

8.2 Estimation methods (MLE, method of moments)

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Every model on a desk has parameters nobody handed you. A volatility, a correlation, a default intensity, a mean-reversion speed: none of these is observed, all of them are inferred from data, and the inference procedure is not a technicality. It determines what the number means, how wrong it can be, and whether two people quoting "the volatility" are quoting the same thing.

This section is about the two procedures that do almost all the work. Their names sound academic. What they actually decide is the number on the screen.

Estimator, estimate, and why the distinction is the whole subject

An **estimator** is a rule: a function of the data. An **estimate** is the number that comes out when you feed it a particular sample. The rule is the object worth thinking about, because a rule has properties and a number does not.

KEY IDEA

An estimator is a random variable. Feed it a different sample from the same world and it gives a different answer, so it has a distribution, a mean and a variance of its own. Almost every mistake in applied statistics comes from treating the estimate as a fact rather than as one draw from that distribution.

Three properties are worth having names for. An estimator is **unbiased** if its average over all possible samples equals the true value. It is **consistent** if it converges to the true value as the sample grows. It is **efficient** if, among reasonable estimators, it has the smallest variance. These are separate: an estimator can be biased and still consistent, and a slightly biased estimator with low variance is often preferable to an unbiased one that is all over the place.

Method of moments

The oldest idea, and the simplest. The distribution's theoretical moments are functions of its parameters. The sample has its own moments. Set them equal and solve.

If the model has one parameter, match the mean. Two parameters, match the mean and the variance. And so on, one moment per parameter.

> WORKED EXAMPLE
>
> Daily returns are modelled as normal with unknown mean μ and variance σ^2. The first theoretical moment is μ and the second central moment is σ^2, so matching gives
>
> $$\hat{\mu} = \bar{x}, \qquad \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2.$$
>
> Which is exactly what anyone would have written down without the theory. That is the point: for the normal, the method of moments produces the obvious answer, which is why it is worth seeing a case where it does not.

Maximum likelihood

The other idea. Ask, for each candidate parameter value, how probable the data you actually observed would have been. Choose the value that makes the observed data most probable.

Formally, for independent observations $x_1, \ldots, x_n$ with density $f(\cdot\,; \theta)$, the likelihood is

$$L(\theta) = \prod_{i=1}^{n} f(x_i; \theta), \qquad \ell(\theta) = \log L(\theta) = \sum_{i=1}^{n} \log f(x_i; \theta),$$

and $\hat{\theta}_{\mathrm{MLE}}$ maximises either. We work with the log because it turns a product into a sum, which is both easier to differentiate and numerically survivable: the product of a thousand small densities underflows to zero, the sum of a thousand logarithms does not.

> INTUITION
>
> Likelihood runs the probability question backwards. Probability asks: given the parameter, how likely is this data? Likelihood fixes the data, which you have, and asks which parameter would have made it least surprising. It is not the probability of the parameter, and calling it that is the classic slip, since the parameter is not random in this framework.

The one result everyone should be able to derive

For a normal sample, maximise the log-likelihood and you get

$$\hat{\mu} = \bar{x}, \qquad \hat{\sigma}^2_{\mathrm{MLE}} = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

The mean estimator is the sample mean, as expected. The variance estimator divides by n, and it is **biased**:

$$\mathbb{E}\big[\hat{\sigma}^2_{\mathrm{MLE}}\big] = \frac{n-1}{n}\,\sigma^2.$$

The reason is that $\bar{x}$ was itself estimated from the same data, so the deviations $x_i - \bar{x}$ are systematically smaller than the deviations from the true mean. One degree of freedom has been spent, and dividing by $n-1$ instead of n puts it back.

KEY IDEA

Maximum likelihood does not give the $n-1$ estimator. The familiar $n-1$ in every spreadsheet's standard deviation function is the *unbiased* estimator, not the maximum likelihood one. They are different rules and they answer different questions: one is unbiased, the other is the most likely explanation of the data. Neither is wrong; quoting a number without saying which one you used is.

Where the two methods separate

For the normal they agree. Take a case where they do not.

WORKED EXAMPLE

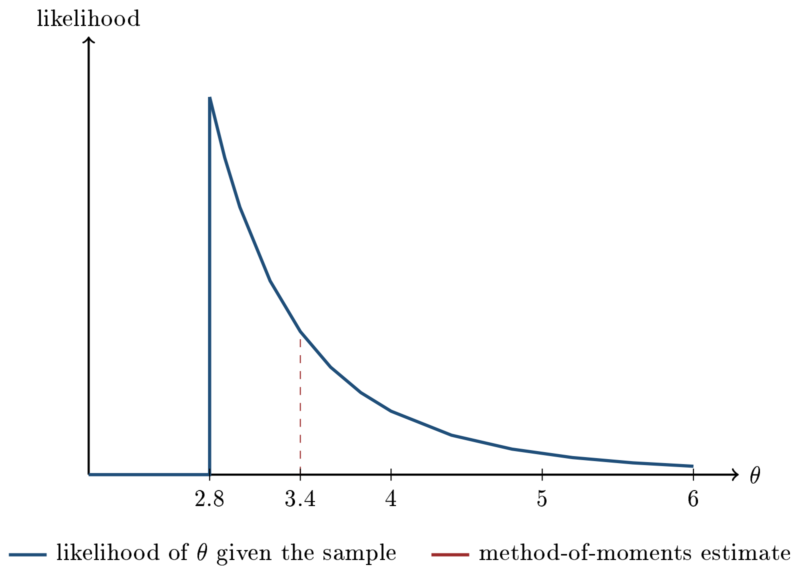
Observations are uniform on $[0, \theta]$ and you must estimate θ.

Method of moments. The mean of a uniform on $[0, \theta]$ is $\theta/2$, so matching gives $\hat{\theta}_{\mathrm{MoM}} = 2\bar{x}$.

Maximum likelihood. The density is $1/\theta$ on $[0, \theta]$ and zero outside, so the likelihood is θ^{-n} if θ is at least as large as every observation, and exactly zero otherwise. Since θ^{-n} decreases in θ, the likelihood is maximised at the smallest feasible value: $\hat{\theta}_{\mathrm{MLE}} = \max_i x_i$.

Take the sample $0.3,\ 1.1,\ 1.9,\ 2.4,\ 2.8$. The mean is 1.7, so the method of moments gives $\hat{\theta} = 3.4$, while maximum likelihood gives $\hat{\theta} = 2.8$. Different answers from the same data, and neither is a mistake.

Now take the sample $0.1,\ 0.2,\ 0.3,\ 0.4,\ 3.0$. The mean is 0.8, so the method of moments gives $\hat{\theta} = 1.6$. But one of the observations was 3.0, and under $\theta = 1.6$ that observation was *impossible*. The method of moments has produced an estimate the data rules out. Maximum likelihood cannot do this, because a parameter that makes the data impossible has likelihood zero by construction.



The likelihood is flat at zero until the largest observation, jumps, and then decays. The maximum sits at the jump, which is why the maximum likelihood estimate is the sample maximum and why it is at a boundary rather than at a point where a derivative vanishes. The dashed line marks where the method of moments landed: feasible here, but visibly not where the data is most consistent.

Why maximum likelihood is the default

Three properties, and the third is the one people forget.

- **Consistency and asymptotic normality.** As the sample grows, the estimator converges to the truth and its error becomes normal, which is what lets you put a confidence interval around it at all.
- **Asymptotic efficiency.** In large samples it attains the smallest variance any unbiased estimator can have, the Cramer-Rao bound. Nothing does better.
- **Invariance.** If $\hat{\theta}$ is the maximum likelihood estimate of θ, then $g(\hat{\theta})$ is the maximum likelihood estimate of $g(\theta)$, for any function g. So the maximum likelihood estimate of volatility is the square root of the maximum likelihood estimate of variance, with nothing further to prove. This is not true of unbiasedness: the square root of an unbiased variance estimator is *not* an unbiased volatility estimator, which is why the standard deviation reported by every piece of software is slightly biased even when the variance it came from was not.

Against that, it needs a distributional assumption. The method of moments needs only the moments to exist, so it survives on data whose distribution you would not want to commit to, and it is often used to produce a starting value for the numerical search that maximum likelihood requires.

Estimation is not calibration

The distinction that separates a statistician from a quant on a derivatives desk, and it is worth stating precisely because the algebra looks identical.

Estimation fits parameters to *historical data*, under the real-world measure. It answers: what has this asset actually done? **Calibration** fits parameters to *current market prices*, under the risk-neutral measure. It answers: what set of parameters reproduces what people are paying right now?

Both minimise a distance. They produce different numbers for the same parameter, and the gap between them is not error, it is a risk premium. Historical volatility estimated from returns and implied volatility calibrated from options differ persistently and for good reasons, which is the subject of implied versus realized volatility. The same split reappears as historical against risk-neutral default probability in default risk. The mechanics of fitting a model to market prices are in model calibration.

> INTERVIEW TRAP
>
> Using an estimated parameter where a calibrated one is required, or the reverse. Pricing a derivative off historical volatility means your price is not consistent with the market you have to trade in. Forecasting realised risk off implied volatility means your forecast carries a premium that is not a forecast of anything. The two numbers answer different questions and substituting one for the other is a category error, not an approximation.

Interview angle

REAL INTERVIEW QUESTION (Barclays · EQD)

What formula do you use to compute historical volatility?

WHAT THEY TEST / ANSWER

Write it down, then defend every choice in it, because the question is really about the choices.

The formula: take log returns $r_i = \ln(S_i/S_{i-1})$ over n periods, then

$$\hat{\sigma} = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(r_i - \bar{r})^2} \times \sqrt{252}$$

for daily data. Say why log returns rather than simple returns: they add across time, so the scaling by $\sqrt{252}$ is legitimate.

Then the three choices an interviewer will probe. **Why $n-1$**: the sample mean was estimated from the same data, so one degree of freedom is gone and dividing by n would bias the variance downward. Add the sharper version if you want the point: the maximum likelihood estimator *is* the one with n, and it is biased, so $n-1$ is the unbiased choice rather than the likelihood choice. **Whether to subtract the mean at all**: on daily equity data many desks do not, setting $\bar{r}$ to zero, because the true daily drift is tiny relative to the noise and estimating it costs more variance than it removes. **The window**: a short window responds fast and is noisy, a long one is stable and stale, and there is no correct answer, only a stated one.

Show the size of the choices with numbers rather than asserting they matter. On five daily returns of $+1.2\%, -0.8\%, +0.5\%, -1.5\%, +1.1\%$, the annualised estimate is **17.0%** dividing by n, **19.0%** dividing by $n-1$, and **17.1%** with the mean forced to zero. Two full volatility points between the first two, on a definitional choice. Then say the thing that closes it: the ratio between the two is $\sqrt{n/(n-1)}$, which is 1.118 at five observations and 1.002 at a year of daily data, so the choice is decisive on a short window and irrelevant on a long one. Knowing when a detail stops mattering is worth as much as knowing the detail.

REAL INTERVIEW QUESTION (Bank of America · Exo)

What is the correlation between rates and equity indices at the moment? Could you estimate the correlation over the past year for me?

WHAT THEY TEST / ANSWER

Two questions again, and the second is the one being marked. On the first, give the current regime rather than a number to two decimals: for most of the post-2000 period the correlation between bond prices and equities was negative, so bonds hedged equities, and that relationship inverted during the 2022 inflation shock when both fell together. Say which regime you think we are in and what would flip it.

On the second, describe the procedure out loud, because that is what is being tested. **Choose the variables**: index log returns against changes in yield, not the level of the yield, since correlating a return with a level is meaningless. **Choose the frequency and the window**: daily over one year is about 252 observations, which is a reasonable trade-off, and say that weekly would reduce the effect of asynchronous closes across markets. **Compute it**: the sample correlation is the method of moments estimator, sample covariance over the product of sample standard deviations. Then quote the number as a range rather than a point.

The part that earns the question is what comes after the procedure. A correlation estimate is a **random variable**, and on 252 observations its standard error is roughly $(1-\rho^2)/\sqrt{n}$, so at a true correlation of 0.3 the estimate wobbles by about 0.06 from sample to sample before anything in the world has changed. Anyone reading a move from 0.30 to 0.36 as a regime change is reading noise. And correlation is not stable across regimes: it is exactly in a crash, when the estimate is most needed, that the historical number is least informative. Say both and the question is finished.

INTERVIEW TRAP

Four errors. (1) Calling the likelihood the probability of the parameter. (2) Assuming maximum likelihood is unbiased: for the normal variance it is not, and the correction is the familiar $n-1$. (3) Reporting an estimate without a sample size, which is reporting a number without its uncertainty. (4) Estimating a parameter from history and then using it to price a derivative, when pricing needs a calibrated risk-neutral parameter and not a historical one.

Where this goes next

Deciding whether an estimate is far enough from a hypothesised value to matter is hypothesis testing; estimating relationships between variables rather than parameters of one is linear and multiple regression. The most heavily used maximum likelihood application in finance is fitting volatility models, which is volatility modeling, and its risk-neutral counterpart is model calibration.

8.3 Hypothesis Testing

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Every number a desk produces is an estimate from a finite sample, and the only question that matters about it is whether it is real or whether it is noise. A strategy made money last year: skill or luck? A risk model was breached four times: broken or unlucky? A hedge ratio came out at 0.83: different from 1 or not? Hypothesis testing is the machinery for answering that class of question, and the reason it appears in interviews is that almost everyone can run the test and almost nobody states correctly what the answer means.

Descriptive statistics summarises a sample; estimation produces a number from it. This section is about the third step, deciding whether that number is distinguishable from nothing.

The four objects

A test has exactly four moving parts, and naming them in order is most of the answer to any interview question on the topic.

1. The **null hypothesis** H_0, the boring explanation: the strategy has zero edge, the model is correctly calibrated, the coefficient is zero. The null is what you assume while you compute.
2. The **alternative** H_1, what you would conclude if the null fails. One-sided or two-sided, and you must fix this *before* seeing the data.
3. The **test statistic**, a number computed from the sample whose distribution *under the null* you know. Usually an estimate divided by its own standard error, which turns any quantity into something comparable to a normal or a t.
4. The **decision rule**, a threshold, expressed either as a critical value or through the **p-value**: the probability of seeing a statistic at least as extreme as the one you got, if the null were true.

> KEY IDEA
>
> A p-value is $P(\text{data at least this extreme} \mid H_0)$. It is *not* $P(H_0 \mid \text{data})$, and it is not the probability that your strategy works. Inverting the two is exactly the error Bayes' theorem exists to prevent: to get from one to the other you need a prior on how likely it was that the strategy had edge in the first place, and if you tested a thousand ideas that prior is low. Say this sentence in an interview and you are in the top decile of candidates on the topic.

The two errors, and which one a desk fears

	H_0 **true**	H_0 **false**
Reject H_0	Type I error (α)	correct
Fail to reject	correct	Type II error (β)

A **Type I** error is a false alarm: declaring an edge that is not there. A **Type II** error is a miss: failing to detect an edge, or a broken model, that is there. For a fixed sample size you trade one against the other, and the choice of which to protect is a business decision rather than a statistical one.

INTUITION

A researcher pitching a strategy is protecting against Type I: they set α at 5% because the cost of allocating to a fake edge is real money. A risk manager backtesting a VaR model has the opposite loss function. The expensive error is failing to notice that the model understates risk, so regulators deliberately set a trigger that fires reasonably often on a *correct* model, accepting false alarms to buy sensitivity. When you are asked which error matters more, the answer is always "it depends on which mistake costs more", followed by the specific cost in the situation you were given.

Power, and the brutal arithmetic of sample size

Power is $1 - \beta$, the probability of detecting an effect that is really there. It is the part everyone skips, and in finance it is usually the binding constraint, because financial signals are small relative to their own noise.

Take the cleanest case. A strategy has annualised Sharpe ratio SR, and you want to know whether its mean return is distinguishable from zero. With T years of data the t-statistic on the mean is approximately

$$t \;\approx\; SR\sqrt{T},$$

so to clear the usual two-sided 5% bar you need $SR\sqrt{T} \geq 1.96$, that is

$$T \;\geq\; \left(\frac{1.96}{SR}\right)^2 \text{ years.}$$

Annual Sharpe	Years to significance at 95%
0.3	42.7
0.5	15.4
1.0	3.8
2.0	1.0

Read the first row and let it land. A genuinely good long-only equity strategy runs at a Sharpe near 0.3, and proving statistically that it beats zero takes longer than most careers. This is not a defect of the test; it is the actual information content of the data. It also explains a great deal of market behaviour: since a three-year track record cannot distinguish skill from luck at anything below a Sharpe of about 1.1, allocators who claim to be doing so on three years of returns are doing something other than statistics.

REAL INTERVIEW QUESTION (Société Générale · desk not recorded)

What is the Sharpe ratio?

WHAT THEY TEST / ANSWER

Give the definition first, in one line: excess return per unit of volatility,

$$SR = \frac{\mathbb{E}[R] - r_f}{\sigma(R)},$$

annualised by multiplying a daily Sharpe by $\sqrt{252}$, since the mean scales with time and the standard deviation with its square root.

Then add the two things that separate a good answer from a definition. The first is what it ignores: it is a two-moment measure, so it is blind to skewness and kurtosis, and a strategy that sells options has an attractive Sharpe and a terrible tail, which is precisely why the ratio is popular with people selling such strategies. That link is developed in descriptive statistics.

The second is the one that belongs here: a Sharpe ratio is an *estimate*, with a standard error of roughly $1/\sqrt{T}$ in years, so its own t-statistic is about $SR\sqrt{T}$. A fund quoting 0.8 over three years has a t of 1.39, which is not significant at any conventional level. Saying this shows you treat performance numbers as samples rather than facts, and it is the answer an interviewer on a quantitative desk is listening for.

Multiple testing, or why backtests look so good

Run one test at $\alpha = 5\%$ on a strategy with no edge and you have a 5% chance of a false positive. Run one hundred and the expected number of false positives is five, and the probability of at least one is $1 - 0.95^{100} = 99.4\%$. A researcher who tries a hundred parameter combinations and reports the best one has not found an edge; they have found the maximum of a hundred draws of noise, and the p-value printed next to it is meaningless because it was computed as though the test had been chosen in advance.

The fixes are all forms of paying for the search. **Bonferroni** divides the threshold by the number of tests, so one hundred tests at an overall 5% requires each to clear 0.0005, a z of 3.48 rather than 1.96, which kills almost every candidate strategy and is meant to. **Out-of-sample testing** holds back data the search never touched, and is the only honest defence when the number of things tried is unknown, which in practice it always is. The right interview sentence is that the *denominator matters*: a p-value is only interpretable if you know how many hypotheses were examined to produce it.

VaR backtesting: the test a desk actually runs

The cleanest applied hypothesis test in a bank is the backtest of a value-at-risk model. It is a binomial test, and it is worth carrying the numbers because the framework is regulatory and therefore fixed.

WORKED EXAMPLE

A desk runs a one-day 99% VaR. Over 250 trading days, each day is a Bernoulli trial: the loss either exceeds the VaR (an **exception**) or it does not.

Null hypothesis: the model is correct, so the exception probability is $p = 1\%$ and the count is Binomial$(250, 0.01)$.

Under the null: expected exceptions $250 \times 0.01 = 2.5$, with a standard deviation of $\sqrt{250 \times 0.01 \times 0.99} = 1.57$.

The distribution, which is the whole point:

Exceptions	Probability	Cumulative
0	8.11%	8.11%
1	20.47%	28.58%
2	25.74%	54.32%
3	21.49%	75.81%
4	13.41%	89.22%
5	6.66%	95.88%
6	2.75%	98.63%
7	0.97%	99.60%

The Basel decision rule, which is exactly a rejection region:

- **Green**, 0 to 4 exceptions. Occurs 89.22% of the time if the model is correct. No action.
- **Yellow**, 5 to 9. Probability under the null 10.76%, so a correct model lands here roughly one year in nine. The capital multiplier is increased on a sliding scale.
- **Red**, 10 or more. Probability under the null 0.025%. The model is rejected.

The observation that makes the test honest: the yellow trigger fires on a correct model more than one year in ten, and the regulator knows it. That is a deliberately large Type I rate, bought to get any power at all. And the power is still poor. If the model is really running at a 2% exception rate, meaning the stated 99% VaR is closer to a 98% VaR, the 5-exception trigger catches it only 56.1% of the time in a given year, and the red zone catches it 3.0% of the time. Twelve months of data cannot reliably distinguish a good VaR model from one that is wrong by a factor of two on its tail probability. This is the sample-size arithmetic again, in the place where a bank cares about it most.

REAL INTERVIEW QUESTION (CACIB · Fixed Income)

What is backtesting?

WHAT THEY TEST / ANSWER

Define it, then show you know what it is a test *of*, because the word means two different things depending on which side of the floor asked.

Risk backtesting: comparing a risk model's predictions against realised outcomes. For VaR, counting exceptions over 250 days and asking whether the count is consistent with the stated confidence level. Give the numbers: at 99% you expect 2.5 exceptions a year, the Basel traffic light puts 0 to 4 in green, 5 to 9 in yellow and 10 or more in red, and that is a binomial hypothesis test with the model as the null. Add the refinement if you have room: exceptions should also be *independent* in time, so a complete backtest tests clustering as well as frequency, since a model that fails on three consecutive days in a crisis is broken in a way a count alone will not show.

Strategy backtesting: running a trading rule over historical data to estimate its performance. Here the answer the interviewer wants is the list of ways it lies. Look-ahead bias, using information not available at the time. Survivorship bias, testing on the index members who survived. Ignoring transaction costs and market impact, which is what usually turns a profitable backtest into a losing strategy. And overfitting through repeated search, where the reported p-value is invalid because the number of hypotheses tried is not in the denominator.

Close on the point that unites them: a backtest is a hypothesis test on a finite sample, so the first question about any backtest result is how large the sample was and how many variants were tried before this one was shown to you.

INTERVIEW TRAP

1. **Reading a p-value backwards.** "p = 0.03, so there is a 97% chance the strategy works" is wrong and is the single most common error on this topic. The p-value is computed *assuming* the null and says nothing on its own about how likely the null is.

2. **Treating "not significant" as "no effect".** Failing to reject is not accepting. With a Sharpe of 0.3 and five years of data you will fail to reject almost regardless of the truth, because the test has no power. Absence of evidence, in a low-power test, is barely evidence of absence.

3. **Choosing the alternative after seeing the data.** A one-sided test picked because the estimate came out positive is a 10% test wearing a 5% label.

4. **Confusing statistical and economic significance.** A hedge ratio significantly different from 1 at $t = 2.1$ may be 0.997, which no desk would act on. Significance is about whether an effect is detectable, not whether it is large enough to matter.

5. **Forgetting that returns are not independent.** The $t \approx SR\sqrt{T}$ formula assumes independent returns. With autocorrelation, common in illiquid or smoothed marks, the true standard error is larger and the honest sample size is smaller than the number of observations suggests.

VISUAL / CODE TO BUILD: Interactive

A significance machine for track records. The user sets a true Sharpe ratio and a track-record length; the page simulates a few thousand independent histories and plots the distribution of *realised* Sharpe ratios, shading the region that would clear a t of 1.96. Two moments are the point. Holding the true Sharpe at 0.5 and dragging the length from twenty years down to three shows the shaded region shrinking to almost nothing, which is statistical power made visible. Then setting the true Sharpe to *zero* and adding a slider for "number of strategies tested" shows the best-of-N realised Sharpe climbing steadily above 1 with nothing behind it, which is the multiple-testing problem in one picture.

A second panel runs the VaR backtest: a slider for the model's true exception rate, a bar chart of the binomial distribution over 250 days, and the green, yellow and red zones drawn as vertical bands, so the reader can drag the true rate from 1% to 2% and watch how much of the distribution still lands in green.

8.4 Linear and multiple regression

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Regression is the one statistical tool you will actually touch on a desk. When a trader says a stock has "a beta of 1.3 to the index", when a risk manager attributes yesterday's PnL to spot, vol and carry, when a quant builds a factor model or a hedge ratio for a basis trade, the machinery underneath is always the same: fit a straight line through a cloud of points and ask how much of the noise that line explains. Interviewers like it because it is impossible to bluff. Either you can write down the estimator, state the assumptions, and say what breaks in financial data, or you cannot.

Starting from what you know

You have two columns of numbers observed together: a variable you want to explain, y, and a variable you think explains it, x. In markets these are almost always **returns** rather than prices (we will see below why that matters). Say y is the monthly return of a single stock and x is the monthly return of the index. Plot the n pairs (x_i, y_i) as a scatter. If the cloud slopes upward, the stock tends to rise when the index rises. Regression turns "tends to" into a number.

INTUITION

Imagine stretching a rigid steel rod through the cloud of dots, and attaching each dot to the rod by a small vertical rubber band. Each band pulls on the rod with a force that grows with how far the dot sits from it. Release the rod and it settles in the position where the total stored energy is smallest. That resting position is the regression line. Two things follow immediately and are worth remembering: the bands are *vertical*, so the line is chosen to explain y from x and not the other way round, and the energy grows with the *square* of the stretch, so one far-away dot can drag the whole rod.

The simple linear model

We assume the data were generated by

$$y_i \;=\; \alpha \;+\; \beta x_i \;+\; \varepsilon_i, \qquad i = 1, \ldots, n,$$

where α (the **intercept**) and β (the **slope**) are unknown constants and ε_i is an unobservable **error**: everything moving y that x does not capture. Note the vocabulary carefully, because interviewers do. The error ε_i belongs to the true model and is never observed. What you can compute, once you have estimates $\hat\alpha$ and $\hat\beta$, is the **residual**

$$e_i \;=\; y_i - \hat\alpha - \hat\beta x_i.$$

Ordinary least squares (OLS) picks the pair $(\hat\alpha, \hat\beta)$ that minimises the sum of squared residuals

$$\mathrm{SSR}(\alpha, \beta) \;=\; \sum_{i=1}^{n} \big(y_i - \alpha - \beta x_i\big)^2.$$

Differentiate with respect to α and β and set both derivatives to zero. The first condition, $\sum_i (y_i - \alpha - \beta x_i) = 0$, says the residuals must average to zero and gives $\hat\alpha = \bar y - \hat\beta \bar x$: the line passes through the centre of gravity $(\bar x, \bar y)$ of the cloud. Substituting that back into the second condition, $\sum_i x_i (y_i - \alpha - \beta x_i) = 0$, gives the estimator you must be able to write from memory:

$$\boxed{\hat\beta \;=\; \frac{\sum_i (x_i - \bar x)(y_i - \bar y)}{\sum_i (x_i - \bar x)^2} \;=\; \frac{\widehat{\mathrm{Cov}}(x, y)}{\widehat{\mathrm{Var}}(x)}} \qquad \hat\alpha \;=\; \bar y - \hat\beta\, \bar x.$$

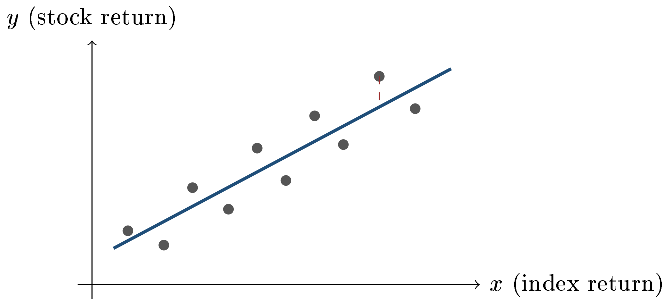
> **KEY IDEA**
>
> The slope is a covariance divided by a variance. Written with the correlation ρ and the two standard deviations it becomes $\hat\beta = \rho\,\dfrac{\sigma_y}{\sigma_x}$, which is the single most useful identity in this section. Correlation is a *unitless* measure of how tightly the points hug a line; beta is a *scaled* measure of how much y moves per unit of x. A stock can be perfectly correlated with the index ($\rho = 1$) and still have a beta of 0.5 if it is half as volatile.

How much did the line explain? R^2

Decompose the variation in y into the part the line reproduces and the part it does not. Writing $\hat y_i = \hat\alpha + \hat\beta x_i$ for the fitted values,

$$\underbrace{\sum_i (y_i - \bar y)^2}_{\text{TSS, total}} \;=\; \underbrace{\sum_i (\hat y_i - \bar y)^2}_{\text{ESS, explained}} \;+\; \underbrace{\sum_i e_i^2}_{\text{SSR, residual}}, \qquad R^2 \;=\; \frac{\mathrm{ESS}}{\mathrm{TSS}} \;=\; 1 - \frac{\mathrm{SSR}}{\mathrm{TSS}}.$$

The cross-term vanishes exactly because of the two first-order conditions above, which is why the decomposition is clean only for OLS with an intercept. So R^2 is the fraction of the variance of y explained by x, it lies in $[0, 1]$, and in a simple regression it is nothing more than ρ^2. In equity terms: R^2 is the share of the stock's variance that is **systematic**, and $1 - R^2$ is the share that is **idiosyncratic**.



A worked example: estimating a stock's beta

WORKED EXAMPLE

Five monthly returns, in percent. Index x, stock y:

x_i	2	-1	3	0	1
y_i	3	-3	5	1	4

Means: $\bar{x} = 5/5 = 1$ and $\bar{y} = 10/5 = 2$. Deviations $x_i - \bar{x} = (1, -2, 2, -1, 0)$ and $y_i - \bar{y} = (1, -5, 3, -1, 2)$.

Cross-products: $1(1) + (-2)(-5) + 2(3) + (-1)(-1) + 0(2) = 1 + 10 + 6 + 1 + 0 = 18$.
Squares of x deviations: $1 + 4 + 4 + 1 + 0 = 10$.
Squares of y deviations: $1 + 25 + 9 + 1 + 4 = 40$ (this is TSS).

So $\hat{\beta} = 18/10 = 1.8$ and $\hat{\alpha} = 2 - 1.8(1) = 0.2$. The fitted line is $\hat{y} = 0.2 + 1.8x$.

Check with the identity $\hat{\beta} = \rho\,\sigma_y/\sigma_x$. Here $\sigma_y/\sigma_x = \sqrt{40/10} = 2$ and $\rho = 18/\sqrt{10 \times 40} = 18/20 = 0.9$, so $\rho\,\sigma_y/\sigma_x = 0.9 \times 2 = 1.8$. It matches, as it must.

Residuals, from $\hat{y}_i = (3.8, -1.6, 5.6, 0.2, 2.0)$: $e_i = (-0.8, -1.4, -0.6, 0.8, 2.0)$. They sum to zero, the first sanity check of any fit. Then $\text{SSR} = 0.64 + 1.96 + 0.36 + 0.64 + 4.00 = 7.60$ and

$$R^2 \;=\; 1 - \frac{7.60}{40} \;=\; 1 - 0.19 \;=\; 0.81 \;=\; \rho^2.$$

WORKED EXAMPLE

Continuing the same five observations: **is the beta statistically there?** The residual variance is estimated with $n-2$ degrees of freedom (two parameters were fitted): $s^2 = \mathrm{SSR}/(n-2) = 7.60/3 = 2.533$. The standard error of the slope is

$$\mathrm{se}(\hat\beta) = \sqrt{\frac{s^2}{\sum_i (x_i-\bar x)^2}} = \sqrt{\frac{2.533}{10}} = \sqrt{0.2533} = 0.503, \qquad t = \frac{1.8}{0.503} = 3.58.$$

The two-sided 5% critical value of a t distribution with 3 degrees of freedom is 3.182, so we just reject $\beta = 0$. An R^2 of 0.81 looks impressive, yet with five observations the slope is barely significant. That contrast is the whole lesson: *fit is not evidence.* State it in an interview and you sound like someone who has actually run regressions.

Using it. If you are long €1,000,000 of the stock and want to neutralise index risk, short $\hat\beta$ times the notional, so €1,800,000 of the index (via futures, in practice). What remains is the idiosyncratic 19% of variance, plus the very real risk that $\hat\beta$ itself moves tomorrow.

Multiple regression

One explanatory variable is rarely enough. A stock is exposed to the market, but also to a sector, to rates, to a value or momentum factor. With k regressors,

$$y_i \;=\; \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik} + \varepsilon_i,$$

which is far easier to handle stacked into matrices. Let y be the $n \times 1$ vector of observations and X the $n \times (k+1)$ **design matrix** whose first column is a column of ones (that column is what produces the intercept). Then minimising $\|y - X\beta\|^2$ gives the **normal equations** $X^\top X\hat\beta = X^\top y$, hence

$$\boxed{\hat\beta \;=\; (X^\top X)^{-1}X^\top y}$$

whenever $X^\top X$ is invertible, that is whenever no regressor is an exact linear combination of the others. Geometrically, $\hat y = X\hat\beta$ is the orthogonal projection of y onto the space spanned by the columns of X, and the residual vector is what is left perpendicular to it. That single picture explains why the residuals are orthogonal to every regressor and why adding a regressor can never increase the SSR.

Interpretation changes in an important way. In a multiple regression, β_j is the effect of x_j on y **holding the other regressors fixed**. The market beta of a stock estimated alone and its market beta estimated alongside a sector index are different numbers answering different questions, and confusing the two is a classic interview trap.

Two practical consequences:

- **Multicollinearity.** If two regressors are nearly redundant (a sector index and the broad index, or the 5y and 7y swap rate), $X^\top X$ is close to singular. The coefficients stay unbiased but their standard errors explode: they flip sign from one sample to the next and neither is "significant", even though jointly they explain a lot. The fix is to drop one, or to rebuild the regressors as orthogonal blocks (regress on the index and on the *sector minus index* spread, which is what desks usually do).
- **Adjusted R^2.** Because R^2 mechanically rises with every regressor added, even a random one, compare models with $\bar R^2 = 1 - (1-R^2)\dfrac{n-1}{n-k-1}$, which charges a price for each parameter. With n small and k large, R^2 approaching 1 means nothing.

The assumptions, and which ones matter

Say these in the right order, separating what you need for *unbiasedness*, what you need for *efficiency*, and what you need only for *exact inference*. That structure is the answer interviewers are grading.

1. **Linearity in the parameters.** The model is linear in β, not necessarily in the raw data: x^2 or $\log x$ as a regressor is still OLS.
2. **Exogeneity**, $\mathbb{E}[\varepsilon \mid X] = 0$. The one that really matters. If it fails (an omitted variable driving both y and x, or y feeding back into x), $\hat{\beta}$ is **biased** and no amount of data fixes it.
3. **No perfect multicollinearity**, so $X^\top X$ is invertible. Needed for the estimator to exist at all.
4. **Homoskedasticity**, $\text{Var}(\varepsilon_i \mid X) = \sigma^2$ for all i.
5. **No autocorrelation**, $\text{Cov}(\varepsilon_i, \varepsilon_j) = 0$ for $i \neq j$.
6. **Normal errors.** *Not* required for OLS to be unbiased or efficient. It is required only for the t and F statistics to be exactly t and F in a small sample; in large samples the central limit theorem covers you.

Assumptions 1 to 3 give unbiasedness, $\mathbb{E}[\hat{\beta}] = \beta$. Adding 4 and 5 gives the **Gauss-Markov theorem**: OLS is BLUE, the Best Linear Unbiased Estimator, meaning smallest variance among all linear unbiased estimators. Note the two qualifiers, *linear* and *unbiased*: a biased or non-linear estimator (ridge, shrinkage towards 1 for equity betas) can beat OLS on mean squared error, which is exactly why practitioners use shrunk betas.

Now the part that separates a finance candidate from a statistics student: **in market data, 4 and 5 fail almost always**. Volatility clusters, so residual variance is anything but constant, which is the subject of ARCH and GARCH modelling. Residuals of overlapping or illiquid-asset returns are serially correlated, see autocorrelation. When 4 or 5 fail, the coefficients are still unbiased but the standard errors are wrong, usually too small, so you overstate significance. The desk fix is not to abandon OLS but to correct the covariance matrix: White (heteroskedasticity-consistent) or Newey-West (also autocorrelation-consistent) standard errors.

> INTERVIEW TRAP
>
> The two errors that end a technical round. First, **spurious regression**: regress the *level* of one non-stationary price on another and you will get $R^2 = 0.95$ and a huge t statistic between two totally unrelated series. This is why desks regress returns, not prices, and why stationarity is tested first. Second, confusing **correlation with causation**, or in the regression dialect, forgetting that exogeneity is an assumption you cannot test from the residuals alone. A close third: saying "OLS requires normally distributed errors". It does not, and the interviewer is waiting for that one.

REAL INTERVIEW QUESTION (Credit Suisse · desk not recorded)

What are the assumptions of linear regression?

WHAT THEY TEST / ANSWER

They are testing whether you can organise knowledge, not recite it. Give the six assumptions grouped by what each one buys you: linearity, exogeneity and no perfect collinearity for unbiasedness; homoskedasticity and no autocorrelation for efficiency, that is Gauss-Markov and BLUE; normality only for exact small-sample t and F inference. Then volunteer the finance twist without being asked: in returns data homoskedasticity and independence both fail, so the coefficients survive but the standard errors do not, and you switch to White or Newey-West. Candidates who list nothing but "linear, independent, normal" sound like they learned it from a slide.

REAL INTERVIEW QUESTION (Société Générale · desk not recorded)

Define beta and illustrate it. What does a beta above or below 1 mean?

WHAT THEY TEST / ANSWER

Beta is the slope of the regression of the asset's returns on the market's returns, $\beta = \mathrm{Cov}(r_a, r_m)/\mathrm{Var}(r_m) = \rho\,\sigma_a/\sigma_m$. It is the expected move in the asset for a 1% move in the market, and equivalently the hedge ratio: to neutralise market risk on €1 of the asset, short €β of the index. $\beta > 1$ means amplification, typically cyclicals, high-leverage or long-duration growth names; $\beta < 1$ means damping, typically utilities and staples; $\beta \approx 0$ means no linear market exposure, which is not the same as no risk. Illustrate with the scatter and the fitted line, and add the two details that show experience: beta says nothing about the $1 - R^2$ of idiosyncratic variance you keep, and estimated betas are unstable, which is why they are shrunk towards 1 and re-estimated on a rolling window.

REAL INTERVIEW QUESTION (Natixis · QIS)

How would you build a portfolio with low beta risk (market-neutral)?

WHAT THEY TEST / ANSWER

Answer as a process, not a formula. Estimate each name's beta by regressing its returns on the chosen market factor, pick a horizon and frequency you can defend (for example two years of daily or five years of weekly returns, weekly reducing the asynchronous-trading bias in daily data), then size the short index leg so the notional-weighted betas cancel, $\sum_i w_i \beta_i = 0$. Then say what the naive answer misses, which is the real test: beta-neutral is not risk-neutral. You are still exposed to the residual factors, sector, size, momentum and rates, so a serious implementation regresses on several factors and neutralises each one, which is precisely a multiple regression. Add that beta drifts, so the hedge has to be rebalanced, and that rebalancing costs money, so the choice of estimation window is a trade-off between stale betas and noisy betas.

REAL INTERVIEW QUESTION (Société Générale · EQD)

What is the relationship between stochastic volatility and local volatility? What is the beta of a linear regression?

WHAT THEY TEST / ANSWER

A two-part question, and the second half is the free marks. The beta is the slope coefficient, $\hat{\beta} = \widehat{\mathrm{Cov}}(x, y)/\widehat{\mathrm{Var}}(x)$, the average change in y per unit change in x, which for asset returns is the market sensitivity and the hedge ratio. Say it in one clean sentence, give the covariance-over-variance form, and do not drift into R^2 unless asked. For the first half, the link between local and stochastic volatility is the subject of local volatility and stochastic volatility models: local vol is a deterministic function of spot and time calibrated to today's surface, while stochastic vol gives volatility its own random driver, and the local vol is recovered as a conditional expectation of the stochastic variance. The lesson for the interview itself is to answer both halves and to signal which one you are answering, rather than letting the easy half disappear.

VISUAL / CODE TO BUILD: Interactive

A regression sandbox. The reader drags points in a scatter plot and watches $\hat{\alpha}$, $\hat{\beta}$, R^2, $\mathrm{se}(\hat{\beta})$ and the t statistic update live, with the residual vertical bars drawn. Three preset buttons make the teaching points that static text cannot: **(i) outlier**, drag one point far away and watch the slope swing while R^2 collapses; **(ii) small sample**, delete points and watch R^2 stay high while the t statistic dies; **(iii) spurious**, switch the axes from returns to cumulated prices on two independent random walks and watch R^2 jump above 0.9. Backed by a few lines of NumPy; also expose the fitted numbers as a table for accessibility.

Where this goes next

The inference machinery used above (t statistics, critical values, what a p value does and does not say) is developed in hypothesis testing, and the covariance and correlation inputs come from independence and correlation. Regression with time on the right-hand side, or with the past of y itself, is the gateway to time series analysis and the AR and MA family, where the no-autocorrelation assumption is not a nuisance but the object of study. Modelling the residual variance directly leads to ARCH and GARCH. On the desk, the same least-squares logic reappears as the attribution step of PnL explain and as the factor mapping behind Value at Risk.

8.5 Time series analysis

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Almost every number a desk looks at is a time series. Prices, rates, spreads, volumes, volatilities, P&L: all of them are sequences indexed by time, and in every one of them the order of the observations carries information. Shuffle a cross-section of survey responses and you lose nothing. Shuffle a price history and you have destroyed the only thing that made it a price history.

That single fact breaks most of the statistics built for independent samples, and repairing it occupies the rest of this chapter. This section sets out what breaks, what the standard repairs are, and where each one is developed.

What order-dependence destroys

The classical toolkit assumes observations are **independent and identically distributed**. A financial time series usually violates both halves.

- **Independence fails.** Today's observation is related to yesterday's. That relationship is measured by the autocorrelation function, developed in autocorrelation, and modelled in AR, MA, ARMA and ARIMA models.
- **Identical distribution fails.** The mean, the variance or both change over time. A series whose distribution does not drift is called **stationary**, and it is the property everything else needs, which is why it has its own section in stationarity and its own tests in unit root tests.

> KEY IDEA
>
> The effective sample size of a dependent series is smaller than the number of observations, sometimes far smaller. A thousand daily returns during one long trend is not a thousand independent pieces of evidence about the trend. Every standard error computed as if the data were independent is too small, and every confidence interval built from it is too narrow.

The one transformation that fixes most of it

Prices are not stationary. A price is a random walk plus a drift: it has no level it returns to, and the variance of where it will be grows without limit. Its sample mean estimates nothing, because there is nothing to estimate.

Returns are approximately stationary. Differencing the log price removes that problem, and the transformed series has a stable mean near zero and a variance that moves but stays finite. This is why financial analysis almost always begins by turning prices into returns, and why skipping that step is the commonest error in applied work.

> WORKED EXAMPLE
>
> Take a random walk $S_t = S_{t-1} + \varepsilon_t$ with ε_t having standard deviation σ. Then
>
> $$\mathrm{Var}(S_t) = t\,\sigma^2, \qquad \mathrm{sd}(S_t) = \sigma\sqrt{t},$$
>
> so the spread of possible values grows without bound and there is no long-run mean. The first difference $S_t - S_{t-1} = \varepsilon_t$ is white noise: constant mean, constant variance, no autocorrelation. One subtraction has turned an unusable series into a usable one.
>
> Note what this also says: the $\sqrt{t}$ scaling that lets you annualise a daily volatility by multiplying by $\sqrt{252}$ is exactly the random-walk property. If returns were autocorrelated, that scaling would be wrong, which is why the two facts belong together.

Why this matters more than it sounds: spurious regression

Here is the concrete cost of ignoring non-stationarity, and it is worse than most people expect.

Take two series that are completely independent of each other. Regress one on the other. If both are stationary, you reject the null of no relationship about 5% of the time, as a 5% test should. If both are random walks, you reject it most of the time.

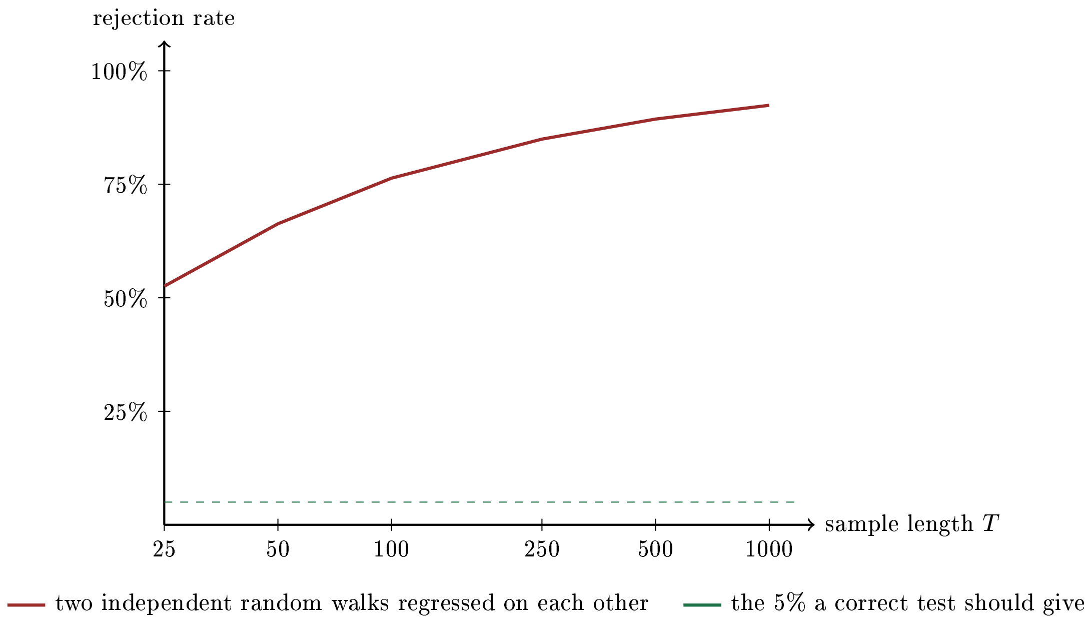
WORKED EXAMPLE

Generate pairs of *independent* random walks, regress one on the other, and count how often the slope comes out statistically significant at the 5% level. Two hundred thousand simulations at each length, so the figures below are the true rates rather than one noisy run:

Length T	Rejections at 5%	Mean R^2
25	52.6%	0.24
50	66.3%	0.24
100	76.4%	0.24
250	85.0%	0.24
500	89.4%	0.24
1000	92.4%	0.24

There is no relationship in the data at all. The correct answer is 5%. At a year of daily data, $T = 250$, the test declares a relationship **85%** of the time, with an average R^2 of 0.24, which on a printed regression output looks like a finding.

The last column is the part to internalise: R^2 does not fall as the sample grows. And the rejection rate *rises*. Collecting more data makes the illusion stronger, which is the exact opposite of the intuition every other part of statistics gives you.



The gap between the two lines is the whole argument for differencing before regressing, and it is why unit root tests exist as a routine step rather than as a formality.

Decomposition: four things in every series

The classical way to look at a series is as a sum of four components.

- **Trend**: the slow direction.
- **Seasonal**: a pattern that repeats on a fixed calendar.
- **Cycle**: a repeating pattern with no fixed period, such as the business cycle.
- **Irregular**: what is left, and the part that is modelled as random.

Seasonality is easy to dismiss as an economics-textbook concern, and in finance it is real and specific. Natural gas and heating oil have genuine annual demand cycles. Equity index volumes have day-of-week and turn-of-month patterns, and expiry weeks have their own. Earnings arrive quarterly and implied volatility rises into them and collapses after. Government bond issuance follows a published calendar. Anything a desk measures over time should be checked against the calendar before it is called a signal.

The stylised facts, and the one that surprises people

Financial returns behave the same way across markets and decades, which is why the same three observations appear in every treatment.

1. **Returns are almost unautocorrelated.** Knowing yesterday's return tells you almost nothing about today's sign. This is what market efficiency looks like in the data.
2. **Squared and absolute returns are strongly autocorrelated.** Big moves follow big moves. This is **volatility clustering**, and it is the reason volatility is forecastable when returns are not.
3. **Returns have fat tails.** Extreme moves happen far more often than a normal distribution allows.

WORKED EXAMPLE

Two simulated series of 5,000 daily returns. The first is independent normal draws. The second has clustered volatility, built so that its long-run level is about 22% annualised. Lag-one autocorrelations:

Series	of r_t	of r_t^2	of $\lvert r_t \rvert$
Independent normal	$+0.000$	$+0.010$	$+0.017$
Clustered volatility	-0.002	$+0.147$	$+0.150$

Read the first column: the two series are indistinguishable. Both have essentially zero return autocorrelation, so a test built on returns alone cannot tell them apart. Read the second and third: they are completely different, and the difference does not disappear at longer lags. The autocorrelation of squared returns decays from its lag-one value, but it decays *slowly*, and it is still clearly positive five and ten days out.

That is the whole reason volatility modelling is a separate discipline. The structure in financial data is not in the returns, it is in their magnitude, you have to square or absolute them before it becomes visible, and the slow decay is what makes it forecastable over horizons that matter. Modelling it is volatility modeling.

Interview angle

REAL INTERVIEW QUESTION (Morgan Stanley · QIS)

How do you set the date as the index?

WHAT THEY TEST / ANSWER

The literal answer is one line of code, so give it and then show that you know why anyone would ask. Parse the date column into a real datetime type, set it as the index, and sort. Then say what that actually buys you, because the question is a proxy for whether you have handled time series or only read about them.

A proper time index gives you four things a plain integer index does not. **Ordering**, which is not automatic: files arrive in whatever order they were written, and a series that is out of order silently corrupts every lag, every return and every rolling window. **Alignment**: two series indexed by date can be joined on the date rather than on position, which matters the moment two markets have different holidays, and a positional join across an unmatched holiday shifts one series by a day and manufactures a correlation that does not exist. **Resampling**: converting daily to weekly or monthly, which needs to know what a week is. **Slicing by period** without hand-written masks.

Then volunteer the traps, because they are what a desk actually loses money to. **Time zones and stamping conventions**: a European close and a US close on the same calendar date are seven or eight hours apart, so a same-date join of two indices is not a same-time join and it biases the measured correlation downward. **Duplicate and missing dates**: duplicates from a bad vendor feed, gaps from holidays, and whether a gap should be forward-filled depends entirely on what you are computing, since forward-filling a price is defensible and forward-filling a return invents a zero. **Look-ahead**: an index built from the date a value refers to, rather than the date it became known, will let a backtest use information it could not have had.

Say the last one out loud even if it was not asked. Getting the index right is the whole of the difference between a backtest that means something and one that does not.

REAL INTERVIEW QUESTION (Tikehau Capital · Market Risk)

Do you know RiskMetrics?

WHAT THEY TEST / ANSWER

Yes, and the reason to know it is that it is the simplest useful time series model of volatility and the ancestor of everything that came after it.

RiskMetrics was J.P. Morgan's risk methodology, published in the mid-1990s and made freely available, which is a large part of why value-at-risk became an industry standard rather than one bank's tool. Its estimator is an **exponentially weighted moving average** of squared returns:

$$\sigma_t^2 = \lambda\,\sigma_{t-1}^2 + (1-\lambda)\,r_{t-1}^2,$$

with $\lambda = 0.94$ for daily data and 0.97 for monthly. Explain what the choice does: weights decay geometrically into the past, so recent observations matter more, and λ sets how fast the memory fades. At 0.94 the half-life is about **11 days** and the effective sample size is $1/(1-\lambda) \approx 17$ days, which is why the estimate responds to a volatility spike within a fortnight instead of a quarter.

Two things elevate the answer. First, the connection: EWMA is exactly a GARCH(1,1) with the constant term set to zero and the two coefficients summing to one, so it is a GARCH with *no mean reversion*. That is its main limitation: it will happily forecast today's volatility forever rather than pulling it back toward a long-run level, which is precisely what a proper GARCH adds. See volatility modeling. Second, the critique: the whole framework assumes normally distributed returns and a covariance matrix estimated from a short recent window, so it underestimates tails and updates its correlations only after a regime has already changed. Those two weaknesses are most of the case for the post-crisis move to stressed calibrations and expected shortfall, which is value at risk.

INTERVIEW TRAP

Four errors that survive into production systems. (1) Regressing levels on levels, when the table above shows an 85% false-positive rate at a year of daily data. (2) Using overlapping windows, monthly returns sampled daily for instance, and then treating the observations as independent, which understates every standard error by roughly the square root of the overlap. (3) Look-ahead bias: using a value on the date it refers to rather than the date it was published, which quietly makes any backtest work. (4) Testing many specifications on one history and reporting the best, which is a search over noise and not a result.

Where this goes next

The property everything needs is stationarity, and how to test for it is unit root tests. Measuring the dependence is autocorrelation and modelling it is AR, MA, ARMA and ARIMA models. The structure that actually exists in financial data, in the magnitudes rather than the signs, is volatility modeling, and its most-used application is value at risk.

8.6 Stationarity

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Stationarity is the assumption underneath every statistic a desk quotes, and it is almost never stated. A mean, a standard deviation, a correlation, a beta: each of these is a property of a distribution, and quoting one for a time series presumes that there *is* a fixed distribution to have a property. If the process wanders, the number you computed describes the sample you happened to see and nothing else. Everything in this chapter depends on getting this right, which is why it sits before autocorrelation and the ARMA family rather than after them.

The definition, in the version people actually use

A process $\{X_t\}$ is **strictly stationary** if the joint distribution of any collection of observations is unchanged by shifting all of them forward in time. That is a strong condition and almost never testable, so in practice everyone uses the weaker one.

> **KEY IDEA**
>
> $\{X_t\}$ is **weakly**, or **covariance**, stationary if three things hold:
>
> 1. $\mathbb{E}[X_t] = \mu$, the same for all t;
> 2. $\mathrm{Var}(X_t) = \sigma^2 < \infty$, the same for all t;
> 3. $\mathrm{Cov}(X_t, X_{t+k})$ depends on the lag k only, not on t.
>
> The third condition is the one that does the work: it says the relationship between today and k days from now is the same relationship whenever you look. Under normality the two definitions coincide, since a joint normal is fully described by its first two moments.

Prices are not, returns roughly are

This is the single fact to carry out of the section, and it explains a convention that otherwise looks arbitrary.

An equity index at 4,000 today was at 400 thirty years ago. There is no μ: the mean of the level is a function of the window you chose. The variance grows with the horizon rather than settling. And a shock is **permanent**, because a level that falls twenty percent does not come back to where it was by construction. All three stationarity conditions fail.

Take log differences and the picture changes. Daily returns have a mean near zero that does not drift with the sample, a variance that is finite and roughly stable over moderate windows, and an autocovariance structure that looks similar in 1998 and in 2024. That is why every model in finance is written on returns, why descriptive statistics of a price series is a category error, and why geometric Brownian motion models the log of the price rather than the price.

The honest caveat, and the one that gets you the follow-up: returns are only *approximately* stationary. Volatility clusters, so the second moment is itself time-varying, which is the whole reason GARCH exists, and it is the same phenomenon as the gap between implied and realised volatility regimes in implied versus realised.

The knife edge: random walk against AR(1)

The cleanest way to hold stationarity in your head is one equation with one parameter,

$$X_t \;=\; \phi X_{t-1} + \varepsilon_t, \qquad \varepsilon_t \text{ independent, mean zero, variance } \sigma^2.$$

For $|\phi| < 1$ the process is stationary. It has a long-run mean of zero, a finite long-run variance $\sigma^2/(1-\phi^2)$, and shocks that decay geometrically. For $\phi = 1$ it is a **random walk**: the variance after T steps is $T\sigma^2$, there is no long-run mean, and every shock is permanent because it is simply added to the level and never removed.

INTUITION

The single number worth quoting is the **half-life** of a shock, $\ln(0.5)/\ln(\phi)$, the time for a disturbance to decay to half its size. At $\phi = 0.9$ that is 6.58 periods, and the process visibly reverts. At $\phi = 0.99$ it is 69.0 periods, and over any sample shorter than a few hundred observations the series is statistically indistinguishable from a random walk even though it is, formally, stationary. That gap is why mean-reversion trades are hard: a spread can be genuinely mean-reverting and still take longer to revert than your risk limit allows.

Contrast the two dispersions directly. At $\phi = 0.9$ the stationary variance is $\sigma^2/(1-0.81) = 5.26\,\sigma^2$, and it stops there no matter how long you wait. At $\phi = 1$ the variance after a trading year is $252\,\sigma^2$, a standard deviation of $15.87\,\sigma$, and it keeps going. A bounded process and an unbounded one, separated by one tenth in a coefficient, and at $\phi = 0.99$ the bound is still there, at $50.3\,\sigma^2$, one hundredth from the edge.

Spurious regression, or why this is not pedantry

The practical damage from ignoring stationarity is that ordinary statistical tests stop working, and they fail in the direction that flatters you. Regress one random walk on an independent one and you reject the null of no relationship most of the time rather than 5% of the time, with a respectable R^2 and a large t-statistic, on data built to have no relationship at all. Worse, the false-positive rate *rises* with sample length, so the usual reassurance that more data fixes things is exactly backwards. The simulation that demonstrates this, and the numbers, are in time series analysis.

What belongs here is the mechanism, because it is what the follow-up asks and because it is a fact about stationarity rather than about regression. Two random walks both wander, and over any finite sample a straight line fits whatever wandering happened. The residuals of that regression are themselves a random walk, which violates the independence assumption the standard errors are built on, so the standard errors are far too small and the t-statistic they produce is meaningless. This is a hypothesis testing failure with a specific cause, and it is the reason regression on financial levels is almost always wrong. The fix follows directly from the diagnosis: regress the differences, which are $I(0)$, and the test recovers its nominal size.

What to do about it

- **Difference.** A series that becomes stationary after d differences is **integrated of order** d, written $I(d)$. Prices are $I(1)$, returns are $I(0)$, and taking log differences is the whole of the fix for most of finance.
- **Test rather than assume.** The formal check is a unit-root test, developed in unit-root tests. Note the awkward null: the standard test has non-stationarity as H_0, so failing to reject is not evidence of stationarity, merely absence of evidence against a random walk, and with $\phi = 0.99$ you will fail to reject almost regardless of the truth.
- **Cointegration**, the exception that pays. Two $I(1)$ series can have a linear combination that is $I(0)$: each wanders, but they wander together and the spread does not. That is the

statistical content of a pairs trade, of a cash-and-carry basis, and of a swap-spread position. The trade is not a bet on either level; it is a bet that the stationary combination reverts. When it stops being stationary, which is what a structural break means, the trade becomes a directional position nobody sized for.

REAL INTERVIEW QUESTION (JP Morgan · desk not recorded)

Do you believe in the random walk?

WHAT THEY TEST / ANSWER

This is an opinion question with a right shape of answer: separate the statistical claim from the economic one, and refuse the yes or no.

As a statistical description, largely yes, and usefully so. Price levels behave like an $I(1)$ process, daily return autocorrelations are tiny and unstable, and the increments are close enough to unpredictable that treating the level as a random walk is a good working model. It is also the model with real teeth: it is what makes Brownian motion the right object for option pricing, and it is why the best forecast of tomorrow's price is today's price.

As a literal description, no, and say precisely where it breaks. Volatility is strongly autocorrelated even when returns are not, so the squared returns are predictable when the returns are not, and that is the GARCH observation. Returns have fat tails and negative skew, which a Gaussian walk does not produce. And some spreads are genuinely mean-reverting, which is a direct violation of the random walk on the object being traded.

Then resolve the tension, which is what the question is really for. The random walk is close enough to true that beating it is hard, and the deviations from it are small, unstable and expensive to trade. So the sophisticated position is that markets are *nearly* efficient: the predictable component is real but is roughly the size of the cost of capturing it, which is why it survives. Saying you flatly believe in the random walk makes you sound like you have never looked at data; saying you flatly do not makes you sound like you think you can beat the market with a regression.

INTERVIEW TRAP

1. **Computing a correlation between two price series.** Almost every such number is spurious. Correlate returns, always, and if someone shows you a correlation of 0.9 between two levels, ask what it is on differences.
2. **Reading "failed to reject a unit root" as "it is a random walk".** The test is low-powered against near-unit-root alternatives by construction, so this is the Type II error from hypothesis testing in its natural habitat.
3. **Assuming stationarity is permanent.** Regimes break. A volatility mean that was stable for five years moves after a structural change in the market, and the model calibrated on the old regime keeps producing confident numbers.
4. **Differencing something already stationary.** Over-differencing introduces a spurious negative autocorrelation at lag one and inflates the variance. Test first; do not difference reflexively.
5. **Confusing stationary with independent.** A stationary series can be strongly autocorrelated, which is exactly what an AR(1) with $\phi = 0.9$ is. Stationarity constrains the moments, not the memory.

VISUAL / CODE TO BUILD: Interactive

One panel, one slider: ϕ from 0 to 1.02. The page draws a long simulated path of $X_t = \phi X_{t-1} + \varepsilon_t$ from a fixed seed, so only the parameter changes as the reader drags, together with a shaded band at $\pm 2\sigma/\sqrt{1-\phi^2}$ where that is finite. Three moments are the point. At $\phi = 0.5$ the path visibly rattles inside a fixed band. As ϕ approaches 1 the band widens toward the edges of the plot and the path starts to look like a price chart. At exactly 1 the band disappears and the path wanders off, and above 1 it explodes, which shows that the stationary region has an edge rather than a gradient.

A second panel shows the residuals rather than the fit, which is the part this section owns: regress one drawn walk on the other and plot the residual series beneath the scatter, so the reader sees that the residuals themselves wander instead of scattering about zero. That is the assumption the standard errors rest on, visibly broken. Toggling to differences makes the residuals flatten into noise in the same frame. The live spurious-regression experiment itself belongs to time series analysis.

8.7 Autocorrelation

sales · trading · structuring | core | cross-asset

Correlation between two different assets is familiar. Autocorrelation asks a stranger question: how correlated is a series with *its own past*? The answer decides whether prices are predictable, whether momentum and mean-reversion strategies have anything to trade, whether the $\sqrt{252}$ you use to annualise a daily volatility is legitimate, and whether the Sharpe ratio in a fund's pitch book is real. Interviewers reach for it because one clean formula, the variance of a sum, connects all of those at once.

From correlation to autocorrelation

Take a series observed through time, $X_1, X_2, \dots, X_n$: daily returns of an index, changes in the 10y swap rate, the daily PnL of a desk. Pair each observation with the one k periods later and compute the correlation of that paired sample. That is the autocorrelation at **lag** k.

> INTUITION
>
> Autocorrelation is the **echo** in a series. Clap once in a padded studio and the sound dies instantly: no echo, no memory, each instant tells you nothing about the next. That is what a return series with zero autocorrelation looks like. Clap in a stone church and the sound rolls on, each instant an attenuated copy of the last: positive autocorrelation, a trend, momentum. There is also a stranger room, where every echo comes back *inverted*, so a loud instant is followed by a quiet one: negative autocorrelation, mean reversion. The lag k is simply how long you wait before listening for the echo.

For this to be a well-posed question, the statistical properties must not themselves drift through time. We therefore assume the series is **weakly stationary** (constant mean, constant variance, and an autocovariance that depends only on the lag k and not on where you are in the sample), which is developed in stationarity. Under that assumption define the **autocovariance** and **autocorrelation** functions

$$\gamma_k \;=\; \mathrm{Cov}(X_t,\, X_{t+k}), \qquad \rho_k \;=\; \frac{\gamma_k}{\gamma_0}, \qquad \gamma_0 = \mathrm{Var}(X_t).$$

So $\rho_0 = 1$ by construction, $\rho_k \in [-1, 1]$, and $\rho_{-k} = \rho_k$. The whole collection $\{\rho_k\}_{k\geq 1}$ is the **autocorrelation function**, or **ACF**, and plotting it against k is the first thing anyone does with a new time series. From a sample of length n with mean $\bar{X}$, the usual estimator is

$$\hat{\rho}_k \;=\; \frac{\sum_{t=1}^{n-k}(X_t - \bar{X})(X_{t+k} - \bar{X})}{\sum_{t=1}^{n}(X_t - \bar{X})^2}.$$

Note the asymmetry of the sums: the numerator has $n-k$ terms and the denominator n. That is deliberate, it shrinks $\hat{\rho}_k$ towards zero at long lags and keeps the estimated ACF from implying a variance that cannot exist. It also means high-lag estimates are heavily biased downward, so never read much into $\hat{\rho}_{40}$ from 50 observations.

> KEY IDEA
>
> A process with $\rho_k = 0$ for all $k \geq 1$ is **white noise**: no linear memory. Prices being a **random walk** is exactly the statement that their *returns* are white noise. Zero autocorrelation is a statement about *linear* dependence only, and this is the distinction that earns marks: squared returns can be violently autocorrelated while returns themselves are not. Uncorrelated does not mean independent.

Why it matters: the variance of a sum

Here is the formula that turns autocorrelation from a diagnostic into money. For a weakly stationary series, the variance of the n-period sum $S_n = X_1 + \cdots + X_n$ is

$$\mathrm{Var}(S_n) \;=\; n\gamma_0\Big[\,1 + 2\sum_{k=1}^{n-1}\Big(1 - \tfrac{k}{n}\Big)\rho_k\Big].$$

Every cross-term $\mathrm{Cov}(X_i, X_j)$ appears once for each of the $n-k$ pairs at distance k, twice by symmetry, which is where the $2(1-k/n)$ weight comes from. Now read the consequence. If and only if all ρ_k vanish does this collapse to $\mathrm{Var}(S_n) = n\gamma_0$, that is

$$\sigma_{n\text{-period}} \;=\; \sqrt{n}\,\sigma_{1\text{-period}}.$$

The square-root-of-time rule is not a law of nature. It is a corollary of zero autocorrelation. Annualising a daily volatility with $\sqrt{252}$, scaling a one-day VaR to ten days with $\sqrt{10}$, and multiplying a monthly Sharpe ratio by $\sqrt{12}$ are all the same assumption wearing three hats.

With positive autocorrelation the bracket exceeds 1, so true long-horizon risk is *larger* than the scaled number and you are under-reporting risk. With negative autocorrelation it is smaller. For the important special case where the ACF decays geometrically, $\rho_k = \rho^k$ with $|\rho| < 1$ (an AR(1), see the AR and MA family), the bracket tends to a memorable limit as n grows:

$$1 + 2\sum_{k=1}^{\infty} \rho^k \;=\; 1 + \frac{2\rho}{1-\rho} \;=\; \frac{1+\rho}{1-\rho}.$$

WORKED EXAMPLE

A fund with a smooth track record. Monthly returns show $\hat{\rho}_1 = 0.3$, plausible for a credit or private-asset book where positions are marked from stale or model prices. Assume an AR(1) shape, so the long-horizon variance inflation factor is

$$\frac{1+\rho}{1-\rho} \;=\; \frac{1.3}{0.7} \;=\; 1.857, \qquad \sqrt{1.857} \;=\; 1.363.$$

The honest annualised volatility is 36.3% *higher* than the one obtained by multiplying the monthly figure by $\sqrt{12}$. Since the annualised return is unaffected, the reported Sharpe ratio is overstated by the same factor: a headline 2.0 is really about $2.0/1.363 = 1.47$. Nothing was falsified. Someone simply scaled by $\sqrt{12}$ without checking one number.

WORKED EXAMPLE

The alternating series, worked in full. Eight daily returns, in percent:

$$r = (+1,\, -1,\, +1,\, -1,\, +1,\, -1,\, +1,\, -1).$$

The mean is 0, so with the zero-mean convention $\hat{\gamma}_0 = \frac{1}{8}\sum_t r_t^2 = \frac{8}{8} = 1$ and

$$\hat{\gamma}_1 = \frac{1}{8}\sum_{t=1}^{7} r_t r_{t+1} = \frac{7\times(-1)}{8} = -0.875, \qquad \hat{\rho}_1 = \frac{-0.875}{1} = -0.875.$$

Extreme mean reversion, as the pattern makes obvious. Now check the variance of the sum against the formula. Perfect alternation means $\rho_k = (-1)^k$, so with $n = 8$

$$\sum_{k=1}^{7}\left(1 - \tfrac{k}{8}\right)(-1)^k = \frac{-7+6-5+4-3+2-1}{8} = \frac{-4}{8} = -0.5,$$

and the bracket is $1 + 2(-0.5) = 0$, giving $\mathrm{Var}(S_8) = 0$. That is right: the eight returns sum to zero every time. Meanwhile the naive scaling would have predicted $\sqrt{8}\times 1\% = 2.83\%$ of eight-day volatility out of thin air.

Testing for it

Three tools, in increasing order of seriousness.

Bartlett bands. Under the null that the series is white noise, $\hat{\rho}_k$ is approximately $N(0, 1/n)$ for each fixed k. So a single lag is significant at 5% if $|\hat{\rho}_k| > 1.96/\sqrt{n}$. With $n = 1000$ daily returns that threshold is $1.96/31.6 = 0.062$. Any lag inside ± 0.062 is noise, and this is what the dashed lines on every ACF plot are.

Ljung-Box. Testing twenty lags one at a time guarantees a false positive: that is multiple testing. The joint test uses

$$Q_m \;=\; n(n+2)\sum_{k=1}^{m}\frac{\hat{\rho}_k^2}{n-k} \;\sim\; \chi^2_m$$

under the white-noise null, with the degrees of freedom reduced by the number of parameters if you are testing the residuals of a fitted model. One number, one p value, all m lags at once.

Durbin-Watson. The classical statistic for lag-1 autocorrelation in regression residuals (see linear and multiple regression):

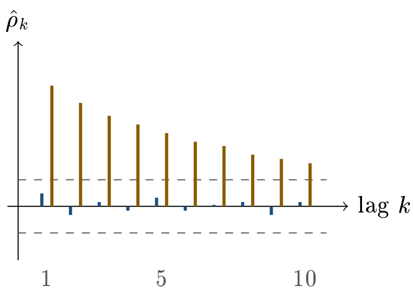
$$DW \;=\; \frac{\sum_{t=2}^{n}(e_t - e_{t-1})^2}{\sum_{t=1}^{n} e_t^2} \;\approx\; 2\big(1-\hat{\rho}_1\big).$$

It runs from 0 to 4: about 2 means no autocorrelation, near 0 means strong positive, near 4 strong negative. On the alternating series above, $\sum_{t=2}^{8}(e_t - e_{t-1})^2 = 7 \times 2^2 = 28$ and $\sum_t e_t^2 = 8$, so $DW = 28/8 = 3.5$, while the approximation gives $2(1 + 0.875) = 3.75$. The gap is a useful reminder that the $2(1 - \hat{\rho}_1)$ form is an approximation which ignores the endpoints, and it matters most in short samples.

What the data actually looks like

Two facts, both robust across markets and decades, and both worth stating in an interview because together they are the whole story.

1. **Returns are very close to white noise.** Daily equity index return autocorrelations sit within a few hundredths of zero, mostly inside the Bartlett band. This is weak-form market efficiency stated statistically: if a reliable positive ρ_1 existed, it would be arbitraged into non-existence, and the act of arbitraging it is what removes it.
2. **Squared and absolute returns are strongly and persistently autocorrelated.** Turbulent days cluster with turbulent days, quiet with quiet, and the ACF of r_t^2 decays slowly over weeks. This is **volatility clustering**: the sign of the next move is unpredictable, but its *size* is quite predictable. It is why volatility is forecastable at all, hence the entire ARCH and GARCH family, and why variance is a tradable asset.



ACF of returns r_t — ACF of squared returns r_t^2 — Bartlett band $\pm 1.96/\sqrt{n}$

Where does non-zero return autocorrelation come from in practice? Four sources, and being able to name them is what separates a candidate who has looked at data from one who has read a textbook.

- **Stale and illiquid marks.** A bond or a fund NAV that is not really re-priced every day drifts towards its true value over several days, spreading one economic shock across consecutive returns. Positive ρ_1, understated volatility, inflated Sharpe. This is the mechanism behind the previous example.
- **Bid-ask bounce.** Transaction prices alternate between the bid and the offer even when the mid never moves, injecting a spurious *negative* ρ_1 in high-frequency returns. Use mid prices, or sample less often.
- **Overlapping windows.** Computing weekly returns every day means consecutive observations share four days of data, which mechanically creates autocorrelation with nothing economic behind it. This is the classic reason to use Newey-West standard errors on overlapping-return regressions.
- **Genuine slow information diffusion.** Small caps and cross-listed names lag the index. This is real, it is also small, and it is the thin edge that systematic momentum books try to harvest net of costs.

Is the random walk true? The question is stationarity's, and the answer belongs there, but the autocorrelation half of it belongs here and is worth stating plainly. Prices follow a random walk if returns are white noise, $\rho_k = 0$ for all $k \geq 1$. Measured autocorrelations of liquid index returns do sit inside the Bartlett band, and that is not a coincidence: it is the fingerprint of competition, since any exploitable linear pattern gets traded away. What the band does *not* say is that returns are independent. Squared returns are strongly autocorrelated, which is volatility clustering and the subject of ARCH and GARCH, so a series can pass every test in this one and still be highly predictable in its second moment. That gap is the whole reason derivatives desks make their living on volatility rather than on direction.

REAL INTERVIEW QUESTION (Société Générale · Volatility)

Historical volatility: definition? Comparison with implied vol? You have a stock that does $+1\%$ one day and -1% the next, what is its volatility?

WHAT THEY TEST / ANSWER

Two parts, and the numerical part is a trap about conventions rather than about arithmetic. Historical (realised) volatility is the annualised standard deviation of past returns, and how it compares with implied volatility is developed in equity volatility. For the number, state your convention before computing, because that is what is being graded. With the desk convention of zero assumed mean, the daily volatility is $\sqrt{\left((1\%)^2 + (1\%)^2\right)/2} = 1\%$, and annualising gives $1\% \times \sqrt{252} = 15.9\%$. If instead you subtract the sample mean, which is zero here, and divide by $n - 1 = 1$, you get a daily $\sqrt{2} \times 1\% = 1.41\%$ and 22.5% annualised. Say both, say which one desks use and why, then add the observation that makes the answer stand out: two observations are far too few to estimate anything, and the alternation here is an accident of a tiny sample rather than a pattern. If it were the pattern, annualising at all would be the wrong operation, which is what the alternating-series question in this section turns on.

REAL INTERVIEW QUESTION (Société Générale · Volatility)

What is the volatility of an underlying that does +2 then −2 alternately every day?

WHAT THEY TEST / ANSWER

Give the number, then refuse to stop there, because the number is the easy half. The estimator itself and the choices inside it (log returns, the divisor, whether to subtract the mean, the window) are settled in estimation methods, so state your convention and compute: with the desk convention of zero assumed mean, $\hat{\sigma}_{\text{daily}} = \sqrt{\frac{1}{n}\sum_t r_t^2} = 2\%$, annualised as $2\% \times \sqrt{252} = 31.7\%$.

Then say what is wrong with that number, which is what is being listened for here. The $\sqrt{252}$ holds *only because* returns are assumed serially uncorrelated, and this series is the most serially correlated one there is: the eight-point version worked earlier in this section has $\hat{\rho}_1 = -0.875$, and perfect alternation gives $\rho_k = (-1)^k$, which drives the bracket in $n\gamma_0[1 + 2\sum_k (1 - k/n)\rho_k]$ to zero. So the underlying has 31.7% of one-day volatility and essentially *none* over even horizons: it goes nowhere. Quoting 31.7% as the volatility of this asset is the trap, and naming the horizon is the answer. In general the naive scaling is too small when $\hat{\rho}_1$ is positive and too large when it is negative, and here it is negative and extreme.

REAL INTERVIEW QUESTION (Société Générale · Volatility)

What is a Brownian motion? Give us the 5 points?

WHAT THEY TEST / ANSWER

They want the defining properties recited cleanly, since everything in pricing rests on them. For a standard Brownian motion W_t: (1) $W_0 = 0$; (2) increments are **independent**, so $W_t - W_s$ is independent of anything before s; (3) increments are stationary and Gaussian, $W_t - W_s \sim N(0, t-s)$, hence variance growing linearly in time and volatility in $\sqrt{t}$; (4) paths are continuous, so no jumps; (5) paths are nowhere differentiable and have infinite total variation, which is why we need Ito calculus rather than ordinary calculus. The connection to make here, and the reason this question sits in a section on autocorrelation, is that point (2) plus point (3) are precisely the zero-autocorrelation and constant-variance assumptions of this section: independent increments are what give $\text{Var}(W_t) = t$ and therefore the $\sqrt{t}$ scaling of volatility. Point (4) is what real markets break with jumps, and volatility clustering is what breaks the constant-variance part.

INTERVIEW TRAP

Three traps. First, **concluding independence from zero autocorrelation**: it rules out linear memory only, and financial returns are the textbook counterexample since r_t is white noise while r_t^2 is not. Second, **measuring the ACF of a non-stationary series**: run it on prices rather than returns and you get $\hat{\rho}_1$ near 0.99 at every lag, which tells you about the unit root and nothing about memory, see unit root tests. Third, **data-mining the ACF**: with twenty lags and a 5% threshold you expect one significant lag under pure noise, so use the joint Ljung-Box test rather than celebrating the one bar that pokes through the band.

VISUAL / CODE TO BUILD: Interactive

An ACF explorer with two linked panels: the series on top, its ACF with Bartlett bands below. A slider sets the true ρ of a simulated AR(1) from -0.9 to $+0.9$, and the reader watches the path change character from jagged mean reversion to smooth trending while the ACF changes shape. Two toggles carry the section's lessons: **squared returns**, which switches the ACF to r_t^2 on a simulated GARCH path so the reader sees a flat ACF for returns and a decaying one for squares at the same time; and **scale to horizon**, which plots true versus $\sqrt{n}$-scaled volatility as n grows so the gap opens visibly as ρ moves away from zero. Show $\hat{\rho}_1$, DW and the Ljung-Box p value live as numbers too.

Where this goes next

Autocorrelation is the diagnostic; modelling it is the next step. Choosing a functional form for the ACF and fitting it is exactly what AR, MA, ARMA and ARIMA models do, and the partial autocorrelation function is the tool that tells an AR from an MA. Everything here presumes weak stationarity, established in stationarity and tested in unit root tests. Autocorrelation in the *squares* rather than the levels leads to ARCH and GARCH. On the desk the same formula reappears whenever a risk number is scaled across horizons, in Value at Risk, and whenever regression standard errors have to survive serially correlated residuals, in linear and multiple regression.

8.8 AR, MA, ARMA and ARIMA models

trading · structuring | core | cross-asset

Autocorrelation measures that yesterday matters. These models are what you write down once you want to say *how*. They are the small grammar of time series: two building blocks, their combination, and one operation that makes the whole thing work on data that would otherwise be unusable.

They also come with a warning that belongs at the top rather than the bottom. Fitted to equity returns they find almost nothing, and that is not a failure of the models. It is what an efficient market looks like when you point a forecasting tool at it. The places they do work are listed at the end, and they are not the places a beginner expects.

AR: the series remembers where it was

An autoregressive model of order one says today is a fraction of yesterday plus a shock:

$$X_t = c + \phi X_{t-1} + \varepsilon_t, \qquad \varepsilon_t \text{ white noise.}$$

Everything about it is in ϕ.

- $|\phi| < 1$: the series is **stationary** and mean-reverting, with long-run mean $\mu = c/(1-\phi)$ and unconditional variance $\sigma^2/(1-\phi^2)$.
- $\phi = 1$: a **random walk**. No mean to revert to, variance growing without limit, and the whole apparatus of unit root tests exists to detect this case.
- $|\phi| > 1$: explosive, and not a model of anything financial.

The single most useful number to extract is the **half-life**: how long a shock takes to decay to half its size. Since a shock decays as ϕ^h,

$$\text{half-life} = \frac{\ln 0.5}{\ln \phi}.$$

WORKED EXAMPLE

On daily data:

ϕ	Half-life
0.50	1.0 day
0.90	6.6 days
0.95	13.5 days
0.99	69.0 days
0.999	692.8 days

Two things to take from the table. The relationship is violently non-linear near one: moving ϕ from 0.99 to 0.999 multiplies the half-life by ten. And that is exactly the region where estimating ϕ is hardest, so a mean-reverting trade sized on an estimated half-life of ten weeks may really be sitting on one of two years. When someone says a spread "mean-reverts", the only question worth asking is over what horizon.

INTUITION

An AR(1) is a spring. ϕ near zero is a stiff spring that snaps back immediately; ϕ near one is a slack one that takes forever; ϕ equal to one is no spring at all, and the thing drifts wherever the shocks push it. Every mean-reversion trade in markets is a bet on the spring being stiffer than the price implies.

MA: the series remembers the shock

A moving-average model of order one says today is this period's shock plus a fraction of last period's:

$$X_t = \mu + \varepsilon_t + \theta\, \varepsilon_{t-1}.$$

The structural difference from AR is the memory. An MA(q) remembers exactly q periods and then forgets completely, because ε_{t-q-1} simply does not appear in the equation. An AR(1) never forgets: it remembers geometrically, forever.

An MA is always stationary, whatever θ is, since it is a finite sum of white noise terms. The condition it needs instead is **invertibility**, $|\theta| < 1$, which is what allows it to be rewritten as an infinite AR and therefore what allows the unobservable shocks to be recovered from the observable data. Without it you can write the model but you cannot estimate it from the series alone.

KEY IDEA

AR is infinite memory that fades; MA is finite memory that stops. That single contrast is what the identification procedure below detects, and it is the answer to almost every conceptual question about these models.

Reading the order off the data

The contrast shows up directly in the autocorrelation function and the partial autocorrelation function, both defined in autocorrelation. The signatures are the reason those two functions are computed as a pair rather than separately.

Model	ACF	PACF
AR(p)	decays geometrically	cuts off after lag p
MA(q)	cuts off after lag q	decays geometrically
ARMA(p, q)	decays	decays
White noise	zero at every lag	zero at every lag

The mirror symmetry is not a coincidence: an AR can be written as an infinite MA and an invertible MA as an infinite AR, so each looks like the other seen through the other function. The practical procedure, the Box-Jenkins method, is three steps repeated: identify candidate orders from these two plots, estimate the model, then check that the residuals look like white noise. If they do not, the model has missed structure and the orders change. Where two candidates both pass, choose by an information criterion, which trades fit against parameter count, with BIC penalising extra parameters more heavily than AIC and therefore usually choosing the smaller model.

INTERVIEW TRAP

Adding parameters until the fit is good. Any ARMA can be made to fit a finite sample arbitrarily well by raising p and q, and the result forecasts worse than a smaller model because it has fitted noise. The discipline is that a model earns a parameter only if it improves an out-of-sample criterion, and on financial data the honest answer is usually a very small model or none.

The I in ARIMA

ARMA needs a stationary series, and most financial series are not stationary. ARIMA adds one step in front: difference the series d times until it is, fit ARMA to the result, then integrate the forecasts back.

For financial prices $d = 1$ almost always, because the first difference of a log price is a log return and returns are approximately stationary, as set out in time series analysis.

Two identities worth knowing, because they connect the notation to things you already understand:

- **ARIMA(0,1,0) is exactly a random walk.** Difference once, and what is left is white noise with no structure at all. So the efficient-market benchmark is a particular ARIMA model, and every richer specification has to beat it.
- **ARIMA($p, 1, q$) on log prices is ARMA(p, q) on log returns.** There is no third thing being modelled. The differencing step is the prices-to-returns conversion everyone does anyway, written in the model's own notation.

Forecasting, and how fast the forecast becomes useless

The h-step forecast from an AR(1) is

$$\hat{X}_{t+h} = \mu + \phi^h (X_t - \mu),$$

which decays geometrically toward the mean, and the variance of the forecast error is

$$\mathrm{Var} = \sigma^2 \frac{1-\phi^{2h}}{1-\phi^2},$$

which rises to the unconditional variance. Both facts together are the whole story of what a stationary model can and cannot do.

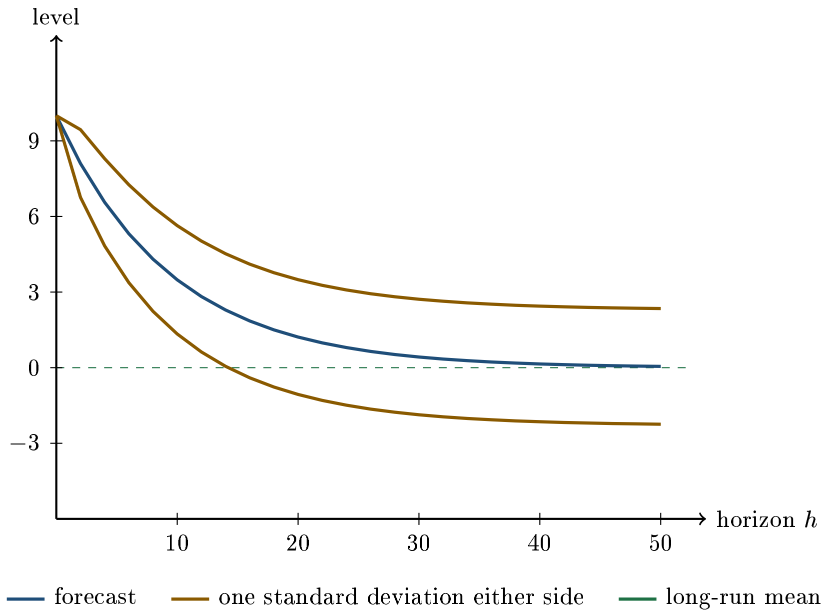
WORKED EXAMPLE

$\phi = 0.9$, $\mu = 0$, $\sigma = 1$, and the series currently sits at $X_t = 10$. The unconditional standard deviation is $\sigma/\sqrt{1-\phi^2} = 2.294$, so that is about four and a third unconditional standard deviations above its mean.

Horizon h	Forecast	Forecast standard deviation
1	9.000	1.000
2	8.100	1.345
5	5.905	1.851
10	3.487	2.150
20	1.216	2.277
50	0.052	2.294
∞	0.000	2.294

At one step the forecast is 9 with an error standard deviation of 1, so the model is saying something. By twenty steps the forecast is 1.2 with an error standard deviation of 2.3: the signal has decayed below the noise, and the model is saying nothing you did not already know from the unconditional mean.

That crossover is the useful horizon of the model, and it arrives at roughly two to three half-lives, not at the horizon anybody wants a forecast for. A stationary model forecasts usefully for about as long as the mean reversion is fast, and then it stops.



The forecast falls toward the mean while the band widens toward the unconditional spread, and the two meet. Everything to the right of where the band swallows the forecast is a model that has run out of information.

Where these models actually work

Fitted to equity index returns, an AR(1) returns ϕ indistinguishable from zero, and that is the correct answer rather than a disappointing one. The series where they earn their keep share one property: they are *differences or ratios* that something forces back together.

- **Spreads and bases**: a bond against its swap, a future against its cash index, one maturity against another. These are held together by arbitrage, so they mean-revert by construction and an AR(1) half-life is the natural way to describe how fast.
- **Volatility proxies**: the log of realised volatility is well described by an AR with a high ϕ, which is the discrete statement of volatility clustering and the direct ancestor of volatility modeling.
- **Macro and flow series**: inflation, order flow, inventory, all of which have genuine persistence.
- **Short rates**: the mean-reverting short-rate models used in rates are the continuous-time version of an AR(1), which is why they are written with a pull toward a long-run level and a speed of adjustment. The continuous-time object is in stochastic differential equations.

Interview angle

When the drift stops being negligible against the noise is a question about moments, and it is answered in descriptive statistics. What matters for specification here is narrower: an ARMA model is defined on a mean-adjusted series, so the drift is not something the model estimates around, it is something removed before fitting. Leave a trend in and the autocorrelation function decays slowly enough to look like a unit root, which is why the differencing decision in ARIMA and the drift decision are the same decision made twice.

> **INTERVIEW TRAP**
>
> Four errors. (1) Fitting ARMA to equity returns, finding a small coefficient, and trading it: the coefficient is usually inside the bid-offer even when it is statistically significant. (2) Fitting to levels rather than differences, which produces the spurious results in time series analysis. (3) Quoting a mean-reversion speed without its uncertainty, when a ϕ estimated near one has a half-life whose confidence interval spans an order of magnitude. (4) Choosing orders by in-sample fit, which always prefers the bigger model.

Where this goes next

Applying exactly this machinery to squared returns rather than returns is volatility modeling, which is where the structure in financial data actually is. Deciding how many times to difference is unit root tests, the property being tested for is stationarity, and the continuous-time version of a mean-reverting AR(1) is in stochastic differential equations.

8.9 Volatility modeling (ARCH, GARCH)

trading · structuring · sales | advanced | cross-asset · equity

Stationarity left one loose end: returns pass the test on their mean but not cleanly on their variance, because volatility moves in regimes. This section is the loose end, and it is a rare case where a piece of econometrics earns its place on a trading floor, because the model it produces has a term structure and desks trade term structure.

The fact the model exists to capture

Daily returns are close to unpredictable in sign. Their *magnitudes* are not. Large moves are followed by large moves and quiet days by quiet days, so the autocorrelation of r_t is near zero while the autocorrelation of r_t^2 is strongly positive and decays slowly.

> KEY IDEA
>
> **Volatility clustering** is the observation that returns are serially uncorrelated but not independent. Uncorrelated kills the linear predictability of direction; not independent leaves the predictability of size. Every model in this section is a way of writing that one sentence as an equation, and it is also why a sample standard deviation computed over a fixed window is a bad estimate of tomorrow's volatility: it weights a move from six months ago exactly as heavily as yesterday's.

ARCH, then GARCH

Engle's **ARCH(q)** makes today's variance a function of recent squared shocks,

$$\sigma_t^2 \;=\; \omega + \sum_{i=1}^{q} \alpha_i \varepsilon_{t-i}^2.$$

It captures clustering but needs a long lag structure to do it, because the memory in volatility is long and each lag costs a parameter.

Bollerslev's **GARCH(1,1)** fixes that by letting yesterday's variance stand in for the whole history:

$$\sigma_t^2 \;=\; \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2.$$

Three parameters, and each has a job worth naming out loud:

- α is the **reaction**, how much a new shock moves the variance estimate.
- β is the **memory**, how much of yesterday's estimate survives into today.
- ω anchors the process, and with $\alpha + \beta < 1$ the **long-run variance** is $V = \omega/(1 - \alpha - \beta)$.

That last condition is the stationarity condition from the previous section wearing different clothes: $\alpha + \beta$ is the persistence, and it plays the role ϕ played in the AR(1). Below one, variance mean-reverts to V; at one, shocks to volatility are permanent and there is no long-run level to revert to.

Persistence, and why the forecast has a shape

Everything useful follows from one recursion. Iterating the equation forward,

$$\mathbb{E}_t[\sigma_{t+n}^2] \;=\; V + (\alpha+\beta)^n \left(\sigma_t^2 - V\right).$$

The forecast decays geometrically from where you are today toward the long-run level, at rate $\alpha + \beta$. Nothing else in the model matters for forecasting.

WORKED EXAMPLE

A shock, and its whole forward curve. Take the parameter values a daily equity series typically produces, $\alpha = 0.08$ and $\beta = 0.90$, so persistence is 0.98 and the half-life of a volatility shock is $\ln(0.5)/\ln(0.98) = \mathbf{34.3}$ trading days. Set the long-run level at 16% annualised, so $V = 0.16^2/252 = 1.016 \times 10^{-4}$ per day and $\omega = V(1 - 0.98) = 2.03 \times 10^{-6}$.

Now suppose something happens and today's volatility estimate is **40%** annualised, a daily variance of 6.349×10^{-4}. The single-day forecasts decay like this:

Horizon	$(\alpha+\beta)^n$	**Forecast vol for that day**
1 day ($n = 1$)	0.9800	39.66%
1 month ($n = 21$)	0.6543	33.69%
3 months ($n = 63$)	0.2801	25.15%
1 year ($n = 252$)	0.0062	16.26%

An option does not care about the variance on one future day; it cares about the *average* variance over its life. Averaging the path above gives what the model says a straddle should be worth:

Option maturity	**Model volatility**
1 month	36.61%
3 months	31.76%
1 year	22.71%

That is a downward-sloping volatility term structure, produced by nothing but a shock and a persistence parameter. It is also, qualitatively, what the implied surface does after a selloff: the front end spikes and the back end barely moves, because the back end is mostly V. A candidate who can say *why* the term structure inverts in a crisis, rather than only that it does, is using this model whether or not they name it.

EWMA, the special case you will meet first

Set $\omega = 0$ and $\alpha + \beta = 1$ and the recursion collapses to

$$\sigma_t^2 \;=\; (1-\lambda)\,\varepsilon_{t-1}^2 + \lambda\,\sigma_{t-1}^2,$$

an exponentially weighted moving average. This is the RiskMetrics estimator, conventionally with $\lambda = 0.94$ for daily data, which is a half-life of 11.2 days and an effective window of about 17 days.

It is a GARCH sitting exactly on the non-stationary boundary, and that is its one real defect: with no ω there is no long-run variance, so the n-day-ahead forecast is a flat line at today's level. EWMA is a perfectly good *filter* of current volatility and a bad forecaster of the term structure, which is why it survives in one-day VaR and does not survive in option pricing.

REAL INTERVIEW QUESTION (Credit Suisse · Volatility)

What is GARCH and why do we use GARCH to simulate volatility?

WHAT THEY TEST / ANSWER

Answer in three moves: the fact, the equation, and the reason it beats the obvious alternative.

The fact. Volatility clusters. Returns are serially uncorrelated but their squares are strongly autocorrelated, so a constant-volatility model is contradicted by the data on the first thing you check.

The equation. GARCH(1,1) makes the conditional variance $\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$: a long-run anchor, a reaction to the last shock, and a memory of the last estimate. Persistence is $\alpha+\beta$, typically around 0.98 on daily equity data, and the long-run variance is $\omega/(1-\alpha-\beta)$.

Why simulate with it. Give three reasons and you have answered better than the question asked.

First, it produces **realistic paths**. Simulating with a constant volatility gives you thin tails and no crises; GARCH generates fat unconditional tails and clustered quiet and violent stretches out of Gaussian innovations, purely because the variance moves. That matters for anything tail-sensitive, which is most of risk.

Second, it produces a **term structure**. The forecast decays from today's level to the long-run level at rate $\alpha + \beta$, so the model knows that a shock today raises one-month volatility far more than one-year volatility. A flat historical estimate cannot say that, and the shape it gives you is qualitatively the shape the implied surface takes after a shock.

Third, it is **cheap and estimable**. Three parameters by maximum likelihood, as in estimation methods, on data everyone already has.

Then close by naming the limitation, because the follow-up is coming anyway. GARCH volatility is *deterministic* given the past: today's variance is known at the start of today, driven by realised returns rather than by its own random shock. That is why option desks reach instead for a stochastic volatility model such as Heston, where volatility has its own Brownian motion and can therefore be correlated with spot, which is what generates skew. GARCH is the right tool for forecasting and simulating under the real-world measure; it is not a pricing model.

INTUITION

The distinction that separates a good answer from a memorised one: GARCH is an estimator of *realised* volatility dynamics, fitted to returns, under the real-world measure. An implied volatility surface is a set of prices, under the risk-neutral measure, and includes a variance risk premium. They are not two measurements of the same quantity, and GARCH saying 22.7% for the year while the one-year implied prints 25% is not an arbitrage. Treating the gap as a signal is a separate trade with its own justification.

INTERVIEW TRAP

1. **Using GARCH to price options.** It forecasts under the physical measure. Pricing needs the risk-neutral one, and the difference is the variance risk premium, not a mispricing.
2. **Ignoring the asymmetry.** Plain GARCH squares the shock, so a down 3% day and an up 3% day raise the variance identically. Equity volatility rises far more on the way down. The standard fix is a model with a leverage term, GJR-GARCH or EGARCH, and knowing that plain GARCH *cannot* produce skew is the point worth carrying.
3. **Fitting $\alpha + \beta$ at or above one** and not noticing. The long-run variance $\omega/(1-\alpha-\beta)$ blows up or turns negative, the forecast stops mean-reverting, and the model is outside the stationary region from stationarity.
4. **Reading persistence as a horizon.** $\alpha + \beta = 0.98$ does not mean shocks last 98 days; the half-life is $\ln(0.5)/\ln(0.98) = 34.3$ days. Quote the half-life, since it is the number that means something.
5. **Estimating on too little data.** Three parameters on a year of daily returns gives unstable estimates of β in particular, and the usual symptom is a persistence that jumps around between refits.

VISUAL / CODE TO BUILD: Interactive

Two linked panels driven by sliders for α, β and the long-run volatility, with the constraint $\alpha + \beta < 1$ enforced visibly rather than silently.

The upper panel simulates a return path from a fixed seed and plots the conditional volatility beneath it, so the reader can watch a large return push the volatility line up and then watch it decay. Raising β alone makes the decay visibly slower without changing how far the line jumps; raising α alone does the opposite. That separation of reaction from memory is the thing the two parameters mean and the thing a table of numbers cannot show.

The lower panel plots the forecast term structure from the current state, average variance against maturity, converted to volatility. Dragging today's volatility above and below the long-run level flips the curve between downward-sloping and upward-sloping, and pushing $\alpha + \beta$ toward 1 flattens it into the horizontal EWMA line, which shows what the missing ω costs.

8.10 Unit Root Tests (ADF)

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Stationarity said to test rather than assume, and pointed here. This is the test. It is a short section by design: the mechanics take a page, and the useful content is almost entirely in the two things that make it behave unlike any other test you will run, namely that its critical values are not the ones printed in a t table and that it has very little power against exactly the alternatives you care about.

The regression

Start from the AR(1), $y_t = \phi y_{t-1} + \varepsilon_t$, and subtract y_{t-1} from both sides:

$$\Delta y_t \;=\; \gamma\, y_{t-1} + \varepsilon_t, \qquad \gamma = \phi - 1.$$

The unit root $\phi = 1$ has become $\gamma = 0$, so the hypothesis is now about a regression coefficient and can be tested with a t-ratio:

$$H_0 : \gamma = 0 \text{ (unit root, non-stationary)}, \qquad H_1 : \gamma < 0 \text{ (stationary)}.$$

The test is one-sided and the rejection region is in the left tail, because $\phi > 1$ is explosive rather than an alternative anyone entertains.

KEY IDEA

The null is **non-stationarity**. That is backwards from most tests and it drives everything: the burden of proof is on stationarity, so a series is treated as a random walk unless the data insists otherwise. "Failed to reject" therefore means "no evidence against a random walk", never "the series is a random walk", which is the Type II error from hypothesis testing sitting in the one place it does the most damage.

The **augmented** Dickey-Fuller test adds lagged differences,

$$\Delta y_t \;=\; \alpha + \delta t + \gamma\, y_{t-1} + \sum_{i=1}^{p} \psi_i\, \Delta y_{t-i} + \varepsilon_t,$$

and the augmentation exists for one reason: the t-ratio is only valid if the residuals are serially uncorrelated, so the lagged differences are there to soak up any short-run autocorrelation in the increments. They are nuisance terms. Nobody interprets ψ_i, and p is chosen by an information criterion or by adding lags until the residuals are clean.

Why the critical values are different

Under the null, y_{t-1} is itself a random walk, so the regressor is non-stationary and the usual asymptotics fail. The statistic does not converge to a normal or a t; it converges to a distribution defined in terms of Brownian motion, and it is shifted well into the left tail.

There is no formula, so the values are tabulated by simulation. Below are the ones I obtained directly, by generating 20,000 random walks of length 500 under H_0 and taking the quantiles of the resulting t-ratios:

Deterministic terms	**1%**	**5%**	**10%**
none	−2.60	−1.97	−1.63
constant	−3.45	−2.86	−2.58
constant and trend	−3.95	−3.40	−3.13

Read the middle row against the number you would otherwise have used. The 5% critical value with a constant is −2.86, not −1.65. Using the normal table would reject a true unit root far more often than 5% of the time, so the mistake produces spurious *stationarity*, which then licenses the regression on levels that stationarity warned about.

INTERVIEW TRAP

The deterministic terms are a modelling choice, not boilerplate, and the table shows what they cost. Including a trend moves the 5% threshold from -2.86 to -3.40, so the same series can pass one specification and fail the other. Choose from what the series is: a spread that should sit around zero takes no constant, a series with a non-zero mean takes a constant, and a trend term belongs only when you genuinely want to test stationarity *around* a deterministic trend. Fitting all three and reporting the one that agrees with you is the multiple testing problem with three tests and no correction.

Power, which is the whole practical story

The test is consistent, so with enough data it works. The question is how much data, and the answer is the reason a great deal of published mean-reversion is not real. I simulated AR(1) series at several persistence levels and sample sizes and counted rejections at the 5% level using the constant specification:

True ϕ	$T = 250$	$T = 1{,}000$	$T = 2{,}500$
0.90 (half-life 6.6 d)	97.0%	100%	100%
0.95 (half-life 13.5 d)	45.8%	100%	100%
0.99 (half-life 69.0 d)	7.4%	29.8%	96.5%

The size check confirms the test is calibrated: at $\phi = 1$ and $T = 250$ it rejects 5.5% of the time, which is the 5% it should be. So the third row is not a bug. A series that reverts with a **69-day half-life** is strongly mean-reverting by any trading standard, and one year of daily data detects it 7.4% of the time, barely distinguishable from the false-positive rate. Four years gets you to 29.8%. You need about a decade before the test reliably notices.

INTUITION

State the consequence rather than the statistic. A unit-root test on one year of daily data can tell you a fast mean-reverting spread is mean-reverting, and it cannot tell you anything at all about a slow one. Since the slow ones are where the money and the risk both live, the correct professional habit is to treat a failure to reject as uninformative, and to argue for a spread's stationarity from its *economics* first, that there is a mechanism forcing the two legs together, and use the test as a consistency check rather than as evidence.

Two variants worth naming

Phillips-Perron tests the same null and fixes the same autocorrelation problem non-parametrically, correcting the standard error rather than adding lags. It uses the Dickey-Fuller critical values above, and it gives a useful robustness check: if ADF and PP disagree, the lag choice is doing the work and neither result is solid.

KPSS reverses the hypotheses, taking *stationarity* as the null. That is worth more than it sounds, because two tests with opposite nulls give four outcomes rather than two, and the honest ones are the two where they disagree with each other:

- ADF rejects, KPSS does not: stationary, and both tests agree.

- ADF does not reject, KPSS rejects: a unit root, and both agree.
- Neither rejects: the sample is too short to say. This is the common case in finance and it is a legitimate answer.
- Both reject: the series is neither, which usually means a structural break or long memory, and the model is wrong rather than the test.

The failure mode nobody tests for

A structural break makes a stationary series look like a random walk. If a spread mean-reverted around 20 basis points until a regime change and around 60 afterwards, the pooled series has a mean that moved, the ADF test sees a level that wandered and does not reject, and the conclusion drawn is that the spread has a unit root when in fact it has two perfectly stationary regimes and one break.

The practical instruction is to plot the series before running anything and to look for a break, because the test cannot distinguish these two stories and you usually can. It is also the reason a cointegration relationship that tested well historically can stop working without any of the diagnostics changing until well after the money is lost.

VISUAL / CODE TO BUILD: Interactive

Two panels sharing one control, a slider for ϕ from 0.90 to 1.00 and a second for the sample length.

The left panel draws the simulated series and reports its ADF statistic against the three critical values as horizontal reference lines, so the reader can watch a series that is visibly reverting fail to clear the threshold as ϕ approaches 1.

The right panel is a power curve: rejection rate against ϕ, one line per sample length. The intended moment is seeing all the lines collapse toward the 5% size as $\phi \to 1$, which is what "the test has no power against a near unit root" means as a picture rather than a sentence. A toggle adds a structural break in the middle of the sample and shows the rejection rate collapse even at $\phi = 0.9$, which is the last subsection made visible.

Chapter 9

Financial Mathematics & Stochastic Calculus

9.1 Stochastic processes

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A random variable is one draw. A stochastic process is a whole history: a collection of random variables indexed by time, $\{X_t\}$, describing not just what a quantity might be but how it might get there. That distinction is the reason this chapter exists. A price is not a number with error bars, it is a path, and almost everything a derivatives desk cares about, whether a barrier was touched, how much variance was realised along the way, when an autocall triggered, depends on the path and not on the endpoint.

This section sets up the vocabulary and the families. Brownian motion itself is Brownian motion, the fair-game property is martingales, and the calculus for manipulating these objects is Ito's lemma.

Four questions that classify any process

- **Is time discrete or continuous?** A daily closing price is a discrete-time process; a quoted price that can move at any instant is continuous-time. The choice is modelling convenience, not a fact about the world.
- **Is the state discrete or continuous?** A credit rating takes finitely many values; a price takes any positive number. Discrete state means you can write the dynamics as a table of transition probabilities, which is often far easier.
- **Does the past matter beyond the present?** If not, the process is **Markov**.
- **Are the increments independent, and is their distribution stable over time?** Independent increments means each step is unrelated to the last; stationary increments means a one-day move has the same distribution whenever you take it.

KEY IDEA

The **Markov property** says that the future depends on the past only through the present: to forecast tomorrow, today's value is a sufficient summary and the route to it adds nothing. It is an enormous simplification, and it is the reason most derivatives can be priced at all, since it means the model carries a state rather than a history. It is also the assumption that fails first: implied volatility surfaces move in ways that depend on the recent path, and any model with memory, including stochastic volatility and rough volatility, is a statement that price alone is not a sufficient summary.

A related idea worth separating from Markov, because they are routinely confused, is the **filtration**, written $\mathcal{F}_t$: the information available at time t. It grows with time and never shrinks. Almost every statement in this chapter is really a statement about a filtration, for example that a quantity is known by time t, or that its conditional expectation given $\mathcal{F}_t$ equals its current value, which is the definition in martingales.

REAL INTERVIEW QUESTION (Bank of America · desk not recorded)

What is a stochastic process and what is Brownian motion? Explain their principles.

WHAT THEY TEST / ANSWER

Two definitions, and the interviewer wants to hear that you know which is the general concept and which is one example of it.

A stochastic process is a family of random variables indexed by time, $\{X_t\}$, which is to say a random *path* rather than a random number. Specifying one means specifying the joint behaviour of all those variables, not just each one separately. Then give the vocabulary that shows you can classify one: discrete or continuous time, discrete or continuous state, whether it is Markov, and whether its increments are independent and stationary. Those four answers pin down most processes used in practice.

Brownian motion is then the specific process that plays the role of a building block, and the five properties are worth listing crisply: it starts at zero; it has independent increments; those increments are normally distributed with variance equal to the elapsed time, so $W_t - W_s \sim N(0, t-s)$; it has continuous paths; and it is Markov. Everything else follows, including the fact that it is nowhere differentiable, which is why ordinary calculus does not apply and Ito's lemma exists. The detail is in Brownian motion.

Then connect them to finance, because the definitions alone are not the answer. Say that a price is modelled as a process rather than as a distribution because the payoffs that matter are path dependent, and that Brownian motion is used as the driver because independent normal increments make the mathematics tractable and reproduce the one property everyone agrees on, that variance grows with time. Then say plainly what it gets wrong: real returns have fat tails, volatility clusters rather than being constant, and prices jump, none of which a Brownian driver produces. That closing qualification is usually what the question is fishing for.

The families you actually meet

- **Random walk**, the discrete-time ancestor: each step is an independent draw added to the last. Everything else in this list is a variation on it.
- **Geometric Brownian motion**, where the *proportional* change rather than the absolute

change is the random part, so the level cannot go negative and it is the *log* returns rather than the prices that are normally distributed. This matters: a simple return cannot be normal on a variable floored at -100%. This is the default for equities and the engine of the Black-Scholes world.

- **Mean-reverting processes**, where the drift pulls the level back towards a long run value at a speed κ. Used for rates, spreads, volatility and commodity prices: anything with an economic anchor that a random walk would not respect.
- **Jump processes**, where a Poisson-type arrival adds a discrete move. Needed because a continuous process assigns probability zero to a jump, and jumps happen.
- **Markov chains**, a finite set of states with a table of transition probabilities. The natural model for anything that takes discrete values, ratings most obviously.

INTUITION

The single most useful distinction in this list is between a process whose variance grows without limit and one whose does not. A random walk or a geometric Brownian motion wanders: the uncertainty about where it will be keeps growing with the horizon. A mean-reverting process settles: uncertainty grows at first and then flattens out at a level set by the tension between the volatility pushing it away and the reversion pulling it back. Ask which of those two behaviours the quantity in front of you actually has, and half the modelling choice is made. An equity index is the first. A credit spread, a rate and a volatility are all the second, which is why modelling them as random walks produces nonsense at long horizons.

WORKED EXAMPLE

Mean reversion is usually parameterised by a speed κ, and the intuitive way to quote it is the **half-life**, the time taken for half of a deviation from the long-run level to be worked off:

$$t_{1/2} = \frac{\ln 2}{\kappa}.$$

With $\kappa = 2$ per year, $t_{1/2} = 0.693/2 = 0.347$ years, about **4.2 months**. So a spread trading 100 bp above its long-run level is expected, absent new shocks, to be about 50 bp above it four months later.

Quote the half-life rather than κ when speaking to anyone. Half of the argument about whether a mean-reverting model is calibrated sensibly disappears once the parameter is stated in months.

WORKED EXAMPLE

A Markov chain makes an awkward credit question easy. Take a two-state chain, healthy and defaulted, where default is absorbing, and suppose the one-year default probability is 2%.

Survival for two years is $0.98^2 = 0.9604$, so the cumulative two-year default probability is $1 - 0.9604 = 3.96\%$, not $2 \times 2\% = 4\%$. Over five years, survival is $0.98^5 = 0.9039$ and the cumulative default probability is 9.61%, not 10%.

The gap is small here and grows with the hazard rate and the horizon, and the reason for it is structural rather than numerical: **you can only default once**, so the second year's default probability applies only to the fraction that survived the first. Multiplying survival probabilities is right; adding default probabilities is wrong. Default modelling proper is default risk; the point here is that the chain does the bookkeeping automatically.

In the language of this section, the **random walk hypothesis** is the claim that a price is a Markov process with zero conditional drift, so the best forecast of tomorrow's price is today's. Two of the three processes above are enough to see why it is only an approximation: volatility clusters, so $|r_t|$ is strongly autocorrelated while r_t is not, and spreads, rates and volatility mean revert rather than wander. Whether to believe the hypothesis, and the evidence on both sides, belongs to stationarity, which owns the question and carries the interview answer.

REAL INTERVIEW QUESTION (Barclays · Cross Asset)

We know there is going to be a jump in a stock. What strategy do you adopt?

WHAT THEY TEST / ANSWER

Before choosing a position, say what the premise does to the model, because that is the insight and it changes the answer.

A continuous process cannot represent this. Geometric Brownian motion has continuous paths, so it assigns probability zero to a discontinuous move, and any price it produces is therefore wrong for this situation in a specific direction: it understates the value of anything convex in the underlying. The correct model adds a jump component, with a jump intensity and a distribution of jump sizes, and the immediate consequence is that the implied volatility smile steepens, because out-of-the-money options gain far more from a jump than at-the-money ones do.

Then the position, stated as an exposure rather than as a product. If you expect a large move of unknown sign, you want to be **long convexity and long the wings**: the payoff should be convex in the underlying so that magnitude pays regardless of direction, and it should be concentrated where a diffusion model thinks nothing can reach, since that is where the mispricing is. If you have a view on the sign, the same reasoning applies on one side only. The option structures that express that exposure belong to the strategies chapter rather than here; what this section owns is the reason a diffusion model prices them wrongly in the first place.

Then the two qualifications that make it a desk answer. First, if the jump is *scheduled*, an earnings date or a referendum, everyone else knows too, and the options already carry an elevated implied volatility for that date; you are not buying a jump the market has ignored, you are taking a view on whether the priced jump is large enough. Second, delta hedging does not protect you across a jump. Hedging works because the move is small enough for the delta to be locally accurate, and a gap breaks that, which is why jump risk is carried rather than hedged and why it is sized as a scenario loss.

INTERVIEW TRAP

Three errors. (1) Saying a stochastic process is "a random variable that changes over time", which loses the point: it is the joint behaviour of all times together, and that joint behaviour is what path-dependent payoffs read. (2) Assuming Markov because everything you have been taught is Markov. Volatility clustering alone shows prices are not, and every model with a stochastic volatility state is an admission of it. (3) Adding default probabilities across years instead of multiplying survival probabilities.

Where this goes next

The process that drives almost all of the others is Brownian motion, the fair-game condition that makes risk-neutral pricing work is martingales, and the calculus for transforming these objects is Ito's lemma. Writing the dynamics down formally is stochastic differential equations, and simulating them is Monte Carlo methods. The credit application of the chain used above is default risk, and the economic reason a random walk is a good first approximation is price discovery.

9.2 Brownian Motion

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Brownian motion is the object every diffusion model in finance is built on. It is the continuous-time limit of a random walk, and its properties are not a mathematical formality: the square-root-of-time scaling is why volatility is quoted annualised, the quadratic variation is why Ito's lemma has an extra term that ordinary calculus does not, and the non-differentiability of its paths is why a delta hedge has to be rebalanced rather than solved once.

The definition, in the form interviews ask for

REAL INTERVIEW QUESTION (Société Générale · Volatility)

What is a Brownian motion? Give us the five points.

WHAT THEY TEST / ANSWER

The question asks for a list, so give the list cleanly and then say which entries are doing the work.

A standard Brownian motion W_t satisfies:

1. $W_0 = 0$. A normalisation, nothing more.
2. **Independent increments**: for $s < t < u < v$, the increment $W_t - W_s$ is independent of $W_v - W_u$. The past tells you nothing about the next move.
3. **Stationary Gaussian increments**: $W_t - W_s \sim \mathcal{N}(0, t - s)$. The distribution of a move depends only on how long you wait, and its variance is exactly the elapsed time.
4. **Continuous paths**. It does not jump.
5. **Nowhere differentiable**, with quadratic variation $[W]_t = t$. The path is continuous and has no slope at any point.

Then say which ones matter and why, which is what separates a list from an answer.

Point 3 is the whole of volatility scaling: the standard deviation over horizon t is $\sqrt{t}$, which is why a 1% daily volatility is $1\% \times \sqrt{252} = 15.9\%$ annualised, and why every quoted volatility carries an implicit square root. Worth adding that the rule is a consequence of *independence*, point 2, rather than of the model: it holds exactly here because increments are independent, and it fails in real data to the extent they are not, which is autocorrelation.

Points 2 and 3 together are what make it a martingale and therefore the right driver for a no-arbitrage model: the best forecast of W_{t+s} given everything up to t is W_t itself.

Point 5 is the strangest and the most consequential. Continuous but nowhere differentiable means the path has infinite length in any interval, however short, so there is no velocity to speak of. But its *quadratic* variation, the sum of squared increments, converges to t rather than to zero, which is the statement usually written $(dW)^2 = dt$ and is the reason stochastic calculus is not ordinary calculus.

Close with the finance version. Prices are not modelled as Brownian motion directly, since that would allow negative prices; the log of the price is, which is geometric Brownian motion. Saying that shows you know the difference between the object and its use.

Square root of time, and where drift takes over

KEY IDEA

Because the variance of an increment is the elapsed time, the standard deviation grows as $\sqrt{t}$ while any drift grows as t. Over short horizons the noise dominates, and over long horizons the drift does. The crossover is where they are equal,

$$\mu T = \sigma\sqrt{T} \quad \Longrightarrow \quad T = \left(\frac{\sigma}{\mu}\right)^2,$$

which is $1/SR^2$, the same quantity that appears in stationarity as the sample size needed to detect an edge. They are the same statement in different clothes.

WORKED EXAMPLE

When does an equity index's return exceed its noise? With an expected excess return of 6% and volatility of 20%,

$$T = \left(\frac{0.20}{0.06}\right)^2 = \mathbf{11.1} \text{ years.}$$

It takes eleven years before the accumulated drift equals a single standard deviation of the accumulated noise. That number explains a great deal: why equity risk premia are hard to measure, why a three-year track record carries almost no information, and why short-dated options are almost entirely a volatility bet while very long-dated ones start to be a directional one. It is also why option pricing can ignore the real-world drift entirely over the horizons options actually trade.

Geometric Brownian motion, and the term everyone loses

Prices are modelled as

$$dS_t = \mu S_t\,dt + \sigma S_t\,dW_t,$$

so that returns rather than absolute moves are what behaves like Brownian motion. Applying Ito's lemma to $\ln S$ gives the solution

$$S_T = S_0 \exp\left[\left(\mu - \tfrac{1}{2}\sigma^2\right) T + \sigma W_T\right].$$

INTUITION

The $-\frac{1}{2}\sigma^2$ is the single most examined detail in this chapter, and the way to remember it is that it is *not* a correction to the expected price. The expected price still grows at μ. What the term does is separate the **mean** from the **median**: the log of the price drifts at $\mu - \frac{1}{2}\sigma^2$, so the median path grows more slowly than the average path, because a lognormal distribution is skewed and the mean is pulled up by a thin right tail.

Concretely, at $\mu = 10\%$ and $\sigma = 30\%$ over one year, the expected price is $e^{0.10} = 1.1052$ times spot while the median is $e^{0.10-0.045} = 1.0565$ times spot. The mean is **4.6%** above the median, and the gap $\frac{1}{2}\sigma^2 = 4.5\%$ a year is what practitioners call **volatility drag**: the typical outcome is worse than the average outcome, and the gap widens with the square of volatility.

That is why a leveraged product compounding a volatile index underperforms the index times the leverage, and it is the same arithmetic as the $(1+x)(1-x) = 1 - x^2$ round trip.

REAL INTERVIEW QUESTION (Société Générale · EQD)

Tell me about Brownian motion.

WHAT THEY TEST / ANSWER

An open question, so impose a structure: what it is, why finance uses it, and what it gets wrong. The third part is the one that distinguishes you, because everyone does the first two.

What it is. The continuous-time limit of a random walk, with the five properties above. The one to lead with is that increments are independent and normally distributed with variance equal to elapsed time, because everything else follows from it.

Why finance uses it. Three reasons, in order.

It makes prices *unpredictable* in exactly the sense a no-arbitrage model needs: the increments are unforecastable, so the discounted price is a martingale and there is no strategy that makes money on direction alone.

It produces **lognormal** prices when applied to the log, so prices cannot go negative and returns rather than levels are the stationary quantity, which matches the data far better than the alternative.

And it is *tractable*. The distribution at any horizon is known in closed form, which is why Black-Scholes has a formula at all rather than requiring simulation.

What it gets wrong, which is the real content.

No jumps. Paths are continuous by assumption, so gaps cannot happen, and gaps are exactly where a delta-hedged book loses money.

Constant volatility. Real volatility clusters and is itself stochastic, which is what stochastic volatility models exist to repair.

Tails too thin. A Gaussian assigns essentially zero probability to moves that occur every few years, and the 1987 crash was about eighteen standard deviations under this assumption.

Independent increments. Returns are close to unforecastable but squared returns are not, so the independence assumption is false in the second moment even where it holds in the first.

Close on the professional position, which is neither defending the model nor dismissing it: Brownian motion is the coordinate system the market quotes in rather than a belief about reality. Implied volatility is a price expressed in Black-Scholes units, and the smile in implied versus realised is precisely the market correcting these assumptions strike by strike. Nobody believes the model; everybody uses its language.

INTERVIEW TRAP

1. **Saying prices follow Brownian motion.** The *log* of the price does. Brownian motion itself goes negative.
2. **Treating $-\frac{1}{2}\sigma^2$ as a haircut on returns.** The expected price still grows at μ. The term separates the median from the mean.
3. **Scaling volatility linearly in time.** It scales with $\sqrt{t}$, and getting this wrong is the fastest way to be visibly wrong on a trading floor.
4. **Forgetting that the path has no derivative.** There is no instantaneous velocity, which is why hedging is a repeated discrete act rather than a solved problem, and why rehedging frequency determines the variance of your P&L.
5. **Believing quadratic variation is zero because increments are small.** Squared increments sum to t, not to zero, and that single fact is what makes stochastic calculus different.
6. **Defending the Gaussian tails.** They are wrong and everyone knows it. The interesting answer is what the market does about it, which is the smile.

VISUAL / CODE TO BUILD: Interactive

A path panel with a horizon slider. Several Brownian paths are drawn from a fixed seed with a $\pm\sqrt{t}$ envelope over them, so the reader can watch the cone widen as the square root rather than linearly. A toggle overlays a drift line at μt, and dragging the horizon out shows the straight line eventually escaping the cone, with the crossing point printed and matching $(\sigma/\mu)^2$. At the section's numbers it lands at eleven years, which is usually much further out than readers expect.

A second panel demonstrates non-differentiability by zooming: the reader picks a point on a path and repeatedly magnifies it, and the path never straightens. Beside it a counter shows the total path length growing without bound as the resolution increases while the sum of squared increments converges to t. Seeing one quantity diverge and the other converge, on the same path, is the clearest possible statement of what quadratic variation means and why $(dW)^2 = dt$.

A third draws the lognormal distribution at a chosen horizon with the mean and median marked separately and the gap labelled as volatility drag. Increasing volatility with the drift held fixed pushes them apart quadratically while the mean stays put, which is the section's most misunderstood point made visual.

9.3 Martingales

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A martingale is the mathematical formalisation of **a fair game**, and it is the single most useful abstraction in derivatives pricing. It gives precise content to two statements that sound like folklore: that you cannot beat a fair game with a clever betting rule, and that the price of a derivative is an expected value. It also settles a family of interview puzzles that are designed to look paradoxical, and the settling is always the same move.

What you need first: information, and best guesses

Two ideas have to come before the definition.

A **filtration** $(\mathcal{F}_t)$ is a growing family of information sets: $\mathcal{F}_t$ is everything you know at time t. It grows because information is not forgotten. If a quantity is known at time t, it is said to be **$\mathcal{F}_t$-measurable**, and a process whose value at each time is known then is **adapted**.

The **conditional expectation** $\mathbb{E}[X \mid \mathcal{F}_t]$ is your best guess of X given what you know at t. Two properties are used constantly. **Known things pass through**: if Y is known at t then $\mathbb{E}[YX \mid \mathcal{F}_t] = Y\,\mathbb{E}[X \mid \mathcal{F}_t]$. And the **tower property**: for $s < t$, $\mathbb{E}\big[\mathbb{E}[X \mid \mathcal{F}_t] \mid \mathcal{F}_s\big] = \mathbb{E}[X \mid \mathcal{F}_s]$, which says that your best guess today of your best guess tomorrow is just your best guess today. See conditional probability.

The definition

A process (M_t) is a **martingale** with respect to $(\mathcal{F}_t)$ if it is adapted, integrable ($\mathbb{E}|M_t| < \infty$), and

$$\mathbb{E}[M_t \mid \mathcal{F}_s] \;=\; M_s \qquad \text{for all } s < t.$$

If the equality is replaced by $\geq$ the process is a **submartingale** (it drifts up), and by $\leq$ a **supermartingale** (it drifts down). The naming trips everyone up, so remember it by the finance reading: a *sub*martingale is the good one to own.

> **INTUITION**
>
> Take the sentence literally: **whatever has happened so far, your expected future position is your current position.** That is a fair game. It does not say the process is constant, or that it has no variance, or that it cannot trend for years by luck. It says only that the *expected* increment, conditional on everything you know, is zero. A casino game with a house edge is a supermartingale for the player. A stock with a positive risk premium is a submartingale. And the central trick of derivatives pricing is to change the probability measure until the thing you care about becomes an exact martingale, because then its expected value is trivially its current value.

Three examples worth carrying:

- A **symmetric random walk** $S_n = X_1 + \cdots + X_n$ with independent mean-zero steps is a martingale, since $\mathbb{E}[S_{n+1} \mid \mathcal{F}_n] = S_n + \mathbb{E}[X_{n+1}] = S_n$.
- **Brownian motion** W_t is a martingale, for the same reason in continuous time, see Brownian motion.
- $W_t^2 - t$ is a martingale, which is less obvious and much more useful. Write $W_{t+s} = W_t + (W_{t+s} - W_t)$ and square: the cross term has zero mean and the new increment contributes variance s, so $\mathbb{E}[W_{t+s}^2 \mid \mathcal{F}_t] = W_t^2 + s$ and therefore $\mathbb{E}[W_{t+s}^2 - (t+s) \mid \mathcal{F}_t] = W_t^2 - t$. This is the statement that the variance of Brownian motion accumulates at exactly rate one per unit time, and it is the seed of quadratic variation and of Ito's lemma.

You cannot beat a fair game: the martingale transform

Suppose M is a martingale and you trade it with a strategy H_k, where H_k is the position you hold over step k and must be chosen using only information available *before* that step (such a process is called predictable). Your cumulative gains are

$$G_n \;=\; \sum_{k=1}^{n} H_k\,(M_k - M_{k-1}).$$

Then G is itself a martingale, because $\mathbb{E}[H_k(M_k - M_{k-1}) \mid \mathcal{F}_{k-1}] = H_k\,\mathbb{E}[M_k - M_{k-1} \mid \mathcal{F}_{k-1}] = 0$, using that H_k is known at $k-1$ so it passes through.

KEY IDEA

No strategy based on past information turns a fair game into a favourable one. That single theorem is why technical trading rules cannot create expected return from a martingale price process, and it is the mathematical backbone of no-arbitrage pricing in no-arbitrage pricing. Note carefully what the proof needs: the position must be chosen *before* the move, and it must be finite. Every apparent counterexample violates one of those two conditions, and finding which one is the whole skill.

WORKED EXAMPLE

The doubling strategy, and where the free money goes. Bet €1 on a fair coin. If you lose, double the stake; keep doubling until you win, then stop. When you finally win, you recover all previous losses plus €1: after k losses you have staked $1+2+\cdots+2^{k-1} = 2^k - 1$ and the winning bet of 2^k returns 2^{k+1}, leaving you exactly €1 ahead. Since a fair coin comes up heads eventually with probability one, this appears to be a guaranteed profit from a fair game.

It is not, and the resolution is a capital constraint. Surviving n consecutive losses requires a bankroll of $2^n - 1$. Suppose you have exactly that and must stop if it is exhausted. Then you win €1 with probability $1 - 2^{-n}$, and with probability 2^{-n} you lose the entire $2^n - 1$. The expected value is

$$\big(1 - 2^{-n}\big) \times 1 \;+\; 2^{-n} \times \big(-(2^n - 1)\big) \;=\; 1 - 2^{-n} - 1 + 2^{-n} \;=\; 0,$$

exactly zero for every n. With a €1,023 bankroll you win a euro about 999 times out of 1,000 and lose €1,023 once, and the arithmetic closes precisely. The unbounded version escapes the theorem only by requiring an unbounded position, which is the condition the martingale transform assumes away, and the formal statement of when you may stop is the optional stopping theorem in stopping times.

The reason this matters beyond puzzles: a doubling strategy is the mathematics of **selling a deep tail to buy a high win rate**. It produces a track record of small steady gains and one catastrophic loss, which is the P&L signature of a short-gamma or short-volatility book, and it is why win rate on its own tells you nothing about whether a strategy makes money.

Why pricing is an expectation

The link to derivatives is one sentence long, and the rest of the chapter makes it precise. Under suitable conditions, **no arbitrage is equivalent to the existence of a probability measure, equivalent to the real one, under which discounted asset prices are martingales.** That measure is called risk-neutral, and once you have it, the price of a claim is just the discounted expectation of its payoff, because a martingale's expected future value *is* its current value.

Two consequences that answer common confusions. The risk-neutral measure is not a forecast: it is a pricing device chosen so that discounted prices are fair games, which is why expected returns under it are the risk-free rate for every asset regardless of risk. And the construction of that measure, and the price you pay for changing it, is the subject of change of measure and the Girsanov theorem.

REAL INTERVIEW QUESTION (Credit Suisse · desk not recorded)

You have a fair coin and start with a dollar: if you toss heads your position doubles, if you toss tails your position halves. What is the fair ticket price to play this game?

WHAT THEY TEST / ANSWER

Compute the expectation first, since that is what "fair price" asks for, then deliver the nuance that is the actual point of the question. **Per round**, the expected multiplier is $\frac{1}{2}(2) + \frac{1}{2}(0.5) = 1.25$, so one round on a starting dollar is worth \$1.25 and n rounds are worth 1.25^n under a pure expected-value criterion. Say that clearly, then say what is strange about it. The **geometric** mean per round is $\sqrt{2 \times 0.5} = 1$, and equivalently $\mathbb{E}[\log X] = \frac{1}{2}\log 2 + \frac{1}{2}\log 0.5 = 0$, so log **of your wealth is a martingale while your wealth itself is a submartingale**. In words: the typical path goes nowhere while the average grows at 25% a round, because the average is carried by a shrinking set of very large outcomes. After many rounds the median outcome is still about one dollar and the probability of being below your starting point tends to one half, while the mean diverges.

That gap, 1.25 against 1, is exactly **volatility drag**, the wedge between arithmetic and geometric returns, which is why a leveraged or rebalanced product can have a positive expected return and a negative median outcome. So the honest answer is two-part: the risk-neutral, expected-value price is 1.25 per round, and no real participant with finite capital should pay anything close to that for many rounds, because they experience the geometric mean rather than the arithmetic one. If they push, name the criterion you would actually use, maximising expected log wealth or a utility function, and note that this is the same structure as the St Petersburg problem: an expectation can be dominated by outcomes too rare to be experienced.

REAL INTERVIEW QUESTION (Barclays · Credit)

At roulette: you bet 100 euros on 1 number, what is the expected gain? You bet 100 euros on 2 numbers, what is the expected gain?

WHAT THEY TEST / ANSWER

State your convention first, because the answer depends on it: assume a European wheel with 37 pockets (0 to 36) and a single number paying 35 to 1. **One number**: you win 3,500 with probability 1/37 and lose 100 with probability 36/37, so

$$\mathbb{E} \;=\; \tfrac{1}{37}(3{,}500) + \tfrac{36}{37}(-100) \;=\; \tfrac{3{,}500-3{,}600}{37} \;=\; -\tfrac{100}{37} \;\approx\; -€2.70.$$

Two numbers, splitting the same €100 as €50 on each: if either comes up, which has probability 2/37, the winning €50 returns $35 \times 50 = 1{,}750$ and the other €50 is lost, so the net is +1,700; otherwise you lose €100. Then $\mathbb{E} = \frac{2}{37}(1{,}700) + \frac{35}{37}(-100) = \frac{3{,}400-3{,}500}{37} = -\frac{100}{37}$, the **same** −€2.70.

That coincidence is the answer they are looking for. Spreading the stake changes the variance and not the expectation, because **every bet on the wheel carries the same house edge**, one pocket in thirty-seven, or 2.70% of the stake. Say the two things that make it a professional answer. The edge comes entirely from the zero: the payout is calculated as though there were 36 pockets while there are 37, and an American wheel with a double zero has 38 pockets and therefore double the edge at 5.26%. And in the language of this section, your bankroll at roulette is a **supermartingale**, so no combination or sequence of bets can make it a fair game: by the martingale transform, a strategy on an unfavourable game stays unfavourable, which is precisely why the doubling system does not work.

REAL INTERVIEW QUESTION (BNP · EQD)

If I offered you to play your internship on a coin flip, would you accept?

WHAT THEY TEST / ANSWER

This is a risk-preference question and there is a right answer, but the marks are in the reasoning rather than the choice. **Decline**, and say why in the language of the job. The bet is *fair in expectation* and that is not the relevant criterion, because my utility over this outcome is strongly concave: the internship is close to a one-off, indivisible, career-relevant payoff, so losing it costs me far more than doubling it would gain me. A fair game with a non-linear payoff is not an acceptable game, which is the same reason a trader with a good franchise does not bet the book on a coin flip even at fair odds, and the same reason position limits exist. Then show you understand the other side, since answering only "no" sounds timid. I would take the bet in the situations where the structure changes: if it were repeatable and small relative to my total, since many independent small fair bets are survivable while one large one is not, which is the Kelly and diversification argument; or if the odds were sufficiently in my favour and the downside bounded to something I can absorb. Close on the distinction this section makes precise: a martingale is a fair game, but fairness is a statement about the *mean*, and what a risk-taker manages is the distribution, which is why sizing rather than edge is what keeps people in the seat. Say it with a bit of humour and without hedging, because the interviewer is also checking whether you can hold a position under mild pressure.

INTERVIEW TRAP

Four traps. First, **thinking a martingale cannot trend**: it can rise for years by chance, and the condition is only that the *expected* next increment is zero given what you know. Second, **mixing up sub and super**: a submartingale drifts up, which is counter-intuitive from the names alone. Third, **believing a betting system beats a fair game**: it cannot, and every apparent counterexample either uses future information or requires an unbounded position, so find which. Fourth, **treating the risk-neutral measure as a forecast**: it is a pricing device under which discounted prices are martingales, not anyone's view of the world.

VISUAL / CODE TO BUILD: Interactive

A fair-game laboratory in three tabs. **Paths**: simulate many symmetric random walks with the running mean overlaid, so the reader sees individual paths trending strongly while the average stays flat, which is the martingale property made visible and kills the "cannot trend" misconception. **Doubling**: play the doubling strategy against a fair coin with a bankroll slider, tracking the equity curve and a running histogram of outcomes, so the reader watches a long sequence of small wins punctuated by one ruin and sees the mean sit at exactly zero however the bankroll is set, reproducing the worked example. A toggle switches the coin to a house edge so the same strategy on a supermartingale becomes visibly negative. **Arithmetic versus geometric**: simulate the doubles-or-halves game with the mean, the median and the full distribution plotted on a log axis, so the 1.25 against 1.00 wedge appears as a widening gap between mean and median and volatility drag becomes an observation rather than a formula.

Where this goes next

The processes this definition is applied to are in stochastic processes and Brownian motion, and the question of when you are allowed to stop, which is what resolves the doubling paradox formally, is stopping times. Turning a price process into a martingale by changing the probability measure is change of measure and the Girsanov theorem, and the pricing theorem they deliver is used in no-arbitrage pricing. The puzzles in this style, and how to attack them under time pressure, are in brain teasers.

9.4 Stopping times

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Almost every decision a desk automates is a rule for *when* to act, not what to buy. An autocallable redeems the first time the underlying closes above its trigger. A knock-out dies the first time spot touches the barrier. A stop-loss fires the first time the book is down more than the limit. A margin call arrives the first time collateral falls short. All of those are **stopping times**, and the mathematics of stopping times exists to answer two questions that come up constantly: is this rule actually executable, and what is a game worth if you are allowed to choose when to walk away.

The one property that matters: no peeking

We work with a process (X_n) and a **filtration** $(\mathcal{F}_n)$, the formal record of what is known at each date, introduced in stochastic processes. Read $\mathcal{F}_n$ as "everything an observer could have seen up to and including time n".

> **INTUITION**
>
> Imagine you are watching a screen and you have a big red button. A stopping rule is legitimate if, at every instant, you can tell from the screen alone whether to press the button now. It is illegitimate if deciding requires knowing something that has not happened yet. "Sell the first time the stock falls 10% below where it opened the year" is a rule you can execute. "Sell at the high of the year" is not a rule at all, it is a wish, because you only learn where the high was in December. Everything in this section is bookkeeping around that one distinction.

Definition

A random time τ taking values in $\{0, 1, 2, \dots\} \cup \{\infty\}$ is a **stopping time** for the filtration $(\mathcal{F}_n)$ if

$$\{\tau \le n\} \in \mathcal{F}_n \quad \text{for every } n.$$

In words: whether or not you have already stopped by time n is decidable from the information available at time n. In continuous time the same statement is $\{\tau \le t\} \in \mathcal{F}_t$ for every $t \ge 0$.

The standard example: a hitting time. For a level b, define the **first passage time**

$$\tau_b = \inf\{\, n \ge 0 : X_n \ge b \,\},$$

with the convention $\tau_b = \infty$ if the level is never reached. This is a stopping time, because $\{\tau_b \le n\}$ is the event that at least one of $X_0, \dots, X_n$ reached b, and you can check that from the path so far. Hitting times are the reason this whole topic is on a derivatives desk: a barrier event, an autocall trigger and a default in a structural credit model are all first passage times.

The standard non-example. The time at which X attains its maximum over $\{0, \dots, N\}$,

$$\sigma = \arg\max_{0 \le n \le N} X_n,$$

is *not* a stopping time. To know at time n whether the maximum has already occurred you would have to know the rest of the path. This is not a technicality: it is the precise reason that a backtest which sells at the peak is worthless, and the precise reason that "buy low, sell high" is not a strategy.

> **KEY IDEA**
>
> Stopping times are closed under the operations you would want. If σ and τ are stopping times then so are $\sigma \wedge \tau$ (the earlier of the two), $\sigma \vee \tau$ (the later), and $\sigma + \tau$. This is what lets you build realistic products: an autocallable stops at the first observation date on which the trigger is met *or* at maturity, whichever comes first, and that is a minimum of two stopping times, so it is one.

Stopping a martingale: the optional stopping theorem

Martingales, the formalisation of a fair game, are covered in martingales. The one-line recall is $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n$: your expected wealth tomorrow, given everything you know today, is your wealth today.

The natural question is whether you can beat a fair game by choosing a clever moment to quit. The answer, under conditions, is no.

KEY IDEA

Optional stopping theorem. If (M_n) is a martingale and τ is a stopping time, then $\mathbb{E}[M_\tau] = \mathbb{E}[M_0]$, provided at least one of the following holds:

- τ is bounded ($\tau \leq N$ for some fixed N); or
- τ is finite almost surely and M is bounded up to τ; or
- $\mathbb{E}[\tau] < \infty$ and the increments $|M_{n+1} - M_n|$ are bounded by a constant.

No stopping rule turns a fair game into a favourable one.

The conditions are not decoration, and the classic counter-example is the betting system that gave martingales their name. Bet \$1 on a fair coin; if you lose, double the stake and bet again; stop at your first win. You stop with probability one, and whenever you stop you are up exactly \$1, so $\mathbb{E}[M_\tau] = 1 \neq 0 = \mathbb{E}[M_0]$. Nothing is broken: $\mathbb{E}[\tau] = 2$ is finite, but the increments double every round and are therefore unbounded, so the third condition fails. The financial translation is exactly the one a risk manager cares about: a strategy that appears to make money with certainty is really a strategy that requires unbounded funding, and the losing tail it hides is the part that ends careers. Doubling down after a loss is the oldest way of turning a fair game into a ruinous one.

WORKED EXAMPLE

Gambler's ruin, solved twice with the same tool. A symmetric random walk starts at 0 and steps ± 1 with probability $\frac{1}{2}$. Let τ be the first time it reaches $+a$ or $-b$, with $a, b > 0$ integers.

Where does it stop? S_n is a martingale and τ is finite with finite mean, with bounded increments, so optional stopping applies:

$$0 = \mathbb{E}[S_0] = \mathbb{E}[S_\tau] = a\,p + (-b)(1-p) \quad \Longrightarrow \quad p = \mathbb{P}(\text{hit } +a \text{ first}) = \frac{b}{a+b}.$$

How long does it take? Use the second martingale $S_n^2 - n$, which is a martingale because each step has variance 1. Optional stopping gives $\mathbb{E}[S_\tau^2] = \mathbb{E}[\tau]$, and

$$\mathbb{E}[S_\tau^2] = a^2 \frac{b}{a+b} + b^2 \frac{a}{a+b} = \frac{ab(a+b)}{a+b} = ab \quad \Longrightarrow \quad \mathbb{E}[\tau] = ab.$$

Sanity check the two results. With $a = b$ the walk is symmetric and $p = 1/2$, as it must be. With $a = 1$ and $b = 100$, you take the small profit first with probability $100/101 \approx 99\%$, and the expected time to resolution is 100 steps. That asymmetry, frequent small wins and rare large losses, is what every short-volatility and short-gamma strategy looks like from the inside, and why the hit rate of a strategy tells you almost nothing on its own.

Optimal stopping: what a game is worth when you can quit

The optional stopping theorem says you cannot beat a *fair* game. Optimal stopping asks the other question: given a process whose value you can lock in whenever you like, what is the best rule, and what is the game worth? Write V_n for the value with n decisions still available. The recursion is

$$V_n = \mathbb{E}\big[\max\big(\text{take it now},\ V_{n-1}\big)\big],$$

which is the discrete **Snell envelope**: at every point you take the larger of the immediate payoff and the continuation value, and the optimal rule is to stop the first time the immediate payoff wins.

WORKED EXAMPLE

The dice game. Roll a fair die. You may keep the number showing, or reroll once and be stuck with the second roll. What is the game worth?

Work backwards. With no rerolls left, the value is $\mathbb{E}[\text{die}] = 3.5$. So on the first roll you keep anything strictly above 3.5, that is 4, 5 or 6, and reroll on 1, 2 or 3. The threshold *is* the optimal stopping rule. Its value:

$$V_2 = \underbrace{\tfrac{1}{2}}_{\{4,5,6\}} \times \underbrace{5}_{\mathbb{E}[\text{die}|\geq 4]} + \underbrace{\tfrac{1}{2}}_{\{1,2,3\}} \times 3.5 = 2.5 + 1.75 = 4.25.$$

Now the follow-up they always ask: two rerolls. The continuation value is now 4.25, so you keep only 5 or 6:

$$V_3 = \tfrac{1}{3} \times 5.5 + \tfrac{2}{3} \times 4.25 = 1.8\overline{3} + 2.8\overline{3} = 4.6\overline{6} = \tfrac{14}{3}.$$

Two things to say out loud. The value rises but with diminishing returns, 3.5 then 4.25 then 4.67, because extra optionality is worth less each time you add it. And the threshold *ratchets up* as more rerolls remain, from 3.5 to 4.25, because a better fallback makes you pickier. That second observation is exactly the exercise boundary of an American option: the level at which you exercise moves with time to maturity.

REAL INTERVIEW QUESTION (JP Morgan · EQD)

How much is this game worth: you roll a die, a value appears, and you can either take that value or reroll once and take the new value (in your head)?

WHAT THEY TEST / ANSWER

They are testing backward induction and whether you can hold three numbers in your head under mild pressure, not whether you know the phrase "optimal stopping". Say the reasoning as you go: the reroll is worth 3.5, so keep 4, 5 or 6 and reroll otherwise; that is $\frac{1}{2} \times 5 + \frac{1}{2} \times 3.5 = 4.25$. Deliver the answer, then immediately volunteer the structure, that the optimal rule is a threshold and the threshold is the value of continuing. That single sentence is what separates a candidate who computed it from a candidate who understood it, and it sets you up for the two-reroll follow-up ($14/3 \approx 4.67$, keeping only 5 or 6) that they will ask next. More of this style in brain teasers.

Where this lives on the desk

American and Bermudan options. An American option is literally an optimal stopping problem: the holder chooses a stopping time $\tau \le T$ to maximise the discounted expected payoff, and the price is $\sup_\tau \mathbb{E}^{\mathbb{Q}}[e^{-r\tau}\,\text{payoff}(S_\tau)]$. The "take it now versus continue" recursion above is what a binomial tree computes at each node when you compare the intrinsic value with the discounted continuation value. It is also why a closed-form Black-Scholes price does not exist for an American put, and why Monte Carlo needs an extra device, the regression of continuation values, covered in Monte Carlo methods. The option types themselves are in options.

Barriers and autocallables. A knock-out pays only if the hitting time of the barrier exceeds maturity; its price therefore depends on the whole distribution of the first passage time, not just on the terminal distribution. An autocallable is a chain of stopping times, one per observation date, and the product terminates at the first one that triggers. See exotic options, path-dependent products and autocallables.

Credit. In a structural default model the firm defaults at the first time its asset value crosses a debt boundary, which is a first passage time. That is the whole modelling content of the structural approach, described in structural versus reduced-form models and default risk.

Risk controls. Stop-losses, position limits and margin calls are stopping rules imposed on a trader, and the gambler's ruin arithmetic above is exactly the calculation that tells you how likely a limit is to be hit before a target is reached. See intraday hedging.

REAL INTERVIEW QUESTION (Exane · Exo)

In what case do you exercise an American option early?

WHAT THEY TEST / ANSWER

The structure they want is: exercising early means giving up the remaining optional value, so you only do it when something else is worth more than that. Then the two cases. For a **call on a non-dividend-paying stock**, never: you would be paying the strike sooner for no reason and destroying time value, so the American call is worth the same as the European one. Introduce a dividend and it can become optimal to exercise immediately before a large ex-dividend date, because the dividend is cash you can only capture by owning the share. For an **American put**, early exercise can be optimal when it is deep in the money, because exercising turns the position into cash on which you earn interest, and there is very little optional value left to sacrifice when the payoff is nearly linear. Finish with the framing that scores: it is an optimal stopping problem, so the answer is a boundary in spot that moves with time to maturity, and you exercise the first time spot crosses it.

REAL INTERVIEW QUESTION (HSBC · Structured Products)

How does the delta behave at the barrier?

WHAT THEY TEST / ANSWER

They are checking whether you have thought about what a hitting time does to a hedge. Pick the case that actually bites, a **reverse** knock-out, say an up-and-out call with the barrier above the strike. Just below the barrier the option is well in the money and one touch away from being worth nothing, so as spot rises *towards* the level its value *falls*: the delta is large and *negative* exactly where a plain call's would be close to +1, and gamma goes with it. A down-and-out call struck above its barrier is the mild case by comparison, since the option is out of the money at the level anyway and its value runs smoothly down to zero. Be precise about the discontinuity too, because it is a follow-up: with continuous monitoring the value is zero *at* the barrier and the problem is the slope, while with discrete monitoring the value really does jump across an observation, which is worse. Either way the hedge ratio flips violently on tiny spot moves right where liquidity is worst and everyone else's barriers sit at the same round number. Name the practical responses: trade a barrier shift or an overhedge so the risk is smoothed before you reach the level, widen the price for the gap risk, and monitor the concentration of barriers across the book. The underlying point is that pricing depends on a first passage time, and hedging a first passage time is fundamentally harder than hedging a terminal payoff. Convexity mechanics are in Gamma and convexity.

INTERVIEW TRAP

Four traps. First, calling a hindsight rule a strategy; the time of the maximum is not a stopping time and any backtest that uses it is broken. Second, quoting the optional stopping theorem without its conditions, and then being unable to explain the doubling strategy, which is the counter-example every interviewer keeps in reserve. Third, assuming a first passage time is finite; a random walk with negative drift may never reach an upper level, so τ can be infinite with positive probability and $\mathbb{E}[\tau]$ can be infinite even when τ is finite almost surely. Fourth, confusing continuous monitoring with discrete monitoring: a barrier checked on daily closes is a different, and strictly less likely, event than one checked continuously, and the price difference is real money.

VISUAL / CODE TO BUILD: Interactive

Two linked panels. The left one simulates a random walk and lets the reader drag an upper level $+a$ and a lower level $-b$; it draws the stopped path, and a running tally compares the empirical hit frequency and average duration against the theory, $b/(a+b)$ and ab. The right one is a discrete optimal-stopping trainer: the reader plays the dice game with a chosen number of rerolls, the panel shows the optimal threshold at each stage against the reader's actual choice, and it tracks the value given up. The intended "aha" is that the optimal rule is always a threshold, and that the threshold is exactly the continuation value.

Where this goes next

Stopping times are the last piece of the language before the calculus starts. Combined with martingales they give the tools for change of measure and the risk-neutral pricing framework. Applied to Brownian motion they give first passage distributions in continuous time, which is what closed-form barrier formulas are built from. And the free-boundary problem that an American option becomes is the optimal stopping story rewritten as a partial differential equation, in PDEs in finance.

9.5 Change of measure

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Option pricing has one genuinely strange feature: the expected return of the underlying does not appear in the answer. Two traders who violently disagree about where a stock is going will price the same option identically, provided they agree about its volatility. A change of measure is the machinery that makes this true, and understanding it properly means understanding what has been changed and, just as importantly, what has not.

> **INTUITION**
>
> Changing measure does not change the world. It changes the probabilities you attach to the states of the world, keeping the same set of possible states. Nothing that was possible becomes impossible and nothing impossible becomes possible; the outcomes are merely reweighted. Prices are unaffected because a price is a fact, not a probability.

Equivalent measures and the object that connects them

Two probability measures $\mathbb{P}$ and $\mathbb{Q}$ on the same space are **equivalent** when they agree about which events have probability zero. That is a weak requirement and a crucial one: it says the two measures disagree about odds but not about possibilities.

When they are equivalent, a single random variable connects them: the **Radon-Nikodym derivative** $Z = d\mathbb{Q}/d\mathbb{P}$. For any payoff X,

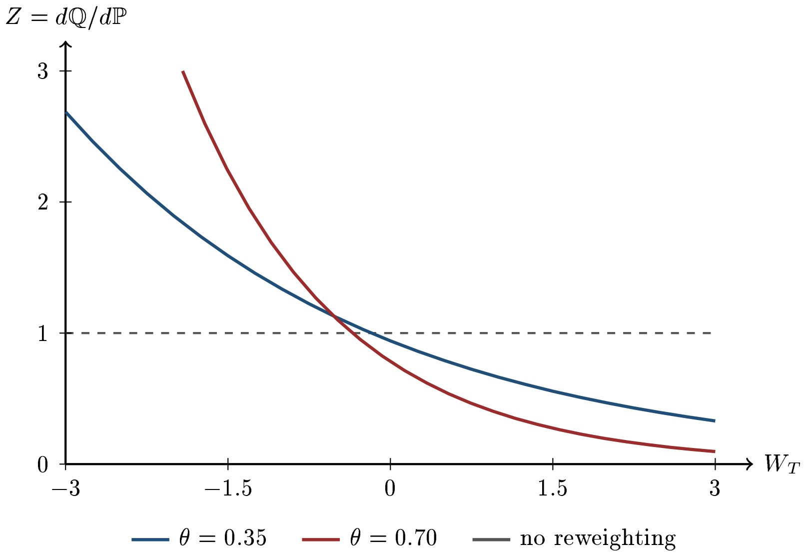
$$\mathbb{E}^{\mathbb{Q}}[X] \;=\; \mathbb{E}^{\mathbb{P}}[X\,Z].$$

So an expectation under $\mathbb{Q}$ is just a $\mathbb{P}$-expectation of a reweighted payoff. Z is strictly positive, which is equivalence again, and $\mathbb{E}^{\mathbb{P}}[Z] = 1$, which is what makes $\mathbb{Q}$ a probability measure at all.

For a Brownian setting the reweighting takes the explicit form

$$Z = \exp\!\left(-\theta W_T - \tfrac{1}{2}\theta^2 T\right),$$

and the theorem that says what this does to the process is Girsanov's theorem. What belongs here is the object itself and what it is for.



the new measure adds weight to bad outcomes and removes it from good ones, and never reaches zero

The picture is the whole idea. Bad outcomes, on the left, are given more weight under $\mathbb{Q}$ than under $\mathbb{P}$; good outcomes are given less. The tilt is steeper when θ is larger, and the curve never touches the axis, which is exactly the statement that the two measures remain equivalent no matter how hard the reweighting pulls.

Why any of this is legitimate

Two results justify the whole construction, and they are worth being able to state.

The first fundamental theorem. A market is free of arbitrage if and only if there exists a measure, equivalent to the real-world one, under which every asset price discounted by the money-market account is a martingale. Absence of arbitrage and existence of a pricing measure are the same statement.

The second fundamental theorem. That measure is *unique* if and only if the market is complete, meaning every payoff can be replicated. Uniqueness is what turns "a price consistent with no arbitrage" into "the price".

Together these say something worth internalising: the risk-neutral measure is not an assumption bolted onto a model. It is what no-arbitrage looks like when written in the language of probability. The one-period version of this argument, where the algebra is visible, is in no-arbitrage pricing.

The market price of risk, and what ignoring it costs

WORKED EXAMPLE

A stock at 100, volatility 20%, expected return 10%, with the risk-free rate at 3%. The quantity that governs the change of measure is

$$\theta = \frac{\mu - r}{\sigma} = \frac{0.10 - 0.03}{0.20} = \mathbf{0.35},$$

the **market price of risk**, which is the Sharpe ratio of the underlying. It is the excess return demanded per unit of volatility, and the change to $\mathbb{Q}$ is precisely the reweighting that removes it.

Under $\mathbb{P}$ the stock is expected to be at $100e^{0.10} = 110.52$ in a year. Under $\mathbb{Q}$ it is expected to be at $100e^{0.03} = 103.05$. The distributions are otherwise identical: same volatility, same shape, different centre.

Now price a one-year at-the-money call both ways. Under $\mathbb{Q}$ the Black-Scholes price is

$$d_1 = 0.2500, \quad d_2 = 0.0500, \quad C = 100(0.598706) - 100e^{-0.03}(0.519939) = \mathbf{9.4134}.$$

Do it the intuitive and wrong way instead: take the expected payoff under the *real-world* measure and discount it at the risk-free rate. That gives $\mathbb{E}^{\mathbb{P}}[(S_T - K)^+] = 14.6653$, and discounting at 3% gives **14.2318**.

That is 51.2% too expensive. And the error has an obvious sign: you have taken credit for the equity risk premium in the numerator without paying for the risk in the denominator. Quote that price and you will buy every option anyone offers you.

The change of measure is what stops this. It moves the drift from μ to r so that discounting at r is consistent, and it does so at the cost of a reweighting whose steepness is exactly θ.

KEY IDEA

Changing measure changes the drift and leaves the volatility alone. That single asymmetry is why option prices depend on volatility and not on expected return, why two traders with opposite views on a stock agree on the price of an option on it, and why the entire industry argues about volatility rather than about direction.

Changing the numeraire, which is the general version

Discounting by the money-market account is a convention, not a requirement. **Any strictly positive traded asset can serve as numeraire**, and each choice comes with its own martingale measure under which prices divided by that asset are martingales. Choosing well is the main practical use of this material, because the right numeraire makes the awkward term disappear.

- **The money-market account** gives the familiar risk-neutral measure. It is the wrong choice when interest rates are stochastic, because the discount factor is then random and correlated with the payoff.
- **A zero-coupon bond maturing at T** gives the **T-forward measure**, under which the price today is simply the bond price times the expected payoff, with no discounting inside the

expectation. This is the reason forward rates are martingales under the right measure and it is what makes Black's formula for caps and swaptions legitimate rather than a convenient fiction. When a rate is observed or paid under a different measure from the one it is a martingale under, the correction is a convexity adjustment, which is where the ones in forward rates come from.

- **The stock itself** gives a measure that simplifies the first term of many option formulas, which is why $N(d_1)$ and $N(d_2)$ are probabilities under two *different* measures. They are not two estimates of the same thing, and treating them as such is a standard error.
- **A foreign money-market account** is what a quanto requires, and the drift adjustment that appears in quanto pricing is nothing more than the price of switching between domestic and foreign measures.

> **KEY IDEA**
>
> The choice of numeraire is free and the price is not. Whatever numeraire you pick, the value you compute is the same; only the difficulty of the computation changes. Picking the numeraire that makes your payoff simple is one of the few genuinely free lunches in derivatives pricing.

Interview angle

> **REAL INTERVIEW QUESTION** (Barclays · EQD)
>
> *What is the risk-neutral model? What is it for? How do you interpret it?*
>
> **WHAT THEY TEST / ANSWER**
>
> Three clauses, and the third is the one that separates answers. Take them in order.
>
> **What it is**: not a model of the world but a change of probability measure. Same states, same volatility, reweighted odds, chosen so that discounted traded prices are martingales. Nothing has been assumed about anybody's preferences, and the phrase "we assume investors are risk-neutral" is the standard wrong answer; nobody assumes that.
>
> **What it is for**: it makes the expected return of the underlying drop out of the price. Under $\mathbb{P}$ the stock in the example above drifts to 110.52; under $\mathbb{Q}$ it drifts to 103.05, and only the second is consistent with discounting at the risk-free rate. Take the real-world expected payoff and discount it at 3% and you get 14.2318 against a correct 9.4134, **51.2% too expensive**, because you have claimed the equity risk premium in the numerator without paying for the risk in the denominator. That is why two traders who disagree about direction still agree on the option price.
>
> **How you interpret it**, which is the third clause and the real question. $\mathbb{Q}$-probabilities are *prices*, not beliefs: a $\mathbb{Q}$-probability is the cost today of one unit paid in a state, not an estimate of that state occurring. The gap between the two is the risk premium, and it is systematically large. Read the implied distribution off index option prices and the risk-neutral probability of a large fall is far above the historical frequency of such falls, which is not a market forecast of disaster but the price of insurance against one. Anyone quoting a $\mathbb{Q}$-probability as a real-world likelihood, in a risk report or a client pitch, is quoting an insurance premium and calling it a weather forecast.
>
> If pushed for the deepest version: the risk-neutral measure is not an assumption bolted onto a model, it is what absence of arbitrage looks like written as a probability, which is the first fundamental theorem stated above.

INTERVIEW TRAP

Four errors. (1) Saying the market "assumes investors are risk-neutral"; the measure is a change of variable, not a claim about anybody's preferences. (2) Quoting a risk-neutral probability as a real-world likelihood, which reports an insurance premium as a forecast. (3) Believing a change of measure can alter volatility; it cannot, and that invariance is the whole reason the machinery is useful. (4) Treating $N(d_1)$ and $N(d_2)$ as probabilities under the same measure when they belong to different numeraires.

Where this goes next

The theorem that says exactly what the reweighting does to a Brownian motion is Girsanov's theorem. The one-period argument where the pricing measure can be seen directly is no-arbitrage pricing, and the continuous-time model it produces is the Black-Scholes model; those two own the risk-neutral-pricing questions themselves, where the binomial and Black-Scholes settings make the algebra visible. Simulating under the pricing measure rather than the real-world one is the first thing to get right in Monte Carlo methods, and the convexity adjustments that arise when a rate is paid under the wrong measure are in forward rates.

9.6 Girsanov's Theorem

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Change of measure explains why we want to reweight probabilities at all. Girsanov's theorem is the specific result that says what happens to a Brownian motion when we do, and it has one consequence that the whole of derivatives pricing rests on.

KEY IDEA

Under an equivalent change of measure from $\mathbb{P}$ to $\mathbb{Q}$ generated by a process λ_t, the object

$$\widetilde{W}_t \;=\; W_t + \int_0^t \lambda_s\, ds$$

is a Brownian motion under $\mathbb{Q}$.

In words: **changing measure changes the drift and leaves the volatility alone.** That is the entire content of the theorem, and almost everything a desk does with it follows from those nine words.

Why that one fact is worth a chapter

Under the real-world measure a stock follows

$$dS_t = \mu S_t\, dt + \sigma S_t\, dW_t.$$

Choose

$$\lambda = \frac{\mu - r}{\sigma},$$

the market price of risk defined in change of measure, and substituting $dW = d\widetilde{W} - \lambda\, dt$ gives

$$dS_t = r S_t\, dt + \sigma S_t\, d\widetilde{W}_t.$$

The expected return has been replaced by the risk-free rate and nothing else has moved. Three consequences follow, and they are the reason the theorem matters rather than the theorem itself.

Pricing does not require the real drift. μ has vanished, which is fortunate, because μ is the one parameter nobody can estimate. Two traders who disagree completely about a stock's expected return must still agree on its option prices, and that is not a coincidence, it is Girsanov.

Volatility is measure-invariant. σ appears identically under both measures. This is why you can calibrate volatility from option prices, which live under $\mathbb{Q}$, and then use the same number to hedge in the real world, which lives under $\mathbb{P}$. Without this, implied volatility would not be a meaningful quantity at all, and comparing implied with realised would be comparing objects from different universes.

The risk premium is exactly what was removed. The difference between the two worlds is $\lambda\sigma = \mu - r$, no more and no less. The risk-neutral measure is not a claim that investors are indifferent to risk; it is the real measure with the risk premium taken out and put into the discounting instead.

REAL INTERVIEW QUESTION (Barclays · EQD)

What is the real world? How is it different from the risk-neutral model?

WHAT THEY TEST / ANSWER

Answer in three moves: what each measure is for, what differs between them, and what does *not*. The third move is the one that earns the question.

What each is for. The **real-world** measure $\mathbb{P}$ is what actually happens: it is what you simulate for risk, for VaR, for a capital projection, and its drift is the expected return investors demand including a risk premium. The **risk-neutral** measure $\mathbb{Q}$ is a pricing device: under it every tradeable asset drifts at the risk-free rate, so a discounted price is a martingale and today's price is simply a discounted expectation.

What differs: only the drift. Under $\mathbb{P}$ the stock grows at μ; under $\mathbb{Q}$ at r. That is the whole difference, and Girsanov's theorem is the statement that such a change is legitimate and what it costs.

What does not differ: the volatility, and the set of possible paths. Both are essential and both get missed.

The volatility is identical under both measures, which is why an implied volatility extracted from a price is usable as a hedging parameter.

And the two measures are **equivalent**, meaning they agree on which events are impossible. You may reweight probabilities but you may not create or destroy outcomes: nothing that can happen in the real world has probability zero under $\mathbb{Q}$, and nothing impossible becomes possible. That is what makes the reweighting a change of *view* rather than a change of *world*.

Then the practical statement, which is where a desk answer lands. Use $\mathbb{Q}$ to price and $\mathbb{P}$ to manage risk, and never mix them: a VaR computed under the risk-neutral measure understates risk because it has removed the risk premium by construction, and a price computed under the real-world measure is arbitrageable. The common confusion is treating a risk-neutral probability as a forecast, when the gap between the two is the risk premium rather than an error in either number.

The density, and how far apart the two worlds are

The change of measure is implemented by a **Radon-Nikodym derivative**, the random variable that reweights each path:

$$\frac{d\mathbb{Q}}{d\mathbb{P}} = \exp\left(-\int_0^T \lambda_s \, dW_s - \tfrac{1}{2}\int_0^T \lambda_s^2 \, ds\right),$$

which for constant λ is $\exp(-\lambda W_T - \frac{1}{2}\lambda^2 T)$. It is a positive random variable with expectation one under $\mathbb{P}$, which is what makes $\mathbb{Q}$ a probability measure, and the $-\frac{1}{2}\lambda^2 T$ is the same normalisation as the $-\frac{1}{2}\sigma^2$ in geometric Brownian motion, for the same lognormal reason: it is the Itô correction that keeps the exponential a martingale rather than letting it drift.

WORKED EXAMPLE

How different are the two measures? The density has mean one by construction, so the question is entirely about its *dispersion*: how hard any individual path is being reweighted. For constant λ over a horizon T the density is lognormal and

$$\mathrm{sd}\left(\frac{d\mathbb{Q}}{d\mathbb{P}}\right) = \sqrt{e^{\lambda^2 T} - 1}.$$

At an equity-index market price of risk of 0.35 over one year that is $\sqrt{e^{0.1225} - 1} = \mathbf{0.361}$. Typical paths are therefore reweighted by tens of percent: the two measures assign materially different probabilities to the same outcomes without either being wrong.

The dispersion grows faster than λ does, which is the practical warning:

Market price of risk λ	Standard deviation of $d\mathbb{Q}/d\mathbb{P}$
0.20	0.202
0.35	0.361
0.50	0.533
1.00	1.311

At a market price of risk of one, the reweighting has a standard deviation larger than its mean, so a handful of paths carry most of the risk-neutral probability. Anyone who quotes a risk-neutral probability as though it were a forecast is ignoring a reweighting of that size.

Three uses beyond pricing a stock

Change of numéraire. The same theorem is what licenses switching the reference asset away from the money market account, and the construction and its consequences are in change of measure.

Importance sampling. In a Monte Carlo pricing of a deep out-of-the-money option almost every path expires worthless and contributes nothing but compute. Deliberately shift the drift so that paths reach the region that matters, then divide by the Radon-Nikodym derivative to undo the bias. The estimate is unchanged in expectation and its variance can fall by orders of magnitude. This is Girsanov used as an engineering technique rather than as a theory.

Reading a risk premium. Because the difference between the measures *is* the risk premium, comparing a risk-neutral quantity with its real-world counterpart measures the premium directly. Implied volatility above realised volatility is the variance risk premium; a risk-neutral default probability above the historical rate is the credit risk premium. Both are instances of the same subtraction.

INTERVIEW TRAP

1. **Reading a risk-neutral probability as a forecast.** It contains a risk premium. A CDS-implied default probability is systematically above the historical default rate and neither number is wrong.

2. **Thinking risk-neutral means investors are risk-neutral.** It is a change of variables. The premium has been moved into the discounting, not assumed away.

3. **Believing volatility changes with the measure.** It does not, and that invariance is what makes implied volatility a usable hedging input rather than a quoting convention.

4. **Computing risk under $\mathbb{Q}$.** A risk-neutral VaR has the risk premium removed by construction and therefore understates the loss distribution. Price under $\mathbb{Q}$, manage under $\mathbb{P}$.

5. **Forgetting equivalence.** You may reweight what can happen; you may not make an outcome impossible. Any construction that assigns zero probability to a real event has left the theorem's scope.

6. **Applying it where the drift is not the difference.** Girsanov changes drift. A model whose two measures differ in volatility or in jump intensity needs more than Girsanov, which is why jump models have an extra layer of machinery.

VISUAL / CODE TO BUILD: Interactive

A two-measure panel. The same set of simulated paths is drawn once, and a slider for the market price of risk reweights them: each path's line thickness or opacity is proportional to its Radon-Nikodym weight. The paths themselves never move, which is the point, and the terminal distribution drawn beside them slides from the real-world one to the risk-neutral one as the slider advances.

Watching identical paths produce two different distributions purely through reweighting is the clearest statement of what a change of measure is, and the fact that no path ever disappears is equivalence made visible.

A second panel quantifies it: a histogram of the density $d\mathbb{Q}/d\mathbb{P}$ across paths, with its mean pinned at one and its standard deviation printed, and the reader can push λ up and watch the distribution become so skewed that a small number of paths dominate the total weight. That is the visual argument for why a risk-neutral probability is not a forecast.

A third demonstrates importance sampling on a deep out-of-the-money digital: the naive estimator's standard error against the shifted one at the same path count, with the shift under the reader's control. The variance reduction is dramatic and the estimate does not move, which is Girsanov paying for itself.

Where this goes next

The reason a risk-neutral measure exists to change to at all is no-arbitrage pricing, and the drift substitution above, applied to a single lognormal stock, is exactly the step that produces Black-

Scholes.

9.7 Stochastic integration

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The object $\int_0^T \Delta_t \, dS_t$ is the profit and loss of a position that holds Δ_t units of an asset at each instant. That is all a stochastic integral is: the accumulated gains of a trading strategy. Everything technical in this section exists because the obvious way to define that sum does not work when S is driven by Brownian motion, and the repair that does work carries a piece of financial meaning that is easy to miss and worth more than the mathematics.

Why the ordinary definition fails

An ordinary integral $\int f \, dg$ is built by chopping the interval, multiplying f by the change in g on each piece, and adding up. It converges provided g does not wiggle too much, formally that g has finite total variation, meaning the sum of the absolute sizes of its increments stays bounded as the partition is refined.

Brownian motion fails that test. Over n steps of length t/n, each increment has typical size $\sqrt{t/n}$, so the sum of absolute increments is of order $n \times \sqrt{t/n} = \sqrt{nt}$, which grows without bound. **A Brownian path has infinite total variation**, so the Riemann-Stieltjes construction has nothing to converge to.

The same counting shows what survives. The sum of *squared* increments is of order $n \times (t/n) = t$, a finite number, and in fact it converges exactly to t.

> **KEY IDEA**
>
> Brownian motion has **infinite total variation but finite quadratic variation**, and that single sentence is the whole reason stochastic calculus is a separate subject. First order sums blow up, so the ordinary integral does not exist. Second order sums do not vanish, so the second order term never disappears from a Taylor expansion, which is where the extra term in Ito's lemma comes from. The integral has to be constructed in the L^2 sense, using the finiteness of the squares rather than of the absolute values.

The choice that carries the finance

In ordinary calculus it makes no difference whether you evaluate the integrand at the left end, the right end or the middle of each small interval, because the differences wash out in the limit. For Brownian motion they do not: the increment is of the same order as the square root of the interval, so where you stand matters and different choices give genuinely different answers.

Ito's construction takes the **left endpoint**: on the interval $[t_i, t_{i+1}]$ the integrand is evaluated at t_i, before the increment $W_{t_{i+1}} - W_{t_i}$ is revealed.

INTUITION

That is not a technical preference, it is the only choice a trader could implement. Holding Δ_{t_i} over the next instant means **deciding the position before seeing the move**. Evaluating at the right endpoint would mean choosing the hedge with knowledge of the change you are about to profit from, which is not a strategy, it is hindsight. The formal name for the requirement is that the integrand be *adapted* to the filtration, and its financial content is that you cannot trade on information you do not yet have.

The alternative convention, the Stratonovich integral, evaluates at the midpoint and has the pleasant property that ordinary calculus rules apply with no correction term. It is used in physics for that reason and is almost useless in finance, because a midpoint rule peeks at the future. **The awkward extra term in Ito calculus is the price of causality.**

Two consequences follow immediately and are worth stating in an interview.

- **The Ito integral has zero expectation.** Each increment is independent of the position held going into it, so each term contributes Δ_{t_i} times something with mean zero. $\int_0^T f_t\,dW_t$ is a martingale, in the sense of martingales, and this is the mathematical statement that **you cannot make money on average by trading a driftless asset**, however clever the strategy, as long as it is adapted *and the position stays bounded*. That second condition is not decoration: adaptedness alone permits a doubling strategy, which reaches any target almost surely by staking ever more after each loss, and it is ruled out by the admissibility requirement that losses be bounded below rather than by any property of the integral.
- **Its variance is computable**, by the **Ito isometry**:

$$\mathbb{E}\left[\left(\int_0^T f_t\,dW_t\right)^2\right] = \mathbb{E}\left[\int_0^T f_t^2\,dt\right].$$

  A hard object, the square of an integral against a random driver, becomes an ordinary integral of a square. This is the workhorse identity of the subject.

WORKED EXAMPLE

Take $f_t = t$ on $[0,1]$, so the strategy holds a position that grows linearly with time. The integral $\int_0^1 t\,dW_t$ has zero mean, and by the isometry its variance is

$$\int_0^1 t^2\,dt = \tfrac{1}{3},$$

so the outcome is $N(0,1/3)$, with standard deviation $1/\sqrt{3} = 0.577$.

Compare with holding one unit throughout: $\int_0^1 1\,dW_t = W_1 \sim N(0,1)$, standard deviation 1. The linearly increasing position ends up with less risk because it was small for most of the period, and the isometry quantifies exactly how much less without simulating anything.

REAL INTERVIEW QUESTION (BNP · Fixed Income)

Compute $\int dX$, and then $\int X\,dt$, where X_t is lognormal.

WHAT THEY TEST / ANSWER

The question is checking that you notice these are two different kinds of integral. Say that first, then do both.

Set up. Lognormal X_t means $dX_t = \mu X_t\,dt + \sigma X_t\,dW_t$, so $\log X_t$ is normal and X_t is positive.

The first integral is trivial and you should say so quickly.

$$\int_0^T dX_t = X_T - X_0.$$

Integrating a differential telescopes, exactly as in ordinary calculus. No Ito correction appears because nothing is being transformed; the correction shows up when you integrate a *function* of X, which is Ito's lemma.

The second is the real question. $\int_0^T X_t\,dt$ is a dt integral, not a dW integral: it is an ordinary Riemann integral computed path by path, and it is simply the area under the path. It is random because the path is, but it involves no stochastic integration at all. Its expectation is easy, since $\mathbb{E}[X_t] = X_0 e^{\mu t}$:

$$\mathbb{E}\left[\int_0^T X_t\,dt\right] = \int_0^T X_0 e^{\mu t}\,dt = X_0\,\frac{e^{\mu T}-1}{\mu},$$

which tends to $X_0 T$ as $\mu \to 0$, as it must.

Then the point that makes it a markets answer. The expectation is easy and the *distribution* is not: a sum of lognormals is not lognormal, and $\int_0^T X_t\,dt$ has no closed form. That is not a curiosity, it is why **arithmetic-average Asian options have no Black-Scholes formula** and are priced by moment matching or by Monte Carlo, while geometric-average Asians, whose average is an integral of $\log X$ and therefore normal, do have one.

If you want to close strongly, contrast the two integrals directly: $\int X_t\,dt$ has non-zero expectation and is a smooth average, whereas $\int X_t\,dW_t$ has zero expectation and is a martingale. Mixing them up is the error the question is designed to catch.

WORKED EXAMPLE

Put numbers on it. With $X_0 = 100$, $\mu = 5\%$ and $T = 2$:

$$\mathbb{E}\left[\int_0^2 X_t\,dt\right] = 100\,\frac{e^{0.10}-1}{0.05} = 210.34,$$

so the expected *average* level over the two years is $210.34/2 = 105.17$.

Note it is not $X_0 e^{\mu T/2} = 100\,e^{0.05} = 105.13$, the value at the midpoint. The two differ because the exponential is convex, so averaging over the path picks up slightly more than the midpoint value. Four hundredths of a point on this example, but the same convexity is what separates an average-rate payoff from a fixing at the midpoint, and it is not zero.

The integral you actually trade

Write the value of a self-financing portfolio holding Δ_t units of an asset:

$$V_T = V_0 + \int_0^T \Delta_t \, dS_t.$$

Self-financing means no money is added or removed, so every change in value comes from the market moving against the position held. That equation is the formal statement of replication: if a strategy Δ_t exists whose integral reproduces an option's payoff, the option's price must be V_0, or there is an arbitrage.

KEY IDEA

Read as a piece of financial writing, $V_T = V_0 + \int_0^T \Delta_t \, dS_t$ says that **a derivative's price is the cost of the trading strategy that manufactures it**. The theory is not asserting what an option is worth; it is asserting what it costs to build, and those are the same number only because a mismatch would be arbitraged. Every practical qualification a trader makes, that hedging is discrete rather than continuous, that there is bid-offer, that the volatility used to compute Δ_t is not the volatility that realises, is a statement about the gap between this integral and its real-world approximation.

The Gaussian integral $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$, and the polar-coordinates trick that evaluates it, belong to limits, continuity, derivatives and integrals, which owns definite-integral computation. It is worth knowing here only for where it lands: it is the normalisation that puts the $\sqrt{2\pi}$ into the Gaussian density, and hence into every Black-Scholes formula. The integral in this section is a different object. Where a Riemann integral sums a function against dt, the Ito integral sums against dW_t, and W itself has paths of unbounded variation, so the integral is defined as an L^2 limit of the integrals of simple predictable integrands rather than pathwise.

INTERVIEW TRAP

Three errors. (1) Saying Brownian paths are "too rough to integrate". They are too rough for the ordinary construction; the correct statement is infinite total variation but finite quadratic variation, and the second half is what makes the theory possible. (2) Treating the left-endpoint rule as a convention. It is the adaptedness requirement, and giving it up means assuming knowledge of the next move. (3) Confusing $\int X \, dt$ with $\int X \, dW$. The first has non-zero expectation and needs no stochastic calculus; the second is a martingale.

Where this goes next

The driver being integrated against is Brownian motion, the zero-expectation property is martingales, and the transformation rule that the finite quadratic variation forces is Ito's lemma. Writing dynamics in this language is stochastic differential equations, approximating these integrals numerically is Monte Carlo methods, the financial reading of the integral is replication strategies, and the product that trades quadratic variation directly is variance swaps.

9.8 Itô's lemma

trading · structuring | core | equity · cross-asset

This is the single most useful formula in quantitative finance, and it is asked in interviews more often than any other piece of stochastic calculus. Itô's lemma is the chain rule for random processes. It answers a question every desk faces: if the stock follows a random process, what process does an *option on that stock* follow? Without it there is no pricing equation, no hedging argument, and no explanation of why a trader gets paid for convexity. With it, all three fall out in half a page.

Why the ordinary chain rule breaks

We assume you have met Brownian motion W_t and the Itô integral. The one-line recall: W_t has independent increments, $W_{t+\Delta} - W_t \sim \mathcal{N}(0, \Delta)$, and its paths are continuous but nowhere differentiable.

That last property is the whole story. In ordinary calculus, when you expand $f(x + \Delta x)$ you keep the first-order term and throw away $(\Delta x)^2$ because it is negligible. For a smooth path, where $\Delta x \sim \Delta t$, the square is of order $(\Delta t)^2$ and genuinely vanishes. For a Brownian path it does not: the increment scales like $\sqrt{\Delta t}$, so its square scales like Δt, which is exactly the order of the term you are keeping. The second-order term survives, and Itô's lemma is the bookkeeping that keeps it.

> **INTUITION**
>
> Picture a car on a smooth road versus a car on a violently corrugated one. Over a short stretch, the smooth road's wiggle is negligible compared with the distance travelled, so you can pretend the road is a straight line: that is the ordinary chain rule. The Brownian road is so jagged that over any stretch, however short, the up-and-down wiggle is comparable in size to the forward progress. You cannot ignore it, and if your car's suspension responds non-linearly to bumps (that is, if f is curved), the wiggle produces a systematic effect that does not average out. The extra term in Itô's lemma is that systematic effect, and on a trading desk it has a name: Gamma PnL.

The multiplication table

Everything follows from one fact, the **quadratic variation** of Brownian motion, which is established in Brownian motion. The calculation is worth one repetition here, because the lemma is unusable without it. Chop $[0, T]$ into n equal pieces and add the squared increments. Each $(\Delta W_i)^2$ has mean $\Delta t = T/n$ and variance $2(\Delta t)^2$, so the sum has mean

$$n \cdot \frac{T}{n} = T \qquad \text{and variance} \qquad n \cdot 2\left(\frac{T}{n}\right)^2 = \frac{2T^2}{n} \xrightarrow[n\to\infty]{} 0.$$

The sum of squared increments converges to T with *no randomness left*. This is why $\sum(\Delta W)^2$ behaves like an ordinary time integral, and it is summarised by the rules every quant memorises:

$$(dW_t)^2 = dt, \qquad dW_t\, dt = 0, \qquad (dt)^2 = 0.$$

Contrast this with $\sum(\Delta t)^2 \to 0$: time has zero quadratic variation, a Brownian path has quadratic variation T. That single difference is the entire gap between ordinary and stochastic calculus.

The lemma

Let X_t solve the stochastic differential equation

$$dX_t = \mu(t, X_t)\, dt + \sigma(t, X_t)\, dW_t,$$

and let $f(t,x)$ be twice continuously differentiable. Then

KEY IDEA

Itô's lemma.

$$df(t, X_t) = \underbrace{\left(\frac{\partial f}{\partial t} + \mu\frac{\partial f}{\partial x} + \tfrac{1}{2}\sigma^2\frac{\partial^2 f}{\partial x^2}\right)dt}_{\text{drift}} + \underbrace{\sigma\frac{\partial f}{\partial x}\,dW_t}_{\text{diffusion}}.$$

The ordinary chain rule would give every term except $\frac{1}{2}\sigma^2\partial_{xx}f$. That term is the Itô correction, and it is non-zero whenever f is curved and X is random.

Where it comes from. Expand f to second order in Δt and ΔX, which is the Taylor expansion:

$$\Delta f \approx \partial_t f\,\Delta t + \partial_x f\,\Delta X + \tfrac{1}{2}\partial_{xx} f\,(\Delta X)^2 + \tfrac{1}{2}\partial_{tt} f\,(\Delta t)^2 + \partial_{tx} f\,\Delta t\,\Delta X.$$

Substitute $\Delta X = \mu\Delta t + \sigma\Delta W$ and apply the multiplication table. The last two terms die, and $(\Delta X)^2 = \mu^2(dt)^2 + 2\mu\sigma\,dt\,dW + \sigma^2(dW)^2$ collapses to $\sigma^2\,dt$. Collecting what survives gives the lemma. This is a heuristic argument, not a proof, but it is exactly the argument to give in an interview because it shows you know *which* term survives and why.

Worked example: geometric Brownian motion

WORKED EXAMPLE

The standard equity model is

$$dS_t = \mu S_t\,dt + \sigma S_t\,dW_t,$$

which says returns, not prices, are what behave additively. To solve it, apply Itô to $f(x) = \ln x$. Here $\partial_t f = 0$, $\partial_x f = 1/S$ and $\partial_{xx} f = -1/S^2$, so

$$d(\ln S_t) = \left(\mu S\cdot\frac{1}{S} + \tfrac{1}{2}\sigma^2 S^2\cdot\left(-\frac{1}{S^2}\right)\right)dt + \sigma S\cdot\frac{1}{S}\,dW_t = \left(\mu - \tfrac{\sigma^2}{2}\right)dt + \sigma\,dW_t.$$

Notice what the lemma has bought us. The left-hand side now has constant coefficients, so an equation that was multiplicative in S has become additive in $\ln S$ and can be integrated directly. That is the step this section exists to teach. The resulting solution, the lognormal distribution it implies, and the separation between the mean $S_0e^{\mu T}$ and the median $S_0e^{(\mu-\sigma^2/2)T}$ that the $-\sigma^2/2$ term produces, are developed in stochastic differential equations, and what that term means for a compounding investor is in Brownian motion.

REAL INTERVIEW QUESTION (BNP · Fixed Income)

Itô's lemma.

WHAT THEY TEST / ANSWER

Deliberately bare, and what they are timing is how fast you get to the correction term. Write $dX_t = \mu\, dt + \sigma\, dW_t$, state the lemma, and say the one sentence that matters: the extra $\frac{1}{2}\sigma^2 \partial_{xx} f\, dt$ appears because $(dW_t)^2 = dt$ rather than being negligible, which itself follows from Brownian motion having non-zero quadratic variation. Then, without being asked, apply it to $f = \ln S$ under geometric Brownian motion and produce $d(\ln S) = (\mu - \sigma^2/2)dt + \sigma dW$. That is the standard follow-up, so pre-empting it buys you the rest of the interview. If they push further, give the trading translation: the second-derivative term is Gamma, so Itô is the formal statement that a convex position earns money from movement regardless of direction.

Worked example: an integral ordinary calculus gets wrong

WORKED EXAMPLE

Compute $\int_0^T W_t\, dW_t$. Ordinary calculus would answer $\frac{1}{2}W_T^2$. Apply Itô to $f(x) = x^2$ with $X_t = W_t$ (so $\mu = 0$, $\sigma = 1$): $\partial_x f = 2x$, $\partial_{xx} f = 2$, hence

$$d(W_t^2) = 2W_t\, dW_t + dt.$$

Integrating from 0 to T and using $W_0 = 0$ gives $W_T^2 = 2\int_0^T W_t dW_t + T$, so

$$\int_0^T W_t\, dW_t = \tfrac{1}{2}W_T^2 - \tfrac{1}{2}T.$$

Ordinary calculus was too big by $T/2$. Check it against a property you already know: the Itô integral of an adapted, square-integrable process has zero mean, and indeed $\mathbb{E}[\frac{1}{2}W_T^2 - \frac{1}{2}T] = \frac{1}{2}T - \frac{1}{2}T = 0$, since $\mathbb{E}[W_T^2] = T$. The naive answer $\frac{1}{2}W_T^2$ would have had mean $T/2 \neq 0$, so it cannot be a stochastic integral. That consistency check is worth knowing: it catches sign and factor errors in seconds.

REAL INTERVIEW QUESTION (BNP · Fixed Income)

Compute the integral of dX, then the integral of $X\, dt$, where X_t is lognormally distributed.

WHAT THEY TEST / ANSWER

They want to see that you distinguish the three kinds of object: an integral against dX, an integral against dW, and an ordinary integral against dt. Set $dX_t = \mu X_t dt + \sigma X_t dW_t$, so X is lognormal. The first is a telescoping sum and needs no cleverness: $\int_0^T dX_t = X_T - X_0$. The second is not a stochastic integral at all, it is a pathwise Riemann integral of a random function, so what you can give is its expectation, and here Fubini is the tool: $\mathbb{E}\big[\int_0^T X_t dt\big] = \int_0^T \mathbb{E}[X_t]dt = \int_0^T X_0 e^{\mu t} dt = X_0(e^{\mu T} - 1)/\mu$. Say explicitly that $\int X\, dt$ has no closed form pathwise, which is precisely why Asian options are priced numerically. If they then ask for $\int X\, dW$, the answer is that it has mean zero and you would characterise it through its variance by the Itô isometry.

From Itô to a pricing equation

This is what the lemma was built for, and it is the most frequently asked derivation in derivatives interviews. The full model, its assumptions, its closed-form solution and its limits belong to the Black-Scholes section; what follows is only the Itô step, kept deliberately short.

Let $V(t, S)$ be the value of a derivative on S, with $dS = \mu S dt + \sigma S dW$. Itô gives

$$dV = \Big(\partial_t V + \mu S\, \partial_S V + \tfrac{1}{2}\sigma^2 S^2 \partial_{SS} V\Big) dt + \sigma S\, \partial_S V\, dW.$$

That is the Itô step, and it is where this section stops. Everything that follows, building the hedged portfolio so the dW terms cancel, imposing no-arbitrage on the resulting riskless position, and rearranging into the Black-Scholes partial differential equation, is carried out in the Black-Scholes section.

What is worth saying here, because it is Itô's payoff rather than the model's, is how a trader reads the second-order term. $\partial_{SS} V$ is Gamma and $\partial_t V$ is Theta, and the $\frac{1}{2}\sigma^2 S^2 \partial_{SS} V$ that Itô produced is precisely the money a convex position earns from movement, regardless of direction. Gamma and Theta are therefore two sides of one trade, which is developed in Gamma and convexity and Theta as the cost of convexity.

REAL INTERVIEW QUESTION (Société Générale · EQD)

How do you obtain the PDE for the European call in Black-Scholes?

WHAT THEY TEST / ANSWER

The derivation is a short sequence of steps and the grading is about structure, so walk through it in order rather than jumping to the answer. The step to nail, and the only one that belongs to this section, is applying Itô to $V(t, S)$ to obtain dV: be explicit that the $\frac{1}{2}\sigma^2 S^2 \partial_{SS} V$ term is there because $(dW)^2 = dt$ rather than being negligible, since that is the single line that distinguishes a candidate who understands the derivation from one who has memorised its conclusion. The remaining steps, forming the hedged portfolio, imposing no-arbitrage, and rearranging to the PDE, together with the punchline that μ drops out, are set out in the Black-Scholes section.

More than one source of risk

With n correlated drivers, $dW^{(i)} dW^{(j)} = \rho_{ij}\, dt$, and the lemma picks up every cross term:

$$df = \partial_t f\, dt + \sum_i \partial_{x_i} f\, dX^{(i)} + \tfrac{1}{2} \sum_{i,j} \rho_{ij}\, \sigma_i \sigma_j\, \partial_{x_i x_j} f\, dt.$$

Those cross terms are where multi-asset pricing lives: the cross-Gamma of a basket, the quanto correction when a payoff in one currency references an underlying in another, and the spot-vol cross term in a stochastic volatility model. See equity correlation, stochastic volatility models and first and second-order Greeks.

INTERVIEW TRAP

Five recurring errors. First, applying the ordinary chain rule and losing the correction term; if your answer has no $\frac{1}{2}\sigma^2$ anywhere, it is wrong. Second, dropping the $\frac{1}{2}$. Third, applying the correction where there is nothing to correct: if f is linear in x then $\partial_{xx} f = 0$, the Itô term vanishes, and stochastic and ordinary calculus agree, which is why a forward needs no Itô argument and an option does. Fourth, confusing the real-world drift μ with the risk-free rate r; they coincide only after the change of measure covered in change of measure, not before. Fifth, applying Itô to a function that is not twice differentiable, such as the payoff $\max(S-K,0)$ at the strike, where the kink means you need a more general result rather than the naive formula.

VISUAL / CODE TO BUILD: Interactive

A convergence sandbox for the one fact the lemma rests on. The reader simulates a Brownian path and drags a slider for the number of subdivisions n. Two traces plot against t: the running sum of squared increments $\sum(\Delta W)^2$, and the running sum of squared time steps $\sum(\Delta t)^2$. As n grows, the first settles onto the straight line $y = t$ with the visible noise shrinking like $\sqrt{2/n}$, while the second collapses to zero. A readout gives the empirical variance of the sum against the theoretical $2T^2/n$. A second panel makes the consequence concrete: the reader picks a function f from a short list (linear, x^2, $\ln x$, a call payoff) and the panel plots the realised change in $f(X_t)$ against what the ordinary chain rule predicts, so the gap is exactly the accumulated Itô term and is visibly zero for the linear choice. The intended "aha" is that the correction is not a mathematical formality but a measurable quantity that grows with curvature.

Where this goes next

Itô's lemma is the engine; the rest of the chapter is what you drive with it. Solving the resulting equations is stochastic differential equations; turning the pricing PDE into something a computer solves is PDEs in finance and Euler and Milstein schemes; simulating the solution instead is Monte Carlo methods. The step that replaces μ with r is change of measure. And the statistical objects the lemma keeps producing, means, medians and higher moments of a lognormal, are in expectation, variance and higher moments. The quadratic variation that started the whole thing is itself tradable, as variance swaps.

9.9 Stochastic differential equations

trading · structuring · sales | advanced | cross-asset

Every pricing model on a desk is a stochastic differential equation plus a choice of what to do with it. So the practical skill is not deriving them, it is recognising them: knowing which SDE a market uses, what its solution looks like, which of its properties are features and which are bugs, and what happens when you simulate it. This section is that catalogue, built up from what an SDE actually says.

What an SDE says

An ordinary differential equation such as $dX_t = a\,dt$ says the state changes at a known rate. A **stochastic** differential equation adds a random term driven by Brownian motion:

$$dX_t \;=\; a(X_t,t)\,dt \;+\; b(X_t,t)\,dW_t.$$

The first term is the **drift**, the expected direction of travel per unit time. The second is the **diffusion**, the size of the random buffeting, and dW_t is a Brownian increment with mean zero and variance dt. That variance-proportional-to-time is the whole reason this is a different subject from calculus: over a short interval the random part is of size $\sqrt{dt}$, which is *larger* than dt, so second-order terms cannot be discarded. That is what Ito's lemma exists to handle.

> INTUITION
>
> Read an SDE as a **recipe for the next instant**: take where you are, add a nudge in the drift direction proportional to the time elapsed, then add a random kick whose size is set by the diffusion. Two things distinguish one model from another. Whether the drift *pulls back* towards a level, which gives mean reversion, and whether the diffusion *scales with the state*, which decides whether the process can go negative and whether percentage or absolute moves are the natural unit.

Strictly, the differential notation is shorthand: what it means is the integral equation $X_t = X_0 + \int_0^t a\,ds + \int_0^t b\,dW_s$, and the second integral is the object constructed in stochastic integration. A **strong** solution is a process satisfying that equation path by path for the given Brownian motion; a **weak** solution only reproduces the right distribution, which is all that pricing needs.

The zoo, with solutions

1. Arithmetic Brownian motion. $dX_t = \mu\,dt + \sigma\,dW_t$, solved by simply integrating: $X_t = X_0 + \mu t + \sigma W_t$. Gaussian at every horizon, absolute moves of constant size, and it can go negative. Natural for spreads, for basis, and as the base for Gaussian rate models.

2. Geometric Brownian motion, the workhorse. $dS_t = \mu S_t\,dt + \sigma S_t\,dW_t$, that is, *proportional* drift and diffusion, so the natural unit is a percentage. Its solution is

$$S_t \;=\; S_0 \exp\Big(\big(\mu - \tfrac{1}{2}\sigma^2\big)t + \sigma W_t\Big),$$

and the $-\frac{1}{2}\sigma^2$ is not a typo, it is the Ito correction: the log of the process drifts more slowly than the process itself. Consequences: S is lognormal, it stays strictly positive, and

$$\mathbb{E}[S_t] = S_0 e^{\mu t}, \qquad \text{median}(S_t) = S_0 e^{(\mu - \sigma^2/2)t}.$$

This is the model behind Black-Scholes.

> WORKED EXAMPLE
>
> **Mean against median under GBM.** Take $S_0 = 100$, $\mu = 8\%$, $\sigma = 20\%$, one year. The expected price is $100\,e^{0.08} = 108.33$. The median price is $100\,e^{0.08-0.02} = 100\,e^{0.06} = 106.18$, because $\frac{1}{2}\sigma^2 = \frac{1}{2}(0.04) = 0.02$. So the average outcome is €2.15 above the typical one, and the gap is entirely the variance drain $\sigma^2/2$, which grows with volatility and not at all with the drift. It is the same arithmetic-versus-geometric wedge met in martingales, and it is why a high-volatility asset can have an attractive expected return and an unattractive median outcome.

3. Ornstein-Uhlenbeck, or Vasicek in rates. $dX_t = \kappa(\theta - X_t)\,dt + \sigma\,dW_t$. The drift now *pulls back*: when X is above θ the drift is negative. Two numbers characterise it, and both are worth being able to state. The **half-life** of a shock is $\ln 2/\kappa$, and the **stationary standard deviation** is $\sigma/\sqrt{2\kappa}$, so faster mean reversion means both quicker decay and a tighter long-run distribution. The process is Gaussian and can go negative, which was a fatal objection to Vasicek for decades and became a feature after 2014.

WORKED EXAMPLE

Reading a mean-reverting fit. Suppose a spread is fitted with $\kappa = 2$ per year and $\sigma = 2\%$. The half-life is $\ln 2/2 = 0.347$ years, so a dislocation is expected to decay by half in about four months. The stationary standard deviation is $0.02/\sqrt{4} = 1\%$, so the spread should spend most of its time within a couple of points of θ. Those two numbers are what turn a fitted parameter set into a trade: they tell you the horizon you must hold for and the size of move you should expect to tolerate on the way.

4. Cox-Ingersoll-Ross. $dX_t = \kappa(\theta - X_t)\,dt + \sigma\sqrt{X_t}\,dW_t$. The $\sqrt{X}$ makes the diffusion vanish as the state approaches zero, so the process stays non-negative, and strictly positive when the **Feller condition** $2\kappa\theta \geq \sigma^2$ holds. Used for short rates and for the variance process in Heston.

5. Coupled systems. Stochastic volatility models are two SDEs, one for the asset and one for its variance, correlated through their Brownian drivers, which is what generates a smile. See stochastic volatility models.

6. Jumps. Adding a compensated Poisson term produces discontinuous paths, which is what a diffusion cannot do and what real markets do regularly, see jump-diffusion models.

KEY IDEA

Choose the SDE from the **behaviour** you must reproduce, not from familiarity. Does the quantity have to stay positive? Does it mean-revert, and on what horizon? Are the natural moves absolute or proportional? Does it gap? Does its volatility need its own randomness? Every model in the list above is the minimal answer to one of those questions, and most modelling errors on a desk are a mismatch between the question and the answer, such as putting a lognormal process on a quantity that can be negative.

Simulating them

The general recipe is **Euler-Maruyama**: advance by the drift times Δt and add the diffusion times $\sqrt{\Delta t}\,Z$ with Z a standard normal draw. That $\sqrt{\Delta t}$ is the only thing distinguishing it from the deterministic Euler scheme, and it is exactly the variance-proportional-to-time property this section opened with.

Two traps are worth knowing before writing any code, because both follow from the solutions above rather than from numerical analysis. A GBM should be simulated *exactly* by stepping its log, since its solution is closed form, rather than approximately by Euler on S. And a naive Euler step on CIR can jump below zero where the square root is undefined, so a process that is positive in theory is not automatically positive in code. The schemes themselves, their convergence properties and the treatment of those boundaries belong to numerical schemes.

REAL INTERVIEW QUESTION (HSBC · Rates)

What are the pros and cons of Black-Scholes, Monte Carlo and geometric Brownian motion for pricing a simple cap?

WHAT THEY TEST / ANSWER

Separate the three, because the question deliberately mixes a model, a numerical method and a process. **Black's formula**, which is the Black-Scholes framework applied to a forward rate, is the market standard for a cap: a cap is a strip of caplets, each is priced as an option on its own forward rate assumed lognormal, and the strip is summed. The pros are that it is closed form, instant, and it is the convention in which the market actually quotes volatility, so calibration is direct. The cons are real: each caplet is priced under its own measure with its own volatility, so the approach does not give you an internally consistent dynamic model of the whole curve, it cannot price anything path-dependent or callable, and a lognormal rate assumption breaks outright at zero or negative rates, which is why the market moved to shifted-lognormal or normal quoting conventions.

Geometric Brownian motion on the rate is the wrong process for rates as such: it has no mean reversion, so it lets rates diffuse without bound, and it says absolute volatility scales with the level, which the data does not support across regimes and which fails completely near zero. For a single caplet, the lognormal forward assumption is a market convention rather than a dynamic claim, and that distinction is the sophisticated point to make.

Monte Carlo is unnecessary for a plain cap, since a closed form exists and simulation would only add noise and time, but it is the right tool the moment the product becomes path-dependent, callable, or written on several rates at once, and it is the natural engine for a term-structure model.

What you would actually use: Black or a shifted variant for quoting and hedging a vanilla cap, and a proper mean-reverting term-structure model such as Hull-White, or a market model, calibrated to the cap surface, whenever consistency across the curve or optionality on the exercise decision matters.

REAL INTERVIEW QUESTION (Société Générale · Credit)

A study of the Hull-White model: long rates, short rates, how do the different economic aspects appear in the model?

WHAT THEY TEST / ANSWER

Write the SDE and then read each parameter as an economic statement, which is exactly what the question asks for. Hull-White is

$$dr_t \;=\; \big(\theta(t) - a\,r_t\big)\,dt \;+\; \sigma\,dW_t,$$

an Ornstein-Uhlenbeck process for the short rate with a time-dependent drift level.

Reading the parameters. $\theta(t)$ **is calibration, not economics**: it is chosen so that the model reproduces *today's* entire yield curve exactly, which is what makes the model arbitrage-free with respect to observed bond prices, and it is the whole reason Hull-White is used in practice rather than plain Vasicek. a **is mean reversion**, and it does two economic jobs at once: it sets the half-life of a rate shock at $\ln 2/a$, and it controls how much short-rate volatility survives out along the curve. σ **is the level of rate volatility**, and being additive rather than proportional it means the model is Gaussian and rates can go negative, which was considered a defect before 2014 and a necessity afterwards.

The long-versus-short question has a clean answer in this model, and giving it is what distinguishes the response. The volatility of the τ-maturity zero-coupon yield is the short-rate volatility scaled by

$$\frac{1 - e^{-a\tau}}{a\tau},$$

which decreases in τ, so **long rates are less volatile than short rates, and mean reversion is the reason**. With $a = 0.05$, the factor is 0.79 at ten years and 0.52 at thirty, so the model predicts the thirty-year yield moves roughly half as much as the very short rate for a given shock, and the half-life of a shock is $\ln 2/0.05 \approx 14$ years. Then the honest limitations: one factor means all rates move together, so the model cannot produce a curve twist and cannot price a steepener option sensibly, which is what motivates two-factor and market models; and the Gaussian form gives tractable analytics, including closed-form bond options, at the cost of realistic tails.

INTERVIEW TRAP

Four traps. First, **forgetting the $-\frac{1}{2}\sigma^2$** in the GBM solution, which is the single most common slip in this material and turns a median into a mean. Second, **putting a lognormal process on something that can be negative**, such as a rate or a spread, which was the industry's actual problem in 2014. Third, **using Euler on a process whose solution you already know**: simulate the log of a GBM exactly instead of introducing bias for nothing. Fourth, **quoting mean reversion without a horizon**: κ on its own is not interpretable, so convert it to a half-life of $\ln 2/\kappa$ and check whether that horizon is one you could actually hold.

VISUAL / CODE TO BUILD: Interactive

An SDE playground with one panel per process and a shared control strip. The reader picks a model (arithmetic Brownian motion, GBM, Ornstein-Uhlenbeck, CIR), sets its parameters and sees simulated paths together with the analytic mean, median and stationary band drawn over them, so for GBM the gap between mean and median widens visibly as volatility rises, and for OU the half-life appears as the time a displaced path takes to close half the distance to θ. A second tab is a discretisation laboratory: the same GBM simulated by Euler on S and exactly on $\log S$, with the bias plotted against step size so the reader sees one scheme converge and the other stay wrong, plus a CIR path with the step size raised until Euler pushes it below zero. A third tab overlays the Hull-White yield-volatility factor $(1 - e^{-a\tau})/(a\tau)$ against maturity for several values of a, connecting one parameter to the shape of the volatility term structure.

Where this goes next

The integral that gives these equations meaning is stochastic integration, and the rule for transforming them is Ito's lemma, with the fair-game property of the driver in martingales. The equations themselves reappear as Black-Scholes, stochastic volatility models and jump-diffusion models, and in rates as the term-structure machinery behind yield curves. Discretising them for simulation is numerical schemes, with the wider simulation framework in Monte Carlo methods, and implementing and testing them is Python for finance and pricing libraries.

9.10 Partial Differential Equations in Finance

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There are two ways to price a derivative and they give the same answer. One is to take a discounted expectation under the risk-neutral measure, which leads to simulation. The other is to write down a partial differential equation that the price must satisfy and solve it backwards from expiry, which leads to a grid. This section is the second route, and the theorem that connects them.

INTUITION

Before any algebra, here is what the equation says, because it is a picture rather than a formula.

An option's value **diffuses** from its payoff. Stand at expiry, where the value is the payoff exactly, a kinked or stepped line with corners in it. Now step backwards in time. At each step the value spreads sideways and smooths, the way heat spreads along a bar, because uncertainty about where the stock will end up blurs a sharp payoff into a rounded one.

Everything below is that sentence made precise. The equation is the heat equation in disguise; solving it is stepping backwards from the payoff; and the reason gamma is largest at the strike is that the strike is where the payoff is most kinked, so it is where the smoothing has the most work to do.

Deriving the equation

REAL INTERVIEW QUESTION (Société Générale · EQD)

How do you obtain the PDE for a European call in Black-Scholes?

WHAT THEY TEST / ANSWER

This is a derivation question, so give the four steps in order and name what each one does. Most candidates can produce the algebra; far fewer can say what the hedging step accomplishes.

Step one, write the dynamics. $dS = \mu S\,dt + \sigma S\,dW$.

Step two, apply Ito to the option price $V(S,t)$:

$$dV = \left(\frac{\partial V}{\partial t} + \mu S\frac{\partial V}{\partial S} + \tfrac{1}{2}\sigma^2 S^2\frac{\partial^2 V}{\partial S^2}\right)dt + \sigma S\frac{\partial V}{\partial S}\,dW.$$

The second-order term is there because $(dW)^2 = dt$, which is the whole difference from ordinary calculus.

Step three, build a hedged portfolio and choose the hedge to kill the randomness. Take $\Pi = V - \Delta S$. Then

$$d\Pi = (\cdots)\,dt + \sigma S\left(\frac{\partial V}{\partial S} - \Delta\right)dW,$$

so setting $\Delta = \partial V/\partial S$ removes the dW term entirely. The portfolio is now riskless over the next instant.

Step four, invoke no arbitrage. A riskless portfolio must earn the risk-free rate, so $d\Pi = r\Pi\,dt$. Substituting and rearranging gives

$$\frac{\partial V}{\partial t} + \tfrac{1}{2}\sigma^2 S^2\frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV \;=\; 0.$$

This is the Black-Scholes equation. Its closed-form solution for a European call, and the assumptions that have to hold for any of it to be true, are the Black-Scholes model; what follows here is the other thing you can do with the equation, which is solve it numerically for products that have no closed form.

Then say the thing the derivation is for, because this is where the marks are. μ **has disappeared.** It was eliminated at step three by the choice of hedge, not by an assumption about investor preferences. So the option price does not depend on the expected return of the underlying, and two people who disagree about where the stock is going must still agree on the option's value. That is the same conclusion Girsanov reaches by changing measure, and noting that the two arguments meet is worth a sentence: one removes the drift by hedging, the other by reweighting, and they agree because they are the same statement about replication.

Finish with the terminal condition, since the PDE alone does not identify a call: $V(S,T) = \max(S-K,0)$, plus $V(0,t) = 0$ and $V \to S - Ke^{-r(T-t)}$ as $S \to \infty$. The equation is the physics; the conditions are the product.

KEY IDEA

That last point generalises and is the most useful structural fact in the section. **A call, a put, a digital, a barrier and an autocallable all satisfy the same PDE.** They differ only in the terminal condition and the boundary conditions.

A knock-out barrier is $V = 0$ imposed along the barrier. A digital is a step function at expiry. An American option adds the inequality $V \geq$ intrinsic at every node. So a PDE solver is a general pricing engine into which products are plugged as boundary data, which is exactly the instrument-model-method separation that pricing libraries is built on.

Feynman-Kac: why the two routes agree

The bridge between the PDE and the expectation is the **Feynman-Kac** theorem, which says that the solution of a parabolic PDE of this form can be written as

$$V(S,t) \;=\; e^{-r(T-t)}\,\mathbb{E}^{\mathbb{Q}}[\,\text{payoff}(S_T) \mid S_t = S\,]\,.$$

Read it in both directions, because each is useful. Left to right: if you have a PDE you may solve it by simulating paths, which is what makes Monte Carlo legitimate rather than a heuristic. Right to left: if you have an expectation you may instead solve a deterministic equation on a grid, which is faster and gives you the Greeks for free.

That is the theoretical content of the choice between engines. The practical content, which dimension and which features push you to one or the other, is in pricing libraries.

It is the heat equation in disguise

Substituting $x = \ln S$ and $\tau = \frac{1}{2}\sigma^2(T-t)$, and absorbing the remaining first-order and zeroth-order terms by a further exponential rescaling, turns the Black-Scholes equation into

$$\frac{\partial u}{\partial \tau} = \frac{\partial^2 u}{\partial x^2},$$

the heat equation, run forward in τ and therefore backward in calendar time.

Two things follow. Every result in a century of numerical analysis of the heat equation applies directly to option pricing, which is why the schemes below are not invented for finance. And it is what licenses the diffusion picture this section opened with: the value curve really is a smoothed version of the payoff, and gamma really is largest where the payoff is most kinked, because those are statements about the solution of a heat equation rather than analogies.

The three schemes, and the one that bites

- **Explicit.** Each new value is computed directly from the previous time level. Trivial to implement and *conditionally* stable: the step sizes must satisfy roughly $\Delta\tau \leq \frac{1}{2}(\Delta x)^2$.
- **Implicit.** Each time step requires solving a linear system, which for the tridiagonal matrix that arises is cheap. *Unconditionally* stable, so the time step is chosen for accuracy rather than for survival.
- **Crank-Nicolson.** The average of the two, second-order accurate in time rather than first, and the default choice. Its defect is specific and important: it is stable but not *monotone*, so a discontinuous payoff produces spurious oscillations near the discontinuity. The standard repair is Rannacher smoothing, a few fully implicit steps at the start before switching over.

WORKED EXAMPLE

What conditional stability costs. The explicit scheme's constraint $\Delta\tau \leq \frac{1}{2}(\Delta x)^2$ ties the time step to the *square* of the space step:

Space step Δx	**Largest stable** $\Delta\tau$	**Time steps over** $\tau = 1$
0.020	2.0×10^{-4}	5,000
0.010	5.0×10^{-5}	20,000
0.005	1.25×10^{-5}	80,000

Halving the space step *quadruples* the number of time steps, so the total work rises by a factor of eight for a factor of four in spatial accuracy. That is why nobody uses the explicit scheme in production despite its simplicity, and why the extra effort of a linear solve at each step is worth it: the implicit and Crank-Nicolson schemes have no such constraint and let you choose $\Delta\tau$ for accuracy alone.

Note also what violating the condition does. It does not merely degrade accuracy; the solution oscillates with exponentially growing amplitude and produces obvious nonsense, which is at least a loud failure rather than a quiet one.

Beyond one factor

REAL INTERVIEW QUESTION (Barclays · Exo)

Pricing a swap and a swaption by PDE, in Hull-White.

WHAT THEY TEST / ANSWER

Take the two instruments separately, because one of them does not need a PDE at all, and noticing that is the first mark.

The swap does not need one. A vanilla swap is linear in the underlying rates, so it is priced by discounting its cash flows off the curve. No model and no PDE. Say this immediately: reaching for machinery that the product does not require is the error the question is set up to catch.

The swaption does. Hull-White is a one-factor short-rate model,

$$dr = (\theta(t) - ar)\,dt + \sigma\,dW,$$

mean-reverting at speed a with a time-dependent $\theta(t)$ chosen to fit today's curve exactly. Being able to say why θ is time-dependent is worth a sentence: it is the calibration handle that makes the model arbitrage-free against the initial term structure rather than a description of how rates behave.

The PDE. Under the risk-neutral measure the price $V(r,t)$ of any derivative in this model satisfies

$$\frac{\partial V}{\partial t} + (\theta(t) - ar)\,\frac{\partial V}{\partial r} + \tfrac{1}{2}\sigma^2\frac{\partial^2 V}{\partial r^2} - rV \;=\; 0,$$

which is the same structure as Black-Scholes with r as the state variable instead of S, and with the discounting term now carrying the state variable itself. Solve it backwards from the swaption's exercise date, where the terminal condition is the swap's value at that date if positive and zero otherwise.

Then the three things that make this a real answer.

One factor means one dimension, so the grid is cheap and a Bermudan swaption, which needs early exercise at every call date, is natural on it: you compare continuation value against exercise value at each node, which is free on a backward solve and awkward in simulation.

Hull-White is Gaussian, so rates can go negative. That was considered a defect and then became a feature, and either way it is the honest thing to say about the model.

One factor means all rates move together, so the model cannot represent a change in the shape of the curve. It prices a swaption adequately and cannot price a product whose payoff depends on the spread between two points of the curve, which needs two factors and therefore a two-dimensional grid or a simulation.

Close by naming the alternative: the same swaption can be priced analytically in Hull-White through the Jamshidian decomposition, so the PDE is what you reach for when the product has early exercise or a feature the closed form cannot carry.

INTERVIEW TRAP

1. **Thinking the PDE identifies the product.** It does not. The terminal and boundary conditions do, and the same equation prices everything in the chapter.
2. **Losing the point of the hedging step.** μ disappears because the hedge removes it, not because anyone assumed risk neutrality.
3. **Using Crank-Nicolson on a discontinuous payoff without smoothing.** It oscillates near the discontinuity, which is exactly where a digital or barrier lives. Rannacher steps first.
4. **Ignoring the explicit scheme's stability condition.** The time step scales with the *square* of the space step, and violating it produces exponentially growing nonsense.
5. **Taking a PDE into high dimensions.** Cost grows exponentially with the number of state variables, and beyond three or four the answer is simulation.
6. **Pricing a linear product with a model.** A vanilla swap is a discounting exercise. Reaching for Hull-White is a signal that the product was not understood.

VISUAL / CODE TO BUILD: Interactive

A backward-diffusion animation. The payoff is drawn at expiry as a kinked or stepped line, and the reader steps backwards in time watching it diffuse into the smooth value curve, with the option's gamma shown beneath as the curvature. Seeing the kink round off, and seeing the curvature concentrate exactly where the kink was, makes both "value is a smoothed payoff" and "gamma is largest at the money" a single observation about the heat equation.

A stability panel sits beside it: sliders for Δx and $\Delta \tau$ with the ratio $\Delta \tau/(\Delta x)^2$ printed and shaded green below one half and red above. Pushing into the red and running the explicit scheme produces the oscillating blow-up on screen, which is a far more memorable statement of the condition than the inequality.

A third panel switches the scheme between explicit, implicit and Crank-Nicolson on a *digital* payoff, so the reader can see Crank-Nicolson's spurious oscillation near the strike and watch it disappear when Rannacher smoothing is toggled on. That is the failure most likely to be met in practice and the least likely to be recognised.

Where this goes next

The equation solved here came from a diffusion, and the general question of what dynamics you are allowed to write down is stochastic differential equations. Feynman-Kac says the other route is legitimate, and the errors you make walking it are the discretisation errors of Euler and Milstein, which are the simulation counterpart of the stability question above: both are about what a finite step does to a continuous process.

The reason to want a grid rather than a formula is the products that have no formula, and the payoffs that push you there are barriers, digitals and Asians, each of which enters the solver purely as boundary data. The by-product is that a grid gives you the Greeks by finite differencing the same solution you already computed, which is a real reason to prefer this route over Monte Carlo in low dimension. And the closed form that the grid reproduces in the one case where it exists is

Black-Scholes, which is the best regression test a solver has.

9.11 Monte Carlo methods

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Monte Carlo is the method you use when the integral you need cannot be done. Since a derivative's price is an expectation, and an expectation is an integral over the possible futures, pricing anything without a formula reduces to estimating an integral in a large number of dimensions. Simulation does that by sampling: generate futures, evaluate the payoff in each, average. Its appeal is that the difficulty of the payoff barely matters; its cost is an error that shrinks embarrassingly slowly, and most of the craft is in fighting that error.

The estimator and its error

The price of a derivative is $\mathbb{E}^{\mathbb{Q}}[\text{discounted payoff}]$. Draw N independent paths, compute the discounted payoff Y_i on each, and take the average $\bar{Y}$. The law of large numbers says $\bar{Y}$ converges to the price. The central limit theorem says more, and it is the more useful statement:

$$\text{standard error} = \frac{\sigma_Y}{\sqrt{N}},$$

where σ_Y is the standard deviation of the payoff across paths, not of the underlying.

KEY IDEA

Two facts follow, and everything practical about Monte Carlo follows from them.

One: the error falls as $1/\sqrt{N}$. Halving the error costs four times the paths. One extra decimal digit costs a hundred times the paths. This is slow, and it does not improve with better computers so much as it forces you to be cleverer.

Two: the rate does not depend on the dimension. A ten-underlying basket converges at the same $1/\sqrt{N}$ as a single stock. Every grid-based method, trees and finite differences alike, degrades exponentially with dimension. **That single asymmetry is why Monte Carlo exists in derivatives.** Below about three factors, use a grid; above it, you have no choice.

WORKED EXAMPLE

The classic illustration is estimating π by throwing darts uniformly at a square and counting the fraction that land inside the inscribed quarter circle. That fraction converges to $\pi/4 = 0.7854$, so $\hat{\pi} = 4k/N$.

Each dart is a Bernoulli draw with $p = 0.7854$, so its standard deviation is $\sqrt{p(1-p)} = \sqrt{0.7854 \times 0.2146} = 0.4105$, and the standard error of $\hat{\pi}$ is $4 \times 0.4105/\sqrt{N} = 1.642/\sqrt{N}$.

With $N = 10{,}000$ darts the standard error is 0.0164, so the 95% interval is $\pi \pm 0.032$, that is [3.109, 3.174]: the second decimal is still moving, so you have π to **one** decimal place. With $N = 1{,}000{,}000$ darts, a hundred times the work, the standard error is 0.00164 and the interval is [3.1384, 3.1448], every point of which rounds to 3.14: you have bought exactly one more decimal place. **A million samples for three correct digits** is the honest summary of unaided Monte Carlo, and it is why the rest of this section is about variance reduction rather than about running more paths.

REAL INTERVIEW QUESTION (CACIB · Exo)

Tell me about Monte Carlo. What is the principle, what are the steps, paths, payoffs, parameters, final price, and why does it give the right answer?

WHAT THEY TEST / ANSWER

Answer in the order asked: principle, steps, then the justification, which is the part most candidates skip and the reason the question ends the way it does.

The principle. A derivative's price is the expectation of its discounted payoff under the risk-neutral measure. An expectation is an integral, and when the integral has no closed form you estimate it by sampling: simulate many possible futures, compute what the contract would pay in each, and average.

The steps, concretely. *One, fix the model and its parameters*: the dynamics of the underlyings, the volatility or the volatility surface, rates, dividends or repo, and the correlations if there is more than one underlying. *Two, generate paths* on the dates the contract cares about. This means drawing random numbers, correlating them where needed, and stepping the state forward. *Three, evaluate the payoff on each path.* This is where simulation earns its keep, because the payoff can read the whole path: a barrier that was touched, an average, an autocall that triggered. *Four, discount and average*, and report the standard error alongside the price, because a Monte Carlo price without an error bar is an incomplete answer.

Why it gives the right answer, in two parts, and say both. *The financial part*: the risk-neutral pricing theorem says the price is that expectation, so if you can estimate the expectation you have the price. Simulating in the real-world measure and averaging would give a different and wrong number. *The statistical part*: the law of large numbers guarantees the average converges to the expectation, and the central limit theorem tells you how fast, at $\sigma_Y/\sqrt{N}$, which lets you state a confidence interval rather than a bare number.

Then the honest closing. Say that convergence is slow, so production use always involves variance reduction; that the estimator is unbiased for the model but says nothing about whether the model is right; and that discretising the dynamics introduces a separate bias distinct from the sampling error, which is the subject of numerical schemes. Distinguishing **sampling error from discretisation bias** is the mark of someone who has actually run one.

Making it converge faster

There are only two ways to reduce $\sigma_Y/\sqrt{N}$: raise N, which is linear in money and square-root in benefit, or lower σ_Y, which is free once you have thought of it. The whole discipline is the second.

REAL INTERVIEW QUESTION (Axa IM · Credit)

What are the Monte Carlo acceleration methods?

WHAT THEY TEST / ANSWER

Group them, because there are four families and they work for different reasons.

Antithetic variates. For every draw Z, also use $-Z$. The two paths are negatively associated, so their payoffs are negatively correlated and the average of the pair has lower variance than two independent draws. It is nearly free to implement. Say the caveat, because it is what shows understanding: this works when the payoff is **monotone** in the driver. For a symmetric payoff such as an at-the-money straddle, $f(Z)$ and $f(-Z)$ are *positively* correlated and antithetic sampling makes the variance worse, not better.

Control variates, the most powerful of the four when one is available. Find a related quantity whose true value you know analytically, price it in the same simulation, and correct your estimate by the error the simulation made on it. The canonical case is an arithmetic-average Asian option controlled by the geometric-average Asian, which has a closed form because the geometric average of lognormals is lognormal. The two payoffs are highly correlated, so most of the noise cancels.

Importance sampling. If the payoff is concentrated in a region the model rarely visits, a deep out-of-the-money option or a default, most paths contribute nothing and all the variance comes from the few that do. Sample from a shifted distribution that visits that region often, then reweight by the likelihood ratio to remove the bias. This is the only technique that helps materially with genuinely rare events.

Quasi-Monte Carlo. Replace pseudo-random numbers with a low-discrepancy sequence such as Sobol, which fills the space more evenly than randomness does. The error improves from roughly $1/\sqrt{N}$ towards something closer to $1/N$, at the cost of losing the clean statistical error bar and of degrading when the effective dimension is high. It is standard for structured products, where the effective dimension is usually far lower than the nominal number of time steps.

Close with the practical hierarchy: a good control variate beats everything else and costs almost nothing per path; quasi-random sequences are the default for path-dependent equity products; antithetics are worth having as a habit but only for monotone payoffs; importance sampling is specialist and reserved for tails. And note that all of these reduce *variance*, so none of them repairs a discretisation bias or a wrong model.

WORKED EXAMPLE

The leverage in variance reduction is worth quantifying, because the numbers are not intuitive.

Suppose an option's discounted payoff has standard deviation $\sigma_Y = 15$ across paths. With $N = 10{,}000$ the standard error is $15/\sqrt{10{,}000} = 0.15$, giving a 95% confidence interval of roughly $\pm 1.96 \times 0.15 = \pm 0.29$. To reach a standard error of 0.015 you need $N = 1{,}000{,}000$, a hundred times the work.

Now suppose a control variate cuts the payoff's standard deviation by a factor of ten, to 1.5. The same standard error of 0.015 now needs N such that $1.5/\sqrt{N} = 0.015$, that is $N = 10{,}000$.

A tenfold reduction in σ_Y is a hundredfold reduction in computation, because the error depends on σ_Y linearly and on N as a square root. That asymmetry is why a day spent finding a control variate beats a month of hardware.

Where simulation is the wrong tool

REAL INTERVIEW QUESTION (Edmond de Rothschild · Sales)

Why use Monte Carlo rather than Black-Scholes for an American option?

WHAT THEY TEST / ANSWER

Correct the premise gently first, because it contains the answer.

There is no Black-Scholes formula for an American option to reject. The closed form prices a European exercise only. The single exception worth naming is an American call on a non-dividend-paying stock, which is never optimally exercised early and therefore equals the European call. Once there is a dividend, or for any American put, no closed form exists.

But Monte Carlo is not the natural replacement either, and saying so is the answer that separates candidates. Early exercise is a **backward** problem: at each date you compare the immediate exercise value with the value of continuing, and the continuation value depends on the future. Monte Carlo runs **forward**, path by path, so at the moment you would decide, the estimator does not know the continuation value. Methods that go backwards, binomial trees and finite difference solutions of the pricing equation, handle early exercise naturally because they compute exactly that comparison at every node. For a single-underlying American option, a tree or a PDE is the right tool, which is partial differential equations in finance.

So when is Monte Carlo used for early exercise? When the dimension is too high for a grid: several underlyings, or a stochastic volatility or stochastic rate state variable alongside the spot. Grids degrade exponentially with dimension and simulation does not, so above roughly three factors there is no alternative. The standard technique is least-squares Monte Carlo, which estimates the continuation value at each exercise date by regressing realised discounted continuation payoffs on functions of the current state, and then applies the resulting rule going forward.

Finish with the bias point, which is the trap. Naive fixes are not neutral. Deciding exercise on each path using that path's own future is perfect foresight and gives a price that is **too high**. Applying any fixed suboptimal rule gives a price that is **too low**. Those two facts are useful rather than annoying: together they bracket the true value, which is how practitioners sanity-check a least-squares implementation.

INTERVIEW TRAP

Three errors. (1) Quoting a Monte Carlo price without a standard error. The number alone does not say whether the third decimal is real. (2) Confusing sampling error with discretisation bias. More paths shrink the first and leave the second exactly where it was, so a price can converge tightly to the wrong answer. (3) Saying Monte Carlo handles American options well. It handles them, at the cost of an approximation, and only because grids fail worse in high dimension.

Where this goes next

Stepping the dynamics forward, and the bias that introduces, is numerical schemes. The backward alternative is partial differential equations in finance, and the factorisation that turns independent draws into correlated ones is matrix decompositions. Getting the parameters that go in is model calibration, the payoffs that make simulation necessary are exotic options, and the risk application of the same machinery is value at risk.

9.12 Numerical schemes (Euler, Milstein)

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A stochastic differential equation describes a process in continuous time. A computer cannot do continuous time, so every simulated path is an approximation on a grid, and the choice of approximation introduces an error that has nothing to do with how many paths you run. Knowing which of your two errors dominates is the whole practical skill here, and most people optimise the wrong one.

INTUITION

There are two independent errors in any Monte Carlo price. **Discretisation bias** comes from stepping through time in jumps, and shrinks with smaller steps. **Sampling error** comes from using finitely many paths, and shrinks with more of them. They are not interchangeable, and spending on the wrong one buys nothing.

Euler-Maruyama

For an SDE $dX_t = a(X_t)\,dt + b(X_t)\,dW_t$, the obvious scheme is to freeze the coefficients over each step:

$$X_{t+\Delta} \;=\; X_t + a(X_t)\,\Delta + b(X_t)\,\sqrt{\Delta}\,Z, \qquad Z \sim \mathcal{N}(0,1).$$

This is **Euler-Maruyama**, and it is the deterministic Euler method with a random increment bolted on. It is trivial to implement, it works for essentially any SDE, and it is what almost everything is done with in practice.

Its weakness is visible in the derivation it skips. Freezing $b(X_t)$ over the step ignores the fact that b itself moves during the step, and because the driving increment is of order $\sqrt{\Delta}$ rather than Δ, that omission is larger than the intuition from deterministic numerics suggests.

Milstein

Expanding $b(X_s)$ inside the step with Ito's lemma and keeping the leading correction gives the **Milstein** scheme:

$$X_{t+\Delta} \;=\; X_t + a\,\Delta + b\,\Delta W \;+\; \tfrac{1}{2}\,b\,b'\,\big((\Delta W)^2 - \Delta\big).$$

The extra term costs one derivative of the diffusion coefficient and almost no computation, since $(\Delta W)^2$ is already in hand. Note that it vanishes when b is constant: for additive noise Euler is already as good as Milstein, which is why the distinction only bites for multiplicative noise, meaning almost everything in finance.

Strong and weak convergence are different questions

This is the distinction that separates people who have read about schemes from people who have used them.

Strong convergence measures the pathwise error, $\mathbb{E}\left|X_T^{\text{scheme}} - X_T^{\text{exact}}\right|$, against the exact path driven by the *same* Brownian increments. It is order $\frac{1}{2}$ for Euler and order 1 for Milstein.

Weak convergence measures the error in an expectation, $\left|\mathbb{E}[f(X_T^{\text{scheme}})] - \mathbb{E}[f(X_T^{\text{exact}})]\right|$. It is order 1 for both.

KEY IDEA

A price is an expectation, so **pricing a European payoff is governed by weak convergence, where Euler and Milstein are the same order**. The extra term does not improve the weak order, and in the experiment below it makes the weak error an order of magnitude *worse*: equal orders do not mean equal constants. Strong convergence matters when the individual path matters: barriers, lookbacks, Asians, any early-exercise rule, and any variance reduction that pairs a coarse simulation against a fine one. Reaching for Milstein to price a vanilla is a misunderstanding of which error you are fighting.

WORKED EXAMPLE

Measured, not quoted. Geometric Brownian motion with $S_0 = 100$, drift 5%, volatility 30%, one year, priced as a **European call struck at 100 and discounted at 5%**, simulated with `numpy` over two million paths per step size. The exact terminal value is available in closed form from the same Brownian increments, so both errors are measured against the exact path rather than estimated.

That coupling matters more than it sounds. Estimating the weak error as the difference of two independently simulated prices leaves it buried under a Monte Carlo standard error of 0.0225, which is larger than most of the signal. Differencing the payoffs *path by path* instead cancels almost all of that noise and brings the standard error down to the figures in brackets below, at which point the structure is unmistakable.

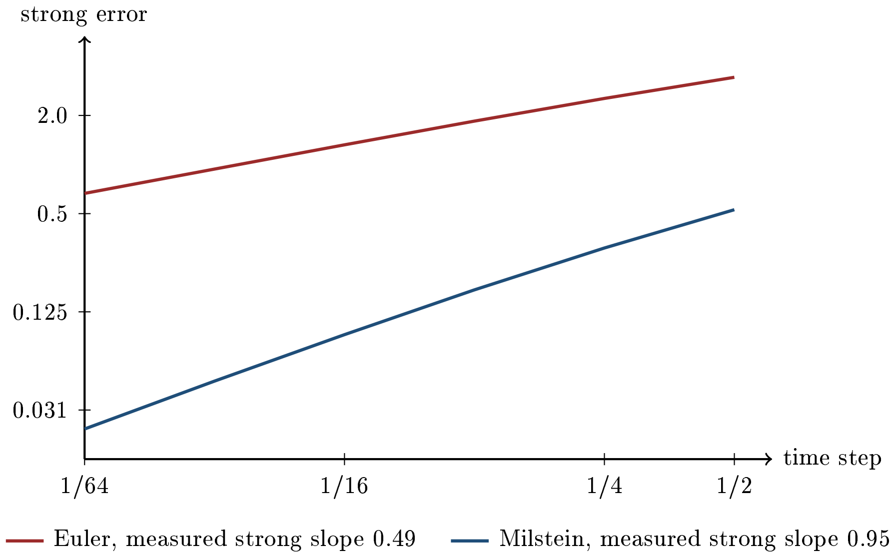
Time step	Euler strong	Milstein strong	Euler weak (se)	Milstein weak (se)
1/2	3.4267	0.5283	−0.0388 (0.0027)	−0.2795 (0.0005)
1/4	2.5414	0.3078	+0.0141 (0.0019)	−0.1506 (0.0003)
1/8	1.8454	0.1704	+0.0123 (0.0014)	−0.0782 (0.0001)
1/16	1.3207	0.0906	+0.0092 (0.0010)	−0.0398 (0.0001)
1/32	0.9384	0.0470	+0.0041 (0.0007)	−0.0201 (0.0000)
1/64	0.6653	0.0239	+0.0020 (0.0005)	−0.0101 (0.0000)

Fitting a slope to the strong errors over the finer steps gives **0.491** for Euler and **0.945** for Milstein, against theoretical orders of $\frac{1}{2}$ and 1. The theory is confirmed by measurement.

The weak columns are where the surprise is, and they behave completely differently from each other. Milstein's halves at every refinement, from −0.2795 down to −0.0101, a fitted order of **0.962**: textbook order one, and resolved far outside the standard errors in brackets. Euler's is an order of magnitude smaller throughout, changes sign between the first two rows, and never rises above 0.014 after that.

So the extra Milstein term makes the weak error worse, not better. Both schemes are weak order one, as the theory says, but the constants are not remotely equal and Milstein's is the unfavourable one for this payoff. Its bias is also consistently negative, so it systematically under-prices the call. Buying a better *strong* order cost accuracy on the number actually being computed.

Then the budgeting question, which is the practical lesson. Be careful which standard error you compare against. The one on the *price* with two million paths is **0.0159**; the 0.0225 quoted above is the standard error of a *difference* of two independent runs, which is $\sqrt{2}$ times larger. Against 0.0159, the Euler discretisation bias is already the same size as the noise at **four steps a year**, where the two cannot be told apart, and is clearly below it by **eight**, so refining the grid past that buys nothing you can measure and the next unit of effort belongs in paths, not steps. With Milstein the bias does not clear it until **sixty-four steps**. Work out which of your two errors is binding before spending on either.



both axes are logarithmic, so a convergence order is a slope

The point of plotting this on log axes is that a convergence order becomes a slope you can read off. Milstein's line is twice as steep, which is the entire content of "order one versus order one half", and the vertical gap between the lines is the accuracy you are buying with one extra term.

The case where the error is zero

Geometric Brownian motion has a feature worth exploiting: applying Ito's lemma to $\ln S$ gives

$$d(\ln S_t) = \left(\mu - \tfrac{1}{2}\sigma^2\right)dt + \sigma\, dW_t,$$

whose coefficients are *constant*. A process with constant coefficients has no discretisation error at all, so stepping the logarithm and exponentiating is **exact at any step size**. In the experiment above this scheme reproduced the closed-form terminal value to machine precision at every step size tested.

> **KEY IDEA**
>
> Do not discretise a process you can simulate exactly. Simulating S directly with Euler and then complaining about the bias is a self-inflicted problem when the log transform removes it entirely, and it also removes the possibility of a simulated price going negative. The general lesson is to look for a transformation with constant or simpler coefficients before reaching for a finer grid.

Where the schemes actually break

- **Processes that must stay positive.** Euler applied to a square-root variance process will produce negative variances, at which point the square root fails. The standard fixes are full truncation, reflection, or a specialised scheme; the mechanics of the models involved are in stochastic volatility models.
- **Barriers and other path features.** A grid only observes the path at grid points, so a simulated barrier is systematically less likely to be breached than a continuously monitored one, and the price is biased. This is the same discrete-versus-continuous monitoring problem, with the same correction, as in path-dependent products, and it is a case where a finer grid genuinely does help.

- **Coefficients that are not smooth.** Milstein needs b' to exist. Where the diffusion is not differentiable the extra term is not available and the promised order does not appear.
- **Jumps.** Adding a jump component changes the analysis entirely, since the error is then dominated by how jump times are placed on the grid rather than by the diffusion approximation.

Interview angle

REAL INTERVIEW QUESTION (Natixis · QIS)

How do you simulate your paths in Monte Carlo?

WHAT THEY TEST / ANSWER

An implementation question, so answer like someone who has written the code rather than read about it. Go in the order you would actually make the decisions.

Start from the SDE and ask whether you need a scheme at all. For geometric Brownian motion you do not. Apply Ito's lemma to $\ln S$, get constant coefficients, and step the logarithm: $S_{t+\Delta} = S_t \exp\big((\mu - \sigma^2/2)\Delta + \sigma\sqrt{\Delta}Z\big)$. That is exact at any step size and it cannot go negative. Saying this first tells the interviewer you know discretisation is a cost to be avoided rather than a ritual.

If the process is not exactly simulable, name the scheme and its order. Euler-Maruyama freezes the coefficients over the step, and it is strong order one half, weak order one. Milstein adds $\frac{1}{2}bb'((\Delta W)^2 - \Delta)$ and lifts the strong order to one.

Then the discriminating point, which is which order you actually need. A European price is an expectation, so it is governed by *weak* convergence, where the two schemes are the same order. Strong order matters for barriers, lookbacks, Asians, early exercise, and for pairing coarse and fine simulations. If you have measured it you can go further: equal weak orders do not mean equal weak errors, and on a one-year vanilla the Milstein bias came out an order of magnitude larger than the Euler one and consistently negative. Choosing Milstein to price a vanilla is not merely unnecessary, it can cost accuracy on the number you are computing.

Finish on how you would set the grid, because that is the real question underneath. Compare the discretisation bias against the Monte Carlo standard error and stop refining once the bias is below the noise. In a test I ran on a one-year vanilla with two million paths the standard error on the price was 0.0159, and with Euler the bias was level with it at four steps a year and below it by eight, so refining further bought nothing measurable. Add the caveat that you can only make that comparison if you measure the bias with *paired* paths; estimated from independent runs it disappears into the standard error and you will conclude there is no bias to see.

If there is time, add the practical detail: use the same random numbers across scenarios so that differences are signal rather than noise, and place grid points on every date the payoff looks at, since a barrier or fixing observed between grid points is simply not observed. The estimator itself, and how to shrink its standard error, is Monte Carlo methods.

INTERVIEW TRAP

Five errors. (1) Reaching for Milstein to price a European payoff, where the weak orders are equal and the extra term can make the weak error worse. (2) Reading equal convergence orders as equal errors, when the constants differ by an order of magnitude here. (3) Discretising geometric Brownian motion when the log transform is exact. (4) Refining the time grid while the Monte Carlo standard error is larger than the bias, which improves nothing you can measure. (5) Measuring a weak error from independent runs rather than paired paths, which hides a real order-one bias under the sampling noise and invites the conclusion that there is no bias at all.

Where this goes next

The estimator these paths feed, its $1/\sqrt{N}$ convergence and the techniques for shrinking it are Monte Carlo methods. The models whose positivity constraints break a naive scheme are stochastic volatility models, the discrete monitoring bias and its correction are in path-dependent products, and fitting the model parameters these schemes then simulate is model calibration. The vectorisation that makes a million paths tractable at all is NumPy and pandas.

9.13 Model Calibration

trading · structuring | advanced | cross-asset

Calibration is choosing a model's parameters so that it reproduces the prices the market is currently quoting. It is the step between having a model and being able to use one, and it is where most of the practical difficulty in derivatives modelling actually lives, because the mathematics of the models is settled and the fitting of them is not.

KEY IDEA

Calibration is not estimation. Estimation fits parameters to *historical data* and produces a description of the real world under $\mathbb{P}$: that is what a GARCH fit or a covariance matrix is. Calibration fits parameters to *today's prices* and produces an object consistent with the risk-neutral measure $\mathbb{Q}$ under which the market is quoting.

They answer different questions and routinely disagree, and the gap between them is the risk premium, which is exactly what Girsanov says it should be. A calibrated volatility above a historically estimated one is not an error in either.

The purpose of calibrating is narrower than it looks. You are not trying to learn the truth about the market; you are building a device that *interpolates* between liquid quotes to price the illiquid product a client has asked for, and that produces hedge ratios you will actually trade. Those two purposes pull in different directions, which is the tension the rest of this section is about.

The objective, and why the weights matter more than the solver

Calibration minimises a distance between model and market:

$$\min_{\theta} \sum_{i} w_i \left(\text{model}_i(\theta) - \text{market}_i\right)^2,$$

and three choices in that expression matter more than which optimiser is used.

Fit implied volatilities, not prices. A one-cent error on a deep out-of-the-money option is a large error in volatility terms and a negligible one in price terms, so a price-based objective ignores the wings entirely. Since the wings are where exotic products live, this single choice changes what the model is good for.

Weight by liquidity. An instrument with a wide bid-ask carries less information than one quoted tightly, and a fit that matches a stale quote exactly at the cost of missing a liquid one has optimised the wrong thing. Weighting by inverse bid-ask width is the standard crude fix.

Include only what you will hedge with. Calibrating to instruments you cannot trade produces a model that is consistent with prices you cannot realise.

Root-finding: the calibration everybody does daily

The simplest calibration is one parameter to one price: extracting an implied volatility. It happens millions of times a day and it is the case where the numerical method is worth knowing in detail.

REAL INTERVIEW QUESTION (Société Générale · EQD)

Explain the Newton-Raphson algorithm.

WHAT THEY TEST / ANSWER

Give the method, the reason it converges fast, the finance instance, and then where it breaks, which is the part that matters on a desk.

The method. To solve $f(x) = 0$, take the tangent to f at the current guess and jump to where the tangent crosses zero:

$$x_{n+1} \;=\; x_n - \frac{f(x_n)}{f'(x_n)}.$$

Near a simple root it converges **quadratically**, meaning the number of correct digits roughly doubles each iteration, which is why it is the default when a derivative is available.

The finance instance is implied volatility, and the pleasant part is that the derivative is a quantity you already have. With $f(\sigma) = \mathrm{BS}(\sigma) - P_{\mathrm{market}}$, the derivative $f'(\sigma)$ is **vega**, so the iteration is

$$\sigma_{n+1} \;=\; \sigma_n - \frac{\mathrm{BS}(\sigma_n) - P_{\mathrm{market}}}{\mathrm{vega}(\sigma_n)}.$$

Show it converging. A one-year at-the-money option on a spot of 100 at zero rates, quoted at 8.00, started deliberately far away at 30%:

Iteration	σ	Model price	Error
0	0.30000000	11.923538	3.9×10^{0}
1	0.20053881	7.986955	-1.3×10^{-2}
2	0.20086744	8.000000	-1.1×10^{-7}
3	0.20086744	8.000000	-1.5×10^{-14}

Three iterations from a 50% relative error to machine precision, and the doubling of correct digits is visible in the error column: 10^{0}, 10^{-2}, 10^{-7}, 10^{-14}. That is what quadratic convergence looks like.

Then where it fails, which is what an interviewer is really asking and what separates this from a textbook answer.

Vega goes to zero in the wings. Deep in or out of the money, the denominator collapses and the step f/f' explodes, throwing the guess to a negative or absurd volatility. This is not hypothetical: it is the standard failure of a naive implied-volatility routine on the wings of a surface, which is exactly where you most want a number.

It needs a derivative. If the model has no closed-form vega, use the secant method, which approximates it, at the cost of slower convergence.

It is not globally convergent. A bad starting point can diverge or cycle.

So the production answer is a hybrid: bracket the root first, then run Newton inside the bracket and fall back to bisection whenever a Newton step would leave it. That is a **safeguarded Newton** method, and its derivative-free cousin, which combines bisection with interpolation instead, is **Brent's method**. One or the other is what every library actually ships, and knowing that the answer is “Newton with a safety net” rather than “Newton” is the point of the question.

Where multi-parameter calibration goes wrong

Fitting three or five parameters to a whole surface introduces failures that one-dimensional root-finding does not have, and they are the reason calibration is a craft.

Non-uniqueness. The objective frequently has long flat valleys: many different parameter sets fit the observed prices almost equally well. In a stochastic volatility model, raising the volatility of volatility and lowering the correlation can leave the fitted surface nearly unchanged. The consequence is not a bad fit, it is that the parameters are not identified, so any quantity that depends on them individually rather than on the fitted surface is unreliable.

Instability. If the objective is flat, small changes in the market move the optimiser a long way along the valley. Parameters then jump from day to day, and *hedge ratios jump with them*, so a desk rebalances for reasons that have nothing to do with the market. The standard repair is to regularise: add a penalty on the distance from yesterday's parameters, which trades a slightly worse fit for a usable hedge, and that trade is almost always worth making.

Overfitting. A model with enough freedom will fit stale and illiquid quotes exactly, and it will be fitting noise. The symptom is a fit that is excellent in sample and produces implausible prices between the quoted strikes.

> INTUITION
>
> The deepest tension is between **fitting today** and **having sensible dynamics**, and it is the reason two families of model coexist.
>
> Local volatility can fit today's surface *exactly*, by construction. Its forward smile dynamics are unrealistic: it predicts that the smile flattens as spot moves, which is not what happens.
>
> Stochastic volatility fits today's surface only approximately, and its dynamics are far more plausible.
>
> A perfect fit is therefore not the objective, and a candidate who says so is ahead of one who reports a smaller calibration error. The right question is whether the model reprices the hedging instruments well enough and produces Greeks you are willing to trade on, and the honest test of a calibration is not the residual but the stability of the hedge it implies.

Practical rules

- Calibrate to what you can trade, and weight by liquidity.
- Fit in volatility space, not price space.
- Start from yesterday's parameters, and penalise moving far from them.
- Check the Greeks, not only the fit. A calibration that reprices every instrument and produces a delta that jumps has failed.
- Recalibrate on a schedule, not continuously, unless you want your hedges to inherit the optimiser's noise.
- Keep the calibration inputs versioned alongside the price, because a price cannot be reproduced without them, which pricing libraries makes the same point about.

INTERVIEW TRAP

1. **Confusing calibration with estimation.** One fits prices under $\mathbb{Q}$, the other fits history under $\mathbb{P}$, and their disagreement is a risk premium rather than an error.
2. **Running Newton without a bracket.** Vega vanishes in the wings and the step explodes. Bracket first, fall back to bisection.
3. **Minimising price error.** It ignores the wings, which is where the exotic risk is. Minimise volatility error.
4. **Treating a small residual as success.** A flat valley gives a small residual with unidentified parameters, and the hedge is what reveals it.
5. **Recalibrating too often.** Jumping parameters produce jumping hedges and real transaction costs paid for optimiser noise.
6. **Wanting a perfect fit.** An exact fit to today buys unrealistic dynamics, which is the local versus stochastic volatility trade-off and cannot be optimised away.

VISUAL / CODE TO BUILD: Interactive

A Newton panel. The function $\mathrm{BS}(\sigma) - P$ is drawn against σ with the reader's starting guess marked, and each click draws the tangent and jumps to its root, with the error printed at each step. Starting at 30% reproduces the section's table in three clicks. A strike slider then pushes the option into the wings, where the curve flattens, the tangent goes nearly horizontal and the next iterate is thrown off the chart, which is the vega-to-zero failure shown rather than described. A toggle adds the bisection fallback and the same starting point now converges.

A second panel is a calibration-surface explorer for a two-parameter model: the objective drawn as a contour map over the two parameters, with the minimum marked. The instructive setting is one where the contours form a long narrow valley rather than a bowl. The reader perturbs the market data slightly and watches the located minimum slide a long way along the valley floor while the fitted surface barely changes, and a second readout shows the implied hedge ratio moving with it. That is non-identification and hedge instability shown as one phenomenon, which is what they are.

Chapter 10

Equity Markets

10.1 Equity market structure

sales · trading · structuring core equity

Every equity derivative ends up as an order in the cash equity market. The option is priced in a model, but the delta is bought and sold in a real order book, at a real time of day, on one of several competing venues, and settles on a date that is not today. That plumbing decides what a hedge actually costs and when it can be done at all, which is why an equity derivatives interviewer will happily spend ten minutes on it before touching a Greek.

What an exchange is, how a limit order book works and why a market maker quotes two prices are general facts about all markets and live in exchanges, market microstructure and the bid-ask spread. This section covers what is specific to shares: how a company's size is measured, how the equity trading day is shaped by its two auctions, why one share trades in several places at once, and when the trade actually settles.

Price is not size: market capitalisation and free float

A share price on its own says nothing about how big a company is, because it depends entirely on how many pieces the company was cut into. The number that matters is the **market capitalisation**:

$$\text{market cap} \;=\; \text{share price} \times \text{shares outstanding}.$$

WORKED EXAMPLE

Company A trades at 25 euros with 800 million shares outstanding. Company B trades at 300 euros with 20 million shares outstanding.

$$\mathrm{A} = 25 \times 800\,\mathrm{m} = 20\ \mathrm{bn}, \qquad \mathrm{B} = 300 \times 20\,\mathrm{m} = 6\ \mathrm{bn}.$$

The 300 euro share belongs to the company that is worth a third as much. A share price is only comparable to its own history, never to another company's.

Market cap is still not the number a trader uses. Part of the register is held by founders, families, states or strategic partners who do not trade, and those shares are unavailable to the market. Strip them out and you get the **free float**, and multiplying the float by the price gives the **free-float market capitalisation**. If 30% of Company A above is locked up, its float cap is $0.70 \times 20 = 14$ billion euros.

KEY IDEA

Free float, not market cap, is what drives the three things a desk cares about: the weight the stock gets in an index, how much of it can be borrowed to short, and how much can be traded in a day without moving the price. Two companies of identical market cap can be completely different trades if one of them is 80% owned by its founder.

Index construction uses the float-adjusted number and is developed in indices; the borrow consequence is in stock borrow and lending.

The equity trading day

Unlike a currency, a share trades for a fixed session and then stops. Take Euronext Paris as the reference: orders can be entered from early morning into an **opening auction** that uncrosses at 09:00 Paris time, continuous trading runs until 17:30, and a **closing auction** then collects orders for five minutes and uncrosses at around 17:35. United States cash equities run continuously from 09:30 to 16:00 New York time with an opening and a closing auction at each end. The exact minutes differ by venue; the three-part shape does not.

The two auctions are not a curiosity. In continuous trading, buyers and sellers arrive one at a time and each trade prints at its own price. In an auction, all the interest is collected first and then executed *at one single price*, which is why an auction is the natural way to start a session (there is no price yet) and to end it (everyone wants the same official number).

INTUITION

Continuous trading is a market stall: people turn up one by one, and each one haggles their own price. An auction is a single sealed-bid round: everybody writes down what they are willing to pay, the auctioneer finds the one price that clears the most goods, and then everybody who trades trades at that price, including the person who would happily have paid more.

How an auction uncrosses

The rule is mechanical. For each candidate price, count how many shares would trade: demand is every buy order with a limit at or above that price, supply is every sell order with a limit at or below it, and the executed volume is the smaller of the two. The auction price is the one that maximises executed volume. If several prices tie, the tie-breaks run in order: smallest remaining imbalance, then the side with the surplus, then the last traded price.

WORKED EXAMPLE

The closing auction book on a share, with limit orders only:

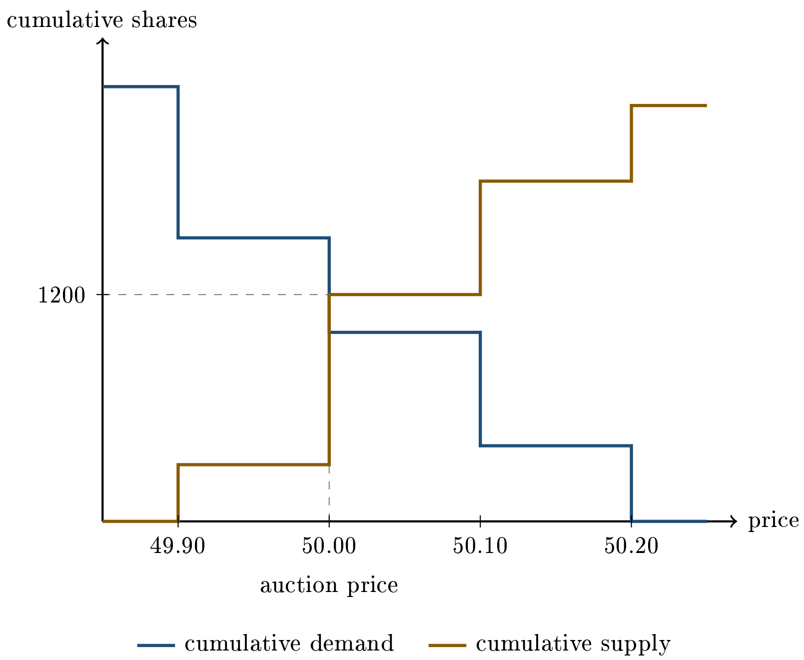
Buy orders		Sell orders	
Size	Limit	Limit	Size
400	50.20	49.90	300
600	50.10	50.00	900
500	50.00	50.10	600
800	49.90	50.20	400

Cumulate each side and take the minimum at every candidate price:

Price	Cumulative demand	Cumulative supply	Executed
49.90	2300	300	300
50.00	1500	1200	**1200**
50.10	1000	1800	1000
50.20	400	2200	400

The maximum is 1200 shares at **50.00**, and it is unique, so no tie-break is needed. Every one of those 1200 shares trades at 50.00, including the buyer who had been willing to pay 50.20 and the seller who had been willing to sell at 49.90. Both of them are filled in full, at a better price than their own limit, which is the whole point of a single-price uncrossing.

Who is left over? On the sell side, the 300 at 49.90 and the 900 at 50.00 fill completely, which is exactly the 1200. On the buy side, the 400 at 50.20 and the 600 at 50.10 fill completely, and the remaining 200 come out of the 500 posted at 50.00, so that order is filled 200 out of 500. The lesson to keep: at the auction price itself you may well be filled only in part, and on Euronext the allocation there runs by time priority, so the order entered earliest is served first.



The two curves cross at 50.00, and the executed volume is the height of the lower curve at that point. Everything to the left is a price where sellers are scarce, everything to the right a price where buyers are scarce.

Why the close is the most important minute of the day

The closing auction produces the **official closing price**, and an enormous amount of the financial system is written to reference precisely that number. Funds strike their net asset value on it. Index levels that matter are computed from it. Structured products fix their initial and final levels on it. Benchmarked orders are measured against it. So flow concentrates there: on an ordinary day, something between a fifth and a third of the volume in a European large cap prints in the closing auction, and on an index review date it can be most of the day's volume.

For a derivatives desk this is not trivia but a scheduling constraint. If the payoff you sold fixes on the official close, then hedging at 16:00 leaves you exposed to whatever the auction does; the hedge belongs in the same auction as the fixing. This is the practical face of the fixing risk described in hedging structured products, and it is the reason a delta that looks continuous in the model is executed in discrete daily bursts in practice, on top of the general argument in delta neutrality and dynamic hedging.

> **INTERVIEW TRAP**
>
> Assuming the last continuous trade is the close. It is not: the official close is the auction uncrossing price, which can differ from the 17:29 print by a visible amount when the auction is imbalanced. A P&L reconciled against the last trade rather than the official close will show a phantom gain or loss every single evening.

One share, several venues

A share is listed on one exchange but it does not only trade there. In Europe, the same stock trades on its primary listing venue and simultaneously on competing multilateral trading facilities, on bank systematic internalisers, and in dark venues where orders are not displayed before execution. In the United States the same is true across a larger number of exchanges and off-exchange venues. The consequence is that the best price available in the market is generally not the best price on the primary book, and an execution that looks at only one venue is systematically paying too much.

WORKED EXAMPLE

Displayed offers on one share, across three venues, and you need to buy 1000 shares.

Venue	Offer	Size
MTF A	50.01	300
Primary	50.02	400
MTF B	50.02	200
MTF A	50.03	500
Primary	50.03	900

Sweeping the consolidated book cheapest first takes 300 at 50.01, then 400 on the primary and 200 on MTF B at 50.02, then 100 at 50.03:

$$300 \times 50.01 + 600 \times 50.02 + 100 \times 50.03 = 15\,003 + 30\,012 + 5\,003 = 50\,018,$$

an average of **50.018**. Going to the primary venue alone takes 400 at 50.02 and 600 at 50.03:

$$400 \times 50.02 + 600 \times 50.03 = 20\,008 + 30\,018 = 50\,026,$$

an average of **50.026**. The difference is 8 currency units on 50 018, which is $8/50\,018 = 0.016\%$, or **1.6 basis points**. That sounds like nothing until you scale it: on a hedge of 100 million euros, 1.6 bp is about 16 000 euros, on a single trade, for nothing but where the order was pointed.

Two consequences follow. First, size in the market is larger than size on any one screen, so a book that looks thin may not be. Second, comparing a fill against the primary venue's price alone is the wrong benchmark; the honest one is the consolidated best bid and offer at the moment of execution. What "liquidity" means once it is split this way is developed in liquidity.

Settlement, and why the calendar matters

Trading a share and owning it are separated by the **settlement cycle**: cash and title change hands a fixed number of business days after the trade. European markets settle on T+2, and the United States and Canada moved to T+1 in 2024, with Europe scheduled to follow. Two practical consequences:

- **Entitlement follows settlement, not the trade.** Whether you receive the dividend depends on whether your purchase settles before the record date, which is why the ex-dividend date sits a settlement cycle ahead of it. The mechanics are in corporate actions and dividends.
- **Selling something you have not received is a short.** If the shares have not settled into your account, delivering them requires a borrow, which costs a fee and can be recalled. That is short selling and stock borrow and lending.

Different countries also settle on different cycles and trade in different currencies and time zones, which is precisely the friction that ADRs and GDRs exist to remove.

When the market stops

Equity venues interrupt trading when a price moves too fast. Euronext applies static and dynamic price collars around a reference price and pauses a stock into a short re-opening auction when a print would fall outside them. United States venues apply limit up / limit down bands per stock, and market-wide circuit breakers halt everything if the S&P 500 falls 7%, 13% or 20% in a day. Trading halts are also imposed around company announcements.

INTERVIEW TRAP

A halt does not pause the option, only the hedge. The stock stops trading, but the derivative you sold on it keeps accruing time and keeps its exposure, and when the stock re-opens it may re-open with a gap. A halted underlying is the cleanest real-world case of a delta hedge that cannot be rebalanced, which is one of the assumptions Black-Scholes quietly makes.

Interview angle

REAL INTERVIEW QUESTION (Société Générale · EQD)

What is the Société Générale share price?

WHAT THEY TEST / ANSWER

This is a filter, not a finance question, and it is asked because it cannot be prepared from a textbook. Say the number, to two significant figures, as of that morning, and if you are genuinely unsure say so and give the range you last saw rather than inventing a level. Nothing damages a candidate faster in an equity interview than a confident wrong price on the bank's own stock.

Then do the thing that turns a checkbox into a conversation: add the number that actually means something. A share price is meaningless without the share count, so give the market capitalisation alongside it, and if you can, one structural fact about the line: whether it has recently been through a buyback that reduced the share count, or where it sits relative to its book value, which is the ratio the equity market watches for banks. The point you are demonstrating is that you know the price is cap/shares and not a measure of anything on its own.

The natural follow-up is "and is that expensive?", which is a trap if you answer from the price. It cannot be answered from the price at all, only from a ratio, and for a bank that ratio is price to book.

REAL INTERVIEW QUESTION (Société Générale · EQD)

What is Société Générale's market capitalisation?

WHAT THEY TEST / ANSWER

What is being tested is whether you can build the number rather than recall it. State the method out loud: market cap is share price times shares outstanding, so you need two inputs and you should say which one you are less sure of.

Do it as an estimate in the open. A large European bank has a share count in the hundreds of millions to low billions and a share price in the tens of euros, so the product lands in the tens of billions. If you take a share price of 25 euros and 800 million shares, that is $25 \times 800\,\text{m} = 20$ billion euros. Give the estimate with its inputs so the interviewer can correct the input rather than the answer, which is the behaviour they are looking for.

Two refinements score the point. First, the number you have just built is the full market cap, and the number an index uses is the free-float cap, which is smaller whenever a large stake is held by someone who does not trade. Second, market cap is the equity value only: it is not what it would cost to buy the business, because that also requires taking on the debt. Bring both up unprompted and the question is finished.

REAL INTERVIEW QUESTION (Société Générale · EQD)

The largest market capitalisation in the CAC, and the second largest?

WHAT THEY TEST / ANSWER

The honest answer is that this changes, and the interviewer knows it changes; what they are testing is whether you looked this week. So give the current top of the index and say explicitly that the ordering moves. In recent years the top of the CAC 40 by market capitalisation has been contested between the luxury names, principally LVMH and Hermès, with the large energy and pharmaceutical names next. Check it the morning of the interview rather than relying on a figure learned months earlier.

Then add the structural observation that makes the answer worth more than a name. The CAC 40 is capitalisation-weighted on free float, so the largest company is also the largest weight, and a small number of names drive most of the index's daily move. That matters directly to a derivatives desk: an index option is not a diversified basket in practice when the top handful of constituents dominate the weight, and index-versus single-stock correlation trades exist because of exactly that concentration. Weighting and capping rules are in indices, and the concentration consequence is in equity correlation.

Be ready for the immediate follow-up, "and what is the index level?". Treat these questions as one block and know the level of the CAC 40, the Euro Stoxx 50 and the S&P 500 to two significant figures before you walk in.

REAL INTERVIEW QUESTION (Credit Suisse · FX)

What is the quote on the Credit Suisse share?

WHAT THEY TEST / ANSWER

This one carries a date stamp: Credit Suisse shares stopped trading when UBS completed the acquisition in 2023, so treat it as the template it is and substitute the bank whose desk you are sitting on. Nothing the question tests depends on the name.

Answer the level first, then hear the word that was actually chosen. The question says *quote*, not *price*, and a quote in a traded market is two numbers, not one: a bid at which you can sell and an offer at which you can buy. Giving a single number is answering a slightly different question, and saying "bid at X, offered at Y, so around X and a half" shows you think like someone who has to trade rather than someone who reads a closing level in a newspaper.

The second thing worth adding is the currency and the venue, because the question was asked on an FX desk. A Swiss share is quoted in Swiss francs, so a euro-based investor holding it owns two exposures, the stock and CHF, and the return in euros is not the return in francs. That is the natural bridge into quanto and composite structures, where the difference between "the payoff in the stock's currency" and "the payoff in mine" has to be priced explicitly.

The width of the two numbers is not decoration either: it is the cost of doing anything, and it is the subject of the bid-ask spread.

INTERVIEW TRAP

The three market-structure answers that cost marks. (1) Comparing two companies by share price. (2) Saying a stock "trades on Euronext" as though that were the only place it trades, when in Europe a large share of its volume is elsewhere. (3) Treating the trade date as the ownership date, which quietly produces the wrong answer to every question about dividends, borrow and short selling.

Where this goes next

How the individual names are assembled into the benchmarks everyone quotes is indices; what happens when a company splits, merges or pays out is corporate actions. The mechanics of being short the market described above are short selling and stock borrow and lending, and the cross-border version of a listing is ADRs and GDRs.

10.2 Indices (Construction, Weighting, Rebalancing)

trading · structuring · sales | core | equity

KEY IDEA

An index is **a rule, not a portfolio**. It is a published recipe with four parts: which securities are in it, how they are weighted, when and how that is revisited, and what happens to dividends and corporate actions. Everything interesting about indices, and every interview question about them, comes from one of those four choices. Once you see an index as a rule, "what happens when a constituent splits" stops being a fact to memorise and becomes a question you can answer.

Weighting, which is the choice that matters most

Free-float market capitalisation is the default. Each constituent is weighted by its market value, adjusted to exclude shares that are not actually available to trade: government stakes, founder holdings, cross holdings. The free-float adjustment matters because an index is meant to be replicable, and a tracker cannot buy shares nobody will sell.

Price weighting weights by share price alone, which is an accident of history rather than a design: the Dow and the Nikkei do it. A stock trading at 500 has ten times the influence of one at 50 regardless of the companies' sizes, and a share split changes the index's composition without anything economic happening.

Equal weighting gives every constituent the same weight. It tilts toward smaller companies, and because prices drift it requires constant rebalancing to stay equal, so its turnover and costs are much higher.

Factor and fundamental weighting uses something else entirely, earnings, volatility, dividends. These are the systematic indices the question below is about.

REAL INTERVIEW QUESTION (Société Générale · QIS)

What is a market-cap index, and how do you construct the weights for such an index?

WHAT THEY TEST / ANSWER

What it is: an index in which each constituent's weight is proportional to its market value, so the index tracks the aggregate value of the companies in it.

The weights: for constituent i with price P_i, shares outstanding N_i and a free-float factor f_i,

$$w_i = \frac{f_i N_i P_i}{\sum_j f_j N_j P_j}.$$

Say the free-float factor out loud, because omitting it is the most common incomplete answer and it is the part with economic content: the index is meant to be investable, so it counts only the shares that genuinely trade. Most providers also cap individual weights, so that a single very large company cannot dominate an index that funds are required to track under diversification rules.

Then the property that makes it the default, and it is the answer to "why is market cap so popular": a market-cap index is **self-rebalancing**. If a stock doubles, its weight in the index rises automatically and so does its weight in a portfolio holding it, so a tracker needs to trade nothing. Every other weighting scheme drifts away from its target and must be traded back, which costs money. Market cap is the only weighting a passive fund can hold with essentially zero turnover, and that is why the entire index-fund industry is built on it.

The trade-off, if they ask: it is momentum by construction. You own most of whatever has gone up most, which is a feature in a trending market and the reason index concentration rises before it falls.

The divisor, which absorbs everything else

An index is not simply a sum. It is

$$\text{Index} = \frac{\sum_i f_i N_i P_i}{D},$$

and the **divisor** D is the mechanism that keeps the index continuous when the rule changes something. Whenever a constituent is added or removed, shares are issued, or a corporate action alters a price without altering value, the divisor is reset so that the index level is *unchanged at that instant.*

WORKED EXAMPLE

Three constituents, each with a free-float market value of 100, so the total is 300. If the index is to read 100, the divisor is $300/100 = 3$.

Now a constituent worth 100 is replaced by one worth 150. The new total is 350. Left alone the index would jump from 100 to $350/3 = 116.7$ for no economic reason, so the divisor is reset to $350/100 = \mathbf{3.5}$ and the index still reads 100. From the next tick it moves with the new basket.

The same mechanism handles a share split: the price halves and the share count doubles, the market value is unchanged, and nothing needs adjusting at all. But in a *price-weighted* index the split halves the constituent's contribution, so there the divisor must move. That difference is a good test of whether someone understands the divisor or has just heard of it.

Dividends, and the three versions of every index

Every index exists in three variants, and confusing them is expensive.

- **Price return**: dividends are ignored. The index falls mechanically on each ex-date. This is the variant nearly every headline quotes and nearly every option and future references.
- **Total return (gross)**: dividends are reinvested at the gross amount.
- **Net return**: dividends are reinvested net of a standard withholding tax assumption, so it sits between the two.

Which variant a derivative references is not a detail. A long futures or long call position on a price index is short the dividends, because the forward is priced off them and a higher dividend lowers it, which is futures and dividends.

Rebalancing

REAL INTERVIEW QUESTION (Macquarie · QIS)

How is a systematic index constructed: selection rules, weighting and rebalancing?

WHAT THEY TEST / ANSWER

The question hands you the three headings, so use them, and add the fourth that it omits.

Selection. Define an eligible universe (listing, country, sector, free-float and liquidity minima), then a rule that picks from it. The rule must be mechanical and published in advance: a systematic index differs from an actively managed fund precisely in that no discretion is exercised at the selection date. State the liquidity screen explicitly, because an index nobody can replicate is not an index.

Weighting. Market cap, equal, or by the factor the index exists to express, with caps on individual weights and often on sector weights. Say what problem the cap solves: concentration limits in the funds that will track it.

Rebalancing. Frequency (quarterly is the norm), the observation date, the implementation date, and the gap between them. Two design points worth volunteering: buffer rules, which stop a constituent near the boundary from entering and leaving every quarter and generating turnover for nothing; and staggered implementation, which spreads the trade over several days rather than concentrating it in one close.

The fourth heading, which the question left out: the treatment of dividends and corporate actions, plus the divisor mechanics that keep the index continuous through them. An index rule that does not specify these is incomplete, and it is where most of the real complexity in an index methodology document lives.

Close on the point that makes this an interview answer rather than a recitation: every rule is a trade-off against **replicability**. More frequent rebalancing tracks the concept better and costs more to follow, narrower selection makes a purer index and a less liquid one. A systematic index is designed backwards from what a fund can actually trade.

The other consequence of a published rule is that everyone knows in advance what will be bought. Index funds must trade at the rebalance, so the trade is predictable, large and concentrated in one moment, which is the classic **index effect**: added names rise before inclusion and give some of it back afterwards. The execution problem is microstructure, and it is why staggering and buffers exist.

Decrement indices

REAL INTERVIEW QUESTION (Barclays · EQD)

What is a decrement index? In what economic context would it be harmful for an investor to hold a decrement index? (detail what happens in a falling market and in a rising market)

WHAT THEY TEST / ANSWER

What it is: an index built from the *total return* version, with a fixed synthetic dividend subtracted continuously, either as a percentage per year or as a fixed number of index points per year. The investor gets the index's performance with dividends reinvested, minus a predetermined amount.

Why they exist, which you should say before the drawback: they remove dividend risk from the product. A desk selling an option on an ordinary index is short the actual dividends and must hedge them out to five years or more in instruments that stop being liquid, as hedging structured products sets out. On a decrement index the dividend is a contractual constant, so that risk disappears, the reserve against it disappears, and the desk can quote a better coupon or a higher participation. That improvement is real and it is paid for by the investor taking the dividend risk instead.

When it hurts the investor: whenever realised dividends come in **below the decrement**. Compare it with the ordinary price index: the price index drops by the actual dividend, the decrement index drops by the fixed one, so the investor is long actual dividends and short the fixed decrement. A decrement of 5% against a market yielding 3% costs the investor 2% a year against simply holding the price index.

The two market states the question asks for.

- **Falling market**: the worst case, and not only for the obvious reason. Companies cut dividends in a downturn, so realised dividends fall while the decrement keeps being deducted in full. The investor takes the market loss *and* the dividend shortfall, exactly when they can least afford it. This correlation is the real complaint about the structure.
- **Rising market**: much less harmful, and possibly not harmful at all. Dividends tend to grow with earnings, so realised dividends can catch up with or exceed the decrement, and the drag is a smaller fraction of a larger gain.

So the honest summary: a decrement index converts an uncertain dividend into a certain one, which is genuinely valuable to both sides, and it hands the investor a short dividend position that is correlated with everything else going wrong. Whether it is a good trade depends entirely on whether the decrement is set above or below what the market will actually pay.

REAL INTERVIEW QUESTION (Morgan Stanley · Structured Products)

What is the difference between a points index and a percentage index?

WHAT THEY TEST / ANSWER

Both are decrement indices; they differ in whether the deduction scales with the index level, and the whole answer is in what that does when the market moves.

Percentage decrement: subtract a fixed *percentage* of the level each year, say 5%. The drag is proportional, so it is 5% of whatever the index is worth, always.

Points decrement: subtract a fixed *number of index points*, say 5 points on an index that started at 100. The drag is constant in absolute terms, so as a percentage it moves inversely with the level.

Make it concrete, because the numbers land the point immediately. Take both calibrated to be identical at inception, an index at 100 with either a 5% or a 5-point decrement:

- Index at **100**: 5% costs 5 points, and 5 points costs 5%. Identical.
- Index at **50**: 5% costs 2.5 points, but 5 points costs **10%**. The points version is twice as heavy.
- Index at **200**: 5% costs 10 points, but 5 points costs **2.5%**. The points version is half as heavy.

So a **points decrement is the more punitive structure in a falling market and the more generous one in a rising market**, while a percentage decrement is neutral to the level. Since the products built on these indices are typically ones where the investor is already short the downside, a points decrement stacks another downside exposure on top of it, and that is the sentence the question is looking for.

If they push on which is more common, the honest answer is that both are used and the choice is negotiated per product: the two are calibrated to agree on day one, so what is really being chosen is who benefits from the divergence afterwards.

INTERVIEW TRAP

1. **Treating an index as a portfolio.** It is a rule, and every question about it is a question about the rule.
2. **Forgetting the free float.** Weights use tradeable shares, not shares outstanding, and the difference is large for state-controlled and founder-controlled companies.
3. **Confusing price, total and net return.** Almost every listed derivative references the price version, so a call holder is short dividends whether they meant to be or not (and a put holder is long them).
4. **Assuming a split changes a market-cap index.** It does not. It changes a price-weighted one, which is a good way to show you understand the divisor.
5. **Reading a decrement as a fee.** It is a synthetic dividend, so what matters is its level relative to what the market will actually pay, not its size.
6. **Treating points and percentage decrements as equivalent.** They agree only at the inception level and diverge in both directions after that.

VISUAL / CODE TO BUILD: Interactive

An index construction sandbox. The reader is given a small universe of companies with prices, share counts and free floats, and chooses a weighting scheme; the resulting index is computed and plotted, and the weights are drawn as a bar chart that updates as prices move, so the market-cap version visibly stays on target while the equal-weighted one drifts and has to be traded back. Corporate action buttons apply a split, a rights issue and a constituent replacement, showing the divisor recomputing and the index level staying continuous through each one. A second panel overlays the price, total return and decrement versions of the same index on one chart, with sliders for the realised dividend yield and the decrement in both points and percentage form, so the reader can find the yield at which the decrement index crosses the price index and watch that crossing move as the level changes.

10.3 Corporate actions

sales · trading · structuring | core | equity

Every other section of this book assumes the underlying is a fixed, well-behaved object called "the share". It is not. Companies split their shares, merge with each other, hand out free shares, sell new ones cheaply to existing holders, spin off divisions, buy their own stock back, and occasionally stop existing. Each of those events is a **corporate action**, and each one changes the thing your contract points at while the contract itself sits unchanged in the book.

That is why corporate actions get their own section rather than a footnote. If the adjustment is wrong, nobody sees a trade, nobody sees a market move, and yet the desk is silently long or short something it never chose to trade. This is also close to the top of the list of things that go wrong in real life, which is why interviewers on equity derivatives and Delta One desks like the topic: it separates candidates who have only priced options from candidates who have imagined owning one for a year.

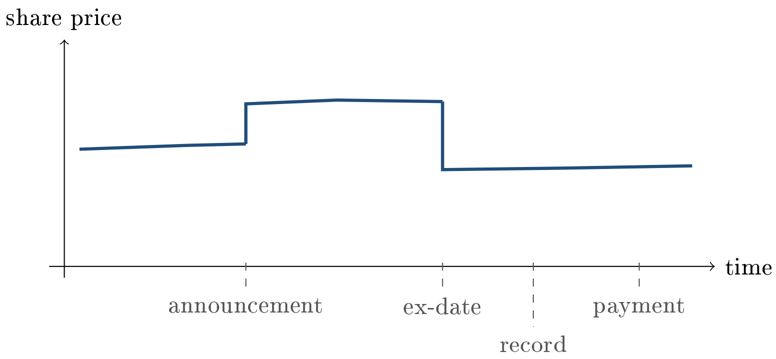
The calendar: four dates, one that matters

A corporate action has a life cycle, and the vocabulary is fixed.

- **Announcement date**: the company tells the market. Prices move here, because this is when the information arrives.
- **Ex-date**: the first day the share trades *without* the entitlement. Buy on or after this day and you do not get the dividend, the free share, or the right. The quoted price drops mechanically to reflect what has left the share.
- **Record date**: the day the company reads its share register to see who is entitled. To appear on that register you must have *settled* by then, so the interval back to the ex-date is one business day under T+2 settlement, and zero under T+1, where the ex-date and the record date fall on the same day. Either way it is the ex-date, not the record date, that drives the price.
- **Payment or effective date**: cash arrives, or new shares appear.

KEY IDEA

The **ex-date** is the only one of the four that moves the quoted price for mechanical rather than informational reasons. On the announcement date the price moves because opinions changed. On the ex-date the price falls (or is restated) because value physically left the share, and no trader made that happen. Every contract adjustment is built to be neutral *across* the ex-date.



quoted share price: an informational jump at announcement, a mechanical drop at the ex-date

A taxonomy you can recite

Sort corporate actions by what they do to the value of what you hold. This is the classification that tells you immediately whether a contract needs adjusting.

1. Purely cosmetic (no value transfer, units change). A **stock split** ("2 for 1": each share becomes two, each worth half) and a **reverse split** (ten shares become one, worth ten times as much) do nothing whatsoever to anybody's wealth. Neither does a **bonus issue** or **scrip issue**, where the company hands existing holders free shares out of reserves: more slices, same cake. Companies do this to keep the share price inside a range that retail investors and option markets find convenient. Contracts must be re-expressed in the new units, and that is all.

2. Value leaves the share (holders are compensated). An ordinary **cash dividend**, a **special dividend**, a **return of capital**, a **rights issue** (existing holders may buy new shares below market price), a **spin-off** (a division is separated and its shares are handed to holders). Here the share price drops on the ex-date and the holder receives something in return. Derivative holders received nothing, so the contract must be adjusted, with one important exception discussed below.

3. The share itself changes or disappears. **Mergers and takeovers** (paid in cash, in stock, or a mix), **delisting**, **nationalisation**, **bankruptcy**. Contracts are not merely rescaled; they are converted, settled early, or, in the ugliest cases, left pointing at something illiquid.

4. The company transacts in its own shares. **Buybacks** and **share issuance**. These are ordinary market flows, not adjustment events: a buyback is the company acting as a large buyer, which matters for borrow and liquidity but does not touch the terms of any contract.

A second, orthogonal axis is who decides. **Mandatory** actions happen to you (splits, ordinary dividends). **Voluntary** actions require you to elect (tender offers). **Mandatory with choice** gives you a default plus options (scrip dividend: take cash or take shares). Operationally, voluntary events are the dangerous ones, since a missed election deadline is a permanent, unhedgeable loss.

The adjustment principle

There is only one idea behind every adjustment formula, and if you can state it you can rederive any of them under pressure.

INTUITION

Imagine freezing the world one second before the ex-date and one second after. The company is worth the same in both frames; only the accounting units changed, or some value moved from the share into the holder's pocket. Someone holding a derivative was not in the room for that transfer. So the exchange rewrites the contract terms with exactly the ratio that leaves that person's economic position identical across the freeze. An adjustment is a change of units, never a gift and never a haircut.

Formally, define the **adjustment ratio**

$$R \;=\; \frac{\text{theoretical price immediately after the event}}{\text{price immediately before the event}},$$

and then apply

$$K_{\text{new}} = R \cdot K_{\text{old}}, \qquad N_{\text{new}} = \frac{N_{\text{old}}}{R},$$

where K is the strike (or barrier, or autocall trigger, or coupon threshold) and N is the number of underlying shares per contract. Strike down, size up, by the same factor, so that $N \cdot (S - K)$ is untouched. The same R rescales every level written into the contract, which is the part candidates forget: a barrier and an autocall observation level are strikes too.

WORKED EXAMPLE

A 2-for-1 split. A stock trades at €80. You are long a call struck at €80 on 100 shares.

After the split each share becomes two, so the theoretical price is €40 and $R = 40/80 = 0.5$. The contract becomes a call struck at $0.5 \times 80 =$ €40 on $100/0.5 = 200$ shares.

Check it. Suppose the stock later trades at €50 post-split, which is €100 in old money. Old contract: $100 \times (100 - 80) =$ €2,000. New contract: $200 \times (50 - 40) =$ €2,000. Identical, as required. A 2-for-1 split has no effect on your PnL, on your delta in cash terms, or on your vega. Only the units moved.

REAL INTERVIEW QUESTION (BNP · EQD)

What is a stock split, and what is the effect of a 2-for-1 split on your P&L?

WHAT THEY TEST / ANSWER

The answer is "none", and the whole test is whether you say it confidently and then prove it. A split multiplies the share count and divides the price by the same factor, so market capitalisation, your position value, and everybody's wealth are unchanged. Options are adjusted by halving the strike and doubling the contract size, so payoffs match to the euro. Then add the two remarks that show desk experience. First, the effect is not exactly nil in practice: a lower price often means better liquidity and a finer tick relative to the price, which slightly tightens spreads, and options market makers get a denser strike ladder to work with. Second, the real danger is the *data*: if your price history is not split-adjusted, the ex-date shows a fake -50% return, which poisons any realized volatility or correlation you compute from it (see implied versus realized volatility). Interviewers on Delta One desks are usually fishing for exactly that second point.

Rights issues and the theoretical ex-rights price

The rights issue is the adjustment every interviewer reaches for when they want to see whether you understood the principle or memorised the split case. The company offers existing holders the right to buy new shares below the market price. That discount is real value handed to holders, so the share price must fall, and the level it falls to has a name: the **theoretical ex-rights price**, or TERP.

Write N_o for the number of old shares at the cum-rights price P_c, and N_n for the number of new shares issued at the subscription price P_s. Then

$$\text{TERP} = \frac{N_o P_c + N_n P_s}{N_o + N_n}, \qquad R = \frac{\text{TERP}}{P_c}.$$

It is just a weighted average: the company is worth what it was worth plus the cash it just raised, spread over more shares.

WORKED EXAMPLE

A 1-for-4 rights issue. A company has 100m shares trading at €20, so it is worth €2,000m. It offers one new share for every four held, at a subscription price of €10.

Holders subscribe 25m new shares and pay in $25\text{m} \times 10 = €250\text{m}$. The company is now worth $2{,}000 + 250 = €2{,}250\text{m}$ across 125m shares, so

$$\text{TERP} = \frac{2{,}250}{125} = €18.00, \qquad R = \frac{18}{20} = 0.9.$$

The share "falls" from €20 to €18 on the ex-date and nobody lost anything: each old share now carries €2 of value in the form of the right itself, which trades separately. The standard shortcut confirms it: the value of one right per existing share is $(20 - 10)/(4 + 1) = €2$.

A call struck at €20 on 100 shares becomes a call struck at $0.9 \times 20 = €18$ on $100/0.9 = 111.\overline{1}$ shares. Test it at a later price of €27, which is €30 in pre-issue money since $30 \times 0.9 = 27$: old basis $100 \times (30 - 20) = €1{,}000$, new basis $\frac{100}{0.9} \times (27 - 18) = \frac{100 \times 9}{0.9} = €1{,}000$. Neutral.

In practice exchanges round the adjusted terms according to their rulebook, so the neutrality holds up to a rounding convention rather than exactly. Know that the rounding exists; do not try to quote a specific exchange's rule from memory.

KEY IDEA

Ordinary dividends are the exception, and it is not arbitrary. Listed option contracts are generally *not* adjusted for ordinary, expected dividends, because those dividends were already inside the forward price used to value the option when it was traded. Adjusting for them would pay the holder twice. Adjustments are triggered by the *unexpected* part: special dividends, returns of capital, and dividends above a threshold set in the rulebook. The valuation of ordinary dividends lives in dividends, and the forward that embeds them in forwards.

Mergers, takeovers and delisting

A takeover is where the tidy ratio logic runs out, because the underlying stops being a random variable.

In an all-**cash** offer at a fixed price, a successful bid converts each share into a known amount of cash. From the moment the outcome is close to certain, the share trades in a narrow band just under the offer price, and its volatility collapses. Options on it lose almost all their time value: an out-of-the-money call struck above the bid becomes close to worthless whatever its remaining maturity, and an in-the-money option becomes an almost deterministic cash claim. Exchanges typically convert outstanding contracts into a right on the cash amount, or accelerate them. The trap for a long-volatility book is that the position was long vega and is now long nothing at all.

In an all-**stock** offer, shares of the target are exchanged for shares of the acquirer at a fixed exchange ratio. Contracts are then re-expressed on the acquirer's stock through that ratio, which is the adjustment principle again with R read off the offer terms rather than off a price.

Delisting and bankruptcy are the genuinely unpleasant cases. There may be no reliable price to settle against, borrow can be recalled without warning, and what happens next is governed by contract documentation rather than by a formula. The lesson for a candidate is to know the boundary of the mechanical answer: adjustment ratios handle events that preserve a tradable underlying, and everything else is a legal and operational question.

Why this lands on structured products hardest

A vanilla option has one strike. An autocallable has an autocall trigger, a coupon barrier and a capital-protection barrier, each observed on specific dates, often on several underlyings at once. Every one of those levels is a strike for adjustment purposes, and every underlying can have its own corporate action calendar. This is why the operational risk in structured products is concentrated here rather than in the pricing, and why the topic recurs in autocallables and in PnL explain, where an unexplained residual on an ex-date is almost always a missed adjustment.

The Covid period gave the market a live demonstration of exactly the boundary this section is about. Under regulatory and political pressure in 2020, European banks and many corporates cancelled or deferred dividends the market had treated as near certain. The corporate-actions point is narrow and it is the one to carry away: those were *ordinary* dividends, which are precisely the ones a rulebook does not adjust for, so no contract was rewritten and no adjustment ratio rescued anybody. The loss landed in full on whoever held the dividend exposure, with no mechanical relief. Who that was, and why, is developed in dividends.

REAL INTERVIEW QUESTION (Barclays · Structured Products)

When you are long an autocall, are you long or short dividend?

WHAT THEY TEST / ANSWER

Short, in one line: dividends pull the forward down, a lower forward makes the autocall trigger harder to reach and the downside barrier easier to breach, and both legs hurt the investor. Equivalently, the investor is short a down-and-in put and a long put is long dividend. The full derivation, with the sensitivity priced out on a worked note, belongs to autocallables.

What this section adds is the mirror and its consequence. The **issuer is long** forward dividends, in size, accumulated across years of issuance, and that is the position that lost money in 2020 when distributions were cancelled. The reason it hurt *here* rather than somewhere else is a corporate-actions reason, and it is the one to give on this desk: an ordinary dividend cut triggers **no contract adjustment**. Unlike a split or a rights issue there is no ratio that restores anybody's economics, so nothing in the rulebook absorbs any part of the move. It lands in full on whoever holds the exposure. Add that the exposure sits in long-dated dividend expectations, where dividend futures and swaps are thinnest, so the hedge is most expensive exactly when it is needed. If they push on the valuation side rather than the event side, that is dividends.

VISUAL / CODE TO BUILD: Interactive

A corporate action sandbox. The reader picks an event from a menu (2-for-1 split, 1-for-4 rights issue at a chosen subscription price, special dividend of a chosen size, cash takeover at a chosen premium) and a starting position (long stock, long call, long autocallable with a trigger and a barrier). The panel then shows, side by side, the pre-event and post-event contract terms, the computed ratio R, and a payoff diagram of the position in both frames overlaid so the reader can see them coincide. A deliberate "break it" toggle applies the event to the price but *not* to the contract, so the payoff curves visibly separate and a running counter shows the euro value of the missed adjustment. Backed by a small Python adjustment engine; expose the terms table for accessibility.

INTERVIEW TRAP

The five that cost money or offers. (1) **Unadjusted price history.** A split shows up as a -50% return in raw close prices, which wrecks realized volatility, betas and correlations. Always ask whether a series is adjusted, and for what. (2) **Adjusting for ordinary dividends.** They are already in the forward. Adjusting again pays the holder twice, and saying so out loud is one of the fastest ways to look like you have read a rulebook. (3) **Adjusting the strike and forgetting the barrier.** Every level in the contract takes the same ratio, including autocall triggers and coupon thresholds. (4) **Confusing the record date with the ex-date.** The price moves on the ex-date. The record date only determines who is on the register. (5) **Assuming a corporate action is PnL-neutral for your *hedge*.** The contract adjustment is neutral, but your delta is now expressed on a different number of shares, and if you do not re-express the hedge you are directionally exposed the next morning.

Where this goes next

The market plumbing these events flow through is equity market structure. The index-level version, where an index committee rather than a company forces the change, is indices. Valuing the dividend stream itself, rather than adjusting for it, is dividends. The reason a corporate action can make a borrow disappear is in stock borrow and lending. And the products where a missed adjustment does the most damage are autocallables.

10.4 Dividends

trading · structuring · sales | core | equity

To an investor a dividend is income. To an equity derivatives desk it is a scheduled leak of value out of the underlying, forecast years ahead, hedged, traded as its own asset class, and responsible for some of the largest losses the business has taken. It is also, by volume, one of the most asked topics in the interview database. This section is about the derivatives view.

The mechanical fact that everything rests on, that the share price drops by approximately the dividend on the ex-date, is established in equities and is not repeated here. What follows is what that fact does to a book.

Dividends live in the forward

An option holder does not own the shares, so they do not receive the dividends, but the price they will pay or receive at expiry is affected by every dividend paid before then. That enters through the forward. With a continuous dividend yield q and rate r,

$$F = S\,e^{(r-q)T},$$

and with discrete known cash dividends,

$$F = \big(S - \mathrm{PV}(\text{dividends})\big)\,e^{rT}.$$

Dividends push the forward *down*. Everything else in this section is a consequence of that sentence, and the forward itself is developed in forwards.

KEY IDEA

Three modelling conventions coexist and they are not interchangeable. **Continuous yield** treats dividends as a constant proportional leak, which is convenient, wrong in detail, and fine for an index over a long horizon. **Discrete cash** places known amounts on known dates, which is what a desk uses for single names over the next year or two, and it matters because a fixed cash amount is a larger proportional hit when the stock has fallen. **Proportional discrete** places dividends as a percentage of the prevailing spot on each date, which avoids the absurdity of a cash dividend exceeding the share price in a deep sell-off. Real books use cash for the near dates, where dividends are announced or reliably forecast, and drift towards yield or proportional for the far ones. Being asked "cash or yield?" and having an answer is a genuine signal of desk exposure.

Which way does it move, and by how much

REAL INTERVIEW QUESTION (BNP · Structured Products)

When I am long a call, am I long or short dividends?

WHAT THEY TEST / ANSWER

Short dividends, because you are long the forward and dividends lower it. Give the chain in one breath: a call pays off against the forward, a dividend takes cash out of the company and therefore out of the share price on the ex-date, so a higher dividend means a lower expected price at expiry, a lower payoff, and a cheaper call. If the expected dividend rises, your call loses value, and a position that loses when dividends rise is short dividends.

Then complete the set, because the follow-up is always one of the others. Long a **put** is *long* dividends, by the same argument in reverse. Long the **stock** is long dividends, since you actually receive them, which is the distinction the question is really probing: the option holder bears the effect of the dividend without receiving it. And long an **autocall** is short dividends, because higher dividends lower the forward, which makes early redemption less likely and the embedded short put more likely to bite, in autocallables.

Close with the size, because the sign alone is half an answer. The sensitivity of a call to the dividend is roughly $-\Delta$ times the present value of the dividends over its life, so it is largest for a deep in-the-money option, whose delta is near one, and smaller for an out-of-the-money one. That is the opposite ordering to vega, and it is the point of the next question.

REAL INTERVIEW QUESTION (Société Générale · EQD)

We have an at-the-money call. How does its price move if the company announces dividends 10% lower than expected?

WHAT THEY TEST / ANSWER

Up, and you should be able to size it in your head. Lower dividends mean a higher forward, and a call is long the forward.

Put numbers on it: spot 100, one year, 25% volatility, zero rates, an expected dividend yield of 3%. The call is worth 8.39. Cut the dividend by 10%, so the yield becomes 2.7%, and it is worth 8.54: a gain of 0.15, about 1.75% of the option's value for a 0.3% of spot change in the forward. The mental version is the first-order estimate, $\Delta\times$ the change in the present value of dividends, which with $\Delta \approx 0.5$ and a change of about 0.3 gives roughly 0.15, matching.

Then the two remarks that make it an answer rather than a calculation. This is an *announcement*, so a second effect runs alongside the mechanical one: a dividend cut is usually bad news about the business, which can move the spot and the implied volatility, and the question of which effect dominates is empirical rather than mechanical. Say which you are answering. And note that the desk's exposure is to the *expected* dividend, not the paid one: the mark moves when the forecast changes, which is often months before any cash moves, so a dividend book is revalued off a forecast curve that itself trades.

REAL INTERVIEW QUESTION (Barclays · Structured Products)

Compare the relative effect of dividends and volatility on two deep in-the-money calls.

WHAT THEY TEST / ANSWER

Dividends win, and the reason is that the two sensitivities have opposite profiles across moneyness.

Deep in the money, the call is almost a forward: delta approaches one, so the dividend sensitivity, which is roughly $-\Delta$ times the present value of dividends, approaches its maximum. Meanwhile vega collapses, because there is almost no probability of the option ending out of the money, so extra volatility changes almost nothing.

The numbers are in the example below: on a strike of 50 the dividend effect is roughly fifty times the volatility effect, and this option's vega is about sixty-six times smaller than the at-the-money vega.

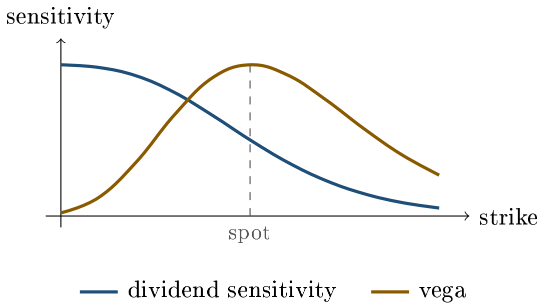
Then say the general rule they are testing: **deep in-the-money options are dividend and rate instruments, not volatility instruments**, which is why a desk holding in-the-money strikes worries about the dividend forecast and the financing rather than about the volatility surface, and why in-the-money structures are a common place for dividend risk to accumulate unnoticed.

WORKED EXAMPLE

Spot 100, one year, 25% volatility, zero rates. Compare a deep in-the-money call struck at 50 with an at-the-money one. The volatility rows are measured with no dividend, so that each column isolates one effect at a time.

	Strike 50	Strike 100
Value with no dividends	50.01	9.95
Value with a 3% dividend yield	47.07	8.39
Effect of the dividend	−2.95	−1.55
Effect of raising volatility 25% to 30%	+0.06	+1.98
Vega per volatility point	0.006	0.396

On the deep in-the-money strike the dividend effect is about fifty times the effect of five whole volatility points, and its vega is sixty-six times smaller than the at-the-money vega. At the money the ordering reverses. The two sensitivities are not just different sizes, they have *opposite shapes across moneyness*, which is what the figure below draws.



each curve normalised to its own maximum; one rises monotonically into the money, the other peaks at the money and decays both ways

REAL INTERVIEW QUESTION (Morgan Stanley · QIS)

What is the price of a call on an underlying that pays 100% dividends?

WHAT THEY TEST / ANSWER

Zero, and the point is to reason to it rather than guess.

If the company distributes its entire value as dividends before the option expires, the share price falls to zero on the ex-date and stays there, so the payoff $\max(S_T - K, 0)$ is zero for any positive strike. In continuous-yield notation the same statement is $F = Se^{(r-q)T} \to 0$ as q grows, and the call price falls with it: on a one-year at-the-money call at 25% volatility with zero rates, a yield of 10% gives 5.46, 50% gives 0.16, and 100% gives essentially zero.

Then give the two remarks that show it is not a trick question. This is the limiting case of a real phenomenon: a large **special dividend** genuinely can be a substantial fraction of the share price, which is why exchanges adjust option strikes for extraordinary distributions rather than letting the holder absorb the drop, covered in corporate actions. And the answer flips for a **put**, which becomes worth its full discounted strike, since the holder ends up with the right to sell a worthless share. Contract adjustment rules exist precisely so that neither of those outcomes falls on a holder by accident.

Dividends as a traded asset

Because the risk is unavoidable for anyone selling equity derivatives, a market exists in the dividends themselves.

REAL INTERVIEW QUESTION (Goldman Sachs · EQD)

How do dividend swaps function, what purpose do they serve, and why would an investor engage in one?

WHAT THEY TEST / ANSWER

Describe the contract, then the flow that created the market, then the two sides.

The contract. A dividend swap exchanges a fixed amount, agreed today, against the *realised* dividends actually paid by an index or a single name over a defined period, usually a calendar year. Nothing is exchanged upfront and the notional is expressed in dividend points. The listed cousin is the **dividend future**, which is the same exposure in exchange-traded form and is the more liquid instrument on major indices.

Why the market exists. It exists because the sell side is structurally **long** dividend risk. Banks issue enormous quantities of structured products in which clients are effectively **long** the forward, and being long the forward is being short dividends, so the bank takes the other side and is left holding long dividend exposure it did not choose. Dividend swaps and futures are how that inventory is laid off, so the market's shape is a consequence of the retail structured-products business rather than of investor demand.

The two sides. A **buyer** of dividends takes a view that companies will pay more than the forward implies, gets an exposure with no direct equity delta, and finds it attractive because dividends are historically less volatile than prices, so it is a way to be long equity fundamentals without being long the market. A **seller**, typically the dealer, is hedging. Add the two honest caveats: the implied dividend curve usually trades at a discount to consensus forecasts precisely because of that structural supply, so it is not an unbiased forecast, and the risk is severely asymmetric, since dividends can be cut to zero overnight while they rise only gradually. Which is exactly what happened next.

REAL INTERVIEW QUESTION (Goldman Sachs · Structured Products)

What was the problem for exotic traders during the Covid period? Dividends.

WHAT THEY TEST / ANSWER

They were long dividends, in size, and the dividends were cancelled outright.

The mechanism, stated in order. Years of autocall and structured-product issuance had left dealers structurally long future dividends, hedged where possible with dividend futures and swaps. In the first months of 2020 European companies suspended or cancelled distributions, in many cases under regulatory pressure on banks and insurers rather than because each company had chosen to, so the cancellations arrived together and were not diversified away. Implied dividend levels for the following years collapsed, and the hedges did not help because the hedges were the same exposure.

The compounding effects are what make it a good answer. The losses arrived at the same time as a volatility spike and a delta-hedging crisis, so the same books were losing on several axes at once. Cancellation was a **regulatory and political** event rather than an economic one, which no dividend model built on payout ratios and earnings was calibrated for. And the risk turned out to be far more correlated across names than assumed, which is the general lesson: a risk that is diversified in normal times and common in a crisis is the one that breaks a book.

Close with what changed afterwards, because that is what a structuring interviewer wants: dividend risk is now marked and limited far more conservatively, structures that concentrate it are priced with an explicit charge, and the episode is the standard example of a hedge that fails because it is the same trade as the exposure. See embedded optionality.

INTERVIEW TRAP

Four errors. (1) Assuming a call is unaffected by dividends because its holder does not receive them: they do not receive them, which is exactly why they suffer from them through the forward. (2) Using a continuous yield for a single name with a large discrete dividend just before expiry, which misprices the option and gets the early exercise decision on an American call wrong. (3) Treating the implied dividend curve as a forecast, when it carries a structural discount from dealer hedging supply. (4) Forgetting that the exposure is to the *expected* dividend, so a book moves on a forecast revision with no cash having been paid.

Where this goes next

The ex-date mechanics and the equity view are in equities, the adjustments for extraordinary distributions in corporate actions, and the forward that carries the effect in forwards. The other component of that forward is equity financing, and the products where dividend risk accumulates are autocallables and embedded optionality.

10.5 Short Selling

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Selling something you do not own sounds like a trick and is a piece of plumbing. This section is the plumbing, its economics and its asymmetries. Where the shares come from is stock borrow and lending, and the same arrangement seen from the cash side is repo.

KEY IDEA

A short sale is not the mirror image of a long position, and treating it as one is the mistake everything else in this section is about. Three asymmetries: the loss is **unbounded** while the gain is capped at 100%; the position **grows as it moves against you**, since a rising price means a larger short exposure, whereas a falling long shrinks; and you are **dependent on a lender** who can ask for the shares back at the worst possible moment. A long position has none of these properties.

REAL INTERVIEW QUESTION (Société Générale · Volatility)

What is short selling?

WHAT THEY TEST / ANSWER

Give the mechanic, the economics and the risk, in that order. The mechanic is thirty seconds and everyone has it; the other two are where the answer is won.

The mechanic. You borrow the shares from a holder, sell them in the market, and hold the cash. Later you buy the shares back and return them. If the price fell, you buy back for less than you sold and keep the difference. You never owned the shares at any point; you owed them.

The economics, which is the part usually skipped. Three cash flows sit alongside the price move. You pay a **borrow fee** to the lender, quoted per annum on the value of the loan, small for a liquid name and very large for a crowded one. You post **collateral**, usually the sale proceeds, and receive a rebate on it, so in a positive-rate world you earn interest on cash you are holding anyway. And you must make a **manufactured dividend** payment: the lender is entitled to any dividend the shares pay, and since the buyer of the shares receives the real one, you pay the lender an equivalent amount out of your own pocket. A short is therefore short the dividend as well as short the price.

The risks, and naming them is what separates the answer. The loss is unbounded, because there is no ceiling on a share price. The position grows against you as it moves against you. And the borrow is not permanent: the lender can **recall** the shares, and if you cannot find them elsewhere you are **bought in**, meaning your broker closes the position at whatever price the market offers, which is precisely the moment you would least choose.

Then close on the point an interviewer wants to hear: most short selling is not a bet that a company will fail. It is hedging. A market maker who has sold a call is short delta and buys stock; one who has sold a put is long delta and *sells* it short, as hedging sets out. Convertible arbitrage, index arbitrage and every delta hedge on the street rests on being able to borrow shares, which is why a ban on short selling widens option spreads within hours.

WORKED EXAMPLE

A short, priced. Sell 100 shares at 50, so you hold 5,000 in proceeds. Hold it for three months, at a borrow fee of 3% a year, and the company pays a dividend of 1.00 per share during that period. Read the 3% as the all-in net cost: the rebate earned on the cash collateral is netted into it rather than shown separately, which keeps the arithmetic to the two frictions that decide the trade.

- **Buy back at 40**: the gross gain is $5{,}000 - 4{,}000 = 1{,}000$. Deduct the borrow fee, $5{,}000 \times 3\% \times \frac{1}{4} = 37.50$, and the manufactured dividend, $100 \times 1.00 = 100$. Net: **862.50**, so the frictions took nearly 14% of a correct call.
- **The stock rallies to 75 instead**: the loss is $5{,}000 - 7{,}500 = -2{,}500$, and the same 137.50 of costs takes it to $-2{,}637.50$. You were wrong by 50% and lost 53% of the proceeds, and there is nothing in the arithmetic to stop it.

The comparison worth drawing: a long position of the same size loses at most 5,000 and pays no fee to hold. The short's carry is negative and its tail is open, which is why short books are run smaller and turned over faster than long ones.

Squeezes, and why they are a funding event

A **short squeeze** is what happens when a crowded short position is forced to cover into a rising market. It is worth understanding as a mechanism rather than as a story, because the mechanism is the same one that appears in liquidity risk: a price rise raises the mark-to-market loss and therefore the margin requirement, which forces buying, which raises the price. The short seller's position size grows at the same time, so the pressure compounds from both directions at once.

Two conditions turn a bad short into a squeeze. **Crowding**, measured as short interest relative to free float or to average daily volume, which tells you how many days of normal trading it would take everyone to get out. And **scarce borrow**, which means recalls arrive at the same time, converting a voluntary decision to cover into an involuntary one.

Regulation, which exists and moves

Short selling is regulated in a way long buying is not, and the specifics matter less than knowing the categories.

- **Naked shorting**, selling without having arranged a borrow, is generally prohibited. The locate requirement is the rule that enforces it.
- **Disclosure** of net short positions above a threshold, to the regulator and above a higher threshold to the market, so large shorts are public.
- **Circuit-breaker rules** restricting shorting in a name that has already fallen sharply on the day.
- **Temporary bans**, imposed on financial stocks in 2008 and on various markets in March 2020. The evidence that they support prices is weak; the evidence that they widen spreads and damage option markets is not.

That last point is the one to have a view on, because it is the natural follow-up. Restricting short selling removes the participants who provide liquidity on the offer side and who hedge derivative books, so the visible effect is wider spreads and thinner markets rather than higher prices. A candidate who can say that, and say why, has understood that short sellers are mostly not speculators.

INTERVIEW TRAP

1. **Treating a short as a negative long.** Unbounded loss, a position that grows against you, and a lender who can recall. None of these has a long-side analogue.
2. **Forgetting the manufactured dividend.** A short pays the dividend, so a high-yield name has a meaningful negative carry before the borrow fee is counted.
3. **Ignoring the borrow cost when it matters.** It is negligible on a large liquid name and can exceed twenty percent a year on a crowded one, which is often exactly the name you wanted to short.
4. **Assuming the borrow is permanent.** Recall risk is highest precisely when the trade is working, because that is when the position is crowded.
5. **Believing short selling is mostly speculation.** Most of it is hedging derivative books and arbitrage, which is why restricting it hits option markets first.

VISUAL / CODE TO BUILD: Interactive

A short position tracker. The reader opens a short at a chosen size and watches a price path, with three panels updating together: the profit and loss, the *position size* in cash terms, and the margin requirement. The intended lesson is visible in the second panel, which grows as the price rises, so the reader sees the exposure increasing exactly when it is hurting, and can compare it against the same chart for a long position where it shrinks. Sliders for the borrow fee and the dividend yield show the carry draining the position over time even when the price does nothing. A squeeze mode raises the borrow fee and fires a recall as the price rises, forcing the reader to cover at the worst level, which is the difference between being wrong and being closed out.

10.6 Stock borrow and lending

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You cannot sell what you do not own, so before anyone shorts a share they have to borrow it from someone who does. That plumbing is invisible in a textbook and unavoidable on a desk: it sets the cost of every short position, it is one of the three components of an equity forward, and it is where a hedged book quietly leaks money when the assumption is wrong.

The transaction, from both ends

REAL INTERVIEW QUESTION (CACEIS · Market Risk)

What is securities lending?

WHAT THEY TEST / ANSWER

Define it as a collateralised loan of a security, not a sale, and then say who is on each side and what each one gets.

A holder of shares, the **lender**, transfers them temporarily to a **borrower** in exchange for collateral, cash or other securities, worth more than the shares, the excess being the **haircut**. The borrower pays a fee, quoted as an annual rate on the value of the loaned stock, and returns equivalent shares later. Legal title passes, which matters: the borrower can sell the shares, and the lender's economic exposure to the price is preserved contractually rather than by ownership.

Who lends and why. Long-term holders with no intention of selling: index funds, pension funds, insurers, custodians acting for them. The fee is incremental revenue on a portfolio they were holding anyway, which is why a physically replicating fund can charge a lower management fee than its costs suggest, in ETFs and index products.

Who borrows and why. Short sellers, market makers who have sold stock they do not hold, arbitrageurs, and desks hedging derivatives. The borrow is what makes the short deliverable at settlement.

Close with the risks, since the question comes from a risk function. **Counterparty risk**, mitigated by collateral marked daily and by the haircut. **Collateral risk**, since the collateral itself can fall in value or be reinvested badly, which is how securities lending caused losses in 2008 for lenders who took cash collateral and reinvested it in instruments that stopped being liquid. **Recall risk**, since most loans are open and the lender can demand the shares back, typically to vote or because they have sold. And the **corporate action and dividend** treatment: the borrower owes the lender a manufactured payment equal to any dividend, and the voting right goes with the shares, which is why lenders routinely recall around contentious votes.

REAL INTERVIEW QUESTION (JP Morgan · Delta One)

What is the point of borrowing stocks?

WHAT THEY TEST / ANSWER

Four reasons, and only the first is the one people think of.

To short, whether as a directional view, as one leg of a relative-value trade, or to hedge a long position elsewhere. Without the borrow, the sale fails at settlement.

To hedge a derivative, which is the reason a bank does most of its borrowing. A desk that has sold a put, or bought a call, or issued almost any structured product, ends up needing to be short delta, and the borrow is what makes that possible. This is the largest and least discussed source of borrow demand.

To settle, covering a fail or a temporary shortfall so a delivery obligation is met on time.

To capture a corporate or tax event, moving shares to a holder who can use a dividend or a right more efficiently, which is a real and heavily scrutinised part of the business.

Then say what makes it expensive, because that is the follow-up: the fee is set by supply and demand for that specific name, so an **easy-to-borrow** large cap costs a few basis points a year while a crowded short in a small float can cost tens of percent, and in a squeeze the constraint becomes availability rather than price, at which point positions are recalled and the short is forced to buy back. That is the mechanism behind every squeeze story, and it is why a borrow assumption is a risk, not a parameter.

Where the borrow shows up: the forward

The borrow fee is not a back-office cost, it sits inside every equity forward and therefore inside every option price. Writing the equity forward with all three components,

$$F = S\, e^{(r - q_{\text{div}} - q_{\text{repo}})\, T},$$

where q_{repo} is the rate the holder of the shares earns by lending them. The economics is an arbitrage: someone holding the stock to deliver a forward funds the purchase at r, receives the dividends, and earns the lending fee, so all three enter the carry with the signs shown. Dividends and the borrow both push the forward *down*, and the rate pushes it up. The derivation is in forwards and the dividend leg in dividends.

KEY IDEA

On a desk the equity "repo rate" means the stock lending rate, and it is not the bond repo rate of repo markets, which is a financing rate against collateral. They share a name and a mechanism and they answer different questions: the bond repo rate is the cost of cash, and the equity repo rate is the price of scarcity in one specific share. Confusing the two is a common interview slip.

The market quotes a **borrow curve** by maturity, and it is usually upward sloping, because a lender guaranteeing availability for a year has given up the option to recall and is paid for it. It can invert sharply around a known event, a vote, an index rebalance, a dividend date or a merger, when short-term demand spikes.

REAL INTERVIEW QUESTION (Commerzbank · EQD)

You are short a call and delta hedged. You priced the call with a repo of 1% but you realise the true repo is 10%. How does your profit and loss evolve, and what do you do?

WHAT THEY TEST / ANSWER

Work out the direction first, because the sign is where candidates fall over.

A higher borrow rate lowers the forward, and a call is long the forward, so the call is worth *less* than you thought. You are short it, so you sold above fair value and the mark moves in your favour. Quantify: spot 100, strike 100, one year, 25% volatility, zero rates and no dividends. At a 1% repo the call is worth 9.41; at 10% it is worth 5.46. You sold at 9.41 something worth 5.46, a gain of about 3.95 per unit.

Confirm it through the hedge, which is the answer they actually want, because the two routes must agree. Being short a call, you delta hedge by *buying* shares. Owning shares, you can lend them and earn the borrow, and the true borrow is nine points higher than assumed. Over the year that extra carry is roughly $\Delta \times S \times 9\%$, around 4.8 at the initial delta of 0.53, in the same direction and of the same order. The two numbers differ because the delta is not constant and because the higher repo itself lowers the delta, from 0.53 to 0.35, which is the second-order detail worth naming.

What you do. Reprice the book off the correct borrow curve rather than a single number, since it is term dependent. Lock the borrow if the position is long dated, because the gain is only real if you can actually lend at 10% for the life of the option, and an open loan can be recalled. Check whether the 10% is a permanent scarcity or an event-driven spike, since the two have completely different implications for a one-year option. And mirror the exercise, since it is the dangerous direction: had you been *long* the call and short the stock, the same discovery would be a loss of comparable size, and you would be paying the higher fee rather than receiving it, with recall risk on top.

The lender's side of the trade

REAL INTERVIEW QUESTION (Amundi · Delta One)

How can securities lending be used to replicate an index?

WHAT THEY TEST / ANSWER

Answer at two levels: what lending does for a physical replicator, and how it can be part of the replication itself.

As a revenue offset. A fund that physically holds the index constituents lends them out and earns fees, which reduces its net tracking difference against the index. It is the reason a fund with a 20 basis point fee can track within 12, and it is why tracking difference rather than the headline fee is the right comparison, as set out in ETFs and index products. Lending revenue is concentrated in the names that are hard to borrow, so it is not spread evenly across the portfolio and a small part of the index can produce most of it.

As a replication tool. Because a borrowed share carries the full economic exposure, lending and borrowing let a manager hold the index exposure without holding every share continuously: names can be lent out and recalled around rebalances, positions can be sourced through the lending market when a constituent is hard to buy, and a combination of borrowed stock and futures can reproduce the index while the physical inventory is optimised for lending revenue.

Then the constraints, which is what a portfolio manager would actually be judged on. Lending introduces counterparty and collateral risk into what the client was told was a passive fund, so most managers cap the proportion on loan and share the revenue with the fund by policy. Loans must be recalled to vote, which creates a governance question the manager has to answer publicly. And recall takes time, so a fund needs the shares back before an index rebalance forces it to sell them, which is an operational constraint rather than a financial one and is the part that actually goes wrong.

INTERVIEW TRAP

Four errors. (1) Treating the borrow as a fixed cost: it is a market price on a specific name, it moves, and it can multiply in days. (2) Confusing the equity repo rate with the bond repo rate, which are different quantities sharing a name. (3) Ignoring recall risk, so a hedge that depends on staying short is priced as though the borrow were guaranteed when most loans are open. (4) Forgetting that the borrower owes a manufactured dividend and gives up the vote, which is where lending interacts with corporate events rather than being purely financial.

Where this goes next

The position this plumbing makes possible is short selling, the wider funding of an equity book is equity financing, and the other leg of the same forward is dividends. The fixed income instrument that shares the name and the mechanism is repo markets, and the funds on the lending side are in ETFs and index products.

10.7 Equity Financing

trading · sales · structuring core equity

Two clients walk onto a Delta One desk. One owns a portfolio of shares and wants cash without selling them. The other wants exposure to those shares without paying for them in full. Equity financing is the business of standing between the two, and it is one of the largest and least discussed activities on an equity floor.

Where the shares themselves come from is stock borrow and lending, the cash-market analogue is repo, and the collateral machinery is collateral and haircuts. What is owned here is the equity financing trade itself.

> **KEY IDEA**
>
> Every financing trade answers the same question twice: **who holds the asset** and **who has the economics**. In a *physical* trade the client keeps the shares and borrows cash against them, so ownership and economics stay together and the lender holds collateral. In a *synthetic* trade the bank buys the shares and passes the economics to the client through a swap, so the client has the exposure and never the asset. The two are economically similar and legally, operationally and prudentially very different, which is why both exist.

The synthetic route: total return swaps

A **total return swap** exchanges the entire economic performance of an asset for a financing payment. The client receiving the total return gets the price appreciation and the dividends, and pays a floating rate plus a spread on the notional. The bank takes the other side and hedges by simply buying the shares, which is why it can quote the trade tightly: its hedge is the underlying itself, and the contract is a swap like any other.

For the client the appeal is leverage and access. They post initial margin rather than the full notional, they gain exposure to markets where they may not have custody or tax status, and they avoid the operational burden of holding hundreds of lines of stock. For the bank, the appeal is a spread earned on a hedged position.

REAL INTERVIEW QUESTION (JP Morgan · Delta One)

What happens if you are long a TRS and the index falls?

WHAT THEY TEST / ANSWER

You lose twice, and saying so in that form is the answer.

Being long a total return swap means *receiving* the total return and *paying* the financing leg. So if the index falls, the return leg is negative and you pay it away: a receiver of a negative return is a payer. On top of that you continue to pay the floating rate plus the spread on the full notional, which does not shrink because the index fell. Your funding cost is unchanged and your asset leg has gone against you.

Then the mechanics that make it a Delta One answer rather than a definition. The loss is settled in cash at the reset, not at maturity, so it arrives as a payment you have to fund. And because the swap is margined, a fall also triggers a **variation margin call** and, if the position has become riskier, a higher initial margin requirement. So the sequence in a sharp fall is: mark-to-market loss, cash out of the door, and a larger collateral demand, all at once, and all while the financing leg keeps running.

Close on the comparison, because it is what the question is set up for: this is exactly the position of a leveraged long in physical shares bought on margin, which is the point. A total return swap is a financing trade wearing a derivative's clothes, so it behaves like leverage in a falling market, and the risk that matters is not the index level but whether you can meet the calls.

REAL INTERVIEW QUESTION (JP Morgan · Delta One)

Are you exposed to interest-rate risk in a TRS?

WHAT THEY TEST / ANSWER

Yes, through two channels, and naming both is the whole answer since most candidates find one.

The funding leg, which is the obvious and usually the larger one. The financing payment is a floating rate plus a spread, so if you receive the total return you pay floating and are hurt when rates rise. That is a genuine interest rate exposure and it can be measured as a DV01 on the remaining life of the swap. If the swap resets quarterly the exposure is small between resets and reappears at each one, so the sensitivity is to the *expected path* of the rate rather than to today's fixing.

The equity leg, which is the subtler one. The value of an equity forward depends on the cost of carry, so a change in rates moves the forward price of the underlying itself. In a total return swap this effect is largely neutralised, because the bank is hedged in physical shares and the financing leg is passing the actual carry through, which is precisely why the structure is priced off a floating rate rather than a fixed forward level. Say that, because it shows you know why the swap is built the way it is.

Then the practical close: the exposure is to your financing spread as much as to the underlying rate, and the spread is reset at rollover rather than being contractually fixed for the life of the relationship. A client who has locked a rate but not a spread has hedged half of a risk, and in stressed markets the spread moves more than the rate does.

What actually sets the price

The spread over the funding rate is not a number the desk chooses freely, and being able to list what determines it is what separates a Delta One answer from a textbook one.

- **The bank's own funding cost**, which is the floor.
- **Balance sheet and capital**, since the hedge sits on the bank's books and consumes leverage ratio and risk-weighted assets. On many desks this is the largest single component.
- **Collateral quality and the haircut**, which determine how much of the exposure is genuinely covered.
- **The client's credit**, and whether the documentation is a two-way margined agreement or something weaker.
- **Whether the collateral can be re-used**, which converts a cost into a revenue and is often the difference between a competitive quote and an uncompetitive one.
- **Offsetting inventory**, since a bank that is already long the stock for another reason can finance it at a much better level than one that must buy it.

REAL INTERVIEW QUESTION (Société Générale · Delta One)

Simulation: a client wants to finance a \$100m equity portfolio. What questions do you ask them, and what price (TRS) do you propose?

WHAT THEY TEST / ANSWER

This is a structured conversation, not a number, so lead with the questions and let the price fall out of them. Ask five things, in this order, because each one changes the quote.

What is in the portfolio? Large-cap liquid names or a long tail of mid-caps. This drives the haircut, the hedge cost and whether the desk can exit if it has to. A hundred million of index constituents and a hundred million of illiquid small caps are not the same trade.

How long, and can it be terminated? Term financing costs more than open financing, and the right to break is worth money to whoever holds it.

How much margin will you post, and in what? Cash, government bonds and equities are three different levels of collateral, and re-usability matters as much as quality.

Who are you, and under what documentation? A margined agreement with daily variation margin and independent amount is a different credit exposure from a weekly one, and the difference is priced.

Is it directional or is it a hedge? A portfolio that offsets what the desk already holds finances better than one that adds to it.

Then the quote. Something of the shape: floating reference plus a spread, say SOFR + 60 basis points on a diversified large-cap book with 20% initial margin. On \$100m that is an all-in cost near 4.9% a year at a 4.3% reference, so \$4.9m of financing cost to the client of which **\$600,000** is the desk's spread revenue, and the 20% margin gives the client five times leverage.

Size the margin out loud, because that is the part being marked. At 25% volatility a five-day move is about 3.5% at one standard deviation and about 8% at a 99% confidence level, so 20% initial margin covers about five and a half standard deviations of move over the period it would take to liquidate. That is the right calculation, and the right caveat is that it assumes you can liquidate in five days, which is exactly what fails in the event you are margining against.

Close on the risk you are taking, which is what a good interviewer is waiting for: this is not a market risk trade, it is a credit trade with a market-risk hedge attached, and the exposure is that the client defaults *and* the collateral has fallen, which are the same event.

Pitching the swap itself, and the client list that goes with it, belongs to swaps. One point from that conversation is specifically a financing point and is worth keeping here, because it is what a financing desk gets wrong: **these are not one product, they are several conversations.** The hedge fund cares about the financing spread to a basis point, the asset manager cares about tracking error and tax leakage, the corporate cares about accounting treatment and disclosure, and the insurer cares about capital. The spread this section is about is the whole trade to one of them and a rounding error to another, so leading with leverage to a pension fund, or with capital treatment to a hedge fund, is how the call gets ended early.

INTERVIEW TRAP

1. **Calling it a derivative trade.** It is a credit and balance sheet trade with a market hedge attached. The market risk is hedged; the counterparty risk is the position.
2. **Ignoring wrong-way risk.** The client defaults when the portfolio has fallen, so the collateral is worth least exactly when it is needed, treated in counterparty risk.
3. **Assuming margin is a function of the client alone.** It is sized off the liquidation horizon of the *collateral*, which is why a concentrated position attracts far more than a diversified one of the same value.
4. **Forgetting the dividend leg.** A total return receiver is long the dividends, so the tax treatment of those dividends is part of the economics and often part of the reason the trade exists.
5. **Quoting a spread without knowing the balance sheet cost.** On many desks that is the largest component, and it moves with the bank's own constraints rather than with the market.
6. **Treating one client's price as the market.** The same portfolio finances at very different levels depending on documentation, re-use rights and what the desk already holds.

VISUAL / CODE TO BUILD: Interactive

A financing quote builder. The reader assembles a portfolio from a menu of names with different liquidity and volatility, then sets the tenor, the margin percentage, the collateral type and the documentation strength. The page builds the quote as a stack of components, base funding, balance sheet charge, credit charge, collateral benefit and margin, so the reader sees which lever actually moves the price and by how much. A second panel runs a stress: a market fall of a chosen size, with the resulting variation margin call, the new initial margin requirement and the collateral shortfall shown together on a timeline, so the reader watches a well-margined trade become an under-margined one in three days. A concentration toggle replaces the diversified book with a single name of the same value and re-runs everything, which is the clearest possible demonstration of why the margin is sized off liquidation and not off notional.

10.8 ADRs and GDRs

sales · trading · structuring | core | equity · fx

A US pension fund wants to own Toyota. Toyota's shares trade in Tokyo, in yen, settle under Japanese rules, and pay dividends in yen with Japanese withholding tax. The fund's mandate, custodian and accounting system all speak dollars. The instrument that resolves this, without anyone opening a Tokyo account, is the **depositary receipt**.

The topic is small but it earns a section because it sits exactly where three things a front-office candidate is expected to hold together meet: an equity, a currency, and a no-arbitrage relation that a Delta One desk trades for a living.

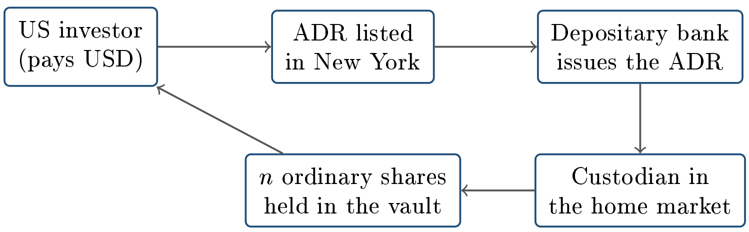
What the instrument actually is

INTUITION

A bank in Tokyo locks n real Toyota shares in a vault. A bank in New York writes a certificate saying "the bearer of this paper owns those n shares", and that certificate is listed and traded in New York, in dollars, on US settlement rules. Nothing about Toyota changed. What changed is the wrapper: the paper you hold is a domestic instrument, and the foreign share is the collateral sitting behind it.

An **American Depositary Receipt** is a negotiable certificate issued by a US **depositary bank** (in practice a small club: BNY Mellon, Citi, JPMorgan, Deutsche Bank) representing a fixed number of shares of a non-US company. Those underlying shares are held by a **custodian** in the company's home market. The ADR trades in USD on a US venue, settles US-style, and pays its dividends in USD after conversion, net of home-country withholding tax and of a depositary fee.

A **Global Depositary Receipt** is the same machine pointed at a different audience: typically issued outside the United States, historically listed in London or Luxembourg, and heavily used by emerging-market issuers wanting international capital without a US listing. You will also meet EDRs (European) and the generic term DR. The mechanics below are identical for all of them; only the venue, currency and regulatory wrapper change.



one ADR is a claim on n real shares locked with a custodian in the home market

Two vocabulary points interviewers use as filters. **Sponsored** means the company itself set the programme up with the depositary and cooperates on reporting; **unsponsored** means a depositary created it without the issuer, which is why unsponsored receipts sit on the over-the-counter market with thin information. And the programme **levels**: Level I trades over the counter with minimal reporting, Level II is exchange-listed but raises no new capital, and Level III is exchange-listed *and* raises capital through a public offering, with the heaviest disclosure burden. Rule 144A and Regulation S programmes place receipts privately with institutions instead.

The ratio and the parity relation

Each programme fixes a **ratio**: one ADR equals n ordinary shares, where n can be a whole number, a fraction, or something awkward like 0.4. The ratio is not economics, it is cosmetics: the depositary picks it so the receipt lands in a price range that US investors and option markets find comfortable, typically a few tens of dollars.

KEY IDEA

Write P_{ord} for the ordinary share price in the local currency, X for the exchange rate expressed as **USD per unit of local currency**, and n for the ratio. Then no-arbitrage forces

$$P_{\text{ADR}} \;=\; n \cdot P_{\text{ord}} \cdot X.$$

This is the law of one price with a currency conversion bolted on (see no-arbitrage pricing). Everything else in this section is either the mechanism that enforces this equation or a reason it temporarily fails.

WORKED EXAMPLE

Reading a receipt. A Japanese company's ordinary share trades at ¥3,000 in Tokyo. The programme ratio is 1 ADR = 5 ordinary shares. USD/JPY is quoted at 150, meaning 150 yen per dollar, so $X = 1/150$ dollars per yen. Fair value of the receipt is

$$P_{\text{ADR}} = 5 \times 3{,}000 \times \frac{1}{150} = \frac{15{,}000}{150} = \$100.00.$$

Suppose the ADR trades at \$100.60, a 0.60% premium. Before calling it an arbitrage, subtract the frictions: the depositary charges an issuance or cancellation fee, capped in practice at a few cents per receipt, which on a \$100 receipt is a few basis points; then add the local settlement cycle, the FX transaction, any local financial transaction tax, and the borrow cost of the leg you have to short in the meantime (see stock borrow and lending).

Then ask the question that actually decides it: *when did Tokyo last trade?* If the Tokyo close was six hours ago and Nikkei futures have risen 0.7% since, the "premium" is not a mispricing at all. It is the ADR doing its job as the only live price on the name, against a stale local reference.

Creation, cancellation, and who enforces the parity

The parity relation is not a convention that market participants politely respect. It is enforced by a physical conversion mechanism, and knowing the mechanism is what separates a candidate who has read a definition from one who could sit on the desk.

If the ADR is expensive relative to the local shares, an arbitrageur buys n ordinary shares in the home market, delivers them to the custodian, and instructs the depositary to **issue** (create) one new ADR, which is then sold in New York. If the ADR is cheap, the trade runs backwards: buy the ADR, instruct the depositary to **cancel** it, receive n ordinary shares from the custodian, and sell them locally. Either way the arbitrageur must also trade the currency to close the loop.

Three practical consequences follow, and each is a good interview answer on its own.

- **Conversion is slow and not free.** It spans two settlement cycles, a depositary fee and an FX trade. So the parity holds inside a band, not to the basis point, and the band widens with the frictions of the home market.
- **The band can break entirely.** If the shares cannot legally leave the home market, or the currency cannot be converted, the mechanism is switched off and the receipt is free to trade anywhere. Capital controls and sanctions do exactly this, and the receipt then prices the *expectation of ever getting the shares* rather than the shares. This is the deep connection to convertibility and FX crises, and it is the reason emerging-market DRs are not the same risk

object as a developed-market ADR.

- **Liquidity flows to where the price is discovered.** For many emerging-market names the DR is more liquid than the local line, so the causality can run backwards from the textbook: the receipt leads and the local share follows.

The currency exposure you did not ask for

A US investor holding an ADR is running two risks, not one. The parity relation makes this precise: since $P_{\text{ADR}} = nP_{\text{ord}}X$, the receipt's return is (to first order) the sum of the local equity return and the currency return. So its variance is not the equity variance:

$$\sigma_{\text{ADR}}^2 \;\approx\; \sigma_S^2 + \sigma_X^2 + 2\rho\,\sigma_S\sigma_X,$$

which is the variance-of-a-sum formula applied to returns. The cross term is the whole story, and it can cut either way.

WORKED EXAMPLE

The correlation decides everything. Take a local equity volatility of $\sigma_S = 25\%$ and a currency volatility of $\sigma_X = 10\%$.

If $\rho = +0.3$: $\;0.25^2 + 0.10^2 + 2(0.3)(0.25)(0.10) = 0.0625 + 0.0100 + 0.0150 = 0.0875$, so $\sigma_{\text{ADR}} = \sqrt{0.0875} = 29.6\%$. The dollar investor holds something *more* volatile than the local share.

If $\rho = -0.3$: $\;0.0625 + 0.0100 - 0.0150 = 0.0575$, so $\sigma_{\text{ADR}} = \sqrt{0.0575} = 24.0\%$. The dollar investor holds something *less* volatile than the local share, because the currency systematically moves against the equity and damps it.

Same company, same two volatilities, a swing of more than five volatility points purely from the sign of a correlation. This is also why a receipt is not a currency-protected instrument: it converts at whatever the rate happens to be, which is a *composite* exposure. An instrument that pays the foreign return at a rate fixed in advance is a quanto, a genuinely different and more expensive object, priced on exotics desks.

REAL INTERVIEW QUESTION (UBS · Structured Products)

Do you know what a quanto compo option is?

WHAT THEY TEST / ANSWER

What they are testing is whether you can keep two currency conventions apart under pressure, because candidates routinely blur them. A **composite** (compo) option is written on a foreign underlying but struck and settled in your currency, converting at the *prevailing* FX rate: the payoff is $\max(S_T X_T - K, 0)$ with K in your currency. You are therefore exposed to the product of the two, which means you are exposed to equity volatility, FX volatility *and* their correlation. A **quanto** option pays the foreign payoff $\max(S_T - K, 0)$, with K in the foreign currency, converted at a **fixed**, predetermined rate. The FX level then drops out of the payoff, but it does not drop out of the pricing: hedging a fixed-rate conversion of a random foreign quantity leaves a residual exposure to the equity-FX correlation, which is why the quanto price contains a correlation-driven adjustment to the drift and why quantos live on exotics desks. Land the punchline they are waiting for: an ADR is the cash-equity analogue of a *composite*, not of a quanto. It converts at the spot rate of the day, so it gives you the currency whether you wanted it or not. Correlation itself is covered in independence and correlation, and FX volatility in FX options.

REAL INTERVIEW QUESTION (RBC · Exo)

How has the correlation between the Nikkei and the dollar (and the euro) evolved over the last few years?

WHAT THEY TEST / ANSWER

The honest answer starts by refusing the premise that there is a stable number. The long-standing pattern was a *negative* correlation between Japanese equities and the yen: the yen is a funding and safe-haven currency, so in risk-off episodes it appreciated while the Nikkei fell, and a weak yen supported the earnings of Japan's exporters and lifted the index. For an unhedged dollar investor that is the damping case in the variance formula above, which is exactly why currency-hedged and unhedged Japan trackers behaved so differently. What the question is really probing is whether you know that the sign is a regime, not a constant: the relationship has been unstable, and periods of sharp monetary-policy divergence have driven the yen and the index in the same direction for long stretches, flipping the sign. So the answer to give is structural. State the mechanism, state the historical sign, state that it is regime-dependent and has flipped, and then say what a desk does about it, which is to hold the equity-FX correlation as an explicit risk with its own limit rather than assuming it. If you have a recent number, quote it as an estimate over a stated window, never as a constant. The related equity-to-equity version is in equity correlation.

Dividends, corporate actions and the small print

Dividends reach the receipt holder in dollars, but only after the home country has taken its withholding tax and the depositary has taken its fee, so the realised yield on an ADR is structurally below the yield on the local share. Reclaiming withholding under a tax treaty is possible and slow, and whether it happens depends on the holder, which is one reason two investors can value the identical receipt differently. The dividend itself is valued in dividends.

Corporate actions pass through to the receipt, and the depositary applies the same adjustment logic described in corporate actions. There is one extra degree of freedom worth knowing: the depositary can also change the **ratio** itself, which is a receipt-level event with no counterpart in the home market. A ratio change from 1:5 to 1:1 moves the receipt price by a factor of five and, exactly like a split, is economically neutral and lethal to an unadjusted price series.

Finally, a receipt holder is not on the company's share register. Voting is exercised through the depositary and is often restricted or simply not passed through, which matters for anyone whose mandate requires stewardship.

VISUAL / CODE TO BUILD: Interactive

A parity monitor. Two live-ish inputs, the local share price and the FX rate, plus a ratio selector, drive a computed fair value for the receipt; the reader drags the quoted ADR price and the panel shows the premium or discount in basis points against a shaded no-arbitrage band whose width the reader sets from three sliders (depositary fee, borrow cost, settlement friction). A clock toggle marks the home market as open or closed, and when it is closed the panel greys the local price and displays "stale reference" over the premium reading. The intended "aha" is that most apparent ADR premia sit inside the band or vanish once the local market is marked stale. A second tab reuses the same inputs to plot the receipt's dollar volatility as a function of the equity-FX correlation, sweeping ρ from -1 to $+1$.

INTERVIEW TRAP

(1) **Forgetting the ratio.** Comparing an ADR price directly to the local share price without multiplying by n produces a nonsense premium. Always ask for the ratio first. (2) **Inverting the FX quote.** USD/JPY at 150 means 150 yen per dollar, so the conversion divides. Getting this backwards is the single most common way to fail this question, and it is entirely avoidable by writing the units next to the number. (3) **Calling a stale-close gap an arbitrage.** If the home market is closed, the receipt is the only live price, and the apparent premium is price discovery. (4) **Treating an ADR as currency-hedged.** It is a composite exposure. You own the currency whether you wanted it or not. (5) **Assuming the conversion mechanism always works.** Capital controls, sanctions and issuer restrictions can suspend creation and cancellation, and the receipt can then trade at a large and persistent discount to a share nobody can deliver.

Where this goes next

The venue and settlement machinery a receipt lives on is equity market structure. The conversion leg that makes the arbitrage work runs through FX spot markets. The reason a widening band can become uncrossable is a funding and liquidity story, in liquidity risk. And the general principle the whole section is an instance of is no-arbitrage pricing.

10.9 Equity volatility

trading · structuring · sales | core | equity

Volatility is the number that equity derivatives are actually quoted in, so before any of the pricing machinery it is worth knowing what it measures, how it is computed, and what equity volatility in particular does that a textbook random walk does not. The general comparison between implied and realized volatility is its own section; this one is about the equity market's version of the quantity.

Measuring it

Volatility is the annualised standard deviation of returns. Historically, from a series of n daily log returns r_i,

$$\sigma_{\text{daily}} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} r_i^2}, \qquad \sigma_{\text{annual}} = \sigma_{\text{daily}} \times \sqrt{252},$$

where the mean is conventionally taken as zero, because over short horizons the drift is negligible against the noise and estimating it adds more error than it removes. The $\sqrt{252}$ is the number of trading days in a year, and the square root comes from variance adding over independent periods while standard deviation does not.

KEY IDEA

Learn the **rule of 16**. Since $\sqrt{252} = 15.87 \approx 16$, a daily move of 1% corresponds to about 16% annualised volatility, and the conversion runs both ways: a stock at 32% volatility is expected to move about 2% a day, and a 20% VIX implies roughly a 1.25% daily move in the index. Desks convert between the two constantly and in their heads.

REAL INTERVIEW QUESTION (Société Générale · Volatility)

Define historical volatility. How does it compare with implied volatility? And if a stock does +1% one day and −1% the next, what is its volatility?

WHAT THEY TEST / ANSWER

Definitions. Historical, or realized, volatility is measured from past prices: the annualised standard deviation of returns actually observed. Implied volatility is extracted from an option price: the number that, put into a pricing model, reproduces what the option trades at. One looks backwards at what happened, the other is the market's forward-looking price for what will happen, and the two are different objects that happen to share a unit.

The computation. Daily returns of +1% and −1%. With the zero-mean convention,

$$\sigma_{\text{daily}} = \sqrt{\frac{(0.01)^2 + (-0.01)^2}{2}} = 0.01 = 1\%,$$

so annualising, $1\% \times \sqrt{252} = 15.9\%$, or "about 16%" by the rule of 16.

Then deliver the observation the question is built around, because it is the whole point: the stock is essentially unchanged, since $1.01 \times 0.99 = 0.9999$, a loss of one basis point, and yet it has 16% annualised volatility. **Volatility measures the path, not the destination.** That is why a delta-hedged option position makes or loses money on movement regardless of direction, and why two stocks with identical yearly returns can be worth completely different amounts of option premium.

Close with the caveats an interviewer will probe: two observations is far too small a sample for any real estimate, the convention matters, so say whether you are using log or simple returns and 252 or 260 days, and overnight gaps versus intraday ranges give materially different measurements of the same week.

What equity volatility does that a random walk does not

Four stylised facts hold across equity markets and across decades, and every serious volatility model exists to reproduce them.

- **Clustering.** Large moves follow large moves. Volatility is highly autocorrelated even though returns are close to unpredictable, which is the empirical fact that makes volatility forecastable at all.
- **Mean reversion.** It does not wander indefinitely: high volatility decays towards a long-run level and low volatility eventually rises. This is why the implied volatility term structure is usually upward sloping when spot volatility is low and inverted when it has spiked, in the term structure of Greeks.
- **The leverage effect.** Volatility rises when prices fall, and the correlation between spot and volatility is strongly negative. Two explanations are usually offered, and a good answer gives both: mechanically, a falling equity value raises the firm's leverage and therefore the riskiness of the residual claim, which is the option-on-the-firm picture from equities; behaviourally, investors are long the market and buy protection when it falls, which pushes implied volatility up through demand. This asymmetry is the reason the equity smile is a downward-sloping *skew*, in skew and smile.
- **Fat tails.** Returns have far more extreme observations than a normal distribution allows, which is the same fact as the smile seen in the distribution rather than in the price.

Two more facts are specific to the cross-section rather than to time. **Index volatility is lower than the average volatility of its constituents**, because the constituents are not perfectly correlated and the diversification is real; the gap between the two is the substance of equity correlation and of dispersion trading. And **single-name volatility is dominated by scheduled events**, above all earnings dates, so the implied volatility of an option spanning an earnings announcement is not comparable to one that does not, and desks decompose the surface into a diffusive part and an event part precisely for this reason.

REAL INTERVIEW QUESTION (Natixis · EQD)

For the same market opportunity, you compare an option strategy on Total, with 40% volatility, and on Danone, with 10%. Which will cost more, and why?

WHAT THEY TEST / ANSWER

Total, by roughly a factor of four, and then refuse the implication hidden in the question.

The arithmetic. At the money the premium is close to proportional to volatility: the standard approximation is $0.4 \times \sigma \times S \times \sqrt{T}$, so a one-year at-the-money call costs about $0.4 \times 40\% = 16\%$ of spot on the first name and $0.4 \times 10\% = 4\%$ on the second. Four times the volatility, four times the premium, and the same ratio holds for a straddle or any at-the-money structure.

The point. A higher premium is not a worse trade. What you pay for is a wider distribution, and you receive proportionally more payoff in the scenarios you are buying. The only comparison that means anything is between the implied volatility you pay and the volatility you expect the name to *realise*: buying 40% on a stock that will realise 50% is a better trade than buying 10% on one that will realise 8%, and the absolute premium tells you nothing about which is which.

Two refinements finish it. If the budget is fixed rather than the notional, then the cheaper name buys more notional, which changes the whole shape of the position and is usually the real constraint for a client. And the two names' *skews* will differ as well as their levels, so a strategy involving more than one strike is not simply scaled by the volatility ratio, which is where the naive proportionality breaks.

The volatility index

REAL INTERVIEW QUESTION (Bank of America · Exo)

Do you know what the VIX is? What is its level?

WHAT THEY TEST / ANSWER

Define it precisely, because most candidates define it loosely and the precision is the mark. The VIX is a **30-day forward-looking implied volatility of the S&P 500**, quoted in annualised percentage points. Crucially it is *not* the implied volatility of an at-the-money option: it is computed from a strip of out-of-the-money puts and calls across strikes, weighted so that the result is the fair variance of the index over the next 30 days, which is the same replication that prices a variance swap. That construction is why the VIX is sensitive to the whole skew and not only to the level of at-the-money volatility.

On the level, do the honest thing: give the number as of the day you are interviewing, and if you are unsure, give the regime bands and say you would check. A calm market sits in the low to mid teens, a long-run average is around the high teens, a stressed market runs 30 to 40, and the extremes are rare: the March 2020 peak was above 80, comparable only to 2008. Attach the rule of 16 so the number means something: a VIX of 20 is the market pricing roughly a $20/16 = 1.25\%$ daily move in the index.

Then add the two things that make it a conversation. The VIX is a **measure, not an asset**: you cannot hold it, and the tradable instruments are VIX futures and options, whose term structure is usually in contango, so a long position bleeds on the roll, which is why long-volatility exchange-traded products decay over time. And the VIX is **mean reverting and spikes asymmetrically**, rising far faster than it falls, which is the leverage effect visible in one number and the reason it is used as a fear gauge rather than as a forecast.

INTERVIEW TRAP

Four errors. (1) Quoting a volatility without a convention: log or simple returns, 252 or 260 days, zero or sample mean, and whether it includes overnight gaps. (2) Comparing a single-name implied volatility across an earnings date with one that avoids it, as though they were the same quantity. (3) Assuming index volatility is the average of its constituents': it is lower, and the difference is correlation. (4) Treating the VIX as a forecast rather than as a price that includes a risk premium, or as something you could hold, when only its futures are tradable.

Where this goes next

The shape of volatility across strikes is skew and smile, the reason index volatility is below single-name volatility is equity correlation, and the trade between the two kinds of volatility is implied versus realized volatility. The instrument that isolates realized variance is variance swaps, and what a book's exposure to all of this feels like is vega exposure and volatility regimes.

10.10 Skew and Smile

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What a volatility smile is, and how a surface is built from quoted options, is the volatility smile and skew; the sensitivities to it are skew and smile sensitivities and trading it is skew strategies. This section owns one thing: the *equity* shape, why it is that shape and not another, and what it does to prices.

KEY IDEA

Equity index volatility **skews, it does not smile**. Plotted against strike, implied volatility slopes downward almost monotonically: low strikes carry higher implied volatility than high strikes, and the curve is noticeably steeper on the downside than it is anywhere else. That is different from FX, where the shape is closer to a symmetric smile, and it is different from what Black-Scholes says, which is a flat line. The asymmetry is the fact; the interesting content is why.

REAL INTERVIEW QUESTION (RBC · Structured Products)

What is the shape of equity skew? Why?

WHAT THEY TEST / ANSWER

The shape, in one sentence: implied volatility falls as the strike rises, so a 90% put trades at a higher implied volatility than a 110% call, and the slope is steepest in the downside wing. Describe it as a downward slope rather than as a smile, because on a single-stock name there is often some upturn in the far upside wing while an index rarely shows one.

Why, and give more than one reason, because no single explanation accounts for the whole shape and a good candidate says so.

- **The leverage effect.** As a company's equity falls, its debt is unchanged, so its leverage rises and the equity becomes mechanically more volatile. Volatility and spot are therefore negatively related for a structural reason, not a behavioural one.
- **Crash risk, and the demand for protection.** Returns are negatively skewed with fat left tails, so downside options are worth more than a lognormal says. On top of the true distribution, investors who own equities systematically buy puts and rarely buy calls, so the downside is bid and the upside is offered.
- **Structured product flow**, which is the one that marks you out on this desk. Retail structures are overwhelmingly yield products in which the investor sells a downside put, so dealers are systematically *long* the downside wing and short the body, and they hedge that inventory by pushing the wing's implied volatility around. In Europe this flow is large enough to be a first-order determinant of index skew, as hedging structured products describes.
- **Volatility feedback.** Falling markets raise realised volatility, which raises demand for protection, which steepens the skew, so the effect reinforces itself in stress.

Close on the observable consequence, which is what turns a description into an answer: the skew steepens when the market falls and flattens in calm rallies, so skew is itself a risk to be managed, not a static input.

Why the shape has to be there at all

There is a deeper answer worth having ready. The implied volatility surface is not a set of opinions, it is a re-parameterisation of option *prices*, and those prices imply a distribution for the terminal spot. A downward-sloping skew is exactly what you observe when the market's implied distribution has a fatter left tail and a thinner right tail than a lognormal. So the skew is not a correction bolted onto Black-Scholes; it is the market telling you it does not believe the lognormal, expressed in the only language the model provides.

REAL INTERVIEW QUESTION (Morgan Stanley · Exo)

What is skew? Where does it come from?

WHAT THEY TEST / ANSWER

What: the variation of implied volatility with strike at a fixed maturity. If Black-Scholes were true it would be a horizontal line, since the model has one volatility parameter for all strikes. It is not, so the skew is the visible residue of everything the model leaves out.

Where it comes from, in two layers, and giving both is the answer.

The mathematical layer: implied volatility per strike is a bijection with option price per strike, and the set of option prices across strikes determines the implied terminal distribution. A downward skew corresponds to a distribution with more left-tail mass and less right-tail mass than a lognormal. So the skew exists because the true return distribution is skewed and fat-tailed, and the model is not.

The economic layer: the leverage effect makes volatility rise as spot falls; investors buy protection and sell upside, so supply and demand are asymmetric; and dealers carry inventory from structured product issuance that leaves them holding one wing. Each of these is real and none is sufficient alone.

Then the sentence that shows you know why the question is asked on an exotics desk: because the skew is what makes a model choice matter. Any model can be calibrated to today's smile; they differ in how they say that smile will *move* when spot moves, which is the difference between local volatility and stochastic volatility, treated in local volatility. A digital, a barrier or a cliquet is priced almost entirely off that difference, which is why exotics desks care about skew dynamics rather than about today's skew level.

Across maturity, and how it moves

Skew is not constant in maturity. It is typically **steepest at the front and flattens as maturity extends**, because a crash is a short-horizon event and over long horizons the terminal distribution is smoothed by the central limit theorem. The two-dimensional object, skew by strike and by maturity, is the surface.

How it *moves* is a separate question and the one that determines hedging. The two idealised rules that bracket it, **sticky strike** (the curve stays put in strike space, so spot slides along a fixed curve and the Black-Scholes delta is the right hedge) and **sticky delta**, or sticky moneyness (the curve is a fixed function of K/S and travels with spot, so at-the-money volatility does not move but a fixed strike's does, and the hedge ratio picks up a correction), are defined in the volatility smile and skew.

What belongs here is the equity answer. Neither rule is exactly true: in equity indices the observed behaviour is closer to sticky strike over small moves in quiet regimes and closer to sticky delta over large ones, so a book hedged under one fixed assumption accumulates a directional bias in exactly the regime it can least afford one. The practical consequence is a **skew-adjusted delta** that differs from the model delta, which is skew and smile sensitivities' territory.

What skew does to a price

Any structure priced off two strikes is priced off the *difference* between two points on the curve, so **a downward skew makes anything long a low strike against a short high strike more expensive**, and anything the other way round cheaper. The worked case, a 100/130 call spread costing 6.96 at a flat 20% against 7.45 once the 130 strike is marked at 17%, together with its digital limit, is in the volatility smile and skew.

What is specific to an equity desk is how much of the book that sweeps up. Call spreads and risk reversals are skew trades whether or not they were intended as such, so pricing one off a single volatility does not merely give a slightly wrong number, it removes the exposure the trade actually has. It is also why a client comparing a bank's price to a Black-Scholes calculation on one input will always conclude the bank is expensive on call spreads and cheap on put spreads: the difference is the skew, not the margin. On an index, where the skew is steeper than on most single names, that gap is wide enough to dominate the conversation.

REAL INTERVIEW QUESTION (BNP · EQD)

Questions on skew and smile, and the differences between equities and foreign exchange.

WHAT THEY TEST / ANSWER

Answer by contrast, since that is what the question sets up, and give the reason for the difference rather than only the shapes.

Equity: a downward *skew*. Low strikes are bid, the curve slopes down almost monotonically, and it steepens in stress. The reason is that equity risk is directional and asymmetric: a company can halve overnight and cannot double overnight, leverage rises as the price falls, and everyone holding equities wants the same protection at the same time.

FX: closer to a symmetric *smile*, treated in the FX smile. The reason is structural rather than empirical: an exchange rate is a relative price, so a fall in EUR/USD is a rise in USD/EUR, and the same option is a call to one party and a put to the other. There is no privileged direction, so there is no reason for the surface to be asymmetric *by construction*. What asymmetry does exist reflects a genuine market view about which currency is riskier, and it shows up as a risk reversal that changes sign over time, which an equity index skew essentially never does.

The consequence for conventions, which is worth volunteering. FX quotes the surface as an at-the-money volatility plus a risk reversal and a butterfly, in delta terms, precisely because the shape is naturally described as a symmetric part plus a tilt. Equity quotes by strike or by moneyness, because the tilt *is* the shape.

If they extend it to a third asset, commodities usually smile with the tilt pointing *upward*, since the feared event is a supply shock that sends the price up rather than down, which is the cleanest way to show that the shape follows the direction of the risk rather than any universal law.

INTERVIEW TRAP

1. **Calling the equity shape a smile.** It is a skew. Using the words interchangeably is the fastest way to sound like you have read about it rather than seen it.
2. **Giving one cause.** Leverage, protection demand, structured flow and volatility feedback all contribute, and no single one accounts for the observed steepness.
3. **Pricing a spread off one volatility.** It gets the number wrong and, worse, removes the skew exposure the trade actually carries.
4. **Assuming the surface is static.** Skew steepens in sell-offs, so a skew position is a risk that changes with the market that moves it.
5. **Ignoring how the smile moves.** Sticky strike and sticky delta give different deltas for the same option, and the difference is a hedging error rather than a modelling nicety.
6. **Assuming equity skew generalises.** FX smiles for a structural reason and commodities often tilt the other way. The shape follows the direction of the feared event.

VISUAL / CODE TO BUILD: Interactive

A surface explorer with the causes attached. The reader sees a live equity index skew and can toggle each driver on and off, watching the curve respond: a leverage slider that links volatility to spot, a protection demand slider that bids the downside wing, and a structured issuance slider that shows dealer inventory pushing one region of the surface. A second panel shows the same option priced across the curve, so the reader can drag a call spread's two strikes and see the cost split into a Black-Scholes part and a skew part, reproducing the 6.96 against 7.45 in the text. A third panel animates a spot move under sticky strike and under sticky delta side by side, with the resulting delta of a fixed option displayed under each, so the reader sees that choosing a rule is choosing a hedge.

10.11 Equity correlation

trading · structuring · sales | core | equity

An index is a portfolio, and a portfolio's volatility is not the average of its parts. The gap between the two is correlation, and in equity derivatives it is not a statistical footnote: it is a quoted, traded quantity with its own supply and demand, its own risk premium, and its own way of failing at the worst moment.

The identity everything rests on

For a basket with weights w_i, the variance is

$$\sigma_B^2 \;=\; \sum_i w_i^2\sigma_i^2 \;+\; \sum_{i\neq j} w_i w_j\,\rho_{ij}\,\sigma_i\sigma_j.$$

The first sum is what you would have if the names moved independently; the second is everything correlation adds. Assume for a moment a single correlation ρ between every pair, equal weights and a common single-name volatility $\bar{\sigma}$, and it collapses to something you can hold in your head:

$$\sigma_B^2 = \bar{\sigma}^2\left[\rho + \frac{1-\rho}{n}\right] \xrightarrow[n \text{ large}]{} \rho\,\bar{\sigma}^2, \qquad \text{so} \qquad \sigma_{\text{index}} \approx \sqrt{\rho}\,\bar{\sigma}.$$

KEY IDEA

Index volatility is roughly the square root of correlation times average single-name volatility. Diversification does not remove idiosyncratic risk gradually, it removes it almost entirely once n is large, and what survives is the common factor, whose weight is ρ. This one formula converts between the two markets a desk quotes in, and inverting it gives the **implied correlation**

$$\rho_{\text{implied}} \approx \left(\frac{\sigma_{\text{index}}}{\bar{\sigma}}\right)^2,$$

which is what is actually traded.

WORKED EXAMPLE

Fifty names, each at 30% volatility, equally weighted, with a uniform pairwise correlation of 0.36. The index volatility is $0.30 \times \sqrt{0.36 + 0.64/50} = 18.3\%$, essentially the $\sqrt{0.36} \times 30\% = 18\%$ the approximation gives.

Run it backwards, which is what a desk does every morning: observe an index implied volatility of 18% and an average single-name implied of 30%, and the market is pricing a correlation of $(18/30)^2 = 0.36$.

Note how much correlation matters relative to n. At $\rho = 0$ the same fifty names give an index volatility of 4.2%, and at $\rho = 1$ they give 30%. Almost the whole range is driven by correlation, not by the number of constituents, which is why a diversified portfolio is not as diversified as its name count suggests.

Which way each payoff leans

REAL INTERVIEW QUESTION (Société Générale · EQD)

What parameters does the price of a basket option on equities depend on?

WHAT THEY TEST / ANSWER

List the single-name parameters, then give the one the question is fishing for, which the phrasing usually flags.

For each constituent: **spot**, **dividend** forecast and **borrow**, which together set that name's forward, and its **implied volatility**, by strike and maturity rather than as a single number. Then, at basket level: the **weights**, the **rate** for discounting and the **maturity** and **strike** of the option itself.

And then the parameter that has no single-name analogue: the **correlation matrix**. It does not appear anywhere in the individual names, it cannot be observed from any one option, and it changes the answer enormously. Give the size with the identity: at 30% single-name volatility, a fifty-name basket has a volatility of 4% if the names are uncorrelated and 30% if they move as one, so the basket option's price varies by nearly an order of magnitude over the correlation range with every other input unchanged.

Finish with the practical point that separates a structurer: correlation is not observable and barely hedgeable, so it is marked with a conservative bid-offer, frequently as a single number rather than a full matrix for a large basket, and it is usually the largest source of model risk in a multi-asset product. That is also why correlation-sensitive structures are where a desk warehouses risk it cannot lay off, as in common structuring mistakes.

REAL INTERVIEW QUESTION (Morgan Stanley · Exo)

For a call on a basket, are you short or long correlation, and why? How does the basket's price compare with a worst-of and a best-of?

WHAT THEY TEST / ANSWER

Take the sign first. A **basket call** pays on the average of the constituents. Higher correlation raises the variance of that average, by the identity above, so the option is worth more: you are **long correlation**.

Now the ordering, which is the better half of the question, and it follows from a pointwise inequality rather than from any model. On every scenario,

$$\min_i S_i \ \leq \ \text{average} \ \leq \ \max_i S_i,$$

so with the same strike, **best-of** $\geq$ **basket** $\geq$ **worst-of** for calls, always, whatever the correlation. Say that it is an arbitrage relation rather than a modelling result, because that is what is being checked.

Then the correlation signs, which do *not* all run the same way and are the trap. The **worst-of call** is **long** correlation: higher correlation means less dispersion, so the worst performer is less bad, and the option pays more. The **best-of call** is **short** correlation, by the mirror argument: less dispersion means the best performer is less good. And the extremes make it obvious, which is the fastest way to answer any correlation-sign question under pressure: at $\rho = 1$ all three payoffs coincide, since every name moves identically, so the three prices converge, and any deviation from that at lower correlation tells you the direction each one moves in.

REAL INTERVIEW QUESTION (Citi · EQD)

How do correlation and dispersion between the underlyings affect the price of a worst-of put?

WHAT THEY TEST / ANSWER

A worst-of put is **short correlation**, equivalently **long dispersion**, and the two phrasings are the same statement because dispersion is what low correlation produces.

The mechanism, in one line: the put pays on the worst performer, low correlation means the names scatter, and a scattered basket has a worse worst. So falling correlation raises the value of a worst-of put, and the holder gains.

Then say who holds it, because that is why the question is asked on an equity derivatives desk. In a structured note the *client* is short this option, so the client is long correlation and loses when the names disperse, while the issuer holds the put and is short correlation and must hedge it. That is the exposure the whole retail autocall market generates, and it is developed in autocallables.

Two refinements finish it. The sensitivity is far larger for a worst-of on a handful of names than for one on a broad basket, since with many names the worst is already extreme and adding one more changes little, so the correlation exposure per name is concentrated in small baskets. And the exposure is to *implied* correlation, which for these products trades above realized because dealers are structurally short it and must buy it back: correlation carries a risk premium exactly as volatility does.

Correlation is a traded quantity, not an estimate

REAL INTERVIEW QUESTION (Barclays · Structured Products)

When you are long dispersion, are you long or short correlation?

WHAT THEY TEST / ANSWER

Short correlation. The two are the same trade under two names, and the answer should show why rather than assert an equivalence.

Long dispersion means being positioned for the constituents to move differently from the index: the standard implementation is short index options and long options on the individual names. If correlation *falls* with single-name volatility unchanged, the index realises less than it was sold at while the single names realise what was paid for them, so the gain comes from the short index leg and the long single-name legs simply pay for themselves. Making money when correlation falls is the definition of being short correlation.

The identity ties it together: $\sigma_{\text{index}} \approx \sqrt{\rho}\,\bar{\sigma}$, so buying single-name volatility and selling index volatility is precisely a bet on ρ with the volatility levels netted out.

Two things earn the follow-up. **Why the trade has historically paid**: implied correlation sits above subsequently realized correlation, because dealers are structurally short correlation from issuing worst-of and autocall products and must buy it back, so a short-correlation position collects a risk premium. **Why it is dangerous**: correlation is bounded above by one and rises sharply in a crash, so the trade's worst days are exactly the market's worst days, and it is short the tail it most needs protection against. The construction, the vega weighting and the practical hedging of the position are in dispersion strategies.

Regimes, and the way it fails

Three empirical facts complete the picture and every one of them is a reason a model built on a constant ρ disappoints.

Correlation rises in falling markets. In a sell-off the common factor dominates and everything moves together, so the diversification that a portfolio was built on weakens exactly when it is needed. This is the correlation analogue of the leverage effect in equity volatility, and it is the single most important fact in the section.

There is a correlation skew. Implied correlation is not one number across strikes: it is higher for downside strikes than for upside ones, which is the same phenomenon priced into the surface, and it means a product sensitive to downside scenarios carries a different correlation than its at-the-money mark suggests.

The matrix is not observable. With n names there are $n(n-1)/2$ pairwise correlations and nothing like that many liquid instruments, so the matrix is estimated, smoothed, often reduced to one number, and must be kept positive semi-definite when it is adjusted, which is a real constraint rather than a technicality.

INTERVIEW TRAP

Four errors. (1) Treating index volatility as the average of constituent volatilities: it is roughly $\sqrt{\rho}$ times it, and the difference is large. (2) Assuming every basket payoff has the same correlation sign, when a basket call and a best-of call have opposite ones. (3) Using historical correlation to price, when the traded quantity is implied correlation and it carries a premium. (4) Sizing a diversified position on normal correlations, which is the assumption that breaks precisely in the scenario the diversification was bought for.

Where this goes next

The volatility levels this section relates are equity volatility, the strike dimension is skew and smile, and the statistical object underneath is independence and correlation. The trade built directly on this quantity is dispersion strategies, and the products that generate the market's structural position are autocallables.

Chapter 11

Fixed Income & Rates

11.1 Bond Pricing

sales · trading · structuring | core | rates · cross-asset

Rates questions come up in almost every desk interview, not just Rates ones, because every discounted cash flow in finance, from a bond to a swap to a structured note, uses the same machinery you build here. A bond is the simplest possible instrument to learn it on: a known set of future cash flows, no optionality, no path dependency. Get comfortable pricing a bond and you have the building block for the rest of the chapter.

Starting from nothing

A **bond** is a loan in disguise. A government or a company (the **issuer**) borrows money from investors today and promises to pay it back according to a fixed schedule: a series of periodic interest payments called **coupons**, plus the original amount borrowed, the **face value** or **principal** (usually normalised to 100), paid back at the final date, the **maturity**. If you buy the bond you are the lender, not the borrower.

INTUITION

Think of a bond as a **fruit tree with a fixed harvest schedule**. Every year it drops a known amount of fruit (the coupon), and on one specific future date it also delivers the tree itself, worth a known amount (the face value). The price you should pay for the tree today is simply the value, in today's money, of everything it will ever give you. That single idea, *discount every future cash flow back to today and add them up*, is all of bond pricing.

The formal definition

Consider a bond with face value F, annual coupon rate c (so it pays $C = c \times F$ once a year), and T years to maturity. If every cash flow is discounted at the same annual rate y, the bond's price is

$$P = \sum_{t=1}^{T} \frac{C}{(1+y)^t} + \frac{F}{(1+y)^T}.$$

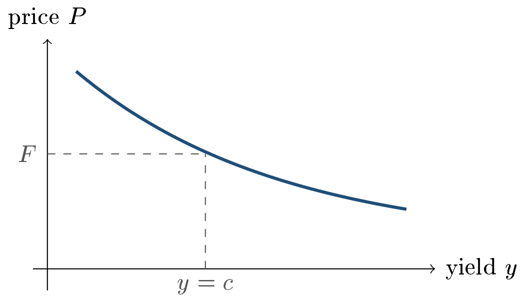
y here is the bond's **yield to maturity** (YTM): the single constant discount rate that makes the sum of discounted cash flows equal the market price. This is a simplification, stated openly: real curves discount each cash flow at its *own* maturity-specific rate rather than one flat y. That more accurate construction, and the zero-coupon building blocks behind it, is exactly what the yield curve and zero-coupon bonds develop next; here we isolate the pricing logic first, with one rate, before adding that layer.

Price, yield, and the coupon: three regimes

Rearranging the formula shows price and yield move in opposite directions: raise y and every term in the sum shrinks, so P falls. This gives three named cases, each worth recognising on sight:

- $c = y$: the bond trades at **par**, $P = F$;
- $c > y$: the bond trades **above par** (a premium), because it pays more than the market currently requires;
- $c < y$: the bond trades **below par** (a discount).

A newly issued bond is typically priced close to par, with c set near the prevailing market yield. As market yields move afterward, the price adjusts so that a buyer at the new yield still earns a fair return; the coupon itself never changes.



Two more things about this curve matter for an interview: it is downward sloping (the inverse relationship above) and it is **convex**, curving more steeply on the low-yield side than the high-yield side. The slope of this curve, divided by the price, is minus the bond's **modified duration**; the curvature, normalised the same way, is its **convexity**. Both are developed properly, with the calculus behind them, in duration and convexity; the only thing to carry from here is that the relationship is not a straight line, and a linear approximation of it will misprice large yield moves.

Clean price versus dirty price

Bonds trade continuously, not only on coupon dates, so a buyer mid-period owes the seller the coupon that has already accrued since the last payment. The **dirty price** (or *full price*) is what the buyer actually pays: $P_{\text{dirty}} = P_{\text{clean}} +$ accrued interest. Quoted prices on screens are almost always the **clean price**, precisely because it strips out this sawtooth accrual pattern and moves smoothly with the yield, which is what the formula above computes.

WORKED EXAMPLE

A 2-year bond, face value 100, annual coupon 5% ($C = 5$), yield to maturity 3%. Its price is

$$P = \frac{5}{1.03} + \frac{5 + 100}{1.03^2} = 4.854 + 98.973 = 103.83.$$

The coupon (5%) exceeds the yield (3%), so, as expected, the bond trades above par: an investor buying it today is happy to pay more than 100 because it delivers more income each year than the market currently demands.

REAL INTERVIEW QUESTION (Citi · EQD)

How do you price a bond? What is its sensitivity, via duration?

WHAT THEY TEST / ANSWER

This tests whether you can state the discounting logic cold and then know where it stops: price is the sum of each cash flow (coupons plus face value) discounted at the appropriate rate, $P = \sum_t C/(1+y)^t + F/(1+y)^T$. The "sensitivity via duration" part is a second question in disguise, do not try to derive duration from scratch here; say in one line that duration is the (negative, first-order) sensitivity of price to yield, point to the fuller treatment, and move on. Interviewers reward candidates who scope their answer correctly rather than rambling into a topic covered elsewhere.

REAL INTERVIEW QUESTION (Société Générale · Rates)

What is the relationship between interest rates and the price of a bond?

WHAT THEY TEST / ANSWER

What they are really testing: do you instinctively know the sign, and can you explain *why*, not just recite it. Rates up, discount factors shrink, every future cash flow is worth less today, so price falls; rates down, price rises. The reason to give out loud: the coupon and face value are fixed by contract, so the only thing that can move to reflect a new market rate is the price. A sharp follow-up: which bond moves more for the same rate change, a short-maturity or a long-maturity one? Longer maturity, because more cash flows sit further out where discounting bites harder, which is the intuition behind duration.

INTERVIEW TRAP

Three slips that come up constantly: (1) quoting the *clean* price when asked for what is actually paid at settlement, which is the *dirty* price; (2) treating the price-yield relationship as linear, when it is convex, so a 100bp up-move and a 100bp down-move do not change the price by the same amount; (3) confusing the **coupon rate** (fixed at issuance, determines the cash flow) with the **yield to maturity** (moves with the market, determines the price). These are different numbers that happen to coincide only when the bond trades at par.

Where this goes next

The single flat discount rate y used here is a simplification. The more rigorous approach discounts each cash flow at a rate specific to its own maturity, taken from the yield curve, built up from zero-coupon bonds and bootstrapped from market instruments. The slope and curvature of the price-yield relationship sketched above are formalised in duration and convexity, which is also how a trading desk actually hedges rate risk.

11.2 Yield Curves

sales · trading · structuring | core | rates · cross-asset

"What is the shape of the yield curve?" is one of the most-asked Rates questions in this database, across essentially every bank. It is also a question you can only half-answer once you know what a bond is: you need to know that there is not *one* interest rate, but a whole curve of them, one per maturity, and that its shape is read by every desk on the floor as a signal about growth, inflation and central bank policy.

Starting from what you already know

Bond pricing discounted every cash flow at the same rate y, on purpose, to isolate the pricing logic first. In reality, borrowing money for one year and borrowing money for ten years are different transactions with different prices: lenders usually want more compensation for tying up their money for longer. The **yield curve** (or **term structure of interest rates**) is simply the function that maps each maturity t to its own market rate $r(t)$.

> **INTUITION**
>
> Think of it as a **price list for fixing a rate**, the way a bank quotes a mortgage. A 2-year fixed-rate mortgage is usually quoted *cheaper* than a 10-year fixed-rate one on the same property, because the bank is giving up more flexibility, and taking on more uncertainty about inflation and its own funding costs, the longer it locks the rate in for. The yield curve is that same idea applied to the market as a whole: the "price of money," quoted separately for every length of time you might want to borrow or lend it, usually rising the longer you commit.

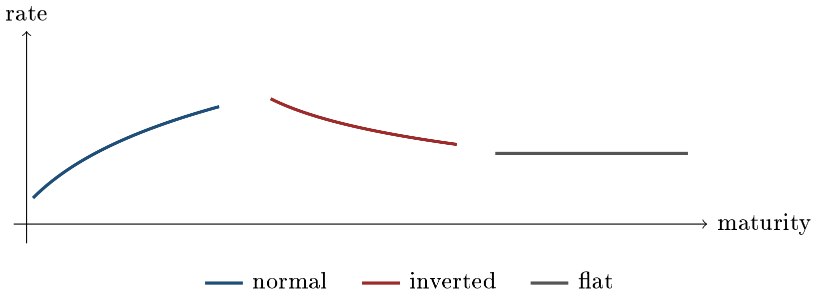
Spot curve versus par curve

There are two curves people mean when they say "the yield curve," and mixing them up is the single most common slip in this topic.

- The **spot curve** (or **zero curve**) gives the pure discount rate $r(t)$ for a single cash flow paid at time t alone. It is the rate you would earn buying a zero-coupon bond of maturity t; the **discount factor** $D(t) = 1/(1 + r(t))^t$ it defines (see zero-coupon bonds for the concept itself) is what should really be used to price any cash flow, replacing the single flat y of bond pricing.
- The **par curve** gives, for each maturity, the coupon rate a bond would need in order to trade exactly at par (price 100) today. It is what you read off a table of "current market yields," but each par yield is really a blend of the spot rates at every one of that bond's coupon dates, not a single point on the spot curve.

The two coincide only when the curve is flat. On an **upward-sloping** curve the par yield of a coupon bond sits slightly *below* the pure spot rate of the same maturity; on a **downward-sloping** curve, above it. The reason is direction-specific: part of a par bond's value comes back early, at the shorter (and, on an upward-sloping curve, lower-rate) maturities, which pulls its single blended yield down relative to the pure T-maturity zero rate. Building the spot curve out of market instruments is its own skill, covered in bootstrapping; here we only need the direction of the effect to be concrete, which the example below makes precise.

Shapes, and what desks read into them



- **Normal / upward-sloping**: the common case; longer maturities pay more, compensating lenders for locking up money longer and for the extra uncertainty (about inflation, credit, and reinvestment) that time brings.
- **Inverted**: short rates sit above long rates, typically because the central bank has pushed short-term rates up to fight inflation while the market expects growth (and future rates) to fall later. It has preceded most recent US recessions, which is why it gets so much attention, but it is a historically correlated signal, not a mechanical trigger: treat "inverted curve causes recession" as an overclaim in an interview.
- **Flat** or **humped**: transition shapes, often seen around policy turning points, when the market is genuinely unsure whether the next move is up or down.

Why the curve takes a given shape is explained by competing, only partially reconciled theories (the market's expectation of future rates, a term premium demanded for locking up money longer, and preferred-habitat segmentation across investor types); state which one you are leaning on rather than presenting any single theory as settled fact.

WORKED EXAMPLE

A curve reading $r(1) = 2\%$, $r(2) = 2.5\%$, $r(5) = 3\%$, $r(10) = 3.3\%$ is upward-sloping (normal). Pricing a 2-year, 5% annual-coupon bond *correctly* uses each cash flow's own spot rate, not one flat y:

$$P = \frac{5}{1.02} + \frac{5 + 100}{1.025^2} = 4.902 + 99.940 = 104.84,$$

a small but real difference from what a single blended yield would give, and the gap widens the steeper the curve is. The same two discount factors, $D(1) = 1/1.02 = 0.980392$ and $D(2) = 1/1.025^2 = 0.951814$, also pin down the 2-year *par* coupon: solving $c\,D(1) + (100 + c)\,D(2) = 100$ for c gives

$$c = \frac{100 - 100\,D(2)}{D(1) + D(2)} = \frac{4.8186}{1.932206} = 2.494\%,$$

below the 2.5% two-year spot rate, exactly as the direction rule above predicts.

REAL INTERVIEW QUESTION (Deutsche Bank · Rates)

What is the shape of the yield curve?

WHAT THEY TEST / ANSWER

A market-color question disguised as a technical one: they want to see that you know curves have shapes with names (normal, inverted, flat, humped), that you can state which one currently applies and to which market, and that you can explain *why* in one or two sentences (central bank stance, growth and inflation expectations) without pretending to certainty about the future. Precision on vocabulary (spot vs par, which segment of the curve you mean) separates a strong answer from a vague one.

REAL INTERVIEW QUESTION (Société Générale · Credit)

How does the absence of an arbitrage opportunity show up in the yield curve?

WHAT THEY TEST / ANSWER

Tests whether you understand that a curve is not just a set of independently quoted numbers, it is a system that must be internally consistent. If it were not, you could borrow cheaply at one maturity and lend at another to lock in a riskless profit; market participants would trade on that until the mispricing disappeared. Concretely, this consistency requirement is what pins down implied forward rates from the spot curve, developed fully in forward rates; here the point to make is qualitative: no-arbitrage is the reason the curve behaves as one coherent object rather than a list of unrelated rates.

INTERVIEW TRAP

Two recurring mistakes: (1) quoting "the par yield" when a spot rate was asked for, or vice versa, they agree only on a flat curve; (2) talking about "the interest rate" as if a single number described the whole market, when in fact every maturity has its own. A third, softer trap: overclaiming that an inverted curve mechanically *causes* a recession rather than correlating with one.

Where this goes next

The spot curve is built, in practice, from zero-coupon bonds and market instruments, through bootstrapping. The rates implied *between* two points on the curve are forward rates, and the sensitivity of a bond or portfolio to the whole curve moving is duration. Since 2008, banks discount collateralised cash flows off an OIS-based curve rather than Libor, a distinction covered in OIS, LIBOR, SOFR and €STR.

11.3 Zero-Coupon Bonds

sales · trading · structuring | core | rates · cross-asset

Its definition and formula are asked, almost word for word, across several banks in this database, because the zero-coupon bond is the atom everything else in fixed income is built from. Once you can price one in your head, you have, without realising it, the exact building block behind the discount factor, the yield curve, and every present-value calculation you will ever do on the desk.

Starting from what you already know

A coupon bond is a stream of cash flows: a coupon every period, plus the face value at maturity. A **zero-coupon bond** (ZCB) strips that down to the simplest possible case: *a single* cash flow, the face value F, paid once, at maturity T, with no coupons in between. It is literally what its name says: a bond with a coupon rate of zero.

> INTUITION
>
> It is an **IOU with one line on it**: "I owe you €100, in exactly one year, nothing before." You would not pay €100 today for that note, because you could instead put your €100 in a bank account today and have *more* than 100 in a year. So the note trades below its face value; how far below is exactly the interest rate the market demands for tying up money for that long. A zero-coupon bond's price is nothing more than that one discounting step, in isolation.

The formal definition

With annual compounding at rate r for maturity T,

$$P \;=\; \frac{F}{(1+r)^T}.$$

Equivalently, $P = F \times D(T)$, where $D(T) = 1/(1+r)^T$ is the **discount factor**: the price today of €1 received at T. This $D(T)$ is exactly the object the spot curve is a family of, one $D(t)$ per maturity, and it is the correct multiplier for *any* future cash flow, not only a zero-coupon bond's.

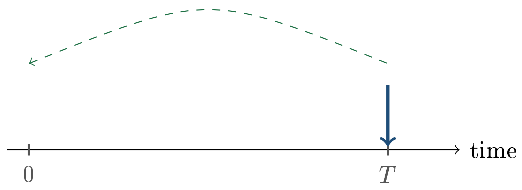
Because a ZCB has only one cash flow, its **yield to maturity** and its **spot rate** for that maturity are one and the same number: there is no reinvestment of intermediate coupons to complicate the picture, unlike a coupon bond, whose YTM implicitly (and, in general, incorrectly) assumes every coupon is reinvested at that same single rate.

A coupon bond is a portfolio of zero-coupon bonds

This is the idea that ties this whole opening arc of the chapter together. Rewrite the coupon bond formula using maturity-specific discount factors instead of one flat y:

$$P_{\text{bond}} \;=\; \sum_{t=1}^{T} C \cdot D(t) \;+\; F \cdot D(T).$$

Every term is just the price of a small zero-coupon bond, one per coupon date, and the coupon bond's price is simply their sum. This is exactly the statement that ZCBs are the atomic unit and coupon bonds are molecules built from them, and it is also, in reverse, how a bank actually gets long-dated zero rates when few ZCBs trade directly at that maturity: it strips or bootstraps them out of coupon-bond prices, the mechanics of which are covered in curve bootstrapping (in the US this literal stripping is done on Treasuries and the resulting securities are called STRIPS).



the single cash flow, face value F paid at T — discounting it back at r
price today $P = F/(1+r)^T$

WORKED EXAMPLE

Invest today at $r = 2\%$ for one year: $P = 100/1.02 = 98.04$. Run it the other way: if a 1-year ZCB trades at $P = 97$, its implied rate solves $97 = 100/(1+r)$, so $r = 100/97 - 1 = 3.09\%$. The two calculations are the same formula, read forwards for price and backwards for rate; being able to do both instantly, in your head, for round numbers is exactly what desks test.

REAL INTERVIEW QUESTION (BNP · Structured Products)

Give the definition of a zero-coupon bond, and its pricing formula.

WHAT THEY TEST / ANSWER

A pure definitional check, but interviewers grade it on precision: a single cash flow (the face value) at maturity, no interim coupons, priced by discounting that one cash flow, $P = F/(1+r)^T$. The strong answer adds, unprompted, that this price equals the discount factor times face value, because that is the detail that shows you see the bigger picture, not just a memorised formula.

REAL INTERVIEW QUESTION (Bank of America · Exo)

What is the zero-coupon rate if I have a bond priced at 97% with a 1-year maturity?

WHAT THEY TEST / ANSWER

Tests the same formula run in reverse, and whether you can do it without a calculator: $97 = 100/(1+r) \Rightarrow r = 100/97 - 1 \approx 3.09\%$. A fast mental route worth having ready: $100/97 - 1$ is identically $(100 - 97)/97$, so the rate is *exactly* the points below par divided by the price, here $3/97 = 3.09\%$. That is an algebraic identity, not an approximation, so the same trick prices any discount instrument in your head: points below par over price. The approximation only enters if you divide by par instead of price ($3/100 = 3\%$), which understates the rate, always.

The duration of a zero is the one case worth stating here, because it is where the general formula collapses: with a single cash flow there is nothing to average, so a zero's Macaulay duration is exactly its maturity. That is the cleanest sanity check on the general definition, and it is the reason a zero is the purest duration instrument available. Duration itself, the modified form $D/(1+y)$, and the convexity correction that makes a long zero fall by less than duration predicts when yields rise, and rise by more than it predicts when they fall, are developed in duration and convexity, which owns them.

INTERVIEW TRAP

The recurring confusion: "zero-coupon" describes the *coupon*, not the *rate*. A zero-coupon bond priced below par still earns its holder a positive return, entirely through price appreciation up to face value; it is not a bond with a 0% interest rate. A second, quieter mistake is applying simple (linear) rather than compound discounting; state the compounding convention you are using, here annual, since money markets and bond markets both use several different ones in practice.

Where this goes next

Every maturity's zero-coupon price, taken together, *is* the yield curve. Building that curve from liquid market instruments, most of which are coupon-bearing, is bootstrapping. The special-case duration result above generalises to any bond in duration and convexity.

11.4 Coupon Bonds

sales · trading · structuring core rates · credit · cross-asset

The coupon bond is the instrument that actually trades. Governments and companies issue it by the trillion, and it is the reference object behind swaps, repo, credit and the funding leg of every structured note. You already know how to discount its cash flows; what desks test here is everything the textbook formula quietly assumes, namely how often the coupon is paid, how it accrues between payment dates, what the quoted yield really means, and when the promised yield is not the return you get.

Starting from what you already know

Bond pricing gave you the machinery: a bond is worth the present value of its cash flows, and zero-coupon bonds showed that each of those cash flows is really a small bond of its own. A **coupon bond** is the general case sitting between them: periodic interest payments plus the face value at maturity. This section does not re-derive the price. It builds the layer above it, the conventions and the yield vocabulary that a rates or credit interviewer expects you to have at your fingertips.

The anatomy of a bond

Every fixed-rate bond is fully described by a short list of terms, and being able to recite that list cleanly is itself an interview question:

- **Issuer**: who owes the money (a sovereign, an agency, a bank, a corporate). This determines the credit risk, developed in corporate bonds.
- **Face value** (par, nominal, principal): the amount repaid at maturity, normally quoted per 100.
- **Coupon rate** c: the annual interest rate, as a percentage of face value, fixed at issuance and unchanged for the bond's life.
- **Coupon frequency** m: how many payments per year. Each payment is $c \cdot F/m$.
- **Day-count convention**: the rule that converts calendar days into a fraction of a coupon period, used for accrual.
- **Maturity**: the final date, when the last coupon and the face value are paid together.

- **Seniority and covenants**: where the holder ranks if the issuer defaults, and what the issuer promises not to do.

REAL INTERVIEW QUESTION (HSBC · Fixed Income)

What are the main components of a bond?

WHAT THEY TEST / ANSWER

A deliberately open opener, and what it tests is whether you think in terms of a contract or only in terms of a formula. List issuer, face value, coupon rate, coupon frequency, day count, maturity and seniority, then add the one sentence that shows you understand the split: the issuer fixes all of these at launch and none of them ever move, so the only thing that can adjust when market rates or credit quality change is the *price*. Candidates who answer with the discounting formula alone have answered a different question.

Coupon frequency, and why the yield quote depends on it

With m coupons per year, a maturity of T years and a yield y quoted with the same compounding frequency m, the price is

$$P = \sum_{k=1}^{mT} \frac{cF/m}{(1+y/m)^k} + \frac{F}{(1+y/m)^{mT}}.$$

The important part is not the extra index. It is that y is a **quote**, not a physical quantity, and its meaning is tied to m. A yield of 6% quoted semi-annually means 3% per half-year, which compounds to $(1.03)^2 - 1 = 6.09\%$ per year. The same economic return quoted annually would be 6.09%. Two bonds are only comparable on yield once the quotes are on the same basis.

Market practice is not uniform, and desks expect you to know their own market:

- US Treasuries, UK gilts and JGBs: **semi-annual** coupons;
- Eurozone government bonds (Bund, OAT, BTP): **annual** coupons;
- most US corporate bonds: semi-annual;
- floating-rate notes: usually quarterly, tracking a 3-month rate.

KEY IDEA

A yield number is meaningless without its convention. Always ask, or state, three things: the compounding frequency, the day count, and whether the price it refers to is clean or dirty. Most "the two banks disagree on the yield" stories are one of these three, not a pricing error.

Day counts and accrued interest

Bond pricing established that a mid-period buyer pays the dirty price, that is the clean price plus accrued interest. What it did not say is how the accrual is measured, and that is a convention chosen at issuance:

- **30/360**: every month counts as 30 days, every year as 360. Standard for US corporate and municipal bonds. Convenient because it makes accrual perfectly regular.
- **Actual/actual**: real calendar days over the real length of the coupon period. Standard for government bonds in the US and the euro area.

- **Actual/360**: real days over a 360-day year. The money-market convention in USD and EUR, so it shows up on the floating leg of swaps and in repo, not usually on bond coupons.
- **Actual/365**: the sterling money-market equivalent.

The accrued interest is then

$$\text{AI} \;=\; \frac{cF}{m} \times \frac{\text{days from last coupon to settlement}}{\text{days in the full coupon period}},$$

with both day counts measured under the bond's convention.

INTUITION

Think of the coupon as **rent on a shared flat**. The landlord pays the whole month's rent to whoever holds the lease on payment day, so if you take over the lease halfway through the month, you owe the previous tenant their half. Accrued interest is that settling-up. It has nothing to do with the bond being cheap or expensive, which is exactly why screens quote the clean price: it strips out a sawtooth that everyone already knows how to compute.

WORKED EXAMPLE

A corporate bond, face 100, coupon 6% paid semi-annually, day count 30/360. The last coupon was paid on 15 March; settlement is 15 July. The clean price is 98.50.

Each coupon is $6/2 = 3.00$. Under 30/360, 15 March to 15 July is four months of 30 days, so 120 days, and a full semi-annual period is 180 days. Therefore

$$\text{AI} = 3.00 \times \frac{120}{180} = 2.00,$$

and the buyer wires $98.50 + 2.00 = 100.50$ per 100 of face value. Note that had the convention been actual/actual, the numerator would have been the real day count from 15 March to 15 July, which is 122 days, and the denominator the real length of the March-to-September period, 184 days, giving $3.00 \times 122/184 = 1.99$. Small in this case, but it is a real cash difference on a notional of several hundred million.

The par bond, and why $c = y$ forces $P = F$

Bond pricing stated the three regimes: par, premium and discount. The par case is worth one line of algebra, because it turns a fact you memorised into something you can reconstruct under pressure. Set $F = 100$ and $c = y$ with annual coupons. The coupon stream is an annuity, so

$$P \;=\; 100y \sum_{t=1}^{T} \frac{1}{(1+y)^t} \;+\; \frac{100}{(1+y)^T} \;=\; 100y \cdot \frac{1-(1+y)^{-T}}{y} \;+\; 100\,(1+y)^{-T}.$$

The y cancels in the annuity term, leaving

$$P \;=\; 100\left[1-(1+y)^{-T}\right] + 100\,(1+y)^{-T} \;=\; 100.$$

The discount factor term cancels exactly, for any maturity T. That is the whole content of "a bond priced at par yields its coupon": the coupon stream is precisely large enough to pay the market rate on 100 of principal each year, so no adjustment to the principal is needed.

Three different yields, and only one of them is a return

The word "yield" is overloaded, and separating the three meanings is a reliable way to sound like someone who has sat on a desk:

- **Coupon rate** c: fixed at issuance, a property of the contract, not of the market.
- **Current yield**: annual coupon divided by clean price. A quick income measure that ignores the pull to par entirely, so it overstates the return on a premium bond and understates it on a discount bond.
- **Yield to maturity** y: the single discount rate that reproduces the market price. It does account for the pull to par, and it is the number quoted on screens.

Yield to maturity is a promise with a condition attached, and the condition is the classic trap. Solving for the y that equates price and discounted cash flows implicitly assumes every coupon received along the way is **reinvested at y itself** until maturity. If rates fall after you buy, your coupons are reinvested at less than y and your realised compound return over the full holding period is below the yield you were quoted. This is **reinvestment risk**, and it is precisely the risk a zero-coupon bond does not have, since it pays nothing to reinvest.

> **KEY IDEA**
>
> A higher coupon is not automatically better. It front-loads your cash flows, which reduces price sensitivity to rates but increases the share of your return that depends on unknown future reinvestment rates. Price risk and reinvestment risk move in opposite directions when rates change, and the horizon at which they cancel is the bond's duration, developed in duration and convexity.

Doing it in your head

Rates interviews are full of quick estimates, and there are two approximations worth owning cold. Going from yield to price, use the first-order rule

$$P \;\approx\; 100 + (c - y) \times T \quad \text{(in points of par)},$$

which is just the duration approximation with maturity standing in for duration. Going the other way, from price to yield, use

$$y \;\approx\; \frac{C + (F - P)/T}{(F + P)/2},$$

the annual coupon plus the annualised pull to par, divided by the average capital employed.

REAL INTERVIEW QUESTION (Commerzbank · Fixed Income)

For a 4% coupon and a 2-year maturity, what is the price of the bond, and then its yield? (Approximation: Price $= 100 + (Coupon - Yield) \times years$*.)*

WHAT THEY TEST / ANSWER

They have handed you the rule of thumb, so the test is whether you can use it and then say honestly where it breaks. With a yield of 5%, the rule gives $100 + (4 - 5) \times 2 = 98$. The exact price is $4/1.05 + 104/1.05^2 = 3.81 + 94.33 = 98.14$. The 14 cent gap is not sloppiness, it is the rule using *maturity* where it should use *modified duration*, which for this bond is 1.87 rather than 2 (the Macaulay duration is 1.96, and dividing by 1.05 gives the modified figure). Since duration is always below maturity for a coupon-paying bond, the rule systematically overstates the price move, and the error grows with maturity and with the size of the yield gap. Deliver the estimate in three seconds, then volunteer the caveat.

REAL INTERVIEW QUESTION (BNP · Fixed Income)

How do you estimate in your head the yield to maturity of a bond quoted at 85% of par?

WHAT THEY TEST / ANSWER

Do not attempt to invert the pricing formula. Use the approximation above: annual coupon plus the annualised pull to par, over average capital. Take a 5-year bond with a 4% coupon quoted at 85. The pull to par is $(100-85)/5 = 3$ per year, average capital is $(100+85)/2 = 92.5$, so $y \approx (4+3)/92.5 = 7.6\%$. The true yield is 7.7%, so you are within about 15 basis points, which is all a mental estimate needs to be. Then name the bias: the method assumes the pull to par arrives evenly rather than being discounted, so it slightly understates the yield on a deep discount bond. Finish by noting what a price of 85 is telling you, namely that the coupon is far below the market rate the issuer would pay today, whether because rates rose or because its credit spread widened.

When the coupon is not fixed: floating-rate notes

A **floating-rate note** (FRN) pays a coupon that resets every period to a reference rate plus a fixed spread, for example three-month SOFR or €STR plus 40 basis points. The coupon is set at the start of each period and paid at the end.

The consequence is the one that gets asked. Because the coupon re-adjusts to the market rate at every reset, the bond stops being a stream of fixed promises and becomes, at each reset date, worth approximately par again. Its price therefore barely moves with the level of rates, and what rate risk it does carry is only the exposure accumulated since the last reset. Its rate sensitivity is that of an instrument maturing at the *next reset date*, not at final maturity. What an FRN still carries in full is credit risk: if the issuer's spread widens above the fixed 40 basis points written into the contract, the note trades below par regardless of where rates are.

REAL INTERVIEW QUESTION (Tikehau Capital · Fixed Income)

What is the duration of a floating-rate bond?

WHAT THEY TEST / ANSWER

Approximately the time to the next coupon reset, so at most one period, and close to zero right after a reset. The reasoning matters more than the number: at each reset the coupon becomes the current market rate, so the instrument re-prices to roughly par, and only the accrual since the last fixing is exposed to rate moves. The follow-up trap is to conclude that an FRN is riskless. It is not. Its *spread* duration is the full maturity, because the credit spread is fixed for the bond's whole life, which is why FRN prices moved sharply in the 2008 crisis even though rates were falling. Duration itself is treated properly in duration and convexity.

INTERVIEW TRAP

Four errors that come up constantly. (1) Comparing a semi-annually quoted yield with an annually quoted one without converting, which manufactures a 9 basis point difference out of nothing on a 6% bond. (2) Quoting current yield when asked for yield to maturity, which hides the pull to par entirely. (3) Claiming yield to maturity is the return you will earn if you hold to maturity, when it is only that if every coupon is reinvested at the same yield. (4) Saying an FRN has no interest-rate risk and stopping there, without mentioning that its spread risk runs to final maturity.

VISUAL / CODE TO BUILD: Interactive

A coupon-bond workbench. The reader sets face value, coupon rate, frequency, maturity and yield, and the panel shows (i) the cash flow ladder as bars, with each bar's discounted value shaded inside it, (ii) the running price, split into the coupon annuity and the principal, and (iii) a settlement-date slider that sweeps through a coupon period so the clean price stays smooth while the dirty price walks up the sawtooth and drops on payment day. A toggle switches the day count between 30/360 and actual/actual so the reader sees the accrual line change slope. The intended realisation: the coupon rate sets the cash flows, the yield sets the price, and the settlement date sets only the accrual.

Where this goes next

The fixed coupon stream you priced here is exactly the fixed leg of an interest rate swap, and a swap is best understood as a par coupon bond exchanged for a floating-rate note, which is why the two halves of this section are worth keeping side by side. The single flat yield y is still a simplification, and replacing it with maturity-specific rates from the yield curve, recovered by bootstrapping, is the next step. The sensitivity of this price to rates is duration and convexity, and the part of the yield that compensates for the issuer possibly not paying is the credit spread.

REAL INTERVIEW QUESTION (UBS · Market Risk)

Which is more sensitive, a zero-coupon bond or a coupon bond?

WHAT THEY TEST / ANSWER

The zero-coupon bond, for the same maturity. Give both the mechanical reason and the market reason. Mechanically, all of a zero's value sits at the final date, so its duration equals its maturity, whereas a coupon bond returns cash earlier and its duration is strictly shorter, which is exactly the front-loading argument from earlier in this section. In market terms, the zero has no reinvestment risk but maximum price risk, and the coupon bond trades the one for the other. The natural follow-up, if two bonds from the same issuer with the same maturity are compared, is that the zero should carry the higher yield: with an upward-sloping curve its yield is the full long-maturity spot rate, while the coupon bond's yield is a weighted average that pulls in shorter, lower rates, and the zero also concentrates all its credit exposure at the single final date.

11.5 Duration and Convexity

sales · trading · structuring | core | rates · credit · cross-asset

This is the most heavily asked topic in the whole fixed income chapter, and it is asked well beyond rates desks: credit, market risk, structuring and cross-asset all use it. The reason is that pricing a bond tells you what it is worth today, which is useful once, while duration and convexity tell you what happens to that value when the market moves, which is what a trader needs every minute of every day. Everything here is one idea developed twice: a first-order sensitivity, then the correction to it.

Why a sensitivity, and not just a price

Bond pricing showed that price falls when yield rises, and that the relationship is a curve, not a line. Knowing the direction is not enough. If you are long \$100 million of bonds and the market sells off 25 basis points before lunch, the desk needs a number, not a sign. Duration is that number, and convexity is the correction that makes it accurate for large moves.

INTUITION

Picture the bond's cash flows as **weights placed along a plank**, one weight per payment date, each as heavy as that cash flow's present value. The plank is huge at the far end, where the face value sits, and light near you, where the coupons are. **Duration is the balance point**: the single distance out along the plank where you could put one finger and hold the whole thing level. A bond whose weight sits far out is long and unwieldy, and a small tilt in rates swings it a long way. A bond whose weight sits close in barely moves. That is the entire concept, and the two formulas below are just the arithmetic of finding the balance point and then measuring how hard it swings.

REAL INTERVIEW QUESTION (BP · Fixed Income)

Explain duration to me simply.

WHAT THEY TEST / ANSWER

They are testing communication, not maths, and they will interrupt you the moment you start writing a sum. Give the two-sentence version: duration is the average time you wait to get your money back, weighted by the present value of each payment, and because of that it also measures how much the price moves for a small change in rates. Then anchor it with one number: a bond with a duration of 5 loses roughly 5% of its value if yields rise by 1%. If the desk is a sales desk, stop there. If it is a trading or risk desk, add the distinction between Macaulay duration in years and modified duration as a percentage sensitivity, which is the next thing they will ask anyway.

Macaulay duration: the balance point

Take a bond with cash flows CF_t at times $t = 1, \ldots, T$ and yield y, priced as in bond pricing. Write the present value of each cash flow as $PV_t = CF_t/(1+y)^t$, so that $P = \sum_t PV_t$. The **Macaulay duration** is

$$D_{\text{mac}} = \sum_{t=1}^{T} t \cdot \frac{PV_t}{P} = \frac{1}{P} \sum_{t=1}^{T} \frac{t \cdot CF_t}{(1+y)^t}.$$

The weights PV_t/P are non-negative and sum to one, so this really is a weighted average of the payment times, measured in years. It is the balance point of the intuition box, made precise.

Modified duration: the sensitivity

Duration in years is a nice picture but not directly usable. Differentiate the price with respect to the yield:

$$\frac{dP}{dy} = -\sum_{t=1}^{T} \frac{t \cdot CF_t}{(1+y)^{t+1}} = -\frac{1}{1+y} \sum_{t=1}^{T} \frac{t \cdot CF_t}{(1+y)^t} = -\frac{P \cdot D_{\text{mac}}}{1+y}.$$

Define the **modified duration** as the percentage price change per unit of yield:

$$D_{\text{mod}} = -\frac{1}{P}\frac{dP}{dy} = \frac{D_{\text{mac}}}{1+y}, \qquad \text{so} \qquad \frac{\Delta P}{P} \approx -D_{\text{mod}}\,\Delta y.$$

With m coupons per year and a yield quoted on the same basis, the divisor becomes $1 + y/m$. That single division is the whole difference between the two durations, and mixing them up is the most common error in this topic.

KEY IDEA

Macaulay duration is a **time**, measured in years, and it answers “when, on average, do I get paid?”. Modified duration is a **sensitivity**, measured in percent per unit of yield, and it answers “how much do I lose per basis point?”. They differ by the factor $1 + y/m$, which is close to one, so the numbers look similar and the concepts are not. Say which one you mean.

DV01: the same thing in currency

Traders do not risk-manage in percentages, they risk-manage in cash. The **DV01** (dollar value of a basis point, also called PV01 or BPV) is the change in the position's value for a one basis point move in yield:

$$\text{DV01} \;=\; D_{\text{mod}} \times P \times 0.0001,$$

where P is the full market value of the position, not the price per 100. A desk quotes its rate risk as a DV01 ladder, so many thousands of dollars per basis point at each maturity bucket, and hedges by trading whatever instrument offsets that number. This is also the language of risk limits: a limit is almost never expressed in duration, it is expressed in DV01.

REAL INTERVIEW QUESTION (HSBC · Rates)

What is the DV01 of a 10-year annual swap with a notional of 100 million?

WHAT THEY TEST / ANSWER

The mechanics of a swap belong to interest rate swaps, but the risk answer is the one they want here, and it is short: the value of a payer swap moves with the market swap rate through the **annuity**, that is the sum of the discount factors on the fixed leg, so $\text{DV01} = N \times A \times 0.0001$. With a flat 3% curve, the 10-year annual annuity is $A = (1 - 1.03^{-10})/0.03 = 8.53$, giving $100{,}000{,}000 \times 8.53 \times 0.0001 \approx \$85{,}000$ per basis point. The number to carry around is that a 10-year swap runs at roughly \$8.5k of DV01 per 10 million of notional, so you can scale it instantly for any size. Note that this annuity is *exactly* the modified duration of a 10-year par bond at the same yield, and that is an identity rather than a coincidence: a payer swap is a short fixed par bond against a long floating-rate note, the FRN carries almost no rate duration, and the two principal legs cancel, so the swap inherits the par bond's DV01. It is also why swap risk is quoted in DV01 rather than in duration, since there is no principal to normalise by.

A full worked example

WORKED EXAMPLE

A 5-year bond, face 100, annual coupon 10%, trading at par so $y = 10\%$. Build the table once and everything follows from it.

t	CF_t	PV_t	$t \cdot PV_t$	$t(t+1)\,PV_t$
1	10	9.0909	9.0909	18.1818
2	10	8.2645	16.5289	49.5868
3	10	7.5131	22.5394	90.1578
4	10	6.8301	27.3205	136.6027
5	110	68.3013	341.5067	2049.0404
		100.0000	**416.9865**	**2343.5694**

The prices column sums to 100, confirming the bond is at par. Then

$$D_{\text{mac}} = \frac{416.9865}{100} = 4.17 \text{ years}, \qquad D_{\text{mod}} = \frac{4.1699}{1.10} = 3.79.$$

So a 1% rise in yields costs about 3.79% of value, and the DV01 per 100 of face is $3.79 \times 100 \times 0.0001 = 0.0379$, that is \$37,908 per basis point on a \$100 million position.

Note that the duration is 4.17 years while the maturity is 5 years. The coupons pull the balance point in, which is always true for a coupon-paying bond.

REAL INTERVIEW QUESTION (Tikehau Capital · Fixed Income)

What is the duration of a 5-year bond with a 10% annual coupon?

WHAT THEY TEST / ANSWER

They want to see you set up the weighted average without panicking, and they will accept an estimate. Assume par pricing unless told otherwise, so $y = 10\%$, and the answer is about 4.2 years Macaulay, 3.8 modified. If you cannot do the sum in your head, bound it out loud and you will still score: it must be below 5 years because the coupons arrive before maturity, and it must be well above the middle of the range because the final payment of 110 dominates every other cash flow, so "a bit over 4" is the right instinct. Do not forget to say which duration you quoted.

What moves duration

Four rules cover almost every follow-up, and each has a one-line reason rooted in the balance-point picture:

- **A zero-coupon bond has Macaulay duration exactly equal to its maturity**, because there is only one weight and it sits at the end. This makes zeros the maximum-duration instrument for a given maturity, as noted in zero-coupon bonds.
- **A higher coupon lowers duration.** More weight arrives early, so the balance point moves in.
- **A higher yield lowers duration.** Discounting bites hardest on the most distant cash flows, so the far weights shrink relative to the near ones.

- **Longer maturity generally raises duration**, though not without limit: for a deep discount bond duration can peak before maturity, because the far coupons are discounted almost to nothing.

REAL INTERVIEW QUESTION (CACIB · Fixed Income)

What is the duration of a 30-year zero-coupon Bund?

WHAT THEY TEST / ANSWER

30 years, Macaulay. This is a one-word answer and the entire test is whether you volunteer the second half: the *modified* duration is $30/(1+y)$, so with Bund yields around 3% it is about 29.1, meaning the first-order estimate of the price move for a 100 basis point shift is about 29%. Saying "30" and stopping leaves the interviewer to ask the follow-up that separates candidates. The reason it is exactly the maturity is that a zero has a single cash flow, so the weighted average of payment times has only one term with weight one. If they push, add that on an instrument this long the first-order number is badly off for a move that size: repricing from 3% to 4% actually costs about 25%, not 29%, because a 30-year zero is extremely convex. That is the cleanest possible illustration of why duration alone is not enough.

Convexity: the correction

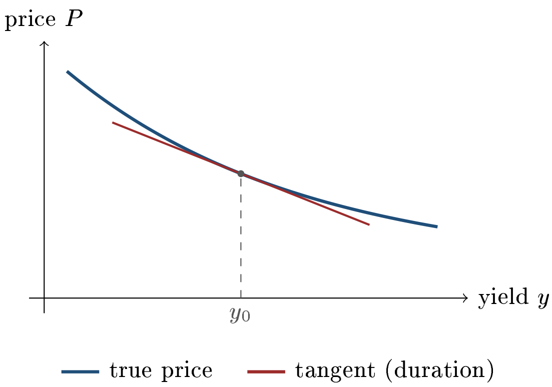
Duration is a straight line drawn tangent to a curve, so it is only right in the limit of small moves. The price-yield relationship curves upward, and that curvature is **convexity**:

$$C \;=\; \frac{1}{P}\frac{d^2P}{dy^2} \;=\; \frac{1}{P}\sum_{t=1}^{T}\frac{t(t+1)\,CF_t}{(1+y)^{t+2}}.$$

Expanding the price to second order in the yield move gives the working formula:

$$\frac{\Delta P}{P} \;\approx\; -D_{\text{mod}}\,\Delta y \;+\; \tfrac{1}{2}\,C\,(\Delta y)^2.$$

Because $(\Delta y)^2$ is positive whichever way rates move, and $C > 0$ for an ordinary bond, the convexity term always *adds* value. That is the punchline: a long bond position gains more from a rate fall than it loses from an equally sized rate rise.



The tangent line sits *below* the curve on both sides. That is the geometric statement that the duration-only estimate understates the price after any move, whether up or down, and convexity is exactly the size of that understatement.

REAL INTERVIEW QUESTION (Société Générale · Fixed Income)

What is convexity? Explain it.

WHAT THEY TEST / ANSWER

Convexity is the curvature of the price-yield relationship, the second derivative of price with respect to yield, normalised by price. What they want to hear is not the formula but the consequence: duration alone assumes a straight line, and because the true relationship bends upward, a positive convexity position makes more on a rally than it loses on an equivalent sell-off. Then close the loop, since convexity is not free: two bonds with the same duration but different convexity will not have the same yield, and the more convex one trades at a lower yield, because the market charges for that asymmetry. Candidates who say "convexity is good" without saying "and therefore you pay for it" have given half the answer.

First and second order, on the same bond

WORKED EXAMPLE

Take the 5-year 10% bond from above: $P = 100$, $D_{\text{mod}} = 3.7908$, and from the last column of the table,

$$C = \frac{2343.5694}{100 \times 1.10^2} = \frac{2343.5694}{121} = 19.37.$$

Now shock yields by ± 100 basis points.

	$\Delta y = +1\%$	$\Delta y = -1\%$
Duration only	96.21	103.79
Duration + convexity	96.31	103.89
Exact reprice	96.30	103.89

Reading the table is the lesson. The duration-only estimate is wrong by about 0.10 in both directions, and it is wrong in the *same* direction both times, always too low. Adding the convexity term $\frac{1}{2} \times 19.37 \times (0.01)^2 = 0.0968\%$ of price recovers essentially all of it. And the asymmetry is now visible in the exact row: the position loses 3.70 when rates rise but gains 3.89 when they fall, a difference of 0.19, which is twice the convexity term.

REAL INTERVIEW QUESTION (CACIB · Fixed Income)

How do you estimate the loss or gain on a \$100 million bond investment if rates rise by 100 basis points, using duration?

WHAT THEY TEST / ANSWER

Give the arithmetic first and the caveat second. Loss $\approx N \times D_{\text{mod}} \times \Delta y$, so with the bond above, \$100m $\times$ 3.79 $\times$ 0.01 = \$3.79 million. Then correct yourself before they do: 100 basis points is a large move, duration is a linear approximation, and the convexity term $\frac{1}{2} \times 19.37 \times (0.01)^2$ gives back about \$97 thousand, so the corrected estimate is \$3.69 million against an exact repricing loss of \$3.70 million. That progression, fast estimate, then stated limitation, then refined number, is exactly the answer shape market risk and rates desks are grading for. If they push further, the next limitation is bigger than convexity: this assumes the whole curve moved by 100 basis points in parallel, which it never does.

Portfolios, hedging, and immunisation

Duration aggregates linearly in market value. For a portfolio of positions with values V_i and modified durations D_i,

$$D_{\text{portfolio}} = \sum_i \frac{V_i}{\sum_j V_j} D_i,$$

and DV01s simply add, which is why desks aggregate in DV01 rather than in duration. A short position contributes a negative DV01, so hedging a rate exposure means adding an offsetting instrument, usually a future or an interest rate swap, sized so the total DV01 is zero.

The related idea a portfolio manager will raise is **immunisation**. When rates rise, a bond portfolio loses on price but gains on the rate at which coupons are reinvested, which is the reinvestment risk of the previous section seen from the other side. Those two effects cancel at exactly one holding horizon, and that horizon is the Macaulay duration. Matching the duration of your assets to the date you need the cash is therefore a way of locking in a return regardless of what rates do in between, at least to first order.

Where duration breaks

Duration is a genuinely limited tool, and knowing its limits is the mark of a candidate who has used it rather than read about it:

- **It assumes a parallel shift.** The entire framework prices a single move in one yield, whereas the curve steepens, flattens and twists. The fix is **key rate duration**, a separate sensitivity to each maturity bucket of the curve, which is how the DV01 ladder is built.
- **It is local.** It is a derivative at the current yield, so it is only valid for small moves, which is exactly what convexity patches.
- **It assumes fixed cash flows.** If the issuer can call the bond, or a homeowner can prepay a mortgage, the cash flows themselves change when rates move. A callable bond gets capped near par as rates fall, so its price-yield curve bends the wrong way and it exhibits **negative convexity**: it gains less on a rally than it loses on a sell-off. Mortgage-backed securities are the extreme case.
- **It says nothing about credit.** A corporate bond also moves with its credit spread, and the sensitivity to that is spread duration, a different exposure hedged with different instruments.

REAL INTERVIEW QUESTION (RBC · Fixed Income)

What is the duration of a bond? What is its formula? What are its limits?

WHAT THEY TEST / ANSWER

This is the full sweep, and the third part is where the marks are. Define it as the present-value-weighted average time to receive the cash flows, give $D_{\text{mac}} = \frac{1}{P} \sum_t t \cdot CF_t/(1+y)^t$ and the modified version divided by $1 + y$, then spend most of your time on the limits: parallel shifts only, valid only locally, and useless where the cash flows themselves depend on rates. Naming key rate duration as the answer to the first limit, convexity as the answer to the second, and callable bonds and mortgage-backed securities as the case where the third bites, turns a textbook answer into a desk answer.

INTERVIEW TRAP

Four traps, all seen regularly. (1) Quoting Macaulay duration when asked for a price sensitivity, so your risk number is off by the factor $1+y$, which matters when yields are high. (2) Saying a bond with duration 5 loses exactly 5% for a 1% rate move; it loses *approximately* that, and the error is the convexity you just failed to mention. (3) Claiming higher convexity is unambiguously better while ignoring that you pay for it in yield. (4) Applying duration to a callable bond or an MBS without flagging negative convexity, which is the single most expensive mistake in this list and the one that blew up plenty of real portfolios.

VISUAL / CODE TO BUILD: Interactive

A price-yield curve with two overlays. The reader picks coupon, maturity and yield, and sees the true price-yield curve, the tangent line (duration) and the fitted parabola (duration plus convexity), with a draggable yield shock that reads out all three prices and the error of each approximation. A second toggle switches the instrument to a callable bond so the curve visibly flattens above par and the tangent line ends up *above* the curve on the rally side, making negative convexity something the reader sees rather than something they memorise. A side panel shows the cash flow bars with their present values and the balance point marked, so the duration number and the picture stay linked.

Where this goes next

The parallel-shift assumption is relaxed by working directly with the curve and its bootstrapped points, which is what turns a single duration into a DV01 ladder. The instrument a desk actually trades to move its DV01 is the interest rate swap. Duration also feeds directly into value at risk, where it is the sensitivity that converts a distribution of rate moves into a distribution of profit and loss.

REAL INTERVIEW QUESTION (CACIB · Cross Asset)

Draw the parallel between delta and duration, and between gamma and convexity.

WHAT THEY TEST / ANSWER

This is a cross-asset interviewer checking that you have one mental model rather than two vocabularies. Duration is the first derivative of price with respect to yield, exactly as delta is the first derivative of an option price with respect to spot; both are the hedge ratio, and both are only valid locally. Convexity is the second derivative with respect to yield, exactly as gamma is the second derivative with respect to spot; both describe how the first-order sensitivity changes as the market moves, both are positive for the natural long position, a long bond and a long option respectively, and both are paid for, convexity through a lower yield and gamma through time decay. The one difference worth naming: duration carries a minus sign by convention, since price falls when yield rises, whereas delta does not. The Greeks themselves are developed in Greek definitions and intuition.

11.6 Curve Bootstrapping

sales · trading · structuring | advanced | rates · cross-asset

Everything in this chapter so far has assumed you already had a curve. You do not. Nobody quotes a discount factor. What the market quotes is a short list of tradeable instruments with their own conventions, and the curve has to be extracted from them. Bootstrapping is that extraction, and it is one of the few genuinely algorithmic things a rates interviewer can ask you to walk through, which is precisely why they do.

The problem

To price anything you need $D(t)$, the discount factor for every date t, as defined in zero-coupon bonds. What you can actually observe is perhaps twenty prices: a few overnight and short-dated deposit rates, a strip of futures or FRAs, and par swap rates at 2, 3, 5, 7, 10, 15, 20 and 30 years. That is twenty numbers, and you need a continuum. The gap between the two is the whole subject.

INTUITION

Think of **building a ladder one rung at a time**. You cannot reach the five-year rung directly, because the five-year swap rate is a blend of every year's discounting up to five. But the one-year instrument involves only one unknown, so you can solve it outright. Once that rung is nailed down, the two-year instrument contains only one *remaining* unknown, so it too can be solved. Each instrument you add gives you exactly one new equation and exactly one new unknown, and you climb. That recursive climb is bootstrapping, and the name comes from the same image of pulling yourself up by your own bootstraps: the curve is built out of pieces of itself.

Which instruments, and why

The curve is stitched together from different markets, chosen for liquidity at each maturity rather than for elegance:

- **Short end, out to a few months**: overnight index rates and short deposits, plus overnight index swaps. Simple interest, money market day counts (see coupon bonds).
- **Front and belly, a few months to two or three years**: exchange-traded interest rate futures and FRAs, which are liquid and directly give a rate over a future period. Futures need a small *convexity adjustment* before being used as forwards, because a future is margined daily and a forward is not, so the two are not the same contract.
- **Long end, two years and beyond**: par interest rate swap rates, the deepest liquidity in rates by a wide margin.

A government curve is built the same way from bills at the short end and on-the-run par bonds further out. The algorithm does not care which market supplies the quote; it cares that each instrument, in maturity order, introduces one unknown.

	$D(1)$	$D(2)$	$D(3)$
1y quote	**solve**		
2y quote	known	**solve**	
3y quote	known	known	**solve**

Each quote, taken in maturity order, adds exactly one new unknown.

The recursion

Take annual par swap rates s_T for $T = 1, 2, 3, \dots$. A par swap is quoted at the fixed rate that makes it worth zero today. Under a single curve, a floating leg that pays the reference rate on a notional of 1 and returns that notional at T is worth exactly 1 today, because it resets to the market rate every period. Stripping out the notional, the floating coupons are worth $1 - D(T)$, so the par condition is

$$s_T \sum_{t=1}^{T} D(t) \;=\; 1 - D(T).$$

The left-hand side is the fixed leg, the sum $\sum_t D(t)$ being the **annuity**. The mechanics of the swap itself belong to interest rate swaps; here it is just the equation that closes the system. Everything in it is known except $D(T)$, so solve:

$$\boxed{D(T) \;=\; \frac{1 - s_T \sum_{t=1}^{T-1} D(t)}{1 + s_T}}$$

Run it for $T = 1$, then $T = 2$, and so on. The zero rate at each node follows immediately from $D(T) = (1 + z_T)^{-T}$.

WORKED EXAMPLE

Annual par swap rates: $s_1 = 3.00\%$, $s_2 = 3.40\%$, $s_3 = 3.75\%$.

Step 1. No prior sum, so $D(1) = 1/1.03 = 0.970874$, and $z_1 = 3.000\%$.

Step 2. The prior sum is $D(1) = 0.970874$, so

$$D(2) = \frac{1 - 0.0340 \times 0.970874}{1.0340} = \frac{0.966990}{1.0340} = 0.935194,$$

and $z_2 = (1/0.935194)^{1/2} - 1 = 3.407\%$.

Step 3. The prior sum is $0.970874 + 0.935194 = 1.906068$, so

$$D(3) = \frac{1 - 0.0375 \times 1.906068}{1.0375} = \frac{0.928522}{1.0375} = 0.894961,$$

and $z_3 = (1/0.894961)^{1/3} - 1 = 3.768\%$.

Two sanity checks. First, the zero rates sit slightly *above* the par rates at every node beyond the first (3.407 against 3.400, 3.768 against 3.750), which is the direction yield curves predicts for an upward-sloping curve. Second, plug the curve back into the 3-year swap: the annuity is $0.970874 + 0.935194 + 0.894961 = 2.801029$, the fixed leg is $0.0375 \times 2.801029 = 0.105039$, and the floating leg is $1 - 0.894961 = 0.105039$. They match to the last digit, as they must.

KEY IDEA

A bootstrapped curve **reprices its own inputs exactly, by construction**. That is not a quality check on the curve, it is the definition of it. So "is my curve right?" is never a question about the algorithm, it is a question about which instruments you fed it, whether their conventions were handled correctly, and what you assumed between the nodes. This is also the practical content of no-arbitrage pricing in rates: the curve is the object that makes every quoted instrument simultaneously fair.

REAL INTERVIEW QUESTION (ING · Rates)

How do you build the discounting and forward curves?

WHAT THEY TEST / ANSWER

This is the flagship question of the section, and it wants a process, not a definition. Give it in four beats. One, choose instruments by maturity bucket: overnight and short deposits at the front, futures or FRAs in the belly with a convexity adjustment, par swaps at the long end. Two, sort by maturity and solve recursively, since each instrument in turn adds exactly one unknown discount factor, which is the boxed formula above. Three, choose an interpolation rule for the dates between nodes and say which one, because it changes the forwards. Four, and this is what separates a candidate who has built a curve from one who has read about it, say that discounting and projection are two *different* curves since 2008, so the build is simultaneous rather than sequential. Then stop and let them pick the thread they want to pull.

Between the nodes: interpolation

The recursion gives you the curve at the instrument maturities and says nothing about anything in between, yet most cash flows you need to discount fall between nodes. You must choose a rule, and the choice is not cosmetic:

- **Linear on zero rates**: simple and stable, but produces forward rates that jump at every node.
- **Linear on the log of discount factors**: equivalent to assuming the forward rate is constant between nodes, so forwards are piecewise flat rather than oscillating, and discount factors stay strictly positive. A common desk default. It does not force forwards to be positive: where the quoted instruments imply negative forwards, as EUR curves did for years, the interpolation reproduces them.
- **Cubic splines on zero rates or on forwards**: smooth forwards, which matters for products sensitive to the shape of the forward curve, at the cost of local stability. Moving one input can ripple into neighbouring segments and even produce forward rates that oscillate implausibly.

The trade-off is always the same, smoothness against locality, and there is no universally correct answer. What is not acceptable is not knowing which rule your curve uses, because forward rates come straight out of the interpolation, and forwards are what your hedge ratios are computed from.

One curve is not enough: OIS discounting

Before 2008, a single Libor-based curve was used both to project the floating coupons and to discount them. Two things broke that. Libor-OIS spreads blew out, showing that unsecured interbank lending carried real credit and liquidity risk. And collateral agreements became universal, so that a derivative's cash flows are backed by collateral posted daily.

The second point is the decisive one, and the argument is short. If a position is fully collateralised under a CSA, the collateral you receive earns the overnight rate, so the cost of carrying that position is the overnight rate, not Libor. The correct discount rate for a collateralised cash flow is therefore the **OIS rate**, built from overnight index swaps on the same currency and, if the CSA specifies it, the same collateral currency.

That splits the curve in two:

- the **discount curve**, bootstrapped from OIS instruments, which answers "what is a future euro worth today?";
- the **projection curve** (or forecast curve), one per reference rate and tenor, which answers "what will the floating coupon fix at?".

A single-curve world had one object doing both jobs. A multi-curve world has one discount curve and a family of projection curves, with a visible **tenor basis** between them, which is what basis swaps trade. The reference rates themselves, and the transition from Libor to SOFR and €STR, are covered in OIS, LIBOR, SOFR and €STR.

REAL INTERVIEW QUESTION (ING · Rates)

How do you distinguish collateralised (OIS) discount curves from uncollateralised ones?

WHAT THEY TEST / ANSWER

Anchor the answer on the funding argument rather than on convention. A collateralised trade is funded at the rate paid on the collateral, which under a standard CSA is the overnight rate, so its cash flows are discounted off the OIS curve. An uncollateralised trade is funded on the bank's own unsecured curve, so it must be discounted at a higher rate, and the difference in value between the two is exactly what funding and credit valuation adjustments are there to capture. Two details will impress. First, the relevant curve depends on the CSA terms, in particular the eligible collateral currencies, because a counterparty who may deliver in the cheapest of several currencies has an option that has to be valued. Second, this is why the same trade can be worth two different numbers to two banks: it is not a pricing error, it is a different funding assumption.

REAL INTERVIEW QUESTION (ING · Rates)

What is dual-curve bootstrapping, and how do you implement it?

WHAT THEY TEST / ANSWER

It is the multi-curve build done properly. Step one, bootstrap the discount curve from OIS instruments alone, exactly as in the recursion above; that part is still purely sequential. Step two, bootstrap the projection curve for a given tenor from the swaps that reference it, but discount every leg with the OIS curve you just built rather than with the curve you are solving for. The par condition is no longer the tidy $s_T \sum_t D(t) = 1 - D(T)$, because the floating leg no longer resets to par once projection and discounting differ, so you write the fixed leg against the projected floating coupons, both discounted on OIS, and solve for the last forward. Say the honest practical part too: in production this is usually posed as a global calibration solved numerically, since basis swaps couple the tenor curves to each other, and the clean rung-by-rung recursion is the pedagogical version. Knowing where the recursion stops being exact is the point of the question.

REAL INTERVIEW QUESTION (Bank of America · Rates)

What is the swap rate?

WHAT THEY TEST / ANSWER

The fixed rate that makes a new swap worth zero at inception, so it is a *par* rate, not a zero rate. That distinction is the whole reason this question sits next to bootstrapping: the swap rate is an average of the market's view over the life of the trade, blended through the annuity, and extracting the individual zero rates hidden inside it is exactly what the recursion above does. Add that the swap curve is the deepest rates market there is, which is why it, and not the government curve, is what most derivative pricing discounts against, and today specifically its OIS leg, for the collateral reason set out above. The instrument itself is developed in interest rate swaps.

INTERVIEW TRAP

Four errors, in rough order of how often they appear. (1) Treating the T-year swap rate as the T-year zero rate; they coincide only for $T = 1$, and the gap widens with the slope of the curve. (2) Forgetting the convexity adjustment when using futures as forwards, which quietly biases the belly of the curve. (3) Ignoring day-count and business-day conventions, so instruments from different markets are stitched together on inconsistent time measures. (4) Presenting a curve as “the” curve without saying which one, when a modern book carries an OIS discount curve plus a projection curve per tenor, and the same trade prices differently on each.

VISUAL / CODE TO BUILD: Interactive

A curve builder. The reader edits a small table of market quotes (a deposit, two futures, then par swaps at 2, 3, 5, 10 and 30 years) and the panel solves the recursion live, showing three linked plots: discount factors, zero rates and implied forwards. Two controls make the lessons visible. An interpolation selector (linear on zeros, linear on log discount factors, cubic spline) leaves the node values untouched while the forward curve visibly changes shape, sometimes into implausible oscillation on the spline. And a toggle between single-curve and OIS-discounted mode redraws the projection curve so the tenor basis appears as the gap between them. A readout confirms that every input instrument reprices to its quote in all configurations, which is the invariant the reader should walk away with.

Where this goes next

The curve you just built is the input to almost everything else. The rates implied between its nodes are forward rates, and they are what interpolation quietly determines. The instrument that supplies most of the long end, and the one desks trade to move their risk, is the interest rate swap. The overnight rates underpinning the discount curve are covered in OIS, LIBOR, SOFR and €STR, and the collateral machinery that makes OIS the right discount rate in the first place is collateral and haircuts.

11.7 Forward Rates

sales · trading · structuring | core | rates · cross-asset

A yield curve does not only tell you what rates are today for each maturity. It also pins down, exactly and with no room for opinion, the rate you can lock in today for a loan that starts in the future. Those are forward rates, and they are the single most reliable source of quick mental arithmetic in a rates interview: several banks in this database ask for one and explicitly forbid the calculator. This section is about getting them right in ten seconds and, just as importantly, about what they do and do not mean.

Note the vocabulary trap up front. A forward *rate* is an interest rate for a future period, which is what this section covers. A forward *contract* on an asset, with its cost-of-carry pricing, is a different object covered in forwards.

The idea: two routes to the same place

INTUITION

You want to put money away for five years. There are two honest ways to do it. **Route one**: buy the five-year rate today and forget about it. **Route two**: buy the two-year rate today, and at the same time agree *today* on the rate at which you will reinvest for the remaining three years. Both routes are risk-free and both tie up the same money for the same five years, so they must return exactly the same amount. If they did not, you would borrow through the cheap route and lend through the expensive one and pocket the difference with no risk and no capital.

That single equation, with one unknown in it, defines the forward rate. It is not a model, it is not a forecast, and there is nothing to estimate. It is arithmetic forced by no-arbitrage.

The formula

Write z_T for the annually compounded spot rate to maturity T, as built in bootstrapping, and f_{T_1,T_2} for the forward rate covering the period from T_1 to T_2. Equating the two routes:

$$(1+z_{T_2})^{T_2} \;=\; (1+z_{T_1})^{T_1}\left(1+f_{T_1,T_2}\right)^{T_2-T_1},$$

so

$$f_{T_1,T_2} \;=\; \left[\frac{(1+z_{T_2})^{T_2}}{(1+z_{T_1})^{T_1}}\right]^{\frac{1}{T_2-T_1}} - 1.$$

In discount factors, which is how a system actually stores it, the same statement is much cleaner:

$$\left(1+f_{T_1,T_2}\right)^{T_2-T_1} \;=\; \frac{D(T_1)}{D(T_2)}.$$

For a money-market period of length τ quoted with simple interest, as a FRA would be, the same ratio appears without the compounding:

$$f \;=\; \frac{1}{\tau}\left(\frac{D(T_1)}{D(T_2)} - 1\right).$$

And with continuous compounding the whole thing collapses to a weighted difference, which is the form worth memorising:

$$f_{T_1,T_2} \;=\; \frac{z_{T_2}T_2 - z_{T_1}T_1}{T_2 - T_1}.$$

KEY IDEA

The mental version of every forward-rate question is $f \approx (z_2T_2 - z_1T_1)/(T_2 - T_1)$: **multiply each rate by its own maturity, subtract, divide by the gap**. It is exact under continuous compounding and accurate to within a basis point or two under annual compounding at normal rate levels, which is far more precision than any mental estimate needs. Learn this one line and most of the questions in this section become instant.

REAL INTERVIEW QUESTION (CACIB · Rates)

If the 5-year spot rate is 3% and the 2-year spot rate is 2%, what is the 2y-5y forward rate? Compute it without the formula and without a calculator.

WHAT THEY TEST / ANSWER

Weight each rate by its maturity, subtract, divide by the three-year gap: $(3\times 5-2\times 2)/3 = (15-4)/3 = 11/3 = 3.67\%$. The exact annually compounded answer is 3.672%, so the mental version is right to about half a basis point here. The reasoning to say out loud while you compute is the one that earns the mark: five years at 3% must equal two years at 2% followed by three years at the forward, so the forward has to carry all the extra return that the five-year rate has over the two-year rate, compressed into three years rather than five. That is why the forward, 3.67%, sits well above the 3% five-year rate, and it is the sanity check that saves you if the arithmetic slips.

Worked examples

WORKED EXAMPLE

A FRA quote. A 12x24 FRA covers the twelve-month period starting in twelve months, so it is the 1y1y forward. With $z_1 = 1\%$ and $z_2 = 2\%$:

$$f_{1,2} = \frac{(1.02)^2}{1.01} - 1 = \frac{1.0404}{1.01} - 1 = 3.01\%,$$

and the mental rule gives $2\times 2 - 1\times 1 = 3\%$ directly, since with $T_2 - T_1 = 1$ the division does nothing.

Now shift both rates down by one point, $z_1 = 0\%$ and $z_2 = 1\%$:

$$f_{1,2} = \frac{(1.01)^2}{1.00} - 1 = 2.01\%, \qquad \text{mental: } 2\times 1 - 1\times 0 = 2\%.$$

The second case is asked for a reason. Both spot rates moved down by a full point, and the forward moved down with them by the same amount, so in both cases it still sits the same distance *above* the two-year rate: one point on the mental rule, 1.01 points on the exact figures. That gap is set by the *slope* of the curve, not by its level: since $f = 2z_2 - z_1$, a parallel shift passes straight through to the forward while the distance between forward and spot depends only on $z_2 - z_1$. Change the slope and the forward moves twice as fast as the spot rate does.

WORKED EXAMPLE

A longer gap. What is the 7-year rate starting in 5 years, given $z_5 = 3\%$ and $z_{12} = 4\%$? Mental: $(4\times 12 - 3\times 5)/7 = (48-15)/7 = 33/7 = 4.71\%$. Exact: $\left[1.04^{12}/1.03^{5}\right]^{1/7} - 1 = 4.720\%$. The approximation is within a basis point again, and note how far above the 4% twelve-year rate the forward sits: a curve that rises by only 100 basis points between year 5 and year 12 still implies a forward fully 72 basis points above its own long end, because the forward has to make up that difference over the back seven years alone.

REAL INTERVIEW QUESTION (Société Générale · Rates)

Estimate the 12x24 FRA given that the 1-year rate is 1% and the 2-year rate is 2% per year; then with the 1-year rate at 0% and the 2-year rate at 1% per year.

WHAT THEY TEST / ANSWER

3% and 2%, in about five seconds each, using $2z_2 - z_1$. The exact figures are 3.01% and 2.01%, and you should say so rather than pretend the approximation is exact. The second half of the question is the whole point, so do not answer it mechanically. Both spot rates dropped by a full point and the forward dropped with them, but it still sits one point above the two-year rate in both cases, because both curves have the same one-point slope. Spell that out: a parallel shift passes straight through to the forward, whereas the gap between forward and spot is a statement about the *slope*. That is why a steepening trade and a rally are completely different exposures even though both involve rates moving.

REAL INTERVIEW QUESTION (CACIB · Rates)

How do you compute a forward rate? For example, knowing the 5-year rate and the 12-year rate, what is the 7-year rate in 5 years?

WHAT THEY TEST / ANSWER

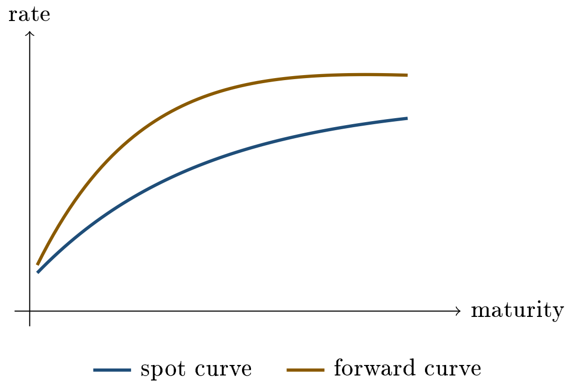
State the no-arbitrage identity first, since that is what is being tested: investing for 12 years must equal investing for 5 and rolling into the forward for the remaining 7, so $(1 + z_{12})^{12} = (1 + z_5)^5(1 + f)^7$, giving $f = [(1 + z_{12})^{12}/(1 + z_5)^5]^{1/7} - 1$. Then show you can do it without a calculator with the weighted-difference form $f \approx (12z_{12} - 5z_5)/7$, and produce a number from whatever levels they give you. If they supply no levels, invent plausible ones out loud and compute anyway; a candidate who says "it depends on the inputs" and stops has failed a question that was really about method.

The forward curve

Do this for every pair of adjacent maturities and you get the **forward curve**, the set of short-dated rates the market has already locked in for every future start date. Its relationship to the spot curve is mechanical:

- an **upward-sloping** spot curve implies a forward curve *above* it, because each new maturity's higher rate must be delivered by the marginal period alone;
- a **flat** spot curve implies forwards equal to spot;
- an **inverted** spot curve implies forwards *below* spot, which is how the market expresses expected cuts.

The spot rate at T is an average of the forwards up to T (exactly an arithmetic average under continuous compounding, a geometric one otherwise), so the forward curve is always the more extreme of the two, in the same way that a marginal cost curve is more extreme than an average cost curve.



REAL INTERVIEW QUESTION (Deutsche Bank · Fixed Income)

What is the forward curve, and what does it look like for the TEC10?

WHAT THEY TEST / ANSWER

Two questions in one, and both halves matter. Definition: the forward curve is the set of rates implied today for future periods, derived from the spot curve by no-arbitrage, and it lies above the spot curve when the spot curve slopes up. Market colour: the TEC10 is the French constant-maturity 10-year government rate, so its forward curve tells you what the market has priced for the 10-year OAT yield at future dates, and the honest answer describes its current slope and admits when you last checked rather than inventing a level. Then add the sentence that turns it into a trading answer: the forward curve is what you are implicitly selling when you go long a bond, so it is the benchmark your view has to beat, not a prediction you have to agree with.

A break-even, not a forecast

This is the conceptual point of the section, and it is where most candidates slip.

KEY IDEA

The forward rate is the level at which you are **indifferent** today. It is the break-even: if rates in one year turn out exactly at the forward, your rolling strategy and your long strategy produce identical returns. Forwards are therefore what you *trade against*, not what you *expect*. Saying "the market expects rates to be 4% in two years because the forward is 4%" is the classic overclaim.

Why it is an overclaim: a forward equals the market's expectation of the future rate *plus* a term premium, the extra return investors demand for locking money up, and a small technical convexity term. Historically the term premium has usually been positive, so upward-sloping curves have implied forwards that overpredicted realised rates. Both parts are unstable and neither is directly observable, so the correct statement is that the forward is the market-clearing break-even, and the split into expectation and premium is a matter of estimation and debate. The competing theories of curve shape sit in yield curves.

This is also the cleanest way to understand **carry and roll-down**. If the curve is upward sloping and, one year from now, it has not moved at all, then a bond you hold has aged into a lower point on the same curve, its yield has fallen, and it has gained in price. That gain is roll-down, and it is added to the coupon carry. The forward rate is precisely the future rate at which that whole pickup would be exactly wiped out. So "the forwards are realised" and "carry and roll-down earn zero" are the same statement.

INTERVIEW TRAP

Four traps. (1) Forgetting to annualise, or annualising over the wrong window: a 2y5y forward is compounded over 3 years, not 5. (2) Mixing compounding conventions between the two spot rates, which produces errors larger than the effect you are trying to measure. (3) Reading forwards as the market's forecast; they are break-evens, and the difference is a term premium. (4) Since 2008, asking for "the" forward without saying which curve: a forward projected off a 3-month tenor curve and one off the OIS curve are different numbers, which is what multi-curve bootstrapping is about.

VISUAL / CODE TO BUILD: Interactive

A two-panel forward explorer. The top panel shows the spot curve with draggable node points; the bottom panel redraws the implied forward curve live as the reader bends the spot curve, so the amplification effect becomes physical: a small steepening at the front sends the forwards sharply higher. A pair of handles selects T_1 and T_2, and a readout gives the exact forward, the weighted-difference approximation and the error between them, which teaches the mental rule by showing where it stays accurate. A third control drags the spot curve into inversion so the reader watches the forward curve cross below it.

Where this goes next

The tradeable contract on a single forward rate is the FRA, and a strip of forward rates chained together is what the floating leg of an interest rate swap pays, which is why a swap rate is essentially an annuity-weighted average of forwards. Which curve those forwards are projected off, and how the interpolation between nodes silently determines them, is curve bootstrapping. The same no-arbitrage argument applied across two currencies instead of two dates produces covered interest parity and the FX forward, and applied to an asset rather than a rate it produces the cost-of-carry formula in forwards.

11.8 Interest Rate Swaps

sales · trading · structuring | core | rates · cross-asset

The interest rate swap is the largest derivatives market in the world and the instrument every other part of this chapter has been quietly leaning on. The curve is built from swap rates. Rate risk is hedged with swaps. And in interviews the swap is asked about relentlessly, from "explain it to a child" all the way to the shape of its counterparty exposure profile. The good news is that it is one object seen from two angles, and once you have both, every question in the database falls out.

What it is

INTUITION

Two people have loans they are unhappy with. One pays a **fixed** rate and wishes she paid a floating one; the other pays a **floating** rate and would sleep better paying fixed. They cannot swap the loans themselves, because each bank lent to a specific borrower and neither wants to renegotiate. So instead they agree, privately, to pay each other's interest bills. Nobody moves the borrowed money, nobody's original loan changes; only the interest payments are exchanged. That agreement is a swap, and the sum they compute their payments on is called the notional precisely because it is notional: it never changes hands.

Formally, a **plain vanilla interest rate swap** is an agreement between two counterparties to exchange, on a fixed schedule until maturity:

- a **fixed leg**, paying an agreed rate s on the notional N;
- a **floating leg**, paying a reference rate that resets each period, on the same notional N.

The notional is never exchanged, and on each date only the *net* difference is actually paid. The reference rate used to be an interbank term rate such as Euribor or Libor, and is now usually a compounded overnight rate such as SOFR or €STR, a transition covered in OIS, LIBOR, SOFR and €STR.

— payer pays fixed s on N — receiver pays the floating reference rate on N

the notional N is never exchanged; only the net difference settles

The naming convention is fixed-leg-centric and catches people out. The **payer** pays fixed and receives floating. The **receiver** receives fixed and pays floating. A payer gains when rates rise, because it has locked in paying the old, lower fixed rate while collecting the new, higher floating one. So "long the swap" and "long rates" both mean payer, and if there is any ambiguity in an interview, say "payer of fixed" and the ambiguity disappears.

REAL INTERVIEW QUESTION (Société Générale · FX)

Explain to me what an interest rate swap is, as if I were a schoolchild.

WHAT THEY TEST / ANSWER

The test is whether you can strip the jargon without losing the substance, which is a sales skill and is graded as one. Something like: "You and I both borrowed money. Your loan has a payment that changes every three months, and you hate not knowing your bill. Mine is the same every year, but I think rates are going to fall and I would rather have the changing one. We cannot trade loans, so instead we promise to cover each other's interest payments. We never move the borrowed money itself, we just settle the difference." Then stop. Do not add notional, day counts or discounting unless they ask. The failure mode here is not being wrong, it is being unable to resist the technical vocabulary, and the interviewer is watching for exactly that.

Two ways to see it, and both are useful

View one: a pair of bonds. The fixed leg is exactly the coupon stream of a fixed-rate bond without the principal. The floating leg is exactly the coupon stream of a **floating-rate note** without the principal. Since the notionals cancel, a payer swap is economically *long an FRN and short a fixed coupon bond*:

$$V_{\text{payer}} \;=\; PV(\text{floating leg}) \;-\; PV(\text{fixed leg}).$$

This view is the fastest route to the risk: a payer is short a bond, so it makes money when yields rise, and its rate sensitivity is essentially that of the fixed-rate *bond* itself, principal included, because the floating-rate note it is paired against reprices to par at every reset and carries almost no rate duration. That is why swaps hedge bond portfolios so cleanly: per unit of notional the two carry the same DV01, even though the swap never exchanges principal. Note that it is the *bond's* duration, not the coupon stream's: the principal legs cancel between the two bonds, they do not drop out of the sensitivity.

View two: a strip of forwards. Each floating payment is a bet on one future fixing, and forward rates tell you exactly what the market has locked in for it today. So a swap is a portfolio of single-period forward contracts on rates, one per period, all settled against the same fixed rate. This view is the fastest route to pricing, and it is the one that survives the multi-curve world where the floating leg no longer conveniently prices to par.

Valuation and the par rate

Discount with the curve from curve bootstrapping and write $D(t)$ for the discount factor to payment date t. Under a single curve, the floating leg (excluding notional) is worth $1 - D(T)$ per unit of notional, so with annual payments

$$V_{\text{payer}} \;=\; N\Big[\big(1 - D(T)\big) \;-\; s\sum_{t=1}^{T} D(t)\Big].$$

The sum $A = \sum_t D(t)$ is the **annuity**, and it is the single most useful quantity in the swap market. Setting $V = 0$ gives the **par swap rate**, the rate quoted in the market:

$$\boxed{s_T \;=\; \frac{1 - D(T)}{\sum_{t=1}^{T} D(t)} \;=\; \frac{1 - D(T)}{A}}$$

The general, multi-curve version replaces the tidy numerator with the projected forwards discounted properly, $s = \sum_i D(t_i)\tau_i f_i / A$, which makes the "annuity-weighted average of forwards" reading explicit.

KEY IDEA

Two facts do most of the work in interviews. First, **a swap is worth zero at inception**, by construction: the fixed rate is chosen to make it so, which is why no money changes hands on day one. Second, once the market moves, the value of an existing swap to the payer is $V = N\,(s_{\text{market}} - s_{\text{contract}}) \times A$, where A is today's annuity. That one line prices any seasoned swap, and its derivative gives you the DV01, $N \times A \times 0.0001$, which is the risk number the desk actually manages.

WORKED EXAMPLE

Use the three-year curve built in curve bootstrapping: $D(1) = 0.970874$, $D(2) = 0.935194$, $D(3) = 0.894961$, so $A = 2.801029$. The three-year par swap rate is

$$s_3 = \frac{1 - 0.894961}{2.801029} = \frac{0.105039}{2.801029} = 3.750\%,$$

which recovers the quote the curve was built from, as it must.

Now trade it: pay fixed at 3.75% on a notional of \$100 million, and suppose the whole par curve immediately rises by 25 basis points, so the three-year swap now quotes 4.00%. Rebuilding the curve at the new quotes gives $D'(1) = 0.968523$, $D'(2) = 0.930679$, $D'(3) = 0.888492$ and $A' = 2.787694$. The swap is now worth, to the payer,

$$V = 100{,}000{,}000 \times (0.0400 - 0.0375) \times 2.787694 = \$696{,}924.$$

Checking it the long way gives the same number: per unit of notional the floating leg is worth $1 - D'(3) = 0.0400 \times 2.787694$ and the fixed leg 0.0375×2.787694, so the difference is $(0.0400 - 0.0375) \times 2.787694 = 0.006969235$, which on \$100 million is \$696,924 again. The two routes agree exactly because the rebuilt curve was constructed to quote the three-year swap at 4.00%, which is precisely the statement $1 - D'(3) = 0.0400\,A'$.

The DV01 is $100{,}000{,}000 \times 2.787694 \times 0.0001 = \$27{,}877$ per basis point, and indeed 25 basis points times \$27,877 is roughly \$697,000, the number above. On a three-year swap the annuity barely moves for a 25 basis point shift, so the linear estimate is almost exact here; on a thirty-year swap it would not be.

REAL INTERVIEW QUESTION (HSBC · Rates)

What is an interest rate swap, and how do you price it?

WHAT THEY TEST / ANSWER

Give the definition in one sentence (an exchange of a fixed rate against a floating rate on a notional that is never exchanged, net-settled on a schedule) and then price it in two moves, because that is what separates this from the schoolchild version. Move one: value each leg off the discount curve, so the swap is worth $PV(\text{float}) - PV(\text{fixed})$ to the payer. Move two: note that the quoted rate is the one that sets that to zero, giving $s = (1 - D(T))/\sum_t D(t)$. Then volunteer the modern caveat, that projection and discounting use different curves since 2008, so the neat $1 - D(T)$ shortcut is the single-curve special case and the general answer projects each forward and discounts it on OIS. Candidates who give the textbook formula and never mention the curve split are dating themselves.

REAL INTERVIEW QUESTION (Société Générale · Fixed Income)

How much is the fixed coupon of a swap worth? Give me the equation.

WHAT THEY TEST / ANSWER

They want the boxed formula and the reasoning behind it, not a paragraph. Say: the fixed coupon is set so the swap has zero value at inception, so fixed leg equals floating leg, so $s\sum_t D(t) = 1 - D(T)$ and therefore $s = (1 - D(T))/\sum_t D(t)$. Then give the interpretation, which is the part they are actually grading: the numerator is the value of the floating leg and the denominator is the annuity, so the swap rate is a *present-value-weighted average of the forward rates* over the life of the trade. That is why a swap rate is a par rate and not a zero rate, and why extracting the zero rates hidden inside it requires bootstrapping.

REAL INTERVIEW QUESTION (Société Générale · Fixed Income)

How much is a swap worth at inception?

WHAT THEY TEST / ANSWER

Zero, and the follow-up is inevitable, so answer both at once. It is zero because the fixed rate is quoted precisely as the level that equates the two legs; if it were not zero, one side would be paying the other to enter a contract that is supposed to be fair. Then add the two qualifications that show you have actually traded one. It is zero on a mid-market basis, but the client pays the bid-offer spread, so the trade is worth something negative to the client and something positive to the dealer the moment it is done. And it is only zero at inception: from the next tick onward it marks to market at $N(s_{\text{market}} - s_{\text{contract}})A$, which is why swaps generate counterparty exposure even though nothing was exchanged up front.

Why anyone does this

Swaps exist because the market where you can raise money cheaply is often not the market where you want your risk:

- a corporate issues a fixed-rate bond because that is where the investor demand is, then swaps to floating because its revenues track short rates;
- a bank funds itself at overnight rates but lends at fixed rates on mortgages, and swaps to close the mismatch;
- a pension fund with long-dated fixed liabilities receives fixed in very long swaps to match them, which is a large part of why the far end of the curve trades where it does;
- a trader with a view on rates uses a swap instead of a bond because it requires no funding of a cash position and no repo to finance it.

REAL INTERVIEW QUESTION (BMCE · Rates)

You have issued a bond on which you pay floating-rate interest. Do you buy or sell a swap?

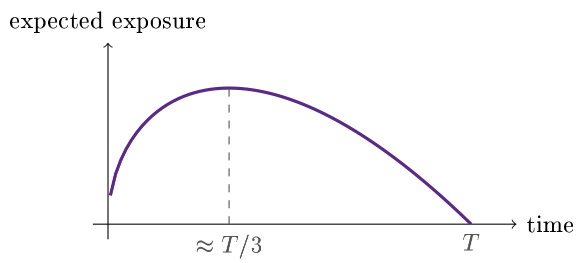
WHAT THEY TEST / ANSWER

You want to become a fixed-rate payer, so you enter a **payer** swap: pay fixed, receive floating. The received floating leg cancels the coupon you owe on your bond, and you are left paying a known fixed rate. Draw it if you can, three arrows: bondholders receive floating from you, the swap counterparty pays you floating and you pay them fixed, and the floating arrows cancel. Two details make the answer strong. Say why you would do it: you have removed the risk that short rates rise and your funding cost explodes. And name the residual risks rather than claiming a perfect hedge: the reference rate on your bond may not match the swap's exactly, which is basis risk, the payment dates may not line up, and you have replaced market risk with counterparty risk to the swap dealer.

Counterparty exposure, and why it is hump-shaped

Because no principal is exchanged, the amount you can lose if your counterparty defaults is not the notional, it is only the mark-to-market of the swap, and only when that mark is in your favour. That makes the credit exposure on a swap far smaller than on a bond of the same size, and it also gives it a distinctive shape through time.

Two forces fight each other. **Diffusion**: the further into the future you look, the more rates could have wandered, so the potential mark-to-market grows, roughly with the square root of elapsed time. **Amortisation**: as the swap ages, fewer payments remain, so the annuity shrinks and each basis point of rate move is worth less, driving the exposure to exactly zero at maturity. Early on diffusion dominates and late on amortisation does, so the expected exposure profile rises, peaks, and falls back to zero, typically peaking around a third of the way into the trade.



In practice this exposure is heavily mitigated: standardised swaps are centrally cleared with daily margin, and bilateral ones sit under a collateral agreement, both covered in collateral and haircuts. What remains is priced as CVA, and the broader framework is counterparty risk.

REAL INTERVIEW QUESTION (BNP · Rates)

What does the exposure profile of an interest rate swap look like, and how do you explain it (diffusion and amortisation effects)?

WHAT THEY TEST / ANSWER

Draw the hump. Then explain it as the product of two opposing terms. Diffusion: uncertainty about where rates will be grows with the square root of time, pushing potential exposure up. Amortisation: the remaining annuity shrinks as payments are made, so the sensitivity to any given rate move falls, and it must reach zero at maturity when nothing is left. The product of a rising and a falling function peaks in the middle, and for a vanilla swap it lands near a third of the maturity. Two refinements will land well: the profile for an amortising swap peaks earlier because the notional decays too, and a cross-currency swap does *not* look like this, because the final principal exchange means its exposure keeps climbing to maturity, which is exactly why those trades consume so much more credit line.

REAL INTERVIEW QUESTION (CACIB · Rates)

Which is riskier, a bond or a swap?

WHAT THEY TEST / ANSWER

Separate the two risks before answering, because the question is designed to catch people who conflate them. On **market risk** they are comparable: a 10-year receiver swap and a 10-year bond of the same notional have similar rate sensitivity, so per unit of notional neither is obviously worse. On **credit risk** the bond is far riskier: you have handed over the full principal, so a default costs you the whole investment less recovery, whereas a swap never exchanges principal and the most you lose is a mark-to-market that is usually a low single-digit percentage of notional, collateralised daily on top. The sharp closing point is that the swap carries a risk the bond does not: it is a leveraged position requiring no capital up front, so the same credit line supports a far larger notional, and that is where the real danger lives. "The bond is riskier per euro invested, the swap is riskier per euro of credit line" is the answer that lands.

INTERVIEW TRAP

Four recurring errors. (1) Getting payer and receiver backwards; anchor on "payer pays fixed" and derive the rest. (2) Saying the notional is exchanged in an interest rate swap; it is not, unlike a cross-currency swap, where it is, which is the whole reason their risk profiles differ. (3) Quoting $s = (1 - D(T))/A$ as universally true rather than as the single-curve case. (4) Claiming that swapping your floating bond to fixed leaves you with no risk, when it leaves you with basis risk, date mismatch and counterparty risk.

VISUAL / CODE TO BUILD: Interactive

A swap blotter. The reader sets notional, maturity, payment frequency and direction, and the panel shows the cash flow ladder with the fixed leg as level bars and the floating leg as bars at the projected forwards, plus a running net. Dragging the curve up, down or into a steepener re-projects the floating bars live and updates the mark-to-market, the annuity and the DV01, so the reader sees that a parallel shift moves the mark through the fixed versus floating gap while a steepener moves it through the far forwards only. A second tab animates the expected exposure profile as the trade ages, with separate toggles for the diffusion and amortisation components so the hump can be decomposed into the two curves that produce it.

Where this goes next

A single-period swap is a FRA, and a swap on two different floating references is a basis swap, which is where the tenor basis introduced in bootstrapping actually trades. Add a second currency and its principal exchange and you get the cross-currency swap in FX forwards and swaps. The rate that the floating leg references, and the market's move away from Libor, is OIS, LIBOR, SOFR and €STR.

11.9 OIS, LIBOR, SOFR and the Euro Short-Term Rate

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Every floating leg references one of these. Every discount curve is built from one of these. Every time a pricing formula says "the risk-free rate", one of these is what is actually meant. And because the whole market changed which one it used between 2018 and 2023, this is a topic where a candidate can sound either current or ten years out of date within a single sentence. It is also the home of one of the most-loved questions in rates interviews, the Libor-OIS spread.

Two families of rate

Everything in this section belongs to one of two families, and almost every confusion comes from mixing them up.

Term interbank rates (the IBORs) answer the question "what does it cost a bank to borrow, unsecured, from another bank, for three months?" They are quoted *in advance* for a period, they exist at several tenors, and because the borrower is a bank that might fail, they contain bank credit risk.

Overnight risk-free rates (the RFRs) answer "what did money actually cost overnight, last night?" They are single-tenor, published after the fact from real transactions, and because a one-day loan is nearly riskless, they contain almost no credit risk.

INTUITION

The difference is the difference between a **quoted price** and a **recorded price**. An IBOR was a panel of banks each saying what they thought they would have to pay, a forward-looking opinion. An RFR is the volume-weighted average of what was genuinely traded overnight, a backward-looking fact. Once you see that one is an estimate about the future and the other a measurement of the past, every practical difference follows: which one can be manipulated, which one is known in advance, and which one you can build a term structure out of directly.

The IBORs: LIBOR and EURIBOR

LIBOR was set daily by asking a panel of banks at what rate they believed they could borrow unsecured, across five currencies and seven tenors. Its fatal flaw was structural, not moral: after the crisis, banks barely lent to each other unsecured for three months, so the submissions were increasingly expert judgement about a market that had stopped existing. That, plus the manipulation scandals of 2012, made it unsustainable. Most settings ceased at the end of 2021, and the remaining US dollar settings ceased at the end of June 2023, with a short-lived "synthetic" version kept alive only so legacy contracts could run off.

EURIBOR did not die, which is the contrast worth knowing. It was reformed onto a hybrid methodology that uses real transactions where they exist and falls back to a defined waterfall of related market data where they do not. So the euro market runs both a term rate (EURIBOR) and an overnight RFR (€STR) side by side, while the dollar market moved almost entirely to SOFR. That asymmetry is a good thing to be able to state.

EONIA, the old euro overnight rate, was recalibrated as €STR plus a fixed 8.5 basis point spread and then ceased publication in January 2022.

REAL INTERVIEW QUESTION (BRED · Rates)

What are EONIA and EURIBOR, and where are they currently?

WHAT THEY TEST / ANSWER

Half of this is knowledge and half is market awareness, and you should answer both halves explicitly. Knowledge: EURIBOR is the euro term interbank rate, quoted at several tenors, unsecured, still published today after a methodology reform; EONIA was the euro overnight rate, was recalibrated as €STR plus 8.5 basis points in 2019 and stopped being published in January 2022, so quoting it as a live rate is the trap in the question. Market awareness: give the current 3-month EURIBOR level and say when you last checked. Never guess a level. A candidate who says "3-month EURIBOR was around X when I checked this morning, and EONIA no longer exists" has answered better than one who recites a definition perfectly and cannot produce a number. Refresh these before every interview: the desk assumes you watch screens.

The risk-free rates

Each major currency replaced its IBOR with an overnight rate computed from actual transactions:

- **SOFR** (USD), published by the New York Fed, based on overnight Treasury repo transactions, so it is **secured**;

- **€STR** (EUR), published by the ECB, based on banks' overnight unsecured borrowing from financial counterparties, so it is **unsecured** but effectively riskless at one-day tenor;
- **SONIA** (GBP), **TONA** (JPY) and **SARON** (CHF) fill the same role in their currencies, SARON being secured like SOFR.

Note that the two biggest, SOFR and €STR, are not the same kind of animal: one is a repo rate and one is a deposit rate. That matters in stress, when collateral scarcity can push a secured rate around for reasons that have nothing to do with credit.

Separately, the **Fed funds rate** is the unsecured overnight rate at which US banks lend reserves to each other. The FOMC sets a target *range* for it, and the effective fed funds rate is the volume-weighted median of actual trades. It is a policy instrument first and a benchmark second, and dollar OIS has largely migrated from it to SOFR.

The overnight index swap

An **OIS** is simply an interest rate swap whose floating leg is the overnight rate compounded over each period, rather than a term rate fixed in advance. Nothing about the swap machinery changes; only what the floating leg references.

Because the floating leg is the realised overnight rate, the fixed rate on an OIS is the market's break-even for the average overnight rate over the life of the trade. That gives the instrument two jobs, and both come up in interviews:

- **Reading policy expectations.** A 1-year OIS rate is essentially where the market thinks the average policy rate will sit over the next year, so the OIS curve is how desks price in central bank meetings. The macro side of this is monetary policy.
- **Discounting.** Collateral posted under a standard agreement earns the overnight rate, so the funding cost of a collateralised derivative is the overnight rate, which makes the OIS curve the correct discount curve. That is the argument developed in curve bootstrapping.

REAL INTERVIEW QUESTION (CACIB · Rates)

What is the OIS?

WHAT THEY TEST / ANSWER

One sentence for the instrument, then two for why anyone cares, which is what they are really asking. Instrument: a swap exchanging a fixed rate against the compounded overnight rate over each period, with no exchange of notional and typically a single net payment at maturity for short tenors. Why it matters: first, the fixed rate is the market's expected average overnight rate, so the OIS curve prices central bank policy directly, and a desk reads rate-cut probabilities off it; second, since collateral earns the overnight rate, OIS is the discount curve for collateralised trades, which is why the entire market rebuilt its curves around it after 2008. If they want more, name the currency's underlying rate, SOFR in dollars and €STR in euros, since "the OIS" means a different index in each.

The Libor-OIS spread

Put the two families side by side over the same three months. Term Libor is what a bank pays to borrow unsecured for three months, so it carries three months of bank credit and liquidity risk. Three-month OIS is the expected average overnight rate over the same window, essentially free of both. The difference between them therefore isolates exactly one thing.

KEY IDEA

The **Libor-OIS spread** is a clean read on perceived bank credit and funding stress, because the two legs cover the same period and differ only in whether the money is genuinely at risk for the whole three months. In calm markets it sat in single-digit to low-double-digit basis points. In October 2008 it blew out above 350 basis points, which is why it became the headline fear gauge of the crisis. It widened again, far less dramatically, in March 2020.

REAL INTERVIEW QUESTION (BMCE · Rates)

What does the spread between Libor and OIS mean?

WHAT THEY TEST / ANSWER

Say what each leg is before saying what the difference is, because that is the reasoning being tested. Libor is three-month unsecured interbank borrowing, so it embeds the risk that the borrowing bank does not survive three months, plus a liquidity premium for tying money up. OIS over the same window is the expected average overnight rate, which carries essentially neither. Subtract and you are left with the market's price of bank credit and funding stress, which is why the spread is a stress indicator rather than a rate view: it can widen with policy rates completely unchanged. Give the historical anchor, above 350 basis points in October 2008 against a handful of basis points in calm periods. Then close with the modern footnote, which is the part that shows you are current: with Libor gone, the same signal is now read off spreads such as SOFR against fed funds, or commercial paper and credit-sensitive indices against SOFR, because the pure Libor-OIS series no longer exists.

The practical headache: compounding in arrears

An overnight rate has one operational disadvantage that a term rate does not. If a coupon period runs for three months and the rate is the compounded overnight rate over those three months, then the coupon is not known until the period has almost ended. Borrowers who need to book a cash payment in advance find that unhelpful.

The compounded rate over a period is

$$R = \left[\prod_{i=1}^{n}\left(1 + r_i \frac{d_i}{360}\right) - 1\right] \times \frac{360}{D},$$

where r_i is the overnight fixing for business day i, d_i the number of calendar days it covers (three over a weekend), and D the total days in the period. The 360 is the act/360 convention used by SOFR and €STR; sterling runs act/365 fixed, so replace both 360s by 365 for SONIA. The market solved the timing problem with conventions rather than with new maths: a **payment delay**, a **lookback** that observes rates a few days earlier than it applies them, an **observation shift**, or a forward-looking **term SOFR** published from SOFR futures and derivatives, whose use is deliberately restricted so the market does not drift back to a thin, quotable term rate.

WORKED EXAMPLE

Suppose the overnight rate sits at 5.30% every day for a 90-day period, act/360. Simple interest would give $5.30\% \times 90/360 = 1.325\%$ for the period. Compounding daily gives

$$\left(1 + \frac{0.0530}{360}\right)^{90} - 1 = 1.33372\%,$$

which annualised is $1.33372\% \times 360/90 = 5.3349\%$. So daily compounding adds about 3.5 basis points a year at this rate level. Small, but on a \$1 billion notional that is about \$349,000 a year, and it is precisely the kind of detail an operations or product-control interviewer will probe. The gap grows roughly with the square of the rate level, so it was negligible at zero rates and is not now.

Which one is "the risk-free rate"?

REAL INTERVIEW QUESTION (Société Générale · Index)

What is the risk-free rate? Give me an example.

WHAT THEY TEST / ANSWER

The honest and strongest answer is that there is no single one, and the right choice depends on what you are pricing. For discounting a collateralised derivative, it is the overnight index rate in the collateral currency, so SOFR in dollars or €STR in euros, because that is what the collateral actually earns. For a textbook equity option, people generally use the OIS or government yield at the option's maturity. Then give the qualifications, since that is where the marks are. Government bonds are usually called risk-free in their own currency, but they carry sovereign credit risk, which euro area spreads demonstrated in 2011 and 2012. SOFR is a *secured* rate, so it moves with collateral scarcity as well as with policy. And no traded rate is truly risk-free; "risk-free" is a modelling idealisation we approximate with the least risky instrument available for the job at hand.

REAL INTERVIEW QUESTION (BRED · Rates)

If you quote LIBOR plus 0.30%, what determines that 0.30%?

WHAT THEY TEST / ANSWER

Break the spread into its parts rather than answering "the credit risk", which is only the largest piece. The borrower's credit quality is the first component, and it is why the same structure prices differently for two issuers with identical cash flows. Then the tenor and liquidity premium: a longer or less liquid commitment costs more. Then the lender's own cost, namely its funding and the regulatory capital the loan consumes, plus its target margin. Then the structure itself: security, seniority, covenants and any collateral all compress the spread. And finally supply and demand, since a spread is a negotiated price, not a computed one. The modern addition worth volunteering: in a legacy contract migrated off Libor there is also a fixed credit adjustment spread bridging the old term rate to the new overnight one, which exists precisely because Libor contained bank credit risk and the replacement does not.

INTERVIEW TRAP

Four errors that instantly date or undermine an answer. (1) Talking about Libor in the present tense; it stopped being published, and EURIBOR did not, so be precise about which market you mean. (2) Treating SOFR and €STR as interchangeable when one is secured and the other unsecured. (3) Calling OIS "the risk-free curve" without noting it is the right discount curve because of *collateral*, not because of some abstract purity. (4) Claiming an overnight rate and a term rate of the same tenor should be equal; the whole content of the Libor-OIS spread is that they are not, and the gap is a price, not an error.

VISUAL / CODE TO BUILD: Interactive

A benchmark comparison panel. One chart overlays a term rate and the corresponding OIS rate over the same tenor, with the spread plotted beneath and crisis periods shaded, so the reader watches the gap breathe with credit stress rather than with the level of rates. A second view is a compounding calculator: the reader enters a path of overnight fixings, or drags a policy meeting into the middle of the period, and sees the compounded coupon build day by day, with switches for lookback, observation shift and payment delay so the reader can see exactly which days each convention captures. A ticker of the current RFR in each major currency, with its publication time and source, makes the point that these are measured, not quoted.

Where this goes next

The gap between a term rate and an overnight rate, and between two different tenors of the same family, is traded as a basis swap, and it is the same object that appears as tenor basis when bootstrapping more than one curve. SOFR's foundation in secured lending connects it directly to repo markets, and the collateral rules that make the overnight rate the correct discount rate are collateral and haircuts. What central banks are actually doing to the front end of all these curves is monetary policy.

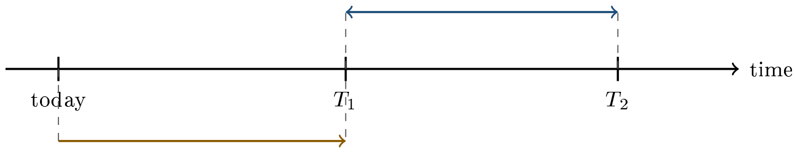
11.10 Forward Rate Agreements

sales · trading · structuring core rates

A forward rate is a number implied by the curve. A **forward rate agreement** is the contract that lets you actually trade it. It is the simplest interest rate derivative there is, and it is worth knowing precisely for two reasons: it is the single building block that an interest rate swap is made of, and its payoff contains one small twist that interviewers use as a filter.

The contract

A FRA is an agreement, struck today, to exchange a fixed rate K against the reference rate that will fix for a specified future period, on an agreed notional. The notional is never lent; only the interest difference settles.



— is the waiting period, from the trade date to the fixing at T_1: N months. — is the interest period the contract is written on, T_1 to T_2, which ends at month M. For a 3x6 the amber leg is three months and the blue leg is three months. The rate fixes at T_1 and settlement also happens at T_1, discounted, even though the interest it stands in for would only have been earned at T_2. That mismatch is the twist developed below.

FRAs are quoted as $N \times M$, where N is the number of months until the period starts and M the number of months until it ends. So a **3x6** FRA covers the three-month period starting in three months, and a **12x24** covers the twelve-month period starting in a year. The buyer of the FRA is the one who *pays fixed and receives floating*, which is economically a borrower locking in a rate, so the buyer gains if rates rise. That is the same convention as the payer of a swap, which is not a coincidence.

KEY IDEA

A FRA is a **one-period swap**. Everything you know from swaps carries over: it is worth zero at inception, its fair fixed rate is the forward rate implied by the curve, and the buyer is long rates. Conversely, a swap is a strip of FRAs sharing one fixed rate. Being able to say both directions of that sentence is worth more than memorising the payoff formula.

REAL INTERVIEW QUESTION (Natixis · Fixed Income)

What is a FRA?

WHAT THEY TEST / ANSWER

Definition, notation, direction, then the link. It is an over-the-counter agreement to exchange a fixed rate against a future fixing of a reference rate over a defined period, on a notional that is never exchanged. Quoted $N \times M$, so 3x6 is the three-month period starting in three months. The buyer pays fixed and receives floating, so a borrower who fears rates rising buys a FRA. Then the sentence that gets you past the definition: its fair rate is precisely the forward rate implied by the curve, and a swap is just a strip of these sharing a single fixed rate. If they want more, mention that with the move to overnight reference rates, term-IBOR FRAs have largely gone in dollars, where the front-end risk is now traded in SOFR futures and short OIS, while EURIBOR FRAs still trade.

The payoff, and the twist

At the fixing date, which is the *start* of the interest period, the reference rate fixes at R. The economic gain to the buyer over the period is $N(R-K)\tau$, where τ is the accrual fraction under the relevant day count. But that money is earned at the *end* of the period, whereas a FRA settles at the start. So the payment is discounted back over the period at the rate that just fixed:

$$\text{Settlement to the buyer} \;=\; \frac{N\,(R-K)\,\tau}{1+R\,\tau}.$$

That denominator is the twist, and it is the whole reason this question gets asked. Everything else about a FRA is intuitive; the discounting is the bit candidates drop.

WORKED EXAMPLE

A 3x6 FRA on a notional of \$100 million, struck at $K = 4.00\%$, on a period of 91 days with an act/360 day count, so $\tau = 91/360 = 0.252778$. At fixing the reference rate comes in at $R = 4.50\%$.

The interest saved over the period is

$$100{,}000{,}000 \times (0.0450 - 0.0400) \times 0.252778 = \$126{,}389,$$

but that is a gain measured at the end of the period. Settling at the start means discounting it at the rate that just fixed:

$$\frac{126{,}389}{1 + 0.0450 \times 0.252778} = \frac{126{,}389}{1.011375} = \$124{,}967.$$

Dropping the discounting overstates the settlement by \$1,422 on this trade. Had the fixing instead come in at 3.50%, the buyer would have paid \$125,281, and note that this is not the mirror image of the gain: the discount factor differs because it uses the realised fixing, so the payoff is very slightly asymmetric in R.

REAL INTERVIEW QUESTION (Bank of America · Rates)

What is the payoff of a FRA?

WHAT THEY TEST / ANSWER

Write $N(R-K)\tau/(1+R\tau)$ and then explain both halves, because the formula alone reads as memorisation. The numerator is the plain interest difference over the period, which is what any beginner would write. The denominator exists because a FRA settles at the beginning of the interest period while the interest itself would have been earned at the end, so the cash flow is discounted back over the period, at the fixing that just happened. Getting the numerator right and the denominator wrong is the single most common answer, and volunteering the denominator unprompted is the fastest way to signal you have priced one rather than read about one. Finish with the direction check: the buyer pays fixed, so the payoff is positive when $R > K$, meaning the buyer is long rates.

Pricing and mark-to-market

Since a FRA is worth zero at inception, its quoted rate must be the forward rate from the curve, $K = f_{T_1,T_2}$, computed exactly as in forward rates. Once the market moves, the FRA marks to market at the present value of the expected settlement: with f the *current* forward for the same period,

$$V \approx N\,(f - K)\,\tau \times D(T_2),$$

discounting from the end of the period on the curve from bootstrapping. This is the same $N \times$ (market rate − contract rate) × annuity shape as a swap, with an annuity of a single period.

FRA versus interest rate future

Both let you trade the same forward rate, and they are not the same instrument. An interest rate future is exchange-traded, standardised onto fixed quarterly dates and contract sizes, quoted as

100 minus the rate, and **margined daily**. A FRA is over-the-counter, any date and size you like, and settled once.

Daily margining is not a technicality, it changes the fair rate. A trader who is long the *rate* through futures receives cash on the days rates rise, and can reinvest that cash at the new, higher rate; when rates fall they pay out, and fund the payment at the new, lower rate. Both sides of that work in their favour, so being long rates via futures is strictly better than via a FRA, and the market compensates by setting the futures rate slightly *above* the forward rate:

$$\text{forward rate} \;=\; \text{futures rate} \;-\; \text{convexity adjustment}.$$

The adjustment is a couple of basis points at the front and grows with maturity and volatility, which is exactly why bootstrapping applies it before using futures in the belly of the curve.

REAL INTERVIEW QUESTION (CACIB · Rates)

What is the difference between a forward and a future?

WHAT THEY TEST / ANSWER

On a rates desk, give the generic contract differences briefly and then go straight to the one with a pricing consequence. Generic: futures are exchange-traded, standardised in size and date, and cleared, so counterparty risk is largely removed; forwards and FRAs are bilateral and bespoke, with counterparty risk managed through collateral. The pricing consequence: futures are margined daily, so cash moves in and out over the life of the trade, and that cash flow is correlated with the rate itself. Being long rates through futures means receiving cash exactly when reinvestment rates are high and paying it when they are low, which is worth something, so the futures rate sits above the forward rate by a convexity adjustment. A candidate who names the convexity adjustment and says which way it goes has answered the question the interviewer was actually asking. The general contract comparison, on assets rather than rates, is in futures.

INTERVIEW TRAP

Four traps. (1) Forgetting the $1/(1 + R\tau)$ discounting in the settlement. (2) Reading "3x6" as a six-month period, when it is the three-month period running from month three to month six. (3) Getting the direction backwards; anchor on "the buyer of a FRA is a borrower hedging a rise", so the buyer pays fixed and gains when rates rise. (4) Feeding futures rates into a curve as if they were forwards, with no convexity adjustment, which silently biases every forward you derive from the belly of that curve.

Where this goes next

Chain FRAs across consecutive periods against one common fixed rate and you have the interest rate swap. The rate a FRA is struck at comes from forward rates, and the curve those come from is bootstrapped partly out of the futures whose convexity adjustment this section explains.

VISUAL / CODE TO BUILD: Interactive

A FRA timeline you can drag. The reader sets N and M with two handles on the axis and the quote label updates live, so 3x6, 6x12 and 1x4 are the same picture at different widths rather than three things to memorise. Below the axis, three markers show what happens when: trade date, fixing at T_1, settlement also at T_1, and the notional interest date at T_2 greyed out to make the point that nothing changes hands there. A second panel takes a notional, a strike and a realised fixing and computes both the undiscounted interest difference and the discounted settlement, showing the gap between them and letting the reader sweep the fixing to see the payoff's slight asymmetry in R, which is the one thing about a FRA that is not obvious from the description.

11.11 Basis Swaps

sales · trading · structuring | advanced | rates · credit · cross-asset

A **basis** is the spread between two things that ought to be the same and are not. Three-month money against six-month money. An overnight rate against a term rate. A bond's yield against the swap curve. Before 2008 these gaps were rounding errors and most desks ignored them. Then they blew out, stayed out, and every one of them became a traded market with its own curve. Knowing where the bases are, and why they are not zero, is a large part of what separates a rates candidate from someone who has read a textbook written in 2006.

What a basis swap is

An ordinary interest rate swap pays fixed against floating. A **basis swap** pays *floating against floating*: two different reference rates, on the same notional, over the same period. Since neither leg is fixed, the price of the trade is a spread added to one of the legs, and that spread is the quoted market.

INTUITION

Two runners who should finish together keep finishing a few seconds apart. The basis swap is the bet on that gap, and the quoted spread is the handicap you have to give the slower runner to make the race fair. Nobody is trading the runners' speed, which is the level of rates; they are trading the *difference*, which is a much smaller and much more stable number, and which usually moves for reasons of credit, funding or regulation rather than of monetary policy.

Tenor basis

The most common one. A EUR 3s6s basis swap exchanges three-month EURIBOR, paid quarterly, against six-month EURIBOR, paid semi-annually, with a spread added to the three-month leg.

Why is it not zero? Borrowing unsecured for six months exposes the lender to a failure of the borrower over six months; rolling three-month loans twice lets the lender walk away halfway through. Those are genuinely different risks, so the six-month rate must exceed the market's expectation of two consecutive three-month rates, and the difference is a real premium for credit and liquidity, not an arbitrage. The spread therefore goes on the shorter, safer leg, to bring it up.

Before 2007 this was a basis point or less and was ignored. During the crisis it widened by tens of basis points, because the risk it prices is exactly the risk that was in question. It has never gone back to zero.

> **KEY IDEA**
>
> Tenor basis is the reason a modern desk carries **one projection curve per tenor** rather than one curve. That plurality, and the OIS discount curve alongside it, is what multi-curve bootstrapping exists to handle. Basis swaps are the instruments that pin the curves to each other, so they are inputs to the calibration, not an afterthought.

OIS versus IBOR basis

Push the tenor argument to its limit. On one side, an overnight rate compounded over three months, carrying one day of credit risk at a time. On the other, a three-month term rate carrying the full quarter. The basis swap between them is the traded version of the Libor-OIS spread, and it moves with perceived bank credit and funding stress rather than with the level of rates.

Since the move to overnight benchmarks, much of this basis has been absorbed into the fallback spreads written into legacy contracts, but the economics did not disappear: wherever a credit-sensitive term rate still trades against an overnight one, the spread between them still prices bank credit.

Cross-currency basis

Swap three-month dollars against three-month euros and, by textbook covered interest parity, the spread should be zero. It is not, and it has not been since 2008. Dollar funding scarcity, regulatory balance sheet costs and one-directional hedging demand from non-US issuers all push it around, and it widens sharply in stress precisely when everyone wants dollars at once. This one is developed in cross-currency basis, since it is inseparable from FX; the thing to carry from here is that it belongs to the same family and exists for the same kind of reason.

Asset swaps

An **asset swap** is the basis trade between a specific bond and the swap curve. The investor buys the bond and simultaneously enters a swap that pays away the bond's fixed coupons and receives a floating rate plus a spread. The package converts a fixed-rate bond into a floating-rate asset, and the spread is what the investor earns over the reference rate for taking that issuer's credit.

In the standard **par asset swap**, the investor pays 100 for the package whatever the bond's market price, and the difference is worked back into the spread. Setting the value of the package to zero gives

$$s_{\text{ASW}} = (c - s_{\text{swap}}) + \frac{100 - P}{100\,A},$$

where c is the bond's coupon rate, s_{swap} the par swap rate to the bond's maturity, P the bond's clean price per 100 of face and A the swap annuity from interest rate swaps. The first term is the coupon pickup over the swap curve; the second spreads the bond's discount or premium over the life of the trade.

WORKED EXAMPLE

A 5-year bond, 4% annual coupon, trading at 98. The 5-year swap rate is 3%, and on a flat 3% curve the annuity is $A = 4.5797$.

$$s_{\text{ASW}} = (4\% - 3\%) + \frac{100 - 98}{100 \times 4.5797} = 1.0000\% + 0.4367\% = 143.7 \text{ bp}.$$

Cross-check it against the cruder measure: the bond's yield to maturity at a price of 98 is 4.455%, so its spread over the 3% swap rate is 145.5 basis points. The two numbers are close but not equal, and the gap is the point. The yield spread discounts everything at one blended rate and quietly assumes coupon reinvestment at that rate; the asset swap spread is a genuine tradeable cash flow, computed off the actual curve. When a desk says "the bond is 144 over", it means the asset swap spread.

REAL INTERVIEW QUESTION (Natixis · Fixed Income)

Define an asset swap for me.

WHAT THEY TEST / ANSWER

Package first, purpose second. The package: buy the bond, and at the same time enter a swap paying the bond's fixed coupons away and receiving a floating reference rate plus a spread, on the bond's notional and to its maturity. The purpose: it strips the interest rate risk out of a bond and leaves you with a floating-rate asset whose only remaining exposure is the issuer's credit, so the asset swap spread is a clean price of that credit. Add the structural detail if they want depth: in a par asset swap the buyer pays 100 regardless of the bond's market price, so a discount or premium is amortised into the spread rather than paid up front, which is why the ASW spread is not simply the yield minus the swap rate.

REAL INTERVIEW QUESTION (Natixis · Fixed Income)

What are the two main advantages of an asset swap?

WHAT THEY TEST / ANSWER

They have told you they want two, so give exactly two, cleanly separated. **One, isolation of risk**: the swap removes the duration, leaving a floating-rate asset exposed only to the issuer's credit, which is what a bank treasury or a credit investor actually wants to own since it is funded at floating rates. **Two, comparability**: the asset swap spread puts bonds with different coupons, maturities and prices onto a single like-for-like scale over the swap curve, so it is the standard relative value measure in credit, and comparing it to the same issuer's CDS gives the CDS-bond basis. If you want a third, funding: the package can usually be repo-financed, so the investor can hold credit risk without tying up much cash.

REAL INTERVIEW QUESTION (Barclays · Rates)

How do you trade an asset swap?

WHAT THEY TEST / ANSWER

Walk the mechanics as an execution, since this is a trading desk asking. You buy the bond in the cash market and simultaneously execute an interest rate swap to the bond's maturity, paying fixed at the bond's coupon and receiving floating plus the asset swap spread, matched to the bond's notional and coupon dates so the fixed flows cancel exactly. You finance the bond in repo. Then name what can go wrong, which is what they are really testing: the swap is to maturity but the bond may be called or default, in which case you are left with a live swap and no asset, so the hedge is imperfect from day one; the repo may not roll at the rate you assumed; and if the bond is illiquid the spread you print is not the spread you can unwind at. Say you would quote the package as one price, not two legs, because the client is buying a spread.

Basis risk, as a general idea

Everything above is a special case of one thing. **Basis risk** is the risk you carry when your hedge is not the thing you are hedging: correlated, but not identical. It never shows up in the headline exposure and it is where hedges fail.

REAL INTERVIEW QUESTION (Allianz Global Investors · Market Risk)

What is basis risk in hedging corporate credit risk with CDS?

WHAT THEY TEST / ANSWER

Define it generally, then apply it, then say why it matters. Generally, it is the residual risk left when the hedge instrument differs from the exposure, so the two move together but not exactly. Applied here: you own a bond and hedge with a CDS on the same issuer, but they are not the same contract. Reference obligations and deliverable baskets may differ, the CDS maturity rarely matches the bond exactly, the CDS carries counterparty and now clearing considerations, documentation defines default events that do not map one-to-one onto your bond's terms, and the two markets have separate liquidity so they can dislocate. The CDS spread minus the bond's asset swap spread is the CDS-bond basis, so a negative basis means CDS trading tighter than cash, which has been the familiar state since 2008. It is a traded number, not an error. Why it matters: a position marked as hedged can still lose money, and it typically does so in stress, which is exactly when the risk report says your exposure is flat. Credit spreads and CDS are developed properly in the credit chapter.

INTERVIEW TRAP

Four errors. (1) Assuming a basis should be zero and calling it an arbitrage; each one prices a real difference in credit, funding or regulation, and it persists because the trade that would close it consumes balance sheet. (2) Putting the tenor basis spread on the wrong leg; it goes on the shorter, safer one. (3) Quoting yield minus swap rate as the asset swap spread; they are close, they are not the same, and the desk means the latter. (4) Describing a bond hedged with CDS as risk-free, when the CDS-bond basis is precisely the risk you kept.

VISUAL / CODE TO BUILD: Interactive

A basis dashboard. One panel plots several bases on a common basis-point axis over time, tenor basis, OIS-IBOR and cross-currency, with crisis windows shaded, so the reader sees that they widen together and for related reasons. A second panel is an asset swap calculator: the reader sets coupon, maturity, price and swap rate and sees the ASW spread decompose into the coupon pickup and the price-amortisation term, side by side with the simple yield-minus-swap-rate number so the gap between the two is visible and changes as the bond is dragged from deep discount to deep premium. A third view overlays an issuer's ASW spread and CDS spread with the basis shaded between them.

Where this goes next

The multiple curves that tenor basis forces you to carry are built in curve bootstrapping, and the reference rates whose differences create the bases in the first place are OIS, LIBOR, SOFR and €STR. The cross-currency member of the family is cross-currency basis. The credit that an asset swap spread is really pricing, and the CDS it is compared against, are credit spreads and credit default swaps.

11.12 Inflation-Linked Products

sales · trading · structuring | advanced | rates · cross-asset

Inflation is the one macroeconomic variable that has its own derivatives market. You can buy it, sell it, cap it and floor it, and a whole segment of the government bond market is indexed to it. What makes this a good interview topic is that almost every question reduces to one distinction, between **nominal** and **real**, and candidates who have that distinction firmly in place answer everything, while those who do not produce confident-sounding answers with the sign wrong. The macro drivers of inflation itself belong to inflation; this section is about the instruments.

Real and nominal

A **nominal** rate is what the contract pays in currency. A **real** rate is what it pays in purchasing power, after inflation has taken its cut. The Fisher relation links them:

$$(1+n) \;=\; (1+r)(1+\pi), \qquad \text{so} \qquad n \;\approx\; r+\pi$$

for small numbers, where n is nominal, r real and π inflation.

INTUITION

You lend someone enough money to buy 100 loaves of bread and they pay you back enough for 103 loaves. Your **real** return is 3%, and it does not care what a loaf costs. If bread also got 2% more expensive over the year, the number of euros you got back grew by about 5%, and that is your **nominal** return. Everything in this section is bookkeeping around which of those two numbers a contract fixes and which it leaves floating.

A conventional bond fixes the nominal and leaves the real return exposed: if inflation comes in higher than expected, you still get your fixed coupons, and they buy less. An inflation-linked bond does the opposite. It fixes the *real* return and lets the nominal cash flows float with the price index.

Inflation-linked bonds

A **linker** indexes its principal to a published price index. The coupon is a fixed *real* rate applied to that indexed principal, so both coupons and redemption grow with the index. Mechanically, everything runs off the **index ratio**, the reference index level on the payment date divided by the level at issue.

The main issues are US **TIPS** (indexed to CPI-U), French **OAT€i** (euro area HICP excluding tobacco) and **OATi** (French CPI excluding tobacco), Italian **BTP€i**, and UK index-linked gilts (historically RPI, with an announced move to CPIH from 2030). Three features recur and are worth knowing:

- **Indexation lag.** The index used for a given date is typically the published level about three months earlier, because the statistics office needs time to publish. So the last few months of inflation before a payment do not affect it, and the bond carries a small residual exposure to that stub.
- **Deflation floor.** Most linkers guarantee the original par at redemption even if the index has fallen since issue, so the investor is long an embedded option on cumulative deflation. Coupons are generally not floored, only the redemption amount.
- **Quoting convention.** Linkers are quoted on a **real yield**, and their price behaviour is driven by that real yield, not by inflation directly. This is the single most important sentence in the section.

KEY IDEA

A linker's price moves with the **real** yield exactly as a conventional bond's price moves with the nominal yield: same duration mechanics, different input. Inflation feeds the *cash flows*, which accrue over time; the real yield sets the *discounting*, which reprices instantly. That is why a linker can lose money in an inflation shock, and it is the trap in half the questions on this topic.

REAL INTERVIEW QUESTION (Natixis · Fixed Income)

Why do OATi bonds still lose value in a period of very high inflation?

WHAT THEY TEST / ANSWER

Because their price is a function of the *real* yield, and a big inflation shock usually pushes real yields up. Spell out the sequence: inflation surges, the central bank responds by tightening, the market prices higher policy rates for longer, and real yields rise across the curve. The linker's indexation is accruing in its favour, but that accrual arrives slowly over the remaining life, while the discount rate repriced this morning, and on a long-dated bond the duration effect dominates by a wide margin. Then add the two supporting reasons. The indexation lag means the most recent inflation is not yet in the cash flows. And breakevens can compress even as spot inflation prints high, if the market believes the tightening will work, which hurts the linker relative to the nominal. The sharp closing line: a linker protects your *redemption purchasing power* if you hold it to maturity, it does not protect your *mark-to-market* along the way. 2022 was the live demonstration.

Breakeven inflation

Put a nominal bond and a linker of the same maturity and issuer side by side. The **breakeven inflation rate** is the level of inflation at which the two deliver the same return:

$$1 + \text{breakeven} \;=\; \frac{1 + y_{\text{nominal}}}{1 + y_{\text{real}}}, \qquad \text{approximately} \qquad \text{breakeven} \;\approx\; y_{\text{nominal}} - y_{\text{real}}.$$

> WORKED EXAMPLE
>
> A 10-year nominal yield of 3.20% against a 10-year real yield of 0.90% gives a breakeven of $3.20 - 0.90 = 2.30\%$ on the quick subtraction, and $1.0320/1.0090 - 1 = 2.2795\%$ done exactly. The 2 basis point difference is the cross term, and quoting the subtraction while knowing it is an approximation is exactly the right level of precision for a desk conversation.
>
> Reading it: if inflation over the next ten years averages more than 2.30%, the linker wins; below that, the nominal bond wins. Buying the linker and selling the nominal is therefore a long position in breakeven: matched by duration, the common move in yields cancels and what is left is the spread between them.

> KEY IDEA
>
> Breakeven is **not** a forecast of inflation, for the same reason a forward rate is not a forecast of rates. It is expected inflation *plus* an inflation risk premium, which investors demand for bearing inflation uncertainty, *minus* a liquidity premium, since linkers trade far less than nominals and cheapen in stress. In March 2020 breakevens collapsed partly because linkers were being dumped, not because the market suddenly forecast deflation.

Inflation derivatives

The bond market is the visible part; the swap market is where the price is actually made.

Zero-coupon inflation swap (ZCIS). The benchmark instrument, and the one breakevens are quoted from. A single exchange at maturity: one side pays a fixed compounded rate, the other pays realised cumulative inflation. Per unit of notional the settlement to the inflation receiver is

$$\frac{I_T}{I_0} \;-\; (1+b)^T,$$

where b is the traded breakeven and I the index. One cash flow, no intermediate payments, which is why it is the cleanest expression of a view on inflation over a horizon.

Year-on-year inflation swap (YoY). Pays the annual inflation rate each year against a fixed rate. It matches the cash flow profile of a liability that reprices annually, such as an indexed pension or a rent, so it is what corporates and pension funds usually want. Note that a YoY swap is not simply a chain of ZCIS: converting between them requires a convexity adjustment, because the payoff depends on the ratio of two index levels and the correlation between inflation and rates matters.

Inflation caps and floors. Options on the index, typically on the year-on-year rate. Floors struck at zero are in structural demand, and it is worth being precise about who is short them: holding a linker makes the *investor* long the embedded deflation floor, so the short sits with whoever wrote deflation protection, a dealer who put a zero floor in an inflation-linked note or who gave a pension fund capped-and-floored indexation. That short has to be bought back in the option market.

Asset swaps and repacks. A linker can be asset-swapped so the investor keeps the credit and the yield but hands the inflation flows to the dealer, or the reverse.

WORKED EXAMPLE

A 5-year ZCIS on a notional of \$100 million, traded at a breakeven of $b = 2.30\%$. The fixed leg accrues to $(1.023)^5 = 1.120413$. Suppose realised inflation averages 2.60% per year, so the index ratio at maturity is $(1.026)^5 = 1.136938$. The single settlement to the inflation receiver is

$$100{,}000{,}000 \times (1.136938 - 1.120413) = \$1{,}652{,}498.$$

Note how leveraged the number is to a small error in the rate: 30 basis points a year of inflation, over five years, is worth 1.65% of notional. That compounding is why breakevens are quoted and hedged to a basis point.

REAL INTERVIEW QUESTION (BNP · Fixed Income)

What are the inflation derivatives?

WHAT THEY TEST / ANSWER

Give the taxonomy in order of importance and say what each is *for*, since a bare list is a weak answer. Zero-coupon inflation swaps first: one exchange at maturity, fixed compounded rate against cumulative realised inflation, and this is where the breakeven curve is actually quoted. Year-on-year swaps second: annual inflation against a fixed rate, matching liabilities that reprice each year, which is what pension funds and inflation-linked lease payers need. Then options, caps and floors on the year-on-year rate, with structural demand for zero floors from the dealers who have written deflation protection into notes and indexed liabilities. Then the cash side: linkers themselves, and asset swaps or repacks that separate their credit from their inflation. Finish with the point that shows depth: the market is *one-directionally supplied*, since governments and utilities are natural payers of inflation and pension funds are natural receivers, so dealers sit structurally on one side and the risk premium in breakevens reflects that imbalance rather than pure expectation.

REAL INTERVIEW QUESTION (CACIB · Cross Asset)

What is an inflation-linked bond (a linker)? Why are they so often the ones used in repacks?

WHAT THEY TEST / ANSWER

Definition first: a bond whose principal is indexed to a price index, paying a fixed *real* coupon on that indexed principal, quoted on a real yield, usually with a three-month indexation lag and a deflation floor at redemption. On the repack question, explain the mechanism rather than asserting a reason. A repack wraps a bond in a special purpose vehicle and swaps its cash flows into whatever profile the end investor wants. That is worth doing precisely when a bond's natural cash flows are awkward, and a linker's are: they are index-dependent, unpredictable in nominal terms, and the secondary market is far thinner than for the equivalent nominal bond. So an investor who wants the sovereign credit and yield but not the inflation exposure, or who wants the inflation in a different shape, is naturally served by a repack, and the dealer captures the linker's liquidity concession in the process. The wrapper mechanics are in structured notes.

REAL INTERVIEW QUESTION (CACIB · Rates)

What is the difference between inflation and core inflation?

WHAT THEY TEST / ANSWER

Headline inflation is the change in the full consumer price basket. Core strips out the most volatile components, food and energy, and sometimes also tobacco and administered prices. Give the reason for the distinction, which is the actual question: energy and food prices are driven by supply shocks that central banks cannot influence and that tend to reverse, so core is a better read on the underlying, persistent trend that monetary policy is trying to steer. Then the market-relevant twist that separates a good answer: linkers and inflation swaps settle on *headline*, not core, because that is what protects purchasing power and what the index series publish. So a trader can be right about the policy path, which follows core, and still lose on a breakeven position, which pays on headline. In the euro area the relevant series is HICP excluding tobacco.

REAL INTERVIEW QUESTION (CACIB · Cross Asset)

What product could you build to give inverse exposure to inflation, and what would be the point?

WHAT THEY TEST / ANSWER

The cleanest answer is the simplest: *receive* the fixed leg on a zero-coupon inflation swap, so you pay realised inflation and are paid $(1+b)^T$, or equivalently sell breakeven by buying the nominal bond and selling the linker. Get the leg the right way round, because the market convention is that the *payer* of the fixed leg is the inflation buyer, which is the opposite of what is wanted here. Both leave you gaining if inflation undershoots the level priced today. In note format for a client, structure a note whose coupon is a fixed rate minus realised inflation, floored at zero so the investor cannot owe money, which the desk hedges with a ZCIS plus an inflation cap. On the "what would be the point" half, name real users rather than speculation: an entity whose revenues are fixed in nominal terms while its costs are indexed is naturally short inflation and may want to double down cheaply, or more commonly an investor who thinks breakevens have overshot and wants to fade a level that already embeds a risk premium. Then state the risk honestly: compounding makes the payoff concave in inflation, so the position is short convexity, and it carries negatively in exactly the case that hurts it, when realised inflation runs *above* the breakeven you sold. Inflation shocks are the fat-tailed event that punishes it.

INTERVIEW TRAP

Four traps. (1) Saying a linker protects you from inflation full stop; it protects the redemption purchasing power if held to maturity, while the price is driven by real yields and can fall hard in an inflation shock. (2) Reading breakeven as the market's inflation forecast, when it is expectation plus an inflation risk premium minus a liquidity premium. (3) Ignoring the indexation lag, which matters for short-dated linkers where the stub is a large share of the remaining life. (4) Assuming linkers settle on core inflation; they settle on headline.

VISUAL / CODE TO BUILD: Interactive

A nominal-versus-real workbench. The reader sets a nominal curve and a real curve and sees the breakeven curve computed between them, exactly and by subtraction, with the cross-term error displayed so the approximation's accuracy is visible rather than asserted. A scenario panel then shocks inflation and real yields independently and shows a linker's price, its accrued indexation and its total return decomposed into the two effects, which makes the "loses value in an inflation shock" result something the reader produces rather than reads. A third view lays a ZCIS and a YoY swap side by side on the same inflation path so their very different cash flow timings are obvious, with the convexity adjustment between them shown as a residual.

Where this goes next

The macro question of what actually drives the price index is inflation, and the central bank reaction function that turns an inflation print into a real yield move is monetary policy. The swap machinery used throughout is interest rate swaps, the real-yield duration that drives linker prices is duration and convexity, and the repack wrapper is structured notes.

11.13 Repo Markets

sales · trading · structuring core rates · equity · cross-asset

Repo is the plumbing. It is how a bond desk pays for its inventory, how a hedge fund gets leverage, how a short seller obtains the security to deliver, and how central banks push liquidity into the system. It is also the reason a rates candidate and an equity derivatives candidate can both be asked about "repo" and mean different things by it, so this section covers both and is explicit about which is which.

What a repo is

A **repurchase agreement** is the sale of a security today combined with an agreement to buy it back at a fixed price on a fixed future date. Legally it is a sale and a repurchase. Economically it is a **collateralised loan**: the seller borrows cash, the buyer lends it, and the security sits with the lender as collateral. The difference between the two prices is the interest, expressed as the **repo rate**.

The same trade has two names depending on which side you are on. If you deliver the bond and take the cash, you are doing a **repo**. If you put up the cash and take the bond, you are doing a **reverse repo**.

> INTUITION
>
> It is a **pawn shop for institutions**. You hand over your watch, take the cash, and agree to buy the watch back next week for slightly more. The shop is not really interested in your watch; it wants the interest, and it holds the watch only so it cannot lose. The difference from a pawn shop is that the watch here is a government bond, so it is worth almost exactly what everyone agrees it is worth, which is why the interest rate charged is far lower than on an unsecured loan.

The practical mechanics matter and are asked about:

- **Haircut**: the cash lent is less than the collateral's market value, so the lender has a buffer. Deeper detail in collateral and haircuts.
- **Term**: overnight, term (a fixed period), or open (rolling daily until either side stops).
- **Margining**: positions are marked daily and margin flows both ways, which is what keeps the exposure near zero.
- **Coupons**: if the bond pays a coupon during the repo, it is passed back to the original owner as a manufactured payment. Economic ownership never moves, only legal title.
- **Documentation and venue**: bilateral under a GMRA, or tri-party with an agent bank handling collateral selection and substitution.

REAL INTERVIEW QUESTION (Société Générale · Volatility)

Explain repo to me.

WHAT THEY TEST / ANSWER

Lead with the economic substance and only then the legal form, because that ordering is what shows you understand it. Substance: a repo is a collateralised loan; one side borrows cash and posts a security as collateral, and the repo rate is the interest. Form: it is documented as a sale plus a forward repurchase, which is what gives the lender clean title to the collateral if the borrower fails, and is precisely why the rate is below the unsecured rate. Then give one use case so it is not abstract: a bond desk buys a bond and repos it out overnight to fund the purchase, so it holds the position without tying up its own cash. If they are on an equity or volatility desk, add the second meaning immediately, that in equity derivatives "repo" means the borrow cost of the stock and it enters the forward price, since that is almost certainly why they asked.

General collateral and specials

Two different reasons to do a repo produce two different markets.

A **cash-driven** trade is about the money: the borrower wants funding and does not much care which bond it delivers, and the lender will accept anything from an agreed basket. This is **general collateral** (GC), and the GC rate tracks the overnight index rate closely, since both are essentially the secured cost of short-term money. SOFR itself is built from exactly this kind of transaction, as noted in OIS, LIBOR, SOFR and €STR.

A **collateral-driven** trade is about the bond: someone specifically needs *that* security, usually to deliver against a short. They must lend cash to get it, and if demand is strong enough they will accept a below-market return on the cash to secure the bond. That bond is **on special**, and its repo rate trades below GC, sometimes deeply negative. The gap is called **specialness**.

KEY IDEA

Special means cheap funding for the holder, expensive for the short. The repo rate on a special is *low*, because the cash lender is being compensated in bond access rather than in interest. If you own that bond you are being paid to lend it out, and if you are short it your financing bleeds. Newly issued on-the-run government bonds are the classic specials, because everyone shorts the current benchmark.

REAL INTERVIEW QUESTION (JP Morgan · Delta One)

What does a repo curve look like?

WHAT THEY TEST / ANSWER

Answer in two layers. The GC curve first: it is a short-dated funding curve, anchored at the front to the central bank's policy rate and to the overnight index rate, and generally sloping with expectations of policy, so it looks like a very short piece of the money-market curve. Then the distortions, which is where the marks are, because a repo curve is much lumpier than a swap curve. Quarter-ends and especially year-ends show sharp spikes or drops as banks manage their published balance sheets. Central bank asset purchases remove collateral from the market and push GC below the policy rate, which is why euro repo traded deeply negative for years. And overlaying all of it, individual bonds trade on **special** well below the GC curve, so "the" repo curve is really a GC curve plus a scatter of issue-specific points underneath it. If they push, mention that repo is where funding stress shows up first, as in September 2019 when US overnight repo spiked far above policy.

Financing a bond: carry

The most common practical use is to buy a bond and fund it in repo. The position then earns the bond's coupon accrual and pays the repo rate, and the difference is the **carry**.

WORKED EXAMPLE

Buy a 10-year bond with a 3% coupon and finance it in repo at 3-month Libor plus 50 basis points, with Libor at 50 basis points. From 1 January 2017 to 31 March 2017 is 89 days.

The financing rate is $0.50\% + 0.50\% = 1.00\%$. On a money-market act/360 basis, the repo cost over 89 days is $1.00\% \times 89/360 = 0.2472\%$. The bond accrues its coupon on act/365, so it earns $3.00\% \times 89/365 = 0.7315\%$. Net carry is

$$0.7315\% - 0.2472\% = 0.4843\% \text{ of notional,}$$

or €484,285 on a €100 million position. The quick desk version, $(3.00\% - 1.00\%)/4 = 0.50\%$, gets you within 2 basis points in seconds.

What this calculation is *not*: the total profit and loss. It is the carry only. The position also marks to market on the bond's price, which can easily swamp 48 basis points of carry in a quarter.

REAL INTERVIEW QUESTION (BMCE · Rates)

You finance a bond with a 3% coupon and a 10-year maturity using a repo at 3-month Libor plus 50 basis points. What is your P&L if Libor is at 50 basis points from 01/01/2017 to 31/03/2017?

WHAT THEY TEST / ANSWER

Do the carry in your head and then correct the question, because the correction is the point. Carry: you earn 3%, you pay 1%, so roughly 2% a year, and one quarter of that is about 50 basis points of notional, or €500,000 per €100 million. Refine it with the day counts if you want the exact number, 0.4843% over 89 days, since the bond accrues on act/365 and the repo on act/360.

Then say what they left out: that is only the carry. Your actual P&L over the quarter is carry *plus* the change in the bond's price, and a 10-year 3% bond at par has a modified duration of about 8.5, so a 10 basis point move in yields is worth about 85 basis points of notional, roughly 1.8 times the entire quarter's carry. So a candidate who answers "50 basis points" and stops has computed the smaller half of the number. Add the residual risks too: the repo has to be rolled at each maturity, so you carry funding risk if repo widens, and the bond may go special or cheapen in repo against you.

Repo in equity derivatives

On an equity desk, "repo" usually means the cost of borrowing the stock, sometimes called the borrow rate. It matters because it enters the forward price. Ignoring the details of compounding conventions, the equity forward is

$$F \;=\; S\,e^{(r-q-\mathrm{repo})T},$$

where r is the funding rate, q the dividend yield and repo the rate a holder of the stock can earn by lending it out. The full derivation of the forward and its no-arbitrage argument is in forwards; what belongs here is why the repo term sits there with a minus sign.

REAL INTERVIEW QUESTION (JP Morgan · Delta One)

You are long rates, short dividends and short repo. Explain economically where that comes from.

WHAT THEY TEST / ANSWER

Build the replication and read the signs off it. To be long a forward synthetically you borrow cash and buy the stock today. You pay the funding rate on the borrowed cash, so a higher r makes the replication more expensive and pushes the forward up: you are **long rates**. While you hold the stock you collect its dividends, which reduce your net cost, so a higher dividend yield pushes the forward down: you are **short dividends**. And you can lend the stock out in the borrow market and collect the repo rate, which reduces your cost the same way, so a higher repo pushes the forward down: you are **short repo**. Every sign in $F = Se^{(r-q-\mathrm{repo})T}$ falls straight out of that one sentence about carrying the stock, which is why this is a better answer than reciting the formula. Add that on hard-to-borrow names the repo term is large and moves, so a Delta One book has real repo exposure, not a rounding error.

REAL INTERVIEW QUESTION (Commerzbank · EQD)

What is the impact of repo on options and on forwards?

WHAT THEY TEST / ANSWER

Say the route first, because it is the whole answer: repo never touches an option directly, it reaches it through the forward. On the forward the effect is mechanical, $F = Se^{(r-q-\text{repo})T}$, so a higher repo means a **lower** forward: the holder of the shares can lend them out and be paid for it, which reduces the net cost of carrying them to the delivery date.

On options everything then follows from that one move. A call is long the forward, so a higher repo makes it *cheaper*; a put is short the forward, so it makes it *dearer*. Put-call parity holds throughout, since both are quoted against the same forward. Delta is computed off the forward too, so a lower forward means a lower $N(d_1)$ and a smaller delta on the call, which is a rehedge with no spot move at all. Name the exposure rather than describing it: this is **repo rho**, a real Greek on an equity book, alongside the ones in the Greeks.

Close with why the desk cares. On an index or a large liquid name the repo term is a few basis points and nobody argues about it. On a hard-to-borrow single name it can be tens of percent, it moves, and the borrow underlying it can be recalled, so it is a live risk rather than a pricing constant, and a book priced off a stale borrow curve is mismarked. The equity-borrow side of that, with the numbers, is stock borrow and lending.

Why repo breaks

Repo looks safe because it is collateralised, and it is safe most of the time, which is exactly what makes its failures violent:

- **Run risk**. Repo is overwhelmingly short dated, so a borrower funding long-dated assets with overnight repo must roll every day. Lenders can simply decline, and a firm that looked solvent on Friday cannot fund on Monday. That is essentially what happened to Bear Stearns and Lehman.
- **Haircut procyclicality**. Haircuts rise when collateral gets volatile, which is when borrowers can least afford to post more, forcing sales that make the collateral more volatile still.
- **Wrong-way risk**. Collateral correlated with the counterparty, an Italian bank posting Italian government bonds, is worth least exactly when it is needed.
- **Collateral scarcity**. When a central bank has bought a large share of the free float, GC can trade far below policy, and squeezes in individual issues become routine.

The systemic dimension is liquidity risk.

INTERVIEW TRAP

Four traps. (1) Getting specialness backwards; a special bond has a *low* repo rate, and being short it is expensive. (2) Confusing repo with securities lending; they are economically similar, but securities lending is collateral-driven, often against non-cash collateral, and documented and fee-quoted differently. (3) Answering a financing question with carry alone and forgetting the mark-to-market, which is usually the larger number. (4) On an equity desk, treating repo as a constant; on hard-to-borrow names it is large, volatile and a genuine source of P&L.

VISUAL / CODE TO BUILD: Interactive

A repo desk simulator. Panel one shows a GC curve with individual bonds plotted as points beneath it, coloured by specialness, and a slider for central bank holdings that drains free float and watches GC sink below policy while specials deepen. Panel two is a carry calculator: the reader sets coupon, repo spread, day counts and holding period, and sees carry accumulate as a line, with a second line for mark-to-market driven by a yield slider, so the relative size of the two is immediate rather than argued. Panel three switches to equity mode and plots the forward and the call delta as the repo rate is dragged from 0% to 15%, making the Commerzbank question something the reader can replay.

Where this goes next

The haircuts and margin rules that govern how much cash a given piece of collateral raises are collateral and haircuts. On the equity side, the borrow market this section touches is developed in stock borrow and lending, and the forward whose repo term it feeds is forwards. The secured overnight rates built out of repo transactions are OIS, LIBOR, SOFR and €STR.

11.14 Collateral and Haircuts

sales · trading · structuring | advanced | rates · credit · cross-asset

Collateral is the machinery that turned derivatives from a credit business into a market business. It is also why the OIS curve is the right discount curve, why a swap's counterparty exposure is small, and why several otherwise unrelated crises all ended the same way, with someone unable to meet a margin call. This section is the mechanics: what a haircut is, what sets it, how margin actually flows, and why the whole apparatus is stabilising in normal times and destabilising in bad ones.

What collateral actually does

If you owe me money and I hold nothing, I carry your credit risk in full. If you post me a government bond worth more than you owe, and we true that up every day, I no longer really carry your credit risk. What I carry instead is smaller and different: the risk that the collateral falls in value, that I cannot sell it quickly, that the paperwork fails, or that its value collapses precisely because you did.

KEY IDEA

Collateral does not delete risk, it **transforms** it. Counterparty credit risk becomes market risk on the collateral, liquidity risk in selling it, legal and operational risk in claiming it, and wrong-way risk if the two are correlated. Every question in this topic is really asking whether you know that the transformation is not free.

The haircut

A **haircut** is the discount applied to collateral when working out how much it covers. With a haircut h, collateral of market value C supports only $C(1-h)$ of exposure, so to cover exposure

E you must post

$$C = \frac{E}{1-h}.$$

The reciprocal is the more intuitive reading: $1/h$ is the maximum leverage the haircut permits.

> **WORKED EXAMPLE**
>
> Borrow €100 million in repo against government bonds with a 2% haircut. You must deliver $100/0.98 =$ €102.04 million of bonds, so €2.04 million of your own money is tied up and your leverage is $1/0.02 = 50$ times.
>
> Now the market turns and the haircut goes to 5%. Two ways to see the same squeeze. To keep borrowing €100 million you now need $100/0.95 =$ €105.26 million of bonds, so you must find €3.22 million more collateral. Or, holding the same €102.04 million of bonds, the cash they raise falls to $102.04 \times 0.95 =$ €96.94 million, leaving you €3.06 million short *today.* A three point change in a number nobody was watching just created a multi-million euro funding hole, and the only fast way to fill it is to sell bonds, which is what everyone else is doing at the same time.

What sets the level of a haircut is essentially a risk calculation over the time it would take to get out:

- **Price volatility** of the collateral, which is why equities are haircut far more than short-dated government bonds;
- **Liquidity**, since the relevant question is the move over the period needed to liquidate, not over one day;
- **Maturity and duration** of the collateral, a 30-year bond moving far more than a 2-year one for the same yield shock;
- **Currency mismatch** between collateral and exposure;
- **Wrong-way risk**, the correlation between the collateral's value and the counterparty's own default.

Formally a haircut is close to a value at risk on the collateral over the liquidation horizon, at a high confidence level.

REAL INTERVIEW QUESTION (ECB · Market Risk)

How does the ECB manage its risk (counterparty selection, collateral, haircuts)?

WHAT THEY TEST / ANSWER

Answer as three concentric defences, because that is how a central bank actually thinks about it. **Counterparty selection**: only sound credit institutions subject to the Eurosystem's reserve requirements and supervisory standards can access the operations at all, so the first control is who is allowed through the door. **Collateral eligibility**: all lending is fully collateralised against a published list of eligible assets, government and corporate bonds, covered bonds, asset-backed securities and, under national frameworks, some credit claims, each subject to minimum credit-quality thresholds under the Eurosystem credit assessment framework. **Valuation and haircuts**: assets are marked daily, and haircuts vary by liquidity category, residual maturity, coupon structure and credit quality, so a long-dated ABS takes a far larger haircut than a short government bond, with margin calls when the marked value drifts.

Then make the point that shows judgement rather than recall: this framework is deliberately *broad*, because a central bank's job in a crisis is to lend against collateral that is fundamentally sound but that the market has temporarily stopped accepting. That is Bagehot's rule, and state it in full if you invoke it: lend freely, against good collateral, at a penalty rate. So the ECB accepts a wider collateral set than a commercial counterparty would, and manages the resulting risk through haircuts and eligibility rules rather than through refusal. The tension between that role and prudent risk management is the interesting part of the question.

How margin actually flows

Between two dealers, collateral is governed by a **credit support annex** (CSA) attached to the master agreement. Two flows matter, and mixing them up is a reliable way to fail a risk interview.

Variation margin is the daily exchange of the change in mark-to-market. Its purpose is to reset current exposure to zero every day, so whoever is in the money holds cash or securities equal to that gain. It is a transfer of value that already exists.

Initial margin is a buffer posted on top, sized to cover the move that could happen between the last variation margin exchange and the moment the position is finally closed out after a default. That window is the **margin period of risk**, set at ten business days for non-cleared trades under the uncleared margin rules and shorter for cleared ones. Initial margin is *not* a transfer of value: it is posted by both sides and, under the uncleared margin rules, must be segregated so it cannot be reused.

Around those two sit the terms that determine how much actually moves:

- **Threshold**: an amount of exposure left uncollateralised; zero for dealer-to-dealer, often large for a corporate client.
- **Minimum transfer amount**: a floor that stops trivial daily calls.
- **Eligible collateral schedule**: which currencies and securities may be delivered, each with its own haircut.
- **Remuneration**: cash collateral earns an agreed rate, normally the overnight index rate, which is the fact that makes OIS the correct discount curve in curve bootstrapping.
- **Rehypothecation**: whether the receiver may reuse the collateral, which is efficient and is also how collateral chains propagate stress.

KEY IDEA

When a CSA allows several eligible currencies or securities, the poster holds a **cheapest-to-deliver option**: they will post whatever is cheapest to fund at the time. That option has value, it moves with rate and basis differentials, and it has to be priced, which is one of the reasons a "CSA discount curve" is not a single obvious object. Naming this is a reliable way to show you have seen a real collateral agreement.

Cleared versus bilateral

For standardised products, a central counterparty steps in between the two sides, so each faces the CCP rather than each other. The CCP collects variation and initial margin daily and stands behind a **default waterfall**: the defaulter's own margin first, then the defaulter's contribution to a mutualised default fund, then the CCP's own capital, then the surviving members' contributions. That concentrates risk into a heavily regulated node rather than eliminating it, which is the nuance worth stating: clearing makes the system more transparent and the node more systemically important at the same time. Which products are *obliged* to clear, and the rules that made it so, are mandatory clearing. Bespoke trades stay bilateral under a CSA, with uncleared margin rules imposing broadly similar discipline.

REAL INTERVIEW QUESTION (Louis Dreyfus · Commodity)

How can you mitigate your counterparty risk?

WHAT THEY TEST / ANSWER

Give a layered answer, from cheapest to most structural, and note what each layer leaves behind. **Selection and limits**: trade with better credits, and cap exposure per counterparty. **Netting**: a master agreement with close-out netting turns many gross exposures into one net one, which is often the single largest reduction available. **Collateral**: a CSA with daily variation margin resets exposure to zero each day, and initial margin covers the gap until close-out. **Clearing**: novate to a CCP so you face a member-backed institution under a default waterfall. **Transfer**: buy protection, a CDS or a guarantee, or price the residual as CVA. **Structural**: shorten tenors, use break clauses, require prepayment or letters of credit, which is common in commodities where counterparties are corporates rather than banks.

Close with the honest caveat, since a commodities desk will expect it: none of this removes risk, it converts it. Collateral leaves you with market and liquidity risk on the collateral, clearing leaves you exposed to the CCP, and a physical commodity counterparty can fail to *deliver* rather than fail to pay, which no amount of cash margin fixes.

Why this is procyclical

The entire apparatus works by demanding more cash exactly when cash is scarce. Volatility rises, so haircuts and initial margin rise, so borrowers must post more at the same moment their assets are falling, so they sell, so volatility rises further.

This is not theoretical. It is the mechanism by which margin calls forced liquidation at LTCM in 1998, drove the collateral spirals of 2008, produced the March 2020 dash for cash, and in 2022 forced UK liability-driven investment funds to sell gilts into a falling market to meet margin on their gilt and swap positions, which pushed gilt yields higher and generated more margin calls. The same loop each time, with different collateral.

REAL INTERVIEW QUESTION (BNP · desk not recorded)

How do you hedge against collateral risk (haircut risk)?

WHAT THEY TEST / ANSWER

First reframe it, because "hedging a haircut" is not a clean market trade. There is no liquid instrument on haircut levels, so the answer is management, not hedging, and saying that is better than inventing a product. The levers are: hold a **liquidity buffer** of unencumbered high-quality assets sized to a stressed haircut scenario rather than today's; **term out** the financing so haircuts reset less often, which converts overnight repo risk into a longer commitment at a higher rate; **diversify** collateral and counterparties so one asset class being re-graded does not hit everything at once; negotiate **fixed haircuts** contractually for the life of the trade where you can; and reduce **wrong-way risk** by not posting collateral correlated with your own or your counterparty's credit.

Then the modelling answer: stress test it. Run haircuts and initial margin to crisis levels and check the resulting funding need against your buffer, which is exactly what liquidity coverage requirements are designed to force. If they want a market hedge, the closest honest answer is buying optionality on the collateral itself or on volatility, since haircuts are essentially a value-at-risk number and rise with volatility, but say plainly that it is a proxy and a poor one.

INTERVIEW TRAP

Four traps. (1) Saying collateral eliminates counterparty risk; it converts it into market, liquidity, legal and wrong-way risk. (2) Confusing variation margin (a transfer of realised mark-to-market) with initial margin (a segregated buffer against future moves during close-out). (3) Forgetting that collateral is remunerated, which is the entire reason OIS is the discount curve. (4) Treating haircuts as fixed parameters; they are risk measures that rise in stress, which is what makes them a systemic amplifier rather than a safety feature.

VISUAL / CODE TO BUILD: Interactive

A margin spiral simulator. The reader holds a leveraged bond portfolio financed in repo and controls a volatility slider. Raising volatility increases the haircut through a value-at-risk rule, which generates a funding shortfall, which forces sales, which feed back into price and volatility, so the reader can watch the loop run and see how much starting leverage the position survives. A side panel breaks the daily flows into variation margin and initial margin so the difference is visible rather than described, and a toggle switches between a bilateral CSA and a cleared setup so the reader can compare the two margin profiles on the same scenario.

Where this goes next

The residual counterparty risk that collateral does not remove is priced as CVA, DVA and FVA and managed under counterparty risk. The funding side of the spiral is liquidity risk. The market where haircuts are set and cash is actually raised is repo markets, and the discounting consequence of collateral remuneration runs back through curve bootstrapping to the overnight rates collateral earns.

Chapter 12

Credit Markets

12.1 Corporate bonds

sales · trading · structuring core credit

A government bond can be treated, near enough, as a pure interest-rate instrument. A corporate bond cannot, because the borrower can fail. Everything that makes credit a separate business from rates comes out of that one extra possibility: a second risk factor to be paid for, a ranking that decides who gets paid first when it happens, a document that says what the borrower may and may not do in the meantime, and a market that trades by appointment rather than in a continuous book.

How a bond's cashflows are discounted, what accrued interest and day counts are, and how duration and convexity work are rates mechanics and are not repeated here: see bond pricing, coupon bonds and duration and convexity. This section is about what changes once the issuer is a company.

A bond is a loan cut into tradeable pieces

A company that needs money can borrow it from a bank, or it can issue a bond. The cashflows can be made identical. What differs is everything around them.

- **One lender or many.** A loan is an agreement with a bank or a small syndicate. A bond is sold in standardised denominations to hundreds of investors, none of whom needs to know the others.
- **Transferable or not.** A bond is a security, designed to be sold on. A loan can be transferred, but slowly and with consent. This is the reason a bond has a market price at all.
- **Who watches you.** A loan comes with maintenance covenants tested every quarter and a relationship banker who will renegotiate. A bond typically comes with weaker, incurrence-based covenants and a dispersed set of holders who cannot easily renegotiate anything.
- **Fixed or floating.** Loans are usually floating-rate; public bonds are usually fixed-rate. An issuer who wants the other one swaps it, which is the single largest source of corporate flow into interest rate swaps.

> KEY IDEA
>
> The bond exists to convert one negotiated, illiquid, closely-monitored obligation into many identical, transferable, loosely-monitored ones. The issuer buys size, tenor and freedom from oversight, and pays for it with a public price that everyone can see going wrong.

Reading the term sheet

A corporate bond is defined by a short list of terms, and an interviewer can test all of credit by asking you to read one. Each line changes the price.

- **Issuer and guarantor.** You lend to a legal entity, not to a brand. A bond issued by a finance subsidiary and guaranteed by the parent is a different risk from one issued by an operating company that owns the assets. When they differ, the difference is called structural subordination and it is worth real basis points.
- **Ranking.** Secured, senior unsecured, subordinated, hybrid. This is the subject of the next subsection.
- **Size and denomination.** Benchmark deals in Europe are typically 500 million or more; the minimum denomination (often 100 000) is what keeps a bond out of retail hands and changes which rules apply to it.
- **Maturity and amortisation.** Most corporate bonds are **bullets**: coupons along the way, the whole principal at the end.
- **Coupon type.** Fixed, floating over a reference rate (see OIS, LIBOR, SOFR and €STR), or step-up, where the coupon rises if the rating falls.
- **Call schedule.** The issuer's right to repay early. A call is an option the investor has sold, so it must be paid for in yield, and it caps the upside when spreads tighten. The resulting price behaviour is negative convexity, which is developed in duration and convexity.
- **Covenants.** What the company promises not to do: pledge assets to someone else, sell the crown jewels, lever beyond a threshold. Plus, usually, a change-of-control put letting holders sell the bond back at 101 if the company is taken over.

INTERVIEW TRAP

Treating the call date as the maturity, or the maturity as the call date. Neither is safe. A callable bond trading above its call price is priced to the call; the same bond after spreads widen is priced to maturity, and the switch between the two happens without anything being traded. Quote yield-to-worst, which is the lower of the two, and say which one you are on.

Seniority: who gets paid, and in what order

If the company defaults, the proceeds of whatever is left are distributed down a strict ladder. Each level is paid in full before the next one receives anything.

1. **Secured debt**, backed by specific assets it can seize.
2. **Senior unsecured** bonds, the standard investment-grade instrument.
3. **Subordinated** debt, contractually behind the senior claims.
4. **Hybrids**, which are deeply subordinated, very long or perpetual, and can defer their coupons. Bank versions, contingent convertibles, can be written down or converted into equity while the bank is still alive.
5. **Equity**, which receives what is left, and usually that is nothing.

WORKED EXAMPLE

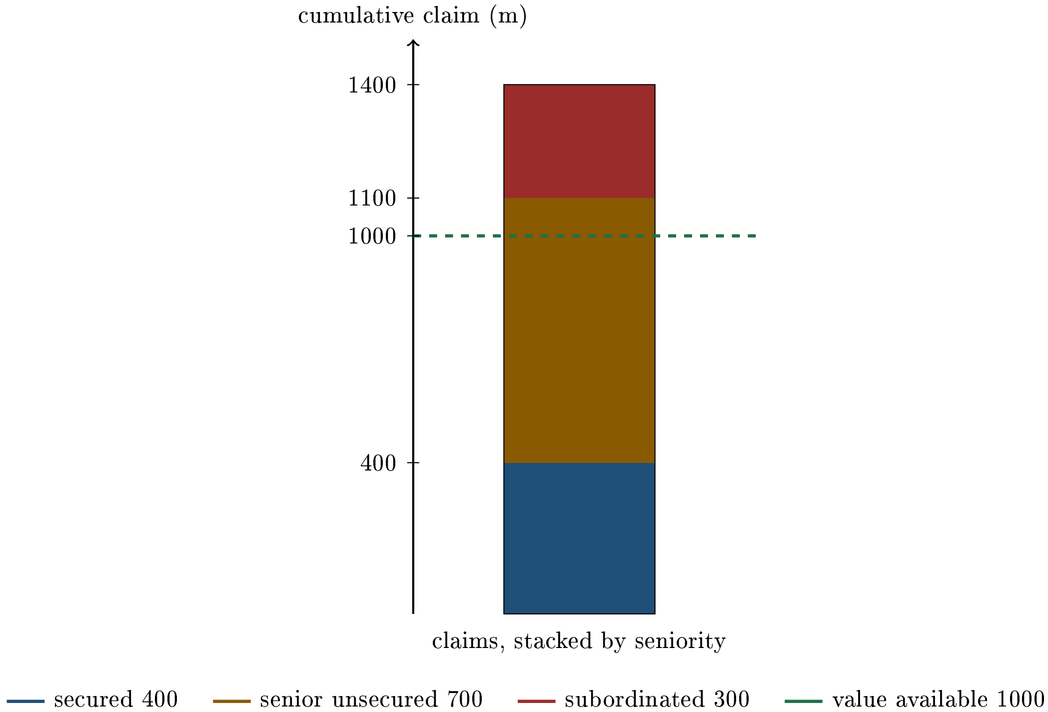
A company defaults. Liquidating the business realises **1000** million. The claims are 400 million secured, 700 million senior unsecured and 300 million subordinated, 1400 million in total.

Work down the ladder. The secured lenders take their 400 in full, leaving 600. The senior unsecured holders are owed 700 and receive the remaining 600:

$$\frac{600}{700} = 85.7\%.$$

The subordinated holders and the shareholders receive nothing.

So one default produces four different recovery rates on the same company at the same moment: 100% secured, 85.7% senior unsecured, 0% subordinated, 0% equity. Note also how sharp the edges are. If the business had realised 1100 instead of 1000, senior unsecured would have been made whole and subordinated would still have received nothing; if it had realised 900, senior would drop to 500/700 = 71.4%. A 10% error in the recovery value moves the senior recovery by roughly 14 points, which is why recovery assumptions are argued about so fiercely. How they are actually estimated is recovery rates.



The dashed line is where the money runs out. Everything below it is paid, everything above it is not, and the tranche the line cuts through is the one whose recovery is uncertain. That single picture is most of what seniority means.

Investment grade and high yield

The two halves of the corporate bond market are separated by the rating boundary, and they behave like different asset classes.

	Investment grade	High yield
Rating	BBB− / Baa3 and above	BB+ / Ba1 and below
Quoted in	spread over a curve	outright price
Driven mainly by	rates, spread second	the credit, rates second
Structure	bullet, light covenants	callable, incurrence covenants
Investor base	insurers, pensions, index funds	credit and cross-over funds
Typical size	benchmark, 500m and up	smaller, less frequent

Three consequences are worth carrying.

First, the boundary is mechanical, and that makes it dangerous. Index membership and many investment mandates are written against it, so a downgrade from BBB− to BB+ forces a set of holders to sell regardless of what they think, which is why a **fallen angel** usually cheapens before the downgrade is announced and can rebound after it. The rating scale itself is rating agencies.

Second, an investment-grade bond is mostly a rates instrument wearing a credit hat. Its spread is a small part of its yield, so most of its daily P&L comes from the curve, and a credit desk that is long IG bonds and does not hedge the rate risk is running a rates book by accident.

Third, high yield is priced on price, not on spread. Once a bond trades at 70, the distinction between coupon and principal stops mattering and the question becomes what the company is worth in a restructuring, which is the recovery analysis above rather than a spread calculation.

How a bond is born: the primary market

A corporate bond does not appear on a screen; it is built in a day. The sequence is always the same, and knowing it is what separates a candidate who has read about debt capital markets from one who has watched it.

1. **Mandate.** The issuer picks banks to run the deal.
2. **Announcement.** The deal is announced in the morning with an indication of size, maturity and **initial price thoughts**, a deliberately generous starting spread.
3. **Book building.** Investors place orders through the day. The order book grows and the syndicate revises the spread tighter in one or two steps.
4. **Pricing.** The final spread is fixed, the coupon is set so the bond prices at or just below par, and bonds are allocated. Oversubscribed books are cut back, which is why investors inflate orders.
5. **Break.** The bond starts trading in the secondary market. If it tightens, the deal is said to have performed.

WORKED EXAMPLE

A large corporate issues 500 million of 10-year senior unsecured debt.

The 10-year mid-swap rate is 2.50%. The syndicate announces at initial price thoughts of mid-swaps plus 120 basis points. The book reaches 1.9 billion, which is 3.8 times the deal size, so the spread is revised to plus 95 area and finally set at **mid-swaps plus 90**. The all-in yield is therefore

$$2.50\% + 0.90\% = 3.40\%.$$

The coupon is set at **3.375%**, the nearest eighth below the yield, so the bond prices slightly below par. Discounting ten annual coupons of 3.375 and 100 of principal at 3.40% gives a reoffer price of

$$P = \sum_{t=1}^{10} \frac{3.375}{1.034^t} + \frac{100}{1.034^{10}} = 99.791.$$

Now the number the desk actually argues about. The issuer's existing bonds of similar maturity trade at mid-swaps plus 82. The new bond came at plus 90, so the issuer paid a **new issue premium of 8 basis points** to get 500 million done in one afternoon. If the new bond converges to its curve, an investor who bought at reoffer gains that 8 basis points of spread. At a modified duration of 8.37, tightening by 8 basis points is worth

$$8.37 \times 0.0008 \times 99.791 \approx 0.67 \text{ points},$$

and the exact repricing at 3.32% is 100.462, a gain of **0.671**. On 500 million of bonds that is about 3.4 million, which is the entire reason investors chase new issues and the entire reason issuers resent paying the premium.

INTUITION

The new issue premium is the discount you offer to sell a lot of something at once, exactly like a block trade in equities. Initial price thoughts are the asking price you never expect to get, the revisions are the haggle, and a book that is four times covered means the first asking price was too generous. A deal that prices with no premium and then trades below reoffer has been squeezed too hard, and the buyers remember.

The secondary market is not an order book

Corporate bonds trade over the counter. A large issuer may have fifteen bonds outstanding, each a separate line, most of which do not trade on a given day. There is no central limit order book to look at; an investor asks a set of dealers for a price, in competition, and the dealer who wins takes the bond onto its own balance sheet.

Three practical consequences. Investment-grade bonds are quoted in **spread** rather than price, because the rate component is not what is being negotiated. Bid-offer is wide and varies enormously with the name, the size and the day, far more than in equities. And the last traded price can be days old, so a portfolio is marked from dealer quotes and model curves rather than from trades, which is a genuine source of disagreement about what a credit book is worth. What liquidity means when it is structured this way is liquidity.

What you are actually risking

Owning a corporate bond is owning at least five distinct exposures, and a candidate who lists them cleanly has answered most of a credit interview.

- **Interest rate risk.** The discount curve moves. Measured by duration, hedged with swaps or futures, and for an IG bond it is usually the largest daily driver.
- **Credit spread risk.** The market's required compensation for this issuer moves, without any default occurring. This is the dominant mark-to-market risk in high yield and the subject of credit spreads.
- **Default risk.** The issuer fails and you recover a fraction. See default risk and recovery rates.
- **Liquidity risk.** You can be right and still be unable to sell at anything near the mark, which is a different risk from the first three and the one that dominates in a crisis.
- **Optionality and event risk.** The call, the change-of-control put, and the leveraged buyout or debt-funded acquisition that widens your spread without any deterioration in the business.

Interview angle

REAL INTERVIEW QUESTION (Bank of America · Rates)

What is the difference between issuing a bond and taking out a loan?

WHAT THEY TEST / ANSWER

The trap is to answer "none, both are debt", which is true of the cashflows and misses everything the question is about. Say that the cashflows can be made identical and that the differences are structural, then give them in pairs.

Lenders: one bank or a syndicate against hundreds of anonymous investors. **Transferability**: a bond is a security designed to be traded, a loan is an agreement designed to be held, which is why only one of the two has a public price. **Monitoring**: a loan has maintenance covenants tested every quarter and a banker who will sit down and renegotiate, while a bond has looser covenants and a holder base that cannot easily be assembled to agree anything. **Rate**: loans float, public bonds are usually fixed. **Speed and cost**: a loan can be drawn quickly and quietly, a bond needs documentation, a rating and a marketing process, but reaches far more money and longer maturities.

Then the point that makes it a real answer: the choice is not either-or, and companies run both deliberately. A revolving credit facility is kept undrawn as insurance and as the thing that lets you issue a bond at all, since it is what convinces investors you can survive a closed market. And the flexibility runs the other way too: a loan can be repaid whenever you like, whereas repaying a bond early requires a call feature you paid for at issue.

REAL INTERVIEW QUESTION (Société Générale · Fixed Income)

How do you tell apart two bonds issued by two different companies with the same coupon, the same maturity and so on?

WHAT THEY TEST / ANSWER

The question is checking whether you understand that everything you have been given is deliberately irrelevant. Coupon and maturity describe the cashflow schedule. What distinguishes the two bonds is the probability that the schedule is honoured, and what you get if it is not.

Answer in that order. **Price and therefore yield**: if the two bonds trade at different prices, the market is already telling you which is riskier, and the difference in yield over the same risk-free curve is the credit spread. **Seniority**: senior secured against senior unsecured against subordinated changes the recovery, and therefore the spread, on the same company with the same coupon. **Issuing entity**: an operating company against a holding company changes who is structurally subordinated to whom. **Covenants and optionality**: a call, a change-of-control put, a step-up coupon. **Currency and documentation**. **Liquidity**: a benchmark-size bond in an index trades far better than an orphan line of the same credit, and that alone is worth basis points.

The follow-up is usually "so how do you compare their prices fairly?", and the answer is that you do not compare prices at all, you compare spreads over a common curve, because a price mixes the rate view and the credit view together. Which spread measure to use is credit spreads.

REAL INTERVIEW QUESTION (CACIB · Cross Asset)

What are your risks on a bond?

WHAT THEY TEST / ANSWER

This is a listing question with a hidden ordering test: name the risks, but also say which one dominates and when, because a list in any order sounds memorised.

Start with **interest rate risk**, measured by duration, and note that for an investment-grade bond this is usually the largest source of daily P&L. Then **credit spread risk**, the repricing of the issuer without any default, which dominates in high yield. Then **default risk** proper, which is the tail and is characterised by a probability and a recovery rather than by a sensitivity. Then **liquidity risk**: you may be unable to exit near the mark, and this correlates with the others exactly when you need it not to. Then the smaller ones that still matter: **optionality**, if the bond is callable, **currency**, if it is not in your base currency, **inflation**, if you care about the real value of a fixed coupon, and **reinvestment** of the coupons.

Finish with the observation that separates the risks properly: the first two are sensitivities you can hedge, with swaps for the rate and CDS for the spread, whereas the last ones are not hedgeable so much as sized. Saying that shows you know a risk you can hedge and a risk you can only limit are managed by different people in different ways.

REAL INTERVIEW QUESTION (HSBC · Credit)

What is the difference between IG and HY bonds?

WHAT THEY TEST / ANSWER

Give the definition first, in one line: investment grade is BBB− / Baa3 and above, high yield is BB+ / Ba1 and below, and the boundary is a rating, not a yield level. Then show that you know why the boundary matters more than the ratings do.

The behavioural differences follow. An IG bond is quoted and risk-managed in spread over a curve, and most of its daily move comes from rates; a HY bond is quoted in price, and most of its move comes from the credit. IG bonds are typically bullets with light covenants, HY bonds are typically callable with incurrence covenants, so a HY investor is short an option and needs to be paid for it. The investor bases differ, and so do the mandates: a great deal of money is contractually forbidden from holding sub-investment grade.

That last point is the answer to the follow-up, which is what happens at the boundary. A downgrade through it forces holders to sell whether or not they agree, so the price move around a fallen angel is not purely a repricing of risk, it is a repricing plus a forced flow. It also runs the other way for a rising star. The candidates who score here are the ones who add that the market anticipates the downgrade, so the bond often cheapens before the announcement and stabilises after it.

REAL INTERVIEW QUESTION (Natixis · Credit)

Total calls you to issue 500 million euros of 10-year debt. What do you do?

WHAT THEY TEST / ANSWER

This is a process question, and the interviewer wants the day, in order, not a definition of a bond. Say what you check, what you propose, and what you actually do.

Before anything, establish the facts: rating, existing curve, whether there is an outstanding bond near ten years, what the documentation programme allows, and whether the company wants fixed or floating in the end, because if it wants floating you are also selling a swap. **Then price it**: take the issuer's own secondary curve as the anchor, interpolate to ten years, cross-check against comparable issuers in the same sector and rating, and add a new issue premium for size. That gives you the spread you believe, and you announce wider than that.

Then run the deal: announce in the morning with initial price thoughts, build the book through the day, revise the spread tighter once or twice as the book grows, price at a level that leaves a small premium so the bond performs, set the coupon so the bond comes at or just below par, and allocate. **Then defend it**: how it trades in the first week is what determines whether the issuer calls you next time.

The judgement question underneath is the one to volunteer: how hard do you squeeze? Price too wide and the issuer overpaid and knows it. Price too tight and the bond breaks below reoffer, the investors who supported the book lose money, and the next deal is harder to sell. The job is to leave a few basis points on the table on purpose, and being able to say that out loud is what the question is testing.

INTERVIEW TRAP

Four answers that lose marks. (1) Saying a corporate bond is "a government bond plus a spread" and stopping: the spread is not a constant, it is the whole risk. (2) Quoting a single recovery rate for a company, when recovery is a property of the tranche and the example above produced four at once. (3) Calling covenants protection: incurrence covenants restrict new actions, they do not force anything when the business simply deteriorates. (4) Assuming the bond you can see is the one you can buy: most lines do not trade on most days, and a quote is not a fill.

Where this goes next

The compensation for taking the risk, and how it is measured, is credit spreads. Who assigns the letters that move the boundary is rating agencies. Turning a spread into a probability is default risk, what you get afterwards is recovery rates, and the instrument that lets you trade the credit without owning the bond is credit default swaps.

12.2 Credit spreads

sales · trading · structuring | core | credit · rates

A rates trader quotes a yield. A credit trader quotes a **spread**, and does it without ever mentioning the price. That is not a stylistic quirk. A corporate bond's price moves for two entirely different reasons, because the risk-free curve moved or because the market's opinion of the issuer moved, and only the second one is the credit desk's job. The spread is the instrument that separates them.

Get comfortable here and a large part of the credit chapter becomes bookkeeping. This section owns the definition, the family of spread measures and the difference between them, what is actually inside a spread, and how a spread move turns into PnL. Default probability and recovery, the two ingredients that a spread is supposed to compensate you for, have their own homes in default risk and recovery rates.

The idea in one picture

INTUITION

Two bonds mature on the same day and promise the same coupons. One is issued by a government that prints the currency, the other by a car manufacturer. Nobody pays the same price for both, and the price gap, expressed as extra yield per year, is the spread. It is the annual rent the market charges the borrower for the possibility that the promise is not kept, plus a fee for the trouble of holding something harder to sell.

The unit is the **basis point**, one hundredth of a percent. A bond "trading at 150 over" pays 1.50% a year more than the reference curve. A spread that **widens** means credit got worse and the bond price fell; a spread that **tightens** means the opposite. Learn to hear widening as bad news immediately, because the entire market speaks that way and hesitating on it in an interview is expensive.

REAL INTERVIEW QUESTION (Tikehau Capital · Credit)

What is the spread?

WHAT THEY TEST / ANSWER

Deliberately open, and what they are testing is whether you say *spread over what.* A one-line definition earns nothing. Answer in three beats. First, the concept: the extra yield a risky bond offers over a risk-free or reference rate of the same maturity, quoted in basis points, which is the market price of that issuer's credit risk. Second, the immediate qualification: "over what reference, and measured how", because a spread to governments, a spread to swaps, a Z-spread and an asset swap spread on the same bond are four different numbers. Third, what it is compensating: expected loss, which is default probability times loss given default, plus a risk premium, plus a liquidity premium. If you want to close it properly, add the direction convention, that widening means deterioration and a lower price, and note that a credit desk quotes and risk-manages in spread precisely so that a move in the underlying rates curve does not show up as a credit view.

The family of spread measures

"The spread" is a family, not a number. They differ in what they are measured against and in how carefully they treat the shape of the curve. In rough order of increasing sophistication:

- **G-spread (nominal spread).** The bond's yield to maturity minus the yield of an interpolated government bond of the same maturity. One number minus one number. Easy to compute, easy to quote, and it quietly assumes that a single yield summarises a whole schedule of cash flows.
- **I-spread.** The same idea against the interpolated swap rate rather than the government curve. Used where the swap curve, not the government curve, is the relevant funding benchmark.
- **Z-spread (zero-volatility spread).** The constant amount added to *every point* of the reference zero curve that makes the present value of the bond's cash flows equal its market price. Cash flow by cash flow, so it respects the shape of the curve instead of collapsing it into one yield. This is the standard analytical measure.
- **Asset swap spread (ASW).** The spread over the floating index earned on an actually traded package: buy the bond, pay its fixed coupons into a swap, receive floating plus a spread. Its distinguishing feature is that it is a *traded* number, whereas a Z-spread is a computed one.
- **Option-adjusted spread (OAS).** For bonds with embedded optionality (callables, putables, mortgage-backed securities), the Z-spread minus the value of the option, obtained from a model. It is the only member of the family that is comparable across bonds with and without embedded options, and the only one whose value depends on a model choice.
- **CDS spread.** The premium on a credit default swap on the same issuer. It is a spread on a derivative rather than on a cash bond, and the gap between the two is the **CDS-bond basis**, developed in credit default swaps.

KEY IDEA

The Z-spread is defined implicitly. Given a bond with cash flows C_k at times t_k, a market price P, and reference zero rates $z(t_k)$, the Z-spread is the number s solving

$$P \;=\; \sum_k \frac{C_k}{\big(1 + z(t_k) + s\big)^{t_k}}.$$

There is no closed form: you solve it numerically. Note what this buys you over a G-spread. Each cash flow is discounted at its own maturity's rate, so a steep or kinked curve, or an unusual coupon profile, no longer contaminates the answer.

WORKED EXAMPLE

Z-spread and G-spread on the same bond. Reference annual zero rates are 2.00% at 1 year, 2.50% at 2 years, 3.00% at 3 years. Consider a 3-year bond paying an annual coupon of 5 on a face of 100, so cash flows of $5, 5, 105$.

At a Z-spread of 150 bp the price is

$$\frac{5}{1.0350} + \frac{5}{1.0400^2} + \frac{105}{1.0450^3} = 4.8309 + 4.6228 + 92.0111 = 101.4648.$$

Now compute the G-spread on the same bond at that same price. Its yield to maturity is the single y solving $5/(1+y) + 5/(1+y)^2 + 105/(1+y)^3 = 101.4648$, which gives $y = 4.4675\%$. The 3-year government par yield implied by the zero curve above is 2.9803%. So

$$\text{G-spread} = 4.4675\% - 2.9803\% = 148.7 \text{ bp}, \qquad \text{Z-spread} = 150.0 \text{ bp}.$$

The bond has not changed and neither has its price. The 1.3 bp gap is produced entirely by the upward slope of the curve, which the yield-to-maturity calculation flattens away. On a steeper curve, a longer bond, or a low-coupon bond the gap grows quickly, which is the whole argument for the Z-spread.

REAL INTERVIEW QUESTION (RBC · Fixed Income)

What type of investor uses the Z-spread? And yield to maturity?

WHAT THEY TEST / ANSWER

The test is whether you understand that a spread measure is chosen for a purpose, not ranked by correctness. Yield to maturity is a **quoting and comparison** tool: it is a single number, everyone can reproduce it from price alone with no curve, and it answers "what do I earn if I buy this and hold it to maturity". That suits real-money and retail investors, buy-and-hold portfolios, and anyone comparing two bonds quickly. Its defects are exactly its convenience: it implicitly assumes reinvestment at the same yield and squashes the whole curve into one point. The Z-spread is a **relative-value and risk** tool: it needs the zero curve, it discounts each cash flow at the right maturity, and it isolates the issuer-specific component from the level and shape of rates. That suits credit traders, relative-value and arbitrage desks, and risk systems, because it makes two bonds with different coupons and maturities genuinely comparable and it is what you hedge. The complete answer names the boundary case too: as soon as the bond has an embedded call, neither is enough and you need an OAS.

The asset swap is the one member of the family that is a package trade rather than a calculation, so it has mechanics of its own: which leg pays what, why the resulting floating-rate exposure is what a credit investor wants to own, and how the spread is struck. Those belong to the instrument, not to the measure, and are developed in basis and asset swaps. The swap leg itself is covered in interest rate swaps and the floating benchmarks in OIS, LIBOR, SOFR and €STR.

What is actually inside a spread

A candidate who says "the spread compensates for default risk" and stops has given a third of the answer. Decompose it:

- **Expected loss**: default probability times loss given default. This is the piece an actuary would charge.
- **Credit risk premium**: compensation for bearing the *uncertainty* around that expected loss, and in particular for the fact that defaults cluster in bad states of the world, exactly when investors can least afford them.
- **Liquidity premium**: compensation for holding something that is slow and costly to sell, which is a large and time-varying component (see liquidity risk).
- **Technical and regulatory residue**: capital treatment, index membership, tax, collateral eligibility, and plain supply and demand.

WORKED EXAMPLE

The credit spread puzzle. Suppose an investment-grade name trades at a 5-year spread of 150 bp. Long-run rating-agency default studies put 5-year cumulative default rates for the BBB / Baa category at roughly 2%, and average senior unsecured recovery near 40%.

Turn that into an annual number. A 2% cumulative probability over five years is an annualised default rate of $1-(1-0.02)^{1/5} = 0.403\%$. With a loss given default of $1-0.40 = 60\%$, the expected loss is

$$0.403\% \times 60\% = 0.242\% = 24.2 \text{ bp per year.}$$

That is about 16% of the 150 bp spread. The other 84% is risk premium, liquidity and technicals. This gap is well documented and has a name, the "credit spread puzzle", and it is the single best answer to "why can't I just back out default probabilities from spreads?" You can, but what you get is a *risk-neutral* default probability, which is systematically larger than the historical one. Treat the two as different objects, a distinction developed in default risk. The figures above are round orders of magnitude, not a quotable statistic: rating-agency default rates and recoveries vary a lot by cohort, seniority and vintage.

From spread move to PnL: spread duration and CS01

The reason a desk cares about the number is that it converts directly into money. To first order,

$$\frac{\Delta P}{P} \approx -D_s \cdot \Delta s,$$

where D_s is the **spread duration**, the sensitivity of the price to a parallel shift in the spread. For a plain fixed-rate bullet bond, spread duration is essentially the modified duration, so the machinery in duration and convexity carries over unchanged. The dealer version of the same number is the **CS01** (also written DV01 of credit): the PnL for a 1 bp widening, quoted in currency rather than in percent, which is what actually appears on a risk report.

WORKED EXAMPLE

A widening. A 5-year bond paying a 4% annual coupon and priced at par has a Macaulay duration of 4.630 years, so a modified duration of $4.630/1.04 = 4.452$. If its spread widens by 50 bp with the risk-free curve unchanged,

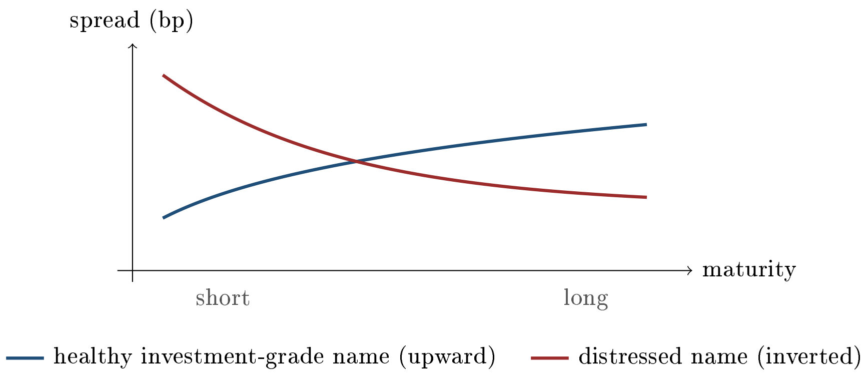
$$\frac{\Delta P}{P} \approx -4.452 \times 0.0050 = -2.23\%.$$

On a €10m position that is roughly €223,000 of mark-to-market loss, and the CS01 is about €4,450 per basis point. Note what the approximation hides: it is linear, so for a large move you need the convexity term, and it assumes a *parallel* spread shift, which is precisely what does not happen when a name gets into trouble.

The credit curve, and why it can invert

An issuer does not have one spread, it has a term structure of them. For a healthy investment-grade name the credit curve slopes **upward**: lending for ten years is riskier than lending for one, since there is more time for things to go wrong.

For a distressed name the curve **inverts**, and the reason is worth being able to explain cold. The market's fear is concentrated in the near term: if the company survives the next twelve months, it has probably refinanced or restructured and is much safer afterwards. So the short-dated protection is the expensive part, and the curve slopes down. An inverting credit curve is one of the cleanest distress signals in markets, and it usually precedes the rating downgrade rather than following it.



The sovereign case, where the two bonds share a currency and the spread carries redenomination risk on top of credit and liquidity, is developed in sovereign debt.

VISUAL / CODE TO BUILD: Interactive

A spread decomposition workbench. The left panel takes a reference zero curve (draggable at 1y, 2y, 5y, 10y) and a bond definition (coupon, maturity, price) and computes G-spread, I-spread and Z-spread side by side, with a live readout of the gap between them so the reader can steepen the curve and watch the G-spread drift away from the Z-spread. The right panel takes a spread level and splits it into expected loss (from a default probability and a recovery slider) versus residual premium, drawn as a stacked bar, so the reader can hunt for the default probability and recovery that would make expected loss explain the whole spread and discover that they have to assume implausible numbers. A third control widens the spread by a chosen number of basis points and prints the resulting PnL and CS01 for a user-set notional.

INTERVIEW TRAP

(1) **Quoting a spread without saying over what.** G, I, Z, ASW and CDS spreads on the same bond are different numbers. Naming the reference is half the answer. (2) **Reading spreads as historical default probabilities.** Spreads embed a risk premium and a liquidity premium; what you extract from them is a risk-neutral probability, which is systematically higher. (3) **Sign confusion.** Widening means worse credit and a *lower* price. Getting this backwards under pressure is fatal on a credit desk. (4) **Using a Z-spread on a callable.** If there is an embedded option the Z-spread silently contains its value. Use an OAS, and say which model. (5) **Assuming a spread move is issuer news.** A sovereign spread can widen because the benchmark rallied on a flight to quality. Always ask which leg moved. (6) **Forgetting the spread duration weighting.** A 50 bp move on a 2-year bond and on a 10-year bond are not the same trade. Compare CS01, not basis points.

Where this goes next

The instrument the spread is quoted on is corporate bonds. The two ingredients a spread is meant to compensate for are default risk and recovery rates. The derivative that lets you trade the spread directly, without owning the bond or its interest-rate risk, is the credit default swap, and its portfolio version is CDS indices. The rating that many investors use as a shorthand for a spread level, and the reasons that shorthand misleads, is rating agencies.

12.3 Rating Agencies

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A credit rating is an opinion about relative creditworthiness, published on an ordinal scale, by a firm paid by the entity being rated. Every part of that sentence matters, and most of the interview questions on this topic are really about one of them.

> **KEY IDEA**
>
> A rating is **ordinal, not cardinal**. It says that a BBB is riskier than an A and less risky than a BB; it does not say by how much, and it is not a probability. It is also a view on *credit* alone: it says nothing about whether the bond is cheap, whether it is liquid, or what it will do if rates move. A candidate who says "AAA means a 0.02% default probability" has confused the rating with the historical default study that the agency publishes alongside it.

The scales, which you should simply know

Two notations for the same idea. **Moody's** runs Aaa, then Aa1 through Aa3, A1 through A3, Baa1 through Baa3, then Ba1 through Ba3, B1 through B3, Caa1 through Caa3, Ca and C. **S&P and Fitch** run AAA, then AA+ / AA / AA−, A+ / A / A−, BBB+ / BBB / BBB−, then BB+ / BB / BB−, B+ / B / B−, CCC+ / CCC / CCC−, CC, C and D for default.

The subdivisions are **notches**, and moves are quoted in them: a two-notch downgrade from A− to BBB is a different event from a one-notch move within a category, particularly near the boundary below.

REAL INTERVIEW QUESTION (Natixis · Credit)

What is the last investment-grade rating before crossing into speculative?

WHAT THEY TEST / ANSWER

BBB− at S&P and Fitch, Baa3 at Moody's. One notch below is BB+ or Ba1, which is the first speculative-grade rating, also called high yield or, less politely, junk. Give both notations, because the question is checking whether you know the scales are two names for one thing.

Then say why that particular line matters more than any other on the scale, because that is the real question. The investment-grade boundary is **hardwired into rules rather than into risk**. Bank capital requirements, insurance solvency treatment, central bank collateral eligibility, bond index membership and a very large number of fund mandates all change behaviour at exactly that point. Nothing about the issuer changes discontinuously when it crosses; the demand for its paper does.

So a downgrade from BBB− to BB+ is a **cliff**. The issuer leaves the investment-grade indices, index-tracking funds must sell, mandate-constrained holders must sell, and the buyers on the other side are high-yield funds who know all of the above. That forced-seller dynamic is why a **fallen angel** typically widens far more than the change in its fundamental risk would justify, and why the spread usually starts moving months before the agency acts.

If they push for the trade, the answer is that this is a well-known and therefore crowded one: the anticipation is often priced, and the money is in distinguishing issuers that will be downgraded from issuers that already trade as if they had been.

What is being rated, and what it measures

REAL INTERVIEW QUESTION (Natixis · Credit)

What are the two broad types of rating found on bonds?

WHAT THEY TEST / ANSWER

The issuer rating and the issue rating, and the distinction is more useful than it sounds.

An **issuer rating** (also called a long-term counterparty or corporate family rating) is an opinion on the entity's overall creditworthiness: can this company service its obligations at all.

An **issue rating** applies to a specific instrument and can differ, sometimes by several notches, from the issuer's. The reason is that a default is an event of the issuer, but a *loss* is an event of the instrument, and instruments rank differently. A secured bond sits above an unsecured one, which sits above a subordinated one, which sits above hybrid capital. So the agency **notches** the issue rating up or down from the issuer rating according to seniority, collateral, structural position in the group, and expected recovery, which is recovery.

Give the practical consequence: two bonds from the same company can carry different ratings and trade at very different spreads with identical default probability, because the difference is entirely in what you get back. That is why a credit analyst reads the documentation and not only the rating.

If the question intended a different pair, the two other splits worth naming in a clause each are **long-term against short-term** ratings, which use separate scales and matter for commercial paper, and **solicited against unsolicited**, which is about who asked and who paid.

One further distinction is worth carrying because it separates the two largest agencies. **Moody's rates expected loss**, which combines the probability of default with the severity given default. **S&P rates the probability of default** primarily, treating recovery through a separate recovery rating. So the two firms can legitimately disagree on the same instrument without either being wrong: a bond likely to default but likely to recover well scores better on Moody's measure than on S&P's.

Outlook, watch and the direction of travel

Ratings come with modifiers that carry real information. An **outlook** (positive, stable, negative) is a view on the likely direction over one to two years. A **watch** or **review** is shorter and sharper, signalling that a change is likely within months, usually triggered by an event such as an acquisition. Markets react to these more than to the rating changes themselves, because by the time the downgrade arrives it has usually been anticipated.

REAL INTERVIEW QUESTION (Moody's · Credit)

How do you assess the credit risk of financial institutions in Italy (e.g. MPS)?

WHAT THEY TEST / ANSWER

Asked by an agency about a bank in a stressed sovereign, so the answer has three layers and the third is what the question is really for.

The standalone bank. Asset quality first, since that is what actually kills Italian banks: the stock of non-performing exposures, the coverage ratio against them, and the trend rather than the level. Then capital, meaning the CET1 ratio and, more informatively, the buffer above the regulatory requirement, because that buffer is what determines whether a loss becomes a resolution. Then funding and liquidity: the deposit base and how sticky it is, reliance on wholesale funding and central bank facilities, and the loan-to-deposit ratio. Then earnings, which for Italian banks has been the chronic weakness rather than the acute one, since a bank that cannot generate capital internally cannot grow out of a problem.

The sovereign link, which is the layer specific to this question. Italian banks hold large quantities of their own government's bonds, so bank and sovereign credit are mechanically correlated: a sovereign widening damages the banks' capital directly through the mark on those holdings, while a banking problem becomes a fiscal problem. This is the doom loop, and it is why bank ratings are usually constrained at or near the sovereign ceiling regardless of the standalone analysis.

Support and resolution. A rating on a bank's senior debt is not only an opinion about the bank. It is an opinion about what happens when it fails: whether the state would step in, and where in the creditor hierarchy the instrument sits once bail-in rules apply. Post-resolution frameworks have removed most of the assumed government support from senior ratings and pushed the analysis onto the loss-absorbing stack, which is why the same bank's instruments now span a wide range of ratings.

Close on the honest observation: for a name like this, the market has an opinion long before the agency changes one, so the useful work is comparing the standalone trend against what the spread and the equity price already imply, which is credit spreads.

REAL INTERVIEW QUESTION (Natixis · Credit)

Name three rating agencies.

WHAT THEY TEST / ANSWER

S&P Global Ratings, Moody's and Fitch, the three that between them hold the overwhelming majority of the market.

Then add something, because a three-word answer wastes the opening. Worth saying in a clause each: the concentration is itself a policy concern and a reason for the registration regimes that now exist in the European Union and the United States; there are credible smaller agencies, DBRS Morningstar being the largest, alongside regionally important ones such as Scope in Europe and JCR and R&I in Japan; and in some frameworks a rating counts only if it comes from a registered agency, which is why the identity of the agency, and not only the rating, appears in documentation.

If you want to end well, note that many large investors also run **internal ratings** mapped to the same scale, and that regulators increasingly prefer them, which tells you the direction the industry is moving in without you having to editorialise.

INTERVIEW TRAP

1. **Treating a rating as a default probability.** It is ordinal. The probabilities come from the agencies' historical default studies, which are a different publication.
2. **Confusing the issuer and the issue rating.** Two bonds from one company can be several notches apart, entirely because of seniority.
3. **Assuming the two big agencies measure the same thing.** Moody's rates expected loss and S&P rates default probability, so disagreement is not necessarily error.
4. **Expecting ratings to lead.** Spreads move first, and the downgrade usually confirms what the market already did.
5. **Underestimating the BBB− cliff.** The risk does not jump at the boundary; the forced selling does.
6. **Ignoring outlook and watch.** They carry most of the forward-looking content, and markets price them.

VISUAL / CODE TO BUILD: Interactive

A migration and cliff explorer. One panel shows a rating transition matrix as a heat map, with the reader able to pick a starting rating and a horizon and see the distribution of where issuers of that rating ended up, so the ordinal scale acquires an empirical shape without ever being presented as a probability attached to a letter. A second panel plots spread against rating for a live universe of issuers, with the enormous overlap between adjacent categories visible as scatter rather than asserted, and a slider that shows how much of the BBB− to BB+ spread jump is explained by the ratings and how much by index exclusion. A third mode simulates a fallen angel: the reader watches the spread widen over the months before the downgrade, then the forced index selling on the rebalance date, and can toggle the index-eligibility rule off to see how much of the move it accounts for.

12.4 Default risk

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Everything in credit is a bet on one binary event that has not happened. Making that tractable requires three things: a definition of the event precise enough to settle a contract, a way of describing its probability through time, and an honest account of which probability you are actually looking at, because there are two and they are not close to each other.

What counts as a default

"The company went bust" is not a definition a contract can settle on. In the derivatives market the trigger is a **credit event** as defined in the standard documentation, and the main ones are worth knowing by name:

- **Bankruptcy**: insolvency, or the formal opening of a protection procedure.

- **Failure to pay**: a missed payment beyond any grace period and above a materiality threshold, which is the most common trigger in practice.
- **Restructuring**: a change to the terms of the debt that is adverse to creditors, such as a reduction in coupon or principal, a maturity extension or a subordination. This is the contentious one, since a distressed exchange can be agreed rather than imposed, and it is why documentation distinguishes several restructuring conventions by region.
- For sovereigns, **repudiation, moratorium and obligation acceleration** are added, because a state does not go bankrupt in the corporate sense.

The reason this matters beyond legal detail is that the definition determines both whether a hedge pays and when. A bond holder who is economically wiped out by a restructuring but whose contract does not recognise it has a hedge that does not fire, which is the basis risk at the heart of credit hedging.

Describing the probability through time

Default is not a single number, it is a curve, and three different quantities are all called "the default probability". Keeping them apart is most of the skill.

- The **cumulative** probability of defaulting at any point before t.
- The **marginal**, or unconditional, probability of defaulting during a particular year, seen from today.
- The **conditional**, or forward, probability of defaulting during that year *given survival* until its start. This is the hazard rate, and it is the natural object because it is what a spread for that period prices.

The standard model makes the link. Let λ be a constant **hazard rate**, the instantaneous conditional default intensity. Then the **survival probability** and cumulative default probability are

$$S(t) = e^{-\lambda t}, \qquad \mathbb{P}(\tau \leq t) = 1 - e^{-\lambda t},$$

with τ the default time. Default is the first arrival of a Poisson process: the company survives each instant with a fixed chance, and survival compounds down.

> WORKED EXAMPLE
>
> Take $\lambda = 2\%$. Cumulative default probabilities are 1.98% at one year, 9.52% at five and 18.13% at ten: notice they grow but less than linearly, because you can only default once and the paths that have already defaulted are not available to default again.
>
> Now separate the three quantities in year five. The **conditional** probability of defaulting during year five, given survival to the start of it, is $1 - e^{-0.02} = 1.98\%$, the same as year one, because the hazard is constant. The **marginal** probability seen from today is $e^{-0.08}\,(1 - e^{-0.02}) = 1.83\%$, lower, because you first have to survive four years to get there. Confusing these two is the single most common error in a credit interview.

Real hazard curves are not flat. An investment-grade issuer typically has an upward sloping curve, since the near term is comfortable and uncertainty accumulates, while a distressed issuer's curve inverts: if it survives the next eighteen months it has probably refinanced, so the near-term hazard is the highest. An inverted credit curve is therefore a distress signal, and it is often the earliest one available.

Getting a number

REAL INTERVIEW QUESTION (Bank of America · Credit)

How do you calculate a probability of default?

WHAT THEY TEST / ANSWER

There are three routes, they answer different questions, and naming all three is the answer.

From market prices, the risk-neutral route. Strip the credit spread out of a bond or take it from a CDS, then use the credit triangle: over a period, the spread compensates for the default probability times the loss given default, so $\lambda \approx s/(1-R)$. A 120 basis point spread with a 40% recovery assumption implies a hazard rate of $0.012/0.6 = 2\%$ a year, and the survival curve follows as $e^{-\lambda t}$. Say immediately that this is a *risk-neutral* probability and that it depends entirely on the recovery assumption, since the two cannot be separated from a single instrument. The spread side of this is in credit spreads and the recovery side in recovery rates.

From history, the actuarial route. Take the issuer's rating and apply the observed historical default frequency for that rating over the horizon, from a transition matrix. This gives a real-world probability. It is backward looking, coarse, since every issuer in a rating bucket gets the same number, and slow, since ratings move after spreads do.

From a model, the structural route. Treat equity as a call option on the firm's assets struck at the debt, and default as the assets finishing below the debt, which gives a probability from the equity price and its volatility, in structural versus reduced-form models. Its appeal is that it updates continuously with a liquid market and needs no rating.

Then deliver the comparison that shows you understand what you have computed: **risk-neutral probabilities are systematically higher than historical ones**, often by a multiple rather than a margin, because the spread also pays for liquidity, for the risk premium investors demand on a loss that arrives in bad times, and for the fact that defaults cluster. So the market-implied number is the right one for *pricing and hedging*, and the historical one is the right starting point for *expected loss and capital*, and quoting one where the other belongs is a real error rather than a philosophical one.

REAL INTERVIEW QUESTION (Moody's · Credit)

How do you assess the credit risk of financial institutions in Italy, for example MPS?

WHAT THEY TEST / ANSWER

Say first why a bank is not assessed like an ordinary company, because that framing is the answer. A bank's balance sheet is its business, it is opaque by construction, it is leveraged an order of magnitude beyond an industrial firm, and it can fail from a funding run while remaining solvent on paper. So the ordinary cash-flow analysis is not the starting point.

Work through four blocks. **Capital**: the loss-absorbing buffer, its quality, and the headroom above regulatory requirements, since it is the distance to a breach rather than the absolute level that matters. **Asset quality**: the composition of the loan book, non-performing exposures, provisioning, and concentration by sector and by geography, which for an Italian bank means sovereign exposure specifically, since the domestic sovereign and its banks are one risk rather than two. **Funding and liquidity**: the deposit base and how sticky it is, reliance on wholesale funding, maturity mismatch, and the liquidity buffer, because this is the axis along which banks actually fail and fail quickly. **Earnings**: whether the franchise generates enough to rebuild capital after losses, which is what separates a recoverable problem from a terminal one.

Then the two things specific to this sector. **Sovereign linkage** runs both ways: the state supports the banks and the banks hold the state's debt, so the two credits are correlated and neither can be assessed alone. And **seniority is everything**: resolution frameworks are explicitly designed to impose losses on subordinated and senior unsecured holders before any public money is used, so "the credit risk of the bank" is not one number but a different number per layer of the capital structure. Close on the market cross-check: compare your conclusion to the CDS level and the equity price, and if they disagree with your analysis, find out why before trusting either.

The investment grade boundary is where a rating and a hazard rate are most often compared, and the scale itself, along with the cliff effects written across it, belongs to rating agencies and the credit asset class. What belongs here is what the boundary says about *default probability*. A rating is a through-the-cycle opinion moving in discrete steps, while a spread reprices continuously, so the two default probabilities they imply routinely disagree, and the spread moves first. A bond trading at high-yield spreads while still rated investment grade is common and informative: it means the market's implied hazard rate has already crossed a line the rating has not. The consequence for the issuer is not only its spread but a smaller investor base and often covenants or collateral triggers tied to the rating, so a downgrade raises the default probability it is reporting on.

INTERVIEW TRAP

Four errors. (1) Confusing the conditional default probability for a year with the unconditional one seen from today: they differ by the survival factor, and the gap grows with horizon. (2) Quoting a market-implied default probability as though it were a forecast, when it contains a large risk premium. (3) Extracting a default probability without stating the recovery assumption, since only the product of the two is identified by a single spread. (4) Assuming default risk is idiosyncratic and diversifiable: defaults cluster in time and by sector, which is precisely why the premium exists and why credit correlation is its own subject.

Where this goes next

The price of this risk is credit spreads, what is left after it happens is recovery rates, and the instrument that isolates it is credit default swaps. The two families of model that produce these probabilities are structural versus reduced-form models, the opinion-based assessment is rating agencies, and the fact that defaults arrive together is credit correlation.

12.5 Recovery rates

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Default is not the end of the story. A creditor of a defaulted issuer usually gets something back, and how much decides whether a default is an inconvenience or a catastrophe. The recovery rate is that fraction, and it is the second of the two quantities a credit spread pays for. It also turns out to be the harder of the two to pin down, for a reason that is structural rather than practical.

Definition and the two conventions

The **recovery rate** R is the fraction of face value a creditor ultimately receives, and the **loss given default** is its complement,

$$\mathrm{LGD} = 1 - R.$$

Two conventions coexist. **Recovery of face value** expresses R as a percentage of notional, which is the market standard and the one used in CDS. **Recovery of market value** expresses it against the price just before default, which appears in some models but is not how contracts settle. Say which one you mean.

For CDS the recovery is not assumed at settlement, it is *observed*: after a credit event an industry auction determines a final price for the defaulted obligations, and the protection buyer receives $1-R$ per unit of notional in cash, which is $100-R$ expressed in points of face. So the conventional recovery used in pricing is a market convention, while the recovery used in settlement is an auction outcome, and they routinely differ.

What actually determines it

- **Seniority**, which dominates everything else. The waterfall pays secured creditors first, then senior unsecured, then subordinated, then hybrid capital, and equity last. Each layer is paid in full before the next receives anything, so a small change in the recoverable value can move a junior layer from partial recovery to nothing.
- **Security and collateral**. A claim secured on specific assets recovers from those assets, which is why secured bank debt and covered bonds recover far more than senior unsecured paper of the same issuer.
- **The nature of the business**. A firm whose value is tangible and redeployable, property, aircraft, infrastructure, recovers well; one whose value was people, brand or software recovers badly, because the assets walk out of the building.
- **Jurisdiction**. Bankruptcy regimes differ in how much value they preserve and how long they take, and the delay itself is a cost since recovery arrives years later and undiscounted headline figures overstate it.
- **The cycle**, and this is the one to volunteer: recoveries are *lower* when default rates are higher. Distressed assets are being sold into a market full of other distressed assets and few buyers. Default probability and loss given default are positively correlated, so portfolio losses in a bad year are worse than multiplying two independent averages suggests.

The market conventions worth knowing are 40% for senior unsecured corporate debt, which is the standard assumption in CDS pricing, materially lower for subordinated debt, and lower again for emerging market sovereigns, where there is no collateral to seize and the outcome is a negotiation.

REAL INTERVIEW QUESTION (BNP · Credit)

How do you calculate the recovery rate and the expected loss?

WHAT THEY TEST / ANSWER

Separate the two, because one is an estimate and the other is a formula.

The recovery rate is not calculated, it is estimated or observed. Three routes: from the **auction** after an actual credit event, which is the only observation of the true number; from **historical studies** by seniority, sector and jurisdiction, which is what feeds a conventional assumption; or from **a model** of the firm's asset value against its liability stack, which gives a recovery per layer of the capital structure. In day-to-day pricing, none of these is used directly: a conventional figure is assumed, usually 40% for senior unsecured, and the reason for that is the next question.

Expected loss is a formula, and stating it with its three components is the answer:

$$\mathrm{EL} = \mathrm{PD} \times \mathrm{LGD} \times \mathrm{EAD} = \mathrm{PD} \times (1 - R) \times \mathrm{EAD},$$

with EAD the exposure at default. On a 100 exposure with a 2% annual default probability and 40% recovery, the expected loss is $2\% \times 60\% \times 100 = 1.2$ a year.

Then the three qualifications that separate a good answer. Which **probability** you are using matters: a risk-neutral PD gives a price, a historical one gives an actuarial expected loss, as in default risk. **Exposure at default is not always the notional**: on a derivative it is the mark-to-market at the time of default, which is itself random and possibly correlated with the default, which is the whole subject of CVA, DVA and FVA. And expected loss is a *mean*, so it is the right input for pricing and provisioning and the wrong one for capital, which is sized on the unexpected loss in the tail.

The identification problem

KEY IDEA

A credit spread pays for the product of two things, not each separately: $s \approx \lambda\,(1-R)$. From one instrument you can identify only that product, so **recovery and default probability cannot be separated from a spread alone**. Take a 240 basis point spread: with $R = 40\%$ the implied hazard rate is 4%, with $R = 20\%$ it is 3%, and with $R = 60\%$ it is 6%. The default probability doubles across a plausible recovery range while the market price does not move at all.

This is why the market fixes a conventional recovery and quotes the spread. It is also why the convention does little harm for pricing similar instruments, since a wrong R produces a compensating wrong λ and roughly the right price, and real harm for instruments whose sensitivity to the split differs: a digital CDS, which pays a fixed amount regardless of recovery, a recovery swap, which trades the quantity directly, or a tranche, whose attachment points sit at particular loss levels.

REAL INTERVIEW QUESTION (JP Morgan · Credit)

What would you quote for a five-year CDS on Ford, given a recovery rate of 40?

WHAT THEY TEST / ANSWER

They have handed you the recovery, which means they want the triangle used in reverse and, above all, they want to hear your assumption stated before your number.

The relation is $s \approx \lambda\,(1-R) = 0.6\,\lambda$, so all you need is a view on the hazard rate, and the honest way to form one is through a cumulative default probability you can defend. Say what you think the credit is: an automaker is cyclical, capital intensive and carries a large finance arm, and has traded around the boundary between investment grade and high yield. So put a range on the five-year cumulative default probability and convert.

If you believe roughly a 10% chance of default over five years, then $\lambda = -\ln(0.9)/5 = 2.1\%$ and the spread is $0.6 \times 2.1\% \approx 126$ basis points. If you think it is closer to 20%, then $\lambda = 4.5\%$ and the spread is about 268 basis points. So quote a range, something like 150 to 300 depending on where in the cycle you are, and say which end you would lean to and why.

Three points finish it. Give a **two-way price**, since you were asked to quote: name a bid and an offer, and make the spread wide enough for a name whose credit can move quickly. Note that recovery on an automaker is plausibly *above* the 40% convention, given tangible assets and a securitisable receivables book, which would mean the same spread implies a higher default probability than the standard assumption suggests. And say that you would check the traded level and the bond spreads before committing, because the point of the exercise is the reasoning, and nobody expects a memorised level.

When recovery is written into the contract

REAL INTERVIEW QUESTION (Tikehau Capital · Credit)

Define contingent convertible bonds.

WHAT THEY TEST / ANSWER

A **CoCo** is a subordinated bank bond that absorbs losses automatically while the bank is still a going concern, either by converting into equity or by being written down, when a predefined trigger is breached, typically a common equity capital ratio falling below a threshold. The main regulatory form is **Additional Tier 1**, which is also perpetual with issuer call dates and has fully discretionary, non-cumulative coupons that can be cancelled without triggering a default.

The reason it belongs in a discussion of recovery is the point to make: for a CoCo the recovery is **contractual rather than judicial**. An ordinary creditor's recovery is decided years later by a bankruptcy process; a CoCo holder's is written into the terms and can be applied while the bank continues to operate, at a moment defined by a ratio rather than by insolvency. A write-down CoCo can go to zero with the bank still trading, which is what makes it different in kind from subordinated debt and not merely in degree.

Three risks to name, and they are not the usual credit risks. **Trigger risk**, the distance of the capital ratio to the trigger, which is the real variable and is not a spread. **Coupon cancellation risk**, since coupons can be switched off without default, so the instrument is closer to a preferred equity than to a bond. **Extension risk**, since the market prices these to the first call and a non-call is a repricing event. Add that the instrument is deliberately designed to be loss absorbing, so an investor is being paid a high coupon precisely to stand in front of a loss the state does not intend to absorb, and that the 2023 episode in which AT1 holders were written down while shareholders received value is the standard reminder that the waterfall you assume is the one in the documents rather than the one in the textbook.

INTERVIEW TRAP

Four errors. (1) Quoting a default probability without the recovery assumption behind it, when only the product is identified by the spread. (2) Assuming 40% everywhere: it is a convention for senior unsecured corporates and is wrong for secured debt, for subordinated debt and for sovereigns. (3) Treating recovery as independent of default rates, when the two are correlated and both move against you in the same year. (4) Comparing headline recovery statistics without asking whether they are discounted and how long the process took, since an undiscounted 40% received after four years is worth considerably less than 40%.

Where this goes next

The other half of the same spread is default risk, the price that contains both is credit spreads, and the instrument that settles against an observed recovery is credit default swaps. The reason recovery and default move together is credit correlation, the instruments whose value depends on the loss level directly are tranches, and exposure at default on a derivative is CVA, DVA and FVA.

12.6 Credit Default Swaps (CDS)

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A credit default swap is insurance on a borrower, written as a swap. One side pays a regular premium; the other pays out if a defined credit event occurs. That is the whole instrument, and everything difficult about it is in the definitions of "credit event" and "pays out".

INTUITION

Think of it as insurance on a house you may not own. The protection buyer pays a premium each quarter and receives a payment if the house burns down. The two features that make it unlike ordinary insurance are that you do not need to own the house, so the market can be larger than the debt it references, and that the payout is settled by auction rather than by assessing your actual loss.

REAL INTERVIEW QUESTION (Société Générale · Credit)

What is a CDS, what is its use, how is it quoted and how does it move?

WHAT THEY TEST / ANSWER

Four questions, so four short paragraphs, in the order asked.

What. A bilateral contract on a reference entity. The protection buyer pays a periodic coupon on a notional until maturity or until a credit event; if a credit event occurs, the contract terminates and the protection seller pays the notional less the recovery value of the reference obligation, determined at an industry auction. Standard maturities are one to ten years with the five-year the benchmark.

Use. Three, and give all three because most candidates give one. *Hedging*: a lender or bondholder buys protection to remove default risk while keeping the asset. *Expressing a view*: buying protection is the cleanest way to be short a credit, because shorting a corporate bond is operationally painful and often impossible. *Relative value and funding*: the CDS market is often more liquid than the underlying bonds and requires no funding, so it is where credit views are actually expressed.

Quoting. In basis points per annum of the notional. Since the standardisation that followed 2009, contracts trade with a **fixed coupon**, conventionally 100 basis points for investment grade and 500 for high yield, plus an **upfront payment** that reconciles the fixed coupon with the market spread. Quoting the running spread is still how people talk, but the cash flows are coupon plus upfront, which matters for netting and clearing.

How it moves. It widens when the market's assessment of default risk worsens, and it is far more responsive than a rating. It also moves with recovery expectations, since the payout is one minus recovery, and with the general risk appetite of the market rather than only with the issuer's own news. Add the point that gets the follow-up: buying protection is a long-volatility, long-crisis position that carries negatively, so it bleeds for long periods and pays in a few days, which is why it is a difficult position to hold and an easy one to be right about too early.

Credit events and settlement

The contract is only as good as its definitions. The standard credit events for a corporate are **bankruptcy**, **failure to pay** and **restructuring**, the last being the contentious one, since a negotiated maturity extension can trigger protection without anything resembling a bankruptcy. Different markets use different restructuring conventions, which is why the same name can trade at two spreads under two documentation standards.

Determination is made by a committee rather than by the counterparties, and settlement is by **auction**: the market agrees a single recovery price for the deliverable obligations, and every contract settles in cash against it. That mechanism exists because the notional outstanding can exceed the deliverable debt, which produced short squeezes in the era of physical settlement.

Pricing

REAL INTERVIEW QUESTION (Bank of America · Credit)

How do you price a CDS?

WHAT THEY TEST / ANSWER

Two legs, set equal, and the whole answer is that structure plus the one approximation everyone actually uses on a desk.

The premium leg is the present value of the coupons you expect to pay, which is not simply an annuity because payments stop at default. So it is discounted *and* weighted by the survival probability:

$$\mathrm{PV}_{\mathrm{prem}} \;=\; s \sum_i \Delta_i \, D(t_i) \, Q(t_i),$$

where D is the discount factor and Q the survival probability. That sum is the **risky annuity** or RPV01, and it is the single most useful quantity in credit trading.

The protection leg is the present value of the expected payout:

$$\mathrm{PV}_{\mathrm{prot}} \;=\; (1-R) \int_0^T D(t) \, \big(-\mathrm{d}Q(t)\big),$$

the loss given default times the probability of defaulting in each interval, discounted.

The fair spread sets them equal. Under a constant hazard rate λ and continuous approximation everything collapses to the **credit triangle**:

$$s \;\approx\; \lambda \, (1-R),$$

which is the relation to have memorised. A five-year spread of 200 basis points at a 40% recovery assumption implies a hazard rate of $0.02/0.6 = \mathbf{3.33}\%$ a year, so a five-year survival probability of $e^{-0.0333\times 5} = \mathbf{84.6}\%$.

Then the part that matters in practice, which is marking an existing position. A CDS is a swap, so its value is the difference between the contract spread and the market spread, annuitised:

$$\mathrm{MTM} \;=\; (s_{\mathrm{market}} - s_{\mathrm{contract}}) \times \mathrm{RPV01} \times \mathrm{notional}.$$

Bought protection at 100 basis points, the name now trades at 250, five years, and the risky annuity is about 4.05: the position is worth $0.015 \times 4.05 \times 10\mathrm{m} =$ **€607,500** per 10 million of notional. The same arithmetic gives the standardised upfront payment, since that is exactly the difference between the market spread and the fixed coupon annuitised.

Close on the honest caveat: the recovery rate is assumed rather than known, so the hazard rate and the recovery are not separately identified from a single spread. Desks fix recovery by convention (40% for senior unsecured corporates) and let the hazard rate absorb everything, which is fine for marking and wrong for a distressed name, where recovery is the whole question.

REAL INTERVIEW QUESTION (BNP · Credit)

Why do we need CDS if we already know the expected loss?

WHAT THEY TEST / ANSWER

Because knowing an expected loss and being able to *transfer* it are different things, and the question is testing whether you see that.

Expected loss is an average; risk is the distribution around it. An expected loss of 1% on a portfolio does not mean losing 1%. It means losing nothing most of the time and a great deal occasionally. Nobody hedges an average; they hedge the outcome that breaches their capital. A CDS transfers the realisation, not the expectation.

You can compute it, but you cannot act on it. The expected loss is a number in a model. A CDS is a contract with a counterparty who will pay you. Turning an analysis into a position requires an instrument, and for credit, shorting the underlying bond is impractical: the borrow is thin, the bond may not exist at the maturity you want, and it must be funded. A CDS requires no funding and exists at standard maturities.

The market price is information you did not have. Your expected loss is your estimate; the CDS spread is the price at which someone will take the other side, and it includes a risk premium, a liquidity premium and everyone else's estimate. Historically the spread is several times the realised expected loss, and that gap is not an error, it is the compensation for bearing a risk that arrives when everything else is going wrong.

And it separates credit from everything else. A corporate bond bundles credit risk, interest rate risk and funding. A CDS isolates the credit. That is the deepest answer, and the one a credit desk wants: the instrument exists to make credit a tradeable factor in its own right rather than a component of a bond, which is what allows relative value between the two, treated in credit spreads.

The hedge is never exact

A CDS hedge of a corporate credit exposure is never exact, and basis swaps takes up the question of what basis risk *is*. What belongs here is the part that comes from the CDS contract itself, since these are mismatches between a contract and an exposure rather than between two rates.

- **Reference entity mismatch.** You lend to a subsidiary and hedge with the parent's CDS, or you hedge a name with an index. The legal entities differ, and in a restructuring that difference is the whole outcome.
- **Obligation mismatch.** The CDS settles against deliverable obligations at an auction recovery. Your exposure is a specific loan or bond with its own seniority and security, so your realised loss and the auction recovery need not match.
- **Maturity mismatch.** Your exposure runs to an odd date and the liquid CDS is the five-year. You are hedged for the wrong horizon and exposed to the shape of the credit curve.
- **Documentation mismatch.** Restructuring conventions differ, so a credit event under your loan may not be one under the contract.
- **Counterparty risk on the hedge.** Your protection is only worth the seller's ability to pay, and their ability to pay is correlated with the event, which is wrong-way risk and is counterparty risk.

The residual is not noise. Every one of these mismatches widens when credit deteriorates, so the hedge underperforms precisely when it is called upon, which is why the residual has to be sized explicitly and held capital against rather than booked as flat.

INTERVIEW TRAP

1. **Treating a CDS as a bond short.** It isolates credit and carries no funding or rate exposure, so the two do not offset each other exactly. The difference is the basis.
2. **Ignoring the restructuring clause.** The same name trades at different spreads under different conventions, and the difference is the definition of the event.
3. **Forgetting that recovery is assumed.** A single spread does not identify the hazard rate and the recovery separately; the 40% convention is a convention.
4. **Quoting the running spread as the cash flow.** Contracts pay a standard coupon plus an upfront, and the upfront is the spread difference annuitised.
5. **Marking without the risky annuity.** A spread move of 150 basis points is worth roughly four times that in present value on a five-year, not one times.
6. **Assuming protection always pays.** It pays if the seller can, and the seller is least able exactly when the event happens.

VISUAL / CODE TO BUILD: Interactive

A CDS pricer with both legs exposed. The reader sets the spread, the recovery assumption and the curve, and the page draws the premium leg and the protection leg as stacked bars per period, weighted by the survival probability, so the risky annuity is visible as an area rather than as a symbol and the fair spread is found by dragging until the two areas match. A second panel takes a position entered at one spread and marks it as the market spread moves, decomposing the profit into the spread change and the annuity change so the reader sees why a widening on a short-dated contract is worth so much less than the same widening on a ten-year. A third mode turns recovery into a slider with the spread held fixed, showing the hazard rate moving to compensate, which is the cleanest demonstration that a single quote does not identify the two.

Where this goes next

The spread this contract quotes, and its relationship to the bond it hedges, is credit spreads. The two inputs the credit triangle cannot separate have their own sections: default risk for the hazard rate and recovery rates for the assumption the auction eventually replaces. The portfolio version of the same contract is CDS indices, and slicing that portfolio by seniority is tranches. The margin and clearing arrangements that make the counterparty leg tolerable are collateral and haircuts, and the ratings this instrument is usually contrasted with are rating agencies.

12.7 CDS indices

sales · trading · structuring | core | credit

If you want to be long or short credit risk right now, in size, at a price you can see, you do not buy corporate bonds and you do not trade single-name protection. You trade a CDS index. It is the most liquid instrument in credit by a wide margin, it is what the market means when it says "credit widened today", and it is the building block on top of which tranches and index options are constructed.

What a single-name credit default swap is, how its two legs work and what counts as a credit event are in credit default swaps, and the spread measures used below are in credit spreads. This section is about what changes when you package many of them into one contract.

An index is a portfolio, not an average

A CDS index is a fixed list of N reference entities, equally weighted, traded as a single contract. Buying 100 million of protection on a 125-name index is economically the same as buying 0.8 million of protection on each of the 125 names, in one trade, at one price, on one confirmation.

> **KEY IDEA**
>
> An index is a portfolio of protection, not an average of spreads. That one sentence answers most index questions, because everything that surprises people about indices (why the index does not equal the mean of its constituents, why a default shrinks the contract rather than ending it) follows from it.

> **INTUITION**
>
> Think of it as buying insurance on 125 houses at once rather than negotiating 125 policies. You pay a single premium, and if one house burns down you are paid for that house and keep insuring the other 124. The contract does not end; it gets one house smaller.

The families

Two families dominate, one European and one North American, each split by credit quality.

Index	Universe	Names	Standard coupon
iTraxx Europe (Main)	European investment grade	125	100 bp
iTraxx Crossover	European sub-investment grade	75	500 bp
iTraxx Europe Senior Financials	European bank senior debt	30	100 bp
CDX NA IG	North American investment grade	125	100 bp
CDX NA HY	North American high yield	100	500 bp

The five-year maturity carries most of the liquidity; three, seven and ten years are quoted and trade far less. Main and Crossover are the two numbers a credit person quotes when asked how the market is: Main for the investment-grade tone, Crossover for risk appetite, and the ratio between them for how nervous people are about the lower end.

Series and the roll

The constituent list is not permanent. A new **series** of each index is published every six months, on 20 March and 20 September. Names that no longer belong (downgraded out of the investment-grade universe, acquired, no longer liquid enough) are replaced, and the new series becomes **on-the-run**.

Three things follow.

- **Liquidity migrates immediately.** Almost all trading moves to the new series within days. Older series keep trading, more thinly, until the positions in them run off, which is the credit version of the on-the-run and off-the-run distinction familiar from government bonds.
- **The new index is not five years long.** Credit default swaps mature on the standard dates of 20 March, June, September and December. A five-year index rolled on 20 March matures on 20 June five years later, so at inception it has **five years and three months** to run, and it shortens through the six months until the next roll.
- **The roll is a real risk.** A hedge in the old series and a position in the new one are not the same instrument: different names, different maturity, different liquidity. Rolling a hedge costs the bid-offer twice a year, and positions kept in an old series can become genuinely hard to close.

Fixed coupon, traded spread, upfront

Indices trade with a **fixed** coupon, set at launch and never changed: 100 basis points for the investment-grade indices, 500 for the high-yield ones. The market price moves, so the difference between where the index trades and what the contract actually pays is settled as a cash **upfront** at the start.

WORKED EXAMPLE

You buy 10 million of protection on iTraxx Main. The contract pays a coupon of 100 basis points. The index is quoted at 65 basis points.

You are contracted to pay 100 for protection the market says is worth 65, so you are overpaying by 35 basis points a year for five years, and the seller compensates you at the start. The compensation is the present value of that 35 basis points, discounted for both interest rates and the chance that the contract is not still running to receive it, which is the **risky annuity**. At a 3% discount rate, a 40% recovery assumption and a 65 basis point spread, the five-year risky annuity is 4.4996, so

$$\text{upfront} = (100 - 65)\ \text{bp} \times 4.4996 \times 10\,\text{m} = 0.0035 \times 4.4996 \times 10\,\text{m} = 157\,485,$$

which is 1.57% of notional, received by you, the protection buyer.

The sign is the part candidates get wrong. Coupon above the traded spread means the buyer is overpaying on the running leg and therefore *receives* the upfront. Coupon below the traded spread, as on Crossover when it trades wider than 500, means the buyer *pays* it.

The reason for this design is fungibility. Because every trade in a given series has the same coupon and the same dates, two offsetting trades cancel exactly, and positions can be netted and cleared centrally. A market where each trade carried its own negotiated running spread could not do that.

Why the index is not the average of its names

Take the naive view: the index has N names, so its spread should be the average of their spreads. It is not, and the gap is a real trading position.

The reason is that an index spread is the single running rate that makes one premium leg worth the same as one protection leg over the whole portfolio. Each name contributes its protection value, but it also contributes its own risky annuity, and a wider name has a *smaller* annuity, because there is less chance of surviving to collect the later premiums. So the fair index spread is the annuity-weighted average of the constituent spreads, and since the wide names carry the smallest weights, it sits **below** the arithmetic average.

WORKED EXAMPLE

A five-name equally weighted index. Constituent spreads of 100, 150, 250, 400 and 700 basis points. Assume a 3% discount rate, a 40% recovery and quarterly premiums over five years, which gives risky annuities of 4.43, 4.34, 4.16, 3.92 and 3.48 respectively.

The arithmetic average is

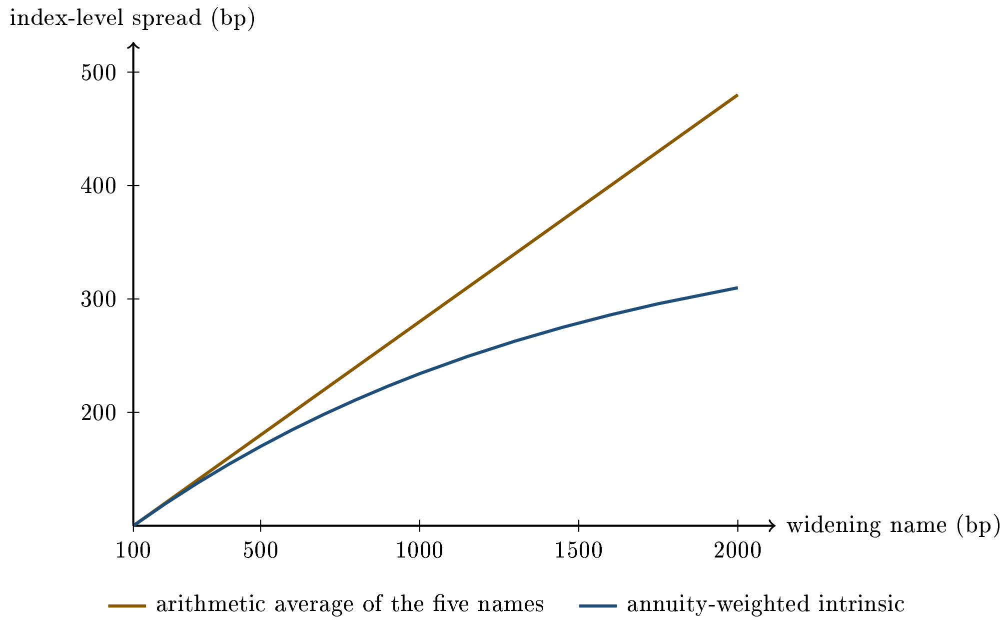
$$\frac{100 + 150 + 250 + 400 + 700}{5} = 320 \text{ bp.}$$

The annuity-weighted average, which is the intrinsic value of the index, is

$$\frac{\sum_i s_i A_i}{\sum_i A_i} = 301.8 \text{ bp.}$$

The gap is **18 basis points**, and it comes entirely from the 700 name being down-weighted by its short annuity.

The effect scales with dispersion, not with level. Put four names at 100 and let a fifth widen: at 100 for all five both numbers are 100, at 600 for the fifth the average is 200 against an intrinsic of 185, and at 2000 for the fifth the average is 480 against an intrinsic of 310. A single distressed name pulls the arithmetic average anywhere and the index barely follows it.



Four names are held at 100 basis points and the fifth is widened along the horizontal axis. The arithmetic average is a straight line by construction. The intrinsic bends away from it and flattens, because the widening name loses weight exactly as fast as it gains spread.

The traded index does not sit on the intrinsic either. The difference between the two is called the **skew** or the index basis, and it is a position people run deliberately. It appears because the index is far more liquid than its constituents, so a wave of macro hedging pushes the index without moving 125 single names, and because demand for tranches leaves dealers with index risk to recycle. When the skew gets large enough to cover the transaction costs, arbitrageurs trade the index against the full basket of single names and close it.

What happens when a name defaults

The contract does not terminate. The defaulted name is settled and removed, and the index continues on the remaining names with a reduced notional. The scaling factor is the **index factor**: the fraction of the original names still standing.

> WORKED EXAMPLE
>
> You hold 10 million of protection on CDX NA HY, which has 100 names, so each name carries $10\,\mathrm{m}/100 = 100\,000$ of notional. One name defaults and the auction settles at a recovery of 30%.
>
> You receive
>
> $$(1 - 0.30) \times 100\,000 = 70\,000,$$
>
> that name is removed, and the index factor becomes $99/100 = 0.99$. Your position is now 9.9 million of protection on 99 names, and the coupon you pay from the next date onwards is calculated on 9.9 million rather than on 10 million.
>
> Two consequences to state without being asked. The quoted spread you see afterwards refers to the surviving names only, so an index can look tighter after a default purely because the worst name has been taken out of it. And a position is not described by its original notional: two people holding "10 million of Series 41" hold different amounts of risk if one of them bought before a default and the other after. Always quote the factor.

How the auction determines that 30% is in recovery rates, and the settlement mechanism itself is in credit default swaps.

What the index is for

- **Macro hedging.** A portfolio of a hundred corporate bonds can be hedged approximately in one trade. The hedge is imperfect: the index names are not your names, and the residual is basis risk, which is the price you pay for being able to do it at all and in an afternoon.
- **Expressing a view on credit as an asset class**, without picking issuers. Selling protection on Main is the cleanest long-credit position available.
- **Relative value.** Index against the basket of its constituents (the skew above), Main against Crossover, Europe against North America, five years against ten.
- **Cash against synthetic.** The bond market and the CDS market disagree about the same credit surprisingly often, and the gap between them is the basis, covered in credit spreads.
- **As a building block.** Index tranches slice the index by loss level, which is tranches, and options on the index give convexity on credit itself.

Interview angle

REAL INTERVIEW QUESTION (Barclays · EQD)

What is the iTraxx index and what does its spread measure?

WHAT THEY TEST / ANSWER

Two halves, and the second one is where the marks are.

The first half is definitional: iTraxx is a family of credit default swap indices on European reference entities, the main ones being iTraxx Europe with 125 investment-grade names and iTraxx Crossover with 75 sub-investment-grade names, each an equally weighted portfolio of single-name protection traded as one contract, refreshed into a new series every six months, with the five-year point carrying the liquidity.

The second half is the trap. The spread does *not* measure the average spread of the constituents. It is the single running premium that makes the premium leg worth the protection leg for the whole portfolio, which makes it an annuity-weighted average of the constituent spreads. Wider names have shorter risky annuities, so they carry less weight, so the index sits below the arithmetic mean, and the gap grows with dispersion rather than with level. If you can say that, you have answered a question most candidates answer with "it is the average".

What it measures economically is the market price of diversified corporate credit risk in Europe: how much you must be paid to insure a broad basket against default. That is why it is quoted as the credit market's temperature, and why an equity derivatives desk watches it. Credit and equity are two claims on the same companies, so Crossover widening while equity volatility does nothing is a genuine signal that one of the two markets is wrong.

REAL INTERVIEW QUESTION (Natixis · Credit)

What is a CDS? How is the CDS market doing at the moment?

WHAT THEY TEST / ANSWER

Answer the first part in two sentences, since the interviewer is clearing the ground, not testing it: a credit default swap is a contract where the buyer pays a running premium and receives compensation for the loss if the reference entity has a credit event, which makes it insurance against default that you can trade without owning the debt. The full mechanics are in credit default swaps.

The second part is the real question, and the way to answer it is with index levels, because that is how the market talks about itself. Quote iTraxx Main and iTraxx Crossover, give the direction over the last few weeks, and say what has been driving it. Adding CDX for the North American comparison shows you look at both regions.

Then give the interpretation, which is what turns a quote into an answer. The **ratio** of Crossover to Main is the risk-appetite indicator: when it compresses, investors are reaching for yield, and when it widens, they are retreating to quality faster than the headline index suggests. Say whether the move has been spread-driven or rate-driven, since a credit portfolio can lose money in a rally. And mention issuance, because a heavy new issue calendar drags secondary spreads wider for purely technical reasons and is the most common explanation for a move nobody can pin on news.

Never answer this one with "the market is volatile at the moment". Give numbers, give a direction, give a cause.

REAL INTERVIEW QUESTION (JP Morgan · Rates)

Exercise on a basket credit-linked note: know the iTraxx, and the attachment and detachment points.

WHAT THEY TEST / ANSWER

The exam here is whether you can say what the investor actually owns. Build it in layers.

The underlying is a CDS index, so start there: an equally weighted portfolio of single-name protection, quoted as one spread, with a factor that falls as names default. Then the slicing. Rather than taking the whole portfolio's losses, the note references a **tranche** defined by two numbers, the attachment and the detachment point, expressed as percentages of the portfolio. The investor absorbs losses only between them: nothing until portfolio losses exceed the attachment, everything between the two, and nothing above the detachment because by then the tranche is wiped out. So a 3 to 6% tranche is untouched until losses pass 3% of the portfolio and gone once they reach 6%.

Then the wrapper. A credit-linked note is that exposure funded: the investor pays cash up front, receives a coupon, and gets back principal reduced by the tranche losses. Two points to volunteer, because they are what the question is really probing. The investor is **selling** protection, so they are long credit even though the instrument is called a note. And they are exposed to the issuer of the note as well as to the portfolio, so a note from a weak issuer referencing a strong portfolio is not a strong investment.

Finally, name the sensitivity: how the losses fall across tranches depends on how the defaults cluster, so tranche value depends on default correlation and not only on the level of spreads, with the junior and senior tranches taking opposite sides of that exposure. Tranche mechanics are developed in tranches and the correlation position in credit correlation.

INTERVIEW TRAP

Four index errors. (1) Calling the index spread the average of its constituents. (2) Getting the upfront sign backwards: when the traded spread is below the fixed coupon, the protection buyer overpays on the running leg and *receives* cash at the start. (3) Thinking a default ends the contract, when it removes one name, pays that name's loss and continues on a smaller notional. (4) Quoting a position as a notional without its factor, which overstates the risk of any index that has taken a default.

Where this goes next

Slicing the index by loss level is tranches, and what decides how those slices are priced is credit correlation. The single-name contract that the index is built from is credit default swaps, the spread measures used to compare it against cash bonds are credit spreads, and the recovery that turns a default into a payment is recovery rates.

12.8 Tranches

sales · trading · structuring | advanced | credit

Take a hundred risky loans. Individually, none of them is investment grade and no pension fund is allowed near them. Pool them, then slice the pool into layers ranked by who absorbs losses first, and the top layer can be rated AAA. That trick is **tranching**, and it is simultaneously one of the most elegant ideas in financial engineering and the mechanism at the centre of the 2008 crisis.

Both halves of that sentence are true, and an interviewer wants to hear that you know why. The elegance is real: tranching genuinely creates a safe claim out of risky ones, in the same way that a senior mortgage on a house is genuinely safer than the house. The danger is equally real, and it is not that the maths was wrong. It is that the safety of the senior layer depends almost entirely on one unobservable input, and that input was assumed rather than measured.

The waterfall

INTUITION

Picture the pool's cash as water poured into a tower of buckets, and the pool's losses as a fire that burns from the bottom up. Cash fills the top bucket first and overflows downward; losses destroy the bottom bucket first and work upward. The investor in the bottom bucket is paid a lot, because they get burned first. The investor at the top is paid very little, because the fire has to consume everything below them before it reaches them at all. Nobody's risk disappeared. It was sorted.

The layers, from the bottom:

- **Equity** (or first-loss) tranche: absorbs the first losses, receives the highest coupon, typically unrated and often retained by the arranger.
- **Mezzanine**: the middle layers, absorbing losses once equity is gone.
- **Senior** and **super-senior**: paid last in losses, first in cash, thin coupon, and historically where the AAA ratings lived.

Attachment, detachment, and the loss function

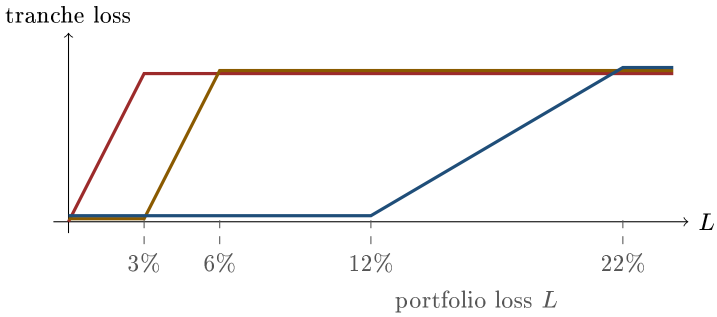
A tranche is defined by two numbers, both quoted as a percentage of the pool's notional: the **attachment point** A, where it starts to lose money, and the **detachment point** D, where it is completely wiped out. The tranche's own notional is $(D - A)$ times the pool notional. A "3-6% tranche" attaches at 3 and detaches at 6.

Let L be the pool's *cumulative* loss as a fraction of pool notional. The loss suffered by the tranche, in currency, is

$$\mathrm{Loss}_{[A,D]}(L) \;=\; \min\big(\max(L-A,\,0),\; D-A\big) \;=\; \max(L-A,\,0) \;-\; \max(L-D,\,0).$$

KEY IDEA

Read that second form again. **A tranche is a call spread on the cumulative portfolio loss**, long the strike at A and short the strike at D. Every intuition you have about call spreads from standard payoff profiles transfers directly, including the crucial one: the value of a call spread depends on the whole *distribution* of the underlying, not merely on its mean. Two pools with identical expected loss can produce wildly different tranche prices. What separates them is the shape of the loss distribution, and what drives that shape is correlation.



each curve is a call spread struck at its own A and D

The standardised index tranches are worth memorising because they come up by name. On iTraxx Europe the conventional slices are 0-3, 3-6, 6-9, 9-12 and 12-22%; on CDX North America investment grade they are 0-3, 3-7, 7-10, 10-15 and 15-30%. The indices themselves are covered in CDS indices.

WORKED EXAMPLE

Counting defaults through the stack. Take a synthetic pool built on an index of 125 equally weighted names, €10m of notional each, so a pool of €1.25bn. Assume a recovery rate of 40%, so a loss given default of 60%.

Each name is $1/125 = 0.80\%$ of the pool, so one default destroys $0.80\% \times 60\% = 0.48\%$ of pool notional. Everything else is arithmetic.

- The 0-3% equity tranche has a notional of $3\% \times 1.25\text{bn} =$ €37.5m. It is exhausted once cumulative loss reaches 3%, which takes $3/0.48 = 6.25$ defaults. So after 6 defaults the loss is 2.88% and the equity holder has lost $2.88/3 = 96\%$ of their notional; the 7th default finishes them.
- That same 7th default takes the loss to 3.36% and is the first euro of pain for the 3-6% mezzanine tranche, which loses $(3.36 - 3)/3 = 12\%$ of its notional.
- The 12-22% tranche, notional $10\% \times 1.25\text{bn} =$ €125m, is untouched until cumulative loss reaches 12%, which needs $12/0.48 = 25$ defaults. **One name in five must default before this tranche loses anything**, and the 26th default costs it only $0.48/10 = 4.8\%$ of its notional.

That last line is the whole sales pitch for senior tranches, and also the whole problem with them. Everything depends on whether 25 simultaneous defaults out of 125 investment-grade names is unthinkable or merely unlikely, and that is not an arithmetic question.

REAL INTERVIEW QUESTION (Société Générale · Credit)

What is a CDO, define each tranche, and what is the difference with a CLO?

WHAT THEY TEST / ANSWER

Structure the answer in three parts and do not skip the last one. A **collateralised debt obligation** is a vehicle that holds a pool of credit exposures and issues notes ranked by seniority, so that losses on the pool are allocated bottom-up through a waterfall. Then define the layers by their attachment and detachment points, not by adjectives: equity or first-loss (0-3% on the standard index conventions), mezzanine (the middle slices), senior and super-senior, with coupon falling and rating rising as you climb. Then the distinction they actually asked for: a **CLO** is a CDO whose collateral is specifically *leveraged loans*, usually senior secured and floating rate, *actively managed* by a collateral manager during a reinvestment period, and funded in cash. A generic CDO can hold bonds, asset-backed securities, or nothing at all, since a *synthetic* CDO holds no assets and takes its exposure through a portfolio of CDS. Three details separate a strong answer: CLO collateral is secured and historically recovers far more than unsecured bonds, which is why the whole capital structure attaches differently; the CLO manager's trading is itself a risk you are taking; and CLOs largely survived 2008 while ABS CDOs did not, which is the fact that stops the two words being interchangeable.

REAL INTERVIEW QUESTION (Natixis · Credit)

Pricing CDOs and CDS. Tell me about the mezzanine tranche.

WHAT THEY TEST / ANSWER

The mezzanine is the interesting one precisely because it is the least stable, and saying so is the answer. Start from the payoff: mezzanine is the call spread struck between the equity detachment and the senior attachment, so it only starts losing once the first-loss cushion is gone, and it is fully destroyed long before the senior layer is touched. Then give the three properties that make traders wary of it. First, **cliff risk**: it is thin, so the distance between "money good" and "wiped out" can be a handful of defaults, which makes its value highly non-linear in the loss level. Second, **ambiguous correlation sensitivity**: equity is long correlation and senior is short it, and mezzanine sits in between, so its correlation exposure is small, changes sign depending on where the pool's loss distribution sits, and cannot be signed from memory. Third, **model dependence**: because its value lives on the shoulder of the loss distribution rather than in the body or the extreme tail, it is the tranche most sensitive to the copula and correlation assumptions, which is why the standard market approach quotes it through base correlations on equity-style tranches rather than directly. If they push on pricing, the honest one-line answer is that you price the whole loss distribution, then value every tranche as a call spread on it, and the modelling argument is entirely about that distribution, not about the tranche. The correlation machinery is developed in credit correlation.

Correlation: the input that decides everything

The structural fact this section owns is the one established above: a tranche is a call spread on cumulative loss, so its value is a functional of the *whole* loss distribution and not of its mean. Push that one step further and the sensitivity falls out of the payoff algebra by itself.

KEY IDEA

Net the call spread down for the first-loss piece. The equity investor absorbs $\max(L, 0) - \max(L - D, 0)$, and because cumulative loss cannot be negative the first leg is simply L: linear, and therefore carrying no optionality at all. What survives is that the investor holds a **long call on the loss struck at the detachment point** D, plus a linear short position in L.

This is worth spelling out because "holds the first-loss piece and yet is helped by dispersion" sounds like a contradiction until you notice that the zero-strike leg has no convexity to lose. Anything that widens the loss distribution, holding its mean fixed, raises that call.

Correlation is exactly such a thing: holding expected pool loss fixed, it moves probability mass out of the middle and into both tails. So the equity tranche is long correlation, the senior tranche is short it, and the mezzanine sits between with a small, sign-unstable exposure. Those signs, the reason expected pool loss is correlation-independent in the first place, and the base-correlation convention desks actually quote through, are developed in credit correlation. What matters here is only that the tranche payoff is what makes the whole question well posed.

Cash, synthetic, and the squared version

A **cash** CDO actually buys the bonds or loans, funds them by issuing notes, and pays investors out of the collateral's cash flows. A **synthetic** CDO owns no assets: it sells protection through credit default swaps on a reference portfolio and passes the premium through the same waterfall. The synthetic version is faster to assemble, needs no funding for the unfunded tranches, and can be built on any reference portfolio someone will quote, which is exactly why the synthetic market grew to a multiple of the underlying bonds outstanding.

A **CDO-squared** takes tranches of other CDOs as its collateral. Each layer of tranching amplifies the correlation sensitivity, so a structure two layers deep is almost entirely a bet on a correlation number nobody can observe. If a candidate can say only one thing about CDO-squared, this is the thing.

REAL INTERVIEW QUESTION (Axa IM · Credit)

Cash CDOs, synthetic CDOs.

WHAT THEY TEST / ANSWER

An open prompt, so impose a structure: say what each one *holds*, how it is *funded*, and what that changes. A **cash** CDO buys the collateral outright, bonds or loans, funds the purchase by issuing rated notes, and pays investors from the collateral's own cash flows. A **synthetic** CDO holds no assets at all: it sells protection through credit default swaps on a reference portfolio and runs the premium through the same waterfall.

Then give the consequences, which is where the marks are. *Funding*: a cash deal must raise the full notional, so it is slow and balance-sheet heavy, whereas synthetic tranches can be left unfunded, which is why the synthetic market grew to a multiple of the bonds outstanding. *Supply*: a cash deal needs bonds that actually exist and can be sourced, while a synthetic deal can reference any portfolio someone will quote, so it has no natural size limit, which is the single most consequential difference and the one that mattered in 2008. *Counterparty risk*: the synthetic version replaces the issuer's collateral with an exposure to protection sellers, so it carries counterparty risk a cash deal does not. *Basis*: the synthetic tranche prices off CDS and the cash deal off bonds, so the two disagree by the CDS-bond basis. *Management*: cash structures, especially CLOs, usually have an active collateral manager and a reinvestment period, so you also take manager risk. Close by noting that a partially-funded structure sits between the two, since the tranches nearest the equity are usually funded even in a synthetic deal.

Why the senior tranches were the problem

The honest post-mortem is not "the maths was wrong", and interviewers reward candidates who resist that easy line.

A super-senior tranche pays a very thin spread for a payoff that is fine in almost every state of the world and catastrophic in a small set of them. That is the profile of a *sold deep out-of-the-money option*, and it carries all the familiar pathologies: the premium looks like free carry for years, the risk is invisible in any short historical sample, and the loss when it arrives is not proportional to the premium collected. Three things then compounded it. The correlation input was calibrated on a benign sample, a failure whose anatomy belongs to credit correlation. Ratings mapped a tranche's expected loss onto a scale built for single-name bonds, which says nothing about the shape of the loss distribution behind that expectation (see rating agencies). And CDO-squared structures levered the same assumption a second time. The narrative is developed in the 2008 financial crisis, and the regulatory response to securitisation capital in Basel III and IV.

VISUAL / CODE TO BUILD: Interactive

A tranche laboratory. The reader sets the pool (number of names, recovery rate) and a pairwise default correlation on a slider, and the panel draws the resulting portfolio loss distribution as a histogram with the standard attachment points drawn on it as vertical lines. Beside it, a bar chart shows each tranche's expected loss and fair spread, updating live. The intended "aha" is delivered by a single readout pinned at the top: *pool expected loss*, which the reader watches stay perfectly constant while the correlation slider moves and every tranche spread swings violently. A second mode drips defaults into the pool one at a time and animates the loss climbing through the waterfall, so the reader sees the equity tranche die on the 7th default in the worked example above. Backed by a one-factor Gaussian copula simulation, with the loss distribution and tranche table also exposed as text for accessibility.

INTERVIEW TRAP

(1) **Saying tranching removes risk.** It redistributes it. The sum of the tranches is the pool, always. (2) **Getting the correlation signs backwards.** Equity long, senior short. Rebuild it from the payoff rather than memorising it: net the call spread down and ask what widening the loss distribution does to the surviving option. The signs are developed in credit correlation. (3) **Forgetting recovery when counting defaults.** A default does not destroy a full name's notional, it destroys the loss given default. Skipping the 60% in the worked example above overstates the damage by two-thirds. (4) **Confusing pool expected loss with tranche expected loss.** The first is correlation-independent, the second is almost purely a correlation statement. (5) **Treating a rating as a risk description.** A AAA senior tranche and a AAA sovereign can share an expected loss and have nothing else in common. (6) **Using CDO and CLO interchangeably.** Different collateral, different seniority, different recovery, and a very different track record.

Where this goes next

The correlation models that turn a pool into a loss distribution, and the base correlation convention the market actually quotes, are in credit correlation. The two ingredients of every loss number here are default risk and recovery rates. The frameworks used to generate correlated defaults in the first place are in structural versus reduced-form models. And the single-name instrument the whole synthetic market is built out of is the credit default swap.

12.9 Credit Correlation

trading · structuring · sales | advanced | credit

Once a product references more than one name, the individual default probabilities stop being sufficient. What matters is whether the defaults arrive *together*, and that is a separate parameter which nobody can observe directly. This section is about that parameter: what it does, who is long it, and how it is modelled. The tranche structures themselves are tranches.

KEY IDEA

Correlation does not change the **expected loss** of a portfolio; it redistributes *where* that loss falls. The sum of the individual default probabilities is fixed, so the average number of defaults is fixed too. What correlation changes is the shape of the distribution around that average: low correlation concentrates outcomes near the mean, high correlation produces a barbell of "almost nothing happens" and "everything happens at once". Since a tranche is a slice of that distribution, every tranche has a correlation exposure even though the pool does not.

WORKED EXAMPLE

Two names, and the whole idea. Take two issuers, each with a 10% probability of defaulting over the horizon. The expected number of defaults is 0.2 whatever the correlation.

- **Independent.** $P(\text{at least one}) = 1 - 0.9^2 = \mathbf{19}\%$, and $P(\text{both}) = 0.1^2 = \mathbf{1}\%$.
- **Perfectly correlated.** They default together or not at all, so $P(\text{at least one}) = \mathbf{10}\%$ and $P(\text{both}) = \mathbf{10}\%$.

Read the two rows as two products. A **first-to-default** basket pays on the first event, so its risk *falls* from 19% to 10% as correlation rises: it is long correlation. A **second-to-default** basket needs both, so its risk *rises* from 1% to 10%: it is short correlation. The expected number of defaults never moved. That is the entire mechanism, and every tranche result is this example with more names.

Who is long and who is short

REAL INTERVIEW QUESTION (Bank of America · Credit)

Equity tranche of a CDO: long or short correlation?

WHAT THEY TEST / ANSWER

Long correlation, and the reason is worth giving because the answer alone is a coin flip.

The equity tranche absorbs the first losses, so what it wants is **zero defaults**. Raising correlation makes the "nothing happens" outcome more likely, because the names now survive together as well as fail together, and that is exactly the scenario the equity tranche needs. So raising correlation reduces the equity tranche's expected loss and increases its value: an investor holding it, meaning one who has sold protection on it, is long correlation.

The limiting cases make it obvious and are the fastest way to say it. At **zero correlation** with a large pool, the law of large numbers applies and the realised loss is almost exactly the expected loss, so a thin equity tranche is wiped out with near-certainty. At **correlation one**, the pool behaves like a single name: it either survives entirely, in which case equity loses nothing, or it defaults entirely. Equity strictly prefers the second world.

The senior tranche is the mirror: it only takes losses in a catastrophe, and catastrophes require many simultaneous defaults, so raising correlation hurts it. A senior investor is short correlation.

Then the detail that separates a good answer: **the mezzanine tranche is ambiguous**. It sits between the two effects and its sign depends on the attachment points, the pool's default probability and where correlation currently is, so it can flip. That is why mezzanine is the hardest tranche to hedge and why it caused the most trouble in 2005 and again in 2008: desks that were hedged on a correlation number found the sign of their exposure had changed underneath them.

REAL INTERVIEW QUESTION (CACIB · Credit)

Correlation positioning for credit derivative products.

WHAT THEY TEST / ANSWER

Give the map rather than one product, since the question is open.

Long correlation: equity tranches, first-to-default baskets, and in general any structure that is hurt by *idiosyncratic* risk and helped by everything moving together. The buyer of protection on a senior tranche is also long correlation, being the other side of a short.

Short correlation: senior and super-senior tranches, second-to-default and n-th-to-default baskets for n above one, and protection sold on them. Anyone who is fine with a few defaults and ruined by many is short correlation.

Ambiguous: mezzanine, as above.

Then two points that make it a desk answer rather than a taxonomy.

The street is structurally on one side. Investors want yield, so they buy mezzanine and equity risk, which leaves dealers holding senior and super-senior. The market is therefore systematically short correlation at the top of the structure, and that concentration is what turned a correlation move into a solvency problem in 2008, as the crisis describes.

Correlation is not hedgeable in the ordinary sense. There is no correlation instrument; you hedge a correlation position by taking an offsetting tranche position, which requires someone to want the other side. When correlation moves sharply, everyone needs the same offsetting trade at once and it is not available. That is the same structural problem as dispersion in equities, where the dealer community is also systematically on one side.

How it is modelled

REAL INTERVIEW QUESTION (CACIB · Credit)

What is a copula? What is it for?

WHAT THEY TEST / ANSWER

What it is: a function that joins marginal distributions into a joint distribution. Sklar's theorem says any joint distribution can be written as a copula applied to its marginals, so the copula *is* the dependence structure with the individual behaviour stripped out.

Why that is useful in credit: because the two halves of the problem come from different places. The marginal default probability of each name is observable, since it is implied by that name's CDS spread. The dependence between names is not observable at all. A copula lets you take the marginals straight from the market and then impose a dependence structure separately, rather than needing one model that produces both.

The standard implementation is the **Gaussian copula**: give each name a latent variable, correlate those variables with a single correlation parameter, and declare a name defaulted when its variable falls below a threshold calibrated to its CDS spread. One number then controls the entire dependence structure, which is why it became the market standard for tranches.

Then say what is wrong with it, because that is what the question is for. The Gaussian copula has **no tail dependence**: however high you set the correlation short of one, extreme joint events remain far less likely than they are in reality. Credit defaults cluster in exactly the tail the model thins. It also compresses the whole dependence structure into one parameter, which is why the market had to invent **base correlation** and ended up with a **correlation smile**, different implied correlations for different tranches of the same pool. A single parameter that has to take different values depending on which slice you look at is a model telling you it is misspecified.

The alternatives, in one clause each: the Student-t copula, which has tail dependence and is the cheapest fix; Archimedean copulas such as Clayton, which are asymmetric so that dependence is stronger in the lower tail; and random-factor-loading or stochastic-correlation models that let correlation itself rise in stress. Naming a fix as well as a flaw is what separates a criticism from an answer, and the general statistical background is independence and correlation.

Base correlation and the smile

Because the Gaussian copula cannot fit all tranches of a pool with one number, the market quotes **base correlation**: the correlation that reproduces the market price of an equity tranche running from zero to each detachment point. Any tranche is then the difference between two base tranches, and the quoting convention becomes internally consistent even though the model is not.

The resulting curve slopes upward: higher detachment points require higher correlations. That upward slope is the credit analogue of the equity volatility skew, and it exists for the same reason, which is that the market prices more joint-tail risk than the model does.

INTERVIEW TRAP

1. **Thinking correlation changes the pool's expected loss.** It does not. It only moves risk between tranches.

2. **Getting the equity tranche sign backwards.** Equity is long correlation, because it needs zero defaults and correlation makes zero defaults more likely.

3. **Assuming mezzanine has a stable sign.** It does not, and that is the tranche that has caused the most losses.

4. **Treating one correlation number as the truth.** A model that needs a different parameter for each tranche of the same pool is misspecified, and the base correlation curve is the evidence.

5. **Trusting a Gaussian copula in the tail.** It has no tail dependence, which is precisely where credit losses actually arrive.

6. **Assuming correlation is stable.** It rises in stress, so a position hedged at a calm correlation is unhedged in the event it was hedged against.

VISUAL / CODE TO BUILD: Interactive

A loss distribution laboratory. The reader sets a pool size, a common default probability and a correlation, and the page draws the distribution of portfolio losses, with the tranche attachment points marked as vertical lines so each tranche's expected loss is a visible area. Dragging the correlation slider leaves the *mean* of the distribution fixed while the shape moves from a tight bell to a barbell, which makes the section's central claim something the reader watches rather than accepts, and the tranche values update alongside so the equity and senior signs appear without being asserted. A second panel plots each tranche's value against correlation on one chart, so the equity line slopes up, the senior line slopes down, and the mezzanine line visibly changes sign as the reader moves the attachment points. A third mode fits a base correlation curve to a set of market tranche prices and displays the resulting smile, with a toggle to a Student-t copula that flattens it.

12.10 Structural vs reduced-form models

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There are two ways to model a default, and they disagree about something fundamental: whether default is the predictable consequence of a firm running out of value, or an unpredictable event that simply arrives. Both are used, by different people, for different purposes, and knowing which is which is a standard credit interview question.

Structural models: default has a cause

The founding idea is Merton's, and it is one of the most elegant results in finance. Suppose a firm's assets have value V_t and follow a lognormal diffusion with volatility σ_V, and that the firm has issued a single zero-coupon debt of face value D maturing at T. At maturity there are two cases: if $V_T > D$ the debt is repaid and the shareholders keep $V_T - D$; if $V_T \leq D$ the firm defaults and the creditors take whatever there is.

So the shareholders hold $\max(V_T - D, 0)$. **Equity is a call option on the firm's assets, struck at the face value of its debt.** And by the same accounting, the creditors hold a risk-free bond minus a put on the assets: lending money is selling insurance on the firm's value. The correspondence is the one already used in equities, seen from the other side.

Everything follows from Black-Scholes. The risk-neutral default probability is

$$\mathbb{P}(\text{default}) = \mathbb{P}(V_T < D) = N(-d_2), \qquad d_2 = \frac{\ln(V/D) + (r - \frac{1}{2}\sigma_V^2)T}{\sigma_V\sqrt{T}},$$

and d_2 has a name of its own: the **distance to default**, the number of asset standard deviations between the firm's current value and its liabilities.

WORKED EXAMPLE

Assets 100, debt with face value 70 maturing in one year, asset volatility 20%, zero rates. Then $d_2 = 1.68$, so the distance to default is 1.68 standard deviations and the default probability is $N(-1.68) = 4.6\%$.

The equity, as a call, is worth 30.25, so the debt is worth $100 - 30.25 = 69.75$ against a face value of 70. That gives a yield of 0.36%, a **credit spread of 36 basis points** over the risk-free rate, produced entirely by an option calculation with no credit input at all.

Now run the maturity down and watch what happens: at five years the spread is 131 basis points, at one year 36, at six months 6, and at three months 0.2. The spread collapses to zero as maturity shortens, and that is not a feature of these particular numbers.

KEY IDEA

The vanishing short-dated spread is structural models' defining flaw. Because assets follow a continuous diffusion, a firm that is solvent today cannot become insolvent in the next instant, so short-dated default probability goes to zero faster than time does and so does the spread. The market says otherwise: short-dated credit spreads are positive and can be large. The fix is to allow the asset value to **jump**, which restores a non-zero instantaneous default probability and, not coincidentally, is the same repair that fixes the short-dated equity smile in the Black-Scholes model. The two markets are showing the same missing ingredient.

Two extensions matter. **First-passage models**, in the Black-Cox tradition, let default happen at any time the asset value touches a barrier rather than only at maturity, which is more realistic since covenants and refinancing failures do not wait for a maturity date. And the commercial implementations, of which the KMV framework is the best known, turn the distance to default into an empirically calibrated default frequency rather than trusting the normal distribution in the tail. They also measure the distance with the asset's actual expected return in place of r, so their distance to default is a real-world quantity, while the d_2 above, driven by the risk-neutral drift, is not. The two are the same formula answering different questions.

Reduced-form models: default just happens

The other family refuses to explain default and concentrates on pricing it. Default is the first jump of a point process with **intensity** λ_t, so over a short interval the conditional probability of defaulting is $\lambda_t\,dt$, and the survival probability is

$$S(t) = \mathbb{E}\Big[\exp\Big(-\int_0^t \lambda_u\,du\Big)\Big],$$

with λ allowed to be stochastic and correlated with rates. The credit spread follows as $s \approx \lambda(1-R)$, which is where the triangle of default risk comes from.

The philosophical difference is sharp. In a structural model, default is a *predictable* event: you can watch the asset value approach the barrier. In an intensity model it is *inaccessible*: it arrives without warning, which is a better description of how defaults actually look to the market, where a name trades normally and then does not.

The practical difference is that the intensity is not derived from anything, it is **calibrated to prices**. Feed in the CDS curve and out comes a hazard curve that reproduces it exactly. That is the family's strength and its weakness in one sentence: it will always fit, and it never tells you why.

REAL INTERVIEW QUESTION (Bank of America · Credit)

What are the main credit risk models, structural versus intensity based?

WHAT THEY TEST / ANSWER

Give the two families, the mechanism of each, then the comparison, which is what is being marked.

Structural, after Merton: model the firm's asset value, treat equity as a call on the assets struck at the debt, and let default be the assets ending below the liabilities. Output: a default probability from the equity price and its volatility, and a distance to default. **Reduced-form**, or intensity based: do not model the firm at all, treat default as the first arrival of a process with hazard rate λ, and calibrate λ to observed credit prices.

Structural strengths: an economic story, a link between the equity and credit markets that makes capital-structure arbitrage possible, sensible comparative statics since more leverage or more asset volatility raises the default probability, and no need for a liquid credit market to produce a number, which is why it is used for private firms and for counterparty risk. **Weaknesses**: the asset value and asset volatility are not observable and must be backed out of equity, the real capital structure is nothing like one zero-coupon bond, and the model produces short-dated spreads near zero, which the market contradicts.

Reduced-form strengths: it fits market prices by construction, it is tractable, it handles a term structure of credit and correlation with interest rates naturally, and it is what actually prices and hedges credit derivatives. **Weaknesses**: no economic content, so the intensity is a fitted parameter rather than an explanation, no link to the equity, and no guidance at all when there is no market to calibrate to.

Close with the sentence that decides which to use: **structural models are for understanding and for risk, reduced-form models are for pricing**. A rating or capital framework, a counterparty exposure model or an equity-credit relative value trade uses the first; a CDS curve, a bond mark or a tranche uses the second. Hybrid models exist, letting the intensity depend on a firm-value state variable, and they are an attempt to get the economics of one and the calibration of the other.

Where the models meet the market

REAL INTERVIEW QUESTION (HSBC · Rates)

Determine the spread of a CDS, and discuss the dependence between the spread and interest rates.

WHAT THEY TEST / ANSWER

The spread comes from equating the two legs. The protection buyer pays s on the notional until default or maturity, and receives $(1-R)$ at default. Writing both as expectations under survival, the fair spread is the ratio of the expected loss leg to the risky annuity, and with a flat hazard and constant recovery it reduces to the relation worth quoting from memory,

$$s \approx \lambda\,(1-R),$$

which is the reduced-form model in one line. Say that in practice a curve of hazard rates is bootstrapped from CDS quotes at several maturities, exactly as a zero curve is bootstrapped from swaps.

The dependence on rates has three layers, and giving all three is the answer. *Mechanically*, in the pricing formula, rates enter only through discounting both legs, and because they discount the fee leg and the protection leg similarly, the effect on the fair spread is second order. That is the answer to the narrow question, and it surprises people.

Economically, rates and credit are strongly linked, but through the state of the world rather than through the formula. Higher rates raise refinancing costs and pressure leveraged issuers, which widens spreads; but rates usually rise because growth is strong, which narrows them. The net sign depends on which force is driving the move, which is why the correlation between rates and spreads flips regime by regime and is negative in a flight to quality, when government yields fall and spreads widen at the same time.

Structurally, in a firm-value model the risk-free rate is the drift of the assets, so a higher rate raises the expected asset value, lowers the default probability and narrows the spread. That is a real channel and it is the one most often quoted in interviews, so name it and then note that it is usually swamped by the economic channel above.

Close on the practical consequence: this is why a credit desk hedges rate risk separately, with swaps, rather than assuming the spread absorbs it, and why the asset-swap spread exists as a way of stripping the rate exposure out of a bond and leaving the credit.

INTERVIEW TRAP

Four errors. (1) Quoting Merton default probabilities as real-world numbers: they are risk-neutral and, in the tail, badly dependent on the lognormal assumption. (2) Forgetting that a structural model needs asset value and asset volatility, neither of which is observable, so the equity-implied inputs carry all the model risk. (3) Claiming a reduced-form model "explains" a spread: it reproduces it, which is a different claim. (4) Assuming rates and spreads are independent because rates drop out of the spread formula, when the two are linked through the economy and the correlation is regime dependent.

Where this goes next

The probability these models produce is default risk, the other input to every spread is recovery rates, and the instrument that both families are calibrated to is credit default swaps. Extending either family to several names at once is credit correlation, and the option-pricing machinery they both borrow is the Black-Scholes model.

12.11 CVA, DVA, FVA

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Every pricing formula in this book quietly assumes the person on the other side pays. If they might not, the contract is worth less than the formula says, and the difference is a real number that appears in a bank's accounts. The family of adjustments that repairs this assumption is collectively the XVAs, and since 2008 they have become a desk in their own right.

The core idea: a derivative contains a short option on your counterparty

Consider an uncollateralised swap that is currently in your favour by 5. If the counterparty defaults tomorrow you do not lose the swap's notional, you lose that 5, recovering only a fraction of it. If instead the swap is against you by 5, their default costs you nothing: you still owe your side.

So your loss depends on the *positive part* of the mark-to-market at the default time. You are, in effect, **short a series of options on your own exposure**, knocked in by the counterparty's default. That asymmetry is the whole subject.

> KEY IDEA
>
> **Credit valuation adjustment** is the market price of that short option:
>
> $$\text{CVA} \;=\; (1-R)\int_0^T \text{EE}(t)\, d\,\mathbb{P}(\tau \le t)$$
>
> discounted, where $\text{EE}(t)$ is the **expected exposure**, the expected positive mark-to-market at t, and τ is the counterparty's default time. Three ingredients, each from a different market: the *exposure* from the derivative's own model, the *default probability* from the counterparty's credit spread, and the *recovery* from convention. The price you actually charge a client is the risk-free price minus the CVA.

WORKED EXAMPLE

An uncollateralised five-year swap on a notional of 100, whose expected positive exposure averages 2 over its life. The counterparty's five-year CDS trades at 150 basis points, and recovery is taken at 40%, so the hazard rate is $0.015/0.6 = 2.5\%$ and the cumulative default probability over five years is $1 - e^{-0.125} = 11.75\%$.

Then

$$\text{CVA} \approx (1 - R) \times \overline{\text{EE}} \times \mathbb{P}(\text{default}) = 0.6 \times 2 \times 0.1175 = 0.14,$$

so 14 basis points of notional. The desk rule of thumb gives the same order instantly: $\text{CVA} \approx \text{EPE} \times s \times T = 2 \times 1.5\% \times 5 = 0.15$. That is the number a salesperson has to build into the quoted rate, and it is why an uncollateralised client is quoted a worse level than a collateralised one on an identical trade.

Exposure is the hard part

The credit inputs are read off a market. The exposure is not: it has to be simulated, because it is the future value of the derivative, which is random.

REAL INTERVIEW QUESTION (BNP · Market Risk)

You have a payer swap with a counterparty. Explain how it works, who is exposed to movements in rates, and does that imply a movement in counterparty risk?

WHAT THEY TEST / ANSWER

The mechanics. In a payer swap you pay a fixed rate and receive floating on an agreed notional and schedule. At inception the fixed rate is set so the swap is worth zero to both sides.

Who is exposed to rates. Both, in opposite directions. If rates rise, the floating leg you receive is worth more, so the swap moves *in your favour* and against the counterparty. If rates fall, the reverse. Say it in terms of the position: a payer swap is short duration, so it gains when rates rise.

And yes, that moves counterparty risk, which is the point of the question. Counterparty exposure is the positive part of the mark-to-market, so it is not symmetric. When rates rise the swap is in your favour, your exposure to *their* default grows, and your CVA charge grows with it; when rates fall you have no exposure at all and they have exposure to you, which is their CVA and your DVA. So a single rate move transfers counterparty risk from one side to the other, and this is precisely why counterparty risk cannot be assessed from the trade alone: it depends on the market.

Then describe the **exposure profile** (defined formally, with EE, EPE and PFE, in counterparty risk), which is what a counterparty risk interviewer wants to hear. It starts at zero, since the swap is at par. It then rises, because rates diffuse away from the initial level and the potential mark-to-market widens. But it eventually falls back to zero at maturity, because payments have been made and fewer cash flows remain to be exposed. The result is a **hump**, peaking somewhere around a third of the way through the life for a standard swap, where the diffusion effect stops outrunning the amortisation effect. Contrast it with a cross-currency swap, whose notional is exchanged at maturity, so the exposure keeps growing to the end and is far larger.

Close on the mitigants, since they are what a bank actually does: **netting** across all trades in the same agreement, so offsetting positions reduce the exposure rather than adding to it; a **collateral agreement** with daily margin, which cuts the exposure to the gap risk between margin calls; and **clearing**, which replaces the counterparty with a central counterparty, in clearing houses.

The rest of the family

REAL INTERVIEW QUESTION (Société Générale · Rates)

What are CVA, DVA, FVA and KVA?

WHAT THEY TEST / ANSWER

Four adjustments to the same risk-free price, each answering a different question about who bears a cost the textbook ignored. Give each one in a sentence and then say the thing that ties them together.

CVA, credit valuation adjustment: the expected loss from *your counterparty* defaulting while the trade is in your favour. It reduces the value of your position and is charged to the client.

DVA, debit valuation adjustment: the mirror image, the expected gain from *your own* default while the trade is against you. It is the counterparty's CVA seen from your side, so bilaterally the two net. It is also deeply uncomfortable, since your own credit deteriorating produces a profit, and it cannot be hedged except by selling protection on yourself, which is why it is excluded from regulatory capital even though it is required by accounting.

FVA, funding valuation adjustment: the cost of funding an uncollateralised position. If you hedge a client trade in the cleared market you post margin there and receive none from the client, so you fund the difference at your own cost, above the risk-free rate. FVA prices that funding gap, and its overlap with DVA was one of the more serious arguments in derivatives valuation, since both partly reflect your own credit.

KVA, capital valuation adjustment: the cost of holding regulatory capital against the trade for its life. Capital is not free, so a trade that consumes a lot of it must earn a return on it, which is why long-dated uncollateralised trades became so much more expensive after the reforms.

Then the sentence that shows you understand the direction of travel: **MVA** for initial margin completes the set, and together they are the reason a modern derivatives price is the risk-free price minus a stack of adjustments computed centrally by an XVA desk rather than by the trading desk. And name the driver: almost every one of these exists because 2008 demonstrated that the counterparty, the funding and the capital were not free, and the pricing had been assuming they were.

REAL INTERVIEW QUESTION (ING · Rates)

What is CVA, and what is CVaR? What is the difference between the two?

WHAT THEY TEST / ANSWER

They sound alike and they are unrelated, which is why the question exists.

CVA is a *valuation* adjustment: the expected loss from counterparty default, computed under the risk-neutral measure, that reduces the fair value of a derivative. It is a price. It sits on the balance sheet, it is hedged with CDS and with the underlying market factors, and it is the responsibility of an XVA desk.

CVaR, conditional value at risk, also called expected shortfall, is a *risk measure*: the average loss in the worst tail of a portfolio's profit and loss distribution beyond a confidence level, computed under the real-world measure. It is not a price, it is not on the balance sheet, it says nothing about any counterparty, and it is the responsibility of market risk. It is covered in expected shortfall.

Give the three-way contrast to close: **purpose**, price against risk limit; **measure**, risk-neutral against real-world; **object**, the counterparty's default against the portfolio's own loss distribution. Then note the one genuine connection, which shows you are not just reciting definitions: CVA is itself a mark-to-market that moves, so a bank computes a value at risk or an expected shortfall *on its CVA*, which is where the two ideas meet and where CVA volatility became a capital charge in its own right.

Wrong-way risk

The single most dangerous assumption in this section is that the exposure and the counterparty's default are independent. **Wrong-way risk** is the case where they are positively correlated, so the exposure is largest exactly when the counterparty is most likely to fail. It is treated in full as a risk concept in counterparty risk; what matters here is what it does to the price.

The examples are memorable and worth having ready. Buying protection on an Italian bank from an Italian bank: the reference credit and the counterparty deteriorate together, so the protection is worth most when it is least likely to be paid. Trading an oil-linked derivative with an oil producer whose creditworthiness depends on the oil price. Taking collateral in the counterparty's own shares. The mirror case, right-way risk, exists as well and reduces CVA, but it is far less common than counterparties claim.

INTERVIEW TRAP

Four errors. (1) Computing exposure as the notional: it is the positive part of the mark-to-market, and for an interest rate swap it is a small fraction of notional. (2) Applying CVA trade by trade when a netting agreement exists, which massively overstates it, since CVA is a property of the netting set and not of the trade. (3) Assuming independence between exposure and default, which is exactly what wrong-way risk breaks. (4) Confusing CVA with CVaR, a price with a risk measure.

Where this goes next

The default probabilities feeding these formulas are default risk and recovery rates, the instrument used to hedge the credit leg is credit default swaps, and the trades that generate the exposure are interest rate swaps. The risk-management view of the same subject is counterparty risk, the tail measure it is often confused with is expected shortfall, and the institution that removes most of it is clearing houses.

Chapter 13

Foreign Exchange Markets

13.1 FX spot markets

sales · trading · structuring core fx

Foreign exchange is the largest market in the world and the one candidates most often fumble, for a reason that has nothing to do with difficulty. There is no hard mathematics in spot FX. There is a **convention**, and if you have the convention backwards then every subsequent sentence you say is wrong, confidently and in the opposite direction. Interviewers on FX desks know this and use it as a one-question filter.

So this section is mostly about reading a quote correctly, and then about the plumbing that makes the market work: what a pip is, how crosses are built, when the money actually moves, and what makes spot go up and down.

The scale, and the shape

In the BIS Triennial Survey of April 2022, global foreign exchange turnover was about \$7.5 trillion *per day*, of which roughly \$2.1 trillion was spot and the majority of the rest was FX swaps. The US dollar was on one side of close to 90% of all trades, and EUR/USD alone was the single biggest pair. The survey runs every three years, so treat those as the shape of the market rather than as current statistics, and quote the latest survey if you are asked for a number.

The structure follows from a single fact: there is no exchange. FX is over-the-counter and decentralised, so "the price" is a set of prices from different dealers on different venues, tied together by arbitrage rather than by a central book. Around that sit interdealer platforms, single-dealer platforms, ECNs, prime brokerage relationships that let smaller players trade on a bank's credit, and retail aggregators.

It also trades continuously from the Sydney open on Monday to the New York close on Friday, handing off Sydney to Tokyo to London to New York. London is by a distance the largest centre, and the London and New York overlap in the early afternoon London time is when liquidity is deepest and spreads are tightest. Corporate and index-tracking flow concentrates in the late-afternoon London fix, which is why that window has its own microstructure and its own compliance history.

Reading a quote, which is the whole game

KEY IDEA

A currency pair is written **BASE/QUOTE**, and the number is **how many units of the quote currency buy one unit of the base**. So EUR/USD at 1.0850 means one euro costs 1.0850 dollars. The pair *is* the price of the base currency.

Everything follows:

- "Buying EUR/USD" means buying euros and selling dollars.
- The pair going **up** means the **base** strengthened, the quote currency weakened.
- USD/JPY at 150.00 means one dollar costs 150 yen. USD/JPY rising is the *yen* weakening, not the dollar, so the intuitive reading is inverted relative to EUR/USD.

Which currency gets to be the base is pure market convention, and the convention is an ordering: EUR outranks GBP, which outranks AUD, then NZD, then USD, then everything else. That is why you say EUR/USD and GBP/USD but USD/JPY, USD/CHF and USD/CAD. Learn the ordering rather than the individual pairs, because it also tells you how a cross will be quoted. The trading floor also keeps nicknames: cable for GBP/USD, Aussie, Kiwi, Loonie, Swissy.

A **pip** is the conventional last decimal of the quote: the fourth decimal for most pairs, the second for yen pairs. Dealers now quote a fifth decimal (a "pipette" or fractional pip), which is where the actual competition happens in liquid pairs.

WORKED EXAMPLE

What a pip is worth. You are long €10m against USD at 1.0850, and the pair moves to 1.0853, three pips. Each pip on €10m is $10{,}000{,}000 \times 0.0001 = \$1{,}000$, so the move is worth $3,000.

Note where the PnL lands. Your position is in euros and your profit is in dollars, because it is the quote currency that measures the price. This is exactly the bookkeeping that catches people out under pressure: a pip's value in the quote currency is fixed by the base notional and does not depend on the level of the pair.

REAL INTERVIEW QUESTION (CACIB · FX)

You are an American importer: are you happy or not about the move in EUR/USD?

WHAT THEY TEST / ANSWER

This is a convention test wearing a business costume, and the only wrong move is to answer before you have said which way EUR/USD went. Do it in two steps, out loud. First, fix the convention: EUR/USD is dollars per euro, so EUR/USD *rising* means the euro is stronger and the dollar is weaker. Second, fix the exposure: an American importer buys goods abroad, so they hold dollars and must pay euros, which makes them **short EUR and long USD** economically. Put the two together: a rising EUR/USD makes their European purchases more expensive in dollars, so they are unhappy; a falling EUR/USD is good for them. An exporter is the mirror. Then add the two things that turn a correct answer into a good one. The importer's real problem is not the level but the *uncertainty*, since they have committed to a price in euros and will pay later, which is a forward or option hedging question, covered in FX forwards and swaps. And if the interviewer has not told you the direction of the move, ask; answering a directional question about an unspecified move is the trap.

Crosses and triangular arbitrage

Most currency pairs are not quoted directly against each other. A **cross** is a pair that does not involve the dollar, and it is constructed from the two dollar legs. If both pairs share the dollar in the right position, you multiply; if the dollar is on the same side of both, you divide.

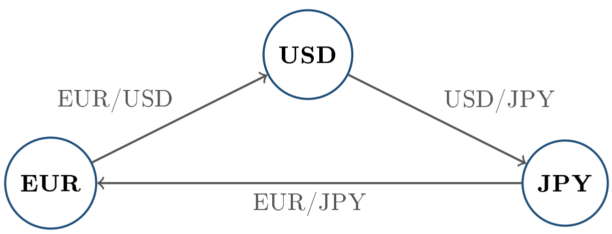
WORKED EXAMPLE

Building EUR/JPY. EUR/USD trades at 1.0850 (dollars per euro) and USD/JPY at 150.00 (yen per dollar). The units cancel:

$$\frac{\text{USD}}{\text{EUR}} \times \frac{\text{JPY}}{\text{USD}} = \frac{\text{JPY}}{\text{EUR}}, \qquad \text{EUR/JPY} = 1.0850 \times 150.00 = 162.75.$$

Always write the units out. Doing that makes the multiply-or-divide question answer itself and removes the single most common arithmetic error in an FX interview.

If a dealer quotes EUR/JPY away from that computed level, the three prices are inconsistent and a round trip through all three makes money. This is **triangular arbitrage**, and it is the mechanism that keeps a decentralised market coherent.



the three legs must be consistent: any round trip that returns more than it started with is an arbitrage

WORKED EXAMPLE

Sizing the arbitrage. Keep EUR/USD at 1.0850 and USD/JPY at 150.00, so the cross should be 162.75, and suppose someone quotes EUR/JPY at 163.00. That quote overvalues the euro in yen, so sell euros there and buy them back through the dollar.

Start with €1. Sell EUR/JPY at 163.00 and receive ¥163.00. Convert yen to dollars at 150.00: $163.00/150.00 = \$1.086667$. Convert dollars back to euros at 1.0850: $1.086667/1.0850 =$ €1.0015361.

The round trip nets 0.15361%, or about 15.4 basis points, which is exactly $163.00/162.75 - 1$. In liquid pairs this opportunity is measured in microseconds and in fractions of a pip, and the calculation above is really a check that your units are right, not a trade idea. In illiquid crosses and stressed markets the gap can be real, but so is the cost of crossing three bid-ask spreads, and that cost is worth pricing rather than assuming. Half a pip on each leg costs

$$\frac{0.00005}{1.0850} + \frac{0.005}{150.00} + \frac{0.005}{162.75} \approx 1.1 \text{ basis points},$$

so a 15 basis point dislocation would be worth more than ten times the friction, which is exactly why it cannot survive in the majors. It takes something like seven pips on every leg, the sort of spread you see only in a thin cross or a stressed market, before the three crossings swallow a gap that size.

Value dates, and the risk of settling

Spot is not instant. The market convention is settlement on **T+2**, two business days after the trade, with USD/CAD settling T+1. "Business days" means good business days in *both* currency centres plus, usually, New York, which is why holidays in one country shift value dates for pairs that country is not obviously part of. The date the money moves is the **value date**, and it is what a forward is defined relative to (see FX forwards and swaps).

That two-day gap creates a specific and famous hazard. In an FX trade each side pays in a different country, in a different time zone, so one leg can settle hours before the other. If your counterparty fails in between, you have paid away one currency and received nothing. This is **settlement risk**, still called **Herstatt risk** after the German bank whose 1974 closure, mid-settlement-day, left counterparties who had already paid Deutsche marks with no dollars coming back. The industry's answer is payment-versus-payment settlement through CLS, which settles both legs simultaneously or neither. Bring this up when asked about FX risk: most candidates list market risk and stop, and settlement risk is the one an operations-aware interviewer is listening for. The general framework sits in counterparty risk.

What actually moves spot

INTUITION

A currency's price is the outcome of everyone who needs to convert. Exporters and importers convert because of trade; investors convert because of where they want their capital; central banks convert to lean against either. So spot moves when the attractiveness of holding a currency changes, whether through the return it pays, the growth it buys, or the safety it offers when things go wrong.

The drivers a candidate should be able to list, roughly in order of how often they dominate:

- **Interest rate differentials and the policy path.** Not the level of rates but the *expected* path relative to what is already priced. A rate rise that was fully anticipated moves nothing (see monetary policy).
- **Growth and inflation surprises**, which move that expected path.
- **Risk sentiment.** A handful of currencies (USD, JPY, CHF) strengthen in stress regardless of their own fundamentals, because capital comes home or goes to safety.
- **Terms of trade**, which link commodity currencies (AUD, CAD, NOK, BRL) to the prices of what they export.
- **Positioning and flows.** Crowded carry trades unwind violently, and the unwind is a much faster move than the accumulation was.
- **Official action**: intervention, capital controls and regime choices, covered in central banks and FX policy and pegged currencies and FX crises.

REAL INTERVIEW QUESTION (Société Générale · FX)

Where is EUR/USD trading today, and what factors move it?

WHAT THEY TEST / ANSWER

Two questions in one, and both are graded. The level is a preparation check: know where the major pairs are on the morning of your interview, to the figure and ideally the pip, along with a rough range for the year, and say it without hedging. Never guess a level, and never claim a precision you do not have. The second half is the real question, and the structure to give is: it is a *relative* price, so every driver has to be framed as a differential. Relative expected policy paths first, because the euro-dollar rate is mostly a Fed-versus-ECB story and what matters is the change in expectations rather than the level of either policy rate. Then relative growth and inflation surprises, which move those expectations. Then the dollar's role as the reserve and safe-haven currency, which means EUR/USD tends to fall in global risk-off episodes for reasons that have nothing to do with Europe. Then energy terms of trade, which mattered enormously for the euro after 2022. Finish with the positioning point, that a level can persist well past what fundamentals justify because everyone is already on one side, and that is what makes the reversals so fast.

REAL INTERVIEW QUESTION (JP Morgan · Rates)

You have funds in a country where you fear a depreciation of the currency. What can you do to avoid losing money? What is the consequence for the currency, and for rates, if every investor behaves the same way?

WHAT THEY TEST / ANSWER

Answer the mechanics first, then the reflexivity, because the second half is the part they are actually testing. The individual answer: sell the local currency forward against a hard currency for the horizon of your exposure, which locks a rate today without needing to move the underlying assets; or buy a put on the local currency if you want to keep the upside and are willing to pay a premium; or, if the currency is not deliverable, use a non-deliverable forward that cash-settles the difference against a fixing. If you can also move the capital, converting to the hard currency outright is the cleanest hedge and the one that actually leaves the country.

Now the second half. If everyone does this at once, the hedge is itself the depreciation. Every investor selling the local currency forward pushes the banks who take the other side into selling it spot to hedge, so the aggregate hedging flow produces exactly the move it was defending against. That is a self-fulfilling run, and it is the standard anatomy of an emerging-market currency crisis. Rates then move in two ways at once: the domestic policy rate is usually forced sharply higher, both to defend the currency by making it expensive to be short and because the central bank is losing reserves, and the local forward points and the cost of hedging blow out because nobody will lend the local currency to a market that wants to short it. The authorities' remaining options, intervention, rate defence or capital controls, are covered in pegged currencies and FX crises.

REAL INTERVIEW QUESTION (JP Morgan · desk not recorded)

Why does the FTSE fall when sterling appreciates?

WHAT THEY TEST / ANSWER

Because the FTSE 100 is not really a bet on the British economy. A large majority of its constituents' revenues are earned outside the United Kingdom, in dollars and other currencies, by oil majors, miners, and global consumer and pharmaceutical companies. Those foreign earnings are translated back into sterling for reporting, so a stronger pound mechanically shrinks them, and the index falls even if nothing at all happened to the underlying businesses. There is a second, competitiveness channel for genuine exporters, whose goods become dearer abroad, but the translation effect is the dominant one and is the answer they want. The two additions that show real understanding: this makes the FTSE 100 and sterling structurally negatively correlated, so a dollar-based investor holding an unhedged FTSE position has an exposure that partly cancels, which is the same composite-exposure arithmetic as in ADRs and GDRs; and the effect is much weaker for the FTSE 250, which is far more domestically focused, so a Brexit-style shock that hits sterling can send the two UK indices in opposite directions on the same day. That divergence is the detail that ends the question.

VISUAL / CODE TO BUILD: Interactive

A triangular arbitrage trainer. Three live-editable quotes (EUR/USD, USD/JPY, EUR/JPY) with the implied cross computed from the first two and shown against the quoted third, plus a basis-point readout of the gap. The reader clicks a direction to walk one unit of currency around the triangle and the panel animates each conversion with the units written out at every step, so the multiply-or-divide decision is visible rather than memorised. A friction toggle applies a configurable bid-ask spread to each leg and shows the profit collapsing, so the reader discovers for themselves how wide the three legs have to be quoted before an apparent dislocation stops being a trade. A second tab overlays a 24-hour clock with the Sydney, Tokyo, London and New York sessions and a typical spread profile, so the reader can see where the cheap hours are.

INTERVIEW TRAP

(1) **Reading the pair backwards.** EUR/USD up means a stronger euro; USD/JPY up means a *weaker* yen. Say which currency is the base before you say anything directional. (2) **Multiplying when you should divide.** Write the units as fractions and cancel them. It takes three seconds and it never fails. (3) **Saying "the dollar went up" without a counterpart.** An exchange rate is a relative price. Against what? (4) **Thinking spot settles today.** Spot is T+2, T+1 for USD/CAD, and value dates skip holidays in both currency centres. (5) **Listing FX risks and omitting settlement risk.** The Herstatt story and CLS are what an experienced interviewer is waiting for. (6) **Quoting a level you are unsure of.** An FX desk will forgive "around 1.08, I have not seen a screen this morning". It will not forgive a confident wrong figure.

Where this goes next

The same trade done for a future value date, and the interest rate arithmetic behind it, is FX forwards and swaps, formalised as interest rate parity. Optionality on the pair, and the conventions that make FX options quoted differently from every other asset class, are FX options and delta conventions. The no-arbitrage logic that holds the triangle together is no-arbitrage pricing. And the wider place of the asset class in a bank sits in foreign exchange.

13.2 FX Forwards and Swaps

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An FX forward is an agreement to exchange two currencies at a fixed rate on a future date. An FX swap is two of them in opposite directions at different dates. Between them they account for the majority of turnover in the largest market in the world, and neither is a directional instrument, which is the first thing to understand about both.

Why the forward rate is what it is, as a no-arbitrage result, is interest rate parity, and the deviation from it is cross-currency basis. What is owned here is the construction, the quoting and the use.

Building a forward out of things that already exist

REAL INTERVIEW QUESTION (Société Générale · FX)

How is a currency forward constructed?

WHAT THEY TEST / ANSWER

Build it rather than quote a formula, because the construction *is* the proof and it survives any pair or tenor.

Suppose you want to buy euros against dollars in one year. You can achieve that today with three spot transactions:

- **Borrow dollars** for one year.
- **Buy euros spot** with those dollars.
- **Deposit the euros** for one year.

In a year you hold euros plus euro interest, and you owe dollars plus dollar interest. You have converted a known amount of dollars into a known amount of euros at a rate fixed today, without ever trading a forward. So the forward rate can only be that rate, or one of the three legs is mispriced.

Writing S as the spot in quote currency per unit of base, and r_q and r_b for the two money-market rates over the period τ:

$$F \;=\; S\,\frac{1+r_q\tau}{1+r_b\tau}.$$

The base currency's rate is in the denominator, which is the sign everyone gets wrong: **the higher-yielding currency trades at a forward discount**. It has to, because otherwise you could earn its higher interest and sell it forward at the same price, which is free money.

Then two points of craft. Use *money-market* rates and the correct day count for each currency, not swap rates, because the replication above is built from deposits. And note that this is the construction, not the market practice: nobody executes three legs, they trade the forward directly with a dealer who does exactly this internally, and the price they quote is *forward points* added to spot rather than an outright rate.

Forward points, report and déport

REAL INTERVIEW QUESTION (Société Générale · FX)

Define forward premium and forward discount points.

WHAT THEY TEST / ANSWER

They are the two directions of the same quantity, the **forward points**, which is the difference between the forward rate and the spot rate expressed in pips.

Report (a forward premium) means the forward is *above* spot, so the points are positive and you add them to spot. **Déport** (a forward discount) means the forward is below spot and you subtract them.

The direction is not a market opinion, it is arithmetic on the two interest rates. From $F = S(1 + r_q\tau)/(1 + r_b\tau)$, the base currency is at a **déport** whenever its own rate is the higher of the two, and at a **report** when it is the lower. Say it as a rule you can apply rather than a fact you recall: *the currency you earn more on is worth less forward*, because the market takes the interest advantage away from you in the forward rate.

Put a number on it. With EUR/USD spot at 1.1000, one-year dollar rates at 5% and euro rates at 3%:

$$F = 1.1000\,\frac{1.05}{1.03} = 1.121359,$$

so the forward points are +213.6 pips. The euro is at a report of 214 points against the dollar, because dollar rates are higher. The quick mental version, $S(r_q - r_b)\tau$, gives 220 pips, close enough to quote from and wrong by the compounding term, which is worth mentioning as the reason traders use the exact formula on long tenors.

Two practical notes worth adding. Points scale roughly with the tenor, so a three-month version of the same pair is near 55 pips, a quarter of the one-year. And the sign convention on a broker screen is that a **negative points quote is a déport**, which is why the two-sided quote sometimes reads with the larger number first; misreading that is a real dealing error rather than a theoretical one.

The FX swap

REAL INTERVIEW QUESTION (Société Générale · FX)

What is an FX swap?

WHAT THEY TEST / ANSWER

Two legs in opposite directions. You exchange two currencies at spot today and simultaneously agree the reverse exchange at a fixed rate on a future date. Both legs are agreed now, so there is no open currency position at any point.

The consequence to state immediately, because it is what the question is for: **an FX swap has essentially no FX risk**. If spot moves, both legs move together and offset. What you have transacted is not a view on the currency but an exchange of *funding* in two currencies for a period. The near leg gives you one currency and takes the other, and the far leg gives it back.

So what is it economically? **A secured loan.** You borrow one currency and lend the other, and each side's cash is the other side's collateral, which is why it is the same idea as repo with a currency in place of a bond. The price of the loan is the difference between the two legs, which is the forward points, which is the interest rate differential.

Then give the uses, which is where the volume comes from.

- **Funding**: a bank with euro deposits and dollar assets swaps euros into dollars for the term it needs.
- **Rolling a hedge**: an existing forward approaching maturity is extended by swapping it out to a new date, which is the single largest source of FX swap volume.
- **Liquidity management**: moving cash between currencies for days at a time without taking currency risk.

Close on the fact that surprises people and that follows from all of the above: FX swaps are the largest single instrument by turnover in the FX market, larger than spot. That is not because anyone has a view; it is because every hedge has to be rolled and every treasury has to fund itself in more than one currency.

REAL INTERVIEW QUESTION (BNP · FX)

How do you price an FX swap?

WHAT THEY TEST / ANSWER

You price the **points**, not the two rates, and saying so first shows you know how the instrument actually trades.

The mechanic. The near leg is done at spot, or at a spot reference agreed at the time. The far leg is that same spot plus the forward points. So the only thing being negotiated is the points, and the swap's value is entirely in them:

$$\text{points} \;=\; S\left(\frac{1+r_q\tau}{1+r_b\tau}-1\right) \;\approx\; S\,(r_q-r_b)\,\tau.$$

Using the numbers above, a one-year EUR/USD swap prices at about +214 pips. Note that spot itself almost cancels: it sets the scale of the points and nothing else, which is why the near leg's exact spot level is a convention rather than a negotiation.

The inputs, in the order they matter. The two money-market curves out to the tenor, with each currency's own day count. The **cross-currency basis**, which is the amount by which the market's actual points differ from the parity calculation, and which is not a rounding error: for many pairs it is the dominant term at longer tenors and it is its own subject. And the credit and collateral terms, since an uncollateralised swap with a weak counterparty prices differently from a cleared one.

What you are really quoting. Because the swap is a secured loan, the points are a funding price. So when dollar funding is scarce the points move against anyone borrowing dollars, regardless of where the two central banks have set their policy rates. Saying that is the answer to the usual follow-up about why the points do not match the rate differential: the mismatch is the price of dollar funding, and it widens in every stress episode.

If they push on a non-deliverable currency, add that the same structure exists as an NDF, which settles the difference in dollars against a fixing because the restricted currency cannot be delivered, as the asset class overview sets out.

INTERVIEW TRAP

1. **Getting the sign backwards.** The higher-yielding currency trades at a forward *discount*. Derive it from the replication rather than recalling it.

2. **Thinking an FX swap is a currency position.** Both legs are fixed at inception, so there is no open FX risk. It is a funding trade.

3. **Using swap rates instead of money-market rates.** The replication is built from deposits and their day counts.

4. **Ignoring the basis.** The parity calculation is a starting point; the traded points include a basis that is the dominant term at long tenors and in stress.

5. **Confusing an FX swap with a cross-currency swap.** An FX swap has two cash exchanges and no interim coupons; a cross-currency swap exchanges interest throughout its life.

6. **Assuming spot matters to the swap.** It scales the points and otherwise cancels, which is why the near-leg rate is a convention.

VISUAL / CODE TO BUILD: Interactive

A forward points calculator built as the replication rather than as a formula. The reader enters spot, two interest rates and a tenor, and the page animates the three legs of the construction as cash flows on a timeline, showing the borrowed currency, the spot conversion and the deposit maturing, with the implied forward appearing at the end as the ratio of the two terminal amounts. The points are then displayed in pips with the sign labelled report or déport, so the direction is attached to the rates rather than memorised. A second panel plots the points curve across tenors and overlays the market's actual quotes, with the gap shaded as the cross-currency basis, so the reader sees where parity stops being a sufficient description. A stress toggle raises the basis to its 2008 and March 2020 levels and re-prices a rolling hedge, which turns an abstract basis into a funding cost the reader can measure.

13.3 Interest rate parity

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Interest rate parity is the one result in FX that does not depend on a model, a forecast, or anybody's view. It says that the forward exchange rate is not a prediction of the future spot rate: it is the spot rate adjusted for the two interest rates, and it is pinned there by arbitrage. Everything else in FX is opinion. This is arithmetic.

It is also the most reliably asked FX interview question, in two forms: derive the forward, and then explain why the high-yielding currency is not free money.

Notation

Write the pair as EUR/USD, the base currency first and the quote currency second, so that the rate is the number of units of the quote currency per one unit of the base. If EUR/USD is 1.0800, one euro buys 1.08 dollars. Quoting conventions in full are in FX spot markets.

Let S be the spot rate, F the forward rate for maturity T, r_b the interest rate on the base currency and r_q the rate on the quote currency, both simple rates over the period.

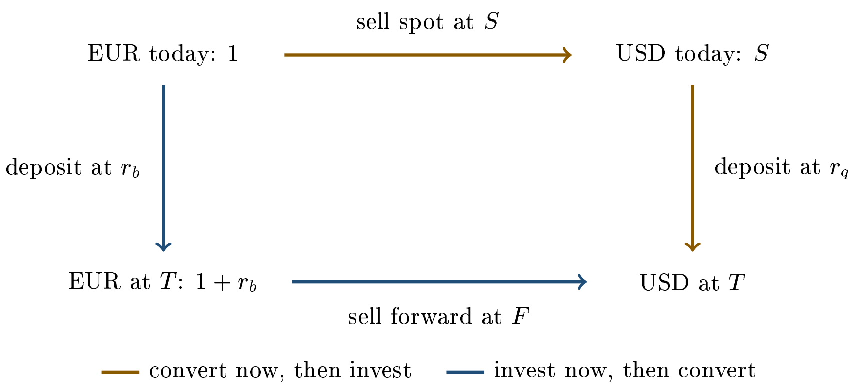
The derivation, which is just a square

You have one euro and you want dollars in one year. There are exactly two ways to get them, and both are risk-free today, so they must cost the same.

1. **Convert now, then invest.** Sell the euro at S for S dollars and deposit them at r_q. You end with $S(1 + r_q)$ dollars.
2. **Invest now, then convert.** Deposit the euro at r_b to get $1 + r_b$ euros, and sell those euros forward today at F. You end with $F(1 + r_b)$ dollars.

Both routes are locked in at the start and neither has any exposure to where spot finishes, so if they gave different amounts you could run one and reverse the other for a certain profit. Therefore

$$F(1 + r_b) = S(1 + r_q) \qquad \Longrightarrow \qquad \boxed{F = S\,\frac{1 + r_q}{1 + r_b}}$$



Both paths start at the same corner and end at the same corner, so they must arrive at the same number of dollars. That is the whole theorem.

> **KEY IDEA**
>
> The forward rate contains no forecast. It is spot compounded forward at the rate differential, and nothing else. A trader who tells you the forward is "the market's expectation of spot" has said something the arbitrage argument forbids.

Which currency is at a premium

Read the formula. If the quote-currency rate is higher than the base-currency rate then $F > S$: it takes more dollars to buy a euro forward than spot. The dollar, the high-interest currency, is **weaker forward**. That is the general rule, and it is not a coincidence.

INTUITION

Holding the high-interest currency pays you more interest, so if the forward did not take it back you would be paid twice for doing nothing. The forward market removes exactly the extra interest, no more and no less. The high-yielding currency is always expected to weaken in the forward market by precisely the amount of its yield advantage, which is another way of saying that the carry is not free.

WORKED EXAMPLE

Spot EUR/USD is 1.0800. The one-year dollar rate is 4.50% and the one-year euro rate is 2.50%.

$$F = 1.0800 \times \frac{1.045}{1.025} = 1.101073,$$

so the one-year forward is quoted at **1.1011**. The difference from spot, $1.101073 - 1.0800 = 0.021073$, is **210.7 forward points** (a point, or pip, being the fourth decimal). The euro is at a forward premium against the dollar because dollar rates are the higher of the two.

A useful desk shortcut is the linear version, $F \approx S\big(1 + (r_q - r_b)T\big)$, which here gives $1.0800 \times 1.02 = 1.1016$, or 216.0 points. That is 5.3 points too wide. Fine for a mental check on the sign and the order of magnitude, useless for a price, and the error grows with the rate differential and the maturity.

What happens if the forward is wrong

The relation is enforced, not hoped for. If a dealer quotes a forward away from parity, the trade that closes the gap is mechanical.

WORKED EXAMPLE

Same market as above, so the fair one-year forward is 1.1011. Suppose someone quotes it at **1.1050**: the euro is too expensive forward. Sell the euro forward and manufacture it synthetically.

Borrow 10 million dollars at 4.50%. Buy euros spot at 1.0800, giving $10\,\text{m}/1.0800 = 9\,259\,259$ euros. Deposit them at 2.50%, so at maturity you hold $9\,259\,259 \times 1.025 = 9\,490\,741$ euros. You had already sold that amount forward at 1.1050, which delivers

$$9\,490\,741 \times 1.1050 = 10\,487\,269 \text{ dollars.}$$

Repay the dollar loan, $10\,\text{m} \times 1.045 = 10\,450\,000$. What is left is

$$10\,487\,269 - 10\,450\,000 = 37\,269 \text{ dollars,}$$

locked in on day one, with no position in anything. Every leg was contracted today, so there is no market risk at all.

That is why the relation holds: this trade requires no view and no skill, only balance sheet, and it is done in size the moment the gap covers transaction costs.

The uncovered version, and why it fails

Covered parity is arithmetic. **Uncovered** interest rate parity is an economic hypothesis, and it says something much stronger: that the expected future spot rate equals the forward rate, so that a currency with a higher interest rate is expected to depreciate by exactly its yield advantage and no strategy in FX earns anything on average.

It is not arbitrage, because nothing is locked in. You would have to leave the position unhedged and take the spot risk. And empirically it does not hold. High-yielding currencies have historically depreciated by *less* than the forward implied, so borrowing the low-yielding currency and lending the high-yielding one has made money on average. This is the **forward premium puzzle**, and the trade is the **carry trade**.

> **INTERVIEW TRAP**
>
> Concluding from that record that carry is a free lunch. It is not. The returns are paid for in shape rather than in average: long periods of small steady gains from the rate differential, interrupted by sharp losses when positions unwind together and the funding currency rallies violently. The pattern is a short-volatility payoff written on FX, and it behaves like one in a crisis, which is exactly when the investor holding it needs it not to.

Interview angle

> **REAL INTERVIEW QUESTION** (Société Générale · Fixed Income)
>
> *Can you give me the derivation of an FX forward? Do it with EUR/USD.*
>
> **WHAT THEY TEST / ANSWER**
>
> This is a derivation question, so the marks are for the argument, not for the formula. Do not write the answer down and then justify it. Build it.
>
> State the setup in one line: EUR/USD at S dollars per euro, a euro rate r_b, a dollar rate r_q, horizon T. Then say the sentence that is the whole proof: "I have a euro today and I want dollars at T, and there are two risk-free ways to do it, so they must cost the same."
>
> Route one, convert then invest: sell the euro spot for S dollars, deposit at r_q, end with $S(1+r_q)$. Route two, invest then convert: deposit the euro at r_b, end with $1+r_b$ euros, and sell those forward today at F, giving $F(1+r_b)$ dollars. Nothing in either route depends on where spot ends up, because every leg is contracted today. Equating them gives $F = S(1+r_q)/(1+r_b)$.
>
> Then make it concrete, because the question asked for EUR/USD specifically. Spot 1.0800, dollars at 4.50%, euros at 2.50%, so $F = 1.0800 \times 1.045/1.025 = 1.1011$, and the euro is 211 points at a premium because dollar rates are higher.
>
> Close with the interpretation, which is the part that separates candidates: the forward is not a forecast. It is spot plus carry, enforced by the arbitrage above; if it were set by a forecast instead, it would drift off parity and the risk-free trade in the previous example would be sitting there for the taking. Expect the follow-up "so why does the forward move when the market changes its mind about the euro?" The answer is that it moves because spot and the two rate curves move, which is all it is made of.

REAL INTERVIEW QUESTION (Société Générale · Fixed Income)

The rate is EUR/USD. If US rates are positive and EUR rates are negative, will the forward rate be greater or smaller than 1?

WHAT THEY TEST / ANSWER

Answer precisely, because the question as posed contains a trap worth pointing out politely: the rate differential tells you where the forward sits relative to *spot*, not relative to 1. Whether the forward is above or below 1 depends on where spot happens to be trading, and that is a separate fact.

So give both. Relative to spot: EUR/USD is quoted as dollars per euro, so the dollar is the quote currency and $F = S(1 + r_{USD})/(1 + r_{EUR})$. With US rates positive and euro rates negative, the numerator exceeds the denominator, so $F > S$. The forward is above spot, the euro is at a forward premium, and equivalently the dollar, which is the higher-yielding currency, is weaker forward. Relative to 1: if spot is 1.08 then the forward is above 1.08 and therefore above 1; if the euro had been trading at 0.95 the forward would be above 0.95 and could easily still be below 1.

Then give the economic sentence, because that is what the question is really probing: the currency that pays the higher interest rate always trades at a forward discount, by exactly the amount of its yield advantage, otherwise you could earn the differential with no risk. Negative rates change nothing about the mechanism; they are just a smaller number in the denominator.

The follow-up is usually about signs under the other quoting convention. If asked about USD/EUR instead, the roles swap and the ratio inverts, so say which currency is the base before you answer anything.

REAL INTERVIEW QUESTION (BNP · FX)

You win one million euros on the lottery and want to invest risk-free, but the rate in Europe is 0%. You can invest in Turkey at 10% with an exchange rate of 1 for 1. Is that attractive? Describe the operation, given that you invest for one year and want euros at the end.

WHAT THEY TEST / ANSWER

The question is engineered so that the greedy answer is obviously wrong once you say the words "and I want euros at the end". Say those words first.

Describe the operation completely. Convert 1 million euros into 1 million lira at the spot rate of 1. Deposit at 10%, which gives 1.1 million lira in a year. Now the crucial step: you owe yourself euros, not lira, so to be risk-free you must sell the 1.1 million lira forward today. By covered interest rate parity the one-year forward is

$$F = 1 \times \frac{1.10}{1.00} = 1.10 \text{ lira per euro},$$

so the forward converts your 1.1 million lira into $1.1\,\mathrm{m}/1.10 = 1\,000\,000$ euros. Exactly the million you started with, and exactly what leaving it in euros at 0% would have given you. The 10% was never a yield; it was compensation the forward market takes straight back.

Then answer the actual question: no, it is not attractive, and it could not have been, because a genuinely risk-free 10% in a market with a 0% risk-free rate is an arbitrage, and the reason you cannot find one is that the forward closes it.

Finish by saying what you would be doing if you skipped the hedge, since that is the follow-up. Unhedged, you are no longer investing, you are taking a directional lira position, and the breakeven is the forward: you keep the 10% only if the lira falls by less than $1-1/1.10 = 9.1\%$ over the year. That is a view, it may be a good one, but it is not "investing risk-free" and calling it that is how people lose money in emerging market currencies.

REAL INTERVIEW QUESTION (BNP · FX)

Do you know the carry trade? Explain it, give an example, and say what your risk is.

WHAT THEY TEST / ANSWER

Three parts asked explicitly, so answer in three parts and do not skip the third, which is where the question is aimed.

What it is: borrow in a low-interest-rate currency, convert, and lend in a high-interest-rate currency, leaving the FX exposure unhedged. You earn the interest differential for as long as spot does not move against you. It is precisely the trade of the previous question with the forward hedge removed, which is the cleanest way to describe it, because it makes clear that the profit comes from declining to hedge rather than from any yield pickup.

Example: borrow yen, where rates have been near zero for decades, and buy a higher-yielding currency such as the Australian dollar, holding AUD/JPY long. The funding currencies have historically been the yen and the Swiss franc, and the target currencies the commodity and emerging-market ones. Uncovered interest rate parity says this should earn nothing on average, since the high-yielder should depreciate by the differential; historically it has not depreciated by that much, which is the forward premium puzzle and the reason the trade exists.

The risk, which is the answer they want: you are short the exchange rate, and the breakeven is the forward, so you lose as soon as the target currency weakens by more than the carry. Two features make it worse than that sounds. The distribution is skewed: small steady gains and rare very large losses, so the position looks safe on every measure that assumes normality right up until it is not. And it is a crowded trade funded in the same few currencies, so the unwind is correlated across everyone holding it, and the funding currency rallies hardest exactly when other risk assets are falling. Saying that carry is short volatility, and that it therefore fails when you most need diversification, is the sentence that finishes this question.

INTERVIEW TRAP

Four ways to get parity wrong on a whiteboard. (1) Inverting the ratio, which is what happens when you have not said out loud which currency is the base. Sanity-check the sign against the rule that the higher-yielding currency is weaker forward. (2) Using annual compounding on a three-month forward: money-market rates are simple and usually on ACT/360, so a 92-day forward uses a factor of $r \times 92/360$, and getting the day count wrong is worth more than most of the spread. (3) Using the wrong rates: the relevant ones are the rates at which you can actually borrow and lend the two currencies over that horizon, which since 2008 has not been the same thing as the published deposit curves. (4) Saying the forward predicts spot. It does not, and the sentence marks you out.

When parity itself breaks

Since 2008 covered interest rate parity has not held exactly, even between major currencies. The gap is small in normal conditions and widens at quarter-ends and in stress. It persists because the arbitrage in the example above is not actually free: it consumes balance sheet, capital and credit lines, and after the post-crisis regulatory changes those are scarce enough that a small deviation is no longer worth closing. The residual has a name and a market of its own, the cross-currency basis, and it is cross-currency basis.

Where this goes next

The instruments in which all of this is actually quoted and traded, forward points and FX swaps, are FX forwards and swaps. The breakdown of the parity relation is cross-currency basis. Which rates go into the formula in practice is OIS, LIBOR, SOFR and €STR. And what happens when a central bank tries to hold the spot rate against the differential is central banks and FX policy and pegged currencies and FX crises.

13.4 Cross-currency basis

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Covered interest parity says the forward exchange rate is pinned by the two interest rates, because any deviation is a riskless arbitrage. It is derived in interest rate parity and it is one of the cleanest no-arbitrage relations in finance.

It has not held since 2008. The gap is called the cross-currency basis, it is quoted, traded and hedged, and explaining why a textbook arbitrage persists for a decade is one of the better questions a rates or FX interviewer can ask.

Where the basis sits

The instrument that quotes it is the **cross-currency basis swap**: both parties exchange notionals at the start, pay each other floating interest in their respective currencies over the life, and re-exchange the notionals at the end at the original spot rate. In market convention the dollar leg pays its reference rate flat and the other leg pays its own reference rate *plus a spread*, and that spread is the **basis** b.

For most currencies against the dollar, b is **negative**. A negative basis means the non-dollar side accepts less than its own reference rate in order to obtain dollars, which is another way of saying **dollars are expensive to borrow synthetically** relative to what the two interest rates alone imply.

> WORKED EXAMPLE
>
> Spot EUR/USD at 1.10, one-year rates of 3% in euro and 5% in dollar. Parity gives a one-year forward of $1.10 \times 1.05/1.03 = 1.1214$.
>
> Now put a basis of -20 basis points on the euro leg. The euro side effectively earns 2.80% rather than 3% when swapping into dollars, so the forward implied by the traded market is $1.10 \times 1.05/1.028 = 1.1235$, about 19 basis points away from the parity value. The forward market is not mispriced; parity is simply not the price.
>
> The consequence for a real investor is direct. A euro-based fund buys a US corporate bond yielding 5% and hedges the currency back to euro. It gives up the rate differential of 2%, and it pays the basis of 0.20%, so the hedged yield is about 2.80% against 3% available at home. **The hedged pickup is negative**, and the entire reason is the basis. This calculation is run daily by every insurer and pension fund investing across currencies, and in some periods it has closed the US bond market to Japanese and European buyers entirely.

Why the arbitrage does not close

The trade that should close it is simple: borrow euro, swap into dollars through the basis swap, lend dollars, and pocket the difference. It does not close because the trade is not free.

- **Balance sheet is scarce.** The arbitrage requires borrowing and lending simultaneously, which inflates the balance sheet. Post-crisis leverage ratios charge for that regardless of how riskless the position is, so the arbitrageur needs a return on capital rather than a positive spread.
- **Structural dollar demand.** Non-US banks and corporates fund dollar assets without a dollar deposit base, and foreign investors in dollar bonds hedge back. Both are one-way flows into paying the basis, and there is no matching flow the other way.
- **Counterparty and funding costs.** The swap is long dated and creates exposure, so it carries the adjustments of CVA, DVA and FVA, which eat a spread of a few basis points before anyone makes money.
- **Reference rate mismatch.** The two legs pay different rates with different credit and liquidity content, so part of what looks like a basis is a comparison of things that were never identical, in OIS, LIBOR, SOFR and ESTR.

The observable fingerprint of the balance-sheet explanation is **turn-of-quarter and turn-of-year spikes**: the basis widens sharply over reporting dates and normalises immediately afterwards, which is not a story about interest rates at all.

REAL INTERVIEW QUESTION (Barclays · Rates)

Where is the one-year EUR/USD cross-currency basis? How does the basis move if all markets sell off? And the EUR/PLN basis?

WHAT THEY TEST / ANSWER

Three parts, and only the first needs a number you would check on the day.

Level. Say you would quote it from the screen, then give the shape rather than a stale figure: the EUR/USD basis has been negative for most of the post-crisis period, typically a modest number of basis points in calm markets, widening substantially in stress, and it is more negative at the short end around reporting dates. Negative means paying to obtain dollars.

In a sell-off, it becomes more negative, and you should give the mechanism. A risk-off move produces a scramble for dollar funding, since dollar liabilities are global; simultaneously the balance sheet available to intermediate the arbitrage shrinks, because the banks that would supply it are managing their own risk; and hedging demand rises as investors move to hedge foreign holdings. Demand up, supply down, so the price of synthetic dollars rises and the basis widens. The 2008 and March 2020 episodes are the standard examples, and both were addressed by central bank swap lines rather than by private arbitrage.

EUR/PLN, and this is the discriminating part of the question, does not behave like EUR/USD. It is an emerging-market cross with a much smaller and less liquid basis market, so the level is driven by local factors, the structural direction of local funding demand, and any regulatory or capital-control friction, and it can sit on either side of zero. It is also far less liquid and gaps more in stress, so a position in it is not a scaled version of the EUR/USD trade. Saying that the two are different animals, rather than quoting a number for both, is what the question is testing.

INTERVIEW TRAP

Four errors. (1) Treating covered interest parity as an identity: it is an arbitrage relation, and the arbitrage costs balance sheet, so it holds only approximately. (2) Quoting a hedged foreign yield as the rate differential alone, omitting the basis, which is the difference between a positive and a negative pickup. (3) Forgetting the notional exchange in a cross-currency swap, and so understating the counterparty exposure. (4) Assuming every currency's basis behaves like the dollar's: the dollar basis reflects global dollar demand, and a small-currency basis reflects local conditions.

Where this goes next

The relation this section documents the failure of is interest rate parity, and the instruments that quote it are FX forwards and swaps. The single-currency cousin is interest rate swaps, the reference rates whose differences muddy the measurement are OIS, LIBOR, SOFR and ESTR, and the official-sector backstop is central banks and FX policy.

13.5 FX Options

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Everything in options applies to FX options. This section is about the ways in which it does not, and there are more of them than people expect, all traceable to one fact.

KEY IDEA

An FX rate is a **ratio**, so every FX option is two options at once. A contract giving the right to buy euros with dollars is simultaneously a **EUR call** and a **USD put**: exercising it means acquiring one currency and delivering the other, and there is no way to say which of the two is "the underlying". Every peculiarity below follows from that. There is no equity analogue, because a share is not priced in another share.

REAL INTERVIEW QUESTION (Société Générale · FX)

What differentiates FX options from options on other underlyings?

WHAT THEY TEST / ANSWER

Five differences, all downstream of the ratio point, and giving them in that order shows they are one idea rather than a list.

One, every option is two options. A EUR call is a USD put. That means put-call parity is almost trivial here, and it means the same contract is described in two different ways by the two counterparties.

Two, there are two interest rates, not a rate and a dividend. The pricing model is Garman-Kohlhagen, which is Black-Scholes with the foreign rate playing exactly the role a dividend yield plays in equities:

$$F \;=\; S\, e^{(r_d - r_f)T},$$

so the forward, not the spot, is what the option is really written on. Both rates matter and each one has a rho.

Three, the notional exists in two currencies. A trade is €10 million against roughly \$11 million, and which of the two is the fixed notional changes the payoff slightly. Related, the **premium can be paid in either currency**, and paying it in the base currency changes the delta, because part of your exposure is the premium itself. That is what premium-adjusted delta is, and it is delta conventions.

Four, the market quotes volatility rather than price, and quotes it by delta rather than by strike. A 25-delta risk reversal is a standard quoted instrument; nothing equivalent exists in equities, where the strike is the coordinate. The surface is therefore expressed as an at-the-money level plus a risk reversal and a butterfly.

Five, the smile is roughly symmetric rather than skewed, for the structural reason that follows from the first point: a fall in EUR/USD is a rise in USD/EUR, so there is no privileged direction to be afraid of. The asymmetry that does exist is a genuine view rather than a permanent feature, which is why an FX risk reversal changes sign over time and an equity skew essentially never does. That contrast is drawn in equity skew, and the FX shape itself is the FX smile.

Close on the operational one if there is time: FX options are over-the-counter, twenty-four hours, with expiry times specified by city cut, so a "one-month option" has a precise expiry moment that is part of the contract rather than an exchange convention.

Moneyness runs backwards depending on which side you name

REAL INTERVIEW QUESTION (Société Générale · FX)

What is a dollar call / euro put that is OTM, ATM and ITM (with the thresholds inverted: if ATM is 1.07 then ITM is above 1.07 and OTM below 1.07)?

WHAT THEY TEST / ANSWER

The question contains its own answer and is testing whether you can explain *why* the thresholds invert, so lead with the reason.

A dollar call and euro put is the right to **sell euros and receive dollars** at the strike. Quoted in the market convention, EUR/USD is dollars per euro, so this option pays when euros are worth *less* than the strike.

Now hold the market rate fixed and vary the strike, which is what the question is doing. Being able to sell euros at a *higher* strike is more valuable, so:

- **ATM** at 1.07 is the strike at the current level (or at the forward, depending on which at-the-money convention is meant, and it is worth asking).
- **In the money**: strikes **above** 1.07, because you may sell euros above the market.
- **Out of the money**: strikes **below** 1.07.

That is the inversion. It is the exact mirror of a EUR call, where lower strikes are in the money, and the two are the same statement once you see where the flip comes from: naming the option by the other currency turns a call into a put, since a EUR call *is* a USD put, and calls and puts always run opposite ways in the strike. So a dollar call's moneyness has to invert relative to a euro call's, even though both are being read off the one EUR/USD quote.

Then volunteer the check that prevents the error in practice: **name the currency you are buying, and ask whether you would rather buy it cheaply or dearly**. You want to buy dollars cheaply, meaning you want euros expensive, meaning a high EUR/USD strike. Reasoning from the transaction rather than from the word "call" is what stops you inverting it under pressure, and interviewers on an FX desk ask this precisely because so many candidates invert it.

WORKED EXAMPLE

An FX option priced. EUR/USD spot 1.1000, one year, dollar rates 5%, euro rates 3%, volatility 8%.

The forward is $1.1000\, e^{(0.05-0.03)} = 1.1222$. An at-the-money-forward EUR call struck there is worth

$$e^{-r_d T}\big[F\, N(d_1) - K\, N(d_2)\big] = 0.03406$$

dollars per euro, quoted as **340.6 USD pips**, or about **3.10%** of the euro notional. On a €10 million trade that is $340,600 of premium.

Two conventions visible in that one number. The premium was quoted in *dollars per euro of notional*, which is one of four possible conventions, and the strike was the *forward* rather than the spot, which is the usual meaning of "at the money" on an FX desk and not the usual meaning on an equity desk. Getting either wrong misprices the trade by more than the spread.

Using them

REAL INTERVIEW QUESTION (Union Capital · FX)

If I buy a EUR call, what is my risk?

WHAT THEY TEST / ANSWER

Deceptively simple, and the good answer names three different risks because the question does not say which one it wants.

The obvious one: your maximum loss is the premium. You have bought an option, so there is no obligation and no margin call. Say that first and briefly, because it is the answer they expect.

The one they are testing: which currency is that premium in, and which currency is your *exposure* in? A EUR call is a USD put, so you are long euros and short dollars, and your profit and loss is naturally measured in dollars. If your reporting currency is euros, then even the premium itself carries currency risk, and if you paid the premium in euros the option's delta is not the textbook delta because part of your euro exposure is the premium you already spent. That is premium-adjusted delta.

The ones that are not about direction at all, and volunteering them is what marks the answer. You are **long volatility**, so you lose if implied volatility falls even with spot unchanged. You are **short theta**, so you bleed with time. And you have exposure to **both** interest rates through the forward, since a change in either curve moves the forward the option is written on, which on a long-dated FX option is a material rho rather than a footnote.

Close on the practical framing a salesperson would use: buying a EUR call converts an unlimited downside into a known cost, and the honest question is not whether the risk is capped but whether the premium is worth paying relative to simply hedging with a forward, which costs nothing upfront and gives up the upside.

REAL INTERVIEW QUESTION (CACIB · FX)

How do I hedge when I sell dollars (i.e. sell dollar so buy euro, buy EUR call / USD put)?

WHAT THEY TEST / ANSWER

Establish the exposure before choosing an instrument, because the question has already done half of it and is checking that you follow.

Selling dollars to buy euros means you will need euros in the future, so your risk is that **euros become more expensive**, meaning EUR/USD rises. The hedge must gain when EUR/USD rises: a **EUR call, which is a USD put**, exactly as the question states, or a long forward.

Then give the choice properly, since that is the actual content.

- **Forward**: costs nothing upfront and fixes the rate completely. You are protected and you also give up any favourable move. For a known cash flow on a known date, this is usually the right answer and the option needs to justify itself against it.
- **Bought EUR call**: protects the downside and keeps the upside, at the cost of a premium paid today. Right when the cash flow is *uncertain*, because a forward on a flow that does not materialise leaves you with an unwanted position, whereas an unexercised option leaves you with nothing.
- **Collar or risk reversal**: buy the EUR call and sell a EUR put below, so the premium nets toward zero. You keep protection above and give up favourable moves below a level. This is what corporates actually do, because the treasurer's constraint is often zero premium rather than optimal economics.
- **Participating forward or knock-out**: cheaper still, and cheaper because the protection is conditional. Worth naming, and worth naming the condition being sold.

Close on the two questions that decide it, which is the answer a sales desk wants: **is the exposure certain**, which chooses between forward and option, and **what is the budget**, which chooses between paying premium and selling something to fund it. Everything else is implementation.

INTERVIEW TRAP

1. **Saying "call" without naming the currency.** A call on what? Every FX option is a call on one currency and a put on the other, and the ambiguity is a real dealing error.
2. **Inverting the moneyness.** Reason from the transaction, not the word: which currency are you buying, and would you rather buy it cheaply or dearly?
3. **Treating the foreign rate as a detail.** It is the dividend yield of this market, it sits in the forward, and both rates carry a rho.
4. **Assuming at-the-money means spot.** On an FX desk it usually means the forward, or the delta-neutral straddle strike, and the three differ.
5. **Ignoring the premium currency.** It determines the delta, and a premium-adjusted delta is not the Black-Scholes delta.
6. **Expecting an equity-style skew.** The FX smile is roughly symmetric for a structural reason, and its tilt changes sign over time.

VISUAL / CODE TO BUILD: Interactive

A two-sided option viewer. Every contract is displayed simultaneously in both descriptions, EUR call and USD put, with the payoff drawn twice, once in EUR/USD and once in the inverted quote, so the reader watches the same position become a call and a put by changing the axis rather than the trade. Dragging the strike moves both pictures at once and the moneyness labels flip in opposite directions, which is the inversion made mechanical instead of memorised. A pricing panel implements Garman-Kohlhagen with separate sliders for the two interest rates, showing the forward moving and the premium following, and a selector cycles the premium through all four quoting conventions on the same trade so the reader sees four different numbers describing one price. A hedging tab takes a stated future cash flow and compares forward, bought call, collar and knock-out on one payoff chart with their premiums listed, which is the conversation the CACIB question is really about.

13.6 FX volatility smile

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What a volatility smile *is*, and why implied volatility varies by strike at all, is built in the volatility smile and skew. This section does not repeat that. It answers a narrower and more interesting question: why the FX smile has a different *shape* from the equity one, and why FX is the only asset class that quotes its smile as a set of instruments rather than as a curve over strikes.

Both answers come from the same source, and it is worth stating up front because it organises everything else. **A currency pair is a ratio, and which currency sits on top is a convention.** An equity can go to zero and cannot go to minus one; there is an asymmetry built into the object. EUR/USD has no such preferred direction, because EUR/USD falling *is* USD/EUR rising, and neither is more natural than the other.

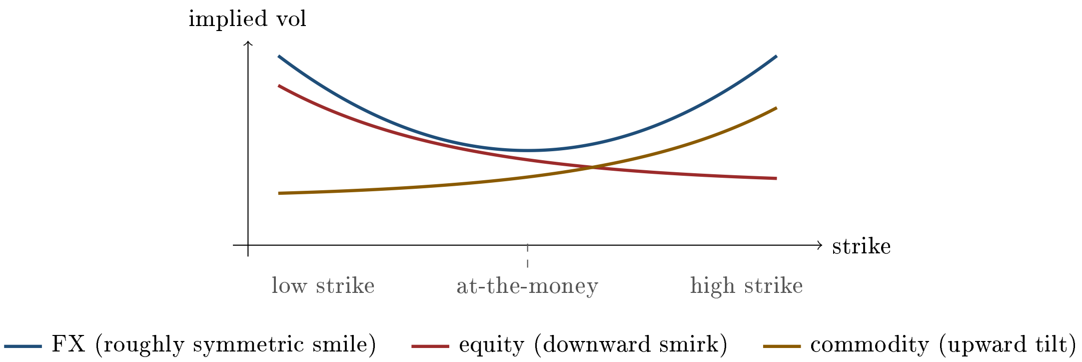
Why the FX smile is a smile and the equity smile is a smirk

> **INTUITION**
>
> Ask an equity investor what keeps them awake and they say a crash. Ask them what a 30% rally would do and they shrug. That asymmetry of fear is why equity implied volatility rises as you go down in strike and falls as you go up: a one-sided slope, a smirk.
>
> Now ask a euro-dollar trader the same question. "A crash in what?" A big move down in EUR/USD is a big move up in USD/EUR, and there is a bank somewhere with the mirror position who is equally frightened of it. Both tails are somebody's disaster, so both wings of the FX smile lift. What you get is a genuine smile: a roughly symmetric curve, tilted rather than sloped.

Make that precise. Write the pair as S and the inverted pair as $1/S$. A call on S struck at K is, exactly, a put on $1/S$ struck at $1/K$, rescaled. Since both quotes describe the same market, no-arbitrage forces their volatility surfaces to be images of each other under inversion. There is no consistent way for the FX market to price one wing systematically higher than the other *for reasons of direction alone*, in the way equity markets do. So the leading term of the FX smile is curvature (both wings up together), and asymmetry is a second-order tilt on top of it.

The equity smirk, by contrast, has three drivers that FX simply does not have: the leverage effect, since a falling share price mechanically raises a firm's debt-to-equity ratio and its equity volatility; persistent one-way demand for downside protection from institutions that are structurally long the market; and the empirical fact that equity returns are negatively skewed, with crashes faster than rallies. The asset-specific equity shape belongs to equity skew and smile.

Commodities are the third shape and are often the reverse of equities: for electricity, gas, or a crop facing a bad harvest, the frightening direction is *up*, since a supply shock spikes the price, and consumers who need the physical good bid for calls. So the smile tilts upward in strike. The asset class is covered in commodities.



REAL INTERVIEW QUESTION (MUREX · Volatility)

Draw the volatility smile for an equity, a currency and a commodity.

WHAT THEY TEST / ANSWER

Draw three axes and three shapes, and then say the one sentence that makes them a system rather than three memorised pictures: *the smile shows you which direction the market is afraid of.* Equity is a downward smirk, highest at low strikes, because the fear is a crash and because institutions are structural buyers of puts. Currency is a roughly symmetric smile with both wings up, because a currency pair is a ratio and each tail is somebody's crash, so there is no privileged direction. Commodity is often the mirror of equity, tilting up in strike, because for an energy or agricultural consumer the disaster is a supply shock and a price spike, so the demand is for calls. Two additions land the answer. First, name the exception rather than pretending the FX rule is universal: an emerging-market pair against the dollar is asymmetric, because the local currency can devalue violently and cannot appreciate violently, so it behaves much more like an equity. Second, note that all three curves get flatter as maturity extends and that most of the interesting curvature lives in the short end.

REAL INTERVIEW QUESTION (BNP · FX)

Why does the volatility smile have this shape?

WHAT THEY TEST / ANSWER

Answer in two layers, the model layer and the market layer, because a candidate who gives only one sounds like they learned it from one book. The **model** layer: Black-Scholes assumes lognormal returns with constant volatility, so a flat surface, and every deviation of the real return distribution from that assumption shows up as a strike-dependent implied volatility. Real distributions have fatter tails than lognormal, which lifts both wings, and volatility is itself stochastic and correlated with the spot, which tilts the curve. That is precisely what stochastic-volatility models such as SABR formalise: the volatility of volatility controls the curvature, and the spot-volatility correlation controls the tilt. The **market** layer: the smile is also a supply-and-demand object. Who is structurally long and short each wing determines where the price settles, which is why equity puts are permanently bid and why the shape survives even when a model says it should not. Then give the FX-specific point they are fishing for on an FX desk: because a pair is a ratio, the smile must be consistent under inverting the pair, so FX gets curvature first and asymmetry second, and the asymmetry is quoted separately as the risk reversal.

The FX quoting convention: by delta, and as three numbers

Every other asset class publishes a volatility surface against strike. FX does not. It quotes a small number of *instruments*, at fixed *deltas*, and the surface is reconstructed from them.

There are two good reasons. Deltas are comparable across pairs and across maturities in a way strikes never are, since a 25-delta option is roughly the same distance out of the money whether the pair is at 1.08 or at 150, and it stays that distance as spot moves. And the three quoted instruments map exactly onto the three things a trader wants to control: the level, the tilt and the curvature.

KEY IDEA

The FX smile is quoted as three numbers per maturity:

- σ_{ATM}, the at-the-money volatility, which sets the **level**;
- $\text{RR}_{25} = \sigma(25\Delta \text{ call}) - \sigma(25\Delta \text{ put})$, the 25-delta risk reversal, which sets the **tilt**;
- $\text{BF}_{25} = \frac{1}{2}\big(\sigma(25\Delta \text{ call}) + \sigma(25\Delta \text{ put})\big) - \sigma_{\text{ATM}}$, the 25-delta butterfly, which sets the **curvature**.

Inverting those two definitions gives the wings back:

$$\sigma(25\Delta \text{ call}) = \sigma_{\text{ATM}} + \text{BF}_{25} + \tfrac{1}{2}\text{RR}_{25}, \qquad \sigma(25\Delta \text{ put}) = \sigma_{\text{ATM}} + \text{BF}_{25} - \tfrac{1}{2}\text{RR}_{25}.$$

Notice the structure: the butterfly is the *symmetric* part and the risk reversal is the *anti-symmetric* part. That decomposition is exactly the inversion symmetry established at the top of this section, made operational.

WORKED EXAMPLE

Rebuilding the wings. A EUR/USD 3-month surface is quoted as $\sigma_{\text{ATM}} = 8.00\%$, $\text{RR}_{25} = -0.50\%$ and $\text{BF}_{25} = 0.25\%$. Then

$$\sigma(25\Delta \text{ call}) = 8.00 + 0.25 - 0.25 = 8.00\%, \qquad \sigma(25\Delta \text{ put}) = 8.00 + 0.25 + 0.25 = 8.50\%.$$

Check both definitions: the difference is $8.00 - 8.50 = -0.50\%$, matching the risk reversal, and the average of the wings is 8.25%, which is 0.25% above the at-the-money, matching the butterfly.

Read the signs. A *negative* risk reversal means the 25-delta put is more expensive in volatility terms than the 25-delta call, so the market is paying up for protection against the base currency falling. Here the base is EUR, so the market is bid for euro downside. A *positive* butterfly, which is the normal state, simply means the wings are above the at-the-money, which is the smile being a smile.

Two cautions before you use these numbers on a desk. The quoted butterfly is conventionally a *market strangle* rather than the smile-consistent “smile strangle”, and the two differ once the smile is steep, so a production system calibrates rather than applying the formula above literally. And 25Δ is itself ambiguous until you say which delta convention is meant, since FX supports spot and forward deltas, premium-adjusted or not, with the convention varying by pair and by maturity. That is the subject of delta conventions, and the two instruments themselves are developed in risk reversals and butterflies. Treat the arithmetic above as the first-order reconstruction that gets you through an interview and gets a trader in the right ballpark, not as a pricing library.

REAL INTERVIEW QUESTION (CACIB · FX)

What does the volatility smile look like on a currency pair, and how do you interpret it?

WHAT THEY TEST / ANSWER

Give the shape, then the parameterisation, then the reading, in that order. The shape: roughly symmetric, both wings above the at-the-money, because a pair is a ratio and there is no privileged crash direction, unlike equities. The parameterisation: FX does not publish a curve over strikes, it quotes at-the-money volatility for the level, the 25-delta risk reversal for the tilt and the 25-delta butterfly for the curvature, all at fixed deltas rather than fixed strikes so the quotes stay comparable across pairs and across time. The reading, which is what "interpret" means and where most candidates stop too early: the risk reversal is a directional sentiment indicator, telling you which side of the pair the market is paying up to protect, and it is one of the cleanest positioning signals in markets because it is a price rather than a survey. The butterfly is a tail indicator: it widens when the market fears a large move without knowing the direction, which is exactly the state before a referendum or a central bank meeting with a genuinely uncertain outcome. Finish with the exception, that on emerging-market pairs against the dollar the smile is strongly asymmetric because devaluation risk is one-sided, so the symmetry argument is a property of two comparable free-floating currencies and not a law.

When the symmetry breaks

The inversion argument says the FX smile cannot be one-sided *for reasons of direction alone.* It says nothing about the case where the two currencies are genuinely different objects, and that case is common.

- **Emerging market against the dollar.** A currency that can devalue by 30% overnight and cannot appreciate by 30% overnight is asymmetric in substance. The smile for such a pair looks much more like an equity smirk, with the wing that pays out on local-currency weakness heavily bid.
- **Safe havens.** USD/JPY and USD/CHF carry a persistent risk reversal toward yen and franc strength, because those currencies rally in global stress. The tilt is the market pricing a correlation with risk appetite, not a view on Japan.
- **Carry pairs.** A high-yielding currency bought against a low-yielding one grinds up slowly and falls violently when the trade unwinds, so the smile tilts toward protection on the high-yielder, which is the option market pricing the skewness of the carry trade's return.
- **Pegs and managed regimes.** This is the extreme case. A credible peg suppresses at-the-money volatility to almost nothing while the wing that pays if the peg breaks stays expensive, because the peg can only fail in one direction. The resulting smile is grotesque by free-float standards and is the single best illustration that an implied volatility is a price, not a forecast. See pegged currencies and FX crises.

Term structure, and where the shape lives

The smile is not one curve but a surface, and its behaviour across maturity is as informative as its behaviour across strike. Curvature is concentrated in the short end: a one-week butterfly is far larger than a one-year butterfly, because a single jump matters enormously over a week and averages out over a year. As maturity extends, the smile flattens toward the at-the-money level.

The term structure also carries **event risk** explicitly. FX options spanning a central bank meeting, an election or a referendum price a discrete jump, so the at-the-money term structure develops a visible kink at the event date and the butterfly for maturities just after the event jumps. Traders read the difference between the maturity just before and just after the event as the market's price of the event itself, which is one of the few genuinely clean event-risk readings available anywhere.

REAL INTERVIEW QUESTION (BNP · Exo)

What is the level of 3-month at-the-money EUR/USD volatility, and what does the smile look like?

WHAT THEY TEST / ANSWER

A level question and a structure question stapled together, and both are graded. For the level, know it on the morning of the interview and quote it with its units: FX volatility is quoted in annualised percentage points, EUR/USD is one of the least volatile major pairs, and its 3-month at-the-money volatility has historically spent most of its time in the mid single digits to low teens, spiking in crises. Say the number you actually saw, say when you saw it, and if you have not seen a screen say so and give the range instead. Never invent a decimal. Then the structure half, which is where the marks are: describe it as three quotes rather than a curve, at-the-money plus a 25-delta risk reversal plus a 25-delta butterfly, note that EUR/USD is close to the symmetric case so the butterfly is modestly positive and the risk reversal is small and flips sign with the macro regime, and reconstruct the wings out loud with the formula if they want a number. If you can add the current *driver* of the tilt, whether the market is paying for euro downside or dollar downside and why, you have turned a memory question into a market conversation, which is the actual point of asking it.

VISUAL / CODE TO BUILD: Interactive

A three-slider smile builder. The reader sets σ_{ATM}, RR_{25} and BF_{25} and the panel draws the reconstructed smile against delta, with the 25Δ put, at-the-money and 25Δ call points marked and their volatilities printed. A second panel shows the same surface against strike, so the reader can watch the delta-space picture stay put while the strike-space picture slides as they move spot, which is the clearest possible demonstration of why FX quotes in deltas. A preset menu loads characteristic shapes: EUR/USD (near-symmetric), USD/JPY (safe-haven tilt), an emerging-market pair (equity-like smirk) and a pegged pair (suppressed at-the-money with one fat wing). A third tab plots the term structure of the three parameters and lets the reader drop a central bank meeting onto the calendar to watch the kink appear.

INTERVIEW TRAP

(1) **Importing the equity skew story into FX.** The leverage effect and crash-o-phobia are equity arguments. Quoting them on an FX desk signals that you have only ever read about equities. (2) **Saying the FX smile is symmetric, full stop.** It is symmetric for two comparable free floats and strongly asymmetric for emerging markets, safe havens, carry pairs and pegs. State the rule with its exceptions. (3) **Confusing the sign of the risk reversal.** It is the call volatility minus the put volatility, on the *base* currency. Say which currency the base is before you interpret the sign. (4) **Quoting a delta without a convention.** Spot or forward, premium-adjusted or not: in FX these are different options, and the difference is not negligible in high-yielding currencies. (5) **Applying the butterfly formula as if it were exact.** The quoted butterfly is a market strangle by convention, so the reconstruction is a first-order approximation and a real system calibrates. (6) **Reading implied volatility as a forecast.** A pegged currency's wing is the cleanest counterexample there is: it prices a scenario, not an expectation.

Where this goes next

The two instruments the surface is quoted in are risk reversals and butterflies, and the ambiguity in the word "25-delta" is resolved in delta conventions. The general theory of why a smile exists at all is the volatility smile and skew, the models that fit it are local volatility and stochastic volatility, of which SABR is the FX and rates workhorse. The Greeks that measure your exposure to the shape itself are in skew and smile sensitivities.

13.7 Delta conventions

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Every other option market quotes by strike. FX quotes by **delta**, and it means several different things by that word depending on the currency pair. This looks like trivia until you realise that two desks agreeing on "25 delta" while using different conventions are agreeing on different strikes, and that the gap can be worth real money on a large ticket. This is the section that stops that happening.

Why FX quotes in delta at all

An equity option market has a fixed strike ladder and a spot that moves slowly relative to it. A currency pair has no natural ladder, and a broker's quote has to remain meaningful as spot moves through the day and across years.

Quoting by delta solves that. A **25-delta call** is a specific distance out of the money *measured in the market's own terms*, since delta already contains the volatility and the maturity. It is the same object today and next year, on EUR/USD and on USD/JPY, which is what allows one surface to be compared with another. The surface is therefore stored as a level, a risk reversal and a butterfly at fixed deltas, in risk reversals and butterflies, and strikes are computed from the deltas rather than the other way round.

The four deltas

Two independent choices give four conventions, and a pair's market convention specifies which one is in use.

Choice one: spot or forward. The **spot delta** is the sensitivity to the spot rate, so it is the amount of spot to trade to hedge, and for a call it is $e^{-r_f T}N(d_1)$. The **forward delta** is the sensitivity to the forward, so it is the amount of *forward* to trade, and it is $N(d_1)$ without the discount factor. The two differ by exactly the foreign discount factor, multiplicatively rather than by subtraction, so the gap matters for long-dated options and hardly at all for a one-week trade. Spot delta is conventional at short tenors and forward delta from around one year outward, since beyond that the foreign discount factor is too large to leave inside a hedge ratio; many emerging market pairs use forward delta at every tenor.

Choice two: premium adjusted or not. This one is specific to FX and it is the one interviewers use to separate candidates. In FX the option premium is itself an amount of currency, and depending on the pair's convention it is paid in the *base* currency rather than the quote currency. If you pay the premium in the same currency you are exposed to, then the premium is part of your currency position, and the correct hedge is smaller by the premium. Formally, for a call,

$$\Delta_{\text{pa}} \;=\; \Delta - \frac{V}{S} \;=\; e^{-r_d T}\,\frac{K}{S}\,N(d_2),$$

where V is the premium in quote-currency terms. The adjustment subtracts the premium from the delta *whatever the option is*, so it moves delta **down algebraically** rather than towards zero: it shrinks a call's delta and *enlarges* a put's. At the parameters below the call goes from $+0.5345$ to $+0.4556$ while the put goes from -0.4556 to -0.5345. The put version of the closed form is $\Delta_{\text{pa}} = -e^{-r_d T}\,\frac{K}{S}\,N(-d_2)$. The gap in both cases is the premium expressed in the base currency, V/S, which equals V only when spot is 1.0000 as it happens to be in the example.

WORKED EXAMPLE

A one-year at-the-money-forward EUR call, spot 1.0000, domestic rate 4%, foreign rate 1%, volatility 20%, **with the premium paid in EUR, the base currency**, so the premium-adjusted convention applies. The forward is 1.03045 and the strike is set there. Read the pair as a constructed one: real EURUSD pays premium in USD and is *not* adjusted, as the next paragraph sets out.

The premium currency is not a detail you may assume. The rule is that you adjust when the premium is paid in the **base** currency, because only then is the premium itself part of the exposure being hedged. In practice EURUSD, GBPUSD and AUDUSD pay premium in USD, the *quote* currency, so they are **not** premium adjusted; USDJPY, USDCHF and most pairs with USD as the base, including the bulk of emerging market crosses, pay premium in USD as the *base* currency and **are** adjusted. Confirm it per pair rather than reasoning from the majority.

Convention	Delta
Spot delta	0.5345
Premium-adjusted spot delta	0.4556

The premium is 0.0789, or 7.9% of spot, and the two deltas differ by exactly that: $0.5345 - 0.0789 = 0.4556$. **Eight delta points**, on a plain vanilla option, purely from a quoting convention. On a 100 million ticket that is 8 million of spot hedge either present or missing, which is why this is not a detail.

REAL INTERVIEW QUESTION (Société Générale · FX)

What distinguishes FX options from options on other underlyings?

WHAT THEY TEST / ANSWER

Five things, and the conventions are the ones that surprise people, so lead with the structural point and land on them.

Every FX option is two options, since a EUR call is a USD put on the same contract, so put-call terminology means nothing until you say which currency you are calling. That, and the Garman-Kohlhagen pricing in which the foreign rate plays the role of a continuous dividend yield, are set out in FX options; the forward they imply is interest rate parity.

Symmetry, and a smile rather than a skew. An equity market has a natural direction, down is bad, so its surface is a downward-sloping skew. A currency pair has no natural direction, since one side's fall is the other's rise, so the surface is more nearly symmetric and closer to a genuine smile, developed in the FX volatility smile.

It is quoted in volatility and in delta, not in price and strike: the market trades an at-the-money volatility, a risk reversal and a butterfly at fixed deltas, and the strikes are derived.

And the conventions bite, which is the closing point. The premium can be paid in either currency, which changes what delta means; the at-the-money strike can be the forward or the delta-neutral straddle strike; and delta can be spot or forward based. Naming those and saying that you would confirm the pair's convention before quoting is what an FX interviewer is listening for.

Which strike is "at the money"

Even the middle of the surface is a convention. Three candidates are used.

- **ATM spot**: strike equal to the spot rate. Intuitive, and almost never the market convention, because it makes the at-the-money point depend on the interest rate differential.
- **ATM forward**: strike equal to the forward. This is the natural centre, since it is the strike at which a call and a put have the same value, by put-call parity.
- **Delta-neutral straddle**: the strike at which a straddle has zero delta, which is the most common convention in interbank FX. It is not the forward: for a non-premium-adjusted spot delta it sits at $K = F\,e^{\sigma^2 T/2}$, so on the example above it is 2.0% above the forward, and a straddle struck at the forward has a delta of $+0.079$ rather than zero.

REAL INTERVIEW QUESTION (Société Générale · Structured Products)

What is the delta of an at-the-money spot call, and what is it for an at-the-money forward call?

WHAT THEY TEST / ANSWER

Both are “about a half”, and the whole point is why they are not the same half.

At-the-money forward, meaning struck at the forward, the delta is $e^{-r_f T} N\left(\frac{1}{2}\sigma\sqrt{T}\right)$: just above 0.5 before discounting, because d_1 is positive by half a volatility unit. With one year, 20% volatility and a 1% foreign rate that is 0.534.

At-the-money spot, meaning struck at the spot, the delta also picks up the rate differential, since the forward sits above the spot when domestic rates are higher, so the option is already in the money relative to the forward. With a 3% differential the delta rises to 0.593. So the two differ by six delta points at these parameters, and the gap widens with the rate differential and with maturity.

Then the two refinements that make it an FX answer rather than a generic one. The strike at which the delta is exactly 0.5 is neither of these; and the strike at which a *straddle* is delta neutral is $Fe^{\sigma^2 T/2}$, above the forward, which is the interbank at-the-money convention. And in a premium-adjusted pair every one of these numbers shrinks by the premium, which here is nearly 8 delta points. The right answer therefore ends by asking which convention is meant, because “the ATM delta” is not a number until that is settled.

Which currency you are paid in

REAL INTERVIEW QUESTION (CACIB · FX)

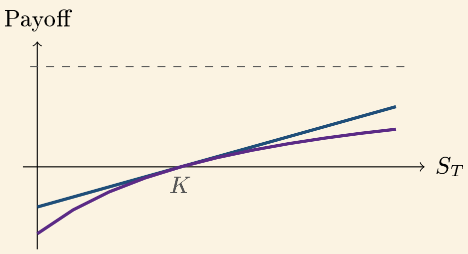
How does the currency used to pay an FX forward contract impact the evolution of the payoff? (make a payoff graph)

WHAT THEY TEST / ANSWER

It changes the payoff from linear to curved, and drawing that is the answer.

Take a forward to buy euros at K, with spot S expressed as dollars per euro. Settled in **dollars**, the payoff per euro of notional is $S_T - K$: a straight line, the familiar picture, and every dollar of gain is worth a dollar.

Settled in **euros**, you receive the same economic gain but converted at the prevailing rate, so the payoff is $(S_T - K)/S_T$ euros. That is *not* linear. It rises but flattens as S_T increases, approaching a horizontal asymptote at 1 euro per euro of notional however far the rate goes, and it steepens on the downside, since losses are converted at an ever weaker rate. Drawn against S_T, it is a concave curve rather than a line.



settled in dollars, $S_T - K$, linear — settled in euros, $1 - K/S_T$, concave — asymptote at 1 euro

The two curves are tangent at $S_T = K$ and separate in both directions, which is the whole content of the picture: the euro-settled payoff gives up gains on the upside and takes larger losses on the downside than the dollar-settled one.

The consequence is the point being tested: **a position that is linear in one currency is non-linear in the other**, so it has convexity, and convexity has to be priced and hedged. This is the same effect that makes premium-adjusted delta necessary, since a premium paid in the currency you are exposed to is itself part of the exposure; it is the reason a quanto product, which fixes the conversion rate, is a genuinely different instrument with correlation risk; and it is why the first question on any FX trade is which currency the notional and the settlement are expressed in. Answer that before drawing anything, and say that a payoff graph is only defined once the vertical axis has a currency on it.

INTERVIEW TRAP

Four errors. (1) Quoting a delta without its convention, when four exist and they differ by several delta points. (2) Assuming the at-the-money strike is the spot, or even the forward, when interbank convention is usually the delta-neutral straddle. (3) Forgetting that the premium adjustment moves delta *down*, not towards zero, so the unadjusted number over-hedges a call and under-hedges a put. (4) Drawing an FX payoff without stating the settlement currency, since the same trade is a line in one and a curve in the other.

Where this goes next

The instruments quoted at these deltas are risk reversals and butterflies, and the surface they build is the FX volatility smile. The forward the strikes are measured against is FX forwards and swaps, and the hedging activity that gives delta its meaning is delta neutrality and dynamic hedging.

13.8 Risk reversals

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The risk reversal is the instrument FX uses to say one thing: **which side of the market is more afraid**. It is simultaneously a structure a client trades and a number the market quotes, and almost every confusion about it comes from not noticing which of the two is being discussed.

As a **structure** it is long an out-of-the-money call and short an out-of-the-money put, or the reverse, on the same pair and the same expiry, conventionally struck at 25 delta on each side. As a **quote** it is the difference between those two implied volatilities, expressed in volatility points. The structure is what you trade; the quote is what the market charges for the asymmetry.

The three numbers that build an FX surface

FX volatility is not quoted by strike. It is quoted as three numbers per expiry, and this is the convention that makes FX different from equity derivatives.

- **ATM**, the at-the-money volatility, which sets the *level* of the smile.
- **Risk reversal**, $\sigma_{25\Delta\text{call}} - \sigma_{25\Delta\text{put}}$, which sets its *tilt*.
- **Butterfly**, the average of the two wings less the ATM, which sets its *curvature*. That one is butterflies.

Level, tilt and curvature. Three numbers, and the wings follow:

$$\sigma_{25\Delta\text{call}} = \text{ATM} + \text{BF} + \tfrac{1}{2}\text{RR}, \qquad \sigma_{25\Delta\text{put}} = \text{ATM} + \text{BF} - \tfrac{1}{2}\text{RR}.$$

WORKED EXAMPLE

Take a EURUSD expiry quoted at ATM = 8.0, 25 delta butterfly = 0.25, and 25 delta risk reversal = −0.60.

$$\sigma_{\text{call}} = 8.0 + 0.25 + \tfrac{1}{2}(-0.60) = 8.25 - 0.30 = \mathbf{7.95}$$

$$\sigma_{\text{put}} = 8.0 + 0.25 - \tfrac{1}{2}(-0.60) = 8.25 + 0.30 = \mathbf{8.55}$$

Both identities check out: the difference is $7.95 - 8.55 = -0.60$, the risk reversal; and the average less the ATM is $8.25 - 8.0 = 0.25$, the butterfly.

Read what that says. The EUR put is trading 0.60 volatility points above the EUR call, so the market is paying more, in volatility terms, to protect against the euro falling than against it rising. **The risk reversal is the market's directional fear, priced.**

KEY IDEA

Sign conventions in FX cause more errors than the mathematics does, so state them explicitly rather than assuming. The quote is conventionally the call on the **base** currency minus the put on it, so for EURUSD a positive number means EUR calls are bid and a negative number means EUR puts are bid. Because that is easy to invert, brokers say it in words, quoting for example “0.6 for the EUR put” rather than a bare signed number.

Always confirm which currency the reversal favours before acting on one. A sign error here is not a rounding issue, it reverses the direction of the position, and it is precisely the error this convention exists to prevent.

Two more conventions matter and belong elsewhere. What “25 delta” means depends on whether spot or forward delta is used and whether the premium is included, which differs by currency pair and is set out in delta conventions. And why the smile has the shape it does at all is the FX volatility smile.

REAL INTERVIEW QUESTION (Credit Suisse · FX)

What is a risk reversal?

WHAT THEY TEST / ANSWER

Give both meanings, because the word does double duty, then say what it is used for.

As a structure, a risk reversal is long an out-of-the-money option on one side and short an out-of-the-money option on the other, same pair, same expiry, conventionally 25 delta on each. Buying the EURUSD risk reversal means buying the 25 delta EUR call and selling the 25 delta EUR put.

As a quote, it is the difference between the two implied volatilities, $\sigma_{25\Delta\text{call}} - \sigma_{25\Delta\text{put}}$, in volatility points. It is one of the three numbers an FX volatility surface is quoted with, alongside ATM and the butterfly, and it is the one that sets the tilt of the smile.

What it tells you. The sign says which side the market is paying up for. A negative EURUSD risk reversal means EUR puts are richer, so the market is more worried about the euro falling than rising. This is the FX equivalent of equity skew, with the difference that in equities the skew is almost permanently negative because everyone is structurally long the market and buys downside protection, whereas in FX **the sign genuinely moves** and can flip with the macro narrative. Every currency pair has two sides and there is no natural long, so the risk reversal is a real sentiment indicator rather than a structural constant.

What it is used for, and give both. *As a hedge*, the zero-cost version: a corporate buys the protective option and sells the one on the other side, choosing strikes so the premiums net to zero. Protection with no upfront cost, at the price of giving up the favourable move beyond the short strike. *As a trade*, it is the clean way to be long or short skew. Because the two legs are struck at similar deltas their vegas largely offset, so the position is close to neutral to the *level* of volatility and exposed to the *difference* between the two wings.

Then the caveat that shows you would actually run the position. A risk reversal is long a call and short a put, which is a synthetic long in the underlying, so it carries substantial delta. To hold it as a pure skew position you must delta hedge it; unhedged, you have taken a directional view and called it a volatility trade. That distinction is the one an interviewer is listening for.

The zero-cost collar, which is where clients meet it

Most risk reversal volume is not a skew trade. It is a corporate or a real-money client hedging a currency exposure and refusing to pay premium.

> **INTUITION**
>
> A hedger who buys protection outright pays a premium and their treasurer has to justify it, and justifying an insurance premium against a risk that did not materialise is an annually recurring argument. Selling the option on the other side removes the premium and removes the argument. In exchange the client gives up the move beyond the strike they sold, which is usually a move they were happy to forgo, since a favourable currency move is a windfall rather than the point of the business.
>
> So the zero-cost collar is not a clever structure, it is a budget constraint expressed as a payoff. **The skew decides how good a deal it is**: if the option you are selling is the expensive side, you can place it further away and keep more of the upside, and if it is the cheap side you cannot. In the example above, a client selling the EUR put is selling the leg that is 0.60 volatility points rich, which buys them a better strike than a symmetric market would.

REAL INTERVIEW QUESTION (CACIB · FX)

How do I hedge when I sell dollars, that is, sell dollars so buy euros, buying a EUR call and a USD put?

WHAT THEY TEST / ANSWER

Confirm the direction first, because the sentence contains the trap. Selling dollars to buy euros means you are exposed to the euro strengthening before you transact, so the protection you want is the right to buy euros at a known rate. That instrument is a **EUR call, which is the same contract as a USD put**: one option, two names, because every FX option is a call on one currency and a put on the other simultaneously. Saying that unprompted is the first mark.

Then set out the choices, cheapest to most protective.

Do nothing and take the rate on the day. Not a hedge, but it is the baseline every other answer is measured against and worth naming.

Forward. Fix the rate outright. Zero premium, complete certainty, and complete loss of any favourable move. Right when the exposure is certain and you have no view.

Buy the EUR call outright. Protection with participation: you exercise if EUR has strengthened and let it lapse if it has weakened, so you keep the good outcome. The cost is the premium, and it must be paid whether or not the risk occurs.

Risk reversal, the zero-cost collar. Buy the EUR call for protection and sell a EUR put to fund it, strikes chosen so the net premium is zero. You are protected above the call strike, you participate down to the put strike, and below that you are obliged to buy at the put strike. This is where most corporate hedging actually sits, and the risk reversal quote is exactly what determines how attractive the two strikes can be.

Then the questions you would ask before recommending anything, because a hedging recommendation with no questions is not advice. Is the exposure certain or contingent, since a contingent exposure hedged with a forward creates a new risk if the underlying transaction does not happen? What is the horizon and is it a single date or a stream? What is the hedge policy, because most treasuries have a mandated ratio and the answer must fit it? And what is the accounting treatment, since hedge accounting eligibility often decides the structure more than the economics do.

Close on the honest trade-off. There is no free hedge. The forward costs you the upside, the option costs you premium, and the collar costs you the tail of the favourable move. Which is right depends on the client's constraint, not on which has the best payoff diagram, and saying that is worth more than naming a structure.

INTERVIEW TRAP

Three errors. (1) Quoting a risk reversal without saying which side it favours. The convention is base currency call minus base currency put, but brokers name the currency for a reason and you should too. (2) Calling a risk reversal a volatility trade while leaving it unhedged. It is a synthetic long or short in the underlying and carries real delta; only the delta-hedged version is a skew position. (3) Assuming FX skew behaves like equity skew. Equity skew is persistently negative because the customer base is structurally long; in FX both sides are someone's home currency, so the sign moves with the macro story.

Where this goes next

The shape this number tilts, and why FX smiles differ across pairs, is the FX volatility smile, and the third quote that completes the surface is butterflies. What 25 delta means, and why that is a harder question in FX than anywhere else, is delta conventions. The instruments themselves are FX options, the general treatment of the phenomenon is volatility smile and skew, and trading it as a position is skew trading.

13.9 Butterflies

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The word butterfly means two different things in this book, and an FX desk means the second one. In options strategy it is a three-strike payoff, long the wings and short two of the body, that pays when the underlying finishes near the middle strike. In FX it is a **volatility quote**: the amount by which the average of the two wing volatilities exceeds the at-the-money volatility.

$$\mathrm{BF} = \frac{\sigma_{25\Delta\mathrm{call}} + \sigma_{25\Delta\mathrm{put}}}{2} - \sigma_{\mathrm{ATM}}.$$

Same name, different objects. One is a position with a payoff diagram; the other is a number on a broker screen. Noticing which is meant is the first thing this section is for.

What the number measures

An FX surface is quoted per expiry as three numbers: the ATM sets the level of the smile, the risk reversal sets its tilt, and the butterfly sets its **curvature**. Those are treated together in risk reversals, and this is the third of them.

> **KEY IDEA**
>
> Curvature is a price for **fat tails**. If returns were exactly lognormal, every strike would price off the same volatility, the smile would be a flat line and the butterfly would be zero. It is not zero, and it is reliably positive, because the market believes large moves are more likely than a lognormal distribution says. The butterfly is therefore the price of excess kurtosis, in volatility points, and a desk long the butterfly is long the tails of the distribution and short the middle.
>
> The risk reversal and the butterfly are asking genuinely different questions. **The risk reversal asks which direction the market fears; the butterfly asks how much it fears extremes in either direction.** A pair can have a large butterfly and a risk reversal of zero, meaning big moves are expected but with no view on the side, and that combination is common ahead of a binary event.

WORKED EXAMPLE

Continuing the surface from the previous section: a EURUSD expiry with ATM = 8.0, and wings of $\sigma_{25\Delta\text{call}} = 7.95$ and $\sigma_{25\Delta\text{put}} = 8.55$.

$$\text{BF} = \frac{7.95 + 8.55}{2} - 8.0 = 8.25 - 8.0 = \mathbf{0.25}.$$

The wings sit a quarter of a volatility point above the at-the-money on average, and that quarter point is the whole of what the market is paying for the distribution having heavier tails than a lognormal one.

Note how little of the surface the butterfly is here: 0.25 against a level of 8.0, about 3%. For a liquid G10 pair the curvature is genuinely a small correction. For an emerging market pair with a managed exchange rate it is not, because a currency that is quiet until it is not has a distribution nothing like a lognormal, and the wings can trade several points over the at-the-money. Comparing the butterfly across pairs is a fast read on which currencies the market thinks are prone to discontinuous moves, and the models built for that are jump diffusion models.

The distinction that catches people out

There are two objects called the butterfly, or equivalently the strangle, and they are not the same number. This is the genuine subtlety of FX volatility quoting.

- The **market strangle** is what brokers actually quote: a single volatility applied to *both* wings such that the resulting strangle has the correct market **price**. It is defined by a price, not by a shape.
- The **smile strangle** is what the formula above computes once you have a calibrated smile: the average of the two wing volatilities read off that smile, less the at-the-money.

INTUITION

These differ because pricing both wings at one volatility is not the same as pricing each wing at its own. The market strangle is a statement about a package's total cost; the smile strangle is a statement about the shape of a curve. Whenever the smile is tilted, the two diverge, and the divergence grows with the size of the risk reversal.

A correctly built surface must **reprice the market strangle**, because that is the instrument that actually trades. So the simple decomposition $\text{ATM} + \text{BF} \pm \frac{1}{2}\text{RR}$ is the right way to *read* a surface and an approximation when *building* one, and the difference between those two uses is exactly where a candidate who has calibrated a surface separates from one who has only read about it. Say which of the two you mean and you will be ahead of most of the room.

REAL INTERVIEW QUESTION (HSBC · desk not recorded)

What is a straddle, a strangle and a butterfly, or fly?

WHAT THEY TEST / ANSWER

Define all three as structures, then say the thing the question does not ask for and that makes the answer: in FX these words name volatility quotes rather than payoffs.

Straddle. A call and a put at the same strike, usually at the money, same expiry. Maximum sensitivity to volatility and to movement, no directional view: you profit if the underlying moves far enough in either direction to cover the two premiums. It is the purest long volatility position available in vanilla options, which is why the at-the-money volatility of a surface is effectively the straddle's price.

Strangle. The same idea with the two strikes moved apart, both out of the money. Cheaper than the straddle because an option is worth most at the money and both legs have been moved away from it, and it requires a larger move to pay, so it is a long volatility position with a higher threshold and lower cost.

Butterfly. Long the two wings and short two of the middle strike, which is the strangle against the straddle. The payoff peaks at the middle strike and is bounded on both sides, so it is a bet that the underlying stays near a level, or equivalently that realised movement is small. It is cheap, its maximum loss is the premium, and its maximum gain is capped.

Then the point that lifts it. On an equity or rates desk those are payoff structures. On an FX desk, **straddle, strangle and butterfly are the names of quotes**: the ATM is the straddle's volatility, and the butterfly is how far the average of the wings sits above it. So if an FX trader says the fly is 0.25, they are not describing a position they have on, they are telling you the curvature of the smile. Recognising which meaning is in play is the difference between following the conversation and guessing at it.

If you want one more sentence, add the relationship between the two meanings, because they run in opposite directions. **The structure** is long the wings and short the body in *options*, but its payoff peaks at the middle strike and is zero beyond either wing, so its price is essentially the risk-neutral probability of finishing near the middle: the holder is long the middle of the distribution and short the tails. **The quote** runs the other way, since being long the FX butterfly is being long the two wing volatilities against the at-the-money, which is a statement about kurtosis. Same word, opposite side of the distribution, which is exactly why the two are worth separating. The payoff structure itself is range-bound strategies; the quote convention is FX's own.

Why the number is reliably positive

INTUITION

A trader who sells the wings is short the tails, and the tails are exactly where hedging fails: a gap move cannot be delta hedged through, so the loss on a far out-of-the-money option sold is not bounded by any hedging programme. Being paid for that risk requires a premium above the at-the-money volatility, and that premium is the butterfly.

So the butterfly is positive for the same reason insurance costs more than its expected payout: the seller is taking a risk they cannot lay off continuously, and they charge for it. Any explanation of a positive butterfly that relies only on the statistical distribution is incomplete, because it does not say why the market's price should exceed the statistical fair value, and **the reason it does is that the seller's hedge breaks in the region they are selling**.

INTERVIEW TRAP

Three errors. (1) Using butterfly to mean the payoff structure while talking to an FX desk, or the quote while talking to an equity desk. Both are correct usages of the word and they refer to different things. (2) Treating the market strangle and the smile strangle as interchangeable. They coincide only when the smile is untilted, and the gap grows with the risk reversal. (3) Explaining a positive butterfly purely as fat tails in the data. That explains why it should not be zero; it does not explain why the market charges more than the statistics, which is that selling the wings is selling a risk that cannot be hedged continuously.

Where this goes next

The other two quotes that complete the surface are the at-the-money level and risk reversals, and the shape all three describe is the FX volatility smile. What the delta in "25 delta" means is delta conventions, the instruments are FX options, the general treatment of the phenomenon is volatility smile and skew, and the models built to reproduce a curved smile are jump diffusion models.

13.10 Central banks and FX policy

sales · trading · structuring | core | fx · rates

FX is the only major market in which one participant can create one side of the trade at will. A central bank that wants to sell its own currency never runs out of it. That asymmetry, and the fact that the same institution also sets the interest rate that FX is priced off, is why the FX desk spends more time on central banks than any other desk in the building.

Who the major central banks are and what their mandates say is major central banks; how policy works on the economy in general is monetary policy and monetary transmission mechanisms. This section is about the exchange-rate channel specifically, in both directions: policy that moves the currency as a by-product, and policy aimed at the currency on purpose.

Direction one: rates move the currency

Start from the parity relation in interest rate parity. It fixes the *forward* given spot and the two rate curves; it does not tell you where spot goes. So the naive chain "the central bank hikes, so the currency is worth more, so it rises" is not an argument, it is a slogan, and interviewers use it to separate candidates.

The honest version has three steps.

1. Spot responds to changes in the **expected future path** of rates, not to today's level. Today's level is already in the price.
2. Therefore only the **surprise** matters: the difference between what the central bank does and what was priced before it did it.
3. And the surprise that matters most is usually in the guidance, not in the decision, because the decision is one meeting and the guidance is the next eight.

KEY IDEA

A rate decision moves the currency to the extent that it was not already expected. A hike that was fully priced does nothing. A hike delivered alongside a statement implying fewer hikes ahead usually sends the currency *down*, which is the single most common way a beginner gets an FX question wrong.

WORKED EXAMPLE

The current policy target midpoint is 4.375%. Interest rate futures imply an expected post-meeting rate of 4.525%. A hike would be 25 basis points, taking the rate to 4.625%.

The market-implied probability p of the hike solves

$$4.375 + 0.25\,p = 4.525 \qquad \Longrightarrow \qquad p = \frac{0.150}{0.25} = 0.60,$$

so the market is 60% priced for a hike. If the hike is delivered, the rate lands at 4.625% against an expectation of 4.525%, a surprise of **plus 10 basis points**. If the bank holds, the rate stays at 4.375% against 4.525%, a surprise of **minus 15 basis points**. Check that the expectation was consistent:

$$0.60 \times (+10) + 0.40 \times (-15) = 6 - 6 = 0.$$

So a hike here is worth 10 basis points of repricing, not 25. And that is only the front end. If the accompanying statement drops a hike from the expected path further out, the two-year rate can fall on a day when the policy rate rose, and the currency will follow the two-year rate, not the headline.

Real rates, not nominal rates

The second refinement that separates answers. What an investor earns holding a currency is the rate net of the inflation that erodes it, so the differential that matters is the **real** one.

WORKED EXAMPLE

The United States is at 5.00% nominal with 3.00% inflation. Japan is at 2.00% nominal with 0.00% inflation. The nominal gap is 300 basis points, which sounds decisive. The real rates are

$$\frac{1.05}{1.03} - 1 = 1.94\% \qquad \text{against} \qquad \frac{1.02}{1.00} - 1 = 2.00\%,$$

a real differential of -6 basis points. To first order both are simply 2%. On the nominal comparison the dollar looks overwhelmingly attractive; on the real comparison there is nothing in it.

This is why a currency can fall on a rate hike delivered to fight inflation: if the inflation forecast rises by more than the policy rate, the real rate has gone down.

Beyond rates, spot also responds to the terms of trade (a commodity exporter's currency moves with its export price), to growth differentials, and to risk sentiment, which is why the yen and the Swiss franc rally when equity markets fall regardless of what their central banks are doing.

Direction two: the currency is the target

Sometimes the exchange rate is not a by-product of policy but the object of it. The toolkit runs from cheap to expensive.

- **Verbal intervention.** An official says the move is "excessive and disorderly" and that they are "watching with a high sense of urgency". Costs nothing, works briefly, and stops working if it is never followed by anything.
- **Actual intervention.** The bank trades: selling reserves to buy its own currency, or creating its own currency to buy foreign assets.
- **Rates set for the currency.** Moving the policy rate primarily to affect the exchange rate rather than domestic demand.
- **Capital controls.** Restricting who may convert, and how much.

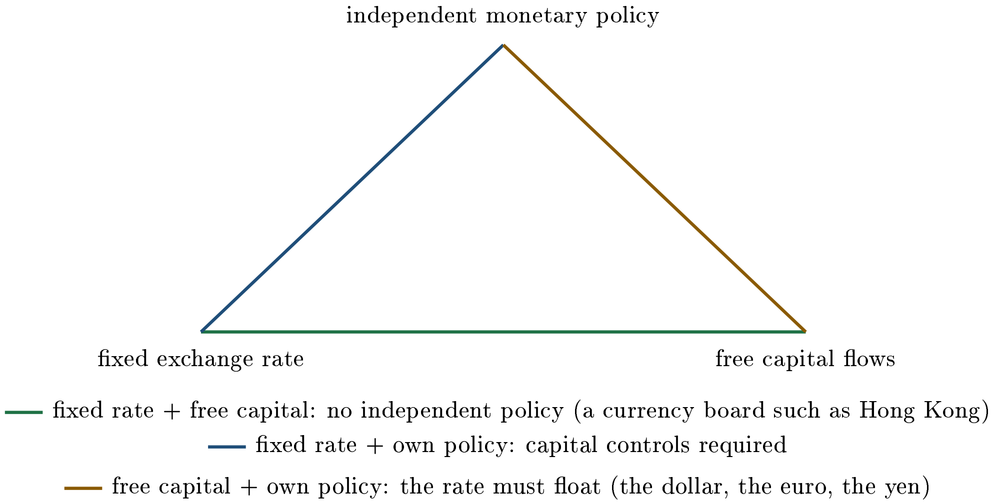
KEY IDEA

Intervention is deeply asymmetric. To *weaken* your currency you sell it, and you can create it without limit, so that side of the trade can be sustained essentially forever, at the cost of an ever-growing balance sheet. To *strengthen* it you must buy it with foreign reserves, and reserves are finite. Defending a weak currency has a deadline. Capping a strong one does not.

A second distinction that gets asked. **Unsterilised** intervention leaves the resulting change in domestic money supply in place, so it is a monetary easing or tightening as well as an FX trade, and it has a real effect. **Sterilised** intervention offsets it with a domestic operation so the money supply is unchanged, which makes it a pure portfolio operation and, on most evidence, far weaker and shorter-lived. When a bank intervenes and the effect fades in a day, sterilisation is usually why.

The impossible trinity

Behind all of it sits one constraint. A country can have a fixed exchange rate, free movement of capital, and an independent monetary policy, but only two of the three at once. Choose any two and the third is determined.



Each edge joins the two things you keep, and its legend line names what you have given up to keep them. Every FX regime in the world is one of those three edges, and every FX crisis is a country discovering that it was trying to sit at all three corners at once.

A case worth knowing: the Swiss franc floor

In September 2011, with the franc appreciating hard as investors fled the euro area crisis, the Swiss National Bank announced it would not tolerate a euro-franc rate below 1.20 and would enforce that floor by buying foreign currency in unlimited quantities.

The instructive part is that this was the sustainable side of the asymmetry: the SNB was selling francs, which it can create, so it never faced a hard constraint. What it faced instead was a balance sheet that kept growing, from foreign currency reserves of roughly 200 billion francs before the floor to roughly 500 billion by the end of 2014, with the mounting risk of a very large loss on those holdings if the franc ever did appreciate. Late in 2014, with the pressure building again, the SNB added a second instrument and took its deposit rate negative, to -0.25%, to make holding francs expensive in the way the floor could not.

On 15 January 2015 the floor was abandoned without warning, and the same announcement cut the deposit rate to -0.75%: the rate was the cushion for the removal, not a support for the floor, which is the detail candidates usually get the wrong way round. Euro-franc fell from 1.20 to below parity within minutes, touching roughly 0.85 at the extreme before closing the day near 1.01, so the franc appreciated by something like a fifth on the day and by far more than that at the low. Several retail brokers were bankrupted by clients whose losses exceeded their accounts, because the stop-loss orders that were supposed to cap those losses could not be filled at anything near their level.

> **INTERVIEW TRAP**
>
> Assuming a defended level is a safe level. A floor or a ceiling makes the distribution of outcomes deeply one-sided rather than tight: nothing happens for years, and then everything happens in one print with no liquidity in between. Options on a defended pair look absurdly cheap on realised volatility precisely because realised volatility is not measuring the risk, and the smile prices that (see the FX volatility smile). Selling volatility on a managed currency is picking up pennies in front of something you cannot see.

Interview angle

REAL INTERVIEW QUESTION (Société Générale · Rates)

If the Fed raises rates, what happens to the dollar, and what is your view on where the dollar goes?

WHAT THEY TEST / ANSWER

Two questions. The first has a right answer and a trap; the second has no right answer and is testing whether you can hold an opinion.

On the first, resist the reflex. The textbook chain is that higher rates attract capital and lift the currency, and it is a reasonable first-order statement, so say it. Then immediately qualify it in the way that gets the marks: what moves spot is not the level of the policy rate but the change in the *expected path*, so a hike that was fully priced moves nothing, and a hike delivered with guidance implying fewer hikes ahead routinely sends the dollar down on the day. Point at what you would actually look at: the two-year yield and its differential against the other currency, not the policy rate. Add that the real rate is what matters, so a hike that follows an even larger jump in the inflation forecast is a real easing.

On the second, give a view with a structure rather than a forecast with a number. Say what would make you bullish the dollar (the Fed repricing hawkish relative to others, the growth differential widening in America's favour, a risk-off episode, since the dollar is where people run) and what would make you bearish (other central banks catching up, a soft landing that lets the Fed cut into a stable economy, fiscal concerns). Then commit to one, briefly, and name the level or event that would tell you you were wrong.

The failure mode is a candidate who recites both sides and refuses to choose. The desk does not need a summary of the debate; it needs to know whether you can hold a position and say what would change your mind.

REAL INTERVIEW QUESTION (BNP · FX)

How do interest rates affect exchange rates? Link it to current events and to the rate rises by the Fed and the ECB.

WHAT THEY TEST / ANSWER

Structure it as three layers, because the question runs from mechanism to markets to news.

Mechanism: covered interest rate parity ties the forward to spot through the rate differential, exactly and by arbitrage. What it does not do is pin spot. Spot moves when the expected path of rates changes, because capital is priced off the whole curve rather than off today's overnight rate. So the working variable is the **two-year rate differential**, and the real differential rather than the nominal one, since inflation erodes what you earn.

Markets: the currency of whichever central bank is repricing hawkish relative to the other strengthens, and the important word is relative. Both banks hiking together can leave the pair unchanged. Two additional channels are worth naming so the answer is not purely about carry: the growth differential, and risk sentiment, which sends the dollar, the yen and the franc higher in a sell-off regardless of rate levels.

News: talk about the Fed and the ECB as a *gap* rather than as two separate stories. Say which is expected to move first and by how much, whether the market has already priced it, and where the two-year differential sits relative to recent history. The tightening cycle that began in 2022 is the canonical illustration: the Fed moved earlier and faster than the ECB, the differential widened, and the euro weakened against the dollar to the point of touching parity in 2022, before recovering as the ECB caught up. That is one clean sentence of history that shows the mechanism working, and it is worth more than a list of policy rates.

Prepare the levels before the interview: the two policy rates, the direction of the next move for each, and roughly where EUR/USD is trading. Being vague about the current level undoes a good explanation.

REAL INTERVIEW QUESTION (Credit Suisse · FX)

What is the SNB's monetary policy, and what is its impact for Swiss companies?

WHAT THEY TEST / ANSWER

The SNB is the best possible example of a central bank whose monetary policy *is* exchange rate policy, so say that first and then justify it. Switzerland is a small, extremely open economy, so the exchange rate transmits to inflation and to corporate profits far faster than domestic demand does. The franc is also a safe-haven currency, so every crisis somewhere else pushes it up and imports deflation. That combination means the SNB has consistently used the currency as its main instrument, through intervention and through very low, at times deeply negative, policy rates.

Give the floor as the case study: from September 2011 the SNB enforced a minimum of 1.20 euro-franc with unlimited purchases of foreign currency, which it could sustain because it was selling a currency it creates, and it accumulated an enormous reserve portfolio doing so. It abandoned the floor in January 2015 and the franc jumped violently. The lesson to state is the asymmetry: capping a strong currency is unlimited in principle but builds a balance sheet risk that eventually becomes the binding constraint, whereas defending a weak one is limited by reserves from day one.

Impact on Swiss companies, which is the half candidates skip. A strong franc is a direct hit to exporters and to the tourism industry, because their costs are in francs and their revenues are not. Pharmaceuticals, machinery and watches are the classic exposures. Companies respond by hedging forward, by moving costs into the currencies they sell in, and by relentless cost control, which is part of why Swiss industry is so productive. Importers and consumers gain, since imported goods get cheaper, which is also why a strong franc imports deflation and hands the SNB its problem back.

The follow-up worth pre-empting: if a Swiss exporter hedges forward, they do not escape, they only postpone. The forward already contains the interest rate differential, so with Swiss rates below foreign rates the exporter is selling foreign currency forward at a worse rate than spot. Hedging converts an uncertain loss into a certain cost, and that cost is the parity relation, not a fee.

INTERVIEW TRAP

Three FX-policy errors. (1) Saying a hike lifts the currency without asking what was priced. (2) Comparing nominal rates across countries with very different inflation, when the real differential can point the other way. (3) Treating intervention as symmetric: selling your own currency is unlimited, buying it is not, and almost every currency collapse in history is on the second side of that sentence.

Where this goes next

What happens when the defence fails, and the anatomy of currency regimes and crises, is pegged currencies and FX crises. The parity relation the whole chapter rests on is interest rate parity, and the way the option market prices the one-sided risk described above is the FX volatility smile.

13.11 Pegged Currencies and FX Crises

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Not every currency floats, and the ones that do not are where the most dramatic events in this market happen. The macro taxonomy of regimes is exchange rate regimes and the individual episodes are major FX crises. What is owned here is the mechanics: how a peg is held, why it breaks, and what the payoff of trading one looks like.

The constraint every peg lives under

KEY IDEA

The **impossible trinity**, set out in central banks and FX policy: a country may have free capital movement, a fixed exchange rate, and an independent monetary policy, but only two of the three. The consequence that matters here is that every currency crisis in history is a country discovering it was attempting all three, and the market finding out first.

The regimes are a spectrum rather than a binary: free float, managed float, crawling peg, conventional peg with a band, hard peg, currency board, and finally abandoning the currency altogether through dollarisation or monetary union. Moving right along that list buys credibility and sells flexibility, and the crises happen where a country is further right than its economy can support.

Defending a peg, and why each defence runs out

A central bank holding a rate against selling pressure has three instruments, and all three have a limit.

- **Sell reserves**, buying its own currency. This works until the reserves run out, and the market can see the reserve position, so the endgame is visible in advance.
- **Raise interest rates**, making it expensive to be short. This works until the domestic economy cannot bear the rates, which is why the defence fails fastest when unemployment is high or the banking system is weak. The market's question is never "can they raise rates" but "for how long will they".
- **Impose capital controls**, which is the trinity's third option and works, at the cost of the country's access to foreign capital for years afterwards.

INTUITION

The payoffs are wildly asymmetric, and that asymmetry is the whole reason pegs get attacked. The speculator shorting a pegged currency risks the *carry*, a few percent a year, and stands to make twenty or thirty percent in days if the peg goes. The central bank risks its reserves and its credibility to make the speculator lose a small carry. That is an option with a very attractive payoff ratio, and it means an attack does not need the country's fundamentals to be genuinely unsustainable. It only needs enough people to believe the defence will be abandoned, which is why these crises are self-fulfilling and why they arrive suddenly rather than gradually.

A peg that does hold is usually one where the defence is automatic rather than discretionary. A **currency board** is the clearest case: the monetary base is fully backed by foreign reserves and issuance is mechanically tied to them, so the authority has no discretion to run out of and no policy to be tempted away from.

REAL INTERVIEW QUESTION (Société Générale · Delta One)

What do you know about Hong Kong?

WHAT THEY TEST / ANSWER

Open question on a Delta One desk, so lead with the thing that makes Hong Kong structurally different from any other major market: **the currency board**.

The mechanism. The Hong Kong dollar is pegged to the US dollar within a narrow band, and the peg is maintained not by discretion but by a rule: the monetary base is fully backed by US dollar reserves, and the authority stands ready to convert at the band edges. Because issuance is tied mechanically to reserves, the arrangement does not depend on anyone choosing to defend it.

The consequence, and the part that matters to a trading desk. Hong Kong therefore *imports United States monetary policy.* It has free capital movement and a fixed rate, so by the impossible trinity it has no independent monetary policy: when the Federal Reserve raises rates, Hong Kong rates follow, regardless of whether the local economy wants them to. That has produced long periods of policy that suited the anchor and not the territory, which shows up in property prices rather than in the currency.

Then the market points. It is a major regional equity and derivatives centre, so an index desk cares about it directly: it is the listing venue for a large part of the mainland Chinese corporate universe, which is why the index is not a proxy for the local economy. And the peg's persistence means the currency itself is a very low-volatility instrument with a fat tail, which is exactly the option-pricing problem described below.

If they push on whether the peg will hold, the honest answer is the analytic one rather than a forecast: it has survived repeated pressure precisely because the defence is mechanical and the reserve backing is large relative to the monetary base, and the thing to watch is not sentiment but that backing ratio and the rate differential that opens when local and US policy diverge.

REAL INTERVIEW QUESTION (Credit Suisse · FX)

What is the SNB's monetary policy, and what is its impact on Swiss companies?

WHAT THEY TEST / ANSWER

Two halves, and the first should be told as the story of a peg, because that is what makes the second half concrete.

The policy. The Swiss National Bank's problem is a permanently strong currency: the franc is a haven, so every episode of global stress pushes it up, which imports deflation and hurts an export-driven economy. Its response has been unusually aggressive for a small central bank, running policy rates below zero for years and intervening in the currency on a scale that has made its balance sheet enormous relative to Swiss output.

The episode to know is the floor. For several years the SNB committed to defending a minimum EUR/CHF level, buying euros without limit to hold it, and in January 2015 it abandoned that commitment without warning and the franc appreciated violently in minutes. The lesson is the important part and it is not the usual one: **the SNB was not attacked and did not run out of anything**. Defending a *floor* means selling your own currency, which you can print without limit, so the constraint was never reserves. It was that holding the floor required accumulating foreign assets without bound, and the balance sheet risk of that eventually outweighed the policy benefit. A floor fails from unwillingness; a ceiling fails from inability. Knowing which side you are on tells you which failure to expect.

The impact on Swiss companies. Exporters and the tourism sector carry a structural competitiveness problem and hedge heavily, which is why Swiss corporate FX hedging programmes are long-dated and unusually systematic. Firms with foreign revenue and Swiss costs see margins compressed whenever risk aversion rises anywhere in the world, so their currency exposure is correlated with global equity drawdowns rather than independent of them. And the 2015 episode taught a specific lesson about hedging policy: several companies were hedged with structures that assumed the floor would hold, and a hedge conditioned on a central bank's promise is not a hedge.

Being paid to stand in front of it

REAL INTERVIEW QUESTION (BNP · FX)

You win one million euros on the Euromillions and want to invest it risk-free, but the rate in Europe is 0%. You can invest in Turkey at 10% with an exchange rate of 1 for 1: is that attractive? Describe the operation, given that you invest for one year and want euros at the end.

WHAT THEY TEST / ANSWER

The words *sans risque* are the trap, and the answer is that the trade is not risk-free at all: you have swapped interest rate risk for currency risk, which is a much larger one.

Do the arithmetic first. Convert €1m into 1m lira at par, earn 10%, and finish with 1.1m lira. Converting back at a rate of X lira per euro gives $1.1/X$ million euros, so you break even at $X = 1.10$. The lira must weaken by only $1 - 1/1.1 = \mathbf{9.1}\%$ over the year for the whole 10% to be wiped out.

Then the point that makes it a market answer. That break-even is not a coincidence: by interest rate parity the one-year forward is exactly $1.0 \times 1.10/1.00 = 1.10$. The market is already offering to lock in that rate, and it is offering it because that is the rate at which the trade earns nothing. So the trade is not "10% risk-free", it is **a bet that the lira falls by less than the forward says**, and if you wanted the 10% with no currency risk you could sell the lira forward and discover you have exactly 0%.

And the tail. High-yield emerging currencies do not depreciate smoothly at the forward rate; they hold and then move a long way at once. A 30% fall takes the position to $1.1 \times 0.70 = 0.77$ million euros, a **23% loss**. A 50% fall leaves **0.55 million**, a 45% loss, and the historical record of this particular currency includes moves of that size. You have written a large amount of tail insurance to collect 10%.

Close by naming what the trade actually is, which is the answer they want: this is a **carry trade**, the oldest position in foreign exchange, and it is short a tail rather than long a yield. The correct response to being offered 10% for nothing is to ask what you are being paid to accept, and here it is the risk of a devaluation.

Pricing options on a currency that does not move

A pegged currency produces a distribution nothing like a lognormal: almost all the probability sits within a narrow band, with a small chance of a very large jump. Realised volatility is therefore extremely low, and a model calibrated to it will price crash protection at nearly nothing.

The consequences are worth knowing because they recur wherever a price is administered rather than traded. Implied volatility trades far above realised, and correctly so. The smile is violently asymmetric in the direction of the break, which is the one case where an FX surface looks like an equity skew. And options on the peg are the efficient way to express the view, because the cost is bounded and the payoff is the jump, whereas a spot short pays the carry every day for an uncertain wait.

INTERVIEW TRAP

1. **Assuming a fixed rate means no currency risk.** It means the risk is not visible day to day and arrives all at once.
2. **Reading low realised volatility as low risk.** On a peg the two are unrelated, and the model that equates them is the one that misprices the tail.
3. **Treating a floor and a ceiling as symmetric.** Defending a ceiling can exhaust reserves; defending a floor cannot, so a floor ends by choice and a ceiling by force.
4. **Believing fundamentals decide.** A defence can fail purely because enough participants expect it to, which is what makes these events self-fulfilling and sudden.
5. **Ignoring the impossible trinity.** If a country has a fixed rate and open capital markets, its interest rates are set somewhere else, and every domestic consequence follows from that.
6. **Hedging with an instrument that assumes the peg holds.** A hedge conditioned on a central bank's promise is a bet on that promise.

VISUAL / CODE TO BUILD: Interactive

A peg defence simulator. The reader plays the central bank: a band is set, a speculative flow arrives that scales with the market's belief that the peg will break, and the reader chooses each week between selling reserves, raising rates and imposing controls. Reserves, the domestic rate and an unemployment proxy update, and the market's belief responds to those choices, so a defence that is visibly running down reserves attracts more pressure and the reader experiences the self-fulfilling dynamic from the inside. A second mode flips sides and lets the reader run the attack with a stated size, showing the carry bleeding daily against the payoff if the peg goes, with the break-even holding period displayed, which is the asymmetry of the trade made concrete. A third panel prices a one-month option on the pegged rate under a lognormal calibrated to realised volatility and under a jump model, and displays the two premiums side by side, which is the clearest possible demonstration of why realised volatility is the wrong input here.

Chapter 14

Commodities Markets

14.1 Physical vs derivative commodities

sales · trading · structuring core commodities

Every other asset class in this book is a promise written on paper. A share is a claim, a bond is a schedule of payments, a swap is a contract. A commodity is a *thing*: a barrel of crude that has to sit somewhere, a tonne of copper that has to be shipped, a megawatt-hour that must be consumed the instant it is produced. That single fact is the source of everything that makes commodities strange, and it is what a commodity interview is really probing. If you can hold both worlds in your head at once, the physical one where oil rots in a tank and the paper one where a futures contract is just a number on a screen, the rest of the chapter follows.

Three layers of the same market

When a market participant says "the oil price", they could mean any of three different things, and confusing them is the most common beginner mistake.

1. **The physical (or "wet") market.** An actual cargo, of an actual grade, loading at an actual terminal in an actual date window. It is bought and sold bilaterally between producers, refiners and merchants, and it is settled by moving the molecules. Prices here are quoted as a **differential** to a benchmark, for example "dated Brent minus $1.50", not as an outright number.
2. **The OTC derivative market.** Bilateral forwards and swaps that reference a published benchmark price, with no intention of delivery. This is where an airline hedges jet fuel and a mining company hedges copper, because they need exactly their own volume, tenor and index rather than a standardised contract.
3. **The listed futures market.** Standardised contracts on an exchange, centrally cleared, marked to market daily and traded in enormous volume, mostly by participants who will never see the commodity. This is the "paper" market, and it is the price you see on a screen.

INTUITION

Think of a **concert**. The physical market is the venue: a fixed number of real seats, on one night, in one city, and someone has to actually sit in them. The paper market is the ticket-resale platform, where the same seat changes hands twenty times before the doors open, by people who have no intention of attending. The resale price is what everybody quotes, and it is genuinely informative, because it aggregates everyone's view. But on the night of the concert the two must meet, since the last holder of the ticket has to either sit down or find someone who will. **Delivery is what welds the paper price to the physical one**, and the moments when the venue is full and nobody can sit down are the moments commodity markets do things no financial market ever does.

Why a commodity is not a financial asset

The pricing of a forward on a stock rests on a simple argument: buy the stock, hold it, and you have replicated the forward, so the forward price is the spot compounded at the financing rate. Try it with crude oil and four things break.

- **Carrying it costs real money.** Storage, insurance, freight, evaporation and spoilage, all of which are physical, lumpy, and capacity-constrained. See storage costs.
- **Holding it yields a non-financial benefit.** A refiner with a full tank can keep running when supply is disrupted; nobody can pay you to be a share. That benefit is the **convenience yield**, developed in convenience yield.
- **You cannot short it by borrowing it.** You can borrow a share, sell it, and buy it back. You cannot borrow ten million barrels of oil that does not exist, which is why the upside of a physical squeeze is essentially unbounded while the downside is limited by the fact that consumption never stops.
- **It is not fungible across time, place or quality.** A share today and a share next month are identical. Crude at Cushing in June and crude at Rotterdam in September are three different commodities: different grade, different location, different date. Every commodity price is a triple, and the differences between those triples are what commodity desks actually trade.

KEY IDEA

The one-line version, and the answer to the most common opening question of a commodities interview: **a commodity price is not a number, it is a grade, a location and a delivery date**. Everything a commodity desk does is trading the spreads between those coordinates: the time spread (the forward curve), the location spread (the basis), and the quality spread (the crack, the crush, the grade differential).

Benchmarks, grades and how contracts settle

Because the physical market is fragmented, it needs reference prices. A handful of grades became the benchmarks that everything else is quoted against, and the two crude benchmarks illustrate the physical-versus-paper distinction better than any definition.

Contract	Venue	Size	Settlement
WTI crude	NYMEX (CME)	1,000 barrels	physical, at Cushing, Oklahoma
Brent crude	ICE	1,000 barrels	cash, against a published index
Henry Hub gas	NYMEX (CME)	10,000 MMBtu	physical, at Henry Hub, Louisiana
Gold	COMEX (CME)	100 troy ounces	physical, approved vaults
Copper	LME	25 tonnes	physical, LME-approved warehouses

Two details in that table repay attention. First, **WTI is physically delivered at a single inland point** while **Brent is cash settled** against an index of North Sea cargoes that can move by sea to any refinery in the world. Second, on the published specifications WTI is the *better* crude, lighter and lower in sulphur than Brent, so it is "sweeter" and cheaper to refine into petrol. And yet WTI has traded at a persistent discount to Brent for much of the past fifteen years. The reason is purely physical: American shale production overwhelmed the pipeline capacity out of the mid-continent, so barrels piled up at a landlocked delivery point with no easy access to the sea. A better barrel in the wrong place is worth less than a worse barrel in the right one. That sentence is the whole section in miniature.

Delivery, and what happens when the venue is full

Fewer than one percent of listed commodity futures are typically held to delivery, which tempts people to treat futures as pure paper. That is a mistake, because the *threat* of delivery is exactly what anchors the price. As expiry approaches, a long who does not want 1,000 barrels arriving in Oklahoma must sell, and a short who does not own any must buy. If the physical infrastructure cannot absorb the imbalance, the paper price is forced to whatever level makes someone accept the physical consequence.

WORKED EXAMPLE

20 April 2020: the day oil traded below zero. The expiring May contract of NYMEX WTI settled at $-\$37.63$ per barrel. Nothing was broken and no model failed. Covid had destroyed demand, OPEC and Russia had briefly increased supply, and storage at Cushing (working capacity on the order of 75 to 80 million barrels) was effectively full or already booked. Longs facing physical delivery the next day had no tank to put the oil in and no legal way to refuse it, so the only way out was to pay someone else to take it. A negative price is not a broken market: it is the correct price of an obligation to receive something that costs money to hold and that you cannot store.

The tell that separates candidates is the follow-up. **Brent never went negative** in the same week. Not because Europe was less oversupplied, but because ICE Brent is cash settled against a seaborne index, so no holder ever faces a truck arriving. Same commodity, same glut, different contract design, different price. This is the single best illustration in market history that *settlement mechanics are part of the price.*

Who is actually in the market

Four groups, with genuinely different objectives, and knowing them lets you answer almost any "why did the price move" question sensibly.

- **Producers** (oil majors, miners, farmers), structurally long the commodity, natural sellers of forwards to lock in revenue.
- **Consumers** (refiners, airlines, utilities, food processors), structurally short, natural buyers of forwards or of caps.

- **Physical merchants and trade houses**, who own tanks, ships and pipelines and whose business is precisely the spreads: buy here, sell there, store now, deliver later. They are the arbitrageurs who force paper and physical to agree.
- **Financial participants**: banks intermediating and structuring, index and ETF investors seeking exposure, and speculators providing liquidity. They carry price risk the physical players want to shed, and their rolling of positions is itself a recurring flow that the market front-runs.

A bank commodity desk sits mostly in the third and fourth groups. It can quote a swap on an index without ever owning a molecule, but if it also handles physical it needs tanks, shipping and a credit department that understands demurrage. This is why commodity structuring interviews mix pricing questions with startlingly concrete ones about barrels and grades: the desk genuinely needs both.

Basis risk: the price of hedging a real thing with paper

Since the physical article and the listed contract are never the same grade, place and date, a hedge is never perfect. What is left is **basis risk**, and it is the main risk on a physical book.

WORKED EXAMPLE

A trade house buys a 500,000 barrel cargo at a fixed price. It is now long physical oil and hedges by selling ICE Brent futures. At 1,000 barrels per contract that is 500,000/1,000 = 500 contracts sold.

Over the next week flat price falls by \$4.00 per barrel, and the cargo's own differential to Brent *also* weakens, from Brent minus \$1.50 to Brent minus \$2.20.

- Futures leg (short): gains $\$4.00 \times 500{,}000 = \$2{,}000{,}000$.
- Physical leg (long): loses the flat price move plus the differential move, $(\$4.00 + \$0.70) \times 500{,}000 = \$2{,}350{,}000$.
- Net: $-\$350{,}000$, which is exactly the \$0.70 basis move on 500,000 barrels.

The flat price risk, the big number, was hedged away perfectly. What remains is a \$350,000 loss on a risk the futures contract simply cannot express. Say this out loud in an interview: *the purpose of a commodity futures hedge is to convert flat price risk into basis risk*, because basis is smaller and more forecastable, not because it is zero.

Basis decomposes into three pieces, and naming them is worth marks: **location** (Cushing versus the Gulf Coast versus Rotterdam), **quality** (API gravity, sulphur content, protein content, metal purity), and **timing** (the cargo loads in a window, the contract expires on a date). Each has its own market, and the time piece is the forward curve itself, covered in forward curves and contango and backwardation.

Units, because they will ask

Commodity desks live in physical units and interviewers use them as a cheap filter for genuine interest. Learn a handful cold: one barrel is 42 US gallons, close to 159 litres; one metric tonne of crude is roughly 7.3 barrels, varying with density; one troy ounce is 31.10 grams, and precious metals are always quoted per troy ounce, never per the 28.35 gram avoirdupois ounce; natural gas trades in MMBtu in the United States and MWh in Europe; power trades in MWh with a delivery profile (baseload, peakload) attached, because a megawatt-hour at 3am is not the same product as one at 7pm.

REAL INTERVIEW QUESTION (BP · Commodity)

Do you know any commodity derivatives?

WHAT THEY TEST / ANSWER

An invitation to show breadth in a structured way rather than to list names at random. Sort your answer by instrument and give a commodity-specific example of each: listed **futures** (NYMEX WTI, ICE Brent, LME copper) and their **options**; **swaps**, where a fixed price is exchanged against the average of a published index over a month, the workhorse of corporate hedging; **Asian options**, which dominate here rather than being an exotic curiosity, precisely because physical exposure accrues daily so a monthly average is the economically relevant reference; **calendar spreads**, trading the shape of the curve rather than its level; **crack, spark and crush spreads**, which are options on a processing margin rather than on a price, see crack spreads and spark spreads; and physically-flavoured structures such as storage and transport options, where the underlying is capacity rather than a commodity. Finish with the point that identifies someone who has thought about the asset class: the average-price and spread structures dominate because the physical exposures being hedged are themselves averages and spreads.

REAL INTERVIEW QUESTION (HSBC · desk not recorded)

Why did the price of oil go negative?

WHAT THEY TEST / ANSWER

They want a causal chain, not a Covid anecdote. Build it in four links. Demand collapsed suddenly in the spring of 2020 while supply was slow to adjust, so inventory built fast. Storage is finite and lumpy, and Cushing, the single inland delivery point of the NYMEX WTI contract, approached its working capacity, with much of the remainder already leased. The expiring May contract obliged longs to take physical delivery within days, and a long with no tank cannot refuse the barrels, so the only escape was to pay someone else to take the obligation. Hence a settlement of −\$37.63 per barrel on 20 April 2020. Then deliver the two closing observations that separate a prepared candidate. Cash-settled Brent did not go negative in the same week, which isolates contract design rather than the glut as the proximate cause. And a negative price is economically coherent rather than a market failure: when the cost of storing or disposing of a thing exceeds its value, its price is negative, which is also true of nuclear waste and of European power on a windy Sunday.

REAL INTERVIEW QUESTION (BP · Commodity)

What is Cushing?

WHAT THEY TEST / ANSWER

A three-word answer wastes the opportunity. Cushing, Oklahoma, is the delivery point of the NYMEX WTI futures contract: a landlocked tank farm at a pipeline crossroads, historically called the pipeline crossroads of the world, and the physical location that gives the WTI contract its meaning. Then say why anyone cares. Because delivery is physical and at that one point, the WTI price is the price of oil *there*, so Cushing inventory and spare capacity drive the front of the WTI curve, weekly EIA stock numbers for Cushing are a traded data release, and pipeline constraints in and out of it explain the persistent WTI discount to Brent despite WTI being the lighter, sweeter grade. It is also why the negative print of April 2020 happened on WTI and not on Brent. Being able to link a place name to a contract specification to a price relationship is exactly what a commodity desk is listening for.

REAL INTERVIEW QUESTION (BP · Commodity)

What is the capacity of a barrel?

WHAT THEY TEST / ANSWER

42 US gallons, which is about 159 litres. Answer instantly, because this question is not a knowledge test but a filter: it separates candidates with genuine interest in the physical market from those who prepared only the pricing theory. Since you have five spare seconds, spend them on a second unit to show it was not a lucky guess: a metric tonne of crude is roughly 7.3 barrels depending on density, and precious metals are quoted per troy ounce of 31.10 grams. The 42-gallon figure is historical, inherited from the tierce used in the Pennsylvania oil fields of the 1860s, and the standard survived because everyone had already written contracts in it.

REAL INTERVIEW QUESTION (BNP · Commodity)

What variables influence prices in the physical markets?

WHAT THEY TEST / ANSWER

Answer with a framework, because a list will sound like guessing. Split the drivers into four groups. **Supply**: production and OPEC or cartel policy, spare capacity, outages and maintenance, harvests and weather, mine grades, geopolitics and sanctions. **Demand**: the industrial cycle, seasonality (heating, driving, cooling, planting), substitution between fuels, and efficiency trends. **Inventories and logistics**, the piece purely financial candidates forget: storage levels and remaining capacity, pipeline and port constraints, freight rates, and the location differentials they create. **Financial conditions**: the financing rate for carry, the dollar, since most commodities are dollar-priced, and positioning and index rolls. Then close on the mechanism that ties them together: in the physical market these variables act through *inventory*, since anything that tightens the balance drains stocks, and a tight stock position is what raises the convenience yield and pulls the front of the curve above the back, developed in convenience yield and contango and backwardation.

INTERVIEW TRAP

Three errors that mark you as a purely financial candidate. First, applying the financial-asset forward formula and concluding the curve must be upward sloping: commodities have storage costs and convenience yields, so the curve can slope either way, and a backwardated curve is not an arbitrage. Second, saying "the oil price" without a grade, a location or a month, then being unable to answer which one you meant. Third, treating a futures hedge as perfect: flat price risk is hedged, basis risk is not, and basis is where physical books actually lose money.

VISUAL / CODE TO BUILD: Interactive

A "one commodity, three markets" panel for a single crude grade. Left, a map with the delivery points (Cushing, US Gulf, Rotterdam, Singapore) and the current location differentials on the links between them. Centre, the forward curve for the listed contract with a draggable expiry marker. Right, a hedge simulator: the reader sets a cargo size and a differential, sells the implied number of futures, then shocks flat price and basis independently with two sliders and watches the two legs and the net PnL update, reproducing the worked example above. The intended realisation is that the flat-price slider moves both legs and nets to zero, while the basis slider moves only one. A toggle for "storage full at Cushing" caps the deliverable volume and lets the front contract print negative.

Where this goes next

The mechanics of the contracts used here, margining, marking to market and the difference between a forward and a future, belong to futures. The costs of the physical leg are developed in storage costs, the benefit of holding inventory in convenience yield, and the two combine into the shape of forward curves and the vocabulary of contango and backwardation. The sector-specific physical detail follows in energy markets, metals and agricultural commodities, while the location and freight leg of the basis is shipping and freight.

14.2 Seasonality

trading · structuring · sales | core | commodity

Commodities are the one asset class where the calendar is a price factor in its own right. Nobody expects a share to be worth more in January than in July for reasons of weather, but gas is, and reliably. That single difference reshapes the forward curve, changes what a calendar spread means, and creates the most common trap in commodity interviews.

Where it comes from

Seasonality is not one phenomenon. It arrives from either side of the market and the two behave differently.

Demand seasonality is the familiar one. Natural gas and heating oil are consumed for heat, so demand peaks in winter. Gasoline demand peaks in the northern-hemisphere summer driving season. Electricity peaks twice a year in most temperate markets, for heating and for air conditioning, and its daily shape peaks twice a day.

Supply seasonality is the agricultural case and it works the other way. A crop arrives in a harvest window and is then consumed over twelve months, so supply is a spike and demand is a flat line. Prices are therefore lowest at harvest and rise through the crop year as inventory is drawn down. Hydroelectric power has the same shape driven by snowmelt.

> **KEY IDEA**
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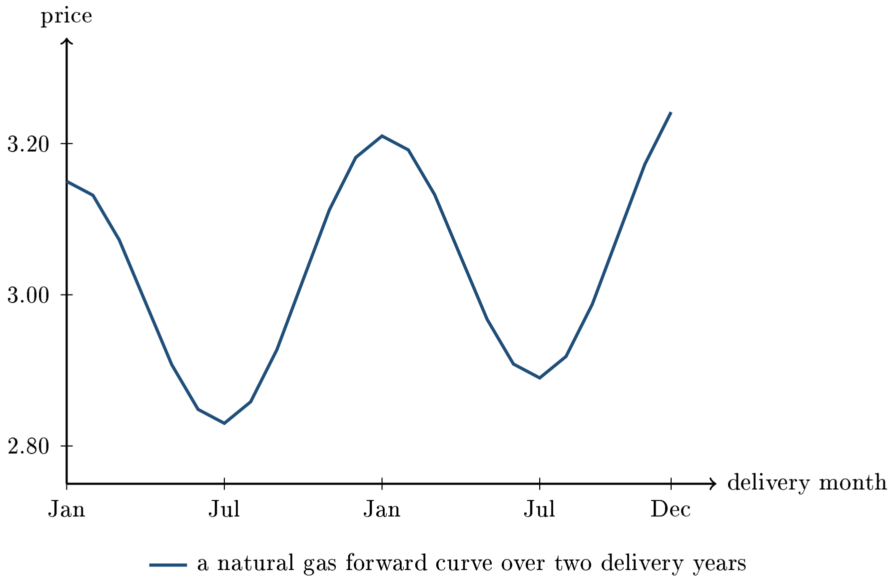
> Demand seasonality makes the high-price months the consumption months. Supply seasonality makes the low-price month the production month. Both produce a repeating annual pattern in the forward curve, but they peak at opposite ends of the physical cycle, so knowing which one you are looking at tells you which way the curve should lean.

What a seasonal forward curve looks like

For most assets the forward curve is a smooth line, sloping up or down. A seasonal commodity's curve is a repeating wave: each delivery month is priced separately, and the pattern repeats every twelve months on top of whatever overall slope the market has.



Two features are worth naming. The winter contracts are priced above the summer contracts on *both* sides of the picture, so this is a shape rather than a trend. And the second winter is slightly above the first, which is the ordinary carry slope sitting underneath the seasonal wave. A commodity curve is usually both things at once, and separating them is the first thing a commodity analyst does. The slope itself is contango and backwardation, and the general construction of the curve is forward curves.

The trap: seasonality is priced, not forecast

Here is the mistake, and it is made in interviews constantly. "Gas is cheap in summer and expensive in winter, so buy gas in summer and sell it in winter."

You cannot. The January contract already costs more than the July contract, today, on the screen. The winter premium is not a forecast waiting to come true; it is the current price of January delivery. Buying the July contract and holding it does not turn it into a January contract: it settles in July, at the July price.

KEY IDEA

A known, repeating seasonal pattern is worth nothing on its own, because everybody knows it and it is already in every quoted month. The tradeable quantity is the **difference between the seasonal shape the curve is pricing and the one you think will realise**. Seasonal strategies are bets on the amplitude being wrong, not on the season arriving.

This is the commodity version of a general principle. A forward price is not a prediction, it is a price, and anything predictable is already inside it.

Storage is what caps seasonality

The reason the winter premium cannot be arbitrarily large is that somebody can buy summer gas, store it, and deliver it in winter. That trade puts a ceiling on the seasonal spread, and the ceiling is the full cost of carry: physical storage, insurance, losses, and the financing of the inventory. The costs themselves are in storage costs.

WORKED EXAMPLE

July gas trades at \$2.80 and January at \$3.15, a seasonal spread of \$0.35. Storage costs \$0.04 per unit per month and money costs 5% a year.

Buy in July, store for six months, deliver in January:

- Storage: $0.04 \times 6 = \$0.24$.
- Financing the \$2.80 tied up for half a year: $2.80 \times 0.05 \times 0.5 = \$0.07$.
- Total carry: **\$0.31**.

The spread is \$0.35 and the carry is \$0.31, so the trade earns \$0.04. Thin, but positive, so it gets done until the buying of July and selling of January compresses the spread to about \$0.31 and the profit disappears. **That is why the observed seasonal spread sits at roughly the cost of carry: storage is the arbitrage that enforces it.**

Now suppose a cold forecast pushes January to \$3.60 while July stays at \$2.80. The spread is \$0.80 against a carry of \$0.31, an apparent free \$0.49. It is not free, and the reason is the sentence that answers most commodity questions: **storage is finite**. If the tanks and caverns are already full, nobody can put the trade on, the arbitrage that normally holds the spread down is unavailable, and the spread stays wide until either the weather changes or the season passes.

Seasonal spreads therefore behave in two regimes. When storage has room, they sit near the cost of carry and are boring. When storage is full or nearly empty, they detach and can move violently, which is where almost all the money in seasonal commodity trading is made and lost.

What happens when you cannot store

Electricity is the extreme case, and the cleanest way to see what storage was doing.

Power cannot be stored at scale, so there is no trade linking one delivery period to another. Nothing caps the difference between a summer afternoon and a winter evening, and nothing caps the difference between 6pm and 3am on the same day. The consequences are visible in the prices:

electricity has by far the strongest seasonal and daily shape of any traded commodity, its volatility is an order of magnitude above other markets, prices spike enormously on a hot still afternoon, and they can go **negative** when inflexible generation cannot be turned off and demand is low.

> **KEY IDEA**
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> Storability is the variable that decides how seasonal a market looks. A perfectly storable commodity has its seasonality flattened to the cost of carry. A non-storable one has none of that discipline, so each delivery period is effectively its own market with its own supply-and-demand balance. Everything between those two poles, and most commodities are between them, is a question of how much storage there is relative to the seasonal swing.

Volatility is seasonal too

The pattern is not only in the price level. In weather-driven markets the *uncertainty* is seasonal as well, and it has to be, because winter gas demand depends on temperatures that are themselves far more variable than summer ones. So implied volatility on winter gas contracts trades well above summer contracts, and options on different delivery months of the same commodity are not options on the same volatility.

Agricultural markets show the same thing around their own calendar: implied volatility rises into pollination and harvest windows, and into scheduled government crop reports, then falls once the number is known. That is the same event-volatility pattern as equity earnings, run off a different calendar.

> **INTERVIEW TRAP**
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> Quoting one volatility for a commodity. There is no such thing: there is a volatility per delivery month, and the term structure of a seasonal commodity is not monotone in maturity, it is periodic. A December contract can have a higher implied volatility than a June contract expiring later, which breaks every intuition built on equity term structures.

Interview angle

REAL INTERVIEW QUESTION (Louis Dreyfus · Commodity)

Explain the principle of a calendar spread with an example.

WHAT THEY TEST / ANSWER

Define it, work an example, and then say what it is really a bet on, because the third part is what distinguishes a commodity answer from a generic derivatives one.

Definition: buy one delivery month and sell another of the same commodity. You are long the difference between two dates rather than long the commodity, so a parallel move in the whole curve leaves you flat and only the *shape* matters.

Example: buy January gas at \$3.15 and sell July at \$2.80, so you have paid \$0.35 for the January premium. If by expiry that premium has widened to \$0.50 you have made \$0.15 per unit whether the outright level went to \$5 or to \$1. Be explicit about which leg you bought, because the same two prices describe the opposite trade if you reverse them.

What it is a bet on, which is the answer they want. It is not a bet that winter gas is more expensive than summer gas, because that is already in the two prices you traded. It is a bet on the spread against the **cost of carry**. Work it: storage at \$0.04 a month for six months is \$0.24, financing \$2.80 at 5% for half a year is \$0.07, so full carry is \$0.31 against a spread of \$0.35. Anyone with storage capacity can buy July, store, and deliver January for a profit of \$0.04, and doing that pushes the spread back toward \$0.31.

Finish with the two things that break the arbitrage, since that is where the trade actually lives. **Storage capacity is finite**, so when tanks are full nobody can put the trade on and the spread can sit far above carry. And a spread can go the other way entirely: if the market is backwardated, the near month is above the far one and no amount of storage helps, because you cannot store backwards. The premium for holding the physical rather than the paper in that situation is the convenience yield.

What a physical trader actually watches. The full list of variables that move a physical commodity price, supply, demand, logistics and quality, belongs with physical versus derivative commodities. One of the four groups is this section's own: **inventory and storage**. The stock level relative to normal is what tells you how much of a shock the system can absorb, and storage capacity relative to the seasonal swing is what decides whether the seasonal spread sits at carry or detaches from it. That is the variable this section has spent its length on, and it is the one a seasonal trade is really a bet about.

INTERVIEW TRAP

Three seasonal errors. (1) Treating a known seasonal pattern as a trading signal, when it is already in every quoted month. (2) Annualising or comparing prices across delivery months as though they were the same asset, when a January contract and a July contract are different goods delivered at different times. (3) Forgetting that the arbitrage which flattens seasonality is capacity-constrained, so the relationship that has held for years fails exactly when storage fills, which is exactly when the position is largest.

Where this goes next

The cost that sets the ceiling on a seasonal spread is storage costs, and the premium that appears when the physical is scarce is convenience yield. How the whole curve is built and read is forward curves and contango and backwardation. The two markets where seasonality dominates everything else are energy markets and agricultural commodities.

14.3 Storage costs

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A share costs nothing to keep. A million barrels of crude costs money every single day it exists, and someone has to pay it. Storage is therefore not a footnote to commodity pricing, it *is* the pricing mechanism: it is the physical operation that connects the price today to the price in three months, it sets a hard ceiling on how steep a forward curve can get, and when it runs out the ceiling disappears and prices do things that look insane from a screen. Every commodity interview eventually asks a version of one question: given the cost of storing this, is the calendar spread worth trading?

What storage actually costs

Start with the intuition, because candidates who only memorise a formula miss half the components.

> **INTUITION**
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> You are a **warehouse landlord** with a tank that holds a million barrels. What do you charge someone to keep their oil in it for a month? You need the rent on the tank, because the space has an alternative use. You need the insurance, because tanks leak and burn. You need to be paid for the handling, since pumping oil in and out costs energy and staff. If it is *your* oil, you also need the interest on the cash you tied up buying it, which is usually the largest single item. And you need compensation for the barrels that quietly disappear, through evaporation, through settling, through the water and sediment that accumulate at the bottom of the tank. Add those up and you have the cost of carry. The forward price cannot exceed today's price plus that number by much, because if it did, everyone with a tank would do the trade until it stopped being there.

Formally, the carry stack has five components. Only the first is purely financial, which is why purely financial candidates under-count it.

- **Financing.** Interest on the cash locked up in the inventory, at the rate you can actually borrow at, secured on the commodity. For a \$80 barrel at 5% this is about \$0.33 per barrel per month, and it rises and falls with the policy rate, which is why high rates make holding inventory expensive and physically drain stocks.
- **Rent.** The tank, warehouse or silo, quoted in currency per unit per month. This is a market price with its own supply and demand, and it is the item that explodes in a glut.
- **Insurance.** Small in normal times, non-trivial for high-value metals and for anything sitting in a war-risk zone.
- **Handling and throughput.** Fees to pump or lift in and out, usually a one-off per cycle rather than a monthly charge. It matters because it makes short storage cycles disproportionately expensive.

- **Loss and degradation.** Evaporation for light products, boil-off for LNG, moisture and insect damage for grain, and for crude the slow accumulation of water and sediment that downgrades the quality. This is a physical leak in the position, and for perishable agricultural products it can dominate everything else.

> **KEY IDEA**
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> Storage cost as a *percentage of value* is the number that determines a commodity's behaviour. Custody on gold runs a few tenths of a percent a year, so gold's carry is essentially just financing and its curve is almost always in orderly upward slope. Crude at \$80 with roughly \$0.80 a barrel a month of all-in carry costs about 1% of its value *per month*, near 12% a year, so oil's carry is enormous and its curve flips shape regularly. Cheap-to-store relative to value means a well-behaved curve; expensive-to-store means a volatile one.

The cash-and-carry bound

Here is the arbitrage that makes storage the pricing mechanism. Suppose the forward price F for delivery at T is high relative to spot S. Then today you can:

1. borrow S and buy one unit of the physical commodity at S;
2. pay to store it until T, at a total cost U expressed in time-T money;
3. simultaneously sell one unit forward at F;
4. at T, deliver the unit you have been holding, receive F, repay $S(1 + rT)$ and the storage bill.

Nothing here is uncertain: the profit is locked at inception. This is the forward cost-of-carry relation with one commodity-specific change, a storage term that a financial asset does not have. So no-arbitrage requires

$$F \;\leq\; S\,(1 + rT) \;+\; U,$$

or in the continuous-compounding form desks usually write, with u the storage cost as a continuous rate,

$$F \;\leq\; S\,e^{(r+u)T}.$$

The right-hand side is called **full carry**. A market trading at that level is said to be "at full carry", and it can be no higher without handing out free money.

> **KEY IDEA**
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> This bound is **one-sided**, and that asymmetry is the most important structural fact in commodities. To arbitrage the other direction, a forward that is too *cheap* relative to spot, you would have to short the physical commodity: borrow a million barrels, sell them, and buy them back at T. There is no reliable borrow market for barrels. So the reverse trade is unavailable, F can fall arbitrarily far below S, and a steeply downward-sloping curve is not an arbitrage but a signal about physical scarcity. The gap between full carry and the actual price is what the convenience yield measures; the shapes themselves are contango and backwardation.

Deciding whether to store: the calendar spread trade

WORKED EXAMPLE

Should you buy the calendar spread and store? Crude, three-month horizon.

Spot price S	\$80.00 / bbl
Financing at 5% p.a. for 3 months: $80.00 \times 0.05 \times 0.25$	\$1.00
Tank rent at \$0.35 / bbl / month for 3 months	\$1.05
Insurance at \$0.05 / bbl / month for 3 months	\$0.15
Throughput in and out, one-off	\$0.20
Total cost of carry, 3 months	**\$2.40**
Full-carry 3-month forward	**\$82.40**

The fifth component, loss and degradation, is set to zero here: over three months in a well-run crude tank it is small enough to round away. On a grain or a light product it would be a line of its own, and on a perishable it would be the largest line.

The market's three-month futures is at \$83.00, so the calendar spread (three-month minus prompt) is \$3.00 against a carry of \$2.40. The trade is on: buy the prompt barrel, sell the three-month, store, deliver, and bank

$$\$3.00 - \$2.40 \;=\; \$0.60 \text{ per barrel,}$$

locked at inception. On a 500,000 barrel tank that is $0.60 \times 500{,}000 = \$300{,}000$.

WORKED EXAMPLE

The same trade, two refinements. First, the breakeven, which is what a trader actually quotes. Divide the carry by the horizon: $\$2.40/3 = \$0.80$ per barrel per month. You need at least 80 cents a month of contango to make storing worthwhile. Quote the breakeven as a monthly spread and you have answered the question in one number.

Second, full cost versus marginal cost. Suppose the three-month is at \$82.00 instead, a \$2.00 spread against \$2.40 of carry, so on full cost the trade loses 40 cents. If you have *already leased* the tank, the rent is sunk, and the decision should use only the avoidable costs: financing \$1.00, insurance \$0.15 and throughput \$0.20, which is \$1.35. Against a \$2.00 spread you make 65 cents by filling a tank you are paying for anyway. Full cost decides whether to *lease* capacity; marginal cost decides whether to *use* it. Confusing the two is the classic trading-house interview trap.

The tank is an option, not a cost line

Storage capacity is not a fixed expense, it is the right to buy at one date and sell at another whenever the spread pays. That right has no obligation attached: if the curve is backwardated you simply leave the tank empty. So a lease on storage is economically a **strip of calendar-spread options**, and its fair value is the expected payoff of exercising the carry trade optimally, which rises with the *volatility* of the spread and not only with its current level. Trading houses value tanks and pipelines exactly this way. It also explains a fact that puzzles newcomers: storage can be worth leasing at a rent higher than today's spread justifies, because you are buying the option on the spread widening. The industry name for the systematic version is "storage optionality" and it is the same reasoning as an option on a spread, priced with the machinery of options.

When storage runs out

The cash-and-carry bound holds only if there is somewhere to put the commodity. When tanks fill, the arbitrage cannot be executed at any size, the ceiling stops binding, and the contango can widen past full carry into what the market calls **super-contango**. This happened in 2008 to 2009 and again in the spring of 2020.

Two things follow, and both are excellent interview material. First, the marginal storage facility becomes a **ship**. A very large crude carrier holds roughly two million barrels, so a time charter at \$50,000 a day implies

$$\frac{\$50{,}000}{2{,}000{,}000 \text{ bbl}} = \$0.025 \text{ / bbl / day} \;\approx\; \$0.75 \text{ / bbl / month},$$

which is in the same range as land storage and therefore a genuine substitute. In the 2020 glut, VLCC rates were quoted above \$200,000 a day, which is \$0.10 per barrel per day, about \$3.00 a barrel a month, so floating storage only paid because the contango had widened even further. The freight market and the oil curve are joined at the hip, developed in shipping and freight.

Second, at the very limit the price of the expiring contract can go negative, because a holder obliged to take delivery with nowhere to put the barrels must pay to be relieved of them. That is the mechanism behind the April 2020 WTI settlement, told in full in physical vs derivative commodities.

Not everything can be stored

The whole apparatus above assumes storability, and how well a commodity stores is what gives each market its character.

- **Electricity** is essentially non-storable at scale. There is no cash-and-carry, so no arbitrage links one delivery period to the next, and power forward prices are driven by expected supply and demand in each hour rather than by carry. This is why power has the wildest curve shapes and the largest spikes of any market, see energy markets.
- **Natural gas** stores well but with physical injection and withdrawal limits and a strong seasonal pattern, so the summer-to-winter spread is close to a pure storage economics trade, see seasonality.
- **Metals** store almost indefinitely and cheaply relative to their value, so LME curves sit near full carry most of the time. The frictions here are institutional rather than physical: warehouse queues and the warrant system can make metal that legally exists practically unavailable.
- **Agricultural products** degrade, and the crop cycle resets supply once a year, so their curves reset at each harvest and old-crop and new-crop contracts are really different commodities.

REAL INTERVIEW QUESTION (Louis Dreyfus · Commodity)

Given insurance and storage costs over 3 months, when is it worth buying the calendar spread and storing?

WHAT THEY TEST / ANSWER

They are testing whether you can run the cash-and-carry decision as a trader, in a number, not recite a formula. Do it in three steps. Build the full carry over the horizon: financing on the cash tied up in the cargo, plus rent, plus insurance, plus throughput in and out, plus any expected loss. Convert it to a *per unit per month* breakeven by dividing by the horizon, since the market quotes you a monthly spread. Then compare: if the calendar spread exceeds the breakeven, buy the prompt, sell the deferred, store and deliver, and the profit is locked at inception because both legs and all costs are known today. With \$80 oil, 5% financing, \$0.35 a barrel a month of rent, \$0.05 of insurance and \$0.20 of throughput, the carry is \$2.40 over three months, so you need more than 80 cents a month of contango. Then volunteer the two refinements that identify a real candidate. If the tank is already leased the rent is sunk, so the operating decision uses only avoidable costs while full cost governs whether to lease at all. And the risks that remain are not price risk but basis and operational risk: the grade you store must be the grade you deliver, the delivery windows must line up, and financing has to be committed for the whole period.

REAL INTERVIEW QUESTION (BP · Commodity)

What is the current problem with Cushing?

WHAT THEY TEST / ANSWER

A live-market question, so answer with the structure and be explicit that the level needs checking on the day. Cushing, Oklahoma, is the physical delivery point of the NYMEX WTI contract and a landlocked tank farm, so its *utilisation rate* is the binding constraint on the front of the WTI curve, and the recurring problem is that it sits too close to one of its two limits. Too full, as in the spring of 2020, and longs facing delivery have nowhere to put barrels, storage rents spike, the contango widens past full carry, and the expiring contract can trade to extreme or even negative levels. Too empty, as in several recent years, and you approach tank bottoms, the minimum volume needed to keep pipelines and tanks operable, which is functionally unavailable inventory: the deliverable supply becomes thin, the front tightens into backwardation and the contract is vulnerable to a squeeze. Then say the operationally correct thing: quote the weekly EIA Cushing stock number and the working-capacity utilisation, note the direction of the last few prints, and say you would read it against the shape of the front spread rather than in isolation.

REAL INTERVIEW QUESTION (BP · Commodity)

In a tanker, are there barrels or tanks?

WHAT THEY TEST / ANSWER

Tanks. A crude carrier is a hull subdivided into large cargo tanks, typically with segregated ballast tanks, and the oil is pumped in and out as bulk liquid. Nobody has shipped crude in physical barrels for well over a century: the barrel survives purely as a *unit of account*, 42 US gallons, not as a container. Say that in one breath, because the question is a check on whether your interest in the physical market is real or decorative. Then convert the answer into something useful by linking it to storage economics: because a vessel is a set of tanks, a chartered ship is a floating storage facility, and dividing the daily charter rate by the roughly two million barrels a VLCC holds gives its storage cost per barrel per day. That is the number a trader compares against land tank rent and against the contango when deciding whether to store at sea.

INTERVIEW TRAP

Four traps. First, **forgetting financing**: candidates quote the tank rent and omit the interest on the cash, which is usually the biggest line and the reason a rate rise drains inventories. Second, **treating the arbitrage as two-sided**: storage gives an upper bound on the forward price only, since you cannot borrow the physical to short it, so backwardation is information rather than free money. Third, **mixing full and marginal cost**: sunk rent should not enter the decision to fill a tank you already pay for. Fourth, **assuming the bound always holds**: when storage is physically full there is no trade to do at size, and the curve can and does go past full carry.

VISUAL / CODE TO BUILD: Interactive

A storage-economics calculator with a curve overlay. Inputs: spot price, financing rate, tank rent, insurance, throughput and expected loss, plus a horizon slider. Outputs: the carry stack as a stacked bar, the full-carry forward price, and the breakeven contango per month, all live. The overlay plots the observed forward curve against the full-carry line so the reader sees at a glance where the market is inside the bound (store), at the bound (full carry), above it (storage full, super-contango) or below it (backwardated, no trade available). A “tank already leased” toggle switches the breakeven from full to marginal cost and shows the decision flipping, and a “floating storage” toggle swaps land rent for a VLCC day rate divided by two million barrels.

Where this goes next

Storage is one half of the cost of carry. The other half, the benefit of actually holding the physical commodity, is convenience yield, and the two together determine the shape of forward curves and the vocabulary of contango and backwardation. The seasonal storage cycle is seasonality, the marine leg is shipping and freight, and the market structure that makes delivery and therefore storage matter at all is physical vs derivative commodities.

14.4 Convenience Yield

trading · structuring core commodities

A financial forward is fully determined by arbitrage: buy the asset, fund it, carry it, deliver it. Commodities break that, and the convenience yield is the single term that measures by how much. It is the reason chapter 14 exists as something other than an appendix to forwards, and it is the reason an oil curve can slope downward for years without anyone being able to arbitrage it.

What it is

KEY IDEA

The **convenience yield** is the benefit of holding the physical commodity rather than a claim on it. It is a yield in the sense that it accrues to the holder of the barrels and not to the holder of the contract, and it is not observable: it is defined as whatever makes the cost-of-carry equation hold.

$$F = S\,e^{(r+u-y)T},$$

where r funds the position, u is the storage and insurance cost as a rate, and y is the convenience yield. Set $y = 0$ and you have the financial forward formula, which is exactly what a commodity is not.

What the holder is actually buying with it is **optionality on scarcity**. A refiner with tanks full of crude can keep running when a pipeline goes down, can meet a contract when the spot market is bid, and can sell into a squeeze. Someone holding a December future can do none of those things in July. That flexibility has a value, it varies with how tight the market is, and it is what y measures.

The arbitrage bound goes one way only

This is the mechanical heart of the section, and it is where most candidates stop one step short.

If the forward is *too high*, above full carry, the arbitrage is real: buy the physical, fund it, store it, sell the forward, deliver. Anyone with storage can do it, so the market cannot stay there. That gives a hard ceiling,

$$F \le S\,e^{(r+u)T}.$$

If the forward is *too low*, the mirror trade would be to sell the physical short, invest the proceeds and buy the forward. But you cannot short a barrel you do not have. Borrowing physical crude is not a market in the way borrowing stock is, and if you already hold inventory, selling it to capture the spread means giving up precisely the flexibility that made the curve backwardated in the first place. So there is **no floor**, and no-arbitrage gives an inequality rather than an equation.

INTUITION

Convenience yield is what fills the gap left by the missing half of the arbitrage. On an equity you can borrow the stock, so both bounds bind and the forward is pinned. On oil only the upper bound binds, and everything below it is a matter of physical supply and demand. That asymmetry, in one sentence, is why commodity curves have shapes that financial curves cannot have.

Backing it out of a real curve

Since y is not quoted, the working method is to invert the equation: take the observed forward, compute what full carry would have been, and call the difference the implied convenience yield.

WORKED EXAMPLE

Brent, implied convenience yield from two points on the curve. Spot at \$80.00 per barrel, financing at 5% a year, storage and insurance at \$0.35 per barrel per month.

One month. Financing is $80.00 \times 0.05/12 = \$0.333$ and storage is \$0.35, so full carry would put the one-month forward at

$$80.00 + 0.333 + 0.350 \;=\; \$80.68.$$

The one-month forward is trading at **\$79.20**. The implied convenience yield is therefore $80.68 - 79.20 = \$\mathbf{1.48}$ per barrel per month, which annualises to $1.48 \times 12/80 = \mathbf{22.2}\%$.

Six months. Financing $80.00 \times 0.05 \times 0.5 = \$2.00$, storage $6 \times 0.35 = \$2.10$, so full carry is \$84.10. The six-month forward is at **\$76.50**, so the implied convenience yield is $84.10 - 76.50 = \$7.60$ per barrel over six months, or **19.0%** annualised. In continuous terms, with storage as a rate, $u = 0.35 \times 12/80 = 5.25\%$, so $r + u = 10.25\%$ and $y = r + u - \ln(F/S)/T = 19.2\%$, which is the same answer to two tenths of a point.

Two things to say out loud about this. First, the convenience yield is *lower* at six months than at one, which is the normal shape: scarcity is a now problem, and the market does not believe it persists. Second, and this is the check that shows you understand the bound, the market being \$7.60 below full carry is not a mispricing and there is no trade in it. Selling the barrels you hold to capture it means giving up the very flexibility the \$7.60 pays you for.

Roll yield: the convenience yield as it reaches your P&L

Nobody receives a convenience yield as a cash flow. It shows up as the **roll**, and this is where the concept becomes a number a desk actually books.

Hold a futures position rather than physical and you must roll it before expiry, selling the expiring contract and buying the next one. When $y > r + u$ the curve is downward sloping, the deferred contract is cheaper than the one you are leaving, and a *long* futures position rolls into a lower price and earns. When $y < r + u$ the curve is upward sloping and the same long position pays to roll. Chapter 14 gives the shapes their names in contango and backwardation; what this section adds is that the roll yield is not a separate phenomenon but the same y arriving as P&L, which is why a long commodity index in a contangoed market bleeds even when the spot price never moves.

REAL INTERVIEW QUESTION (BP · Commodity)

I am long Brent. How do I hedge?

WHAT THEY TEST / ANSWER

Answer the mechanics in one line and then spend the time on what actually goes wrong, because the mechanics are not what is being tested.

The hedge is to sell futures, most simply the front-month ICE Brent contract, sized to your barrels. If you hold physical, the hedge is short futures against long cargo; if you are long a paper position, it is the offsetting contract. That is the easy part.

Then the four things that make it imperfect, which is the answer.

Basis. The futures track dated Brent, and what you own is a specific grade at a specific place on a specific date. Quality differentials and location differentials move independently of the flat price, so you have converted price risk into basis risk rather than eliminating risk. State that trade explicitly.

The roll. A hedge held past the front contract has to be rolled, and the roll is not free. Short futures against physical in a backwardated market means buying back the expiring contract high and selling the next one lower, so the hedge bleeds every month. That bleed is the mirror of the convenience yield your barrels are earning, so it is a real cost of the hedge and not an accident of execution, and it is the number to quote if asked how expensive the hedge is.

Maturity mismatch. If your exposure is a year out and the liquidity is in the front months, you either stack and roll, which concentrates the roll risk, or you use a less liquid deferred contract and pay a wider spread.

Margin. The futures leg is margined daily and the physical is not, so a rally in the price you are hedged against produces immediate cash calls on a position that is economically flat. Naming this is what separates someone who has worked near a commodity book from someone who has read about one.

Close with the choice you did not make: you could hedge with a swap and avoid the roll and the margin at the cost of a bank credit line and a wider price, or buy puts and keep the upside at the cost of premium. Saying which one you would pick and why makes it an answer rather than a list.

How much of it there is, by commodity

The size of y is a good proxy for how physical a market is.

- **Gold** has a convenience yield of essentially zero. It is cheap to store, above-ground stocks are enormous relative to consumption, and it can be lent, so both arbitrage bounds bind and the gold forward is a financial forward. Gold is almost always in contango, and that is not a market view.
- **Crude and refined products** carry a large and violently variable convenience yield, because storage is finite, disruptions are real and a refinery that stops is expensive to restart.
- **Natural gas and power** take it furthest. Power cannot be stored at all, so the cost-of-carry relation does not link one delivery period to another, and each maturity is close to a separate asset. Gas storage is seasonal, which is why the shape belongs with seasonality as much as with carry.

- **Agriculturals** are dominated by the harvest cycle: the convenience yield collapses just after harvest when everything is available and rises through the season as stocks draw down.

INTERVIEW TRAP

1. **Treating the convenience yield as an observable.** It is a residual, computed from a forward price and an assumed storage cost. Get u wrong and the whole error lands in y, so quote it as implied and say what storage assumption produced it.
2. **Claiming an arbitrage in backwardation.** There is no floor, because you cannot short the physical. This is the most common way to fail the follow-up question.
3. **Assuming a downward-sloping curve forecasts a falling price.** It does not. It says carrying inventory is currently valuable. The curve is a statement about the present state of the physical market, not a forecast, and forward curves makes the same point from the other direction.
4. **Using one convenience yield for a whole curve.** It has a term structure, usually declining, as the worked example shows. A single number fitted to the front and applied at two years will misprice everything in between.
5. **Forgetting it can be negative.** If storage is full, holding the physical is a burden rather than a convenience, and that is the state that let front-month WTI trade below zero in April 2020.

VISUAL / CODE TO BUILD: Interactive

A carry calculator with the curve drawn from it. Inputs are spot, the financing rate, storage per unit per month and a convenience yield; the output is the forward curve out to twenty-four months, plotted against the full-carry curve computed with $y = 0$ so the gap between the two lines *is* the convenience yield, made visible as an area rather than a number.

The intended sequence is: start with gold-like inputs, cheap storage and zero y, and see the two lines coincide in a gentle contango. Then raise y and watch the traded curve pivot below the full-carry line and go into backwardation, with the crossover happening exactly at $y = r + u$, which is the condition worth remembering.

A second control lets the user give y a term structure, high at the front decaying to a long-run level, and the curve develops the concave shape real oil curves have. Dragging y negative pushes the front of the curve above full carry and the panel labels the state "storage full".

14.5 Forward curves

trading · structuring · sales core commodity

Ask what an equity is worth and there is one answer. Ask what oil is worth and the honest answer is a question: for delivery when, and where. A commodity's price is not a number, it is a **curve** of prices for delivery at each future date, and almost everything a commodity desk does is expressed in the shape of that curve rather than in its level.

What the curve actually is

A commodity forward curve is a strip of separately traded contracts, one per delivery month, each specifying a quantity, a grade and a delivery location. It is not a discount curve bootstrapped from instruments; each point is a price somebody is quoting for a physical obligation on a particular date.

That has an immediate consequence. Every point on the curve is its own contract with its own liquidity, so the front months trade tightly and the back of the curve can be thin enough that its quoted prices are model output rather than trades. Never treat a five-year commodity forward as though it were as real a price as the front month.

The cost-of-carry relation, and why it only works one way

The relation linking a forward price to the spot is

$$F_T \;=\; S\,e^{(r+u-y)T},$$

where r is the financing rate, u the storage cost rate, and y the **convenience yield**, the benefit of holding the physical rather than a claim on it. The storage side is developed in storage costs and the convenience yield in convenience yield.

For a financial forward, the equivalent relation is enforced from both sides: if the forward is too high you buy the asset and sell it forward, and if it is too low you short the asset and buy it forward. Both trades are available, so the forward is pinned.

For a commodity, only one of them is.

- **Forward too high**: buy the physical, store it, sell it forward. Available to anyone with storage, so it caps the forward at $S\,e^{(r+u)T}$. This bound is real but it is *capacity-constrained*: it stops working when the storage is full.
- **Forward too low**: you would need to sell the physical you do not have and buy it forward. That requires borrowing the commodity, and there is no deep lending market for crude oil or wheat. So there is no lower bound at all.

> **KEY IDEA**
>
> A commodity forward curve is bounded above by the cost of carry and unbounded below. This single asymmetry explains most of what makes commodity curves different from every other forward curve in finance: they can sit in deep backwardation indefinitely with no arbitrage available to close it, the convenience yield is whatever residual is needed to make the equation balance rather than an independently observed quantity, and the upper bound itself fails when storage fills.

Reading the curve as an inventory signal

Because the upper bound comes from storage and the lower bound does not exist, the slope of the curve is a statement about the physical balance rather than about expectations.

- **Upward sloping**, with the far months above the near ones, says holding the physical is not worth much: inventory is comfortable and the market is paying you to store. A *steep* upward slope says storage is close to full, because the spread has pushed past the normal cost of carry and the arbitrage that would flatten it has run out of capacity.

- **Downward sloping**, with the near months above the far ones, says the physical is scarce now: whoever needs it needs it today, and no amount of storage economics can create supply that does not exist.

The names for those two shapes, and the debate about what else drives them, are in contango and backwardation. The point here is narrower and more useful: the slope is the market's inventory report, published continuously, and a commodity analyst reads it before reading anything else.

Roll yield: the return nobody expects

Here is the practical fact that separates commodities from every other asset. **You cannot hold a commodity by holding a contract.** Futures expire, so a position held over time must be repeatedly closed in the expiring month and reopened in the next one. That operation is the **roll**, and it has a return of its own that has nothing to do with whether the commodity went up.

If the curve slopes down, you sell the expensive near contract and buy the cheap far one, so each roll is a gain. If it slopes up, each roll is a loss.

WORKED EXAMPLE

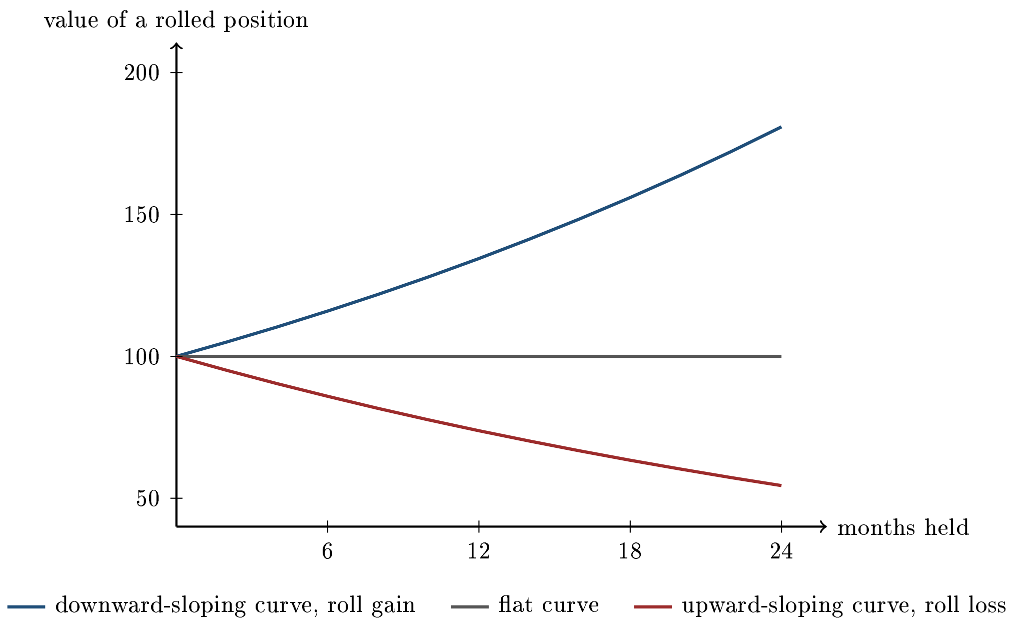
Front-month crude at \$80, second month at \$78, so the curve is downward sloping by \$2 a month. Suppose the whole curve stays exactly where it is, so the commodity itself does nothing.

Each month you sell the front at \$80 and buy the next at \$78, and a month later that contract has become the front and is worth \$80 again. You gain \$2 measured against the \$80 front price, **2.5% per month**. (Measured against the \$78 you paid it is 2.56%; quote which base you are using, because on a wide spread the two differ visibly.) Compounded, that is

$$1.025^{12} = 1.345,$$

a **34.5%** annual return from a market that did not move.

Reverse the curve, front at \$80 and second at \$82, and the identical logic runs backwards: you lose 2.5% a month, and $0.975^{12} = 0.738$, a **26.2%** annual loss from a market that did not move.



All three lines describe a commodity whose price never changes. After two years the rolled positions are worth 181, 100 and 54. That divergence is entirely the roll, and it is why a commodity index fund can lose most of its value in a market that ended where it started.

> **INTERVIEW TRAP**
>
> Calling roll yield free money. It is not a profit, it is a payment: in a downward-sloping market the roll gain is what you are paid for bearing the risk of holding a claim on something that is currently scarce, and it disappears the moment the curve flattens. Any strategy sold on its historical roll return is short exactly that scarcity, and it has been wiped out more than once when the curve flipped.

When the upper bound broke

On 20 April 2020 the expiring May contract for West Texas Intermediate settled at **minus $37.63** a barrel. Holders were paying to give oil away.

Nothing about it is mysterious once the asymmetry above is understood. WTI settles by physical delivery at Cushing, Oklahoma, and Cushing's tanks were effectively full as the pandemic collapsed demand. A holder of the expiring contract therefore had to take delivery of barrels with nowhere to put them, and the cost of not being able to store was unbounded, so the price that made someone willing to accept them was negative.

The lesson is not that prices can be negative, though they can. It is that the storage-based arbitrage which normally disciplines a commodity curve is a *capacity*, and every relationship built on it holds only while that capacity has room. That is the same constraint that caps seasonal spreads in seasonality, arriving here at the front of the curve instead of across the year.

Interview angle

REAL INTERVIEW QUESTION (CACIB · Commodity)

Is the oil price elastic?

WHAT THEY TEST / ANSWER

No, and saying why is the whole answer, because inelasticity is what makes the market behave the way it does.

Demand is inelastic in the short run. People drive to work and heat their houses whatever the price does this week; the vehicle fleet, the heating systems and the industrial plant are all fixed capital that takes years to change. **Supply is inelastic in the short run too**, and this is the half people forget: a well already drilled produces at close to capacity almost regardless of price, and bringing new production on takes months or years. So both curves are close to vertical over the horizon a trader cares about.

The consequence is the interesting part. When two nearly vertical curves have to be made to intersect, a small shift in either one produces a very large move in price. That is why a supply disruption of a few percent of world output can move the price by tens of percent, and it is the structural reason commodity volatility is high in a way that has nothing to do with anybody's model.

It also explains the curve. Because quantities cannot adjust in the short run, the adjustment falls on **inventory**, and the forward curve is the price of inventory across time. That is why a stock draw shows up immediately as backwardation and a stock build as contango: the curve is where the inelasticity is absorbed.

Finish with the long run, which is the follow-up. Elasticity rises with the horizon: high prices eventually bring in new supply and demand substitution, so the far end of the curve is far more anchored than the front. That is another way of saying that the back of the curve moves much less than the front, which is a fact a trader uses every day.

INTERVIEW TRAP

Four curve errors. (1) Quoting "the oil price" without a month, when the front and the twelve-month can differ by a fifth. (2) Assuming a commodity forward is arbitrage-pinned in both directions, when only the upper bound exists and even that is capacity-limited. (3) Treating a historical roll return as a property of the asset rather than as a payment for a scarcity that may not persist. (4) Hedging a long-dated physical exposure with the front month and calling it hedged, when what is left is a curve position that can be larger than the one you removed.

Where this goes next

The two components of the carry relation are storage costs and convenience yield, and the names and interpretations of the curve's slope are in contango and backwardation. The repeating shape that sits on top of the slope is seasonality, and the market where all of this is most visible is energy markets.

14.6 Contango and backwardation

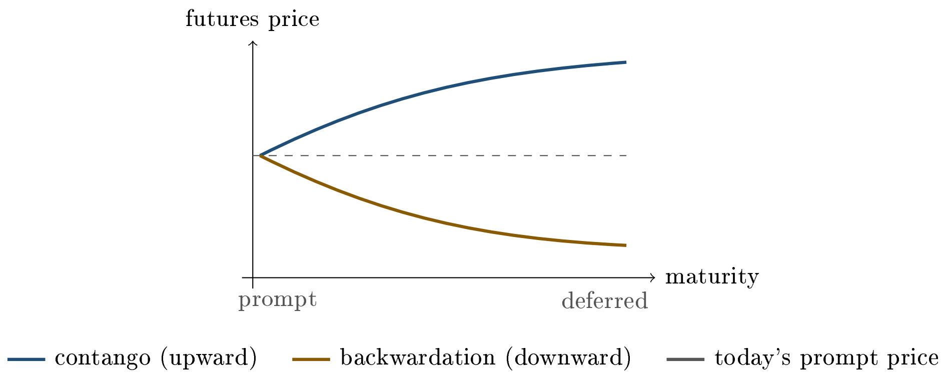
sales · trading · structuring | core | commodities

These two words are the vocabulary test of every commodities interview, and almost every candidate passes the definition and fails the follow-up. The definition takes ten seconds. What matters is the second layer: the shape of a commodity curve tells you the state of the physical market, it decides whether a long futures position quietly bleeds or quietly earns, and it is the reason an investor can be right about the oil price and still lose most of their money.

The two shapes

Take the futures prices of one commodity for a sequence of delivery months and plot them against maturity. That is the forward curve, built in forward curves. Two names describe its slope:

- **Contango**: the curve slopes *upward*, far months trade above near months. Deferred oil is more expensive than prompt oil.
- **Backwardation**: the curve slopes *downward*, near months trade above far months. Prompt oil is more expensive than deferred oil.



Note the wording carefully, because it is where marks are lost. Contango and backwardation describe the *observable slope of the curve*, nothing else. They are not forecasts, and a contangoed curve does not mean the market expects prices to rise. It usually means the opposite of drama: it means the physical market is comfortable.

Where the shape comes from

The carry relation assembled in storage costs and convenience yield is

$$F(T) \;=\; S\,e^{(r+u-y)T},$$

with r the financing rate, u the storage cost rate and y the convenience yield, the non-financial benefit of physically holding the commodity. Read the sign of the exponent and both shapes fall out immediately:

- $r + u > y$, so the cost of holding exceeds the benefit: **contango**. This is the state of a well-supplied market with comfortable inventories. Nobody needs to own the barrel today, so you have to be paid to store it, and that payment is the upward slope.
- $y > r + u$, the benefit of holding exceeds the cost: **backwardation**. This is a tight market. Inventory is scarce, being the one with physical product has real value, and buyers pay a premium for prompt delivery.

KEY IDEA

The curve is an inventory gauge. Contango means ample stocks, backwardation means scarcity. And there is an asymmetry that follows directly from the one-sided cash-and-carry bound: contango is capped at full carry, because beyond that anyone with a tank arbitrages it away, while backwardation has *no bound at all*, because you cannot borrow a physical commodity to sell it short. That is why commodity curves flip suddenly and violently into backwardation during a squeeze and only ever grind slowly into contango.

Contango is not "normal backwardation"

A genuine differentiator, and a trap set deliberately by interviewers who know their history. **Normal backwardation** is a different statement from backwardation, made by Keynes in the 1930s, and it is not about the slope of the observed curve at all.

Backwardation compares two things you can see: the futures price and today's spot price. Normal backwardation compares the futures price with something you cannot see, the *expected* spot price at delivery, and claims

$$F(T) \ < \ \mathbb{E}\big[S_T\big].$$

The argument is about risk premia rather than carry: producers are structurally long the commodity and want to hedge, so they are natural sellers of futures. Speculators take the other side, and being asked to hold price risk they demand compensation, which they receive as a futures price set below the expected future spot, so that the position drifts up towards the spot on average. Hicks made the symmetric point for markets where consumers dominate the hedging flow, which gives "normal contango".

So a market can be in contango (upward-sloping curve) and in normal backwardation (futures below expected spot) at the same time, with no contradiction. One statement is about carry, the other about who needs the hedge and what the risk premium is. Say that sentence in an interview and you have separated yourself from the field.

Roll yield: the shape becomes a cash flow

Futures expire, so any investor who wants continuous exposure must repeatedly sell the expiring contract and buy a later one. That operation is the **roll**, and it is where the shape of the curve turns into real money.

Decompose the total return of a collateralised long futures position into three pieces:

$$\underbrace{\text{collateral return}}_{\text{cash earns } r} + \underbrace{\text{spot return}}_{\text{the flat price move}} + \underbrace{\text{roll return}}_{\text{what the shape does to you}} .$$

Only the third piece is new. In contango you sell the cheap expiring contract and buy the expensive deferred one, so you are systematically buying high: the roll return is **negative**. In backwardation you sell the expensive prompt and buy the cheaper deferred: the roll return is **positive**. Equivalently, and this is the cleaner way to see it, a futures price must converge to spot at expiry, so a deferred contract bought above spot has to fall towards it if the spot does not move.

WORKED EXAMPLE

The cost of being right about oil. Spot crude is \$80.00, the one-month futures is \$81.00, and suppose that shape is stable: the spot stays at \$80 and the one-month contango stays at \$1.00 every month.

You buy the one-month contract at \$81.00. A month later it expires at the spot of \$80.00, so you have lost \$1.00. You roll into the new one-month at \$81.00 and repeat. Per month that is

$$\frac{\$1.00}{\$80.00} = 1.25\%,$$

quoted against spot as the market does. Measured against the \$81.00 you actually paid it is 1.235%, which is the more honest denominator and the one to name if pressed. Compounded over a year, $0.9875^{12} = 0.860$, so a drag of about 14% a year. The spot price of oil did not move a cent over the whole year, you were never wrong about direction, and you lost a seventh of your money to the shape of the curve.

Flip the curve. If the one-month traded at \$79.00 instead, a \$1.00 backwardation, the same mechanical roll *earns* 1.25% a month on a flat spot. This is why systematic commodity investors care more about curve shape than about their price view, and why carry strategies in commodities are built on ranking markets by the steepness of their front spread.

KEY IDEA

Roll yield is not free money, and claiming it is will end the conversation. It is only the third term of a decomposition whose second term is the spot move, and the two are linked. A market that stays in steep contango indefinitely with an unchanged spot is paying storage to someone, and in equilibrium the persistent negative roll of the long investor is the mirror image of a risk premium earned by the physical player who has the tank. Harvest the roll and you are being paid to provide something real, either storage capacity or the willingness to bear price risk when hedgers want out. That is a respectable trade, not an arbitrage.

Why an oil ETF can lose money while oil rises

This is the practical consequence a sales or structuring interviewer will push you towards. An exchange-traded product that promises "oil exposure" usually holds front-month futures and rolls them monthly. In steep contango that roll bleeds, so over any long horizon the product substantially underperforms the spot price it appears to track. In the spring of 2020, with the front of the WTI curve in the deepest contango on record (the gap between the prompt contract and the one a year later widened to well over ten dollars a barrel), this stopped being a technicality: the largest US oil ETF had to spread its holdings across several delivery months rather than concentrating on the prompt, and it carried out a reverse share split after its price collapsed. Retail investors who correctly called the rebound in crude still lost money, because they had bought the roll, not the barrel.

The lesson generalises beyond commodities: whenever a product delivers exposure through a rolling derivative rather than through the asset itself, ask what the roll costs. See ETFs and index products.

Trading the shape rather than the level

Because the slope is informative and mean-reverting in a way the flat price is not, desks trade it directly.

- **Calendar spreads** express a view on the shape with the flat price largely netted out, so margin is lower and a crash in the outright price need not hurt.
- **The physical arbitrage**, buying the prompt and storing it against a sold deferred contract, is the same trade with the tank attached, and it is what enforces the full-carry ceiling. Priced in storage costs.
- **Roll timing.** Index products roll on published schedules, which is a large predictable flow, so the spreads they trade tend to be pushed around on those dates. So-called roll-enhanced or deferred-roll indices exist precisely to avoid being the predictable side of that trade.
- **Seasonal shapes.** In natural gas and agriculture the curve is not a simple slope but a repeating annual pattern, and treating a summer-to-winter spread as "contango" misses the point entirely. See seasonality.

REAL INTERVIEW QUESTION (UBS · Commodity)

Can you talk about contango and backwardation in commodity curves?

WHAT THEY TEST / ANSWER

An open invitation, so the grading is on structure and on how many layers you can reach unprompted. Go in four. Definitions: contango is an upward-sloping curve with deferred above prompt, backwardation the reverse. Cause: from $F = Se^{(r+u-y)T}$, contango means financing plus storage exceed the convenience yield, which is a comfortable well-stocked market, while backwardation means the convenience yield dominates, which is physical tightness. Asymmetry: contango is capped at full carry by the cash-and-carry trade while backwardation is unbounded because the commodity cannot be borrowed and sold short, which is why curves snap into backwardation and grind into contango. Consequence: the shape is a cash flow through the roll, negative for a long in contango and positive in backwardation, which is why an oil ETF can lose money in a year when oil is flat. If you still have their attention, add that contango is not the same as normal backwardation, since one is about carry and the other about the risk premium relative to the expected spot.

REAL INTERVIEW QUESTION (BNP · Commodity)

What is contango? Backwardation? Right now, what does the oil market look like?

WHAT THEY TEST / ANSWER

Two definitions and a live-market question, and the live part is the real test since it checks whether you follow the market or only studied it. Give the definitions in one sentence each with the inventory reading attached: contango is deferred above prompt and signals ample stocks, backwardation is prompt above deferred and signals scarcity. For the current state, never guess. Say which observables you would quote and what they imply: the front spread on Brent and WTI in dollars per barrel and its sign, the twelve-month spread for the shape further out, OECD and Cushing inventories against their five-year range, and OPEC spare capacity. Then give the conditional reasoning that shows you can think rather than recite: a market in backwardation with low inventories and little spare capacity is one where a supply disruption moves the front violently and the back barely at all, whereas a contangoed market with full storage absorbs shocks in the front and needs a demand recovery rather than a supply cut to fix it. Finish by stating what the levels were the last time you looked and when that was. Being explicit about the as-of date reads as discipline, not ignorance.

REAL INTERVIEW QUESTION (Unicredit · EQD)

What is backwardation?

WHAT THEY TEST / ANSWER

Asked from an equity-derivatives desk, so the expected answer is short, correct and aware that the concept travels across asset classes. Backwardation is a downward-sloping forward curve: prompt trades above deferred. In commodities it means the convenience yield exceeds financing plus storage, so the physical market is tight, and it cannot be arbitraged away because the commodity cannot be borrowed and sold short. Then make the cross-asset link that explains why an EQD desk is asking. The same word is used for the *volatility* term structure: the VIX curve or an implied-volatility term structure is in backwardation when short-dated implied vol trades above long-dated, which is the signature of a stress episode, since panic is priced as immediate and assumed to fade. The economics are different, the geometry is identical, and knowing both is what the question is fishing for.

REAL INTERVIEW QUESTION (UBS · Commodity)

What kind of shapes do commodity markets have, and what are the specificities you learned about them?

WHAT THEY TEST / ANSWER

They want a taxonomy plus evidence of genuine exposure to the asset class. Name four shapes: contango, backwardation, seasonal (a repeating annual pattern rather than a monotone slope, as in natural gas or agriculture), and humped or kinked, where a near-term disruption or a harvest boundary distorts one part of the curve while the rest is unaffected. Then the specificities that distinguish commodities from financial curves. The curve is bounded above by full carry but unbounded below. Its shape reads inventory, so it is information rather than a forecast. It is not a single curve but a family, since each grade and each location has its own, so the location and quality basis matters as much as the time dimension. Non-storable commodities such as power have no carry arbitrage at all, so their curves are pure expectations of supply and demand hour by hour and can be spiky in ways no financial curve is. And the front is far more volatile than the back, since the far end anchors to a long-run marginal cost of production while the prompt clears against physical constraints. Close with a concrete case you understand properly, such as the 2020 super-contango and the storage limit that caused it.

INTERVIEW TRAP

Four traps, in the order interviewers spring them. First, saying **contango means the market expects higher prices**: the shape is set by carry against convenience yield, not by a forecast, and under normal backwardation the futures can sit below the expected spot in a contangoed market. Second, confusing **backwardation** with **normal backwardation**. Third, calling roll yield **free money** instead of compensation for storage or risk-bearing. Fourth, getting the **sign of the roll backwards**: in contango a long sells low and buys high, so the roll costs the long money. Work through which leg you buy every single time rather than trusting memory.

VISUAL / CODE TO BUILD: Interactive

A curve-shape laboratory. Panel one: a draggable forward curve with a full-carry ceiling line computed from user-set financing and storage inputs, so the reader can try to pull the curve above the ceiling and see the arbitrage flag light up. Panel two: a rolling investment simulator that takes the curve shape and a spot path and plots three cumulative series, collateral return, spot return and roll return, alongside the total, reproducing the worked example where a flat spot and a stable \$1 monthly contango produce a 14% annual loss. Panel three: a historical replay of the WTI curve through 2008, 2020 and a backwardated year, animated month by month, with Cushing inventory plotted underneath so the reader watches the shape and the stock level move together.

Where this goes next

The two ingredients that set the slope are developed in storage costs and convenience yield, while the construction and interpolation of the curve itself is forward curves. Repeating annual shapes are seasonality. The market structure that makes delivery, and therefore the whole carry argument, binding is physical vs derivative commodities, and the investor-facing consequence of the roll is in ETFs and index products.

14.7 Energy Markets

trading · sales core commodities

Energy is the part of commodities that interviewers actually ask about, and the questions are unusually concrete: name the benchmarks, explain a spread, tell me who sets supply, tell me how big the market is. This section is built to answer those, and the organising idea is that oil is not one price but a *grid* of prices differing by quality, by location and by date. Every energy question is about one of those three axes.

The two benchmarks, and what actually separates them

Brent is a light sweet crude from the North Sea, priced free on board and **waterborne**, which is the property that matters: it can be loaded onto a tanker and sold anywhere, so it prices against the global market. Something like two thirds of internationally traded crude is priced off it.

WTI is a light sweet crude delivered at **Cushing, Oklahoma**, which is landlocked. Its price is the price of a barrel sitting in the middle of a continent, and getting it to the coast costs money and pipeline capacity. Physical versus derivative covers what Cushing is; what matters here is that the delivery point is the whole story of the spread.

Commodities owns the question of what explains the Brent-WTI spread, and the answer there is the right one: not quality, since both are light and sweet, but location. What this section adds is the consequence. The spread is an arbitrage bounded by infrastructure, so it is only as tight as the pipelines and export terminals allow. WTI traded slightly *above* Brent while the US was a net importer; shale then produced barrels faster than pipelines could move them, with a crude export ban still in force, and the discount went double-digit; repealing the ban and building pipelines brought it back to roughly transport cost. If the pipeline to the Gulf runs $1.50 a barrel and the Atlantic voyage $2.00, WTI has to sit about **$3.50** under Brent before US barrels clear to Europe, and that is where a normal market puts it. Cushing inventories, shale volumes, export capacity and freight rates are the four things that move it, and each of them is a piece of infrastructure rather than a view.

The supply side

REAL INTERVIEW QUESTION (BP · Commodity)

Tell me about OPEC. Who leads it?

WHAT THEY TEST / ANSWER

Facts first, briefly, then the part that has content.

The facts. OPEC was founded in 1960 in Baghdad by five countries, Iran, Iraq, Kuwait, Saudi Arabia and Venezuela, and is headquartered in Vienna. Membership has moved around, with Qatar, Ecuador and Angola all having left in recent years, so give the founding five rather than reciting a current list you may have wrong. Since 2016 the relevant body is **OPEC+**, which adds Russia and several non-OPEC producers, because OPEC alone is no longer a large enough share of supply to move the price.

Who leads it. There is a Secretary General in Vienna and a rotating conference presidency, but those are administrative. The answer they want is **Saudi Arabia**, and the reason is worth stating: Saudi Arabia is the only member holding meaningful *spare capacity*, which means the ability to raise or cut output quickly at will. Spare capacity is what makes a swing producer, and a cartel without one cannot defend a price.

Then the analytical point, which is what separates this from a Wikipedia answer. OPEC's power is limited by two things. It is a cartel, so every member has an incentive to cheat on its quota and most historically have. And US shale has a much shorter investment cycle than conventional oil, months rather than years, so when OPEC cuts to raise the price, shale responds within a year and takes the market share. That is why OPEC decisions move the front of the curve much more than the back: the market believes the cut and does not believe it lasts. Finish there, and you have told them something about the curve rather than about an organisation.

The cost curve, and two different breakevens

Oil supply is a cost curve. At any price, everything cheaper than that price produces and everything dearer does not, which is why the price gravitates toward the cost of the marginal barrel rather than toward an average.

REAL INTERVIEW QUESTION (BP · Commodity)

What is the estimated minimum oil price needed to stay profitable?

WHAT THEY TEST / ANSWER

The question is ambiguous on purpose, and the whole answer is noticing that there are **two** breakevens which differ by a factor of several.

Production breakeven, the price at which producing a barrel covers its cost. This is a curve, not a number, and the shape is what to convey. Saudi conventional onshore is at the bottom, on the order of $10 a barrel or less to lift. US shale sits in the middle, roughly $40 to $60 full cycle including the cost of drilling the well, but far lower once the well is drilled, since the capital is sunk and only operating cost matters. Deepwater and oil sands are at the top, roughly $50 to $70 and rising with project complexity. Quote these as approximate ranges, because they move with technology and with service costs, and being precise about them is a tell that you have memorised rather than understood.

Fiscal breakeven, the price a producing *state* needs to balance its budget. For Saudi Arabia this is in the region of $80, roughly eight times its production cost, because the budget is built on oil revenue rather than on oil profit.

Why the distinction is the answer. It explains behaviour that looks irrational otherwise. Saudi Arabia can produce profitably at $30 and still cut output hard to defend $80, because the constraint that binds is fiscal, not economic. And a shale producer with a drilled well keeps pumping well below its full-cycle breakeven, because the alternative is zero revenue against sunk capital, which is why supply is much stickier on the way down than the cost curve suggests.

Close on the market consequence: the short-run supply curve is steep, so demand shocks move the price violently, while the long-run curve is much flatter because shale can be switched on within a year. That is the structural reason oil volatility is high at the front and lower in the back of the curve.

The size of it

REAL INTERVIEW QUESTION (SCB Group · Commodity)

How many barrels of oil are produced per day in the world?

WHAT THEY TEST / ANSWER

This is an estimation question wearing a fact question's clothes. Give the number, then show you could have derived it, because the second part is what is being marked.

The number is about **100 million barrels a day**, and that has been the right order of magnitude for years. Quoting "about a hundred million" is better than quoting a precise figure that is out of date.

The derivation, if you did not know it. Eight billion people consuming an average of roughly four and a half barrels of oil a year, a figure anchored by the fact that a US resident uses around twenty and an Indian resident around one, gives

$$8 \times 10^9 \times 4.5 \;=\; 36 \times 10^9 \text{ barrels a year,}$$

which divided by 365 is **98.6 million barrels a day**. Landing within a few percent of the true figure from two remembered anchors is exactly the skill the question tests.

Then make it a market. At \$80 a barrel that is $100 \times 10^6 \times 80 = \$8$ billion of crude changing hands every day, close to **\$2.9 trillion** a year in physical crude alone, before refined products and before the paper market, which is several times larger again. That framing is what a commodity desk means when it says oil is the largest commodity market in the world.

Gas and power, which behave differently

The interview database is almost entirely about oil, but a complete answer on energy needs one paragraph on the other two, and the distinguishing feature is storability.

- **Natural gas** can be stored, but only in limited and seasonal capacity, so it injects in summer and withdraws in winter. The forward curve is therefore shaped by the calendar rather than by carry alone, which is seasonality's subject. Gas is also regional in a way oil is not, since moving it requires a pipeline or liquefaction, so European, US and Asian prices can diverge by multiples rather than by freight cost.
- **Power** cannot be stored at all. That removes the cost-of-carry link between delivery periods entirely, so each hour is close to a separate commodity, prices can spike by orders of magnitude within a day and can go negative when generation cannot be turned off. The link between gas and power is the spark spread, which is the generator's margin and the analogue of the refiner's crack spread.

INTERVIEW TRAP

1. **Explaining the Brent-WTI spread by quality.** They are both light and sweet. The spread is location, and saying so immediately is the difference between a good and a bad answer.
2. **Quoting a price level as a fact.** Levels move. Know roughly where oil is, say "around" and give the level, and move immediately to structure. An interviewer asking for a level is checking that you look at screens, not testing recall.
3. **Confusing the two breakevens.** Production and fiscal breakevens differ by a factor of several for the same country, and conflating them makes OPEC policy look irrational when it is not.
4. **Treating OPEC as a price setter.** It is a cartel with a cheating problem facing a competitive fringe with a short investment cycle. It influences the front of the curve much more than the back.
5. **Assuming oil intuition transfers to power.** Non-storability breaks the carry relation, so nothing about convenience yield or contango applies to an hourly power curve in the way it does to crude.

VISUAL / CODE TO BUILD: Interactive

A supply-curve panel: cumulative world production on the horizontal axis and production cost on the vertical, drawn as a staircase from Saudi conventional at the left through Russian and other conventional, then US shale, then deepwater and oil sands at the right. A draggable horizontal line is the price. The area to the left of where it cuts the staircase is production, and the reader can drag the price down and watch which producers switch off in order.

The intended moment is dragging the price from \$80 to \$40 and seeing that almost nothing at the left switches off, which is the steep short-run supply curve made visible, and then toggling a "sunk capital" switch that drops shale's step to its operating cost and watching even less switch off, which is why supply is stickier on the way down than the curve suggests.

A second panel overlays fiscal breakevens for the major producing states as markers on the same price axis, so the gap between a country's production cost and the price it needs politically is a visible distance rather than an assertion.

14.8 Metals

trading · structuring · sales core commodity

"Metals" is one word covering two markets that have almost nothing in common. Copper is an industrial input whose price is set by how much of it is being consumed this quarter. Gold is a monetary asset whose price is set by what else you could have done with the money. They sit in the same chapter, trade on some of the same exchanges, and require opposite reasoning.

The variable that separates them is not chemistry. It is how much of the stuff already exists relative to how much is produced each year.

Stock to flow: the number that decides everything

Most commodities are consumed. Oil is burned, wheat is eaten, and what exists at any moment is a few weeks of consumption sitting in tanks and silos. Gold is not consumed. Almost every ounce ever mined still exists, in vaults and jewellery, and could in principle be sold tomorrow.

WORKED EXAMPLE

Roughly 216,000 tonnes of gold have been mined in all of history, and annual mine production is roughly 3,600 tonnes. So

$$\frac{216{,}000}{3{,}600} = 60 \text{ years of production already above ground.}$$

Copper is consumed at roughly 26 million tonnes a year, and commercial and exchange inventories together typically amount to a few weeks of that. Take three weeks:

$$\frac{3}{52} = 0.058 \text{ years of consumption in stock.}$$

The two ratios differ by a factor of about a thousand. Every behavioural difference between the two metals follows from that one comparison.

KEY IDEA

When the stock-to-flow ratio is large, this year's mine supply is a rounding error and the price cannot be set by the flow balance. It is set by whether existing holders want to keep holding, which makes gold behave like a currency rather than like a commodity. When the ratio is small, inventory can genuinely run out, so the price is set by the physical balance and can move violently when supply is interrupted.

What that does to the two forward curves

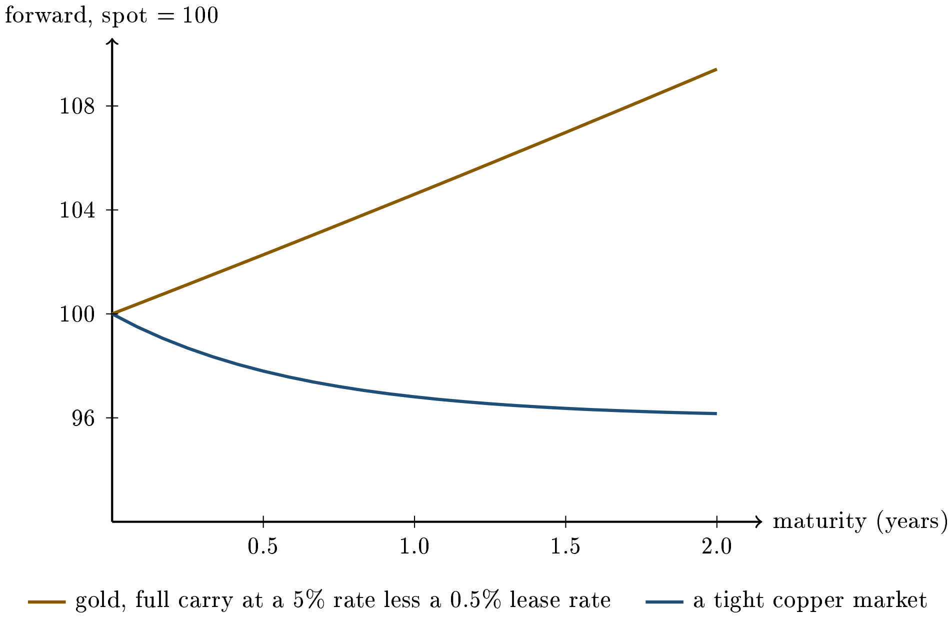
Recall from forward curves that a commodity forward is capped above by the cost of carry and has no lower bound, because you cannot borrow the physical.

Gold breaks that rule, and it is the only major commodity that does. There is a real lending market: central banks and vaulted holders lend gold, and the rate they charge is the **lease rate**. That makes the reverse arbitrage available, so the gold forward is pinned from both sides at

$$F_T = S\, e^{(r-\ell)T},$$

with ℓ the lease rate. Because the lease rate is small, gold sits in near-perfect contango at essentially the financing cost, and it stays there. Gold does not go into backwardation in any sustained way, because there is no scarcity to price: there are sixty years of it above ground.

Copper does the opposite. There is no meaningful copper lending market, inventories are thin, and when the physical is tight the front of the curve rises above the back and stays there for as long as the tightness lasts.



The gold curve is a clean exponential at the financing rate, because that is all it is. The copper curve is backwardated and flattens out, because the tightness is a present condition that the market does not expect to last two years. Neither shape is a forecast of the price; both are statements about availability. The vocabulary for the two shapes is in contango and backwardation.

Gold: what actually moves it

If gold is priced by holders' willingness to keep holding, then what matters is the opportunity cost of holding it. Gold pays no coupon, so its competitor is a safe asset that does.

- **Real interest rates** are the dominant driver, and the relationship is negative. When real yields rise, the income you give up by holding an asset that pays nothing rises with them, and gold falls. This is more reliable than the relationship with inflation, which is what most people quote.
- **The dollar**, negatively, and largely mechanically, since gold is quoted in dollars, so a stronger dollar makes it more expensive in every other currency and damps demand.
- **Crisis demand**, positively and unpredictably, since gold is the asset people buy when they distrust the alternatives.
- **Central bank buying**, which is a slow but large flow, and a policy decision rather than a market one. See central banks and FX policy.

Notice what is missing from that list: mine supply and jewellery demand. Both are real, and both are trivial against sixty years of above-ground stock. An analyst who forecasts the gold price from mine output has misunderstood the asset.

Copper: the opposite instrument

Copper is the industrial metal with the widest use, in construction wiring, motors, grid infrastructure and vehicles, and demand is closely tied to industrial and construction activity. That is why it is treated as a macro indicator, sometimes called the metal with an economics degree.

Two properties make it a good one and one property makes it a worse one than the nickname suggests.

Good: demand is broad rather than concentrated in one industry, and it is early in the construction cycle, because wiring goes in before a building is finished. **Also good**: supply is very slow to respond, since a new mine takes the better part of a decade, so demand surprises show up in the price rather than in output.

Less good: a large share of demand is concentrated in China, so the copper price is often reporting on one country's construction and grid investment rather than on the world economy. The copper-to-gold ratio, which pairs the industrial metal against the monetary one, is used as a cleaner growth signal for exactly that reason, and it tracks real yields reasonably well.

The exchange matters more here than anywhere else

Base metals trade principally on the London Metal Exchange, and the LME's conventions are genuinely unlike other exchanges. It trades **daily prompt dates** out to three months rather than monthly contracts, because it grew out of financing physical shipments that arrived on particular days. It operates a network of **approved warehouses**, so an LME position can end in metal in a shed, and it retains open-outcry ring sessions that set reference prices. The exchange itself is described in exchanges.

Those features matter because they make squeezes possible. Deliverable stock sits in known warehouses in known quantities, so a participant who controls enough of it and enough of the front-month open interest can force shorts to buy back at whatever price they choose.

WORKED EXAMPLE

In March 2022 the LME nickel price roughly doubled in a matter of hours, at one point trading above $100,000 a tonne, as a very large short position held against physical production ran into a market that could not deliver against it. The exchange suspended trading and then **cancelled several hours of executed trades**, reversing them.

The lesson is not about nickel. It is that on a physically settled exchange with a warehouse system, the ultimate risk in a short position is not the price, it is *deliverability*, and the ultimate backstop is not the clearing house, it is the exchange's discretion to change the rules. Both are risks that no model contains and that a candidate who has only studied financial derivatives will not think to name.

Precious metals beyond gold

Silver has a foot in both camps: it is a monetary metal with a long history and an industrial input, notably in solar panels and electronics. That dual identity makes it more volatile than gold, since it responds both to gold's monetary drivers and to industrial demand, and the gold-to-silver ratio is watched as a gauge of which of the two is in charge.

Platinum and palladium are essentially industrial, dominated by autocatalyst demand, and their supply is highly concentrated by geography. The markets are small enough that a supply disruption or a substitution decision by carmakers moves them enormously, which is why the two have repeatedly traded above and below each other as the automotive industry switched between them.

Interview angle

REAL INTERVIEW QUESTION (Société Générale · Commodity)

Tell me a bit about gold.

WHAT THEY TEST / ANSWER

Open with what makes it different rather than with its uses, because the entire answer follows from one structural fact.

Gold is not consumed. Roughly 216,000 tonnes have ever been mined and roughly 3,600 tonnes are mined each year, so there are about sixty years of production already above ground. Annual supply is therefore about 1.7% of the stock. That means the flow balance cannot set the price: what sets it is whether existing holders want to keep holding, which is a monetary question and not an industrial one.

So what moves it. Real interest rates, negatively, and this is the relationship to lead with because gold pays no income and a rising real yield raises the cost of holding it. The dollar, negatively, since gold is quoted in dollars. Crisis and distrust, positively. Central bank buying, which is large and slow. Mine supply and jewellery demand barely feature, and saying so is the point.

Two technical facts that mark you out. First, gold is the one commodity with a real lending market, so it has a **lease rate** and its forward is pinned from both sides at $S\,e^{(r-\ell)T}$ rather than only capped from above like every other commodity. That is why gold sits in stable contango at roughly the financing cost and does not go into sustained backwardation. Second, that puts gold much closer to a currency than to copper, which is why it is often traded off an FX desk rather than a commodity desk.

Expect the follow-up "so is it a good inflation hedge?", and answer honestly: over very long horizons it holds purchasing power, but over the horizons anyone actually invests over it tracks real rates far more reliably than it tracks inflation, and there have been long stretches where inflation rose and gold fell.

REAL INTERVIEW QUESTION (Huxley · Commodity)

Tell me about copper.

WHAT THEY TEST / ANSWER

Set it against gold, because the contrast is the fastest way to show you understand both.

What it is: the most widely used industrial metal, in electrical wiring, motors, grid infrastructure and vehicles, valued for conductivity and ductility. Consumption is roughly 26 million tonnes a year, and inventories amount to a few weeks of that, so unlike gold it is genuinely consumed and can genuinely run out.

What that means for the price: it is set by the physical balance. Demand tracks industrial and construction activity, supply is extremely slow to respond because a new mine takes close to a decade, and the two together mean a demand surprise lands almost entirely in the price. That is also why the curve can be backwardated for extended periods, which gold's never is.

Why anyone outside the metal cares: copper is used as a growth indicator, because its demand is broad and it goes in early in the construction cycle. Give the caveat unprompted, because it is the part that shows judgement: a large share of demand is Chinese, so copper is often reporting on one country's property and grid investment rather than on global growth. That is why the copper-to-gold ratio is a preferred signal, since it divides the industrial metal by the monetary one and tracks real yields more cleanly than either alone.

The structural story if they push further: electrification is copper-intensive, since grids, renewables and electric vehicles all use several times the copper of what they replace, while new supply is constrained by permitting, declining ore grades and the lead time on new mines. That is the long-run bull case, and the honest counterweight is substitution to aluminium where conductivity per kilogram matters less than cost.

REAL INTERVIEW QUESTION (Huxley · Commodity)

What is the correlation between the dollar and gold?

WHAT THEY TEST / ANSWER

Negative, typically, and the answer is worth more if you separate the mechanical part from the economic part instead of just giving the sign.

The mechanical part: gold is quoted in dollars, so if nothing about gold changes and the dollar strengthens against everything else, the dollar price of gold falls, because it has become more expensive for every non-dollar buyer. That much is close to an accounting identity and it guarantees some negative correlation.

The economic part: both are driven by the same underlying variable. When US real yields rise, the dollar attracts capital and strengthens, while gold, which pays nothing, becomes less attractive and falls. So the negative correlation is largely two assets responding to one driver rather than one causing the other, and **real rates** are the variable to name.

Then the caveat, which is what the question is really probing. The relationship is not stable and it breaks in the episodes that matter most. In a global crisis both the dollar and gold can rally together, because both are safe havens and the flight to safety overwhelms the rate channel. It has also weakened in periods of heavy central bank gold buying, where the buyer is diversifying away from the dollar and is deliberately indifferent to the price. Say that a correlation estimated over a calm sample will mislead you in exactly the scenario you bought gold for, and quantify the estimate rather than asserting it: over a year of daily data a sample correlation carries a standard error of a few points, so distinguishing -0.3 from -0.4 is not a distinction the data supports.

INTERVIEW TRAP

Four metal errors. (1) Reasoning about gold from mine supply and jewellery demand, which are each under 2% of the above-ground stock. (2) Expecting gold to go into backwardation, when its lending market rules that out. (3) Reading copper as a global growth signal without saying how much of the demand is one country's. (4) Treating an LME short as a purely financial position, when it is an obligation to deliver metal to a warehouse and the risk is deliverability rather than price.

Where this goes next

The curve mechanics both metals sit inside are forward curves, with the cost side in storage costs and the scarcity premium in convenience yield. The exchange conventions referred to above are in exchanges, and the other half of the commodity complex is energy markets.

14.9 Agricultural commodities

sales · trading · structuring | core | commodities

Agriculture is where the physical logic of commodities is at its purest. Supply arrives once a year and then only decays, quality is measured in several dimensions and each one has a price, the same product is worth different amounts in different countries on the same day, and a government can destroy or create demand with a single decree. The market is dominated not by banks but by trade houses, and their interviews are unusually concrete: expect to be handed yields, freight rates and prices and asked whether the trade makes money. This section gives you the framework and the arithmetic to answer that.

The three families

Family	Examples	Typical quotation
Grains and oilseeds	corn, wheat, soybeans	US cents per bushel (CBOT)
Oilseed products	soybean meal	$ per short ton (CBOT)
	soybean oil	US cents per pound (CBOT)
Softs	sugar, coffee, cocoa, cotton	cents per pound or $ per tonne (ICE)
Livestock	live cattle, lean hogs	US cents per pound (CME)

Units matter here more than anywhere else in commodities, and they are a standard filter question. A **bushel** is a unit of volume that was long ago fixed by weight per crop, so one tonne is a different number of bushels for each grain: corn and sorghum weigh 56 pounds per bushel, giving $2{,}204.62/56 = 39.37$ bushels per tonne, while wheat and soybeans weigh 60 pounds, giving $2{,}204.62/60 = 36.74$. Soybean meal trades in dollars per *short* ton of 2,000 pounds while soybean oil trades in cents per pound, so any crush calculation needs a conversion before it needs an opinion. Raw sugar trades in cents per pound and white sugar in dollars per metric tonne, which is exactly why the premium between them has to be computed rather than read off a screen.

What makes agriculture different

INTUITION

Oil is produced continuously from a tap that can be opened and closed. A crop is a **single annual delivery into a warehouse that leaks**. Once the harvest is in, the total supply for the year is known and fixed, and from then on it can only be consumed or spoil: no producer can respond to a price rise by growing more this year. All they can do is plant more *next* year, which arrives too late to help. That is why agricultural prices spike so violently on a weather shock and then collapse a season later.

Five structural features follow, and each has a market consequence.

- **The crop year, and old crop versus new crop.** Contracts before the harvest are drawn on this year's fixed supply; contracts after it are drawn on a crop that does not exist yet. So the curve does not simply slope, it *breaks* at the harvest boundary. The old-crop and new-crop contracts are effectively different commodities, no storage arbitrage links them (you cannot store what has not grown), and their spread is a pure supply-expectation trade. Within a crop year the carry logic of storage costs applies normally.
- **Weather is the supply variable**, and it is unforecastable beyond a few weeks. Since major producers sit in both hemispheres, the calendar has two harvests for some crops (North American soybeans harvested around September to November, South American around February to May), which smooths supply and means the market spends half the year watching one continent's rainfall and half watching another's.
- **Quality is multi-dimensional and priced.** Protein in wheat and in soybean meal, oil content in oilseeds, moisture, foreign material, damaged kernels, polarisation in raw sugar, grade and screen size in coffee. Contracts specify a standard and a schedule of premiums and discounts, so the physical price is a base plus a quality adjustment.

- **Government sits inside the market.** Export bans and quotas, import tariffs, minimum support prices, and above all biofuel mandates, which convert a food crop into an energy product by regulation and permanently link corn to ethanol and vegetable oils to diesel. A policy announcement can move an agricultural market more than any weather event.
- **Substitution is easy on the demand side.** Animal feed rations are formulated to hit a protein and energy target, so corn, wheat and barley compete directly, as do the vegetable oils. This caps how far any one crop can dislocate from its neighbours and creates a family of tradeable inter-commodity spreads.

Stocks-to-use: the one number to know

The standard measure of tightness is the **stocks-to-use ratio**: ending stocks divided by annual consumption, usually expressed as a percentage, and published for every crop and country in the balance sheets that the USDA releases monthly.

> KEY IDEA
>
> The price response to stocks-to-use is strongly **convex**. Between comfortable levels, moving the ratio a point or two barely moves price: there is plenty of grain and buyers are relaxed. Below a critical level, roughly the amount of pipeline inventory the system physically needs to keep functioning, the same one-point change produces a violent move, because someone has to be rationed out of the market and only price can do the rationing. This is the agricultural version of the mechanism behind convenience yield and it is why ag price distributions are so right-skewed: the downside is bounded by the cost of production, while the upside is bounded only by demand destruction.

The same logic explains the curve. A comfortable balance sheet gives an orderly contangoed curve near full carry, and a tight one flips the front into backwardation, as in contango and backwardation. Because the storage cost of grain includes real spoilage, ag full carry is wider than for metals.

Basis: the price where the crop actually is

An ag futures contract is deliverable at named locations under a defined specification. Everything real trades at a differential to it, called the **basis**:

$$\text{local cash price} = \text{futures} + \text{basis},$$

where the basis bundles freight from the farm to the delivery point, local storage availability, the quality of that particular parcel against contract standard, and local supply and demand. Basis is quoted directly ("December corn plus 20") and traded as its own market: a farmer can price the basis and leave the futures leg open, or vice versa. On a physical book, basis is the residual risk left after hedging flat price, exactly as in physical vs derivative commodities, and in agriculture it also embeds the export freight leg covered in shipping and freight.

Processing margins: the crush and the refining spread

Most agricultural trading is not a bet on a price, it is the operation of a plant that buys one thing and sells two or three others. The margin between them is the business, and computing it under a set of given yields and prices is the standard trade-house interview exercise. The general treatment of a processing margin as a spread, and as an option on that spread, is in crack spreads; here we do the agricultural arithmetic.

WORKED EXAMPLE

An oilseed crush margin. A plant processes 400 tonnes of beans a day, and you are told the process yields 56% meal and 42% biodiesel by weight. Those are the figures the exercise hands you, not the industry standard: a real soybean crush is closer to 79% meal and 19% oil, with the oil then esterified into biodiesel. Work with the numbers given and say so, which is the point of the exercise. Processing costs are \$30 per tonne of beans crushed. Output volumes are fixed by the yields:

$$\text{meal} = 0.56 \times 400 = 224 \text{ t}, \qquad \text{biodiesel} = 0.42 \times 400 = 168 \text{ t}.$$

Scenario A: beans \$450/t, meal \$400/t, biodiesel \$900/t.

- Revenue: $224 \times 400 + 168 \times 900 = 89{,}600 + 151{,}200 = \$240{,}800$.
- Cost: $400 \times 450 + 400 \times 30 = 180{,}000 + 12{,}000 = \$192{,}000$.
- Gross margin: \$48,800 a day, or $48{,}800/400 = \$122$ per tonne crushed.

Scenario B: beans \$520/t, meal \$360/t, biodiesel \$780/t.

- Revenue: $224 \times 360 + 168 \times 780 = 80{,}640 + 131{,}040 = \$211{,}680$.
- Cost: $400 \times 520 + 12{,}000 = \$220{,}000$.
- Gross margin: $-\$8{,}320$ a day, or $-\$20.80$ per tonne crushed.

Always report the margin *per tonne of input* as well as per day, because that is the number comparable across plants and across the forward curve, and it is what tells you whether to keep crushing. A negative gross margin means the variable economics do not work and the plant should idle, whatever the fixed costs are, since running only increases the loss.

WORKED EXAMPLE

The white sugar premium, and whether to refine. Raw sugar is quoted in cents per pound and white sugar in dollars per metric tonne, so start by putting them in the same unit. With one tonne equal to 2,204.62 pounds, raw sugar at 20.00 cents per pound is

$$0.20 \times 2{,}204.62 \;=\; \$440.92 \text{ per tonne.}$$

If white sugar trades at \$520.00 per tonne, the **white premium** is $520.00 - 440.92 = \$79.08$ per tonne.

Now the refining decision. Refining loses weight, so take a yield of 0.93 tonnes of white per tonne of raw. To produce one tonne of white you must buy $1/0.93 = 1.0753$ tonnes of raw, costing $1.0753 \times 440.92 = \$474.12$. Add a refining cost of \$65 per tonne of white and your total is \$539.12 against a sale price of \$520.00: a loss of \$19.12 per tonne.

Read the same result as a breakeven, which is how the market talks. The premium must cover two things: the refining cost of \$65, and the value of the weight lost, which is $474.12 - 440.92 = \$33.20$. So the breakeven white premium is $65.00 + 33.20 = \$98.20$, against an observed \$79.08. The premium is \$19.12 too thin, so you do not refine, and if you own a refinery you either idle it or wait for the premium to widen. That consistency check, computing the answer twice by different routes and getting the same number, is worth doing out loud in an interview.

REAL INTERVIEW QUESTION (Louis Dreyfus · Commodity)

From soybean you get 56% meal and 42% biodiesel with a daily input of 400mt; given prices for 4 scenarios, what is the gross margin for each scenario?

WHAT THEY TEST / ANSWER

A pure execution test: they want to see you set up a repeatable calculation, run it without arithmetic errors, and interpret the sign. Build the frame once and reuse it for every scenario. Output volumes are fixed by the yields, so $0.56 \times 400 = 224$ tonnes of meal and $0.42 \times 400 = 168$ tonnes of biodiesel a day. Then for each scenario, gross margin equals meal tonnes times meal price, plus product tonnes times product price, minus input tonnes times bean price, minus processing cost times input tonnes. Report it *per tonne of beans crushed* as well as per day, since that is the comparable unit. With beans at \$450, meal at \$400, biodiesel at \$900 and \$30 of processing, that is \$240,800 of revenue against \$192,000 of cost, so \$48,800 a day or \$122 a tonne. Then add the judgement they are listening for: state your convention explicitly (weight basis, and that yields summing to 98% implies the remainder is loss), and if the yields you are handed differ from the industry standard, which for a soybean crush is closer to 79% meal and 19% oil, use the numbers given and say so in one sentence rather than silently substituting your own. Close by naming what the margin is worth commercially: it is the crush spread, it can be hedged by selling the products and buying the beans in the futures market, and a negative gross margin is a signal to idle the plant.

REAL INTERVIEW QUESTION (Louis Dreyfus · Commodity)

You must order two kinds of soybean meal (from Argentina and Brazil) with different costs, transport costs and protein values: which one do you choose?

WHAT THEY TEST / ANSWER

The trap is comparing the two prices, or even the two landed prices. The buyer of meal is not buying tonnes, they are buying **protein**, because a feed ration is formulated to a protein specification. So the correct comparison is the landed cost per unit of protein delivered. Take an illustration: origin A at \$380 a tonne with \$30 of freight and 46% protein lands at \$410, which is $410/46 = \$8.91$ per protein point, while origin B at \$370 with \$45 of freight and 44% protein lands at \$415, which is $415/44 = \$9.43$. Origin A wins despite quoting a higher price. Then show you know the adjustments a real trade needs. Contracts carry premium and discount schedules for protein, moisture and fibre, so the differential is contractual rather than notional. If the low-protein cargo forces the customer to add another protein source, the cost of that substitution belongs in the comparison. And the non-price terms often decide it: laycan and delivery timing against the customer's consumption, credit and country risk, GMO or sustainability certification where the buyer requires it, and vessel size against the draft of the discharge port. Finish with the general principle, since that is what transfers: in physical trading you always normalise competing offers to the unit the *end user* actually values, whether that is a protein point, an energy unit, or a barrel of refined product.

REAL INTERVIEW QUESTION (Louis Dreyfus · Commodity)

What is the difference between raw and white sugar (the white premium)?

WHAT THEY TEST / ANSWER

Answer in three layers. The products: raw sugar is unrefined cane sugar, trading as ICE No. 11 in cents per pound at a standard polarisation, while white (refined) sugar trades as ICE No. 5 in dollars per metric tonne. The premium: the white premium is the difference between the two once they are in the same unit, so you must convert the raw price at 2,204.62 pounds per tonne before subtracting. At 20 cents a pound raw is \$440.92 a tonne, so a \$520 white price is a \$79.08 premium. The economics, which is the real question: the premium is the market price of *refining capacity*, so it is bounded below by nothing in particular but pinned above by the cost of refining plus the value of the weight lost in the process, since above that level idle refineries switch on and compete it away. Add the two observations that mark someone who has traded it: the premium is seasonal and regional because refining capacity and consumption sit in different places, so it is really a spread between two locations as much as between two products, and the destination matters, since many importing countries protect domestic refiners with tariffs that fall on white sugar and not on raw.

REAL INTERVIEW QUESTION (Louis Dreyfus · Commodity)

Given refinery costs, is it worth transforming your raw sugar into white sugar?

WHAT THEY TEST / ANSWER

Turn it into a breakeven and compare, and be explicit that the answer is a number and not a view. Work per tonne of *output*, because that is what you sell. You need 1/yield tonnes of raw for each tonne of white, so at a 0.93 yield and raw at \$440.92 a tonne the input costs \$474.12. Add the refining cost, say \$65, for \$539.12 all in. Refine only if the white price exceeds that. Equivalently, and more usefully at a desk, the breakeven *premium* is the refining cost plus the value of the yield loss, $65.00 + 33.20 = \$98.20$, so with the premium at \$79.08 you are \$19.12 short and you do not refine. Then add the three refinements that show operational understanding. Use avoidable costs for the run-or-idle decision, since a refinery you already own has sunk capital and possibly contracted labour, and full costs only for the decision to build or lease capacity. Both legs can be locked in the futures market, No. 11 against No. 5, which converts a price view into a captured margin. And remember the by-product and logistics terms, molasses revenue, energy cost, and the freight and duty differential between shipping raw or white to the final destination, all of which enter the same comparison.

INTERVIEW TRAP

Four errors that give away a candidate who has only read about agriculture. First, **comparing prices in different units**: cents per pound against dollars per tonne, or short tons against metric tonnes, which is the single most common arithmetic slip in trade-house interviews. Second, **ignoring the yield loss** in a processing calculation, which understates the required margin by exactly the value of the lost weight. Third, **applying storage arbitrage across the harvest boundary**: old crop and new crop are not linked by carry, so their spread is not bounded by the cost of storage. Fourth, **treating a quality differential as noise**: protein, moisture and polarisation are contractual and priced, and in agriculture the quality basis is often larger than the move in the futures you hedged with.

VISUAL / CODE TO BUILD: Interactive

Two linked calculators, both driven by editable inputs so the reader can rerun the section's examples and then break them. First, a **processing-margin sheet**: enter input volume, yields, input and output prices and processing cost, and it returns margin per day and per tonne of input, flags a negative variable margin as an idle signal, and plots the margin against one price on a slider so the breakeven is visible. Second, a **unit converter and premium calculator** for sugar: raw in cents per pound to dollars per tonne, the observed white premium, the breakeven premium given refining cost and yield, and the refine-or-not verdict, with the yield-loss component broken out separately. Add a third panel plotting a crop's stocks-to-use ratio against price history, so the convexity below the critical level is something the reader sees rather than something they are told.

Where this goes next

The annual cycle that gives agriculture its shape is developed in seasonality, the carry arithmetic within a crop year in storage costs, and the resulting curve shapes in contango and backwardation. The general theory of a processing margin, including its treatment as an option on a spread, is crack spreads. The freight leg that sits inside every ag basis is shipping and freight, and the physical market structure underneath all of it is physical vs derivative commodities.

14.10 Shipping and Freight

trading · sales core commodities

Freight is the commodity that connects all the others. Every location spread in the asset class, the Brent-WTI discount in energy markets, the regional gas differentials, the premium one iron-ore port pays over another, is bounded above by what it costs to move the cargo. When freight is cheap those spreads compress toward zero; when freight is expensive they widen, and the arbitrage that was supposed to hold them together stops working.

It also has an unusual market structure worth understanding on its own terms: supply is close to perfectly inelastic in the short run, because a ship takes two to three years to build and thirty years to wear out, so essentially all of the short-run price movement comes from demand. That is why freight rates are among the most volatile prices in finance.

The size classes are named after chokepoints

Vessel classes look like arbitrary jargon until you notice they are almost all named after a physical constraint, either a canal or a port.

REAL INTERVIEW QUESTION (BP · Commodity)

What is the capacity of a MEGAMAX? Of a PANAMAX? When would you choose one over the other?

WHAT THEY TEST / ANSWER

Answer the sizes, then the principle behind the naming, then the trade-off, because the third part is the only one that is about shipping rather than about vocabulary.

The sizes. A **Panamax** is defined by the original Panama Canal locks: roughly 32 metres of beam and about 12 metres of draft. In dry bulk that is a vessel of about 60,000 to 80,000 deadweight tonnes; as a container ship it is around 5,000 TEU. A **Megamax** is at the other end entirely, the largest container class, roughly 24,000 TEU on a hull about 400 metres long and 61 metres in the beam. The 2016 canal expansion added a **Neopanamax** class at about 14,000 TEU, which is worth naming because it shows the classes track infrastructure and move when the infrastructure does.

The principle. The names are constraints, not sizes. **Capesize** carriers are called that because they are too large for the canals and must sail around the Cape of Good Hope, and **Suezmax** is the same idea applied to the Suez Canal. **Aframax** is the one to be careful with, since it looks like it belongs on that list and does not: the name comes from the Average Freight Rate Assessment, a tanker rate scale, not from a waterway, even though the class is in practice bounded by the ports and terminals it serves. Say this and you have told the interviewer you understand why the classes exist, and that you know which name is the exception.

The trade-off. Bigger is cheaper per tonne and less flexible. A Megamax has by far the lowest cost per container, and it earns that only on a high-volume trunk route between deep-water ports that can physically handle it, Asia to Northern Europe being the archetype. It cannot transit Panama at all. A Panamax carries far less but can call at many more ports and take many more routings, so it earns its keep on thinner or more variable trades.

The choice is therefore a question about *the route and the ports*, not about the cargo. Big ship on a fixed high-volume lane where you can guarantee it sails full; smaller ship where the volume is uncertain, the ports are shallow or the routing may change. A half-empty Megamax is worse than a full Panamax, which is the whole answer in one sentence.

The Baltic indices

REAL INTERVIEW QUESTION (BP · Commodity)

Do you know the BDI, the Baltic Dry Index? Where is it today and what is it used for?

WHAT THEY TEST / ANSWER

Three things: what it measures, where it is, and why anyone outside shipping watches it.

What it measures. The Baltic Dry Index is published daily by the Baltic Exchange in London and is a composite of freight rates for **dry bulk** carriers, assessed across a basket of standard routes and weighted across the Capesize, Panamax and Supramax classes. Be precise about the word dry: it covers iron ore, coal and grain, and it does *not* cover oil. Tankers have their own assessments, the dirty and clean tanker indices, and getting this wrong is the standard way to lose the question.

Where it is. Give a range and a shape rather than a number you might have stale. It is an index without units, so the level only means something relative to its own history: it has printed above 11,000 at the 2008 peak and below 300 at the 2016 trough, and it more usually sits in the low thousands. Say roughly where you last saw it and say when you last looked, because the honest version of that answer is much better received than a confident wrong figure.

Why it is watched. This is the part with content. Dry bulk cargo is almost entirely *raw material inputs* to industrial production, so demand for the ships is demand for the stuff you make things out of, ahead of the making. And because ship supply is fixed in the short run, the index moves almost purely on that demand. So the BDI is read as a leading indicator of global industrial activity, and it is the rare macro indicator that cannot be revised, smoothed or manipulated, because it is a set of actual transacted charter rates.

Then the caveat that makes it a good answer. The signal is noisy in a specific way: it is dominated by a small number of trades, above all Chinese iron-ore imports, and its violent moves are often about vessel availability, port congestion or weather rather than about world demand. It is a real indicator with a wide error bar, not a forecast, and treating a doubling of the BDI as a doubling of global growth is exactly the error the question is checking for.

How freight is quoted, and what a voyage earns

Two conventions are worth being able to name.

Worldscale is the tanker quoting convention. A published schedule gives a nominal flat rate in dollars per tonne for every route, called WS 100, and the market then trades as a percentage of it: a quote of WS 60 is 60% of the flat rate for that route. It exists so that one number can be compared across routes of very different lengths.

Time charter equivalent, or TCE, is how an owner measures earnings. It converts a voyage into dollars per day: take the gross freight, subtract the voyage costs the owner bears, bunkers, canal dues and port charges, and divide by the total days the voyage occupies the ship. It is the only number that compares a short expensive voyage with a long cheap one.

WORKED EXAMPLE

A VLCC voyage, and what the freight costs the cargo. A very large crude carrier lifts about 2.0 million barrels. At \$80 a barrel that is a **\$160 million** cargo on one hull, which is why the counterparty and insurance questions in this business are as large as the price questions.

Take an Arabian Gulf to China voyage of about 6,500 nautical miles at 12 knots. Time at sea is

$$\frac{6{,}500}{12 \times 24} = 22.6 \text{ days each way,}$$

so a round trip with four days of loading and discharging occupies the ship for about **49 days**.

Suppose the voyage is fixed at \$8.0 million of gross freight and the owner bears \$3.2 million of bunkers, dues and port costs. Then

$$\text{TCE} = \frac{8.0 - 3.2}{49} \approx \$98{,}000 \text{ per day,}$$

which is the number an owner compares against laying the ship up, and against the time charter rate for handing it to someone else.

Now read the same voyage from the cargo's side, which is what a commodity desk cares about. \$8.0 million over 2.0 million barrels is **\$4.00 per barrel**, or **5%** of the value of the cargo. That \$4.00 is the width of the arbitrage window between the Gulf and China: any price difference smaller than it is not a trade, and any difference larger than it pulls ships onto the route until it closes. This is the same mechanism that put the Brent-WTI clearing level at roughly \$3.50 in energy markets, and it generalises to every location spread in the chapter.

A detail worth getting right

REAL INTERVIEW QUESTION (BP · Commodity)

On a tanker, are there barrels or tanks?

WHAT THEY TEST / ANSWER

Tanks. The answer is one word and the question is testing whether you know that the barrel is a *unit of account* rather than an object. Crude moves in bulk, pumped into and out of the vessel's cargo tanks; nobody has shipped oil in physical barrels at scale since the nineteenth century. The barrel survives as 42 US gallons, about 159 litres, which physical versus derivative commodities covers.

Then use the remaining twenty seconds to show you know what is actually on the ship, because a one-word answer wastes the opening. A crude tanker is divided into a number of separate cargo tanks rather than one hold, which lets it carry different grades without mixing them and limits how much sloshes about as the ship rolls. Modern tankers are double-hulled and carry segregated ballast, both of which are regulatory responses to spills, and that segregation is why ballast water no longer shares a tank with cargo. Add that the tanks are inerted, kept under low-oxygen flue gas, so the vapour space cannot ignite. None of this is exotic knowledge, and offering it turns a trivia question into a demonstration that you have thought about the physical business.

INTERVIEW TRAP

1. **Thinking the BDI covers oil.** It is dry bulk only. Tankers have separate dirty and clean assessments.
2. **Reading the BDI level as a level.** It is a unitless index, so it means something only against its own history, and its range across a cycle is close to a factor of forty.
3. **Forgetting that freight sets the width of every location arbitrage.** A spread between two regions that is smaller than the freight between them is not a trade, however large it looks.
4. **Assuming a bigger ship is always cheaper.** It is cheaper per tonne only if it sails full and the ports can take it. Cost per tonne on a half-empty vessel is worse than on a smaller one that is full.
5. **Ignoring the supply lag.** High rates cause orders, orders deliver two to three years later, and the delivery usually lands into a weaker market. That lag is why shipping cycles overshoot in both directions and why the industry destroys capital so reliably.

VISUAL / CODE TO BUILD: Interactive

An arbitrage-window panel. Two regional prices sit on a vertical axis with a shaded band between them whose width is the freight rate, and a slider sets that rate. When the price difference is inside the band the panel labels the state "no trade"; when it exceeds the band the excess is shaded in a second colour and labelled with the profit per tonne. Dragging freight up until a profitable trade turns unprofitable is the point of the whole section, in one gesture.

A second panel plots the classic shipping cycle as a loop: rates on one axis, the orderbook on the other, with a delivery lag of two to three years built into the animation. Starting from high rates, the reader steps forward and watches orders rise, then deliveries arrive into a market that has already turned, then rates collapse and ordering stops, then the fleet ages out and rates recover. Running it once explains more about why freight is volatile than any statistic about the standard deviation of the index.

14.11 Crack spreads

trading · structuring · sales | core | commodity

A refinery does not speculate on the price of oil. It buys crude, converts it into gasoline and diesel, and sells those. Whether it makes money depends on the *difference* between what it sells and what it buys, and almost not at all on the level of either. That difference is the **crack spread**, and because it is what a refiner actually earns, it is traded directly rather than being left as an accounting residual.

The name comes from catalytic cracking, the process that breaks heavy hydrocarbon molecules into lighter ones. Nothing about the trade requires knowing that, but interviewers ask.

The definition, and the ratios

A crack spread is the value of the refined products minus the cost of the crude used to make them, expressed per barrel of crude.

The complication is units. Crude is quoted in dollars per barrel; gasoline and heating oil are quoted in dollars per gallon, and there are 42 gallons in a barrel. Every crack calculation begins by multiplying the product prices by 42.

The second complication is proportions. A barrel of crude does not become a barrel of gasoline; it becomes a slate of products in roughly fixed proportions. The market therefore quotes standard ratios that approximate a typical refinery's yield.

- **3-2-1**: three barrels of crude produce two of gasoline and one of distillate. The standard, and a reasonable approximation to a US Gulf Coast refinery.
- **5-3-2**: five crude into three gasoline and two distillate, used where the gasoline weighting is lower.
- **1-1**: a single-product crack, used to isolate one product's margin.

WORKED EXAMPLE

Crude at \$80.00 a barrel, gasoline at \$2.40 a gallon and distillate at \$2.60 a gallon.

First convert the products to barrels:

$$2.40 \times 42 = \$100.80 \text{ per barrel}, \qquad 2.60 \times 42 = \$109.20 \text{ per barrel}.$$

Then take the 3-2-1 combination, two barrels of gasoline and one of distillate against three of crude, and divide by three so the answer is per barrel of crude processed:

$$\frac{2 \times 100.80 \ + \ 1 \times 109.20 \ - \ 3 \times 80.00}{3} = \frac{201.60 + 109.20 - 240.00}{3} = \frac{70.80}{3} = \$\mathbf{23.60}.$$

So the 3-2-1 crack is \$23.60 a barrel. Note what has happened to the crude price: it appears in the calculation but a \$10 move in crude that carried products with it one for one would leave the crack unchanged. The refiner is exposed to the spread, not to the level.

Which way a refiner trades it

A refiner is **naturally long the crack**: it holds crude it has bought and will sell products it has not yet made, so its margin rises when products outrun crude and falls when they do not.

To lock the margin in, it does the opposite in the futures market: **sell the crack**, meaning sell product futures and buy crude futures, in the ratio that matches its own yield. Having done that, its realised margin is fixed regardless of what happens to either leg.

A trader with no refinery takes the other side. Buying the crack is a bet that refining margins will widen, which is a bet on product demand or on refining capacity being tight, and it has nothing to do with a view on the oil price.

KEY IDEA

A crack spread is a bet on **processing capacity** rather than on a commodity. It widens when there is more crude available than there is capacity to refine it, and narrows when refineries are underutilised and competing for barrels. Reading it as a view on oil is the standard misunderstanding.

The run decision

The same arithmetic answers a question a refiner faces every day: should the plant run at all?

The rule is the one from microeconomics, and getting it right is a genuine test. In the short run, **fixed costs are sunk** and do not enter the decision. The plant should run whenever the margin covers its *variable* cost, because every unit produced above variable cost contributes something toward fixed costs that would otherwise be lost entirely.

WORKED EXAMPLE

A plant processes 3,000 tonnes a day, with a variable cost of $20 a tonne and a fixed cost of $15,000 a day.

At a processing margin of $18 a tonne, each tonne loses $2 before fixed costs, so running makes things worse: **shut down** and lose the $15,000. At $25 a tonne the contribution is $5 a tonne, which over 3,000 tonnes is $15,000 a day, exactly covering the fixed cost, so the plant breaks even overall. At $30 it contributes $30,000 against $15,000 of fixed cost and makes $15,000.

Two thresholds, and confusing them is the classic error. **$20 is the shutdown point**, below which running destroys value. **$25 is the breakeven point**. Between $20 and $25 the business loses money but should still run, because losing $15,000 by producing nothing is worse than losing less by producing something. In practice the shutdown decision is even stickier than that, because stopping and restarting a refinery costs money and takes days, so plants run through short periods of negative margin rather than cycling.

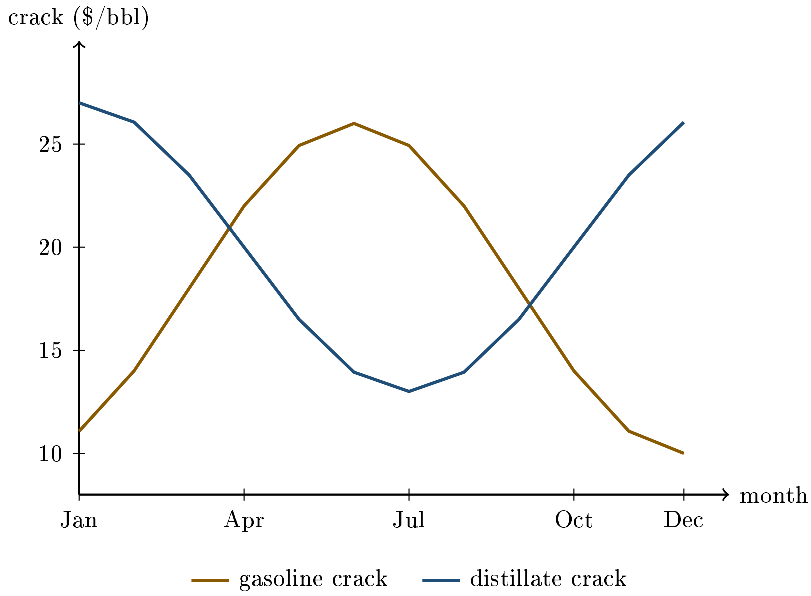
Why the crack is not exactly the margin

Four gaps between the traded spread and the refiner's real economics, and naming them is what separates someone who has priced one from someone who has read the definition.

- **Yield is not fixed.** A refinery can shift its cut between gasoline and distillate within limits, so a 3-2-1 ratio describes an average rather than a constraint. A refiner whose actual yield differs from the ratio is left with a residual exposure after hedging.
- **Crude grades differ.** The crack is quoted against a benchmark, usually a light sweet crude, and a refinery running heavier or more sour barrels buys them at a discount and gets a different product slate. That discount is its own traded spread and a large part of a complex refinery's profitability.
- **Energy is an input.** Refineries consume large amounts of natural gas and power, so a gas price spike compresses real margins while leaving the quoted crack untouched. The same idea applied to power generation is the spark spread, in spark spreads.
- **Location.** Crude bought at one point and products sold at another carry freight and pipeline costs, and the crack ignores all of it.

The crack is seasonal, and the two products are out of phase

Gasoline demand peaks in the northern-hemisphere summer driving season; distillate demand peaks in winter for heating. So the two component cracks move on opposite calendars, and a refinery's total margin is smoother than either of them because it makes both.



The two peak roughly six months apart, which is why refiners shift their cut through the year and why a hedge placed on the calendar-year average leaves seasonal risk in place. The general mechanics of commodity seasonality are in seasonality.

The same idea in other markets

A crack spread is one instance of a **processing spread**: the margin of a plant that converts one commodity into others. The family is worth recognising because the reasoning transfers exactly.

- **Crush spread**: soybeans into meal and oil, worked in full in agricultural commodities.
- **Spark spread**: natural gas into electricity, in spark spreads; the coal version is the dark spread.
- **Frac spread**: natural gas into natural gas liquids.

Every one of them is an option in disguise. The plant owner is not obliged to run: they run when the spread is positive and stop when it is not, so what they own is closer to a strip of call options on the spread than to a forward on it. That is why real processing assets are valued with option models rather than with a discounted margin, and why the value rises with the *volatility* of the spread even when its average does not move.

Interview angle

REAL INTERVIEW QUESTION (Louis Dreyfus · Commodity)

With a variable cost of \$20 a tonne and a fixed cost of \$15,000, how do you decide whether to run a refinery given the market price?

WHAT THEY TEST / ANSWER

Say the rule first and then be precise about which cost enters it, because the entire question is a test of whether you confuse the two.

The rule: in the short run, run whenever the processing margin exceeds the *variable* cost. Fixed costs are sunk. They are incurred whether or not you run, so they cannot influence the decision, and the only question is whether each tonne processed contributes something rather than nothing.

Two different thresholds, and naming both is the answer. The **shutdown point** is a margin of \$20 a tonne: below that, each tonne processed destroys value and you should stop. The **breakeven point** depends on throughput, since fixed cost has to be spread across it. At 3,000 tonnes a day the fixed cost is \$5 a tonne, so you break even overall at a margin of \$25. Between \$20 and \$25 the plant loses money and *should still run*, because it recovers part of a fixed cost it would otherwise pay in full for nothing.

Then the reality checks, which is where the answer earns its marks. Stopping and restarting a refinery is expensive and takes days, so the true shutdown threshold is below the variable cost once restart costs are amortised over the expected shutdown, and plants routinely run through short negative-margin periods. Some units cannot be turned down below a technical minimum. And the relevant margin is forward-looking, not today's: the decision is made against the crack for the period the plant would be running, which is exactly why the crack is traded as a forward curve.

Close with the connection to the previous part: this is the same arithmetic as the crack spread, which is why a refiner can lock its margin by selling the crack and then make the run decision against a known number instead of a moving one.

REAL INTERVIEW QUESTION (Louis Dreyfus · Commodity)

What would you consider if you wanted to open another refinery?

WHAT THEY TEST / ANSWER

The trap is to answer with today's crack. A refinery is a thirty-year asset and the crack you can see is a spot number, so the first move is to say what you are actually buying: a strip of options on the crack spread, one for every day the plant could run, for decades.

Put a number on it, using this section's plant. At a crack of \$30 a tonne the plant contributes \$10 a tonne over 3,000 tonnes, or \$30,000 a day, and nets \$15,000 after the \$15,000 of fixed cost. At 340 running days a year, allowing for turnaround, that is \$5.1 million a year, and at a 10% required return you could pay at most \$51 million for it.

Now show what that valuation is standing on. At a \$25 crack the contribution is \$15,000 a day, exactly the fixed cost, the plant nets zero and the same asset is worth nothing. At \$35 it nets \$30,000 a day, \$10.2 million a year, and is worth \$102 million. A \$5 move in the crack, under a fifth, takes the valuation from zero to twice the base case, because the fixed cost does not move. **A refinery is a levered claim on the crack**, and any single-number answer to this question is really a forecast dressed as a valuation.

Then the things that change the crack you personally capture, which is the part that separates an operator's answer from a screen trader's:

- **Crude slate.** A complex refinery can run discounted heavy and sour crude, so it earns a wider spread than the benchmark 3-2-1 on light sweet. Complexity is what you are paying the capital for.
- **Product slate against local demand.** You want to be long the barrel the region is short of, which is why a diesel-weighted plant and a gasoline-weighted plant are different businesses even at the same headline crack.
- **Logistics.** You earn the delivered spread, not the screen spread: freight and pipeline access on crude in and product out can be worth more than the refining margin.
- **Reliability.** The option is only worth something on days you can run, which is why the 340 above is an assumption and not a detail.

Finish on the asymmetry, because it is the real answer. You can shut a refinery down, which is the option you own, but you cannot un-build it. New capacity compresses the very crack it was built for, so the industry reliably builds at the top of the cycle. And the crack curve is liquid perhaps two or three years out while the asset lives thirty, so you are taking a thirty-year view with a three-year hedge. That gap, not the arithmetic, is what the decision turns on.

INTERVIEW TRAP

Four crack errors. (1) Forgetting the 42-gallon conversion, which puts the answer out by a factor of 42. (2) Reading a wide crack as bullish oil, when it is a statement about refining capacity and can widen while crude falls. (3) Treating the quoted crack as the refiner's margin, when grade, location, yield flexibility and energy costs all sit between them. (4) Using fixed costs in a run decision.

Where this goes next

The power-sector version of the same trade is spark spreads. The markets these spreads sit inside are energy markets and agricultural commodities, the seasonal pattern in the two component cracks is seasonality, and the curve each leg is quoted on is forward curves.

14.12 Spark spreads

sales · trading · structuring advanced commodities

A power station is not an asset that produces electricity. It is a machine that converts one commodity into another, and it will only be switched on when the conversion is profitable. The margin it earns per unit of output is the **spark spread**, and once you see that the operator can always refuse to run, the whole of power trading turns into option theory. This section builds the spread from physical units, prices it with real numbers, and then explains why a power plant is worth more than the spread it is earning today.

The spread, from the physics up

A gas-fired plant buys natural gas, burns it, and sells electricity. Its gross margin per megawatt-hour of electricity produced is

$$\text{spark spread} = P_{\text{power}} - H \times P_{\text{gas}},$$

where P_{power} is the electricity price per MWh, P_{gas} the gas price in whatever unit gas trades in, and H is the **heat rate**: the amount of fuel energy needed per unit of electrical energy. The heat rate is just thermal efficiency turned upside down, and everything else in this section is bookkeeping around those two numbers.

Two conventions coexist and mixing them is the classic unit error.

- **The American convention.** Gas trades in MMBtu, power in MWh. Since one kilowatt-hour is 3,412 Btu, one MWh of electricity is 3.412 MMBtu of energy, so a plant of efficiency η has a heat rate of $3.412/\eta$ MMBtu per MWh. A 50% plant is 6.82, a modern combined-cycle plant at 55% is 6.20, and a best-in-class unit at 60% is 5.69.
- **The European convention.** Both gas and power trade in MWh, so the heat rate is dimensionless and simply equals $1/\eta$. A 55% plant burns 1.818 MWh of gas per MWh of electricity.

A second unit trap sits underneath: efficiencies in Europe are usually quoted on the **lower** heating value of gas and American heat rates on the **higher** heating value, and for natural gas the higher value exceeds the lower by roughly 11%. A number that looks like an 11% pricing edge is very often just this.

Clean spreads: carbon is a fuel

The raw spark spread above ignores two real costs. Add them and you get the number traders actually quote.

$$\text{clean spark spread} = P_{\text{power}} - H \times P_{\text{gas}} - E \times P_{\text{CO}_2} - \text{variable O\&M},$$

where E is the emissions intensity in tonnes of CO_2 per MWh of electricity and P_{CO_2} the price of an emission allowance. The word *clean* means "net of the carbon cost", and in any market with a carbon price a spread quoted without it is close to meaningless.

The emissions intensity also comes from physics. Burning natural gas releases about 0.202 tonnes of CO_2 per MWh of fuel energy, so a plant of efficiency η emits $0.202/\eta$ per MWh of electricity: 0.367 tonnes for a 55% combined-cycle unit. Coal releases about 0.341 tonnes per MWh of fuel energy, so at a typical 38% efficiency a coal plant emits 0.897 tonnes per MWh of electricity, nearly two and a half times as much. The equivalent margin for a coal plant is called the **dark spread**, and net of carbon the **clean dark spread**.

KEY IDEA

Three spreads, one idea. **Spark** is gas to power, **dark** is coal to power, and the **crack** of crack spreads is crude to refined products. In every case you are long the output and short the inputs, the conversion ratio is a physical constant of the plant, and the owner holds the right, not the obligation, to convert. That last clause is the whole valuation problem.

Worked example: run the plant or not

WORKED EXAMPLE

A European combined-cycle plant, front-year baseload. Inputs: power at €90.00 per MWh, gas at €35.00 per MWh thermal, carbon at €70.00 per tonne, efficiency 55%, variable operating cost €3.00 per MWh.

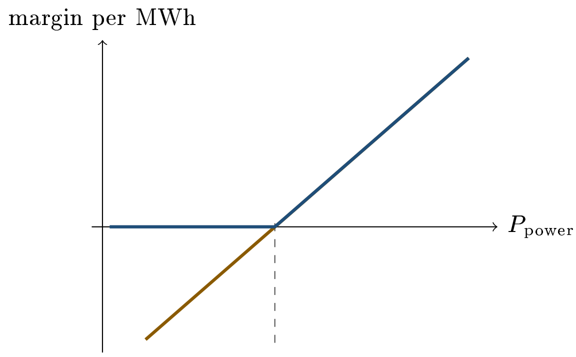
Heat rate $= 1/0.55 = 1.818$, emissions $= 0.202/0.55 = 0.367$ t per MWh.

- Fuel: $35.00 \times 1.818 =$ €63.63
- Carbon: $70.00 \times 0.367 =$ €25.69
- Variable O&M: €3.00
- Total cost: €92.32 against a power price of €90.00

The clean spark spread is −€2.32 per MWh. The plant is *out of the money* and should not run: producing destroys €2.32 of value for every MWh generated, so the rational operator sells the gas instead of burning it.

Now move gas to €30.00. Fuel becomes $30.00 \times 1.818 =$ €54.54, total cost €83.23, and the clean spark spread is +€6.77 per MWh. The same plant, the same power price, and a five-euro move in gas has flipped the decision.

That sensitivity is the point. Because the heat rate multiplies the gas price, a plant at 55% efficiency has a gas exposure of 1.818 MWh per MWh of output, so the spread is almost twice as sensitive to gas as it is to power. A power trader who hedges only the electricity leg has hedged the smaller risk.



— plant that can choose not to run (dispatch option) — plant contractually forced to run dashed: total variable cost, the strike

The switching price: where gas beats coal

Because gas and coal plants compete for the same demand, the carbon price determines which is cheaper to dispatch, and the level at which they are equal is a number the market watches by name.

WORKED EXAMPLE

Coal-to-gas switching price. Keep the gas plant above (fuel €63.63 and €3.00 of O&M, emitting 0.367 t per MWh). For the coal plant, take coal at €100 per tonne with an energy content of 6.97 MWh per tonne, so fuel energy costs $100/6.97 = €14.35$ per MWh thermal, and at 38% efficiency the fuel cost is $14.35/0.38 = €37.76$ per MWh of electricity, with €4.00 of O&M and 0.897 t of emissions per MWh.

Set the two total costs equal and solve for the carbon price C:

$$63.63 + 3.00 + 0.367\,C \;=\; 37.76 + 4.00 + 0.897\,C,$$

$$\Rightarrow \quad 0.530\,C \;=\; 24.87 \quad \Rightarrow \quad C \;=\; €46.92 \text{ per tonne.}$$

Above roughly €47 a tonne, gas is the cheaper way to make a MWh despite its more expensive fuel, because coal pays for two and a half times as much carbon. Below it, coal wins. At the €70 carbon price of the previous example the gas plant's clean spark spread was −€2.32 while the coal plant's clean dark spread was −€14.55, so gas is indeed ahead, and consistent with the switching level. This single calculation is why a carbon market changes the generation mix at all: it does not ban anything, it just moves one number through a switching threshold.

Why the plant is worth more than the spread

Electricity is the one commodity that essentially cannot be stored, as set out in energy markets. Every delivery hour is therefore its own separate commodity, cleared by its own supply and demand, which is why power prices are the most volatile in any market and can spike by orders of magnitude or go negative within a single day.

Now combine non-storability with the operator's freedom to refuse to run. The value of one hour of plant capacity is not the spread, it is

$$\mathbb{E}\big[\max(\text{spark spread, } 0)\big],$$

and since $\max(\cdot, 0)$ is convex, Jensen's inequality makes that at least $\max(\mathbb{E}[\text{spread}], 0)$, and strictly greater whenever the spread can land on either side of zero. Uncertainty alone is not enough: a plant permanently in merit is worth its expected spread and a plant that never runs is worth nothing, so the optionality premium lives exactly where the sign is in doubt. So:

KEY IDEA

A power plant is a strip of spark spread call options, one per delivery hour, struck at its own variable cost. It has value even when the current spread is negative, exactly as an out-of-the-money option does, and that value is *extrinsic*: it comes from the volatility of the spread. Valuing a plant on its average expected margin, an error committed regularly in project finance, systematically undervalues it, and the error is largest for inefficient plants that only run in scarcity hours, where almost all the value is optionality.

Three consequences a power desk lives with:

- **Correlation is the key parameter.** The volatility of a spread depends on the volatilities of power and gas and on the correlation between them, and higher correlation means a *less* volatile spread and therefore a cheaper option. Since gas often sets the marginal cost of power, that correlation is high and unstable, and it is what makes these options hard. The mechanics of variance of a difference are in independence and correlation, and the option machinery in options. Closed forms for spread options exist only as approximations, of which Kirk's is the standard desk choice.
- **Physical constraints truncate the option.** Start-up costs, minimum run times, ramp rates and minimum stable generation mean you cannot exercise hour by hour independently, so a real plant is worth somewhat less than a clean strip of hourly options. Outages remove the option entirely for the duration.
- **The spread is traded as a package.** Buy gas, sell power, buy allowances, in the ratios the heat rate and emissions intensity dictate. A **tolling agreement** does it contractually: the counterparty delivers gas and takes power, paying the plant a fee for the conversion, which converts a merchant plant's option into a fixed cash flow and transfers the spread risk to whoever bought the toll.

The run-or-idle rule itself, that the decision depends only on *avoidable* cost while the fixed cost answers the different question of whether the plant should exist at all, is derived in crack spreads and applies here unchanged. What is specific to power is the *frequency*: because electricity cannot be stored, the decision is retaken every settlement period rather than once for a production run, which is precisely what turns a single run-or-idle choice into the strip of hourly options described above, and what makes start-up costs and minimum run times bite hard enough to be worth modelling.

REAL INTERVIEW QUESTION (Louis Dreyfus · Commodity)

The refinery shuts down after 2 days and repairs take two weeks: what is the cheapest way to solve the problem?

WHAT THEY TEST / ANSWER

They are testing whether you reach for the market rather than for the workshop. Restate the problem correctly first: an unplanned outage does not create an engineering problem, it creates a **short position**. You have sales commitments in refined product you can no longer produce, and feedstock arriving you can no longer process. So there are two legs to fix and each has a market solution. On the output side, compare the cost of buying the equivalent product in the physical market for the affected dates, including freight to your customers' delivery points, against the contractual cost of failing to deliver, which may be a penalty, a claim, or a lost relationship, and buy the product if it is cheaper, which it almost always is. On the input side, resell or divert the incoming cargoes, or store them if the curve makes storage cheap, using the carry arithmetic of storage costs. Then unwind or resize the paper hedge, since you were short the products and long the feedstock as a hedge on a margin you are no longer earning, and leaving that hedge in place converts an operational problem into an outright price bet. Close with the structural insight: the outage is exactly the loss of two weeks of processing options, which is why refiners and power generators buy outage insurance or hold spare capacity, and why the market price of the spread over those dates is the correct measure of what the breakdown cost you.

INTERVIEW TRAP

Five traps, all of them about arithmetic rather than theory. First, **mixing units**: MMBtu against MWh, or a higher heating value efficiency against a lower one, which is worth about 11% on gas. Second, **quoting a dirty spread in a market with a carbon price**, which for a coal plant misstates the margin by tens of euros a MWh. Third, **forgetting that the heat rate multiplies the fuel leg**, so gas is the larger exposure and hedging only power leaves most of the risk on the book. Fourth, **valuing a plant on the average spread**, which throws away the dispatch option and undervalues exactly the plants whose value is mostly optionality. Fifth, **treating power as storable** and applying a cost-of-carry argument across delivery hours, which does not exist here: there is no arbitrage linking 3am to 7pm.

VISUAL / CODE TO BUILD: Interactive

A dispatch and valuation cockpit. Top panel: sliders for power price, gas price, carbon price, efficiency and variable O&M, with the clean spark spread displayed live, the cost stack shown as a stacked bar against the power price, and a run or idle verdict. Second panel: the same for coal, plus a solved coal-to-gas switching carbon price that updates as the fuel inputs move, reproducing the €47 calculation. Third panel: the point static text cannot make. Feed in an hourly power price shape for a week and a gas price, then show two cumulative margins side by side, one for a plant that must run continuously and one that dispatches only when the hourly spread is positive. The gap between them is the option value, and a spread-volatility slider makes it grow visibly. Add a toggle for start-up cost and minimum run time so the reader watches the option value shrink as physical constraints are switched on.

Where this goes next

The non-storability of electricity and the market design that follows from it are developed in energy markets. The same long-output-short-input structure applied to crude and refined products is crack spreads. The option machinery this section leans on is options, the correlation input is independence and correlation, and the storage arithmetic used when a plant reroutes its feedstock is storage costs. The reason any of it prices at all, that delivery is physical and enforceable, is physical vs derivative commodities.

Chapter 15

Derivatives Fundamentals

15.1 The Call Option

sales · trading · structuring | core | equity · cross-asset

Almost every conversation in an equity-derivatives, structuring or exotics interview eventually comes back to one object: the option. If you understand the call option deeply, not just its definition but why it is priced the way it is and how a trader lives with it, you already hold the thread that runs through the Greeks, volatility, structured products and most brain teasers on the desk. So we start here, and we start from zero.

Starting from nothing

A **stock** (or *share*) is a small piece of ownership in a company. If the company is worth more tomorrow, the piece is worth more; its price moves up and down every second the market is open. Call today's price S_0 and tomorrow's unknown price S_T.

Now suppose you believe the stock will rise, but you do not want to buy it outright: maybe you cannot afford it, or you only want to risk a small, known amount. This is exactly the need a call option fills.

INTUITION

Think of a **deposit on an apartment**. You pay the landlord €500 today for the *right* to rent the flat next month at a price fixed now. If a better flat appears, you simply walk away and lose your €500, nothing more. If this flat turns out to be a bargain, you exercise your right and take it at the old, cheap price. A call option is that deposit, written on a stock: a small sum paid today for the right (never the obligation) to buy later at a price agreed now.

The formal definition

A **European call option** gives its holder the *right, but not the obligation*, to buy one unit of an underlying asset at a fixed price K (the **strike**) on a fixed future date T (the **expiry** or **maturity**). For this right the buyer pays a price today called the **premium**.

At expiry the holder acts rationally:

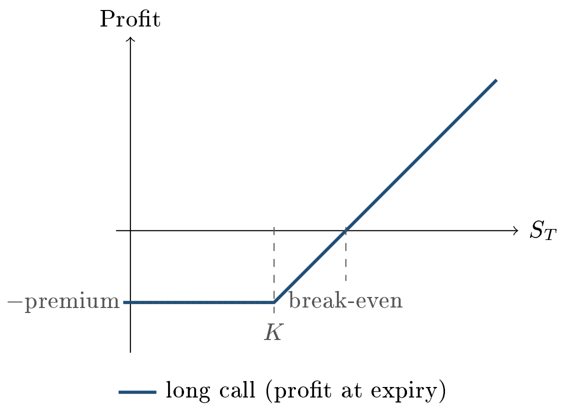
- if $S_T > K$, buy at K, sell at S_T, and pocket $S_T - K$;

- if $S_T \le K$, do nothing, and the option expires worthless.

So the value of the call *at expiry* is the now-famous payoff

$$\text{Call payoff} \;=\; \max(S_T - K,\; 0).$$

The buyer is **long** the call; whoever sold it to them is **short** the call and has the mirror-image payoff, $-\max(S_T - K, 0)$. The short's best case is to keep the premium; their loss is theoretically unlimited as S_T rises. This asymmetry (a capped, small loss for the buyer against a capped gain for the seller) is the single most important picture to have in your head.



REAL INTERVIEW QUESTION (Société Générale · Cross-asset)

Give the payoffs of a call and a put, long and short.

WHAT THEY TEST / ANSWER

Draw the four hockey sticks without hesitating: long call $\max(S_T - K, 0)$, short call $-\max(S_T - K, 0)$, long put $\max(K - S_T, 0)$, short put $-\max(K - S_T, 0)$. Label the axes (S_T horizontal, payoff vertical), mark K, and give the detail that separates a prepared candidate: shift each line down (long) or up (short) by the premium if they ask for *profit* rather than payoff.

Intrinsic value, time value, and moneyness

Before expiry the option is worth more than its payoff, because there is still time for the stock to move in your favour. We split its price into two parts:

$$\underbrace{\max(S_0 - K, 0)}_{\text{intrinsic value}} + \underbrace{(\text{the rest})}_{\text{time value}}.$$

Intrinsic value is what you would get by exercising *now*; time value is what you pay for the remaining optionality. Time value is largest when the stock sits right at the strike, and it decays to zero at expiry.

We describe where the stock sits relative to the strike with **moneyness**:

- **ITM** (in-the-money): $S_0 > K$, the call already has intrinsic value;
- **ATM** (at-the-money): $S_0 \approx K$, maximum time value;
- **OTM** (out-of-the-money): $S_0 < K$, pure time value, a bet on a move.

What actually drives the price?

Six inputs move a call's premium. Feel the *direction* of each; the pricing machinery (Black-Scholes) comes in the Black-Scholes section, and the sensitivities themselves (the Greeks) in the Greeks chapter. Here we only build intuition.

> **KEY IDEA**
>
> A call is worth **more** when: the spot S_0 is higher, the strike K is lower, there is more **time** to expiry, and (the one beginners forget) **volatility** is higher. It is worth *less* when the underlying pays large **dividends** before expiry (that cash leaks out of the stock the holder does not own yet). Rates play a smaller, positive role for a call.

Volatility deserves the spotlight. Because your downside is capped at the premium but your upside is not, a *wilder* stock is unambiguously good for the option holder: it raises the chance of a big favourable move without worsening the fixed downside. This is why option traders often say they "trade volatility" rather than direction, a theme we develop in the Greeks and advanced options.

> **VISUAL / CODE TO BUILD: Interactive**
>
> A single-option cockpit. The reader sets S_0, K, T and drags a **volatility slider**; the panel live-updates (i) the premium, (ii) the payoff-vs-profit diagram, and (iii) a small bar of the Greeks $(\Delta, \Gamma, \mathcal{V}, \Theta)$. The intended "aha": push volatility up and watch the premium and Vega grow while the payoff shape is unchanged, so price is about the *journey*, not just the destination. Backed by a Black-Scholes call in Python; expose the same numbers as a table for accessibility.

First contact with Delta

If the stock moves up by €1, how much does the call gain? That number is the **Delta**. For a call it lives between 0 and 1: a deep-OTM call barely reacts ($\Delta \to 0$), a deep-ITM call moves almost one-for-one with the stock ($\Delta \to 1$), and an at-the-money call sits around 0.5. Delta is also the trader's **hedge ratio**: to neutralise the directional risk of one call, short Δ units of the stock. We treat Delta and its cousins properly in the Greeks chapter; for now, keep the one-line intuition, that Delta is roughly how much of the stock the option currently behaves like.

> **REAL INTERVIEW QUESTION** (Quanteam / Société Générale · EQD · Structured Products)
>
> *What is the Delta of an at-the-money call? And if a call has $\Delta = 0.5$, what is its moneyness?*
>
> **WHAT THEY TEST / ANSWER**
>
> Around 0.5, so a $\Delta = 0.5$ call is roughly at-the-money. The sharp answer adds the nuance interviewers are fishing for: it is at-the-money *forward* (struck at the forward price), and it sits slightly above 0.5 for an ATM-spot call because the forward drifts above spot. Do not just say "0.5"; say "about 0.5, ATM-forward, and here is why it is not exactly 0.5."

REAL INTERVIEW QUESTION (BNP Paribas · EQD)

Roughly what is a 1-year at-the-money call worth if volatility is 16%?

WHAT THEY TEST / ANSWER

Use the desk rule of thumb: an ATM call is worth about $0.4 \times \sigma \times S_0 \times \sqrt{T}$. With $\sigma = 16\%$ and $T = 1$, that is $0.4 \times 0.16 \approx 6.4\%$ of spot. This approximation (from a Taylor expansion of Black-Scholes at the money) is exactly the kind of fast mental estimate desks want to see: deliver the number in seconds, then note that it ignores rates and dividends.

INTERVIEW TRAP

Three classic slips: (1) confusing *payoff* (at expiry, ignores premium) with *profit* (net of premium); (2) claiming higher volatility helps the seller (it does not: the buyer's asymmetry means vol always adds value to a long option); (3) forgetting that dividends push a *call* down and a *put* up. Any of these, and a sharp interviewer will keep pulling the thread.

Where this goes next

Everything heavier is a variation on what you just built. Put options are the mirror ($\max(K - S_T, 0)$) and are tied to calls by put-call parity. The sensitivities (Delta, Gamma, Vega, Theta) are covered in the Greeks chapter. The pricing engine that produces the premium is Black-Scholes. And the moment you add a **barrier** (the option dies if the stock touches a level), you step into exotics, covered in advanced options, where the very first question is usually the one below.

REAL INTERVIEW QUESTION (Société Générale · EQD)

Is a barrier option more or less expensive than a vanilla one?

WHAT THEY TEST / ANSWER

A knock-*out* call is *cheaper* than the vanilla: you keep the same upside but forfeit it in the scenarios where the barrier is touched, so you are buying strictly less. The teaser they follow with, what happens to the price as the barrier moves toward or away from spot, is developed in the exotics section; the anchor to remember is that fewer paths pay out, so the price is lower.

15.2 Forwards

sales · trading · structuring | core | equity · rates · fx · commodity · cross-asset

The forward is the simplest derivative there is, and it is asked about more often than almost anything else in this database, across equity, FX, commodities and Delta One desks. The reason is not that it is hard. It is that the forward is where a candidate first demonstrates whether they can *derive* a price from an arbitrage argument rather than recall it, and because everything heavier is priced off the forward rather than off the spot. Get this section right and the Greeks, Black-Scholes and structured products all become easier.

Note the vocabulary, since forward rates flagged it from the other side: a forward *rate* is an interest rate for a future period, while a forward *contract*, the subject here, is an agreement to trade an asset at a future date.

The contract and its payoff

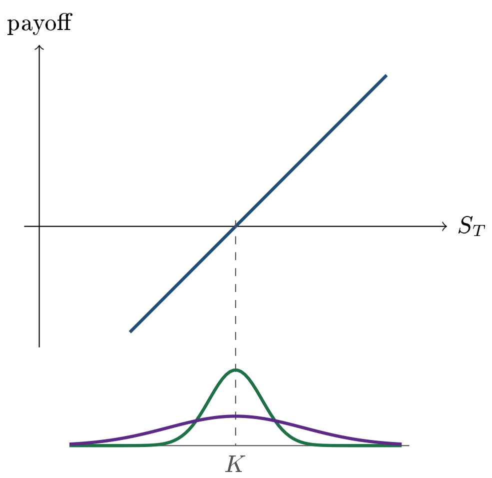
A **forward contract** obliges the buyer to purchase an asset at a fixed price K on a fixed date T, and the seller to deliver it. Both sides are obliged: unlike an option there is no choice, which is why a forward costs nothing to enter and an option costs a premium.

At maturity the long position is worth

$$\text{Payoff} \;=\; S_T - K,$$

a straight line, positive above K and negative below. The short has the mirror image. That linearity is not a detail, it is the property that drives almost every result in this section.

> **INTUITION**
>
> A farmer will have wheat to sell in six months and does not want to find out what it is worth then. A baker will need wheat in six months and has the opposite fear. They agree today on a price for a trade that happens later. Neither pays the other anything now, because neither is doing the other a favour: they are simply removing the same uncertainty from opposite directions. That is the whole contract, and the only interesting question left is what price they should agree on.



long forward payoff — low-volatility distribution — high-volatility distribution, same mean

both distributions are drawn with equal area, so they differ in width and not in probability mass

The price, from arbitrage alone

Start with an asset paying no income. You can reach the same place two ways: agree a forward today, or borrow cash now, buy the asset, and hold it. The second route costs you S_0 today and S_0e^{rT} at maturity, and leaves you holding exactly the asset the forward would have delivered. Both routes must therefore cost the same, so

$$F_0 \;=\; S_0\, e^{rT}.$$

The argument is worth stating as a trade, because that is how it is asked. If $F_0 > S_0e^{rT}$: sell the forward, borrow S_0, buy the asset, hold it. At T you deliver the asset, receive F_0, repay S_0e^{rT} and keep the difference, with no capital and no risk. If $F_0 < S_0e^{rT}$: do the reverse, short the asset, invest the proceeds, buy the forward. Either way the trade is riskless, so neither inequality can survive.

Now add everything the asset does while you hold it. Income reduces your cost of carrying it, and costs increase it:

$$\boxed{F_0 \;=\; S_0\, e^{(r-q-\text{repo}+u)T}}$$

where q is the dividend yield, repo is the rate you earn by lending the asset out (see repo markets), and u is any storage or insurance cost. The exponent is the **cost of carry**, and it is the single idea unifying every asset class:

- **Equities**: $r - q -$ repo, so a high-dividend or hard-to-borrow stock has a forward below spot.
- **FX**: the carry is the interest rate differential between the two currencies, which is covered interest parity, developed in interest rate parity.
- **Commodities**: $r + u - y$, where y is the convenience yield of physically holding the good, which is what produces contango and backwardation in contango and backwardation.

For **discrete** dividends, which is what a single stock actually pays, do not use a yield. Subtract the present value of the known cash instead:

$$F_0 \;=\; \big(S_0 - PV(\text{dividends})\big)e^{rT}.$$

The two forms answer the same question and are used in different places: a yield for an index, discrete cash flows for a single name where the dates and amounts are known.

REAL INTERVIEW QUESTION (BNP · Delta One)

Give the equation of the forward, including dividends and repo, and justify it with an arbitrage.

WHAT THEY TEST / ANSWER

Write $F = Se^{(r-q-\text{repo})T}$ and then *earn* it, because the justification is the question. Say: to replicate a long forward I borrow S at rate r and buy the stock today. While I hold it I collect its dividends q and I can lend it out in the borrow market and collect the repo rate. Both of those reduce my net cost of carrying the position, so the fair delivery price must be lower by exactly that amount, which is why they sit in the exponent with minus signs while r sits there with a plus. Then close the arbitrage: if the market forward is above that level I sell the forward, borrow, buy the stock and lend it out, and lock a riskless profit; if it is below I do the reverse. Interviewers on a Delta One desk are checking that the signs come from the replication rather than from memory, so narrate the carry, not the formula.

REAL INTERVIEW QUESTION (Goldman Sachs · Structuring)

What is the simple forward formula? With continuous dividends? With discrete dividends?

WHAT THEY TEST / ANSWER

Three formulas, and say why the third is different rather than just quoting it. Simple, no income: $F = Se^{rT}$, or $S(1+r)^T$ with annual compounding. Continuous dividend yield: $F = Se^{(r-q)T}$, appropriate for an index where dividends are spread across hundreds of names and are well approximated as a smooth stream. Discrete dividends: $F = (S - PV(D))e^{rT}$, where $PV(D)$ is the present value of the actual cash amounts on their actual dates. The reason for the distinction is the one to volunteer: for a single stock the dividend is a known lump on a known date, so treating it as a yield misprices the forward and, worse, misdates it, which matters enormously for anything with a barrier or an early exercise feature near an ex-dividend date. In structuring specifically, the dividend assumption is a real risk the desk takes, not an input it looks up.

WORKED EXAMPLE

Continuous yield. A stock trades at 50, pays a 5% continuous dividend yield, and the risk-free rate is 0. Then

$$F = 50\,e^{(0-0.05)\times 1} = 50 \times 0.951229 = 47.56.$$

The forward is *below* spot, because holding the stock pays you 5% while financing it costs nothing, so nobody will agree to buy it later at today's price.

Discrete dividend. A stock at 100, rate 3%, one year, with a single dividend of 2 paid in six months. The present value of the dividend is $2e^{-0.03\times 0.5} = 1.9702$, so

$$F = (100 - 1.9702)\,e^{0.03} = 98.0298 \times 1.030455 = 101.02.$$

Without the dividend the forward would have been $100e^{0.03} = 103.05$, so the 2 of dividend removed 2.03 from the forward: the cash itself, plus the interest you no longer earn on it over the remaining half year.

REAL INTERVIEW QUESTION (Natixis · EQD)

Take Total. The market rate is 2% and Total pays 5% of dividends over one year. What is Total's forward price in one year?

WHAT THEY TEST / ANSWER

The carry is $r - q = 2\% - 5\% = -3\%$, so the forward sits about 3% below spot: $F = Se^{-0.03} = 0.9704\,S$, or simply $0.97\,S$ if you are doing it in your head. Give the number and then the interpretation, which is what they are testing. A dividend yield above the financing rate makes the forward curve slope *downward*, so the stock is in backwardation, and that is the normal state of affairs for high-dividend European names in a low-rate environment. Two follow-ups to pre-empt: this says nothing about whether the stock will rise, since the forward is a financing calculation and not a forecast; and in practice you would use Total's actual discrete dividend schedule rather than a flat 5%, plus its repo, which for a large liquid name is small but not zero.

Forward price versus forward value

Two different objects share the word "forward", and separating them is a standard test.

The **forward price** F_t is the delivery price that would make a *new* contract worth zero today. It moves every second with spot, rates, dividends and repo.

The **value** of a forward you already hold is what your existing contract, struck at K, is worth now. Since your contract differs from a new one only in the delivery price,

$$V_t \;=\; (F_t - K)\, e^{-r(T-t)}.$$

At inception K is set equal to F_0, so $V_0 = 0$, which is exactly why no money changes hands when a forward is agreed.

WORKED EXAMPLE

You entered a one-year forward at $K = 100$. Six months later the forward price for the same maturity is 105, and rates are 3%. Your contract is worth

$$V = (105 - 100)\, e^{-0.03\times 0.5} = 5 \times 0.98511 = 4.93.$$

Not 5: the gain is realised at maturity, so it is discounted back.

REAL INTERVIEW QUESTION (Société Générale · FX)

What is the value of a forward contract at inception, and why?

WHAT THEY TEST / ANSWER

Zero, and the "why" is the whole question. The delivery price is chosen precisely as the level at which neither side would pay the other to enter, which is the no-arbitrage forward price; if it were anything else, one side would demand cash up front. Then extend it before they ask, because the follow-up is always the same: it is zero only at inception. From the next tick the contract has value $(F_t - K)e^{-r(T-t)}$, which is why forwards generate counterparty exposure over their life and why they are collateral managed. And note the contrast with an option, which costs a premium on day one precisely because its payoff is asymmetric, while a forward's is not.

Does the forward depend on volatility?

No, and understanding why is worth more than the fact.

KEY IDEA

The forward price is fixed by a **static replication**: borrow, buy, hold. That portfolio is set up once and never touched, and it delivers the asset at T whatever path the price took. Since the hedge never needs rebalancing, no assumption about how the price moves ever enters, and volatility is an assumption about how the price moves. This is also the cleanest illustration of no-arbitrage pricing: the answer comes from a portfolio, not from a probability distribution.

The graphical version, which is what the interviewer usually wants drawn, is in the figure above. The payoff is a **straight line**. Put any distribution of S_T underneath it and take the expectation: because the line has no curvature, the expected payoff depends only on the mean of that distribution, never on its width. Widen the distribution, keeping its mean fixed, and the expected payoff does not move at all. An option's payoff is kinked, so the same experiment does change its value, and that difference, curvature, is exactly what volatility is paid for.

Be honest about the boundaries, since a sharp interviewer will probe them. The result holds for a forward on a traded asset with deterministic rates, dividends and repo. Once any of those becomes stochastic and correlated with the asset, or once the payoff is converted into another currency as in a quanto, convexity reappears and a volatility or correlation assumption is needed. That is also why a futures price is not identical to a forward price, which is developed in futures.

REAL INTERVIEW QUESTION (Société Générale · EQD)

Does the forward depend on volatility? Why? Explain graphically.

WHAT THEY TEST / ANSWER

No. Give the replication argument first: the forward is priced by borrowing cash and buying the asset today and holding it untouched to maturity, a hedge that is set once and never rebalanced, so nothing about how the price travels can enter the price. Then draw it, since they asked for the graph. Sketch the payoff as a straight line through K, and beneath it two distributions of S_T with the same mean and different widths. Because the payoff is linear, the expectation is the same under both, so volatility cannot affect the value. Then contrast with an option: draw the kinked payoff and the same two distributions, and the wider one visibly picks up more value, which is the curvature you pay for in the premium. Finish with the boundary condition to show depth: this breaks for a quanto forward, or when rates or repo are stochastic and correlated with the asset, which is also why futures and forwards diverge.

INTERVIEW TRAP

Four traps. (1) Assuming the forward is the market's forecast of the future price; it is a financing calculation, and treating it as a prediction is the same error as treating a forward rate as a rate forecast. (2) Confusing the forward price (moves constantly, makes a new trade worth zero) with the value of your existing forward (zero only at inception). (3) Using a dividend yield for a single stock when the dividend is a discrete lump on a known date. (4) Forgetting repo, which on a hard-to-borrow name can move the forward more than the interest rate does.

VISUAL / CODE TO BUILD: Interactive

A forward pricer with two panels. The left panel takes spot, rate, dividend (togglable between a continuous yield and a table of discrete dates and amounts), repo and maturity, and plots the whole forward curve out to two years, with the discrete-dividend version showing the visible steps at each ex-date that the smooth yield version misses. The right panel is the volatility demonstration: a payoff diagram with a switch between forward and call, and a distribution of S_T whose width the reader can drag, with a live readout of the expected payoff. The intended experience is that dragging the volatility slider does absolutely nothing in forward mode and moves the number immediately in call mode.

Where this goes next

The exchange-traded, daily-margined version of this contract, and why its price differs slightly, is futures. Add the right to walk away and you have an option, whose payoff is kinked rather than linear and which therefore does depend on volatility. A forward can also be built out of options, a long call and a short put at the same strike, which is synthetic replication. And the sensitivity language for all of it starts in the Greeks, where a forward is the instrument with a delta of essentially one, zero gamma and zero vega.

15.3 Futures

sales · trading · structuring | core | equity · rates · commodity · cross-asset

A future is a forward that has been industrialised. Same economics, same cost-of-carry pricing, but standardised into a contract anyone can trade, guaranteed by a clearing house, and settled in cash every single day. Those three changes are the entire content of this section, and each one produces a question that this database asks repeatedly.

What standardisation and clearing actually change

- **Standardised terms.** Contract size, delivery months, settlement method and, for physicals, deliverable grade are all fixed by the exchange. You choose only how many contracts and at what price. An index future is defined by a **multiplier**: the Euro Stoxx 50 and CAC 40 futures are €10 per index point, the DAX future is €25, and the E-mini S&P 500 is $50, so one contract on an index at 5,000 represents €50,000 of exposure at a €10 multiplier.
- **A clearing house in the middle.** The exchange's CCP novates the trade so each side faces the clearing house rather than each other. Counterparty risk essentially disappears and is replaced by margin requirements and membership rules, covered in collateral and haircuts.
- **Daily settlement.** This is the one with real consequences. A future is marked to market every day and the change in value is paid in cash immediately as variation margin. A forward accumulates its whole gain silently until maturity; a future pays it out in daily instalments.

> INTUITION
>
> A forward is a **private promise**; a future is a **ticket**. The promise is negotiated once, between two named people, and settled at the end. The ticket is identical to every other ticket of its type, so it can be handed to anyone, and the exchange settles up with every ticket holder every evening rather than waiting. That daily settling-up is why nobody needs to trust anybody, and it is also why the future's price is not quite the forward's.

Pricing: the same carry, plus a footnote

The fair price of a future is the cost-of-carry price derived in forwards:

$$F_0 \;=\; S_0\, e^{(r-q-\text{repo})T}, \qquad \text{or} \qquad F_0 \;=\; \big(S_0 - PV(\text{dividends})\big)e^{rT}$$

for discrete dividends. Nothing about listing a contract on an exchange changes the arbitrage that pins it.

WORKED EXAMPLE

A stock trades at 60, the rate is 2%, and a dividend of 2.00 goes ex in three months. The six-month future is worth

$$F = \left(60 - 2.00\, e^{-0.02\times 0.25}\right) e^{0.02\times 0.5} = (60 - 1.9900) \times 1.01005 = 58.59.$$

Without the dividend it would be 60.60, so the dividend costs the future just over 2 points: the cash itself, adjusted for the carry between the ex-date and maturity.

REAL INTERVIEW QUESTION (Société Générale · EQD)

What are the five determinants of a future's price?

WHAT THEY TEST / ANSWER

They have asked for exactly five, so count them out: **spot**, **interest rate**, **dividends**, **repo or borrow cost**, and **time to maturity**. Then give the direction of each in one breath, since a list without signs is half an answer: the future rises with spot and with rates, and falls with dividends, with repo and, if the net carry is negative, with time. Say where they come from rather than reciting, namely that every one of them is a term in the cost of carrying the stock to maturity. Two refinements will separate you: on a single name the dividend must be the actual expected cash on its actual ex-date, not a smoothed yield, and for a real desk the *dividend tax* treatment matters too, because the future is priced off the dividend a hedger actually keeps after withholding, not the gross amount, which is developed in dividends.

REAL INTERVIEW QUESTION (Société Générale · Exo)

A future matures in June 2021 and a dividend is paid in December 2021. It is announced that dividends will increase. What happens to the future? And if the maturity is March 2022 instead of June 2021?

WHAT THEY TEST / ANSWER

Nothing, then something, and the reason is the same in both cases. A future maturing in June 2021 is priced off the cash flows that occur *before* June 2021. A dividend paid in December 2021 is after the contract has already died, so the holder was never going to receive it or forgo it, and raising it changes the June future by exactly zero. That is the trap: the instinct "dividends up, future down" is a reflex, and the question checks whether you look at the dates.

Change the maturity to March 2022 and the December 2021 dividend now falls inside the contract's life, so it matters. Raising it lowers the future by roughly the size of the increase, carried forward from the ex-date to maturity: a 0.50 increase with rates at 2% and three months of carry left lowers the future by $0.50 \times e^{0.02\times 0.25} = 0.5025$. Two sentences to add if they are still listening: the March 2022 future was already exposed to that dividend before the announcement, so what moved is the *expectation*, which is exactly why dividend risk is a traded exposure on a Delta One book; and this is also why dividend futures and swaps exist, developed alongside dividends.

Is a future more expensive than a forward?

Theoretically they are the same price when interest rates are deterministic. Once rates are stochastic, daily margining introduces an asymmetry.

The holder of a long future receives cash on days the price rises and pays cash on days it falls. If the underlying is *positively* correlated with interest rates, then the days you receive cash tend to be days when you can reinvest it at a higher rate, and the days you pay tend to be days when you fund the payment cheaply. That is a genuine advantage, so the futures price must be set slightly *above* the forward price to compensate the seller. Negative correlation reverses it; zero correlation gives equality.

For equity index futures the correlation with rates is weak and the maturities are short, so the difference is negligible in practice. For interest rate futures the underlying *is* the rate, the correlation is by construction perfect, and the effect is the convexity adjustment developed in FRAs.

> **REAL INTERVIEW QUESTION** (Société Générale · Delta One)
>
> *Which is more expensive, a future or a forward?*
>
> **WHAT THEY TEST / ANSWER**
>
> Refuse the false binary and give the condition, because that is the answer. With deterministic interest rates the two are identical, since they promise the same thing and are pinned by the same cash-and-carry arbitrage. With stochastic rates the difference comes from daily margining: a long future banks its gains in cash as they happen, so if the underlying is positively correlated with rates you receive cash precisely when reinvestment is attractive, which is worth something, so the future trades above the forward. Negative correlation and the future trades below. Then land it with magnitudes: on equity index futures this is a rounding error, on short-dated interest rate futures it is the convexity adjustment and is several basis points and worth money. Finish with the practical point a Delta One desk cares about: the two also differ in credit and funding, since a forward carries counterparty risk and consumes credit line while a future consumes margin, and on some names that difference dominates the theory.

Basis, convergence and the roll

The **basis** is the gap between the future and the spot, $F_t - S_t$. Because the cost of carry shrinks as maturity approaches, the basis converges to zero at expiry: on the last day the future *is* the spot, and any gap would be an immediate arbitrage. Everything a future does between now and then is carry.

Since each contract dies, holding a permanent position means **rolling**: closing the expiring contract and opening the next one, usually in the days around expiry when liquidity concentrates there. The cost of the roll is the calendar spread between the two contracts, which is just the carry over the extra period, so a roll is cheap when carry is small and expensive when it is not. In commodities the roll is a first-order part of the return, which is what makes contango and backwardation matter so much there.

REAL INTERVIEW QUESTION (Barclays · Structured Products)

What is the difference between owning a share and owning a future?

WHAT THEY TEST / ANSWER

Group the differences rather than listing them, since there are many. **Economically** they are nearly the same exposure: both give you roughly one unit of price risk. **Financially** they are not. Owning the share requires you to pay for it and then collect the dividends; the future requires only margin, so it is a funded versus unfunded distinction, and the financing and dividends you would have had are already baked into the future's price rather than received. **Legally** the share makes you an owner, with voting rights and a claim in a liquidation, while the future makes you a party to a contract with a clearing house. **Practically** the future expires and must be rolled, trades in fixed contract sizes, and is marked and margined daily, so it can generate cash calls that the share never does. Close with when each is preferred: a future for cheap, short-horizon, leveraged or hedging exposure, and the share for a long-term holding, for the voting rights, or where the dividend or tax treatment favours the physical.

REAL INTERVIEW QUESTION (BNP · Delta One)

How do you hedge if you buy a future on Sanofi?

WHAT THEY TEST / ANSWER

Sell the stock, in the quantity the contract represents, and then say what you have left, because "short the stock" alone is the incomplete answer. You are long the future, so you short the shares to flatten the price risk, and the delta is essentially one per unit of the contract, adjusted for the carry factor to maturity.

What you have not hedged is the interesting part. You are now long the future's implied dividend and short the actual dividend, so you carry **dividend risk**: if Sanofi raises its dividend for an ex-date before maturity, the future falls and your short stock does not compensate you. You are also short the stock, which you must **borrow**, so you carry repo risk for the life of the trade, and repo can move against you or the borrow can be recalled. And you carry funding and margin flow risk, since the future margins daily in cash while the stock position does not. That is precisely the risk set a Delta One book is paid to manage, so naming those three is the answer they are looking for.

REAL INTERVIEW QUESTION (Société Générale · EQD)

Pitch me the sale of a future to someone who understands nothing about the product.

WHAT THEY TEST / ANSWER

This is a sales exercise, and the grading is on restraint. Lead with the need, not the instrument: "you want exposure to the index, but you do not want to tie up the cash today, and you want to be able to get out instantly." Then the product in one sentence: "a future is a standard contract to buy the index at a set date, it trades on the exchange all day, and you post a small deposit rather than the full amount." Then the honest risks, which is what actually sells: "because you only post a deposit, your gains and losses are magnified, the exchange will ask you for cash daily if it moves against you, and the contract expires, so if you want to stay invested we roll it." Then stop and ask a question. The candidates who fail this are the ones who explain cost of carry to someone who does not know what an index is.

INTERVIEW TRAP

Four traps. (1) Assuming a dividend always lowers the future without checking whether its ex-date falls inside the contract's life. (2) Saying a future is always more expensive than a forward; it depends on the correlation between the underlying and rates, and is identical under deterministic rates. (3) Forgetting the multiplier, so a position is mis-sized by a factor of ten or fifty. (4) Treating a future as a cash-free position; it is unfunded but not cash-free, and daily variation margin has bankrupted hedgers whose physical asset was worth more than ever.

VISUAL / CODE TO BUILD: Interactive

A futures cockpit. One panel plots spot and the future together as maturity approaches, with the basis shaded between them so convergence to zero is something the reader watches rather than reads, and with sliders for rate, dividend and repo that visibly tilt the curve. A second panel is a dividend timeline: the reader drags ex-dividend dates along an axis and drags the contract's maturity, and the future's price only moves when an ex-date crosses inside the contract's life, which makes the Société Générale trap above self-evident. A third panel simulates a margin account day by day against a price path, showing variation margin flows and triggering a margin call, so the difference between unfunded and cash-free is concrete.

Where this goes next

Give the buyer the right to walk away rather than an obligation and you have an option. Interest rate futures and the convexity adjustment that separates them from forwards are in FRAs. The margin machinery behind daily settlement is collateral and haircuts, the borrow that a futures hedge relies on is repo markets, and the roll dynamics that dominate commodity futures returns are contango and backwardation.

15.4 Options

sales · trading · structuring | core | equity · rates · fx · commodity · cross-asset

This is the section everything else in the book leans on. The Greeks, the volatility surface, exotics and every structured product are variations on one object, and interviewers know it, which is why "what is a call, what is a put" opens more front-office interviews than any other question in this database. The bar is not knowing the definition. It is being able to draw the payoffs without hesitating, decompose the premium, derive put-call parity on the spot, and say what changes when the holder can exercise early.

The contract

An **option** gives its holder the *right, but not the obligation*, to trade an asset at a fixed price on or before a fixed date. A **call** is the right to buy; a **put** is the right to sell. The fixed price is the **strike** K, the fixed date is the **expiry** T, and because the holder gets a right while the seller takes on an obligation, the holder pays a **premium** up front.

That is the whole difference from a forward, and it is the difference that creates the entire options business. A forward obliges both sides, so it costs nothing and its payoff is a straight line. An option obliges only one side, so it costs a premium and its payoff is bent.

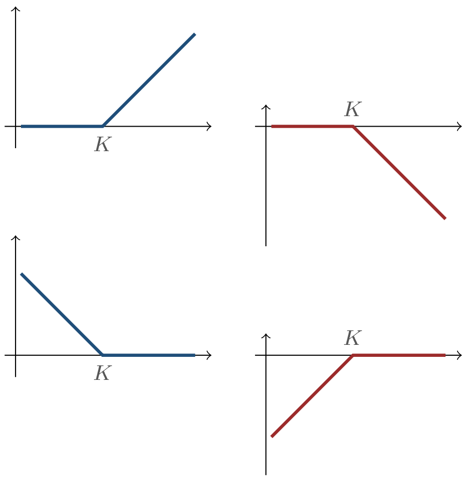
> **INTUITION**
>
> An option is **insurance**, and it is worth thinking of it that way from the start. You pay a small premium now so that if something specific happens you are protected, and if it does not happen you simply lose the premium and nothing worse. The buyer of a put on a stock is buying insurance against the stock falling. The seller is the insurance company: they collect a steady premium and, most of the time, keep it, in exchange for accepting a large loss occasionally. Every asymmetry in this section, capped losses for buyers, capped gains for sellers, and the fact that the premium is paid up front, is already present in that analogy.

Payoffs

At expiry the holder does whatever is profitable and nothing otherwise, so the four payoffs are:

$$\text{long call} = \max(S_T - K, 0), \qquad \text{long put} = \max(K - S_T, 0),$$

with the short positions being the exact negatives. Drawing these four "hockey sticks" fluently is the single most tested motor skill in this chapter.



Top row: —— long call, then —— short call. Bottom row: —— long put, then —— short put.
In all four panels the horizontal axis is the spot at expiry S_T and the vertical axis is the payoff; K is marked on the horizontal axis.

The asymmetry is the point. A buyer's loss is capped at the premium and their gain is not (unbounded for a call, bounded only by the strike for a put). A seller's gain is capped at the premium and their loss is not. Note that these are **payoff** diagrams, ignoring the premium. Shift each line down by the premium (long) or up by it (short) and you have the **profit** diagram, which is what a question about break-even is asking for.

REAL INTERVIEW QUESTION (Morgan Stanley · EQD)

What is an option?

WHAT THEY TEST / ANSWER

Deceptively simple, and the failure mode is being vague. Give the contract precisely in one sentence: the right, not the obligation, to buy (call) or sell (put) an underlying at a fixed strike on or before a fixed date, for which the buyer pays a premium. Then immediately add the two things that show you understand rather than recall. First the asymmetry: the buyer's loss is capped at the premium, the seller's is not, which is exactly why the premium exists. Second the consequence: because the payoff is bent rather than straight, the option's value depends on how much the underlying is expected to *move*, not just on where it ends up, which is why volatility is the thing options traders actually trade. Stop there and let them choose the follow-up.

REAL INTERVIEW QUESTION (CACIB · Delta One)

What is the payoff of a call? Give the graphical representation, then the mathematical one.

WHAT THEY TEST / ANSWER

They asked for graph first, so draw first and talk while you draw. Axes labelled: S_T horizontal, payoff vertical. A flat line along zero up to K, then a 45-degree line rising after it. Mark K on the horizontal axis. Then write $\max(S_T - K, 0)$, and say the sentence connecting the two: the flat part is where you walk away and the sloped part is where you exercise and capture the difference. Score the extra points by volunteering the distinctions they were going to test next: this is the *payoff*, so shift it down by the premium to get *profit*, and the break-even is $K + \text{premium}$; and the short call is the mirror image about the horizontal axis, capped above at the premium and unbounded below.

REAL INTERVIEW QUESTION (Société Générale · Delta One)

I am short a call or a put on SG at 5. What is my maximum loss and my maximum gain?

WHAT THEY TEST / ANSWER

Answer both instruments separately, because their asymmetries differ. **Short call**: maximum gain is the premium, collected if the stock finishes below 5. Maximum loss is theoretically unlimited, since the stock can rise without bound and you must deliver at 5. **Short put**: maximum gain is again the premium. Maximum loss is large but *bounded*, because the worst case is the stock going to zero and you being obliged to buy at 5, so you lose 5 minus the premium per share. Making that distinction, unbounded versus bounded at the strike, is the whole test. Two sentences to add: in practice a naked short call is rarely left unhedged precisely because of that unbounded tail; and a short put has the same payoff as a covered call, which is why it is so common in yield-enhancement structures.

Moneyness, intrinsic value, time value

Before expiry an option is worth more than it would pay if exercised now, because there is still time for the underlying to move. Split the premium:

$$\text{premium} = \underbrace{\max(S-K,0)}_{\text{intrinsic value}} + \underbrace{\text{the remainder}}_{\text{time value}}$$

for a call, with $\max(K - S, 0)$ for a put. **Moneyness** describes where spot sits relative to strike: **in the money** (positive intrinsic), **at the money** (spot at strike), **out of the money** (zero intrinsic, pure time value).

Time value is largest at the money, where the outcome is most uncertain, and decays to zero at expiry, where there is no more time to be paid for.

REAL INTERVIEW QUESTION (BNP · Structured Products)

Decompose the premium into intrinsic value and time value, and explain.

WHAT THEY TEST / ANSWER

Give the split, then the shape, then the use. Split: intrinsic value is what you would collect by exercising immediately, $\max(S-K,0)$ for a call, and time value is everything else you paid. Shape: time value peaks at the money and falls away in both directions, because deep out of the money the option probably expires worthless and deep in the money it behaves like the underlying itself, so in both extremes there is little optionality left to pay for; and it decays to zero at expiry. Use, which is the structuring answer they want: when you sell an option to a client you are mostly selling time value, that is what decays in your favour, and it is the part that volatility moves. Intrinsic value is not a view on anything, it is arithmetic.

What moves the price

Six inputs. Learn the *directions*, since that is what gets asked; the machinery is Black-Scholes and the sensitivities are the Greeks.

Increase in	Call	Put
Spot S	up	down
Strike K	down	up
Volatility σ	up	up
Time to expiry T	up	usually up
Interest rate r	up	down
Dividends q	down	up

Volatility is the only input that *unambiguously* helps both a call and a put, and the reason is the asymmetry again: your downside is capped at the premium while your upside is not, so a wilder underlying is good for anyone who is long an option, whichever side they are long. That single observation is why traders say they trade volatility rather than direction.

The rate and dividend rows are the ones candidates fumble. Both act through the forward: from forwards, $F = Se^{(r-q)T}$, so a higher rate lifts the forward and helps calls, while a higher dividend lowers it and helps puts.

REAL INTERVIEW QUESTION (UBS · EQD)

What is the impact on a call and a put of a change in interest rates, the strike, volatility, maturity or the underlying?

WHAT THEY TEST / ANSWER

Go through them in a fixed order and give a reason for each rather than a table read from memory. Underlying up: call up, put down, trivially. Strike up: call down, put up, again by definition of the payoff. Volatility up: *both* up, and say why, because the loss is capped at the premium while the gain is not, so more dispersion can only help a long option. Maturity up: usually both up, since more time means more chance to move, and flag the exception, that a deep in-the-money European put can be worth *less* with more time because you are waiting longer to receive a nearly certain strike. Rates up: call up, put down, and give the mechanism rather than the fact, that rates lift the forward and a call is long the forward. If they want depth, add that dividends do the opposite for the same reason.

Put-call parity

Two portfolios, one identity, and it is asked by almost every bank in this database.

Portfolio A: buy a call and sell a put, both struck at K with expiry T. Portfolio B: buy the underlying and borrow Ke^{-rT}.

At expiry, A pays $\max(S_T - K, 0) - \max(K - S_T, 0) = S_T - K$ whichever way the market went, and B pays $S_T - K$ as well. Identical payoffs in every state of the world, so identical prices today:

$$\boxed{C - P \ = \ S_0 - Ke^{-rT}}$$

With a continuous dividend yield the spot leg is discounted too, giving $C - P = S_0 e^{-qT} - Ke^{-rT}$; with discrete dividends, subtract their present value from S_0, exactly as in forwards.

Note what is *not* in the identity: volatility, drift, and any distributional assumption. Parity is pure arbitrage, which is why it holds in every model and why breaking it is a genuine arbitrage rather than a model disagreement.

WORKED EXAMPLE

$S_0 = 100$, $K = 100$, $T = 1$, $r = 5\%$, and the call trades at 10.45. Then $Ke^{-rT} = 95.1229$ and

$$P = C - S_0 + Ke^{-rT} = 10.45 - 100 + 95.1229 = 5.57.$$

The call is worth about twice the put at the same strike, purely because the forward sits above spot at a positive rate.

REAL INTERVIEW QUESTION (UBS · EQD)

What is put-call parity?

WHAT THEY TEST / ANSWER

State it, derive it in two lines, and say what it is for. Statement: $C - P = S - Ke^{-rT}$ for European options on the same underlying, strike and expiry. Derivation, which is the part they are grading: a long call plus a short put pays $S_T - K$ in every state, which is exactly a forward, so that combination must cost the same as buying the stock and borrowing the present value of the strike. Uses: it lets you price a put from a call instantly, it pins the relationship between implied volatilities so a call and put at the same strike must trade on the same implied vol, and it is a live arbitrage check on a screen. Two caveats worth volunteering: it holds for *European* options, since early exercise breaks the argument, and you must adjust the spot leg for dividends.

REAL INTERVIEW QUESTION (Société Générale · Structured Products)

For a one-year at-the-money call and put on a non-dividend-paying stock with zero rates, which costs more? And if rates are at 10%?

WHAT THEY TEST / ANSWER

Use parity and do not reach for Black-Scholes, which is the point of the question. With $K = S$ and $r = 0$, parity gives $C - P = S - K = 0$, so **they cost exactly the same**. That is a strong answer because it needs no model, no volatility and no calculation.

At $r = 10\%$: $C - P = S - Se^{-0.10} = S(1 - 0.9048) = 0.0952\,S$, so the **call is more expensive**, by about 9.5% of spot. The intuition to say out loud: at a positive rate the forward sits above spot, the strike is struck at spot, so the call is effectively in the money relative to the forward while the put is out of it. Equivalently, buying the call rather than the stock lets you keep your cash and earn interest on it, and that benefit is worth exactly the discount on the strike.

No-arbitrage bounds

Before any model, arbitrage alone pins the price into a band. For a European call on a non-dividend-paying stock,

$$\max\big(S_0 - Ke^{-rT},\, 0\big) \;\le\; C \;\le\; S_0,$$

and for the put,

$$\max\big(Ke^{-rT} - S_0,\, 0\big) \;\le\; P \;\le\; Ke^{-rT}.$$

The upper bounds say an option can never be worth more than what it lets you obtain: a call is never worth more than the stock, a put never more than the present value of the strike. The lower bounds are the forward again. Everything a model does is choose a point inside this band; see no-arbitrage pricing.

The put's lower bound has a consequence worth pausing on. Deep in the money, that bound is $Ke^{-rT} - S_0$, which is *below* the intrinsic value $K - S_0$. So a deep in-the-money European put can trade below its intrinsic value, meaning its time value is **negative**. With $K = 100$, $S_0 = 20$, $T = 1$ and $r = 5\%$, the put is worth about $95.12 - 20 = 75.12$ against an intrinsic value of 80, a time value of -4.88. You are simply waiting a year to receive a strike that is already effectively certain, and waiting costs interest.

Exercise style, and when to exercise early

A **European** option may be exercised only at expiry. An **American** option may be exercised at any time up to expiry. A **Bermudan** option may be exercised on a set of specified dates, and is common in rates and structured products.

An American option is worth at least as much as the European equivalent, since it offers everything the European does and more. The interesting question is when that extra right is actually worth using.

KEY IDEA

It is **never** optimal to exercise an American call early on a non-dividend-paying stock. Exercising throws away the remaining time value and pays the strike sooner than necessary, so you would always do better to sell the option instead. It *can* be optimal to exercise an American put early, when it is deep in the money, because that is exactly the case above where waiting costs you interest on a strike you are nearly certain to receive. And it can be optimal to exercise an American call early *just before an ex-dividend date*, when the dividend you would capture by owning the stock outweighs the time value you give up.

REAL INTERVIEW QUESTION (Exane · Exo)

In which cases do you exercise an American option early?

WHAT THEY TEST / ANSWER

Give the three cases and the single principle behind them. The principle: exercise early only when holding the option costs you more than the optionality is worth. Case one, an American **call on a non-dividend-paying stock**: never, because you would be discarding time value and paying the strike early, and selling the option always dominates. Case two, an American **call just before an ex-dividend date**: possibly yes, if the dividend captured by owning the stock exceeds the time value surrendered, which is why American calls on high-dividend names are exercised in clusters around ex-dates. Case three, a **deep in-the-money American put**: often yes, because the payoff is nearly certain and waiting means forgoing interest on the strike, which is the same fact as a deep in-the-money European put having negative time value. If they push on pricing, note that early exercise is precisely why American options generally need a numerical method such as a tree or least-squares Monte Carlo, since there is no closed form.

INTERVIEW TRAP

Four traps. (1) Drawing the payoff when asked for the profit, or the reverse; the difference is the premium and the break-even. (2) Saying more time always makes an option worth more, when a deep in-the-money European put is the counterexample. (3) Applying put-call parity to American options, where early exercise breaks the replication and it becomes an inequality. (4) Saying higher volatility helps the buyer and hurts the seller as if these were separate facts; they are the same fact, and it is the payoff's curvature that causes it.

VISUAL / CODE TO BUILD: Interactive

An option laboratory. The reader picks call or put, long or short, and sets strike, spot, volatility, maturity, rate and dividend. The main panel shows the payoff at expiry, the profit line offset by the premium, and the current value curve, which visibly relaxes onto the payoff as maturity is dragged to zero, making time decay something the reader watches rather than reads. A decomposition bar splits the premium into intrinsic and time value live. A parity tab shows the call, the put and the synthetic forward built from them, with a running check that $C - P - S + Ke^{-rT}$ stays at zero as every input is dragged, and a deliberate mode where the reader can break parity and see the arbitrage trade it implies.

Where this goes next

Combining options into structures with a view is standard payoff profiles, and using them to manufacture other instruments is replication strategies. The sensitivities of the premium to each of the six inputs are the Greeks, and the model that turns the inputs into a number is Black-Scholes. The dividend assumptions that shift both the forward and the early-exercise decision are dividends.

15.5 Swaps

sales · trading · structuring | core | equity · rates · fx · credit · cross-asset

"Swap" is not an instrument, it is a *shape*. Once you see the shape, the dozen products carrying the name stop being a list to memorise and become one idea applied to different cash flows. That is what this section builds. The interest rate swap has its own full treatment in interest rate swaps; here we do the general principle and then the equity family, which has no other home.

The shape

A **swap** is an agreement to exchange two streams of cash flows, each computed on an agreed notional, on an agreed schedule, until an agreed maturity. That is the entire definition. What varies between products is only what the two streams are computed from.

> **INTUITION**
>
> Two people swap incomes. One has a salary that changes every month and wants a steady one; the other has a fixed salary and would rather have the variable one. Neither hands over their job, and neither hands over the capital that generates the income. They just agree to pay each other's paycheques, and in practice only the *difference* changes hands each month. Every swap in finance is that arrangement, applied to some pair of cash flow streams: fixed against floating, one currency against another, a stock's return against a financing cost.

Two structural readings do all the work, and each is easier for different products.

A swap is a portfolio of forwards. Each exchange date is a single forward contract on whatever the floating leg references, all struck at the same fixed rate. Value the whole thing by valuing each period against today's forward and adding up. This reading is the right one when the floating leg is awkward or the schedule is irregular.

A swap is a pair of legs. Value each stream as an instrument in its own right and subtract:

$$V = PV(\text{leg received}) - PV(\text{leg paid}).$$

This reading is the right one when both legs are recognisable objects, as in an interest rate swap, where they are a fixed bond and a floating one.

KEY IDEA

Two facts hold for essentially every swap and carry most interview answers. First, a swap is worth **zero at inception**: the fixed level is quoted precisely to make it so, which is why no premium is paid, in sharp contrast to an option. Second, from that moment it marks to market as the difference between the two legs, so it generates counterparty exposure over its life even though nothing was exchanged up front, which is why swaps live under collateral agreements (see collateral and haircuts).

REAL INTERVIEW QUESTION (Marex Solutions · Structured Products)

What is a swap?

WHAT THEY TEST / ANSWER

Define the shape, not an example, then give an example. Shape: an agreement to exchange two streams of cash flows on a notional over a schedule, with the notional itself usually not exchanged and only the net difference settling. Example: the vanilla interest rate swap, fixed against floating. Then add the three things that show structure rather than recall. It is worth zero at inception, because the fixed level is quoted to make it so. It can be seen either as a portfolio of forwards or as a pair of legs, and both give the same value. And the family is defined by what the legs reference, so the same machinery gives you rate, currency, equity, dividend, commodity, variance and credit swaps. That last sentence is usually what turns a definition into a good answer.

The family

Swap	Legs exchanged	Developed in
Interest rate	fixed rate vs floating rate	interest rate swaps
Cross-currency	rate in one currency vs another, *plus* principal exchange	FX forwards and swaps
FX swap	spot exchange reversed at a forward	FX forwards and swaps
Equity / TRS	total return of an asset vs financing	this section
Dividend	realised dividends vs a fixed amount	this section
Inflation	realised inflation vs a fixed rate	inflation-linked products
Variance	realised variance vs a fixed strike	variance swaps
Credit default	periodic premium vs default protection	credit default swaps

REAL INTERVIEW QUESTION (CACIB · Rates)

What is a swap, cross-currency or rates? What is the difference between a currency swap and a rates swap?

WHAT THEY TEST / ANSWER

Give the shared definition once, then isolate the single structural difference, because everything else follows from it. Shared: an exchange of two cash flow streams on a notional over a schedule. Difference: in an interest rate swap both legs are in the **same currency**, so the notionals cancel and are never exchanged; in a cross-currency swap the legs are in **different currencies**, so the notionals do *not* cancel and are genuinely exchanged, normally at both the start and the maturity.

Everything an interviewer wants next comes off that. Because principal is exchanged, a cross-currency swap carries FX risk on the principal and a much larger counterparty exposure, which keeps growing to maturity instead of humping in the middle like a rate swap's. It therefore consumes far more credit line and is priced with a cross-currency basis on top of the two rate curves. Uses differ too: a rates swap converts fixed funding into floating, a cross-currency swap converts funding in one currency into another, which is why it is the natural tool for an issuer who borrows where demand is and needs cash where its business is.

Equity swaps and total return swaps

An **equity swap** exchanges the performance of an equity underlying against a financing leg. In its most common form, the **total return swap** (TRS), one side receives the asset's *total* return, price appreciation plus dividends, and pays a floating reference rate plus a spread on the same notional.

The receiver has the economics of owning the asset with borrowed money, without owning it. The payer has the economics of being short it, and is typically a dealer who hedges by actually buying the stock, which is why the financing spread is really the dealer's cost of balance sheet plus margin.

WORKED EXAMPLE

A TRS on €100 million of a stock, quarterly, financing at the reference rate plus 40 basis points. Over the quarter the stock rises 3% and pays 0.5% in dividends, and the reference rate is 4%.

The total return leg pays $(3\%+0.5\%)\times$ €100m $=$ €3.5 million to the receiver. The financing leg pays $(4\% + 0.4\%) \times \frac{1}{4} \times$ €100m $=$ €1.1 million to the payer. Net, the receiver collects €2.4 million, having posted only margin rather than €100 million of cash. If instead the stock had fallen 3%, the total return would be $-3\% + 0.5\% = -2.5\%$, so the receiver would owe $2.5 + 1.1 =$ €3.6 million, which is the leverage working in the other direction. Note that the dividend is still received: a total-return receiver is long the dividends whichever way the price goes, which is the feature that separates a TRS from a price-return swap.

REAL INTERVIEW QUESTION (Société Générale · EQD)

What is a TRS?

WHAT THEY TEST / ANSWER

Contract, economics, then why anyone uses it. Contract: one leg pays the total return of an asset, price plus dividends, and the other pays a floating reference rate plus a spread, on a notional that is never exchanged. Economics: the total-return receiver has a synthetic leveraged long position, and the payer, usually a dealer, hedges it by buying the asset outright, so the spread is the dealer's financing and balance sheet cost. Uses: leverage without funding the cash position, access to a market where the client cannot easily hold the stock directly, and keeping the position off the client's own balance sheet. Then name the risks, since a one-sided answer is the weak version: counterparty risk on both sides, financing risk if the spread resets or the trade is unwound, dividend risk if actual dividends differ from those assumed, and the concentration problem that made Archegos a case study, where several dealers each saw only their own slice of one client's leverage. The financing angle sits alongside equity financing.

Selling an equity swap is a different exercise from pricing one: the interviewer who asks you to pitch it is testing whether you can name which client type buys it and why each one does, which is client segmentation rather than instrument mechanics. That question, and the client map it turns on, is developed in flow versus real money clients.

Dividend swaps

A **dividend swap** exchanges the dividends actually paid by a stock or index over a period against a fixed amount agreed today. It isolates dividend risk from price risk entirely: no delta, just a view on the cash companies will distribute.

The reason this market exists at the size it does is a hedging imbalance rather than investor enthusiasm. A dealer who has sold autocallables and similar retail structures ends up short the forward, and since a higher dividend lowers the forward, that leaves the dealer structurally *long* dividends. To flatten it they need to sell dividend risk, and the standard market explanation for why listed dividend futures have often traded below consensus dividend forecasts is exactly this one-directional dealer supply meeting limited natural demand.

REAL INTERVIEW QUESTION (Goldman Sachs · EQD)

How do dividend swaps function, and what purpose do they serve in equity markets? Discuss the motivations for investors to engage in dividend swaps and the potential risks.

WHAT THEY TEST / ANSWER

Mechanics: one leg pays the dividends actually declared and paid on a stock or index over a defined period, the other pays a fixed amount fixed at trade date, settled on the difference, with no exposure to the level of the index itself. Purpose: it separates dividend risk from price risk, which nothing else does cleanly.

Motivations, split by who is trading. Dealers hedge: having sold retail structured products they are structurally long dividends and need to sell them, which is the supply side of this market. Investors take the other side because dividends are far more stable and forecastable than prices, so a long dividend position is a relatively high-carry, low-volatility trade, and because the structural dealer supply has historically left dividend futures cheap to consensus forecasts.

Risks, and give the real ones. Dividends can be cut, and they are cut exactly in a crisis, which is when the position is already losing, so it is short a fat tail: 2008 and the 2020 European bank dividend suspensions are the case studies. It is a crowded trade with a structural seller, so liquidity is thin when everyone wants out at once. Index dividend points depend on index composition and weights, so corporate actions create basis. And it is unfunded and collateralised, so it carries counterparty and margin risk like any swap. The underlying corporate behaviour is developed in dividends.

INTERVIEW TRAP

Four traps. (1) Saying the notional is exchanged in every swap; it is not exchanged in an interest rate or equity swap, and it *is* in a cross-currency swap, which is exactly why their risk profiles differ. (2) Calling a cap or a swaption a swap; they are options and cost a premium. (3) Treating a TRS as risk-free financing for the dealer, when it leaves dividend, borrow and concentration risk. (4) Describing dividend swaps as a safe carry trade without saying that dividends are cut precisely in the scenarios that hurt.

VISUAL / CODE TO BUILD: Interactive

A swap builder. The reader picks what each leg references from a menu (fixed rate, floating rate, equity total return, realised dividends, realised variance, a second currency) and the panel assembles the cash flow ladder, labels the resulting product with its market name, and prices it to zero by solving for the fixed level. Switching one leg from a same-currency rate to a foreign-currency rate makes the principal exchange arrows appear at both ends, so the single structural difference between a rate swap and a cross-currency swap is something the reader triggers rather than reads. A second tab runs a TRS through a price path with a margin account, showing how a notional the client never posted still generates daily cash calls.

Where this goes next

The full mechanics, valuation and risk of the most-traded member of the family are in interest rate swaps. The cross-currency and FX members are FX forwards and swaps, the volatility member is variance swaps, and the credit member is credit default swaps. The collateral machinery that makes all of them workable is collateral and haircuts.

15.6 Standard Payoff Profiles

sales · trading · structuring | core | equity · fx · rates · cross-asset

A handful of option combinations come up so often that they have names, and being able to draw any of them in five seconds is a genuine interview skill: "draw me a straddle, a strangle and a butterfly" appears verbatim in this database. This section is the vocabulary and the geometry, the shapes and how to build them. *Choosing* which one to trade for a given market view, and how each behaves through time and in different regimes, is the subject of payoff design, and is deliberately not repeated here.

One principle

> KEY IDEA
>
> **Payoffs add.** The payoff of a portfolio at expiry is the sum of the payoffs of its legs, computed point by point along the same S_T axis. Every structure below is that sum, and every one of them can be drawn by sketching each leg faintly and adding the heights. There is no second technique to learn.

The building blocks are the four positions from options, long and short call and put, plus the underlying itself (a 45-degree line) and cash (a horizontal line). To keep every number below comparable, all examples use one consistent set of market prices on a stock at 100, with zero rates:

$$C_{90} = 14.00, \quad C_{100} = 8.00, \quad C_{110} = 4.00,$$

and, from put-call parity at zero rates, $P_{90} = 4.00$, $P_{100} = 8.00$, $P_{110} = 14.00$.

These are not flat-volatility prices. Taking one year to expiry and backing out an implied volatility strike by strike gives roughly 21.2%, 20.1% and 19.2% at 90, 100 and 110, a downward tilt that is the ordinary equity skew. The set is deliberately built that way, because a flat-volatility example would make every structure below look symmetric when the market never is.

Vertical spreads

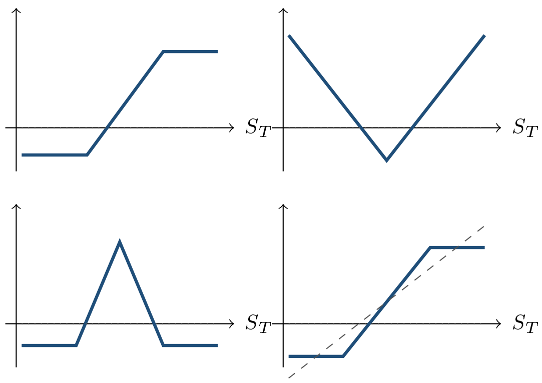
A **vertical spread** is long one option and short another of the same type and expiry, at a different strike. Buying the lower strike and selling the higher gives a **call spread** (bullish); buying the higher put and selling the lower gives a **put spread** (bearish). Both cost a net premium, and both cap the payoff.

WORKED EXAMPLE

Call spread 100/110. Buy the 100 call at 8.00, sell the 110 call at 4.00. Net cost 4.00.

- Below 100: both expire worthless, loss 4.00.
- Above 110: the spread pays its full width of 10, profit $10 - 4.00 = 6.00$.
- Break-even: $100 + 4.00 = 104.00$.

So you have paid 4.00 rather than 8.00 for upside exposure, at the cost of giving away everything above 110. That trade-off, cheaper entry against a capped payoff, is the entire reason spreads exist.



Left to right, top row then bottom row: call spread, straddle, butterfly, collar. In the last panel is the underlying alone, for comparison.

REAL INTERVIEW QUESTION (BNP · Structured Products)

Draw me a call spread and a put spread. How do you build them?

WHAT THEY TEST / ANSWER

Draw first, narrate while drawing. Call spread: buy the lower-strike call, sell the higher-strike one, same expiry. The payoff is flat and negative below the lower strike, rises at 45 degrees between the strikes, and is flat again above the upper strike. Put spread: buy the higher-strike put, sell the lower one, giving the mirror image, flat above, falling to the left, flat below.

Then say the three numbers that define either one without being asked, because that is what separates a drawing from an answer: the **cost** is the net premium, the **maximum payoff** is the distance between the strikes, and the **break-even** sits one net premium above the long strike for a call spread, below it for a put spread. Close with the purpose: you are selling away the tail you do not believe in to pay for the part you do, which is why call spreads are the workhorse for adding leverage inside structured notes.

REAL INTERVIEW QUESTION (Morgan Stanley · EQD)

What is a call spread, and what is the maximum gain?

WHAT THEY TEST / ANSWER

Definition in one line: long a call at K_1, short a call at a higher K_2, same expiry and underlying. Maximum gain: the width of the spread minus the net premium paid, $(K_2 - K_1) - \text{cost}$, achieved for any S_T at or above K_2. Give a number to prove you can compute it: a 100/110 spread costing 4.00 pays at most $10 - 4.00 = 6.00$. Then give the maximum loss unprompted, which is just the net premium, 4.00, and the break-even, 104.00. Finish with the observation an interviewer is listening for: the payoff is bounded on both sides, so unlike a naked call this is a position with a defined risk *and* a defined reward, and the ratio between them is what you are really choosing when you pick the strikes.

Straddles and strangles

A **straddle** is a call and a put at the *same* strike. A **strangle** is a call and a put at *different* strikes, the call above and the put below. Both are long volatility: they pay if the underlying moves far enough in either direction, and lose the premium if it sits still.

WORKED EXAMPLE

Straddle at 100: buy the 100 call at 8.00 and the 100 put at 8.00, cost 16.00. It profits below 84 or above 116, and loses the full 16.00 if the stock finishes exactly at 100.

Strangle 90/110: buy the 110 call at 4.00 and the 90 put at 4.00, cost 8.00. Break-evens at 82.00 and 118.00.

The comparison is the point. The strangle costs exactly half the straddle, so its maximum loss is far smaller, but it needs a much bigger move to pay, and it loses its whole premium anywhere in a twenty-point window rather than at a single point.

REAL INTERVIEW QUESTION (Marigny Capital · Structured Products)

Draw the payoffs of various vanilla strategies (straddle, strangle, butterfly), explaining their different elements.

WHAT THEY TEST / ANSWER

Draw all three on the same axis scale so they can be compared, and for each one name four things: construction, shape, break-evens and what you paid. **Straddle**: call plus put at the same strike, a V with its bottom at the strike, break-even one total premium either side, the most expensive of the three. **Strangle**: call and put at different strikes, the same V but with a flat-bottomed trough between the strikes, cheaper, needs a bigger move. **Butterfly**: long a low strike, short two at the middle, long a high strike, giving a tent peaked at the middle strike, very cheap, and it is the mirror image of the other two since it pays when nothing happens.

Then add the sentence that shows you see them as one family rather than three recipes: a butterfly is exactly a long call spread plus a short call spread stacked at the middle strike, and it is the payoff you get by selling the straddle and buying back the tails, which is why long butterfly and long straddle are opposite views. The Greeks behind that are in the Greeks and the trading logic in volatility strategies.

Butterflies and condors

A **butterfly** is long one option at K_1, short two at K_2, long one at K_3, with the strikes equally spaced. It pays most when the underlying finishes exactly at the middle strike and nothing at the wings, a tent shape.

WORKED EXAMPLE

Butterfly 90/100/110 with calls. Buy the 90 call at 14.00, sell two 100 calls at 8.00, buy the 110 call at 4.00. Cost $14.00 - 16.00 + 4.00 = 2.00$. Maximum payoff is 10 at $S_T = 100$, so maximum profit $10 - 2.00 = 8.00$, maximum loss 2.00, break-evens at 92.00 and 108.00.

Note that the cost, 2.00, is exactly $C_{90} - 2C_{100} + C_{110}$, a discrete second difference of the call price in strike. It must be positive, otherwise you would be paid to hold a structure that can never pay out less than zero, which is a free lunch. That is the arbitrage constraint that call prices must be **convex in strike**, and the butterfly is how you trade it.

A **condor** is the same idea with four strikes instead of three, so the peak is a plateau between the two middle strikes rather than a point. It costs more and is less demanding about exactly where the underlying lands.

Structures built against a position

The remaining names are combinations that include the underlying itself, and they are the ones a sales desk actually shows to clients.

- **Covered call**: long the stock, short a call. You keep the dividend and the premium and give away the upside above the strike. The payoff is identical to a short put, which is worth noticing.
- **Protective put**: long the stock, long a put. Insurance, with the premium as its cost. The payoff is identical to a long call.

- **Collar**: long the stock, long a put below, short a call above. A floor and a cap, with the call premium paying for the put. When the two premiums match exactly it is a **zero-cost collar**, which is how it is almost always sold.
- **Risk reversal**: long a call and short a put (or the reverse), *without* the stock. Its payoff is a synthetic forward that is flat between the strikes. In FX the same words also name a volatility quote, the implied vol of a 25-delta call minus that of a 25-delta put, developed in risk reversals.
- **Ratio spread**: a vertical spread with unequal quantities, for example long one call and short two higher-strike calls, which cheapens the structure but reintroduces an unbounded loss.
- **Calendar spread**: same strike, different expiries, long the far and short the near. It is a trade on time and on the term structure of volatility rather than on direction.

WORKED EXAMPLE

Collar. Long the stock at 100, buy the 90 put at 4.00, sell the 110 call at 4.00. The premiums match, so this is the textbook **zero-cost collar**: your value at expiry is floored at 90 and capped at 110, and it costs nothing to put on. That the two premiums happen to be equal here is the skew doing the work, since the put you buy is ten points out of the money and the call you sell is ten points out on the other side, yet they price the same. Moving either strike breaks the match, and choosing where to break it is what a client conversation actually consists of: how much upside will you give up for how much protection.

REAL INTERVIEW QUESTION (Nexo Capital · Structured Products)

What is the difference between a collar and a bull spread?

WHAT THEY TEST / ANSWER

The difference is whether you own the underlying. A **bull spread** (call spread) is options only: long a low-strike call, short a high-strike one, a defined-risk bet that costs a premium and pays only if the underlying rises. A **collar** is built *around a position you already hold*: long the stock, long a put for protection, short a call to pay for it, so it is a risk-management overlay rather than a directional bet, and it is usually structured to cost nothing.

Then show they are more related than they look, which is the good version of this answer. Both produce a capped-and-floored payoff between two strikes. The collar's floor comes from the put and its exposure below the floor is removed, while the bull spread's floor is simply the premium you paid. And by put-call parity the option overlay in a collar, long put against short call, is a *short* risk reversal, so a collar is a bull spread's cousin with the stock and a synthetic short forward bolted on. Close with the client framing: a bull spread is bought by someone who wants exposure, a collar by someone who already has it and wants to sleep.

INTERVIEW TRAP

Four traps. (1) Drawing the payoff when asked for profit, or vice versa; the structures above are drawn net of premium where a cost is quoted, and you must say which you are showing. (2) Forgetting that a covered call is a short put and a protective put is a long call; interviewers use those equivalences to check you can see through the packaging. (3) Calling a ratio spread a defined-risk trade; the extra short leg reopens the unbounded tail. (4) Quoting a butterfly at a negative cost, which is an arbitrage, since call prices must be convex in strike.

VISUAL / CODE TO BUILD: Interactive

A payoff constructor. The reader drags option legs (type, strike, quantity, long or short) onto a canvas and sees each leg drawn faintly with the net payoff drawn boldly on top, so the "payoffs add" principle is literally the interaction. A toggle switches between payoff and profit, shifting every line by its premium, and a readout gives cost, maximum gain, maximum loss and break-evens live. Preset buttons load each named structure so the reader can see, for instance, that loading a covered call produces exactly the same bold line as loading a short put, and that a butterfly is a call spread plus an inverted call spread. A price-consistency check flags any leg set that would make a butterfly cost less than zero.

Where this goes next

These shapes are the alphabet. Choosing which one expresses a given market view, and how each behaves as spot moves, time passes and volatility changes, is payoff design and the strategy sections that follow it, in particular directional, volatility and range-bound strategies. Using these combinations to manufacture something else entirely, such as replicating a digital with a tight call spread, is replication strategies.

15.7 Greeks (First and Second Order)

sales · trading · structuring | core | equity · fx · rates · cross-asset

An option's price depends on six things, so it has six sensitivities, and those sensitivities have their own sensitivities. Those are the Greeks. This section is the *map*: what a Greek is, the full taxonomy at first and second order, the sign of each for the four basic positions, and the quoting conventions that trip people up in practice. How each one behaves as spot, time and volatility move, and how a desk actually hedges them, is the whole of Greek definitions and intuition and the sections that follow it, and is not repeated here.

One equation organises all of them

A Greek is nothing more exotic than a partial derivative of the option's value with respect to one input. Expand the value V to second order in the things that move:

$$dV \approx \underbrace{\Delta\, dS}_{\text{delta}} + \underbrace{\tfrac{1}{2}\Gamma\,(dS)^2}_{\text{gamma}} + \underbrace{\mathcal{V}\, d\sigma}_{\text{vega}} + \underbrace{\Theta\, dt}_{\text{theta}} + \underbrace{\rho\, dr}_{\text{rho}} + \cdots$$

KEY IDEA

Every Greek is one term in that expansion. First-order Greeks are the straight-line response to a single input moving. Second-order Greeks are the corrections that matter because the first-order ones do not stay put. If you remember only one thing from this section, remember that the Greeks are not a list to memorise but the terms of a Taylor expansion, which is why the signs and the relationships between them are derivable rather than arbitrary.

First order

Greek	Derivative	Measures
Delta Δ	$\partial V/\partial S$	price move per unit of spot; the hedge ratio
Vega $\mathcal{V}$	$\partial V/\partial \sigma$	sensitivity to implied volatility
Theta Θ	$\partial V/\partial t$	value lost as time passes
Rho ρ	$\partial V/\partial r$	sensitivity to interest rates
Dividend sens.	$\partial V/\partial q$	sensitivity to the dividend assumption
Repo sens.	$\partial V/\partial \text{repo}$	sensitivity to the borrow cost

The last two are absent from most textbooks and present on every equity desk, because both act through the forward, as established in forwards, and both are genuinely uncertain and genuinely traded. Note also that "vega" is not a Greek letter; it is a trader coinage, which is why it is sometimes written κ or, as here, $\mathcal{V}$.

Second order, and the third-order ones desks still quote

Greek	Derivative	Measures
Gamma Γ	$\partial^2 V/\partial S^2$	how delta changes as spot moves
Vanna	$\partial^2 V/\partial S\partial\sigma$	how delta changes with vol (or vega with spot)
Volga (vomma)	$\partial^2 V/\partial \sigma^2$	how vega changes with vol
Charm	$\partial^2 V/\partial S\partial t$	how delta changes with time
Veta	$\partial^2 V/\partial \sigma\partial t$	how vega changes with time
Colour	$\partial^3 V/\partial S^2\partial t$	how gamma changes with time
Speed	$\partial^3 V/\partial S^3$	how gamma changes with spot

The last two, colour and speed, are third derivatives rather than second; they sit in this table because they belong to the same family and appear on the same risk screens, not because they are of the same order. Gamma is the one that matters most and has its own section, gamma and convexity. The cross-Greeks, vanna and volga, are what make a smile-aware book different from a Black-Scholes book, and are developed in cross-Greeks.

REAL INTERVIEW QUESTION (CACIB · Cross Asset)

Greeks: what are they, which ones do you know, and what do they represent?

WHAT THEY TEST / ANSWER

Do not begin by listing. Begin with the definition, because it makes the list derivable: a Greek is the partial derivative of the option's price with respect to one input, so there is one Greek per input and then second-order ones for how those change. Then list in order of how much a desk cares: delta, the sensitivity to spot and the hedge ratio; gamma, how fast delta moves; vega, sensitivity to implied volatility; theta, the cost of time; rho, sensitivity to rates; and on an equity book, sensitivity to dividends and to repo, which are real risks even though textbooks omit them. Mention the important second-order ones by name, vanna and volga, and say what they are for, namely that they are what a book carries once you admit the volatility surface is not flat. Finish by framing them as a Taylor expansion of the price, which is the sentence that tells a cross-asset interviewer you understand the structure rather than the vocabulary.

REAL INTERVIEW QUESTION (BNP · Exo)

You are long a call. What are your sensitivities to volatility, delta, dividends, repo and rates?

WHAT THEY TEST / ANSWER

Go through them with a sign and a one-line reason for each, because the reasons are the answer. **Volatility**: long, vega positive, since your loss is capped at the premium and your gain is not, so more dispersion can only help. **Delta**: positive and between 0 and 1, rising toward 1 as the option goes in the money. **Dividends**: short, because dividends lower the forward and a call is long the forward. **Repo**: short, for exactly the same reason, since a higher borrow rate lowers the forward. **Rates**: long, because rates raise the forward, and note that the rho of an equity option is small relative to the others at short maturities but is not small on a multi-year structured product.

The unifying sentence to close with is that dividends, repo and rates are not three separate risks but one, the forward, entering through $F = Se^{(r-q-\text{repo})T}$, so their signs are locked together and you never have to memorise them separately.

Signs

For the four basic positions, the first-order signs are fixed:

	Delta	Gamma	Vega	Theta
Long call	$+$	$+$	$+$	$-$
Short call	$-$	$-$	$-$	$+$
Long put	$-$	$+$	$+$	$-$
Short put	$+$	$-$	$-$	$+$

Two patterns carry the whole table. Gamma and vega always share the sign of the *option position*, not its direction: anyone long an option, call or put, is long gamma and long vega. And theta is always the opposite sign to gamma, which is the trade-off developed in theta as the cost of convexity: you pay for the right to profit from movement, and you pay for it in time.

One honest exception, consistent with the negative time value shown in options: when rates are positive, a deep in-the-money European put can have *positive* theta, because its value is already below intrinsic and time passing brings the near-certain payoff closer. The condition matters: it is discounting that puts the value below intrinsic in the first place, so at zero rates the exception disappears.

REAL INTERVIEW QUESTION (BNP · Structured Products)

Explain the Greeks and give them for a long and short call, and a long and short put.

WHAT THEY TEST / ANSWER

Give the table, but derive it out loud rather than reciting, since that is what is being checked. Start with the two rules: gamma and vega follow whether you are long or short the *option*, not whether you are bullish, so a long put is long gamma and long vega just like a long call. And theta always carries the opposite sign to gamma, because convexity is paid for in time decay. From those two rules plus the obvious delta signs, positive for a long call and short put, negative for a long put and short call, the entire four-by-four table follows without memorisation.

Then add the qualification that distinguishes a structuring candidate: these signs are for vanilla options, and they can invert for exotics. A digital's delta and gamma flip sign across the strike, and a barrier option's gamma explodes and changes sign near the barrier, which is why the risk of a structured book is not the sum of textbook signs.

Conventions, or why your number does not match theirs

The raw derivatives are almost never what is quoted. Desks rescale them into units that correspond to a realistic move:

- **Vega** is quoted per *one percentage point* of implied volatility, so the raw $\partial V/\partial\sigma$ is divided by 100.
- **Theta** is quoted per *calendar day*, so the annual derivative is divided by 365 (some desks use business days, which is a real source of disagreement).
- **Rho** is quoted per 1%, so again divided by 100.
- **Delta** may be quoted per share, as a percentage, or as a cash amount; **gamma** is often quoted as the cash change in delta for a 1% move in spot rather than for one unit.

WORKED EXAMPLE

A one-year at-the-money call, spot 100, implied volatility 20%, zero rates. The raw vega is $S\varphi(d_1)\sqrt{T} = 39.70$, so the *quoted* vega is 0.397: a one point rise in implied vol, from 20% to 21%, is worth about 0.40 on an option priced at 7.97.

The raw theta is -3.97 per year, so the quoted theta is $-3.97/365 = -0.0109$ per day. Sanity check the pair against the standard at-the-money approximation $C \approx 0.4\,\sigma S\sqrt{T}$, which follows from $N(d_1) - N(d_2) \approx \varphi(0)\,\sigma\sqrt{T}$ with $\varphi(0) = 1/\sqrt{2\pi} \approx 0.399$: it gives 8.00 against the 7.97 the model produces. A factor of 100 in vega, or of 365 in theta, is the single most common reason two people quoting the same book disagree.

REAL INTERVIEW QUESTION (CACIB · Delta One)

Define vomma and colour.

WHAT THEY TEST / ANSWER

Both are third-tier Greeks and the question is really testing whether you can construct a definition from the naming logic rather than having memorised a glossary. **Vomma** (also written volga) is $\partial^2 V/\partial\sigma^2$, the sensitivity of *vega* to volatility, so it measures convexity in volatility. It is near zero for an at-the-money option, where vega peaks and is locally flat, and positive in the wings, which is exactly why out-of-the-money options are the instrument of choice for trading volatility-of-volatility and why they are the ones a smile model has to get right. **Colour** is $\partial\Gamma/\partial t$, the decay of gamma through time, so it tells you how your gamma profile will look tomorrow rather than today. It matters most near expiry and near a strike, where gamma is both large and changing fastest, and it is what a gamma-hedging desk uses to anticipate how much its hedge ratio will need to move overnight rather than reacting after the fact.

INTERVIEW TRAP

Four traps. (1) Saying a long put is short vega or short gamma; both follow the option position, not the direction. (2) Comparing Greeks without comparing conventions; vega per point versus per unit differs by 100. (3) Assuming theta is always negative for a long option, when a deep in-the-money European put is the counterexample. (4) Applying vanilla sign rules to digitals and barriers, where gamma can flip sign and blow up.

VISUAL / CODE TO BUILD: Interactive

A Greek explorer with two linked views. The top view plots the option value against spot, with the tangent line (delta) and the fitted parabola (delta plus gamma) drawn over it, so the Taylor expansion at the head of this section is literally what the reader sees. The bottom view plots any chosen Greek against spot, with sliders for volatility, maturity and rate, and a toggle for call or put and long or short that flips the curve so the sign table becomes something the reader verifies rather than memorises. A units switch re-expresses every readout between raw derivatives and desk conventions (vega per point, theta per day), displaying both simultaneously so the factor of 100 or 365 is visible rather than mysterious.

Where this goes next

This was the taxonomy. The behaviour starts at Greek definitions and intuition, where each Greek gets its own treatment: how gamma and convexity really work, why theta is the price of gamma, how vega changes across the surface, and how the Greeks move as spot, volatility and time do. Turning those numbers into trades is delta, gamma and vega hedging, and the model that produces the formulas behind every derivative in this section is Black-Scholes.

15.8 Implied vs Realised Volatility

sales · trading · structuring | core | equity · fx · rates · cross-asset

Two numbers are both called volatility and they are not the same thing. One is measured from what the underlying actually did; the other is backed out from what options cost. Nearly every volatility question in this database is checking whether a candidate keeps them apart, can compute either one under pressure, and understands that the gap between them is itself a tradeable object.

Two definitions

Realised volatility (also historical volatility) is a *measurement*. Take the log returns $r_i = \ln(S_i/S_{i-1})$ over n periods and compute

$$\sigma_{\text{realised}} = \sqrt{\frac{A}{n}\sum_{i=1}^{n} r_i^2},$$

where A is the number of periods in a year, conventionally 252 trading days. This zero-mean form is the market convention, and it is what a variance swap settles on.

Implied volatility is an *inference*. It is the single number you must feed into Black-Scholes to make its output equal the option's market price. Nothing is measured: it is the market's price, re-expressed in volatility units because that is a more comparable scale than a premium.

INTUITION

Realised volatility is the **speedometer reading you recorded on the drive you just took**. Implied volatility is the **insurance premium the company is quoting for tomorrow's drive**. The first is a fact about the past. The second is a price, and like any price it contains a forecast plus a margin plus whatever supply and demand happens to be doing that morning. Confusing them is like assuming your insurer's quote is a measurement of your driving.

KEY IDEA

Implied volatility is a **price**, not a forecast. It sits systematically *above* subsequent realised volatility most of the time, a gap known as the **volatility risk premium**, because option sellers are taking on an unbounded, crisis-correlated risk and demand to be paid for it. That premium is the reason systematically selling volatility makes money most months and occasionally loses everything at once.

REAL INTERVIEW QUESTION (UBS · Structured Products)

What is the difference between implied volatility and realised volatility?

WHAT THEY TEST / ANSWER

Define both precisely, then give the relationship, which is the actual content. Realised is computed from the underlying's historical returns, annualised by the square root of time; implied is extracted from an option's market price by inverting Black-Scholes. So one is a statistic and one is a price. Then the three things worth adding. Implied is forward-looking over the option's remaining life while realised is backward-looking over whatever window you chose, so they are not even measuring the same period. Implied exceeds subsequent realised on average, the volatility risk premium, which is why option sellers earn a carry and why that carry is short a fat tail. And the difference is directly tradeable: a delta-hedged long option makes money exactly when realised exceeds implied, which is what variance swaps package cleanly.

Annualisation, and the square root of time

Volatility scales with the square root of time because *variance* scales linearly with time. Independent returns have variances that add, so $\sigma_{\text{annual}} = \sigma_{\text{daily}}\sqrt{252}$, and since $\sqrt{252} \approx 15.87$, the desk shortcut is that a daily move of about 1% corresponds to an annual volatility of about 16%.

WORKED EXAMPLE

A stock rises 1% one day and falls 1% the next, repeatedly. Its daily returns have a root-mean-square of essentially 1.00%, so

$$\sigma_{\text{annual}} = 1.00\% \times \sqrt{252} = 15.87\% \approx 16\%.$$

This is the number worth committing to memory in both directions: 16% annual volatility means roughly 1% daily moves, and 32% means roughly 2%.

REAL INTERVIEW QUESTION (Société Générale · Volatility)

Historical volatility: definition? Comparison with implied volatility? I give you a stock that does $+1\%$ one day and -1% the next, what is its volatility?

WHAT THEY TEST / ANSWER

Three parts, answered in order and quickly. Definition: the annualised standard deviation of log returns, conventionally $\sqrt{(252/n)\sum r_i^2}$. Comparison: realised is measured from the past, implied is inverted out of an option price, and implied usually exceeds subsequently realised because it contains a risk premium. Number: daily moves of 1% give $1\% \times \sqrt{252} \approx 16\%$ annualised, and say the shortcut out loud, that $\sqrt{252} \approx 16$, so 1% daily is 16% annual.

Then add the detail that shows you thought rather than pattern-matched: a strictly alternating $+1\%, -1\%$ path has perfect negative autocorrelation, so its variance measured over long horizons barely grows with the horizon: the price is pinned near where it started however many days you look at. The standard estimator sums squared daily returns and ignores autocorrelation entirely, so it still reports 16%. That gap between "how much it wiggles" and "how far it travels" is exactly what a gamma-hedged position monetises and why realised volatility estimators are sensitive to sampling frequency.

REAL INTERVIEW QUESTION (BNP · EQD)

A stock does -2% then $+2\%$ every day for three years, starting at 100. What is its volatility over one year? And what do you pay for an at-the-money call on it?

WHAT THEY TEST / ANSWER

Volatility first: daily moves of 2% give $2\% \times \sqrt{252} = 31.75\%$, call it 32%. Then the call, using the standard at-the-money approximation $C \approx 0.4\,\sigma S\sqrt{T}$ derived in vega exposure, so $0.4 \times 0.32 \times 100 = 12.80$ for one year, ignoring rates and dividends and saying so.

Then volunteer the observation the question is really fishing for. A stock that alternates -2% and $+2\%$ does *not* return to where it started: $0.98 \times 1.02 = 0.9996$, so it loses 4 basis points every two days. Over three years, about 375 such pairs, the spot is down to roughly 86. That is **volatility drag**, the arithmetic fact that $\mathbb{E}[\ln S] < \ln \mathbb{E}[S]$, and it is the same $\sigma^2/2$ term that appears in the drift of geometric Brownian motion. Noticing it unprompted is what separates a good answer here.

Variance adds, volatility does not

This is the single most useful computational fact in the topic. Over non-overlapping periods, variances add:

$$\sigma_{\text{total}}^2 T \;=\; \sigma_1^2 T_1 + \sigma_2^2 T_2.$$

Volatilities themselves never add, and averaging two volatilities is almost always wrong.

WORKED EXAMPLE

A stock realises 20% annualised over the first six months of a year, and 40% annualised over the full year. What did it realise in the second half?

Work in variance:

$$0.40^2 \times 1 \;=\; 0.20^2 \times \tfrac{1}{2} \;+\; \sigma_2^2 \times \tfrac{1}{2}$$

$$0.16 = 0.02 + 0.5\,\sigma_2^2 \;\Longrightarrow\; \sigma_2^2 = 0.28 \;\Longrightarrow\; \sigma_2 = 52.9\%.$$

The naive answer, that the second half must have been 60% because the average of 20 and 60 is 40, is wrong, and it is wrong in the direction that matters: the true figure is 53%, materially lower, because the big number dominates a variance average far more than it dominates an arithmetic one.

REAL INTERVIEW QUESTION (BNP · EQD)

Over six months a stock realised 20% annualised. Over the full year it realised 40%. What did it realise over the last six months?

WHAT THEY TEST / ANSWER

Say "variance adds, volatility does not" before doing anything else, because that sentence is the answer and the arithmetic is a formality. Then: $0.40^2 = \frac{1}{2}(0.20^2) + \frac{1}{2}\sigma_2^2$, so $\sigma_2^2 = 2(0.16) - 0.04 = 0.28$ and $\sigma_2 = 52.9\%$, call it 53%. Pre-empt the trap by naming it: the intuitive answer of 60%, from averaging volatilities, is wrong, and the correct figure is lower because variance weights the larger number more heavily. If they push, the same principle is why you cannot average implied volatilities across maturities to get a forward volatility, and why forward variance is the object that actually interpolates on a term structure.

Trading the gap

If implied and realised differ, someone is wrong, and there is a well-defined way to collect on it. Buy an option, delta-hedge it continuously, and your profit and loss over the life of the trade is, approximately,

$$\text{P\&L} \;\approx\; \tfrac{1}{2}\int_0^T \Gamma_t\, S_t^2\, \big(\sigma^2_{\text{realised}} - \sigma^2_{\text{implied}}\big)\, dt.$$

You paid for volatility at the implied level and you are re-hedging at the realised level, so you make money precisely when the underlying moves more than you paid for. Note the weighting: it is not the raw difference in variance but the difference weighted by gamma and by spot squared, so *when* the movement happens matters as much as how much of it there is. This is developed in delta, gamma and vega hedging; the instrument that strips out the gamma weighting and pays the clean variance difference is the variance swap.

REAL INTERVIEW QUESTION (BNP · Trading)

How does the price of a straddle evolve if you priced it with a volatility of 2.5% and the realised volatility turns out to be 5%?

WHAT THEY TEST / ANSWER

Separate the two channels, because the question is ambiguous on purpose and a good answer says so.

Mark-to-market channel: if implied volatility re-marks toward the realised level, the straddle's value roughly doubles. An at-the-money straddle is worth about $0.8\,\sigma S\sqrt{T}$, which is close to linear in volatility, so implied moving from 2.5% to 5% roughly doubles it. You bought cheap and it re-priced.

Delta-hedging channel: if you delta-hedge, you do not need implied to move at all. You collect the gamma P&L from the realised moves, which is the integral above with realised variance at 0.05^2 against implied at 0.025^2. Realised variance is four times implied, so the gamma P&L is proportional to $\sigma_R^2 - \sigma_I^2 = 3\sigma_I^2$. Convert that into multiples of the premium, since that is the units the desk thinks in. With an at-the-money straddle gamma of $\Gamma \approx 0.8/(S\sigma_I\sqrt{T})$ and a premium of $0.8\,\sigma_I S\sqrt{T}$, the integral collapses to

$$\frac{\text{gamma P\&L}}{\text{premium}} \approx \frac{1}{2}\,\frac{\sigma_R^2 - \sigma_I^2}{\sigma_I^2} = \frac{3}{2},$$

so you collect about one and a half times what you paid, on top of getting the premium itself back. That beats the mark-to-market channel, which only doubles your money, and the reason is exactly that gamma pays in variance while the premium is quoted in volatility. Quoting the ratio, rather than asserting "three times", is what a trading desk wants.

Then the caveat that shows judgement: if you are long the straddle and do *not* hedge, your P&L depends on where spot actually finishes, not on realised volatility, so a stock can realise 5% and finish exactly at the strike and you lose the entire premium. Realised volatility only becomes your P&L once you are hedging it.

REAL INTERVIEW QUESTION (BNP · EQD)

Is it better to use implied or historical volatility to estimate future volatility?

WHAT THEY TEST / ANSWER

Implied, with a correction, and the reasoning matters more than the verdict. Implied is forward-looking and aggregates every market participant's view including information not yet in the price history, such as a scheduled earnings date, an election or a pending deal, none of which a historical window can know about. Empirically, implied volatility is the better predictor of subsequent realised volatility than a historical window is.

The correction is the part that gets you the mark: implied is a *biased* predictor, because it contains the volatility risk premium, so it overpredicts on average. The practical answer is therefore to use implied as the base and adjust it downward for that premium, and to use the historical series to understand the regime and the shape of the distribution rather than to forecast the level. Finish by naming the practitioner's version, that you would look at the whole surface rather than one number, since the implied vol relevant to your question depends on which strike and which maturity you care about.

INTERVIEW TRAP

Four traps. (1) Averaging volatilities instead of variances, which is wrong whenever the two numbers differ and badly wrong when they differ a lot. (2) Comparing a 30-day realised volatility to a one-year implied and concluding one is cheap; they cover different horizons. (3) Treating implied volatility as the market's forecast rather than as a price containing a risk premium. (4) Saying "the volatility of the stock" as if there were one number, when there is a whole surface across strike and maturity.

VISUAL / CODE TO BUILD: Interactive

A two-line chart of implied volatility and subsequently realised volatility over the same forward window, with the gap shaded so the volatility risk premium is visible as an area that is usually positive and occasionally violently negative. Beneath it, a delta-hedging simulator: the reader sets an implied volatility to buy at, then draws or randomises a price path with a chosen realised volatility, and the panel accumulates the gamma P&L step by step, so the reader watches the integral in this section fill up and sees that the result depends on *where* the moves happened, not only how big they were. A third control resamples the same path at daily, hourly and five-minute frequency to show the realised volatility estimate changing with sampling.

Where this goes next

Implied volatility is not one number but a surface across strike and maturity, which is the smile and skew. The mechanics of monetising the implied-realised gap are delta, gamma and vega hedging, and the instrument that isolates it is the variance swap. How a book's volatility exposure behaves across regimes is vega exposure and volatility regimes.

15.9 Volatility smile and skew

sales · trading · structuring | core | equity · fx · commodity · cross-asset

"Draw me the smile" is probably the single most frequent whiteboard request in an equity-derivatives, exotics or volatility interview. It is asked because it separates two kinds of candidate in about thirty seconds: the one who has memorised that implied volatility "makes a U shape", and the one who can say what is on each axis, why the shape is not flat, why equity and FX look different, and what it costs you on a real trade. This section builds the second kind.

The observation that breaks the model

Recall the one definition everything here rests on: the **implied volatility** of a traded option is the single number σ you must feed into the Black-Scholes formula to make it return the option's market price. It is a price quoted in different units, not a forecast. That distinction, and what it means for a trader, is developed in implied versus realized volatility; the pricing formula itself is derived in the Black-Scholes section.

Now the awkward fact. Black-Scholes assumes the underlying has *one* volatility, a single constant σ that describes the asset itself. If that were true, every option on the same underlying, whatever its strike and whatever its maturity, would imply exactly the same number. Take a real option chain on any liquid underlying and back out the implied volatility strike by strike, and you get a curve that is emphatically not a flat line. That curve is the **smile**, and the whole apparatus of modern volatility trading exists because it is not flat.

> **INTUITION**
>
> Think of insurance premiums per euro of cover. A policy against your car being scratched is cheap per euro insured. A policy against your house burning down costs far more per euro insured, even though burning down is far less likely, because everyone wants that cover, few want to sell it, and the insurer cannot diversify it away (houses tend to burn in the same fire). Deep out-of-the-money puts are the house-fire policy of the equity market. Implied volatility is the price per euro of cover, so it is higher exactly where the disaster protection sits.

Definitions: surface, smile, skew, term structure

Write $\sigma_{\text{imp}}(K, T)$ for the implied volatility of the option struck at K expiring at T. This function of two variables is the **implied volatility surface**. Two slices of it have names.

- A **smile** is a slice at fixed maturity: σ_{imp} plotted against strike (or against moneyness). The word describes the convex, roughly symmetric U shape typical of currency pairs, where both far strikes are dearer than the middle.
- A **skew** (also called a *smirk*) is the same slice when it is strongly tilted rather than symmetric: in equities it falls monotonically as the strike rises, so low strikes carry much higher implied volatility than high strikes. In everyday desk speech "the skew" also means the *slope* itself, the number you are long or short.
- The **term structure of volatility** is the orthogonal slice: fix the moneyness, vary T. It is typically upward sloping in calm markets (long options carry more uncertainty about the future volatility regime) and inverts in a panic, when the immediate month prices a shock that the market expects to fade. Term structure is a maturity effect and has nothing to do with the smile, which is a strike effect. Confusing the two is a standard interview casualty.

The x-axis matters as much as the shape, and interviewers check it. Four conventions are common: raw strike K, simple moneyness K/S_0, log-moneyness $\ln(K/F)$ where F is the forward, and *delta*, which is the market standard in FX. Plotting against log-moneyness or delta is what makes smiles for different spots and maturities comparable, because it removes the trivial dependence on the level of the underlying.

> **REAL INTERVIEW QUESTION** (Exane · Exo)
>
> *What are the smile and the skew?*
>
> **WHAT THEY TEST / ANSWER**
>
> They test whether you know that implied volatility is a function of strike, not a property of the asset. Give the definition in one breath: plot implied volatility against strike for a fixed maturity, and you do not get a flat line; a symmetric convex shape is called a smile, a tilted monotone one is called a skew or smirk, and equity indices show a pronounced downward skew while G10 currency pairs are closer to a genuine smile. Then say the sentence that shows you have thought about it: this is not a defect of the market, it is the market telling you that the lognormal assumption behind a single Black-Scholes volatility is wrong, and the smile is the correction it applies.

Why the surface cannot be flat

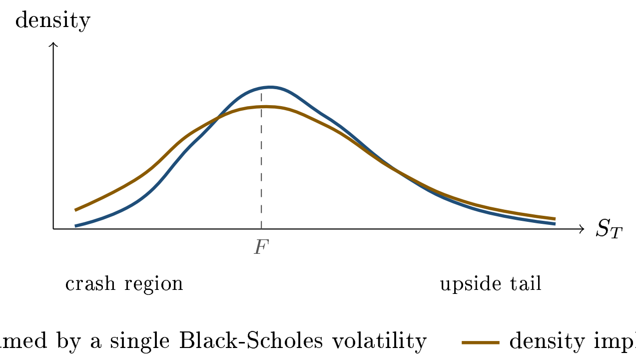
There are two layers to the answer, and strong candidates give both. The first is distributional and applies to every asset class. The second is structural and explains why each asset class has its own characteristic shape.

Layer one: the real distribution is not lognormal. Black-Scholes assumes returns are normal, so the terminal price is lognormal: no jumps, and a volatility that never changes. Actual returns have fat tails (extreme moves happen far more often than a normal law allows) and, in equities, a pronounced negative asymmetry (the big moves are disproportionately downward). An option struck far from the money pays only in the tail, so its value is almost entirely a statement about how heavy that tail is. If the true tail is heavier than lognormal, the option must be worth more than the flat-volatility formula says, and the only way to express "more" inside that formula is to raise the volatility you feed it. Hence a higher implied volatility away from the money.

That argument can be made exact. Option prices do not merely hint at the distribution, they contain it. Let $C(K, T)$ be the price today of a call struck at K maturing at T, and let q be the risk-neutral density of S_T. Then

$$\frac{\partial C}{\partial K} \;=\; -e^{-rT}\,\mathbb{Q}(S_T > K), \qquad q(K) \;=\; e^{rT}\,\frac{\partial^2 C}{\partial K^2}.$$

This is the **Breeden-Litzenberger** result, and it is the cleanest way to see what the smile is. The second derivative of the call price in strike is, up to discounting, the probability density the market is using. Concretely, a tight butterfly (long one call at $K-h$, short two at K, long one at $K+h$) costs approximately $h^2 e^{-rT} q(K)$, so butterfly prices are a direct read of the density. A flat implied volatility across strikes would force q to be exactly lognormal. The market quotes a non-flat surface precisely because it does not believe that.



Layer two: supply and demand for each tail. Distributions explain why the surface bends; flows explain which way it tilts.

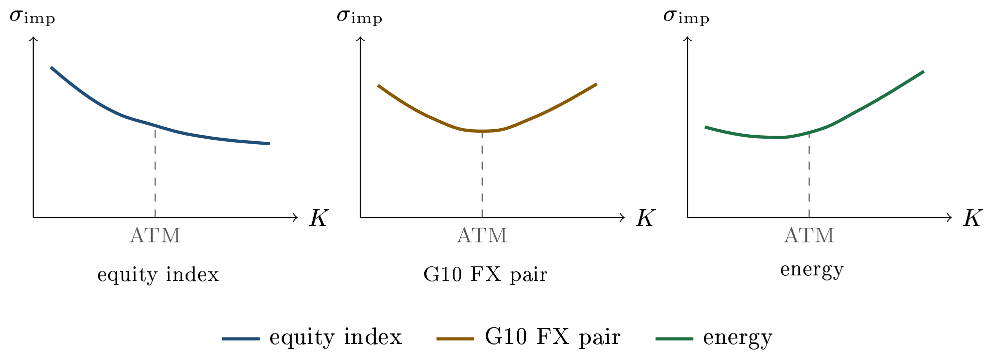
- **Leverage.** When a company's equity falls, its debt is unchanged, so its leverage rises and the equity becomes mechanically riskier. Falling spot and rising volatility are therefore linked at the level of the firm, not just by sentiment. This is the classic explanation for a negative equity skew.
- **Crash memory.** Before October 1987 the equity index smile was close to flat. It has been persistently down-sloping ever since. Markets price the crash they have seen.
- **Structural hedging demand.** Institutions are structurally long equities, so they buy downside puts and sell upside calls to finance them. That is a one-way order flow that bids up low strikes and offers high strikes, and no arbitrage forces it back, because a dealer who sells the puts has to carry the risk.
- **Issuance flow.** Retail structured-product issuance leaves dealers with systematic, one-directional exposures across the strike range. In an autocallable it is the investor who sells the downside

put, so the street ends up *long* that put and has to recycle the risk in the listed market, and because the autocall feature makes the position's vega and skew flip as spot moves, the hedging flow itself pushes the surface around; see autocallables for the mechanics.

- **Index versus single name.** Index skew is steeper than the skew of the average constituent, because in a crash correlation rises and the index realises more than the diversification of a normal day implies. That gap is the raw material of dispersion trading.

> **KEY IDEA**
>
> The smile is not a model. It is a *quoting language*: the market takes a formula everyone agrees is wrong, Black-Scholes with one constant volatility, and restores the correct price by using a different volatility for every strike and maturity. Implied volatility is therefore a coordinate, not a forecast, and the shape of the surface is a picture of the risk-neutral distribution the market is actually using.

One shape per asset class



Equities skew downward, steeply and persistently, for the four reasons above. The asymmetry is the point: the crash is what people insure against, and nobody insures against a rally. The shape of the equity curve in detail is treated in equity skew and smile.

G10 currencies are much closer to a symmetric smile. The reason is structural rather than statistical: a currency pair is a ratio, and a call on EUR/USD is a put on USD/EUR. There is no natural direction in which "everyone is long", so no one tail attracts all the hedging demand. Wings are still bid (jumps happen in both directions) which gives convexity, but the tilt is modest. The tilt becomes large as soon as the symmetry breaks: an emerging-market or pegged currency can devalue violently and revalue only gently, so its smile develops a strong one-way skew. The FX-specific treatment lives in the FX volatility smile.

Commodities vary by product, and energy is the instructive case. Physical supply can be disrupted (a pipeline, a cold snap, an embargo) but rarely over-delivered overnight, so the violent moves are upward. Energy options therefore often show an *upside* skew, the mirror image of equities, with calls carrying the higher implied volatility. See energy markets for the physical drivers.

REAL INTERVIEW QUESTION (MUREX · Volatility)

Draw the volatility smile for a stock, a currency and a commodity.

WHAT THEY TEST / ANSWER

They are testing whether your knowledge is a memorised picture or a causal story. Draw three panels with the same axes (strike or moneyness horizontal, implied volatility vertical, ATM marked), then justify each shape in one sentence: equity slopes down because of leverage, crash memory and one-way put demand; a G10 currency pair is nearly symmetric because a call on one currency is a put on the other, so neither tail owns the hedging flow; energy tilts up because supply shocks move prices up, not down. Volunteering that an emerging-market currency breaks the FX symmetry and skews toward devaluation is the detail that ends the question in your favour.

REAL INTERVIEW QUESTION (Société Générale · Volatility)

What is the smile effect? Plot it, specify the axis and say whether it is symmetrical.

WHAT THEY TEST / ANSWER

The trap is the last clause. Label the axes explicitly (strike or moneyness on the x-axis, implied volatility on the y-axis, never price), then answer the symmetry question by asset class rather than in general: broadly symmetric for a G10 currency pair, clearly *not* symmetric for an equity index, where it is a downward skew. Add why the axis choice matters: quoting in log-moneyness or in delta is what lets you compare two maturities or two underlyings on the same picture.

REAL INTERVIEW QUESTION (RBC · Structured Products)

What does the skew look like in equity? Why?

WHAT THEY TEST / ANSWER

Downward sloping in strike, and steeper for indices than for single names. Give the causal chain rather than one reason: leverage rises mechanically as equity falls; 1987 taught the market that gaps are downward; investors are structurally long and therefore structurally buyers of puts and sellers of calls; and index correlation rises in a sell-off, which steepens index skew beyond single-stock skew. A structuring interviewer will then ask what that does to the coupon they can offer, so be ready to connect the skew to the price of the downside optionality they are selling.

The trap: the smile is across strikes, not across call and put

At one strike there is exactly one implied volatility, shared by the call and the put. This follows from put-call parity, which for a European pair is model-free:

$$C - P \;=\; e^{-rT}\,(F - K).$$

The right-hand side contains no volatility at all. Black-Scholes satisfies this identity for *any* σ you plug in, so the σ that reproduces the market call price necessarily reproduces the market put price too. If a market ever quoted different implied volatilities for a call and a put at the same strike and maturity, that would not be skew, it would be an arbitrage in the parity relation. Put-call parity itself is established in options.

REAL INTERVIEW QUESTION (CACIB · FX)

Take a call struck at 100 with the underlying at 105: is the implied volatility of the put with the same strike higher, lower or equal to that of the call? Why?

WHAT THEY TEST / ANSWER

Equal, and the reason is put-call parity, not a convention. They deliberately made the call in-the-money to tempt you into saying "the put is out-of-the-money, and by the skew out-of-the-money puts trade at a higher volatility". That confuses two different axes: the skew compares *different strikes*, whereas here the strike is the same and only the option type changes. Write $C - P = e^{-rT}(F - K)$, point out that volatility does not appear, and conclude that one strike carries one implied volatility. This is a favourite because it catches candidates who learned the smile as a picture instead of as a function of K.

What the skew actually costs you

The smile is not decoration. Any structure whose payoff depends on more than one strike is priced off the *difference* between two points on the curve, so its value moves with the slope. The call spread is the cleanest case, and it is asked constantly.

WORKED EXAMPLE

Take $S_0 = 100$, one year to maturity, and ignore rates and dividends ($r = q = 0$, so the forward equals spot). Price a 100/130 call spread, that is long the 100 call and short the 130 call, two ways.

Flat Black-Scholes, one volatility of 20% for both strikes.

$$C(100) = 7.9656, \qquad C(130) = 1.0089, \qquad \text{call spread} = 6.9567.$$

(The 100 strike is at-the-money, and the standard at-the-money approximation $0.4\,\sigma S_0\sqrt{T} = 0.4 \times 0.20 \times 100 = 8.0$, developed in vega exposure, recovers 7.97, a useful consistency check.)

With an equity skew: 20% at the 100 strike, 17% at the 130 strike. Only the short leg changes:

$$C(130) = 0.5129, \qquad \text{call spread} = 7.9656 - 0.5129 = 7.4527.$$

Using the skew makes the same structure **0.50 more expensive**, about 7% of its flat-volatility price, and every cent of that comes from the leg you are short. The first-order estimate agrees: the vega of the 130 call is roughly 0.19 per volatility point, so lowering its volatility by 3 points removes about 0.57 of premium. The exact figure, 0.50, is smaller because vega itself falls as the volatility drops, which is the second-order effect (volga) covered in skew and smile sensitivities.

REAL INTERVIEW QUESTION (Natixis · EQD)

Take a 100-130 call spread. Will it cost more if you price it with the Black-Scholes model or if you use different volatilities per strike (according to the skew)?

WHAT THEY TEST / ANSWER

More expensive with the skew, provided the flat volatility you are comparing against is the at-the-money one, which is the implicit assumption; say that assumption out loud, because it is half the answer. Reasoning: with an equity skew the 130 strike carries a lower implied volatility than the 100 strike, so the call you are *short* is worth less, so the spread you paid for is worth more. They are checking that you know a call spread is a two-strike product and therefore a skew position, not merely a directional one. The follow-up is usually the digital, since a call spread scaled by the strike gap converges to one.

The digital shows the same effect in closed form and is worth carrying in your head. A digital call paying 1 if $S_T > K$ is the limit of a tight call spread, so its price is $-\partial C/\partial K$. Differentiating through the smile, where the volatility used at strike K is itself a function of K:

$$D = -\frac{dC}{dK} = \underbrace{-\frac{\partial C_{\mathrm{BS}}}{\partial K}}_{\text{flat-vol digital}} - \mathcal{V}\,\frac{\partial \sigma_{\mathrm{imp}}}{\partial K},$$

where $\mathcal{V}$ is the vega of the vanilla call. With a downward-sloping equity skew, $\partial\sigma_{\mathrm{imp}}/\partial K < 0$ and vega is positive, so the correction term is positive: the digital is worth *more* than the naive flat-volatility price. Quoting a digital off a single volatility is one of the classic ways to lose money on a first structuring job. Barrier and digital products are developed in exotic options.

REAL INTERVIEW QUESTION (Leonteq · Structured Products)

If you are short a call spread, are you long or short skew?

WHAT THEY TEST / ANSWER

Short skew. Fix the convention first, because the word is ambiguous and saying so earns the point: being "long skew" means profiting when the curve steepens, that is when low-strike volatility richens relative to high-strike volatility. A long call spread is long the low strike and short the high strike, so a steepening lifts what it owns and cheapens what it owes: long call spread equals long skew. Flip it, and a short call spread is short skew. Cross-check it against the price: we just saw that introducing the skew makes the call spread *more* expensive, so the holder gains from skew and the seller loses. What they are really testing is whether you think in terms of the volatility differential between two strikes rather than in terms of a single vega number.

How the surface is quoted, and how it moves

Desks do not quote a strike-by-strike list. They quote three numbers per maturity: the at-the-money volatility (the level), a *risk reversal* (the tilt, the volatility of an out-of-the-money call minus that of the symmetric out-of-the-money put), and a *butterfly* (the convexity, the average of those two wings minus the at-the-money level). In FX these are standardised at the 25-delta and 10-delta points and traded as instruments in their own right; the conventions have real subtleties, including the difference between the strangle brokers actually trade and the strangle implied by the fitted smile. They are covered in risk reversals and butterflies, with the delta conventions themselves in the FX volatility smile.

The second question a trader cares about is how the curve moves when spot moves, since that determines the hedge. Two idealised rules bracket reality:

- **Sticky strike**: $\sigma_{\text{imp}}(K)$ stays put as spot moves. The at-the-money point then slides along a fixed curve, so a falling market mechanically raises at-the-money volatility, which matches what is usually observed in quiet regimes. Under this rule the Black-Scholes delta is the right hedge.
- **Sticky delta** (or sticky moneyness): the curve is a fixed function of K/S and travels with spot, so at-the-money volatility does not move with spot. Under this rule a fixed-strike option's volatility does change when spot moves, adding a term to the hedge ratio, $\Delta_{\text{total}} = \Delta_{\text{BS}} + \mathcal{V}\,\partial\sigma/\partial S$. With a downward-sloping equity skew, a rise in S lowers K/S and therefore raises σ for that fixed strike, so the correction is positive and the hedge ratio exceeds the Black-Scholes delta.

Real markets sit between the two and switch regime under stress. The Greek machinery for all of this, vanna and volga, is the subject of skew and smile sensitivities, and the question of how a whole book behaves as volatility regimes change is in vega exposure and volatility regimes.

INTERVIEW TRAP

Four ways candidates lose the smile question. (1) Saying the put has a higher implied volatility than the call at the same strike: put-call parity forbids it. (2) Merging skew (across strikes) with term structure (across maturities) into one hand-wave; they are perpendicular slices of the surface and have different causes. (3) Answering "because of fat tails" and stopping there: it is true and incomplete, since it explains convexity but not the direction of the tilt, which is a supply-and-demand story. (4) Treating a call spread, a digital or an autocallable as a pure directional or pure vega trade after learning that skew exists; the moment a payoff references two strikes, it is also a skew position.

VISUAL / CODE TO BUILD: Interactive

A surface explorer with three linked panels. Panel A: the implied volatility curve for one maturity, with draggable control points for level, tilt and convexity (equivalently at-the-money volatility, risk reversal, butterfly), plus presets for equity index, G10 FX and energy. Panel B: the risk-neutral density obtained live from the curve by Breeden-Litzenberger, $q(K) = e^{rT}\partial^2 C/\partial K^2$, so the reader sees the left tail fatten as they steepen the skew. Panel C: a price table recomputing the 100/130 call spread and the digital at 130 under the current curve versus a flat at-the-money volatility, showing the gap in cash and in percent. The intended realisation: the shape of the smile and the shape of the distribution are the same object seen twice, and the difference between them is what you are paid or charged. Backed by a Black-Scholes pricer with a spline-interpolated smile; expose the raw numbers as a table for accessibility, and enforce a no-arbitrage check (call prices convex in strike) on the draggable points.

Where this goes next

The smile poses a modelling problem it does not solve: Black-Scholes with a different volatility per strike is an internally inconsistent way to price anything path dependent, because the model cannot hold several volatilities for the same underlying at once. The two standard repairs are local volatility, which makes volatility a deterministic function of spot and time fitted exactly to today's surface, and stochastic volatility, which gives volatility a life of its own and generates the smile endogenously. Trading the slope rather than the level is skew trading. The asset-specific shapes are developed in equity skew and smile and the FX volatility smile. And the question of whether the level of the surface is rich or cheap against what the underlying actually delivers is the subject of implied versus realized volatility.

15.10 Delta, Gamma and Vega Hedging

sales · trading · structuring | core | equity · fx · cross-asset

A dealer who sells you an option does not want your view. They want the spread, and then they want the risk gone. Hedging is how it goes. This section is the mechanics: what it means to neutralise a Greek, which instrument can hedge which, and why gamma and vega can only be separated by maturity. How the hedge behaves dynamically as spot moves and time passes is delta neutrality and dynamic hedging, which picks up exactly where this leaves off.

What hedging a Greek means

To hedge a Greek is to add a position whose exposure to that input is equal and opposite, so the sum is zero. Being **delta-neutral** means the book's value does not change for a small move in spot. It does not mean the book has no risk: it means the *first-order* spot risk is gone, and every other risk is untouched.

> KEY IDEA
>
> **The underlying can only hedge delta.** A share has a delta of one and a gamma, vega, theta and rho of exactly zero, so no quantity of it can ever offset an option's gamma or vega. Gamma and vega can be hedged *only with other options.* That single fact drives the structure of every options book: delta is cheap and continuous to manage, while gamma and vega must be traded against other options, at a bid-offer, with someone who also wants to be on the other side.

This produces a natural hierarchy. Delta is hedged constantly and almost mechanically, in the underlying or a future. Gamma and vega are managed at the book level over days, by trading options, and are the exposures a trader actually has a view on. Rho, dividend and repo exposures are usually managed periodically or passed to another desk.

REAL INTERVIEW QUESTION (ECB · Market Risk)

How does delta hedging work?

WHAT THEY TEST / ANSWER

Give the mechanic, then the purpose, then the catch. Mechanic: compute the position's delta, then take the opposite position in the underlying, so if you are short a call with a delta of 0.5 on 1,000 shares you buy 500 shares. Purpose: the combined position no longer moves for a small change in spot, so you have isolated everything that is not directional, which for an option means volatility and time. The catch, and this is the part being tested: delta is not constant. It changes as spot moves (that is gamma), as time passes (charm) and as volatility moves (vanna), so the hedge must be *re-done continuously*, and in reality it is re-done discretely, which leaves residual risk and costs transaction fees. So delta hedging does not remove risk, it converts directional risk into gamma and rehedging risk, which is precisely the risk an options desk chooses to hold.

REAL INTERVIEW QUESTION (BNP · EQD)

I sell a call. What do I do to hedge myself? And if I sell a put?

WHAT THEY TEST / ANSWER

Sell a call: you are short delta, so you **buy** the underlying, in the quantity given by the option's delta times the notional. Sell a put: you are long delta (a short put profits when spot rises), so you **sell** the underlying. Say the sign reasoning rather than reciting, because the follow-up is always "and if spot moves?".

Then answer that follow-up before it comes. In both cases you are **short gamma**, so your delta moves against you: as spot rises you need to buy more, as it falls you need to sell, meaning you are systematically buying high and selling low. That bleed is what your theta is paying for. And note the asymmetry a desk cares about: a short call leaves you buying into a rally, an unbounded exposure, while a short put leaves you selling into a decline, bounded at zero but correlated with everyone else doing the same thing.

The gamma-theta trade-off, in numbers

For a delta-hedged position, the daily profit and loss is essentially two terms fighting each other:

$$\text{P\&L}_{\text{day}} \approx \underbrace{\tfrac{1}{2}\,\Gamma\,(\Delta S)^2}_{\text{gamma, from movement}} + \underbrace{\Theta\,\Delta t}_{\text{theta, from time}} .$$

Long options means positive gamma and negative theta: you profit if the underlying moves enough and bleed if it does not. Short options is the mirror.

Setting those two terms equal defines the **break-even move**: the daily move at which gamma exactly pays for theta. Its derivation, the cash gamma units desks actually quote it in, and the resulting break-even table belong to theta as the cost of convexity. The one result needed here is that at an implied volatility σ the break-even is $\sigma/\sqrt{252}$ per day, which is the statement that a hedged option position makes money exactly when realised volatility beats the implied volatility it was bought at.

WORKED EXAMPLE

That trade-off, seen from the short side. You sell a call on a stock with a volatility of 16% and delta-hedge it, earning a theta of €1,000 a day. The break-even daily move at 16% volatility is $16\%/\sqrt{252} = 1.008\%$, so on a 1% day you make nothing and keep your theta only in the sense that it exactly offsets the gamma loss.

Now the spot moves 2% in one day. Gamma P&L scales with the *square* of the move, so the loss is $(2/1.008)^2 \approx 3.9$ times the break-even amount:

$$\text{P\&L} \;=\; \underbrace{+1{,}000}_{\text{theta}} \;-\; \underbrace{3{,}938}_{\text{gamma}} \;=\; -€2{,}938,$$

or about €3,000 lost if you do the arithmetic with a round 1% break-even. A move of double the break-even costs you roughly four times a day's theta, which is the asymmetry that makes short gamma dangerous: your gains accumulate linearly in time and your losses grow quadratically in the move.

Hedging gamma and vega

Because only options carry gamma and vega, hedging them means trading options against options, and the two exposures cannot be chosen independently of each other by accident: any option you add to fix vega also changes gamma and delta.

The lever that separates them is **maturity**. For an at-the-money option,

$$\Gamma \;\propto\; \frac{1}{\sigma\sqrt{T}}, \qquad \mathcal{V} \;\propto\; \sqrt{T}.$$

Short-dated options are almost all gamma and very little vega; long-dated options are almost all vega and very little gamma. So a book's gamma is managed at the front of the curve and its vega at the back, and a calendar spread is the instrument that separates them.

REAL INTERVIEW QUESTION (Société Générale · Rates)

How do you delta-hedge and gamma-hedge?

WHAT THEY TEST / ANSWER

Two different problems, and saying so is the first mark. **Delta hedge**: take the opposite delta in the underlying or a future, and re-do it as spot moves. It is cheap, liquid and continuous, and the underlying has exactly the property you need, a delta of one.

Gamma hedge: you cannot use the underlying at all, because its gamma is zero, so you must trade another option, typically one near the strike and maturity where your gamma is concentrated. And then say what makes this hard in practice: the hedging option brings its own delta, which you re-neutralise with stock, and its own vega, which changes your volatility exposure. So gamma hedging is a simultaneous problem, not a sequential one, and in practice a desk solves it as a small system, choosing quantities in two or more options so that gamma and vega land where the trader wants and then delta-hedging the remainder with the underlying. Close with the reason a desk bothers: being gamma-flat means your delta hedge stops costing you money every time the market moves, which is what lets you hold a large book without rehedging constantly.

REAL INTERVIEW QUESTION (BNP · EQD)

How do you get delta-neutral, vega-neutral and gamma-positive on a stock where you expect a move but not its direction, using options?

WHAT THEY TEST / ANSWER

This is the classic three-constraint problem and it has a clean answer: **buy short-dated options and sell long-dated ones**, in the ratio that zeroes vega, then delta-hedge the residual with stock.

The reasoning is the maturity scaling. At the money, gamma scales like $1/\sqrt{T}$ and vega like $\sqrt{T}$. So if you buy N_1 options at maturity T_1 and sell N_2 at a longer T_2, vega neutrality requires $N_1\sqrt{T_1} = N_2\sqrt{T_2}$, meaning you buy proportionally more of the near-dated ones. Plug that ratio into the gamma and the near leg dominates, because the same trade that balances $\sqrt{T}$ leaves $1/\sqrt{T}$ unbalanced in favour of the short maturity. So the position is automatically gamma-positive. Finally, buy or sell stock to flatten the delta.

You have built a reverse calendar spread, and you have expressed exactly the view stated: you want movement now (long gamma), you do not want to pay for a view on the long-dated volatility level (vega-flat), and you have no directional opinion (delta-flat). Say what it costs you: the position is short theta and you will bleed if the move does not come.

INTERVIEW TRAP

Four traps. (1) Saying a delta-neutral book is riskless; it is neutral to small spot moves and fully exposed to gamma, vega, time and everything else. (2) Trying to hedge gamma or vega with the underlying, which has neither. (3) Forgetting that gamma P&L scales with the *square* of the move, so one large day dominates many small ones and a short-gamma book's risk is concentrated in tails. (4) Hedging vega without noticing the hedge changed your gamma and delta; the Greeks are hedged as a system.

VISUAL / CODE TO BUILD: Interactive

A hedging simulator. The reader takes a position in an option, chooses a rehedging frequency (continuous, daily, only when delta drifts past a band) and a transaction cost, then runs a price path. The panel accumulates the realised P&L and decomposes it live into gamma, theta and hedging-cost contributions, so the reader sees the break-even move as the level where the first two cancel, and sees rehedging cost eat the edge as frequency rises. A second control sets the volatility the option was priced at independently of the volatility of the generated path, making the implied-versus-realised result of the previous section something the reader can reproduce rather than accept. A third tab gives the reader two options at different maturities and asks them to hit vega-neutral and gamma-positive, scoring the resulting exposures.

Where this goes next

How the delta hedge behaves as spot moves through a strike and as expiry approaches is delta neutrality and dynamic hedging, with the convexity itself in gamma and convexity and the cost of holding it in theta as the cost of convexity. How a book's vega behaves across volatility regimes is vega exposure and volatility regimes, and the instrument that gives you the implied-versus-realised exposure without any of this rehedging machinery is the variance swap.

15.11 No-Arbitrage Pricing

sales · trading · structuring | core | equity · rates · fx · cross-asset

Every price in this book has been derived the same way, and it is time to say so explicitly. We never once asked what the underlying was likely to do. We built a portfolio that reproduced the payoff and said the two must cost the same. That principle, and the strange-looking probabilities that fall out of it, is what this section is about. It is also the single most conceptually tested idea in derivatives interviews, because a candidate who has genuinely understood it answers a dozen questions with one sentence.

The principle

An **arbitrage** is a portfolio that costs nothing (or less) to set up, never loses, and sometimes gains. If one exists, someone scales it up without limit, so in a functioning market it cannot persist. The pricing consequence is the **law of one price**: two portfolios with identical payoffs in every future state must have identical prices today.

KEY IDEA

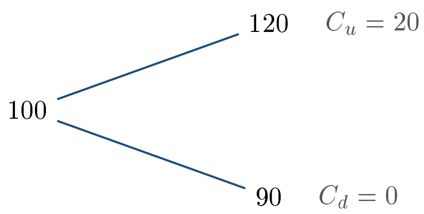
Replication is pricing. If you can build a portfolio of traded instruments whose value equals a derivative's payoff in every state of the world, then the derivative's price is the cost of that portfolio. Not approximately, and not on average: exactly, because otherwise the difference is a riskless profit. Everything else in derivatives pricing is machinery for finding the replicating portfolio.

You have already seen this twice. The forward price came from borrowing cash and buying the asset. The put-call parity identity came from two portfolios with the same payoff. Neither derivation contained a probability.

The one-period binomial

The smallest model where an option can be replicated, and the one every interviewer reaches for.

Let the stock be at 100 and, over one period, go either to 120 or to 90. Take zero rates for clarity. Price a call struck at 100, which pays 20 in the up state and 0 in the down state.



The one-period lattice. —— are the two branches the stock can take, ending at the values shown; to the right of each is the payoff of a call struck at 100 in that state. Two states, two instruments, so the payoff can be matched exactly, and that is the whole reason a price exists.

Step one, replicate. Hold Δ shares and borrow B. Match both states:

$$120\Delta - B = 20, \qquad 90\Delta - B = 0.$$

Subtracting gives $30\Delta = 20$, so

$$\Delta = \frac{20 - 0}{120 - 90} = \frac{2}{3}, \qquad B = 90 \times \tfrac{2}{3} = 60.$$

Step two, price it. The replicating portfolio costs

$$\Delta S_0 - B = \tfrac{2}{3} \times 100 - 60 = 66.67 - 60 = 6.67.$$

That is the call's price. Notice what never appeared: the probability of the stock going up. It is not in the calculation because it is not needed, and that is the whole point.

INTUITION

The probability of the up move is already **inside the stock price**. If the market thought a rise were more likely, the stock would not be at 100. Since your hedge is built out of the stock, you inherit the market's view for free, and asking for it separately would be double counting. This is why a derivatives trader can price an option without having any opinion about where the underlying is going, and why "what do you think the stock will do?" is the wrong question to ask them.

Risk-neutral probabilities

Now rewrite the same answer in a form that generalises. Define

$$p \;=\; \frac{(1+r) - d}{u - d} \;=\; \frac{1 - 0.9}{1.2 - 0.9} \;=\; \frac{1}{3},$$

and compute the expected payoff using p as if it were a probability:

$$\tfrac{1}{3} \times 20 + \tfrac{2}{3} \times 0 = 6.67,$$

the same price. The number p is the **risk-neutral probability**. It has two defining properties: it lies between 0 and 1 exactly when there is no arbitrage ($d < 1+r < u$), and under it the stock's expected value is its forward price, here $\frac{1}{3}(120) + \frac{2}{3}(90) = 100$.

KEY IDEA

The risk-neutral probability is **not a forecast**, and it is not a probability in the ordinary sense of probability either. It is a change of units that turns the replication algebra into something that looks like an expectation. Its content is "the discounted asset price is a martingale", which is just a restatement of no-arbitrage. If the true probability of the up move were 90%, the option would still be worth 6.67, because the hedge would still work identically. Real-world probabilities do not enter the price of a replicable payoff.

REAL INTERVIEW QUESTION (Société Générale · EQD)

Why can we use risk-neutral probabilities to price an option?

WHAT THEY TEST / ANSWER

Give the logic in three steps and resist the temptation to invoke measure theory. One: the option can be replicated by a dynamically adjusted position in the stock and cash, so by no-arbitrage its price must equal the cost of that replicating portfolio. Two: that cost, written out algebraically, can always be rearranged into a discounted expectation, provided you use a specific set of weights, and those weights are the risk-neutral probabilities. Three: therefore risk-neutral pricing is not an assumption about investor preferences, it is a *reformulation* of the hedging argument that happens to be far easier to compute with.

Then say the sentence that proves you understand it: the real-world probability never enters because it is already embedded in the spot price you are hedging with, so using it again would double count. And if they push: risk neutrality is a name, not a claim that anyone is indifferent to risk. The market's risk aversion is fully expressed in where the spot sits.

REAL INTERVIEW QUESTION (Natixis · QIS)

If I give you u and d, what is p? What does p represent? What is the risk-neutral probability?

WHAT THEY TEST / ANSWER

Formula first: $p = \dfrac{(1+r)-d}{u-d}$, or with continuous compounding $p = \dfrac{e^{r\Delta t}-d}{u-d}$. Derive it in one line if you have a board: p is the weight that makes the expected stock value equal its forward, $pu + (1-p)d = 1+r$, and solving gives the formula.

What it represents: not the chance of an up move, but the weight under which the discounted stock is a martingale, so that discounted expected payoffs give arbitrage-free prices. Then add the two facts an interviewer is listening for. First, $p \in (0,1)$ if and only if $d < 1+r < u$, and that condition *is* the absence of arbitrage: if the stock could only beat cash, you would borrow and buy without risk. Second, p does not depend on anyone's risk appetite or on the true probability, which is exactly why two traders with opposite views on the stock still agree on the option's price.

REAL INTERVIEW QUESTION (Natixis · QIS)

Binomial tree: explain the principle. Once you have the payoffs, what do you do?

WHAT THEY TEST / ANSWER

Principle: chop the life of the option into steps, and at each step let the underlying move up by u or down by d, chosen so the tree's variance matches the volatility you are pricing with, conventionally $u = e^{\sigma\sqrt{\Delta t}}$ and $d = 1/u$. That builds a lattice of possible terminal values.

Then the answer to the second half, which is the actual question. Compute the payoff at every terminal node. Then work **backwards**: at each earlier node the value is the discounted risk-neutral expectation of the two nodes it leads to, $e^{-r\Delta t}[pV_u + (1-p)V_d]$. Repeat until you reach today, and that value is the price. Two things to volunteer. The hedge ratio drops out for free at every node as $\Delta = (V_u - V_d)/(S_u - S_d)$, so the tree gives you the Greeks as well as the price. And for an **American** option you add one line to the backward step, taking the maximum of the continuation value and the immediate exercise value, which is precisely why trees are used for early exercise where Black-Scholes has no closed form.

Two worlds, and which one to use

Both measures are real and useful, and confusing them is a classic error.

	Real-world	Risk-neutral
Drift of the asset	the actual expected return μ	the risk-free rate r
Volatility	σ	the same σ
Used for	risk, VaR, scenarios, forecasting	pricing and hedging
Answers	what will probably happen	what it must cost today

The change between them moves the *drift* and leaves the *volatility* untouched. That is why volatility is the parameter you can estimate from history and use in a pricing model, while the expected return is not, and it is why an option price depends on σ but not on μ.

REAL INTERVIEW QUESTION (Barclays · EQD)

What is the real world? How does it differ from the risk-neutral model?

WHAT THEY TEST / ANSWER

Say what each is *for*, since that is the cleanest distinction. The real-world measure describes what the asset is actually expected to do, with its true drift, and it is what you use for risk management: value at risk, stress scenarios, expected shortfall, capital, anything that asks "what will probably happen to me". The risk-neutral measure is a pricing device in which every asset drifts at the risk-free rate, and it is what you use to value and hedge, because it is the reformulation of the replication argument.

Then the two technical points that separate a strong answer. The change of measure alters the drift but not the volatility, which is why implied and realised volatility are even comparable while expected returns never appear in an option formula. And risk-neutral probabilities are *not* predictions: a risk-neutral density backed out of option prices will assign a much higher probability to a crash than the real-world density does, because it embeds the risk premium investors charge for that crash. Reading one as a forecast of the other is a mistake regulators and journalists make constantly.

Where the principle stops working

No-arbitrage gives a unique price only when the payoff can actually be replicated. When it cannot, the market is **incomplete** and no-arbitrage delivers a *range* of prices rather than a point:

- **Jumps.** If the underlying can gap, two hedging instruments cannot span three or more outcomes, so a jump-diffusion model has no perfect hedge. See jump-diffusion models.
- **Stochastic volatility.** If volatility itself is a random factor, hedging it requires a traded volatility instrument, and the price depends on the market price of volatility risk.
- **Transaction costs and discrete hedging.** Continuous rehedging is free in the theory and expensive in reality, which turns the exact price into a bid-offer band.
- **Untradeable underlyings.** Weather, an unlisted asset or an illiquid single name cannot be held, so there is nothing to replicate with.

In all these cases the honest statement is that no-arbitrage bounds the price and a model, plus a risk premium, picks a number inside the bounds. Saying that out loud in an interview is a mark of maturity, not of doubt.

INTERVIEW TRAP

Four traps. (1) Calling the risk-neutral probability the market's forecast; it is a pricing weight and it embeds a risk premium. (2) Claiming risk-neutral pricing assumes investors are indifferent to risk; it assumes nothing about preferences, since they are already in the spot price. (3) Using the real-world drift in a pricing formula, or the risk-free rate in a value-at-risk calculation; each measure has its own job. (4) Asserting a unique arbitrage-free price where the payoff cannot be replicated, which is most exotic and all illiquid situations.

VISUAL / CODE TO BUILD: Interactive

A one-period binomial laboratory. The reader sets S, u, d, r and the strike, and the panel shows three things side by side: the replicating portfolio (shares held and cash borrowed, with both state payoffs verified to match), the price as the cost of that portfolio, and the same price recomputed as a discounted risk-neutral expectation, so the reader sees them agree to the last digit for any inputs. A slider for the *real-world* probability of an up move is deliberately included and deliberately does nothing to the price, while it visibly changes the expected profit of a naked position, which makes the central point of the section impossible to miss. Pushing u or d past the no-arbitrage bound turns p negative and the panel flags the arbitrage explicitly, naming the trade.

Where this goes next

Turning the principle into concrete trades, synthetic forwards, digitals built from call spreads and payoffs assembled from vanillas, is replication strategies. Letting the binomial step shrink to zero produces Black-Scholes, which is the same argument in continuous time. And the real-world measure this section carefully sets aside is the one used throughout value at risk and risk management.

15.12 Replication Strategies

sales · trading · structuring | advanced | equity · fx · cross-asset

No-arbitrage pricing established that if you can replicate a payoff you can price it. This section is the practice: how a desk actually manufactures a payoff out of instruments it can trade, what it costs, and where the replication is imperfect enough that the imperfection becomes the trader's margin. It is the bridge between derivatives theory and the structuring business.

Static and dynamic

KEY IDEA

Static replication is set up once and never touched: a portfolio of instruments held to maturity that reproduces the payoff whatever the path. **Dynamic** replication is rebalanced continuously, as in delta hedging. Static is overwhelmingly preferable when available, because it is *model-free*: it does not care about volatility, jumps, transaction costs or your rehedging discipline. Dynamic replication is only as good as the model and the market's liquidity on the day you need it.

You have already met two static replications. The forward price came from borrowing and buying, and neither leg is ever adjusted. And put-call parity says that a long call plus a short put at the same strike is a forward, which is the **synthetic forward**.

REAL INTERVIEW QUESTION (Barclays · Cross Asset)

How do you replicate a forward with options?

WHAT THEY TEST / ANSWER

Long a call and short a put, same strike and same expiry. Prove it by payoff rather than quoting parity: above K the call pays $S_T - K$ and the put is worthless; below K the call is worthless and you are short a put paying $-(K - S_T) = S_T - K$. Same expression in both regions, so the combination is a forward struck at K in every state of the world. Pricing follows from put-call parity, $C - P = S_0 - Ke^{-rT}$, so choosing K equal to the forward price makes the structure cost zero, which is why a zero-cost synthetic forward is always struck at the forward.

Then say why anyone bothers, since the question is really about use cases. You may want forward exposure where the stock is hard or expensive to borrow so a physical short is impractical; or where you want the exposure in options format for margin, tax or mandate reasons; or as a market-maker arbitrage check, since a mispriced call-put pair against the forward is a genuine arbitrage. And note the risk you inherit: a synthetic forward is delta-one but built from two options, so you carry their dividend and repo assumptions rather than the market's.

Replicating a digital

A **digital** (binary) call pays a fixed amount, say 1, if the underlying finishes above K, and nothing otherwise. It is a step function, and that is exactly the problem: at expiry its delta at the strike is infinite, so it cannot be delta-hedged at all near maturity near the strike. Nobody hedges a digital dynamically. They replicate it with a **call spread**.

Buy a call at $K - \frac{w}{2}$, sell one at $K + \frac{w}{2}$, and hold $1/w$ of them. The result is a ramp that rises from 0 to 1 across a window of width w centred on the strike, and as $w \to 0$ the ramp becomes the step.

WORKED EXAMPLE

Spot 100, three months, implied volatility 20%, zero rates. A digital call struck at 100 paying 1 is worth $N(d_2) = 0.4801$.

Replicate it with centred call spreads of shrinking width:

Width w	Spread value	Error
4.0	0.480456	+0.000395
2.0	0.480161	+0.000099
1.0	0.480086	+0.000025
0.5	0.480067	+0.000006
0.1	0.480061	+0.0000002

The convergence is quick, and the error falls with w^2 almost exactly: dividing each error by w^2 gives about 0.0000249 at every width in the table. By $w = 0.1$ the error has effectively vanished. But look at what you must trade: at $w = 0.5$ you need $1/0.5 = 2$ spreads per unit of digital notional, and at $w = 0.1$ you would need 10. On a €100 million digital that is €1 billion of call spread notional through the market. Accuracy is bought with size.

The version a desk actually uses when it is *short* the digital is the **overhedge**: shift the whole spread down so that neither strike sits above the digital's, buying $K - w$ and selling K. That spread pays at least as much as the digital in every state, so it can never leave you short.

WORKED EXAMPLE

Same market. A $(99, 100)$ spread scaled by $1/1$ is worth 0.5001 against the digital's 0.4801, so overhedging with a one-point gap costs 0.0200, and a two-point gap costs 0.0401. That difference, four cents on a payout of one, is not an error: it is the price of a hedge that cannot fail, and it is where the desk's margin and risk buffer live.

REAL INTERVIEW QUESTION (Morgan Stanley · Exo)

What is a digital or binary option? How do you replicate it? Why do you have to replicate it? Why do you use a call spread?

WHAT THEY TEST / ANSWER

Four parts, and the third is the one being tested. **What**: pays a fixed amount if the underlying is above (or below) a level at expiry, zero otherwise, so the payoff is a step. **Why you must replicate**: because the step means the delta at the strike tends to infinity as expiry approaches, and gamma with it, so no dynamic hedge exists. You would be trying to be either fully long or fully short the underlying depending on which side of a line the spot lands on, and the spot will cross that line repeatedly on the last day. **How**: a call spread straddling or below the strike, in quantity one over the strike gap, which turns the step into a ramp you can actually hedge. **Why a call spread**: because it is a *static* hedge built from liquid listed instruments, so it is model-free and needs no rehedging.

Finish with the trade-off, since it is the natural follow-up: narrowing the gap improves the replication but multiplies the notional you must trade by one over the gap, so the choice is accuracy against liquidity and cost. And when you are short the digital you deliberately overhedge, placing the spread below the strike so it dominates the payoff, and the cost of that conservatism is your margin.

REAL INTERVIEW QUESTION (Société Générale · EQD)

How do you replicate a digital with a call spread? How do you choose the strike gap, and what do you take into account?

WHAT THEY TEST / ANSWER

Mechanics in one line, then spend the answer on the gap, because that is the question. Buy $1/w$ call spreads of width w around or below the digital's strike.

Choosing w is a trade-off between replication error, which falls with the gap, and traded notional, which rises as $1/w$. The inputs to that decision: **listed strike granularity**, since you can only trade the strikes that exist and interpolating means going over the counter; **liquidity of the underlying and its options**, because $1/w$ times your notional has to actually clear without moving the market; **bid-offer**, which you pay on the whole $1/w$ notional, so a tight gap can cost more in spread than it saves in replication error; **time to expiry**, since a wide gap is harmless far from expiry and the residual risk concentrates as expiry approaches; **volatility**, because for the same accuracy a high-volatility underlying tolerates a wider gap; and the **direction of your position**, since if you are short the digital you shift both strikes below the level so the hedge dominates rather than approximates.

Close with the operational point: the gap is not chosen once, it is a running decision, and desks typically start wide and tighten as expiry approaches while the residual risk becomes material.

Digitals, skew, and the price you actually quote

There is a second reason the flat-volatility digital price is not the right one, and it is a favourite follow-up.

A digital is the limit of a call spread, so its price is the negative of the derivative of the call price with respect to strike. But the implied volatility itself depends on the strike, so differentiating properly gives

$$\text{Digital} \;=\; -\frac{dC}{dK} \;=\; e^{-rT}N(d_2) \quad \underbrace{-\,\mathcal{V}\,\frac{\partial\sigma}{\partial K}}_{\text{skew correction}} \;.$$

In equities the smile slopes downward, $\partial\sigma/\partial K < 0$, and vega is positive, so the correction *adds* to the price and the digital is worth *more* than the flat-volatility value $e^{-rT}N(d_2)$. Quoting that flat-volatility value on an equity underlying systematically undersells digitals, and a call spread priced on the correct strike-dependent volatilities automatically picks the skew up, which is another argument for pricing the replication rather than the idealised payoff. The skew itself is traded in its own right.

REAL INTERVIEW QUESTION (Société Générale · EQD)

How do you price a digital? How does the price depend on the call spread gap, and on the trader's margin?

WHAT THEY TEST / ANSWER

Three linked answers. **Price**: do not quote the theoretical $N(d_2)$; quote the cost of the call spread you would actually trade to hedge it, priced with each leg's own implied volatility off the surface. That automatically includes the skew correction, which on an equity underlying makes the digital worth more than the flat-volatility value.

Dependence on the gap: if you are short the digital and overhedge below the strike, the spread's value rises with the gap, roughly linearly for small gaps. With spot at 100, three months and 20% volatility, a one-point overhedge costs about 2 cents on a unit payout and a two-point overhedge about 4 cents, against a theoretical digital value of 48 cents. So the gap is a direct dial on the price you quote.

Dependence on the trader's margin: that difference between the overhedged spread and the theoretical digital *is* the margin, plus a buffer for the residual risk of the ramp. A wider gap means a safer hedge, a higher price and a less competitive quote; a narrower gap means a sharper price and more risk near the strike at expiry. Pricing a digital is therefore a commercial decision as much as a mathematical one, and saying that explicitly is the answer they are looking for.

Replicating any European payoff

The digital and the forward are special cases of a general result worth knowing exists. Any European payoff $f(S_T)$ that is smooth enough can be written as a fixed cash amount, plus a position in the forward, plus a continuum of calls and puts weighted by the payoff's second derivative:

$$f(S_T) \;=\; f(F) + f'(F)(S_T - F) + \int_0^F f''(K)\,(K - S_T)^+ dK + \int_F^\infty f''(K)\,(S_T - K)^+ dK.$$

Read it as a statement about curvature: wherever the payoff bends, you need options struck there, in proportion to how sharply it bends. A linear payoff needs none, which is why a forward is model-free. This is the result that makes the variance swap replicable by a strip of options, and it is the formal backbone of payoff engineering.

INTERVIEW TRAP

Four traps. (1) Saying you would delta-hedge a digital; near the strike at expiry its delta is unbounded and the hedge is undefined. (2) Choosing the strike gap on accuracy alone, ignoring that the notional traded scales as one over the gap. (3) Centring the spread on the strike when you are short the digital, which leaves you underhedged half the time; shift it so the hedge dominates. (4) Quoting the flat-volatility $N(d_2)$ for a digital on an equity underlying, which ignores the skew term and undersells the option.

VISUAL / CODE TO BUILD: Interactive

A replication workbench. The reader picks a target payoff (digital, forward, or a freehand piecewise-linear payoff drawn with the mouse) and the panel solves for the replicating portfolio of vanillas, drawing the target and the replication on the same axes with the residual shaded between them. A gap slider on the digital tightens the call spread and simultaneously shows three readouts moving in opposite directions: replication error falling, notional traded rising as one over the gap, and total bid-offer cost rising with it, so the trade-off is experienced rather than described. A toggle switches from centred to overhedging placement and shows the residual becoming one-signed. A skew switch tilts the volatility surface and shows the quoted digital price moving away from the flat-volatility value.

Where this goes next

Digitals, barriers and the rest of the exotic taxonomy are exotic options, and the strip-of-options replication sketched above is what makes the variance swap work. Assembling replicated building blocks into products a client will buy is payoff engineering, and the dynamic counterpart to everything here is delta, gamma and vega hedging.

Chapter 16

Greeks & Risk Dynamics

16.1 Greek definitions and intuition

sales · trading · structuring core equity · fx · rates · cross-asset

Greeks (first and second order) gave the map: a Greek is a partial derivative, there is one per input, and the sign table and quoting conventions follow. This section is the other half, and it is the half interviews actually dig into. Each Greek gets its closed form, the sentence that explains what it *is* rather than what it differentiates, and a number you can carry in your head so that "the vega looks big" becomes a claim you can check. The taxonomy, the sign table and the per-point and per-day conventions are not repeated here.

Everything below is Black-Scholes, with spot S, strike K, maturity T, implied volatility σ, rate r and dividend yield q. The model itself is derived in the Black-Scholes section. Write N for the standard normal cumulative distribution and φ for its density.

The two numbers everything is built from

Every Black-Scholes Greek is a rearrangement of two quantities:

$$d_1 = \frac{\ln(S/K) + \left(r - q + \frac{1}{2}\sigma^2\right) T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}.$$

Both are the same thing in different clothes: the distance, measured in standard deviations of the log price over the life of the option, between where the forward sits and where the strike sits. That is why they always appear divided by $\sigma\sqrt{T}$, the total volatility of the period.

The difference between them carries real meaning, and interviewers test it relentlessly.

- $N(d_2)$ is the **risk-neutral probability that the option finishes in the money**, that is $\mathbb{Q}(S_T > K)$.
- $N(d_1)$ is *not* a probability. It is the same quantity re-weighted by the size of the payoff, which is exactly what a hedge ratio has to be, and it is the delta.

The gap between them is $\sigma\sqrt{T}$ in the argument, so it widens with volatility and with maturity. For a one-year at-the-money call at 20% volatility, $d_1 = 0.10$ and $d_2 = -0.10$, giving a delta of 0.540 against a genuine exercise probability of 0.460. Calling delta "the probability of exercise" would be wrong here by eight points, and the error grows with maturity.

KEY IDEA

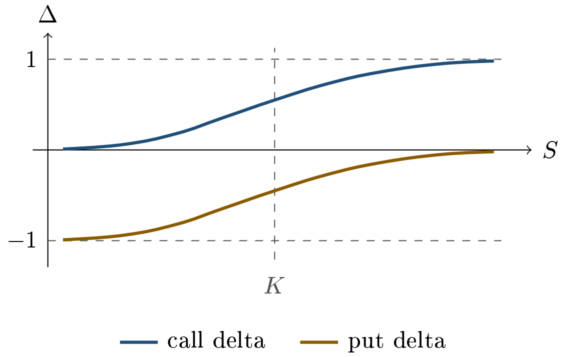
$N(d_2)$ answers "how likely am I to be exercised?". $N(d_1)$ answers "how much stock must I hold to be flat?". They are close for short-dated near-the-money options, which is why the confusion survives, and they diverge exactly where the money is, in long-dated and high-volatility books.

Delta: the hedge ratio

$$\Delta_{\text{call}} = e^{-qT}N(d_1), \qquad \Delta_{\text{put}} = -e^{-qT}N(-d_1) = e^{-qT}\big(N(d_1) - 1\big).$$

Delta is the number of units of underlying that behave like the option right now. Own one call with a delta of 0.54 and, for small moves, you own 0.54 shares. Sell 0.54 shares and your position stops caring about direction, which is the whole of delta hedging.

The bounds follow from the formula and from common sense. A call's delta lives in $[0, e^{-qT}]$, so in $[0,1]$ with no dividends, and a put's in $[-1,0]$. A deep out-of-the-money option barely reacts to spot, so its delta tends to zero; a deep in-the-money one is already effectively the stock (or short the stock), so its delta tends to the extreme. Between the two, delta rises smoothly, and the speed of that rise is gamma.



The vertical gap between the two curves is constant and equal to e^{-qT}, which is 1 when there are no dividends. That is not a coincidence: differentiating put-call parity in S gives $\Delta_{\text{call}} - \Delta_{\text{put}} = e^{-qT}$ directly, so you never need to remember the put formula.

REAL INTERVIEW QUESTION (BNP · EQD)

Delta = ? (= $N(d_1)$), re-derive it.

WHAT THEY TEST / ANSWER

They want to see whether you can differentiate the formula or only quote it. Start from $C = Se^{-qT}N(d_1) - Ke^{-rT}N(d_2)$ and differentiate in S. You get three terms: the obvious $e^{-qT}N(d_1)$, plus $Se^{-qT}\varphi(d_1)\frac{\partial d_1}{\partial S} - Ke^{-rT}\varphi(d_2)\frac{\partial d_2}{\partial S}$. Since $\partial d_1/\partial S = \partial d_2/\partial S = 1/(S\sigma\sqrt{T})$, those last two cancel, because the identity $Se^{-qT}\varphi(d_1) = Ke^{-rT}\varphi(d_2)$ holds exactly (write out both densities and use $d_1^2 - d_2^2 = (d_1 - d_2)(d_1 + d_2) = 2\ln(Se^{(r-q)T}/K)$). So $\Delta = e^{-qT}N(d_1)$, and the point of the exercise is that the messy terms vanish. Close by saying what makes the result non-obvious: $N(d_1)$ looks like a probability but is not, since the probability of exercise is $N(d_2)$. That is the sentence they are waiting for.

REAL INTERVIEW QUESTION (Société Générale · EQD)

Delta of an in-the-money, at-the-money and out-of-the-money call.

WHAT THEY TEST / ANSWER

Near 1, near 0.5, near 0, and then earn the point by refusing to leave it there. At the money the honest answer is "slightly above 0.5": with zero rates and dividends, an at-the-money spot call has $d_1 = \frac{1}{2}\sigma\sqrt{T}$, so $\Delta = N(\frac{1}{2}\sigma\sqrt{T}) > 0.5$, which is 0.54 for one year at 20% volatility. Give the reason rather than the number: the terminal distribution of the price is lognormal and therefore right-skewed, so the upside branch carries more weight. Then note the two extremes are limits, not values: a deep in-the-money call reaches 1 only as the remaining optionality dies, so it is a statement about $\sigma\sqrt{T}$ being small, not about the strike alone.

REAL INTERVIEW QUESTION (Société Générale · Structured Products)

What are the bounds of delta for a call and for a put?

WHAT THEY TEST / ANSWER

$[0, 1]$ and $[-1, 0]$, and then the qualification that separates candidates. Strictly the bounds are $[0, e^{-qT}]$ and $[-e^{-qT}, 0]$, because delta is expressed in shares and a share held to T pays dividends the option holder does not receive; see dividends. Add that the bounds are a vanilla statement and that exotics break them: a digital's delta is unbounded as expiry approaches, and a barrier option's delta can exceed one or flip sign near the barrier, which is exactly why those books are hard to hedge. See exotic options.

REAL INTERVIEW QUESTION (BNP · EQD)

Effect of maturity on the delta of an at-the-money call.

WHAT THEY TEST / ANSWER

It rises with maturity. With zero rates and dividends the at-the-money spot delta is $N(\frac{1}{2}\sigma\sqrt{T})$, so at 20% volatility it is 0.512 at one month, 0.540 at one year and 0.588 at five years, tending to 1 as T grows. The intuition is the one above: more time means more total volatility $\sigma\sqrt{T}$, which stretches the lognormal distribution and pushes more of its mass to the upside branch. Two additions score well. First, positive rates push it higher still, since they lift the forward. Second, a large dividend yield or borrow cost works the other way, and on a long-dated equity option that effect is not small, which is why a five-year call's delta is a statement about the forward curve as much as about the option.

Gamma: the curvature

$$\Gamma = \frac{e^{-qT}\,\varphi(d_1)}{S\,\sigma\sqrt{T}}, \qquad \text{identical for the call and the put.}$$

Gamma is how fast delta moves, so it is the error you make by hedging with a straight line a payoff that is curved. Long gamma means your delta improves by itself: it grows as the market rallies and shrinks as it falls, so a re-hedge is always a sale into strength or a purchase into weakness.

Short gamma is the reverse, and it is the reason losing days on an option book are fast and rarely alone.

The one number to hold: for spot 100, one year, 20% volatility and zero rates, $\Gamma = 0.0198$. A one-point move in spot changes your delta by about two shares per hundred options. Why gamma peaks at the money, how it explodes near expiry, and what a gamma-hedged book actually earns are the subject of gamma and convexity.

Vega: the exposure to the price of volatility

$$\mathcal{V} \;=\; S\,e^{-qT}\varphi(d_1)\sqrt{T}, \qquad \text{identical for the call and the put.}$$

Vega is exposure to the *quoted* volatility, not to what the underlying is about to do; the difference between the two is the trade discussed in implied versus realized volatility. It is positive for any long option because the buyer's loss is capped at the premium while the gain is not, so a wider distribution can only help. That asymmetry, not any formula, is the answer to why vega is positive.

Two properties are worth carrying. Vega scales roughly with $\sqrt{T}$, so long-dated options carry most of a book's vega even though their gamma is small: at spot 100 and 20% volatility, vega is 0.20 per volatility point at three months and 0.40 at one year. And vega is largest around the money and dies in the wings, which is why the wings are traded for their convexity in volatility rather than for vega itself. How vega behaves across the surface and across regimes is vega exposure and volatility regimes, and its behaviour once the surface is not flat is skew and smile sensitivities.

> **REAL INTERVIEW QUESTION** (Société Générale · Index)
>
> *What is the vega of a call, and why is it positive?*
>
> **WHAT THEY TEST / ANSWER**
>
> Definition first: $\mathcal{V} = \partial V/\partial\sigma = Se^{-qT}\varphi(d_1)\sqrt{T}$, quoted per volatility point, so divide by 100. Then the reason, which is what is being tested, and it is not "because the formula is positive". A long option has a capped loss and an uncapped gain, so any increase in dispersion adds value to the good branch without adding to the bad one. Two extras finish the answer: vega is identical for the call and the put at the same strike, which follows from put-call parity because the parity relation contains no volatility, and vega is a sensitivity to the *implied* volatility you are marked at, not to what the stock will realise. A trader can be long vega and still lose money if realised volatility collapses while implied does not move.

Theta and rho

With no dividends and zero rates the theta of both a European call and a put is

$$\Theta \;=\; -\frac{S\,\varphi(d_1)\,\sigma}{2\sqrt{T}},$$

and the general form adds a rate term, $-rKe^{-rT}N(d_2)$ for a call and $+rKe^{-rT}N(-d_2)$ for a put, plus a dividend term. Theta is the value the option loses as the clock runs, all else equal. Rho, $\partial V/\partial r$, is $KTe^{-rT}N(d_2)$ for a call and $-KTe^{-rT}N(-d_2)$ for a put; it is small on short-dated equity options and anything but small on a multi-year structured note.

The relationship worth memorising is not the theta formula, it is what links theta to gamma. Setting $r = q = 0$ in the Black-Scholes equation leaves

$$\Theta \;=\; -\tfrac{1}{2}\,\sigma^2 S^2\,\Gamma.$$

Convexity and time decay are the same quantity with opposite signs. You cannot own one without paying the other, and the exchange rate between them is $\frac{1}{2}\sigma^2 S^2$. That is the entire economics of an option book in one line, and it is developed in theta as the cost of convexity.

WORKED EXAMPLE

One set of numbers, checked against each other, for spot 100, strike 100, one year, $\sigma = 20\%$, $r = q = 0$. Then $d_1 = 0.10$, $d_2 = -0.10$ and $\varphi(d_1) = 0.3970$.

- Price: $C = P = 7.97$. The call and the put are equal because put-call parity gives $C - P = S - K = 0$ here.
- Delta: 0.540 for the call, -0.460 for the put, differing by exactly 1.
- Gamma: 0.0198, the same for both.
- Vega: 39.70 raw, so 0.397 per volatility point, the same for both.
- Theta: -3.97 per year, so -0.0109 per calendar day.

Now verify the identity rather than trusting it: $-\frac{1}{2}\sigma^2 S^2 \Gamma = -\frac{1}{2}(0.04)(10{,}000)(0.019848) = -3.970$, which is the theta. And the exercise probability is $N(d_2) = 0.460$, well below the delta of 0.540. A useful sanity anchor: the at-the-money rule of thumb $0.4\,\sigma S\sqrt{T} = 8.00$ lands within one percent of the exact 7.97.

The relations that save you from memorising

Differentiating put-call parity, $C - P = Se^{-qT} - Ke^{-rT}$, gives every call-put Greek relation at once, since the right-hand side does not depend on σ at all:

$$\Delta_C - \Delta_P = e^{-qT}, \qquad \Gamma_C = \Gamma_P, \qquad \mathcal{V}_C = \mathcal{V}_P.$$

So gamma and vega are properties of the *strike and maturity*, not of the option type. This is the same fact that forces one implied volatility per strike in the volatility smile and skew: if the call and the put share a vega, they must share a volatility.

INTERVIEW TRAP

Three errors that are specific to definitions rather than to dynamics. (1) Saying delta is the probability of finishing in the money: that is $N(d_2)$, and the gap is already eight points on a one-year at-the-money option. (2) Quoting an at-the-money delta as exactly 0.5: it is $N(\frac{1}{2}\sigma\sqrt{T})$ before rates and dividends, so above 0.5 and rising with maturity, and an interviewer who asks "exactly 0.5?" is checking for this. (3) Treating gamma or vega as different for a call and a put at the same strike: put-call parity makes them identical, and claiming otherwise is an arbitrage.

VISUAL / CODE TO BUILD: Interactive

A d_1 versus d_2 viewer, deliberately narrow. The reader sets S, K, T, σ, r, q and sees the lognormal terminal distribution with the region above K shaded. Two readouts sit side by side: $N(d_2)$, labelled "probability of exercise" and equal to the shaded area, and $N(d_1)$, labelled "delta", drawn as the same area re-weighted by the terminal price so the reader watches the second exceed the first. A maturity slider makes the two numbers separate visibly as $\sigma\sqrt{T}$ grows, and a readout of the difference makes the point quantitative. Backed by a Black-Scholes pricer; expose both numbers and d_1, d_2 as a table for accessibility.

Where this goes next

The formulas are static; a book is not. Delta becomes a hedging programme in delta neutrality and dynamic hedging. Gamma becomes profit and loss in gamma and convexity, paid for by theta. Vega becomes a view in vega exposure and volatility regimes, and the cross-Greeks that appear once the surface is not flat are in cross-Greeks and interaction effects. How all of this moves when spot moves is Greeks under spot moves, how it adds up across a book is portfolio-level aggregation, and how it is used to explain yesterday's profit and loss is Greeks and PnL explain.

Outside equities the same first-and-second-derivative structure appears as duration and convexity, and the parallel, together with the sign and quoting conventions that differ between the two, is drawn in duration and convexity.

16.2 Delta Neutrality and Dynamic Hedging

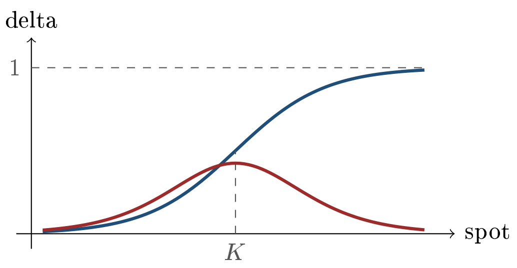
sales · trading · structuring | core | equity · fx · cross-asset

Delta, gamma and vega hedging established what it means to neutralise a Greek and which instrument does it. This section is about what happens *next*: delta neutrality does not stay put, and the business of continually restoring it is where an options trader's profit and loss is actually made and lost. Everything here is a consequence of one fact, that delta moves.

Neutrality is a snapshot, not a state

Being delta-neutral means the book's value is insensitive to a small, immediate move in spot. The instant spot moves, gamma has changed the delta and you are no longer neutral. Nothing has gone wrong; that is simply what gamma is.

> **KEY IDEA**
>
> Delta hedging does not remove risk, it **transforms** it. You give up your exposure to the *direction* of the underlying and take on exposure to how much it *moves*. That is the whole trade: after hedging, you are long or short volatility, and your profit and loss is the difference between what the market realises and the volatility you paid for, developed in implied versus realised volatility.



call delta against spot — its slope, the gamma — dashed: delta ceiling of 1 and the strike

The delta curve is the S-shape, running from 0 far out of the money to 1 far in. Its slope is the gamma, the bell peaked near the strike. Where the delta curve is steep, your hedge goes stale fastest.

REAL INTERVIEW QUESTION (BNP · Delta One)

Explain the delta-neutral strategy.

WHAT THEY TEST / ANSWER

Three beats. **What**: hold a position in the underlying that offsets the option book's delta, so the combined value does not move for a small change in spot. **Why**: because a dealer wants the spread and the volatility exposure, not a directional bet, so removing delta isolates what they are actually paid for. **The catch, which is the answer**: delta is not constant, so neutrality lasts only until the next tick, and maintaining it means trading the underlying continuously as spot moves. That continual rehedging is not an administrative chore, it is where the position makes or loses money: if you are long gamma you rehedge by selling into strength and buying into weakness, which is profitable, and if you are short gamma you do the opposite and it bleeds.

Close by naming what you are left holding: after the delta is gone you still have gamma, vega, theta and the rest, so a delta-neutral book is exposed to volatility and time, not to nothing.

The rehedge, and which way it hurts

The mechanics are mechanical, and the sign is everything.

If you are **long gamma**, your delta rises when spot rises. To restore neutrality you *sell* the underlying after it has gone up, and *buy* it after it has come down. You are systematically selling high and buying low, and each round trip banks a small profit. That is what your theta is paying for.

If you are **short gamma**, everything reverses. Your delta falls as spot rises, so you must *buy* after a rally and *sell* after a sell-off: buying high and selling low, on every move, all the way through. The theta you collect is compensation for accepting that.

REAL INTERVIEW QUESTION (Société Générale · Fixed Income)

How do you delta-hedge when you sell a call? And if the underlying rises by 10%, what do you do?

WHAT THEY TEST / ANSWER

Selling the call leaves you short delta, so you **buy** the underlying, in the amount of the option's delta times the notional. Say the number if they give you one: a 0.5 delta on a 100,000-share call means buying 50,000 shares.

Now spot rises 10%. The call moves toward the money, so its delta rises, say from 0.5 to 0.7, and since you are short the option you are now short delta again. You must **buy more**, another 20,000 shares, at the new higher price. That is the answer, and then give the interpretation, which is what separates it: you have just bought stock after it went up, which you will have to sell lower if it comes back, and that is the short-gamma bleed. Every such rehedge is a small realised loss, and the theta you earn is the premium you were paid for accepting a stream of them. If they push further: the loss on the round trip grows with the *square* of the move, so a single 10% day hurts far more than ten 1% days.

REAL INTERVIEW QUESTION (Société Générale · Volatility)

I sell a put. How do I hedge? And if the spot falls, how do I adjust?

WHAT THEY TEST / ANSWER

Selling a put leaves you **long** delta, since a short put gains when spot rises, so you hedge by **selling** the underlying short, in the amount of the put's delta.

Now spot falls. The put moves into the money, its delta becomes more negative, and being short the put you become more long delta, so you must **sell more**. You are selling into a falling market, which is the short-gamma bleed again and, this time, in the direction that matters systemically: everyone short puts is selling as the market drops, which is part of why sell-offs accelerate. Two things worth adding. Your hedge is a short stock position, so you also carry borrow risk and the possibility of a recall or a rising borrow cost precisely when the name is under stress. And this is the classic retail structured-product exposure: dealers who have sold downside protection are short gamma below the market and must sell into weakness to stay hedged.

How long to recover a bad day

Short gamma means a large move costs you and quiet days pay you back. The arithmetic of that trade is cleaner than it looks.

From the break-even move, the daily theta exactly offsets the gamma loss from a move of $x^* = \sigma/\sqrt{252}$. A move of size x therefore costs $|\Theta|(x/x^*)^2$, since gamma P&L scales with the square of the move. Dividing by the daily theta gives the number of days of theta the move has cost, which is also the elapsed time to recover it since theta accrues on the day of the move too:

$$\boxed{n = \left(\frac{x}{x^*}\right)^2, \qquad x^* = \frac{\sigma}{\sqrt{252}}}$$

Notice that gamma and theta have both cancelled. The answer depends only on the size of the move relative to the volatility you are short.

WORKED EXAMPLE

You are short an option, delta-hedged, and the stock falls 3% in a day. Then it goes flat.

At an implied volatility of 16%, the break-even daily move is $16\%/\sqrt{252} = 1.008\%$, so recovery takes $(3/1.008)^2 = 8.9$ days in total, call it nine: the day of the move itself and about eight further flat days after it.

At 20% volatility the break-even move is 1.26% and recovery takes $(3/1.26)^2 = 5.7$ days. At 30% it takes only 2.5 days. The higher the volatility you sold, the more theta you collect and the faster a given move is repaid, which is exactly as it should be: a 3% move is unremarkable for a 30% volatility stock and a serious event for a 16% one.

REAL INTERVIEW QUESTION (Société Générale · EQD)

I am short an option, delta-hedged. The stock loses 3%. Assuming it stays flat afterwards, how many days do I need to recover my losses?

WHAT THEY TEST / ANSWER

Give the general result first, because it is more impressive than any single number: the answer is $(x/x^*)^2$ where x^* is the break-even daily move, $\sigma/\sqrt{252}$. Gamma and theta both cancel out, so you do not need their values at all, only the volatility.

Then produce a number by assuming a volatility out loud. At 16%, the break-even move is about 1%, so a 3% move is three times break-even and costs $3^2 = 9$ days of theta. At 20% the break-even is about 1.26% and it takes under six days. Finish with the two caveats a trader would add: this assumes your gamma stays put, whereas after a 3% move you are at a different point on the gamma profile and closer to expiry, where gamma is larger; and it assumes genuinely flat, since the recovery is eaten by every subsequent move, so in practice recovery is slower than the formula suggests.

You cannot hedge continuously

The theory assumes continuous rehedging. Reality gives you a choice between hedging often and hedging cheaply.

- **Hedging error.** With n discrete rehedges over the life of the option, the replication is imperfect and the terminal P&L is random around the theoretical value, with a standard deviation that falls roughly like $1/\sqrt{n}$. Doubling your hedging frequency reduces the error by only about 30%.
- **Transaction costs.** These rise with the number of rehedges, so the two effects pull in opposite directions and there is an interior optimum.
- **Hedging bands.** The practical answer is not to hedge on a clock but on a threshold: let the delta drift and rehedge only when it leaves a band around zero. Wider bands mean fewer trades, lower costs and more residual risk. The right width grows with transaction costs and, less obviously, with gamma: a high-gamma book churns its delta so fast that keeping it tightly hedged is ruinously expensive, so you let it drift further. Measured as a move in *spot* rather than in delta, the trigger does tighten as gamma rises, which is the version most people remember and the reason the two sound contradictory.
- **Path dependence.** Because of all this, two traders holding the same option and the same view will realise different P&L on the same path, purely from when they chose to rehedge.

Pin risk

The worst place to be is at the strike on expiry day. Delta there is switching between roughly 0 and roughly 1 on tiny moves in spot, gamma is enormous, and no hedge is stable. Worse, for a physically settled option you do not know whether you will be assigned until after the close, so you do not know whether you will wake up flat or holding the underlying. Desks manage this by reducing positions into expiry, rolling to a different strike, or accepting a deliberately imperfect hedge for the last day rather than churning. It is the practical reason traders dislike large expiries at round-number strikes.

INTERVIEW TRAP

Four traps. (1) Describing a delta-neutral book as hedged, full stop; it is neutral to small spot moves and fully exposed to gamma, vega and time. (2) Forgetting the sign of the rehedge: long gamma sells rallies, short gamma buys them. (3) Assuming more frequent hedging is always better, when the error only falls as $1/\sqrt{n}$ while costs rise with n. (4) Carrying a large position at the strike into expiry and treating pin risk as a rounding error.

VISUAL / CODE TO BUILD: Interactive

A rehedging game. The reader is short or long an option and watches a price path unfold tick by tick, choosing when to rehedge, either manually, on a fixed clock, or by setting a delta band. The panel tracks realised P&L decomposed into gamma, theta and transaction costs, and plots the running delta against the band so the reader sees the trade-off directly. Running the same path repeatedly with different hedging frequencies shows the P&L distribution narrowing as $1/\sqrt{n}$ while the cost line rises linearly, making the optimum visible. A final mode drops the reader at the strike on expiry day with the delta flickering, to make pin risk something they have felt rather than read about.

Where this goes next

The convexity that makes delta move in the first place is gamma and convexity, and what you pay to hold it is theta as the cost of convexity. Attributing the resulting profit and loss to each Greek after the fact is PnL explain, and the desk context in which all of this happens is market making.

16.3 Delta

sales · trading · structuring | core | equity · cross-asset

Delta is the first number an options trader looks at in the morning and the last one they check before the close. Every hedge, every risk limit and a surprising share of interview questions start here. We assume you already know what a call option is; now we ask a sharper question: when the underlying moves, how much does the option move with it?

The intuition: a slope

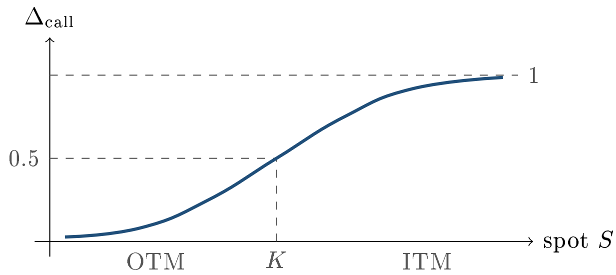
Picture the curve of an option's value against the spot price S. It is not a straight line; it bends. **Delta is the slope of that curve at today's spot**: the amount the option price changes for a small €1 move in the underlying. A gentle slope means the option barely reacts; a steep slope near 45° means it tracks the stock almost one-for-one.

KEY IDEA

$\Delta = \dfrac{\partial V}{\partial S}$, the sensitivity of the option's value to the underlying. For a call it runs from 0 (far out-of-the-money, the option ignores small moves) to 1 (deep in-the-money, the option is effectively the stock). For a put it runs from -1 to 0. Two readings live in the same number: a *sensitivity* (how much I make if spot ticks up) and a *hedge ratio* (how much stock cancels my directional risk).

The shape of the curve

Plot a call's Delta against spot and you get the smooth S-shape below. It is flat and near 0 when the option is far out-of-the-money, rises fastest as the stock approaches the strike, passes through roughly 0.5 at-the-money, and flattens again toward 1 once deep in-the-money.



Under Black-Scholes this curve has a closed form: $\Delta_{\text{call}} = N(d_1)$ and $\Delta_{\text{put}} = N(d_1) - 1$, where N is the standard normal CDF and $d_1 = \big[\ln(S/K) + (r + \tfrac{1}{2}\sigma^2)T\big]/(\sigma\sqrt{T})$. That form assumes *no dividends*: with a continuous dividend yield q it becomes $\Delta_{\text{call}} = e^{-qT}N(d_1)$, $\Delta_{\text{put}} = e^{-qT}\big(N(d_1) - 1\big)$, and d_1 carries $(r - q + \tfrac{1}{2}\sigma^2)$. We keep $q = 0$ throughout this section. You do not need to re-derive any of it here (that lives in Black-Scholes); you need to *read* it: Delta is a probability-like number squashed between the two extremes by the normal CDF.

REAL INTERVIEW QUESTION (Société Générale · EQD)

What is the delta of a call ITM, ATM and OTM?

WHAT THEY TEST / ANSWER

Definition first: the sensitivity of the option price to the underlying, $\partial V/\partial S$, equivalently the hedge ratio. Then the three anchors for a call: deep OTM ≈ 0, ATM ≈ 0.5, deep ITM ≈ 1 (mirror it for a put: 0, -0.5, -1). Say the word "monotonic" and sketch the S-curve; that single picture answers half the follow-ups they have lined up.

Delta as a hedge ratio

Here is why the desk cares. If you are long one call with $\Delta = 0.4$, your position gains and loses like 0.4 shares of stock for small moves. Sell 0.4 shares short and, for a small move, the two legs cancel: you are **delta-neutral**, immune to the *direction* of the next tick. What you are left holding is the option's curvature and its volatility exposure, which is exactly the point, you wanted the volatility bet, not the directional one.

WORKED EXAMPLE

You hold 10 ATM calls, each on 100 shares. How do you hedge? ATM delta is about 0.5, so each call behaves like $0.5 \times 100 = 50$ shares; ten of them, 500 shares. To be delta-neutral you *short 500 shares*. As spot moves the delta drifts (that is gamma), so you re-short or buy back to stay neutral: this continuous adjustment is *dynamic hedging*.

REAL INTERVIEW QUESTION (UBS · Structured Products)

You hold 10 ATM calls. How can you hedge yourself?

WHAT THEY TEST / ANSWER

Short roughly $10 \times \text{delta} \times \text{contract size}$ of the underlying. With ATM delta ≈ 0.5 and 100 shares per contract, that is about 500 shares short. The follow-up they want: it is only a *local* hedge, it decays as spot moves and time passes, so you rebalance, and that rebalancing cost is paid for by theta.

Position delta and cash delta

On a real book you do not track option-by-option; you aggregate. **Position delta** is the sum of (quantity $\times$ delta) across every trade, the number of shares-equivalent you are long or short. Desks usually quote it in money as **cash delta** (or dollar/euro delta) $= \Delta \times S \times$ quantity: the notional of stock your options currently behave like. It answers "how many euros do I make for a 1% move?" in one glance.

WORKED EXAMPLE

Cash delta \$4,500, stock at \$1,000, spot rises 4%. P&L? A cash delta of \$4,500 means you are effectively long \$4,500 of stock, so a +4% move earns about $0.04 \times 4{,}500 = \$180$ (to first order, before gamma). The lesson: once expressed as cash delta, the strike and contract size drop out, P&L is just percentage move $\times$ cash delta.

REAL INTERVIEW QUESTION (UBS · Market Risk)

If I buy a call, what is the sign of the delta, of the gamma, and so on?

WHAT THEY TEST / ANSWER

Long call: $\Delta > 0$, $\Gamma > 0$, $\mathcal{V} > 0$, $\Theta < 0$. You are long direction, long convexity, long volatility, and you pay for all of it through time decay. Being able to reel off the sign table instantly, and explain *why* theta is the rent you pay for gamma, is a standard filter.

The nuance interviewers hunt for: is ATM delta exactly 0.5?

Almost, but not quite, and the gap is the trap. From $\Delta = N(d_1)$, an at-the-money-*forward* call (struck at the forward) has $d_1 = \frac{1}{2}\sigma\sqrt{T} > 0$, so its delta is a touch *above* 0.5. An at-the-money-*spot* call is higher still, because the forward sits above spot by the cost of carry. So the correct answer is "about 0.5, and slightly above it, here is the d_1 reason", not a flat "one half".

REAL INTERVIEW QUESTION (Morgan Stanley · Exo)

What is the delta of an ATM call? What is its true value? Can you prove it?

WHAT THEY TEST / ANSWER

Around 0.5, and the proof is one line of $N(d_1)$. Struck at the forward, $\ln(S/K) + rT = 0$, so $d_1 = \frac{1}{2}\sigma\sqrt{T} > 0$ and $\Delta = N(d_1) > 0.5$: marginally *above* one half, by roughly $\frac{1}{2}\sigma\sqrt{T} \cdot N'(0) \approx 0.2\,\sigma\sqrt{T}$. At $\sigma = 20\%$ and $T = 1$, that is about 0.54. Read backwards, a $\Delta = 0.5$ call is therefore slightly *below* the forward, not exactly at it. Candidates who say a flat "0.5" get pushed; those who separate ATM-spot from ATM-forward and name d_1 close the topic.

VISUAL / CODE TO BUILD: Interactive

A live Delta curve. Axes: Delta vs spot. Sliders for volatility, time to expiry and whether it is a call or a put. As the reader drags volatility up, the S-curve *flattens* (delta becomes less sensitive to spot, deep-ITM delta pulls back from 1); as time to expiry shrinks toward zero, the curve sharpens into a step at K. A marker shows the current delta and the equivalent hedge in shares. Backed by $N(d_1)$ in Python; expose a table of delta at 5 spot levels for accessibility.

Delta moves too: enter gamma

The catch in "delta-neutral" is the word *small.* Delta is only constant for a tick; for a real move it changes, and the rate at which it changes is gamma. A long-option position has positive gamma, so its delta grows as the market rises and shrinks as it falls: you automatically get longer into a rally and shorter into a sell-off, which is the pleasant side of owning options.

REAL INTERVIEW QUESTION (Bank of America · Exo)

Your gamma is \$2m. If your spot rises by 2%, how does your delta change?

WHAT THEY TEST / ANSWER

State the convention first, because desks differ: equity desks quote **cash gamma** as the change in cash delta per +1% move in spot. On that convention a +2% move adds about $2 \times \$2\text{m} = \$4\text{m}$ of cash delta, so if you started flat you are now long \$4m of stock-equivalent. Second-order caveat they want to hear: gamma itself moves along the path, so \$4m is the linear estimate, and the exact answer needs the integral of gamma over the move. The intuition being tested is long gamma $\Rightarrow$ you get longer into a rally and shorter into a sell-off, which is comfortable and is paid for by theta.

INTERVIEW TRAP

Four classic slips. (1) Sign errors: a long put has *negative* delta. (2) Saying ATM delta is exactly 0.5 (it is a shade more, via d_1). (3) Treating delta as a fixed hedge: it drifts with spot (gamma), time and vol, so a static hedge leaks. (4) Calling delta "the probability of finishing in-the-money", it is close to the risk-neutral probability ($N(d_2)$ is the exact one), but they are not the same number, and a sharp interviewer will make you pay for the shortcut.

Where this goes next

Delta is the first of the Greeks; its rate of change is gamma, its decay through time is theta, and its dependence on volatility is vega. When the volatility you hedge at is itself skewed across strikes, delta acquires a correction (the market's "sticky-strike vs sticky-delta" debate), which is where skew re-enters. And the full machine that produces $N(d_1)$ in the first place is Black-Scholes.

16.4 Gamma and Convexity

sales · trading · structuring | core | equity · fx · cross-asset

Gamma is the reason an option is worth more than its forward-implied value, the reason delta hedging is a business rather than a formality, and the single Greek that most often blows a book up. It is also the same idea as convexity in fixed income, one level up the derivative chain, which is why a candidate who understands one can usually be led to the other.

Curvature is value

Gamma is $\partial^2 V/\partial S^2$: how fast delta changes as spot moves, and equivalently how much the value curve bends.

KEY IDEA

A convex payoff **gains from movement in either direction**. Because the gamma term in the P&L is $\frac{1}{2}\Gamma(\Delta S)^2$ and $(\Delta S)^2$ is positive whichever way spot went, a long-gamma position profits from a move up and from a move down. That asymmetry, produced purely by curvature, is what you are buying when you buy an option and what you are selling when you sell one. Everything else in this section is a consequence.

Seen through expectations, this is Jensen's inequality: for a convex function, the expected value of the function exceeds the function of the expected value, and the gap grows with dispersion. A forward has zero curvature, so it has no gamma and no volatility dependence. An option bends, so it has both.

The shape of gamma

For a vanilla option,

$$\Gamma = \frac{\varphi(d_1)}{S\,\sigma\sqrt{T}}.$$

Three readings, each of which is an interview answer:

- **Across strike**: $\varphi(d_1)$ peaks at the money, so gamma is concentrated near the strike and dies away in both tails, where the option behaves like cash or like the underlying and its delta has stopped moving.
- **Across time**: the $\sqrt{T}$ in the denominator means gamma grows without bound as expiry approaches. A short-dated at-the-money option is almost pure gamma.
- **Across volatility**: σ is also in the denominator, so a *low*-volatility underlying produces *higher* gamma for the same maturity, which surprises people. The reason is that low volatility concentrates the terminal distribution near the strike, so delta has to swing across a narrower range of spot.

WORKED EXAMPLE

At-the-money gamma on a spot of 100 at 20% volatility with zero rates, so that $d_1 = \frac{1}{2}\sigma\sqrt{T}$ and the table is reproducible, together with its cash equivalent (the change in cash delta for a 1% move):

Maturity	Γ	Cash gamma per 1%
1 year	0.0198	1.98
6 months	0.0281	2.81
3 months	0.0398	3.98
1 month	0.0691	6.91
1 week	0.1438	14.38

The one-month option has 2.45 times the gamma of the six-month, which is $\sqrt{6} = 2.449$ up to the small $\varphi(d_1)$ correction that also drifts with maturity: gamma scales as $1/\sqrt{T}$ and little else is changing. From one year to one week it multiplies by more than seven.

REAL INTERVIEW QUESTION (Société Générale · Structured Products)

I have an at-the-money spot call expiring in one month and another expiring in six months. Which has the larger gamma?

WHAT THEY TEST / ANSWER

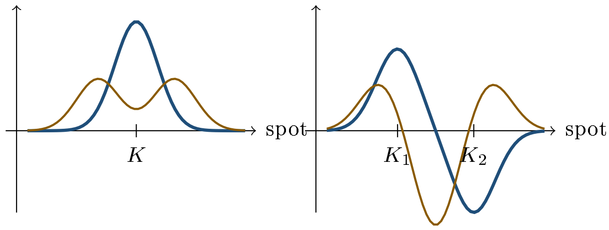
The one-month, and by a specific factor. Gamma at the money scales like $1/\sqrt{T}$, so the ratio is $\sqrt{6/1} = \sqrt{6} \approx 2.45$: the short-dated option has about two and a half times the gamma. Give the mechanism rather than the formula: with one month left the option's delta must travel from near zero to near one over a small range of spot, because there is little time for the stock to wander, so the delta curve is steep. With six months the same journey is spread over a much wider range, so the curve is gentle.

Then volunteer the mirror image, which is what they are usually setting up: **vega** goes the other way, scaling like $\sqrt{T}$, so the six-month option has about 2.45 times the vega. Short-dated options are gamma instruments and long-dated ones are vega instruments, and that separation is how a desk manages the two exposures independently, developed in the term structure of Greeks.

Gamma of structures

Gamma adds across legs, so any structure's gamma profile is the sum of bells with signs and positions given by the legs.

- **Straddle**: two long options at the same strike, so a single bell of double height, centred at the strike. Maximum gamma exactly where you want it if you are betting on a move.
- **Strangle**: two bells at separate strikes, so a flatter, twin-peaked profile with a dip between them.
- **Call spread**: long gamma around the lower strike, short gamma around the upper one, crossing zero between them. Its gamma is *signed by where spot is*, which is why a call spread is a much calmer position than a naked call.
- **Butterfly**: short two options at the middle strike against one at each wing, so gamma is strongly negative at the centre and positive at the wings. A long butterfly is short gamma exactly where it makes its money.



Gamma against spot. Left: —— straddle, one bell of double height at the strike, against —— strangle, two bells with a dip between them. Right: —— call spread, long gamma at the lower strike and short at the upper, crossing zero between them, against —— butterfly, strongly negative at the centre and positive at both wings. Every profile is a sum of bells with signs.

REAL INTERVIEW QUESTION (Barclays · Structured Products)

What does the gamma of a call spread look like?

WHAT THEY TEST / ANSWER

Draw it as the difference of two bells. You are long a call at K_1 and short one at $K_2 > K_1$, so gamma is the bell at K_1 minus the bell at K_2: **positive** when spot is near or below K_1, crossing through zero somewhere between the strikes, and **negative** when spot is near or above K_2. Far outside both strikes it decays to zero, since neither option's delta is moving any more.

Then make the point that matters commercially: because the gamma changes sign inside the structure, a call spread's risk is bounded in a way a naked call's is not, and its gamma is largest in magnitude near whichever strike spot is closest to. As expiry approaches both bells get taller and narrower, so a call spread with spot sitting near one of the strikes on expiry day is a genuinely uncomfortable position, and that is the same discomfort as the pin risk in dynamic hedging.

REAL INTERVIEW QUESTION (Société Générale · Rates)

Draw the gamma curve of a straddle.

WHAT THEY TEST / ANSWER

A single symmetric bell centred on the strike, positive everywhere, twice the height of one option's gamma because a call and a put at the same strike have the *same* gamma and you are long both. Say that equality explicitly, since it is the detail being checked: put-call parity says the call minus the put is a forward, which has zero gamma, so the two options' gammas must be identical.

Then add how the picture moves. As expiry approaches the bell becomes taller and narrower, concentrating all the gamma in a tiny window around the strike. As volatility rises the bell becomes shorter and wider. So a long straddle held into expiry is a position whose risk becomes more and more concentrated at a single point, which is why traders roll or reduce rather than run it to the last day.

REAL INTERVIEW QUESTION (Société Générale · EQD)

What are the vega and gamma positions of a straddle and of a butterfly?

WHAT THEY TEST / ANSWER

Straddle: long gamma and long vega, both peaking at the strike, because you are long two options and every long option is long both. It is the cleanest expression of "I want movement".

Butterfly: the reverse where it counts. Long the wings and short two at the centre, so near the middle strike you are **short gamma and short vega**, and out near the wings you are long a little of both. A long butterfly is a bet that nothing happens, and it is short exactly the two Greeks that pay when something does.

The sentence that ties it together: gamma and vega both peak at the money for a vanilla, so for a single option they move together, and any structure that is short one near a strike is short the other there too. Where they *separate* is across maturity, not across strike, since gamma scales as $1/\sqrt{T}$ and vega as $\sqrt{T}$. That is why you cannot untangle gamma from vega by choosing strikes, only by choosing maturities.

Trading gamma against vega and theta

Because gamma and vega both peak at the money, separating them requires the maturity lever from the hedging section: buy short-dated, sell long-dated, sized to neutralise vega, and the residual is positive gamma.

Gamma against *theta* is harder, because for a vanilla book they are tied together by construction: in the zero-carry Black-Scholes world, $\Theta \approx -\frac{1}{2}\Gamma\sigma^2 S^2$, so positive gamma forces negative theta. That is not a coincidence, it is the statement that you pay rent for convexity, developed in theta as the cost of convexity.

But "impossible" is too strong, and the exception is a good interview answer. That relation ignores carry. Once rates and dividends contribute to theta, an option deep in the money can have *negative time value*, which means positive theta, while still being a long option and therefore long gamma.

WORKED EXAMPLE

Gamma-positive and theta-positive, two ways.

A deep in-the-money European *put*: spot 50, strike 200, one year, rates 5%, no dividend. Its value is 140.25 against an intrinsic value of 150, so its time value is -9.75 and its theta is positive: every day that passes brings the near-certain strike closer. It is a long option, so its gamma is positive. This position is **long dividends**, since dividends push the forward down and help a put.

A deep in-the-money European *call* on a high-dividend name: spot 200, strike 100, one year, rates 1%, dividend yield 6%. Its value is 89.35 against an intrinsic value of 100, a time value of -10.65, so again positive theta and positive gamma. This one is **short dividends**.

So gamma-positive with theta-positive is achievable, and the sign of the dividend exposure can be chosen by picking the put or the call version.

REAL INTERVIEW QUESTION (BNP · EQD)

How do you construct a Gamma+/Vega- position? How do you construct a Gamma+/Theta+ position? What is that position's dividend sensitivity, and can you flip the sign of the dividend sensitivity while staying Gamma+/Theta+?

WHAT THEY TEST / ANSWER

Gamma+ / Vega-: buy short-dated options and sell long-dated ones, in the ratio that makes vega negative overall. Since gamma scales as $1/\sqrt{T}$ and vega as $\sqrt{T}$, the short-dated leg dominates the gamma and the long-dated leg dominates the vega, so the two can be set independently. That is a reverse calendar spread.

Gamma+ / Theta+: say first that this is impossible for a zero-carry vanilla book, since $\Theta \approx -\frac{1}{2}\Gamma\sigma^2 S^2$ ties the signs together. Then give the exception, which is what the question is fishing for: a **deep in-the-money European option with negative time value**. A deep in-the-money put when rates are positive is worth less than its intrinsic value, so time passing helps you, giving positive theta while it remains a long option with positive gamma.

Dividend sensitivity: that put is long dividends, because dividends lower the forward and a put gains. **Flipping the sign**: yes. Use a deep in-the-money *call* on a stock whose dividend yield exceeds the rate. Its time value is also negative, for the symmetric reason that you forgo the dividends while holding the option rather than the stock, so it is gamma-positive, theta-positive and **short** dividends. Both constructions exist; the choice between them is precisely the choice of which side of the dividend you want to be on.

INTERVIEW TRAP

Four traps. (1) Saying higher volatility means higher gamma; the formula has σ in the denominator, so for a given maturity a quieter underlying has more gamma. (2) Assuming gamma and vega always move together; they do across strike and oppositely across maturity. (3) Treating short gamma as a carry trade with a known cost, when its loss grows with the square of the move and is therefore concentrated in exactly the days you did not plan for. (4) Claiming gamma-positive and theta-positive is impossible; it is impossible without carry, and deep in-the-money options with negative time value are the counterexample.

VISUAL / CODE TO BUILD: Interactive

A gamma surface explorer. The main panel plots gamma against spot with sliders for maturity and volatility, so the reader watches the bell grow taller and narrower as maturity shrinks, and shorter and wider as volatility rises, with a numeric readout of the peak. A structure builder adds legs and shows the summed gamma profile, so a call spread's sign change and a butterfly's negative centre appear as the reader constructs them. A second tab overlays gamma and vega on the same strike axis to show they peak together, then switches the x-axis to maturity to show them diverging as $1/\sqrt{T}$ and $\sqrt{T}$, which is the separation the section turns on.

Where this goes next

What you pay to hold gamma, and why the two are the same quantity seen from two sides, is theta as the cost of convexity. How the vega half of the pair behaves as volatility itself moves is vega exposure and volatility regimes, and how both change with maturity is the term structure of Greeks. The same convexity idea, one derivative lower and applied to yields, is duration and convexity.

16.5 Theta as the cost of convexity

sales · trading · structuring | core | equity · fx · rates · cross-asset

A delta-hedged option trader has no view on direction. What they own instead is curvature, and what they pay for it is time. Every morning the same question decides their day: how far does the underlying have to move for the convexity to cover the decay? This section turns that question into one formula, one number per day, and the arithmetic desks actually do in their heads. It is asked in interviews more often than almost anything else in the Greeks chapter, usually as a two-line mental calculation with a trap in the units.

The definition of theta and its closed form are in Greek definitions and intuition, and gamma's own shape and behaviour are in gamma and convexity. Here we only care about the relationship between the two.

INTUITION

Owning an option is renting a machine that converts movement into money. Gamma is how efficiently the machine converts; theta is the daily rent. If the underlying moves enough, the machine earns more than the rent and you keep the difference. If it sits still, you still owe the rent. Nobody hands you a free convexity machine, and the rent is set so that, if the world behaves exactly as the option's implied volatility says it will, you break even to the cent.

The identity behind the whole trade

Take a long option, delta-hedged, and let one short interval δt pass. Expand the option value to second order in the spot move δS, then subtract what the hedge made:

$$\delta\Pi \;\approx\; \underbrace{\Delta\,\delta S}_{\text{cancelled by the hedge}} \;+\; \tfrac{1}{2}\Gamma\,(\delta S)^2 \;+\; \Theta\,\delta t.$$

So the hedged profit and loss over the interval is $\frac{1}{2}\Gamma(\delta S)^2 + \Theta\,\delta t$. Now substitute the relationship established in Greek definitions and intuition, which with zero rates and dividends reads $\Theta = -\frac{1}{2}\sigma^2 S^2 \Gamma$ where σ is the implied volatility the option is marked at:

$$\boxed{\delta\Pi \;\approx\; \tfrac{1}{2}\,\Gamma S^2 \left[\left(\frac{\delta S}{S}\right)^2 - \sigma^2\,\delta t \right].}$$

KEY IDEA

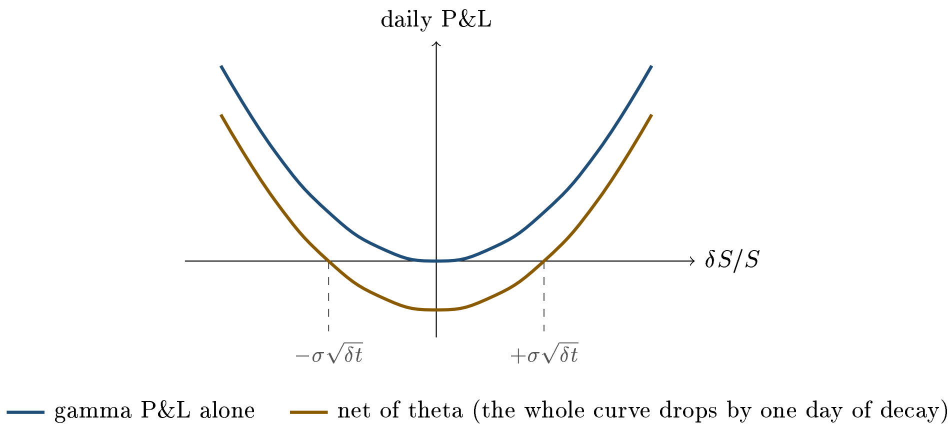
A delta-hedged option book is not a bet on the market going up or down. It is a bet on **realised variance against implied variance**, one interval at a time, with $\frac{1}{2}\Gamma S^2$ as the stake. Gamma and theta are not two risks, they are the two sides of that one bet: gamma collects the realised move, theta pays away the implied one. Everything else in this section is arithmetic on that bracket.

The bracket flips sign exactly when the size of the return equals $\sigma\sqrt{\delta t}$. That is the **break-even move**: the daily move at which the convexity earned exactly pays the day's decay. Over one day,

$$\text{break-even return} \;=\; \frac{\sigma}{\sqrt{252}} \quad \text{(business days)}, \qquad \text{or} \qquad \frac{\sigma}{\sqrt{365}} \quad \text{(calendar days)}.$$

Which convention a desk uses is a real source of disagreement, so state yours. On a business-day basis $\sqrt{252} \approx 15.87$, which is close enough to 16 that the desk shortcut is to divide the annual volatility by 16.

Implied volatility	Break-even daily move (252)	(365)
10%	0.63%	0.52%
16%	1.01%	0.84%
20%	1.26%	1.05%
30%	1.89%	1.57%
40%	2.52%	2.09%



Two features of that picture are the section in one image. The gamma curve is a parabola, so profit grows with the *square* of the move: a move twice as large is worth four times as much, which is why a long gamma book loves a gap and is indifferent to a drift. And theta shifts the entire parabola down by a fixed amount that does not care what happens, which is why a long gamma position loses on every quiet day and a short gamma position collects on every quiet day right up until it does not.

Doing it in cash, the way a desk does

Nobody on a desk works in Γ per share. The quoted number is **cash gamma**, and the most common equity convention is the change in *cash delta* for a one percent move in spot, $\Gamma_{1\%} = 0.01\,\Gamma S^2$. State the convention before you compute, because some desks quote ΓS^2 instead and the two differ by a factor of one hundred.

With that convention the arithmetic needs no calculus. If spot moves x percent, your cash delta changes by $\Gamma_{1\%}\,x$, so the *average* extra delta you carried through the move is half of that, and it earned the move itself, x percent:

$$\text{gamma P\&L} \;=\; \frac{\Gamma_{1\%}\,x}{2} \times \frac{x}{100} \;=\; \frac{\Gamma_{1\%}\,x^2}{200}.$$

That single expression answers most desk questions on this topic in about ten seconds.

REAL INTERVIEW QUESTION (Société Générale · EQD)

I am long a call, delta-hedged, with a cash gamma of 3 million and a theta of 15,000. How far does my underlying have to move for me to make money?

WHAT THEY TEST / ANSWER

One percent. Say the convention out loud first, that cash gamma is quoted as the change in cash delta per 1% move, because the whole computation hinges on it. Then: gamma P&L $= \Gamma_{1\%}x^2/200 = 15{,}000x^2$, set equal to theta, so $x = 1\%$. The follow-up that separates candidates is the sanity check: a 1% daily break-even is $1\% \times \sqrt{252} \approx 16\%$ annualised, so you have just recovered the implied volatility the book is marked at, which is the real content of the question. Add the qualification that this is a break-even on the *size* of the move, not its direction, and that it is per day, so a book that moves 0.5% every day for a week loses five times, while one that gaps 3% once and sits still for four days makes money overall.

REAL INTERVIEW QUESTION (BNP · Trading)

16% volatility, a 2% move in the underlying and 5 million of gamma: what is your P&L? What is your break-even?

WHAT THEY TEST / ANSWER

Break-even first, because it is the number they are really after: $16\%/\sqrt{252} = 1.01\%$, or just $16/16 = 1\%$ using the desk shortcut. Then the P&L. Gamma leg: $\Gamma_{1\%}x^2/200 = 5{,}000{,}000 \times 4/200 = 100{,}000$. Theta leg: the decay is whatever makes the break-even 1.01%, that is $5{,}000{,}000 \times (1.0079)^2/200 \approx 25{,}400$. Net, you are up about **75,000** on the day. Two things earn extra credit. First, name the convention you used for both the cash gamma and the day count, since a 365-day basis gives a 0.84% break-even and a net closer to 82,000. Second, note what you have assumed away: this is the P&L at constant implied volatility, and on a real 2% day the implied would almost certainly have moved too, adding a vega term covered in vega exposure and volatility regimes.

The quiet day

REAL INTERVIEW QUESTION (HSBC · EQD)

A day passes without the underlying moving (same parameters, no delta-neutral hedging): has the straddle made or lost money? Why?

WHAT THEY TEST / ANSWER

Lost, and the amount is one day of theta. The clause about not delta hedging is the distractor: a straddle struck at the money is delta-neutral to begin with, and in any case the underlying did not move, so there was nothing for a hedge to do. The only term left in the expansion is $\Theta\,\delta t$, and for a long option that is negative. Put a number on it to finish: a one-year at-the-money straddle with spot 100 and 20% volatility is worth 15.93, its theta is about -0.022 per calendar day, so a completely still day costs roughly 0.14% of the position. Then add the sentence that shows you understand why: the straddle owner paid for movement and received none, which is exactly the bargain implied volatility describes. The same day is a gain for whoever is short it.

Theta is not only gamma rent

The identity $\Theta = -\frac{1}{2}\sigma^2 S^2 \Gamma$ assumed zero rates and dividends. In general theta also contains carry terms, $-rKe^{-rT}N(d_2)$ for a call plus a dividend term, which is why a deep in-the-money European put can have *positive* theta: it is worth less than its intrinsic value because that value is only receivable at maturity, so time passing brings a near-certain payoff closer. On short-dated equity options the carry part of theta is small and the gamma rent is essentially all of it. On a multi-year structured note it is not small at all.

There is a second, subtler qualification. The identity uses the implied volatility each option is marked at, so it holds *per option*, not with one common σ across a book. That is what makes the following question answerable rather than a trick.

REAL INTERVIEW QUESTION (Bank of America · Exo)

Could you give me a strategy that is long gamma and long theta?

WHAT THEY TEST / ANSWER

The instinctive answer, that it is impossible, is right for a single vanilla and wrong for a book, and saying both is the answer. For one option $\Theta = -\frac{1}{2}\sigma^2 S^2 \Gamma$, so positive gamma forces negative theta. Across two options with *different implied volatilities*, the constraint loosens: net theta is $-\frac{1}{2}S^2 \sum_i \sigma_i^2 \Gamma_i$ while net gamma is $\sum_i \Gamma_i$, so you need $\sum_i \Gamma_i > 0$ with $\sum_i \sigma_i^2 \Gamma_i < 0$. Own gamma where volatility is cheap and sell more expensive gamma against it. Concretely, if the strike you buy is marked at 15% and the one you sell at 30%, the ratio of squared volatilities is four, so any position whose bought gamma is between one and four times the sold gamma is long gamma and long theta at the same time. Then say what you are actually short, because they will ask: you are short the volatility *spread*, so the position is a skew or calendar relative-value trade dressed up as a Greeks question, and it loses if the cheap strike stays quiet while the expensive one moves. The surface that makes this possible is the volatility smile and skew.

REAL INTERVIEW QUESTION (Unicredit · EQD)

An investor asks you to build a strategy with negative theta: what do you do?

WHAT THEY TEST / ANSWER

Negative theta means net long options, so you buy optionality: a straddle, a strangle, a call spread paid, or any structure whose premium is paid rather than received. The interviewer is not testing the trade, they are testing whether you translate a Greek into client language. Negative theta is not a goal anyone has; it is the price tag on a view. So the right response is a question back: what does the client actually want, a directional view with limited downside, exposure to a move whose direction they do not know, or protection? Each answer points to a different structure with negative theta, and each has a different break-even that you should quote in terms of how far the underlying must move, using $\sigma\sqrt{\delta t}$, rather than in terms of the Greek itself. Then note the mirror image: if they want *positive* theta, they are selling optionality, and you must present the tail risk they are taking on, since short gamma is exactly the position that pays every day until it does not.

INTERVIEW TRAP

Four traps, and three of them are units. (1) Mixing cash gamma conventions: per 1% move versus ΓS^2 differ by a factor of 100, and the answer will be off by ten in the break-even. (2) Mixing day counts: $\sqrt{252}$ and $\sqrt{365}$ give break-evens that differ by about 20%. (3) Comparing a daily theta to an annual gamma P&L. (4) The conceptual one: treating theta as a payment you collect for waiting. You collect it only if realised volatility comes in below implied. Being short gamma is not an income strategy, it is a short volatility position with a negatively skewed distribution, and the phrase to avoid in an interview is "picking up theta".

VISUAL / CODE TO BUILD: Interactive

A break-even simulator. The reader sets implied volatility, cash gamma and the day-count convention, and the widget draws the daily P&L parabola with the two break-even moves marked, updating as the sliders move. A second panel runs a path: it simulates or replays a sequence of daily returns at a chosen *realised* volatility, accumulates $\frac{1}{2}\Gamma S^2[(\delta S/S)^2 - \sigma^2\delta t]$ day by day, and plots the running P&L against the straight line the trader would have earned if every day had realised exactly the implied. The intended realisation: the path with the same realised volatility but concentrated in a few large days makes materially more than the smooth one for a long gamma book, because the parabola is convex, which is the practical difference between volatility and variance. Expose the daily table for accessibility.

Where this goes next

The bet described here, realised against implied, is set up in implied versus realized volatility, and the instrument that isolates it without any of the gamma bookkeeping is the variance swap in variance swaps. How the decay itself changes as expiry approaches is Greeks under time decay. Turning the terms of the expansion into an attribution of yesterday's result is Greeks and PnL explain, and how that attribution is used to run a book is PnL explain on the desk.

16.6 Vega Exposure and Volatility Regimes

sales · trading · structuring core equity · fx · cross-asset

Vega is the exposure an options desk is actually paid to run. Gamma is what you monetise day to day by rehedging; vega is the position you hold on the level of implied volatility itself. This section covers what vega looks like, how to compute it in your head, why a single vega number is misleading, and how the whole exposure behaves differently depending on the volatility regime you are in.

What vega is, and why it is always positive

Vega is $\partial V/\partial\sigma$, quoted per one percentage point of implied volatility. For any long vanilla option, call or put, it is positive, and the reason is *convexity* rather than any directional asymmetry. An option payoff is a convex function of the terminal price: flat on one side of the strike, sloped on the other (see options). Widening the distribution while holding the forward fixed raises the expected value of any convex payoff, so more volatility is worth more. State it that way rather than as "your loss is capped and your gain is not", because a put's gain is capped too, at the strike, and its vega is positive all the same.

> **KEY IDEA**
>
> Vega is long for anyone **long an option**, regardless of whether it is a call or a put and regardless of any directional view. This is why an options book's volatility exposure is a separate business from its directional exposure, and why "long volatility" and "bullish" are unrelated statements.

Shape, and the mental formula

For a vanilla, $\mathcal{V} = S\varphi(d_1)\sqrt{T}$.

- **Across strike**: $\varphi(d_1)$ peaks at the money, so vega is largest for at-the-money options and decays into both wings, where the option is nearly certain to expire worthless or nearly certain to be exercised, and a shift in volatility changes little.
- **Across maturity**: the $\sqrt{T}$ means vega *grows* with maturity, the exact opposite of gamma, which shrinks as $1/\sqrt{T}$. Long-dated options are vega instruments.

The desk approximation follows from the at-the-money price rule $C \approx 0.4\,\sigma S\sqrt{T}$, which is itself a one-line consequence of the model: at the money the price is $S\,[N(d_1) - N(d_2)]$, the two arguments straddle zero a distance $\sigma\sqrt{T}$ apart, so the difference is about $\varphi(0)\,\sigma\sqrt{T}$ with $\varphi(0) = 1/\sqrt{2\pi} \approx 0.399$. The 0.4 is a computed constant, not folklore. Differentiating in σ gives

$$\mathcal{V} \;\approx\; 0.4\,S\sqrt{T} \text{ per unit of } \sigma, \qquad \text{or} \qquad \frac{0.4\,S\sqrt{T}}{100} \text{ per volatility point.}$$

REAL INTERVIEW QUESTION (Union Capital · FX)

I buy a EUR 10 million at-the-money one-month call with a volatility of 10%. Roughly what is my vega position?

WHAT THEY TEST / ANSWER

Use the rule of thumb and say you are using it. Vega per volatility point is about $0.4\,S\sqrt{T}/100$, so with a notional of 10 million and $T = 1/12$, $\sqrt{T} = 0.289$, giving

$$\frac{0.4 \times 10{,}000{,}000 \times 0.289}{100} \approx €11{,}500 \text{ per volatility point.}$$

The exact Black-Scholes figure is €11,515 at zero rates, so the approximation is within a third of a percent, which you should mention because it shows you know when the shortcut is safe: it is at its most accurate at the money and degrades in the wings.

Then interpret rather than stopping at the number: an implied volatility move from 10% to 11% is worth about €11,500 to you, and note that this is roughly 0.115% of notional per point, so a one-month option is a small vega instrument. If they wanted real vega they would have asked for a one-year, which would be $\sqrt{12} = 3.5$ times larger.

REAL INTERVIEW QUESTION (Société Générale · EQD)

When is a call's vega at its maximum?

WHAT THEY TEST / ANSWER

Two answers, because "maximum" can mean across strike or across maturity, and a good candidate gives both. **Across strike**, at a fixed maturity, vega peaks at the money, more precisely where $d_1 = 0$, which is a strike just above the forward. The reason is that this is where the outcome is most genuinely uncertain, so widening the distribution changes the value most. Deep in or out of the money the outcome is nearly settled and vega collapses toward zero.

Across maturity, at a fixed moneyness, vega increases with $\sqrt{T}$ over any realistic maturity: a longer option has more vega. Strictly it does turn over, and it is worth knowing where, because it is a good follow-up. Since $\mathcal{V} = S\varphi(d_1)\sqrt{T}$ and d_1 itself grows like $\sqrt{T}$, vega is proportional to $d_1\varphi(d_1)$, which is maximised at $d_1 = 1$. At the money *forward* with zero rates that means $T = 4/\sigma^2$, about a hundred years at 20% volatility, so the monotone statement is safe. At the money *spot* with 5% rates and 20% volatility it is nearer eight years, which is inside the traded range for long-dated equity and hybrid options. Then give the contrast that makes the answer memorable: gamma peaks at the same strike but goes the *opposite* way in maturity, so the same at-the-money strike gives you maximum gamma at the front of the curve and maximum vega at the back. That is the entire basis for separating the two exposures, as in the term structure of Greeks.

REAL INTERVIEW QUESTION (CACIB · Exo)

Two calls, three years and five years. Which is more expensive? Which has the bigger vega, and why? And their gammas?

WHAT THEY TEST / ANSWER

Three quick answers with one reason behind all of them, namely the $\sqrt{T}$ scaling. **Price**: the five-year is more expensive, since it has strictly more optionality, and at the money the price scales roughly as $\sqrt{T}$, so it is about $\sqrt{5/3} = 1.29$ times the three-year, before any dividend or rate effects. **Vega**: the five-year again, and by the same factor of 1.29, because vega scales as $\sqrt{T}$. **Gamma**: the other way round, the three-year has the larger gamma, also by 1.29, because gamma scales as $1/\sqrt{T}$.

Then add the caveat an exotics interviewer is waiting for: on a real underlying the five-year is more exposed to the dividend and repo assumptions, which are far less certain at five years than at three, and on a high-dividend name the longer call can even be worth less than the $\sqrt{T}$ scaling suggests because the forward has decayed further. The clean scaling is a Black-Scholes statement, not a market one.

One vega number is not enough

Adding up the vega of a book gives a single figure, and that figure quietly assumes the entire volatility surface moves in parallel. It does not. A realistic book is long vega at one maturity and short at another, so a parallel-shift number can be near zero while the position has large exposure to the volatility term structure *steepening*.

Desks therefore carry a **vega ladder**: vega bucketed by maturity, and often by strike as well. It is the direct analogue of the DV01 ladder in rates, and it exists for exactly the same reason: the underlying curve does not move in parallel.

REAL INTERVIEW QUESTION (CACIB · Delta One)

Explain vega and its importance.

WHAT THEY TEST / ANSWER

Definition in a line, then importance, which is the real question. Vega is the change in an option's value for a one point change in implied volatility, positive for any long option.

Importance, in three steps. First, it is the exposure a dealer runs deliberately: after delta hedging, a book's value is driven by volatility, so vega is the position rather than a residual. Second, it is the exposure that reprices without anything happening to the underlying, since implied volatility can move on its own and mark your book overnight with spot unchanged, which is exactly what makes it a risk rather than an accounting artefact. Third, and this is the sentence that lands: a single vega figure assumes the whole surface shifts in parallel, and it never does, so desks run a vega ladder bucketed by maturity and strike. Saying that shows you have looked at a real risk report rather than a textbook.

Regimes

The same vega position behaves differently depending on the volatility environment, and this is where the section earns its title.

Calm regime. Implied volatility is low and its term structure slopes *upward*, because the market prices mean reversion back toward a long-run average. Vol of vol is low, so vega is a slow, carry-like exposure. Being short vega earns the volatility risk premium of implied versus realised, quietly and most of the time.

Stressed regime. Implied volatility spikes and the term structure *inverts*, because the market prices a crisis now and a return to normal later. Three things change at once, and naming all three is a strong answer:

- **The front-end vega bucket dominates**, since the short-dated implieds move far more than the long-dated ones, so a book that measured its vega as a parallel number will be badly wrong about the size of the move, in whichever direction it is positioned.
- **Vol of vol rises**, so **volga** matters: your vega itself moves, and wing options gain relative to at-the-money ones.
- **Spot-vol correlation bites**, which in equities is strongly negative: spot falls and volatility rises together. That is a **vanna** exposure, and it means a delta-hedged long-vega position is implicitly a position on the direction of the market, developed in cross-Greeks.

The practical consequence is that short-vega positions are short a fat tail: they earn steadily in the calm regime and lose several years of that carry in a few days when the regime changes, because vega, volga and vanna all move against them at once.

Computing the P&L

WORKED EXAMPLE

Vega alone. A book with a vega of €500,000 and implied volatility rising 3 points makes $500{,}000 \times 3 = €1{,}500{,}000$. Vega is quoted per point, so the computation is a multiplication, and the only way to get it wrong is a units error.

Vega and gamma together. A book that is long €100,000 of vega and €1 million of cash gamma (quoted, as usual, as the change in cash delta per 1% move). The market rises 3% and implied volatility rises 1 point.

$$\text{Vega P\&L} = 100{,}000 \times 1 = €100{,}000$$

$$\text{Gamma P\&L} = \frac{\Gamma_{\text{cash}}\, x^2}{200} = \frac{1{,}000{,}000 \times 3^2}{200} = €45{,}000$$

for a total of about €145,000, assuming the book was delta-hedged and ignoring theta over the period.

REAL INTERVIEW QUESTION (JP Morgan · Delta One)

You are long 100k of vega and 1 million of gamma. The market rises 3% and the underlying's volatility rises by 1 point. What is your P&L?

WHAT THEY TEST / ANSWER

Take the two terms separately and state your conventions, because the gamma one is convention-dependent and an interviewer will accept any convention you name. Vega is quoted per volatility point, so a one point move gives 100,000 × 1 = €100,000. Taking the gamma figure as cash gamma per 1% move, the gamma P&L is $\Gamma_{\text{cash}} x^2/200 = 1{,}000{,}000 \times 9/200 =$ €45,000. Total, about **€145,000**.

Then give the three qualifications that turn a calculation into a trader's answer. This assumes you were delta-hedged, otherwise the delta P&L would dominate both terms. It ignores theta, which over the same period works against you if you are long gamma. And in equities a 3% *rally* with volatility *up* one point is an unusual combination, since spot and vol are normally negatively correlated, so the scenario itself is a vanna event and a real book would want to know whether its vanna helped or hurt.

INTERVIEW TRAP

Four traps. (1) Quoting a single portfolio vega as if the surface moved in parallel; run the ladder. (2) Mixing vega conventions, per point versus per unit of volatility, which is a factor of 100. (3) Assuming vega is stable; in a stress regime volga and vanna move it, which is exactly when you care. (4) Treating short vega as a carry trade with a known cost, when its losses arrive concentrated and correlated with everything else going wrong.

VISUAL / CODE TO BUILD: Interactive

A vega workbench. The main panel plots vega against strike with a maturity slider, and a second axis plots vega against maturity at fixed moneyness, so the reader sees the peak at the money and the monotone $\sqrt{T}$ growth side by side, with gamma overlaid to show it running the other way. A regime switch morphs the volatility term structure between an upward-sloping calm shape and an inverted stressed one, and a book of positions is repriced under each, so the reader sees a book whose parallel vega is near zero posts large gains or losses purely from the term-structure move. A P&L calculator takes vega, cash gamma, a spot move and a vol move and decomposes the result, with a deliberate units toggle that shows what a per-point versus per-unit mistake does to the answer.

Where this goes next

How vega varies across strike rather than maturity, and what that does to a book, is skew and smile sensitivities. The cross-Greeks that make vega itself move, vanna and volga, are cross-Greeks, and the maturity dimension is the term structure of Greeks. The instrument that gives a clean volatility exposure without any of this strike and maturity bookkeeping is the variance swap.

16.7 Skew and smile sensitivities

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Delta, gamma, vega and theta are the Greeks of a world with one volatility. The market does not have one volatility, it has a surface, as the volatility smile and skew establishes. So a real book carries two more risks that a flat-volatility book cannot even express: what happens to your delta when volatility moves, and what happens to your vega when volatility moves. Those are **vanna** and **volga**, and they are the reason a desk talks about being long or short skew rather than simply long or short vega.

This section owns the smile-facing Greeks specifically. The general cross-derivative machinery, charm, veta, colour and speed, lives in cross-Greeks and interaction effects, and the taxonomy of all of them is in Greeks (first and second order).

Vanna: your delta has a volatility exposure

$$\text{Vanna} \;=\; \frac{\partial^2 V}{\partial S\,\partial\sigma} \;=\; \frac{\partial \Delta}{\partial \sigma} \;=\; \frac{\partial \mathcal{V}}{\partial S} \;=\; -\,e^{-qT}\,\frac{\varphi(d_1)\,d_2}{\sigma}.$$

Read it either way and both readings are useful on a desk: it is how much your hedge ratio moves when implied volatility moves, and it is how much your vega moves when spot moves. Same number, two conversations.

The sign is entirely determined by d_2, and d_2 is positive when the strike sits below $Se^{(r-q-\sigma^2/2)T}$, which is marginally below the forward rather than equal to it. So:

- **Low strikes** (out-of-the-money puts, in-the-money calls) have *negative* vanna. Raise volatility and their delta falls.
- **High strikes** (out-of-the-money calls) have *positive* vanna. Raise volatility and their delta rises.
- Vanna passes through zero where $d_2 = 0$, that is at $K = Se^{(r-q-\sigma^2/2)T}$, marginally *below* spot. For spot 100, one year and 20% volatility with zero rates, that is $K = 98.0$, not 100.

The intuition is worth more than the formula. Raising volatility widens the terminal distribution. For a strike far above spot, widening pulls probability mass *up* across the strike, so the option becomes more stock-like and its delta rises. For a strike below spot, widening pulls mass *down* across the strike, so an in-the-money call becomes less certain to be exercised and its delta falls toward one half. Vanna is that redistribution, measured.

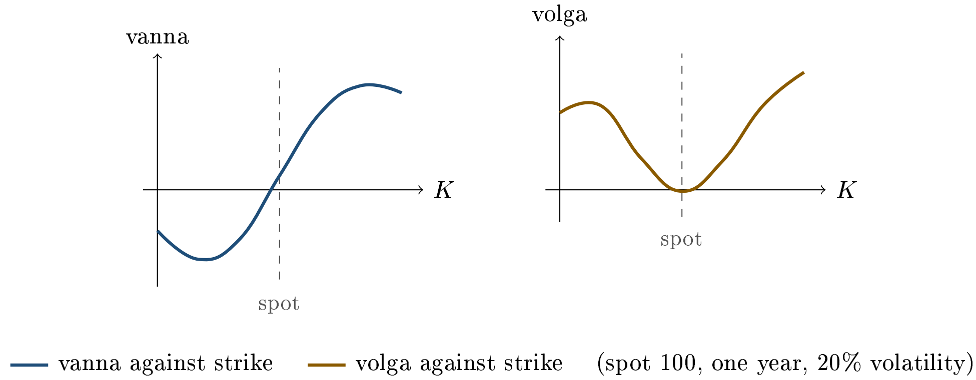
> **KEY IDEA**
>
> Vanna is the Greek that makes "delta neutral" a statement about one particular volatility level rather than a fact. A book that is flat delta at 20% volatility is not flat delta at 22%, and the gap is the vanna. That is why a trader who is long skew and delta-hedged still wakes up with a directional position after a volatility move.

Volga: your vega has a volatility exposure

$$\text{Volga} \;=\; \frac{\partial^2 V}{\partial \sigma^2} \;=\; \frac{\partial \mathcal{V}}{\partial \sigma} \;=\; \mathcal{V}\,\frac{d_1 d_2}{\sigma}.$$

Volga (also written vomma) is convexity in volatility, so it is to vega what gamma is to delta. Its sign is the sign of $d_1 d_2$, which is negative only in the narrow band where $d_1 > 0 > d_2$. That band is $Se^{(r-q-\sigma^2/2)T} < K < Se^{(r-q+\sigma^2/2)T}$, which for the same spot 100, one year, 20% example runs from 98.0 to 102.0. Everywhere else volga is positive and large.

So volga is essentially zero at the money and big in both wings. This is the formal reason wings are the instrument for trading volatility-of-volatility: an at-the-money straddle gives you vega with almost no convexity in it, while a strangle gives you a vega that grows as volatility rises. It is also why a smile model that fits only the at-the-money level will misprice the wings badly, no matter how well it is calibrated in the middle.



(spot 100, one year, 20% volatility)

> **WORKED EXAMPLE**
>
> Spot 100, one year, 20% implied volatility, zero rates and dividends. Raw values, then the desk units, which are per volatility *point* for vanna (divide by 100) and per squared point for volga (divide by 10,000).
>
> | Strike | d_2 | Vega | Vanna (per point) | Volga (per point^2) |
> |---|---|---|---|---|
> | 80 | $+1.016$ | 19.05 | -0.0097 | $+0.0118$ |
> | 90 | $+0.427$ | 32.78 | -0.0070 | $+0.0044$ |
> | 100 | -0.100 | 39.70 | $+0.0020$ | -0.0002 |
> | 110 | -0.577 | 37.16 | $+0.0107$ | $+0.0040$ |
> | 120 | -1.012 | 28.70 | $+0.0145$ | $+0.0118$ |
> | 130 | -1.412 | 19.14 | $+0.0135$ | $+0.0164$ |
>
> Read one row out loud to make it concrete. The 120-strike call has a vanna of 0.0145 per point, so if implied volatility rises from 20% to 21% its delta rises by about 0.014, from 0.209 to 0.223. Own ten thousand of them and you have just become long 145 shares without the market moving at all. The same row shows why the wings matter for volga: the 120 strike has a volga of $+0.0118$ per squared point against -0.0002 at the money, roughly sixty times larger in magnitude and of the opposite sign.

Translating Greeks into skew and smile positions

The point of vanna and volga is not the derivative, it is the position they describe. Two dictionary entries do most of the work.

Vanna is your skew position. A book with negative vanna is net long low strikes against high strikes, which in equity language is long the downside, and that is the position that gains when the skew steepens. A book with positive vanna is the reverse.

Volga is your smile-convexity position. Positive volga means net long wings against the middle, which is long the butterfly and long volatility-of-volatility.

State your convention, because "long skew" is defined differently in different markets and interviewers use the ambiguity to see whether you noticed. In equities the usual reading is that long skew means long low-strike volatility against high-strike volatility, so long skew corresponds to *negative* vanna. In FX the quoted risk reversal is the call volatility minus the put volatility, so a long risk reversal is long the high strike and carries *positive* vanna. The Greek does not change; the sign of the word does. FX conventions are set out in risk reversals and butterflies.

This dictionary is also a pricing method. Since a European option's smile risk is essentially spanned by vega, vanna and volga, an FX desk can price an exotic by taking its Black-Scholes value at the at-the-money volatility and adding a hedge portfolio of the traded at-the-money straddle, risk reversal and butterfly chosen to match those three Greeks. That is the vanna-volga method, and it is a direct application of the two sentences above rather than a separate theory.

REAL INTERVIEW QUESTION (Société Générale · Exo)

Impact of a move in volatility on the delta?

WHAT THEY TEST / ANSWER

Name the Greek in the first sentence: this is vanna, $\partial\Delta/\partial\sigma$, and its Black-Scholes value is $-e^{-qT}\varphi(d_1)d_2/\sigma$. Then give the sign by region rather than a single answer, because there is no single answer. For a strike above the forward, $d_2 < 0$, vanna is positive and a rise in volatility raises the delta: the option becomes more likely to be reached and behaves more like stock. For a strike below the forward the opposite holds, and an in-the-money call loses delta as volatility rises, because certainty of exercise is being taken away from it. Note the crossing point is at $d_2 = 0$, that is $K = Se^{(r-q-\sigma^2/2)T}$, slightly below spot rather than exactly at it. Finish with the consequence, which is what an exotics interviewer is after: a delta-neutral book is only delta-neutral at the volatility it was hedged at, so on any day implied volatility moves you inherit a directional position equal to vanna times the volatility move.

REAL INTERVIEW QUESTION (BNP · Volatility)

I have a delta-neutral book: when I raise volatility by 1% I become short delta. What is my skew position?

WHAT THEY TEST / ANSWER

Work it in three steps and say each one. Step one: delta falling when volatility rises means $\partial\Delta/\partial\sigma < 0$, so the book has **negative vanna**. Step two: vanna is negative for strikes below the forward, so a negative-vanna book is net long low strikes and short high strikes, that is long puts against short calls in the wings. Step three: that is the position that gains when low-strike volatility richens relative to high-strike volatility, so in the equity convention you are **long skew**. Then defend the convention explicitly, because it is the trap: if the desk quotes skew the FX way, as the call volatility minus the put volatility, the same book would be described as short the risk reversal. The Greek is unambiguous, the word is not, so answer with the Greek and then translate. A good cross-check to volunteer: a long call spread is long the low strike and short the high strike, so it has negative vanna and is long skew by the same reasoning, which is why a call spread gets more expensive as the skew steepens.

REAL INTERVIEW QUESTION (BNP · EQD)

How do you get short skew?

WHAT THEY TEST / ANSWER

Any position that is net short low-strike volatility against high-strike volatility, and you should give at least two concrete constructions plus the Greek that unifies them. Constructions: sell a put spread or, more directly, sell the downside put and buy the upside call, which is a short risk reversal in equity terms; or sell a call spread, since a call spread is long the low strike and short the high strike, so selling it flips both legs. Unifying Greek: all of these have *positive* vanna, at any strikes a desk would actually trade. The qualifier is worth knowing rather than saying: vanna peaks a little above the 120 strike in the table above and falls away beyond it, so a spread struck entirely in that far wing reverses the sign. Then say what you are really taking on, because that is the risk-management half of the answer: short skew is short the crash, so it earns a small carry in calm markets and loses badly precisely when the equity market gaps down and low-strike volatility explodes, which is the same tail that structured-product issuance already leaves the street exposed to. Mention that you would size it against a steepening scenario, not against a parallel volatility shift, since a parallel shift is a vega risk and this position is deliberately close to vega neutral. See skew trading for how it is traded properly.

The delta the smile actually implies

If the smile moves when spot moves, the total derivative of value in spot has a second term:

$$\Delta_{\text{total}} \;=\; \Delta_{\text{BS}} \;+\; \mathcal{V}\,\frac{\partial \sigma_{\text{imp}}}{\partial S}.$$

Whether that term is present at all depends on the regime, sticky strike or sticky delta, defined in the volatility smile and skew. Under sticky strike the correction is zero and the Black-Scholes delta is right. Under sticky delta the curve travels with spot, so a fixed strike's implied volatility rises as spot rises given a downward-sloping equity skew, the correction is positive, and the true hedge ratio exceeds the Black-Scholes one.

Two things deserve emphasis. First, the correction is proportional to *vega*, so it is largest for at-the-money and long-dated options, exactly where vega lives, and small for short-dated wings. Second, this is a modelling choice you are making whether you notice or not: a book delta-hedged under the wrong regime accumulates a systematic directional bias, and the profit and loss shows up as an unexplained residual in PnL explain rather than as an obvious error.

REAL INTERVIEW QUESTION (Morgan Stanley · EQD)

Effect of volatility on the delta of a digital?

WHAT THEY TEST / ANSWER

Do not answer with one sign, because the answer changes with moneyness, and saying so is the answer. A cash-or-nothing digital call is worth $e^{-rT}N(d_2)$ and its Black-Scholes delta is $e^{-rT}\varphi(d_2)/(S\sigma\sqrt{T})$, which is a spike centred near the strike. Raising volatility *smooths that spike*: for a digital near the money the delta falls sharply, from about 0.040 at 10% volatility to 0.020 at 20% and 0.010 at 40%, for spot 100 and one year. For a digital well out of the money the effect reverses at low volatility, because raising volatility first brings the strike within reach: at a 120 strike the delta rises from 0.0069 at 10% to 0.0120 at 20% before falling again at higher volatility. Then add the part that makes it an exotics answer: on a real desk you do not price a digital at one volatility at all, because the skew adds a term proportional to vega times the slope of the smile, so the digital's true delta inherits the smile's own dependence on spot. That is why digitals are risk-managed as tight call spreads rather than as their closed form, as covered in exotic options.

REAL INTERVIEW QUESTION (BNP · Exo)

On an accumulator (long 1 call, short 2 puts), give the volatility sensitivity and the smile sensitivity.

WHAT THEY TEST / ANSWER

Set the structure up first: the calls are struck above and the puts below, and the two short puts are what pays for the long call, which is where the leverage and the risk both come from. **Volatility sensitivity**: net vega is one long wing against two short wings, so the position is short vega on any reasonable strike choice, and short gamma below the put strike where the double size bites. **Smile sensitivity**: the short legs are the low strikes, so the position is short low-strike volatility against long high-strike volatility, that is **short skew**, equivalently positive vanna. State the consequence rather than stopping at the label: this is a position that is comfortable in a calm, drifting market and hurts in exactly the scenario where equity markets gap down, since spot falls into the double short puts while low-strike implied volatility richens against you, so the delta, the gamma and the skew all move the wrong way at once. That correlation of losses, not the individual Greeks, is what makes the accumulator a structuring interview favourite.

INTERVIEW TRAP

Four traps. (1) Saying "long skew" without stating the convention: equity and FX sign it oppositely, and the interviewer is often checking exactly that. (2) Believing that delta neutral means delta neutral: it holds at one volatility level, and vanna is the size of the error after a volatility move. (3) Placing the vanna zero and the volga sign change at the money: they sit at $d_2 = 0$ and at $d_1 = 0$, which are just below and just above spot, a distance of order $\sigma^2 T$. (4) Treating volga as negligible because it is near zero at the money: at-the-money is precisely the one place it vanishes, and a book of wings has most of its volatility risk in it.

VISUAL / CODE TO BUILD: Interactive

A vanna and volga surface explorer. Panel A plots vanna and volga against strike for a chosen maturity and volatility, with the two zero-crossings ($d_2 = 0$ and $d_1 = 0$) marked so the reader sees them shift away from spot as $\sigma^2 T$ grows. Panel B is a position builder: the reader assembles a book from vanilla legs (a call spread, a risk reversal, a strangle, an accumulator) and the widget reports net vega, vanna and volga plus a one-line translation into desk language, long or short skew, long or short smile convexity. Panel C shocks the surface, letting the reader apply a parallel volatility shift, a steepening and a wings-up move separately, and shows the resulting profit and loss decomposed into the vega, vanna and volga contributions. The intended realisation: the three shocks hit three different Greeks, which is why one vega number cannot describe a book that lives on a surface.

Where this goes next

The level of the surface and how a book behaves as regimes change is vega exposure and volatility regimes. The other cross-derivatives, and how the Greeks interact rather than simply add, are in cross-Greeks and interaction effects. Trading the slope deliberately rather than inheriting it is skew trading, the asset-specific shapes that generate these exposures are in equity skew and smile and the FX volatility smile, and the products that hand a dealer these Greeks whether they want them or not are autocallables.

16.8 Cross-Greeks and Interaction Effects

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The first-order Greeks are not constants, and the second-order Greeks you have already met, gamma and volga, describe how each one moves when *its own* input moves. The cross-Greeks describe how each one moves when a *different* input moves, and they are where a Black-Scholes book and a real book diverge. They are also, in interviews, the questions used to separate people who have used the Greeks from people who have listed them.

Vanna: delta and volatility

Vanna is $\partial^2 V / \partial S\, \partial\sigma$, and it has two equally useful readings:

$$\text{vanna} = \frac{\partial \Delta}{\partial \sigma} = \frac{\partial \mathcal{V}}{\partial S}.$$

It answers both "what happens to my delta when volatility moves?" and "what happens to my vega when spot moves?", and those are the same number.

For a vanilla option, differentiating $\Delta = N(d_1)$ gives

$$\text{vanna} = -\frac{\varphi(d_1)\, d_2}{\sigma},$$

so the sign is simply the opposite of the sign of d_2:

Option (spot 100, 1y, 20% vol)	d_2	Delta	Vanna
In-the-money call, $K = 80$	$+1.02$	0.888	-0.968
At-the-money, $K = 100$	-0.10	0.540	$+0.198$
Out-of-the-money call, $K = 120$	-1.01	0.209	$+1.452$

KEY IDEA

Raising volatility pulls every delta **toward the middle**. An in-the-money option's delta falls, an out-of-the-money option's delta rises, and an option struck near the money barely moves. The intuition is that higher volatility makes the final outcome less certain, so every option starts to look more like a coin flip. The dividing line is exact: vanna changes sign where $d_2 = 0$, that is at $K = Fe^{-\sigma^2 T/2}$, so every strike below it has negative vanna and every strike above it positive. Read the rule as a statement about ordinary volatility moves rather than a limit: pushed far enough, *every* call's delta eventually rises toward one, since an infinitely volatile call is worth the stock itself.

REAL INTERVIEW QUESTION (BNP · EQD)

An in-the-money call, and volatility rises. What does the delta do?

WHAT THEY TEST / ANSWER

It **falls**. Give the intuition first because it is the memorable part: an in-the-money call is currently expected to be exercised, so its delta is high, near 0.888 in the numbers above. Raising volatility makes it less certain that it will still be in the money at expiry, so the option becomes less stock-like and its delta declines. In the example, taking volatility from 20% to 21% moves the delta from 0.888 to 0.879.

Then formalise it if they want more: this is negative vanna, and vanna $= -\varphi(d_1)d_2/\sigma$, so the sign flips with d_2. In-the-money means $d_2 > 0$ and therefore negative vanna; out-of-the-money means $d_2 < 0$ and positive vanna, so an out-of-the-money option's delta *rises* with volatility.

Do not close by saying the delta converges to one half. It does not, and the shortcut is wrong in both directions. Writing $u = \sigma\sqrt{T}$, $d_1 = \ln(S/K)/u + u/2$ is *minimised* at $u = \sqrt{2\ln(S/K)}$, where it equals $\sqrt{2\ln(S/K)}$, strictly positive for an in-the-money call. So the delta bottoms out above one half and then climbs back toward one, since an infinitely volatile call is worth the stock itself. For the numbers above, $S = 100$ and $K = 80$, the turn is at $\sigma = \sqrt{2\ln 1.25} = 66.8\%$, where the delta is 0.748; by 200% volatility it is back to 0.867. What is true, and is what the question is testing, is that the delta compresses toward the at-the-money-forward level over the range anyone trades. Quote the turning point only if pushed.

REAL INTERVIEW QUESTION (Morgan Stanley · EQD)

What is the impact of volatility on the delta curve?

WHAT THEY TEST / ANSWER

Describe the whole curve, not one point, since they asked for the curve. The delta curve is the S-shape running from 0 far out of the money to 1 far in. Raising volatility **flattens and stretches** it: the transition from 0 to 1 spreads over a wider range of spot, so deep out-of-the-money deltas rise, deep in-the-money deltas fall, and the at-the-money delta is nearly unchanged. Lowering volatility does the reverse, steepening the curve toward a step function, which in the limit of zero volatility is exactly what it becomes.

Two consequences worth volunteering. First, since the slope of the delta curve is gamma, flattening it means **gamma falls as volatility rises**, which is the σ in the denominator of the gamma formula from gamma and convexity. Second, the pivot near the at-the-money point is why vanna changes sign there, and it is why hedging a book through a volatility spike requires rehedging delta even if spot has not moved at all.

REAL INTERVIEW QUESTION (BNP · EQD)

I am long a put struck at 80%, and the price moves to 95%. How does the vega evolve?

WHAT THEY TEST / ANSWER

The vega **rises**. The put was struck well below spot, so it was deeply out of the money and its vega was small; as the underlying falls toward the strike the option moves toward the money, and vega peaks at the money, so it grows substantially on the way.

Then give the cross-Greek reading, which is what the question is testing: this is vanna seen from the other side. Vanna is $\partial\mathcal{V}/\partial S$ as well as $\partial\Delta/\partial\sigma$, and for a downside put it is negative, meaning vega falls as spot rises and therefore rises as spot falls. The two statements are the same number. Finish with why a desk cares: a book short downside puts is short vega that *grows* exactly as the market sells off, so its volatility exposure is largest when volatility is spiking, which is the mechanism behind most structured-product hedging losses.

Because vanna links spot and volatility, and because the sign of a strike's vanna follows the sign of $-d_2$, a book's net vanna is the standard first-order proxy for its exposure to the shape of the volatility surface rather than its level. Reading vanna as a *skew position*, translating it into the equity and FX quoting conventions, and the smile corrections built on it belong to skew and smile sensitivities; what is developed here is the mechanics that reading rests on.

Volga: vega and volatility

Volga (or vomma) is $\partial^2 V/\partial\sigma^2$, the sensitivity of vega to volatility itself. It is near zero at the money, where vega peaks and is locally flat, and positive in both wings.

So a book long wing options is long volga, meaning long *volatility of volatility*: it gains when implied volatility moves a lot, in either direction, over and above what its vega alone would predict. That is why a butterfly in volatility space is a volga trade while a risk reversal is a vanna trade, and why the two together, vanna and volga, are the standard first-order corrections a smile-aware desk applies to a flat-volatility price.

Charm: delta and time

Charm is $\partial\Delta/\partial t$, how delta drifts purely because time passes. As expiry approaches, an in-the-money option's delta climbs toward 1 and an out-of-the-money option's toward 0, so the delta curve steepens into a step. This is why a book can require rehedging over a weekend during which nothing traded, and why desks look at charm on a Friday afternoon.

Charm is also the reason behind a question that looks like it is about delta: the effect of *maturity* on the delta of an at-the-money call. Lengthening the maturity is charm run backwards, and the delta rises, because d_1 grows with $\sqrt{T}$: at a strike set equal to the forward, $d_1 = \frac{1}{2}\sigma\sqrt{T}$. Said out loud: the forward strike is the *mean* of the terminal distribution but sits above its *median*, by a factor $e^{\sigma^2 T/2}$ that widens with the horizon, so the option becomes less likely to finish in the money, $N(d_2)$ falls, and yet it becomes steadily more stock-like, so $N(d_1)$ and the delta rise. At a very long horizon the call is worth almost the stock itself and its delta tends to one. The levels at each maturity are tabulated in Greek definitions, which owns that question.

One correction is worth carrying, because the approximation it replaces is repeated everywhere. Neither the delta nor $N(d_2)$ is exactly one half at the money *forward*, where they are $N(\frac{1}{2}\sigma\sqrt{T})$ and $N(-\frac{1}{2}\sigma\sqrt{T})$, straddling it from either side. Delta is exactly one half at $K = Fe^{\sigma^2 T/2}$, which is the delta-neutral straddle strike quoted in FX, and $N(d_2)$ is exactly one half at $K = Fe^{-\sigma^2 T/2}$, the median of S_T. Three different strikes, and the habit of calling all of them "at the money" is what makes the question a trap.

INTERVIEW TRAP

Four traps. (1) Treating delta as a function of spot alone; it depends on volatility through vanna and on time through charm, so a book can need rehedging with spot unchanged. (2) Getting the vanna sign backwards; it is the opposite of the sign of d_2, so in-the-money is negative and out-of-the-money is positive. (3) Assuming volga matters everywhere, when it is close to zero at the money and lives in the wings. (4) Quoting a skew position without stating the sign convention, since the risk reversal is quoted differently across desks and asset classes.

VISUAL / CODE TO BUILD: Interactive

A cross-Greek explorer. The reader sees the delta curve against spot with a volatility slider, and watches it flatten and steepen live, with the pivot near the at-the-money point made visible by a second trace showing the *change* in delta, which is vanna and which visibly changes sign at the pivot. A maturity slider does the same thing for charm, letting the reader watch the curve steepen into a step as expiry approaches. A second tab plots vega against spot so the reader can confirm that the slope of that curve is the same number as the vanna read off the first tab, making the two readings of vanna one object rather than two facts. A position builder then reports vanna and volga for a book, with preset risk reversals and butterflies so the reader sees which structure isolates which.

Where this goes next

How the smile itself, rather than a single volatility, generates these exposures is skew and smile sensitivities, and how a whole book behaves when the surface shifts is Greeks under volatility shifts. The time dimension that charm opens is Greeks under time decay, and trading the skew these Greeks measure is skew trading.

16.9 Greeks Under Spot Moves

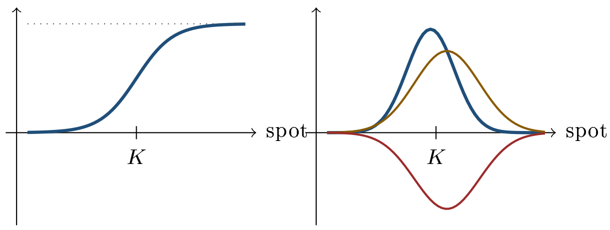
sales · trading · structuring | core | equity · fx · cross-asset

Hold time and volatility still, move spot, and watch what each Greek does. That is the experiment this section runs, and it is the one an interviewer runs on you when they say "the stock is up 10%, now what?". The answers are not memorised facts; they are readings off four curves, and once you can draw the curves you can answer any variant.

The four profiles

Against spot, at a fixed maturity and volatility:

- **Delta** is the S-curve, rising from 0 far out of the money to 1 far in, passing near one half around the strike.
- **Gamma** is the slope of that S-curve, so it is a bell peaked near the strike and decaying to zero in both directions.
- **Vega** is a bell too, and peaked near the strike for the same reason: gamma and vega both measure how much the outcome is still genuinely in doubt. It is the wider of the two, because $\mathcal{V} = \sigma S^2 T\,\Gamma$ exactly, so vega is the gamma bell reweighted by S^2 and its peak sits a little above the strike rather than a little below it.
- **Theta** is that same bell flipped below the axis, most negative near the strike and approaching zero in the wings. With zero rates and dividends $\Theta = -\frac{1}{2}\sigma^2 S^2 \Gamma$, so it is the S^2-weighted gamma again, which is why it tracks vega's width rather than gamma's.



Left, —— delta: the S-curve from 0 to 1, with the dotted line at 1. Right, the three that live at the strike: —— gamma, the slope of that S-curve, peaking just below the strike; —— vega, the same bell reweighted by S^2, so wider and peaking just above the strike; and —— theta, that wider bell flipped below the axis. Heights are not to a common scale, only the shapes and signs are. All four are read against the same spot axis, which is the point: one move relocates you on every one of them.

> KEY IDEA
>
> **Gamma, vega and theta all live where the strikes are.** Move spot far enough away from a strike and all three collapse toward zero, leaving an option that behaves like cash (deep out of the money) or like the underlying itself (deep in the money). So a large spot move does not merely change your delta, it migrates you along every profile at once and can leave a book that was full of risk this morning with almost none, or the reverse.

REAL INTERVIEW QUESTION (BNP · Trading)

How do the delta and gamma of a call evolve with the price of the underlying?

WHAT THEY TEST / ANSWER

Describe both curves and the relationship between them, since that relationship is the answer. **Delta** traces an S: essentially zero when spot is far below the strike, rising through about one half near the strike, and flattening out at one when spot is far above. **Gamma** is the *slope* of that S-curve, so it is a bell: near zero in both tails, maximum near the strike.

Then give the consequence, which is what a trading desk cares about. Where gamma is large your delta is unstable, so you rehedge constantly and your P&L is driven by movement. Where gamma is small your delta barely moves and the position behaves like a static holding of cash or stock. So as spot travels through the strike, an option transforms from something inert into something that demands attention and back again. Add the maturity dimension if they want more: as expiry approaches the S-curve steepens toward a step and the gamma bell becomes taller and narrower, so the same option becomes far more sensitive to exactly where spot is.

Repricing under a spot move

To reprice quickly, expand:

$$V(S+\Delta S) \;\approx\; V + \Delta\,\Delta S \;+\; \tfrac{1}{2}\,\Gamma\,(\Delta S)^2.$$

The first term is what everyone gives; the second is what separates a prepared candidate, and for a small move it is a genuinely small correction.

WORKED EXAMPLE

Spot 125. You are long a three-month call struck at 140, bought at 8, with a delta of 0.4 and gamma assumed constant over the move. Spot rises to 127.

First order: $8 + 0.4 \times 2 =$ €8.80.

The gamma correction: backing out the volatility consistent with a 0.4 delta at these parameters gives about 57% (the €8 is the price paid, not a model-consistent premium; only the *change* is being validated here), and the corresponding gamma is 0.0108, so the second-order term is $\frac{1}{2} \times 0.0108 \times 2^2 =$ €0.02, giving €8.82.

The check is satisfying: repricing the option exactly at 127 with the same volatility moves it by 0.822, against the Taylor estimate of $0.80 + 0.02 = 0.822$. For a move this size the two-term expansion is not an approximation in any practical sense.

REAL INTERVIEW QUESTION (Morgan Stanley · EQD)

The underlying is at 125. You are long a call struck at 140, three months, bought at 8, with a delta of 0.4 and gamma assumed flat. What is the call's premium if the underlying rises to 127?

WHAT THEY TEST / ANSWER

Answer in one line and then improve it. One line: the move is 2 and the delta is 0.4, so the premium rises by about 0.8 to **€8.80**.

Improve it by adding the term they left the door open for. Since gamma is assumed constant, the exact second-order expansion applies: $\Delta V = \Delta\,\Delta S + \frac{1}{2}\Gamma(\Delta S)^2$. You are long the call, so gamma is positive and the true premium is slightly *above* 8.80. At these parameters the gamma term is worth about two cents, giving roughly €8.82.

Then the sentence that shows judgement: gamma matters here only because the move is small. Had they said the stock jumped to 150, the delta-only answer would be badly wrong, because delta itself would have risen substantially on the way, and you would want to reprice rather than expand. Knowing when a Taylor expansion stops being adequate is more valuable than the expansion itself.

Structures move too

The same profiles apply leg by leg, so a structure's delta can change dramatically as spot moves through its strikes.

WORKED EXAMPLE

An at-the-money straddle, spot 100, strike 100, three months, 20% volatility, zero rates. Its delta is $N(d_1) + \big(N(d_1) - 1\big) = 2N(d_1) - 1 = 0.04$, essentially flat, as you would expect from a position with no directional view.

Now the underlying rises 10% to 110. The call delta climbs to 0.84 and the put delta falls to -0.16, so the straddle's delta becomes $2(0.842) - 1 = 0.68$. A position that was delta-neutral this morning is now long the equivalent of two thirds of a share, purely because spot moved.

REAL INTERVIEW QUESTION (HSBC · EQD)

If the underlying rises 10%, how does the delta of a straddle evolve?

WHAT THEY TEST / ANSWER

It becomes substantially **long**. Start from the structure: a straddle is a call plus a put at the same strike, so its delta is $\Delta_{\text{call}} + \Delta_{\text{put}} = 2N(d_1) - 1$, which is near zero at the money. After a 10% rally the call is in the money and the put is out, so the call's delta dominates and the combined delta turns positive. With three months and 20% volatility the delta goes from about 0.04 to about 0.68.

Then say why it matters, which is the point of asking. A straddle is a long-gamma position, so it *automatically* becomes long as the market rises and short as it falls. That is the convexity working for you if you are long it, and it is precisely why a long straddle profits from a move in either direction. If you are delta-hedging it, this is also the rehedge: you would sell stock into the rally to flatten, banking the gain, which is the long-gamma mechanic from dynamic hedging.

REAL INTERVIEW QUESTION (BNP · EQD)

What is the delta of an at-the-money call? Of a put? Of a call spread? Of a straddle at the money? And how does the delta of a deep in-the-money call change when volatility rises?

WHAT THEY TEST / ANSWER

Run through them, each with a reason rather than a number alone. **ATM call**: about 0.5, and strictly a little above it, since $d_1 = \sigma\sqrt{T}/2 > 0$ at the money with zero rates. **ATM put**: about -0.5, from $\Delta_{\text{put}} = \Delta_{\text{call}} - 1$, which itself comes from put-call parity. **Call spread**: the difference of two call deltas, so it is positive, smaller than the long leg's alone, and largest when spot sits between the strikes; deep in or out of the money both deltas converge and the spread's delta goes to zero. **ATM straddle**: about zero, since the call and put deltas nearly cancel, which is what makes it a non-directional position.

The last part is a cross-Greek question: a deep in-the-money call's delta **falls** when volatility rises, because higher volatility makes the outcome less certain and compresses the delta profile across strikes. Say compresses rather than "pulls toward one half": the deltas converge on the at-the-money-forward delta, not on a half, and pushed far enough every call's delta turns and rises toward one. That is negative vanna, developed in cross-Greeks.

The asymmetry of large moves

Two things make big moves qualitatively different from small ones, and both are worth saying unprompted.

First, the expansion breaks down. Delta is only locally valid and gamma is only locally constant, so for a large move you must reprice rather than extrapolate. A book's stress scenarios exist precisely because the Greeks do not survive being extrapolated.

Second, in equities a large *downward* move almost never happens with volatility unchanged. Spot and volatility are negatively correlated, so a sell-off moves you along the spot profiles *and* shifts the volatility surface at the same time, and the two effects interact through vanna. That is why a scenario grid on a real desk is two-dimensional, spot against volatility, rather than a single row.

INTERVIEW TRAP

Four traps. (1) Answering a spot-move question with delta alone when the move is large enough for gamma to matter, or with a Taylor expansion when the move is large enough that neither is valid. (2) Forgetting that vega and theta also migrate, so a book far from its strikes has little volatility exposure regardless of how many options it holds. (3) Saying a straddle is delta-neutral, full stop; it is delta-neutral at the money and directional everywhere else. (4) Shocking spot without shocking volatility, which in equities is not a realistic scenario.

VISUAL / CODE TO BUILD: Interactive

A spot-scan panel. A vertical marker sweeps across the spot axis while four stacked charts show delta, gamma, vega and theta, so the reader watches all four profiles being read at the same spot simultaneously and sees gamma, vega and theta collapse together as the marker moves into the wings. A position builder lets the reader add legs and watch the aggregate profiles develop kinks at each strike. A reprice tool takes a spot shock and reports three numbers side by side: the delta-only estimate, the delta-plus-gamma estimate and the exact reprice, so the reader discovers empirically how large a move has to be before the expansion fails. A second axis adds a volatility shock so the reader can see that a realistic equity sell-off scenario moves both at once.

Where this goes next

The same experiment run against volatility instead of spot is Greeks under volatility shifts, and against time is Greeks under time decay. The interaction terms that couple the spot and volatility axes are cross-Greeks, and doing all of this for a whole book rather than one option is portfolio-level Greeks aggregation.

16.10 Greeks under volatility shifts

trading · structuring · sales | advanced | equity · fx · cross-asset

A Greek is not a constant. It is a number computed at one volatility level, and when that level changes every Greek in the book changes with it, some of them dramatically and in directions most candidates guess wrong. This section takes one book and asks what happens to its whole risk profile when implied volatility moves from 20% to 40%, which is roughly what a crisis does in a week.

Three results carry the section, and they are worth knowing before the derivations. At-the-money gamma is *inversely* proportional to volatility. At-the-money vega is almost *unchanged* by it. At-the-money theta is roughly *proportional* to it. None of those is obvious, and together they explain why an option book feels like a different instrument in a high-volatility regime.

The Greeks themselves are defined in Greek definitions and intuition, and the specific derivatives of delta and vega with respect to volatility, vanna and volga, belong to skew and smile sensitivities. Here we care about the shape of the whole profile, not the individual derivative.

Gamma: volatility spends it

Gamma's own definition and shape belong to gamma and convexity; the result inherited from there is that, with zero rates and dividends, at-the-money gamma is

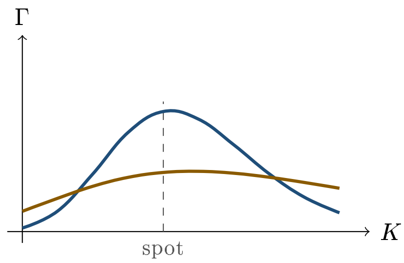
$$\Gamma_{\text{ATM}} \;=\; \frac{\varphi(\tfrac{1}{2}\sigma\sqrt{T})}{S\,\sigma\sqrt{T}} \;\approx\; \frac{0.4}{S\,\sigma\sqrt{T}},$$

so it falls roughly as $1/\sigma$. Double the volatility and you halve the gamma of an at-the-money option. For spot 100 and one year, gamma is 0.0398 at 10% volatility, 0.0198 at 20% and 0.0098 at 40%.

That is not gamma disappearing, it is gamma *spreading out*. The total curvature a strike can offer is finite; raising volatility widens the distribution of where the spot might end up, so the curvature has to be shared over a wider range of strikes. The formal statement is the sign of $\partial\Gamma/\partial\sigma$, sometimes called zomma:

$$\frac{\partial\Gamma}{\partial\sigma} \;=\; \Gamma\,\frac{d_1 d_2 - 1}{\sigma},$$

which is negative wherever $d_1 d_2 < 1$, a band that comfortably contains the money, and positive far out in the wings. So a volatility rise takes gamma away from strikes near spot and gives it to strikes far from spot.



— gamma profile at 20% volatility — the same book at 40% (one year, spot 100, strikes 60 to 150)

Integrated over *all* strikes the two curves enclose almost the same area; inside the plotted range the high-volatility curve looks like the smaller of the two only because its tails carry on past both edges of the picture, which is the redistribution itself showing up as a cropped figure. The low-volatility profile is a tall spike sitting on spot; the high-volatility one is a broad plateau. That single picture explains a great deal of desk behaviour. In a calm market a gamma book is a *pin* risk: almost all your curvature is at one strike, and it matters enormously whether spot is at 100 or 105. In a stressed market the same book has modest gamma spread over a hundred-point range, so individual strikes stop mattering and the position becomes a bet on the level of volatility rather than on where spot pins.

REAL INTERVIEW QUESTION (BNP · Trading)

How does gamma evolve with volatility?

WHAT THEY TEST / ANSWER

At the money it *falls*, roughly as $1/\sigma$, which is the opposite of what most candidates guess, so lead with the number: from 0.0398 at 10% volatility to 0.0198 at 20% and 0.0098 at 40% for a one-year option on a spot of 100. The formula makes it obvious, $\Gamma_{\text{ATM}} \approx 0.4/(S\sigma\sqrt{T})$, but the reason is the part they want: gamma is curvature per unit of spot, and raising volatility widens the range of outcomes over which that curvature is distributed, so it thins out where spot currently sits. Then complete the answer, because "gamma falls" is only true near the money: in the wings gamma *rises* with volatility, since higher volatility brings distant strikes into play. The formal statement is $\partial\Gamma/\partial\sigma = \Gamma(d_1 d_2 - 1)/\sigma$, negative near the money and positive far from it. Finish with the trading consequence, which is what a trading desk is really testing: high volatility does not mean lots of gamma, it means flat, spread-out gamma, so the strike-by-strike pin risk of a calm market becomes a level-of-volatility risk instead.

Vega: almost invariant at the money, convex in the wings

Vega is $Se^{-qT}\varphi(d_1)\sqrt{T}$, and at the money $d_1 = \frac{1}{2}\sigma\sqrt{T}$ is small, so $\varphi(d_1)$ sits close to its maximum 0.399 regardless of the volatility level. The consequence surprises people:

Implied volatility	10%	20%	40%	60%
ATM price	3.99	7.97	15.85	23.58
ATM gamma	0.0398	0.0198	0.0098	0.0064
ATM vega (per point)	0.398	0.397	0.391	0.381
ATM theta (per day)	−0.0055	−0.0109	−0.0214	−0.0313

Across a six-fold change in volatility, at-the-money vega moves by four percent. The price sextuples, gamma falls by a factor of six, theta rises by a factor of six, and vega barely notices. That is the practical meaning of volga being near zero at the money, established in skew and smile sensitivities: an at-the-money option's exposure to volatility is essentially linear in volatility, so its vega is a constant.

Out of the money it is a different story. There volga is large and positive, so vega grows as volatility rises: a wing that carried almost no volatility exposure in a calm market carries real vega once volatility doubles. A book of strangles therefore gets *longer* vega precisely as volatility rises, which is pleasant when you are long and dangerous when you are short.

REAL INTERVIEW QUESTION (Barclays · Exo)

Questions on the Greeks. Vega of an at-the-money-forward straddle. If volatility rises.

WHAT THEY TEST / ANSWER

Two parts, and the second is the trap. **The vega**: a straddle is a call plus a put at the same strike, and put-call parity makes their vegas identical, so the straddle's vega is exactly twice the call's, $2Se^{-qT}\varphi(d_1)\sqrt{T}$. At the money forward $d_1 = \frac{1}{2}\sigma\sqrt{T}$, so the straddle vega is close to $0.8\,S\sqrt{T}$ per unit of volatility, that is about 0.79 per volatility point for spot 100 and one year. It is the largest vega available at that maturity, which is why the at-the-money straddle is the standard volatility instrument. **If volatility rises**: the position gains, but the vega itself barely changes, because volga is essentially zero at the money. Say that explicitly, since the expected wrong answer is that vega grows. If they want you to be long that convexity, you have to go to the wings, where volga is large: a strangle gets longer vega as volatility rises while a straddle does not. Mention that the straddle *does* lose gamma as volatility rises, so a long straddle in a high-volatility regime is much more a vega position and much less a gamma position than the same trade in a calm market.

Delta and theta

Delta flattens. Raising volatility compresses the delta profile across strikes: a deep in-the-money call loses delta, a far out-of-the-money call gains it, and the delta curve against spot becomes a gentler S. The deltas converge on the at-the-money-forward delta rather than on a half, which matters because that anchor is itself above a half and rises with $\sigma\sqrt{T}$. The magnitude is vanna, treated in skew and smile sensitivities; the picture is that high volatility makes every option less certain about what it will become, and delta is a statement about that certainty.

Theta grows in proportion to volatility, which follows directly from the identity $\Theta = -\frac{1}{2}\sigma^2 S^2\Gamma$ of theta as the cost of convexity: gamma falls as $1/\sigma$ and the coefficient rises as σ^2, so the product rises as σ. A long option book in a crisis pays roughly double the daily rent it paid in calm markets, and the break-even move doubles with it. That is the honest reason "buy volatility when there is a crisis" is not a strategy: you are paying the higher rent too.

REAL INTERVIEW QUESTION (BNP · EQD)

In-the-money call, the volatility rises. What does the delta do?

WHAT THEY TEST / ANSWER

It *falls*. Give the mechanism, not just the sign: an in-the-money call's delta is high because exercise is nearly certain, and raising volatility puts that certainty back in doubt, pushing the delta down. In Greek terms the strike sits below the forward, so $d_2 > 0$, so vanna $= -e^{-qT}\varphi(d_1)d_2/\sigma$ is negative. Numbers make it convincing: a one-year call struck at 60 with spot 100 has a delta of 0.996 at 20% volatility, 0.930 at 40% and 0.875 at 60%. Do not extend that into a claim that the delta approaches a half: for this strike it bottoms at 0.844 near $\sigma = 101\%$ and rises back toward one thereafter, because $d_1 = (\ln(S/K) + \sigma^2 T/2)/(\sigma\sqrt{T})$ is eventually increasing in σ. Then flip it to show you know the general rule rather than one case: an out-of-the-money call does the opposite and gains delta as volatility rises, and the crossover sits at $d_2 = 0$, marginally below spot. The follow-up is usually about the hedge, so add that this is why a delta hedge computed at yesterday's volatility is wrong today, and the size of the error is vanna times the volatility move.

When spot and volatility move together

Real days move both. Take a delta-hedged book, so the first-order spot term is already neutralised, and expand to second order in both variables:

$$\delta\Pi \;\approx\; \underbrace{\tfrac{1}{2}\Gamma\,(\delta S)^2}_{\text{gamma}} \;+\; \underbrace{\mathcal{V}\,\delta\sigma}_{\text{vega}} \;+\; \underbrace{\text{vanna}\,\delta S\,\delta\sigma}_{\text{cross}} \;+\Theta\,\delta t \;+\;\cdots$$

The first two are what everyone computes. The cross term is what gets forgotten, and on an equity book it is systematically *not* small, because spot and implied volatility are negatively correlated: down days are volatility-up days, so $\delta S\,\delta\sigma$ has a consistent sign rather than averaging out. A book with the wrong sign of vanna therefore bleeds on a schedule rather than at random.

WORKED EXAMPLE

A book that is long 100,000 of vega (per volatility point) and 1 million of cash gamma (per 1% move). The market rises 3% and implied volatility rises by one point.

- Vega leg: $100{,}000 \times 1 = 100{,}000$.
- Gamma leg, using the cash-gamma arithmetic of theta as the cost of convexity, $\Gamma_{1\%}x^2/200 = 1{,}000{,}000 \times 9/200 = 45{,}000$.
- Theta leg: the question does not give a volatility, so assume the book is marked around 16%, at which one day of decay on that gamma is $1{,}000{,}000 \times (1.008)^2/200 \approx 5{,}100$.

Net, about **140,000**. Two caveats belong in the answer rather than in a footnote. The gamma leg assumes the 3% arrived as a single move; the same 3% arriving as six 0.5% moves and back would earn a fraction of it, since gamma pays on the sum of squared moves. And nothing above includes the cross term, which needs the book's vanna: a spot move of +3% together with a volatility move of +1 point is an unusual combination for equities, where the two normally move in opposite directions, so a book positioned for the usual relationship would lose on the cross term here.

REAL INTERVIEW QUESTION (JP Morgan · Delta One)

If you integrate your gamma from 0 to infinity, what happens? And if you integrate without limits?

WHAT THEY TEST / ANSWER

Ask which variable they mean, because the two integrals are different and both are instructive. **Over spot**, for one option, gamma is the derivative of delta, so integrating it from zero to infinity is just the total change in delta:

$$\int_0^\infty \Gamma(S)\,dS \;=\; \Delta(\infty) - \Delta(0) \;=\; 1$$

for a call, whatever the volatility and whatever the maturity. That is the cleanest statement of what the section above describes: raising volatility cannot create or destroy gamma, it can only redistribute a fixed total of one along the spot axis, which is why the peak collapses exactly as the tails fatten. **Over strike**, holding spot fixed, the answer is nearly but not exactly one:

$$\int_0^\infty \Gamma(K)\,dK \;=\; e^{(r+\sigma^2)T},$$

which is 1.041 at 20% volatility and one year, and 1.174 at 40%. So the two profiles above enclose almost the same area once the entire strike axis is counted rather than only the range that fits in the picture, and the small excess grows with $\sigma^2 T$. Finish with the trading reading, which is what a Delta One interviewer is after: gamma is a conserved quantity you cannot manufacture by changing the volatility you mark at, so a book that needs more gamma has to buy more options or shorter ones, and a book complaining that it has lost gamma in a volatility spike has not lost anything, it has had it spread over a range where spot no longer is.

The two-factor arithmetic in the example above is developed properly, with the vanna term included, in vega exposure and volatility regimes.

The client-facing version

Volatility levels are not only a trading input, they are a pricing input the sales desk has to explain. The rule to internalise is that a structure's terms improve for the investor when the optionality the investor *sells* becomes more expensive, and that is exactly what a high volatility level does.

REAL INTERVIEW QUESTION (Société Générale · Structured Products)

If volatility were higher at issuance, would the coupon be higher or lower, and why?

WHAT THEY TEST / ANSWER

Higher, and the reason is that the investor is a seller of optionality. In a standard yield product the client is short a downside put, and the premium of that put is what funds the coupon. Higher implied volatility at issuance means the put is worth more, so there is more premium to recycle, so the coupon printed on the term sheet is larger. Make the trade-off explicit rather than presenting it as free money, since that is the sales judgement being tested: the coupon is larger because the risk being sold is larger, and a product issued at 40% volatility offers a headline coupon that a product issued at 15% cannot match while also being materially more likely to breach its barrier. Then note the second-order effect that separates a structuring answer: it is the volatility *at the strike of the sold option* that matters, not the at-the-money level, so the skew and its steepness feed directly into the coupon, which is why issuance terms improve when the skew steepens even if at-the-money volatility has not moved. See autocallables for the full mechanics.

INTERVIEW TRAP

Four mistakes. (1) Assuming high volatility means high gamma: at the money it is the reverse, gamma goes as $1/\sigma$. (2) Assuming vega grows with volatility: at the money it is essentially flat, and only the wings have real volga. (3) Forgetting that theta scales with volatility, so the break-even move in a crisis is far larger and a long gamma book is not automatically a winner. (4) Computing a two-factor day as vega plus gamma and dropping the vanna cross term, which on equity books is not noise, since spot and implied volatility are reliably negatively correlated.

VISUAL / CODE TO BUILD: Interactive

A volatility-level slider driving four linked profiles. The reader sets maturity and drags a single volatility slider from 5% to 80%; four panels redraw simultaneously, showing delta, gamma, vega and theta against strike. The intended realisations, in order: the gamma spike collapses and spreads while its area barely changes; the vega curve keeps almost the same height at the money but fattens in the wings; the delta S-curve flattens; and theta scales up visibly. A readout above each panel gives the at-the-money value so the $1/\sigma$, flat, and σ scalings can be read as numbers rather than inferred from the picture. A second mode adds a two-factor P&L calculator with separate spot-move and volatility-move inputs, decomposing the result into gamma, vega, vanna and theta contributions so the cross term is visible rather than implicit.

Where this goes next

The same exercise done in the other variable is Greeks under spot moves, and in time it is Greeks under time decay. How maturity and volatility interact across the surface is the term structure of Greeks. Putting the whole picture together into how a book behaves in calm, stressed and crisis markets is Greeks across market regimes, and the level of volatility as something to have a view on rather than to react to is vega exposure and volatility regimes.

16.11 Greeks Under Time Decay

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The previous experiment held time still and moved spot. This one does the reverse: hold spot and volatility still, let the clock run, and watch the Greeks change. The results are more dramatic than most people expect, because nothing about an option is stable as expiry approaches, and the changes accelerate rather than proceeding smoothly.

Everything sharpens

As T falls toward zero, each profile from the spot section changes shape in a predictable way:

- **Delta** steepens. The S-curve compresses toward a step function at the strike, since there is less and less time for spot to travel anywhere else.
- **Gamma** explodes, scaling as $1/\sqrt{T}$ at the money, and the bell becomes taller and narrower.
- **Vega** decays, scaling as $\sqrt{T}$, so a nearly expired option barely cares about implied volatility at all.
- **Theta** accelerates, scaling as $1/\sqrt{T}$ at the money, so decay is slow at first and brutal at the end.

KEY IDEA

Near expiry, gamma and theta both **blow up** while vega **vanishes**. So a short-dated book is a gamma and theta business with almost no volatility exposure, and a long-dated book is the reverse. That single sentence resolves most questions in this section, and it is why desks manage their gamma at the front of the curve and their vega at the back.

WORKED EXAMPLE

An at-the-money option on a spot of 100 at 20% volatility, with zero rates. Theta per day divides the annual figure by 365 calendar days; the break-even work later in this section uses $\sqrt{252}$ trading days, which is the right convention for a volatility and the wrong one for a decay schedule.

Maturity	Gamma	Theta per year	Theta per day
1 year	0.0198	−3.97	−0.0109
6 months	0.0281	−5.63	−0.0154
3 months	0.0398	−7.97	−0.0218
1 month	0.0691	−13.81	−0.0379
1 week	0.1438	−28.77	−0.0788

Theta at one month is 3.48 times theta at one year, which is $\sqrt{12}$ up to the small $\varphi(d_1)$ correction that also drifts with maturity. Gamma has grown by the same factor over the same period, which is not a coincidence: the two are proportional, and both scale as $1/\sqrt{T}$.

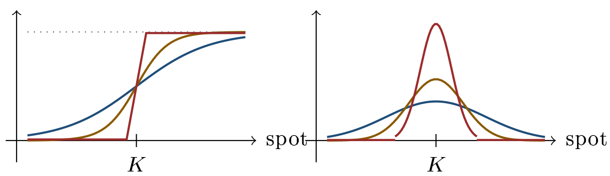
REAL INTERVIEW QUESTION (HSBC · EQD)

Explain why an option's theta increases as maturity approaches.

WHAT THEY TEST / ANSWER

Give the intuition, then the algebra, then the caveat. **Intuition**: time value is payment for the chance that spot moves somewhere useful. With a year left there is plenty of chance and the value declines slowly; with a week left, each day removes a large fraction of the remaining opportunity, so the same calendar day destroys far more value. Decay is not linear, it accelerates. **Algebra**: an at-the-money option is worth roughly $0.4\,\sigma S\sqrt{T}$, and differentiating in time gives a theta proportional to $1/\sqrt{T}$, so it goes to infinity as $T \to 0$. Concretely at 20% volatility on a spot of 100, daily theta runs at about 0.011 with a year to go and 0.079 with a week to go, seven times faster.

Caveat, and this is the part that earns the mark: this is the *at-the-money* story. An option that is well out of the money has almost no time value left to lose, so its theta shrinks toward zero as expiry approaches rather than exploding. Theta accelerating is really theta *concentrating* at the strike, which is the same statement as gamma concentrating there.



The same option as expiry approaches, plotted against spot: —— far from expiry, —— closer, —— close. Left, delta compresses from a gentle S-curve toward a step at the strike, with the dotted line at 1. Right, gamma is its slope, so it becomes taller and narrower at each step; the area under each bell is the same, because delta still travels from 0 to 1 whatever the maturity. Gamma is not created as expiry approaches, it is concentrated.

REAL INTERVIEW QUESTION (JP Morgan · Delta One)

Draw your delta one year before expiry, six months before, and so on, then do the same for gamma.

WHAT THEY TEST / ANSWER

Draw them as a family of curves on one pair of axes, and narrate the trend rather than each curve. **Delta**: at one year the S-curve is gentle, sloping smoothly from 0 to 1 over a wide range of spot. As you step to six months, three months and one month, it steepens progressively, and at expiry it is a vertical step at the strike, 0 below and 1 above. **Gamma**, as the slope of those curves, is a low broad bell at one year that becomes taller and narrower at each step, and at expiry it is a spike at the strike and zero everywhere else.

Then give the two observations that turn a drawing into an answer. The area under the gamma curve is roughly preserved as it narrows, so gamma is not being created, it is being *concentrated*, which is why a large position near a strike near expiry is dangerous while the same position far from the strike is inert. And vega does the opposite, shrinking with $\sqrt{T}$, so the family of vega curves flattens toward zero as the delta curves steepen.

A day passes and nothing happens

REAL INTERVIEW QUESTION (HSBC · EQD)

A day passes without the underlying of a straddle moving, with all other parameters unchanged and no delta hedging. Have you made or lost money, and why?

WHAT THEY TEST / ANSWER

If you are long the straddle you have **lost** money, specifically one day's theta. The reasoning: a straddle's value is almost entirely time value, since at the money there is essentially no intrinsic value, and time value is payment for future movement. A day has gone by and no movement was delivered, so you paid for something you did not receive. Nothing else in the position changed, so the loss is exactly the theta.

Two refinements to add. The "no delta hedging" clause is a hint rather than a complication: an at-the-money straddle is nearly delta-neutral anyway, so not hedging costs you nothing over a day in which spot did not move, and saying that shows you noticed. And the size of the loss depends on how close to expiry you are, since theta accelerates as $1/\sqrt{T}$: the same motionless day costs a one-week straddle about seven times what it costs a one-year straddle. If you were *short* the straddle, you made exactly that amount, which is the compensation for the risk you were running.

Recovering gamma from theta

Because gamma and theta are proportional, either one determines the other, and interviewers use this to check whether the relationship is understood rather than memorised. The cleanest route is through the break-even move of the hedging section: the daily theta is exactly offset by a spot move of $x^* = \sigma/\sqrt{252}$ percent, and the gamma P&L from a move of x percent is $\Gamma_{\text{cash}} x^2/200$, so

$$\Gamma_{\text{cash}} = \frac{200\,|\Theta|}{(x^*)^2}, \qquad x^* = \frac{\sigma}{\sqrt{252}}.$$

WORKED EXAMPLE

An option with one year to expiry, 25% volatility, and a theta of 20,000 per day. The break-even daily move is $25\%/\sqrt{252} = 1.575\%$, so

$$\Gamma_{\text{cash}} = \frac{200 \times 20{,}000}{1.575^2} \approx €1{,}613{,}000 \text{ per } 1\% \text{ move}.$$

Checking backwards: a 1.575% move produces $1{,}613{,}000 \times 1.575^2/200 = €20{,}000$ of gamma P&L, exactly the day's theta, as it must.

REAL INTERVIEW QUESTION (BNP · Volatility)

What is the gamma of a one-year option with 25% volatility and a theta of 20K?

WHAT THEY TEST / ANSWER

Do not reach for the Black-Scholes theta formula. Use the break-even relationship, which is faster and shows you understand why the two Greeks are linked. At 25% volatility the break-even daily move is $\sigma/\sqrt{252} = 1.575\%$, meaning a move of that size produces exactly enough gamma P&L to pay the day's theta. Since gamma P&L is $\Gamma_{\text{cash}}x^2/200$, setting that equal to 20,000 gives a cash gamma of about **€1.6 million per 1% move**.

State your convention as you go, since "gamma" is quoted several ways and the interviewer will accept any named one; here it is the change in cash delta for a 1% move in spot. Then close with the structural point: gamma and theta are proportional through $\Theta \approx -\frac{1}{2}\Gamma\sigma^2 S^2$, so quoting one is quoting the other, and the constant of proportionality is the volatility. That is why the answer needed the volatility and nothing else, and it is the content of theta as the cost of convexity.

Structures decay differently

A spread's theta is the difference of two thetas, which makes it far smaller than either leg's and lets it change sign.

REAL INTERVIEW QUESTION (BNP · EQD)

What is the difference between the theta of a put spread and the theta of a put?

WHAT THEY TEST / ANSWER

Two differences, magnitude and sign. **Magnitude**: a long put spread is long one put and short another, so its theta is the difference of two thetas that largely cancel. A put spread therefore decays far more slowly than an outright put, which is a large part of why clients buy spreads rather than outrights when they want a cheap directional view that they intend to hold.

Sign: an outright long put has negative theta essentially always. A long put spread does not. When spot sits above both strikes the structure is nearly worthless and decaying, so theta is negative; but when spot is below both strikes the spread is worth close to its maximum payout and trades slightly below it, so time passing pulls it *up* toward that maximum and theta is **positive**. The crossover sits between the strikes. Close with the practical reading: a deep in-the-money spread is a position that earns carry simply by waiting, which is exactly the same mechanism as the negative time value of a deep in-the-money European option.

The last week

Near expiry the divergence between at-the-money and wing options becomes extreme, and it is worth stating explicitly because it drives real behaviour:

- **At the money**: gamma and theta are enormous, vega is negligible, and the position needs constant attention.
- **Away from the money**: everything is near zero and the option is effectively dead, whichever direction it is dead in.

- **Delta** becomes bimodal, flipping between roughly 0 and roughly 1 on small moves, which is the pin risk of dynamic hedging.

The consequence is that risk near expiry is concentrated at specific spot levels rather than spread across a range, so a desk's expiry-day exposure is best read as a list of strikes and sizes rather than as an aggregate Greek.

INTERVIEW TRAP

Four traps. (1) Saying theta always accelerates into expiry; that is the at-the-money behaviour, while an out-of-the-money option's theta decays toward zero. (2) Forgetting that vega dies as gamma explodes, and quoting a short-dated book's vega as if it mattered. (3) Assuming a spread decays like its long leg, when the two thetas mostly cancel and the sign can flip. (4) Aggregating Greeks across strikes near expiry, which hides exactly the concentration that matters.

VISUAL / CODE TO BUILD: Interactive

A time-lapse of the Greeks. A single maturity slider drives four stacked charts of delta, gamma, vega and theta against spot, and dragging it from one year to one day animates the whole family at once: the delta curve steepening toward a step, the gamma bell rising and narrowing, the vega bell collapsing, and the theta trough deepening. A checkbox freezes previous curves as faint traces so the reader accumulates the family the JP Morgan question asks them to draw. A second panel plots a fixed out-of-the-money strike alongside an at-the-money one, so the divergence in the final week, one exploding and the other dying, is visible on the same axes.

Where this goes next

Why theta and gamma are proportional at all, rather than merely correlated, is theta as the cost of convexity. Reading these same effects across the maturity dimension for a whole book is the term structure of Greeks, and the volatility axis of the same experiment is Greeks under volatility shifts.

16.12 Term Structure of Greeks

sales · trading · structuring | advanced | equity · fx · cross-asset

A book is not a Greek, it is a *ladder* of Greeks across maturities. The previous sections established how each Greek scales with time; this one puts those scalings side by side, because the differences between them are what let a desk hold one exposure and hedge another, and because several interview favourites are really questions about which scaling dominates.

The scaling laws in one place

At the money, with everything else held fixed. The first two rows are established in gamma and convexity and time decay; what is new here is putting them beside the carry Greeks, which scale differently:

Greek	Scales as	Consequence
Gamma	$1/\sqrt{T}$	lives at the front
Theta	$1/\sqrt{T}$	lives at the front
Vega	$\sqrt{T}$	lives at the back
Option price	$\sqrt{T}$	from $0.4\,\sigma S\sqrt{T}$
Rate, dividend and repo sensitivity	$\approx T$	lives at the back, and grows fastest

KEY IDEA

The three families scale differently and that is the whole point. **Gamma and theta** concentrate at the short end, **vega** at the long end, and the **carry Greeks**, sensitivity to rates, dividends and repo, grow roughly *linearly* in maturity and therefore fastest of all. A five-year book is a carry and vega book that happens to contain options; a one-week book is a gamma and theta book. Any question of the form "which maturity has more of X?" is answered by looking up which column X is in.

The carry Greeks are the ones most often forgotten. They scale with T rather than $\sqrt{T}$ because they act through the forward, and the forward exponent from forwards is $(r - q - \text{repo})T$, linear in time. So doubling the maturity roughly doubles the dividend exposure while multiplying vega by only $\sqrt{2}$.

WORKED EXAMPLE

A one-year at-the-money call on a spot of 100 at 20% volatility with a 2% dividend yield. The dividend assumption is cut by 20 basis points, from 2.00% to 1.80%, which lifts the forward and therefore the call:

Maturity	Price at $q = 2.00\%$	Price at $q = 1.80\%$	Change
1 year	6.936	7.034	+0.099
2 years	9.175	9.368	+0.194
3 years	10.637	10.922	+0.285

The changes are almost exactly in the ratio 1 : 2 : 3, confirming the linear scaling. Note the contrast with the prices themselves, which are in the ratio 1 : 1.32 : 1.53, slower even than the $\sqrt{T}$ that would give 1 : 1.41 : 1.73, because the dividend is decaying the forward at the same time.

REAL INTERVIEW QUESTION (BNP · Exo)

You are long a one-year at-the-money call on a stock. If the dividend falls by 20 basis points, how does the option's price move? And if the maturity is two years, or three?

WHAT THEY TEST / ANSWER

Direction first: a lower dividend raises the forward, and a call is long the forward, so the option is worth **more**. Then the size, and the point of the question is the scaling. Dividend sensitivity is roughly $\partial C/\partial q \approx -S\,T\,N(d_1)$, so it is proportional to *maturity*, not to its square root. On a spot of 100 with a delta near 0.5, a 20 basis point cut is worth roughly $100 \times 1 \times 0.5 \times 0.002 = 0.10$ at one year, so about 0.10% of spot, and roughly 0.20 at two years and 0.29 at three.

Then close with the comparison that shows you understand the term structure: vega grows only as $\sqrt{T}$ while dividend sensitivity grows as T, so on long-dated structures the *dividend* assumption becomes a larger source of P&L uncertainty than the volatility assumption. That is why long-dated structured products are hedged with dividend futures and swaps and why dividend risk is a traded book of its own, developed alongside dividends.

Ladders, not numbers

Because the scalings differ, a single portfolio Greek can be near zero while the book carries large offsetting exposures at different maturities. Desks therefore run **ladders**: gamma and vega bucketed by maturity, exactly as a rates desk runs a DV01 ladder. The vega ladder is developed in vega exposure and volatility regimes; what the differing scalings add here is that the gamma ladder and the vega ladder are concentrated at opposite ends of the same book.

A book that is long one-month options against short two-year options can be vega-neutral overall and still be enormously long gamma and short theta, and it will make or lose money on the *shape* of the volatility term structure rather than on its level. The instrument that expresses that view deliberately is the calendar spread from standard payoff profiles.

REAL INTERVIEW QUESTION (Société Générale · Delta One)

How does an option's price evolve as a function of maturity and strike?

WHAT THEY TEST / ANSWER

Take the two dimensions separately and then say where they interact. **In strike**: a call's price falls monotonically as the strike rises, and it must be convex in strike, since a butterfly can never be worth less than nothing, which is the constraint from payoff profiles. **In maturity**: at the money the price grows roughly as $0.4\,\sigma S\sqrt{T}$, so it increases but with diminishing returns, and doubling maturity multiplies the price by about $\sqrt{2}$ rather than by two.

Then give the interaction, which is what a Delta One interviewer is listening for: those statements assume a flat surface. In reality the implied volatility depends on both strike and maturity, so the price surface is the Black-Scholes shape distorted by the smile and the volatility term structure, and the carry terms tilt it further as maturity grows. The honest version is that price rises with maturity *usually*, and the exception is worth knowing, which is the next question.

Is a call always worth more with more time?

For a European call on a non-dividend-paying stock, yes: more time is strictly more optionality. Once the underlying pays a large dividend yield, no.

WORKED EXAMPLE

An at-the-money European call, spot 100, strike 100, 20% volatility, zero rates.

Maturity	6m	1y	2y	3y	5y
No dividend, $q = 0$	5.64	7.97	11.25	13.75	17.69
$q = 10\%$	3.40	3.75	3.59	3.16	2.27

With no dividend the price rises monotonically. With a 10% dividend yield it peaks near one year and then *falls*: the forward is decaying at 10% a year, so a strike fixed at today's spot moves further and further out of the money in forward terms, and eventually that outweighs the extra time.

REAL INTERVIEW QUESTION (Société Générale · EQD)

Is a call's price always increasing in maturity? If not, in what case?

WHAT THEY TEST / ANSWER

Not always. Give the clean case first: for a European call on a non-dividend-paying stock the price is non-decreasing in maturity, because more time can only add optionality and, with zero or positive rates, the forward does not work against you.

The exception: a **high dividend yield**. The forward is $Se^{(r-q)T}$, so when $q > r$ the forward decays with maturity, and a strike fixed at today's spot becomes progressively further out of the money relative to that forward. Beyond some maturity, that decay outweighs the extra time value and the price falls. With 20% volatility, zero rates and a 10% dividend yield, the at-the-money call peaks near one year at about 3.75 and is down to 2.27 by five years.

Two refinements if they want them. This is a *European* phenomenon: an American call is never worth less with more time, since more time strictly adds exercise opportunities, which is one of the cleanest illustrations of why early exercise has value. And the same logic in reverse means a European *put* can decrease in maturity when rates are high, since discounting the strike further hurts.

Delta across maturity

Delta depends on d_1, which depends on total variance $\sigma^2 T$ and on the forward. That single observation answers every maturity question about delta, and it unifies this section with vanna: time and volatility enter delta the same way, through $\sigma\sqrt{T}$, so more of either has the same effect.

WORKED EXAMPLE

Deltas at 20% volatility with zero rates and no dividend:

Strike	3 months	1 year	3 years	5 years
$K = 60$ (deep ITM)	1.000	0.996	0.950	0.914
$K = 100$ (ATM)	0.520	0.540	0.569	0.588
$K = 160$ (deep OTM)	0.000	0.012	0.118	0.204

The rows converge as maturity grows: the deep in-the-money delta falls from 1.000 toward 0.914, the deep out-of-the-money delta rises from 0.000 toward 0.204, and the at-the-money delta drifts gently upward. More time means more uncertainty about the destination, so the profile across strikes is compressed toward the at-the-money-forward delta, $N(\frac{1}{2}\sigma\sqrt{T})$. Note that this anchor is not one half and does not stay put: it creeps upward with $\sigma\sqrt{T}$, which is why the middle row rises rather than standing still.

REAL INTERVIEW QUESTION (Société Générale · Index)

How does delta vary with maturity, and why?

WHAT THEY TEST / ANSWER

Give the unifying reason first: delta is $N(d_1)$ and d_1 depends on time only through $\sigma\sqrt{T}$ and through the forward, so *more time acts exactly like more volatility*. Raising $\sigma\sqrt{T}$ compresses the whole delta profile across strikes toward the at-the-money-forward delta $N(\frac{1}{2}\sigma\sqrt{T})$, so longer maturity does the same.

Then read off the three cases. A deep in-the-money option's delta **falls** toward the middle, from essentially 1 at three months to about 0.91 at five years, because it is less and less certain to stay in the money. A deep out-of-the-money option's delta **rises**, from essentially 0 to about 0.20 over the same span. An at-the-money option's delta drifts *up* slightly, from about 0.52 to 0.59, because the $\sigma^2 T/2$ term in d_1 grows and, when rates exceed dividends, the forward drifts above spot as well.

Close with the practical consequence: a long-dated option is a much less efficient way to get directional exposure than a short-dated one, since its delta sits closer to a half whatever the strike, and that is why Delta One exposure is taken in forwards and futures rather than in long-dated options.

REAL INTERVIEW QUESTION (Société Générale · Fixed Income)

Two at-the-money calls on the same underlying with the same volatility, one one-year and one two-year. In which case do I need to buy more of the underlying initially?

WHAT THEY TEST / ANSWER

The **two-year**, and the margin is small. To delta-hedge a short call you buy delta units of the underlying, and the at-the-money delta rises with maturity: at 20% volatility with zero rates it is about 0.540 at one year and 0.556 at two, so hedging the two-year requires roughly 3% more stock per unit of notional.

Give the reason rather than the number alone: at the money d_1 grows with $\sigma\sqrt{T}$, which is the whole of the effect in the numbers above, since they are computed at zero rates. With positive rates a second force pushes the same way, because the forward drifts further above spot over a longer horizon, so both terms raise $N(d_1)$. Then the sentence that shows you are thinking about the trade rather than the formula: the initial hedge is nearly the same, but everything afterwards is not. The one-year has substantially more gamma, so it will demand far more rehedging, while the two-year has more vega and much more dividend and repo exposure. Comparing two options by their initial delta alone misses where the actual risk difference lies.

INTERVIEW TRAP

Four traps. (1) Quoting a single portfolio gamma or vega when the book has offsetting exposures at different maturities; run the ladder. (2) Assuming every Greek scales the same way in time, when gamma and vega scale in opposite directions and carry scales faster than either. (3) Saying more time always makes an option worth more, which fails for a European call on a high-dividend name. (4) Ignoring dividend and repo sensitivity on long-dated structures, where they can exceed the vega exposure.

VISUAL / CODE TO BUILD: Interactive

A maturity-axis explorer. The x-axis is maturity rather than spot, and the reader plots any Greek along it, with gamma, vega and dividend sensitivity overlaid on normalised axes so the $1/\sqrt{T}$, $\sqrt{T}$ and T scalings are three visibly different shapes. A ladder builder lets the reader place positions in maturity buckets and reports bucketed gamma and vega alongside the parallel totals, so a book can be constructed that shows near-zero aggregate vega and large bucket exposures. A third panel plots option price against maturity with a dividend-yield slider, and dragging the yield past the rate makes the curve visibly turn over, reproducing the non-monotonicity in this section rather than asserting it.

Where this goes next

Aggregating these ladders across a real book, with many underlyings, is portfolio-level Greeks aggregation. The behaviour of each Greek as time actually passes, rather than across a cross-section of maturities, is Greeks under time decay, and the volatility term structure that sits behind the vega ladder is developed with vega exposure and volatility regimes.

16.13 Portfolio-level Greeks aggregation

trading · structuring · sales | advanced | equity · fx · cross-asset

A trader does not risk-manage one option, they risk-manage a few thousand. The Greeks of that book are what appears on the risk screen every morning, and the whole discipline of aggregation is knowing which of those numbers you may add together and which are lies when you do. Differentiation is linear, so in one sense every Greek adds. The trouble is that adding two Greeks only means something if a single scenario moves both of them the same way, and for vega across maturities, or gamma across underlyings, no such scenario exists.

The individual Greeks are defined in Greek definitions and intuition. This section is about what happens when you put them in a book.

> **KEY IDEA**
>
> A Greek is a derivative with respect to a *named* risk factor. Two Greeks may be added if and only if they are derivatives with respect to the *same* factor, in the same units. That one sentence generates every rule below: delta adds within a name and needs a beta to cross names; vega adds within a maturity bucket and needs a weighting to cross buckets; gamma adds within a name and needs a matrix to cross names.

Delta: cash first, then beta

Adding deltas expressed in shares is meaningless across two underlyings, since one share of a 900 euro name is not one share of a 9 euro name. The first step is always to convert to **cash delta**, $\Delta_i S_i$ per position, which is the currency value of the underlying you effectively hold. Cash deltas add without qualification within a currency, and across currencies once converted at spot, which is itself an FX delta you have just created.

To collapse a multi-name book into one directional number, weight each cash delta by the name's beta to the chosen index:

$$\Delta^{\text{index-equivalent}} = \sum_i \beta_i \, \Delta_i S_i.$$

This is the number a risk manager uses for an index-level limit, and it comes with a health warning that belongs in every interview answer: beta is estimated from history, it is unstable, and in a genuine sell-off single-stock betas converge toward one. A book hedged to zero index-equivalent delta using calm-market betas is therefore not hedged in the scenario the hedge exists for.

Vega: the number that lies when you add it

Vega is where naive aggregation does the most damage, because summing the vegas of a 1-month option and a 5-year option implicitly assumes both implied volatilities move by the same amount. They do not. Short-dated implied volatility is far more volatile than long-dated implied volatility, roughly in proportion to $1/\sqrt{T}$. That is an empirical regularity rather than a theorem, and everything below inherits its accuracy, so it is worth being aware that you are spending an assumption before you spend it.

Desks handle this in two steps.

Bucket first. The vega ladder itself, what it is and why a book needs one, belongs to vega exposure and volatility regimes; taking it as given, the aggregation rule is that vega is reported by maturity bucket (1M, 3M, 6M, 1Y, 2Y, 5Y and beyond), and by strike or moneyness bucket if the book has real skew risk. A ladder is a vector, not a scalar, and it is the honest representation.

Then weight, if you must have one number. Under the square-root-of-time rule, a shock that moves the reference maturity's volatility by h moves maturity T_i by $h\sqrt{T_{\text{ref}}/T_i}$, so the profit and loss is

$$\delta\Pi \;=\; h\sum_i \mathcal{V}_i\sqrt{\frac{T_{\text{ref}}}{T_i}},$$

and the sum is the **weighted vega** in reference-maturity units. With a one-year reference the weights are 3.46 for 1M, 2.00 for 3M, 1.41 for 6M, 1.00 for 1Y, 0.71 for 2Y and 0.45 for 5Y. Treat this as a convention with a rationale rather than a theorem; it approximates how volatility term structures actually move, and desks differ on the exponent.

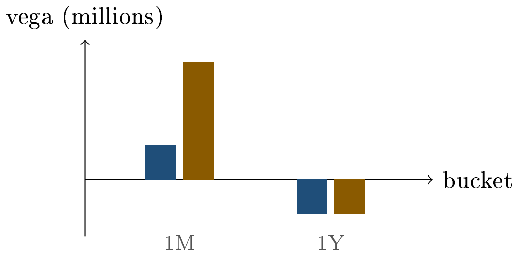
WORKED EXAMPLE

A book is long 1,000,000 of vega in the 1-month bucket and short 1,000,000 in the 1-year bucket. The risk screen shows **net vega zero**, and the book is described as vega neutral.

Apply a realistic shock instead. If one-year implied volatility rises by one point, the square-root-of-time rule puts the one-month move at $\sqrt{12} = 3.46$ points, so

$$\delta\Pi \;=\; 1{,}000{,}000 \times 3.464 \;-\; 1{,}000{,}000 \times 1 \;=\; +2{,}464{,}000.$$

The "vega neutral" book makes two and a half million on a one-point move in the reference volatility. Weighted vega gives the same answer directly: $1{,}000{,}000 \times 3.464 - 1{,}000{,}000 \times 1.00 = 2{,}464{,}000$ of one-year equivalent vega. The raw sum was not merely imprecise, it had the wrong order of magnitude and hid the entire position.



raw vega, which sums to zero — weighted to one-year equivalent, which sums to +2.46

Vega and gamma can point in opposite directions

Once maturities are separated, a fact that confuses candidates becomes obvious: vega lives at the long end and gamma lives at the short end. Vega scales with $\sqrt{T}$, gamma with $1/\sqrt{T}$. So a book can be long one and short the other, and the calendar spread is the clean example.

WORKED EXAMPLE

Spot 100, both legs struck at the money, both marked at 20% volatility, zero rates. Long one one-year call, short one one-month call.

	Vega (per point)	Gamma	Theta (per year)
Long 1Y call	$+0.397$	$+0.0198$	-3.97
Short 1M call	-0.115	-0.0691	$+13.81$
Net	$+0.282$	-0.0492	$+9.84$

The position is **long vega and short gamma**, and long theta as a consequence of the short gamma. It profits if implied volatility rises across the curve and loses if the underlying starts moving in the next month. That is a coherent view, that the market is priced too cheaply for the long run while being quiet in the short run, and it is invisible to anyone reading a single net-vega number.

REAL INTERVIEW QUESTION (Bank of America · EQD)

Hedging a call or a put, delta hedging. A vega-positive, gamma-negative position?

WHAT THEY TEST / ANSWER

A calendar spread: long the far-dated option, short the near-dated one, same strike, both delta-hedged. Justify it with the maturity scaling rather than by memory, since that is the point being tested: vega goes as $\sqrt{T}$ and gamma as $1/\sqrt{T}$, so the long leg dominates the vega and the short leg dominates the gamma. Numbers make it concrete, with spot 100 at 20% volatility: the one-year call has vega 0.40 per point and gamma 0.020, the one-month call has vega 0.12 and gamma 0.069, so long one against short one nets to $+0.28$ of vega and -0.049 of gamma. Then state the risk honestly: short gamma means you are long theta but exposed to a near-term gap, so the position bleeds on one big move even if you are eventually right about the level of long-dated volatility. The mirror trade, long gamma and short vega, is the same structure reversed.

Gamma across underlyings: a matrix, not a number

For a book on a single underlying, cash gamma adds. For a book on several, the second-order term is a quadratic form rather than a sum:

$$\delta\Pi^{(2)} \;=\; \tfrac{1}{2}\sum_{i,j}\Gamma_{ij}\,\delta S_i\,\delta S_j, \qquad \Gamma_{ij} = \frac{\partial^2 V}{\partial S_i \partial S_j}.$$

The diagonal entries are the familiar per-name gammas. The off-diagonal entries are **cross-gammas**, and they exist for any multi-asset payoff: a basket option, a worst-of, a spread option. They are invisible in a per-name Greek report, which is why a book of multi-asset structures cannot be risk-managed one name at a time.

The practical question is not the sign of any single entry but whether the whole matrix is positive semi-definite. If it is, then $\delta S^{\mathsf{T}}\Gamma\,\delta S \geq 0$ for *every* move, so the second-order term never costs you money whatever the underlyings do. A single negative off-diagonal entry does not break that: it only says the position prefers the two names to move in opposite directions. A negative *diagonal* entry, by contrast, guarantees there is a direction that loses.

REAL INTERVIEW QUESTION (Société Générale · EQD)

What is the correlation sensitivity of a call on a basket? Can you make money in gamma if one element of the gamma matrix is negative?

WHAT THEY TEST / ANSWER

Two questions, and the second is the one they care about. **Correlation sensitivity**: positive. The basket's variance is $\sum_{i,j} w_i w_j \sigma_i \sigma_j \rho_{ij}$, which increases in every ρ_{ij}, so higher correlation means a more volatile basket and a more valuable call. Long a basket call is long correlation, which is the other side of dispersion. **The gamma matrix**: yes, you can still make money, and the reason is that the sign of one entry is the wrong thing to look at. The second-order profit and loss is the quadratic form $\frac{1}{2}\,\delta S^{\mathsf{T}} \Gamma\, \delta S$, so what matters is whether Γ is positive semi-definite, not whether some Γ_{ij} is negative. A negative off-diagonal term simply means the position likes the two names to diverge, and a matrix with negative off-diagonals can be positive definite, in which case every possible move makes money. The sign that would genuinely hurt you is a negative entry on the *diagonal*, because that is a direction in which you are short gamma outright. Finish with the practical point: this is exactly why a multi-asset book is stress-tested on joint scenarios rather than by shocking one name at a time, since one-at-a-time shocks only ever touch the diagonal.

Measuring what you cannot add: revalue the book

For higher-order risks there is often no additive Greek worth trusting, and the honest answer is to reprice. Convexity in volatility is the standard case: a book's volga is spread across strikes and maturities, and summing per-option volgas assumes a parallel shift that will not happen. Shocking the book and revaluing does not.

REAL INTERVIEW QUESTION (BNP · Volatility)

How, in two operations, do you determine the volatility convexity of the book?

WHAT THEY TEST / ANSWER

Shift the whole surface up by h and revalue, then shift it down by h and revalue. Those are the two operations, because the third value you need, the book at today's marks, you already have. Convexity in volatility is then the second difference,

$$\text{volga} \;\approx\; \frac{\Pi(\sigma+h)+\Pi(\sigma-h)-2\,\Pi(\sigma)}{h^2},$$

and the same two revaluations give you the vega asymmetry directly, since a book with zero convexity gains and loses the same amount on the two shocks while a convex book gains more than it loses. Say why you do not simply add per-option volgas: they are derivatives at a point, they assume a parallel shift of the surface, and on a book with wings they miss the term that matters. Two further points score well. Choose h large enough to see the convexity, a few volatility points rather than a basis point, since a small bump reproduces the analytic vega and tells you nothing new. And if "operations" means trades rather than revaluations, the answer is the other one: a strangle against a straddle isolates volatility convexity, being long the wings where volga lives and short the at-the-money where it vanishes, as set out in skew and smile sensitivities.

What a risk screen actually shows

Putting it together, a well-built book report is not a row of six numbers. It is:

- cash delta per underlying, plus one index-equivalent aggregate with the beta assumption named;
- cash gamma per underlying, plus the cross-gamma terms for any multi-asset position;
- a vega *ladder* by maturity bucket, with weighted vega as a summary and never as a substitute;
- theta per day, which is the one Greek that genuinely just adds, since time passes at the same rate for every position;
- and a set of full-revaluation scenarios for anything the linear numbers cannot express.

Theta being the only trivially additive Greek is worth noticing, and it is a useful thing to say out loud in an interview: every other line on the screen carries an implicit assumption about which scenario is being applied.

INTERVIEW TRAP

Four aggregation errors that cost real money. (1) Summing vega across maturities and calling the book vega neutral, when a realistic shock moves the front more than the back by a factor of three. (2) Adding deltas in shares rather than in cash, or in cash across currencies without noticing the FX delta that creates. (3) Reading a multi-asset book one name at a time, which shocks only the diagonal of the gamma matrix and never the cross terms. (4) Believing a beta hedge sized on calm-market betas will still be a hedge in a crash, when betas converge toward one exactly then.

VISUAL / CODE TO BUILD: Interactive

A book-level risk screen the reader assembles. They add positions (underlying, strike, maturity, quantity) and the widget maintains a live report: cash delta per name and index-equivalent with an editable beta, a vega ladder drawn as a bar chart by maturity bucket with the weighted-vega summary beside it, and for multi-asset positions the full gamma matrix rendered as a heatmap with its eigenvalues shown, so positive semi-definiteness is something the reader can see rather than assert. A scenario panel applies shocks that naive aggregation gets wrong: a front-end volatility spike with a static back end, a correlated multi-name sell-off, and a single-name gap, reporting the true revalued profit and loss next to the prediction from the summed Greeks. The intended realisation: the gap between those two numbers is the aggregation error, and it is largest exactly in the scenarios a book is supposed to survive.

Where this goes next

How the Greeks vary along the maturity axis, which is what makes vega bucketing necessary in the first place, is the term structure of Greeks. Turning the aggregated numbers into an explanation of yesterday's result is Greeks and PnL explain, and running the book against them day to day is book management. On the risk side, the same aggregates feed aggregation of Greeks and the limit structures in risk limits frameworks. The correlation exposure that appears the moment a book holds multi-asset payoffs is developed in equity correlation and dispersion strategies.

16.14 Greeks and PnL Explain

sales · trading · structuring | core | equity · fx · cross-asset

At the end of the day a number appears, and someone has to say where it came from. The Greeks are the answer: each one is a term in an expansion, and the expansion, evaluated against what the market actually did, reconstructs the day's profit and loss. This section is that decomposition. The desk process built on top of it, daily attribution, sign-off and the treatment of unexplained, is PnL explain in the trading chapter.

The identity

Expand the book's value in everything that moved:

$$\text{P\&L} \;\approx\; \underbrace{\Delta\,\delta S}_{\text{delta}} + \underbrace{\tfrac{1}{2}\Gamma\,(\delta S)^2}_{\text{gamma}} + \underbrace{\mathcal{V}\,\delta\sigma}_{\text{vega}} + \underbrace{\Theta\,\delta t}_{\text{theta}} + \underbrace{\rho\,\delta r}_{\text{rho}} + \underbrace{\text{vanna}\,\delta S\,\delta\sigma + \tfrac{1}{2}\text{volga}\,(\delta\sigma)^2}_{\text{cross terms}} + \varepsilon.$$

KEY IDEA

Every P&L line item is a Greek multiplied by a market move. That is the whole of attribution: the Greeks say how sensitive you are, the market says what happened, and the product is your money. The residual ε, the **unexplained**, is the part the expansion did not capture, and its size is a diagnostic on your risk model rather than a rounding error to be ignored.

For a **delta-hedged** book the first term vanishes by construction, which is exactly why desks hedge delta: it removes the largest and least informative line so that what remains is the business they are actually in.

REAL INTERVIEW QUESTION (Société Générale · EQD)

Give the P&L formula.

WHAT THEY TEST / ANSWER

Write the second-order expansion and label every term:

$$\text{P\&L} = \Delta\,\delta S + \tfrac{1}{2}\Gamma(\delta S)^2 + \mathcal{V}\,\delta\sigma + \Theta\,\delta t + \dots$$

Then improve it in the three ways an interviewer is waiting for. First, specialise: for a delta-hedged book the first term is zero, so P&L $\approx \tfrac{1}{2}\Gamma(\delta S)^2 + \Theta\,\delta t$ over a day with volatility unchanged, which is the working formula on a trading desk. Second, note the units trap: $\delta\sigma$ must be in the same units as the vega quote, per point or per unit, and δt in the same units as theta, usually one day. Third, mention the cross terms, vanna and volga, which are negligible on a quiet day and are exactly what dominates in a crisis when spot and volatility both move hard. Ending on the cross terms signals that you have seen a real attribution report rather than a textbook formula.

The delta-hedged form

Set every Greek but gamma and theta to zero and the identity collapses to the one relation a delta-hedged book lives by. Over a move x the gamma term grows as x^2 while theta accrues regardless, so the two are equal at a single *break-even move*

$$x^* = \frac{\sigma}{\sqrt{252}}, \qquad \text{and the day's P\&L is} \qquad -\,\Theta\left[\left(\frac{x}{x^*}\right)^2 - 1\right].$$

Mind the sign: Θ here is the signed Greek of the identity above, so it is *negative* when you are long gamma, and the leading minus turns the bracket into a gain once the move exceeds the break-even. Short gamma flips both signs and the same expression reads as a loss on a big day. Traders usually quote theta as a positive number and carry the sign in words ("I pay 25,000 a day"), which is where the confusion starts.

Only the ratio of the realised move to the break-even matters: the notional, the gamma and the currency all cancel. A book that moves twice its break-even earns four days of theta and pays one, whatever its size. That is the whole of the gamma-theta trade-off, and it is why a trader quotes a break-even rather than a theta.

The worked case in currency, the desk conventions that make "5 million of gamma" ambiguous, and the daily attribution process built on top of all this belong to PnL explain, which owns them.

Two things the decomposition hides

Vega P&L is not realised. A gamma gain is banked in cash through rehedging: you sold stock high and bought it back low, and the money is in the account. A vega gain is a revaluation, and it can reverse tomorrow if implied volatility comes back. Both appear in the same P&L line, and treating them as equally solid is a classic error, especially on illiquid surfaces where the mark is a model rather than a trade.

The cross terms are not small when it matters. On a quiet day vanna and volga contribute little. In a crisis, when spot falls several percent and implied volatility jumps several points at once, the vanna $\cdot\, \delta S \cdot \delta\sigma$ term can rival the first-order terms, which is why a book that looked balanced in the first-order Greeks can lose badly, as developed in cross-Greeks.

REAL INTERVIEW QUESTION (BNP · Trading)

If you only take into account the volatility and the movement of the underlying, do you make P&L or not?

WHAT THEY TEST / ANSWER

Not reliably, and the question is testing whether you know what is missing. Spot movement plus a volatility level is not enough to determine your P&L for three reasons.

First, **theta**. Movement earns you gamma P&L, but you paid for that gamma, and whether the day was profitable depends on the move relative to the break-even, not on the move alone. At 16% volatility the break-even is 1.008%, so a 1% day is essentially flat, not a gain.

Second, **the path, not just the endpoint**. Gamma P&L accrues through rehedging, so the same closing price reached by a straight line or by a wild round trip produces very different results. Realised volatility, not the net move, is what you are paid on.

Third, **implied volatility**. If the surface moved, you have a vega P&L that is entirely independent of whether the underlying moved at all, plus carry from rates, dividends and repo, which is real money on a long-dated book. So the honest answer is that spot and volatility tell you the two largest terms, and a real attribution needs theta, vega, carry and the cross terms before the number closes.

When the P&L should be zero

Some events change every number on the screen and should change no money at all, and checking that is a genuine part of the job.

REAL INTERVIEW QUESTION (BNP · EQD)

What is a stock split, and what is the effect of a two-for-one split on your P&L?

WHAT THEY TEST / ANSWER

A split multiplies the share count and divides the price by the same factor, so a two-for-one split turns one share at 100 into two shares at 50. Nothing of economic substance changes: market capitalisation, ownership and the value of every position are identical before and after.

So the effect on your P&L should be **exactly zero**, and saying that firmly is the answer. That holds only if every derivative contract is adjusted to match, which is the mechanical half of the answer and belongs to corporate actions.

The half that belongs here is what happens when it is not. If the adjustment is missed, or applied to the position and not to its hedge, or applied on different dates in two systems, the book shows a spurious 50% move overnight. That phantom P&L is one of the most common sources of a large unexplained on a corporate-action date, and it has the signature worth recognising: it appears entirely in the residual rather than in any Greek term, because no risk factor moved. An unexplained that no sensitivity accounts for should send you to the static data before it sends you to the model.

INTERVIEW TRAP

Four traps. (1) Treating vega P&L as realised cash when it is a revaluation. (2) Using the net spot move rather than the realised path, when gamma P&L depends on the path. (3) Ignoring the cross terms, which are negligible exactly when nothing is happening and material exactly when something is. (4) Dismissing unexplained P&L as noise; a persistent unexplained means a Greek is missing from the model or a market is being marked wrongly.

VISUAL / CODE TO BUILD: Interactive

An attribution waterfall. The reader sets a book's Greeks and a day's market moves (spot, implied volatility, time, rates) and sees a waterfall chart build the P&L term by term, each bar labelled with the Greek and the move that produced it, ending with the residual. A scenario selector loads a quiet day, a trending day and a crisis day, so the reader watches the cross terms go from invisible to dominant. A second mode runs a full price path and compares the true rehedged P&L against the attribution computed from opening Greeks alone, so the unexplained appears naturally as the gap and grows with the size of the move, which is the point the section makes about when the expansion is trustworthy.

Where this goes next

The desk process built on this decomposition, running it daily, agreeing it with product control and investigating the unexplained, is PnL explain. The gamma and theta pair is developed in gamma and convexity and its relationship to realised volatility in implied versus realised volatility. The cross terms that dominate in stress are cross-Greeks.

16.15 Greeks across market regimes

trading · structuring · sales | advanced | equity · fx · cross-asset

The rest of this chapter moved one variable at a time: spot, then volatility, then time. Markets do not have that courtesy. On a real bad day spot falls, implied volatility rises, the skew steepens, correlation goes to one, the volatility term structure inverts and the bid-ask doubles, all in the same session and all in the same direction for anyone on the wrong side. This closing section is about the joint behaviour, because a book is only ever tested by joint moves, and because the difference between a candidate who has learned the Greeks and one who has thought about them is exactly here.

> **KEY IDEA**
>
> One-factor Greeks are computed by holding everything else fixed, and the whole point of a regime is that nothing else is fixed. The Greeks are not wrong, but the profit and loss they predict for a crisis is the sum of several terms that all arrive together and that a one-at-a-time sensitivity report never shows you added up.

Four regimes and what each does to a book

	Calm bull	Choppy range	Sell-off	Crisis
Spot	drifting up	sideways	falling	gapping down
ATM volatility	low, drifting down	moderate	rising	very high
Skew	flat-ish	stable	steepening	very steep
Term structure	upward sloping	upward sloping	flattening	inverted
Correlation	moderate	moderate	rising	near one
Realised vs implied	realised below	roughly equal	realised above	far above
Liquidity	good	good	thinning	gapping

The row that decides most outcomes is the last but one. A long gamma book is paid when realised volatility exceeds implied, as established in theta as the cost of convexity, so it bleeds through calm and range regimes and is paid in sell-offs and crises. The mirror is the more dangerous position, because a short gamma book earns a small amount on most days and gives it all back on a few. That distribution, many small wins and rare large losses, is what makes a short volatility position feel like an income strategy right up to the moment it is revealed not to be.

Spot and volatility arrive together

In equities the two are reliably negatively correlated, for the structural reasons set out in the volatility smile and skew: leverage rises as equity falls, hedging demand is one-sided, and the market's memory of gaps is a memory of downward gaps. A working rule of thumb on major equity indices is that a one percent fall lifts at-the-money implied volatility by roughly one point, and by considerably more once a move becomes disorderly. Treat that as a calibration to sanity-check against, not a law.

The consequence for a delta-hedged option book is that the vega leg usually dwarfs the gamma leg on the day of a crash, which is the opposite of what most candidates expect.

WORKED EXAMPLE

A delta-hedged book with 2,000,000 of cash gamma (per 1% move) and 500,000 of vega (per volatility point), marked at 20% volatility. In one session the market falls 5% and implied volatility rises by 5 points.

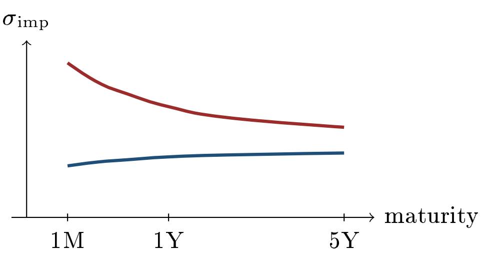
- Gamma leg: $\Gamma_{1\%} x^2/200 = 2{,}000{,}000 \times 25/200 = 250{,}000$, in the cash-gamma convention set up in the joint spot and volatility decomposition, which is where the leg arithmetic itself is derived.
- Vega leg: $500{,}000 \times 5 = 2{,}500{,}000$.
- Theta leg: one day at 20% volatility costs $2{,}000{,}000 \times (20/\sqrt{252})^2/200 \approx 15{,}900$.

Net, roughly **2,730,000**, of which the gamma that everyone talks about contributes under ten percent. The lesson is not that gamma does not matter, it is that on the day of the event a long option book is mostly a vega position, and the gamma only pays if the market keeps moving afterwards. Reverse every sign for the short book, and note that the short book also has to buy back its hedge into a falling market, so its realised loss is worse than these numbers, for the reasons in the liquidity paragraph below.

Two refinements belong in any serious answer. First, the cross term: this scenario is spot down with volatility up, so a book with the wrong sign of vanna loses again, and because equity spot and volatility move together in a consistent direction, that loss is systematic rather than random. Vanna is treated in skew and smile sensitivities. Second, in a sell-off the skew steepens as well as the level rising, so the vega leg above is only right if the book's vega sits at the money; a book whose vega is in the downside wings makes materially more, and one whose vega is in the upside calls makes less.

The term structure inverts

In calm markets implied volatility rises with maturity, because the far date carries uncertainty about which regime will be in force. In a crisis it inverts: the front month prices an event happening now, and the market expects the storm to pass, so long-dated volatility rises much less.



calm regime, upward sloping — crisis regime, inverted

the maturity axis is drawn in $\sqrt{T}$, the usual scale for a volatility term structure, so the tick spacing is not linear in years

Three consequences follow directly. A book long front-end vega and short back-end vega, which looked vega neutral on the naive sum criticised in portfolio-level Greeks aggregation, makes a great deal of money. The calendar spread economics reverse, so a position that was long the back and short the front, and therefore long vega and short gamma, is now on the wrong side of both legs. And the gamma you own gets cheaper to hold in relative terms at the back and much more expensive at the front, since theta scales with volatility.

Correlation goes to one

Diversification is a calm-market property. In a genuine sell-off single-name correlations rise together, so index volatility rises by more than the average constituent's volatility and index skew steepens by more than single-name skew. Any book that is long dispersion, in other words short index volatility against long single-name volatility, is short exactly that move. This is also why a beta hedge shrinks in a crash, as noted in aggregation: betas converge toward one at the same time.

REAL INTERVIEW QUESTION (Société Générale · EQD)

How does the price of a worst-of option change when you increase the correlation between the underlyings? When you increase the volatility?

WHAT THEY TEST / ANSWER

Correlation: a long worst-of *call* is *long* correlation, which is the opposite of what the diversification instinct suggests and is the point of the question. (Sign the payoff, do not memorise: raising correlation lifts the worst performer, so it helps a worst-of call and hurts a worst-of put, which is why the dealer side of an autocallable, long the worst-of put, is short correlation.) With low correlation the names wander independently, so it is very likely that at least one of them is down, and the worst-of pays on the worst one; raising correlation makes the names move together, which removes the chance that one lags badly, so the worst performer is better and the option is worth more. Say the sanity check: at correlation one, with equal volatilities, the worst-of collapses to a single-asset option, which is the upper bound on its value. **Volatility**: two effects pull against each other, and saying so is the answer. Raising one asset's volatility adds optionality, which helps, but it also makes that asset more likely to be the laggard the payoff is measured on, which hurts. The net is that a long worst-of call is long that asset's volatility by far less than a vanilla on it would be, and once one asset's volatility is high enough relative to the others the second effect wins and the sensitivity turns negative, because that asset becomes the near-certain worst performer. Then connect it to the regime, which is what an EQD interviewer will push toward: in a crash correlation rises toward one and every volatility rises together, so the two effects fight, and the net for the dealer depends on which leg dominates. Because the street is structurally *long* the worst-of puts it takes from investors in autocallable issuance, and therefore short correlation, that fight is a systematic street-wide exposure and not an academic question. See dispersion strategies.

The Greeks that are not on the report

A regime change reprices things the standard Greek report does not have a column for.

Dividends and repo. Both enter through the forward, so a change in either moves every option on the book. In an ordinary week they barely move. In March 2020 dividend expectations collapsed within days as companies suspended payouts, and equity derivatives desks discovered that their structured-product books carried a very large dividend exposure that had never been the risk anyone watched. The same applies to borrow costs; see dividends and repo markets.

Liquidity and gap risk. Every gamma calculation in this book assumes you can re-hedge continuously at the mid. In a crisis the bid-ask widens, depth disappears, and a gap means you re-hedge *after* the move rather than through it. For a long gamma book that turns the theoretical profit into a smaller realised one. For a short gamma book it is worse than a haircut: you are forced to sell into a falling market to re-hedge, which is the dynamic that turned portfolio insurance into an amplifier in 1987 and that reappears whenever a large enough cohort is hedging the same way at once.

Crowding. Because dealer books are built from what clients buy, the street tends to hold the same sign of risk at the same time. Retail issuance leaves dealers *long* the downside puts investors sell them, but with that gamma concentrated at the same barriers and triggers across every book, so in a sell-off the whole street needs the same hedge at the same levels and the hedging flow itself moves the market. The crowding is in where the risk sits, not in its sign. The February 2018 episode, in which short-volatility exchange-traded products were destroyed in a single session as their own rebalancing bought volatility into a spike, is the cleanest recent illustration.

The Covid dividend episode is the cleanest illustration. Structured-product books carried a large one-directional exposure to future dividends, and it repriced violently when payouts were suspended in March 2020. The question is asked as a dividends question and belongs there; what is regime-specific, and what belongs here, is why it hurt so much. The risk was not on the main Greek report: everyone watched delta, gamma and vega, and dividend sensitivity was a second-tier line that had never moved much. And it arrived with everything else at once, spot down, volatility up, correlation up, liquidity gone, so the loss landed on a book already being hit on three other axes. **The risk that hurts a structured book is almost never the one being hedged; it is the parameter that stayed stable long enough to stop being watched.**

REAL INTERVIEW QUESTION (BNP · Structured Products)

Am I long or short volatility when I am long an autocall?

WHAT THEY TEST / ANSWER

Short volatility. Decompose the product rather than asserting it: an autocallable pays the investor a coupon and returns capital unless the underlying breaches a downside barrier, so the investor is effectively short a down-and-in put and long a stream of contingent coupons. Being short that put is what pays for the coupon, and it makes the investor short volatility, short skew and long the calm regime. The dealer holds the mirror. Then give the regime consequence, which is what a structured-products interviewer wants: the investor's position performs in exactly the environment described as a calm bull or a range, and it fails in the sell-off and crisis columns, when volatility rises, the skew steepens and the barrier comes into range at the same time. Add the nuance if you want the follow-up to go well: the sensitivity is not constant, because as spot approaches the barrier the product's vega and its gamma both grow sharply, so the investor is most short volatility precisely when volatility is highest. See autocallables.

REAL INTERVIEW QUESTION (Barclays · Cross Asset)

We know there is going to be a jump on a stock: what strategy do you adopt?

WHAT THEY TEST / ANSWER

Buy convexity and be honest about what is already priced. The instinctive answer, buy a straddle, is right in direction and incomplete: if the jump is known to the market, for example an earnings date or a binary regulatory decision, then the front-dated implied volatility is already elevated and you are paying for the event. So the trade is not "buy volatility", it is "buy more convexity than the market is pricing for the size of jump you expect", which usually means the wings rather than the at-the-money, since a strangle is cheaper per unit of tail and carries the volga that pays if the jump is larger than expected. State the second-order points to separate yourself. A pure jump is a gamma event, not a vega event, because it is realised in one move and does not need the volatility level to rise, and after the jump the front implied volatility usually *collapses*, so a long straddle can be right about the move and still lose on the volatility crush. If you have a directional view, express it with the skew: buy the side the market is underpricing rather than paying for both. And if the timing is known, buy the maturity that contains the event and sell the one that does not, so you are not paying for calendar days on either side of it.

INTERVIEW TRAP

Four regime errors. (1) Adding one-factor Greeks and calling it a stress scenario: a real crisis moves spot, volatility, skew, correlation and liquidity together, and the cross terms are where the surprise lives. (2) Expecting gamma to be the main contributor on a crash day, when the vega leg is usually several times larger. (3) Treating short volatility as carry: its distribution is many small gains and rare large losses, so its average day tells you nothing about its risk. (4) Watching only the Greeks that usually move: the parameter that has been stable for years, dividends, repo, borrow, is exactly the one with no hedge on it when the regime changes.

VISUAL / CODE TO BUILD: Interactive

A regime switcher. The reader builds or loads a book, then selects one of four preset regimes (calm, range, sell-off, crisis), each of which applies a *joint* shock: a spot move, a correlated volatility move sized by the spot-volatility rule of thumb, a skew steepening, a term-structure rotation, a correlation shift and a widening of the bid-ask used for re-hedging. The output is a waterfall chart decomposing the profit and loss into delta, gamma, vega, vanna, volga, correlation, dividend and transaction-cost contributions, shown next to the naive prediction from summing one-factor Greeks. A slider lets the reader dial the severity continuously so the non-linearity is visible. The intended realisation: the gap between the joint-shock result and the summed-Greek result grows faster than the shock itself, which is the quantitative content of "the Greeks are computed one at a time and markets are not".

Where this goes next

The formal version of joint shocks is stress testing, and the day-to-day mechanics of hedging when liquidity is thin are in intraday hedging and inventory management. Attributing a bad day's result across all these terms after the fact is Greeks and PnL explain. The products that build the street's structural position, and therefore most of the crowding described here, are in autocallables, and the correlation exposure that comes with them is in equity correlation.

Chapter 17

Options Strategies & Payoff Engineering

17.1 From Market View to Payoff Design

sales · trading · structuring | core | equity · fx · cross-asset

Standard payoff profiles gave you the alphabet. This chapter is about writing with it, and this section is the method: given what someone believes about a market, and what they are willing to pay or give up, produce the structure that expresses it. That is the entire job of a structurer, and it is the question this database asks in a dozen different costumes.

A view has more than one dimension

The mistake beginners make is treating a view as a direction. Almost nobody has a view that is only a direction, and the extra dimensions are what determine the structure.

KEY IDEA

Before designing anything, extract **six** things. **Direction**: up, down, or no opinion. **Magnitude**: how far, since that sets the strikes. **Conviction**: how sure, since that sets how much premium is worth spending. **Horizon**: by when, which sets the maturity. **Volatility**: a separate view from direction, and often the one the client has without realising it. **Constraints**: budget, what they are willing to give up, capital protection, accounting and mandate limits. A structure that gets the direction right and the horizon wrong is a losing trade.

The four levers

Once the view is written down, there are only four things you can adjust:

- **Strikes**: where the payoff turns on, which encodes the magnitude of the view and the level at which the client stops caring.
- **Maturity**: the horizon, and with it the balance between gamma and vega from the term structure of Greeks.
- **Ratios**: quantities per leg, which create leverage, at the cost of reintroducing unbounded risk somewhere, developed in strategy construction with ratios.
- **What you sell**: the financing leg, which is what makes the structure affordable.

KEY IDEA

The financing principle: **buy the part of the distribution you believe in, and sell the part you do not.** A call spread sells the far upside to cheapen the near upside. A collar sells the upside to buy downside protection. A reverse convertible sells the downside to buy yield. Every structure in this chapter is an application of that one sentence, and the design question is always "which part of the outcome space is this client genuinely indifferent to?"

REAL INTERVIEW QUESTION (Exane · Structured Products)

How do you choose which type of structured product to propose, given a market view?

WHAT THEY TEST / ANSWER

Do not jump to a product. Show the process, because the process is what is being graded. First, **decompose the view** into direction, magnitude, conviction, horizon, volatility and constraints, and say out loud that a client who says "I am bullish" has told you one of the six things you need. Second, **map each dimension to a lever**: magnitude sets the strikes, horizon sets the maturity, the volatility view decides whether you are a net buyer or seller of optionality, and the budget decides what you have to sell to finance it. Third, **propose two or three structures** that span the trade-off rather than one, so the client can choose where to give something up.

Then give a concrete map so it is not abstract. Moderately bullish with a budget: a call spread, giving up the far upside. Bullish but needing capital protection: a zero-coupon plus a call, giving up participation. Neutral and wanting yield: a reverse convertible or an autocall, giving up the downside. Expecting movement without direction: a straddle or strangle, paying theta. Finish with the sentence that separates a structurer from a salesperson: whatever you propose, state explicitly what the client is *selling*, because every structure that looks free is financed by something, and the client needs to know what.

REAL INTERVIEW QUESTION (Morgan Stanley · EQD)

Give me an option strategy with a positive P&L whether the underlying rises or falls.

WHAT THEY TEST / ANSWER

A **long straddle**, or a strangle for a cheaper version: buy a call and a put, and you profit from a large move in either direction. Draw the V.

Then immediately supply the qualification, because the question as asked is a trap and the interviewer is waiting to see whether you notice. The payoff is positive on both sides only *beyond the break-evens*, which sit one total premium either side of the strike. Between them you lose, and at the strike you lose the entire premium. So there is no strategy with a positive P&L for every outcome, and there cannot be, because that would be an arbitrage and no-arbitrage forbids it.

Close with the honest restatement of what the client actually wants: a position that profits from *movement* rather than direction, which means being long volatility, which means paying theta and needing realised volatility to beat the implied you paid, as in implied versus realised. If they want positive P&L in all states with no premium at risk, what they want is a capital-guaranteed structure, and that is a different design with a different cost, namely participation.

Designing to a constraint: capital protection

The most common real-world brief is not a view at all, it is a constraint: get my money back, and give me some upside. The construction is mechanical once you see it as a budget: a zero-coupon bond that repays the principal at maturity, and whatever is left over spent on a call. The two-year note at 3% rates leaves about 5.7% of notional for optionality, which buys roughly half a call and therefore roughly 50% participation. The full arithmetic, the strike convention that decides whether the headline rate is 51% or 41%, and the levers that buy participation are developed in leverage control through structure and are not repeated here; what belongs in this section is the design move itself, which is recognising that a constraint plus a budget is a complete brief and that every extra point of participation has to be paid for out of protection, cap or maturity.

REAL INTERVIEW QUESTION (Société Générale · EQD)

Give a strategy and its hedge for a client who wants to place 2 million, get it back in two years, and benefit from a possible rise in the index.

WHAT THEY TEST / ANSWER

Structure first: a **capital-guaranteed note**, built as a zero-coupon bond plus a call. Walk the budget out loud, since that is the answer. At 3% rates the zero-coupon that repays €2 million in two years costs about €1.885 million, leaving roughly €115 thousand, or 5.7% of notional, to spend on optionality. A two-year call struck at the forward costs around 11% of notional at 20% volatility, so the client gets about 50% participation in the rise above that forward. If they want participation from today's level instead, the call is dearer and the rate drops to about 40%, and establishing which one they mean before quoting is the difference between a competitive price and an embarrassing one. Quote it as a participation rate, since that is the number they will compare against other banks.

Then the hedge, which is the half most candidates forget. The bank is short the call it sold, so the desk buys the call in the market or, more usually, delta-hedges it dynamically and manages the residual gamma and vega on the book. The zero-coupon leg is a funding trade and sits with treasury. Name the residual risks the desk keeps: the dividend and repo assumptions over two years, the volatility level it priced the call at, and its own funding spread, since the "zero-coupon" is really the bank's own credit and that is what makes the guarantee only as good as the issuer. Add that if participation is too low to sell, the standard fixes are to cap the upside, lower the protection to 90%, or lengthen the maturity, and each of those is a specific thing the client gives up.

Designing to a payoff: ratios and leverage

Sometimes the client describes the payoff directly, and the job is to reverse-engineer it into tradeable legs.

WORKED EXAMPLE

A client wants a product that loses twice the underlying's fall and gains three times its rise. Written as a payoff on a notional, that is

$$3 \times \max(S_T - K, 0) \; - \; 2 \times \max(K - S_T, 0),$$

which reads directly off the page as **long 3 calls and short 2 puts**.

Now price it. With spot at 100 and the option prices used throughout payoff profiles, three at-the-money calls cost $3 \times 8 = 24$ and two at-the-money puts raise $2 \times 8 = 16$, so the structure costs 8, or 8% of notional. If the client wants it for nothing, move the strikes: three calls struck at 110 cost $3 \times 4 = 12$ against two puts struck at 100 raising $2 \times 8 = 16$, a net *credit* of 4.

The redesign has changed the deal, and saying so is the point. The client now gets triple participation only above 110 rather than above 100, and is still doubly exposed below 100. Zero cost was bought by moving the upside further away, not by financial alchemy.

REAL INTERVIEW QUESTION (Crédit Suisse · Structuring)

How would you structure a product such that if the underlying falls I lose twice the move of the underlying, and if it rises I gain three times as much?

WHAT THEY TEST / ANSWER

Read the payoff into legs, which is the method owned by payoff engineering and is worked through just above: **long three calls and short two puts**, same strike and maturity, on the client's notional. That is the answer to the question as asked, and it takes ten seconds. Everything after that is what the interviewer is actually grading.

Risk. The short put leg is *doubly* levered, so a large fall costs twice the market's move and the loss is bounded only by the underlying going to zero. Say it out loud: this is not a hedged position, it is a leveraged bullish bet with a fat downside tail.

Skew. Not at a common strike. A put and a call struck at the same level and maturity carry the *same* implied volatility, because put-call parity is model-free and any difference would be an arbitrage. Skew enters only once you move the call strike up to reach zero cost, as above: you are then short the 100 puts and long the 110 calls, and the equity skew makes the wing you sold relatively expensive, which helps the cost and worsens the risk. That is why these are easy to sell and dangerous to own, and getting the reason right matters, because the version that invokes skew at a common strike is a claim an interviewer can refute in one line.

Interrogating the brief

Occasionally a client asks for a Greek rather than a payoff, and the right first move is to ask why.

Asking for a Greek is asking for a cost or a payment, so the move is to convert it back into a view. Negative theta means being net long optionality, and since theta is what you pay for gamma and vega, the real question is which of those the client wants: gamma lives at the front of the maturity curve and vega at the back, so the two answers sit at opposite ends and picking the wrong one is how a correct view loses money. The trade-off itself belongs to theta as the cost of convexity. What belongs here is the discipline that follows: quantify the bleed and the horizon before proposing anything, because negative theta with no time limit is how positions get closed at the worst moment.

> **INTERVIEW TRAP**
>
> Four traps. (1) Designing from direction alone and ignoring horizon, which is the single most common way a correct view loses money. (2) Presenting a zero-cost structure as free, when it is financed by selling something the client may care about more than they realise. (3) Adding leverage through ratios without stating that the unbounded tail has come back. (4) Quoting a participation rate without saying what happens if the client wants a higher one, since every route to more participation costs protection, cap or time.

> **VISUAL / CODE TO BUILD: Interactive**
>
> A design bench. The reader enters a view as sliders (direction, magnitude, horizon, conviction, volatility view) and a budget, and the panel proposes two or three candidate structures, drawing each payoff and reporting cost, break-evens, maximum gain and maximum loss side by side so the trade-off is explicit rather than argued. A capital-protection mode takes a notional, a rate and a protection level and solves for participation, with a slider on each input so the reader sees participation rise with rates and fall with volatility. A payoff-to-legs mode lets the reader draw a piecewise-linear payoff freehand and returns the option legs that replicate it, with the running cost, which turns the Crédit Suisse question into something they can build.

Where this goes next

The structures themselves, organised by the view they express, are directional and volatility strategies, with the leverage machinery in construction with ratios. The capital-protected family is developed in capital-protected products, and the errors this method is designed to avoid are collected in common structuring mistakes.

17.2 Directional strategies

sales · trading · structuring | core | equity · fx · cross-asset

Everyone can say "I am bullish, so I buy a call". The interview, and the job, start one question later: bullish by how much, by when, and what are you willing to give up to pay for it. A directional view has at least four common expressions, they cost very different amounts, and each one is optimal for a different version of the same view. Getting that match right is the whole skill, and it is what from market view to payoff design sets up in general terms and this section makes concrete.

We work throughout with the view "the stock rises about 10% over three months", on a spot of 100, at 20% implied volatility, ignoring rates and dividends so the forward equals spot. Flat volatility is assumed for comparability; what the skew does to these prices is taken up at the end.

The four ways to be bullish

The payoff diagrams of all four, and of every other standard combination, are drawn and catalogued in standard payoff profiles. They are not redrawn here. The question this section answers is the one that comes after the shapes are known: given a specific view, which of them should you actually buy, and what are you giving up to pay for it.

1. **Buy the underlying.** Unlimited upside, unlimited downside, full capital outlay, no time decay. The benchmark everything else is judged against.
2. **Buy a call.** Loss capped at the premium, upside uncapped, and you pay theta. You are buying a view on direction *and*, whether you meant to or not, a long volatility position.
3. **Buy a vertical spread** (a call spread, also called a bull spread): buy a call, sell a higher-strike call. You give up everything above the upper strike, and the premium you receive for it cuts your cost sharply.
4. **Buy a collar or risk reversal**: buy an upside call, sell a downside put. You finance the call by selling the downside you claim not to fear, often down to zero net premium.

There is a fifth that is worth knowing precisely because it is *not* an option position: long a call and short a put at the same strike is a **synthetic forward**, whose payoff is a straight line. It has no vega, no gamma and no theta, because the optionality of the two legs cancels exactly. If your view is pure direction with no view on volatility at all, that is the honest structure, and put-call parity guarantees its price matches the forward. The parity argument itself, and the general business of building one instrument out of others, is replication strategies.

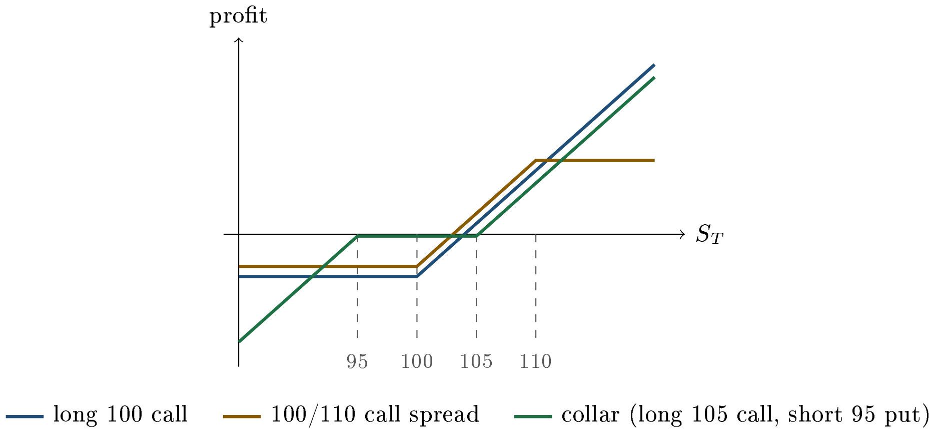
WORKED EXAMPLE

Spot 100, three months, 20% volatility, zero rates and dividends. These are the same four structures priced in standard payoff profiles, re-priced here at a single common maturity so that they can be ranked against one view rather than described one at a time.

Structure	Cost	Break-even	Profit at 110	Max profit
Long stock	100	100.00	+10.00	unlimited
Long 100 call	3.99	103.99	+6.01	unlimited
Long 105 call	2.06	107.06	+2.94	unlimited
100/110 call spread	3.03	103.03	+6.97	+6.97
Collar: long 105 call, short 95 put	0.18	105.18	+4.82	unlimited
Synthetic forward (100)	0.00	100.00	+10.00	unlimited

Read the fourth column against the second and the ranking is not what intuition suggests. For *exactly* this view, the call spread is the best of the option structures: it costs less than the outright call, it breaks even lower, and it makes more at 110. The reason is that you sold the part of the distribution your view says will not happen. Expressed as a return on premium, the call spread turns 3.03 into 6.97, roughly 230%, against 151% for the outright call.

The catch is in the last column. If the stock goes to 130, the outright call makes 26 and the call spread still makes 6.97. You did not buy a cheaper version of the same trade, you sold your own tail.



KEY IDEA

Every directional structure is the outright option *minus* something you sold to pay for it. A call spread sells the far upside. A collar sells the downside. A covered call sells the upside of stock you already own. The design question is never "which is cheapest", it is "which part of the distribution am I genuinely willing to give away", and the answer comes from the view, not from the price.

REAL INTERVIEW QUESTION (JP Morgan · Exo)

Do you know what a call spread is? Draw the payoff, and give me a situation where it is worth using.

WHAT THEY TEST / ANSWER

The drawing is the easy half: buy the K_1 call, sell the K_2 call with $K_2 > K_1$, so the profit is flat at minus the net premium up to K_1, rises one-for-one between the strikes, and is capped at $K_2 - K_1$ minus the premium above K_2. The second half is what the question is for, so give a specific situation rather than "when you are moderately bullish". Three good ones. **A target price**: you think the stock goes to 110 and not much further, so you sell the 110 strike, and for this view the spread strictly dominates the outright call, cheaper, lower break-even and a bigger profit at your target. **Expensive volatility**: the outright call is a long volatility position whether you want one or not, and selling the upper strike cuts the vega, so a call spread is how you express direction when you think implied volatility is rich. **Budget**: a structured note has a fixed premium budget, and capping the payoff is what buys enough participation to be worth showing a client. Then name the cost honestly: you have sold your own tail, so if the stock doubles you keep $K_2 - K_1$ and nothing more. If they push further, add that with equity skew the upper strike carries a lower implied volatility, so the leg you are selling is cheap and the call spread costs more than a flat-volatility model suggests, as shown in the volatility smile and skew.

Collar or spread: two different things you are selling

Both structures cheapen a call. They are not variations of each other, and confusing them is a standard interview casualty.

A **call spread** sells *upside*. Its worst case is unchanged, the premium, and its best case is capped.

Its risk profile stays that of a long option: limited loss, and the position remains long gamma near the lower strike.

A **collar** sells *downside*. Its best case is unchanged and uncapped, and its worst case is now large, because a short put loses with the market all the way down. It is much closer to being long the underlying with a floor removed than to being long an option.

When you already own the asset

Two structures exist mainly for holders rather than speculators, and sales desks quote them constantly.

Covered call (overwriting): long the stock, short a call. You convert uncertain upside into certain premium today. The payoff is identical to a short put, which is worth saying out loud, because it makes the risk obvious: you keep all the downside. Institutions run systematic overwriting programmes on holdings they do not intend to sell, and that persistent selling is one of the flows that shapes the upper part of the volatility surface.

Protective put: long the stock, long a put. Insurance, with the premium as the deductible. Its payoff is identical to a long call at the put's strike, again by put-call parity, which is the fastest way to price it and the fastest way to see that buying protection is not free.

Both are worth internalising as parity statements rather than as separate products: covered call equals short put, protective put equals long call. A candidate who says that has demonstrated the replication reflex that the rest of the chapter runs on.

REAL INTERVIEW QUESTION (Natixis · EQD)

A client holds one million in Total shares. What arguments would you use to move them toward derivatives, and what strategy would you propose?

WHAT THEY TEST / ANSWER

This is a sales question wearing a product question's clothes, so do not open with a structure. Open with what the client already owns and what they are unhappy about, because every answer below is a different complaint.

Establish the position first: they are long a single stock, so they hold direction, volatility, dividend and single-name event risk, all unhedged and all unpaid for. Then propose against the complaint.

If they are **neutral to mildly positive** and want income, sell upside: a covered call on part of the holding turns the upside they do not expect into premium today. Say the risk in the same breath, because it is the whole objection: the payoff is a short put, so they keep every bit of the downside, and if the stock is taken over they are called away below the bid. If they are **worried about a fall** but will not sell, buy a protective put, and price the insurance honestly rather than presenting it as free. If the put is too expensive, that is the moment for a collar, funding the put by selling upside, which is where the conversation usually lands and why the collar is the most-quoted structure on a concentrated holding.

Two arguments that are about the client rather than the product, and that are what the interviewer is really listening for. **Concentration**: a million in one name is an undiversified position, and derivatives let them reduce the risk without a taxable sale or a signal to the market. **Efficiency**: they can keep the exposure and free most of the capital, or keep the capital and reshape the exposure, and which of the two they want tells you the trade.

Then say what you would not do. Do not overwrite a holding the client may need to sell, and do not collar a position whose upside is the entire reason they hold it. A proposal that names its own wrong client is worth more than one that fits everybody.

The Greeks of a spread are not the Greeks of an option

A vertical spread is two positions with opposite signs, so every Greek partly cancels, and the cancellation is what makes the structure behave differently from the outright. How each Greek evolves through a strategy's life is the subject of Greeks evolution within strategies, and the Greeks themselves are defined in Greek definitions and intuition; what follows is only the one consequence that decides how a spread is traded.

What the skew does to all of this

The prices above assume one volatility for every strike, which is false. With a downward equity skew:

- The **call spread** gets more expensive, because the upper strike you are selling carries a lower implied volatility than the flat assumption gives it.
- The **collar** gets cheaper, often to a credit, because the put you are selling carries a higher implied volatility than the call you are buying. A "zero cost" collar in equities is usually struck asymmetrically for exactly this reason.
- The **outright call** is barely affected if it is near the money, and cheaper than a flat model suggests if it is far out of the money.

None of this is a detail. On a three-month structure the skew can move a spread's price by several percent of its premium, and on a multi-year structured note it is the difference between a term sheet that prints and one that does not. The mechanism is in the volatility smile and skew.

INTERVIEW TRAP

Four errors. (1) Pitching a call spread as "a cheaper call" without saying that the upside is sold: it is a different trade, not a discount. (2) Treating a collar as a low-risk structure because its premium is near zero, when its downside is the same as owning the stock below the put strike. (3) Pricing any of these at a single volatility, when every one of them is a two-strike position and therefore a skew position. (4) Forgetting the parity identities, so that a covered call is analysed as something other than a short put and its downside is quietly ignored.

VISUAL / CODE TO BUILD: Interactive

A view-to-structure matcher. The reader enters a view as a distribution rather than a direction: a target price, a confidence interval and a horizon. The widget prices the six structures in this section against the current surface and ranks them by expected profit *under the reader's own stated view*, showing for each one the cost, break-even, profit at target, worst case and maximum. Dragging the target price re-ranks the list live, and the intended realisation is that the ranking flips: the call spread wins for a contained target, the outright call wins once the target moves far enough out, and the collar wins when the reader's downside probability is genuinely low. A second toggle switches between a flat volatility and the live skew so the reader sees which structures the skew helps and which it taxes.

Where this goes next

Views on movement rather than direction are volatility strategies, and views on the market going nowhere are range-bound strategies. Trading the shape of the surface rather than the underlying is skew-based strategies. Breaking the one-for-one leg ratio to change the participation is strategy construction with ratios, and how any of these behave as time passes rather than at expiry is strategy behaviour over time.

17.3 Volatility Strategies

sales · trading · structuring | core | equity · fx · cross-asset

"I think volatility is going up" is not one view, it is two, and the structures that express them are different and sometimes opposite. This section is about making that distinction and then choosing the instrument. The shapes themselves were built in standard payoff profiles; here we choose between them.

Two different volatility trades

KEY IDEA

There are **two** volatility trades and they are not the same. **Realised** volatility is traded through **gamma**: you buy an option, delta-hedge it, and profit if the underlying actually moves more than the implied you paid for. **Implied** volatility is traded through **vega**: you buy an option and profit if the market's *quoted* volatility rises, whether or not anything happens to the underlying. A trader can be right that the market will be turbulent and lose money because implied volatility fell, and vice versa. Ask which one you mean before choosing a structure.

The lever that separates them is maturity, from the term structure of Greeks: gamma scales as $1/\sqrt{T}$ and vega as $\sqrt{T}$. So a realised-volatility view is expressed at the front of the curve and an implied-volatility view at the back.

REAL INTERVIEW QUESTION (Société Générale · Cross Asset)

Are you long or short volatility when you are long a call? Why?

WHAT THEY TEST / ANSWER

Long volatility, and the "why" is the whole question. The reason is the asymmetry of the payoff: your loss is capped at the premium and your gain is not, so a wider distribution of outcomes can only help you. More volatility raises the chance of a large favourable move without worsening the fixed downside, so the option is worth more.

Then add the two refinements that make it a desk answer. First, the same argument applies to a long *put*, so being long volatility has nothing to do with being bullish; it follows from being long the option, not from the direction. Second, "long volatility" is ambiguous and you should say which kind: a long call is long vega, so it gains if implied volatility rises, and it is also long gamma, so if you delta-hedge it you gain when realised volatility beats the implied you paid. Those are different trades in the same position, and separating them is the content of this section.

Long volatility: which structure

- **Straddle**, a call and a put at the same strike. Maximum gamma and vega, concentrated at the strike. The purest expression, and the most expensive.
- **Strangle**, the same with the strikes split. Cheaper, but it needs a bigger move and it loses the full premium over a whole range rather than at a single point.
- **Reverse calendar**, long the near-dated and short the long-dated. Isolates *realised* volatility: long gamma while keeping vega near zero or negative.
- **Variance swap**, which pays the realised-minus-implied difference directly with no rehedging at all, developed in variance swaps.

WORKED EXAMPLE

Using the consistent prices from payoff profiles, spot 100: the at-the-money straddle costs 16.00 with break-evens at 84 and 116, while the 90/110 strangle costs 8.00 with break-evens at 82 and 118.

The strangle is exactly half the price, so its maximum loss is half as large, but it needs a move of 18 rather than 16 to pay, and it loses everything anywhere in a twenty-point window rather than at one point. Which is better depends entirely on how big a move you expect, which is exactly the magnitude dimension from payoff design.

REAL INTERVIEW QUESTION (Natixis · EQD)

Give me examples of positions such as a straddle or a strangle, and explain the relevance of these strategies.

WHAT THEY TEST / ANSWER

Give each one a view rather than a description, since "relevance" is the question. **Straddle**: you expect a large move and have no view on direction, typically around a known catalyst such as earnings, a central bank meeting or a court ruling. Maximum gamma and vega at the strike, and the most expensive way to say it. **Strangle**: the same view but you expect a *very* large move, or you want the same exposure for less premium and accept that a moderate move now loses. **Butterfly or condor**: the opposite view, that nothing will happen, expressed with a defined risk instead of a naked short straddle. **Calendar**: a view on the volatility *term structure* rather than its level.

Then the sentence that shows you are choosing rather than listing: all of these are decisions about where in the distribution you want to be paid, and the cost of each is the premium you spend on the parts you do not believe in. Add the honest caveat: a long straddle only pays if realised volatility beats the implied you bought, so buying one before earnings is not a free bet on a big move, because the implied volatility already prices the event. That last point is where most candidates fall down.

Short volatility: the carry and the tail

Selling volatility earns the volatility risk premium of implied versus realised, which is positive most of the time. Short straddles and strangles, iron condors, covered calls and systematic put selling are all versions of it.

The honest framing is that these are all **short a fat tail**. The strategy earns a small, steady amount most months and loses a large, correlated amount occasionally, and the losses arrive when gamma, vega, volga and vanna all move against you at once, as in volatility regimes. The defined-risk versions, iron condors and butterflies, exist precisely to cap that tail, and they earn less for exactly that reason.

Separating gamma from vega

When the view is specifically about realised rather than implied volatility, the structure has to separate them, and maturity is the only lever that does.

WORKED EXAMPLE

At 20% volatility on a spot of 100, a one-month at-the-money option has a gamma of 0.0691 and a vega of 0.115 per point, while a one-year option has a gamma of 0.0198 and a vega of 0.397.

Sell one of the one-year and buy two of the one-month:

$$\text{vega} = 2(0.115) - 0.397 = -0.167, \qquad \text{gamma} = 2(0.0691) - 0.0198 = +0.118.$$

The position is **long gamma and short vega**. Buying about 3.45 of the one-month instead would make it vega-neutral and even more strongly long gamma, which is the reverse calendar of the previous section. The ratio is the lever: below 3.45 you are short vega, above it you are long.

REAL INTERVIEW QUESTION (Bank of America · Exo)

Could you give me a strategy that is long gamma and short vega?

WHAT THEY TEST / ANSWER

Buy short-dated options and sell long-dated ones, sized so that the vega of the long-dated leg dominates. That is a reverse calendar spread, and you then delta-hedge the residual.

Give the reasoning from the scaling rather than asserting the recipe. Using the maturity scalings quoted above from the term structure of Greeks, the near-dated leg carries most of the gamma and the far-dated leg most of the vega, and the two can be set independently by choosing the ratio. Put a number on it: at 20% volatility, buying two one-month at-the-money options against one one-year gives roughly -0.17 of vega and $+0.12$ of gamma per unit, so it is long gamma and short vega as asked. Buying about 3.45 of them instead would be vega-flat, and more than that would flip you long vega.

Close with the view it expresses and its cost: you think the underlying will move about in the near term while implied volatility falls, which is typically a post-event position, after a catalyst has passed and the term structure is expected to normalise. The cost is heavy theta at the front, so it is a position with a deadline.

How often to rehedge

REAL INTERVIEW QUESTION (JP Morgan · Delta One)

How often do you hedge your delta when you are short gamma? And when you are long gamma?

WHAT THEY TEST / ANSWER

The answer is asymmetric, and saying why is the point.

Short gamma: more often. Your delta moves against you, so an unhedged drift compounds into a loss that grows with the *square* of the move. Rehedging frequently converts one uncertain large loss into many small realised ones. Gross of costs that does not change what you expect to lose, only how tightly the outcome clusters around it: the expected sum of squared moves is $S^2\sigma^2T$ whatever the rehedging frequency, so frequency moves the dispersion and not the mean. What makes frequent hedging expensive is transaction costs, not the rehedging itself. You hedge tightly anyway, because the tail is what kills a short-gamma book.

Long gamma: less often. Your delta moves in your favour, so drift is not dangerous, and waiting is genuinely optional. Wider bands mean fewer transactions and lower costs, and they let larger moves accumulate before you monetise them. You can afford to be patient, and patience is worth money.

Then give the framework rather than a number, since the interviewer will push: hedge on a *band* around zero delta rather than on a clock, with the width set by transaction costs against gamma. Hedging error falls only as $1/\sqrt{n}$ in the number of rehedges while costs rise as $\sqrt{n}$ under proportional transaction costs, or as n if the fee is per ticket, so there is an interior optimum either way, and it sits tighter for a short-gamma book because the loss function is asymmetric. The mechanics are in dynamic hedging.

INTERVIEW TRAP

Four traps. (1) Saying "long volatility" without specifying realised or implied, when the two are traded at opposite ends of the maturity curve. (2) Buying a straddle into a known event and calling it a free bet on a big move, when the implied volatility already prices the event. (3) Describing short volatility as a carry trade without saying it is short a fat tail that arrives correlated with everything else. (4) Hedging a long-gamma and a short-gamma book at the same frequency, when the loss functions are asymmetric.

VISUAL / CODE TO BUILD: Interactive

A volatility-trade chooser. The reader states a view on realised volatility and, separately, on implied volatility, and the panel highlights which structures fit, greying out the ones that express the other view. Selecting a structure draws its payoff, its gamma and vega profiles, and its cost. A simulation tab then runs a price path with independently settable realised and implied volatility, so the reader can construct the case where realised comes in high and implied falls and watch a long straddle lose money while a delta-hedged version makes it, which is the distinction the section opens with. A rehedging-frequency slider on the same path shows the long-gamma and short-gamma books responding asymmetrically to the same choice.

Where this goes next

Volatility varies by strike as well as by level, and trading that dimension is skew-based strategies. The instrument that gives the realised-minus-implied exposure with no rehedging is the variance swap. The Greek behaviour these structures rely on is gamma and convexity and vega exposure and volatility regimes.

17.4 Range-bound strategies

sales · trading · structuring core equity · fx · cross-asset

Directional strategies answered “which way”; volatility strategies answers “how much movement”. This section answers a third and commercially enormous question: what do you trade when your view is that nothing much happens? Range views are the backbone of the structured-products business, because a client who accepts a band in exchange for a coupon is selling exactly the optionality a dealer wants to buy. They are also where retail and junior traders lose the most money, because a range structure feels safe and its risk arrives all at once.

Throughout: spot 100, three months, 20% implied volatility, no rates or dividends, and flat volatility across strikes so the arithmetic is comparable.

Set the band from the distribution, not from a feeling

Before choosing a structure, choose the band, and choose it from what the market is already pricing. One standard deviation of the log price over the life of the option is $\sigma\sqrt{T}$, which here is $0.20 \times 0.5 = 10\%$. So the market's own estimate is that the stock ends between 90 and 110 with a probability of about two thirds, and the exact risk-neutral figure from the option prices is 68.4%.

That single number disciplines everything that follows. A range structure whose band is narrower than one standard deviation is a genuine bet against the market's estimate. One that is wider is closer to a carry trade, collecting a small premium for a risk that is real but infrequent. Saying which of the two you are doing, in those terms, is what a desk means by having a view.

> **KEY IDEA**
>
> Every range structure is short the tails and long the middle, so it is short gamma, short vega and long theta near the centre. It earns a little on most days and pays out on the few days that matter. The design question is not “how much can I collect” but “where do I put the boundary, and do I want a defined loss beyond it or not”.

Two structures, one view

Both shapes are drawn and catalogued in standard payoff profiles; what follows is what each one is for.

The butterfly. Buy the 90 call, sell two 100 calls, buy the 110 call. The payoff is a tent: zero outside the wings, peaking at 10 if the stock finishes exactly at 100. It is a bet on the stock *pinning* a level, and its cost is a direct statement about the probability density at that level, which is the Breeden-Litzenberger reading given in the volatility smile and skew.

The iron condor (again, shape in standard payoff profiles). Sell the 95 put, buy the 90 put, sell the 105 call, buy the 110 call. The payoff is a plateau: you keep the full credit anywhere between

95 and 105, and your loss is capped by the two long wings. It is a bet on a *band* rather than a point, and it is what most people actually mean by a range trade.

The pair is worth studying together because they are the same view expressed with different tolerance for being slightly wrong.

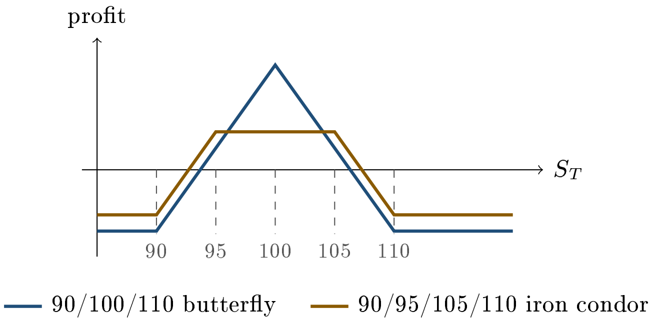
WORKED EXAMPLE

Spot 100, three months, 20% volatility, zero rates.

	90/100/110 butterfly	90/95/105/110 iron condor
Cash flow at inception	pay 3.69	receive 2.29
Best case	+6.31 (at exactly 100)	+2.29 (anywhere in 95 to 105)
Worst case	−3.69	−2.71
Break-evens	93.69 and 106.31	92.71 and 107.29
Chance of any profit	47.2%	53.4%
Chance of the best case	near zero	38.3%

The probabilities are risk-neutral, read off the option prices themselves as differences of $N(d_2)$, so they are the market's numbers and not an opinion.

Two readings. First, neither structure is generous: the butterfly risks 3.69 to make 6.31 with a 47% chance of any profit at all, and the condor risks 2.71 to make 2.29 with a 53% chance. Short-volatility structures are not free money once you price them properly, and a candidate who says so is ahead of one who quotes the credit. Second, the butterfly's best case requires the stock to land on a point, so its realistic outcome is somewhere on the slope; the condor's best case is a band it has a 38% chance of hitting outright. If your view is "quiet" rather than "pinned at 100", the condor matches it and the butterfly does not.



Defined risk is the whole point of the wings

Strip the long wings out of the iron condor and you have a short 95/105 strangle: the same credit-collecting profile in the middle, but with an unbounded loss. The wings cost part of the credit and convert an open-ended risk into a known one.

Whether to buy them is a real decision rather than an obvious one, and the honest framing is that you are choosing where to sit on a spectrum:

- **Short strangle**: maximum credit, unbounded loss, and a margin requirement that grows in exactly the scenario that is hurting you, which is what turns a bad day into a forced unwind.
- **Iron condor**: less credit, known worst case, and no margin spiral. The wings also mean you are long the tails, so a crash that destroys a short strangle merely costs the condor its maximum loss.

For anyone running a book rather than a single trade, the second matters more than the arithmetic suggests, because it is the difference between a loss you sized and a loss that sizes itself.

One consequence of the tent shape is worth stating because it decides how the structure is risk-managed. The Greeks themselves are defined in Greek definitions and intuition and their behaviour through a trade's life in gamma and convexity; the point here is only that a butterfly's are *not one number*. Because gamma and vega are concentrated at the money, the two short centre options dominate when spot is near the centre: on the 90/100/110 example at 20% volatility the position has a gamma of about -0.031 per unit of spot and a vega of about -0.157 per volatility point, which is -15.7 per unit of volatility. Watch the unit, because the two conventions differ by a factor of one hundred and this is the commonest slip in the chapter. But move spot out toward a wing and the long leg there takes over, and the position becomes locally *long* gamma. That sign change across spot is why a butterfly is a bet on the distribution rather than on volatility: short volatility in the middle and long the tails, which is exactly the shape of a bet that the market lands here rather than somewhere else.

REAL INTERVIEW QUESTION (Crédit Suisse · Structuring)

What are a straddle, a strangle and a butterfly spread? Explain their purpose.

WHAT THEY TEST / ANSWER

Define them by what view each one expresses, not by their legs, because the legs are the easy part. **Straddle**: long a call and a put at the same strike. The purpose is a pure bet on movement with no directional view, and it is the most expensive way to buy that because both legs are at the money where premium is concentrated. **Strangle**: the same view with the strikes pushed apart. It is cheaper, needs a bigger move to pay, and carries more convexity in volatility because the wings are where volga lives. **Butterfly**: long both wings and short two centre options, so it is the mirror of the straddle. Its purpose is a bet that the underlying finishes *near* a specific level, and because a butterfly of width h costs about the risk-neutral probability density at that level times h^2, it is the cleanest way to express "the market is pricing too much chance of a big move" with a known maximum loss. Close with the structuring point, which is what the desk is listening for: the first two are how a client buys optionality and the third is how a client sells it with a floor under the loss, so in a note the butterfly appears when you want to recycle premium without handing the investor an unlimited downside. Volatility views themselves are developed in volatility strategies.

Choosing the underlying, not just the strikes

A range view has two inputs, and candidates usually think about only one. The strikes matter, but so does which underlying you sell optionality on, and the two decisions are not independent.

REAL INTERVIEW QUESTION (Leonteq · Structured Products)

You sell me a call and you have the choice among 50 stocks: which one do you take?

WHAT THEY TEST / ANSWER

Do not answer with a name, answer with a criterion, and be explicit that you are choosing on the gap between implied and realised rather than on the level of either. As the seller you want the stock whose implied volatility is highest relative to what you believe it will actually realise over the life of the option, since your profit is that gap collected through theta, as in theta as the cost of convexity. So you screen for a large implied-minus-realised spread, then eliminate the ones where the spread exists for a reason. Name those reasons, because that is the real test: an earnings date or a regulatory decision inside the life of the option, a pending bid or corporate action, a borrow that is tight or expensive so you cannot delta-hedge cleanly, a wide bid-ask that eats the edge, or a small capitalisation where a single piece of news moves the stock 20%. Finish with the risk framing: selling a call is not a range view on its own, it is a short of the upside tail, so the right choice is a liquid, well-covered name with rich implied volatility and no scheduled catalyst, and the wrong choice is the cheap-looking one that is cheap because everyone can see what is coming.

The exotic versions, and what they buy you

Vanilla range structures pay on where the underlying *ends*. Clients often want to be paid on where it *stays*, and that is a different product. A range accrual pays a coupon for each day spot sits inside a corridor; a double no-touch pays a fixed amount only if neither boundary is ever touched. Both are path-dependent, so they are worth more or less than the vanilla version depending entirely on how the underlying travels rather than where it lands, and both are far more sensitive to the shape of the volatility surface than a vanilla condor is. They belong to exotic options, and the general treatment of path dependency is in path dependency and scenario analysis.

The practical point for a range view is that path dependency lowers the probability of the payout, because far more paths touch a boundary than end beyond it. The same premium therefore buys a bigger headline coupon. That is why a double-no-touch coupon looks generous next to a condor's, and it is not a transfer in the dealer's favour but a different and harder condition being paid for. A client should be shown the probability alongside the coupon rather than the coupon alone.

INTERVIEW TRAP

Four traps. (1) Setting the band by eye instead of against $\sigma\sqrt{T}$: at 20% volatility over three months a "wide" 90 to 110 range is only one standard deviation, so the market already thinks there is a one in three chance of leaving it. (2) Selling a naked strangle for the extra credit and discovering that its margin requirement grows fastest in the scenario that is already costing you money. (3) Quoting the credit as the return: the honest number is the credit against the maximum loss and the probability of each, and on the condor above that is 2.29 against 2.71. (4) Treating a butterfly as a volatility trade: it is short volatility in the middle and long it in the wings, so it is a bet on the shape of the distribution, not on its width.

VISUAL / CODE TO BUILD: Interactive

A band designer. The reader drags two boundaries on a chart of the underlying and the widget immediately reports, from live option prices, the risk-neutral probability of finishing inside the band, the best credit available from a condor with those boundaries, the corresponding maximum loss, and the same figures for the naked strangle so the cost of the wings is explicit. A shaded overlay of the implied distribution sits behind the chart with the one-standard-deviation band marked, so the reader sees whether they have chosen a bet or a carry trade. A second panel runs the structure forward through time and through a volatility shock, showing how the profit and loss profile steepens as expiry approaches and how a volatility spike hurts before any spot move has happened at all.

Where this goes next

The mirror view, that the market moves a lot, is volatility strategies, and its extreme form is tail-risk strategies, which is what the wings of a condor are quietly buying. Changing the leg ratios to reshape any of these is strategy construction with ratios. How these structures behave between now and expiry rather than at it is strategy behaviour over time, and matching them to the prevailing environment is strategy selection by market regime.

17.5 Skew-Based Strategies

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Volatility strategies traded the *level* of implied volatility. This section trades its *slope* across strikes. What the skew is and why it exists belongs to the smile and skew; here we take it as given and ask which structures are long it, which are short it, and what that costs.

What a skew position is

Every option you hold sits at a strike, and therefore at a point on the volatility surface. A position spanning two strikes is implicitly a position on the *difference* between the two implied volatilities, whether or not you intended it.

KEY IDEA

In equities the convention is that being **long skew** means being long downside implied volatility relative to upside: you gain when the surface steepens, when puts richen against calls. Being **short skew** is the reverse. State the convention when you use it, because desks and asset classes quote the risk reversal with different signs, and an interviewer is checking whether you know it is a convention rather than a fact.

The Greek behind it is **vanna**, whose closed form and sign are developed in skew and smile sensitivities: strikes below the crossover at $d_2 = 0$, which sits just under the forward, carry negative vanna and strikes above it positive vanna, so vanna is the first-order measure of how much skew you are carrying. How it interacts with the other Greeks is cross-Greeks.

Vertical spreads are skew trades in disguise

This is the point most candidates miss. A call spread is long the lower strike and short the higher one, so it is long the lower strike's implied volatility and short the higher strike's. Since equity skew makes lower strikes richer, a long call spread *gains* when the skew steepens.

Skew makes a long call spread more expensive and a long put spread cheaper, and by enough to matter: on a one-year 100/130 call spread, with the 130-strike volatility marked about three points below a 20% at-the-money, the difference against pricing both legs flat is about seven percent of the premium. The pricing itself is done in the smile and skew, which owns it, and is not repeated here.

> **REAL INTERVIEW QUESTION** (Leonteq · Structured Products)
>
> *On a short call spread, what move in volatility, and in the skew, is favourable to you?*
>
> **WHAT THEY TEST / ANSWER**
>
> Take the two separately, because they are different exposures.
>
> **Volatility level**: you want it *down*. The lower-strike call you are short has more vega than the higher-strike call you are long, since vega peaks at the money, so the structure is net short vega. At 20% volatility on a spot of 100 the one-year vegas are about 0.40 at the 100 strike and 0.19 at the 130 strike, so a short 100/130 spread is short about 0.21 of vega per unit and a general rise in implied volatility hurts.
>
> **Skew**: you want it *flatter*. You are short skew, so you gain when the lower strike's volatility falls relative to the higher strike's.
>
> The nuance worth adding: those two can move in opposite directions, and in equities they usually do. A sell-off typically lifts the whole surface *and* steepens the skew, so both legs of your exposure lose together, which is precisely the correlated-loss pattern that makes short-skew positions uncomfortable in the scenarios that matter.

The trades that express it deliberately, and where they live

If a desk believes the skew is too steep, the trades that express it run from the risk reversal, which is the pure expression, through put spreads against call spreads, to ratio put spreads, which are short skew and reintroduce an unbounded tail. Those are deliberate skew positions and they belong to skew trading, along with the instruments, the sign conventions and the carry.

What this section is for is the other case: **the skew you did not choose**. A short call spread is short skew as a by-product of a directional structure, and a desk that has sold many of them inside products accumulates a position in the slope that nobody ever traded. That is the exposure worth isolating here, because it is the one that is invisible until the surface moves.

The carry and the tail

Skew is priced the way it is for structural reasons, not because the market is wrong: investors buy downside protection, and dealers who have sold retail yield products are long the downside wing from that issuance, so demand for protection runs one way and the dealer inventory runs the other. The residual after vanilla hedging is a separate question, taken up in hedging structured products. The consequence for a strategy book is the one this section cares about: a desk whose inventory is mostly sold call spreads and sold puts is short skew without having taken a view, and it is short it *alongside* everybody else, so the position it never chose is also crowded.

That makes it the same shape of trade as short volatility from volatility strategies: profitable most of the time, and losing in a way that is correlated with everything else in the book. The difference is that skew positions can lose even when the volatility level is flat, because only the *slope* needs to move, which is why they need to be measured separately rather than folded into a single vega number.

INTERVIEW TRAP

Four traps. (1) Pricing a vertical spread off a single volatility, which systematically flatters whichever leg the smile marks down. (2) Quoting a skew position without stating the sign convention, since it differs by desk and by asset class. (3) Treating a call spread as a pure directional trade when it carries a real skew exposure. (4) Describing short skew as a carry trade without saying that the loss arrives in exactly the scenario where the rest of the book is also losing.

VISUAL / CODE TO BUILD: Interactive

A skew workbench. The reader sets a volatility surface with independent controls for level and slope, and builds a position from vanilla legs. The panel reports vega and vanna, and then re-prices the position under a level shift and a slope shift separately, so the reader sees a call spread's value move when only the slope changes with the level fixed. A comparison mode prices any structure twice, once at a single flat volatility and once off the surface, and displays the gap for any strike pair the reader chooses. A scenario tab applies a realistic equity sell-off, level up and skew steeper together, so a short-skew position is seen losing on both exposures at once.

Where this goes next

The shape of the surface itself, and why equities skew the way they do, is the smile and skew. The Greek sensitivities to the surface are skew and smile sensitivities, and the professional version of trading it, including the models that must reproduce it, is skew trading.

17.6 Tail-risk strategies

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Tail hedging is the only strategy in this chapter whose stated purpose is to lose money most of the time. That is not a defect, it is the product: you are buying insurance, and insurance that pays in ordinary years is not insurance. The whole discipline is therefore about a single trade-off, between how much you bleed and how much you collect when the event happens, and about resisting the structures that make the first number look better by quietly destroying the second.

Everything below uses spot 100 and three-month options. Unlike the earlier sections we do *not* assume a flat volatility, because a tail hedge is bought exactly where the skew is steepest. With at-the-money volatility at 20%, the 90-strike put costs 0.71 priced flat against 1.32 priced on the skew, so pretending the surface is flat would understate the hedge by a factor of nearly two. The shape of the surface is in the volatility smile and skew.

What the protection actually costs

WORKED EXAMPLE

Spot 100, three months, at-the-money volatility 20%, and wing volatilities taken from a typical equity skew rather than a flat surface.

Hedge	Implied vol	Premium	As % of notional	Rolled 4x, per year
90 put (10% out)	25%	1.32	1.32%	about 5.3%
80 put (20% out)	30%	0.40	0.40%	about 1.6%

The annualised figures assume spot and the surface are unchanged at each roll, so treat them as an order of magnitude rather than a forecast. They are still the number that decides the argument. A fund hedging its whole equity exposure with rolling 10% out-of-the-money puts is giving up something in the region of five percent a year, which is most of a normal year's equity risk premium. That is why almost nobody hedges their whole book that way, and why anyone who claims to be fully protected is either paying far more than they realise or is protected against much less than they think.

KEY IDEA

The honest framing of tail hedging is not "how do I get cheap protection". It is a choice between two ways of carrying less risk: pay the premium and keep the position, or hold a smaller position and pay nothing. A hedge cheap enough not to hurt the return is usually too small to change the outcome, and recognising that trade-off explicitly is what separates a real answer from a product pitch.

Why it is worth anything at all: convexity on convexity

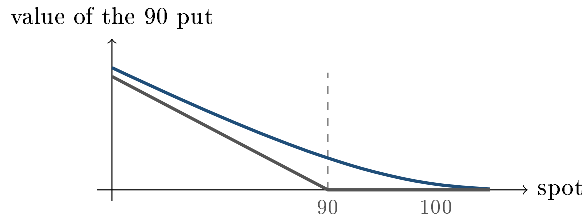
Against that carry sits a payoff that is violently non-linear, and for a reason that goes beyond the option's own gamma. When an equity market falls hard, four things happen to a deep out-of-the-money put at once: spot moves toward the strike (delta), the delta itself grows (gamma), implied volatility rises (vega), and the skew steepens so the low strike you own richens by more than the at-the-money (vanna, in the language of skew and smile sensitivities). All four push the same way. That stacking is what a tail hedge is buying, and it is why the instrument is a deep out-of-the-money put rather than a smaller position in the underlying.

WORKED EXAMPLE

The 90-strike three-month put bought for 1.32. Mark it a few weeks later, with 0.2 of a year (73 days) left, under a spot-down and volatility-up path.

Spot	Implied vol	Value of the put	Multiple of premium
90	35%	5.61	4.3×
85	40%	9.05	6.9×
80	45%	12.94	9.8×
70	55%	21.55	16.3×

Note where the money comes from at the top of the table. With spot at 90 the option is only just at the money and has *no intrinsic value at all*, yet it is worth more than four times what you paid. Every cent of that is time value created by the volatility rise and the steeper skew. A hedge sized on intrinsic value alone would have been sized wrong by a factor of four in the scenario it exists for.



mark-to-market value along a spot-down, volatility-up path — intrinsic value only

The structure that looks clever and is not

The standard way to make protection cheaper is to sell something further out: a 90/80 put spread instead of a 90 put. On the numbers above the spread costs 0.92 against 1.32, a saving of about 30%.

It is also, for this purpose, close to useless. The spread caps its payoff at 10, so in the scenario the hedge exists for it stops working exactly when the loss becomes serious: with the market at 60, the outright put pays 30 and the spread pays 10. You saved 0.40 of premium and gave up 20 of protection.

This is worth stating as a general principle, because the same mistake appears in many disguises: **financing a tail hedge by selling a further tail is not financing, it is cancelling**. The legitimate ways to cheapen protection all give up something *other* than the far tail: sell upside you do not expect (a collar), accept protection that starts further away (a lower strike), accept a shorter maturity, or take a smaller size. Selling the far wing is the one choice that removes the reason you bought the structure.

REAL INTERVIEW QUESTION (Natixis · EQD)

Could the premium not be enough to fund my hedge (on a call)? Why?

WHAT THEY TEST / ANSWER

No, and the reason is the cleanest statement of what a premium is. Under Black-Scholes the cost of delta-hedging a sold option all the way to expiry is, in expectation, exactly the premium you received, because the model prices the option as the cost of its own replication. So the premium funds the hedge *on average and only on average.* It is enough if the underlying realises the volatility you sold it at, insufficient if it realises more, and more than enough if it realises less. Then give the three practical reasons it is systematically insufficient in real life rather than merely uncertain: discrete rather than continuous re-hedging, so you always capture slightly less gamma than the model assumes; transaction costs and bid-ask on every re-hedge; and gaps, where the underlying jumps through your hedge and the loss is realised at a price you never traded. That is why a market maker sells at an implied volatility above their forecast of realised, and the difference is not greed, it is the cost of the frictions the model leaves out. Connect it to this section if they let you: this is also why buying tail protection is expensive, since the seller has to be paid for exactly those frictions in the scenario where they are worst.

Hedging the volatility rather than the spot

Since so much of the tail payoff comes from the volatility leg, an alternative is to buy volatility directly rather than through a strike.

- **Options on a volatility index.** A call on the VIX pays on the volatility spike itself, with no dependence on where the index finishes. It is the purest expression, and it is priced accordingly, since everyone knows what it is for. What the index is and how it is built is equity volatility.
- **Variance swaps.** These pay the difference between realised and implied variance with no strike and no delta hedging to run, and because they pay on *squared* returns they are naturally convex in a crash. See variance swaps.
- **Longer-dated wings.** Out to about six months, pushing a fixed out-of-the-money strike further out in maturity increases volga, so the option gains more as volatility rises, at the price of more premium and slower response. It turns over sooner than people expect: at a strike 20% out with 30% volatility the volga runs 43 at three months, peaks near 53 at six, then falls to 47 at one year and 33 at two, because the option stops being far out of the money measured in standard deviations. Hold the delta fixed rather than the strike and the effect is monotone.

None of these escapes the basic economics. Every one of them is bought from a dealer who knows the same thing you do, and the carry is the price of the convexity.

REAL INTERVIEW QUESTION (Oddo · Volatility)

Which option strategies let you play a rise or a fall in implied volatility?

WHAT THEY TEST / ANSWER

Separate the pure expressions from the ones that come with baggage. **Long implied volatility**: buy a straddle or a strangle and delta-hedge it, so the directional exposure is removed and what is left is vega plus gamma; buy calls on a volatility index for the level without any spot dependence; or buy a variance swap, which is the cleanest since it has no strike and no hedging programme. **Short implied volatility**: the mirror of each, and here you must say the risk out loud, because a short straddle or a short variance swap has an unbounded loss and its worst day is the day everything else is going wrong too. Then add the distinction that gets the follow-up right: a delta-hedged straddle is a bet on *realised* volatility exceeding implied, paid day by day through gamma, whereas a volatility-index call or an untouched straddle is a bet on *implied* volatility rising. Those are different trades and they can win and lose in opposite directions in the same week, which is the point of implied versus realized volatility. Finish with strike selection: for a view on the volatility *level* you want at-the-money, where vega is largest, and for a view on a volatility *spike* you want the wings, where volga is.

Dynamic protection, and why it can fail

Instead of buying an option, you can manufacture one by trading. Sell as the market falls, buy back as it recovers, and you replicate a put. That is the idea behind portfolio insurance and, in its modern form, constant proportion portfolio insurance.

REAL INTERVIEW QUESTION (Allianz Global Investors · Market Risk)

Have you heard of CPPI (Constant Proportion Portfolio Insurance)?

WHAT THEY TEST / ANSWER

Yes. Give the mechanism in one line and then spend the answer on the part that belongs to a tail-risk conversation. **Mechanism**: a floor, a cushion above it, and a fixed multiple of that cushion held in the risky asset, so the rule mechanically sells risk as markets fall and buys as they rise; the floor, cushion and multiplier are set out in capital-protected products. **Failure mode**, which is the tail-risk point: it is dynamic replication, so it works only if you can trade continuously. A gap through the floor leaves the strategy short of the guarantee with no cushion left, which is exactly the gap risk an option seller charges you for. Worse, the rule sells into falling markets, so when enough capital follows the same rule the hedging flow amplifies the move it is reacting to, which is the mechanism that turned portfolio insurance into an accelerant in 1987. Close with the comparison the interviewer wants: a bought put has a known cost and no gap risk, and CPPI has no upfront cost and gap risk, so the choice is whether you would rather pay a premium or carry the tail yourself.

Sizing: by scenario, never by delta

The last discipline is sizing, and delta is the wrong tool for it. A three-month 90-strike put has a delta of about -0.18, so hedging a portfolio's delta with it would demand an enormous notional and an unpayable premium. The correct method is to size on the *scenario*: pick the loss you cannot tolerate, work out what the hedge pays in that scenario including the volatility and skew moves that accompany it, and buy enough to cover the gap.

That method also makes the trade-off explicit, because it produces two numbers at once: the annual carry and the scenario payoff. Presenting both, rather than either alone, is what a risk committee needs and what a client deserves. The formal version is stress testing.

INTERVIEW TRAP

Four errors. (1) Pricing protection at the at-the-money volatility: the strike you want is where the skew is steepest, and the flat-volatility price can be half the real one. (2) Funding a tail hedge by selling a further-out tail, which cancels the protection precisely where it matters. (3) Sizing the hedge on intrinsic value, when most of a tail hedge's payoff in the first weeks of a crisis is time value created by the volatility spike. (4) Presenting the carry without the scenario payoff, or the scenario payoff without the carry: either number alone is a sales pitch rather than an analysis.

VISUAL / CODE TO BUILD: Interactive

A protection cost-benefit calculator. The reader sets a portfolio size, a strike as a percentage out of the money, a maturity and a roll frequency; the widget shows the annual carry drawn as a downward staircase against the portfolio's return, and beside it a scenario panel giving what the hedge pays for a range of market falls, with the volatility and skew moves that historically accompany each fall applied automatically rather than held fixed. A toggle switches the structure between an outright put, a put spread and a collar so the reader can watch the put spread's payoff line flatten exactly where the outright's keeps rising. The intended realisation, made unavoidable by putting both numbers on one screen: the structures that reduce the carry the most are the ones that reduce the payoff the most, and the choice is a preference, not an optimisation.

Where this goes next

The mirror trade, being paid to carry the tail, is what range-bound strategies and every short-volatility structure amount to, and the reason those tails are priced the way they are is the volatility smile and skew. Buying the convexity directly is variance swaps. What actually happens to a book when the tail event arrives is Greeks across market regimes, and the packaged retail version of the same idea is capital-protected products.

17.7 Strategy Construction with Ratios

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Every structure so far has held one option against one option. Break that symmetry, sell two against one, and two things happen at once: the structure gets cheaper or free, and a tail that was bounded becomes unbounded. Ratios are the most powerful lever in payoff design and the one that has caused the most damage, and the two facts are the same fact.

The lever and its price

KEY IDEA

A ratio buys **cheapness with tail risk**. Selling more options than you buy raises more premium, which is why ratio structures can be quoted at zero cost or better. But beyond the short strike the extra short legs are naked, so the payoff turns over and keeps going. The rule for reading any ratio structure: find where the short legs stop being covered, and ask what happens past that point. If the answer is "it keeps losing", you are looking at a leveraged position, however defensive the name sounds.

The one-by-two

The canonical example. Buy one call at K_1, sell two at a higher K_2.

WORKED EXAMPLE

Spot 100, using the prices from payoff profiles: buy one 100 call at 8.00, sell two 110 calls at 4.00 each. You receive 8.00 and pay 8.00, so the structure is **exactly zero cost**. It is free to put on, which is the whole of its appeal.

Now trace the payoff, which at zero cost is also the profit:

S_T	Payoff	Profit
100	0.0	0.0
105	+5.0	+5.0
110	+10.0	+10.0
115	+5.0	+5.0
120	0.0	0.0
130	−10.0	−10.0

Maximum profit is 10 at exactly 110. Above that the second short call is naked, so the position loses one for one, crosses zero at $2K_2 - K_1 + \text{credit} = 120$, and loses without limit thereafter. A structure that was free to enter and looks bullish is short the market above 120.

REAL INTERVIEW QUESTION (Barclays · Structured Products)

What is a ratio call spread, and what is the purpose of this product?

WHAT THEY TEST / ANSWER

What: a call spread with unequal quantities, typically long one call at a lower strike and short two at a higher one, same expiry. **Purpose**: the extra short leg raises enough premium to bring the structure to zero cost or to a credit, so a client gets participation in a moderate rise without paying anything up front. That is genuinely attractive, and it is why these sell.

Then draw the payoff and name the catch, because the catch is the answer. The profile rises from the lower strike, peaks at the upper strike, and then *turns over*, because beyond the upper strike one of the two short calls is uncovered. With the zero-cost 100/110 structure above, the peak is 10 at a spot of 110, the break-even on the way back down is 120, and above that the loss is unlimited. So the client is expressing a view that the market rises *but not too far*, and if they are right about direction and wrong about magnitude they lose money on a correct call. Close with what a desk watches: the position is short gamma above the upper strike, so the hedge destabilises exactly where the risk is, and the structure is **long** skew, because the two legs you are short sit on the upside, where the equity surface marks volatility *down*. That is worth stating as a contrast, because the accumulator below is the mirror image: a ratio that sells the upside is long skew, a ratio that sells the downside is short skew. The practical consequence is that respecting the skew makes the one-by-two *dearer* to put on, not cheaper, since the calls you are selling are marked below the at-the-money volatility.

Accumulators

The same idea, wrapped for clients who want to buy an asset cheaply. An **accumulator** lets the buyer accumulate a position at a discount to spot over a schedule of dates, with two asymmetries: it usually knocks out if the underlying rallies past a level, and the buyer accumulates at *double* the rate if it falls below the strike.

Stripped to its core, that is long one call and short two puts, which is why the market nickname for it is “I-kill-you-later”.

REAL INTERVIEW QUESTION (CACIB · FX)

What is an accumulator?

WHAT THEY TEST / ANSWER

Describe it as the client sees it, then as the risk manager sees it. **Client view**: a schedule of forward purchases of an asset at a fixed price below the current spot, so they accumulate at a discount. If the market rises past a knock-out level the structure terminates, having delivered a modest gain. If the market falls below the strike, they accumulate at double the size.

Risk view: it is a leveraged short-volatility, short-downside position. Long one unit of upside against short two units of downside, with the upside knocked out and the downside geared. So the client is paid a small discount most of the time and takes a doubled loss in the scenario that hurts.

Then give the two things a desk cares about. The payoff is **path-dependent** through the knock-out and the schedule, so it cannot be valued as a single European option. And the structure sells the wing where the skew is steepest, which makes it look cheap on a flat-volatility model and correctly expensive on a surface. Finish with the reputational note: accumulators caused large, well-publicised client losses in 2008, which is why they attract suitability scrutiny and why the honest pitch leads with the doubling clause rather than the discount.

REAL INTERVIEW QUESTION (BNP · Exo)

On an accumulator, long one call and short two puts, give the volatility sensitivity and the smile sensitivity.

WHAT THEY TEST / ANSWER

Take them separately, since they are different exposures with the same cause.

Volatility: **short vega**. You are long one option and short two, so on any reasonable strike configuration the two short puts dominate and the position loses if implied volatility rises. That is the first answer, and it explains why these structures are sold aggressively when volatility is high and disappear when it is low.

Smile: **short skew**, and this is the more interesting half. The two options you are short sit on the *downside*, where the equity and FX smiles are steepest, so you are short two units of the richest part of the surface against one unit of a cheaper part. The vanna sign agrees, and the reading convention is the one used in skew-based strategies.

The consequence to state: pricing this on a flat volatility badly undervalues what you are short, because it marks the two downside legs at the at-the-money level instead of their real, higher implied volatilities. A desk that quotes accumulators off a flat vol will sell them far too cheap, and will discover the error precisely when the market falls and the skew steepens, since both exposures lose together.

Ratios inside structured products

Ratios are how a structured note manufactures participation greater than 100% without spending more premium. The mechanism is always the same: sell more of something to buy more of something else.

- **Participation multipliers**: buy two call spreads instead of one call, financed by capping the upside further.
- **Leveraged downside**: gear the short put so a given fall costs the investor more than the market's move, which raises far more premium than an ordinary put and funds a higher coupon.
- **Ratio collars**: sell two calls against one bought put, which turns a protective structure into a levered short above the call strike.

REAL INTERVIEW QUESTION (Natixis · Structured Products)

Which option strategies are used inside a structured product to leverage the payoff, for example a call spread?

WHAT THEY TEST / ANSWER

Name the mechanism first: leverage inside a note always comes from **selling more optionality to buy more of the payoff you want**, and the structures are the ways of doing it.

Call spreads are the workhorse: capping the upside costs the investor a tail they rarely value highly and releases premium that buys a higher participation rate on the part they do care about. **Ratio spreads** go further by selling more than one unit of the cap. **Geared downside**, a leveraged put or a down-and-in put, raises the most premium of all and is what funds the headline coupons on autocallables and reverse convertibles. **Digitals** concentrate the payoff at a level rather than spreading it, buying a big number at one point for the price of a small number everywhere.

Then say what the client is paying with in each case, because that is the structuring answer rather than the vocabulary answer: with a call spread they pay the far upside, with a ratio they pay an unbounded far upside, with geared downside they pay a magnified loss in a fall, and with a digital they pay everything just below the level. And add the desk-side point: every one of these leaves the issuer short gamma somewhere, usually near the cap or the barrier, which is where the hedging cost that funds the structure actually comes from.

Reading any ratio structure

A short procedure that works on anything with unequal quantities:

1. **Find the covered region.** Between the strikes, the long leg covers the shorts and the payoff behaves like an ordinary spread.
2. **Find where cover runs out.** Past the short strike, count the naked legs. Two short against one long means one naked, so the payoff slopes down at one for one.
3. **Compute the second break-even.** For a one-by-two it is $2K_2 - K_1$, adjusted by the premium.
4. **Ask what the tail is worth.** Unbounded above for calls, bounded at zero for puts but still large.
5. **Check the Greeks in the tail.** Short gamma beyond the short strike means the hedge is unstable exactly where the loss accumulates.

INTERVIEW TRAP

Four traps. (1) Calling a zero-cost ratio structure risk-free or defensive; it is free precisely because it is short a tail. (2) Quoting only the first break-even, when a one-by-two has a second one on the way back down that is the one that matters. (3) Pricing a ratio off a flat volatility, which systematically undervalues the wing you are doubly short. (4) Being right on direction and losing anyway, which is the characteristic failure of these structures and should be said to the client before they buy rather than after.

VISUAL / CODE TO BUILD: Interactive

A ratio builder. The reader sets strikes and a quantity ratio and the panel draws the payoff with the covered region shaded differently from the naked region, marks both break-evens, and reports the premium as it swings from debit to credit as the ratio is raised. A gearing slider on the short leg runs continuously from 1.0 to 3.0 so the reader watches the peak profit rise while the second break-even marches toward the peak. A Greeks overlay shows gamma flipping sign at the short strike. A final panel prices the same structure on a flat volatility and on a skewed surface side by side, so the undervaluation of the doubly-short wing is a number rather than a warning.

Where this goes next

The skew exposure that ratios concentrate is skew-based strategies, and the short-gamma behaviour in the tail is gamma and convexity. Assembling these into products a client buys is payoff engineering, and the specific errors this section is warning about are collected in common structuring mistakes.

17.8 Leverage control through structure

sales · structuring · trading | core | equity · cross-asset

Leverage in derivatives is not borrowed money. It is the ratio between the exposure you control and the cash you put down, and an option delivers it by construction: a small premium buys the behaviour of a much larger position. The structuring job is to dial that ratio to what the client wants, and every notch you turn it up is paid for somewhere. This section is about where.

Three things people call leverage

The word is used for three different quantities, and conversations go wrong when two people mean different ones. Say which you mean.

1. **Premium leverage**, or notional gearing: the notional you control per unit of premium. Buy a call for 4 on a spot of 100 and you control 100 of notional for 4, a gearing of 25.
2. **Elasticity**, written Ω: the percentage change in the option's value per percentage change in the underlying,
$$\Omega = \frac{\Delta\, S}{V}.$$
This is the honest measure of how the position *behaves today*, and it is always smaller than the notional gearing because delta is below one.

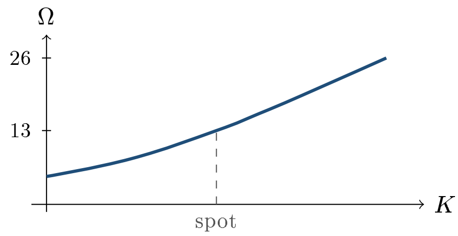
3. **Participation rate**: the structured-products meaning. A note offering 87% participation pays 87% of the index's rise. Here leverage is a term-sheet number that has to be manufactured out of a budget, and manufacturing it is most of the structuring job.

WORKED EXAMPLE

Spot 100, three months, 20% volatility, zero rates. Elasticity by strike:

Strike	Premium	Delta	Elasticity Ω
80	20.04	0.989	4.9
90	10.71	0.865	8.1
100	3.99	0.520	13.0
105	2.06	0.331	16.0
110	0.95	0.183	19.2
120	0.15	0.038	25.9

The at-the-money call moves 13% for every 1% the stock moves. Push the strike out to 120 and that becomes 26%, but you have also cut the probability of ever collecting it. Elasticity and probability trade off against each other along the strike axis, and there is no strike that improves both.



elasticity of a three-month call against strike (spot 100, 20% volatility); the underlying itself has $\Omega = 1$

KEY IDEA

Elasticity is not a constant of the position, it decays as the option works. A one-year at-the-money call has $\Omega \approx 6.8$; the same call once the stock has risen 30% has $\Omega \approx 3.9$, and about 3.3 after a 40% rise. Deep in the money Ω tends to $S/(S-K)$, so it falls through 2 when the stock has doubled and approaches 1 from above thereafter. Leverage is strongest before you are right and weakest after, which is why a leveraged position that has worked is no longer the position you put on and usually needs restriking.

Where participation comes from

A capital-protected note is two instruments in a wrapper. Part of the money buys a zero-coupon bond that matures at the protected amount; whatever is left is the option budget. So participation is a division:

$$\text{option budget} \;=\; 1 - \text{ZC price}, \qquad \text{participation} \;=\; \frac{\text{option budget} - \text{fees}}{\text{option price}}.$$

Everything a structurer does to a term sheet is an attempt to raise the numerator or lower the denominator.

The immediate consequence, and one of the most-asked structuring points, is that **participation depends on interest rates**. High rates make the zero-coupon bond cheap, which leaves a large option budget; near-zero rates leave almost nothing to spend, which is precisely why capital-guaranteed products nearly disappeared during the zero-rate years and came back when rates rose. Nothing about the option changed.

> **REAL INTERVIEW QUESTION** (Société Générale · Structured Products)
>
> *With rates at 5%, a call at 25% and a five-year maturity, how do you build the payoff of a capital-guaranteed product (zero-coupon plus call)?*
>
> **WHAT THEY TEST / ANSWER**
>
> Start by resolving an ambiguity the question creates on purpose. "A call at 25%" alongside "a rate at 5%" most naturally reads as the call's *volatility*, but it can also be read as its price in percent of notional, and the two give different answers. Say which reading you are taking and give both, because a candidate who does that has answered the question either way.
>
> The part that does not depend on the reading: to guarantee 100 at five years you buy a zero-coupon bond worth $100/1.05^5 = 78.35$, so with 100 invested you have 21.65 of option budget. Participation is that budget divided by the cost of one unit of the option.
>
> **Reading it as a price** of 25 per 100 of notional: participation is $21.65/25 = 86.6\%$. **Reading it as a 25% volatility**: the at-the-money call over five years costs roughly $0.4\,\sigma\sqrt{T} = 0.4 \times 0.25 \times \sqrt{5} = 22.4$ of notional by the standard approximation, giving participation of about 97%. Price it properly at a 5% rate with the strike at spot and the forward is well above the strike, so the call costs about 32.5 and participation falls to about 67%. Quote the approximation, then say that the exact answer depends on whether the strike is set at spot or at the forward, which is the follow-up.
>
> State the compounding convention too: annual here, and a continuously compounded 5% would make the bond 77.88 and the budget 22.12. Then add the three qualifications a structurer would. **Fees**: the issuer's margin and distribution costs come out of the same 21.6, so a realistic term sheet shows perhaps 80% rather than 86%. **Credit**: the guarantee is the issuer's promise, not a collateralised one, so the client is taking bank credit risk, and that is what makes the zero-coupon leg cheaper than a government bond. **Rates**: at 1% instead of 5% the zero-coupon costs 95.1 and the budget collapses to 4.9, giving under 20% participation, which is the reason these products live and die with the rate environment.

The levers, and what each one costs

When the budget does not buy the participation the client wants, there are exactly two moves: find more budget, or find a cheaper option. Each lever belongs to one of the two, and each has a price.

More budget

- **Lower the protection**: guarantee 90% of capital instead of 100%. The zero-coupon leg shrinks proportionally and the budget grows fast. This is the single most powerful lever, and the one that most changes the product's risk.
- **Longer maturity**: more discounting means a cheaper zero-coupon. It works when the extra budget, which grows roughly with rT, outpaces the extra option cost, which grows roughly with $\sqrt{T}$. That is why capital-guaranteed notes are long-dated.
- **Accept issuer credit**: a lower-rated issuer's zero-coupon leg is cheaper. Real, commonly used, and a real risk transfer to the client.

Cheaper option

- **Cap the upside**: replace the call with a call spread. Priced at 20% volatility with zero rates, a five-year at-the-money call costs about 17.7 while a 100/130 call spread costs about 9.0, so capping the payoff at +30% roughly doubles the participation the same budget can buy.
- **Average the payoff**: an Asian option on the final year's observations is cheaper because averaging reduces the effective volatility. The client gives up the best single closing level.
- **Choose a cheaper underlying**: a lower-volatility index, or a basket, whose diversification lowers the option's volatility. A worst-of is cheaper still and is where the real risk hides.
- **Add a barrier or a knock-out**: fewer paths pay, so the option is cheaper. See exotic options.

REAL INTERVIEW QUESTION (Kepler · Structuring)

I have 100 euros to invest over five years, I put 90 euros into a zero-coupon. I want to participate in the market rise but the call costs 20 euros. Propose solutions.

WHAT THEY TEST / ANSWER

Frame it as a budget problem first: the zero-coupon costs 90 and returns 100, so the budget is 10 and the call costs 20, giving 50% participation. Note in passing that the 90 also tells you the rate, since 90 growing to 100 over five years is about 2.1% a year, which is worth saying because it shows you read the numbers rather than the words. Then give solutions sorted by what the client gives up. **Cap the upside**: a 100/130 call spread costs roughly half the outright, taking participation to near 100% at the price of no payoff above +30%. **Reduce the protection**: guaranteeing 90 rather than 100 costs 81 of zero-coupon and leaves 19 of budget, so participation rises to about 95% while the client now risks 10% of capital. **Cheapen the option itself**: average the final year, use a lower-volatility index or a basket, or add a knock-out barrier. **Extend the maturity**, so the zero-coupon discount grows faster than the option premium. The answer they are grading is whether you present these as *trade-offs* rather than as a menu of free improvements: each one takes something away from the client, and a good structurer names what it is before the client discovers it.

REAL INTERVIEW QUESTION (Natixis · Structured Products)

How do you boost the coupon of a structured product, and at the price of what risk?

WHAT THEY TEST / ANSWER

The second clause is the question; anyone can list levers. Organise by what is being sold. **Sell more downside**: lower the barrier's protection or raise the multiplier on the embedded put, which raises the coupon and moves the loss closer and makes it steeper. **Sell the upside**: cap or remove participation entirely, as a reverse convertible does, so the client keeps a coupon and none of the rally. **Sell dispersion**: move from a single underlying to a worst-of on three, which raises the coupon dramatically and is the largest source of hidden risk in the retail market, because the client is now short the correlation between names they chose individually. **Sell path**: add autocall triggers and observation schedules, which cheapen the structure by removing paths that pay. **Sell duration or credit**: extend the maturity, or issue off a weaker balance sheet. Then give the sentence that is the actual answer: a higher coupon is never a better product, it is a bigger short position, and the honest presentation names the scenario in which it bites before the client finds it. The classic failure, covered in common structuring mistakes, is a term sheet whose headline coupon comes from a worst-of barrier the client has never priced.

Leverage on the way down

Everything above raises leverage on the upside. The structuring market also sells leverage on the *downside*, and that is where clients are most often surprised, because it is expressed as a multiplier rather than as a risk.

REAL INTERVIEW QUESTION (Morgan Stanley · Sales)

What is a PDI? What is the difference with a leveraged put?

WHAT THEY TEST / ANSWER

A **PDI**, a put down-and-in, is a put that only comes into existence if the underlying trades through a barrier. Until the barrier is touched the client is short nothing; once it is touched the put springs into existence and stays alive for the rest of the trade, even if the underlying recovers well above the barrier. That is the first thing to get right, because the term sheet's language invites the opposite reading. That discontinuity is the point: it lets the term sheet say the client is protected down to the barrier, while the loss once the barrier breaks is measured from the strike rather than from the barrier, so it arrives all at once. A **leveraged put** has no barrier; it is simply a short put in a multiple, say 1.5 or 2 times the notional, so the loss starts immediately below the strike and accumulates at that multiple. The difference to lead with is where the risk sits. The PDI concentrates it in a jump at one level, so it is short a barrier and therefore very sensitive to skew and to spot sitting near the barrier; the leveraged put spreads the same risk linearly, so it is simply more of an ordinary short put. For a given amount of premium raised, the PDI usually looks better on the term sheet and behaves worse in the scenario it exists to survive, which is exactly what a sales interviewer wants you to say out loud. Both are the funding leg of a coupon, so in either case the first thing to tell the client is what the loss looks like at the barrier and 20% below it, not the coupon. See embedded optionality.

INTERVIEW TRAP

Four errors. (1) Quoting notional gearing when the client is asking how the position behaves: elasticity is the number that answers that, and it is much smaller. (2) Treating elasticity as constant, when it halves as the option goes into the money and the leverage you priced is not the leverage you end up carrying. (3) Presenting the levers as improvements: every one of them sells something, and a term sheet that does not name what was sold is not a description, it is a pitch. (4) Forgetting that the guarantee is an issuer promise. The whole structure rests on bank credit, which is also the reason the zero-coupon leg was cheap enough to leave an option budget in the first place.

VISUAL / CODE TO BUILD: Interactive

A term-sheet builder. The reader sets notional, maturity, rate, protection level, underlying volatility and fees, and the widget computes the zero-coupon cost, the option budget and the resulting participation, updating live. Each lever from this section is a control: a cap slider that turns the call into a call spread, an averaging toggle, a barrier level, a protection-level slider, and an issuer-spread input. Every control displays two numbers at once, the participation gained and the client outcome given up, so the trade-off cannot be hidden. A second panel plots the finished payoff against the index alongside a direct investment in the index, with a scenario table for falls of 10, 20 and 40% so the protection and its limits are visible rather than described. A rate slider makes the headline point on its own: hold everything else fixed, move rates from 1% to 5%, and watch participation roughly quadruple.

Where this goes next

Changing leg ratios rather than strikes is strategy construction with ratios, and the directional structures these are built from are in directional strategies. The products themselves are developed in capital-protected products and participation products, with the funding legs in embedded optionality. The zero-coupon arithmetic that sets the budget is in zero-coupon bonds, and the ways this goes wrong are catalogued in common structuring mistakes.

17.9 Payoff Visualization and Intuition

sales · trading · structuring | core | equity · fx · cross-asset

Standard payoff profiles gave you the named shapes and payoff design gave you the reasons to choose between them. This section is narrower and more mechanical: the technique for drawing *any* payoff, named or not, on a whiteboard in under a minute, and for running the same machine backwards to recognise a structure you have only been shown as a picture or a formula. Behaviour through time belongs to strategy behaviour over time and the Greeks inside a structure to Greeks evolution; here the concern is the picture itself.

The skill is worth isolating because the database asks for it constantly and in both directions. *Draw the payoff of this product from my description* is one question. *Here is a formula, identify it* is the same question inverted, and the same three-step method answers both.

Every payoff is three numbers repeated

An expiry payoff is a piecewise linear function of one variable. It has no curvature anywhere, only corners. So it is completely determined by where the corners are, how steep each straight piece is, and how high the whole thing sits.

KEY IDEA

A payoff diagram is fixed by three things, in this order. **Kinks** sit at the strikes, and nowhere else. **Slopes** change only at a kink. **Level** is set by the net premium, and moves the whole picture up or down without changing its shape. Draw them in that order and you cannot produce an inconsistent diagram.

The second step is the one worth having as a reflex, because it collapses calls and puts into a single rule. Start at $S_T = 0$, at the far left of the picture, and walk right:

- **Starting slope** is the net linear delta of what you hold with no optionality: $+1$ per share or long forward, -1 per short, and -1 for every long put, $+1$ for every short put, because a put is the leg that is alive at zero.
- **At each strike**, the slope changes by $+1$ for every long option and -1 for every short option, *whether it is a call or a put.* Multiply by the quantity of each leg.

That every long option kinks the line upward and every short option kinks it downward, regardless of type, is not a coincidence. It is convexity: being long an option is being long a corner that opens upward, and by put-call parity a call and a put struck at the same place differ only by a forward, which is a straight line and cannot kink anything.

WORKED EXAMPLE

The collar, drawn by the rule. Long the stock at 100, long the 90 put at 4.00, short the 110 call at 4.00, using the price set from standard payoff profiles.

- Starting slope: $+1$ for the stock, -1 for the long put, so **0**. The picture starts flat, which is the floor.
- At 90, a long option: $0 + 1 = \mathbf{1}$. The middle segment tracks the stock one for one.
- At 110, a short option: $1 - 1 = \mathbf{0}$. Flat again, the cap.

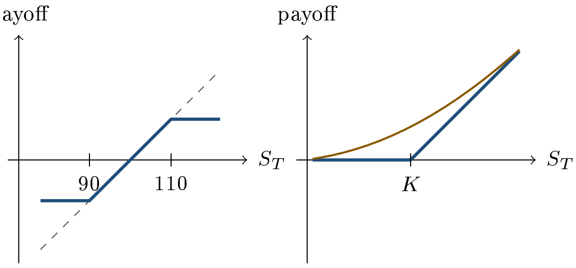
Three numbers, two kinks, and the shape is done before you have thought about a single premium. Then the level: the net premium is $4.00 - 4.00 = 0$, the zero-cost collar, so here the profit line sits exactly on the payoff line. That is a special case and it is worth saying so out loud, because a net premium of zero is the only time the two diagrams coincide. Anywhere a structure carries a net debit or credit, the profit line sits exactly that far below or above the payoff line, and shifting it is the only thing a premium ever does to the picture.

Notice what the rule bought you. You never asked yourself what a collar looks like, and you never needed to have seen one. The same three numbers produce a ratio spread, an eight-leg structured note or something a client invented on the phone.

Payoff or profit: say which

Two different diagrams share the name. The **payoff** is the value of the structure at expiry, ignoring what it cost. The **profit** is the payoff minus the premium paid, or plus the premium received. They are the same shape, shifted vertically, and the shift is the only thing that moves.

This matters because break-evens live on the profit diagram and only there. A long call's payoff is never negative; its profit is negative below $K +$ premium. Read break-evens off the picture as the points where the profit line crosses the axis, rather than solving for them, and you will be faster and less prone to sign errors. Say out loud which of the two you are drawing, because the interviewer cannot see the label in your head.



Left, the collar built by the walking rule: —— is the payoff, flat then slope one then flat, and —— drawn dashed is the underlying alone, the line the middle segment lies on. Right, a long call: —— is the terminal payoff and —— the value curve at an earlier date, above it through the strike and converging onto it in both tails, where there is nothing left to be uncertain about. Both are payoff diagrams, not profit: no premium has been subtracted, and the left panel is measured relative to the spot at inception.

Reading the picture backwards

The inverse skill is turning a formula into a structure, and it needs only a short dictionary. Every payoff written with min and max is options in disguise:

- $\max(S_T - K, 0)$ is a long call struck at K.
- $\max(K - S_T, 0)$ is a long put struck at K.
- $\min(S_T, K) = S_T - \max(S_T - K, 0)$: long the underlying, short the call. A covered call.
- $\max(S_T, K) = K + \max(S_T - K, 0)$: cash K plus a long call. Capital protection with participation.

The mechanical version, when the dictionary does not resolve it in one look, is to split the real line at every strike that appears, evaluate the expression in each region, and read the slopes. A region where the payoff is constant is a region where you own nothing directional; a region of slope one is full participation; a slope of two is a ratio.

REAL INTERVIEW QUESTION (CACIB · Delta One)

What is the payoff of a call? Give its graphical representation, then its mathematical one.

WHAT THEY TEST / ANSWER

Take the order seriously: the question asks for the picture first, and it is asking whether the graph or the formula is the thing you actually think in.

Graphically: flat along the axis at zero for every S_T below the strike, then a straight line at forty-five degrees above it, with the single kink at K. Label the axes, S_T horizontal and payoff vertical, because an unlabelled diagram is the commonest way to lose this question. If you are drawing profit rather than payoff, say so and shift the whole line down by the premium, which puts the break-even at K plus the premium.

Mathematically: $\max(S_T - K, 0)$, also written $(S_T - K)^+$.

Then make the connection explicit, since that is what the question is testing: the $\max(\cdot, 0)$ *is* the kink, and the slope of one above the strike is the delta of one that a deep in-the-money call converges to. The two representations are the same object, and being able to move between them on demand is what lets you identify a structured payoff you have never seen from its term sheet formula alone.

REAL INTERVIEW QUESTION (BNP · Structured Products)

Draw the payoff of a given product from its description.

WHAT THEY TEST / ANSWER

This is the open version of the question, so answer with a method rather than a shape, and then execute it out loud.

First, **find the strikes**. Every percentage in a term sheet, the barrier, the cap, the protection level, the coupon trigger, is a strike, and each one is a kink. Nothing kinks anywhere else. Second, **walk the slopes** from the left: start at the linear exposure, then add one per long option and subtract one per short option at each strike, times the participation rate. Third, **set the level** with the premium, or with the note's issue price if it is a note rather than a package of options.

Two things to volunteer, because they separate a drawing from an understanding. Say whether you are drawing payoff or profit. And say which kinks are *discontinuous* rather than merely sharp: a digital coupon or a European barrier is a jump, not a corner, and drawing it as a corner is the error the question is usually set to catch, since the hedging problem at a jump is a completely different one, treated in replication strategies.

Finish by naming the decomposition rather than only drawing it: "so this is a zero-coupon plus a call spread", which tells the desk what to buy.

REAL INTERVIEW QUESTION (Barclays · Structuring)

Identify this payoff: $\max(\min(S_T, K_2), K_1)$. *How would you hedge such a product?*

WHAT THEY TEST / ANSWER

Work from the inside out, with $K_1 < K_2$, and split the line at the two strikes.

- $S_T < K_1$: $\min(S_T, K_2) = S_T$, and $\max(S_T, K_1) = K_1$. The payoff is K_1, flat.
- $K_1 \leq S_T \leq K_2$: the payoff is S_T, slope one.
- $S_T > K_2$: $\min(S_T, K_2) = K_2$, and the payoff is K_2, flat again.

So it is **floored at K_1, capped at K_2 and fully participating between them**: a collar payoff, and read as a note, a capital-protected product with protection at K_1 and participation capped at K_2.

The algebraic decomposition falls out of the same three regions:

$$\max(\min(S_T, K_2), K_1) = K_1 + \max(S_T - K_1, 0) - \max(S_T - K_2, 0),$$

which you can verify region by region. Cash K_1 plus a K_1/K_2 call spread.

For the hedge, say that the identity *is* the hedge, since you buy exactly what the right-hand side names and hold it, and then hand the follow-up to payoff engineering, which is where turning a desired payoff into tradeable legs and running the resulting book belong.

The line at expiry and the curve before it

Everything above draws the payoff at maturity. Before maturity the position is worth something else, and the second diagram, the value curve at a date t, is a smoothed version of the first sitting above it.

INTUITION

The terminal payoff is what the structure pays. The value curve is what someone will pay you for it today. The vertical gap between them is time value, the curvature of the smoothing is where your gamma is, and the whole curve sagging toward the kinked line as expiry approaches is theta. Three Greeks visible in one picture, which is why the drawing is worth more than the formula when explaining a position to someone who does not price options.

Drawing rules for the curve: it is above the payoff line for a long option position, it is smooth where the payoff has a corner, it hugs the payoff far in and far out of the money where there is nothing left to be uncertain about, and the gap is widest near the strike. Curvature and gap are the same fact seen twice, which is the visual form of the gamma and theta relationship developed in gamma and convexity.

REAL INTERVIEW QUESTION (Société Générale · EQD)

The price of an American call and a European call, with and without dividends: draw the payoff curves at a date t.

WHAT THEY TEST / ANSWER

Four curves, and the interesting content is which of them coincide.

No dividends. Draw the kinked payoff, then the value curve above it in the usual smoothed shape. The American and the European curve are the **same line**: early exercise on a call paying no dividend is never optimal, since exercising throws away the time value you can see as the gap between the two lines, so the American right is worth nothing extra. Say it that way, pointing at the gap, because the picture makes the argument. The result itself belongs to options.

With dividends. The European curve moves *down*, because the forward the option is written on is lower by the discounted dividends. The American curve now separates from it and sits above, because exercising just before an ex-dividend date can be optimal: you give up the remaining time value and collect the dividend instead, and if the dividend is large enough that trade is worth making. So the American curve is bounded below by intrinsic value everywhere, and near a large dividend it can be pushed right down onto the intrinsic line, which is exactly where exercise happens.

The order to state at the end, since it is what the question is checking: without dividends $C_{\text{Am}} = C_{\text{Eu}} >$ intrinsic, and with dividends $C_{\text{Am}} \geq C_{\text{Eu}}$, both still above the payoff line except at the exercise boundary where the American touches it.

REAL INTERVIEW QUESTION (BRED · FX)

The EURUSD forward is at 1.16 and spot is at 1.13. Can you draw the payoff at maturity across different future spot levels?

WHAT THEY TEST / ANSWER

A long EURUSD forward pays $S_T - 1.16$ per euro of notional, so the diagram is a single straight line of slope one, no kinks, and the whole question is where it crosses zero.

It crosses at **1.16, the forward, not 1.13, the spot**. That is the answer being fished for. If the spot is unchanged at maturity, at 1.13, the long forward has lost $1.13 - 1.16 = -0.03$ per euro, three big figures, despite nothing having happened to the market. At 1.20 it makes 0.04, at 1.10 it loses 0.06.

Then explain why the two differ, because the number is not arbitrary. By covered interest parity the forward premium is the rate differential: $F/S = 1.16/1.13 = 1.0265$, so $\ln(F/S) = 2.62\%$ over the tenor. The euro trades at a forward premium of that much, which means **dollar rates are above euro rates** by 2.62% annualised for a one-year tenor. The three big figures you appear to lose by being long the forward at unchanged spot are the interest you gave up by holding euros instead of dollars, so it is not a loss at all once the funding leg is counted.

Close on the practical version: this is why a carry position and a directional position look identical on a payoff diagram and are completely different trades, and why an FX diagram should always be drawn against the forward.

INTERVIEW TRAP

1. **Not saying whether it is payoff or profit.** The two diagrams differ by a vertical shift, and every break-even you quote is wrong if the interviewer assumed the other one.
2. **Drawing a price path.** The horizontal axis is the terminal underlying level, not time. Under pressure people draw a wiggly line through their payoff diagram; it belongs on a different chart, and path-dependent products are the subject of path dependency, not of this one.
3. **Corners where there are jumps.** A digital, a European barrier or a trigger coupon is discontinuous. Drawing it as a kink hides the hedging problem that makes it interesting.
4. **Slopes not drawn to scale.** If a structure participates at two times on the downside and one on the upside, the two segments must visibly differ, or the picture asserts something false about the position. Ratios are covered in strategy construction with ratios.
5. **Assuming the value curve is always above the payoff.** It usually is, but a deep in-the-money European put can trade below intrinsic, because exercise is unavailable and the discounting of the strike works against you. Drawing that one wrong is a good way to reveal you memorised the picture rather than the reason for it.

VISUAL / CODE TO BUILD: Interactive

A two-pane whiteboard trainer. The left pane accepts a term sheet in plain text, the kind a structurer would actually be handed, and the right pane is blank canvas where the reader draws the payoff with the mouse. On submit, the correct payoff is overlaid on the attempt and the differences are called out in the vocabulary of this section: a missing kink, a wrong slope in one region, the right shape at the wrong level, or a corner drawn where the product has a jump. A slider for time to expiry morphs the correct answer between the terminal line and the value curve, so the reader can see the corners round off and the gap close as expiry approaches. Scoring is by region rather than by pixel, since the point is the three numbers and not the draftsmanship.

17.10 Strategy behavior over time

sales · trading · structuring | core | equity · fx · cross-asset

Every payoff diagram in this chapter is a photograph of the last instant of an option's life. It is the right picture for explaining what a structure *is*, and the wrong one for understanding what owning it feels like, because a trader spends the entire position in the frames before it. Those earlier frames are smooth where the final one has kinks, they are worth much less than the final one at the points that matter, and they change shape at a rate that accelerates toward the end. A candidate who can describe that movie, rather than only its final frame, is describing the job.

Throughout: spot 100, 20% volatility, no rates or dividends.

Time value goes as the square root of time

The single most useful fact about decay is that an at-the-money option's time value scales with $\sqrt{T}$. It follows from the standard at-the-money approximation $0.4\,\sigma S\sqrt{T}$, which comes from expanding Black-Scholes around the money and in which T appears only under the square root:

Remaining life	1 year	6 months	3 months	1 month	1 week	1 day
ATM call value	7.97	5.64	3.99	2.30	1.11	0.42

Read the row as fractions of the starting value rather than as prices. With three quarters of the life gone the option still holds *half* its original time value, because $\sqrt{1/4} = 1/2$. Turn that around and you have the rule that governs every long-option strategy: half of what you paid is lost in the final quarter of the life, and a quarter of it in the final sixteenth.

KEY IDEA

Decay is not linear, it is back-loaded, so the calendar position of your option matters as much as its strike. Owning optionality is cheap per day early and expensive per day late; a long strategy that needs time should buy it further out and roll, and a short strategy that wants to collect decay should sit in the last third. The Greek behind this, $\Theta \propto 1/\sqrt{T}$ at the money, is developed in Greeks under time decay; here it is a scheduling rule.

Strategies converge to their payoff, and mostly at the end

The corollary that surprises people is that being *right* does not pay until late. A structure's value approaches its expiry payoff slowly, then all at once.

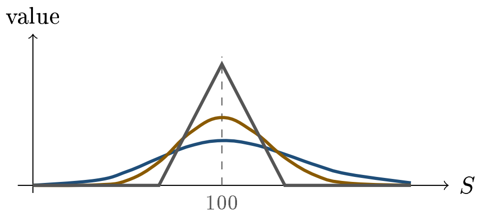
WORKED EXAMPLE

The 90/100/110 butterfly of range-bound strategies, with spot sitting exactly at 100 for its whole life, that is with the view working perfectly from day one:

Remaining life	3 months	2 months	1 month	1 week	1 day	expiry
Value	3.69	4.36	5.59	7.79	9.16	10.00

It was bought for 3.69. After a third of its life with the view perfectly correct it is worth 4.36, a gain of 18% on a trade that is being right every single day. Seventy percent of the total profit arrives in the final month, and a third of it in the final week.

The 100/110 call spread behaves the same way. With spot at 110, which is the best possible outcome, it is worth 6.57 with three months to go, 7.59 with one month, 8.78 with one week, and only 10.00 at expiry. You cannot collect the maximum by being right; you can only collect it by being right *and waiting*.



3 months to expiry — 1 month — at expiry (90/100/110 butterfly)

The picture is the section in one image. Early in the life the butterfly is a low, wide hill: it barely distinguishes between a spot of 95 and a spot of 105, and it is worth a fraction of its maximum even at the peak. As time passes the hill narrows and grows, until at expiry it is the tent of the payoff diagram. Two practical readings follow. Early on, the position is mostly a volatility trade and hardly a view on the level at all. Late on, it is almost pure level, and a three-point move in spot in the final week costs about 1.2, more than the 1.1 that a ten-point move costs with three months left to run.

When to take the money

That convergence profile creates a real decision, and interviewers like it because there is no formula for it.

- **Holding to expiry** collects the whole payoff but takes the whole risk of the final weeks, which is exactly when the position is most sensitive to spot and least forgiving.
- **Monetising early** banks a discount to the maximum but removes a gamma exposure that is about to become severe. On the butterfly above, closing with one month left takes 5.59 of the 10 while eliminating the period in which seventy percent of the total profit, and essentially all of the risk, live.

The useful framing is the one a desk actually uses: ask what you are being paid for the remaining time. Holding the butterfly for the last month earns 4.41 and requires you to be right about a narrow range for four more weeks. If you would not put that trade on fresh today, at that price, you should not be holding it either.

Time value can be negative

One case breaks the usual pattern, and it is asked precisely because it does.

REAL INTERVIEW QUESTION (BNP · EQD)

Define a deep in-the-money put and explain how the put's time value evolves.

WHAT THEY TEST / ANSWER

A deep in-the-money put is one whose strike sits far above spot, so it is nearly certain to be exercised and its delta is close to -1. The counter-intuitive part is that its *time value can be negative*, meaning the European put trades *below* its intrinsic value $K - S$. The reason is discounting, and it is worth stating cleanly: exercising delivers K at maturity, not today, so the most a certain payoff can be worth now is $Ke^{-rT} - S$, which is less than $K - S$ whenever rates are positive. So as expiry approaches the put's value *rises* toward intrinsic rather than decaying toward it, and its theta is *positive*. Two additions finish it. First, this is a European phenomenon: an American put can be exercised immediately, so it is never worth less than $K - S$, and the possibility of early exercise is exactly what that extra value buys, which is why deep in-the-money American puts are exercised early and deep in-the-money American calls on a non-dividend-paying stock are not. Second, note that at zero rates the effect disappears, which is a good way to show you have identified the true driver rather than memorised a special case.

The last week

Two things happen in the final days that no payoff diagram shows.

Gamma concentrates. At-the-money gamma grows as $1/\sqrt{T}$ while the range of strikes carrying any gamma at all shrinks like $\sigma\sqrt{T}$. So the same book's curvature piles into a narrower and narrower band around spot. A structure that spent months behaving smoothly starts behaving like a switch.

Pin risk appears. If the underlying finishes near a strike, the position's delta is neither zero nor one but genuinely unknown until settlement, because whether the option is exercised depends on the closing print. Anyone short options at that strike can be assigned or not, and finds out afterwards. Desks manage this by flattening around large open interest before expiry rather than by hedging it, because there is nothing to hedge: the exposure is discontinuous.

REAL INTERVIEW QUESTION (Goldman Sachs · Structuring)

How do you describe the evolution of gamma through time? Why? How do you explain it mathematically?

WHAT THEY TEST / ANSWER

Answer in three registers, because the question explicitly asks for all three. **Description**: an at-the-money option's gamma grows without bound as expiry approaches, while an option away from the money sees its gamma rise, peak, and then collapse to zero. **Why**: gamma measures how fast delta changes, and at expiry the delta of a call becomes a step function, zero below the strike and one above. Approaching a step means the slope at the step goes to infinity and the slope everywhere else goes to zero, which is exactly the description above. **Mathematically**: $\Gamma = \varphi(d_1)/(S\sigma\sqrt{T})$, so at the money the denominator drives it and $\Gamma \propto 1/\sqrt{T}$, while away from the money d_1 grows in magnitude like $1/\sqrt{T}$ and $\varphi(d_1)$ decays like $e^{-d_1^2/2}$, so the exponential wins and gamma goes to zero. Finish with the consequence for a strategy rather than for a single option, since this is a structuring interview: the width of the region carrying gamma shrinks like $\sigma\sqrt{T}$, so a spread or a butterfly that behaved smoothly for months turns into a near-discontinuous position in its final days, which is why books are flattened around large strikes before expiry instead of hedged through them.

INTERVIEW TRAP

Four errors. (1) Explaining a structure with its expiry diagram and then being surprised that a correct view shows little profit for weeks: convergence is back-loaded, and on the butterfly above more than half the profit arrives in the last month. (2) Assuming decay is linear per day, when it goes as $\sqrt{T}$ and half of an at-the-money premium is lost in the final quarter of the life. (3) Applying "theta accelerates near expiry" to out-of-the-money options, whose theta goes to zero because there is nothing left to lose. (4) Holding a winning short-dated structure to expiry out of habit, without asking whether you would open that same trade today at today's price for the remaining time.

VISUAL / CODE TO BUILD: Interactive

A time-lapse of any structure the reader builds. A single slider runs from inception to expiry and the value curve morphs continuously from smooth to kinked, with the expiry payoff drawn behind it as a fixed reference so the convergence is visible rather than remembered. A second panel plots the position's value through time along a chosen spot path, next to a straight line from today's price to the final payoff, so the back-loading of the profit is explicit. A third readout gives "profit realised so far" against "profit remaining" at every point, which turns the monetise-or-hold decision into a number: it shows what you are being paid for the remaining days and lets the reader compare that against opening the same trade fresh.

Where this goes next

The Greek-by-Greek version of this section is Greeks under time decay, and how each Greek moves inside a multi-leg structure specifically is Greeks evolution within strategies. The economics of paying for time, and the break-even move that justifies it, are in theta as the cost of convexity. The diagrams themselves, and how to read and draw them properly, are in payoff visualization and intuition, and what happens when the path rather than the endpoint determines the payoff is path dependency and scenario analysis.

17.11 Greeks evolution within strategies

trading · structuring · sales core equity · cross-asset

A single option has Greeks that keep their sign for its whole life: a long call is long delta, long gamma, long vega and short theta from the day it is bought to the day it expires. The moment you put two legs together, that stops being true. A call spread is long gamma on one side of the market and short gamma on the other, and it crosses from one to the other while you hold it, without you trading anything. This is one of the most-tested ideas in options-strategy interviews, and it is where candidates who have only memorised the payoff diagrams come apart.

The definitions of the Greeks themselves live in the Greeks chapter, and how each one behaves for a single option is developed in Greeks under spot moves and Greeks under time decay. This section takes them as known and asks a different question: what does a multi-leg structure look like in risk, where do its Greeks change sign, and how does that geography move as time passes.

The one mechanical fact: Greeks add

Every Greek is a derivative of price, and differentiation is linear, so the Greek of a portfolio is the weighted sum of the Greeks of its legs:

$$\Delta_{\text{strategy}} = \sum_i w_i \Delta_i, \qquad \Gamma_{\text{strategy}} = \sum_i w_i \Gamma_i,$$

and the same for vega and theta, with w_i the signed number of each option ($+1$ for a leg you own, -1 for a leg you are short). There is no interaction term and no correction: the risk of a structure is arithmetic on the risk of its pieces.

That sounds trivial and it is the whole engine. Everything interesting comes from one extra observation: each leg's Greeks are peaked at a *different* strike. Gamma, vega and theta of an option are all concentrated near its own strike and vanish far away from it, so when you subtract two legs with different strikes you are subtracting two bumps sitting at different places. The difference of two bumps is positive where the first dominates and negative where the second does. The sign flip is not a subtlety of the model. It is what subtracting two shifted bumps always looks like.

INTUITION

Picture each option's gamma as a small hill centred on its strike. A long 100 call is a hill at 100. A short 110 call is a *hole* at 110. Stand at 95 and you are on the hill: long gamma. Walk up to 115 and you are in the hole: short gamma. Somewhere in between, on the slope, you are flat. You did not change your position, you changed where you were standing.

The call spread, dissected

Take the workhorse: long one 100 call, short one 110 call, three months to expiry, volatility 20%, and (stated explicitly, as always) ignoring rates and dividends, with the same volatility used for both strikes.

WORKED EXAMPLE

Greeks of the 100/110 call spread, spot varying. Vega is per one volatility point, theta is per calendar day, both for one unit of underlying.

Spot	Price	Delta	Gamma	Vega	Theta
90	0.63	0.133	+0.020	+0.082	−0.009
95	1.56	0.243	+0.022	+0.101	−0.011
100	3.03	0.337	+0.013	+0.067	−0.007
104.36	4.59	**0.366**	**0.000**	**0.000**	**0.000**
110	6.57	0.322	−0.014	−0.086	+0.009
115	7.97	0.237	−0.019	−0.123	+0.013
120	8.93	0.148	−0.016	−0.115	+0.013

Read the columns, not the rows. Delta rises, peaks at 0.366, then falls back towards zero: it never reaches 1, because the two legs cancel at high spot. Gamma, vega and theta all change sign, and they all change sign *at the same spot*, 104.36. That coincidence is not luck, and the next box explains it.

Three things in that table are worth saying out loud in an interview.

First, **delta is a hump, not a staircase**. Far below both strikes the spread is worth nothing and moves with nothing. Far above both strikes it is worth its maximum, $K_2 - K_1 = 10$, and again moves with nothing, because it cannot be worth more. A position whose value is bounded above and below *must* have a delta that goes to zero at both ends. The maximum delta of a call spread is always well under 1 (here 0.366), which is the honest answer to "how much market exposure does this give me".

Second, **the peak of delta is exactly where gamma crosses zero**, because gamma is the slope of delta. Those are not two facts, they are one fact stated twice.

Third, the crossing point has a closed form. Gamma flips when the two legs' gammas are equal, and for two calls on the same underlying with the same expiry and the same volatility that happens at

$$S^* \;=\; \sqrt{K_1 K_2}\, e^{-\sigma^2 T/2}.$$

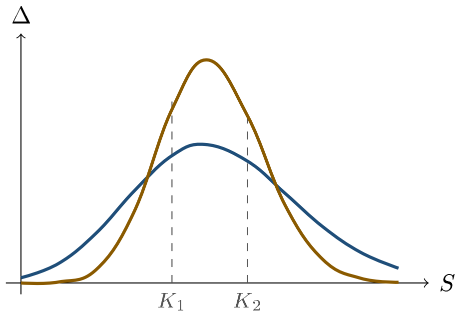
With $K_1 = 100$, $K_2 = 110$, $\sigma = 20\%$ and $T = 0.25$ this gives $S^* = \sqrt{11000}\, e^{-0.005} = 104.88 \times 0.9950 = 104.36$, which is exactly where the table turns. The useful mental version is simpler: *a bit below the geometric midpoint of the two strikes*, and the midpoint is close enough to the arithmetic average (105 here) that you can quote 105 in a live interview and then refine.

KEY IDEA

For a structure whose legs share one expiry and one volatility, with rates and dividends ignored, each leg satisfies $\mathcal{V}_i = S^2\sigma T\,\Gamma_i$ and $\Theta_i = -\frac{1}{2}\sigma^2 S^2\Gamma_i$. The constants do not depend on the strike, so summing over legs gives

$$\mathcal{V}_{\text{strategy}} = S^2\sigma T\,\Gamma_{\text{strategy}}, \qquad \Theta_{\text{strategy}} = -\tfrac{1}{2}\sigma^2 S^2\,\Gamma_{\text{strategy}}.$$

These are per unit of volatility and per year; divide by 100 for the per-point vega and by 365 for the per-day theta quoted in the table above. Gamma, vega and theta of a single-expiry structure are therefore the *same curve* up to a positive constant and a sign. They peak together, they cross zero together, and long gamma always means long vega and short theta. This is why one picture answers three interview questions, and it is also why the only way to break the link is to use more than one expiry.



delta, three months to expiry — delta, one month to expiry

100/110 call spread, volatility 20%, rates and dividends ignored

REAL INTERVIEW QUESTION (Société Générale · Structured Products)

What is the delta of a bull spread when the underlying is well above K_1?

WHAT THEY TEST / ANSWER

They want to hear that it *falls back towards zero*, and to see whether you know why. Well above K_1 but still below K_2 the long leg's delta is approaching 1 while the short leg is still small, so the spread's delta is near its maximum. Push higher, above K_2, and both legs approach delta 1, so the difference collapses: the position is worth its cap of $K_2 - K_1$ and can no longer gain from a rally. State the shape in one sentence, "delta is a hump that peaks a little below the midpoint of the strikes and decays to zero on both sides", then give the peak's location, $\sqrt{K_1K_2}\,e^{-\sigma^2T/2}$, and the fact that the maximum is well below 1. The trap in the follow-up is to ask what happens to that delta as expiry approaches: the hump gets taller and narrower, and right at expiry it becomes a spike, which is the pin risk a trader actually loses sleep over.

REAL INTERVIEW QUESTION (Société Générale · EQD)

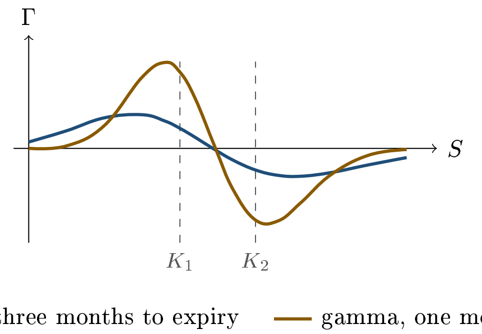
Describe the vega of a call spread: explain the curves, justifying their shape both mathematically and financially.

WHAT THEY TEST / ANSWER

Two justifications, exactly as asked. **Mathematically**: vega is $\mathcal{V} = S\,\varphi(d_1)\sqrt{T}$ per leg, a bump centred near each leg's strike, so the spread's vega is the difference of two bumps and changes sign where they cross, at $S^* = \sqrt{K_1 K_2}\,e^{-\sigma^2 T/2}$. Below that the long lower-strike leg dominates and vega is positive; above it the short upper-strike leg dominates and vega is negative. **Financially**: below the midpoint you own the option that has something to gain from a wider distribution, so more volatility helps you. Above it you are already close to your cap of $K_2 - K_1$, so extra volatility can only drag you back below the cap: it is pure downside, and you are short vega. Finish with the two consequences that show desk sense. The structure is *vega-neutral overall by construction* at one spot only, so calling a call spread "a low-vega trade" is only true near that point. And because the volatility used for the two strikes is not the same in reality, the honest sensitivity of a call spread is to *skew* rather than to a parallel volatility move, which is developed in skew and smile sensitivities and skew trading.

How the profiles sharpen with time

Everything above was a snapshot. Now let time pass, with spot unchanged. Each leg's gamma bump narrows and grows taller as T falls, because the option's remaining uncertainty shrinks around its strike. The consequence for a structure is dramatic: the same position that had a gentle, smeared risk profile at three months becomes a set of sharp spikes of alternating sign in the last week.



same 100/110 call spread; vega and theta are this curve rescaled, per the key idea above

Two numbers make the point. At three months the spread's gamma peaks at about +0.023 near spot 93.5 and bottoms at about −0.019 near 115. At one month those become roughly +0.057 near 98 and −0.050 near 111.5, about two and a half times as large, and they sit much closer to the strikes. The crossing point barely moves, from 104.36 to 104.71, because $e^{-\sigma^2 T/2}$ is close to 1 for any realistic $\sigma^2 T$: the *geography* of the position is stable, its *intensity* is not.

REAL INTERVIEW QUESTION (JP Morgan · Delta One)

Plot your delta one year before expiry, then six months before, and so on, then do the same with gamma.

WHAT THEY TEST / ANSWER

This is a drawing question, and they are watching the *sequence* of drawings, not any single one. Draw the family, one curve per maturity, on the same axes, and narrate the limit. For delta on a long call: a gentle S-curve at one year, steeper at six months, and converging to the step function $\mathbf{1}_{\{S>K\}}$ at expiry. For gamma: a broad low bump at one year, taller and narrower at six months, tending to a spike at the strike, since gamma is the derivative of delta and the derivative of a step is a point mass. Say the conserved quantity out loud, that the area under the gamma curve is essentially fixed while the curve narrows, because that is the sentence that shows you understand rather than remember. Then earn the follow-up by extending it to a structure: on a call spread you draw two families of opposite sign, so what converges is a positive spike at K_1 and a negative spike at K_2, which is exactly why the last days before expiry are the dangerous ones and why desks reduce size or roll before then. The single-option version of this is developed in Greeks under time decay, and what it does to the strategy's profit and loss over its life is in strategy behavior over time.

Straddles and butterflies: the volatility structures

A straddle (long a call and a put at the same strike) stacks two gamma bumps at the same place instead of subtracting them, so nothing changes sign in gamma, vega or theta: it is long all the way. What moves instead is the delta.

REAL INTERVIEW QUESTION (HSBC · EQD)

What is a straddle? What is the delta of an at-the-money straddle?

WHAT THEY TEST / ANSWER

Structure first: long a call and long a put at the same strike and expiry, so you own two gamma bumps stacked at the same place and you are paid for movement in either direction. Delta: *roughly* zero, and the word roughly is the whole question. The straddle's delta is $\Delta_C + \Delta_P = 2N(d_1) - 1$, and at a strike set equal to spot with zero rates $d_1 = \frac{1}{2}\sigma\sqrt{T} > 0$, so the delta is $2N(\frac{1}{2}\sigma\sqrt{T}) - 1 \approx +0.04$ for three months at 20% volatility: slightly *long*, not flat. Say why rather than just quoting it: the terminal distribution is lognormal and therefore right-skewed, so the call leg is worth a little more delta than the put leg gives back. The strike that makes the delta exactly zero is the one solving $d_1 = 0$, namely $K = Fe^{\sigma^2 T/2} = 100.5$ here, which is why a desk asking for a genuinely delta-neutral straddle asks for the *forward*-struck one, not the spot-struck one. Then the structural point this section is about: because the two bumps stack rather than subtract, nothing in a straddle's gamma, vega or theta ever changes sign, unlike every spread in this section. Only the delta moves, and it moves fast, because the straddle's gamma is at its maximum right where you are standing. How far it moves on a given spot shock is Greeks under spot moves, and the re-hedging that monetises it is delta neutrality and dynamic hedging.

A butterfly is the opposite animal. Long one 90 call, short two 100 calls, long one 110 call: at the body you are net short two options against one on each wing, so the middle dominates and the position is **short gamma, short vega, long theta** where it is most likely to sit. Move far enough out and the wings take over and the signs flip back.

Breaking the link: two expiries

Everything so far has tied vega to gamma. A structuring desk often wants them apart: long volatility as a view, without the daily bleed of being long gamma, or the reverse. The only way to do it is to use two maturities, because the constants in the key idea above depend on T.

Take a calendar spread: short the three-month 100 call, long the one-year 100 call, at spot 100 and 20% flat volatility. Net delta is about $+0.02$, net gamma about -0.020, net vega about $+0.198$ per volatility point, and net theta about $+0.011$ per day. The position is **long vega and short gamma at the same time**, and it collects theta rather than paying it. Vega scales with $\sqrt{T}$ while gamma scales with $1/\sqrt{T}$, so the long-dated leg carries most of the vega and the short-dated leg carries most of the gamma, and buying the first while selling the second separates the two. That decoupling is the reason calendar structures exist, and the reason a single "vega" number on a risk report is meaningless without the maturity bucket it sits in. Bucketing by expiry is covered in the term structure of Greeks, and aggregation across a whole book in portfolio-level Greeks aggregation.

INTERVIEW TRAP

Four ways this goes wrong. (1) Quoting a strategy's Greeks as if they were properties of the strategy rather than of the strategy *at today's spot*: "the call spread is long gamma" is only half a sentence. (2) Forgetting that the sign flips happen while you sleep, so a position sized on today's gamma can be the opposite position after a 5% move with no trade. (3) Assuming the sign flip is at the arithmetic middle of the strikes: it is at $\sqrt{K_1 K_2}\, e^{-\sigma^2 T/2}$, close to it but always slightly lower, and the difference matters when the strikes are far apart. (4) Applying the gamma-vega-theta proportionality to a structure with more than one expiry, where it is simply false, or to one priced off a skew, where the legs carry different volatilities and the relation only holds approximately.

VISUAL / CODE TO BUILD: Interactive

A strategy risk cockpit. The reader picks a structure from a list (call spread, put spread, straddle, strangle, butterfly, calendar, ratio spread), sets the strikes, the maturities and the volatility, and sees four stacked panels against spot: payoff, delta, gamma and vega, with theta shown as a rescaling of the gamma panel so the proportionality is visible rather than asserted. A spot marker the reader can drag reports the four live numbers at that point and colours the panel green or red for the sign, so the sign flips are experienced rather than read. A maturity slider replays the whole family of curves from one year down to one day, making the narrowing and the final spikes visible; a separate toggle switches the two legs to different expiries so the vega and gamma panels visibly decouple. Backed by a Black-Scholes engine in Python, with a table view of the same numbers for accessibility.

Where this goes next

The structures themselves are built in directional strategies, volatility strategies and range-bound strategies, and changing the leg ratios rather than the strikes is strategy construction with ratios. What these Greek profiles do to profit and loss along an actual path is path dependency and scenario analysis, and the decomposition of the resulting profit and loss into its Greek contributions is in Greeks and PnL explain. The errors that come from ignoring the sign flips are catalogued in common structuring mistakes.

17.12 Path dependency and scenario analysis

sales · structuring · trading | advanced | equity · fx · cross-asset

Up to here a strategy has been described by where the underlying *ends*. That is enough for a European payoff and misleading for almost everything else a desk actually trades. Two markets can start at 100 and end at 110 and hand you completely different money, and the difference is not a detail: it is the entire product in a barrier, an autocallable or an Asian, and it is most of the profit and loss on a delta-hedged vanilla book.

Once payoff depends on the path, one number per risk factor stops being a description, and the tool that replaces it is scenario analysis. This section is about recognising the first problem and doing the second properly.

Even a vanilla is path dependent once you hedge it

The cleanest example needs no exotic at all. A delta-hedged option earns $\frac{1}{2}\Gamma S^2$ times the *squared* return each step, as theta as the cost of convexity sets out. Sums of squares do not care about direction and care enormously about size, so two paths with the same endpoint pay differently.

WORKED EXAMPLE

Twenty trading days, spot starting at 100 and finishing at 110 on both paths. The book holds 1,000,000 of cash gamma per 1% move, held constant for the illustration.

- **Path A, the grind**: twenty moves of +0.478%. Gamma profit $= 20 \times 1{,}000{,}000 \times 0.478^2/200 \approx \mathbf{22{,}800}$.
- **Path B, the chop**: ten moves of +3.00% and ten of −1.98%, in any order, which also compounds to 110 (exactly, at a down-move of 1.9829%; the rounded 1.98% lands at 110.03). Gamma profit $= 10 \times 1{,}000{,}000 \times (3.00^2 + 1.98^2)/200 \approx \mathbf{646{,}000}$.

Same start, same finish, same position, and twenty-eight times the profit. Nothing about the option changed; the market simply delivered its move in a different shape. Holding gamma constant flatters the arithmetic slightly, since gamma itself moves along each path, but the order of magnitude is real and it is the reason a trader talks about realised volatility rather than about the market being up or down.

KEY IDEA

Path dependency is not a property that exotic products have and vanillas lack. It is a property of *hedged* positions and of any payoff that reads the underlying more than once. The question to ask of any structure is simply: how many times does this contract look at the price? Once means European. More than once means you cannot price it, hedge it or explain it from the terminal distribution alone.

Two degrees of path dependency

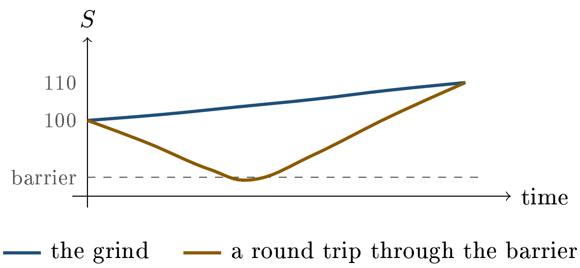
Weak. The payoff depends on the path only through one summary statistic, so the extra state variable is small and tractable.

- **Barriers**: the only thing that matters is whether a level was ever touched. One bit of information about the whole path.
- **Asians**: the payoff uses the average of the observations rather than the final print.

Strong. The payoff needs the trajectory itself, or a sequence of decisions along it.

- **Lookbacks**: the payoff references the maximum or minimum reached.
- **Cliquets and ratchets**: a sequence of returns, each locked in.
- **Autocallables**: a schedule of observation dates, each of which can terminate the product, so the payoff is a stopping time and not a function of S_T at all.

The products themselves belong to exotic options and path-dependent products. What matters here is the consequence for analysis: the moment a payoff is strongly path dependent, no closed form exists, the Greeks stop being stable, and the honest description of the position is a table of scenarios rather than a row of sensitivities.



Both paths start at 100 and finish at 110. A vanilla call pays the same on both. A delta-hedged book earns many times more on the amber one, for the reason the example above quantifies: gamma is paid on a sum of squared moves, and the amber path delivers far more of them. A down-and-out call struck at 100 with that barrier pays the full 10 on the blue path and nothing at all on the amber one, because the trip through the barrier extinguished it, so an identical terminal price hands over the whole payoff in one case and none of it in the other. Any description of the position that cannot distinguish these two pictures is not a description.

Averaging: why an Asian is cheaper

The one piece of path dependency with a clean quantitative answer is averaging, and it is asked constantly.

REAL INTERVIEW QUESTION (Commerzbank · EQD)

Why is an Asian option cheaper than a vanilla option?

WHAT THEY TEST / ANSWER

Because the average of a random walk is less volatile than its endpoint, and an option is worth less on a less volatile underlying. Make it quantitative, because that is what separates the answer: in the continuous-averaging limit the variance of the average of a Brownian path over $[0, T]$ is $\sigma^2 T/3$ rather than $\sigma^2 T$, so the effective volatility is $\sigma/\sqrt{3} \approx 0.58\,\sigma$. At 20% volatility that is 11.5%, and a one-year at-the-money option falls from about 7.97 to about 4.60, roughly 58% of the vanilla, which matches $1/\sqrt{3}$ as it must. Then add the three points that show you understand the product rather than the formula. The discount is smaller in practice because real Asians average over a finite number of dates and often only over the last few months, so the reduction sits between none and the $1/\sqrt{3}$ limit. Arithmetic averaging has no closed form, since a sum of lognormals is not lognormal, so these are priced by Monte Carlo or a moment-matching approximation, while the geometric average does have one. And the commercial reason they exist is not cheapness for its own sake: averaging protects the client against a single bad closing print on the observation date, which is why the final year of a structured note is so often averaged, as in leverage control through structure.

REAL INTERVIEW QUESTION (Société Générale · Exo)

What are the two elements that can be averaged in an Asian option, the spot or the strike?

WHAT THEY TEST / ANSWER

Both, and they give different products. An **average-rate** option averages the *underlying*, so the payoff is $\max(\bar{S} - K, 0)$: the final price is replaced by the average, and this is the common one, used to stop a single print determining the redemption. An **average-strike** option averages to set the *strike*, so the payoff is $\max(S_T - \bar{S}, 0)$: it pays the difference between the final level and the average level over the period, which is what you want if the client is accumulating a position over time and wants to be measured against their own average entry. Say which risk each removes, because that is the real question: the first removes settlement-date risk, the second removes entry-timing risk. Add that both are cheaper than the vanilla, that the average-strike version is genuinely a forward-start-like exposure since its strike is unknown at inception, and that neither has a clean closed form under arithmetic averaging.

Scenario analysis is the substitute for a formula

When the payoff reads the path, the risk report has to as well. Three levels of tool, in increasing cost.

1. A grid. Revalue the position over a matrix of spot and volatility shocks. Cheap, immediately readable, and enough for anything weakly path dependent that is far from its barrier. It fails silently near a barrier, where the true profile is discontinuous and a grid with five-point spacing can step straight over the discontinuity.

2. Named paths. Pick trajectories that correspond to stories rather than to statistics: a slow grind up, a gap down and recover, a gap down and stay, a chop with no net move, a rally into the final observation. Each one is chosen because it breaks a different feature of the structure. This is where barrier products and autocallables reveal themselves, because their behaviour is about *when* and *whether*, not about *how far*.

3. Monte Carlo. Simulate the underlying, apply the contract's rules along each path, and average the discounted payoffs. It is the only general answer for strong path dependency, and it is expensive. The method itself, its convergence and its variance-reduction machinery belong to Monte Carlo methods; what is decided here is only *when* a payoff forces you to reach for it.

REAL INTERVIEW QUESTION (Société Générale · Rates)

How does a Monte Carlo method work to price an option, and when is it necessary?

WHAT THEY TEST / ANSWER

How: simulate a large number of paths for the underlying under the risk-neutral measure, evaluate the contract's payoff along each one, average the results, and discount. The error falls as $1/\sqrt{N}$, which is slow, so in practice you spend effort on variance reduction, antithetic variates, control variates and quasi-random sequences, rather than on raw path count. **When it is necessary** is the better half of the question, and the answer is a rule you can state cleanly: whenever the payoff cannot be written as a function of the terminal value alone. Barriers, Asians, lookbacks, cliquets, autocallables, anything reading the path more than once, plus any multi-asset payoff where the dimension defeats a lattice. **When it is not**: a European payoff on one asset, where a closed form or a one-dimensional integral is faster and exact. Close with the one genuine weakness, since a rates interviewer will ask: plain Monte Carlo handles *early exercise* badly, because the optimal decision at each date depends on a continuation value the simulation has not yet computed, which is why American and Bermudan products need either a lattice or a regression method such as Longstaff-Schwartz.

Reading a term sheet as a set of scenarios

For a sales or structuring candidate this is the single most practical skill in the chapter, and it is examined directly.

REAL INTERVIEW QUESTION (Kepler · Structuring)

Describe the possible scenarios for a Phoenix with a coupon barrier at 70%, a call barrier at 110% and a PDI at 50%.

WHAT THEY TEST / ANSWER

Go through the barriers in the order the product checks them, and be explicit that this is a product observed on a schedule rather than at maturity. **On each observation date**: if the underlying is at or above 110% the product is called, the investor receives capital plus the coupon and the trade ends early; if it is below 110% but at or above 70%, no call but the coupon is paid, and if the product has a memory feature any previously missed coupons are paid too; if it is below 70%, no coupon is paid on that date and the product continues. **At maturity, if it was never called**: if the underlying is at or above the 50% level, capital is returned in full; if it is below 50%, the down-and-in put activates and the investor takes the full fall from the initial level, not from 50%, which is the point clients most often miss and the one you should state without being asked. Then give the scenario summary rather than only the mechanics: this product performs best in a flat-to-mildly-down market, where it collects coupons without being called away early; it performs adequately in a strong rally, where it is called almost immediately and the investor keeps only one coupon while missing the upside; and it fails badly in a severe fall, where coupons stop, the call never triggers and the capital loss is a full equity loss. Say the last one plainly, because the structure hides the three regimes behind three friendly-looking percentages. See Phoenix notes.

REAL INTERVIEW QUESTION (Société Générale · Structured Products)

From a term sheet, explain the mechanism of the product and all the possible scenarios.

WHAT THEY TEST / ANSWER

Use a fixed method rather than reading the document top to bottom, and say that you are doing so. **One**: identify the underlying and, if there are several, whether it is a basket or a worst-of, because a worst-of changes the risk more than any barrier level does. **Two**: list the observation dates, since they determine whether the product is European or path dependent at all. **Three**: list every barrier and what each one switches on or off, coupon, early redemption, or capital protection. **Four**: decompose into building blocks, typically a zero-coupon bond plus a strip of digitals plus a short down-and-in put, so you know which risk the client is being paid for. **Five**: walk the scenarios, up sharply, up mildly, flat, down mildly, down sharply, and state the client's outcome in each, including the timing, since being called in year one is a different outcome from being called in year five. Then close with what the method is for: the purpose is not to explain the product, it is to find the scenario in which the client loses money and be able to describe it in one sentence before they discover it themselves. If you cannot state that scenario, you do not yet understand the term sheet.

INTERVIEW TRAP

Four errors. (1) Reasoning about a hedged book from the net move: gamma pays on the sum of squared moves, so a market that round-trips can pay many times more than one that grinds to the same level. (2) Applying a Greek report to a barrier product near its barrier, where the sensitivities flip sign and blow up and the only honest tool is revaluation. (3) Using a scenario grid coarse enough to step over a discontinuity, which produces a smooth-looking report for a product that is not smooth. (4) Describing an autocallable by its maturity payoff, when the product is a schedule of stopping decisions and its most likely outcome is that it never reaches maturity at all.

VISUAL / CODE TO BUILD: Interactive

A path laboratory. The reader draws or selects a price path (grind up, gap and recover, chop, slow bleed) and the widget evaluates several structures along it simultaneously: a vanilla call, a delta-hedged vanilla, an Asian, a barrier and an autocallable, showing each one's running value and its final outcome. Because every structure sees the same path, the comparison is exact rather than illustrative, and the headline result appears without being argued: two paths with identical endpoints produce identical vanilla payoffs and wildly different everything else. A second mode runs a full Monte Carlo behind the same structures and reports the distribution of outcomes, with the drawn path marked on it, so the reader sees where their intuition sits inside the real distribution. A toggle overlays the scenario grid a risk report would have produced, with the cells that miss the barrier discontinuity flagged.

Where this goes next

The products this section analyses are built in exotic options and path-dependent products, with the retail wrappers in autocallables and Phoenix notes. The formal risk-side treatment of scenarios is scenario analysis and stress testing. How a structure's value moves between now and expiry, which is the same question restricted to a single deterministic path, is strategy behaviour over time.

17.13 Strategy Selection by Market Regime

sales · trading · structuring | advanced | equity · fx · commodity · cross-asset

Payoff design takes a view and produces a structure. It quietly assumes the market will sell you that structure at a sensible price. This section removes the assumption. The families themselves belong to directional, volatility, skew and tail-risk strategies, and how the Greeks themselves behave as conditions change belongs to Greeks across market regimes. What is owned here is the decision in between: given the same view, which member of the family the current market makes it sensible to own.

The distinction is worth stating plainly, because it is the one candidates collapse. A **view** is about the underlying. A **regime** is about the price of the instruments you would use to express it. They are independent, and both are needed before anyone can name a structure.

Four things to read before choosing anything

- **Where implied sits against realised.** Is premium rich or cheap? Measured in implied vs realised volatility; here it decides whether you want to be a net buyer or a net seller of options.
- **The slope of the term structure.** Upward sloping is the ordinary state and makes the front cheap relative to the back; inversion is a stress signal. Vega exposure develops the regimes.
- **The steepness of skew.** It sets the price of the wing you would buy and the wing you would sell, so it decides whether a structure should be financed downside or upside, and it is what the smile measures.
- **The character of realised moves.** Trending or choppy, which is not the same question as large or small, and it decides whether gamma will actually be monetised.

> **KEY IDEA**
>
> The view tells you the *payoff* you want. The regime tells you what that payoff costs, and therefore which *version* of it to buy. Two traders with an identical bullish view should hold different structures in a 20% vol market and a 40% vol market, and if they hold the same one, at least one of them has not looked at the price.

The same view in two regimes

Take a concrete view: spot is 100 and you think it will be near 110 in a year. Below are Black-Scholes values at $S = 100$, $T = 1$, $r = 0$, with a single flat volatility per regime so the comparison isolates the level. Skew is deliberately absent, and belongs to skew-based strategies.

Implied	C_{90}	C_{100}	C_{110}	P_{90}	100/110 spread
20%	13.59	7.97	4.29	3.59	3.67
40%	20.57	15.85	12.11	10.57	3.74

WORKED EXAMPLE

Reading the table as a decision.

- **At 20%**, the outright 100 call costs 7.97 and breaks even at 107.97. Your target of 110 clears it. Buy the call.
- **At 40%**, the same call costs 15.85 and breaks even at 115.85. Your view is 110. The outright call *loses money even if you are exactly right*, which is the whole point: the structure has not changed, the regime has, and the trade is now bad.
- **The 100/110 call spread costs 3.67 at 20% and 3.74 at 40%**. Doubling volatility moved it by less than two percent, because both legs became dearer together and you are long one and short the other. It pays $10 - 3.67 = 6.33$ at 20% and $10 - 3.74 = 6.26$ at 40%, so either way near 6.3.

That single comparison is the section in miniature. Outright options are a bet on the level of implied volatility whether you meant them to be or not; spreads are close to immune to it. When premium is rich you do not abandon the view, you switch to the version of it that does not pay for volatility.

The same logic runs in reverse. When implied is cheap against realised, the spread's short leg gives away most of the upside for a discount that is no longer worth taking, and the outright is the right instrument. Cheap premium is also the regime in which tail hedges are worth carrying, because the carry that normally kills them is small.

Trend is not variance

The fourth state variable is the one that gets skipped, and it produces the most expensive mistake in the list.

INTUITION

A long gamma position is paid on the *sum of squared moves*, not on where the market ends up. A market that grinds up 0.3% a day for a year has more than doubled and delivered a realised volatility of only $0.3\% \times \sqrt{252} = 4.8\%$: the long gamma holder rehedges tiny amounts, banks nothing, and pays theta every day. A market that goes nowhere in six months while moving 2% daily in both directions pays that holder handsomely. Trend and variance are close to unrelated, and only one of them is what you are long.

So a trending regime favours delta: outright directional structures, call spreads, participation. A choppy regime favours gamma: straddles, strangles, and hedging often rather than widely. Buying a straddle because "the market is going to move a lot" without asking whether it will move *in a jagged way* is a directional trade dressed as a volatility one, and common structuring mistakes catalogues the family it belongs to.

Regime	Prefer	Avoid
Implied rich vs realised	spreads, ratios, financed structures	outright premium
Implied cheap vs realised	outright options, tail hedges	selling wings for yield
Term structure steeply upward	owning the front, short the back	long-dated outright
Term structure inverted	selling the front, owning the back	rolling short-dated longs
Skew very steep	put spreads, risk reversals	outright downside puts
Trending, low daily variance	delta structures, participation	long gamma
Choppy, high daily variance	straddles, frequent rehedging	short gamma for carry

REAL INTERVIEW QUESTION (Bank of America · Exo)

What strategy would you propose if you know an event is going to happen today but do not know what effect it will have on your stock (Tesla for example, and the fact that Musk is leaving)?

WHAT THEY TEST / ANSWER

The instinctive answer, buy a straddle, is right in shape and incomplete in substance, and the incompleteness is what is being tested. You know an event is coming. So does everyone else, and the option market has already priced it.

Give the shape first: unknown sign, known occasion, so you want a symmetric long-volatility structure, a straddle or a strangle on the nearest expiry that spans the event. The strangle if you think the move will be violent, since it is cheaper per unit of tail; the straddle if you want the lower break-even.

Then give the part that makes it a trade rather than a reflex. Compare the **implied event move** with your own estimate. The expiry that contains the event trades at an elevated implied volatility, and an at-the-money straddle costs roughly $0.8\,\sigma S\sqrt{T}$, so you can back out the move the market is charging you for and ask whether it is too small. If the market implies a 12% move and you believe 8%, the right trade on a symmetric view is to *sell* the volatility, not buy it, however dramatic the headline. There is no long-volatility edge in an event that is already in the price.

Then the two practical points. Buy the shortest expiry spanning the event, because it carries the most event and the least of everything else. And expect the **volatility crush** immediately afterwards: implied collapses once the uncertainty resolves, so a long straddle that is right about the event but not by enough loses on vega what it made on spot. That crush is why the honest answer to this question is often "nothing", and saying so is stronger than naming a structure.

REAL INTERVIEW QUESTION (Natixis · EQD)

For the same market opportunity, you compare an options strategy on Total (40 % vol) and on Danone (10 % vol): which will cost more, and why (reason through a log-normal law)?

WHAT THEY TEST / ANSWER

Total, by roughly four times, and then immediately explain why that is the wrong way to compare them.

The number first. An at-the-money call is worth about $0.4\,\sigma S\sqrt{T}$, derived in vega exposure, so at $S = 100$ and one year Total's call costs about 16 and Danone's about 4. Exactly, at $r = 0$, Black-Scholes gives 15.85 and 3.99. Premium is close to linear in volatility at the money, so a four-times volatility ratio is a four-times price ratio.

The log-normal reasoning the question asks for is what tells you this is not expensive. Under a log-normal law the dispersion of S_T is governed by $\sigma\sqrt{T}$, so Total's terminal distribution is four times as wide. The straddle break-even is about $\pm 0.8\,\sigma\sqrt{T}$, which is $\pm 32\%$ on Total and $\pm 8\%$ on Danone. Both break-evens sit the same distance out in each name's *own* volatility units, so measured in those units the two options cost *identically*.

So the real answer is that "which is more expensive" has no meaning across two underlyings until you fix the unit. In cash, Total. Per unit of the move it takes to pay off, they are the same, and the only genuine comparison is each name's implied against its own realised volatility, which is the measurement in implied vs realised. Danone at 10% implied against 14% realised is cheap; Total at 40% against 30% is expensive, and the cash premium said the opposite.

REAL INTERVIEW QUESTION (BNP · Commodity)

A European call and a European put have the same strike and the same maturity. The call has a volatility of 30 % and the put 25 %. What strategy would you consider?

WHAT THEY TEST / ANSWER

This is not a regime, it is an arbitrage, and recognising that is the answer. By put-call parity, $C-P = S-Ke^{-rT}$, which contains no volatility at all. The two options are therefore forced to trade at the *same* implied volatility: a European call and put on the same strike and maturity are the same instrument up to a forward. Two different implied volatilities on one strike is a quoting inconsistency, not a market view.

The trade is to buy the cheap volatility and sell the dear one, then remove the direction. Buy the put at 25%, sell the call at 30%, and buy the underlying. That package, long stock, long put and short call at the same strike, pays exactly K at maturity whatever spot does. It is a conversion.

Put numbers on it, at $S = K = 100$, $T = 1$, $r = 0$: the call at 30% is worth 11.924 and the put at 25% is worth 9.948. You collect $11.924 - 9.948 = 1.976$ and pay 100 for the stock, so the package costs 98.024 and settles at 100 with certainty. That is 1.976 of risk-free profit, about 2% of notional, which is far too large to be real.

Which is the honest close. Before trading it, ask what is wrong: a borrow cost or dividend that makes the true forward different from the spot, an American rather than European put, a stale quote on one side, or a mid-price that no one will actually deal on. On a commodity the usual culprit is that the forward is not Se^{rT} at all, because of storage and convenience yield, so the parity relation must be written against the *forward* and the apparent 5 volatility points evaporate. Naming the carry as the suspect is what makes this a commodity desk answer.

Whether a book can be long gamma and long theta at once, and the condition on the two implied volatilities that makes it possible, is theta as the cost of convexity. What belongs to regime selection is where that trade actually lives, because it is not a Greeks curiosity.

It is dispersion. Long cheap single-name gamma against short expensive index gamma satisfies the condition exactly, and the risk you are really taking is *correlation* rather than volatility, which is why the regime that pays it is one where index implied is rich to its constituents and correlation is expected to fall. A steep skew offers the same structure across strikes rather than across names, selling the expensive downside wing against a cheaper body. In both cases you are short a volatility *spread*, so the position loses if the cheap leg stays quiet while the expensive one moves, and that is a different regime call from being long or short volatility outright.

REAL INTERVIEW QUESTION (Unicredit · EQD)

An investor asks you to build a positive-gamma strategy: what do you do?

WHAT THEY TEST / ANSWER

Positive gamma means long options, and everything interesting is in the follow-up questions rather than in that sentence.

First, ask what they want it *for*, because two very different clients ask this. One wants a convex payoff held to maturity, in which case gamma is a description of the shape and the answer is a straddle, a strangle or a capital-protected structure. The other wants gamma to trade, in which case it only pays if it is delta-hedged and rehedged, and the position is really a bet that realised volatility beats the implied volatility they are about to pay. Selling the first client a rehedging strategy, or the second a static structure, is the actual failure mode.

Second, price the theta out loud. Positive gamma is bought with negative theta, so quote both, and give the break-even move $x^* = \sigma/\sqrt{252}$ per day: at 16% implied that is about 1% daily, and anything quieter than that loses money. A client who has not been told the daily bleed has not been told the trade.

Third, and this is the regime part, choose *where* to buy it. Gamma per unit of premium is highest short-dated and at the money, and so is the decay, so short expiries are the aggressive expression and longer ones the patient one. If implied is rich against realised across the surface, buy the gamma in a spread or a ratio so you are not paying the full premium, and if it is cheap, buy it outright. And if the term structure is steeply upward sloping, the front is where the cheap gamma is.

Close on the honest caveat: if implied is well above realised everywhere, the right advice may be that this client should not buy gamma at all.

INTERVIEW TRAP

1. **Naming a structure before looking at the price.** The commonest interview failure in this chapter. "I am bullish so I buy a call" is an answer that has not asked what the call costs.
2. **Trading a regime that is already priced.** Everyone can see the event, the elevated volatility and the steep skew. The trade is in the difference between the priced move and your estimate, not in the existence of the move.
3. **Selling volatility because it is high.** High implied is usually high for a reason and is often still below what will realise. Mean reversion in volatility is real over long horizons and no comfort at all over the two weeks that matter.
4. **Confusing a big move with high variance.** A trend delivers a large displacement and little realised variance. Long gamma is paid on the second, not the first.
5. **Treating regime labels as data.** "Risk-on", "low-vol regime" and "crisis" are named after the fact. The state variables in this section are observable now; the labels are not.
6. **Assuming a regime is global.** Equity skew steepens in stress while FX skew can flip sign with the direction of the carry trade, and commodity term structure moves with storage rather than with fear. A regime read on one asset does not transfer.

VISUAL / CODE TO BUILD: Interactive

A regime dial with a fixed view. The reader states a view once, spot to 110 in a year, and then moves three sliders: implied volatility level, term structure slope and skew steepness. Every candidate structure is repriced live and ranked by profit *at the stated view*, so the ordering visibly reshuffles as the sliders move, the outright call falling below the call spread as implied rises past the point where its break-even passes the target. A second panel plots each structure's cost against the volatility level, making the near-flat line of the spread and the steep line of the outright the picture the reader takes away. A "what would have to be true" readout inverts the question and reports the implied level at which each structure becomes the best choice.

17.14 Common structuring mistakes

structuring · sales · trading | core | equity · cross-asset

Every structure in this chapter was built by adding and subtracting legs until the payoff matched a view. That process has a small number of failure modes, and they repeat: the same five or six mistakes account for most of the bad term sheets, most of the losses that surprise a client, and most of the follow-up questions in a structuring interview. This section is the catalogue. It is deliberately the last section of the chapter, because each mistake is a way of getting one of the earlier ones wrong.

It helps to sort them into three kinds, because the fix is different for each.

1. **Pricing errors**: the number on the term sheet is wrong, usually because the model was fed one volatility where the payoff needs several.

2. **Hedging errors**: the price is right but the risk cannot actually be run, so the desk loses in the very scenario the structure was designed around.
3. **Disclosure errors**: the price is right and the risk is manageable, but the client did not understand which risk they sold, and finds out on the day it costs them.

> KEY IDEA
>
> There is one sentence behind all three. **Every improvement in a payoff is paid for by selling something.** A higher coupon, a higher participation, a lower premium: each is financed by a risk transferred to the client. A structure is well built when you can name that risk in one sentence and say in which scenario it bites. If you cannot, you have not designed a product, you have hidden a short position.

Mistake 1: pricing a multi-strike payoff with a single volatility

The most common quantitative error in the chapter. Black-Scholes takes one σ. A structure with two strikes needs two, because the market quotes a different implied volatility at each strike, which is the skew, described in the volatility smile and skew. Feed the at-the-money volatility to both legs and the answer is not approximately right, it is systematically wrong, and always in the same direction.

> WORKED EXAMPLE
>
> A 100/130 call spread, one year, spot 100, ignoring rates and dividends.
>
> Priced with a single volatility of 20% for both legs: the 100 call is worth 7.97, the 130 call is worth 1.01, so the spread costs 6.96.
>
> Priced with the equity skew, where the higher strike trades at a lower implied volatility, say 20% at 100 and 17% at 130: the 130 call is now worth only 0.51, and the spread costs 7.45.
>
> The flat-volatility price is 7% too cheap. With the 130 leg at 18% instead the error is still 5%. The whole difference sits in the leg you are *short*: you are short a strike whose true volatility is lower than the one you used, so you credited yourself too much premium for selling it.

REAL INTERVIEW QUESTION (Natixis · EQD)

Take a 100-130 call spread. Will it cost more priced with the Black-Scholes model, or by splitting the volatilities according to the skew?

WHAT THEY TEST / ANSWER

More with the skew, and the reason must come out in one step. In equity the implied volatility falls as the strike rises, so the 130 call is worth less than a flat at-the-money volatility says. You are short that call, so paying its true, lower price means you receive less, and the spread as a whole costs more. Put a number on it: at 20% flat the spread is about **6.96**, with the 130 leg marked at 17% it is about 7.45, roughly 7% more expensive. Then add the sentence that shows you know why the question is asked at all. A call spread is not a volatility trade, it is a *skew* trade: its exposure to a parallel move in volatility is small and changes sign, as shown in Greeks evolution within strategies, while its exposure to the *slope* of the volatility surface does not. That is why desks quote spreads off the surface leg by leg and never off a single number, and it is developed in skew trading.

The sign convention is worth fixing once, because it is the form the question usually takes on a desk: a long call spread is long skew and a short call spread is short skew. The structuring mistake is not getting that sign wrong, it is quoting either one off a single volatility afterwards, which is why the sensitivity that matters here is bucketed by strike rather than parallel, as in skew and smile sensitivities. The position itself, and how a desk trades it deliberately, belongs to skew-based strategies.

Mistake 2: replicating a discontinuity and pretending the gap is free

A digital pays a fixed amount if the underlying finishes above a level, and nothing otherwise. Nobody can hedge a jump, so it is sold as a tight call spread instead; the construction and its super-replication argument are exotic options, and all that is needed here is that the spread is struck a width ε below the digital level and sized $1/\varepsilon$. The mistake this section is about is treating ε as a modelling detail. **It is a commercial decision**, and it is where the margin and the risk both live.

WORKED EXAMPLE

A digital paying 1 if the spot finishes above 100, one year, spot 100, 20% volatility, rates and dividends ignored. Its theoretical value is $N(d_2) = 0.4602$.

Gap ε	Super-replicating spread	Cost over theory	Hedge size $1/\varepsilon$
10	0.5624	+22.2%	0.10
5	0.5108	+11.0%	0.20
2	0.4802	+4.4%	0.50
1	0.4701	+2.2%	1.00

The spread struck $(K-\varepsilon, K)$ pays at least as much as the digital in every scenario, so selling the digital and buying it is a safe hedge. Halving the gap halves the cushion and doubles the number of options you must trade.

Read the two columns together and the trade-off is the whole answer. A wide gap is cheap to trade and expensive for the client. A narrow gap is fair to the client and leaves the desk holding a position whose gamma near the strike explodes as expiry approaches, in a size that grows like $1/\varepsilon$. There is no value of ε that is right in the abstract: it is set by the liquidity of the strikes you actually have to trade, the size of the deal, and how close spot is expected to sit to the barrier at expiry.

REAL INTERVIEW QUESTION (Société Générale · EQD)

How do you replicate a digital option with a call spread, how do you choose the strike gap, and what factors do you take into account?

WHAT THEY TEST / ANSWER

Give the construction, then the trade-off, then the factors, in that order. The construction: a digital paying 1 above K is $1/\varepsilon$ call spreads struck $(K-\varepsilon, K)$, which dominates the digital everywhere and therefore super-replicates it; the symmetric spread $(K-\varepsilon/2, K+\varepsilon/2)$ prices almost exactly at the digital's theoretical value but does not dominate it, so a seller who wants to be safe uses the lower pair and the difference is their cushion. The trade-off: the cost of the hedge falls as ε shrinks, from about 22% over theoretical value at $\varepsilon = 10$ to about 2% at $\varepsilon = 1$ for a one-year at-the-money digital at 20% volatility, while the number of options to trade grows like $1/\varepsilon$ and the gamma near the strike grows with it. The factors: the liquidity and the listed strike grid of the underlying, the size of the trade against that liquidity, the skew between the two strikes, since the spread is a skew position and the digital's value is essentially a skew number, and the risk of spot pinning the strike at expiry. Close by saying who bears what: the gap is not an approximation error, it is the desk's compensation for a risk that cannot be hedged, and a client who is quoted a digital without being told the assumed gap has not been quoted a price. The exotic side of this is in exotic options.

Mistake 3: manufacturing a headline number without naming the risk sold

This is the disclosure error, and commercially it is the expensive one. A client asks for a bigger coupon, a bigger participation, a lower entry price. There is always a way to give it to them, and the way is always to sell more of something on their behalf. The mistake is delivering the new number without the new sentence.

INTUITION

A term sheet is a trade, not a product. If the coupon went from 3% to 8% and nothing in the world changed, then the client just sold something worth 5% a year. The only question worth asking is what, and the only failure is not asking.

REAL INTERVIEW QUESTION (Société Générale · Structured Products)

A client holds a Phoenix with a 3% coupon and wants another one with a higher coupon: what do you do?

WHAT THEY TEST / ANSWER

Do not start from the coupon, start from the levers, because the coupon is an output. List them with what each one sells. **Lower the capital barrier or make it continuously observed**: the client is short a bigger, more easily triggered put. **Raise the coupon barrier**: the coupon is paid less often, so each one can be bigger. **Switch the underlying to a more volatile name**: the embedded short put is worth more, so it funds more. **Move from a single underlying to a worst-of basket**: by far the largest uplift, and by far the largest risk transfer. **Lengthen the maturity or remove the memory feature**. Then say the thing they are grading: every one of these raises the coupon by increasing the probability or the severity of the client's loss, so the correct response to the request is a question, which is how much more downside risk they are willing to take, not a new term sheet with a bigger number on it. Finish by checking whether the client actually wants higher coupon or higher *expected* return, since the two are not the same and the second is not improved by any of these levers. The product mechanics are in Phoenix notes and autocallables.

REAL INTERVIEW QUESTION (Kepler · Structuring)

What is the point of adding a worst-of to an autocall?

WHAT THEY TEST / ANSWER

It makes the option package the issuer buys from the client much cheaper, so the coupon can be much higher, and that is the entire point. The mechanism to state: the client is short a put on the *worst* performer of the basket, and the worst of several assets is far more likely to breach a barrier than any single one, so the short option is worth more and pays for more coupon. The exposure that follows is the one they want to hear named, and getting its sign right is the point of the question: a worst-of put is worth *more* when correlation is low, because low correlation means the assets disperse and the worst of them is worse. The client is short that put, so the client is **short dispersion**, equivalently **long correlation**: they lose when correlation falls. The issuer, holding the put, is on the other side and hedges it. Then the judgement, which is what separates a structurer from a reciter: the coupon uplift is real and the risk uplift is larger and much harder for a client to price, so a worst-of is the clearest example in the market of a headline number manufactured out of a risk the buyer cannot value. That is why regulators and internal product committees look hardest at these, and why the honest presentation always shows the scenario where one name falls 45% and the other two are up. Correlation itself is developed in dispersion strategies, and the retail framing in retail versus institutional structuring.

Mistake 4: choosing the underlying by its coupon

Closely related, and worth its own line because it is where the mistake usually enters in practice. Two candidate underlyings produce two different coupons for the same structure, and the temptation is to show the client the bigger one.

REAL INTERVIEW QUESTION (Société Générale · Structured Products)

You have two Athenas, one on Total and one on LVMH. Which do you recommend to your client?

WHAT THEY TEST / ANSWER

Refuse the trap in the question, which is that the answer is a name. Explain what drives the difference first: for a given structure the coupon rises with the underlying's implied volatility, with its dividend yield, since a higher yield lowers the forward and makes the client's short put worth more, and with the steepness of its skew, since the embedded put sits below the money. So work out which of the two has the higher volatility, the higher dividend yield and the steeper skew, and you have worked out which pays more, without needing to memorise either. Then give the actual recommendation, which is a method: the higher coupon is not a better product, it is compensation for a worse outcome in the scenario the client cares about, so you recommend the one whose *risk* the client can carry. Ask what else is in their portfolio, since selling a downside put on a name they already hold concentrates rather than diversifies; ask whether they have a view on the sector; and check the single-name event risk, because an idiosyncratic shock on one stock is exactly what a barrier structure is worst at surviving. A candidate who names a stock in the first sentence has answered the wrong question.

Mistake 5: designing for maturity and ignoring the path

A payoff diagram is a picture of one day, the last one. The client lives with the product every day until then, sees a mark-to-market on a statement, and calls when it moves. A structure designed only against its final payoff will be sold with a story the first monthly statement contradicts.

REAL INTERVIEW QUESTION (Société Générale · Structured Products)

You sold a client a call at 7%, one-year maturity on the EuroStoxx. Six months later the client calls and asks why the call is now worth 4%. How do you justify it?

WHAT THEY TEST / ANSWER

Answer with a decomposition, not an apology, and do the arithmetic in front of them. A one-year at-the-money call priced at 7% of spot implies a volatility of about 17.6%. Time alone accounts for most of the move: at the money the premium scales roughly with $\sqrt{T}$, so six months later, with spot and volatility unchanged, the same call is worth about $7\% \times \sqrt{1/2} = 4.95\%$, and the exact Black-Scholes value is 4.95%. The remaining gap has to come from the other two inputs, so name them and quantify. A fall in implied volatility from 17.6% to about 14.2% takes that 4.95% down to exactly the 4% being asked about, and spot is essentially unchanged, which is why the option is still at the money. That is the whole answer: **theta, then vega, and delta last**, which for an option that has not moved contributes nothing at all. Give the order in that form rather than by habit, because naming delta second when spot has not moved is the tell that you are reciting Greeks instead of reading the position. Two things earn the point. First, that theta is not a surprise but the thing they bought, since an option that did not decay would not be an option, and the decay accelerates into expiry rather than running linearly. Second, that this conversation should have happened at the point of sale, with the six-month value shown next to the payoff diagram. The formal decomposition is Greeks and PnL explain.

Mistake 6: sizing on today's risk

The last one is short because Greeks evolution within strategies does the work. A multi-leg structure changes the sign of its gamma, vega and theta as spot moves, with no trade from anyone, so a position sized to a gamma limit today can be against that limit after a move of a few percent. The two places this bites hardest are the days before expiry, when the Greek profiles of a spread become opposite-signed spikes at the two strikes, and spot sitting on a barrier, where the same effect is a discontinuity rather than a spike. Both are reasons desks reduce size or roll early, and both are the correct answer to "what would worry you about this trade" long before anything about the payoff.

INTERVIEW TRAP

The five sentences that lose the point. (1) "We priced it at the at-the-money volatility": for anything with more than one strike, that is not a price. (2) "The digital is worth $N(d_2)$": it is worth the call spread you can actually trade, and the gap is a decision you have to defend. (3) "This one pays a better coupon": no structure pays a better coupon, it sells a bigger risk. (4) "The client is protected down to the barrier": they are protected until the barrier, after which the loss is measured from the strike, which is the opposite of protection at the worst moment. (5) "It is a one-year product so we look at it in one year": the client marks it monthly, and so does the risk system.

VISUAL / CODE TO BUILD: Interactive

A term-sheet stress viewer, built to make mistakes visible rather than described. The reader loads a structure and sees three linked panels. **Panel one** prices it twice, once off a flat volatility and once off a skewed surface the reader can tilt with a slider, showing the two prices and the error in percent, so mistake one has a number attached to it. **Panel two** takes any digital or barrier in the structure and lets the reader set the replication gap, showing simultaneously the hedge cost, the hedge size and the peak gamma, so the trade-off cannot be avoided. **Panel three** is a scenario grid: for each of a set of paths (steady rise, slow grind down, one gap of -30% on a single name, a fall through the barrier in the final month) it shows the client's outcome and the mark-to-market at each observation date, with the worst-of case highlighted. A "name the risk sold" field asks the reader to write one sentence before the term sheet can be exported, which is the habit the whole section is trying to build.

Where this goes next

The design process these mistakes corrupt is from market view to payoff design, and the levers that get abused are in leverage control through structure. The products where the stakes are highest are in autocallables and embedded optionality, with the desk's side of the same trades in hedging structured products. The distribution context, which decides how much of this a client is expected to understand, is retail versus institutional structuring.

Chapter 18

Advanced Options & Volatility

18.1 Black-Scholes model

trading · structuring · sales | core | equity · cross-asset

Black-Scholes is the most important wrong model in finance. Everyone knows its assumptions fail, everyone uses it anyway, and understanding why both of those are true at once is the difference between a candidate who has memorised a formula and one who understands what a derivatives desk does all day.

The idea: a price is the cost of a hedge

The insight is not a probability calculation. It is that an option can be **manufactured**.

Suppose you have sold a call. You do not know whether the stock will rise, but you know that if it rises you will owe money, and the more the option is in the money the more of the stock you should already own. So hold Δ shares, adjust continuously as the stock moves, and you have a portfolio whose value tracks the option's. If that tracking is exact, the option's fair price is simply the cost of running the portfolio, because otherwise there is an arbitrage between the two.

> **KEY IDEA**
>
> The reason the option's price does not depend on the stock's expected return is that the hedge removes the direction. A delta-hedged position makes and loses money on the *size* of moves, not their sign, so the only thing left to price is volatility. That single sentence explains why μ is absent from the formula, why the world's probabilities can be replaced by risk-neutral ones, and why derivatives desks talk about volatility rather than about the market going up.

The derivation

REAL INTERVIEW QUESTION (JP Morgan · Credit)

Derive Black-Scholes using a replicating portfolio.

WHAT THEY TEST / ANSWER

Five steps. Say them in order and do not skip the reasoning in step three, which is where the whole thing happens.

One, the model for the underlying. Assume the stock follows a geometric Brownian motion,

$$dS_t = \mu S_t\, dt + \sigma S_t\, dW_t,$$

so returns are normal, prices are lognormal and the volatility σ is constant.

Two, the portfolio. Let $V(S,t)$ be the option's value and build $\Pi = V - \Delta S$: long the option, short Δ shares.

Three, apply Itô and kill the risk. Itô's lemma gives

$$dV = \left(\frac{\partial V}{\partial t} + \mu S\frac{\partial V}{\partial S} + \tfrac{1}{2}\sigma^2 S^2\frac{\partial^2 V}{\partial S^2}\right) dt + \sigma S\frac{\partial V}{\partial S}\, dW,$$

so

$$d\Pi = \left(\frac{\partial V}{\partial t} + \tfrac{1}{2}\sigma^2 S^2\frac{\partial^2 V}{\partial S^2} + \mu S\left(\frac{\partial V}{\partial S} - \Delta\right)\right) dt + \sigma S\left(\frac{\partial V}{\partial S} - \Delta\right) dW.$$

Both brackets carry the same factor $\partial V/\partial S - \Delta$, so choosing $\Delta = \partial V/\partial S$ kills **both at once**: the dW term vanishes, and so does the only term containing μ. The portfolio is **riskless over the next instant** and the drift has gone with the risk. That the same single choice does both jobs is the punchline of the derivation, and it is worth saying in exactly those words.

Four, no arbitrage. A riskless portfolio must earn the risk-free rate, so $d\Pi = r\Pi\, dt$. Substituting gives the **Black-Scholes PDE**,

$$\frac{\partial V}{\partial t} + \tfrac{1}{2}\sigma^2 S^2\frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0.$$

Five, boundary condition and solution. Impose $V(S,T) = \max(S-K, 0)$ for a call and solve, giving the formula below.

Close with the two remarks that separate a real answer. The hedge must be rebalanced *continuously* and costlessly, which is the assumption that fails first in practice. And the PDE is the heat equation in disguise, which is why the same numerical machinery prices options and diffuses heat, and why finite-difference methods work at all.

The formula, and what its pieces mean

For a European call on a non-dividend-paying stock,

$$C = S\,N(d_1) - Ke^{-rT}N(d_2), \qquad d_1 = \frac{\ln(S/K) + \left(r + \tfrac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}.$$

WORKED EXAMPLE

Spot 100, strike 100, one year, volatility 20%, rate 5%, no dividends.

$d_1 = (0 + 0.07)/0.20 = 0.350$ and $d_2 = 0.150$, so $N(d_1) = 0.6368$ and $N(d_2) = 0.5596$. Then

$$C = 100 \times 0.6368 - 95.12 \times 0.5596 = 63.68 - 53.23 = 10.45.$$

The put follows from put-call parity without any further work: $P = C - S + Ke^{-rT} = 10.45 - 100 + 95.12 = 5.57$.

Sanity-check it against the desk approximation. The forward is 105.13, so this option, struck at 100, is at the money *spot* and below the forward. An at-the-money *forward* call, struck at 105.13, would be worth about $0.4 \times \sigma \times \sqrt{T}$ of the discounted forward, which is 8.0. Ours is worth more, 10.45, precisely because its strike sits below the forward and it therefore carries intrinsic value against it. Doing that check in ten seconds is worth more in an interview than reproducing the formula.

Now read the two terms, because interviewers ask.

The term $N(d_2)$ is the **risk-neutral probability that the option finishes in the money**. So the second half of the formula, $Ke^{-rT}N(d_2)$, is the present value of paying the strike, multiplied by the chance of having to pay it.

$SN(d_1)$ is the present value of *receiving the stock*, conditional on exercise, and $N(d_1)$ is the **delta**, the hedge ratio. It is not a probability, and the common claim that it is one is wrong: it is a probability under a different measure, the one that uses the stock as the numeraire, which is why $N(d_1) > N(d_2)$ always.

So the formula reads: *what you get, weighted by how likely you are to get it, minus what you pay, weighted by how likely you are to pay it.*

REAL INTERVIEW QUESTION (Société Générale · EQD)

Why can we use the risk-neutral probability to price an option? You are short a put: how do you hedge it? What kind of option do you buy to become vega positive?

WHAT THEY TEST / ANSWER

Three questions, and the first is the conceptual one.

Why risk-neutral pricing works. Because the payoff can be replicated by a self-financing portfolio of the stock and cash. Once that is true, the option's price is the cost of the replicating portfolio, and that cost does not depend on anybody's view of the stock's expected return, since the hedge removes the direction. Formally, no arbitrage implies the existence of a measure under which every traded asset discounted at the risk-free rate is a martingale, and the price is the discounted expectation under *that* measure, which no-arbitrage pricing establishes in general. Say clearly what this is not: it is not a claim that investors are risk neutral, or that the real drift is r. It is a computational device that gives the right price because it gives the right hedging cost.

Hedging a short put. A put has negative delta, so being short it leaves you with *positive* delta: you hedge by **selling** $|\Delta|$ shares, and you rebalance as the market moves. The uncomfortable part, and the part they want to hear, is that you are also short gamma, so the rebalancing loses money: you sell more as the market falls and buy back as it rises, and the theta you collect is the compensation for exactly that, in delta neutrality and dynamic hedging.

Becoming vega positive. Start by noting that the underlying has no vega at all, so no amount of stock changes your vega and it can only be moved with *another option*. Short a put, you are short vega, so you must buy an option with **more vega than the one you are short**, and there are exactly two ways to get it.

Move the strike towards the money. Vega peaks at the money and decays away from it in both directions, so if the put you are short is out of the money, buying the at-the-money option of the same maturity leaves you net long vega.

Move the maturity out. Vega grows roughly with $\sqrt{T}$, so a longer-dated option of the same strike carries more vega than the one you are short, and buying it leaves you long vega even at an identical strike.

Then give the contrast, because it is the trap in the question. Buying the call of the *same* strike and maturity does **not** work. By put-call parity that call has exactly the same vega as your short put, so the two cancel and you are left **vega flat**, holding a synthetic long stock position and no volatility exposure at all. That is the vega-neutral trade, which is the opposite of what was asked. Naming it as the thing you are deliberately not doing is a stronger answer than avoiding it silently.

Where the model is wrong, and why it survives

Each assumption fails in a specific way, and each failure has a known repair.

- **Constant volatility** is the big one. Its fingerprint is the smile: if the assumption held, every strike would trade at the same implied volatility, and none do. The repairs are local volatility, stochastic volatility and jumps, in local volatility models, stochastic volatility models and jump-diffusion models.
- **Lognormal returns** understate the tails. Crashes happen far more often than the model permits, which is the same fact as the smile seen in the distribution rather than in the price.

- **Continuous costless hedging** is impossible. Real hedging is discrete and costs money, so the replication error is a real profit and loss line, and the price a desk charges must cover it.
- **Constant known rates and no dividends** are the easy ones, repaired by carrying a rate curve and a dividend forecast into the forward, in extensions of Black-Scholes.

REAL INTERVIEW QUESTION (Natixis · EQD)

What does the volatility of an underlying actually look like in reality, as opposed to what Black-Scholes assumes?

WHAT THEY TEST / ANSWER

The model assumes one number, constant through time and identical across strikes. Reality contradicts that on three axes, and naming all three is the answer.

Across strikes: implied volatility varies by strike, which is the smile or, in equities, a downward-sloping skew, so low strikes trade at higher volatilities. This follows from the market pricing fatter downside tails and rising leverage in a fall, and it is developed in the volatility smile and skew.

Across maturities: there is a term structure, usually upward sloping when spot volatility is low and inverted after a shock, because volatility mean reverts.

Through time: realized volatility is not constant at all. It clusters, so calm follows calm and violence follows violence, it mean reverts to a long-run level, and it is negatively correlated with the spot, so it rises when the market falls.

Then close on the sentence that makes it a good answer rather than a list of complaints: the market did not abandon the model, it **re-purposed** it. Black-Scholes is used as an invertible function between price and one number, so a trader can quote a strike in volatility instead of currency and compare instruments across underlyings, maturities and time. Each strike is quoted with its own implied volatility, which is precisely an admission that the assumption is false and a way of working despite it.

REAL INTERVIEW QUESTION (Edmond de Rothschild · Sales)

What is the unknown in the Black-Scholes-Merton equation?

WHAT THEY TEST / ANSWER

Volatility, and the answer is worth two sentences of context rather than one word.

Look at the inputs. The spot is on a screen. The strike and the maturity are in the contract. The interest rate comes from the curve and the dividend from a forecast that is itself quoted. Every one of them is observable. The volatility is the only input that cannot be looked up, because it describes the future rather than the present.

That single fact organises the whole market. Since the price is observable in the market and σ is not, the formula is inverted: rather than putting a volatility in to get a price, traders put the traded price in and read the **implied volatility** out. The market therefore quotes and trades in volatility, and Black-Scholes serves as the translator between the two languages.

Add the refinement if there is room: strictly, the volatility is the only unknown *parameter*, and the deeper unknown is the model itself, since the formula assumes a constant σ while the market quotes a different one for every strike. Saying that shows you know the difference between an input you cannot observe and an assumption that is false, in implied versus realized volatility.

INTERVIEW TRAP

Four errors. (1) Calling $N(d_1)$ the probability of exercise: that is $N(d_2)$, and $N(d_1)$ is the delta, a probability only under the stock numeraire. (2) Saying risk-neutral pricing assumes investors are risk neutral: it assumes replication, and the measure is a device. (3) Claiming the expected return is irrelevant "because it cancels": say why it cancels, which is that the hedge removes the direction. (4) Treating the model as a description of reality when the market's own quoting practice, one implied volatility per strike, is an explicit statement that it is not.

VISUAL / CODE TO BUILD: Interactive

A model laboratory in two panels. The **pricing panel** takes the six inputs on sliders and shows the price alongside the two terms $SN(d_1)$ and $Ke^{-rT}N(d_2)$ separately, so the reader can watch which one drives a change, plus the Greeks and the lognormal terminal distribution with the in-the-money region shaded, whose area is literally $N(d_2)$. The **replication panel** is the one that teaches: it simulates a price path, runs a delta hedge at a rebalancing frequency the reader chooses, and plots the hedged portfolio value against the option's payoff. At a very fine frequency the two lines coincide and the derivation is visibly true; coarsen the rebalancing and a gap opens whose size grows with gamma and with realized volatility, which is the replication error a desk actually lives with. A toggle lets the simulated path have a different volatility from the priced one, so the reader watches the profit and loss of being long or short volatility emerge from the hedging itself rather than from a formula.

Where this goes next

The no-arbitrage principle underneath is no-arbitrage pricing and the general replication argument is replication strategies. The sensitivities the formula generates are the Greeks, the hedging it prescribes is delta neutrality and dynamic hedging, and the empirical failure it is most famous for is the volatility smile and skew. The repairs are extensions of Black-Scholes, local volatility models and stochastic volatility models.

18.2 Extensions of Black-Scholes

sales · trading · structuring   advanced   cross-asset

The model itself, its assumptions and its derivation are built in the Black-Scholes model. This section starts from the other end. Every assumption in that model is false. The market knows this, has known it since roughly the morning after the model was published, and uses the model anyway, all day, on every desk. Understanding *how* is the difference between a candidate who can recite the formula and one who could sit next to a trader.

KEY IDEA

Black-Scholes survives because it is used as a **quoting convention**, not as a belief about the world. It is a bijection between a price, which is what the market agrees on, and a single number, the implied volatility, which is what humans can compare and talk about. Once you see it that way, every extension in this section falls into exactly one of two families:

- **Keep the formula, change the input.** The Black-Scholes machinery is preserved and one argument is replaced by something more appropriate. Cheap, exact when it applies, and the origin of most of the named models.
- **Change the dynamics.** The assumption about how the underlying moves is replaced, which generally destroys the closed form and buys a numerical method instead. Expensive, and necessary once you need a smile that is internally consistent.

The assumption failure map

The organised way to hold this in your head, and the answer to most of the interview questions in this area, is a table read from left to right: the assumption, how reality violates it, and what the market does about it.

- **No dividends.** Underlyings pay. *Fix*: replace the spot with its present value net of the income stream. Same formula.
- **Constant risk-free rate, borrow equals lend.** Rates move, and there are several curves. *Fix*: discount off the right curve and, for long-dated products, model rates stochastically.
- **Constant volatility.** Volatility varies with time, with strike and at random. *Fix*: three separate extensions, escalating in cost.
- **Lognormal returns, continuous paths.** Returns have fat tails and prices jump. *Fix*: jump-diffusion or a fat-tailed distribution.
- **Continuous, costless hedging.** You hedge daily and pay a spread each time. *Fix*: an adjusted volatility, and a genuine bid-ask in volatility terms.

- **European exercise.** Most listed single-stock options are American. *Fix*: no closed form, so trees or a Monte Carlo with a regression for the exercise decision.
- **No counterparty or funding cost.** Trades are collateralised, counterparties default, and balance sheet is not free. *Fix*: valuation adjustments layered on top of the model price.

REAL INTERVIEW QUESTION (CACIB · FX)

Which Black-Scholes assumptions do you think are false? Why?

WHAT THEY TEST / ANSWER

The trap is to recite the assumption list, which is the *previous* question. They asked which ones are false and why, so every item must come with a piece of market evidence and a consequence. Give three or four properly rather than eight superficially.

Constant volatility is false, and the proof is on the screen: if it were true, implied volatility would be identical across every strike and maturity, and instead you see a smile and a term structure, which is the market telling you the assumption fails (see the volatility smile and skew). **Lognormal returns with continuous paths** is false, and the evidence is that realised returns have far more kurtosis than a Gaussian and that prices gap overnight, on earnings, and on central bank decisions; the consequence is that short-dated out-of-the-money options are worth much more than the model says. **Continuous costless hedging** is false, and this one is the trader's daily reality: you rebalance discretely and cross a spread each time, so a seller cannot replicate for the model price and there is a genuine bid-ask in volatility. **Constant rates** is false and matters mainly for long-dated products, which is why it is a real problem in structured products and almost none in a 3-month FX option.

Then, since this is an FX desk, add the asset-specific point: in FX the natural version of the model is two-rate, so the "dividend yield" is the foreign interest rate, and the model's inversion symmetry is what makes the FX smile a smile rather than a smirk (see the FX volatility smile). Close with the sentence that reframes the whole question: the assumptions being false is not a scandal, because nobody uses the model as a description of the world; they use it as the map between price and implied volatility.

Family one: same formula, different input

Three of the best-known "models" in finance are Black-Scholes with one argument swapped. Recognising that they are one object saves an enormous amount of memory.

Continuous dividend yield (Merton). If the underlying pays a continuous yield q, the holder of the option does not receive it, so the relevant quantity is the spot stripped of that income. Replace S by Se^{-qT} throughout. The forward becomes $F = Se^{(r-q)T}$, which is exactly the forward from forwards, and dividend valuation itself lives in dividends.

Options on forwards and futures (Black 76). If the underlying of the option is a forward or future F rather than a spot, the forward has zero carry by construction and the formula collapses to

$$C \;=\; e^{-rT}\big[F\,N(d_1) - K\,N(d_2)\big], \qquad d_1 = \frac{\ln(F/K) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}.$$

This is the workhorse for caps, floors and swaptions in rates (see interest rate swaps) and for commodity options.

FX (Garman-Kohlhagen). A currency pays interest in its own right, so the foreign interest rate r_f plays exactly the role of a continuous dividend yield: substitute $q = r_f$. Nothing else changes. See FX options.

KEY IDEA

All three are the same statement. **Black-Scholes prices an option on a forward**, and everything that looks like a different model is just a different way of computing the forward and the discount factor. If you can say that sentence in an interview, you have compressed three formulas into one.

Volatility, in three escalating repairs

The constant-volatility assumption fails in three distinct ways, and there is a distinct fix for each. Candidates routinely blur them.

1. Volatility varies with time but is known in advance. This is the cheapest repair and it costs nothing: Black-Scholes only ever uses volatility through the variance to maturity, so replace $\sigma^2 T$ by $\int_0^T \sigma^2(t)\,dt$. The model is untouched and the number you plug in is the **root-mean-square** volatility over the life of the option. This single observation explains the entire at-the-money term structure without any new mathematics.

WORKED EXAMPLE

The average is not the average. Suppose the market expects volatility of 15% for the next six months and 25% for the six months after that. What is the one-year implied volatility?

Variances add, volatilities do not:

$$\sigma_{1y} = \sqrt{\tfrac{1}{2}(0.15)^2 + \tfrac{1}{2}(0.25)^2} = \sqrt{\tfrac{1}{2}(0.0225) + \tfrac{1}{2}(0.0625)} = \sqrt{0.0425} = 20.62\%.$$

Not 20%. The naive arithmetic average understates it, and the gap widens with the dispersion, which is the same arithmetic-versus-quadratic point made in expectation, variance and higher moments. Run it the other way and you have the standard forward-volatility calculation: given a 6-month vol and a 1-year vol, the implied volatility for the second six months is whatever makes the variances add, and if that number comes out negative your quoted surface is arbitrageable.

2. Volatility varies with strike as well. Fitting a surface exactly with a volatility that is a deterministic function of spot and time is local volatility, Dupire's construction. It reproduces today's smile perfectly by definition, which is its strength, and it predicts a future smile dynamic that is generally wrong, which is its weakness.

3. Volatility is itself random. Making volatility a second stochastic process gives stochastic volatility models, of which Heston and SABR are the two you must know. They fit the smile less exactly than local volatility but produce far more plausible smile dynamics, which is what matters if you are hedging rather than marking.

Separately, allowing the price path to *jump* is jump-diffusion, which is the natural repair for the short-dated wing that no diffusion model can reach, since a continuous process cannot travel far in a week but a jump can.

REAL INTERVIEW QUESTION (Barclays · Exo)

Can you use the Black-Scholes formula with local volatility?

WHAT THEY TEST / ANSWER

The answer is a careful "no, and here is the confusion the question is built on", which is exactly what they are testing. Local volatility $\sigma_{\text{loc}}(S, t)$ is a *state-dependent* function, so the underlying's dynamics are no longer a geometric Brownian motion with a constant coefficient and the Black-Scholes closed form simply does not apply. You price under local volatility by solving the pricing partial differential equation with a state-dependent diffusion coefficient, or by Monte Carlo, not by evaluating $N(d_1)$ and $N(d_2)$.

Then give the two nuances that separate a good answer. First, you *can* always quote a local-volatility price as a Black-Scholes implied volatility, because the formula is a bijection between price and vol: you price properly, then invert Black-Scholes to express the answer. That is quoting, not pricing, and conflating the two is the actual mistake being probed. Second, do not confuse local volatility with implied volatility. They are different objects related by Dupire's formula, and the standard rule of thumb, that near the money the local volatility skew is roughly twice the implied volatility skew, is exactly the kind of detail that shows you have handled a surface rather than read about one. If you want to close it, note the one case where a Black-Scholes-shaped formula does survive: if volatility is a deterministic function of *time alone*, the formula holds with the root-mean-square volatility, and it is the dependence on the spot level that breaks it.

Discrete hedging and transaction costs

The derivation assumes you rebalance continuously and for free. A real desk rebalances at intervals and crosses a spread each time, which has two consequences that show up directly in a trader's PnL.

The first is **hedging error**. Between rebalances your delta is stale, so a gamma position accumulates a residual whose size scales with the square of the move and with the rebalancing interval. This is the mechanism behind the whole gamma and theta story in gamma and convexity, and it is why realised PnL and model PnL differ even when the volatility forecast was perfect.

The second is **cost**, and it is directional. A seller of an option must buy high and sell low as they rebalance, so costs make the position worse and they need to charge a higher volatility. A buyer's costs work the other way. Leland's classical result makes this precise: hedging at intervals δt with a proportional round-trip cost rate k, the seller should price with an adjusted volatility

$$\hat{\sigma}^2 \;=\; \sigma^2 \left[1 + \sqrt{\tfrac{2}{\pi}}\, \frac{k}{\sigma\sqrt{\delta t}} \right],$$

with the sign reversed for the buyer.

WORKED EXAMPLE

What costs are worth in vol points. Take $\sigma = 20\%$, daily rebalancing so $\delta t = 1/252$ and $\sqrt{\delta t} = 0.0630$, and a round-trip cost of 10 bp, so $k = 0.001$. Then

$$\sqrt{\frac{2}{\pi}}\,\frac{0.001}{0.20 \times 0.0630} = 0.7979 \times 0.0794 = 0.0633,$$

so $\hat{\sigma}^2 = 0.04 \times 1.0633 = 0.04253$ and $\hat{\sigma} = 20.62\%$. The seller needs about 0.6 of a volatility point over the frictionless price.

Now make the underlying less liquid, $k = 50$ bp: the bracket becomes 1.3166, $\hat{\sigma}^2 = 0.05267$, and $\hat{\sigma} = 22.95\%$, so nearly three volatility points. That is the honest origin of the bid-ask in volatility, and it explains why a market maker's two-way price widens on an illiquid name even when their view on realised volatility has not moved at all. Note the structure of the correction: it grows as δt shrinks, so hedging more often is not free, and there is an optimal rebalancing frequency trading hedging error against cost.

American exercise, and why it needs a machine

A European option has a closed form because there is nothing to decide. An American option carries an **optimal stopping problem** inside it: at every instant the holder compares the value of exercising with the value of waiting, and the boundary between those two regions is itself part of the unknown. That is why no Black-Scholes-style formula exists.

That places American exercise squarely in the second family of this section: the assumption cannot be repaired by swapping an input, so the closed form goes and a numerical method takes its place. Which method, why plain Monte Carlo cannot handle early exercise at all, and what least-squares Monte Carlo does about it are developed where they belong, in Monte Carlo methods and partial differential equations in finance. The one fact worth carrying out of this section is the exception: an American call on a non-dividend-paying stock is never exercised early, so it equals the European call and Black-Scholes prices it exactly. That is the only case where this assumption failure costs you nothing.

REAL INTERVIEW QUESTION (Barclays · Exo)

Why use a model with both stochastic volatility and stochastic rates, and what does it cost in complexity?

WHAT THEY TEST / ANSWER

This is the logic of this section applied twice, so answer it as a cost-benefit rather than as a description.

Why. Each extension is bought to fix an assumption that has started to matter. Stochastic volatility is bought because constant volatility cannot produce a smile that moves realistically, so it is what you need to price and hedge anything whose value depends on where the smile will be later. Stochastic rates are bought when the option is long-dated enough that uncertainty in the discount factor is comparable to uncertainty in the underlying. Put them together and you also buy the thing neither gives you alone: the **correlation between rates and the underlying**, which is a real price effect on long-dated equity and FX structures and is invisible to either model on its own.

What it costs. Be specific, because "it is more complex" is not an answer. You lose the closed form, so every price is a simulation or a high-dimensional grid, and grids degrade badly as factors are added. Calibration becomes joint and under-determined: the vanilla surface pins down the volatility parameters and the rates market pins down the rate parameters, but the *correlation* between them is barely observable, so it is usually assumed, and the price then depends on a number nobody can mark. Greeks become noisy and expensive because they come from simulation. And the model has to be recalibrated and revalidated, which is an operational cost rather than a computational one.

Then give the judgement, which is what they are testing: you do not buy extensions you cannot pay for. For a three-month vanilla, deterministic rates are fine and the extra machinery is pure cost. For a ten-year autocallable or a long-dated quanto, the rate-underlying correlation is a first-order term and omitting it is not conservatism, it is a mispricing. The right question is always which assumption is currently the binding one.

Rates, discounting, and the adjustments on top

The last group of extensions is the one that has changed most since 2008 and that candidates rarely mention.

Stochastic rates matter when the option's maturity is long enough that uncertainty in the discount factor is comparable to uncertainty in the underlying. For a 3-month FX option this is negligible; for a 10-year structured note it is not, and the correlation between rates and the underlying enters the price.

Discounting is no longer one curve. Collateralised trades discount off the overnight-indexed curve rather than a term rate, which is a change in the meaning of r in the formula rather than a change in the formula. See OIS, LIBOR, SOFR and €STR.

Valuation adjustments. Counterparty default risk, funding cost and capital are priced as explicit additions to the model value, the family of XVAs covered in CVA, DVA and FVA. Black-Scholes prices a claim against a counterparty that cannot fail, funded at a rate nobody actually pays.

REAL INTERVIEW QUESTION (Société Générale · Structured Products)

What are the limits of Black-Scholes?

WHAT THEY TEST / ANSWER

Do not give a list of assumptions. Give a list of *limits*, meaning failures with consequences, and organise them so the interviewer can hear the structure.

Say there are three tiers. **Tier one, cosmetic**: assumptions that are false but repairable inside the same formula, such as dividends, a foreign interest rate, or a volatility that varies deterministically with time. You change an input and the machinery survives. **Tier two, structural**: the constant-volatility assumption, which the existence of the smile refutes directly, and the continuous lognormal path assumption, which fat tails and overnight gaps refute. These require different dynamics, meaning local volatility, stochastic volatility, or jumps, and they cost you the closed form. **Tier three, practical**: continuous costless hedging, which is the one that actually costs a desk money every day and is the origin of the bid-ask in volatility, plus counterparty and funding costs, which the model assumes away entirely.

Then, because this is a structured products desk, land the point that matters to them: for exotics the limits are not academic. A payoff that depends on the whole path, or on where the smile will be in a year, is priced almost entirely by the model's *dynamics*, not by today's marks, so two models that fit today's vanilla surface identically will disagree materially on a barrier or an autocallable. That is model risk, it is a live number on their book, and naming it is what separates this answer from a textbook one. Finish with the defence of the model: it remains the market's language, because implied volatility is a Black-Scholes construct and every one of these richer models is quoted, compared and risk-managed through it.

VISUAL / CODE TO BUILD: Interactive

A model-comparison bench. The reader picks a payoff (vanilla call, digital, up-and-out call, one-year autocallable) and the panel prices it under four engines in parallel: Black-Scholes with a flat volatility, Black-Scholes with a term-structure of deterministic volatility, local volatility, and a stochastic volatility model, all four *calibrated to the same vanilla surface*. A prices table shows them agreeing to the basis point on the vanilla and diverging by percentage points on the barrier and the autocallable, which is model risk made visible in one screen. Two extra controls: a transaction-cost slider that applies the Leland adjustment and shows the resulting two-way volatility price, and a rebalancing-frequency slider that plots simulated hedging PnL distribution against frequency so the reader can find the optimum trading hedging error against cost.

INTERVIEW TRAP

(1) **Answering "what are the limits" with the assumption list.** The limits are the consequences, not the assumptions. (2) **Averaging volatilities.** Variances add, volatilities do not. Two six-month periods at 15% and 25% give 20.62% over the year, not 20%. (3) **Treating local volatility and implied volatility as the same number.** They are different functions linked by Dupire's formula. (4) **Reaching for Monte Carlo on a one-dimensional American option.** A tree or a grid is better, and naive Monte Carlo cannot handle early exercise at all. (5) **Forgetting that an American call on a non-dividend-paying stock equals the European one.** It is the standard follow-up and the cheapest mark available. (6) **Believing the model is dead.** Every alternative model in this chapter is quoted, calibrated and hedged in Black-Scholes implied volatility. It is the coordinate system, whatever the dynamics underneath.

Where this goes next

The three dynamics-level repairs each have their own section: local volatility, stochastic volatility with Heston and SABR, and jump diffusion. Choosing between them, and fitting them to a surface, is model calibration, and the numerical schemes that make them computable are Euler and Milstein. The products where the choice of model actually decides the price are exotic options.

18.3 Local volatility models

trading · structuring | advanced | equity · fx · cross-asset

The market quotes a different implied volatility for every strike. That is not a model, it is a set of inconsistent models: you cannot price a barrier with the 90% strike's volatility and the same barrier with the 110% strike's volatility and expect the two answers to agree. Something is needed that reproduces every vanilla price with a *single* process. Local volatility is the minimal answer to that question, and it is still the workhorse of exotic pricing.

The idea

Keep the Black-Scholes framework and make one change: let the volatility depend on where the spot is and when,

$$dS_t = (r-q)\,S_t\,dt + \sigma_{\text{loc}}(S_t,t)\,S_t\,dW_t.$$

There is still one Brownian motion and no new source of randomness. The volatility is still deterministic, in the sense that if you know the spot and the date you know the volatility. The only difference from Black-Scholes is that the number is now a surface rather than a constant.

Dupire's result is that this is enough, and that the surface is *uniquely determined* by the vanilla prices. If $C(K,T)$ is the market price of a call at strike K and maturity T, then

$$\sigma^2_{\text{loc}}(K,T) \;=\; \frac{\dfrac{\partial C}{\partial T} + (r-q)\,K\,\dfrac{\partial C}{\partial K} + qC}{\frac{1}{2}K^2\,\dfrac{\partial^2 C}{\partial K^2}},$$

which with zero rates and dividends is simply the ratio of the calendar derivative to the butterfly derivative. Both of those are observable: the numerator is the price of an infinitesimal calendar spread, and the denominator is the price of an infinitesimal butterfly, which is the risk-neutral density.

KEY IDEA

Local volatility is not "the volatility if spot goes to K". It is the market's **expected instantaneous variance, conditional on the spot being at K at time T**. That is why implied and local volatility are different objects: an implied volatility is an *average* of local volatilities along all the paths reaching that strike, while a local volatility is the value at one point of the surface. The relationship is the same as between a yield and a forward rate, and it produces the standard rule of thumb: near the money, **the local volatility skew is about twice the implied volatility skew**.

REAL INTERVIEW QUESTION (BNP · Exo)

Explain local volatility models.

WHAT THEY TEST / ANSWER

Four beats: the problem, the fix, the calibration, the use.

The problem. The market gives one implied volatility per strike and maturity, which is a statement that Black-Scholes is wrong. To price anything that is not a vanilla you need one process that is simultaneously consistent with all of those prices.

The fix. Make volatility a deterministic function of spot and time, $\sigma_{\mathrm{loc}}(S, t)$. This keeps the model *complete*, meaning the option is still replicable by trading the underlying alone, so there is a unique price and a well-defined delta.

The calibration. Dupire's formula inverts the vanilla surface into the local volatility surface, and it is exact rather than a fit: by construction the model reprices every listed vanilla. Say what makes this delicate in practice: the formula involves a second derivative of price in strike, so it amplifies noise, and the input surface must first be smoothed and made free of butterfly and calendar arbitrage or the formula returns negative variances.

The use. Price exotics by Monte Carlo or by a partial differential equation with that surface, which guarantees the exotic is consistent with the vanilla market. This matters most for products whose value depends on the whole smile rather than on one strike: barriers, digitals and everything in exotic options.

Close on what it buys and what it does not: it is the cheapest way to be arbitrage-free against vanillas, and it gets the *today* right by construction. Whether it gets tomorrow right is the next question.

REAL INTERVIEW QUESTION (Barclays · Exo)

Can you use the Black-Scholes formula with a local volatility?

WHAT THEY TEST / ANSWER

No, and the reason is worth stating precisely because the question is designed to catch a plausible shortcut.

The Black-Scholes *formula* is the closed-form solution of the Black-Scholes PDE under a **constant** volatility. With σ a function of spot and time, the PDE still holds, with $\sigma_{\text{loc}}(S,t)^2$ in place of σ^2, but it no longer has that closed-form solution. You solve it numerically, by finite differences or by simulating the diffusion.

Then give the three refinements that turn a "no" into a good answer. **You cannot plug a local volatility into the formula as if it were an implied volatility**: they are different objects, one a point value and the other a path average, so substituting one for the other misprices even a vanilla. **You can still quote in implied volatility**: price the instrument under local volatility, then invert Black-Scholes to express the result as a number, which is the model being used as a translator, as in the Black-Scholes model. And **consistency is guaranteed for vanillas**: because the surface was calibrated to them, a vanilla priced under local volatility returns exactly its market implied volatility, so the two agree where they should and differ everywhere else.

The problem: right prices, wrong dynamics

Local volatility fits today's surface perfectly. What it does badly is predict how that surface will move, and for a large class of products that is what matters.

The failure is specific and it is worth being able to state it. Because the volatility is a fixed function of the spot level, the model implies that when the spot moves, the whole smile moves the *wrong way*: as spot falls the model shifts the smile toward *higher* strikes, and as spot rises it shifts it toward lower ones, whereas real equity smiles travel in the same direction as spot. Concretely, the model predicts an at-the-money implied volatility that moves roughly twice as fast with spot as it actually does, which makes the model's delta wrong in a way traders correct by hand.

Three consequences follow.

- **Forward smiles are wrong.** The smile the model implies for an option starting in one year is flatter than the market's, so anything forward-starting, cliquets, ratchets, forward-start options, is systematically mispriced.
- **There is no vol-of-vol.** Volatility is deterministic given the spot, so the model cannot price a product whose payoff depends on volatility being uncertain, and it has no natural volatility-of-volatility parameter to calibrate.
- **Hedge ratios are unstable.** The delta contains the model's view of how the smile moves, and since that view is wrong, the delta is biased, which is a daily profit and loss problem rather than a theoretical one.

REAL INTERVIEW QUESTION (BNP · Hybrid)

What are the differences between local volatility and stochastic volatility, and how do you price with each?

WHAT THEY TEST / ANSWER

Compare on four axes and then say what the market actually does, because the honest answer is that neither is used alone.

Source of randomness. Local volatility has one Brownian motion and a volatility that is a deterministic function of (S, t). Stochastic volatility adds a *second* Brownian motion driving the variance, correlated with the first, as in stochastic volatility models.

Completeness. Local volatility is complete, so the option is replicable with the underlying alone and the delta is unique. Stochastic volatility is incomplete: volatility is not tradable, so you need a second instrument, an option, to hedge, and the price depends on a market price of volatility risk.

Calibration. Local volatility fits the vanilla surface *exactly*, by construction. Stochastic volatility has a handful of parameters and fits it only approximately, though its shape is far more natural.

Dynamics. This reverses the ranking: local volatility gets today right and tomorrow wrong, stochastic volatility produces realistic smile dynamics and a genuine vol-of-vol, which is what forward-starting products need.

How you price with each. Both by Monte Carlo, and by a PDE in low dimension: one space dimension for local volatility, two for stochastic volatility, which is why the second is materially more expensive. Neither has a closed form for exotics, and only some stochastic volatility models have one for vanillas.

Close with the resolution, which is the answer a structuring desk wants: **stochastic local volatility**. Take a stochastic volatility model for the dynamics and multiply it by a deterministic *leverage function* calibrated so the combination still reprices every vanilla exactly. That gives exact calibration and realistic dynamics at once, and it is the industry standard for exotics. It is developed in stochastic volatility models, which owns it, since it is structurally a stochastic volatility model with a leverage function attached rather than a variant of the local volatility one.

REAL INTERVIEW QUESTION (Barclays · Exo)

What is the point of a model combining local volatility with stochastic rates, and for which products?

WHAT THEY TEST / ANSWER

The point is that for long-dated products the interest rate is not a constant, and treating it as one throws away a risk that is comparable in size to the equity risk.

Why it matters. Rate volatility contributes to a payoff's uncertainty in proportion to maturity, so it is negligible for a three-month option and material for a ten-year one. More importantly, the *correlation* between rates and the underlying changes the forward's distribution, and the forward is what an option is written on. A deterministic-rate model cannot express a view on that correlation at all, and cannot price any product whose payoff depends on both.

Which products. Long-dated structured notes and capital-guaranteed products, where the zero-coupon leg is a large fraction of the value and moves with rates; autocalls with long maturities, whose call probability depends on the forward; convertible bonds, which are simultaneously equity and rates instruments; and genuine hybrids such as equity-linked notes with a rate-dependent coupon, or the power-reverse-dual-currency family in FX, which is the textbook case.

The cost. Each additional stochastic factor adds a dimension, so a PDE becomes impractical and you are in Monte Carlo, calibration becomes a joint problem across two markets, and the correlation between rates and spot is a parameter that is barely observable and has to be marked conservatively. Say that last point explicitly: the model does not remove the risk, it makes it explicit and gives it a number, and the number is the least reliable input in the model.

INTERVIEW TRAP

Four errors. (1) Confusing local with implied volatility: implied is a path average, local is a point value, and near the money the local skew is about twice the implied skew. (2) Believing that an exact fit to vanillas means the model is right, when the fitting is the easy half and the dynamics are the half that prices exotics. (3) Applying Dupire's formula to a raw quoted surface without smoothing and arbitrage-checking it first, which produces negative variances. (4) Using pure local volatility for forward-starting products, which is exactly the class it prices worst.

Where this goes next

The model this one repairs is the Black-Scholes model, and the market observation that forces the repair is the volatility smile and skew. The alternative family is stochastic volatility models, with the two standard members in the Heston model and the SABR model, and the third repair is jump-diffusion models. The products that make the choice matter are exotic options and path-dependent products.

18.4 Stochastic volatility models

trading · structuring | advanced | equity · fx · cross-asset

Black-Scholes assumes volatility is a constant. The market quotes a different implied volatility for every strike and every maturity, so the assumption is false in a way you can read straight off a screen. There are two ways to repair it, and the choice between them is one of the defining questions of a derivatives desk.

The first is to keep a single source of randomness and let volatility be a known function of where the spot is and what time it is, $\sigma(S,t)$. That is a local volatility model, and it lives in local volatility models. The second is to give volatility a life of its own: a second random process, driven by its own Brownian motion. That is stochastic volatility, and it is this section.

The general form

Every stochastic volatility model is a version of the same two equations. Write v_t for the instantaneous variance:

$$\frac{dS_t}{S_t} = \mu\,dt + \sqrt{v_t}\,dW_t, \qquad dv_t = a(v_t,t)\,dt + b(v_t,t)\,dZ_t, \qquad d\langle W,Z\rangle_t = \rho\,dt.$$

The first equation is Black-Scholes with the constant σ replaced by a moving $\sqrt{v_t}$. The second is whatever you decide variance does, and the models differ only in the choices of a and b: Heston takes a mean-reverting and b proportional to $\sqrt{v}$, SABR takes volatility lognormal with no mean reversion, and so on. Those live in the Heston model and the SABR model.

What matters at this level is that whatever you choose, you have introduced exactly three new economic quantities, and each one buys you one feature of the observed surface.

New parameter	What it controls	What it fixes
Vol of vol, the size of b	how much variance itself moves	curvature of the smile
Correlation ρ	spot and vol moving together	the tilt, i.e. the skew
Mean reversion in a	how fast vol returns to normal	the term structure

KEY IDEA

Three knobs, three features. Vol of vol makes the smile curved, correlation tilts it, and mean reversion decides how quickly it flattens as maturity grows. If you remember nothing else about stochastic volatility, remember that map, because it is what lets you look at a surface and say which parameter is wrong.

Why random volatility produces a smile at all

This is the argument worth being able to give from scratch, and it needs no stochastic calculus. Suppose volatility is random but independent of the spot. Then the terminal distribution of the stock is not one lognormal; it is a **mixture** of lognormals, one for each possible volatility path, weighted by how likely that path was.

A mixture of lognormals with different widths has *fatter tails* than any single lognormal with the same average variance. Extreme outcomes need only one of the high-volatility scenarios to occur. What is thinned out is not the centre but the *shoulders*: the high-volatility scenarios push their mass out to the tails while the low-volatility ones concentrate theirs near the middle, so the mixture is simultaneously more peaked than a single lognormal and fatter-tailed than one, and thinner in between. Options far from the money pay only in the tails, so they are worth more than a single-volatility model says, so their implied volatilities are higher. That is a smile.

WORKED EXAMPLE

Take the crudest possible stochastic volatility model: volatility is drawn once, and it is either 15% or 35%, each with probability one half. Forward at 100, one year, no rates. To price an option, price it under each volatility and average the two prices, since the draw is independent of the stock.

Strike	Mixture price	Implied volatility
70	31.183	28.43%
80	22.664	26.63%
90	15.403	25.38%
100	9.935	**24.97%**
110	6.309	25.31%
120	4.086	26.13%
130	2.734	27.16%

Read the last column. The implied volatilities are not constant, and they are not even close: they bottom out at the money and rise in both directions, by three and a half points at the 70 strike. A model in which volatility is genuinely constant could never produce that column, and we produced it by allowing exactly one random draw.

Two numbers are worth checking against your intuition. The at-the-money implied volatility is 24.97%, essentially the arithmetic average of 15 and 35. That is not a coincidence: the Black-Scholes price is very nearly linear in volatility at the money, so averaging prices is almost the same as averaging volatilities. Away from the money the price is *convex* in volatility, so averaging two prices gives more than the price at the average volatility, and the implied volatility must rise to make up the difference. The root-mean-square volatility, $\sqrt{(0.15^2 + 0.35^2)/2} = 26.93\%$, is a third number again, which is worth noticing precisely because it is tempting to assume the smile should be flat at that level and it is not flat at any level.

Where the skew comes from

Independence gives a symmetric smile. Equity markets do not have a symmetric smile; they have a downward slope, high implied volatility at low strikes. That asymmetry is what the correlation parameter is for.

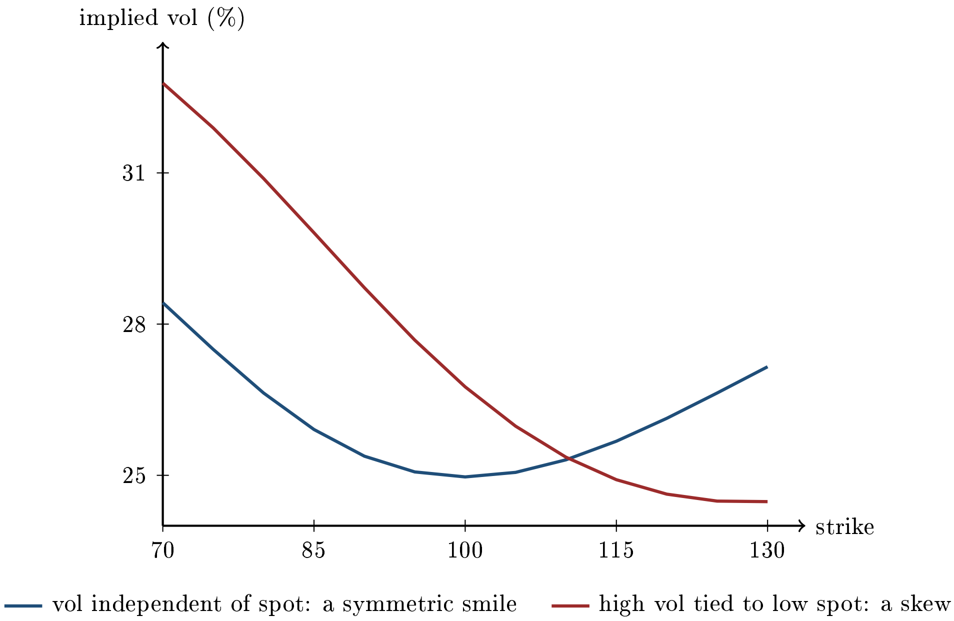
Set $\rho < 0$, so that when the spot falls, volatility rises. Now the mixture is no longer symmetric: the scenarios in which the stock ends up low are also the scenarios in which it was moving violently, so the left tail is fed by the high-volatility paths and the right tail is starved of them. Low-strike puts are worth more and high-strike calls are worth less, which is exactly a downward-sloping skew.

WORKED EXAMPLE

Keep the two-state model and tie the states to the spot. In the bad state, probability one half, the forward is 90 and volatility is 35%. In the good state, probability one half, the forward is 110 and volatility is 15%. The overall forward is still $(90 + 110)/2 = 100$, so nothing has been added to the drift; all that has changed is that the high-volatility state is now the one where the stock is low, which is negative spot-vol correlation in its crudest form.

Strike	Mixture price	Implied volatility
70	31.910	32.78%
85	20.034	29.81%
100	10.642	26.76%
115	4.780	24.91%
130	2.015	24.48%

The column now falls monotonically, from 32.78% at the 70 strike to 24.48% at 130. The symmetric smile has become a smirk, and the only thing that changed was making the high-volatility state coincide with the low-spot state.



Both curves come out of a model with one random draw and no calculus. That is the whole economic content of stochastic volatility: uncertainty about volatility gives you curvature, and correlation between volatility and spot gives you tilt.

The price you pay: the market is no longer complete

In Black-Scholes there is one source of randomness and one instrument to hedge it with, so the option is exactly replicable and its price is forced. Add a second Brownian motion and that stops being true. You cannot hedge two random factors with one asset.

The consequences are not academic.

- **You need a second hedging instrument**, and it has to be another option, because only an option has vega. A delta hedge alone leaves the volatility factor naked.

- **The price is no longer unique from no-arbitrage alone.** It depends on the market price of volatility risk, that is, on what investors demand to bear it. In practice this is not estimated from history; it is calibrated away by fitting the model to traded option prices, which is why every desk's model is anchored to the vanilla surface before it is used for anything else.
- **The delta changes.** If spot and volatility are correlated, then a spot move also implies a volatility move, and the hedge ratio has to account for it. The resulting minimum-variance delta is the Black-Scholes delta plus a vega term multiplied by how much implied volatility moves per unit of spot. The Greek itself is treated in skew and smile sensitivities; the point here is that the correction is a *model output*, not an adjustment bolted on afterwards.

INTERVIEW TRAP

Saying that stochastic volatility "makes the model more realistic" and stopping. It also breaks perfect replication, which is the foundation the entire Black-Scholes argument rests on. A candidate who mentions incompleteness unprompted, and says that it is why the model must be calibrated to traded options rather than estimated from history, has said the one thing about this topic that is hard to fake.

Against local volatility: same fit, different dynamics

Here is the comparison that matters, and it is the reason both model families exist side by side.

A local volatility model, correctly calibrated, reproduces today's vanilla surface *exactly*, which no stochastic volatility model with three or four parameters can do. So on the criterion of fitting today's prices, local volatility wins outright.

But two models that agree on every vanilla price today can disagree completely about everything else, because a vanilla only sees the terminal distribution at one date. Anything that depends on how the surface *moves* is left undetermined by the fit, and that is where the two families part.

	Local volatility	Stochastic volatility
Fit to today's surface	exact	approximate
Number of random factors	one	two
Market complete?	yes	no
Forward smile	flattens too fast	persists
Smile when spot moves	shifts the wrong way	broadly sticky in moneyness
Natural home	vanilla-consistent pricing	forward-starting and vol-sensitive payoffs

The row that decides real trades is the forward smile: the smile the model says will be observed at a future date, seen from today. Local volatility makes it flatten away, because the only thing driving volatility is the spot level and by that future date the spot could be anywhere. Stochastic volatility keeps it, because volatility is still random at that future date. Any product whose value depends on a future smile, cliquets, forward-starting options, anything that resets, is priced on that row and not on today's fit.

KEY IDEA

Calibrating to today's vanilla surface does not pin down a model. It pins down the marginal distribution at each maturity and leaves the dynamics free. Two models can agree on every vanilla price and differ by a large amount on a barrier or a cliquet. Choosing a model is choosing dynamics, not choosing a fit.

The practical resolution is to stop choosing. Production systems for exotics generally run **local-stochastic volatility**: a stochastic volatility engine that supplies the dynamics, multiplied by a local volatility correction, the leverage function, that forces the exact fit to vanillas. You get the surface from local volatility and the behaviour from stochastic volatility, at the cost of a considerably harder calibration.

One limitation of stochastic volatility alone is worth knowing because it is a standard follow-up. A diffusion needs time to generate a wide distribution, so at very short maturities a stochastic volatility model cannot produce a smile as steep as the market's. The market's very short-dated smile is steep because the market prices the possibility of a gap, and reproducing that requires jumps, which is jump-diffusion models.

Interview angle

REAL INTERVIEW QUESTION (Société Générale · Structuring)

What is the difference between local volatility and Heston? What is stochastic volatility?

WHAT THEY TEST / ANSWER

Answer the general question first and the named model second, because that ordering shows you understand the family rather than one member of it.

Stochastic volatility means volatility is a second random process with its own Brownian motion, correlated with the spot's. Write the two equations, say that a model is characterised by what it assumes variance does, and give the three-knob map: vol of vol produces the curvature of the smile, spot-vol correlation produces the tilt, and mean reversion controls how quickly the smile flattens with maturity. Then say why a smile appears at all, which is the part that shows understanding: random volatility makes the terminal distribution a mixture of lognormals, mixtures have fatter tails than any single lognormal, and fatter tails mean out-of-the-money options are worth more than a constant volatility implies.

Local volatility keeps one random factor and makes volatility a deterministic function of spot and time. **Heston** is one specific stochastic volatility model, in which variance mean-reverts and its volatility scales with the square root of variance.

Then the difference that matters, which is not the fit. Local volatility reproduces today's surface exactly and Heston does not, so if fitting vanillas were the criterion local volatility would simply win. It is not the criterion, because two models can agree on every vanilla and disagree about how the surface moves. Local volatility has one factor, so it is complete and the smile it predicts for a future date flattens away; Heston has two, so it is incomplete, it needs a second option to hedge, and it keeps a forward smile. Anything forward-starting or resetting is priced on that difference. Finish with what desks actually do, which is neither: local-stochastic volatility, using a leverage function to bolt the exact vanilla fit onto stochastic dynamics.

REAL INTERVIEW QUESTION (Société Générale · Volatility)

Local vol? Stochastic? When do you use one or the other?

WHAT THEY TEST / ANSWER

This is the applied version of the previous question, so skip the definitions and answer the "when". Give the decision rule first, then the two supporting reasons.

The rule: use local volatility when the payoff's value is determined by the marginal distributions the vanilla surface already pins down, and stochastic volatility when the payoff depends on how volatility itself behaves through time. If the product can be sensibly hedged with a static book of vanillas plus a delta, local volatility is adequate and its exact fit is an advantage. If the product's risk is genuinely in the volatility process, local volatility will price it confidently and wrongly.

Reason one, the forward smile. A local volatility model implies a future smile that is much flatter than today's, because the only thing that drives its volatility is the spot level and by that future date the spot is diffuse. So anything whose payoff is set against a future starting point, a cliquet, a forward-starting option, a ratchet, is mispriced by construction, usually too cheap on the volatility-of-volatility it is short.

Reason two, the dynamics. The two families disagree about how the smile moves when spot moves, and therefore about the delta of an option, not only about its price. A desk that hedges a large book on local volatility deltas in a skewed market is running a systematic hedging error even if every vanilla in the book is marked correctly.

Then the honest answer to what happens in practice: neither on its own. Local-stochastic volatility is the production choice for exotics because it takes the exact vanilla fit from one and the dynamics from the other. And add the limitation that gets asked next: at very short maturities neither diffusion generates enough smile, and you need jumps.

REAL INTERVIEW QUESTION (Société Générale · EQD)

What is the relationship between stochastic volatility and local volatility? And what is the beta of a linear regression?

WHAT THEY TEST / ANSWER

Two questions bolted together, and the second is not a change of subject; it is the mathematical shape of the answer to the first.

The relationship: they are not rivals describing different worlds. For any stochastic volatility model there is a local volatility function that reproduces exactly the same vanilla prices, and it is not the average volatility. It is the *conditional expectation* of the instantaneous variance, given that the spot is at that level at that time. In words: the local volatility a model implies at spot level S and time t is the average of the instantaneous volatilities across all the paths that happen to be at S at time t. So local volatility is a projection of a two-factor model onto one factor, chosen so that all vanilla prices survive the projection.

That projection is exactly what a regression does, which is the bridge to the second question. **The beta of a linear regression** of Y on X is $\beta = \mathrm{Cov}(X,Y)/\mathrm{Var}(X)$, the slope that minimises mean-square error, so it is the projection of Y onto the *linear* functions of X. Widen the projection from linear functions to *all* functions of X and the minimiser becomes $\mathbb{E}[Y \mid X]$, which is exactly what local volatility is. Local volatility is the same operation with Y the instantaneous variance and X the spot: throw away the part of the variance that the spot cannot explain, and keep its conditional mean.

Then say what the projection costs, because that is the point of the question. A projection preserves the marginal distribution at each date and destroys the dynamics. Everything the discarded factor was responsible for, the forward smile, the volatility of volatility, the need for a second hedging instrument, is gone. That is why the two models agree on every vanilla and can disagree materially on a cliquet or a barrier.

INTERVIEW TRAP

Three errors. (1) Claiming stochastic volatility fits the surface better than local volatility: it does not, local volatility fits exactly and that is precisely why it is not enough. (2) Treating vol of vol and correlation as interchangeable smile parameters, when one produces curvature and the other tilt and a surface tells you which is wrong. (3) Forgetting that a stochastic volatility model still needs a second option to hedge, and quoting a delta from it as though the book were complete.

Where this goes next

The two named models are the Heston model, the standard choice in equities, and the SABR model, the standard choice in rates. What you add when a diffusion is not enough is jump-diffusion models. The single-factor alternative is local volatility models, the products that live or die on the difference are in exotic options, and trading the tilt itself is skew trading.

18.5 The Heston Model

trading · structuring | advanced | equity · fx · cross-asset

Stochastic volatility as an idea is its own section. Heston is the specific implementation that became the market standard, and it became so for one reason that is worth stating before any mathematics: it is the stochastic volatility model that can be *calibrated fast enough to use*, because European options have a semi-closed form.

The two equations

$$\mathrm{d}S_t = \mu S_t \,\mathrm{d}t + \sqrt{v_t}\, S_t \,\mathrm{d}W_t^1, \qquad \mathrm{d}v_t = \kappa(\theta - v_t)\,\mathrm{d}t + \xi\sqrt{v_t}\,\mathrm{d}W_t^2,$$

with $\mathrm{d}W^1\,\mathrm{d}W^2 = \rho\,\mathrm{d}t$.

The first is Black-Scholes with the volatility no longer constant. The second is the whole model: **variance**, not volatility, follows a mean-reverting square-root process. Two design choices in that line do all the work. Modelling *variance* keeps the mathematics tractable, since variance is what adds across time. The $\sqrt{v_t}$ in the diffusion term means the variance's own volatility shrinks as variance approaches zero, which is what keeps it from going negative.

KEY IDEA

Each of the five parameters controls a different feature of the surface, and knowing which is which is the entire practical content of the model. v_0 sets the level of short-dated at-the-money volatility. θ sets the long-dated level. κ sets how fast the term structure travels from one to the other. ξ, the volatility of variance, sets the *curvature* of the smile, so it prices the butterfly. ρ, the spot-variance correlation, sets the *tilt*, so it prices the risk reversal, and a negative ρ is what produces the equity skew from skew and smile.

That mapping is why calibration is stable in practice: the parameters are close to orthogonal in their effects, so the fit does not chase itself around.

The Feller condition and the term structure

Variance stays strictly positive if

$$2\kappa\theta \;>\; \xi^2,$$

the **Feller condition**. Violate it and the process can touch zero, which is not fatal but changes the behaviour and makes some numerical schemes misbehave. In practice calibrations to equity markets routinely violate it, because fitting the observed smile curvature demands a large ξ, and that tension is itself a signal that the model is being asked for more than it has.

WORKED EXAMPLE

A parameter set, read off. Take $v_0 = 0.04$, $\theta = 0.0625$, $\kappa = 2$, $\xi = 0.4$, $\rho = -0.7$.

- Spot volatility is $\sqrt{0.04} = \mathbf{20}\%$ and the long-run level is $\sqrt{0.0625} = \mathbf{25}\%$, so this is a calm market expected to normalise upward.
- **Feller**: $2\kappa\theta = 0.25$ against $\xi^2 = 0.16$, so it is satisfied with room. Raise ξ to 0.6 and it is violated.
- **Speed**: the half-life of the reversion is $\ln 2/\kappa = 0.35$ years, about **four months**.
- **The term structure that implies.** Expected variance integrates to $\theta T + (v_0 - \theta)(1 - e^{-\kappa T})/\kappa$, so the at-the-money implied volatility runs **21.2%** at three months, **23.0%** at one year and **24.5%** at five years, converging slowly toward the 25% long-run level.

Five numbers produced an entire upward-sloping term structure, and no part of it was fitted separately. That is what a model buys you over an interpolated surface.

Why it won

The characteristic function of the log price is known in closed form under Heston. European options can therefore be priced by a single numerical Fourier inversion rather than by simulation, which is a difference of several orders of magnitude in speed, and calibration to a whole surface becomes a matter of seconds rather than hours. No competing stochastic volatility model of the same era had that property, and everything else about Heston's popularity follows from it.

REAL INTERVIEW QUESTION (Citi · EQD)

Questions on stochastic volatility. The Heston model?

WHAT THEY TEST / ANSWER

Open, so give the shape of the answer first and let them choose the depth.

What stochastic volatility is: volatility is itself a random process with its own dynamics, rather than a constant as in Black-Scholes or a deterministic function of spot and time as in local volatility. The motivation is empirical, since volatility visibly clusters and mean-reverts, and it is structural, since a model with a second random factor can produce a smile that persists rather than one that has to be re-fitted every day.

What Heston is specifically: variance follows a mean-reverting square-root process correlated with the spot. Then give the five parameters and what each controls, because that is what an EQD desk actually uses: v_0 and θ for the two ends of the term structure, κ for the speed between them, ξ for the smile's curvature and ρ for its tilt.

Why it is the standard: the characteristic function is closed form, so calibration is fast. Say this explicitly, because it is the real answer to "why Heston and not something else".

Where it fails, which is what separates the answer. Heston **cannot fit the short-dated smile**. A diffusion needs time to generate skew, so at a week or two the model's smile is far flatter than the market's, and no parameter choice fixes it. The market's short-dated skew comes from the possibility of a jump, and the standard remedies are therefore to add jumps, giving the Bates model and jump diffusion, or to overlay a local volatility component and run a local-stochastic volatility model, which is what most exotics desks actually use.

Close by saying what the model is *for*, since it is not for pricing vanillas: vanillas are quoted by the market. It is for pricing and hedging the exotics whose value depends on how the smile *moves*, which is the distinction drawn in local volatility.

REAL INTERVIEW QUESTION (Barclays · Exo)

In a model with stochastic volatility and deterministic rates, what risks does the trader carry and how does he hedge them?

WHAT THEY TEST / ANSWER

Enumerate the risks in the order the model creates them, and give the hedge instrument for each, because the question asks for both.

Spot risk (delta): hedged in the underlying or a future, as always, and unchanged in principle by the model.

Variance risk (vega): this is the new one, and it is qualitatively different from Black-Scholes vega. Under Black-Scholes, vega is the sensitivity to a *parameter* you assumed constant; under stochastic volatility it is exposure to a genuine traded risk factor. Because the market is incomplete, **it cannot be hedged with the underlying**. It must be hedged with other options, and that is the whole reason vega hedging means trading options against options.

The cross risks, which follow from the model's own parameters:

- **Vanna**, exposure to ρ, the spot-variance correlation, which is a skew position, hedged with risk reversals.
- **Volga**, exposure to ξ, the volatility of variance, which is a smile-curvature position, hedged with butterflies.

Both are developed in cross-Greeks, and naming them by their hedging instrument rather than by their Greek letter is what an exotics interviewer is listening for.

Then the two risks the model does not contain, and volunteering them is the strongest part of the answer. First, **model risk**: the parameters are calibrated, they move day to day, and a position hedged on yesterday's calibration is not hedged today, so a desk carries an exposure to its own re-calibration. Second, the **market price of volatility risk**: because the market is incomplete there is no unique price, and the model's risk-neutral variance dynamics differ from the real-world ones by a premium that has to be assumed. That premium is not hedgeable at all; it is reserved against.

Close on the practical shape: the delta is hedged continuously in the underlying, the vega and its cross terms are hedged in a listed option book rebalanced over days, and what remains is model and premium risk, which is carried and priced at inception rather than hedged.

INTERVIEW TRAP

1. **Saying Heston models volatility.** It models *variance*, which is what makes the mathematics work and what makes the parameters read the way they do.
2. **Confusing it with local volatility.** Local volatility is deterministic and the market stays complete; Heston adds a second risk factor and the market does not.
3. **Expecting it to fit short maturities.** A diffusion cannot generate enough short-dated skew. That is a structural limitation, not a calibration failure.
4. **Ignoring the Feller condition.** Equity calibrations routinely violate it, and that violation is a signal about the fit rather than a detail.
5. **Treating vega as a parameter sensitivity.** Under stochastic volatility it is exposure to a real risk factor and cannot be hedged in the underlying.
6. **Using it to price vanillas.** The market quotes those. The model exists for the products that depend on how the surface moves.

VISUAL / CODE TO BUILD: Interactive

A five-slider surface. Each Heston parameter gets a control and the implied volatility surface redraws live, with the intended discovery being that the sliders are close to independent: ξ bends the smile without tilting it, ρ tilts it without bending it, and κ moves only the term structure. A Feller indicator turns red when $2\kappa\theta$ falls below ξ^2, so the reader finds out how hard they have to push the curvature before the condition breaks. A second panel overlays the model surface on a real market surface and reports the fit error by maturity bucket, which makes the short-dated failure visible as a systematic gap rather than as an assertion, and a toggle adds a jump component to watch that gap close. A third mode animates a spot move and shows the smile moving under Heston and under local volatility side by side, which is the difference the two models are actually chosen between.

18.6 SABR model

trading · structuring | advanced | rates · fx · cross-asset

SABR is not the most realistic stochastic volatility model, and nobody claims it is. It is the most *useful* one, and it won its position in rates and FX for a reason that has nothing to do with realism: Hagan and co-authors found a closed-form approximation for the implied volatility it produces. That single fact turns a stochastic differential equation into something a trader can put in a spreadsheet, calibrate to five quotes in milliseconds, and recalibrate on every tick across a whole swaption grid.

The general theory of making volatility random is stochastic volatility models, and the equity workhorse is Heston. This section is about the specific model that owns the rates and FX smile, why its four parameters are the right four, and what it gets wrong.

The model

SABR stands for "stochastic alpha, beta, rho", which is a name assembled from its parameters rather than from any economics. It is written on the **forward** F, not on the spot, which is exactly right for its natural habitat: forward swap rates, forward Libor or SOFR rates, and FX forwards.

$$dF_t = \alpha_t\, F_t^{\beta}\, dW_t^1, \qquad d\alpha_t = \nu\, \alpha_t\, dW_t^2, \qquad dW_t^1\, dW_t^2 = \rho\, dt.$$

Read it as two lines. The forward follows a **constant elasticity of variance** process, so its volatility depends on its own level through the exponent β. The volatility parameter α is itself a driftless lognormal process with volatility ν, correlated with the forward through ρ.

KEY IDEA

The four parameters are not interchangeable knobs. Each controls one visible feature of the smile, and that separation is the entire reason SABR is used.

- α sets the **level** of the smile.
- β sets the **backbone**, meaning how the at-the-money volatility moves when the forward moves. It is a choice about dynamics, not about shape.
- ρ sets the **tilt**, the skew.
- ν, the volatility of volatility, sets the **curvature**, the smile.

A trader can therefore look at a market smile and read off, without fitting anything, which parameter is doing what. That diagnostic clarity is worth more on a desk than a better fit.

Hagan's expansion, and its small print

The reason SABR is everywhere is this. For a European option on F struck at K with maturity T, the model's Black implied volatility is approximately

$$\sigma_B(K,F) \;\approx\; \frac{\alpha}{(FK)^{\frac{1-\beta}{2}}\left[1 + \frac{(1-\beta)^2}{24}\ln^2\frac{F}{K} + \frac{(1-\beta)^4}{1920}\ln^4\frac{F}{K}\right]} \cdot \frac{z}{x(z)} \cdot \Big[1 + \Lambda\, T\Big],$$

$$\Lambda = \frac{(1-\beta)^2\alpha^2}{24\,(FK)^{1-\beta}} + \frac{\rho\beta\nu\alpha}{4\,(FK)^{\frac{1-\beta}{2}}} + \frac{2-3\rho^2}{24}\,\nu^2,$$

$$z = \frac{\nu}{\alpha}\,(FK)^{\frac{1-\beta}{2}}\ln\frac{F}{K}, \qquad x(z) = \ln\left[\frac{\sqrt{1-2\rho z+z^2}+z-\rho}{1-\rho}\right].$$

At the money, where $K = F$, the factor $z/x(z)$ tends to 1 and the whole thing collapses to something you can evaluate in your head:

$$\sigma_{\text{ATM}} \;\approx\; \frac{\alpha}{F^{1-\beta}}\,\big[1+\Lambda T\big], \qquad \Lambda = \frac{(1-\beta)^2\alpha^2}{24\,F^{2-2\beta}} + \frac{\rho\beta\nu\alpha}{4\,F^{1-\beta}} + \frac{2-3\rho^2}{24}\,\nu^2.$$

KEY IDEA

This is an **asymptotic expansion**, valid to leading order in $\nu^2 T$ and in the distance from the money. It is not an exact solution. That matters in practice: for long maturities, large vol-of-vol, or strikes far in the wings, the formula can imply a *negative probability density*, which is a genuine arbitrage in your surface rather than a rounding problem. Treat the formula as excellent for the short and medium end and as something to replace, not patch, at the long end.

WORKED EXAMPLE

Alpha is not the at-the-money volatility. Take the lognormal case $\beta = 1$, with $\alpha = 20\%$, $\nu = 40\%$, $\rho = -0.30$ and $T = 1$ year. The $(1-\beta)$ terms vanish, leaving

$$\Lambda = \frac{\rho\beta\nu\alpha}{4} + \frac{2-3\rho^2}{24}\nu^2 = \frac{(-0.30)(0.40)(0.20)}{4} + \frac{2-3(0.09)}{24}(0.40)^2.$$

The first term is $-0.024/4 = -0.006$. The second is $(1.73/24) \times 0.16 = 0.011533$. So $\Lambda = 0.005533$ and

$$\sigma_{\text{ATM}} \approx 0.20 \times (1 + 0.005533) = 20.11\%.$$

The lesson is the one that catches people calibrating SABR for the first time: α is *close to* the at-the-money volatility but is not equal to it, and the gap grows with $\nu^2 T$. In production you do not set α directly, you invert this relation so that the model reproduces the quoted at-the-money volatility exactly, then fit ρ and ν to the wings. Note also that the two corrections here pull in opposite directions: the negative correlation drags the level down and the vol-of-vol pushes it up.

Beta, the backbone, and why it is a hedging choice

β is the parameter candidates understand least and interviewers ask about most, because it is the one that is not about today's smile at all.

Set $K = F$ and ignore the small ΛT correction. Then $\sigma_{\text{ATM}} \approx \alpha / F^{1-\beta}$. That relation is the **backbone**: the path the at-the-money volatility traces out as the forward moves, holding the model parameters fixed.

- $\beta = 1$ (lognormal): $\sigma_{\text{ATM}} \approx \alpha$, so the at-the-money *lognormal* volatility does not move when the forward moves. Natural for FX and equity.
- $\beta = 0$ (normal, Bachelier): $\sigma_{\text{ATM}} \approx \alpha / F$, so the lognormal volatility is inversely proportional to the forward. This is the natural description of rates, where the *absolute* size of moves is more stable than the relative size.
- $\beta = 0.5$: the compromise, and the conventional choice on many rates desks.

WORKED EXAMPLE

What the backbone does to your book. A forward swap rate moves from 3.00% to 4.00% and nothing else changes.

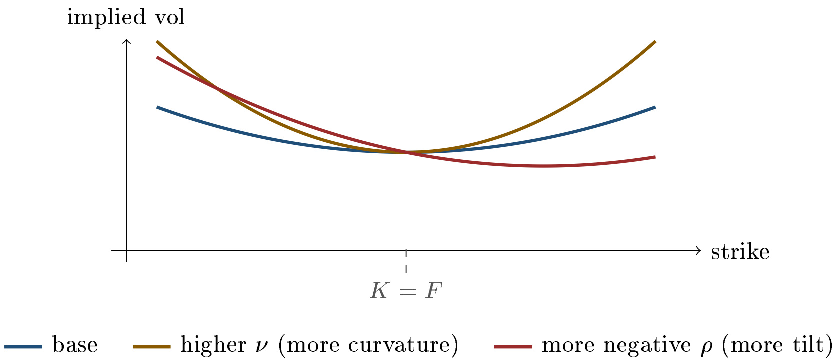
Under $\beta = 1$, the at-the-money lognormal volatility is unchanged. Under $\beta = 0$, it scales by $0.03/0.04 = 0.75$, so it falls by 25%. Same market move, same quoted smile today, and two completely different marks tomorrow.

That difference is not academic, because it goes straight into the delta. Your total sensitivity to the forward is

$$\frac{dV}{dF} = \frac{\partial V}{\partial F} + \frac{\partial V}{\partial \sigma} \cdot \frac{\partial \sigma}{\partial F},$$

and β is precisely what determines the second term. Choosing β is therefore a statement about *smile dynamics*, which is a hedging decision, not a curve-fitting decision.

This is also why β is almost always **fixed by convention rather than fitted**. β and ρ both tilt the smile, so they are close to collinear in a calibration: many pairs (β, ρ) fit today's quotes almost equally well while implying different dynamics. Fitting both produces unstable parameters that jump around day to day. Fix β from the asset class and your view of the backbone, then fit α, ρ and ν. Calibration practice generally is in model calibration.



REAL INTERVIEW QUESTION (ING · Rates)

Which model is used to price caps, floors and rate tunnels?

WHAT THEY TEST / ANSWER

Answer in two layers, because the question has a trivial reading and a real one. The trivial reading: the market convention is **Black's model**, so each caplet or floorlet is priced as a Black option on the corresponding forward rate, and a cap is the sum of its caplets. A tunnel, or collar, is long a cap and short a floor, so it is priced from the same two curves. The real reading, and the one they want: Black is a *quoting* convention that needs one volatility per strike and per maturity, and the market smile is not flat, so the actual question is what produces those volatilities. That is where **SABR** comes in. It is calibrated per expiry to the quoted smile and then used to interpolate and extrapolate across strikes, so the Black volatility you feed into each caplet comes out of SABR. Three additions land it. First, name the practical reason SABR won: Hagan's closed-form implied-volatility approximation makes calibration fast enough to run across an entire grid in real time. Second, mention that since rates went negative the plain model breaks, because F^β is undefined for a negative forward, so desks use **shifted SABR** or the normal ($\beta = 0$) version, and quotes in EUR rates moved to normal volatilities. Third, note that caps and swaptions are quoted on different underlyings, caplets on forward rates and swaptions on forward swap rates, so the smile is calibrated on the swaption cube and the caplet volatilities are stripped from cap quotes. The instruments themselves are in interest rate swaps and FRAs.

REAL INTERVIEW QUESTION (BNP · Fixed Income)

Give the pricing formula for a caplet in Black's model.

WHAT THEY TEST / ANSWER

Write it cleanly and then say what each piece is for. A caplet on a forward rate F with strike K, fixing at T, paying at $T + \delta$ on a notional $\mathcal{N}$ with accrual δ, is

$$\text{Caplet} = \mathcal{N}\,\delta\,P(0,T+\delta)\,\big[F\,N(d_1) - K\,N(d_2)\big], \quad d_1 = \frac{\ln(F/K) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T},$$

where $P(0, T + \delta)$ is the discount factor to the *payment* date. Then make the three points that show understanding rather than recall. First, this is exactly Black 76, so a caplet is a call on a forward rate and the whole formula is Black-Scholes with F in place of S and the discount factor pulled outside; a cap is the sum of its caplets and each is priced independently. Second, be precise about the discounting: you discount to the payment date, not the fixing date, and under a collateral agreement that is the overnight-indexed curve, which is why post-2008 the forward and the discount factor come from different curves. Third, name the assumption doing the work: the forward rate is lognormal under the forward measure associated with $P(0, T + \delta)$, which is what makes each caplet a clean Black option. If they push on where σ comes from, that is the SABR question above, and if rates can go negative, lognormality fails outright and you move to the normal or shifted-lognormal version.

The hedging argument, which is the real selling point

Any model that fits today's smile can produce today's prices. What separates models is what they say about *tomorrow*, and that shows up in the hedge.

> **INTUITION**
>
> Two models agree exactly on every quoted option today. The forward moves one percentage point. Model A says the whole smile shifts sideways with the forward; model B says the smile stays anchored to fixed strikes and the at-the-money point slides along it. Both were calibrated to the same data and they now disagree about your delta. You hedge with the delta, so you lose or make money according to which one was right, even though neither model ever mispriced anything.

SABR's parameterisation makes this explicit and controllable. Because $\sigma_{\text{ATM}} \approx \alpha/F^{1-\beta}$, the model tells you how the smile moves with the forward, so the SABR delta includes the backbone correction rather than assuming the smile is frozen. That is the practical complaint against local volatility, which fits today's surface perfectly and then predicts a smile dynamic that is empirically wrong in the opposite direction. The general treatment of these sensitivities is skew and smile sensitivities, and what a trader carries and hedges under stochastic volatility generally, rather than under SABR specifically, is in the Heston model.

Negative rates, and where the model breaks

SABR was designed for a world of positive rates, and F^β is simply undefined for $F < 0$ unless β is 0 or 1. When European and Japanese rates went negative this stopped being a theoretical concern, and the market settled on two repairs.

Shifted SABR replaces F^β with $(F+s)^\beta$ for a fixed shift s chosen large enough that $F+s$ stays positive, typically a whole number of percent. It preserves the shape of the model at the cost of introducing a parameter that is arbitrary and that everyone must agree on for quotes to be comparable. **Normal SABR**, the $\beta = 0$ case, has no such problem, since a Bachelier process is happy at any level, and this is the main reason EUR rate volatility is now quoted in normal (basis point) terms rather than lognormal ones.

The other well-known failure is the arbitrage in the wings mentioned above. Hagan's formula is an approximation, and at long maturities it can produce an implied density that goes negative in the low-strike region. The serious fixes replace the expansion rather than adjusting it: solving the model's own forward equation numerically, Hagan's later arbitrage-free formulation, or Antonov's exact results for special cases. If asked, the honest summary is that SABR's approximation is the source of both its popularity and its main defect.

REAL INTERVIEW QUESTION (HSBC · Rates)

What is the price of a zero-strike floor on Libor?

WHAT THEY TEST / ANSWER

This is the negative-rate question in disguise, and the whole answer is "it depends what dynamics you assume, and that is the point".

Under **lognormal** dynamics, which is plain Black and plain SABR with $\beta = 1$, the forward rate cannot go below zero, so a floor struck at zero can never pay and is worth **exactly nothing**. Say that first, because it is the answer the model gives and it is the answer that used to be correct.

Then say why it is wrong today. Rates have traded below zero in EUR, CHF and JPY, so a zero-strike floor plainly has value, and any quoted price for one is a direct measurement of how much probability the market puts on negative rates. A model that returns zero for an instrument that trades at a positive price is not conservative, it is broken, and it will also return a nonsensical implied volatility if you try to back one out: there is no lognormal volatility, however large, that reproduces a positive price.

So name the repairs, which are the ones in this section. Move to **normal** (Bachelier) dynamics, or equivalently SABR with $\beta = 0$, where the rate can sit anywhere on the real line and a zero-strike floor is an ordinary option with an ordinary price. Or use **shifted** lognormal dynamics, pricing on $F + s$ with a strike of s instead of F with a strike of zero, which restores positivity of the shifted variable at the cost of a shift everyone must agree on. This is exactly why EUR rate volatility is now quoted in normal, basis-point terms.

The detail that finishes it: quote the strike convention too. Many cap and floor contracts written before 2014 embed an implicit floor at zero on the underlying index, so "a zero-strike floor" can mean either a traded instrument or a contractual feature nobody priced at the time, and the second one is where losses came from.

VISUAL / CODE TO BUILD: Interactive

A four-slider SABR cockpit. The reader moves α, β, ρ and ν and watches Hagan's implied volatility curve redraw against strike, with the at-the-money point marked and its value printed next to α so the reader sees the two are close but not equal. A second panel plots the **backbone** directly: it sweeps the forward across a range and traces the at-the-money volatility, so switching β between 0, 0.5 and 1 visibly rotates that curve, and a delta readout updates so the reader sees the hedge change with a parameter that left today's smile almost untouched. A validity warning banner computes the implied density from the fitted curve and lights up red over any strike region where it goes negative, so the reader can push ν and T up until they break the approximation themselves. A toggle switches to shifted and normal SABR with a negative forward.

INTERVIEW TRAP

(1) **Saying α is the at-the-money volatility.** It is close and it is not equal. In the worked example, $\alpha = 20\%$ gives $\sigma_{\text{ATM}} = 20.11\%$, and the gap grows with $\nu^2 T$. (2) **Fitting β and ρ together.** They are near-collinear on today's smile, so the calibration is unstable. Fix β, then fit the rest. (3) **Treating β as a shape parameter.** It is a dynamics parameter. It barely changes today's fitted smile and it changes your delta a lot. (4) **Trusting Hagan's formula in the wings at long maturity.** It can imply a negative density, which is arbitrage in your own surface. (5) **Using plain SABR with negative rates.** F^β is undefined. Use shifted or normal. (6) **Claiming SABR is a good model of reality.** It is a good *smile interpolator* with interpretable parameters and a fast formula. Those are different claims, and the second is the honest one.

Where this goes next

The family SABR belongs to is stochastic volatility models, and its equity-market sibling is Heston. The alternative approach that fits today's surface exactly and gets the dynamics wrong is local volatility, and the general question of which input to change versus which dynamics to replace is extensions of Black-Scholes. The FX smile that SABR is also routinely used to interpolate is the FX volatility smile, and trading the tilt the model calls ρ is skew trading.

18.7 Jump-diffusion models

trading · structuring | advanced | equity · cross-asset

A diffusion cannot gap. Whatever volatility you choose, a continuous process moves through every intermediate price, so a share cannot be at 100 one moment and 78 the next. Markets do exactly that: on earnings, on a takeover, on a central bank surprise, on a crash. Jump-diffusion models are what you get when you stop pretending otherwise, and the consequences reach well beyond pricing into how a book can be hedged at all.

Two facts a diffusion cannot explain

Short-dated smiles are far too steep. Under a diffusion, the probability of a large move over a very short horizon collapses towards zero much faster than time does, so out-of-the-money options should be nearly worthless at one week and the smile should be nearly flat. Observed one-week smiles are the steepest on the surface. Something must give a non-vanishing probability of a large move over an arbitrarily short interval, and only a jump does that.

Realized returns have fat tails. Daily equity moves of five or ten standard deviations occur many times more often than a normal distribution permits. Stochastic volatility helps, since mixing normals produces fatter tails, but it does not produce them at short horizons, because volatility itself moves continuously.

The model

Merton's construction adds a Poisson jump to the Black-Scholes diffusion:

$$\frac{dS_t}{S_{t^-}} = (\mu - \lambda k)\, dt + \sigma\, dW_t + (J-1)\, dN_t,$$

where N_t is a Poisson process with intensity λ, so jumps arrive at an average rate of λ per year, J is the random jump multiplier, usually lognormal, and $k = \mathbb{E}[J-1]$ is the average jump size. The $-\lambda k$ in the drift is a compensator: it removes the mean effect of the jumps so the total drift is still what it should be.

Merton's insight for pricing is that, *if* the jump risk is diversifiable and therefore carries no risk premium, the option price is a Poisson-weighted average of Black-Scholes prices:

$$C = \sum_{n \geq 0} \frac{e^{-\lambda' T}(\lambda' T)^n}{n!} \, C_{\mathrm{BS}}(\sigma_n, r_n),$$

each term being the Black-Scholes price conditional on exactly n jumps having occurred, with the volatility and rate adjusted accordingly, and $\lambda' = \lambda(1+k)$ the intensity after the change of measure. The formula is exact and converges quickly, and it is the cleanest illustration of what a jump model is: a mixture over how many surprises happened.

WORKED EXAMPLE

Take a diffusive volatility of 15%, jumps arriving at $\lambda = 0.5$ a year with an average size of -10% and a jump volatility of 15%. Write $m = -0.10$ and $\delta = 0.15$ for the mean and the standard deviation of the *log* jump, which is the pair a lognormal jump model is quoted in. That m is not the same number as the k above: $k = \mathbb{E}[J-1] = e^{m+\delta^2/2} - 1 = -8.5\%$, a little smaller in magnitude, and using one where the other belongs is the standard slip.

Total variance per year is $\sigma^2 + \lambda(m^2 + \delta^2) = 0.0225 + 0.5(0.01 + 0.0225) = 0.0388$, so the effective long-horizon volatility is 19.7%, nearly five points above the diffusive part. Over a long horizon the jumps simply look like extra volatility, and that is exactly why they cannot be identified from a long-dated option.

Over a short horizon they look completely different. The probability of at least one jump is $1 - e^{-\lambda T}$: about 39% over a year, 4.1% over a month, and under 1% over a week. A weekly option is therefore priced by a distribution that is 99% a narrow diffusion and 1% a large move, which is a mixture with a small mass far out in the tails, and that mixture is precisely a steep smile. The jump parameters are identified by the short end of the surface and invisible at the long end.

KEY IDEA

Jumps break **completeness**. In Black-Scholes the option is replicable because the only risk is a small continuous move, which delta neutralises. A jump is not small: when the spot gaps, the option's value and the hedge's value change by different amounts, and no choice of delta covers every possible jump size. So jump risk cannot be hedged with the underlying, only with other options, the price is no longer unique, and it depends on a market price of jump risk. This is not a technicality: it is the mathematical statement of why a short-gamma book loses far more in a crash than its gamma predicted.

Variants and what they are for

Merton uses lognormal jump sizes, so the log jump is normal: thin-tailed, and symmetric only when the mean jump is zero, in which case it gives a smile rather than a skew. A negative mean jump, as in the example above, tilts it into a skew, but the two sides of the distribution cannot be shaped independently. **Kou's double-exponential** model puts a two-sided exponential on the log jump with *separate* decay rates up and down, which gives heavier tails, independent control of each wing, and, usefully, closed forms for barrier and lookback options because of the memorylessness of the exponential. **Bates** combines Heston's stochastic volatility with Merton's jumps, which is the standard way to fit both ends of the surface at once: the jumps do the short maturities and the stochastic volatility does the long ones.

That division of labour is the practical summary. Stochastic volatility explains the smile at long maturities and fails at short ones, because volatility cannot move fast enough; jumps explain the short end and become indistinguishable from volatility at the long end. The two are complements rather than rivals, which is why the models that fit whole surfaces contain both, alongside the calibration device of local volatility models.

REAL INTERVIEW QUESTION (BNP · EQD)

Why is implied volatility not symmetric?

WHAT THEY TEST / ANSWER

Answer the mechanism, not the shape, since the shape itself is described in the volatility smile and skew.

The surface is asymmetric because the *risk* is asymmetric, in three compounding ways. **Jump asymmetry**: equity markets fall in gaps and rise gradually, so the downward jump distribution has more mass and larger sizes than the upward one. A jump model with symmetric jumps produces a symmetric smile; one with a negative mean jump produces a skew, which is the direct answer to the question. **Leverage**: a falling equity value raises the firm's leverage and therefore the volatility of the residual claim, so volatility and spot are negatively correlated, and a correlated stochastic volatility model tilts the smile the same way. **Demand**: investors are structurally long the market and buy downside protection, so low strikes carry a persistent risk premium that high strikes do not.

Then the sentence that shows you understand which explanation applies where: the demand and leverage effects act on all maturities, while the jump effect dominates the short end, because that is the only horizon at which a jump is distinguishable from ordinary volatility. So a steep one-week skew is a statement about gap risk, and a steep two-year skew is a statement about leverage and about who is buying protection.

REAL INTERVIEW QUESTION (Barclays · Cross Asset)

We know there is going to be a jump in a stock. What strategy would you adopt?

WHAT THEY TEST / ANSWER

The trap is to answer "buy a straddle" and stop, because if everyone knows the jump is coming then it is already in the price, and the straddle is expensive precisely for that reason. The real question is what shape to trade, and whether the event is priced too richly or too cheaply.

What shape. A jump is not extra volatility spread evenly, it is a lump of probability far from the money. So its value is concentrated in the **wings**, not at the money: a strangle or the wings of a butterfly capture a jump much more efficiently per unit of premium than an at-the-money straddle, and if you have a view on direction, a one-sided position captures it better still. Conversely, if you think the market has overpriced the event, the trade is to *sell* the wings and, if you want to stay long ordinary volatility, buy the at-the-money against them.

Whether it is cheap. Compare the implied volatility of the maturity spanning the event with the maturity just before it. The difference implies a move size for the event, and that number, rather than the volatility level, is what you form a view on. If the implied event move is 6% and you believe the distribution is 4%, you are a seller whatever the headline volatility is doing.

And the hedging point, which is what the interviewer is really testing: a jump cannot be delta hedged. If you are short the wings and the jump is bigger than priced, you lose the whole gap regardless of how well you were hedged the day before, so this is a position that must be sized on the loss in the jump scenario and not on its Greeks. State it plainly: the Greeks describe the diffusion, and the jump is what happens in between the rebalances.

Where jumps show up in practice

- **Barriers.** A continuous-barrier option assumes the spot touches the barrier on its way through; with jumps it can gap past, which changes the price and, more importantly, means the desk's hedge does not unwind cleanly at the barrier. See exotic options.
- **Digitals and cliff payoffs**, where a gap through the strike converts a modelled smooth transition into a discrete loss.
- **Guarantees maintained by trading rules**, such as a CPPI, whose floor breaks only if the market gaps faster than the desk can rebalance. Gap risk is the jump risk retained by the issuer, in capital-protected products.
- **Scheduled events.** Many desks skip the full jump model and simply add a discrete *event variance* on known dates, earnings, elections, central bank meetings, which is a jump model with a known arrival time and is far easier to calibrate.

INTERVIEW TRAP

Four errors. (1) Believing more volatility can substitute for jumps: raising σ fattens a short-dated distribution far too little, which is why the short-dated smile is the diagnostic. (2) Delta hedging a jump: the delta covers the move you did not have, and the residual is the entire gap. (3) Calibrating jump parameters from long-dated options, where they are indistinguishable from volatility, rather than from the short end. (4) Sizing a short-wing position on its Greeks, when the risk it carries is a scenario loss that the Greeks do not describe.

Where this goes next

The model these repair is the Black-Scholes model, the observation that forces the repair is the volatility smile and skew, and the two other families of repair are local volatility models and stochastic volatility models. The instruments most sensitive to gaps are exotic options, and the empirical behaviour being modelled is in equity volatility.

18.8 Variance Swaps

trading · structuring · sales | advanced | equity · cross-asset

Every other way of trading volatility comes with something attached. An option gives you volatility exposure plus a delta you have to hedge, a gamma profile concentrated at one strike, and a vega that changes as spot moves. A variance swap gives you volatility exposure and nothing else. That purity is the reason it exists, and it is bought with an exposure to the tail that several desks discovered the hard way.

The contract

At maturity the buyer receives

$$N_{\text{var}}\left(\sigma_R^2 - K_{\text{var}}^2\right),$$

where σ_R is the realised volatility over the life of the trade and K_{var} is the strike, both quoted in volatility points. The payoff is linear in *variance*, which is what the name says and what everything else follows from.

Because variance points are an awkward unit, the market sizes trades in **vega notional**, the amount you make per volatility point at the strike, and converts:

$$N_{\text{var}} = \frac{N_{\text{vega}}}{2\,K_{\text{var}}}.$$

The factor of two is the derivative of σ^2 with respect to σ, so the conversion is exact at the strike and approximate away from it, which is the convexity discussed below.

REAL INTERVIEW QUESTION (CIC · Fixed Income)

Which volatility is used in variance swaps?

WHAT THEY TEST / ANSWER

Realised volatility, computed to a contractual formula, and the point of the question is that the formula is part of the contract rather than a statistical choice.

The standard convention is:

$$\sigma_R^2 \;=\; \frac{A}{n}\sum_{i=1}^{n}\left(\ln\frac{S_i}{S_{i-1}}\right)^2,$$

with $A = 252$. Three features of that expression are deliberate and each is worth naming.

Log returns, not simple returns, because the replication argument that prices the contract is built on the log contract, so any other definition would break the hedge.

No mean subtracted. The estimator divides by n and never subtracts the average return, which is not the statistical convention. It is done because it makes *total* variance exactly additive in time: $T_2\sigma^2_{[0,T_2]} = T_1\sigma^2_{[0,T_1]} + (T_2 - T_1)\,\sigma^2_{[T_1,T_2]}$. That is what lets a forward-starting variance swap be traded as the difference of two spot-starting ones, and it is why the contract is quotable across tenors at all. Note what it does not say: the annualised numbers themselves average rather than add, so four quarterly realised variances of 0.042, 0.041, 0.031 and 0.034 give a year of 0.037, not 0.148.

A fixed annualisation factor of 252 business days, contractually specified, along with the observation times, the source of the closing prices, and what happens on a disrupted day. All of these are in the term sheet, and disagreements about them are settlement disputes rather than modelling debates.

Then the distinction that the question is really probing, since the word "volatility" is doing double duty: the *strike* is quoted in implied volatility terms and comes from the option surface, while the *settlement* is realised. A variance swap is therefore a direct bet of implied against realised, which is the cleanest expression of the comparison made in implied vs realised.

Why it can be priced at all

REAL INTERVIEW QUESTION (BNP · Trading)

Pricing variance swap

WHAT THEY TEST / ANSWER

The answer is a replication result, and it is one of the most elegant in derivatives, so give the structure before the formula.

The claim: the payoff of a variance swap can be replicated *statically* by a portfolio of European options, plus a dynamic position in the underlying. Static means bought once at inception and held, which is why the fair strike is model-free.

Where it comes from. A delta-hedged option's profit and loss accumulates as $\frac{1}{2}\Gamma S^2\sigma_R^2$ per unit time, from gamma and theta. So a portfolio whose *dollar gamma* ΓS^2 is constant across all spot levels accrues realised variance uniformly, regardless of where the underlying goes. Constructing that portfolio is the entire problem, and the answer is a strip of options weighted by $1/K^2$.

The fair strike is then the cost of that strip:

$$K_{\text{var}}^2 \;=\; \frac{2e^{rT}}{T}\left[\int_0^F \frac{P(K)}{K^2}\,\mathrm{d}K + \int_F^\infty \frac{C(K)}{K^2}\,\mathrm{d}K\right],$$

puts below the forward and calls above, each weighted by $1/K^2$. The general technique is replication.

Then say what that means in practice, because it is what an interviewer is after. The fair strike is *the whole option surface*, not a single implied volatility. It is therefore higher than the at-the-money implied whenever there is skew, since the $1/K^2$ weighting loads heavily on low strikes and those carry the highest implied volatilities. On an equity index the variance swap strike typically prints several volatility points above the at-the-money level, and that gap is a direct reading of the skew from the surface.

And what breaks it, which is the follow-up: the integral runs to infinity and to zero, but only listed strikes exist. The replication is truncated in practice, so a large move outside the traded strike range leaves the hedge short of variance. That truncation is not a technicality; it is the mechanism behind the losses discussed below.

Variance against volatility

Because the payoff is linear in variance, it is convex in volatility. The buyer therefore does better in both directions than the holder of a volatility swap on the same vega notional, which is why the variance strike must sit above the volatility strike, $K_{\text{var}} > K_{\text{vol}}$. The size of that convexity adjustment, why a volatility swap cannot be replicated statically, and the model risk a dealer takes by quoting one all belong to volatility swaps.

What it is for, and how you hedge it

REAL INTERVIEW QUESTION (BNP · EQD)

A variance swap? What is it for, and how do you hedge it?

WHAT THEY TEST / ANSWER

Two halves, and the second is the one that is actually being tested, because anyone can recite the purpose.

What it is for. Three uses, and they are genuinely different trades. It expresses implied against realised without committing to a strike or carrying a delta, which is the pure version of the view in implied vs realised. It hedges a book that is short convexity somewhere else, because variance is the natural hedge for any exposure that grows with the square of the move rather than with the move. And it is a mechanical long-tail position for a macro book: the payoff is quadratic, so it does not need the direction of the crash to be right, only its size.

How you hedge it. A short variance swap is hedged by the strip that prices it, plus a delta.

1. **Buy the $1/K^2$ strip once.** Puts below the forward, calls above, weighted $1/K^2$. That portfolio has constant dollar gamma, so it accrues realised variance at the same rate wherever spot goes. This leg is static, which is the whole point: it is bought at inception and not touched.
2. **Delta-hedge the residual daily.** The strip replicates the log contract, whose delta is $-1/S$ per unit, so the underlying position scales as $1/S$ and has to be rebalanced as spot moves. This is the only dynamic part, and it is where the operational risk sits.

Then say what the hedge does not cover, because that is the follow-up and it is where the answer separates.

- **Truncation.** The integral needs every strike from zero to infinity; the board stops. Below the lowest listed put the hedge accrues no further variance while the swap keeps paying.
- **Jumps.** The gamma argument assumes a continuous path. A gap contributes its full squared return to the settlement but the hedge accrues only the variance it saw, so the two do not match on the days that matter.
- **Discrete rebalancing.** Hedging once a day rather than continuously leaves a residual that is itself proportional to realised variance, so the hedging error is worst exactly when the position is.
- **The strip's own liquidity.** The $1/K^2$ weighting puts most of the position in deep downside puts, the least liquid line on the board, so unwinding is not the reverse of putting it on.

The tail, and why caps exist

The convexity that makes a variance swap attractive to the buyer is exactly what makes it dangerous to the seller. Take a vega notional of €100,000 at a strike of 20, so the variance notional is $100{,}000/(2\times 20) = 2{,}500$ per variance point. A realised volatility of 40 then pays $2{,}500\times(40^2-20^2)$, or **€3 million**: thirty times the sizing unit, from a volatility that merely doubled.

Two things made this worse than the arithmetic alone suggests. The replication is truncated at the lowest listed strike, so in a crash the hedge stops accruing variance precisely when the payoff does not. And the $1/K^2$ weighting means the position was always concentrated in the deep downside puts, which are the least liquid options on the board and the ones that gap. Sellers of index variance discovered both in 2008, and the market's response was the **capped variance swap**, which limits realised volatility to a multiple of the strike, conventionally 2.5 or 3 times.

A cap at three times a strike of 20 still allows a payout of $2{,}500 \times (60^2 - 20^2)$, or **€8 million**, eighty times the vega notional. The cap makes the risk finite; it does not make it small, and any answer that presents a cap as making the product safe has missed the size of the number.

REAL INTERVIEW QUESTION (BNP · EQD)

What is a variance swap?

WHAT THEY TEST / ANSWER

Four sentences, then stop and let them choose the direction, because this is an opener.

The contract: a forward on realised variance. At maturity you exchange the realised variance of the underlying against a fixed strike, sized in vega notional so that the payoff reads in volatility points near the strike.

Why it exists: it is pure volatility exposure. No delta to hedge, no dependence on where spot ends up, and a vega that does not decay or migrate as the underlying moves, unlike an option's.

Why it is priceable: because it replicates statically as a $1/K^2$-weighted strip of options, so the strike is model-free and reads off the whole surface rather than off one implied volatility.

What it costs you: convexity in the wrong direction if you sold it, concentration in the least liquid strikes, and a replication that truncates where the listed strikes stop.

Then offer the follow-ups yourself rather than waiting: the difference from a volatility swap, why the strike sits above the at-the-money implied, and what a cap does. Naming the three directions the conversation can go is a better use of the pause than filling it.

INTERVIEW TRAP

1. **Confusing vega notional with variance notional.** They differ by $2K_{\text{var}}$, and quoting the wrong one misstates the trade by a factor of forty on a typical strike.
2. **Reading the strike as an at-the-money volatility.** It is the cost of the whole strip, so with skew it sits several points above the at-the-money implied.
3. **Treating variance and volatility swaps as interchangeable.** One replicates statically and one does not, which is the difference between a model-free price and a model risk.
4. **Forgetting the estimator subtracts no mean.** That is what makes *total* variance $T\sigma^2$ exactly additive across periods, which is what a forward-starting trade is built on. The annualised figures average rather than add, and confusing the two is a factor of four on a year of quarters.
5. **Assuming the replication is exact.** It truncates at the listed strikes, and it fails in exactly the move that makes the payoff large.
6. **Thinking a cap makes it safe.** A three-times cap on a twenty strike still permits eighty times the vega notional.

VISUAL / CODE TO BUILD: Interactive

A replication demonstrator. The reader builds the option strip by adding strikes one at a time, and a live chart plots the portfolio's dollar gamma against spot: with a handful of strikes it is visibly lumpy, and as more are added weighted by $1/K^2$ it flattens into the constant line that makes the replication work. A truncation slider then removes the strikes below a level and shows the dollar gamma collapsing to zero beyond it, which is the 2008 failure made visible rather than described. A second panel plots the variance swap and volatility swap payoffs against realised volatility on the same axes with a shared vega notional, so the parabola against the line is the visible reason the strikes differ. A third mode overlays a cap and reports the maximum payout in multiples of vega notional as the reader moves it.

Where this goes next

The convexity adjustment and the model risk in the square root are volatility swaps, and restricting the accrual to a spot range, which is how a desk sells the part of the distribution it wants without the tail it does not, is corridor variance swaps.

18.9 Volatility swaps

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A volatility swap is the instrument an investor asks for and a variance swap is the one they get. The payoff is exactly what the name suggests, and the reason the market trades its square instead is one of the cleanest examples in derivatives of a product being shaped by what can be hedged rather than by what is wanted.

The contract

A **volatility swap** pays, at maturity,

$$N_{\text{vega}} \times \left(\sigma_{\text{realized}} - K_{\text{vol}}\right),$$

where σ_{realized} is the annualised realized volatility of the underlying over the life of the contract and K_{vol} is the strike agreed today. It is linear in volatility, so a vega notional of 100,000 per volatility point means exactly that: one point of realized volatility above the strike pays 100,000.

Compare the **variance swap** of variance swaps, which pays on $\sigma^2_{\text{realized}} - K^2_{\text{var}}$. Same underlying quantity, squared, and that squaring changes everything.

> KEY IDEA
>
> The market trades variance rather than volatility because **variance is replicable and volatility is not**. A variance swap can be manufactured from a static portfolio of vanilla options across all strikes plus a delta hedge, so a dealer can price and hedge it without a model. No such static replication exists for volatility, because the square root is a non-linear function of the replicable quantity. A volatility swap therefore has to be hedged approximately, usually with a variance swap plus an adjustment, and its price contains a genuine model-dependent component.

The convexity adjustment

The square root is concave, so by Jensen's inequality $\mathbb{E}\left[\sqrt{V}\right] \leq \sqrt{\mathbb{E}[V]}$: the fair volatility strike is **below** the square root of the fair variance strike. The gap is the **convexity adjustment**, and it grows with the uncertainty of volatility itself.

> WORKED EXAMPLE
>
> Suppose realized volatility over the period will be either 15% or 25%, with equal probability. Nothing else is uncertain.
>
> The fair **volatility** strike is the expected volatility, $(15 + 25)/2 = 20.00\%$.
>
> The fair **variance** strike is the square root of the expected variance, $\sqrt{(0.15^2 + 0.25^2)/2} = 20.62\%$.
>
> So the variance swap trades 0.62 volatility points above the volatility swap on identical underlying uncertainty, purely because of the squaring. That gap is the convexity adjustment: it is zero if volatility is known, and it widens with the volatility of volatility. Since a dealer selling a volatility swap and hedging it with a variance swap is short exactly that gap, the adjustment is what they charge for, and it is the model-dependent part of the price.

REAL INTERVIEW QUESTION (Barclays · Structured Products)

What is the difference between a variance swap and a volatility swap?

WHAT THEY TEST / ANSWER

Four differences, in increasing order of how much they reveal.

Payoff. One is linear in realized volatility, the other in realized variance. So the variance swap's payoff in volatility terms is *convex*: it gains disproportionately when volatility spikes and loses less than proportionally when volatility collapses.

Replication. The variance swap has a static replication in a strip of vanillas, so it can be hedged model-free. The volatility swap has none, so it is priced with a model and hedged approximately.

Price. The variance strike is above the volatility strike by the convexity adjustment, which is a function of the volatility of volatility. Give the illustration: with realized volatility equally likely to be 15% or 25%, the volatility strike is 20.00% and the variance strike is 20.62%.

Risk profile, and this is the one that matters. Quantify it. Take a strike of 20 and a realized volatility of 80, as in 2008. The volatility swap pays $80 - 20 = 60$ per unit of vega notional. The variance swap pays $(80^2 - 20^2)/(2 \times 20) = 150$, two and a half times as much. In general the ratio is $(\sigma_R + K)/(2K)$, so the variance swap's advantage grows without limit as volatility rises. This is exactly why sellers of variance swaps were destroyed in 2008, and why variance swaps are now routinely sold with a **cap**, typically at two and a half times the strike, which turns them back into something closer to a volatility swap in the tail.

Close on who wants which. An investor wanting a clean, intuitive bet on the level of volatility prefers the volatility swap, and a dealer prefers the variance swap because it is the one they can hedge; the convexity adjustment is the price of moving between the two, and the dealer warehouses the volatility-of-volatility risk that comes with it.

What "realized volatility" means in the contract

REAL INTERVIEW QUESTION (CIC · Fixed Income)

What volatility is used in a variance swap?

WHAT THEY TEST / ANSWER

Realized, not implied, and the value of the answer is in the conventions, since the contract is settled on a formula rather than on a market quote.

The standard definition uses **close-to-close log returns** with a **zero mean**:

$$\sigma^2_{\text{realized}} = \frac{252}{n} \sum_{i=1}^{n} \left(\ln \frac{S_i}{S_{i-1}} \right)^2,$$

annualised with 252 trading days. The zero-mean convention is not laziness: it makes the sum exactly the quantity the option replication delivers, so the contract matches its hedge rather than a textbook estimator.

Then name the details that a confirmation actually specifies, because that is what distinguishes someone who has seen one. The **observation schedule**, which exchange's closing prices are used and what happens on a disrupted or non-trading day. Whether the divisor is n or $n-1$, which matters for short-dated contracts. **Dividends**, since a large ex-date produces a return that is not volatility and contracts usually adjust for it. The **cap**, if any. And whether the strike is quoted in volatility points, as it usually is, with the variance notional derived as $N_{\text{vega}}/(2K)$.

Close on the point of the whole instrument: the trade is the *spread* between this realized number and the implied level paid at inception, which is why variance swaps are the cleanest expression of the implied-versus-realized view of implied versus realized volatility, with no delta hedging, no path dependence in the Greeks, and no residual exposure to the direction of the underlying.

The rest of the family

REAL INTERVIEW QUESTION (BNP · EQD)

What is a covariance swap, and why trade it when variance swaps exist?

WHAT THEY TEST / ANSWER

A **covariance swap** pays on the realized covariance between two underlyings,

$$N \times \left(\frac{252}{n} \sum_i r_i^{(1)} r_i^{(2)} - K_{\text{cov}} \right),$$

using the same conventions as a variance swap but multiplying two return series instead of squaring one. A **correlation swap** is the normalised version, paying on realized correlation.

Why it exists: because the risk people actually carry is often a *relationship* rather than a level, and variance swaps cannot isolate it. Say how you would otherwise try. Using the identity

$$\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X,Y),$$

you can synthesise a covariance exposure from three variance swaps: buy the variance of the basket and sell the variance of each component, in the right proportions. That works, and it is the standard construction, but it costs three bid-offers instead of one, it requires a liquid variance market on the basket as well as on the legs, and the weights have to be maintained. A covariance swap does it in a single trade.

Then the reason a desk cares, which is the better half of the answer. Dealers accumulate correlation exposure from issuing multi-asset structured products, worst-of notes and autocalls above all, as set out in equity correlation, and that exposure has no natural offsetting client flow. Covariance and correlation swaps exist so that risk can be recycled directly rather than through three separate variance legs. Add the caveat: because the correlation swap normalises by the two volatilities, it is not the same trade as a covariance swap, and the difference is itself a volatility exposure, which is a standard follow-up.

INTERVIEW TRAP

Four errors. (1) Saying a variance swap and a volatility swap are the same trade in different units: the squaring is a genuine convexity and it is priced. (2) Taking the square root of the variance strike as the volatility strike: it is an upper bound, not the answer. (3) Forgetting the cap when quoting the tail behaviour of a variance swap, which is precisely the regime where the cap binds. (4) Treating realized volatility as a statistic rather than as a contractual formula, when the observation schedule, the mean convention and the dividend treatment all change the settlement.

Where this goes next

The replicable instrument this one is defined against is variance swaps, its range-restricted cousin is corridor variance swaps, and the view these instruments express is implied versus realized volatility. The behaviour of the quantity being traded is equity volatility, the multi-asset version of the same idea is dispersion strategies, and what carrying this risk feels like on a book is vega exposure and volatility regimes.

18.10 Corridor variance swaps

trading · structuring | advanced | equity · cross-asset

A plain variance swap pays on all the variance the underlying realises, wherever it happens. A **corridor** variance swap pays only on the variance realised while the underlying is inside a specified range. One clause changes, and it changes what the instrument is: a plain variance swap is a pure bet on volatility, and a corridor variance swap is a joint bet on volatility *and* on where the market will be while that volatility occurs.

The mechanics of the plain instrument, its replication and its pricing are variance swaps. This section is only about what the corridor does to it.

The convention matters more than the strike

Two products go by names that are used loosely and are not the same. Both accrue only on observations where the underlying sits inside the corridor $[L, U]$; they differ in what they divide by.

- **Corridor variance swap.** Sum the squared returns on the days spot was inside, and divide by the **total** number of observations. Days spent outside the corridor contribute zero to the numerator and still count in the denominator, so they drag the realised number down.
- **Conditional variance swap.** Sum the same squared returns, but divide by the number of observations **inside** the corridor. Days outside are ignored entirely, so the payoff answers the question "how volatile was it *while* we were in this range".

KEY IDEA

The corridor version mixes two effects: how volatile the market was inside the range, and how much time it spent there. The conditional version isolates the first and discards the second. **They are different trades and the difference is often larger than the bid-offer on either.** Confirm which convention a quote refers to before comparing two prices, because a corridor quote will always look cheaper than a conditional one on the same corridor, and that is a definitional artefact rather than a better price.

WORKED EXAMPLE

Take a one-year contract on 252 observations with a strike of 20 volatility points, and suppose the market spends 60% of the observations inside the corridor. Suppose that, measured over those days alone, the annualised realised volatility was 25, so the variance was $25^2 = 625$.

Conditional convention: divide by the days inside, so realised variance is 625 and the realised volatility that settles the contract is **25.0**.

Corridor convention: divide by all 252 observations, so realised variance is $0.60 \times 625 = 375$, and the effective volatility is $\sqrt{375} = \mathbf{19.36}$.

Same market, same days, same corridor. One contract settles at 25.0 against a strike of 20 and pays out; the other settles at 19.36 against the same strike and **loses money**. The convention alone flipped the sign of the trade.

Why anyone wants one

Three reasons, and they are quite different from each other.

- **The replication is truncated, and it survives.** A plain variance swap replicates as a strip of options across *every* strike, weighted by $1/K^2$. The far wings carry small weight individually but they are illiquid, wide and sometimes simply unavailable, and they are where the replication argument is least reliable. The **corridor** variance swap keeps the same model-free argument on a *truncated* strip: only the options struck inside the corridor, same $1/K^2$ weighting, plus a dynamic position in the underlying. So the position a desk actually has to run is bounded and liquid, and it is still priced without a volatility model. That is the reason the product exists rather than a side benefit.
- **It expresses a joint view.** A client who believes the market will be volatile only if it sells off cannot say that with a plain variance swap. A down-variance contract, corridor from zero to a barrier below spot, says exactly that.
- **It is cheaper, under the corridor convention.** It pays in fewer states with an unchanged denominator, so it costs less, and for a client with a specific view that is a better use of premium than paying for variance in states they do not care about. Under the *conditional* convention it need not be: a down-conditional strike in equities usually trades **above** the plain variance strike, because spot-volatility correlation is negative and the volatility realised in the down region is higher than the blended average. Cheapness is a property of the convention, not of the corridor.

INTUITION

The down-variance trade is the one worth understanding, because it isolates something a plain variance swap merely averages over. Equity markets have strongly negative spot-volatility correlation: they are quiet on the way up and violent on the way down. A plain variance swap prices the blend of both regimes. **Down-variance strips out the calm regime and leaves only the crash regime**, which is why it is used both to hedge a book that suffers in a sell-off and to express a view that the skew is under-priced.

The corresponding statement is that a corridor variance swap is **not a pure volatility instrument**. Whether variance accrues depends on where spot is, so the contract has genuine spot exposure: it has delta, and its value changes when spot moves towards or away from the barrier even with volatility unchanged. A plain variance swap has essentially none of this. If someone describes a corridor product as a way to trade volatility without a directional view, they have the instrument the wrong way round. The relationship between the two variables is skew trading.

REAL INTERVIEW QUESTION (BNP · Volatility)

What is the payoff of a call on a variance swap?

WHAT THEY TEST / ANSWER

Build it up in three steps, because the question is really testing whether you keep variance and volatility straight under a non-linear payoff.

Step one, the underlying instrument. A variance swap pays $N_{\text{var}}\left(\sigma^2_{\text{realised}} - K^2\right)$, linear in *variance* and therefore convex in volatility. Desks quote size in vega notional rather than variance notional, with $N_{\text{vega}} = 2K\,N_{\text{var}}$, so that a one point move in volatility near the strike pays approximately the vega notional.

Step two, the option on it. A call struck at K_{opt} pays

$$N_{\text{var}}\max\left(\sigma^2_{\text{realised}} - K^2_{\text{opt}},\ 0\right),$$

so the loss is floored at the premium and the upside is unbounded. Two consequences follow. It is a **one-way** exposure, which is exactly why a client buys it: a variance swap sold in a crisis has an unbounded loss, and a call on variance does not. And it is **convex on top of convex**, since the variance swap was already convex in volatility and the option adds a second layer.

Step three, what makes it hard, which is the real answer. Pricing it requires a view on the *distribution* of realised variance, not merely its expectation. A variance swap can be priced from the option surface today because its fair strike is an expectation, and a strip of vanillas replicates it. A call on variance cannot: no static portfolio of vanillas reproduces a convex function of realised variance, so you need a model of how variance itself moves, meaning volatility of volatility, which is stochastic volatility. **The plain variance swap is model-free and the option on it is not**, and that sentence is the whole point of the question.

If you want to close well, note the family this sits in, and separate the two members carefully. Corridor and conditional variance swaps are the other way of making a variance product non-linear: instead of putting an option on the payoff, they make the accrual conditional on where spot is. But only one of them costs you the model-free price. **The conditional variance swap breaks the replication**, because its denominator is the random number of observations inside the range, so the payoff is a ratio of two random quantities and no static portfolio of vanillas reproduces it. **The corridor version keeps it**, on a strip truncated to the strikes inside the corridor. Knowing which of the two you are being asked about is the difference between a priced product and a modelled one.

For reference, the baseline this contract is measured against pays $N_{\text{var}}\left(\sigma^2_{\text{realised}} - K^2\right)$ with $N_{\text{var}} = N_{\text{vega}}/(2K)$, and it is set out with its convexity in variance swaps and volatility swaps. The corridor-specific point is that **the corridor contract keeps that convexity intact and only restricts the states in which it applies**.

INTERVIEW TRAP

Three errors. (1) Using corridor and conditional interchangeably. The denominator differs, and on a corridor the market sits inside 60% of the time that difference is worth several volatility points. (2) Calling a corridor variance swap a pure volatility product. Accrual depends on spot, so it carries delta and barrier risk that a plain variance swap does not. (3) Assuming the two conventions price the same way. The corridor version keeps a model-free replication, using only the strip struck inside the corridor; the conditional version does not, because its random denominator makes the payoff a ratio and destroys the static replication.

Where this goes next

The instrument this one modifies is variance swaps, and the close relative whose difference from it is itself a convexity story is volatility swaps. The models needed once replication stops being static are stochastic volatility models, the relationship between spot and volatility that down-variance isolates is skew trading, the barrier machinery the corridor borrows is exotic options, and the sensitivity that a variance position is ultimately made of is gamma and convexity.

18.11 Skew Trading

trading · structuring | advanced | equity · fx · cross-asset

Skew and smile explains why the equity surface tilts and skew and smile sensitivities gives the sensitivities. This section is about taking a position in the tilt itself: what the instruments are, what the conventions mean, and what the position costs to hold.

KEY IDEA

A skew position is a bet on the *relative* implied volatility of two strikes, not on the level of either. That means it must be constructed as a **spread**: long one strike against short another, sized so that the level exposure nets out and only the tilt remains. If your position makes money when all volatilities rise together, you are long vega, not long skew, and you have built the wrong thing.

The instruments

The risk reversal is the pure expression: long an out-of-the-money call against short an out-of-the-money put, or the reverse, usually struck at equal deltas so the two vegas roughly offset. It is the standard quoted skew instrument in FX, where the whole surface is expressed as an at-the-money level plus a risk reversal and a butterfly, treated in risk reversals.

Vertical spreads carry skew whether or not that was the intention, which is the point most often missed and is developed in skew-based strategies.

Ratio structures and put spreads express it with a directional tilt, and belong to skew-based strategies as trades rather than as risk expressions.

Variance against volatility. A variance swap's strike loads on low strikes through the $1/K^2$ weighting, so long variance against short at-the-money vega is a long skew position, which is one of the few ways to hold it without a strike-by-strike book. The mechanism is variance swaps.

WORKED EXAMPLE

What the skew is worth, priced. Spot 100, one year, zero rates, and a 90/110 risk reversal: long the 110 call, short the 90 put.

Under a **flat** 20% volatility, the call is worth 4.29 and the put 3.59, so the structure costs a net **0.70** to put on.

Now price it on a realistic equity surface, with the 90 put at 24% and the 110 call at 17%. The put rises to 4.93 and the call falls to 3.20, so the structure now **collects 1.73**.

The difference of **2.43** is the skew, expressed in premium rather than in volatility points. It is also the reason this trade is described as "selling skew": you are being paid to be short the expensive wing and long the cheap one, and the payment is the whole of that 2.43.

Which way round is long

Conventions differ and interviewers know it, so the safe practice is to state the convention you are using and then answer.

In **equities**, the common usage is that **long skew** means long the downside relative to the upside: you own the put wing against the call wing, and you profit when the downside richens further, which happens in a sell-off. In **FX**, the risk reversal is quoted as the call volatility minus the put volatility at a fixed delta, so a positive quote means calls are bid and the sign carries a direction that changes over time. Saying "long skew, meaning long the put wing" removes all ambiguity in one clause and costs nothing.

REAL INTERVIEW QUESTION (BNP · EQD)

How do you get yourself short skew?

WHAT THEY TEST / ANSWER

Define the target before naming a trade, because "short skew" means profiting when the tilt *flattens*: when downside implied volatility falls relative to upside implied volatility.

The direct expression: sell the downside wing and buy the upside wing. Concretely, sell an out-of-the-money put and buy an out-of-the-money call, struck at comparable deltas so the vegas roughly cancel and what remains is the tilt. That is a short risk reversal in the equity convention, and it is the trade in the worked example above, which collects 1.73 on the numbers given.

Sizing it properly, which is the part that separates the answer. Equal notionals do not give equal vegas, because the two strikes have different vegas, so size on **vega** rather than on notional if the aim is a clean skew position. And the residual delta is not zero, since the two legs have different deltas, so hedge it in the underlying or the position is a directional trade wearing a skew label.

Then the two things you have actually taken on, and volunteering them is what an EQD interviewer is listening for.

First, **you are being paid, and you should know what for**. Equity skew is persistently steeper than the skew that realises, so selling the structurally bid wing earns a positive expected return. In the example above the skew itself is worth 2.43, which is why a structure that would have cost 0.70 on a flat surface instead collects 1.73 in cash. This is not a mispricing you have found; it is a risk premium you have decided to sell, and the price of collecting it is the second point.

Second, **the position is short the crash**. You are short a downside put, so in a sell-off you lose on spot, you lose again as the skew steepens, and you lose a third time because your short put's vega grows as it moves toward the money. Those three losses arrive together, which is the definition of a bad tail.

Close on who is naturally on each side: dealers are structurally long the downside wing from *raw* structured product issuance, since the investor in a yield product is the one selling the down-and-in put. The residual after vanilla hedging is a separate question and is taken up there. Either way the street is often trying to do exactly this trade at the same time, which is why the entry level matters more than the idea.

A strategy book therefore accumulates a position in the slope it never traded. That exposure is skew-based strategies, which owns the skew hiding inside ordinary structures; this section owns the trade taken deliberately.

What you are really being paid for

Skew is not free money in either direction, and the reason is worth stating because it decides whether a skew trade is an edge or a premium.

Implied skew is persistently steeper than the skew that realises, in the same way that implied volatility is persistently above realised. Selling it therefore earns a positive expected return most of the time and loses heavily in the episodes that steepen it, which are the episodes where everything else in an equity book is also losing. So the short skew trade is short a risk that is correlated with the rest of the portfolio, and its apparent Sharpe ratio is flattered by exactly the periods that are missing from a short sample.

The honest framing, and the one that survives a follow-up: a skew position is a view on the difference between the *priced* spot-volatility relationship and the one that will realise. If the market prices a correlation of -0.8 and the world delivers -0.5, being short skew wins. Stating it that way makes the trade falsifiable, and it points at the quantity to measure, which is the realised spot-volatility correlation from vanna.

INTERVIEW TRAP

1. **Building a level position by accident.** If your trade makes money when all volatilities rise together, it is long vega and not long skew. Size on vega, not on notional.
2. **Leaving the delta unhedged.** The two legs of a risk reversal have different deltas, so an unhedged skew trade is a directional trade.
3. **Not stating the convention.** Long skew means opposite things to different desks, and FX quotes it with a sign that changes over time.
4. **Getting the carry backwards.** Short skew in equities sells a structurally bid wing, so it *collects* a premium on the roll. It is a risk premium sold rather than a mispricing found, and the cost is the tail below, not the carry.
5. **Ignoring the correlation of the tail.** A short skew position loses in exactly the episodes when the rest of the book does, which is what makes its historical performance look better than its risk.
6. **Overlooking unintended skew.** Vertical spreads and structured product inventory carry skew whether or not anyone chose it.

VISUAL / CODE TO BUILD: Interactive

A skew position builder. The reader assembles legs on a live surface and the page decomposes the resulting exposure into level (vega), tilt (skew) and curvature (butterfly), so a trade intended as pure skew shows visible vega until the sizing is corrected, and correcting it is the exercise. A carry panel rolls the position forward day by day with the surface unchanged, showing the bleed accumulating, which turns the "you are selling a premium" claim into a number. A stress mode applies a historical sell-off with its actual joint move in spot and surface, and reports the loss split into the three components the text names, so the reader sees them arrive together rather than being told they do. A final panel plots implied skew against subsequently realised spot-volatility correlation over a long history, which is the evidence for the risk premium and also for how often it fails.

18.12 Dispersion strategies

trading · structuring | advanced | equity

An index is not a separate asset. It is a weighted sum of its constituents, so its volatility is completely determined by their volatilities and their correlations. Both sides of that identity have their own option market, quoted by different people for different reasons, and they do not always agree. Dispersion is the trade that takes the disagreement.

The identity itself, and the implied correlation you get by inverting it, belong to equity correlation; take $\sigma_{\text{index}} \approx \sqrt{\rho}\,\bar{\sigma}$ and $\rho_{\text{implied}} \approx (\sigma_{\text{index}}/\bar{\sigma})^2$ as given. This section is about the position: how it is built, what it actually earns, and how it loses.

What the trade is

Long dispersion means **sell volatility on the index and buy volatility on the constituents**. You are betting that the names will move around more, relative to the index, than the two option markets currently imply. Since the only thing that converts between the two is correlation, being long dispersion is being **short implied correlation**, and the sign is worked through in equity correlation.

> **INTUITION**
>
> Picture a crowd walking down a street. The index measures how far the crowd's centre of mass moves; the single names measure how far each person moves. If everyone walks in step, the centre moves as much as the individuals and there is no dispersion. If they mill about in different directions, individuals move a lot and the centre barely moves at all. Long dispersion is a bet on milling about.

Why the trade exists at all

Dispersion is not a clever observation somebody had; it is the natural consequence of who buys what.

Institutional investors hedge portfolios, not individual stocks, so demand for index downside is structural and persistent. Structured product issuance runs the other way: autocallables and worst-of notes leave dealers systematically long single-name optionality, which they must sell back into the market (see autocallables). The two flows push index implied volatility up relative to single-name implied volatility, and since correlation is the ratio of the two, they push **implied correlation above the level realised correlation usually delivers**.

> **KEY IDEA**
>
> The gap between implied and realised correlation is a risk premium, not a mispricing. It is paid to whoever is willing to be short correlation, and it is paid because that position loses badly in exactly the scenario everyone else is hedging: a crash, in which every name falls together and correlation goes to one. Long dispersion earns a steady premium and suffers rare large losses, which is the shape of every insurance-selling trade.

Building it in variance swaps

The cleanest implementation, because it removes delta hedging entirely: sell a variance swap on the index, buy variance swaps on the constituents. Recall the convention from variance swaps: a swap quoted at strike K in volatility points with vega notional N_{vega} has variance notional $N_{\text{var}} = N_{\text{vega}}/(2K)$ and pays $N_{\text{var}}\left(\sigma^2_{\text{realised}} - K^2\right)$.

The choice that has to be made is the **weighting**, and it is the whole design of the trade.

- **Vega-neutral**: equal vega notional on both legs. The simplest, and the standard starting point. It is not correlation-neutral in any exact sense; it is just balanced in first-order volatility exposure.
- **Theta-neutral**: size the legs so the daily time decay nets to zero, which means $N_{\text{var}}^{\text{idx}} K_{\text{idx}}^2 = N_{\text{var}}^{\text{sn}} K_{\text{sn}}^2$, so the single-name variance notional is ρ_{implied} times the index one. That is a *smaller* single-name leg than vega-neutral, and what it buys is a purer correlation exposure: the level of realised single-name variance drops out of the P&L and only the ratio is left.
- **Correlation-weighted**: size the single-name *vega* at $\sqrt{\rho}$ times the index vega rather than matching it, so the trade is neutral to a parallel move in all volatilities and responds only to the ratio. Since $K_{\text{idx}}/K_{\text{sn}} = \sqrt{\rho_{\text{implied}}}$, this lands on exactly the variance notionals of the theta-neutral weighting: the two names are two routes to one trade, and desks use them interchangeably.

WORKED EXAMPLE

The index trades at 18% implied and the vega-weighted basket of constituents at 30% implied, so implied correlation is $(18/30)^2 = 0.36$. Put on a vega-neutral long dispersion with 100,000 of vega notional per leg.

Variance notionals:

$$N_{\text{var}}^{\text{idx}} = \frac{100{,}000}{2 \times 18} = 2{,}777.78, \qquad N_{\text{var}}^{\text{sn}} = \frac{100{,}000}{2 \times 30} = 1{,}666.67.$$

Benign outcome. The year passes quietly: the index realises 15% and the average single name realises 28%, so realised correlation was $(15/28)^2 = 0.287$, below the 0.36 implied.

- Index leg (short): $-2{,}777.78 \times (15^2 - 18^2) = -2{,}777.78 \times (-99) = +275{,}000$.
- Single-name leg (long): $1{,}666.67 \times (28^2 - 30^2) = 1{,}666.67 \times (-116) = -193{,}333$.
- Net: $\mathbf{+81{,}667}$.

Notice what happened: volatility fell on *both* legs, and the trade still made money, because what it is short is the ratio and the ratio moved in its favour.

Crash outcome. Instead there is a sell-off. The index realises 40% and the average single name 45%, so realised correlation was $(40/45)^2 = 0.790$.

- Index leg (short): $-2{,}777.78 \times (40^2 - 18^2) = -2{,}777.78 \times 1{,}276 = -3{,}544{,}444$.
- Single-name leg (long): $1{,}666.67 \times (45^2 - 30^2) = 1{,}666.67 \times 1{,}125 = +1{,}875{,}000$.
- Net: $\mathbf{-1{,}669{,}444}$.

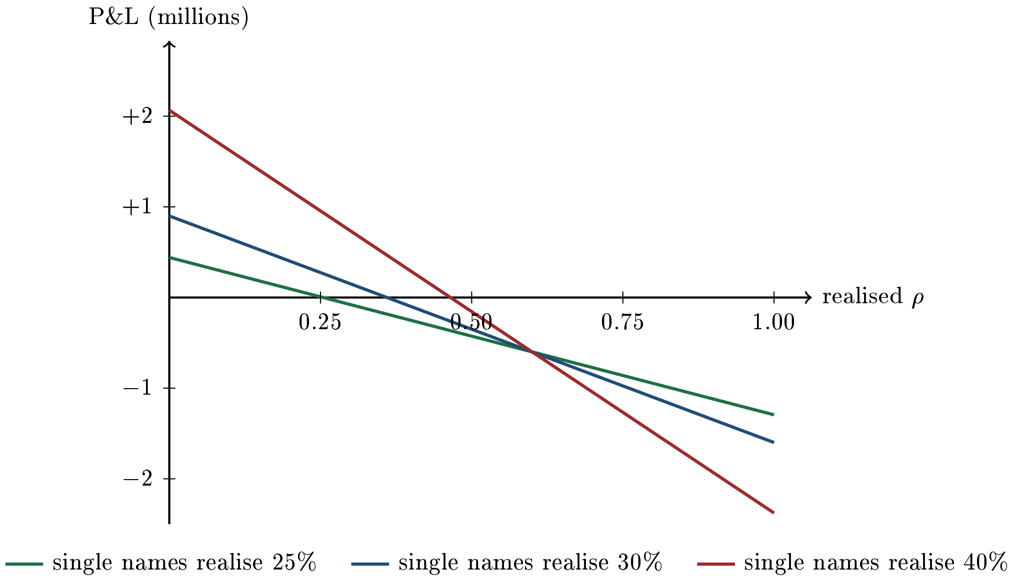
The loss is more than twenty times the gain in the good year, on the same position. That ratio is the entire risk profile of the trade in one comparison, and it is why dispersion books are sized on the crash scenario rather than on the volatility of their P&L.

You are not only short correlation

A vega-neutral dispersion is often described as a pure correlation position. It is not, and seeing why is what separates someone who has traded it from someone who has read about it. Holding the single-name realised volatility fixed at $\bar{\sigma}_r$ and using $\sigma_{\text{idx}}^2 = \rho_r \bar{\sigma}_r^2$, the P&L of the trade above is

$$\text{P\&L}(\rho_r, \bar{\sigma}_r) = -N_{\text{var}}^{\text{idx}} \left(\rho_r \bar{\sigma}_r^2 - 18^2\right) + N_{\text{var}}^{\text{sn}} \left(\bar{\sigma}_r^2 - 30^2\right),$$

a straight line in ρ_r whose slope *and* intercept both depend on the level of single-name volatility.



The three lines fan out rather than lying on top of one another, and that fan is the message. Each line crosses zero at a different correlation: 0.254 when single names realise 25%, 0.360 when they realise 30%, and 0.465 when they realise 40%. So the correlation at which you break even depends on the volatility that shows up, which means the position is a joint bet on correlation and volatility and not on correlation alone.

Two readings follow. The good one: if single-name volatility comes in high, the long leg carries you and the trade tolerates a higher correlation before losing. The bad one: the line is also steeper in that case, so once correlation does go up, the loss accumulates much faster. High volatility with high correlation, which is precisely a crash, is the worst corner of the picture, and it is the corner the two effects lead you into together.

The dispersion swap

Doing all this with individual variance swaps means executing one index trade and fifty single-name trades, paying fifty bid-offers, and then managing fifty positions through corporate actions and index changes. Dealers therefore package the whole thing into a single contract, a **dispersion swap** or correlation swap, whose payoff is written directly on the quantity you wanted:

$$N \times \Big(\sum_i w_i\, \sigma^2_{i,\text{realised}} \;-\; \sigma^2_{\text{index,realised}} \;-\; \text{strike}\Big),$$

or, in the correlation swap version, on realised correlation itself against a strike.

One contract, one confirmation, one bid-offer, and no re-weighting to maintain. What you give up is transparency and flexibility: the strike embeds the dealer's margin in a number you cannot decompose, you cannot restructure one leg without unwinding the whole thing, and you are relying on a single counterparty for a payoff that is largest exactly when their own book is under stress.

How it actually goes wrong

- **Correlation to one.** The tail, quantified in the crash case above. It is not a fat tail in the statistical sense so much as a regime change: the number stops being an average of pairwise relationships and becomes a single common factor.
- **Idiosyncratic events on the long leg, in the wrong direction.** A takeover announcement is a large realised move, so the gap itself is a gain, but the stock then pins to the offer and

realises almost nothing thereafter. The net is a mark-to-market loss on the remaining term of a long single-name variance position, and it has nothing to do with your view.

- **Execution and slippage.** Fifty legs, each with a bid-offer, some of them illiquid. A trade that looks like it captures three correlation points can lose most of that to transaction costs before it starts.
- **Composition drift.** Index constituents change and weights move, so the basket you bought stops being the basket the index contains.
- **The single-name variance leg is genuinely dangerous.** A variance swap pays on squared returns with no cap unless you negotiate one, and a single stock can gap 50% on an announcement. This is why single-name variance swaps are usually capped, typically at 2.5 times the strike, and why quoting them uncapped is a mistake made once.

Interview angle

REAL INTERVIEW QUESTION (BNP · EQD)

What are dispersion strategies?

WHAT THEY TEST / ANSWER

Answer in one sentence, then justify it, then say what the risk is. That three-part shape is what an EQD interviewer is listening for.

The sentence: dispersion is selling volatility on an index and buying volatility on its constituents, which is equivalent to being short implied correlation, because index volatility is roughly $\sqrt{\rho}$ times average single-name volatility and correlation is the only thing that converts between the two markets.

Why it exists: not as a clever idea but as a consequence of flow. Investors hedge portfolios rather than individual stocks, so index puts are structurally in demand, while structured product issuance leaves dealers long single-name optionality that they must sell back. Both flows push index implied volatility rich relative to single-name implied volatility, so implied correlation sits persistently above what tends to realise, and long dispersion collects the difference.

How it is done: cleanest in variance swaps, one short on the index against a weighted basket of longs, because that avoids delta hedging fifty names. Otherwise delta-hedged straddles. Mention the weighting choice, vega-neutral against gamma-neutral, because it is the real design decision and shows you have thought past the concept.

The risk, which is the part that finishes the answer: it is a short correlation position, so it loses when correlation goes to one, and correlation goes to one in a crash, which is exactly when the buyers of index protection you sold to want to be paid. The payoff shape is a small steady premium against a rare large loss. That is not a flaw in the trade, it is what the premium is compensation for, and saying so unprompted is the difference between describing a trade and understanding one.

REAL INTERVIEW QUESTION (Barclays · Structured Products)

What is a dispersion swap?

WHAT THEY TEST / ANSWER

Start with what problem it solves rather than with a definition, because the instrument only makes sense as an answer to a practical difficulty.

Running dispersion in its raw form means one index variance swap against fifty single-name variance swaps: fifty bid-offers to cross, fifty positions to maintain, re-weighting every time the index composition changes, and a corporate action calendar to manage. A dispersion swap collapses all of it into one contract that pays the difference between the weighted sum of realised single-name variances and the realised index variance, against a strike agreed at inception. The correlation swap is the same idea written directly on realised correlation.

Then give the trade-offs, since a one-line definition is not an answer. You gain one execution, one confirmation, one bid-offer, no re-weighting, and no operational tail. You lose the ability to decompose the price, so the dealer's margin sits inside a single strike you cannot check leg by leg; you lose flexibility, because you cannot adjust or monetise one leg without unwinding everything; and you take concentrated counterparty risk on a payoff that is at its largest precisely when the dealer's own book is worst placed.

The follow-up to expect is why realised correlation swaps are harder to price than the variance version. The reason is that correlation appears as a ratio of realised quantities, so the payoff is not a linear function of variances and does not admit the clean static replication that gives a variance swap its price. That is worth knowing because it explains why the dispersion form, written on variances, is the one that actually trades.

REAL INTERVIEW QUESTION (Exane · Exo)

How do you hedge the correlation sensitivity of options on baskets?

WHAT THEY TEST / ANSWER

This is the dealer's side of everything above, and the answer is that dispersion is the hedge. Say that immediately, then build it out.

A dealer who has issued worst-of notes and autocallables is *long* the embedded worst-of puts, because the investor sold that downside in exchange for the coupon. A worst-of put is worth more when the names decouple, since a lower correlation makes the worst performer worse, so the desk is **short correlation**. To reduce that exposure it buys correlation in the vanilla market: buy volatility on the index, sell it on the constituents, which is **short** dispersion. That converts a correlation position that cannot be traded directly into two volatility positions that can, and it is precisely this hedging flow that keeps implied correlation above realised and pays the trader on the other side.

Then be honest about how imperfect it is, because that is the substance of the question. Basket exotics have a **correlation matrix** of exposures, not one number, and they are not equally sensitive to every pair; a dispersion hedge is a single-factor approximation to a many-factor risk. The exposure is also not stable: the correlation sensitivity of a worst-of moves with spot, with time, and with how far apart the names have drifted, so the hedge has to be rebalanced and each rebalance costs bid-offer on every leg. And the correlation sensitivity of an exotic is a second-order quantity computed by revaluing under a bumped matrix, so it is only as good as the model producing it.

Finish with the three practical answers a desk actually uses, in order of preference: warehouse the residual correlation risk and charge for it in the price, which is what correlation margin in structured product pricing is; net it internally against other trades with the opposite sign, which is why desks care about the aggregate book and not the trade; and hedge the remainder with dispersion or a correlation swap when the size justifies the cost. Nobody hedges it perfectly, and saying so is correct rather than evasive.

INTERVIEW TRAP

Four errors. (1) Calling vega-neutral dispersion a pure correlation trade: the figure above shows the breakeven correlation moving from 0.25 to 0.47 as single-name volatility goes from 25% to 40%. (2) Getting the sign backwards: long dispersion is *short* correlation. (3) Ignoring the single-name variance cap, and discovering it matters when one name gaps on an announcement. (4) Judging the trade on its average P&L, when its whole character is a small steady gain against a rare enormous loss, and the average over a quiet sample tells you nothing about the size to trade.

Where this goes next

The identity and the implied correlation the trade is written on are in equity correlation. The instrument the trade is usually built from is variance swaps, with the payoff difference that matters in volatility swaps. The flow that creates the premium in the first place is autocallables, the behaviour of the whole position when correlation goes to one is in Greeks across market regimes, and the other great relative-value trade on the surface is skew trading.

18.13 Exotic Options (Barriers, Digitals, Asians)

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KEY IDEA

An option is **exotic** when its payoff depends on something other than the terminal spot alone. Four things it can depend on instead, and each one defines a family: the **path** the underlying took (barriers), a **discontinuity** in the payoff (digitals), an **average** rather than a point (Asians), or an **extremum** of the path (lookbacks). Naming which of the four a product uses tells you immediately how it prices, how it hedges, and where it will go wrong.

The individual exotic products built from these belong to chapter 19, and the deeper treatment of path dependence is path-dependent products. This section is the taxonomy and the pricing intuition for each family.

Barriers

REAL INTERVIEW QUESTION (Morgan Stanley · EQD)

What are the different types of barrier (in, out, up, down)?

WHAT THEY TEST / ANSWER

Two independent binary choices, so **four combinations**, and saying it that way is faster and more convincing than listing names.

First choice, what the barrier does. A **knock-out** starts alive and dies if the barrier is touched. A **knock-in** starts dead and comes alive if the barrier is touched.

Second choice, where the barrier sits relative to spot at inception: **down** for below, **up** for above.

That gives down-and-out, down-and-in, up-and-out and up-and-in, each of which can be a call or a put, so eight vanilla-barrier combinations in total. Add **double barriers**, with a level on each side, and **window barriers**, live only during part of the life, and you have essentially the whole listed and over-the-counter universe.

Then volunteer the two features that actually matter on a desk, because the list alone is recall.

Monitoring. Continuous, daily-close or at maturity only. The same nominal barrier prices very differently under each, and a European barrier observed only at maturity is a completely different instrument from a continuously monitored one: it can be decomposed statically, whereas the continuous version cannot. That distinction drives everything in autocallables.

Rebates. A knock-out may pay a fixed amount on knocking out, which softens the discontinuity and changes the hedge materially.

Close on the relationship that ties the family together and is the next question anyway: **in plus out equals vanilla**.

That last identity is worth stating on its own because it is exact and model-free. A knock-in and a knock-out with the same strike, barrier, maturity and monitoring cover every path between them exactly once:

$$\text{Knock-in} + \text{Knock-out} = \text{Vanilla}.$$

Either the barrier is touched, in which case the in-option is alive and the out-option is dead, or it is not, and the reverse. So the two must sum to the unconditional payoff, and any barrier price can be checked against the vanilla in one subtraction.

REAL INTERVIEW QUESTION (Société Générale · Structured Products)

Rank the following in increasing order of price and justify it: a down-and-out call struck at 70 with a barrier at 60, a down-and-in call with a barrier at 60, and a vanilla call struck at 70.

WHAT THEY TEST / ANSWER

Lead with the identity, because it answers the question and proves the answer at the same time.

The three instruments satisfy $\mathrm{DI} + \mathrm{DO} = \text{vanilla}$, all at strike 70 with a barrier at 60. Both barrier options are non-negative, so each is worth *less* than the vanilla immediately, and no computation is needed for that half.

So the vanilla is the most expensive, and the only remaining question is which of the two barrier options is cheaper.

The down-and-in is the cheapest, and the reason is the useful part. It only pays if the underlying *first falls to 60 and then finishes above 70.* Those two requirements pull in opposite directions: the path has to go down a long way and then recover past the strike, which is a demanding conjunction. The down-and-out, by contrast, pays whenever the underlying finishes above 70 without ever having visited 60, and since finishing above 70 already makes a visit to 60 unlikely, most of the vanilla's value survives.

Increasing order: down-and-in, down-and-out, vanilla.

Then give the limits, which is the natural follow-up and shows the structure of the family. Push the barrier *down* toward zero and it becomes unreachable: the down-and-in tends to zero and the down-and-out tends to the vanilla. Push it *up* toward spot and the barrier becomes almost certain to be touched: the down-and-in tends to the vanilla and the down-and-out tends to zero. The two prices slide between the same two endpoints in opposite directions, always summing to the vanilla, which is the identity seen as a picture.

Why barriers are hard to run

A barrier's difficulty is concentrated in one place: near the barrier, near expiry. There the payoff has a step, so the delta becomes very large and the gamma changes sign across the level, in the way gamma describes for any discontinuity. A desk cannot rehedge fast enough to follow it, so it prices and hedges a smoothed barrier, which is the same overhedge logic as for a digital in replication, and the difference between the true and the smoothed barrier is a deliberate cost.

Barriers are also the products where the choice of model matters most, because their value depends on the probability of *touching* a level, which is a statement about the dynamics of the surface rather than about its shape today. Local volatility and stochastic volatility calibrate to the same vanillas and disagree about barriers, which is the practical content of local volatility.

Digitals

A digital pays a fixed amount if a condition is met and nothing otherwise, so its payoff is a step. Everything about it follows from that: it is *minus* the derivative of a call price with respect to strike, since $\partial C/\partial K = -e^{-rT}N(d_2)$, its delta is unbounded at the strike as expiry approaches, no dynamic hedge exists, and it is therefore priced and hedged as a tight call spread. The full treatment, including the strike-gap trade-off and the skew correction, is replication strategies, and it is not repeated here.

Asians

The two things an Asian can average, the spot or the strike, and which risk each removes, are in path dependency and scenario analysis. Two implementation details belong here, because a structurer has to specify them and a taxonomy that omits them is incomplete.

Averaging frequency and window. A weekly average over three months is a different instrument from a daily average over the final month, and the difference is not a detail: the shorter and later the window, the closer the product is to a vanilla, and the more a single print can still move the redemption.

Arithmetic against geometric. Arithmetic averaging is what trades; geometric averaging is what has a closed form, because the product of lognormals is lognormal while the sum of them is not. That mismatch is why the standard approach prices the geometric case analytically and uses it as a *control variate* in a Monte Carlo on the arithmetic one, which is the same variance-reduction idea as Monte Carlo generally.

REAL INTERVIEW QUESTION (Natixis · Exo)

Which option is cheaper: a down-and-in or a vanilla? An Asian or a vanilla?

WHAT THEY TEST / ANSWER

Both comparisons go the same way and for two different reasons, and giving the reasons separately is the answer.

The down-and-in is cheaper than the vanilla, and this is immediate from $\mathrm{DI} + \mathrm{DO} =$ vanilla with both terms non-negative. No model, no assumption. The knock-in pays on a strict subset of the paths the vanilla pays on, so it cannot be worth more.

The Asian is cheaper than the vanilla, and here the reason is **averaging reduces volatility**. An average of correlated observations of the path varies less than the endpoint does, and an option is worth more when the underlying is more uncertain. Quantify it, because that is what separates the answer: for a geometric average of a lognormal path over $[0, T]$, the variance of the log average is $\sigma^2 T/3$, so the effective volatility is

$$\sigma_{\text{eff}} = \frac{\sigma}{\sqrt{3}} \approx 0.577\,\sigma.$$

At 20% volatility that is an effective 11.5%, so using the at-the-money approximation an Asian call is worth roughly 0.577 of the vanilla, about 4.6 against 8.0 on a spot of 100 at one year. Arithmetic averaging is slightly different but the order of magnitude is the same.

Then the point that makes it more than a pricing fact: the Asian is cheaper *and* more robust. Because the payoff depends on an average, it cannot be moved by a single print, so it is far harder to manipulate at fixing and far easier to hedge, since the desk's exposure decays smoothly as observations accumulate rather than concentrating at one moment. That is why averaging is standard in commodity structures, where the underlying is illiquid at a point, and why structured products average their final fixings, as hedging structured products sets out. Cheaper and safer at the same time is unusual, and it is the reason Asians are the most-sold exotic in the world.

Lookbacks, and the ordering of the whole family

A lookback pays on the best level the path reached: $\max(S_{\max} - K, 0)$, or a floating-strike version paying $S_T - S_{\min}$. It is the mirror image of an Asian. Where averaging damps the path, a lookback takes its extreme, so a lookback is always *more* expensive than the vanilla, often by a great deal, and it is correspondingly awkward to hedge.

That gives a clean ordering to carry, all with the same strike and maturity:

$$\text{Asian} \;<\; \text{vanilla} \;<\; \text{lookback},$$

with barriers sitting below the vanilla on either side of the in-out split. The single principle behind it: **a payoff that averages the path is worth less than one that samples it, and one that takes the path's extreme is worth more**.

INTERVIEW TRAP

1. **Forgetting in-out parity.** It is exact and model-free, and it answers most barrier ranking questions without any pricing.
2. **Ignoring the monitoring convention.** A European barrier observed once decomposes statically; a continuously monitored one does not, and the two prices differ materially.
3. **Assuming exotic means expensive.** Barriers and Asians are cheaper than the vanilla; only lookbacks and similar extremum payoffs are dearer.
4. **Treating the Asian's discount as a free lunch.** It is cheaper because it pays on less volatility, so it hedges a different risk, and it is the right instrument only if the exposure is genuinely an average.
5. **Pricing barriers off a single volatility.** Their value depends on the probability of touching, so it depends on the dynamics of the surface, and two models fitting the same vanillas will disagree.
6. **Underestimating the barrier gamma.** Near the barrier and near expiry it cannot be hedged faithfully, and the smoothing is a cost to be priced rather than an approximation to be hidden.

VISUAL / CODE TO BUILD: Interactive

A path sampler that prices four payoffs at once. The page simulates paths one at a time and, for each, marks the terminal spot, the running average, the running extremum and whether the barrier was touched, then pays out the vanilla, the Asian, the lookback and both barrier options on that single path. Running many paths builds four price estimates side by side, so the ordering in the text emerges from the simulation rather than being asserted, and a counter shows the in-plus-out sum tracking the vanilla to within Monte Carlo error, which is the identity demonstrated numerically. A slider on the averaging window shows the Asian's price falling as the window lengthens and its effective volatility dropping toward $\sigma/\sqrt{3}$. A final panel drags the barrier level from far below the strike to above it, plotting the knock-in and knock-out prices as two curves that always sum to a flat line at the vanilla's value.

18.14 Path-dependent products

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A vanilla option asks one question at one moment: where is the spot at expiry. Everything that happened on the way there is irrelevant, which is why one number, the terminal distribution, is enough to price it. A path-dependent product asks about the journey. Two paths that end at exactly the same place can pay completely different amounts, and the moment that is true, most of the machinery built for vanillas stops working.

The catalogue of these products, barriers, digitals, Asians, lookbacks, is in exotic options. This section is about what path dependence *does*: to the pricing problem, to the numbers, and to the book.

Three degrees of dependence

The useful classification is not by product name but by how much of the path you have to remember.

- **Path-independent.** The payoff is a function of S_T alone. Vanillas, digitals at maturity, any European terminal payoff. One state variable, closed forms available.
- **Weakly path-dependent.** The payoff needs the spot *and* a small number of extra quantities that can be updated as you go: a running average for an Asian, a running maximum for a lookback, a single flag for whether a barrier has been touched. You do not need the whole path, only a summary of it.
- **Strongly path-dependent.** No finite summary suffices, or the summary is large enough that carrying it is as expensive as carrying the path. Products with many interacting observation dates and memory features live here.

KEY IDEA

The dividing line that matters in practice is whether the path can be summarised by a few numbers that update as time passes. If it can, the product is weakly path-dependent, and the price is a function of spot, time and those extra **state variables**. Each state variable is another dimension in the pricing problem, and that is the entire reason path-dependent products are computationally expensive.

Why closed forms mostly disappear

A vanilla satisfies a partial differential equation in two variables, spot and time (see partial differential equations in finance). Add a running average and the price becomes a function of three variables, so the equation is three-dimensional. Add a running maximum as well and it is four. Solving a grid in n dimensions costs roughly N^n points, so the cost multiplies with each state variable. Two extra variables is usually the practical limit for a grid method.

Monte Carlo has the opposite profile. Its cost per path is essentially unchanged by adding state variables, since you simply carry the running average or the touch flag along the path you were generating anyway, and its error falls as $1/\sqrt{n}$ regardless of dimension. That is why path-dependent products are priced by simulation almost by default, and the mechanics are in Monte Carlo methods.

INTERVIEW TRAP

The trap is the combination. Monte Carlo handles path dependence naturally and early exercise badly, because deciding whether to exercise at a date needs the continuation value, which is a conditional expectation you do not have while running forward. Trees and grids handle early exercise naturally, by comparing at every node, and path dependence badly, because they need the extra dimensions. A product that is *both* path-dependent and early-exercisable is genuinely hard, and needs a regression-based method that estimates the continuation value from the simulated paths themselves.

Averaging: how much cheaper, and why

Averaging is the cleanest case because the effect can be computed rather than asserted. An Asian option pays on the average of the spot over the life rather than on the final spot, and an average is less variable than its final observation, so the option is worth less.

How much less has an exact answer in the continuous limit. If the log-spot follows a Brownian motion with volatility σ, the variance of the average of the log-spot over $[0, T]$ is $\sigma^2 T/3$ rather than $\sigma^2 T$. So a continuously averaged option behaves like a vanilla struck on a quantity with effective volatility

$$\sigma_{\text{eff}} = \frac{\sigma}{\sqrt{3}} \approx 0.577\,\sigma.$$

Averaging also shifts the level the option is struck against, not just its spread. With zero rates the geometric average has $\mathbb{E}[A] = S_0\, e^{-\sigma^2 T/12}$, a little below spot, so the exact continuous price is a vanilla on *that* forward at $\sigma/\sqrt{3}$. Dropping the forward adjustment and pricing a plain vanilla at $\sigma/\sqrt{3}$ is a common shortcut and it overstates the geometric Asian.

WORKED EXAMPLE

Spot 100, strike 100, one year, 20% volatility, zero rates. The vanilla call is worth **7.966**. The continuously averaged (geometric) option prices at $\sigma/\sqrt{3} = 11.55\%$ on a forward of $100\, e^{-0.04/12} = 99.667$, giving **4.432**, about 44% less.

A real Asian averages over a finite number of fixings, so it sits between the two. Pricing the arithmetic average by simulation, with the geometric average as a control variate so the standard error is below 0.001:

Averaging	Call value
Final fixing only (vanilla)	7.966
4 fixings	5.457
12 fixings	4.889
52 fixings	4.669
252 fixings	4.616
Continuous limit (geometric, $\sigma/\sqrt{3}$)	4.432

Two things to read off. The convergence is fast and then very slow: going from 4 fixings to 12 is worth 0.57, going from 52 to 252 is worth 0.05. Beyond weekly observation the number of fixings barely matters, which is why monthly or weekly averaging is the market standard. And the arithmetic values sit above the geometric limit at every count, as they must, because the arithmetic mean is never below the geometric mean.

Monitoring frequency: the number that gets negotiated

Barriers show the same effect with the opposite sign and much larger stakes, because here the frequency is a term in the contract that a client actually argues about.

A knock-out that is checked continuously has more chances to die than one checked daily, which has more than one checked monthly. So a discretely monitored knock-out is always worth *more* than its continuous equivalent, and the gap is largest when observations are sparse.

WORKED EXAMPLE

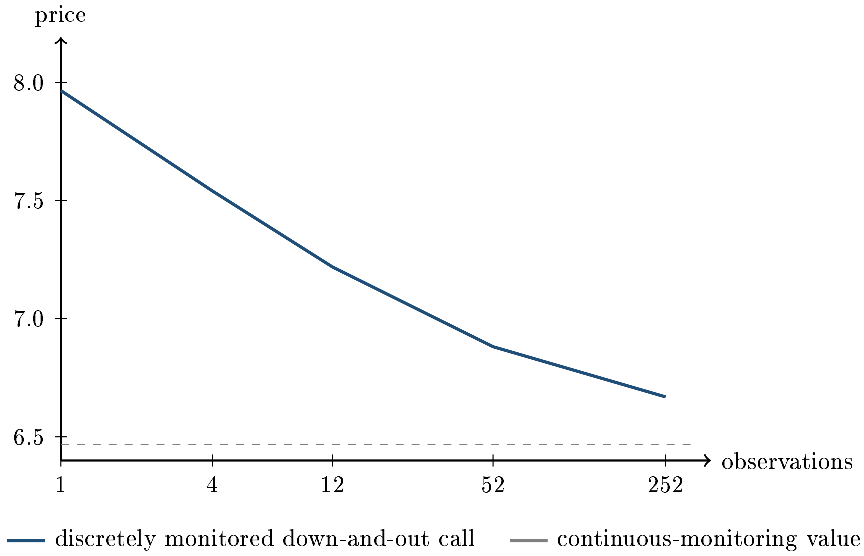
A down-and-out call, spot 100, strike 100, barrier 90, one year, 20% volatility, zero rates. The continuously monitored value is **6.467**; the vanilla is 7.966, so the barrier costs 1.498 of value, which is exactly the down-and-in, as in-out parity requires.

Now monitor discretely. Using the standard continuity correction, which prices a discretely monitored barrier by shifting it away from the spot by a factor $\exp\left(-0.5826\,\sigma\sqrt{T/m}\right)$ for a down barrier with m observations:

Observations m	Effective barrier	Price	Monte Carlo (95% band)
1 (maturity only)	none	7.966	
4 (quarterly)	84.91	7.541	7.575 ± 0.041
12 (monthly)	87.02	7.218	7.237 ± 0.041
52 (weekly)	88.56	6.882	6.919 ± 0.040
252 (daily)	89.34	6.669	
Continuous	90.00	6.467	

The last column is the check, not decoration: I simulated the discretely monitored option directly at 400,000 paths, and the correction sits inside the 95% confidence band at all three frequencies. So the shortcut is not a rule of thumb here, it reproduces the simulation.

The economics: moving from daily to quarterly observation is worth 0.87 on an option worth 6.67, about 13% of its value, for a change that looks like a documentation detail. Anyone who reads the term sheet and misses the observation schedule has mispriced the trade by more than the bid-offer.



The horizontal axis is spaced by the logarithm of the observation count, which is why the curve looks close to a straight line: each halving of the observation interval is worth roughly the same amount, and the process converges to the dashed line only slowly. Note also that the leftmost point, observation at maturity alone, is exactly the vanilla, because a barrier at 90 checked only at expiry cannot matter to a call struck at 100.

What it does to the book

Three consequences that a trader lives with rather than reads about.

The Greeks are discontinuous. Near a barrier the delta of a knock-out can flip sign and its gamma is unbounded, because an infinitesimal move decides whether the option exists. Hedging into that is the hardest routine problem on an exotics desk, and it is why barriers are managed by over-hedging with a vanilla spread rather than by following the model delta. The general behaviour of gamma near a discontinuity is in gamma and convexity.

Realised path becomes a fixed parameter. A fixing that has occurred is no longer risk; it is data. An Asian with eleven of its twelve fixings behind it has almost no volatility exposure left, because eleven-twelfths of the payoff is already determined. So the vega of a path-dependent product decays not with the square root of remaining time, as a vanilla's does, but with how much of the path has already been written. Books of averaging products are systematically less risky than their notional suggests near the end of their life, and systematically more risky than a vanilla comparison suggests near the start.

The past has to be stored, and it is the single most common operational failure. A knock-out that has touched is a different instrument from one that has not, and the record of that touch lives in a database rather than in a market price. Re-marking a book from a clean model without the touch history reprices dead options as live ones.

When the path depends on the model, not just the data

One more layer, and it is where path dependence meets stochastic volatility. A cliquet, or any forward-starting structure, resets its strike at a future date. Its value therefore depends on the volatility smile that will be observed *at that future date*, which today's vanilla prices do not pin down. Two models calibrated identically to every traded vanilla can disagree materially on a cliquet, because they disagree about the forward smile.

This is the sharpest practical reason the model choice in stochastic volatility models is not academic: for a terminal payoff, calibration is nearly enough; for a forward-starting path-dependent payoff, calibration is nowhere near enough.

Interview angle

REAL INTERVIEW QUESTION (Société Générale · Cross Asset)

Between an autocall with an annual observation barrier and one with a quarterly observation barrier, which will be more expensive?

WHAT THEY TEST / ANSWER

Ask yourself which direction the barrier works before answering, because autocalls contain two barriers pointing opposite ways and the question is ambiguous unless you say which one is being observed more often. Say that out loud, then answer both.

For the **autocall trigger**, the upside barrier that redeems the note early: more observation dates means more chances to be called, so early redemption is more likely, the expected life is shorter, and there is less time over which coupons accrue. The investor gets their money back sooner in the good scenarios, which is not what they wanted, so from the investor's point of view frequent observation makes the note less attractive and the issuer can therefore pay a lower coupon.

For the **downside protection barrier**, if it is of the American type observed continuously or often rather than only at maturity: more observation dates means more chances to breach, so protection is weaker, the embedded short put is worth more, and the issuer can pay a *higher* coupon. That is the same effect that makes a discretely monitored knock-out worth more than a continuously monitored one, run backwards.

Then give the general principle, which is what the question is testing: **more observation dates always make a path-dependent condition more likely to be met**, so the direction of the price effect follows entirely from whether meeting the condition is good or bad for the holder. Quantify it if you can: on a one-year down-and-out call struck at the money with a barrier 10% below spot at 20% volatility, moving from daily to quarterly observation is worth about 13% of the option's value, so this is a real number and not a rounding detail. The product itself is in autocallables.

REAL INTERVIEW QUESTION (Natixis · Exo)

For which options do you use binomial trees?

WHAT THEY TEST / ANSWER

The honest answer is a decision rule, not a list, and the rule has two axes: early exercise and path dependence.

Trees are for early exercise. Their virtue is that they step backwards through time, so at every node you can compare the continuation value with the immediate exercise value and take the maximum. That is exactly what an American or Bermudan option needs, and it is the one thing plain forward Monte Carlo cannot do, because running forward you do not know the continuation value at the moment you have to decide. So: American vanillas, Bermudan swaptions, callable structures, and anything with an exercise decision.

Trees are bad at path dependence. Every extra state variable, a running average, a running maximum, a touch flag, adds a dimension to the lattice, and the node count grows geometrically. Two extra state variables is usually the practical ceiling. **Monte Carlo is the reverse**: it carries any number of state variables along a path at essentially no extra cost, with an error that falls as $1/\sqrt{n}$ regardless of dimension, which is why Asians, lookbacks, autocallables and basket products are simulated.

Then name the awkward corner, because that is where the question is going: a product that is both path-dependent *and* early-exercisable belongs to neither method cleanly, and the standard resolution is a regression-based method that estimates the continuation value from the simulated paths themselves. Knowing that this hybrid exists, and why it has to, is what separates a real answer from a memorised one.

Finish with the practical footnote: trees are also the standard teaching device, because they make risk-neutral pricing visible node by node, and a great many people who never use one professionally should still be able to draw a two-step tree on request.

REAL INTERVIEW QUESTION (HSBC · Exo)

Which is more expensive, a plain call or an Asian call?

WHAT THEY TEST / ANSWER

The plain call, and the reason is one sentence: an average is less variable than a final observation, an option is worth more when the underlying quantity is more variable, so averaging makes it cheaper. Say that first.

Then give the number, because the reason alone is the answer most candidates give and the magnitude is what shows you have worked with them. For a continuously averaged option, the variance of the average of the log-spot is $\sigma^2 T/3$ rather than $\sigma^2 T$, so the option behaves like a vanilla with an effective volatility of $\sigma/\sqrt{3} \approx 0.577\,\sigma$. At 20% volatility that is 11.5%, and on a one-year at-the-money option with zero rates the price falls from 7.97 to 4.43, roughly 44% cheaper. Being able to say "about $1/\sqrt{3}$ of the volatility" converts a qualitative answer into a quotable one.

Two refinements to volunteer. The saving depends on the *number of fixings*, and it arrives fast: four fixings already capture most of it, and beyond weekly observation the extra fixings are worth almost nothing, which is why averaging schedules are monthly or weekly rather than daily. And the direction reverses for the *average-strike* version, where the average sets the strike rather than the payoff, so the two must be distinguished before answering.

The natural follow-up is why anyone buys them. Two reasons, and neither is the price: a corporate hedging a stream of purchases has an exposure that genuinely is an average, so an Asian matches the risk better than a vanilla; and averaging makes the payoff far harder to manipulate at a single fixing, which matters on illiquid underlyings. Cheapness is a consequence of matching the exposure, not the reason for it.

INTERVIEW TRAP

Four errors. (1) Reading a term sheet without the observation schedule, when the difference between daily and quarterly monitoring above was 13% of the option's value. (2) Assuming Monte Carlo prices everything, when it handles early exercise badly. (3) Quoting a path-dependent product's vega as though it decayed like a vanilla's, when what governs it is how much of the path is already fixed. (4) Confusing average-rate with average-strike Asians, which move in opposite directions.

Where this goes next

The products themselves are catalogued in exotic options, the most heavily traded path-dependent structure in the retail market is autocallables, and the most heavily traded one on a trading desk is the variance swap in variance swaps, whose payoff is a function of the whole path by construction. The simulation machinery is Monte Carlo methods with the discretisation schemes in numerical schemes, and the model dependence of the forward smile is stochastic volatility models.

Chapter 19

Structuring & Structured Products

19.1 Structured notes

sales · structuring · trading | core | equity · rates · cross-asset

Structuring and structured-products sales are among the largest front-office populations in European equity derivatives, and "what is a structured product?" is the opening question of essentially every structuring and sales interview. The answer most candidates give describes a payoff. The answer a desk wants describes an *instrument*: who issues it, what it is legally, where the money goes, and which of its risks have nothing to do with the underlying at all. This section is that answer; the individual payoffs fill the rest of the chapter.

A note is a bond first

A **structured note** is a debt security issued by a bank whose coupons, redemption amount, or both, are determined by the performance of an underlying: an index, a share, a basket, a rate, a currency, a commodity. The investor pays cash up front, receives a security with an ISIN that settles like any bond, and is repaid according to a formula rather than at par.

The word doing the work in that definition is *security*. It is what separates a note from the derivative inside it, and the distinction is not pedantry: it determines the documentation, the cash flows, the risk the client runs and the regulation that applies.

	Structured note	Derivative contract
Legal form	security, issued under a programme	bilateral contract under an ISDA
Funding	fully funded, cash paid up front	unfunded, margin posted
Credit risk	senior unsecured creditor of the issuer	bilateral, collateralised under a CSA
Access	any client who can buy a bond	counterparties with an ISDA
Secondary	issuer's bid, on request	unwind or novate the contract

REAL INTERVIEW QUESTION (BNP · Structured Products)

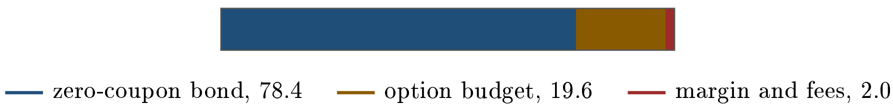
What is the difference between a structured product and a derivative?

WHAT THEY TEST / ANSWER

A derivative is a *contract*; a structured product is a *security that contains* one. Lead with that, then give the three consequences that follow, because the consequences are the answer. **Funding**: the note is bought with cash, so the client hands over the notional and the issuer has that money, whereas a derivative is unfunded and exchanges only margin. **Credit**: the note investor is a *senior unsecured* creditor of the issuing bank, so on a default it recovers the recovery rate on senior unsecured claims, historically a fraction of par and paid years later rather than nothing at all (see recovery rates). A derivative counterparty is collateralised daily *under a CSA*, and the half-sentence worth adding is that an ISDA without a CSA is not collateralised at all, which is exactly the exposure CVA prices. **Access and documentation**: a note is bought by anyone who can buy a bond and comes with a prospectus and a key information document, while a derivative needs an ISDA and a counterparty relationship, which is most of why the retail market is a note market and the institutional market is not. Finish with the sentence that shows you see both sides: from the issuer's point of view, the note is also a *funding instrument*, so a structured note is simultaneously a way to sell a payoff to a client and a way for the treasury to raise money, which is why the desk and the treasury both have a say in the price.

Where the money goes

Every note decomposes into two legs. One is a bond, which delivers whatever capital protection the product promises. The other is a derivative, which delivers the payoff. The client's cash pays for both, and for the bank's margin.



The proportions above are the five-year, 5% rate case worked through in leverage control through structure: the zero-coupon bond that repays 100 costs 78.4, which leaves 21.6, out of which the margin comes before the option budget. The 2.0 drawn here sits at the institutional end; a retail note of the same maturity carries a materially larger cost, as set out below. The whole design problem of the chapter is the size of the amber block and what has to be sold to enlarge it.

The direction of the derivative leg splits the product universe in two, and it is the most useful taxonomy to carry into an interview:

- **The client is long optionality.** Capital-protected notes and participation products: the client buys an option, funded by giving up the coupon on the bond leg. They pay for convexity and their downside is protected. See capital-protected products and participation products.
- **The client is short optionality.** Yield products: reverse convertibles, autocallables, Phoenix notes. The client sells an option, usually a downside put, and the premium becomes the coupon. They are paid for taking risk, and the risk is the product. See reverse convertibles and autocallables.

By volume the second group dominates, which matters for how you talk about the market: the retail structured-products business is largely a machine for transferring downside equity optionality from banks' clients to banks' trading books.

Taking one apart. The two-leg split above is the whole of what a note *is*, and it is also the first move in decomposing one: find the unconditional part, price it as a bond at *the issuer's own funding curve* rather than at the risk-free rate, and whatever remains is a portfolio of derivatives. The general method for that second step, reading a payoff and writing down the portfolio that reproduces it, is payoff engineering, and the replication theory underneath it is replication strategies.

Who does what

A note passes through four desks, and interviewers ask because it tells them whether you understand the job you are applying for.

- **Sales or distribution** owns the client, gathers the view and the constraints, and pitches. In retail this is an external distributor, a private bank or a network, not the issuing bank.
- **Structuring** designs the payoff against the client's view and the current market, prices it, produces the term sheet, and decides which levers buy the headline number.
- **The trading desk** takes the risk onto its book and hedges it. It quotes the internal price and lives with whatever the product leaves it short.
- **Treasury** sets the funding level at which the bond leg is issued, because the note is also debt the bank is raising.

That last one is the piece candidates most often miss. A bank issues structured notes partly because they are a competitive source of term funding, so the level treasury is willing to pay directly changes how good a coupon the client can be shown. When a bank's credit spread widens, its structured notes get *better* for the client, all else equal, which is an uncomfortable fact worth being able to state.

The risks that are not in the payoff

Everything so far is about the derivative. The wrapper carries its own risks, and they are where clients have actually lost money.

Issuer credit. The note is senior unsecured debt. If the issuer fails, the payoff formula is irrelevant and the investor joins the queue of creditors. This is not theoretical: retail investors holding notes issued by Lehman Brothers in 2008 discovered that a "capital-guaranteed" product guarantees nothing beyond the issuer's ability to pay. The guarantee is a promise, and the price of that promise is exactly why the bond leg was cheap enough to leave an option budget. What the queue actually pays out is recovery rates; see also credit risk and credit spreads.

Liquidity. There is no secondary market in the sense of a market. There is the issuer's bid, offered as a service and typically with a wide spread, and the client should assume they are holding to maturity. A product designed for a five-year view and sold to a client who may need the money in two is mis-sold regardless of how the payoff behaves.

Embedded cost. The fee is not charged, it is built into the price, so a note issued at 100 is worth less than 100 the moment it exists. Regulation now requires this to be disclosed, and a candidate should be able to talk about it without embarrassment: it is the price of a bespoke product, and the honest position is that it should be visible rather than absent.

Complexity and suitability. A payoff a client cannot restate in their own words has not been sold, it has been placed. That is both an ethical and a regulatory position, and it is the reason the last interview question in this section exists.

REAL INTERVIEW QUESTION (Société Générale · Structured Products)

What are the disadvantages of a structured product?

WHAT THEY TEST / ANSWER

Separate the disadvantages of the *wrapper* from the disadvantages of the *payoff*, because listing them together is what a memorised answer looks like. **Wrapper**: issuer credit risk, since the client is an unsecured creditor and a guarantee is only as good as the bank; illiquidity, since the secondary market is the issuer's bid and the client should plan to hold to maturity; embedded cost, since the fee sits inside the issue price rather than being charged separately; and complexity, which makes the product hard to compare against alternatives. **Payoff**: the client is usually short an option, so the risk is asymmetric, with a capped gain and a large loss in the tail; and the products that show the best headline numbers are the ones that sold the most, typically a worst-of barrier. Then give the balanced closing, because a pure list of criticisms is not an answer either: the product exists because it delivers a payoff the client cannot easily build themselves, at a size and in a wrapper they can actually buy, and the correct comparison is against the alternative of replicating it, not against a risk-free deposit. The failure mode is not the product, it is selling a five-year view to a client with a two-year horizon.

REAL INTERVIEW QUESTION (Union Capital · Structured Products)

What margin is typically taken when constructing a structured product based on a zero-coupon?

WHAT THEY TEST / ANSWER

Give a range, name the drivers and be explicit that you are describing an order of magnitude rather than a quoted number, since it varies by channel, size, maturity and issuer. Broadly, a retail note carries an all-in cost of roughly one percent per year of maturity, so mid single digits on a five-year, while an institutional ticket priced in competition is a small fraction of that. Then explain the structure of it, which is the part that shows desk knowledge. The margin is *embedded*: it is the difference between the issue price and the value of the legs, so it is paid by reducing the option budget, which is why it shows up as a slightly worse participation or coupon rather than as a fee line. It splits between the structuring margin, the distributor's commission, which is often the larger part in retail, and the trading desk's hedging costs and bid-offer. Two additions land well: because it is funded out of the same budget, a longer maturity or a lower rate environment makes the same margin cost the client proportionally more of their upside; and this is now disclosed in the key information document, so the honest framing in an interview is that it is a cost the client can see and compare, not a hidden one.

The lifecycle

A note is not a trade, it is a process, and knowing the stages is a cheap way to sound like someone who has sat on the desk.

1. **Indication.** Sales brings a view and a budget; structuring returns indicative terms, usually a range, since the market will move before the trade is done.
2. **Marketing period.** The terms are shown to clients while the market moves, so the final coupon or participation is only fixed at the end. Whoever carries that risk in between is a

negotiation.

3. **Strike, or fixing.** The initial level of the underlying is set on a stated date and by a stated method, often a closing price or an average over several days. Everything in the payoff is measured from it, so this date is the single most contested line in the term sheet.
4. **Issuance.** The note is issued under the programme, the client is delivered securities, treasury has the funding and the trading desk has the risk.
5. **Life.** The issuer publishes marks, the desk hedges, observation dates come and go, and the product may redeem early if it has an autocall trigger.
6. **Redemption.** Early or at maturity, on the final fixing, computed the same way as the initial one.

REAL INTERVIEW QUESTION (Société Générale · Structured Products)

How would you explain a structured product to an eighty-year-old who knows nothing about finance?

WHAT THEY TEST / ANSWER

Do not simplify the mechanics, replace them. Use one concrete structure, one analogy and one honest risk, and stop. For example: "You are lending money to the bank for five years. Instead of paying you a fixed interest rate, the bank pays you based on how the stock market does. If the market is flat or up, you get your money back plus a good return. If the market falls a long way, you get back less than you put in, and that is the risk you are taking in exchange for the better return. And because it is a loan to the bank, if the bank itself fails, you could lose your money regardless of the market." Three things make that a good answer rather than a friendly one. It states the downside in the same breath as the upside rather than after it. It names the credit risk, which is the part that is genuinely invisible to the client. And it avoids every word the client would have to take on trust, so "barrier", "autocall" and "volatility" do not appear. Then close by saying what you would do next, since the interviewer is testing judgement and not vocabulary: you would ask them to explain it back to you, and if they cannot, the product is not suitable whatever the term sheet says.

Formats

The same economics can be wrapped several ways, and the choice is driven by tax, regulation and what the client is allowed to hold rather than by the payoff.

- **EMTN**, a note issued off a medium-term note programme. The workhorse.
- **Certificate**, economically similar, often listed, common in some retail markets.
- **Warrant**, a listed option-like security, typically leveraged and unprotected.
- **Deposit**, where the bond leg is a bank deposit, which changes the insurance and credit treatment.
- **Fund or insurance wrapper**, where the note sits inside a vehicle, which is how many retail investors actually own them.

INTERVIEW TRAP

Four errors. (1) Describing a structured product purely as a payoff: it is a security issued by a bank, and half its risks come from that fact. (2) Saying "capital guaranteed" without saying by whom, when the guarantee is the issuer's unsecured promise. (3) Forgetting the funding leg is priced on the issuer's own curve, not the risk-free one, which is why a wider credit spread improves the client's terms. (4) Quoting a coupon or participation without the cost embedded in the issue price, which makes every comparison between two products meaningless.

VISUAL / CODE TO BUILD: Interactive

A note anatomy viewer. The reader picks a product family and a maturity, and the widget draws the client's 100 as a stacked bar splitting into the bond leg, the derivative legs and the margin, with each derivative leg named and priced. Sliders for the issuer's funding spread, the rate level and the underlying's volatility redraw the bar live, so three facts become visible rather than asserted: a wider issuer spread grows the option budget, a lower rate shrinks it, and higher volatility makes the sold options worth more and the bought options cost more depending on which side the client is on. A second panel shows the same product's payoff diagram with the client's outcome at maturity, and a third shows the day-one mark against the 100 paid, so the embedded cost is a number on the screen rather than a discussion.

Where this goes next

The individual families are the rest of this chapter: autocallables, Phoenix notes, reverse convertibles, capital-protected products and participation products. Designing the payoff itself is payoff engineering, what the desk does with the risk once it owns it is hedging structured products, and the options hidden inside the term sheet are catalogued in embedded optionality. How the same economics are sold to two completely different audiences is retail versus institutional structuring, and the arithmetic that sets the headline number is in leverage control through structure.

19.2 Autocallables

sales · structuring · trading | core | equity · cross-asset

The autocallable is the most-sold equity structured product in Europe and the single most-asked product in this database. It is worth learning properly rather than as a definition, because almost every question about it is really a question about the option positions hiding inside it, and a candidate who can name those positions can answer questions they have never seen.

Structured notes covers what a note is and why a bank issues one. This section owns the autocall itself: the mechanism, its decomposition, where the coupon comes from, what moves it, and why the product is difficult to run. The Phoenix variant and the reverse convertible have their own sections; the general hedging apparatus is hedging structured products.

The mechanism

A five-year autocall on an index, with an annual observation, works like this. On each anniversary the index level is compared with the **autocall trigger**. If it is at or above, the note redeems immediately at par plus all accrued coupons and the trade is over. If it is below, nothing happens and everyone waits a year. If the note survives to maturity, the **capital barrier** decides the outcome: above it, the investor gets capital back; below it, the investor takes the full performance of the index.

KEY IDEA

The investor's return is bounded above by the coupons and unbounded below except by zero. That asymmetry is not a flaw in the product, it *is* the product: the investor has sold a put and is being paid for it in instalments. Every autocall question reduces to that sentence.

REAL INTERVIEW QUESTION (Société Générale · EQD)

What is an autocall? How many barriers? At what levels are they? What are they for?

WHAT THEY TEST / ANSWER

Four questions, so give four answers in order.

What. A note that redeems itself early, at par plus a coupon, if the underlying is above a trigger level on an observation date. If it never triggers, it runs to maturity and the investor's capital depends on where the underlying finishes.

How many barriers. Typically **three**, and saying "three, and here is what each one does" is the whole answer. Some products merge two of them, which is worth mentioning because it shows you know they are separate functions rather than a fixed recipe.

Levels and purpose.

- **Autocall trigger**, usually 100% of the initial level, sometimes stepping down a few percent a year. It ends the trade early. Its job is to shorten the expected life, which is what lets the coupon be quoted per annum on a product most buyers expect to last two years, not five.
- **Coupon barrier**, typically 60% to 80%. It decides whether a coupon is paid on a date that did not autocall. An Athena has no coupon barrier at all, since coupons are only paid on call; a Phoenix has one, and that is the difference between the two products.
- **Capital barrier**, typically 50% to 70%, observed at maturity only (European) or continuously (American). It decides whether the investor's capital comes back whole.

Close on the point the question is aiming at: the three barriers do three different jobs, and the investor's risk lives almost entirely in the last one. The first two decide *how much* you get paid, the third decides whether you keep your money.

What the investor is actually holding

REAL INTERVIEW QUESTION (Morgan Stanley · Exo)

Do you know autocalls? What are they made of? Who is long and short of what?

WHAT THEY TEST / ANSWER

Three parts, and the third is the one that matters.

One, a zero-coupon bond issued by the bank. The investor is long the bank's credit and long its funding spread, which is where a piece of the yield comes from and is why two identical-looking notes from two issuers pay different coupons.

Two, a set of conditional coupons. Each one is a digital: it pays a fixed amount if the underlying is above a level on a date. The investor is long these.

Three, a **short down-and-in put** struck at the initial level with the capital barrier as its knock-in. The investor is short it, the bank is long it, and its premium is what funds the coupon. This is the entire economic content of the product.

So the sign sheet. The investor is **short volatility** (short a put), **short the underlying's downside** below the barrier, **long the issuer's credit**, **short dividends** and, on a worst-of, **long correlation**. The bank is the mirror of all of it. Add the observation that finishes the answer: the investor's upside is capped at the coupons whatever the index does, so this is not an equity investment with a safety net, it is a short volatility position wearing one.

The down-and-in put is worth writing out, because it explains where the barrier's value comes from. With a strike at 100 and a European barrier at B observed only at maturity, the payoff $(100 - S_T)$ paid only when $S_T < B$ decomposes exactly:

$$\text{DIP} \;=\; \text{Put}(B) \;+\; (100 - B) \times \text{DigitalPut}(B),$$

a vanilla put struck at the barrier, plus a digital paying the gap between the strike and the barrier. Check it at $S_T = 50$ with $B = 60$: the put pays 10, the digital pays 40, total 50, which is $100 - 50$. And at $S_T = 80$: the put pays nothing, the digital pays nothing, and the down-and-in put is indeed dead. The general technique is replication.

WORKED EXAMPLE

Where the coupon comes from, priced. Five-year note on an index at 100, capital barrier 60% European, issuer funding 3%, dividend yield 2%, volatility 25%.

- **Funding leg.** A five-year zero-coupon at 3% is worth $100e^{-0.15} = 86.07$, so the issuer has $100 - 86.07 = 13.93$ of the investor's money that does not need to be given back at maturity.
- **The put sold.** Put struck at 60 is worth 3.087; the digital paying 40 below 60 is worth $40 \times 0.2019 = 8.077$. The down-and-in put is $3.087 + 8.077 = \mathbf{11.16}$.
- **Coupon budget.** $13.93 + 11.16 = \mathbf{25.09}$ in present value, before the bank's margin and distribution fees.

Undiscounted that is $25.09/0.8607 = 29.15$ at maturity, or 5.83% a year if the note were certain to run the full five. The quoted coupon is higher than that, because most paths autocall early and pay fewer coupons, and turning the budget into a rate needs the distribution of call dates. That is exactly why an autocall is priced by Monte Carlo and not by a formula: the trigger makes it path-dependent.

For scale, a vanilla put struck at 100 would cost 17.30. The barrier is what takes the investor's downside from a 17.30 problem to an 11.16 one, and the coupon is smaller in proportion.

The levers on the coupon

REAL INTERVIEW QUESTION (Goldman Sachs · Structured Products)

How do you increase the coupon on my autocall without touching the barrier?

WHAT THEY TEST / ANSWER

First, pin the constraint, because it changes the answer: I will read "the barrier" as the capital barrier, since that is the one the client cares about and the one that protects them. If every barrier including the autocall trigger is frozen, say so and the list shortens to underlying, maturity, funding and margin.

The coupon is paid for by the put the investor sold, so the question becomes "how do I make that put more valuable without moving its strike or its knock-in". Using the five-year example above, with a 60% barrier:

- **Pick a more volatile underlying.** At 20% volatility the budget is 20.65; at 25% it is 25.09; at 30% it is 29.54. The single largest lever, and the reason the client's "safer" blue chip pays less.
- **Use a worst-of basket.** Three underlyings instead of one, with the payoff on the worst performer. The effective volatility is much higher and the barrier is far more likely to be breached, which is why worst-of coupons dwarf single-underlying ones. It also makes the investor long correlation, treated in dispersion.
- **Pick a higher dividend yield.** Dividends lower the forward and so raise the put. Moving the yield from 2% to 4% takes the budget from 25.09 to 28.10.
- **Lower the autocall trigger**, or step it down year by year, if that counts as available. A lower trigger calls the note away sooner, so fewer coupon periods are paid in expectation and the same budget buys a higher quoted per-annum rate. This is the standard step-down feature, and it is the same mechanism the trigger's description above rests on. It cuts both ways, so say so: calling sooner also retires the sold put sooner, which shrinks the budget, and the net gain is smaller than the headline rate makes it look.
- **Lengthen the maturity**, which grows the funding leg and the put together.
- **Cut the margin.** The two non-market levers are the bank's own fee and the issuer's funding spread, and a cheaper issuer is a worse credit. Say this out loud, because it is the honest part of the answer and it is where a real negotiation ends up.

Then the sentence that separates a structurer from a salesperson: every one of these raises the coupon by making the investor's risk larger. There is no lever that produces yield from nothing, and the client should be told which risk they just bought.

REAL INTERVIEW QUESTION (BNP · Structured Products)

Am I long or short dividends when I am long an autocall?

WHAT THEY TEST / ANSWER

Short dividends, and the reasoning is worth more than the answer.

You are short a put. A dividend lowers the forward, since $F = Se^{(r-q)T}$, and a lower forward makes every put more valuable. The put you are short has become dearer, so you have lost: you are short dividends. In the example above, moving the yield from 2% to 4% raises the down-and-in put from 11.16 to 14.17, and that three points comes out of whoever is short it.

There is a second channel worth adding, because it shows you understand the autocall specifically rather than puts in general: a lower forward also makes the trigger less likely to be reached, so the note lives longer, the capital stays at risk longer, and the coupons you were counting on arrive later or not at all. Both channels point the same way.

The desk's side is the mirror. It is long dividends on every autocall it has sold, in size, out to five years and beyond, and it hedges by selling dividend futures and dividend swaps. That hedge is imperfect at the long end where dividend futures stop trading, which is a real risk on the book and not a theoretical one.

Why it is hard to run

REAL INTERVIEW QUESTION (Société Générale · Exo)

What are the risk-management problems on an autocall?

WHAT THEY TEST / ANSWER

Take them in order of how much money they have actually cost desks.

One, the discontinuity at each observation date. On an autocall date the payoff jumps: a hair above the trigger the note repays par plus coupon and dies, a hair below it lives on. That is a digital, so delta and gamma blow up around the trigger as the date approaches, exactly as in digitals and barriers. The desk cannot hedge it cleanly and overhedges with a spread.

Two, the gamma near the capital barrier. The short down-and-in put gives the investor a cliff and the desk enormous gamma just above the barrier late in the product's life, treated in gamma and convexity. This is where the hedge is most expensive and most likely to fail in a fast market.

Three, the vega sign changes. The book is short volatility overall, but the sign is not uniform across spot: near the trigger and near the barrier the exposure flips, so a desk that manages a single vega number is managing an average of two opposite positions.

Four, dividends and repo out to five years, hedged in instruments that are illiquid past three, as above.

Five, correlation on worst-of, where the desk is structurally short: the put it owns is worth more when the underlyings disperse, so it loses when they move together. Every desk in the market is on the same side of that trade.

Six, and the one that is specific to this product: concentration. Autocalls are issued in enormous size on the same few indices with triggers clustered at the same levels. When the market rallies through them, a large part of the street's book autocalls at once, every desk unwinds the same hedge in the same direction on the same day, and the hedging flow moves the underlying and its volatility surface against them. The risk is not in any one note, it is that everyone owns the same one.

REAL INTERVIEW QUESTION (Société Générale · EQD)

Maximum loss for the holder of an autocall.

WHAT THEY TEST / ANSWER

The entire capital, less any coupons already received. The note is unleveraged, so the loss cannot exceed 100%, and if the underlying goes to zero the investor receives nothing back.

Then add the part the question is testing, which is not the size of the loss but its shape. Below the barrier the investor takes the *full* performance from the initial level, not the performance below the barrier. With a 60% barrier and the index finishing at 59, the investor does not lose 1%, they lose 41%. The barrier is a switch, not a deductible, and that cliff is the single most misunderstood feature of the product among the people who buy it.

Worth closing with the practical asymmetry: the maximum gain is the coupon stream, so a five-year note at 8% that pays all five coupons returns 40% against a possible 100% loss. That ratio is the trade, and stating it is the honest version of the sales pitch.

INTERVIEW TRAP

1. **Calling it capital-protected.** It is *capital-at-risk with a conditional barrier.* The protection disappears entirely the moment the barrier is breached, which is the opposite of how protection normally behaves.
2. **Quoting the coupon as a yield.** It is a per-annum rate on a product whose life is unknown and whose payment is conditional. The expected return is far below the quoted rate, and comparing it with a bond yield is not a comparison.
3. **Thinking a high coupon means a good product.** The coupon is the price of the risk transferred. A higher one always means more risk, never more generosity.
4. **Confusing an Athena with a Phoenix.** Athena pays only on call; Phoenix pays on any date above the coupon barrier and often has memory. The Phoenix section develops it.
5. **Forgetting the European barrier is only observed once.** A European capital barrier is checked at maturity alone, so an index that spent two years below 60 and recovers pays back in full. An American barrier does not forgive that. Get the observation style before answering any question about the risk.
6. **Ignoring the issuer.** The investor is long the bank's credit for the whole life of the note, unsecured. In 2008 that was the part of the product that actually defaulted.

VISUAL / CODE TO BUILD: Interactive

A path simulator with the term sheet exposed. The reader sets the trigger, the coupon barrier, the capital barrier, the volatility, the dividend yield and the funding rate, and the page runs a few thousand paths, drawing a sample of them against the barrier levels and reporting the distribution of outcomes: probability of calling on each date, expected life, expected return, probability of capital loss and its conditional average. A second panel shows the coupon budget decomposed into the funding leg and the down-and-in put, so moving the volatility slider visibly moves the put and therefore the coupon. The intended moment is the one where the reader raises the coupon from 6% to 9% and watches the probability of losing capital rise with it, which is the whole product in one interaction.

19.3 Phoenix notes

structuring · sales · trading | core | equity

Autocallable payoffs dominate equity structured product issuance in Europe, and the Phoenix is the member of that family an interviewer is most likely to hand you and say "walk me through it". It is a variant of the autocallable, whose recall machinery is developed in autocallables, so this section does not rebuild that. It answers one question instead: what is added when you turn an autocall into a Phoenix, why clients buy it, and what they are actually short.

The one idea: a coupon that does not need the market to be up

A plain autocallable pays the client only when the product is recalled: the underlying has to be back above its starting level on an observation date. That is a hard condition, and in a market that drifts sideways below par the client receives nothing for years.

The Phoenix breaks the coupon away from the recall. It sets a **separate, much lower coupon barrier**, and pays the coupon on any observation date where the underlying is above *that* level. So the structure has three levels, and knowing which does what is the whole product:

- the **recall barrier** (typically 100% of the initial level, sometimes higher, sometimes stepping down over time): above it on an observation date, the note redeems early at par plus the coupon and everything stops;
- the **coupon barrier** (typically 60% to 80%): above it on an observation date, the coupon is paid, whether or not the note is recalled;
- the **capital barrier**, the **knock-in level** of the embedded down-and-in put (typically 50% to 65%): if the final level is below it at maturity the put knocks in, and because that put is **struck at 100%**, the client's loss is then measured from the initial level rather than from the barrier.

INTUITION

Think of a tenant who pays rent as long as the building is standing, not only when it is worth more than when you bought it. The Phoenix pays its coupon while the underlying stays above a floor set far below the entry level, so the product keeps producing income through a market that is down 20% or 30%. The price of that is paid at the very bottom, where the client owns all of the loss at once.

REAL INTERVIEW QUESTION (Leonteq · Structured Products)

Why is a Phoenix said to rise from its ashes?

WHAT THEY TEST / ANSWER

Because a missed coupon is not a lost coupon. On an observation date where the underlying sits below the coupon barrier, nothing is paid; but if it comes back above the barrier at a later date, the coupon resumes, and with the memory feature the skipped ones are paid too, so the coupon stream can come back from the dead. Say the mechanism, not just the image, and then add the sentence that shows you know the limit: only the coupon is reborn, the capital is not. The down-and-in put at the bottom is a one-way door, so a name that falls through the capital barrier and stays there hands the client the full loss regardless of how many coupons were paid on the way down. Interviewers use this question to see whether you can talk about a product in a client's language and still hold on to what the risk is.

Reading a term sheet, scenario by scenario

REAL INTERVIEW QUESTION (Kepler · Structuring)

Describe the possible scenarios for a Phoenix with a coupon barrier at 70%, a recall barrier at 110% and a put down-and-in at 50%.

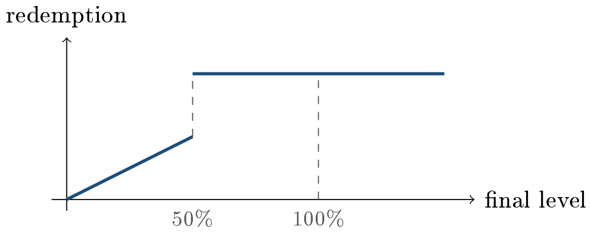
WHAT THEY TEST / ANSWER

Take the observation dates in order and be exhaustive, because that is what is being tested: the ability to enumerate a state machine without missing a branch. Assume quarterly observations, a five-year maturity and a conditional coupon c per period.

On **each observation date before maturity** there are exactly three outcomes. Above 110%: the note is recalled, the client receives 100% plus that period's coupon, plus any coupons in memory, and the trade is over. Between 70% and 110%: the coupon is paid, the note continues. Below 70%: no coupon is paid, the note continues, and with a memory feature the coupon is stored rather than lost.

At maturity, if the note has never been recalled, there are three outcomes again, and the third is the one they are checking. Above 110%: recalled at par plus coupon. Between 50% and 110%: capital is returned in full, and the coupon is paid if the level is also above 70%, so between 50% and 70% the client gets their money back and nothing else. Below 50%: the down-and-in put knocks in, and the client receives the performance of the underlying, so a final level of 40% redeems at 40%.

Then give the two details that separate a good answer. First, the loss is measured **from the strike, not from the barrier**: at 49% the client does not lose 1%, they lose 51%, because the put is struck at 100%. Second, that discontinuity is a cliff: 51% redeems at 100 and 49% redeems at 49, so two points of underlying are worth fifty-one points of redemption. Note the recall barrier at 110% is unusually high here, which means recall is unlikely early on and the client is being paid for a product that will probably run for years, and say whether the barriers are observed only on the observation dates or continuously, because the answer changes the price materially.



capital redeemed at maturity, as a percentage of the initial investment

coupons are paid separately and are not shown; the vertical gap at the barrier is the cliff

WORKED EXAMPLE

How likely are those branches? Take one underlying at 25% volatility, ignoring rates and dividends, with barriers observed only at the date shown, and ignore the recall so the note is still alive. Under the risk-neutral distribution the chance of being above the 70% coupon barrier is 90.3% after one year, 79.7% after two, and 64.0% after five. The chance of finishing below the 50% capital barrier at five years is 16.8%.

Two lessons. The coupon barrier is comfortable early and much less so late, which is the opposite of how the product is usually pitched. And the tail that matters is not negligible: roughly one path in six, on these assumptions, ends with the client taking equity losses measured from 100%. Change the volatility and the picture changes fast: at 20% the five-year breach probability falls to 9.2%, at 30% it rises to 24.3%. These are risk-neutral numbers, not forecasts, and they are the numbers the product is priced off.

The memory effect

REAL INTERVIEW QUESTION (Kepler · Structuring)

What is the memory effect, and what impact does it have on the price?

WHAT THEY TEST / ANSWER

The memory feature, also called a snowball, stores every coupon that was missed because the underlying was below the coupon barrier, and pays all of them on the next date where it is back above. Give the arithmetic: with quarterly coupons of 2%, an underlying below the barrier on observations 5, 6 and 7 and back above on observation 8, the client receives $4 \times 2\% = 8\%$ on observation 8 with memory and 2% without. Then the pricing point, which is the actual question. The memory feature is strictly favourable to the client, since it can only add payments and never removes one, so it has a positive value and the issuer must pay for it by lowering something else: for the same structure, the headline coupon on a Phoenix with memory is lower than on one without. Where the value sits is worth a sentence: memory is worth most when the underlying is volatile enough to cross the coupon barrier back and forth, and worth almost nothing on a name that either stays comfortably above it or collapses through it and never returns. That is a mean-reversion payoff hidden inside an equity note, and it is also why memory makes the product harder to hedge, since its value depends on the path rather than only on the final level. Path dependence in general is path-dependent products.

How it is built, and why you cannot just add up the pieces

REAL INTERVIEW QUESTION (Société Générale · Structured Products)

How do you structure a Phoenix?

WHAT THEY TEST / ANSWER

Build it from the client's side, leg by leg, and say who is long what. The client is **long a zero-coupon bond** issued by the bank, redeeming 100 at maturity, which is where the issuer credit risk enters and where the funding that pays for everything comes from. The client is **long a strip of digital calls**, one per observation date, each struck at the coupon barrier and paying the conditional coupon. The client is **short a down-and-in put** struck at 100% with the knock-in at the capital barrier, which is the leg that funds most of the coupon. And the whole package carries an **automatic early redemption**: above the recall barrier on an observation date, everything terminates.

Then make the point that turns a list into an answer. That last feature is not another option you can price and add, it is a cancellation of all the legs above it, so the value of every coupon depends on whether earlier dates recalled the note. A Phoenix is therefore *path dependent* and cannot be priced by summing closed forms: desks price it by Monte Carlo, or by a backward induction on a grid, and the greeks come out of the same engine. Finish with the budget logic, which is the structurer's actual job: the coupon is whatever the sum of the short put and the funding can buy after fees, so every lever that raises it, a lower capital barrier, a more volatile underlying, a worst-of basket, is a lever that increases what the client is short. That accounting is in leverage control through structure, and the traps in common structuring mistakes.

REAL INTERVIEW QUESTION (Kepler · Structuring)

What is the difference between an Athena and a Phoenix? Which product is more expensive?

WHAT THEY TEST / ANSWER

Same family, one difference: **when the client gets paid**. An **Athena** pays nothing along the way; the coupon accrues and is paid in one lump only if and when the note is recalled, so if the underlying never gets back above the recall barrier the client receives no income at all and simply carries the downside risk to maturity. A **Phoenix** pays a conditional coupon on every observation date where the underlying is above a much lower coupon barrier, recalled or not. So the Athena is a bet on recovering to par, the Phoenix is a bet on not collapsing.

On price, refuse the trick in the phrasing: both are issued at 100, so neither is "more expensive" in the ordinary sense. What differs is the coupon each can pay for the same barriers and underlying, and there the Athena pays more, because its coupon is contingent on a strictly harder event, so each unit of coupon costs the issuer less. Say the exception too, because it is the follow-up: the comparison depends on where the barriers sit, and a Phoenix with a coupon barrier close to the recall barrier converges to an Athena. The right closing sentence is about the client, not the price: an Athena suits someone who expects a rebound and wants maximum payout in that case, a Phoenix suits someone who expects a flat or gently falling market and wants income while they wait.

Where the coupon comes from

REAL INTERVIEW QUESTION (Goldman Sachs · Cross Asset)

If volatility rises on the worst-of underlying your Phoenix autocall, what is the impact on your coupon?

WHAT THEY TEST / ANSWER

Split the question in two, because "the coupon" means different things at issuance and afterwards, and the interviewer wants both.

At **issuance**, a higher volatility means the down-and-in put the client is short is worth more, so it raises more money, and the coupon the issuer can offer goes *up*. That is why the highest-coupon term sheets are always on the most volatile underlyings, and why a client shopping on coupon alone is shopping on risk.

On an **existing note**, a rise in volatility is bad news for the holder: the short put they already own is now worth more against them, so the mark-to-market falls even though the contractual coupon has not changed. Say the sign clearly, that the holder of a Phoenix is short volatility, and then add the two refinements that show depth. The exposure is not uniform: it is concentrated near the capital barrier and near the coupon barrier, where a small move changes a discrete outcome, so the vega profile is lumpy rather than a single number. And on a worst-of the volatility effect comes with a correlation effect, because a fall in correlation disperses the basket and makes the worst performer worse, which hurts the holder for the same reason, covered in dispersion strategies. Vega on these products is developed in hedging structured products.

INTERVIEW TRAP

Four traps, all of them asked. (1) Saying the client is "protected down to 50%": they are protected *until* 50%, after which the loss is measured from 100%, which is the opposite of protection at the worst moment. (2) Confusing the coupon barrier with the capital barrier: they do different jobs and usually sit at different levels. (3) Treating the coupon as a yield: it is a conditional payment financed by a short put, so comparing a 9% Phoenix coupon to a 4% bond yield is comparing a risk premium to a funding rate. (4) Pricing a Phoenix as a sum of closed-form legs: the recall makes it path dependent, so it needs a simulation, and any answer that adds up digitals and a put has quietly assumed the note never gets called.

VISUAL / CODE TO BUILD: Interactive

A Phoenix simulator. The reader sets the three barriers, the maturity, the observation frequency, the coupon and the underlying's volatility, then either drags a price path by hand or draws one from a simulation. The widget shows the observation calendar as a row of markers that light up as recalled, coupon paid, coupon stored or nothing, so the state machine is watched rather than described, with the running total of cash received. A second panel is a histogram over ten thousand simulated paths of the client's total return, with the mass at "recalled in year one", the middle of the distribution and the tail below the capital barrier shaded separately, next to the same histogram for a direct investment in the underlying. A memory toggle redraws both panels so the value of the feature is visible as a shift in the distribution rather than as a sentence. The intended realisation is the shape of the distribution: a Phoenix converts a wide, symmetric equity outcome into a narrow, very likely small gain plus a small, very bad tail.

Where this goes next

The recall mechanics and the family as a whole are in autocallables, and the simplest member of the family, without any recall, is the reverse convertible. The put that funds the coupon is dissected in embedded optionality, the digital and barrier legs in exotic options, and the desk's problem of running the resulting risk in hedging structured products.

19.4 Reverse convertibles

sales · structuring · trading | core | equity

The reverse convertible is the simplest yield product and the right one to learn first, because every other product in this chapter is a variation on the trade it makes: the investor sells a put and is paid the premium as a coupon. Understand where that coupon comes from and the autocallable, the Phoenix and the airbag all become recombinations of parts you already know. Misunderstand it, and you will describe a high coupon as a feature rather than as the price of a risk, which is the single most common failure in a structuring interview.

The structure

A **reverse convertible** is a short-dated note paying a high fixed coupon. At maturity:

- if the underlying is at or above the strike, the investor is repaid par;
- if it is below, the investor is repaid in *shares* instead, delivered at the strike, or in the cash equivalent.

The name is a warning label if you read it: a convertible bond gives the *holder* the right to convert, and a reverse convertible gives it to the *issuer*. The investor is on the wrong end of the option, and that is what pays the coupon.

The decomposition is two lines:

$$\text{reverse convertible} \;=\; \text{zero-coupon bond} \;+\; \text{coupon} \;-\; \text{a put}$$

so that the price paid, par, satisfies

$$100 \;=\; (100 + C)\,\mathrm{DF} \;-\; P, \qquad \text{hence} \qquad C \;=\; \frac{100 + P}{\mathrm{DF}} \;-\; 100,$$

where DF is the issuer's discount factor and P is the premium of the put the investor has sold. Everything commercial about the product is in that second equation: the coupon is the sold put's premium, grossed up for the funding.

KEY IDEA

The coupon of a reverse convertible is not a yield. It is an option premium the investor has been paid in advance, and it is large for exactly the reasons that make the option expensive: a volatile underlying, a strike close to spot, and a long enough maturity. A 15% coupon on a one-year note is not a generous bank, it is a 30% volatility stock.

WORKED EXAMPLE

A one-year reverse convertible on a single stock. Spot 100, strike 100, implied volatility 30%, rate 3%, no dividends, so $\mathrm{DF} = e^{-0.03} = 0.9704$.

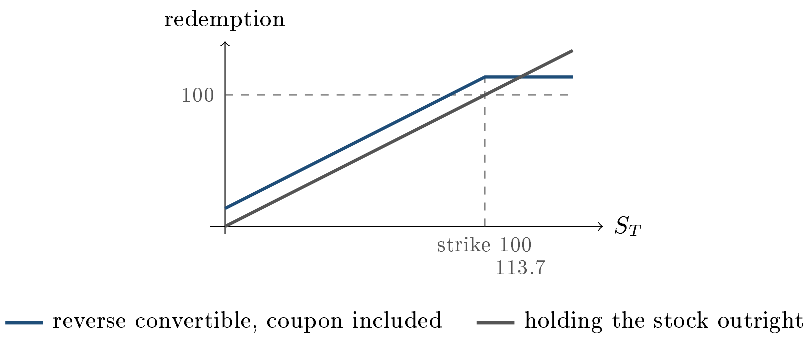
The one-year at-the-money put is worth $P = 10.33$. Therefore

$$C \;=\; \frac{100 + 10.33}{0.9704} - 100 \;=\; \mathbf{13.7}\%.$$

Move the strike down and the coupon falls with it, because the investor is selling a cheaper option. One convention governs the table below. The **conversion ratio** is the denomination divided by the strike, so a 100 note struck at 90 delivers $100/90 = 1.111$ shares rather than one. The note therefore sells 1.111 puts, not one, and the investor's worst case is the full 100 at every strike. A ratio of one would cap the loss below par and is a different product.

Strike	Ratio	Ratio × put premium	Coupon	Investor loses if
100	1.000	$1.000 \times 10.33 = 10.33$	13.7%	the stock ends below 100
90	1.111	$1.111 \times 5.95 = 6.61$	9.9%	the stock ends below 90
80	1.250	$1.250 \times 2.92 = 3.65$	6.8%	the stock ends below 80

There is no free lunch anywhere in that table, and pointing at it is the fastest way to answer any question about making a coupon bigger.



The picture makes the trade explicit. Below the strike the two lines are parallel and the note sits above the stock by exactly the coupon: the investor takes the full fall, cushioned by the premium they were paid. Above the strike the note is flat and the stock keeps rising, so the investor gives up all the upside. The lines cross where the stock has risen by the coupon, here at 113.7. **Above that level the investor would have been better off simply owning the shares**, and that is the sentence a client deserves to hear before they buy one.

REAL INTERVIEW QUESTION (Société Générale · Structured Products)

What is a reverse convertible and what are its components?

WHAT THEY TEST / ANSWER

Define it by the cash flows, then decompose. **Cash flows**: a short-dated note paying a high fixed coupon, repaid at par if the underlying finishes at or above the strike and repaid in shares delivered at the strike if it finishes below. **Components**: a zero-coupon bond of the issuer, which delivers the redemption, plus a fixed coupon, minus a put that the investor has sold, which is what funds the coupon. Write the relation $100 = (100 + C)\,\mathrm{DF} - P$ so the coupon is visibly the put premium grossed up for funding. Then add the two points that mark out a candidate. First, name the risk in the client's language rather than the desk's: the investor is not lending at a high rate, they are being paid to promise to buy the stock at the strike, and they will be, precisely when it has fallen. Second, note that they are also short volatility and short skew, since the put is struck below spot in the barrier version, so the coupon on offer improves when the market gets more frightened, which is when clients most want to buy it and when the risk is highest. See structured notes for the wrapper and embedded optionality for the funding leg in general.

REAL INTERVIEW QUESTION (Leonteq · Structured Products)

Why is a short put used in a reverse convertible?

WHAT THEY TEST / ANSWER

Because it is the only leg that generates cash, and the product's entire purpose is to convert optionality into a coupon the client can be shown. Explain the mechanism rather than restating it: the bond leg alone would pay the issuer's own funding rate, which is unremarkable, so to print an attractive number the structurer has to sell something belonging to the client, and the put is what the client is willing to sell, since a yield-seeking investor is usually someone who would not mind owning the stock lower. That last clause is the honest commercial case for the product and worth saying. Then give the two consequences. The investor is **short volatility**, so the coupon is high exactly when the market is nervous, and **short the tail**, so the loss is unbounded on the downside in the sense that it is a full equity loss below the strike, with the coupon as the only cushion. Why a put and not a call: a short call would cap the upside and raise very little premium on a normal equity skew, whereas the downside is where the premium is, which is a skew statement and ties back to the volatility smile and skew.

Adding a barrier

The commonest variant is the **barrier reverse convertible**, in which the sold put is not a vanilla but a down-and-in: it only comes into existence if the underlying trades through a barrier, typically 60 to 70% of the initial level. If the barrier is never touched, the investor is repaid par whatever the final level, and the note behaves like a straight bond with a high coupon.

That is genuinely better for the investor, and it costs them coupon. On the same one-year, 30% volatility example, replacing the vanilla put with a down-and-in put barriered at 70 and monitored daily reduces the option premium to roughly 6.9, which takes the coupon from 13.7% to about 10.1%. The figure is from a two million path simulation on 252 daily observations rather than a closed form, and the monitoring convention matters, since a continuously-monitored barrier is touched more often than a daily-closing one and is therefore worth more to the buyer of the put.

Two features of the barrier version deserve emphasis because they are where clients are surprised.

The loss is measured from the strike, not the barrier. Breach a 70% barrier and the investor does not lose 30%, they own a put struck at 100, so they take the whole fall from the initial level. The barrier is a switch, not a floor.

The risk is discontinuous. Just above the barrier the note is worth close to par; just below it, much less. That is a gap in value at a single level, and it is why these products are difficult to hedge, as hedging structured products develops, and why the barrier mechanics belong to exotic options.

REAL INTERVIEW QUESTION (Leonteq · Structured Products)

I have 100 euros and I buy a reverse convertible: I am prepared to lose my capital but not more if the stock goes below 90. How do you price it?

WHAT THEY TEST / ANSWER

Take the constraint apart first, because it defines the structure. “Below 90” sets the strike at 90, so the investor is short a 90-strike put and is protected on the first 10% of the fall. “Not more than my capital” rules out *leverage*: a two-times put would lose more than 100 and is excluded. It does not push you away from the standard structure, because at the market conversion ratio of $100/90 = 1.111$ shares the worst case is exactly 100, which is precisely the limit the client named. Say that explicitly, since a candidate who reaches for a one-times put to cap the loss at 90 has solved a constraint the client did not set. **Pricing**: $100 = (100 + C)\,\mathrm{DF} - 1.111\,P(90)$, so $C = (100 + 1.111\,P(90))/\mathrm{DF} - 100$. On a one-year, 30% volatility name at a 3% rate the 90 put is worth 5.95, so the note sells 6.61 of premium and the coupon comes out at about 9.9%, against 13.7% if the strike had been at the money. Then give the two qualifications a desk would want. The put should be priced at the implied volatility of the *90 strike*, not the at-the-money level, and on an equity skew that is materially higher, which raises the coupon. And the quoted coupon must come out net of the issuer’s margin and the hedging cost, so what the client sees is lower than the arithmetic above, as set out in leverage control through structure.

Who it is for, and when it goes wrong

REAL INTERVIEW QUESTION (Société Générale · Structured Products)

Why do a reverse convertible? For what type of investor?

WHAT THEY TEST / ANSWER

For an investor with a specific and quite narrow view: mildly bullish to neutral, expecting the underlying to go sideways or up a little, who wants income rather than growth, and who would be comfortable owning the stock at the strike if it fell. That last condition is the real suitability test, and it is the one to lead with, since the product is a way of being paid to place a limit order you were happy to place anyway. Then name who it is *not* for. It is not for someone who wants equity upside, because the payoff is capped at the coupon and they would have done better owning the shares above the break-even. It is not for someone who cannot tolerate a large capital loss, because below the strike this is a full equity position. And it is not for someone who needs liquidity, since the secondary market is the issuer's bid. Close on why it sells so well, because that is the commercial insight: the coupon is at its most attractive precisely when volatility is high, which is when the market is most frightened and the risk being sold is largest, so the product is easiest to place at exactly the wrong moment. A good sales answer names that tension rather than hiding it.

REAL INTERVIEW QUESTION (Barclays · Structured Products)

What is the difference between a reverse convertible and an autocall?

WHAT THEY TEST / ANSWER

Both pay a coupon funded by a sold downside put, so start with what they share and then separate them on the one axis that matters: the *early redemption feature*. A reverse convertible has a fixed maturity and a known coupon; it runs to the end whatever happens. An autocallable adds observation dates at which, if the underlying is above a trigger, the product redeems early at par plus accrued coupons and the trade is over. Three consequences follow, and giving them is the answer. **For the investor**, the autocall's coupon is usually higher because they have also sold the timing, so they carry reinvestment risk and their best case is a short life rather than a large payout. **For the payoff**, the autocall is path dependent, since its outcome depends on a schedule of observations rather than on the final level, which is why it needs Monte Carlo and a reverse convertible does not; see path dependency and scenario analysis. **For the dealer**, the autocall's Greeks jump around the trigger and its duration is uncertain, so it is far harder to hedge. Finish with the framing: a reverse convertible is one short put, an autocallable is a short put plus a strip of digitals and a knock-out, which is why it belongs in its own section.

INTERVIEW TRAP

Four errors. (1) Calling the coupon a yield. It is a sold option premium, so comparing it with a deposit rate is meaningless without stating the risk that funds it. (2) Reading the barrier as a floor: below it the loss runs from the *strike*, so breaching a 70% barrier is not a 30% loss. (3) Forgetting the upside is capped, so above the break-even, spot plus coupon, the client would have been better off owning the shares. (4) Pricing the sold put at the at-the-money volatility when it is struck below spot, which understates the coupon on an equity skew and is the sort of error that makes a quote uncompetitive.

VISUAL / CODE TO BUILD: Interactive

A coupon builder. The reader sets the underlying's volatility, the maturity, the strike and, optionally, a barrier level, and the widget solves for the coupon that makes the note price at par, showing the sold put's premium beside it so the causal chain is visible. The payoff chart below updates live against a buy-and-hold line, with the break-even marked, so raising the coupon visibly moves the strike up and the break-even with it. A scenario strip reports the investor's outcome for falls of 10, 20, 30 and 50%, in cash rather than in percentages, alongside what holding the shares would have returned in the same scenario. The intended realisation is the one the section is built on: every extra point of coupon is visible as an extra point of loss somewhere in that strip, and the two move together exactly.

Where this goes next

Add an early-redemption trigger and a coupon barrier and the same building blocks become autocallables and Phoenix notes. Turn the sold put into a bought one and the product inverts into capital-protected products. The general catalogue of funding legs is embedded optionality, what the desk does with the resulting risk is hedging structured products, and the same product sold to two different audiences is retail versus institutional structuring.

19.5 Capital-protected products

structuring · sales · trading | core | equity · cross-asset

This is the oldest structured product and still the easiest to explain: get your money back at maturity, plus a share of whatever the market did. It is also the one whose economics are most often misunderstood, by clients and by candidates, because the protection is not free, not certain, and not produced by anything clever. It is produced by lending the bank most of the money.

The budget arithmetic behind these products, the split between a zero-coupon bond and an option and the levers that buy participation, is developed in leverage control through structure and is not repeated here. This section covers the product: what the protection actually is, the two different ways of manufacturing it, the shapes the upside leg can take, and what the client is really exposed to.

Protection is a promise, not a property

The mechanism is one sentence: the issuer takes the client's 100, keeps enough to be able to repay 100 at maturity, and spends the rest on an option. The part kept back is a zero-coupon bond issued by the bank itself, so the "protection" is exactly as good as the bank's ability to pay, and no better.

KEY IDEA

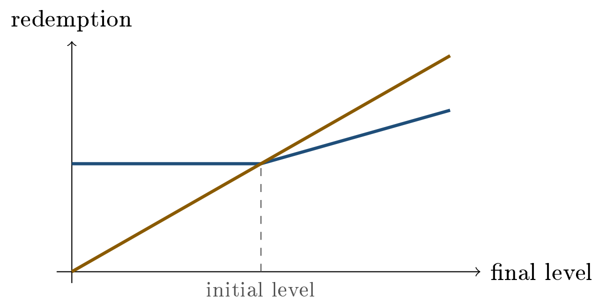
Three consequences follow, and they are the three things an interviewer is listening for. **The guarantee is issuer credit**: if the bank defaults, the client is an unsecured creditor, which is what Lehman note holders discovered in 2008, and it is why the zero-coupon leg is discounted on the bank's own funding curve rather than at the risk-free rate, which adds to the option budget. **The protection applies at maturity only**: sell in the secondary market before then and there is no floor, because the bond leg is marked at the current level of rates and the bank's own credit spread. **The protection is nominal, not real**: getting 100 back after five years of inflation is a loss in purchasing power, and it is also five years of forgone dividends, since the client owns an option on the price, not the shares.

REAL INTERVIEW QUESTION (Natixis · Structured Products)

Give an example of a capital-guaranteed structured product and draw its diagram.

WHAT THEY TEST / ANSWER

Pick the simplest one and draw it cleanly, because the drawing is the test. A five-year note, 100% capital guaranteed at maturity, paying a participation of about 50% in the rise of an index. On the diagram: a flat line at 100 for every final level below the initial one, then a straight line of slope 0.5 above it, with the axes labelled final index level and redemption, and the initial level marked. Then say the three things the picture cannot show. The participation is not a free 50%: it is whatever is left after buying the zero-coupon bond, so it moves with interest rates, not with the issuer's generosity. The flat part is a promise from the issuer, so the product carries bank credit risk. And the comparison line worth drawing next to it is a direct investment in the index, which is above the note everywhere above the initial level and, because of dividends, further above it than the diagram suggests. A candidate who draws the second line unprompted is showing they understand what the client gave up.



capital-protected note, 49.5% participation — direct investment in the index

the note wins only below the initial level, and the direct investment also collects dividends, which are not drawn

Everything depends on rates, nothing depends on the option

Because the budget is what the zero-coupon leg leaves behind, the participation a note can offer is set by the rate environment and by the cost of the option, and a structurer spends most of their time on the second when the first has moved.

WORKED EXAMPLE

Five years, continuously compounded rate 3%, ignoring fees. The zero-coupon bond costs $e^{-0.15} = 86.07$, so the option budget is 13.93.

At 25% volatility and these rates a five-year at-the-money call is worth 28.16 of notional, so the participation is $13.93/28.16 = 49.5\%$. Half the upside, for a guarantee of all the capital.

Now suppose the call cost rises to 50% of notional, because volatility rose or the maturity was shortened. The participation on the same term sheet collapses to $13.93/50 = 27.9\%$.

REAL INTERVIEW QUESTION (Société Générale · Structured Products)

If the cost of the call rises to 50%, how do you adjust the allocation of a capital-guaranteed product so that the client still benefits from the upside?

WHAT THEY TEST / ANSWER

Do the arithmetic first so the answer is a number, not a list. On the budget of 13.93 computed above, a call at 50% of notional buys only 27.9% participation where the 28.16 call bought 49.5%.

To restore the old participation you have to enlarge the budget, and the only way to enlarge it is to shrink the bond. Keeping 49.5% participation against a call costing 50 needs a budget of $0.495 \times 50 = 24.75$, so only 75.25 goes into the zero-coupon, which grows to $75.25 \times e^{0.15} = 87.4$. In other words the guarantee has to fall from 100% to about 87%: the client's capital protection is what pays for the upside, and there is no allocation that preserves both.

Then give the alternative, which is to keep the guarantee and buy a cheaper option rather than a smaller bond, and name the trade-off in each case: a call spread, which caps the payoff, an Asian, which averages away the final level, a digital, which pays a fixed amount rather than a proportional one, or a knock-out. The judgement they are grading is whether you present the choice honestly as protection versus participation versus payoff shape, since the client can have any two of the three but never all three. The full set of levers is in leverage control through structure.

Shaping the upside leg

The bond leg is fixed by the guarantee. Everything creative happens in the option leg, and the standard shapes are worth knowing by name because interviewers ask for them by name.

REAL INTERVIEW QUESTION (Société Générale · Structured Products)

How would you structure a 100% capital-guaranteed five-year product in euros that benefits from a rise in the stock, for example paying +10% if the stock rises by more than 10%, and otherwise the actual performance if it rises by less than 10%?

WHAT THEY TEST / ANSWER

Read the payoff carefully before naming an instrument, because the second clause is the whole question. The client receives the real performance up to 10% and is capped at 10% beyond it, which is min(performance, 10%) on the upside. That is a **capped participation**, so the upside leg is a **100/110 call spread**, *not* a digital. The distinction is the trap: a digital pays a fixed 10% or nothing, so a stock up 5% would pay zero, and this question explicitly says it pays 5%.

So the structure is a zero-coupon bond plus a 100/110 call spread. Price it, because the number is the interesting part. At 25% volatility, a 3% continuous rate and five years, the 100 call is worth 28.16 and the 110 call 24.19, so one call spread costs 3.97 against an option budget of 13.93.

The right answer therefore challenges the term sheet: the budget buys $13.93/3.97 = 3.5$ call spreads, so the same guarantee could pay a cap of about **35%** rather than 10%. A product advertised as "+10% if the stock is up 10%" is either loaded with fees or deliberately conservative, and you should say so.

Then use the digital as the contrast, since it is the distinction being tested. Had the payoff been binary, 10% or nothing, it would be a digital struck at 110, worth about 0.368 per unit of payout, so a 10% payout would cost only 3.7%. Cheaper, but a different product: a digital is a cliff, so 109.9% pays nothing and 110.1% pays everything, which has to be disclosed and which the desk hedges with a call spread around the strike, covered in exotic options. If instead the client wants uncapped upside, the same budget buys about 49.5% participation in a plain call.

REAL INTERVIEW QUESTION (Goldman Sachs · Structured Products)

How is a shark note structured, and can you pitch an interesting shark note or a capital-guaranteed product in the current macro environment?

WHAT THEY TEST / ANSWER

A **shark note**, also called a knock-out note, is a capital-protected note whose upside leg is an **up-and-out call with a rebate**: the client participates in the rise up to a barrier, and if the underlying ever trades through that barrier the call dies and the client receives a fixed rebate instead, on top of the protected capital. Structurally it is a zero-coupon bond, plus an up-and-out call, plus a digital paying the rebate on the knock-out event.

Why it exists: the knock-out makes the call much cheaper than a vanilla, so the same budget buys a far higher participation, often well above 100%. The client is therefore paid more for the moves they think are likely and gives up the large rally entirely, which is a coherent trade for someone expecting a moderate rise, and a bad one for someone expecting a melt-up.

On the pitch, do not name a level you cannot justify; give the logic. Capital-protected products live and die with the rate environment, because the guarantee's cost is the zero-coupon price, so the honest pitch begins with where rates are: when they are high the bond leg is cheap, the budget is large and these products are attractive, and when they are near zero the budget vanishes and the product should not be sold at all. Then match the shape to a view: a shark for a moderate-rally view, a plain participation note for an uncertain but positive view, a digital for a threshold view. Interviewers ask the second half of this question to see whether you can connect a structure to a macro state rather than recite a term sheet.

The other way to protect capital: CPPI

Everything so far is **OBPI**, option-based portfolio insurance: buy the protection once, as an option, and hold it. The alternative manufactures the same promise dynamically, with no option at all.

REAL INTERVIEW QUESTION (Goldman Sachs · EQD)

What is CPPI, and how is it applied to structured products? Illustrate how it dynamically allocates between a risky and a riskless asset.

WHAT THEY TEST / ANSWER

Constant Proportion Portfolio Insurance replaces the option with a trading rule. Define the **floor** as the present value of the amount to be guaranteed, the **cushion** as the portfolio value minus the floor, and allocate

$$\text{risky exposure} \;=\; m \times \text{cushion},$$

with m the **multiplier**, typically 3 to 5. Everything else sits in the riskless asset. Rebalance as the market moves.

Work the numbers. Guaranteeing 100 in five years at 3% means a floor of 86.07, so a portfolio of 100 has a cushion of 13.93; with $m = 4$ the risky allocation is 55.7 and the rest is in the bond. If the risky asset now falls 10%, the portfolio loses 5.57 to 94.43, the cushion shrinks to 8.36, and the rule cuts the risky allocation to 33.4. The mechanism sells into falls and buys into rallies, which is the defining property: CPPI is a *momentum* strategy, and its return profile is convex like an option's, which is the point, but it is manufactured by trading rather than bought.

Then give the two failure modes, because they are why the question is asked. **Gap risk**: the rule only works if the desk can rebalance before the cushion is gone, and an instantaneous fall of more than $1/m$, so 25% when $m = 4$, wipes the cushion out and breaks the floor. The issuer keeps that risk, and it is the reason CPPI guarantees carry a fee that looks like an option premium. **Cash-lock**: once the cushion is near zero the rule allocates almost nothing to the risky asset, and since the cushion can then only grow at the riskless rate, the product is dead for the rest of its life and the client holds an expensive bond. Close with the comparison, which is the real answer: OBPI costs a known premium up front and cannot be knocked out of the market, while CPPI has no option to buy, so it works when options are expensive or when the underlying is a fund or a hedge-fund index with no listed options, at the price of path dependence, gap risk and cash-lock.

REAL INTERVIEW QUESTION (Union Capital · Structured Products)

Draw me the payoff of a knock-in structured product with capital protection.

WHAT THEY TEST / ANSWER

Ask which leg the barrier sits on before drawing anything, because "knock-in with capital protection" describes two different products and the interviewer is watching for the question. If the **upside** leg is a down-and-in call, the picture is a flat line at 100 everywhere, plus participation above the initial level only on the paths where the underlying first traded below the knock-in level: the drawing at maturity is then ambiguous on its own, and you have to say that the payoff depends on the path, not only on the final point. If instead the barrier sits on a **downside** leg, the capital protection is conditional rather than absolute, and the honest drawing is the one that shows the flat line breaking at the barrier and the redemption falling to the level of the underlying below it, which is not a capital-protected product at all but a barrier note.

The point to make out loud is the second one: many products marketed as protected are protected *conditionally*, and the whole difference between them and a genuine guarantee is a knock-in level. A path-dependent leg also cannot be drawn as a single line at maturity, which is why term sheets for these show scenario tables instead of diagrams, and why you should ask for the observation convention, continuous or on fixing dates, before quoting anything.

INTERVIEW TRAP

Four errors. (1) Calling the product risk-free: it is an unsecured claim on the issuer, and that credit spread is part of what funds the option. (2) Believing the protection applies before maturity: the secondary price of a protected note falls when rates rise or the issuer's spread widens, and can trade well below par. (3) Quoting a participation without quoting the rate environment behind it, since the same structure offers a completely different participation at 1% and at 5%. (4) Treating CPPI as equivalent to an option-based guarantee: it carries gap risk and cash-lock risk that an option does not, and the issuer's fee is compensation for exactly those.

VISUAL / CODE TO BUILD: Interactive

A guarantee workshop with two tabs. The **OBPI** tab lets the reader set the maturity, rate, protection level, volatility and fees, and shows the zero-coupon cost, the option budget and the resulting participation live, with a shape selector (call, call spread, digital, up-and-out with rebate) that redraws the maturity payoff against a direct index investment, dividends included, so what the client gives up is visible. A rate slider is the headline control: hold everything else and move rates from 1% to 5% to watch participation multiply. The **CPPI** tab runs the trading rule on a path the reader draws or simulates, plotting the portfolio, the rising floor and the risky allocation over time, with the multiplier as a slider. Two buttons make the failure modes concrete: one injects an instantaneous fall larger than $1/m$ so the floor breaks, and one runs a long drawdown followed by a rally so the product cash-locks and misses the recovery. A side-by-side comparison of the two tabs on the same path is the intended conclusion.

Where this goes next

The upside leg, once the guarantee is relaxed, becomes a participation product, and the same budget spent on selling risk instead of buying it produces the reverse convertible and the autocallable. The wrapper, the issuance process and the credit that makes the guarantee possible are in structured notes, the barrier and digital legs in exotic options, and the ways this product is mis-sold in common structuring mistakes.

19.6 Participation products

sales · structuring · trading | core | equity

Reverse convertibles are what you sell to a client who wants income and will accept equity risk to get it. Participation products are the other half of the chapter: the client wants the equity, and the structure changes the *shape* of that exposure rather than replacing it with a coupon. They are the products people mean when they say a structured product can do something a share cannot.

The organising question is the same one as everywhere else in this chapter, asked in the opposite direction: the client is buying optionality here rather than selling it, so what are they paying with?

> **KEY IDEA**
>
> A participation product is equity exposure with a modified payoff, and the modification is always funded. Three things are available to pay with, and every product in this section uses one or more: the **dividends**, given up by holding a zero-strike call instead of the shares; the **far upside**, given up by capping; and a **barrier**, given up by agreeing that the improvement disappears if the market falls far enough. If you can name which of the three funds a structure, you can price it and you can explain it.

The dividend point deserves emphasis because it is invisible on a term sheet. A **zero-strike call** is a call struck at zero, so it behaves like the share in every respect except one: its holder does not receive dividends, and the option is cheaper by roughly their present value, assuming a deterministic dividend stream and no borrow cost. That discount is the budget for everything else, which is why participation structures are common on high-dividend indices and rare on names that pay nothing. See dividends.

The family

- **Tracker certificate**: the index in a note, 100% participation, no protection and no improvement. It exists for access and tax reasons, not for its payoff.
- **Capital-protected participation note**: a zero-coupon bond plus a call, so the client keeps a floor and participates at less than 100%. The protection dimension belongs to capital-protected products and the participation arithmetic to leverage control through structure.
- **Bonus certificate**: the client is guaranteed at least a bonus level at maturity, provided a barrier is never touched, and participates fully above it.
- **Twin-win**: the client earns the performance whether the market rises or falls, provided the barrier holds.
- **Airbag**: the loss below a barrier is measured from the *barrier* rather than from the initial level, so the first part of a fall is genuinely absorbed.

REAL INTERVIEW QUESTION (Barclays · EQD)

What are the participation structured products?

WHAT THEY TEST / ANSWER

List them by mechanism rather than by name, because the names are regional and the mechanisms are not. Start with the definition that separates the family: the client is *long* optionality and keeps equity upside, as opposed to a yield product where the client is short optionality and receives a coupon. Then: the tracker certificate, which is simply the index in a note; the capital-protected participation note, a zero-coupon bond plus a call, where the client trades the bond coupon for a share of the upside; the bonus certificate, which adds a floor at a bonus level conditional on a barrier holding; the twin-win, which pays the absolute performance so a fall becomes a gain while the barrier holds; and the airbag, which absorbs the first part of a fall by measuring the loss from the barrier rather than from the start. Close with the funding, because that is the structuring content: none of these is free, and each is paid for with the dividends via a zero-strike call, with the upside via a cap, or with a barrier that removes the improvement in a bad enough market. A candidate who names the funding leg for each product has answered the question; one who lists five names has not.

Bonus certificate

The payoff is the clearest statement of "a floor you keep as long as nothing bad happens". If the barrier is never touched, the investor receives at least the bonus level, and the full underlying if it finishes above. If the barrier is touched, the bonus disappears and the investor is left with the underlying.

The decomposition is worth doing slowly, because it is the standard exam question:

$$\max(S_T, \text{bonus}) \;=\; S_T \;+\; \max(\text{bonus} - S_T,\, 0),$$

so, adding the barrier condition, a bonus certificate is a **zero-strike call plus a down-and-out put struck at the bonus level**. The put dies if the barrier is touched, which is exactly the product's promise. The dividends given up by the zero-strike call pay for that put, and the size of the bonus is whatever the dividend stream can buy.

REAL INTERVIEW QUESTION (Goldman Sachs · Structured Products)

What are the most traded products in Switzerland? Develop the bonus certificate.

WHAT THEY TEST / ANSWER

Two parts, and the first is a market-awareness question, so answer it as one: the Swiss market is one of the largest and most retail-oriented structured-products markets, dominated by yield-enhancement products, above all barrier reverse convertibles, alongside participation products such as tracker and bonus certificates, and it is characterised by exchange listing and short maturities rather than the long-dated autocallables that dominate France and Italy. Then develop the bonus certificate properly. **Payoff**: if the barrier is never touched, the investor receives at least the bonus level and participates fully above it; if it is touched, the bonus is gone and they hold the underlying. **Decomposition**: a zero-strike call plus a down-and-out put struck at the bonus level, funded by the dividends the zero-strike call gives up. **Risk**: the barrier is a switch, so the product is worth noticeably less just below the barrier than just above it, and the investor who is watching a market fall toward it has a discontinuity approaching. Finish with what determines the terms, since a structuring interviewer will push there. The bonus is whatever strike the dividend budget can buy, so anything that makes the down-and-out put *cheaper* buys a *higher* bonus. A higher dividend yield enlarges the budget. A *higher* barrier, closer to spot, kills the put more often and so cheapens it, which is why a safer barrier always comes with a smaller bonus. Higher volatility does the same thing, for the same reason.

The trap is that all three cheapening effects work by making the bonus *less likely to be paid*. On a three-year note funded by a 3% dividend yield with the barrier at 70, the affordable bonus rises from about 108 at 15% volatility to 128 at 25% and 154 at 35%, while the probability of knocking out rises from roughly 30% to 55% to 69%. Those figures come from a simulation monitoring the barrier at daily closes; a continuously monitored barrier knocks out more often, so the down-and-out put is cheaper again and the affordable bonus comes out two to three points higher at the top end. The direction is the same either way, which is the only thing the argument rests on. So the frightened market shows the client the best headline bonus and is the least likely to deliver it, which is the sentence worth having ready.

Twin-win

The twin-win pays the absolute performance: up 15% and you make 15%, down 15% and you still make 15%, provided a barrier is never touched. If it is touched, the downside conversion disappears and the investor is simply long the underlying.

The construction follows from writing the payoff down. Being long the underlying gives S_T. To turn a fall of $S_0 - S_T$ into a gain of the same size, you need to add $2(S_0 - S_T)$ on the downside, so:

$$\text{twin-win} \;=\; \text{zero-strike call} \;+\; 2 \times \text{down-and-out put struck at } S_0.$$

With a factor of one instead of two the downside is merely flattened, which is a bonus certificate struck at the initial level; with two it is mirrored. This is why the product is sometimes described as a barrier straddle: the two puts, plus the implicit call inside the long underlying position, give a V-shaped payoff around the initial level, with the barrier switching the left arm off.

REAL INTERVIEW QUESTION (Marigny Capital · Structured Products)

What is a Twin-Win and how is it structured?

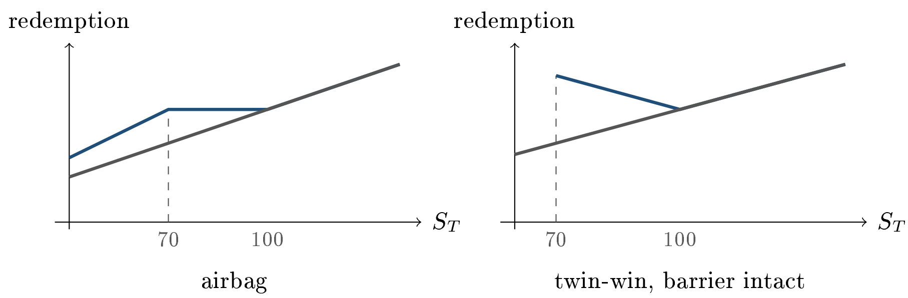
WHAT THEY TEST / ANSWER

What it is: a note that pays the performance of the underlying in absolute value, so the investor gains whether the market rises or falls, conditional on a downside barrier never being touched. If the barrier is touched, the mirror disappears and the investor is left long the underlying, taking the fall like anyone else. **How it is structured**: a zero-strike call, which is the equity exposure funded by giving up the dividends, plus two down-and-out puts struck at the initial level, whose barrier is the product's barrier. Derive the factor of two out loud rather than quoting it: the long underlying already loses $S_0 - S_T$ on a fall, so one put only cancels that loss and gives a flat payoff, and a second put is needed to turn it into a gain of the same size. Then give the honest risk description, which is the point of the question: the client is being shown a product that wins in both directions, and what they have actually bought is a barrier position, so the outcome that matters is not up or down but touched or not touched. A fall that bottoms at 71% of the initial level, one point above the barrier, pays a 29% gain; a fall to 69% breaches it and turns the same view into a 31% loss. Two points of underlying, sixty points of outcome. Path, not direction, decides it, which is path dependency in its purest commercial form.

Airbag

The airbag is the one participation structure whose selling point is genuinely about the *size* of a loss rather than about whether one happens. Below the barrier, the investor's redemption is $100 \times S_T/B$ rather than S_T, so the loss is measured from the barrier and geared by $100/B$.

That gearing is the whole mechanism. With a barrier at 70 the factor is $100/70 = 1.43$, so the investor is short 1.43 puts struck at 70 rather than one put struck at 100. The consequences are worth stating in both directions: a fall to 50 costs the airbag holder $100 - 100 \times 50/70 = 28.6$ against the 50 a plain barrier note would cost them, so the protection is real; but the position is levered, so it converges to a total loss at zero just as fast as the plain structure does, and there is no free protection anywhere in it.



— the structure — holding the underlying (both drawn at 100% participation for shape; a par-priced note buys less)

REAL INTERVIEW QUESTION (Société Générale · Structured Products)

How do you structure an airbag?

WHAT THEY TEST / ANSWER

Build it from three legs and state the gearing explicitly, because that is where the risk is. **Leg one**, a zero-coupon bond of the issuer maturing at 100, which delivers the redemption. **Leg two**, a call struck at the initial level, which delivers the participation in any rise. **Leg three**, short $100/B$ puts struck at the barrier B, which is the funding leg and the source of the airbag effect: below B the investor's redemption is $100 \times S_T/B$, so the loss starts at the barrier rather than at the initial level. With $B = 70$ that is 1.43 puts rather than one. Then solve for the participation that makes the note price at par, since that is what a structurer actually does: on a five-year note at a 3% rate and 25% volatility, the zero-coupon costs 86.07, the levered put package raises 6.01, and the at-the-money call costs 28.16, so the participation is $(100 - 86.07 + 6.01)/28.16 \approx 71\%$. Close with the comparison the client needs: against a plain barrier product, the airbag genuinely reduces the loss for moderate falls, a drop to 50 costing about 29 rather than 50, and it does nothing at all for a catastrophic one, because the gearing takes it to zero at the same place.

REAL INTERVIEW QUESTION (Société Générale · Structured Products)

Are you vega positive or vega negative on an airbag-type structured product, and why?

WHAT THEY TEST / ANSWER

Resist the one-word answer, because the sign genuinely depends on the calibration and saying so with numbers is the strong response. The investor holds two option legs pulling in opposite directions: *long* the participation call, which is long vega, and *short* the levered downside put, which is short vega. On the par-priced five-year example above, 25% volatility, a 70 barrier and 71% participation, the call leg contributes about $+0.54$ of vega per volatility point and the levered put leg about -0.63, so the note is roughly **vega neutral and slightly short**, around -0.09 per point. Move the barrier and the sign moves with it, but it stays modestly negative across the usual range. Then give the answer that is robust to the calibration, which is what they are really after: the investor is short volatility *at the barrier* and long volatility *at the money*, so the position is much more a **skew** exposure than a vega exposure. Steepen the skew, leaving the at-the-money level unchanged, and the note loses, because the leg they are short richens against the leg they are long. That is a vanna statement, developed in skew and smile sensitivities, and it is why an airbag book is managed on the surface rather than on a single volatility number.

INTERVIEW TRAP

Four errors. (1) Describing any of these as free improvements: each is funded by the dividends, a cap, or a barrier, and naming the funding leg is the answer to most questions in this section. (2) Treating a barrier as a floor. In a bonus certificate or a twin-win the barrier switches the benefit *off*, leaving the client long the underlying with nothing to show for it. (3) Reading a twin-win as a directional product: it is a path product, so touched or not touched decides the outcome, not up or down. (4) Assuming an airbag is short vega because it contains a sold put: the participation call is long vega, the net is close to flat, and the durable exposure is to the skew rather than to the level.

VISUAL / CODE TO BUILD: Interactive

A participation designer with a funding meter. The reader picks a family, sets the maturity, the dividend yield, the volatility and the barrier, and the widget solves for whatever the term sheet's headline number would be, the participation, the bonus level or the twin-win factor, showing beside it a bar of where the budget came from: dividends given up, upside capped, barrier sold. Changing the dividend yield visibly moves the bonus level, which is the single fact hardest to convey in words. A payoff panel draws the structure against a buy-and-hold line with the barrier marked, and a toggle switches between the barrier-intact and barrier-touched payoffs so the discontinuity is a thing the reader flips rather than a caveat they read. A scenario strip reports the client's outcome and the buy-and-hold outcome side by side for falls of 10, 20, 30 and 50%, which is where the airbag's advantage and its limit both become obvious.

Where this goes next

The protection dimension of these structures is capital-protected products, and the mirror family in which the client sells rather than buys optionality is reverse convertibles and autocallables. The barrier mechanics that all three of the interesting products here depend on are in exotic options, the general catalogue of funding legs is embedded optionality, and how a payoff is designed from a view in the first place is payoff engineering.

19.7 Payoff engineering

structuring · trading · sales core equity · cross-asset

A client describes what they want in words: money back, twice the rise, nothing above a cap. A structurer's first job is to turn that sentence into a portfolio of instruments that can actually be bought, and the second is to check whether the client can afford it. Both jobs are mechanical. This section is the mechanism.

The good news for anyone learning this is that payoff engineering is not creativity. Almost every payoff a client can describe at maturity is a piecewise-linear function of the final level, possibly with a few jumps, and there is a rule that reads such a function off a drawing and returns the portfolio. Learn the rule and most structuring questions become arithmetic.

The five building blocks

Everything in this chapter is built from five things, and each one has a distinctive signature in the shape of the payoff.

- **Cash**, a zero-coupon bond: a horizontal line. It sets the level.
- **The underlying or a forward**: a straight line of slope 1 through the forward. It sets the tilt.
- **A call** struck at K: adds slope above K and nothing below.
- **A put** struck at K: adds downward slope below K and nothing above.
- **A digital** struck at K: adds a vertical jump at K and no slope.

KEY IDEA

Read a payoff diagram left to right and write down the portfolio as you go.

1. The value of the payoff at zero, or the level of its leftmost segment when that segment is flat, is the **cash** leg. Read it at zero rather than at the lowest flat segment, since a payoff that slopes down on the left has its lowest flat segment somewhere in the middle.
2. The slope of the leftmost segment is the number of **forwards** (or a put position, if the payoff is flat on the left and you prefer to build it with puts).
3. At every **kink** at a level K, the change in slope is the quantity of options struck at K: a slope that increases by q is $+q$ calls at K, a slope that decreases by q is $-q$ calls at K.
4. Every **jump** of size J at a level K is J digitals struck at K.

Kinks become options, slope changes become quantities, jumps become digitals. That is the entire method.

Formally, for any payoff f that is twice differentiable, and any reference level κ,

$$f(S_T) \;=\; f(\kappa) \;+\; f'(\kappa)\,(S_T-\kappa) \;+\; \int_0^{\kappa} f''(K)\,(K-S_T)^+\,dK \;+\; \int_{\kappa}^{\infty} f''(K)\,(S_T-K)^+\,dK.$$

The identity says that any payoff is cash, plus a forward, plus a continuum of puts below κ and calls above it, weighted by the second derivative. For the piecewise-linear payoffs that clients actually ask for, f'' is a sum of point masses at the kinks and the integrals collapse into the finite sum of the rule above. This is also why a variance swap can be replicated by a strip of options, developed in variance swaps, and it is the formal backing for replication strategies.

REAL INTERVIEW QUESTION (Unicredit · EQD)

How would you decompose a structured product?

WHAT THEY TEST / ANSWER

Give a procedure, not an example, because the question is about method. Four steps. **One**: draw the redemption at maturity against the final level of the underlying, and mark every kink and every jump. **Two**: read off the portfolio with the rule, lowest flat segment is cash, leftmost slope is the forward or stock leg, each kink is an option at that strike with quantity equal to the change of slope, each jump is a digital. **Three**: say who is long and who is short each leg, and note that in a note the client is long the wrapper and the bank is the counterparty on every option. **Four**: price the legs and compare the total to the budget, which is the funding the note raises, since a decomposition that does not price out is half an answer.

Then name the boundary of the method, which is what separates a good answer: this works for payoffs that depend only on the *final* level. As soon as the product depends on the path, a barrier, an autocall, an Asian, an averaging period, no finite static portfolio of vanillas reproduces it exactly, and you move to approximate static hedges or to a model and a simulation. Saying where a method stops working is worth more than another worked example.

Reading a payoff, then pricing it

REAL INTERVIEW QUESTION (UBS · Structured Products)

I want a product that returns 100% of the capital at maturity and pays twice the performance of the underlying, struck at 100, but with a performance cap at 150. Decompose it.

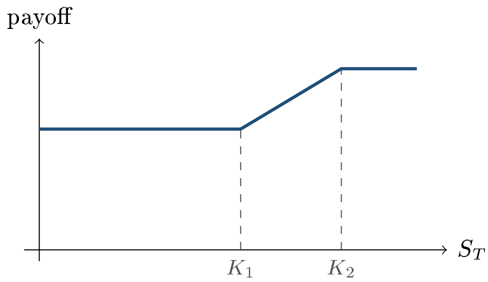
WHAT THEY TEST / ANSWER

Decompose first, then price, then tell them what they cannot have.

Decomposition. The payoff is flat at 100 below the strike, rises with slope 2 between 100 and 150, and is flat again above 150 at $100 + 2 \times 50 = 200$. So: a zero-coupon bond redeeming 100, plus two calls struck at 100, minus two calls struck at 150. Two kinks, two option positions, quantities equal to the slope changes of $+2$ and -2.

Pricing. Take five years, a continuously compounded rate of 3% and 25% volatility. The zero-coupon costs 86.07, so the budget is 13.93. The 100/150 call spread is worth $28.16 - 13.17 = 14.99$, and the client is asking for two of them, which costs 29.98.

The answer they are testing. The product is not affordable: it costs more than twice what the structure can fund. At these parameters the budget buys a participation of about 0.93 on the same capped shape, not 2. So say it plainly and then offer the levers, each with what it takes away: lower the protection to 90%, which lifts the budget from 13.93 to $90e^{-0.15}$ subtracted from 100, that is 22.54, and the participation from 0.93 to about 1.5, still short of the 2 they asked for; bring the cap down from 150, which cheapens the spread; shorten or lengthen the maturity depending on where rates are, choose a less volatile underlying, or accept a lower multiple. A structurer who prices their own decomposition catches this before the client does; one who stops at the decomposition sends out a term sheet that cannot be issued. The budget mechanics are in leverage control through structure.



— $\max(\min(S_T, K_2), K_1)$: flat, then slope one, then flat

two kinks, so two option positions: $+1$ call at K_1 and -1 call at K_2, on top of K_1 in cash

REAL INTERVIEW QUESTION (Barclays · Structuring)

Identify the payoff $\max\big(\min(S_T, K_2), K_1\big)$. *How would you hedge such a product?*

WHAT THEY TEST / ANSWER

Identify it by walking the three regions out loud. Below K_1: $\min(S_T, K_2) = S_T$ and the max floors it at K_1, so the payoff is K_1. Between K_1 and K_2: both operators pass the level through, so the payoff is S_T. Above K_2: the min caps it at K_2, so the payoff is K_2. It is the underlying, floored at K_1 and capped at K_2.

Now the decomposition, which is the answer to the hedging half:

$$\max\big(\min(S_T, K_2), K_1\big) \;=\; K_1 \;+\; (S_T - K_1)^+ \;-\; (S_T - K_2)^+,$$

so it is K_1 in cash plus a K_1/K_2 call spread. Check it at one point in each region rather than asserting it: at S_T below K_1 both options are worthless and the payoff is K_1; at $S_T = K_2$ it is $K_1 + (K_2 - K_1) = K_2$; above K_2 the two calls cancel each other's slope and it stays at K_2.

For the hedge, say that this is a *static* hedge, which is the good news: buy the call spread once, hold it to maturity, and the payoff is matched exactly with no rebalancing and no model risk. The residual risks to mention are the ones that survive a static hedge: the two strikes may not be listed or liquid, so the hedge is approximate in practice; the two legs carry different implied volatilities, so the position is a skew position rather than a volatility position; and the client's version is a note, so the desk still runs the issuer funding leg. Note in passing that the same payoff can be written with puts, $K_2 - (K_2 - S_T)^+ + (K_1 - S_T)^+$, and that the choice between the two is about which strikes trade better, not about the economics.

Reading a portfolio backwards

The rule works in both directions, and interviewers often give you the legs and ask for the picture.

REAL INTERVIEW QUESTION (BNP · Market Risk)

I buy a call at 100, sell two at 110 and buy one back at 120: describe the payoff, and do you pay for this structure?

WHAT THEY TEST / ANSWER

Read the slope changes in order: $+1$ at 100, -2 at 110, $+1$ at 120. Slope zero below 100, slope one from 100 to 110, slope minus one from 110 to 120, slope zero above 120. That is a **butterfly**: a tent peaking at 110 with a maximum payoff of 10, worth nothing outside the 100 to 120 range. Note that the quantities sum to zero, which is exactly why the payoff is flat at both ends.

Yes, you pay for it. The argument to give is a no-arbitrage one rather than a calculation: the payoff is non-negative everywhere and strictly positive on an interval that the underlying can reach, so its value today must be strictly positive, or you would have a free option on a positive payout. Put a number on it to finish: at three months, 20% volatility and ignoring rates, the legs are worth 3.988, 0.954 and 0.147, so the structure costs $3.988 - 2 \times 0.954 + 0.147 = 2.23$ for a maximum payoff of 10. The follow-up is usually what you own in risk terms: you are short volatility and short gamma around 110 and long them near the wings, which is developed in Greeks evolution within strategies.

REAL INTERVIEW QUESTION (Crédit Suisse · Structuring)

How would you structure a product where, if the underlying falls, I lose twice the fall, and if it rises, I gain three times the rise?

WHAT THEY TEST / ANSWER

Translate the sentence into slopes *on the payoff diagram*, and be careful with the sign: losing two for every one the underlying falls means the payoff *rises* at 2 per unit of S_T, so the slope below the current level is $+2$, not -2. Above it the slope is $+3$.

Now apply the rule. Leftmost slope 2 means **two forwards**; the slope change of $+1$ at the money means **one call** struck there. Equivalently, and this is the form to say out loud, **short two puts and long three calls**, all struck at the money, since $2(C-P)+C=3C-2P$ at zero rates. State the convention as you go: ignoring rates and dividends, so the forward equals the spot and "the current level" is unambiguous.

Then price it, because the interesting part is the cost. With zero rates the at-the-money call and put are worth the same by put-call parity, call it V, so the package costs $3V-2V=V$: the client pays exactly one at-the-money option. At one year and 25% volatility that is about 9.9% of notional. Say the two things that follow. This is not a zero-cost structure, and if the client wants it for zero they must accept fewer calls, more puts, or a strike shift, and the cleanest zero-cost version of the same idea is a ratio of the same sign on both wings. And what they have bought is leverage on both sides, so the honest description is not "3 times the upside" but "3 times the upside financed by 2 times the downside", which is the sentence in common structuring mistakes about naming the risk that was sold. One thing not to claim: as built above there is no skew play at all, because every leg sits at the same strike and put-call parity forces the call and the put onto the same implied volatility. The client is simply net long one at-the-money option, so they are long vega and nothing else. Skew enters only once the strikes separate, which is exactly what the zero-cost version does: push the sold puts below the money and an equity skew marks them at a higher volatility, so they raise more premium than a flat surface suggests. That is the one place in this structure where the skew works in the client's favour, and it is worth saying *which* version you mean.

Where static engineering stops

REAL INTERVIEW QUESTION (Barclays · EQD)

How would you structure a Twin Win?

WHAT THEY TEST / ANSWER

Start from what the client is promised: a positive return whether the underlying rises or falls, as long as it never falls too far. So the payoff at maturity is the *absolute* performance, which is a V-shape around the strike, with the downside leg switched off if a barrier below is breached, in which case the client is left with the fall.

Built from the blocks, that is: a zero-coupon bond redeeming the initial level, plus a **call** struck at the initial level for the upside, plus a long **down-and-out put** struck at the initial level with the knock-out at the barrier, which turns a fall into a gain on the paths that never breach, plus a **short down-and-in put** at the same strike and barrier, which is what hands the client the fall on the paths that do. The hint most interviewers are fishing for is that the first three legs are a straddle whose put carries a barrier; the fourth is where the money comes from, since the mirrored downside is expensive and the short knock-in put is what pays for it.

The point to close on is that this is where the static method breaks. The first two legs are static and exact; the barrier leg is not, because a down-and-out put depends on the whole path, not just the final level. So you cannot finish a Twin Win with the kinks-to-options rule: you approximate it, with a reflection-based static hedge when the barrier and strike line up and the skew is mild, or you price and hedge it with a model, which is what desks do. That is also where the risk of the product sits, in the discontinuity at the barrier rather than in the V-shape the client was shown. Barriers are developed in exotic options and path dependence in path-dependent products.

INTERVIEW TRAP

Four traps in decomposition questions. (1) Getting the sign of a slope change wrong: count the change in slope, not the slope itself, and check one point in each region before speaking. (2) Decomposing and stopping: the price against the budget is half the answer, and the half that shows you have done the job. (3) Using the kinks-to-options rule on a path-dependent payoff: barriers, averages and autocalls are not maturity functions of the final level and no finite vanilla portfolio reproduces them. (4) Pricing all the legs off one volatility: a multi-strike decomposition is a skew position, which is the first mistake in common structuring mistakes.

VISUAL / CODE TO BUILD: Interactive

A payoff builder that works in both directions. In **draw mode** the reader clicks to place kinks and jumps on a blank payoff chart and drags the segment slopes; the widget prints the decomposition live, as a list of legs with quantities and strikes, and prices it against a stated rate, volatility and maturity, showing the total next to the option budget of a note of the same maturity, with a red flag when the structure is not affordable. In **build mode** the reader adds legs from a palette (cash, forward, call, put, digital) and watches the payoff assemble, which turns the rule into muscle memory in both directions. A "path-dependent" toggle adds barrier legs and switches the display to a scenario table instead of a single line, making the boundary of static replication visible: the moment a barrier is added, the payoff can no longer be drawn as one curve.

Where this goes next

The blocks themselves are in standard payoff profiles, and the theory that a portfolio of options can reproduce an arbitrary payoff is replication strategies. The finished products this method builds are in capital-protected products, participation products and autocallables, with the funding legs in embedded optionality. Choosing which shape to build in the first place is from market view to payoff design.

19.8 Hedging Structured Products

trading · structuring · sales | advanced | equity · cross-asset

A structured product is designed backwards from what a client wants, which means it is designed without any regard for whether the resulting risk can be traded. Somebody then has to carry it, usually for years. This section is about that job.

Autocallables lists what makes an autocall hard to run; this section owns the apparatus that answers it, and applies to the whole product range. The general Greek mechanics are delta, gamma and vega hedging, the dynamics of a delta hedge are delta neutrality and dynamic hedging, and book-level aggregation is portfolio Greeks. What is specific here is that the position is exotic, one-directional and long dated, and none of those three is true of a textbook option book.

The mirror, and why it cannot be offset

KEY IDEA

The desk holds exactly the opposite of what the client bought, and there is no market in the opposite. Nobody wants to buy a ten-year worst-of down-and-in put on its own, so the desk cannot lay the risk off as a package. It must **replicate** the position out of the instruments that do trade, listed options, futures, dividend swaps and variance, and then manage the difference. Structured hedging is replication, not offsetting, and every difficulty in this section follows from that sentence.

The order of operations on a real desk follows how fast each risk moves and how cheaply it can be traded.

- **Delta**, continuously, in futures on the index. Cheap, deep, and the only exposure that can genuinely be flattened.
- **Gamma and vega**, at the book level over days, by trading listed options. This is where the risk is really taken, because a book the size of a structured desk cannot be vega-flat at every strike and every maturity at once.
- **Dividends and repo**, periodically, in dividend futures and swaps, out to whatever maturity those trade.
- **Correlation and model risk**, not hedged at all. Reserved against, and priced into the product at issue.

REAL INTERVIEW QUESTION (BNP · EQD)

How do you become delta hedged after selling an autocall?

WHAT THEY TEST / ANSWER

Answer in three layers: the mechanic, the sign, and the part that makes it hard.

Mechanic. Compute the product's delta in the pricing model, which for a path-dependent note means a Monte Carlo bump, and trade the index future against it. Rerun and rehedge as spot moves, at least daily and intraday when the market is fast.

Sign. Having sold the note, the desk is long a down-and-in put against the investor and short the coupon stream. Both legs point the same way: the put it owns gains as the market falls, and the coupons it owes are payable when the market is high. So the desk's net delta is *short* the index and it *buys* futures to flatten, which is the same statement as being short the forward and therefore long dividends, below. Say which side you are on rather than quoting a formula, because that is what is being tested.

The hard part. The delta is not stable, and not for the usual gamma reason alone. It jumps at each observation date, because the note either dies at par or survives, and a payoff that jumps is a digital: as an autocall date approaches, the delta near the trigger becomes enormous and then reverses the moment the date passes. So the practical answer is that you delta-hedge an autocall in two modes, ordinary rehedging most of the time, and around observation dates a deliberately smoothed hedge sized off a call spread rather than off the true delta, accepting a known error instead of chasing an unhedgeable one. The technique is digital replication, applied to a note.

Barriers and the discontinuity problem

Every difficulty a structured desk has that an ordinary option desk does not comes from a discontinuity: a barrier, a digital coupon, an autocall trigger. The option itself belongs to barriers and digitals; what belongs here is what a desk does about it.

REAL INTERVIEW QUESTION (BNP · EQD)

How do you hedge barrier options, around the barrier?

WHAT THEY TEST / ANSWER

Start with why the question is hard, because the honest answer is that near the barrier you cannot hedge it properly and the job is to choose which error to take.

At the barrier the payoff is discontinuous, so the delta is theoretically unbounded and the gamma with it. Worse, the delta is *discontinuous* across the barrier on a knock-out: one tick above it you hold a live option with its own delta, one tick below it you hold nothing. On a down-and-out put the delta just above the barrier even carries the *opposite* sign to the vanilla's, because there the option gains value as spot moves away from the barrier. Rehedging that faithfully means trading enormous size on tiny moves, which loses money on bid-offer far faster than the mishedge would.

So the desk does three things.

- **Smooth the barrier.** Price and hedge as if the barrier were a narrow band rather than a level, using a spread of vanillas that straddles it. This is the same overhedge as for a digital, and it is deliberately conservative: you own slightly more than the payoff requires, and the difference is the cost of not being whipsawed.
- **Widen with size and shorten with time.** The band is wide when the position is large or the barrier is far away, and it is tightened as expiry approaches and the uncertainty resolves.
- **Reduce.** The genuinely correct answer when the position is large and the barrier is near expiry is to cut the position, or to buy back part of the structure from the client, rather than to engineer a cleverer hedge.

Then the point that gets the follow-up: the risk is worst *near expiry and near the barrier at the same time*, because that is where both the probability of touching and the size of the jump are meaningful. A barrier far from spot with two years to run is not a hedging problem at all, and saying so shows you know which risks to spend attention on.

REAL INTERVIEW QUESTION (Leonteq · Structured Products)

How can you hedge yourself if you are short a knock-in barrier put?

WHAT THEY TEST / ANSWER

Name the position first: short a down-and-in put is the classic retail side of a barrier note, so this is the client's risk, not the desk's, and the desk is long the same option. Either way the hedging problem is the same one seen from opposite ends.

Static first, dynamic second, because static is better whenever it is available. With a European knock-in observed only at maturity, the option decomposes exactly into vanillas and a digital, as autocallables sets out, so the hedge is a put struck at the barrier plus a digital, and the digital in turn is a tight put spread. That hedge is set once and left, which is the whole advantage.

Dynamic where it must be. With an American or continuously observed knock-in, no exact static hedge exists, and the desk runs a delta hedge in futures plus a vanilla overlay chosen to match gamma and vega near the barrier, rebalanced as spot approaches it.

Then the two risks to volunteer.

- **The gap.** Above the barrier you are short almost nothing; below it you are short a deeply in-the-money put. The exposure appears all at once, so a gap opening through the barrier is a loss no continuous hedge could have prevented.
- **Skew.** A knock-in put's value depends on the shape of the surface between the strike and the barrier, not on one implied volatility, so a hedge built at a single volatility is wrong in a way that grows as skew steepens.

Close with the honest option, which interviewers like because most candidates omit it: the cleanest hedge for a short knock-in put is buying back the knock-in put, and if the position is small the bid-offer on doing that is often less than a year of running an imperfect delta hedge.

Vega, and why one number is not enough

REAL INTERVIEW QUESTION (Morgan Stanley · EQD)

How do you hedge the vega of an autocall?

WHAT THEY TEST / ANSWER

The trap is answering "sell vega". The book's aggregate vega is close to meaningless on this product, for three separate reasons, and naming them is the answer.

One, the sign is not uniform across spot. Near the autocall trigger and near the capital barrier the exposure flips, so a single vega number is the sum of two opposite positions and hedging it flattens neither.

Two, the sign is not uniform across maturity. The note is short volatility at the front, where an early autocall is being priced, and long it at the back through the down-and-in put. A parallel vega hedge in one maturity leaves the whole term-structure exposure untouched, which is why desks manage a **vega ladder** by bucket rather than a total, exactly as a rates book manages a DV01 ladder.

Three, the exposure is to the surface and not to a level. The product's value depends on the skew between the barrier and the strike, so the hedge has to be built out of the strikes that actually matter, which means listed puts near the barrier rather than at-the-money options.

So the practical answer: hedge with listed options bucketed by maturity and strike, concentrating on the region between the barrier and spot; accept that the long end beyond liquid listed maturities cannot be hedged and reserve against it; and expect the hedge to need rebalancing after large spot moves, because the vega itself migrates. Add the flow point that makes it a market problem rather than a book problem: every desk is short the same structure on the same indices, so the hedge everyone needs is the same one, in the same direction, on the same day.

REAL INTERVIEW QUESTION (Bank of America · Exo)

How do you hedge your dividends on an autocall?

WHAT THEY TEST / ANSWER

State the exposure before the hedge. Having sold the autocall the desk is long the down-and-in put, and a put gains when the forward falls, so the desk is **long dividends** and the investor is short them. On a five-to-ten year note on a major index that is a large exposure, and it is one of the few in the book that is genuinely tradeable.

The instruments, in order of preference: **dividend futures** on the index, which are listed and liquid out to about five years; **dividend swaps**, over the counter, for the maturities and single names futures do not cover; and, failing both, a **synthetic** through the put-call parity relationship, since a combination of listed calls and puts and the future prices the forward and therefore the implied dividend.

Then the three things that make it imperfect, which is what the question is really after.

- **Maturity gap.** Autocall issuance runs to ten years and the liquid dividend curve does not, so the long end is hedged with a proxy or not at all, and the residual is reserved.
- **Single names.** A worst-of on three stocks has three dividend exposures, and single-name dividend swaps are far thinner than the index.
- **Basis and correlation with spot.** Dividends are cut precisely when markets fall, so the dividend hedge and the equity hedge fail together, which is what happened across structured books in 2020 when index dividends were suspended by regulators rather than by earnings.

The last point is the one worth ending on: the dividend risk on a structured book is not a smooth sensitivity, it is a jump risk correlated with everything else in the book.

REAL INTERVIEW QUESTION (BNP · Hybrid)

Practical case: Athena 10Y SX5E; PDI 50%; coupon 5%; price = 100. The next day, all else equal, rates rise by 10bp. What is the impact on the price? How will the trader hedge this product?

WHAT THEY TEST / ANSWER

The price falls, and the useful answer is the decomposition rather than the direction, because rates enter this product through three doors pointing two ways.

Down. The note is mostly a zero-coupon bond, and a higher discount rate is worth less. On a ten-year at 3%, the redemption leg falls 0.74 for 10 basis points.

Up. A higher rate lifts the forward, which makes the down-and-in put the investor is short less valuable. Priced at 25% volatility with a 2% dividend yield, the put falls 0.35, and the investor is short it, so that is a gain.

Net, on those two legs, $-0.74 + 0.35 = \mathbf{-0.38}$, an effective duration of about 3.8 rather than the 10 the maturity suggests. Then apply the correction that shows you understand the product: the autocall feature means the note is not expected to live ten years at all, and its expected life of two or three years is what its rate sensitivity is really measured against, so the true number is smaller again, of the order of one or two tenths.

Third door, worth one sentence: the coupons are conditional, and a higher forward makes the trigger more reachable, so the note calls sooner in expectation, which shortens it further and reinforces the same conclusion.

The hedge. Rates risk on a structured book is not managed on the equity desk. The funding leg is passed to treasury or hedged with swaps in the note's currency, sized off the effective duration rather than the legal maturity, and rehedged as the expected life changes with spot. What the equity desk keeps is the part that cannot be transferred: the delta in futures, the vega and skew in listed options, and the dividend leg in dividend futures. The practical point to make is that this note has a rates exposure whose *size depends on the equity level*, which is what makes it a hybrid product and why the question comes from a hybrid desk.

What cannot be hedged

Some of the risk has no instrument, and a desk that pretends otherwise is mismarking rather than hedging.

Correlation, on any worst-of. The desk owns a worst-of put, which is worth more when the underlyings disperse, so it loses when they move together and there is no liquid instrument to buy that back; dispersion is the closest thing and it is a trade, not a hedge. Every desk in the market is on the same side.

The long end, past the maturity at which options, dividend futures and repo stop trading.

Model risk, meaning the difference between the model used to compute the hedge and the market that will actually deliver it.

The answer to all three is the same and it is not clever: a **reserve** taken at issue, released over the life of the trade, sized to the exposure that cannot be traded. That reserve is part of the margin quoted on the product, which is why a worst-of on three names prices with a wider margin than the same structure on one index, and why the quoted coupon and the hedgeability of the risk are the same conversation.

INTERVIEW TRAP

1. **Believing the book can be flattened.** It cannot. The job is to choose which risks to carry, size them, and get paid for them at issue.
2. **Hedging an aggregate Greek.** One vega number on a structured book nets two opposite positions across spot and across maturity. Manage the ladder, not the total.
3. **Chasing the delta into a barrier.** Hedging a discontinuity faithfully costs more in bid-offer than the mishedge it prevents. Smooth, or reduce.
4. **Treating dividend risk as a smooth sensitivity.** It is a jump risk that arrives correlated with the equity loss.
5. **Forgetting the flow is one-way.** Every desk sold the same structure, so the hedge you need is the hedge everyone needs, and the market will move against you while you do it.
6. **Hedging the legal maturity.** A ten-year autocall has the rate and vega sensitivity of a two-to-three-year note, and that expected life moves with spot.

VISUAL / CODE TO BUILD: Interactive

A hedging simulator run as a game. The reader is handed a short autocall book and a set of instruments, index futures, a grid of listed options and dividend futures, and a spot path plays forward day by day with a bid-offer charged on every trade. A live risk panel shows delta, the vega ladder by maturity bucket, dividend exposure and the unhedged residual, and the running P&L splits into hedged and unhedged components so the reader can see which losses were avoidable. The scripted moments are the ones that teach: an observation date approached with spot pinned at the trigger, where the true delta is unhedgeable and the smoothed hedge is the only survivable choice; and a gap down through the barrier, where the loss arrives whatever the reader does and the only decision that mattered was the size taken at issue.

19.9 Embedded optionality

structuring · trading · sales | core | equity · cross-asset

Nobody buys a structured product to sell an option, and yet every client who buys one does. The coupon, the participation, the protection: all of them are paid for by an option the client is short without having asked for it. Embedded optionality is the name for that hidden leg, and understanding it is the difference between describing a product and understanding it.

The term covers two different situations that are worth keeping apart. **Optionality sold by the buyer** is the structured-product case: the note is wrapped so that the client's short position looks like a feature of the payoff rather than a position. **Optionality retained by the issuer** is the debt-market case: a callable bond, a convertible, a mortgage that can be prepaid. In both, someone owns an option that is not priced on any screen, and the whole instrument is mispriced by anyone who ignores it.

The put down-and-in, the funding leg of the equity market

One instrument does most of the work in retail equity structuring, and it comes up in interviews more than any other exotic.

REAL INTERVIEW QUESTION (Société Générale · EQD)

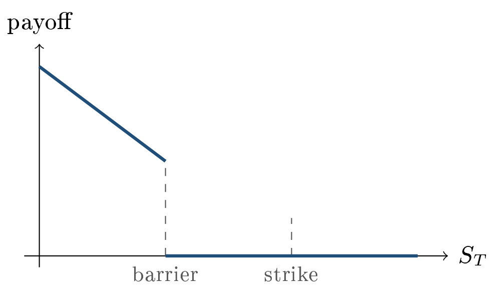
Tell me about the PDI: what is it, and why "down and in"?

WHAT THEY TEST / ANSWER

A **put down-and-in** is a put, usually struck at the initial level of the underlying, that does not exist unless the underlying falls through a barrier set well below it. "Down" is the direction the underlying has to travel; "in" is what the option does when it gets there, namely come into existence. Until then the holder owns nothing, and the client who sold it owes nothing.

Say what it is for, because that is the real question. It is the funding leg of almost every yield-enhancement product: the client sells this option, the issuer pays for it with a coupon, and everything the client receives comes from it. And give the consequence that makes it dangerous, which is that the strike and the barrier are different levels. Once the barrier at, say, 50% is breached, the option that knocks in is struck at 100%, so the loss is measured from the initial level and not from the barrier: a final level of 49% does not cost the client 1%, it costs 51%. That gap between where the risk starts and where it is measured from is exactly why the product can be marketed as "protected down to 50%" while behaving like an outright long position the moment it is not.

Two conventions worth naming before quoting any price: whether the barrier is observed continuously or only at maturity, since the European version is materially cheaper, and whether the underlying is a single name or a worst-of, since the worst-of version is a different risk entirely.



payoff of a long put down-and-in, barrier at 50% and strike at 100%, observed at maturity

the vertical gap at the barrier is the whole product: nothing on one side, half the notional on the other

REAL INTERVIEW QUESTION (Morgan Stanley · Exo)

How do you replicate a PDI with vanillas? Do you take a 45-50 gap or a 50-55 gap, and why?

WHAT THEY TEST / ANSWER

Give the exact decomposition first, for the European case where the barrier is observed only at maturity. A put struck at K knocking in below B pays $K - S_T$ when $S_T < B$, and

$$K - S_T \;=\; (B - S_T) \;+\; (K - B) \quad \text{for } S_T < B,$$

so the PDI is exactly **one put struck at the barrier, plus** $(K - B)$ **digital puts struck at the barrier**. Check it at a point: with $K = 100$, $B = 50$ and a final level of 45, the payoff is 55, and the replication gives $(50 - 45) + 50 = 55$. Above the barrier both legs are zero.

Now the gap, which is what they are actually asking. The digital leg is itself replicated by a put spread, and the choice of gap has a direction as well as a width. Since you are the desk and you are *long* this option, the safe side is the replication that never claims more than you actually own, so you use the put spread that sits *below* the barrier, the 45-50, rather than the 50-55: it under-replicates the digital, so the residual digital − spread is non-negative *for you* in every state and there is no hole in the hedge. Note who bears which side: the client is not a party to the hedge at all, and if anything the conservative mark means the desk pays away a slightly smaller coupon for the same option. A seller of the same option takes the 50-55 for the mirror reason. Then the width: narrower is a closer hedge but requires proportionally more options, $1/\varepsilon$ of them, and concentrates gamma at the barrier as expiry approaches. The choice comes from the listed strike grid, the size, the liquidity and how close spot is expected to sit to the barrier, exactly as for a digital, covered in common structuring mistakes. Finish by saying the honest caveat: this identity is exact only for a maturity-observed barrier, and a continuously observed one needs a reflection argument or a model.

What the PDI is exposed to

WORKED EXAMPLE

A five-year PDI struck at 100% with the barrier at 50%, observed at maturity, on a spot of 100 with a 3% rate and a 2% dividend yield.

Volatility	Put at 50	Digital put	PDI	Vanilla put at 100
20%	0.55	0.065	3.79	13.51
25%	1.45	0.126	7.77	17.30
30%	2.73	0.189	12.19	21.05

Read the last two columns together. Going from 20% to 30% volatility, the vanilla put gains 56% of its value while the PDI more than triples. The barrier option is a leveraged bet on volatility, because volatility does not only change how much the option pays, it changes whether the option exists at all.

REAL INTERVIEW QUESTION (Société Générale · Structured Products)

On a PDI, are you long or short volatility, and long or short dividends?

WHAT THEY TEST / ANSWER

Long both, and the interesting part is by how much. **Volatility**: higher volatility widens the distribution, which both raises the chance of breaching the barrier and deepens the loss once breached, so the holder of the PDI is long vega, and much more so in relative terms than the holder of a vanilla put. On the numbers above, moving from 20% to 30% volatility roughly triples the PDI and adds about half to the vanilla. **Dividends**: dividends lower the forward, which pushes the whole distribution down towards the barrier and the strike, so the holder is long dividends too; raising the yield from 2% to 4% on the same example takes the PDI from 7.77 to 10.26, about a third more, against a fifth more for the vanilla.

Then flip it to the side that matters commercially: the *client* in a structured note is short this option, so they are short volatility and short dividends, which is why a spike in implied volatility marks their note down and why a special dividend can hurt them without the stock having moved at all. Add the refinement that shows desk sense: the exposure is not a single number, because the digital component makes the value sensitive to the slope of the smile right at the barrier as well as to its level, which is why these are marked off a full surface and not off one volatility. Skew sensitivities are in skew and smile sensitivities.

REAL INTERVIEW QUESTION (Société Générale · Structured Products)

How do you reduce the cost of a PDI inside a structured product?

WHAT THEY TEST / ANSWER

Answer the mechanical question and then refuse the premise, because both halves are being tested.

Mechanically, the PDI's value falls when you lower the barrier, when you observe it at maturity instead of continuously, when you shorten the maturity, when you pick a lower-volatility or lower-dividend underlying, and above all when you move the *strike* down to the barrier. That last one is the **airbag**: instead of a put struck at 100% knocking in at 50%, the redemption below the barrier becomes $100 \times S_T/B$, so the client is short $100/B$ puts struck at the barrier. Note the gearing, because it is the whole mechanism and it is easy to drop: with $B = 50$ that is **two** puts, not one, and the leg falls from 7.77 to $2 \times 1.45 = 2.90$, a reduction of about **63%**. The cliff disappears, but the airbag still redeems at zero if the underlying does, so it softens a moderate fall rather than providing protection. The structure itself is developed in participation products; what matters here is only that it is the single most effective lever on the funding leg.

Then refuse the premise: inside a structured product the PDI is not a cost, it is the *funding*. The issuer buys it from the client and pays for it with the coupon, so every one of these levers reduces the coupon by the same amount it reduces the option. There is no version of this where the product gets cheaper for the client and keeps its headline. The right way to present the airbag is therefore not "we made your risk cheaper" but "you now take a smoother, smaller loss and receive a smaller coupon", which is the correct framing and the one a client can act on.

REAL INTERVIEW QUESTION (Goldman Sachs · Structured Products)

Leveraged put versus put down-and-in: which pays better for an autocall with the same characteristics?

WHAT THEY TEST / ANSWER

Define both, then compare on premium raised rather than on how they look. A **leveraged put** is a plain short put in a multiple of the notional, so the loss begins immediately below the strike and accumulates at that multiple. A **PDI** is the barrier version above, where nothing happens until the barrier breaks and then the loss arrives from the strike all at once.

On the numbers, the leveraged put wins on premium and it is not close. With the same five-year parameters, the 50%-barrier PDI is worth 7.77, while a single at-the-money put is worth 17.30, so a 1.5× leveraged put raises about 25.95, more than three times as much, and the multiple can simply be raised further. The PDI's premium is structurally capped by the value of the vanilla put it hides inside, and a distant barrier keeps it far below that.

So say what the PDI is really for, which is the answer: it does not pay better, it *presents* better. It lets the term sheet say the client is protected down to the barrier, and it puts the risk in a place the client considers unlikely, whereas a leveraged put shows the risk starting immediately. For a given amount of premium raised, the PDI concentrates the risk into a jump and is therefore very sensitive to skew and to spot sitting near the barrier, while the leveraged put spreads it linearly and is easier to hedge and to explain. Which one a desk uses is a distribution decision as much as a pricing one, and the comparison of the two risk profiles is in leverage control through structure.

The other family: optionality inside debt

The same idea runs through instruments nobody calls derivatives. A **callable bond** is a straight bond minus a call the investor has sold to the issuer. A **puttable bond** is a straight bond plus a put the investor owns. A **convertible** is a bond plus a call on the issuer's equity. A mortgage is a loan plus a prepayment option the borrower owns. In each case the quoted yield is not comparable to a straight bond's until the option is stripped out.

REAL INTERVIEW QUESTION (Goldman Sachs · EQD)

Callable Convertible Bonds: How do callable convertible bonds work? Explain how the issuer benefits from the call option feature and how the convertible bondholder's interests are protected.

WHAT THEY TEST / ANSWER

Build it in two layers. A **convertible bond** is a corporate bond plus a call on the issuer's own shares held by the investor, which is why it pays a lower coupon than the same issuer's straight debt: the investor has been paid in optionality instead of income. Making it **callable** adds a second option, this time held by the issuer, to redeem the bond early at a set price.

The issuer benefits twice. They can refinance if their credit improves or rates fall, like any callable issuer, and, more importantly for a convertible, the call is the tool that **forces conversion**: once the shares have risen enough that conversion is worth more than the call price, calling the bond leaves the holder choosing between being redeemed at the call price and converting, and they convert. That turns debt into equity at a moment of the issuer's choosing and caps the upside the investor was paid in.

For the holder, the position is long a bond, long a call on the equity, short a call on their own instrument, and the third leg is the one that is easy to forget. It caps the convertible's price near the call level, so the instrument stops behaving like equity exactly when the equity is doing well, and it introduces reinvestment risk at the worst time. That is precisely the harm the question is asking about, so name it before naming the remedies.

How the holder is protected, which is the half of the question with a specific answer. **Call protection**: a hard no-call period, typically the first years of the life, during which the issuer simply cannot call, so the investor is guaranteed the optionality they paid for over that window. **Soft-call triggers**: after that period the call is permitted only if the share price has traded above a threshold, commonly around 130% of the conversion price, for a set number of days in a window, so the issuer cannot call on a brief spike and the investor keeps the upside until conversion is comfortably in the money. **Make-whole provisions**: if the issuer calls early, or on a change of control, the conversion ratio is increased on a contractual schedule so the holder is compensated for the option value taken away. **Dividend and anti-dilution protection**: the conversion ratio adjusts for splits, special dividends and rights issues, so the issuer cannot erode the conversion value through corporate action.

Close on the two points that make it a markets answer. The issuer's call is *rational* rather than automatic, so its value depends on refinancing conditions and not only on the share price. And the usual valuation approach is a model handling both equity and credit together, because the same instrument is equity-like when the shares are up and bond-like when they are down.

INTERVIEW TRAP

Four errors. (1) Confusing the barrier with the strike: the barrier decides *whether* the client loses, the strike decides *how much*, and they are different levels on purpose. (2) Quoting a barrier price without the observation convention, since a continuously monitored barrier is materially more expensive than one checked at maturity. (3) Calling the funding leg a cost: it is what pays the coupon, so reducing it reduces the coupon by the same amount. (4) Comparing the yield of a callable or convertible bond to a straight bond's: the extra yield is an option premium, not a free spread, and the two are only comparable once the option is stripped.

VISUAL / CODE TO BUILD: Interactive

A funding-leg inspector. The reader sets spot, strike, barrier, maturity, volatility, rate and dividend yield, and the widget prices the put down-and-in and displays it three ways at once: as its two replicating legs (a put at the barrier plus digital puts), as a payoff diagram with the cliff drawn, and as a premium bar next to the alternatives that could fund the same coupon, a vanilla short put, a leveraged put and an airbag. Sliders for volatility and dividend yield move all three panels together, so the outsized sensitivity of the barrier option is watched rather than tabulated. A second tab does the same for a callable convertible: the reader drags the share price and sees the instrument's value split into bond floor, conversion value and the issuer's call, with the call visibly flattening the upside once the shares clear the call trigger.

Where this goes next

The products this leg funds are in autocallables, Phoenix notes and reverse convertibles, and the arithmetic that turns its premium into a coupon is in leverage control through structure. The barrier machinery itself is in exotic options, running the resulting risk is hedging structured products, and the ways this leg is under-disclosed are in common structuring mistakes.

19.10 Retail vs institutional structuring

sales · structuring | core | equity · cross-asset

The same autocallable can be sold to an individual through a private bank and to an insurer's treasury desk, and almost nothing about the two transactions is alike: not the size, not the margin, not the documentation, not the sales process, and not what the client is actually buying. Candidates who have read about structured products usually know one of these worlds and assume it is the whole business. Knowing that there are two, and which one the desk you are interviewing with lives in, is worth more than another product definition.

The two worlds

	Retail	Institutional
Who buys	individuals, via a distributor	insurers, funds, asset managers, corporates
Why they buy	a payoff and a story	a mandate, a liability, a balance-sheet need
Ticket	small, aggregated into a tranche	large, one trade
Product	standardised, campaign-driven	bespoke, negotiated
Price formation	margin on a shown price	competitive RFQ against several banks
Margin	percent	basis points
Wrapper	note or certificate, always	note, or an OTC derivative
Protection	KID, target market, suitability	professional client, far less
Flow direction	one-way	two-way

Two rows in that table drive almost everything else.

Price formation. A retail product is *shown* at a coupon or a participation. The client compares it against other products they are shown, not against a fair value they can compute, and the distributor sits between the issuer and the investor taking a commission. An institutional product is *asked for*: the client specifies the payoff and puts it out to three or five banks at once, and the winner is whoever quotes the best level, the mechanics of which are the RFQ. The first market competes on distribution and story; the second competes on price, and the margin difference between them is an order of magnitude rather than a detail. The absolute levels are in structured notes.

Flow direction. This is the one candidates miss, and it has consequences far outside this chapter. Retail flow is overwhelmingly one-directional: retail investors are structurally sellers of downside optionality, because that is what pays the coupon they are shown. Whoever sells, somebody buys: the street ends up collectively **long** the downside puts it took from retail, and long skew from that flow. Desks then try to sell the risk on, which is why sustained retail issuance is generally cited as *depressing* long-dated implied volatility and *compressing* skew rather than as a source of its steepness. The steepness itself is usually attributed to institutional protection buying and to the leverage effect, both in the volatility smile and skew. What does become crowded in a sell-off is the hedging: every desk holds the same gamma concentrated at the same barriers and triggers, so they all need the same delta hedge at the same levels, which is Greeks across market regimes. Institutional flow is two-way: an insurer buying protection can be offset against a fund selling it, so the desk can genuinely intermediate rather than warehouse.

KEY IDEA

Retail structuring is a *distribution* business: the hard part is the product being attractive, explicable and compliant, and the margin pays for reaching the client. Institutional structuring is a *pricing* business: the hard part is being the cheapest bank on a payoff the client has already specified. A candidate who describes the job they are interviewing for in those terms is describing it correctly.

What each client is actually buying

Retail clients buy an outcome they can picture: a coupon that beats a deposit, a way to own an index with some protection, a bet on a market they follow. They rarely have a view on volatility, so the product's job is to convert their directional or income view into something the desk can hedge. Suitability is a real constraint rather than a formality: the product must match a defined target market, a rule that comes from the product-governance regime in MiFID, and a payoff the client cannot restate has not been sold.

Institutional clients buy a solution to a constraint. An insurer hedging a guaranteed minimum on a savings book, a pension fund converting equity exposure into a shape that fits its funding ratio, a corporate hedging a buyback or an employee share plan, an asset manager enhancing yield inside a mandate that forbids naked option selling. The payoff is a means; the constraint is the reason. That changes the sales conversation completely: the useful first question is not "what is your view" but "what are you trying to achieve, and what are you not allowed to do".

REAL INTERVIEW QUESTION (Deutsche Bank · Fixed Income)

How does a private bank work?

WHAT THEY TEST / ANSWER

Describe it as a distribution channel rather than as a building, because that is what makes it relevant to a structuring desk. A private bank manages wealth for individuals and families, and it earns through management fees on assets and through commission on products it places. For structured products it is the *distributor*: it does not manufacture, it selects an issuer, negotiates terms, takes a commission and sells to its own clients, which means the issuing bank is selling to the private bank, not to the end investor. Three consequences worth stating. The private bank's incentives sit between the two, so it cares about the headline number it can show its clients and about the issuer's credit quality, since a default would be its own client problem. It aggregates many small tickets into one tranche, which is what makes a small investor able to buy a bespoke payoff at all. And it carries the suitability obligation, so it enforces the target market and will refuse a payoff it cannot explain to its clients. Finish with the interview-relevant point: if the role is on the manufacturing side, the private bank is your client, and the skill being tested is negotiating and explaining to a professional intermediary, not selling to an individual.

Competing on price

On the institutional side the structurer's day is a sequence of requests to which several banks are responding at once, and losing on price is the normal outcome. Knowing why you lost is the job.

REAL INTERVIEW QUESTION (Commerzbank · Structured Products)

How do you assess whether the decomposition of a structured product, a reverse convertible for example, is competitive in the market, and how do you make it more competitive?

WHAT THEY TEST / ANSWER

Assessing: decompose and price each leg against where you could actually trade it, not against a model. The bond leg goes at your own funding curve, so compare it with where your treasury issues; the option leg goes at the implied volatility and skew you can hedge at, including bid-offer, so compare it with the listed market and your own axe. Then compute the implied margin as the residual between the price you would show and the sum of the legs, and compare that with what the market is printing, which you learn from losing. **Making it more competitive**, in the order a desk actually tries them: check whether your *funding level* is the problem, since on a multi-year note the bond leg is most of the price and a bank with cheaper funding wins before the option is even discussed; look for an *axe*, because if the desk is already short the risk the product creates then taking the other side costs it less than mid, which is the single largest legitimate source of edge; adjust the *underlying*, since a name where you have inventory or a better hedge is cheaper for you than for a competitor; and only then reduce the *margin*. Close with the discipline: the one thing you do not do is win by mispricing the risk, because a structured book that wins every competitive line is usually the one that has mismarked its skew, and it finds out at the first sell-off.

Pitching, in both worlds

REAL INTERVIEW QUESTION (Morgan Stanley · EQD)

Can you pitch one of the products you worked on during your internships, explain the payoff, and say why it would be an interesting product for a client?

WHAT THEY TEST / ANSWER

Answer in three moves and keep the third one longest, because "why interesting for a client" is the actual question and most candidates spend their time on the mechanics. **One, the client**: name a specific type and their constraint, for instance a private bank whose clients want income in a market where deposits pay little and who would accept owning a large index lower. **Two, the payoff**: state it in cash-flow terms, what they receive, in which scenarios, and what they lose in the bad one, in that order and in one breath, without jargon. **Three, why it fits**: connect the product to the client's constraint and then name the scenario in which they lose, before being asked. Two things separate a strong pitch. Say what the client is *giving up*, since every structured product is funded by a sold risk, and a pitch that does not name it is a sales script rather than an argument. And say why *now*, referencing the current level of rates, volatility and skew, since the same product is attractive at one point in the cycle and poor value at another. If you did not work on structured products, pitch what you did work on with the same three moves; the structure of the answer is what is being graded.

REAL INTERVIEW QUESTION (Kepler · Structuring)

In the current environment, which structured product could be interesting?

WHAT THEY TEST / ANSWER

This is a live-markets question, so answer it with a framework plus a current view, and be explicit that the answer changes with the levels. The framework has three inputs. **Rates**: high rates make the zero-coupon leg cheap, so capital-protected and participation products become viable again, which is exactly what happened as rates rose after the zero-rate decade; low rates kill them and push the market toward yield products where the client sells optionality. **Volatility**: high implied volatility makes sold options expensive, so coupons on autocallables and reverse convertibles look generous, and it makes bought options expensive, so participation levels fall. **Skew**: a steep skew improves any product funded by a sold downside put, which is most of them. Then give a view, name the levels you are basing it on, and state the risk: for instance, that with rates meaningfully positive a capital-protected participation note is worth showing again because the client can be given a floor without giving up all the upside, whereas the same product was unsellable when the zero-coupon leg cost 98. Close by saying you would check the current at-the-money volatility and the skew before committing, which is both true and the answer an interviewer wants, because the failure mode here is a candidate confidently recommending a product built for a market that ended two years ago.

What changes for the desk

The same product creates different work depending on where it came from.

- **Warehousing.** Retail flow has to be warehoused, because there is no natural other side. The desk carries the risk and hedges it in the listed market, which is why retail issuance shows up in the volatility surface.
- **Size and timing.** A retail campaign strikes on a known date with an uncertain final size, so the desk hedges a notional it does not yet know. An institutional trade is a known size at a known moment.
- **Lifecycle work.** Retail products come in thousands of small lines that all need marks, corporate-action handling and observation-date processing. Institutional trades are few and individually managed.
- **Reputational asymmetry.** A professional client who loses money on a trade they specified has taken a risk. An individual who loses money on a product they did not understand is a conduct problem, and that asymmetry is why retail documentation and target-market rules exist.

INTERVIEW TRAP

Four errors. (1) Describing structuring as one job: the retail version is a distribution business and the institutional version is a pricing business, and they reward different skills. (2) Forgetting the distributor in the retail chain, so that the issuing bank's client is described as the individual investor when it is the private bank or the network. (3) Assuming margins are comparable: percent against basis points. (4) Getting the sign of one-way retail flow backwards: retail sells the downside, so the street is *long* it, and the consequence is pressure on long-dated volatility plus dealer gamma concentrated at the same barrier levels, not a shared short position.

VISUAL / CODE TO BUILD: Interactive

A two-channel comparison of a single product. The reader configures one autocallable and the widget shows it twice: once as a retail tranche, with a distributor commission, a KID-style cost disclosure, a minimum ticket and the resulting client coupon; and once as an institutional line priced through a competitive RFQ, with a margin slider showing how the coupon improves as the margin compresses toward a few basis points. The same term sheet, two coupons, with the gap attributed explicitly to commission and margin rather than to anything about the payoff. A second panel aggregates a retail campaign into the dealer's book and shows the resulting vega and skew position building up as tickets arrive, so the one-way-flow argument becomes something the reader watches accumulate rather than a claim in a paragraph.

Where this goes next

The product this section keeps using as its example is autocallables, and the funding leg that makes the retail version one-directional is in reverse convertibles. What the accumulated flow does to the market is in the volatility smile and skew and Greeks across market regimes. The wrapper, the fee structure and the credit risk that both channels share are in structured notes, and the broader map of who is on each side of a trade is buy-side versus sell-side.

Chapter 20

Crypto & Digital Markets

20.1 Blockchain fundamentals

sales · trading · structuring core crypto

Banks now run crypto trading, custody and structuring desks, so a front-office candidate is expected to be able to say what the underlying technology actually does. The useful framing for a markets person is not "digital gold" or "the future of money". It is this: **a blockchain is a settlement system.** It is a way to move ownership of an asset from one party to another without a central operator keeping the record. Everything interesting for a desk, the twenty-four-hour trading day, the absence of a clearing house, the bearer-instrument custody risk, follows from that one design choice.

Start from the ledger you already know

When you buy a share, no physical object moves. A chain of record keepers updates their books: your broker, the clearing house, the central securities depository, the custodian. Settlement finishes a couple of days later, and the reason it takes that long is that several institutions have to agree that their separate copies of the record match. The system works because each participant trusts the operator above it, and because someone can reverse an error. See clearing houses for how that machinery is built.

A blockchain replaces that hierarchy with a single ledger that every participant holds a copy of, and a set of rules for agreeing on the next update. There is no operator to trust, and therefore no operator to reverse anything.

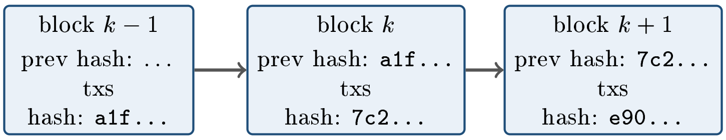
> **INTUITION**
>
> Think of a village with no bank. Instead of one banker holding the book of who owns what, every household keeps an identical copy of the book, read out loud in the square every ten minutes. To spend money you announce the transfer and sign it with a signature nobody can forge. Everyone writes the same line in their own copy. To cheat you would have to convince most of the village to write a different line, and the rules are set up so that doing so is more expensive than whatever you could steal. That is a blockchain. The clever part is not the book; it is the procedure for agreeing on the next page without a chairman.

The three cryptographic pieces

1. Hash functions. A cryptographic hash such as SHA-256 takes any input and returns a fixed-length output, 256 bits written as 64 hexadecimal characters. It has three properties that matter: it is *deterministic* (same input, same output every time), it is *one-way* (you cannot recover the input from the output), and it has the *avalanche* property (change one character of the input and the output is completely different). A hash is therefore a fingerprint of a document: publish the hash and anyone can later check that the document has not been altered by so much as a comma.

2. Digital signatures. Each user has a **private key**, kept secret, and a **public key** derived from it, which is effectively their account address. Signing a transaction with the private key produces a signature anyone can verify against the public key without ever learning the private key. So ownership on a blockchain is not an entry in someone's customer database. Ownership *is* control of a private key. This is the origin of the phrase "not your keys, not your coins", and it is why key management, not counterparty credit, is the dominant operational risk on a crypto desk. See operational risk.

3. Chaining. Transactions are grouped into **blocks**. Each block header contains the hash of the previous block, which is what makes the structure a chain rather than a pile.



Alter a single transaction in block k and that block's hash changes. But block $k+1$ recorded the old hash, so it no longer matches, and neither does anything after it. Tampering with one old entry therefore forces you to rebuild the entire chain from that point forward, faster than everyone else is extending the honest one. That is the whole security model in one sentence.

A refinement worth knowing because it comes up: transactions inside a block are summarised by a **Merkle tree**, a binary tree of hashes whose root sits in the block header. It lets a lightweight client prove that a specific transaction is in a block of n transactions by checking about $\log_2 n$ hashes rather than downloading the block. For a block of a thousand transactions that is roughly ten hashes instead of a thousand.

Consensus: the problem actually being solved

Copying a digital file is free, so the hard problem is not forging a signature, it is **double spending**: sending the same coin to two people and having each believe they were paid. Since there is no central operator to decide which came first, the network needs a procedure for agreeing on ordering. That procedure is the consensus mechanism, and the two live ones are worth being able to contrast.

Proof of work. Miners compete to find a number, the nonce, such that the hash of the block header falls below a target. Because hashes are unpredictable, the only method is trial and error, so finding a block is a lottery in which your chance of winning is your share of total computing power. Bitcoin adjusts the target every 2016 blocks, roughly every two weeks, so that blocks keep arriving about every ten minutes regardless of how much hardware joins. The rule for resolving disagreements is that the chain with the most accumulated work wins. Rewriting history means out-computing everyone else, which is what "51% attack" refers to. The cost is that the security budget is paid in electricity.

Proof of stake. Instead of burning energy, validators post the network's own token as collateral and are selected to propose blocks in proportion to their stake. Misbehaviour, such as signing two conflicting blocks, is punished by destroying part of that collateral, a mechanism called slashing. The security argument shifts from "attacking costs electricity" to "attacking costs you your capital". Ethereum switched from proof of work to proof of stake in September 2022, which cut its energy consumption by roughly three orders of magnitude, and produces a block about every twelve seconds.

KEY IDEA

The practical difference for a trader is **finality**. Proof of work gives *probabilistic* finality: a transaction is never mathematically final, it just becomes exponentially more expensive to reverse as blocks pile on top, which is why exchanges wait for a number of confirmations before crediting a deposit. Proof of stake with checkpointing gives *economic* finality after a fixed number of blocks: reversing it would require destroying an enormous amount of staked collateral. Neither resembles the legal finality that a central securities depository provides by fiat.

WORKED EXAMPLE

How safe are six confirmations? Suppose an attacker controls a fraction q of the hashing power and honest miners control $p = 1 - q$. Once the honest chain is k blocks ahead, the attacker's deficit behaves like a random walk that moves one step toward them with probability q and one step away with probability p. The probability that a walk of this kind, started k behind, ever catches up is

$$\mathbb{P}(\text{catch up}) = \left(\frac{q}{p}\right)^k \qquad \text{for } q < p,$$

which is the gambler's ruin result from stopping times, and it goes to 1 once $q \geq 1/2$. Put numbers on the standard six-confirmation rule:

- $q = 10\%$: $(1/9)^6 \approx 1.9 \times 10^{-6}$, about two chances in a million.
- $q = 30\%$: $(3/7)^6 \approx 0.0062$, about 0.62%, which is not a rounding error on a large transfer.
- $q = 50\%$: certainty, given enough time.

Two honest caveats. This ignores the head start the attacker may build in secret before the merchant starts counting, which is the refinement in the original Bitcoin paper and makes the true figure somewhat worse. And it assumes the attacker's hash share is stable. The takeaway a desk actually uses: the number of confirmations you require should scale with the size of the transfer, exactly the way a settlement limit does.

WORKED EXAMPLE

The other side of the same lottery. A miner with 10% of the network's hashing power on a chain producing one block every ten minutes finds blocks as a Poisson process at rate $0.1 \times 6 = 0.6$ per hour. So the expected wait for their own next block is $1/0.6 \approx 100$ minutes, and the probability of finding at least one block in a given hour is

$$1 - e^{-0.6} = 1 - 0.5488 = 45.1\%.$$

Mining revenue is therefore extremely lumpy for a small operator even though its expectation is smooth, which is the entire commercial reason mining pools exist: they are a variance-reduction arrangement, not a return-enhancing one. That is the same trade a market maker makes when they widen size to smooth PnL.

What changes, and what does not, for a markets desk

Settlement becomes atomic and fast. A transfer either happens completely or not at all, in one operation, in minutes rather than at T+2, and delivery versus payment is a property of the mechanism rather than a service you buy. That removes a genuine category of risk: there is no window in which you have delivered and not been paid.

Settlement also becomes irreversible. There is no error account, no recall, and no chargeback. A payment to a wrong address is simply gone. In traditional markets an operations team fixes fat-finger settlement errors daily; on chain that option does not exist, which is why crypto desks put so much process around address whitelisting and test transfers.

Custody becomes a bearer-instrument problem. Holding crypto is closer to holding physical gold or a bearer bond than to holding a share. Whoever controls the key controls the asset, so the risk is theft and key loss rather than the insolvency of a custodian. This is also why the collapse of centralised venues that commingled customer assets in 2022 pushed institutional flow toward qualified custodians and toward on-chain proof of reserves.

Capacity is small and public. Bitcoin processes roughly seven transactions per second, against thousands per second for a card network, and every transaction is visible to everyone forever. The low capacity is why almost all scaling now happens in a second layer that batches activity and settles periodically to the base chain. The transparency is genuinely unusual for a markets person: flows, holdings and liquidations are observable in a way they never are in equities or FX, which is why on-chain data has become a legitimate input to crypto trading. Market structure is covered in market structure in crypto.

Counterparty risk moves, it does not vanish. Base-layer settlement removes settlement risk against a counterparty. It does nothing about the exchange holding your margin, the stablecoin issuer holding the reserves behind your quote currency, or the smart contract holding your collateral. See stablecoins and DeFi basics.

How this comes up in interviews

A note on sourcing: the question database behind this book contains roughly 3,400 real questions collected from sales, trading and structuring interviews, and none of them are about crypto. That is itself the most useful signal in this section. Crypto is not yet a standard technical round at the banks in the sample, so it appears as a market-culture or motivation topic rather than a pricing one. Expect to be asked what a blockchain does rather than to derive anything, and expect the follow-up to be about risk: what settles, who holds the asset, and what happens when it goes wrong. Prepare a two-minute answer built around settlement and custody rather than around price forecasts, and be able to say plainly which parts of the technology you think are useful to a bank and which are not. Nothing marks out a candidate here faster than either uncritical enthusiasm or blanket dismissal.

INTERVIEW TRAP

Five traps. First, using "blockchain" and "Bitcoin" interchangeably; Bitcoin is one application of the technology. Second, calling transactions anonymous; they are *pseudonymous*, permanently public and highly traceable, which is precisely why chain analytics is a working compliance tool. Third, treating "immutable" as "correct"; the chain guarantees that a record has not been altered, not that it was right when written, and it says nothing about whether the off-chain asset a token claims to represent exists. Fourth, assuming a private permissioned chain inherits the trust-minimisation benefits; once a small consortium controls the validators you have rebuilt a shared database with extra steps, which is often fine, but say so. Fifth, confusing the network with the token: the price of a token is not a measurement of how useful the underlying protocol is.

VISUAL / CODE TO BUILD: Interactive

A two-part sandbox. The first lets the reader type text into a box and watch its SHA-256 hash change completely on every keystroke, then build a three-block chain and edit a transaction in the middle block so that every subsequent block turns red as its stored previous-hash stops matching. The second is a double-spend simulator: sliders for attacker hash share q and number of confirmations k, a live plot of $(q/p)^k$ on a log scale, and a race animation of the honest chain against the attacker's private chain. The intended "aha" is that security is not binary but a cost, and that the required number of confirmations is a risk limit like any other.

Where this goes next

With the settlement layer understood, the rest of the chapter is markets. How the instruments themselves trade is spot versus derivatives in crypto, and the dominant instrument by volume, the perpetual future, together with the funding rate that anchors it to spot. The quote-currency question is stablecoins, the lending and collateral layer is DeFi basics, and the option-relevant behaviour of the underlying is crypto volatility. For where these assets sit next to everything else a bank trades, see cryptocurrencies in the asset class overview.

20.2 Spot vs derivatives in crypto

sales · trading · structuring | core | crypto

If you already know how equities and futures work, most of crypto is familiar and a few things are genuinely different. The differences are not about the blockchain, they are about **market structure**: who holds your assets, what a margin call means, what settles when, and what the dominant risk actually is. Getting those right is what separates someone who can talk about the asset class from someone who has only read the price chart.

Spot: what you are actually holding

Buying a token on a centralised exchange does not usually mean holding the token. It means holding a **claim on the exchange**, recorded in its internal ledger, which it settles on-chain only when you withdraw. That single fact drives most of what follows.

> KEY IDEA
>
> **In crypto, venue risk usually dominates market risk.** On a regulated equity market, client assets are segregated, a clearing house stands between counterparties, and the failure of your broker is inconvenient rather than terminal. On most crypto venues there is no central counterparty, the exchange is simultaneously the venue, the broker, the clearer and the custodian, and an unsecured claim on it is what you own. The 2022 failure of a major exchange was not a market event, it was a custody and segregation event, and it wiped out participants who had no directional position at all. The professional framing is that a crypto trade has two exposures, the asset and the exchange, and you must size and diversify both, as in counterparty risk.

Three more structural features of spot:

- **Self-custody is available, and is a different risk rather than no risk.** Holding your own keys removes the exchange from the picture and replaces it with operational risk: key management, signing procedures, and the fact that an on-chain transfer is irreversible. Most institutions resolve this with qualified custodians and settlement networks so that trading and custody are separated the way they are in traditional markets.
- **The quote asset is often a stablecoin, not a currency.** Trading a token against a dollar-pegged token means you are implicitly long that issuer's credit and its peg, which is a real exposure and not a unit of account, see stablecoins.
- **Liquidity is fragmented across venues** with no consolidated tape and no best-execution obligation of the kind familiar elsewhere, so the same asset has several simultaneous prices and arbitrage between them is capital-intensive rather than free, see market structure in crypto.

Trading is continuous, and settlement is not a cycle

There is no T+2, no exchange holiday and no close. Two consequences a risk manager notices immediately. There is no **weekend gap** in the sense equity desks mean, because the market trades through the weekend, but weekend liquidity is thin, so large moves happen in the worst conditions rather than being deferred to Monday. And the entire apparatus of daily marks, business-day theta accrual and end-of-day risk reporting is built on a market that closes, so running a crypto book on a traditional risk system produces mismatches that are operational rather than conceptual.

On-chain, settlement is final rather than reversible, subject to confirmation. That removes settlement risk of the Herstatt kind and replaces it with a narrow window of confirmation risk plus the irreversibility of a mistake, which is why institutional workflows put controls in front of transfers rather than after them.

The derivatives suite, and why it is bigger than spot

Crypto derivative volumes routinely exceed spot volumes, for the ordinary reasons derivatives exist anywhere: leverage, the ability to short without a borrow market, and capital efficiency. Three products dominate.

- **Perpetual futures**, which have no expiry and use a periodic **funding** payment between longs and shorts to hold the contract near spot. This is the defining crypto instrument and it has its own sections: perpetual futures and funding rates.
- **Dated futures**, which behave like futures anywhere and whose basis to spot is the cleanest window into the market's financing conditions, worked below.
- **Options**, concentrated on a small number of venues, frequently quoted and margined in the coin rather than in dollars, see crypto options.

WORKED EXAMPLE

The crypto basis, and why it is not carry. Suppose spot trades at \$60,000 and the three-month future at \$61,500. The basis is \$1,500, which is

$$\frac{1{,}500}{60{,}000} = 2.5\% \text{ over three months,}$$

or about 10% annualised simply ($2.5\%\times4$), and 10.38% if compounded quarterly (1.025^4-1). A cash-and-carry trade buys spot, sells the future, and delivers into it, locking that return.

Now the interpretation, which is the part worth understanding. In commodities the basis is bounded above by the full cost of carry, and storage is a large part of it, as in storage costs. In crypto, storage is essentially free, so the no-arbitrage basis should be little more than the financing rate. A persistent double-digit annualised basis therefore is *not* carry, it is the price of frictions: arbitrage capital is limited and expensive, the trade requires posting collateral at a venue whose default would destroy the position, moving collateral between venues takes time and creates exposure, and demand for leveraged length is structurally strong. So the basis in crypto is best read as **a measure of leverage demand and of the market's willingness to hold venue risk**, not as a storage cost, and it compresses sharply when speculative positioning unwinds.

Two mechanics with no traditional analogue

Coin-margined (inverse) contracts. A contract may be quoted in dollars but margined and settled in the coin, with a fixed dollar notional per contract. The profit on such a position is linear in the *reciprocal* of the price rather than in the price, so a coin-margined position carries embedded convexity: its dollar exposure changes as the price moves even though the number of contracts is unchanged, and the collateral backing it is the same volatile asset it is exposed to. That last point is the dangerous one, because a falling price simultaneously creates a loss and reduces the value of the margin covering it.

Automatic liquidation and auto-deleveraging. There is no credit officer and no margin call by telephone. When maintenance margin is breached, the exchange's engine liquidates the position mechanically, and cascades of such liquidations in thin conditions are a large part of why crypto moves are so abrupt. If liquidations cannot be filled and the insurance fund is exhausted, many venues apply **auto-deleveraging**, which force-closes the positions of *profitable* counterparties to balance the book. A trader can therefore be right, be solvent, and still have the position taken away, which is a risk with no real equivalent in a centrally cleared market and which must be modelled as a term in the strategy rather than as an operational footnote.

INTERVIEW TRAP

Four traps. First, **treating an exchange balance as ownership**: it is an unsecured claim until you withdraw, and in crypto venue risk usually exceeds market risk. Second, **reading the basis as carry**: storage is nearly free here, so a large basis is a leverage-demand and venue-risk premium, not a cost of storage. Third, **ignoring the currency of margin**: a coin-margined position is collateralised in the asset it is exposed to, so losses and collateral value move together, which is the definition of wrong-way risk. Fourth, **assuming a hedge cannot be taken away**: auto-deleveraging can close a winning position, so a strategy that relies on holding a specific offsetting leg must account for the possibility that the venue closes it for you.

VISUAL / CODE TO BUILD: Interactive

A market-structure comparator plus a basis monitor. The comparator puts three venue types side by side (listed with a CCP, bilateral OTC with a CSA, and a centralised crypto exchange) and lets the reader trace a single trade through each: who the counterparty is after execution, who holds the collateral, what happens on a margin breach, and what happens if the intermediary fails, so the crypto case's missing CCP shows up as a broken link in the chain rather than as an assertion. The basis monitor plots the annualised dated-futures basis against spot over time, with a slider decomposing it into a financing component and a residual, so the reader sees the residual spike with speculative positioning and collapse in deleveraging episodes, reproducing the section's argument. A third panel simulates a coin-margined position: as the reader drags the price, both the dollar exposure and the collateral value update, and a liquidation bar shows the position closing out earlier than a dollar-collateralised equivalent would.

Where this goes next

The instrument that dominates this market is perpetual futures, with the mechanism that anchors it in funding rates. The quote asset's own risk is stablecoins, the venue landscape and liquidity fragmentation is market structure in crypto, and options are crypto options with their volatility behaviour in crypto volatility. The settlement layer underneath everything is blockchain fundamentals. For the traditional contrast, the futures mechanics are futures and the basis logic is contango and backwardation.

20.3 Perpetual futures

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A futures contract works because it expires. The expiry date is what forces the futures price to the spot price, and every statement about basis, carry and convergence rests on that single anchor. A perpetual future removes it. There is no delivery date, no settlement, no convergence, and the contract simply continues. The interesting question, and the one an interviewer will ask, is what stops the price drifting away from the underlying once the anchor is gone.

The answer is that the anchor is replaced by a cash payment. That substitution is the whole design, and it is worth understanding even if you never trade crypto, because it isolates a mechanism that a dated future keeps hidden.

Why an expiry date is doing work

In a dated future, arbitrage enforces the relationship between the futures price and spot: if the future is too rich you sell it, buy the underlying, carry the position to expiry and collect the difference with certainty, because at expiry the two prices are the same number by construction. The cost of carrying the underlying in the meantime, financing minus any income, is what the basis compensates you for. That is the standard argument in futures.

> **KEY IDEA**
>
> Remove the expiry and the arbitrage loses its closing leg. You can still sell the rich contract and buy the underlying, but there is now no date on which the two prices are guaranteed to meet, so you hold an open position of unbounded duration and the trade has no terminal payoff to collect. **Without expiry, the no-arbitrage argument does not bind**, and something else has to.

That something is the **funding rate**: a periodic payment, on major venues every eight hours, made directly between the holders of the contract rather than to the exchange.

- When the perpetual trades **above** its spot index, funding is positive and **longs pay shorts**.
- When it trades **below**, funding is negative and **shorts pay longs**.
- The payment is proportional to the gap, so the further the contract strays, the more expensive it is to be on the crowded side.

> **INTUITION**
>
> Funding does not move the price mechanically. It makes the deviation *cost money to maintain*, which is a different and more robust thing. A crowded long position that pushes the perpetual 1% above spot now pays a stream of cash to whoever is willing to take the other side, and that stream is precisely the incentive needed to bring the other side in. Convergence is bought rather than enforced. **The expiry of a dated future is a promise about a date; funding is a price paid continuously for the same effect.**

This also gives the cleanest way to say what a perpetual is in language from the rest of the book: it is a futures contract whose cost of carry has been **stripped out and paid explicitly in cash**, rather than being embedded in a basis that decays to zero. Economically it sits very close to a rolling contract for difference, or to a total return swap whose floating leg resets every eight hours.

The arithmetic that decides whether a position is viable

Funding rates are quoted per interval, not per year, which is why the cost of holding a perpetual sounds harmless right up until it is annualised.

The size of that carry is what makes the point, and it is larger than most people expect. At the 0.01% per eight-hour interval that venues treat as neutral, the annualised financing cost is about **11%**. The annualisation itself, the correction for compounding the intervals rather than adding them, and the cash-and-carry trade that arbitrages the rate all belong to funding rates and are derived there. What matters here is the consequence for the contract: stripping the expiry out did not strip the cost of carry out, it converted it into a recurring cash payment, and **an 11%**

annual carry is the first thing to say when someone describes a long perpetual as a way to hold the underlying.

> WORKED EXAMPLE
>
> Now add leverage, which is where the number becomes decisive.
>
> A long of \$1m notional at 0.01% per interval pays $\$1{,}000{,}000 \times 0.0001 = \$100$ per interval, so \$300 a day. Funding accrues on **notional**, not on the margin posted.
>
> If that position is held against \$100,000 of equity, ten times leverage, the daily cost is $\$300/\$100{,}000 =$ **0.3% of equity per day**, which annualises to about **109.5%**.
>
> At ten times leverage, neutral funding consumes roughly the whole of your capital in a year while the underlying does nothing. **A levered perpetual position is a decaying asset**, and that decay is a contractual cash payment rather than a modelling artefact. Anyone who describes a levered long perp as equivalent to holding the underlying has not run this division.

The mirror image is the trade that institutions actually put on. Buy the underlying in the spot market, sell the perpetual against it, and collect funding whenever it is positive. The position is flat to the price of the underlying and long the funding stream, which is the same shape as a cash-and-carry basis trade. That flow is the reason funding tends to mean revert: persistently high funding attracts the arbitrage, which sells the perpetual and compresses it.

REAL INTERVIEW QUESTION (HSBC · Structured Products)

What is funding?

WHAT THEY TEST / ANSWER

The word has three distinct meanings in markets, and the right move is to say so and then give whichever one the desk in front of you means.

One, funding as the bank's cost of money. The rate at which the institution borrows to carry a position or to pay away cash. It is what makes an uncollateralised derivative more expensive than a collateralised one, it is the input behind the funding valuation adjustment, and on a rates or treasury desk it is the default meaning of the word.

Two, funding as the financing leg of a position. Holding an asset costs the financing of the cash used to buy it and earns whatever the asset pays. That difference is the carry, and it is why a forward price differs from spot, why a repo rate matters to an equity derivatives desk, and why a total return swap has a floating leg at all.

Three, funding as the periodic payment in a perpetual future, which is the crypto meaning and the newest of the three. Because a perpetual has no expiry to force convergence, the venue makes holders of the rich side pay holders of the cheap side at fixed intervals, in proportion to the deviation from a spot index. Positive funding means longs pay shorts.

Then say the thing that unifies them, because that is the answer rather than the list. **All three are the same quantity: the cost of holding a position over time.** In the first it is a rate the institution pays a lender, in the second it is embedded in the price of the forward, and in the third it is stripped out and paid in cash on a schedule. The crypto version is the most transparent of the three, since the carry is quoted directly rather than being inferred from a basis, and that transparency is why the perpetual is a useful teaching example even for someone who trades none of it. The mechanism and how it is traded are in funding rates.

If asked which one you meant, say you would ask, because using the wrong sense of the word in a client conversation is a real error rather than a pedantic one.

How it differs from a dated future

	Dated future	Perpetual
Expiry	Fixed date	None
Convergence	Forced at expiry	Bought via funding
Carry	Embedded in the basis	Paid in cash, per interval
Roll	Required, at a cost	None
Settlement	Delivery or cash	Never settles
Clearing	Central counterparty	Venue, with its own waterfall
Position closed	By you, or at expiry	By you, or by liquidation

Two rows deserve comment. **No roll** is a genuine advantage: a dated future has to be rolled to maintain exposure, which costs bid-offer and exposes you to the roll's own supply and demand. A perpetual removes that entirely, which is a large part of why the instrument took over its market.

Liquidation is the compensating disadvantage. There is no clearing house of the sort described in the traditional market structure sections, so venues manage credit by closing positions automatically once margin falls below a maintenance level, absorbing the shortfall into an insurance fund, and, when that is exhausted, by auto-deleveraging, meaning **closing profitable positions**

on the other side to balance the book. That last mechanism has no analogue in a cleared market: you can be right, be profitable, and still have the position removed. Sizing must account for it, and that is liquidity risk in an unfamiliar shape.

> **INTERVIEW TRAP**
>
> Three errors. (1) Quoting a funding rate without its interval. "Funding is 0.01%" means nothing until you say per eight hours, and the annualised figure is a thousand times the per-interval one in the sense that matters. (2) Saying funding forces the price back to spot. It makes divergence expensive; convergence is the result of someone choosing to take the payment. (3) Treating a levered long perpetual as equivalent to owning the underlying. At ten times leverage, neutral funding costs roughly the entire equity over a year.

Where this goes next

The mechanism summarised here is set out in full, including how venues compute it and how it is traded as an asset in its own right, in funding rates. The instrument this one is a variation on is futures, the comparison with the underlying market is spot vs derivatives in crypto, the venues and their credit arrangements are market structure in crypto, the asset class itself is cryptocurrencies, and the risk that liquidation represents is liquidity risk.

20.4 Funding Rates

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A perpetual future has no expiry date, which removes the mechanism that normally forces a derivative back to its underlying. A dated future converges to spot because it settles; a perpetual never settles, so something else has to hold it near the index. That something is the **funding rate**, and understanding it is the whole of understanding perpetuals.

> **KEY IDEA**
>
> The funding rate is a periodic cash payment between longs and shorts, typically every eight hours, whose sign depends on where the perpetual trades relative to the spot index.
>
> **Perpetual above spot**: funding is positive and **longs pay shorts**. Holding a long becomes expensive, holding a short is paid, and both pressures push the perpetual back down toward the index.
>
> **Perpetual below spot**: funding is negative and **shorts pay longs**, with the pressures reversed.
>
> There is no settlement and no arbitrageur is required. The tether is a recurring cash flow, and it is paid between market participants rather than by the exchange.

Funding is a financing rate, not a fee

The most useful reframing is that a perpetual long is economically a *leveraged spot position financed at the funding rate*. You have the exposure without the cash outlay, and you pay a carrying cost for it. That is precisely what a repo rate is to a bond position, what the spread over the floating leg is on an equity total return swap, and what the cost of carry is in a commodity forward.

The difference is who sets it. In traditional markets the financing rate is set by a lender against collateral. In perpetuals it is set by the *imbalance of the market itself*: if everyone wants to be long, longs pay, and they pay whatever is required to attract someone to the other side. Funding is therefore a price of positioning as much as a price of money, and that dual nature is what makes it informative.

WORKED EXAMPLE

Small numbers, enormous carry. Funding is quoted per eight-hour period, which is three times a day, so the annualisation multiplies by $3 \times 365 = 1{,}095$. That is simple annualisation, which is how venues quote it; compounded the numbers are larger still, and the direction of the error is against you when you are paying.

Funding per 8h	Per day	Annualised
+0.010%	+0.030%	+11.0%
+0.030%	+0.090%	+32.9%
+0.100%	+0.300%	+109.5%
−0.050%	−0.150%	−54.8%

A funding rate of one basis point per period looks like nothing on a screen and is an 11% annual financing cost. The 0.01% figure is close to the long-run neutral level on major venues, so *ordinary* conditions carry a double-digit annualised cost to being long, and a period of enthusiasm at 0.1% is charging longs over 100% a year. Compounded, those same two rows are 11.6% and 198.8%, so the screen number understates the carry twice over: once by being small and once by being simple.

Read it as a carry trade in the other direction and the same table is an opportunity. Long spot, short the perpetual, and you are flat the price and collect the funding: on \$10 million at 11% that is about \$1.1 million a year, and each individual payment is \$1,000, which is the crypto **cash and carry** and the dominant institutional trade in the asset class.

How the rate is actually computed

The details vary by venue and the shape is common. Funding has two components:

$$\text{funding} \;=\; \text{premium index} \;+\; \text{clamp}(\text{interest rate} - \text{premium index},\ \pm c)\,,$$

where the **premium index** is a time-weighted average of the gap between the perpetual and the spot index over the period, the **interest rate** term reflects the difference in borrowing costs between the two currencies of the pair, and the clamp keeps the correction bounded. Caps on the total rate exist too.

Three practical points follow and are the ones that matter in execution.

Funding is paid on the *notional* of the position, not on the margin posted, so at ten times leverage the cost relative to your capital is ten times the quoted rate.

Funding is charged only to positions open *at the funding timestamp*, which creates a visible pattern of positions being closed just before the mark and reopened after it, and a corresponding liquidity dip.

And funding is a payment between traders, so the exchange is not a counterparty to it. If the venue fails, the payment stream is worth what the venue is worth, which is counterparty risk in a market without the clearing house that a listed future would have.

What funding tells you

INTUITION

Funding is the cleanest positioning indicator in any asset class, because it is a directly observed price of being on one side rather than a survey or an inference from open interest.

Persistently positive funding means longs are paying to stay long, which means the crowd is long and levered. That is not a sell signal by itself, because a trend can persist for a long time while longs pay, but it is a statement about who would be forced to act if the price fell.

Extreme funding is a fragility indicator. When leveraged longs are paying triple-digit annualised rates, a modest price decline liquidates some of them, and forced liquidations sell into the fall, which liquidates more. That is the same mechanism as portfolio insurance and every other forced seller with no natural counterparty, and it is why crypto drawdowns are so often step functions rather than slopes.

Negative funding means shorts are paying, which usually follows a capitulation and is where the cash-and-carry trade reverses: the paid leg is now the long perpetual, so the flat-price version of the trade is short spot against a long perpetual, and it requires somewhere to borrow the coin.

The risks in the carry trade

The basis trade looks riskless and is not, and naming the four ways it breaks is what distinguishes someone who has run it from someone who has read about it.

Funding flips. It is not contractual. A trade entered at 30% annualised can be at zero next week and negative the week after, and the position then bleeds.

Margin and liquidation. The short perpetual leg loses when the price rises, and although the spot leg gains, the two are frequently at different venues with separate margin. A sharp rally can liquidate the short before the spot gain is available to support it, which is a funding-liquidity failure rather than a market one.

Venue risk. Spot on one exchange and the perpetual on another leaves you exposed to both, and the history of the asset class is that exchanges fail. This is the dominant risk in practice and it is not a price risk at all.

Auto-deleveraging. Some venues, when the insurance fund cannot absorb a liquidation, close profitable positions on the other side to balance the book. Your winning hedge can be taken away at exactly the moment it is working, which has no analogue in a cleared market.

INTERVIEW TRAP

1. **Reading funding as a small number.** It is quoted per eight hours. Multiply by 1,095 before deciding it is negligible.
2. **Treating funding as a fee to the exchange.** It is a transfer between traders, and the exchange does not guarantee it.
3. **Computing the cost on margin rather than notional.** Funding is charged on notional, so leverage multiplies its effect on your capital.
4. **Calling the basis trade riskless.** It is short a funding flip, short a liquidation, short two venues, and in some designs short an auto-deleveraging.
5. **Using extreme funding as a timing signal.** It says the crowd is levered and the market is fragile. It does not say when, and being early is indistinguishable from being wrong while you are paying to hold the position.
6. **Forgetting the funding timestamp.** Positions opened and closed around the mark change the effective cost, and the liquidity around those moments is not the liquidity you tested against.

VISUAL / CODE TO BUILD: Interactive

A tether panel. The spot index runs across the chart with the perpetual drawn over it and a funding bar beneath, and the reader can push the perpetual away from the index and watch the funding bar respond and the perpetual drift back. Setting the funding mechanism to "off" and pushing it again shows the price staying dislocated indefinitely, which is the argument for why the mechanism exists rather than an assertion that it works.

A carry calculator sits alongside: notional, leverage and funding per period, with the output shown three ways, per period, per day and annualised, plus the cost as a percentage of *margin* rather than notional. The gap between the last two figures at ten times leverage is usually what surprises people.

A third panel simulates a liquidation cascade. The reader sets the distribution of leverage across longs and applies a price shock; the panel liquidates the positions whose margin is exhausted, feeds those sales through a market-impact function, and repeats. Raising the average leverage turns a shock that damps out into one that does not, and the funding rate is shown rising in the run-up, which makes the link between crowded funding and fragility a mechanism rather than a correlation.

Where this goes next

The asset the funding rate is quoted against is crypto itself, the reason a derivative rather than the underlying carries most of the volume is spot versus derivatives, and the venue fragmentation that turns the cash-and-carry trade into a two-exchange credit position is market structure.

20.5 DeFi basics

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Decentralised finance is a set of familiar financial functions, lending, market making, leverage, rebuilt as programs that run on a public blockchain rather than as contracts enforced by a legal system. The economics are not new. What changes is the enforcement mechanism, and every distinctive feature of DeFi follows from one consequence of that change: **there is no counterparty to sue and no identity to assess**, so every obligation has to be secured by assets the code already holds.

That single constraint explains overcollateralisation, automatic liquidation, and why lending rates are set by a formula rather than by a credit committee. This section covers the two primitives worth understanding, lending and automated market making, and treats them as variations on things the rest of the book already describes.

Lending: securities lending with the trust removed

A DeFi lending protocol is a pool. Lenders deposit an asset and receive a claim that accrues interest; borrowers post collateral and draw the asset out. No lender is matched to a borrower, and no one underwrites anyone.

> KEY IDEA
>
> Because the protocol cannot assess a borrower or pursue one, loans are **overcollateralised**: you post more value than you borrow, typically borrowing 50 to 80% of the collateral's value. And because there is no margin desk to call you, liquidation is **open to anyone**: once the loan breaches its threshold, any participant may repay part of it and seize collateral at a discount, usually 5 to 10%. The margin call has been replaced by a public bounty.
>
> That is the design in one sentence, and it has a strong consequence. DeFi lending cannot extend credit in the economic sense, because credit is precisely the thing you extend to someone whose assets you do not already hold. It provides **liquidity against assets**, which is repo, not lending.

The structure is not new. Its closest traditional analogue is securities lending, where a lender parts with stock only against collateral worth more than the stock, marked daily, and earns the difference between the rate paid on that collateral and the market rate. Why a desk borrows stock in the first place, and the point that the transaction is really a collateralised loan of cash dressed as a loan of a security, belong to stock borrow and lending.

What DeFi changes is only the enforcement. Post collateral worth more than you take, be marked continuously rather than daily, and be closed out automatically rather than after a phone call. The reason these protocols overcollateralise is the reason a securities lender does: with no recourse to a legal claim on the borrower, the collateral is the entire credit assessment.

Automated market makers

The second primitive replaces the order book with a formula. A pool holds two assets, and anyone may swap one for the other at a price the formula determines from the pool's current balances. The dominant design is the **constant product** rule: with x of one asset and y of the other,

$$x \cdot y = k \quad \text{must hold after every trade,}$$

so the marginal price is y/x and buying moves it against you automatically. Nobody quotes, nobody manages inventory, and the pool can never run out, because the price goes to infinity as x approaches zero.

WORKED EXAMPLE

A pool holds 100 units of an asset and 200,000 units of cash, so $k = 20{,}000{,}000$ and the marginal price is $200{,}000/100 = 2{,}000$.

Buy one unit. The pool must end with $x = 99$, so it must end with $y = 20{,}000{,}000/99 = 202{,}020.20$ of cash, which means you pay $202{,}020.20 - 200{,}000 = 2{,}020.20$.

Your effective price is 2,020.20 against a marginal price of 2,000, so the slippage on a 1% trade is $2{,}020.20/2{,}000 - 1 = 1.01\%$. Everything about the execution is known in advance from the formula, with no quote and no discretion, which is the genuine innovation: **the market maker's pricing function is public and identical for everyone**, unlike the process in market makers.

The cost of that simplicity falls on whoever supplies the pool.

INTUITION

A liquidity provider is always selling whichever asset is rising and buying whichever is falling, because the formula forces it: as the price moves up, arbitrageurs buy the asset out of the pool until its price matches the outside market. The provider therefore ends up with less of the asset that went up and more of the one that went down, which is worse than simply holding both.

That gap is called **impermanent loss**, an unfortunate name because nothing about it is impermanent unless the price returns to where it started. The right description is that **providing liquidity to a constant-product pool is a short volatility position**: the provider loses whenever the price moves in either direction, gains nothing from direction, and is compensated by the trading fees, which are the premium. It is a straddle sold in disguise, and it should be evaluated the way any short volatility position is, by comparing the premium earned against the movement realised.

WORKED EXAMPLE

For a constant-product pool, the loss relative to simply holding the two assets depends only on the ratio k by which the relative price has moved:

$$\mathrm{IL} = \frac{2\sqrt{k}}{1+k} - 1.$$

Read off three cases:

- price moves 25%, $k = 1.25$: loss of 0.62%
- price doubles, $k = 2$: loss of 5.72%
- price quadruples, $k = 4$: loss of 20.0%

Two things to notice. The loss is **symmetric in direction**: $k = 0.5$ gives the same 5.72% as $k = 2$, which is the signature of a short volatility position. And it is **convex**, growing far faster than the price move, so a pool that is comfortable through small moves is not comfortable through large ones. A provider earning 20 basis points of fees on turnover needs a great deal of turnover to cover a move that doubles the price.

INTERVIEW TRAP

Three errors. (1) Calling DeFi lending "lending". Without identity or recourse it can only be secured borrowing against assets already posted, which is repo. (2) Treating impermanent loss as a rounding item. It is a short volatility position, it is convex, and it is symmetric in direction, so it should be priced against the fees earned rather than mentioned as a footnote. (3) Assuming a smart contract removes counterparty risk. It replaces a legal claim on a counterparty with a technical dependency on code and on an oracle feeding it prices, and both have failed. The risk moved rather than disappeared.

Where this goes next

The technology underneath, and what a smart contract actually is, is blockchain fundamentals. The unit of account almost every one of these pools is quoted in is stablecoins, the levered instrument that dominates the derivative side is perpetual futures, and the venues and their credit arrangements are market structure in crypto. The traditional counterpart of the pricing function is market makers, and the funding exposure that haircuts create is liquidity risk.

20.6 Stablecoins

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A stablecoin is a token that promises to be worth one dollar. That promise is the entire product, and the interesting question is never whether the price is one dollar today but *what makes it one dollar*, because the answer determines exactly how it fails. Three designs are in use, they fail in three different ways, and only one of them has never had a serious failure.

INTUITION

A peg is not a price. It is a **redemption promise** plus somebody willing to arbitrage against it. If you can always exchange the token for a dollar, then whenever it trades at 0.99 somebody buys it and redeems for a one cent profit, and that flow is what holds the price. Every stablecoin failure is a failure of the redemption promise, not of the price.

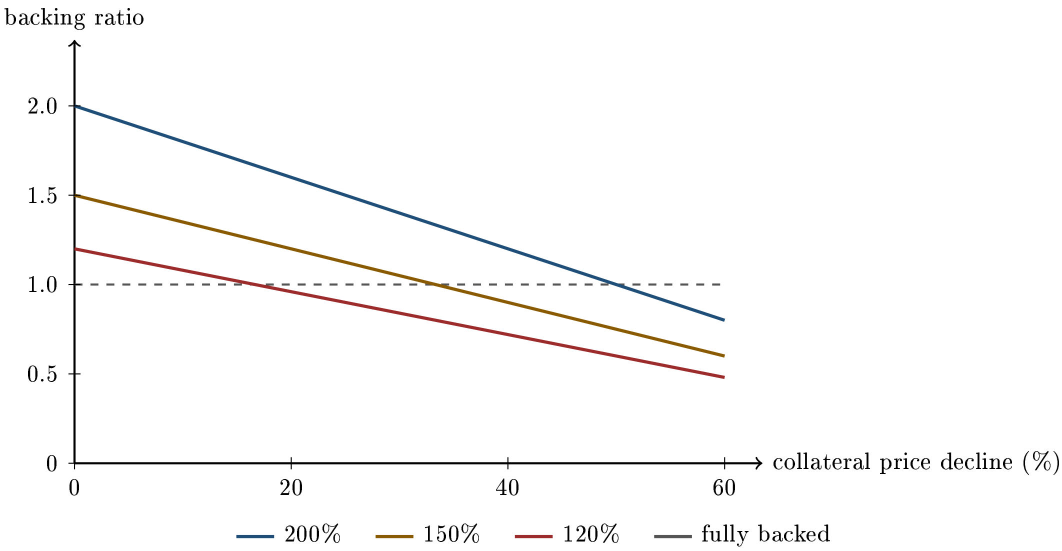
Three designs

Fiat-collateralised. A company holds dollars and short-dated government paper, and issues one token per dollar held. Redemption is a legal claim on the issuer. The peg is as good as the reserve and as good as the issuer's willingness and ability to honour it, which makes this a money market fund with a blockchain distribution channel. Its failure mode is a *run*: doubts about the reserve produce redemptions, and the reserve has to be liquidated into whatever market exists that day.

Crypto-collateralised. A user locks volatile crypto collateral in a smart contract and mints stablecoin against it, with the contract enforcing overcollateralisation and liquidating positions that fall below a threshold. There is no issuer to trust, which is the point, and no issuer to sue, which is the cost. Its failure mode is a *collateral crash faster than the liquidations can clear.*

Algorithmic. No collateral at all. The protocol offers to mint or burn a companion token worth one dollar in exchange for one stablecoin, so arbitrage is supposed to enforce the peg. Its failure mode is *reflexivity*, and it is discussed below because it is the one that matters conceptually.

How much overcollateralisation is enough



a fiat-backed coin sits on the dashed line regardless of this axis, which is the whole difference

The arithmetic is one line. Post collateral worth C times the stablecoin issued, and after a decline d the backing ratio is $C(1-d)$. It reaches one, the point at which the stablecoin stops being fully backed, when

$$d = 1 - \frac{1}{C}.$$

So 200% overcollateralisation survives a 50% fall, 150% survives 33.3%, and 120% survives only 16.7%. Those are not comfortable margins for assets that routinely move 20% in a day, and the buffer has to cover not just the fall but the fall *during the time it takes to liquidate*, on a chain that is congested precisely when everyone is liquidating at once.

Note what the dashed line means for the fiat design: its backing does not depend on this axis at all. That is not a small advantage, and it is why the fiat-collateralised design dominates by size despite requiring exactly the trusted intermediary the technology was meant to remove.

WORKED EXAMPLE

A vault, and what liquidation actually costs. Deposit 10 ETH at $2,000, so $20,000 of collateral. The protocol requires 150% and charges a 13% penalty on liquidation.

Mint the maximum, 20,000/1.5 = $13,333, and you are at the threshold on day one: any fall at all liquidates you. So mint $10,000 instead, a 200% ratio, leaving $10,000 of equity.

Liquidation triggers when collateral falls to $1.5 \times 10{,}000 =$ $15,000, which is ETH at $1,500, a **25% decline**. At that point the position is closed: $15,000 of collateral, less the $10,000 debt, less a $1,300 penalty, leaves **$3,700**.

Your equity went from $10,000 to $3,700, a **63% loss on a 25% price fall**. Fifty points of that is simply the two times leverage; the other thirteen are the penalty. The protocol is fine, the stablecoin stays backed, and the user has lost far more than the market move.

KEY IDEA

Overcollateralisation transfers risk rather than removing it. The stablecoin holder is protected by a buffer that the borrower pays for, and the borrower is paying with a levered position and a punitive liquidation. Ask who holds the risk in any structure that claims to have engineered it away, because it is always somebody.

The reflexive design, and why it cannot work

The algorithmic construction sounds elegant. If the stablecoin trades below a dollar, burn it and receive a dollar's worth of the protocol's companion token, which is minted for the purpose. If it trades above, do the reverse. Arbitrage restores the peg and no collateral is required.

The flaw is that the companion token's value comes from the protocol, whose value comes from the stablecoin being used. The collateral is a claim on the confidence of the people the collateral is supposed to reassure.

Follow it through under stress. Redemptions begin, so companion tokens are minted and sold, so their price falls, so the next redemption requires *more* of them, so more are minted and sold. Each round makes the next round worse, and the process has no equilibrium other than zero. This is not a liquidity event that a backstop can arrest; the mechanism designed to defend the peg is the mechanism destroying it.

KEY IDEA

The test to apply to any stablecoin is simple: **if confidence in this token disappeared entirely, what would the backing be worth?** For a fiat design, the reserve is still there, subject to the issuer being honest. For a crypto-collateralised design, the collateral is still there, subject to it being liquidated in time. For an algorithmic design the answer is zero, by construction, and no amount of mechanism design changes it. The Terra collapse of May 2022 took about a week to run from the first depeg to a companion token worth essentially nothing.

The fiat design has a failure mode too

It is a run, and it happened. In March 2023 a major dollar stablecoin held roughly $3.3 billion of its reserve as deposits at a bank that failed over a weekend. That was under a tenth of the reserve, and the token traded down to around 87 cents.

The arithmetic is worth doing, because it does not match. An impairment of under 10% produced a discount of roughly 13%, and in fact the deposits were made whole and the token returned to par within days. What was being priced was not the expected loss; it was the possibility of loss combined with the fact that redemption was suspended over the weekend. A promise you cannot exercise on Saturday is worth less than one you can.

That is the general point about reserves. What matters is not only the size but the composition and the liquidity: bank deposits are somebody else's credit, commercial paper cannot be sold at par on demand, and only short government paper and central bank balances behave the way holders assume the whole reserve behaves. Regulatory regimes for stablecoins have converged on exactly these questions, requiring segregated high-quality reserves, redemption at par within a fixed window, and public attestation.

Why a market professional cares

Stablecoins are not an investment. Nobody holds one for return; a fiat-backed stablecoin is a dollar that pays you no interest while the issuer earns it. They matter as **plumbing**.

- **They are the settlement rail.** Moving dollars between crypto venues at the weekend is not possible through the banking system, and a stablecoin transfer is.
- **They are the quote currency.** Most crypto derivatives are quoted, margined and settled in stablecoin, so a depeg is a repricing of every position on the venue at once, not an isolated event in one token.
- **They are the funding leg.** The cash-and-carry basis trade in crypto is funded in stablecoin, so stablecoin supply and the basis move together, and the connection to perpetual funding is in cryptocurrencies.
- **They are a genuine short-term paper buyer.** The largest issuers hold enough government bills for their flows to be visible in that market, which is the point at which a crypto question becomes a rates question.

Interview angle

The question database contains no cryptocurrency rows at all, which is the same finding recorded in cryptocurrencies and is informative in itself: this is not yet screened for on the desks in the sample. It comes up as market awareness rather than as a technical test.

If it does come up, the answer that distinguishes you is to refuse the taxonomy and go to the promise. Anyone can list fiat-backed, crypto-backed and algorithmic. What shows understanding is saying that a peg is a redemption promise, that the design determines who honours it and with what, and then applying the confidence test above: if belief in the token vanished, what is the backing worth? That single question sorts the three designs correctly, predicts the Terra outcome, and explains why a bank-deposit concentration in a reserve was worth a 13 cent discount for a weekend.

> **INTERVIEW TRAP**
>
> Four errors. (1) Treating the peg as a property of the token rather than as the output of an arbitrage against redemption. (2) Reading a stablecoin's market price as evidence its reserve is sound; the price holds right up until it does not. (3) Assuming overcollateralisation removes risk, when it moves the risk to the borrower and depends on liquidations clearing during the crash. (4) Calling an algorithmic stablecoin undercollateralised, when the accurate description is that its collateral is a claim on itself.

Where this goes next

The asset class these tokens sit inside, and the perpetual futures they margin, are cryptocurrencies. The settlement layer that makes the transfer work is blockchain fundamentals, the lending and liquidation protocols that mint the collateralised designs are DeFi basics, and the venues where a depeg reprices everything at once are market structure in crypto. The general question of what a peg is and why defending one is hard, in a market with a much longer history, is exchange rate regimes.

20.7 Crypto volatility

trading · structuring · sales | advanced | crypto

Volatility is the parameter everything in derivatives is scaled by, so an asset class whose volatility runs several times that of an equity index is not the same business conducted on a different underlying. The mathematics transfers unchanged from implied versus realised volatility and ARCH and GARCH. What changes is the level, the shape, the conventions used to measure it, and the reasons behind all three.

The convention trap: 365, not 252

Start here, because it is the most common error and it is arithmetic rather than judgement. Annualising a daily volatility means multiplying by the square root of the number of periods in a year, and crypto trades **every day**, with no weekends or holidays, so the factor is $\sqrt{365} = 19.1$ rather than the $\sqrt{252} = 15.87$ used for markets that close.

> WORKED EXAMPLE
>
> A coin moves 4% in a day. Annualised on the crypto convention that is $0.04 \times 19.1 = 76.4\%$; on the equity convention it would be $0.04 \times 15.87 = 63.5\%$. The same observation produces numbers **almost thirteen volatility points apart** purely from the choice of calendar, which is more than enough to turn a profitable quote into a loss. The desk shortcut also changes: the familiar rule that a 16 volatility implies a 1% daily move becomes "divide by 19" here, so a 19 vol is the one that implies a 1% day.

Two more measurement conventions matter and are worth asking about before comparing anyone's numbers. Because there is no close, the **daily cut** is arbitrary, so two desks sampling at different UTC hours compute genuinely different realised volatilities from the same market. And because liquidity is fragmented across venues, the **index construction** behind the price series determines the answer, which is why quoted volatilities are only comparable when they reference the same index and the same cut, see spot versus derivatives in crypto.

What the volatility actually looks like

Stated as regularities rather than levels, since levels date badly.

- **The level is a multiple of equity index volatility**, historically several times it for major coins and higher again for smaller ones, though it has compressed as the market has matured and as institutional participation has grown.
- **Clustering is strong.** Volatility is highly autocorrelated and mean-reverting, so the ARCH and GARCH family applies directly, and the *persistence* of shocks is if anything greater than in equities. The diagnostic is the same one built in autocorrelation: returns near white noise, squared returns strongly autocorrelated.
- **Volatility of volatility is high**, and the term structure inverts sharply in stress, with short-dated implied vol spiking far above long-dated, then normalising quickly.
- **The skew is regime-dependent, and this is the genuine structural difference.** An equity index carries a persistent *put* skew, because crash risk and structural downside hedging demand both sit on the same side, as in skew and smile. Crypto skew **changes sign with the regime**: in strong upward phases, demand for leveraged upside can bid calls above puts, producing call skew that would be extraordinary in an index; in stress it flips to a steep put

skew like everything else. So the sign of the skew is information about positioning rather than a structural constant.
- **The implied-to-realised relationship is familiar**: implied usually sits above realised, so there is a volatility risk premium, sustained by structural yield-seeking through covered calls and yield-enhancement products, and it collapses or inverts violently in a deleveraging episode.
- **Correlation to traditional risk assets is unstable.** It has been near zero for long stretches and has risen sharply in liquidity-driven selloffs, which is the least convenient possible behaviour for anyone holding it as a diversifier.

Why it is so high

Four reasons, and naming them is better than repeating that the asset class is speculative.

1. **No cash-flow anchor.** An equity has earnings and a bond has coupons, so there is a discounted-cash-flow argument that pulls price towards something. A token with no cash flow has no such anchor, so its price is entirely an expectation about future adoption and future demand, and expectations are free to move a long way without any accounting fact contradicting them.
2. **Leverage and forced liquidation.** Positions are margined mechanically and liquidated by an engine rather than negotiated, so a move triggers liquidations that cause the move to continue, which is a reflexive amplifier with no circuit breaker, described in spot versus derivatives in crypto.
3. **Thin depth relative to notional.** Order books are shallow against the size that trades in stress, and because liquidity is split across venues there is no single deep book to absorb a shock.
4. **Policy and headline risk with no lender of last resort.** A regulatory announcement or a venue failure is a genuine step change in the asset's prospects, and there is no central bank whose reaction function dampens it.

Trading it, briefly

Options are concentrated on very few venues, they are frequently **coin-margined**, so vega and premium are denominated in the volatile asset itself, and the resulting position has the embedded convexity described in spot versus derivatives in crypto. Perpetual funding is a partly vol-adjacent carry, since it is bid up by the same leveraged demand that bids up calls, so funding and skew often tell the same story from two directions, see funding rates. The instrument detail is in crypto options.

REAL INTERVIEW QUESTION (BNP · EQD)

Is it better to use implied volatility or historical volatility to estimate future volatility?

WHAT THEY TEST / ANSWER

Answer **implied, as the starting point, with corrections**, and justify it on information content rather than on preference. Implied volatility is a forward-looking price set by participants who must hedge it, so it incorporates known future events, earnings, elections, expiries, in a way a backward-looking sample cannot, and empirically it is the better single predictor of realised volatility over the matching horizon. Then give the two corrections that make the answer honest. Implied contains a **volatility risk premium**, so it is biased *upward* as a forecast, because option sellers require compensation for bearing gap and vega risk; the sensible use is therefore implied minus an estimated premium rather than implied itself. And implied is only available where a liquid option market exists at the strike and maturity you need, so historical volatility, ideally in a model that respects clustering such as an exponentially weighted estimator or GARCH, is what you fall back on and what you use to *sanity check* implied. Say how you would combine them: implied for the level and the term structure, a clustering model for the short-horizon dynamics, and a comparison of the two as a signal in itself, since a large gap is either a positioning opportunity or a warning that you are missing a known event. In crypto specifically, add the two adjustments this section supplies: annualise with $\sqrt{365}$ rather than $\sqrt{252}$ because the market never closes, and treat the implied-to-realised gap with more caution because the option market is concentrated on few venues and its premium is less stable than in an index.

REAL INTERVIEW QUESTION (Natixis · EQD)

What shape does the volatility of an underlying have in reality, contrary to the assumptions of Black-Scholes?

WHAT THEY TEST / ANSWER

Black-Scholes assumes volatility is a single constant. Reality contradicts that in **three separate directions**, and naming all three is the answer. **Across strike**: implied volatility varies by strike, which is the smile or skew, and for an equity index it is a persistent put skew because crash risk and hedging demand sit on the same side, see the volatility smile and skew. **Across maturity**: there is a term structure, usually upward sloping in calm markets and inverting in stress as short-dated implied spikes above long-dated. **Through time**: volatility is stochastic, clustered and mean-reverting, so quiet periods follow quiet periods and shocks decay slowly, which is what ARCH and GARCH models describe in ARCH and GARCH, and which means volatility itself has a volatility. Add that returns are fat-tailed and that prices jump, so the continuous-path assumption fails at exactly the moments that matter. Then say what the industry does about it, which turns a list of criticisms into a professional answer: the surface is treated as an input to be fitted rather than a constant to be estimated, with local volatility, stochastic volatility and jump-diffusion models each repairing a different one of those failures, and Black-Scholes retained as a *quoting convention* that translates a price into a number people can compare. If the conversation is about crypto, the extra observation worth making is that the skew there is not structurally on one side: it flips sign with the regime, bidding calls in leveraged upward phases and puts in stress, so a shape you can rely on in an index is itself a variable in that market.

INTERVIEW TRAP

Four traps. First, **annualising with $\sqrt{252}$**, which understates crypto volatility by about a sixth ($\sqrt{252/365} = 0.83$) for no reason other than habit. Second, **comparing volatilities from different cuts or indices**, which are not the same measurement when there is no official close and liquidity is fragmented. Third, **assuming a persistent put skew**: the sign is regime-dependent here, so a strategy that is implicitly short the equity-style skew shape can be wrong-footed in a bull phase. Fourth, **quoting vega in the wrong currency**: on a coin-margined option the premium and the vega are denominated in the volatile asset, so a hedge sized in dollars drifts as the price moves.

VISUAL / CODE TO BUILD: Interactive

A volatility bench with three panels. **Convention**: a single realised-volatility calculator where the reader toggles the annualisation factor between $\sqrt{252}$ and $\sqrt{365}$ and moves the daily UTC cut, with the resulting volatility displayed for the same underlying price history, so the convention sensitivity is a number rather than a warning. **Clustering**: the return series alongside the autocorrelation function of returns and of squared returns, reproducing the diagnostic from autocorrelation on crypto data so the reader sees the flat ACF and the decaying one side by side. **Surface**: an implied volatility surface over strike and maturity with a regime slider that morphs it between a leveraged-upside state and a stress state, so the skew visibly flips sign and the term structure visibly inverts, with the equity-index surface shown alongside as the contrast. A readout tracks implied minus subsequently realised volatility over history, making the volatility risk premium and its collapses visible.

Where this goes next

The instruments are in crypto options, the leverage mechanism that amplifies the moves is in spot versus derivatives in crypto and funding rates. The general theory transfers from implied versus realised volatility, the volatility smile and skew and ARCH and GARCH, with the equity contrast in equity volatility and skew and smile.

20.8 Crypto options

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This is the corner of crypto where a traditional derivatives skill set transfers almost intact. A call is still a call, Black-Scholes still applies, and delta hedging still works. What differs is a short list of *conventions*, and every one of them changes a number you would otherwise quote wrongly. So this section deliberately explains none of the underlying theory: option payoffs are in options and standard payoff profiles, the Greeks are in Greek definitions, the pricing model is in Black-Scholes, and what a smile is belongs to skew and smile. What follows is only what is specific to this asset class.

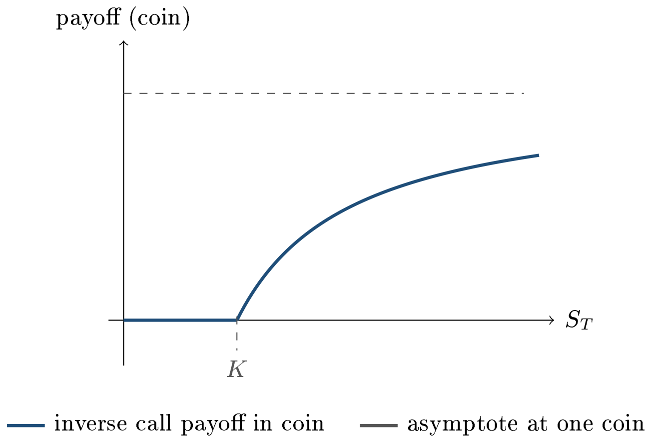
The big one: inverse, coin-margined options

On the venues where most crypto option volume trades, a BTC option is quoted, margined and **settled in BTC**, not in dollars. The premium is paid in coin, the margin is posted in coin, and the payoff arrives in coin. These are called **inverse** or coin-margined contracts, and they are the single most common place a newcomer from an equity desk gets the risk wrong.

A dollar-settled call pays $\max(S_T - K, 0)$ dollars. An inverse call delivers the same dollar value, but denominated in the coin, so the quantity you receive is

$$\text{payoff in coin} \;=\; \frac{\max(S_T - K,\, 0)}{S_T} \;=\; \max\Big(1 - \frac{K}{S_T},\, 0\Big).$$

Look at what that expression does as the spot rises. It does not grow without bound; it converges to 1. **An inverse call's coin-denominated payoff is capped at one coin, no matter how high the price goes.**



WORKED EXAMPLE

BTC spot is \$50,000 and you hold an inverse call struck at \$60,000.

- Expiry at \$80,000: dollar value $80{,}000 - 60{,}000 = \$20{,}000$, so you receive $20{,}000/80{,}000 = 0.25$ BTC.
- Expiry at \$120,000: dollar value \$60,000, so you receive $60{,}000/120{,}000 = 0.50$ BTC.
- Expiry at \$600,000: dollar value \$540,000, so you receive $540{,}000/600{,}000 = 0.90$ BTC.

The dollar payoff grew twenty-seven-fold across those scenarios; the coin payoff went from a quarter of a coin to nine-tenths and can never reach a whole one. If you are short this option and you hedge your coin exposure as though the coin payoff were unbounded, you are massively over-hedged in the tail.

KEY IDEA

There is an exact identity that makes the inverse option easy to think about, and it is the crypto version of a trick every FX trader knows. Since $1 - K/S_T = K\big(\frac{1}{K} - \frac{1}{S_T}\big)$ whenever $S_T > K$,

$$\text{inverse call struck } K \;=\; K \times \Big[\text{put on } \tfrac{1}{S} \text{ struck } \tfrac{1}{K}\Big].$$

An inverse call on the coin is exactly K puts on the *inverse* exchange rate, that is, on the price of a dollar measured in coin. This is a change of numeraire, and the reason it matters is that the quoted volatility, the delta, and even the put-call labelling all depend on which currency you are standing in. The same machinery is in FX options and delta conventions, where a EUR call is a USD put and the market has agreed conventions to stop people confusing the two. Crypto has inherited the problem without fully inheriting the conventions.

The volatility itself

Level. Crypto implied volatilities live in a different range from equity indices. Where a major equity index trades with an at-the-money implied volatility in the teens most of the time, BTC and ETH routinely sit several times higher, and single tokens higher still. Nothing in the maths changes, but everything that scales with volatility does: premiums are a large fraction of spot, the Black-Scholes approximation that an at-the-money option is worth about $0.4 \times \sigma \times S \times \sqrt{T}$ produces numbers that look absurd until you accept the volatility, and second-order effects that are negligible at 15% volatility are not at 80%.

Annualisation. Crypto trades twenty-four hours a day, seven days a week, with no weekend gap and no exchange holidays. Realised volatility is therefore annualised with $\sqrt{365}$, not the $\sqrt{252}$ trading-day convention used for equities. The convention and its consequences for measurement belong to crypto volatility; what matters here is the size of the error it causes when you compare a realised number to an option screen, so it is worth one example.

WORKED EXAMPLE

An asset has a daily return standard deviation of 3%.

$$\sigma_{\text{crypto}} = 0.03 \times \sqrt{365} = 0.03 \times 19.105 = 57.3\%,$$
$$\sigma_{\text{equity}} = 0.03 \times \sqrt{252} = 0.03 \times 15.875 = 47.6\%.$$

Same data, a difference of nearly ten volatility points, purely from the day count. Quote the wrong one against an implied volatility screen and you will conclude a straddle is cheap when it is not. Whenever someone hands you a realised volatility number, ask which day count it uses before you compare it to anything. Implied against realised is treated in implied versus realized volatility.

Smile shape. This is the asset-specific nuance worth knowing, and it is a genuine difference rather than a convention. Equity index skew is persistently downward sloping: puts are bid because the index crashes down and grinds up, and because leverage rises as prices fall. Crypto does not obey that pattern reliably. The distribution is closer to two-sided, with violent upside moves as real a feature as downside ones, so the surface often looks more like a symmetric smile than a skew, and the sign of the risk reversal flips with the regime: calls bid in a bull market as leveraged shorts scramble, puts bid in a deleveraging. A trader carrying the equity assumption that the risk reversal is structurally for puts will be systematically wrong here. Risk reversals as an instrument are in risk reversals, and the empirical behaviour of the underlying is in crypto volatility. The fat tails that produce this shape are the reason jump models earn their keep, as in jump diffusion models.

Carry, hedging and the forward

An equity option's forward is built from the interest rate and the dividend. A crypto option's forward is built from something else entirely, because there is no dividend and, on most venues, no conventional financing market. The hedging instrument of choice is the **perpetual future**, and the cost of carrying a delta hedge is the **funding rate** it pays or receives, which is set by the market's net positioning rather than by a central bank. See perpetual futures and funding rates.

Three practical consequences follow. Funding is volatile and can be large, so the carry component of an option position is itself a risk rather than a known drag. Where a proof-of-stake asset offers a staking yield, that yield behaves like a dividend for forward pricing, and an option holder does not receive it. And because margin is frequently posted in the coin itself and cross-margined across the account, a large adverse move simultaneously moves your position against you and reduces the value of the collateral backing it, which is wrong-way risk built into the market's plumbing. Liquidations then feed the move that caused them, so liquidity and volatility are far more tightly coupled than in listed equity options. See liquidity risk and market structure in crypto.

How this comes up in interviews

As noted in blockchain fundamentals, the question database behind this book contains no crypto questions at all, so this section carries no attributed question rather than an invented one. In practice the crypto question in a derivatives interview is a transfer test, not a knowledge test: they take something you already know and change one convention to see whether you notice. Rehearse three answers. Why is an inverse call's coin payoff capped at one coin. Why does a realised volatility computed on a crypto series need $\sqrt{365}$. Why might a crypto risk reversal be positive when an equity index one is not. Each is a one-minute answer and each demonstrates that you reason from the payoff rather than from memorised market lore.

INTERVIEW TRAP

Four traps. First, pricing an inverse option with a dollar-settled formula; the numeraire change is not a detail, it caps the coin payoff and shifts the delta. Second, mixing day counts when comparing realised to implied volatility. Third, importing the equity assumption that skew is always for puts; in crypto the sign is regime-dependent and you should look, not assume. Fourth, treating the funding rate as a small carry term the way you would treat a repo rate; it is set by positioning, it can be extreme, and it is the dominant cost of a delta hedge in this market.

VISUAL / CODE TO BUILD: Interactive

A payoff and Greeks comparator with a single toggle: dollar-settled versus inverse. The reader sets spot, strike, maturity and volatility, and the panel draws the two payoffs side by side, one unbounded and one asymptoting to a single coin, together with the two deltas so the divergence in the tail is visible rather than asserted. A second control switches the vol-annualisation basis between $\sqrt{252}$ and $\sqrt{365}$ on a realised-volatility readout computed from a live price series, so the reader can watch the same data produce two different numbers. The intended "aha" is that both differences come from conventions, not from new mathematics.

Where this goes next

The behaviour of the underlying that these options are written on is in crypto volatility, and the venues, liquidity and expiry structure that determine whether you can actually trade the surface are in market structure in crypto. For the model extensions that a fat-tailed, jumpy underlying demands, see extensions of Black-Scholes, and for how a vol book behaves when the regime shifts, Vega exposure and volatility regimes. Structured payoffs built on these options reuse the exotics taxonomy in exotic options.

20.9 Market Structure in Crypto

trading · structuring · sales core crypto

The instruments in crypto are familiar. The plumbing is not, and almost every way crypto has lost people money is a plumbing failure rather than a price move. This section is about the structural differences from a traditional market, in the order of how much they matter.

One entity does three jobs

KEY IDEA

In a traditional market the **exchange** matches trades, a **clearing house** stands between the counterparties and guarantees settlement, and a **custodian** holds the assets. Three separate institutions, separately regulated and separately capitalised, and the separation is the point: the failure of one does not take your assets with it.

A crypto exchange typically does **all three**. It matches, it is the counterparty to your derivative, and it holds your coins. So exchange failure is not a settlement inconvenience, it is the loss of the position *and* the collateral *and* the recourse.

Every large loss in the asset class runs through this concentration rather than through the volatility, and if you say only one thing about crypto market structure, say this.

The practical consequence is the phrase **not your keys, not your coins**. Assets held at an exchange are a claim on the exchange, not property you hold, and in an insolvency that claim ranks with everyone else's. Self-custody removes that exposure and replaces it with operational risk: a lost key is a permanent loss with no reissue and no helpdesk. The institutional answer is a qualified custodian with the assets segregated, which reconstructs by contract the separation that traditional markets get by regulation, and it is why custody is the first question an institution asks and the last one a retail participant thinks about. Counterparty risk covers what the exposure is; what is specific here is that it is unavoidable rather than managed.

Trading never stops, and the conventions change with it

Crypto trades twenty-four hours a day, every day. That has three consequences that are easy to state and easy to get wrong.

There is no overnight gap and no protection either. A traditional book carries gap risk across a closed market; a crypto book has no gap and also no pause, so a position that goes wrong at three in the morning goes wrong at three in the morning. Risk processes built around a daily cycle, an end-of-day mark, an overnight batch, a morning limit check, do not fit a market that has already moved twenty percent before the batch runs.

Weekend liquidity is thin and shocks concentrate there. Market makers reduce inventory when they cannot hedge into traditional markets, so the same order moves the price further on a Sunday, and forced liquidations therefore do more damage.

The annualisation convention is different, which is a small technical point that produces visibly wrong numbers when missed.

WORKED EXAMPLE

Two answers for the same volatility. A 3% daily standard deviation annualises as

$$3\% \times \sqrt{252} = \mathbf{47.6}\% \qquad \text{or} \qquad 3\% \times \sqrt{365} = \mathbf{57.3}\%,$$

depending on whether you count trading days or calendar days. Crypto has no non-trading days, so the second is correct, and using the equity convention understates the volatility by a factor of $\sqrt{365/252} = 1.204$, about twenty percent.

The same applies at higher frequency: an equity market is open roughly $6.5 \times 252 = 1{,}638$ hours a year against crypto's $24 \times 365 = 8{,}760$, a factor of **5.35**, so hourly volatility annualises with $\sqrt{8760} = 93.6$ rather than with the equity figure.

Twenty percent is not a rounding error on a volatility used to price an option, and this is the single most common conversion mistake made by someone arriving from a traditional desk.

Fragmentation without a consolidated tape

Traditional equity markets are fragmented too, but they are stitched together by regulation: a consolidated tape, a best-execution obligation, and in the United States a national best bid and offer. Crypto has none of that. Dozens of venues quote the same asset with no obligation to reference each other, so there is no single price, and the dispersion between venues is real rather than a data artefact.

That creates the dominant market-making strategy in the asset class, which is cross-venue arbitrage, and it also bounds it.

INTUITION

An arbitrage between two venues is only available to someone who can move inventory between them, so **the spread is bounded by the cost and time of transfer**, not by the willingness of anyone to trade it. Blockchain confirmation times, withdrawal limits, banking rails for the fiat leg, and capital controls in some jurisdictions all widen that bound.

This is the same structure as the location spreads in freight: a price difference smaller than the cost of moving the goods is not a trade, however large it looks. Persistent premia in particular jurisdictions are the crypto version of a landlocked crude discount, and they persist for exactly the same reason, which is that the arbitrage is bounded by infrastructure rather than by capital.

Liquidation is automatic and mechanical

There is no margin call and no telephone. A position whose margin falls below a maintenance threshold is closed by the venue's engine, immediately, at whatever price the book provides. Three components follow and each is worth knowing.

The liquidation engine sells into the market, which moves the price, which liquidates more positions. The market therefore contains a built-in forced seller whose size is known in advance from the distribution of leverage, which is the mechanism behind funding-driven cascades.

The insurance fund absorbs the shortfall when a liquidation closes below the bankruptcy price, and it is what prevents losses being socialised in ordinary conditions.

Auto-deleveraging is the backstop when the insurance fund is exhausted: the venue closes *profitable* positions on the other side to balance the book. A clearing house has comparable tools at the very end of its recovery waterfall (haircutting variation-margin gains, tearing up contracts), but they are last-resort measures applied to a failing clearer, whereas here the mechanism is routine and automatic. It means a correctly positioned trader can have a winning hedge removed at the worst possible moment. Any strategy relying on a short perpetual leg needs to know whether its venue does this.

What is the same

It is worth being explicit that the economics do not change, because a candidate who treats crypto as a different discipline is as wrong as one who treats it as identical. Market makers earn the spread and manage inventory exactly as in market making; adverse selection works the same way; order book dynamics, queue position and impact behave as microstructure describes. What differs is the absence of the institutional scaffolding, not the presence of new economics.

INTERVIEW TRAP

1. **Treating exchange balances as assets.** They are an unsecured claim on the venue. The separation of exchange, clearer and custodian is what traditional markets provide and crypto does not.
2. **Annualising with 252 days.** Crypto trades 365. The error is about twenty percent on a volatility, and it flows straight into an option price.
3. **Assuming one price.** There is no consolidated tape and no best-execution obligation, so a quoted price is a venue's price.
4. **Believing cross-venue spreads are free money.** They are bounded by transfer cost and time, which is the same constraint that bounds a commodity location spread.
5. **Running a traditional risk cycle.** An end-of-day process on a market that never ends will report yesterday's risk on a book that has already moved.
6. **Ignoring auto-deleveraging.** A hedge that the venue can close for you is not fully a hedge, and the design differs by venue.

VISUAL / CODE TO BUILD: Interactive

A venue-dispersion panel. The same asset's price is drawn on several simulated venues at once, with the spread between the best bid on one and the best offer on another shaded, and a slider for transfer cost and confirmation time drawing a band around zero within which the spread is not tradeable. Widening the transfer cost visibly swallows most of the apparent arbitrage, which is the section's point about bounded arbitrage delivered without an argument.

A second panel is a liquidation cascade with a leverage distribution the reader controls, showing the forced-selling loop and printing how much of the total move was the initial shock and how much was the engine. Setting the leverage distribution to something conservative and then to something aggressive, with the same shock, produces two very different totals.

A third is a clock: a 24/7 timeline with a traditional market's hours overlaid, and a simulated shock the reader can place anywhere on it. Placing it inside market hours and then at 03:00 on a Sunday, with the same shock size, shows the difference in realised impact when the market makers who would absorb it are not there, and the risk process the shock has to wait for.

Chapter 21

Risk Management

21.1 Market Risk

trading · structuring · sales | core | cross-asset

Market risk is the risk of losing money because prices move. That is the whole definition, and its usefulness is in what it excludes: not because a counterparty failed, which is counterparty risk, not because an issuer defaulted, which is credit risk, and not because you could not get out, which is liquidity risk. Market risk is price movement on positions you actually hold.

This section is the chapter's entry point. It sets out what the risk factors are, how a desk thinks about the risk it carries, and the one measurement question that comes up more than any other, which is whether volatility is a good measure of risk. The measures themselves, VaR, expected shortfall and stress testing, each have their own section and are referenced rather than derived here.

Four factors, and everything decomposes into them

Market risk is conventionally split by what moved:

- **Equity risk**, the level of share and index prices.
- **Interest rate risk**, the level and shape of yield curves.
- **FX risk**, exchange rates.
- **Commodity risk**, physical commodity prices.

Two more are usually carved out because they behave differently and need their own hedges: **volatility risk**, since an option book is exposed to implied volatility independently of spot, and **credit spread risk**, since a bond can lose value from widening spreads without anyone defaulting.

> **INTUITION**
>
> The taxonomy is not academic tidiness, it is how a book is hedged and how a loss is explained. Every position gets decomposed into sensitivities to these factors, the sensitivities are aggregated across the desk in Greek aggregation, and at the end of the day the P&L is attributed back to the same factors in P&L explain. If a day's P&L cannot be explained by the factor moves, either the model is wrong or there is a position nobody knew about, and both are worth finding out.

REAL INTERVIEW QUESTION (Natixis · EQD)

What risks does a trader have to manage?

WHAT THEY TEST / ANSWER

Answer with a structure rather than a list, because the structure is what is being tested. Three layers, moving outward from the position.

Layer one, the price risks in the book. Directional exposure to spot through delta, convexity through gamma, implied volatility through vega, the passage of time through theta, and rates and dividends through rho and the dividend sensitivity. Add the cross exposures if it is an exotic book, vanna and volga, and add correlation if it is a multi-asset book. Name these as the things the desk hedges every day, and reference the Greeks rather than defining them.

Layer two, the risks that are not in the Greeks. This is where the answer becomes good. Gap risk, because delta hedging assumes you can trade continuously and a jump means you cannot. Basis risk, because the hedge and the position are rarely the same instrument. Liquidity risk in two senses, being unable to exit a position and being unable to fund it. Model risk, because everything above is computed from a model that is wrong in known ways. And correlation breakdown, since a hedge that relies on a historical relationship stops being a hedge exactly when markets stress.

Layer three, the risks around the trading. Counterparty risk on the people you face, operational risk on booking, settlement and fat fingers, and limit or regulatory risk, since a position that breaches a limit has to be cut whether or not the trader still likes it.

Close on the point that unifies them. The first layer is measured continuously and hedged; the second is what actually causes large losses, because it is exactly what the measurement assumes away. Saying that explicitly is the answer to what is really being asked, which is whether you think risk management is a calculation or a habit of mind.

Is volatility a good measure of risk?

REAL INTERVIEW QUESTION (Allianz Global Investors · Market Risk)

Is volatility a good measure of risk?

WHAT THEY TEST / ANSWER

Give a genuine two-sided answer, because a flat yes or no is wrong in both directions, and then land on the resolution.

Where it is good. It is the only risk number that aggregates properly. Volatility is built from a covariance matrix, so portfolio volatility accounts for diversification automatically, it decomposes into per-position contributions that sum exactly to the total, and it is estimable from data everyone has. Nothing else in risk management has that combination of properties, which is why every framework is built on it whether or not it admits so.

Where it fails, and be specific. Four failures, in order of how much money they have cost.

It is symmetric. Volatility treats an up move and a down move as identical, and nobody has ever complained about upside. What a client cares about is loss, which is why downside deviation and drawdown exist as alternatives.

It only sees two moments. A distribution is not described by its mean and variance unless it is normal, and financial returns are not. Skewness and kurtosis are exactly the features that make a strategy dangerous, and volatility is blind to both, which descriptive statistics develops.

It is backward-looking and it clusters. An estimate from a quiet period says the position is safe precisely because it has been safe, and volatility is autocorrelated, so the calm reading is self-confirming until it is not. That clustering is the subject of GARCH.

It assumes you can hold on. A position with modest volatility that cannot be exited has more risk than the number says, because the risk that matters is the loss you are forced to take, not the one you would have recovered from.

Then the example that proves it, which is the strongest thing you can offer here. Take a strategy that returns $+0.8\%$ in twenty-three months out of twenty-four and -10% in the twenty-fourth. Its mean is 0.35% a month, or 4.2% a year, and its monthly standard deviation is 2.16%, which annualises to $\mathbf{7.5\%}$. A symmetric strategy with the same mean and the same 7.5% volatility has identical risk by this measure. But its skewness is $\mathbf{-4.6}$, and its worst month is -10% against a two-standard-deviation move of 4.3%. Volatility says the two are the same; everything a risk manager cares about says they are not. That is what selling options looks like on a risk report, and it is why the report is not enough.

Resolve it. Volatility is a good *aggregator* and a poor *descriptor*. Use it to combine and allocate risk, and use tail measures and stress tests to describe what can actually happen. A candidate who says both, in that order, has answered the question.

Risk is not weight

The single most useful consequence of measuring risk rather than capital.

WORKED EXAMPLE

A 60/40 portfolio is not 60/40. Take 60% in equities at 18% volatility and 40% in bonds at 6%, correlation 0.2. Portfolio volatility is

$$\sqrt{0.6^2(0.18)^2 + 0.4^2(0.06)^2 + 2(0.6)(0.4)(0.2)(0.18)(0.06)} \;=\; \mathbf{11.52}\%.$$

Now split that between the two holdings. Each position's risk contribution is its weight times its covariance with the portfolio, divided by portfolio volatility, and the two contributions sum exactly to the total:

$$\text{equities: } 10.57\%, \qquad \text{bonds: } 0.95\%, \qquad \text{total } 11.52\%.$$

So a portfolio that is **60/40 by capital is 92/8 by risk**. The bond allocation is doing almost nothing, and an investor who believes they hold a balanced portfolio holds an equity portfolio with a small hedge. This is the entire argument behind risk parity, and it is also the reason a desk sets limits in risk terms rather than in notional: notional is what you bought, risk is what you own.

Where the discipline came from

The discipline has a specific origin worth knowing, because it explains why every bank measures risk the same way. **RiskMetrics** was JP Morgan's market-risk methodology, published openly in 1994 together with the covariance datasets needed to run it, and it grew out of an internal demand for a single daily number showing the firm's total market risk across every desk. Publishing it rather than keeping it proprietary effectively created market-risk management as a common discipline, and the 1996 Basel amendment that let banks use internal models for capital was built on the consensus it produced. Technically it is a parametric variance-covariance approach with an exponentially weighted variance estimate, and time series analysis owns that machinery.

The criticisms are the part that matters here, because they are why the chapter has the shape it does. Joint normality understates tails, a quadratic form cannot represent the convexity of an option book, and the exponential weighting that makes the measure responsive also makes it **procyclical**, cutting measured risk after quiet periods and spiking it after a shock, so every institution using it is pushed to de-risk at the same moment. Historical simulation, expected shortfall and stress testing are all answers to one or another of those three.

Who owns it

Market risk in a bank is organised in **three lines of defence**, and the phrase is worth knowing because it will be used at you.

The **first line** is the desk: the trader owns the risk, hedges it and is accountable for the P&L. The **second line** is the independent risk function: it measures, sets and monitors limits, and it reports through a chain that does not end at the head of trading, which is the entire point. The **third line** is internal audit, checking that the first two are doing what they claim.

INTERVIEW TRAP

1. **Confusing notional with risk.** A €100 million position in a two-year bond and the same notional in a thirty-year bond are not remotely the same risk. Every serious limit is expressed in sensitivities or in volatility terms.
2. **Confusing market risk with credit risk on the same trade.** A corporate bond has both; the spread widening is market risk and the default is credit risk, and they are managed by different teams with different limits.
3. **Believing the risk report.** Every number in it is conditional on a model and on a sample. The purpose of a stress test is precisely to ask what happens outside that sample.
4. **Assuming diversification holds under stress.** Correlations move toward one in a crisis, so the diversification embedded in a covariance matrix is largest exactly when you least have it.
5. **Treating the risk function as an obstacle.** In an interview this reads badly and on a desk it ends careers. The right framing is that limits are the terms on which the firm lends you its balance sheet.

VISUAL / CODE TO BUILD: Interactive

A risk-contribution panel. The user sets two or three asset weights, their volatilities and the correlations between them. One bar shows the split by *capital* and a second, directly beneath it, shows the split by *risk contribution*, with both normalised to the same length so the difference between them is immediately visible.

The intended moment is starting at 60/40 and watching the risk bar sit at roughly 92/8, then dragging the bond weight up and seeing how far it has to go before the risk bar looks balanced, which is a far more convincing argument for risk parity than the arithmetic. A correlation slider then shows the contributions shifting without any weight changing at all.

A second panel addresses the volatility question directly: two return histograms are drawn side by side with identical mean and identical standard deviation, one symmetric and one built from many small gains and one large loss, with the shared volatility printed above both and the skewness, worst month and maximum drawdown printed below each. Seeing one number match and three differ is the argument the section makes in words.

21.2 Credit risk

trading · structuring · sales core credit · cross-asset

Market risk is the risk that a price moves. Credit risk is the risk that someone does not pay, and the two behave so differently that banks manage them in separate departments with separate models and separate capital.

The difference is in the shape. A market position is roughly as likely to gain as to lose, and its distribution is roughly symmetric. A loan cannot do better than being repaid. Almost every day nothing happens, and occasionally you lose most of the principal. That asymmetry is the whole subject.

The credit *market* is covered elsewhere: how spreads are quoted in credit spreads, how default probabilities are extracted in default risk, and what recovery means in recovery rates. This section is about credit as a risk category a bank has to measure, provision for, capitalise and limit.

Four things called credit risk

The word covers several distinct exposures, and a risk manager separates them because they are managed differently.

- **Default risk**: the borrower does not pay. The headline case.
- **Migration risk**: the borrower does not default but is downgraded, so the position is marked lower. This is where most of the actual profit and loss in a credit book comes from, because downgrades are far more common than defaults.
- **Concentration risk**: the portfolio is not diversified, so one name, sector or country can produce a loss out of proportion to its size. Quantified below.
- **Settlement risk**: you have delivered and the other side has not yet. Brief, usually intraday, and occasionally enormous, since the exposure is the full principal rather than a mark-to-market difference.

The related exposure that arises when the amount at risk is itself uncertain, because it depends on where a derivative is marked, is counterparty risk, and it is counterparty risk.

Expected loss: the number you price, not the number you fear

The measurement of credit risk starts with three quantities and multiplies them.

$$\mathrm{EL} \;=\; \mathrm{PD} \times \mathrm{LGD} \times \mathrm{EAD}$$

- **PD**, the probability of default over the horizon, usually one year. Estimated from ratings, from models, or backed out of market prices, which are different numbers for reasons set out in default risk.
- **LGD**, loss given default, the fraction not recovered, so 1 minus the recovery rate. See recovery rates.
- **EAD**, exposure at default, how much is actually owed at the moment it happens. For a drawn loan this is the balance; for an undrawn facility it is more, because borrowers draw down facilities on the way into trouble; for a derivative it is whatever the position is worth then, which is why it needs its own section.

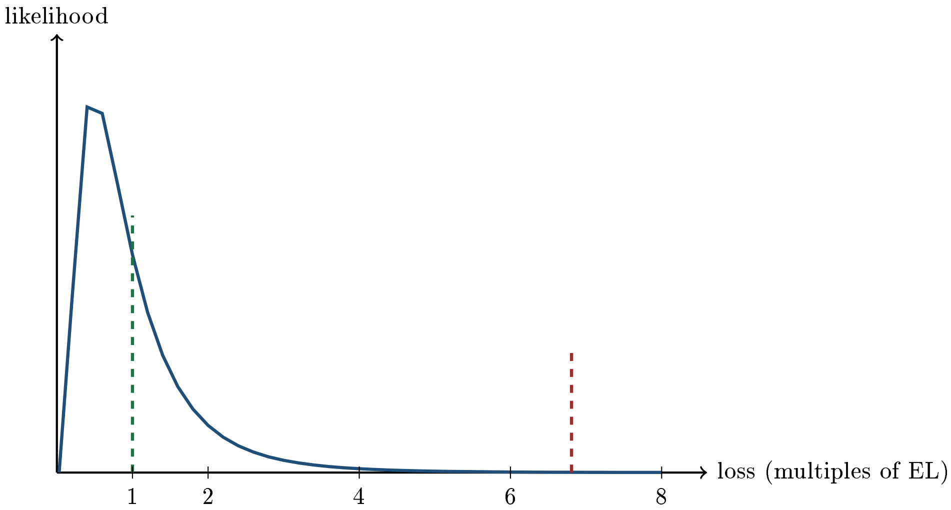
> WORKED EXAMPLE
>
> A £10 million loan to a borrower with a 1% one-year default probability and a 45% loss given default:
>
> $$\mathrm{EL} = 0.01 \times 0.45 \times 10{,}000{,}000 = 45{,}000.$$
>
> That is the number the loan must be priced to cover: 45 basis points a year of pure credit cost before any funding, capital or profit. If the loan is quoted at 40 basis points over funding it is a losing business on average, whatever happens to this particular borrower.

KEY IDEA

Expected loss is a **cost**, not a risk. It is known in advance, priced into the spread, and set aside as a provision. What a bank holds capital against is the amount by which losses in a bad year exceed that expectation, the **unexpected loss**. Confusing the two is the central error in credit risk: EL goes in the income statement, UL goes in the capital calculation.

The loss distribution, and where the capital goes

Put a whole portfolio's losses on one axis and the picture explains the entire framework.



portfolio credit loss distribution — expected loss, covered by provisions — 99.9% quantile, covered by capital

Three features are worth naming, because each one drives a piece of the regulatory machinery.

The mode is below the mean. The single most likely outcome is a loss smaller than the expected loss, which means a credit book looks like it is beating its budget in most years and is not. Any performance measure that rewards a manager for a normal year without charging for the tail is measuring the wrong thing.

The tail is long and one-sided. There is no offsetting good tail, because a loan cannot repay more than it owes. So the distance from the mean to the 99.9% quantile is large, and that distance, not the mean, is what capital is sized against.

The quantile is far out. Regulatory credit capital is calibrated at 99.9% over one year, a once-in-a-thousand-years standard, which is deliberately more extreme than the 99% used for market risk precisely because the distribution has this shape. The framework is in capital requirements.

Concentration: expected loss adds, unexpected loss does not

This is the quantitative heart of credit portfolio management, and it is worth doing with numbers.

Take £10 million of total exposure at a 1% default probability and 45% loss given default. Expected loss is £45,000 however you slice it, because expectation is linear. The standard deviation of loss is not linear, and for a single exposure it is

$$\mathrm{LGD} \times \mathrm{EAD} \times \sqrt{\mathrm{PD}(1 - \mathrm{PD})} = 4{,}500{,}000 \times \sqrt{0.01 \times 0.99} = 447{,}744.$$

Now split the same £10 million across n equally sized borrowers with a uniform pairwise default correlation ρ:

Portfolio	Expected loss	Loss standard deviation
1 borrower	45,000	447,744
10 borrowers, $\rho = 0.2$	45,000	236,924
100 borrowers, $\rho = 0.2$	45,000	204,203
Infinitely many, $\rho = 0.2$	45,000	200,237
100 borrowers, $\rho = 0$	45,000	44,774

Read the last two rows against each other. With independent borrowers, splitting the exposure a hundred ways cuts the risk by a factor of ten. With a correlation of only 0.2, it cuts it by a factor of little more than two, and going from a hundred names to infinitely many buys almost nothing: the standard deviation is already within 2% of its floor.

KEY IDEA

Diversification in a credit portfolio hits a floor at $\sqrt{\rho}$ times the single-name risk, and it gets there fast. Past a few dozen names, adding more borrowers of the same kind barely reduces risk, because what remains is the common factor they all share. Genuine diversification means *different* exposures, to other sectors, other geographies and other cycles, not more of the same ones. The identity behind this is in independence and correlation.

That is also why default correlation is the parameter regulators care most about and estimate most conservatively: it is the difference between the two bottom rows of the table, and it cannot be observed from a period without defaults.

How it is actually controlled

Measurement is only half of it. Credit risk is contained by a limit framework, and the structure is worth knowing because it is what a junior person interacts with daily.

- **Single-name limits**, scaled to the counterparty's rating, so a weaker name gets a smaller line.
- **Sector, country and tenor limits**, which exist precisely because single-name limits do nothing about the concentration shown above.
- **Collateral and netting**, which reduce the exposure rather than the probability, and which are the main tool on the trading side.
- **Hedging** with credit default swaps or index protection, which transfers the risk to someone else at a price.

The general framework these sit inside is risk limits frameworks.

Interview angle

REAL INTERVIEW QUESTION (BNP · Credit)

Why do we need credit default swaps if we already know the expected loss?

WHAT THEY TEST / ANSWER

This is a very good question and it has a clean answer: the expected loss is not the risk. It is the average, and nobody is exposed to an average.

First point, and the one to lead with: expected loss is already a *cost*. It is priced into the spread you charge and provisioned against, so a portfolio that realises exactly its expected loss every year has no problem at all. What a bank cannot survive is the year in which losses come in at five or ten times the expectation, and hedging is bought against that, not against the mean. Knowing the mean of a distribution tells you nothing about its tail, and the credit loss distribution is precisely the one whose tail is nothing like its mean.

Second, the loss does not have to happen to hurt you. A position is marked to market long before any default, so a spread widening or a downgrade produces a real loss on the books while the expected loss calculation has barely moved. A CDS hedges the mark, which the expected loss framework does not address at all.

Third, the expected loss is itself an estimate, built from a PD and an LGD that are uncertain and correlated with each other in the worst way: recoveries are lowest exactly when defaults are highest, so both inputs deteriorate together. A CDS transfers that model risk to somebody else at a known price.

Fourth, the practical reasons that have nothing to do with risk measurement: a CDS frees up a credit line so you can keep doing business with a client you like but are full on; it reduces regulatory capital; it lets you take a negative view without selling a bond you cannot sell or damaging a relationship; and it allows the exposure to be shed at all in markets where the underlying bond is illiquid.

Close with the general statement, since it is the one worth remembering: you provision for the expected and you hedge or capitalise the unexpected. Those are two different jobs, and the expected loss formula only does the first.

REAL INTERVIEW QUESTION (BNP · Credit)

A company wants to trade with you but does not have the funds right now. Do you trade, and what do you look at?

WHAT THEY TEST / ANSWER

Do not answer yes or no. The right answer is that it depends on what you would be taking, and listing what you would look at *is* the answer.

Reframe it first: if they cannot pay now, then trading extends credit, so this is a credit decision rather than a trading one, and it should be priced and limited like one.

What you look at, in order. The **counterparty**: who is the legal entity, what is its rating, where does its CDS or its bonds trade, and is there a parent guarantee? The **exposure**: how large, for how long, and is it a fixed amount or one that grows with the market, because a derivative's exposure is itself uncertain. The **documentation**: is there a signed master agreement, is there netting across the existing book, and is there a collateral annex? The **mitigants**: can you take initial margin, a guarantee or a letter of credit? And the **limits**: is there a line, is there room on it, and who approves an excess?

Then price it. If you extend credit you should be paid the expected loss on it, so the quote widens by roughly the probability of default times the loss given default over the period, plus the cost of the capital the exposure consumes. That charge on a derivative is what CVA is, in CVA, DVA and FVA, and quoting the same price you would show a fully collateralised client is giving away real money.

Finally, the judgement. The commercial answer is often to trade in a smaller size, or to trade against collateral, or to trade with a shorter tenor, rather than to refuse outright. And name the one thing that is not negotiable: you do not quietly exceed a limit because the client is important. You get it approved or you do not do the trade, and saying that is what an interviewer asking this question wants to hear.

INTERVIEW TRAP

Four errors. (1) Treating expected loss as the risk, when it is the cost and the risk is what exceeds it. (2) Believing a portfolio of a hundred similar borrowers is diversified, when the table above shows it is within 2% of the same risk as an infinitely large one. (3) Ignoring migration, which produces far more mark-to-market losses than default does. (4) Estimating default correlation from a benign sample, which is the parameter that decides everything and the one a quiet period cannot reveal.

Where this goes next

When the exposure itself is uncertain because it depends on a derivative's mark, the subject becomes counterparty risk and its pricing is CVA, DVA and FVA. The instrument used to transfer the risk is credit default swaps, the capital rules built on the loss distribution above are capital requirements, and the quantile machinery shared with market risk is value at risk.

21.3 Counterparty risk

sales · trading · structuring | core | cross-asset

A derivative is a promise. If the person who made it disappears, so does your hedge, and you discover that you were never hedged at all: you were long a hedge *and* short the credit of whoever wrote it. That is counterparty risk, and 2008 taught the industry the lesson expensively enough that today an entire apparatus of netting, collateral, clearing and pricing adjustments exists to manage it. Interviews probe it from three angles: can you define it precisely, can you draw an exposure profile, and do you know how a desk actually reduces it.

What makes it different from ordinary credit risk

Counterparty credit risk is the risk that the party on the other side of a derivative or a securities financing transaction defaults before the contract has finished settling, at a moment when they owe you money.

Compare it with lending, covered in credit risk. If you buy a bond for 100, your exposure is 100 from the moment you buy it: known, one-directional and fixed. A derivative is different in three ways that make it harder.

- **The exposure is stochastic.** It is the mark-to-market of the contract, which moves with rates, spot and volatility. You do not know today what you will be owed in two years.
- **It is bilateral.** A swap can be an asset to you today and a liability tomorrow, so both parties are exposed to each other at different times, and both need to worry.
- **It is contingent.** A default only costs you if the contract happens to be in your favour at that moment. If you owe *them*, their default does not release you from paying, so your loss is zero.

> **KEY IDEA**
>
> That third point produces the defining formula. Your exposure at any time t is
>
> $$E(t) \;=\; \max\big(V(t),\,0\big),$$
>
> where $V(t)$ is the value of the netted portfolio to you. **Exposure is a call option on the portfolio value**, struck at zero. Everything that follows, why exposure grows with volatility and with time, why netting helps so much, and why the pricing adjustment is computed with option machinery, is a consequence of that single max.

Two neighbours are worth naming so you do not confuse them. **Settlement risk** is the risk that you pay and the other side fails before paying you, which is a same-day, full principal risk, historically called Herstatt risk after the 1974 failure that created the FX version of it and the reason payment-versus-payment systems exist. **Pre-settlement** or replacement risk is the counterparty risk above: the cost of replacing the contract at market after a default.

Measuring it: the exposure profile

Because exposure is uncertain and time-varying, one number will not do. Desks simulate the portfolio forward over thousands of paths and summarise the distribution at each date:

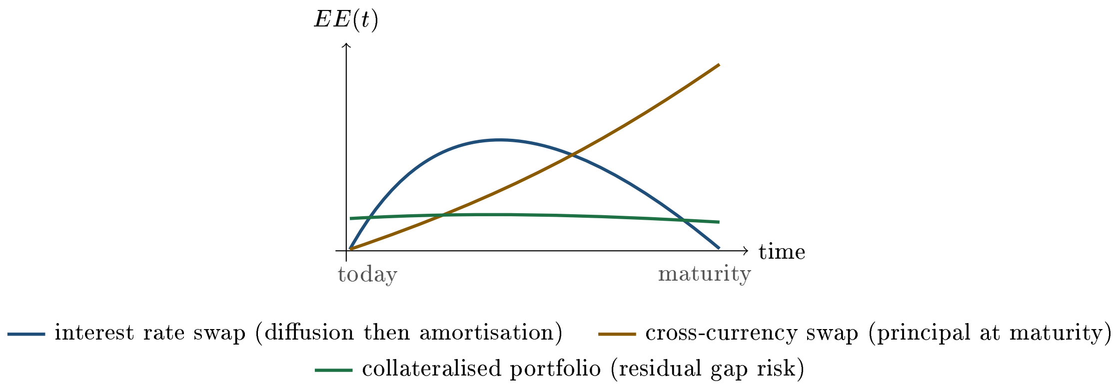
- **Current exposure**: today's $\max(V,0)$ after netting and collateral.
- **Expected exposure**, $EE(t)$: the average of $\max(V(t),0)$ across paths. Averaging $EE(t)$ over the life gives **EPE**, the number used in pricing.
- **Potential future exposure**, $PFE(t)$: a high quantile, typically the 95th or 99th percentile of $\max(V(t),0)$. This is the number used for *limits*, because limits are about bad cases, not average ones. See risk limits frameworks.

The shape of $EE(t)$ over the life of a trade is the thing interviewers ask you to draw, and it is governed by two competing forces.

KEY IDEA

Diffusion versus amortisation. Diffusion pushes exposure *up*: the further into the future you look, the more the underlying can have moved, and uncertainty grows roughly with $\sqrt{t}$. Amortisation pulls exposure *down*: as time passes there are fewer remaining cash flows left to be in the money on, and at maturity there are none. Which force wins where determines the shape.

- **Interest rate swap: a hump.** Early on, diffusion dominates and exposure rises. Later, amortisation dominates and it falls back to zero at maturity. The peak sits somewhere in the first half of the life, commonly around a third of the tenor for a standard fixed-for-floating swap.
- **FX forward or cross-currency swap: monotonically rising.** The principal amounts are exchanged at maturity, so there is nothing to amortise: diffusion runs unopposed and exposure peaks at the end. This is why long-dated cross-currency swaps are such heavy consumers of counterparty limits.
- **A purchased option: one-directional.** You paid the premium, you can never owe anything more, so the value is always non-negative to you and the exposure is simply the option value. Your counterparty, who sold it, has no exposure to you at all after the premium settles.



Reducing it: netting, collateral, clearing

WORKED EXAMPLE

What netting and collateral are actually worth. You have two swaps with one counterparty: trade A is worth +€60m to you, trade B is worth −€40m.

- **No netting.** On default you claim the €60m you are owed as an unsecured creditor, and you must still honour the €40m you owe. Exposure: €60m.
- **With close-out netting** under an ISDA Master Agreement, all trades in the netting set collapse into one net claim: exposure €20m. The €40m liability now works for you rather than against you.
- **With a CSA** carrying a €10m threshold and a €1m minimum transfer amount, the counterparty posts variation margin above the threshold, so you hold €10m of collateral against a €20m net position. Exposure: €10m.

Sixty down to ten, and not one hedge was traded. This is why the legal documentation is part of the risk, not paperwork around it. Whether trades sit inside the same netting set, and whether that netting is enforceable in the counterparty's jurisdiction, cuts the exposure by a factor of three on its own; the CSA on top of it takes the total to six.

The full toolkit, roughly in order of how much it helps:

- **Close-out netting** via an ISDA Master, as above. The single largest lever.
- **Collateral under a CSA.** Variation margin covers today's move; initial margin covers the gap between the last margin call and the close-out of the position. Thresholds, minimum transfer amounts, eligible collateral and haircuts all matter, and are developed in collateral and haircuts.
- **Central clearing.** Novate the trade to a CCP and your counterparty becomes the clearing house, with multilateral netting, standardised margin and a default waterfall behind it. You have not eliminated the risk, you have transformed it into exposure to the CCP and to its margin model. See clearing houses.
- **Contractual features.** Break clauses that let either side terminate at a date, downgrade triggers that require more collateral or termination if a rating falls, and recouponing that resets a swap to par and pays the accumulated value in cash.
- **Limits and diversification.** PFE limits per counterparty, per tenor and per product, monitored against usage in real time.
- **Hedging the credit.** Buy protection on the counterparty in the CDS market, which is how a CVA desk manages what it cannot document away. See credit default swaps.

Pricing it, and wrong-way risk

The expected loss from counterparty default is charged into the price of the trade as **CVA**, the credit valuation adjustment. Its full treatment, with DVA, FVA and the rest of the family, belongs to CVA, DVA and FVA; here we only need the order of magnitude, so that the exposure numbers above connect to money.

WORKED EXAMPLE

A back-of-envelope CVA. An uncollateralised five-year swap with EPE of €10m against a counterparty whose CDS trades at 200bp, with recovery of 40% so $LGD = 60\%$. Ignoring discounting and any correlation between exposure and default.

The quick desk approximation multiplies the average exposure by the spread and the tenor:

$$\text{CVA} \;\approx\; EPE \times s \times T \;=\; 10\text{m} \times 0.02 \times 5 \;=\; €1.0\text{m}.$$

Check it the longer way. A spread of 200bp with 60% loss given default implies a hazard rate $\lambda \approx s/LGD = 0.02/0.60 = 3.33\%$ a year, so the probability of defaulting within five years is $1 - e^{-0.0333 \times 5} = 15.4\%$, and

$$\text{CVA} \;\approx\; LGD \times EPE \times 15.4\% \;=\; 0.60 \times 10\text{m} \times 0.154 \;=\; €0.92\text{m}.$$

The two agree to within about 8%, and the gap is exactly the survival term the linear shortcut ignores. Quote the fast number in an interview, then say what it approximates.

KEY IDEA

Wrong-way risk is the case that breaks every simple calculation: exposure that grows precisely when the counterparty's credit deteriorates. Buying dollars forward against a counterparty's local currency, from a bank in that country, so the devaluation that puts you in the money is the same event that threatens the bank; or buying protection on a market from a counterparty whose own solvency depends on that market, as with the monoline insurers in 2008. When exposure and default are positively correlated, the independence assumption in the CVA above understates the charge badly.

Right-way risk is the mirror image and is a genuine selling point: an airline hedging jet fuel owes you money exactly when fuel is cheap, which is the scenario where it can most easily pay. Note that a plain hedging producer sits on the same side: it owes you when the price rises, which is when its revenue rises. What flips a hedger into wrong-way risk is *leverage* rather than direction, a structure sized well above the underlying exposure, which is how a hedge stops offsetting the business and starts adding to it.

REAL INTERVIEW QUESTION (ECB · Market Risk)

What is counterparty risk?

WHAT THEY TEST / ANSWER

Short question, and the grading is on precision plus one layer of depth. Define it: the risk that the counterparty to a derivative or financing transaction defaults before final settlement, at a time when the contract has positive value to you, so your loss is the cost of replacing it net of recovery. Then immediately give the three properties that distinguish it from lending, because that is the layer they are checking: the exposure is stochastic rather than known, it is bilateral since a swap flips sign, and it is contingent, so your exposure is $\max(V, 0)$ and their default costs you nothing if you are the one who owes. Add that this makes exposure an option on the portfolio value, which is why it grows with volatility and with tenor. If they want more, name the three mitigants in order of power, close-out netting, collateral under a CSA, and central clearing, and note that the residual expected loss is priced as CVA. Distinguishing counterparty risk from settlement risk unprompted is a good final touch.

REAL INTERVIEW QUESTION (BNP · Market Risk)

You have a payer swap with a counterparty: explain how it works, who is exposed to rate moves, and does it imply a move in counterparty risk?

WHAT THEY TEST / ANSWER

Three parts, and the third is the real question. The mechanics: in a payer swap you pay fixed and receive floating on a notional, with no principal exchanged, so the net cash flow each period is the difference between the fixed rate agreed at inception and the floating fix. Who is exposed: you gain when rates rise, since you keep paying the old, now-cheap fixed rate while receiving a higher float, so you are short duration, and your counterparty has the mirror position. Then the link they are testing, which candidates miss: yes, a rate move moves counterparty risk, because exposure is $\max(V, 0)$ and the rate move that makes the swap valuable to you is exactly what creates your exposure to them. So rising rates increase your counterparty exposure on a payer swap and reduce theirs on you, and the market risk and the credit risk of the same trade are two views of one position. Finish with the two consequences a desk cares about: the exposure is not symmetric, so a collateral call is triggered on one side only, and if the counterparty's own creditworthiness happens to be correlated with rates, for example a leveraged borrower in a hiking cycle, then you have wrong-way risk and the simple CVA understates the charge.

REAL INTERVIEW QUESTION (BNP · Market Risk)

A trader calls you while you are alone on the desk about their counterparty limits: what do you do?

WHAT THEY TEST / ANSWER

A judgement question from a control function, and they are testing whether you can be pleasant and immovable at the same time. Say the process, in order. Establish the facts: which counterparty, which trade, what the current PFE usage is against the limit and by how much the trade would breach it. Then the principle: a limit is not yours to waive, whoever is asking and however urgent it sounds, and pressure from a revenue-generating desk while nobody senior is present is precisely the scenario the limit framework exists to survive. Then be useful rather than merely obstructive, because that is the part candidates forget: look for the legitimate routes to the same commercial outcome, such as booking the trade under an existing netted master agreement, adding collateral or moving the trade under a CSA, clearing it, shortening the tenor to cut PFE, using a different entity of the same group that has headroom, or requesting a temporary excess through the proper approver. Escalate immediately to the person with authority, and record the request, the advice given and the outcome in writing whatever happens. Close with the two things that make the answer credible: no, you do not let the trade go through and sort it out later, and yes, you say so to the trader directly and without drama.

INTERVIEW TRAP

Four traps. First, **using the notional as the exposure**: on a swap, no principal is exchanged and the exposure is the replacement value, which is a small fraction of notional. Second, **forgetting the** $\max(\cdot, 0)$: if you are out of the money the counterparty's default costs you nothing, and a candidate who says both parties are always exposed has missed the whole structure. Third, **claiming clearing eliminates counterparty risk**: it concentrates and mutualises it, replacing a bilateral name with the CCP, its margin model and its default waterfall. Fourth, **assuming exposure and default are independent**, which is exactly the wrong-way-risk case that turned monoline exposures, and the leveraged corporate FX hedges that blew up across emerging markets in 2008, into losses far larger than any model had charged for.

VISUAL / CODE TO BUILD: Interactive

An exposure-profile simulator. The reader picks a product (interest rate swap, cross-currency swap, FX forward, bought option), a tenor and a volatility, and the panel Monte-Carlos the value forward and plots $EE(t)$ and $PFE_{95}(t)$ together, so the hump and the rising profile appear from the simulation rather than from an assertion, and a volatility slider scales them. A second layer switches on the mitigants one at a time, netting against a chosen offsetting trade, then a CSA with editable threshold and minimum transfer amount, then clearing, redrawing the profile after each so the reader watches the 60-to-20-to-10 example happen graphically. A CVA readout updates live from the profile and an editable CDS spread, with a correlation slider between exposure and default to make wrong-way risk visible as a number.

Where this goes next

The pricing framework built on the exposure profiles here is CVA, DVA and FVA. The mechanics of the collateral that flattens those profiles are in collateral and haircuts and repo markets, and the institution that mutualises the risk is in clearing houses. Default probability and recovery, the inputs to any CVA, come from credit risk and credit default swaps. How limits are set and policed is risk limits frameworks, and the market-risk side of the same book is Value at Risk.

21.4 Liquidity Risk

trading · structuring · sales | core | cross-asset

Liquidity defines what liquidity is and how it is measured. This section is about it as a *risk*: what goes wrong when it disappears, why that is the failure mode behind almost every institutional collapse, and why it is the one risk that is never adequately captured by the numbers on a risk report.

The reason it deserves its own section is that it is the risk that breaks the others. Market risk measures what a position loses if prices move; liquidity risk is why you cannot get out at the price the screen shows, and therefore why the realised loss is larger than the measured one, systematically and exactly when it matters.

Two different risks with one name

KEY IDEA

Market liquidity risk is the risk that you cannot sell an asset near its marked price, because the market is too thin to absorb your size. **Funding liquidity risk** is the risk that you cannot finance a position you already hold, because lenders raise haircuts, cut lines or refuse to roll. The first is about the asset; the second is about your balance sheet, and an institution can die of either.

They are not independent, and the mechanism that couples them is the reason liquidity crises are violent rather than gradual.

INTUITION

The liquidity spiral. Prices fall, so mark-to-market losses reduce equity and lenders raise haircuts. Higher haircuts force positions to be sold. The selling pushes prices down further, which raises haircuts again. Each turn of the loop is individually rational for every participant and collectively disastrous, and because everyone is running the same risk models on the same assets, everyone is forced to sell the same things at the same time. That is 1998, 2008 and March 2020, and it is the single most useful thing to be able to describe in an interview on this topic.

Market liquidity risk: the cost of getting out

Two numbers matter, and the second is much larger than the first.

WORKED EXAMPLE

A position that is bigger than the market. You hold 5 million shares. Average daily volume is 2 million shares, and the desk's execution policy caps participation at 20% of volume to avoid signalling.

Time. You can sell $0.20 \times 2\text{m} = 400{,}000$ shares a day, so exiting takes

$$\frac{5{,}000{,}000}{400{,}000} \;=\; \mathbf{12.5} \text{ trading days.}$$

A one-day VaR on this position is therefore measuring the wrong horizon. If the risk is roughly proportional to the square root of time, the honest number is about $\sqrt{12.5} = \mathbf{3.5}$ times the one-day figure. This is exactly why regulators moved to horizon-dependent liquidity buckets: the holding period in the risk measure has to be the period you would actually be holding it.

Cost. Now the number people forget. The quoted half-spread might be 10 basis points, and that is what a small order pays. A large order pays **market impact** instead, and the standard empirical rule is that impact scales with the square root of size relative to volume:

$$\text{impact} \;\approx\; \sigma\sqrt{Q/\text{ADV}} \;=\; 2\% \times \sqrt{5/2} \;=\; \mathbf{3.16}\%,$$

at a 2% daily volatility. The cost of exiting is roughly **thirty times** the quoted spread. The bid-ask on the screen is the price of liquidity for someone who does not need it.

The practical consequence is that a mark-to-market price is a price for a normal-sized trade, and a position that is a multiple of daily volume is not worth its mark. Desks handle this with liquidity reserves against concentrated positions, and it is the reason inventory management is a discipline rather than an afterthought.

Hedging an illiquid position is where this stops being a measurement problem and becomes a pricing one. Delta, gamma and vega hedging covers what a desk does when the options it needs are not there: proxy hedge and accept correlation and basis risk, manage the second-order exposure through the underlying instead, or carry the risk and charge for it. What belongs here is the reason the last of those is usually the right answer. An unhedgeable risk is priced, reserved against and limited, not hedged, and the deeper problem is that the mark itself is uncertain. An implied volatility built from few and stale observations is not a market price, so the risk system reports a precision the book does not have, and the first thing a stress does is move the mark to where the market actually is. That gap between the mark and the exit price is liquidity risk in its purest form.

Funding liquidity risk: the haircut is the leverage

A bank borrows short and lends long. That maturity transformation is the business, and it means a bank is never able to meet all its obligations simultaneously, by design. Funding liquidity risk is the risk that it is asked to.

For a trading book, the mechanism is concrete and worth carrying as arithmetic.

WORKED EXAMPLE

A haircut change, and the deleveraging it forces. You fund a bond portfolio in repo at a 5% haircut, so for every 100 of bonds you post 5 of your own money and borrow 95. With $100 million of equity you can hold

$$\frac{100}{0.05} = \$2{,}000 \text{ million of bonds, leverage of } \mathbf{20}\times.$$

Stress arrives and the haircut goes to 15%. Nothing about the bonds has changed and nobody has defaulted. But the same $100 million of equity now supports only

$$\frac{100}{0.15} = \$667 \text{ million, leverage of } \mathbf{6.7}\times,$$

so you must sell **two thirds of the book**, into a market where every other leveraged holder has just been told the same thing.

That is the whole of a funding crisis in one calculation. Note what it does not require: no default, no credit event, no change in the value of the asset. A change in the terms of financing is sufficient.

REAL INTERVIEW QUESTION (HSBC · Structured Products)

What is funding?

WHAT THEY TEST / ANSWER

Start with the plain answer and then show the two levels at which the word is used, because a structured products desk is asking for the second.

Plainly, funding is the cost of the money a position requires. A bank does not have cash sitting idle; every asset it holds has to be financed, either from deposits, from unsecured debt issued by the treasury, or from secured borrowing against the asset itself in repo.

At the desk level, it is a charge. A trading desk is lent balance sheet by the treasury at an internal transfer price, and that charge is a real cost in the P&L of every position that consumes cash: an option the desk has bought, a collateral posting, a stock borrow. It is the reason a trade can be right on the market and lose money.

At the product level, it is a component of the price. A structured note is, from the issuer's side, funding: the bank receives the client's cash and repays it at maturity, so the note competes with the bank's other borrowing and is priced off its own funding curve. This has a consequence worth stating, since it is what the question is testing. A bank with a wider credit spread can offer a *better* coupon on the same structure, because its funding is more expensive and the note is worth more to it. So part of what a client earns on a structured note is the issuer's credit spread, which is also why the guarantee on a capital-guaranteed product is only as good as the issuer.

Close on the risk. Funding is not a fixed cost. It widens in exactly the conditions where the assets it supports are falling, and a business whose model assumes cheap funding is carrying a leveraged position in its own creditworthiness.

How it is regulated

Two ratios came out of 2008, and both are defined in Basel III and IV: the **LCR**, the liquidity coverage ratio, which sets high-quality liquid assets against net cash outflows under a thirty-day stress, and the **NSFR**, the net stable funding ratio, which sets available stable funding against what the balance sheet requires over a one-year horizon. Both must be at least 100%.

What belongs here is why there are two of them, because it maps exactly onto the split this section opened with. The LCR is a *survival* test: can the bank withstand a month-long run without help. The NSFR is a *structural* test: is the maturity transformation that creates the exposure in the first place sustainable at all. A bank can pass either while failing the other, which is precisely why one ratio was not enough.

REAL INTERVIEW QUESTION (BNP · desk not recorded)

How do you hedge against collateral risk, meaning haircut risk?

WHAT THEY TEST / ANSWER

Notice first what the risk actually is, because the question is testing whether you can distinguish it from credit risk. A haircut change is not a default and not a fall in the asset's value; it is a change in the *terms* on which the asset can be financed, and it hits you through your own balance sheet rather than through the position.

Can you hedge it directly? Essentially not. There is no liquid instrument on repo haircuts, and the closest traded proxies, the repo rate itself, cross-currency basis, or the general collateral to special spread, hedge the price of financing rather than its availability. Say this plainly: the honest answer is that haircut risk is managed, not hedged.

So how is it managed? Four ways, and they are the ones a treasury actually uses.

Term out the funding. Term repo or committed facilities fix the haircut for the life of the trade, so the risk becomes a refinancing risk at a known date rather than an overnight risk. This costs money, and paying it is the hedge.

Diversify the counterparties and the currencies. Haircut policies move in unison in a crisis but not perfectly, and a single lender changing terms should not force a sale.

Hold a liquidity buffer. Unencumbered high-quality assets, sized to survive a stressed haircut scenario, are the direct answer to a margin call you cannot predict. That is exactly what the LCR formalises.

Limit leverage in advance. If your leverage assumes today's haircut, you have written a short option on financing conditions. Sizing to a stressed haircut rather than the current one is the cheapest form of the hedge, since it costs only foregone return.

Close with the number, because it makes the point better than the argument. Funding at a 5% haircut is twenty times leverage; at 15% it is under seven, so the same equity supports a third of the position and two thirds of the book has to go. No default is required for that, only a change of mind by the people lending against it.

INTERVIEW TRAP

1. **Treating the bid-ask as the cost of exiting.** It is the cost for a normal-sized order. A position that is a multiple of daily volume pays market impact, which can be an order of magnitude more, as the worked example shows.
2. **Using a one-day risk horizon on a position that takes weeks to sell.** The horizon in the measure has to be the horizon in reality.
3. **Confusing solvency with liquidity.** Institutions fail because they run out of cash, not because a spreadsheet says they are insolvent. A solvent firm that cannot fund itself is finished first.
4. **Assuming liquidity is a property of the asset.** It is a property of the asset *and the moment*. The measured liquidity of a bond in calm markets tells you very little about the liquidity of the same bond in a stress, which liquidity makes the same point about.
5. **Ignoring the correlation between funding and markets.** Funding widens precisely when assets fall. A model that treats the funding cost as a constant has removed the interaction that causes the failure.

VISUAL / CODE TO BUILD: Interactive

A deleveraging panel. Inputs are equity, the current haircut and the asset price. The panel shows the position size and the implied leverage as a bar, and a single slider raises the haircut. As the haircut moves, the bar shortens and the panel prints the notional that has to be sold, so the reader sees that going from 5% to 15% removes two thirds of the book without any change in the asset.

Then a toggle turns on the spiral: forced selling feeds a price decline through a market-impact function, the decline reduces equity, and the reduced equity forces more selling. Running it once with the feedback off and once with it on, from the same starting point, produces two very different endings and is the whole argument of the section.

A second panel plots exit cost against order size as a multiple of average daily volume, with the quoted half-spread drawn as a flat line and the square-root impact curve rising through it. The crossing point, where impact overtakes the spread, is usually far smaller than readers expect, and labelling it is the point of the panel.

21.5 Operational risk

trading · structuring · sales | core | cross-asset

Market risk is being wrong about a price. Credit risk is being wrong about a counterparty. Operational risk is the bank's own machinery failing: a trade booked incorrectly, a system that goes down, a control that somebody worked around, a product sold to a client it was never suitable for.

It is the least glamorous of the three and it has produced several of the largest single-day losses in the history of the industry. It is also the only one where a junior person is not an observer of the risk but part of the control.

The definition, and what it deliberately excludes

The Basel definition is worth knowing verbatim because interviewers quote it: the risk of loss resulting from inadequate or failed **internal processes, people and systems, or from external events**.

Two clarifications sit inside it and both get asked. It **includes legal risk**, meaning fines, settlements and litigation arising from how the bank operated. It **excludes strategic and reputational risk**: choosing the wrong business to be in is a management failure rather than an operational one, and reputational damage is a consequence of losses rather than a loss event itself.

Basel then divides events into seven types, and the list is a better definition than the sentence because it shows the range.

Event type	Typical example
Internal fraud	unauthorised trading, deliberate mismarking
External fraud	payment fraud, cyber intrusion
Employment practices and workplace safety	discrimination and employment claims
Clients, products and business practices	mis-selling, market conduct, suitability
Damage to physical assets	fire, flood, terrorism
Business disruption and system failures	outages, connectivity, failed deployment
Execution, delivery and process management	booking errors, failed settlement, bad data

The last two are where a trading floor lives day to day. The fourth is where the largest fines have come from.

The shape, and why it cannot be modelled the usual way

Operational losses have the same asymmetry as credit losses, only more so: an enormous number of small events, mostly in the last row of the table, and a very small number of catastrophic ones. There is no upside at all. Nothing goes unusually right.

That shape creates a measurement problem that market and credit risk do not have.

> **KEY IDEA**
>
> You cannot estimate a once-in-twenty-years event from your own ten years of data, and if the event has already happened to you once, the sample contains one observation. Operational risk is therefore the one risk category where a bank's own loss history is structurally insufficient to calibrate the number it most needs.

The response was to combine four inputs: internal loss data, external loss data pooled across banks, scenario analysis in which senior people are asked to estimate plausible extreme events directly, and business environment and internal control factors. It produced numbers that varied enormously between banks holding similar risks, which is the fundamental complaint against modelling anything from a sample that does not contain the events of interest.

Basel III's answer was to abandon internal models for operational risk entirely and replace them with a formula. Capital is now driven by a **business indicator**, a size proxy built from income and services, scaled by marginal coefficients, and then adjusted by an internal loss multiplier that reflects the bank's own history.

WORKED EXAMPLE

The business indicator component applies marginal coefficients over three buckets: 12% on the first €1 billion of business indicator, 15% on the portion between €1 billion and €30 billion, and 18% above that.

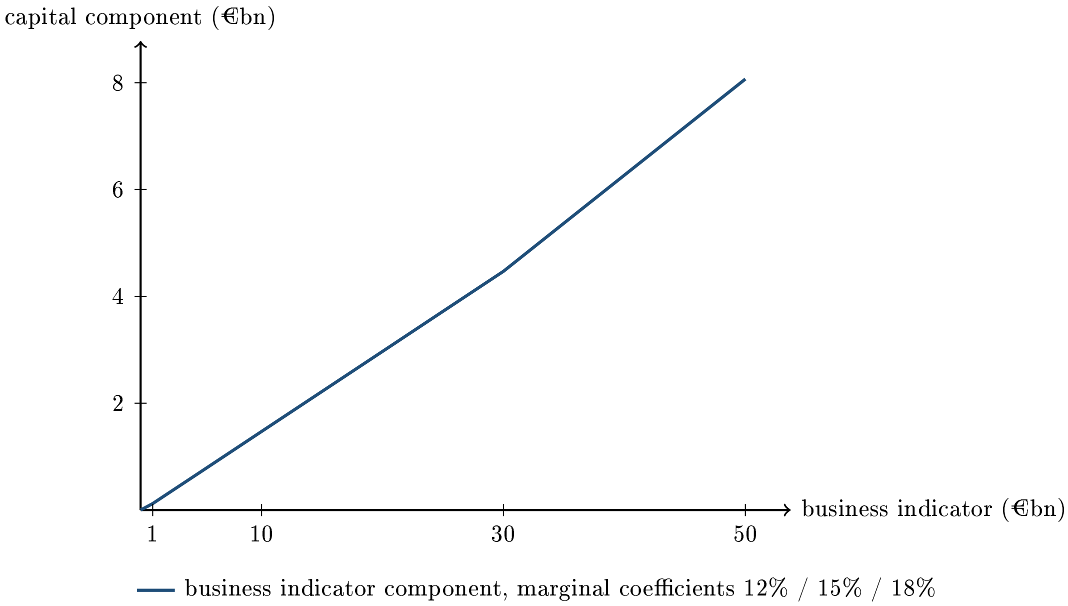
A bank with a business indicator of €5 billion:

$$0.12 \times 1 \ + \ 0.15 \times 4 \ = \ 0.12 + 0.60 \ = \ \text{€}\mathbf{0.72} \text{ billion.}$$

A bank with €40 billion:

$$0.12 \times 1 \ + \ 0.15 \times 29 \ + \ 0.18 \times 10 \ = \ 0.12 + 4.35 + 1.80 \ = \ \text{€}\mathbf{6.27} \text{ billion.}$$

Note what the marginal structure does: the eight-times-larger bank carries roughly nine times the charge, so the schedule is mildly progressive. And note what it does *not* do: it contains no measure of how risky the bank's activities are, only of how large they are. That is the deliberate trade, simplicity and comparability against risk sensitivity, and it is the same trade the whole post-crisis framework made.



The two kinks, at 1 and at 30, are where the coefficient steps up. Everywhere else the line is straight, which is the visual statement that this is a size charge and not a risk charge.

The anatomy of a rogue trader

Four cases are worth knowing, not as trivia but because they share a structure.

- **Barings, 1995**. A trader in Singapore ran both the front office and the settlement function, and used an error account to hide losses that eventually reached around £827 million and destroyed a bank that had existed for two centuries.
- **Société Générale, 2008**. Unauthorised directional positions were concealed behind fictitious offsetting trades, so the risk reports showed a hedged book. The unwind cost roughly €4.9 billion.

- **UBS, 2011**. Around $2.3 billion lost on unauthorised positions hidden using the settlement conventions of exchange-traded funds, which allowed fictitious trades to sit unconfirmed for longer than an ordinary trade would.
- **Knight Capital, 2012**. Not fraud at all: a software deployment left old code running on one of eight servers, and the firm lost about $440 million in some 45 minutes, roughly $10 million a minute, before it could be stopped.

> **KEY IDEA**
>
> Three of the four have the same shape. Somebody understood the *control* process well enough to defeat it, usually because they had worked in it, and the concealment was not in the positions but in the paperwork that was supposed to check them. The fourth is different and worth pairing with them deliberately: no dishonesty, one bad deployment, and a loss of the same order of magnitude in under an hour. Operational risk does not require a villain.

The controls, and why they are what they are

Every control on a trading floor is a response to one of the failures above, and knowing which turns a list of rules into something you can reason about.

- **Segregation of duties**: whoever executes cannot also confirm, settle or value. This is the Barings control, and it is the single most important one.
- **Mandatory block leave**: staff must be away for a continuous period each year, because a concealment that needs daily maintenance collapses while its author is absent. It exists for exactly one reason and it has caught real cases.
- **Independent confirmation and reconciliation**: every trade confirmed with the counterparty by someone who did not do it, and every position reconciled to an external record.
- **Four eyes on anything unusual**: cancellations, amendments, off-market rates, manual overrides. Fraud almost always shows up first as an unusual amendment.
- **Independent price verification**: the marks a desk uses are checked against an independent source, because mismarking is the cheapest way to hide a loss.
- **Change management and kill switches**: the Knight control. A deployment procedure that verifies what is actually running, and an ability to stop a system that is losing money faster than a human can react.

Interview angle

REAL INTERVIEW QUESTION (BNP · Structured Products)

A client calls, nobody else is on the desk, and they absolutely want to trade. What do you do?

WHAT THEY TEST / ANSWER

The interviewer is testing judgement under pressure, and there is a right answer. Do not improvise a price, and do not say you would refuse to engage either, because both are wrong in different directions.

What you actually do, in order. Establish what they want: instrument, size, direction, and whether it is urgent because the market is moving or urgent because they say it is. Be clear about your own position: if you are not authorised to quote that product in that size, you say so, immediately and without embarrassment, because pretending otherwise is how people end up outside their mandate. Then escalate: call the trader who covers it, call the desk head, call whoever is on the escalation list, and if the desk genuinely cannot be reached then somebody senior in the reporting line can be. There is always a list, and the fact that you know to look for it is half of what is being tested.

Meanwhile, keep the client informed. "I am getting you a price, give me two minutes" is a professional answer. Silence is not, and neither is a number you invented.

And what you do not do, which is the part that matters most. You do not quote a price you are not authorised to quote. You do not book a trade to be sorted out later. You do not agree terms verbally and write them up afterwards. The cost of losing one trade is a conversation with your boss; the cost of an unauthorised trade is an incident report, and on a bad day it is the start of one of the cases in this section.

Finish with the frame that shows you understand why the question exists: this is an operational risk question dressed as a client question. Every rogue-trading loss began with somebody deciding that this once, the process could wait. The correct instinct is that the process exists precisely for the moments when it is inconvenient.

INTERVIEW TRAP

Four operational-risk errors. (1) Treating it as a back-office concern, when three of the four largest single-firm trading losses were operational rather than market events. (2) Assuming a hedged-looking risk report means a hedged book, when the Société Générale case was precisely a book that looked hedged because the offsetting trades did not exist. (3) Estimating the tail from internal history, which by construction does not contain the events that matter. (4) Reading the standardised capital charge as a measure of how risky a bank is, when it is a measure of how large it is.

Where this goes next

The limit and escalation machinery this section keeps pointing at is risk limits frameworks, and the floor-level rules are trading floor compliance. The capital framework the standardised approach sits in is capital requirements, the day-to-day book processes whose failures generate most of the loss events are book management and position limits, and the risk category with the most similar loss distribution is credit risk.

21.6 Value at Risk (VaR)

sales · trading · structuring | core | cross-asset

Value at Risk is the most asked risk-management question in front-office interviews, and it is asked because it is the one number that a bank's entire market-risk apparatus is built around: limits, capital, daily reporting to the board and the conversation a trader has when they have used too much of it. It is also the most commonly misquoted concept in finance, which is why interviewers keep asking. The definition takes one sentence, the three calculation methods take five minutes, and the limitations are what separate a candidate who has used VaR from one who has read about it.

The definition, stated properly

KEY IDEA

The **Value at Risk** of a portfolio at confidence level α over horizon h is the loss level that will not be exceeded with probability α over that horizon. In symbols, it is the smallest L such that

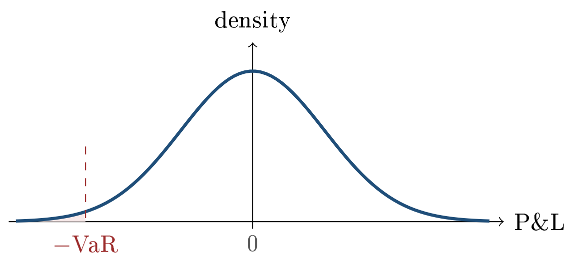
$$\mathbb{P}\big(\text{loss over } h > L\big) \;\leq\; 1-\alpha.$$

It is nothing more and nothing less than a **quantile of the profit-and-loss distribution**. A VaR number is meaningless without all three of its parameters: the confidence level, the horizon, and the currency.

So "the 1-day 99% VaR is €5m" means: on a normal day, there is a 1% chance of losing more than €5m, which is roughly one day in a hundred, or two to three days a year. Say it that way and you have already answered better than most.

INTUITION

VaR is a **flood level**. A town builds its wall at the height that the water has failed to reach on 99 days out of 100. That is a real, useful, decision-relevant number: it lets you compare rivers, budget for the wall, and tell the council how exposed they are. But notice what the wall height does *not* tell you. It says nothing about how deep the water goes on the hundredth day. A wall at three metres is the same number whether the flood behind it is three and a half metres or thirty. That single blind spot is the origin of every criticism of VaR and the reason regulators eventually moved to expected shortfall.



P&L density — the 1% quantile: everything to its left is the tail VaR ignores

The three ways to compute it

Every VaR engine is one of these three, and being able to give the advantages and drawbacks of each is the standard second question.

1. Parametric, also called variance-covariance or delta-normal. Assume the risk factors are jointly normal, map the portfolio to a linear exposure to those factors, and read the quantile off the normal distribution:

$$\mathrm{VaR}_\alpha \;=\; z_\alpha\,\sigma_P\sqrt{h}, \qquad \sigma_P^2 \;=\; w^\top\Sigma\,w,$$

where w holds the money sensitivities to each factor and Σ their covariance matrix. The multipliers are worth memorising, since they are asked directly: 1.28 at 90%, 1.645 at 95%, 2.326 at 99% and 2.576 at 99.5%, all one-tailed. Fast, transparent, and it decomposes cleanly into contributions per desk. It fails on two counts: real returns are fat-tailed, so it understates the tail, and it is linear, so it misprices options unless extended to delta-gamma.

2. Historical simulation. Take the last N days of actual factor moves, typically one to four years, apply each one to today's portfolio, revalue, and sort the N hypothetical P&Ls. The 99% VaR is the empirical quantile: with 500 scenarios it is between the fifth and sixth worst. No distributional assumption, fat tails and observed correlations come free, and it handles non-linear payoffs properly if you fully revalue. Its weakness is that it can only produce crashes that are already in the window: the sample is finite, the tail is estimated from a handful of points, and a quiet year makes the number small precisely when complacency is highest. This is the most widely used method in practice.

3. Monte Carlo. Assume a model for the factor dynamics, simulate many paths, revalue the portfolio on each, take the quantile. The most flexible, since it handles path dependence, non-linearity and any distribution you can specify, including fat-tailed or stochastic-volatility ones. It is also the slowest and the most exposed to model risk, because you now own every assumption you wrote down.

A worked parametric VaR

WORKED EXAMPLE

Two positions, with correlation. A book holds €10m of an equity index with annual volatility 20%, and €5m of a bond with modified duration 5 where the yield has a daily volatility of 5bp. The daily correlation between the equity return and the bond's return is -0.3.

Step 1, put both risks in money terms per day.

- Equity: daily volatility is $20\%/\sqrt{252} = 1.2599\%$, so $\sigma_A = 10{,}000{,}000 \times 0.012599 =$ €125,988.
- Bond: a 5bp yield move changes the price by $5 \times 5\text{bp} = 25\text{bp}$, so $\sigma_B = 5{,}000{,}000 \times 0.0025 =$ €12,500.

Step 2, combine them.

$$\sigma_P^2 = \sigma_A^2 + \sigma_B^2 + 2\rho\,\sigma_A\sigma_B = 125{,}988^2 + 12{,}500^2 + 2(-0.3)(125{,}988)(12{,}500),$$

which is 15,084,354,690, so $\sigma_P =$ €122,818 a day.

Step 3, apply the multiplier.

- 95% 1-day VaR: $1.645 \times 122{,}818 =$ €202,036.
- 99% 1-day VaR: $2.326 \times 122{,}818 =$ €285,676.
- 99% 10-day VaR, scaling by $\sqrt{10}$: $285{,}676 \times 3.1623 =$ €903,385.

Step 4, read the diversification. Adding the two standalone risks gives $125{,}988 + 12{,}500 =$ €138,488, against a portfolio figure of €122,818. The €15,670 difference, about 11.3%, is the diversification benefit created entirely by the negative correlation. Quote that number: it is what makes VaR a *portfolio* measure rather than a sum of positions, and it is why a risk manager will not let you add VaRs together.

Scaling across horizons, and the assumption hidden in it

The $\sqrt{h}$ used above is not a convention, it is a consequence. As shown in autocorrelation, the variance of a sum of h returns equals h times the one-period variance *only if* the returns are serially uncorrelated. So the square-root-of-time scaling of VaR inherits exactly that assumption. If the P&L is positively autocorrelated, because the book holds illiquid positions marked from stale prices, then the true ten-day risk is larger than $\sqrt{10}$ times the one-day figure, and the reported number is too small in precisely the situation where it matters. Volatility clustering, developed in ARCH and GARCH, causes the same problem through a different door: a VaR estimated on a calm window understates risk when the regime turns.

What VaR cannot do

State these as a list, because interviewers count them.

- **It is silent about the tail.** VaR is the threshold, not the severity beyond it. Two books with identical VaR can have wildly different losses on the bad day.
- **It is not sub-additive.** There exist portfolios where $\text{VaR}(A + B) > \text{VaR}(A) + \text{VaR}(B)$, so VaR is not a coherent risk measure and can, pathologically, penalise diversification. The standard counterexample uses two digital or default-type payoffs whose individual losses sit

just outside the quantile but whose combination lands inside it. This is the formal reason regulators moved to expected shortfall.

- **It is backward-looking.** Historical VaR can only reproduce the crises in its window; parametric VaR only the covariances estimated from it.
- **It measures normal markets, not stress.** By construction it says nothing about the 1%, which is what stress testing exists to cover.
- **It is procyclical.** Volatility falls, VaR falls, limits free up, positions grow, and then volatility returns. The measure amplifies the cycle it is meant to control.
- **It can be gamed.** A book can be constructed to look small on a quantile measure while carrying a large tail, for instance by selling far out-of-the-money options, whose losses sit beyond the confidence level almost by design.

Backtesting: is the number honest?

A VaR model makes a testable prediction, which is what makes it respectable. At 99% over 250 business days you expect $250 \times 0.01 = 2.5$ **exceptions**, days when the actual loss exceeded the reported VaR. Counting them is the backtest, and the Basel traffic light reads roughly: zero to four exceptions is the green zone and the model stands, five to nine is amber and the capital multiplier is increased, ten or more is red and the model is presumed broken. The formal version is Kupiec's proportion-of-failures likelihood-ratio test, and a serious backtest also checks that exceptions are *independent* rather than clustered, since three exceptions in one week is a very different failure from three spread across a year.

Regulation followed the criticisms. Basel 2.5 added a **stressed VaR**, calibrated to a crisis period, on top of the ordinary one, and the Fundamental Review of the Trading Book replaced VaR with expected shortfall at 97.5% with liquidity horizons. That confidence level looks arbitrary until you check it: for a normal distribution, expected shortfall at 97.5% is 2.338 standard deviations while VaR at 99% is 2.326, so the switch was calibrated to leave the overall level of conservatism roughly unchanged while removing the blind spot.

REAL INTERVIEW QUESTION (ECB · Market Risk)

Define VaR and give its main calculation methods (historical, parametric, Monte Carlo).

WHAT THEY TEST / ANSWER

A structured recital, and the grading is on precision and completeness rather than insight. Definition first, with the quantifiers: the loss over a given horizon that is not exceeded with a given probability, so a 1-day 99% VaR of €5m means a 1% chance of losing more than €5m in a day, and the number is meaningless without confidence level, horizon and currency. Then the three methods with one advantage and one drawback each. Parametric assumes joint normality and uses $z_\alpha \sigma_P \sqrt{h}$ with $\sigma_P^2 = w^\top \Sigma w$: fast and decomposable, but understates fat tails and mishandles options unless you add gamma. Historical simulation replays actual past factor moves and takes the empirical quantile: no distributional assumption and correct for non-linear payoffs if fully revalued, but limited to the crises inside its window. Monte Carlo simulates from an assumed model: the most flexible for path dependence and exotic payoffs, the slowest, and the most exposed to model risk. Close by saying which is used in practice, historical simulation, and adding one sentence on what all three miss, namely the size of the loss beyond the quantile, which is why the regulatory measure is now expected shortfall.

REAL INTERVIEW QUESTION (Société Générale · Market Risk)

What are the levels of the VaR (confidence level 95%, 99%) expressed in number of sigma?

WHAT THEY TEST / ANSWER

Answer the numbers immediately, then qualify. One-tailed standard normal quantiles: 95% is 1.645 sigma and 99% is 2.326 sigma. Have the neighbours ready too, since they often continue: 1.28 at 90% and 2.576 at 99.5%. Then two qualifications that show you know what you are quoting. These are *one-tailed* numbers, so do not offer 1.96, which is the two-tailed 95% value and the classic trap in this question. And they hold only under the normality assumption of parametric VaR: real P&L distributions are fat-tailed, so the true 99% quantile is further out than 2.326 sigma, which is one reason historical simulation is preferred and why regulators add a stressed calibration on top. A good closing detail is that expected shortfall at 97.5% equals 2.338 sigma under normality, effectively the same level of conservatism as 99% VaR, which is exactly why the FRTB chose that number.

REAL INTERVIEW QUESTION (BNP · Market Risk)

Do we always use the PnL to compute the VaR, even when it is positive?

WHAT THEY TEST / ANSWER

A conventions question that quietly tests whether you understand what the input is. Yes: VaR is computed from the *whole distribution* of hypothetical P&L, gains included, because the gains are part of the distribution whose left tail you are measuring. In historical simulation you generate one hypothetical P&L per scenario, positive and negative alike, sort the entire vector, and read the quantile in the loss tail. Discarding the positive outcomes would not make the calculation more prudent, it would corrupt the distribution and give a meaningless quantile. Then handle the sign convention explicitly, which is what the question is really probing: VaR is reported as a positive number representing a loss, so it is the negative of the low quantile of P&L, and the profitable scenarios simply sit in the right tail and never bind. Two useful additions: the input is the vector of *hypothetical* P&L from revaluing today's portfolio under each scenario, not the realised historical P&L of the desk, which is a different series used for backtesting; and the mean is usually set to zero over a one-day horizon because the expected drift is negligible next to the volatility and assuming it away is the conservative choice.

REAL INTERVIEW QUESTION (Tikehau Capital · Market Risk)

How do you compute the parametric VaR of a portfolio of equities, bonds and options?

WHAT THEY TEST / ANSWER

The answer is the mapping, not the formula, and that is the whole point of the question. Three steps. First, choose risk factors and map every instrument to its money sensitivity to them: equities to index returns and residuals through their betas, bonds to a small set of key-rate durations rather than to a single yield, since curve risk is multi-dimensional, and options to the delta of their underlying. Second, and this is where candidates fail, handle the non-linearity: a delta-only mapping misprices an option book badly, so extend to delta-gamma, where the position's P&L is $\Delta\,\delta S + \frac{1}{2}\Gamma\,\delta S^2$, and add a vega row with implied volatility as an explicit risk factor, because for an option book vega risk is often larger than delta risk. Third, build the covariance matrix across all those factors, compute $\sigma_P = \sqrt{w^\top \Sigma w}$, and multiply by $z_\alpha\sqrt{h}$. Then state the limits honestly, since that is the maturity they are looking for: a quadratic expansion fails for barriers, digitals and anything with discontinuous payoff, the normality assumption understates tails, and the gamma term makes the P&L distribution skewed so a normal quantile is no longer strictly right. For a book with material optionality you would recommend historical simulation with full revaluation, and keep the parametric number for its clean risk decomposition by desk and factor.

REAL INTERVIEW QUESTION (Tikehau Capital · Market Risk)

How do you compute the historical VaR of a portfolio of options?

WHAT THEY TEST / ANSWER

Emphasise **full revaluation**, because that is the answer. Take a window of past daily moves, one to four years, and for each historical day extract the changes in every relevant risk factor, spot, the implied volatility surface, rates and dividends. Apply each day's set of moves to *today's* portfolio and reprice every option with the pricer, not with a Greek approximation, giving one hypothetical P&L per scenario. Sort them and read the quantile: with 500 scenarios, the 99% VaR sits between the fifth and sixth worst loss. Then raise the three issues that make this hard for options specifically, which is what the question is testing. Whether to apply factor moves in relative or absolute terms, which matters enormously for implied volatility and for rates near zero. How to shock a whole surface consistently, since moving one point without the others creates arbitrageable surfaces, so in practice you shock the surface's principal components. And ageing: whether each scenario reprices with today's time to maturity or one day less, which decides whether theta is inside the VaR number. Finish with the honest limitation: the tail is estimated from a handful of observations, so the 99% number is noisy, and the window contains only the crises it happens to contain, which is why it is complemented by stress testing.

REAL INTERVIEW QUESTION (Société Générale · Market Risk)

What is a stress test compared to what the VaR studies?

WHAT THEY TEST / ANSWER

Frame it as a difference in the question being asked, not in technique. VaR is **probabilistic and about normal markets**: it asks what loss is exceeded 1% of the time, estimated from a distribution fitted to recent history, and it returns a number attached to a probability. A stress test is **deterministic and about specific bad worlds**: it asks what happens if this scenario occurs, a historical replay such as 1987, 2008 or March 2020, or a hypothetical construction such as a 200bp parallel rate shock combined with equities down 30% and correlations going to one. It returns a loss with no probability attached, which is a feature rather than a defect, because the events that matter are precisely those whose probability you cannot estimate. Then give the three reasons they are complements. Stress tests reach beyond the historical window, which historical VaR cannot. They break the correlation assumptions, whereas VaR's diversification benefit is exactly what evaporates in a crisis. And they are expressible as a story a board can act on, whereas a quantile is not. Close with the practical point: VaR drives day-to-day limits and capital, stress tests drive scenario limits, contingency planning and the appetite discussion, and a desk is normally constrained by both. The mechanics are developed in stress testing.

INTERVIEW TRAP

Four traps, all common. First, **quoting a VaR without its parameters**: no confidence level, no horizon, no currency, no meaning. Second, saying **"the most I can lose"**: VaR is the threshold of the tail, not the maximum, and the maximum is usually the whole portfolio. Third, offering 1.96 **sigma for** 99%: 1.96 is the two-tailed 95% value, and the correct one-tailed 99% figure is 2.326. Fourth, **adding VaRs across desks**: they combine through the covariance matrix, which is normally sub-additive in practice, and in pathological cases VaR is not sub-additive at all.

VISUAL / CODE TO BUILD: Interactive

A VaR laboratory in three tabs, sharing one portfolio the reader builds from equity, bond and option positions. Tab one, **method comparison**: the same book priced by parametric, historical and Monte Carlo VaR side by side, with the P&L histogram and the quantile line drawn for each, so the reader sees the parametric number sitting inside the historical tail whenever the data are fat-tailed. Tab two, **knobs**: sliders for confidence level, horizon and correlation, with the diversification benefit displayed as a number so the reader can drive it to zero by pushing correlation to one, reproducing the worked example. Tab three, **backtest**: replay a real price history, plot daily P&L against the rolling VaR line, count exceptions, colour the Basel traffic-light zone, and flag clustering, so procyclicality becomes visible as the VaR line falling through a calm period just before a cluster of breaches.

Where this goes next

The measure that fixes VaR's blind spot is expected shortfall, and the deterministic complement is stress testing with scenario analysis. How the number becomes a constraint on a trader's book is risk limits frameworks, and the broader taxonomy it sits inside is market risk. The statistical assumptions it leans on are developed in autocorrelation for the time scaling and ARCH and GARCH for the volatility input, and the sensitivities used in the mapping are the portfolio-level Greeks.

21.7 Expected Shortfall

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Value at risk answers "how bad is a bad day". **Expected shortfall** answers the question VaR refuses to, which is how bad it is when it is worse than that. It is defined as the average of the quantiles beyond the confidence level,

$$\mathrm{ES}_\alpha = \frac{1}{1-\alpha}\int_\alpha^1 \mathrm{VaR}_u(L)\,du,$$

so the 97.5% expected shortfall is the mean of the worst 2.5% of outcomes. It is also called conditional VaR or CVaR, and being able to say that those are the same object under three names is worth a sentence in any interview on the topic.

When the loss distribution is continuous at VaR_α this equals $\mathbb{E}[L \mid L > \mathrm{VaR}_\alpha]$, and that conditional form is the one to quote in an interview because it is the one that explains itself. The distinction bites only when the loss has an *atom* at the quantile, which is exactly the case in the example below: there the worst 5% has to be assembled from the whole 4% default outcome plus 1% of the survival outcome, and it is the averaging form, not the conditional one, that is subadditive. Quoting the conditional form as the definition is a common error and it quietly gives away the property this section is about, because the tail conditional expectation is *not* subadditive on discrete losses.

The section is about two things: the specific mathematical defect in VaR that ES exists to repair, and the cost of the repair, because ES is not strictly better and pretending it is will lose you the follow-up.

The defect: VaR is not subadditive

A risk measure ρ is called **coherent** if it satisfies four axioms. Three are uncontroversial: adding cash reduces risk by that amount, doubling a position doubles its risk, and a position that always loses more is riskier. The fourth is the interesting one.

> KEY IDEA
>
> **Subadditivity**: $\rho(A+B) \le \rho(A) + \rho(B)$. Combining two portfolios cannot create risk. This is not a technicality, it is the mathematical statement that **diversification helps**, and a measure that violates it can tell a firm that splitting a book across two desks reduces its risk, which is an invitation to arbitrage the risk system rather than manage the risk. **VaR is not subadditive. ES is.**

The violation is not exotic, and the cleanest demonstration is a portfolio any credit desk would recognise.

WORKED EXAMPLE

Two bonds, and a measure that punishes diversification. Each bond pays 5 if the issuer survives and loses 100 if it defaults. The default probability is 4% and the two issuers are independent.

One bond, at 95%. The loss is -5 with probability 0.96 and $+100$ with probability 0.04. Since $0.96 \geq 0.95$, the 95th percentile of the loss sits at -5, so

$$\mathrm{VaR}_{95}(\text{one bond}) \;=\; \mathbf{-5},$$

a *negative* risk number. VaR reports the position as risk-free, because the default lives in the 4% it does not look at.

Both bonds. The outcomes are: both survive, loss -10, with probability $0.96^2 = 0.9216$; exactly one defaults, loss $+95$, with probability $2(0.04)(0.96) = 0.0768$; both default, loss $+200$, with probability 0.0016. Now $0.9216 < 0.95$, so the 95th percentile has moved into the one-default outcome:

$$\mathrm{VaR}_{95}(\text{both}) \;=\; \mathbf{+95} \qquad \text{against} \qquad \mathrm{VaR}_{95}(A) + \mathrm{VaR}_{95}(B) \;=\; \mathbf{-10}.$$

Diversifying across two independent issuers has, according to VaR, multiplied the risk. That is the subadditivity violation, in numbers you can recompute in thirty seconds.

The same portfolio measured with ES. For one bond the worst 5% consists of the 4% default outcome and 1% of the survival outcome, so

$$\mathrm{ES}_{95}(A) \;=\; \frac{0.04(100) + 0.01(-5)}{0.05} \;=\; \mathbf{79}.$$

For the pair, the worst 5% is 0.16% at $+200$ and the remaining 4.84% at $+95$:

$$\mathrm{ES}_{95}(A+B) \;=\; \frac{0.0016(200) + 0.0484(95)}{0.05} \;=\; \mathbf{98.36} \;\leq\; 79 + 79 = 158.$$

Subadditivity holds, and the diversification benefit is visible as the gap between 98.36 and 158. Note also that ES priced the single bond at 79 while VaR called it -5: the whole difference is that ES looks inside the tail and VaR only looks at its edge.

The calibration, and why the regulator picked 97.5%

Under a normal distribution ES has a closed form,

$$\mathrm{ES}_\alpha \;=\; \mu + \sigma\,\frac{\varphi(z_\alpha)}{1-\alpha},$$

where φ is the standard normal density and z_α the quantile. That formula explains a choice that otherwise looks arbitrary. The Basel framework replaced 99% VaR with **97.5% expected shortfall**, and the reason is that under normality the two are almost the same number:

Loss distribution	**VaR**$_{99}$	**ES**$_{97.5}$	**ES**$_{97.5}$ / **VaR**$_{99}$
Normal	2.326	2.338	1.005
Student t, $\nu = 6$	2.566	2.659	1.036
Student t, $\nu = 4$	2.650	2.824	1.066

All figures are in standard deviations, with the t distributions standardised to unit variance so the comparison is fair. Read the table as a policy decision. The confidence level was moved down from 99% to 97.5% precisely so that the switch would be roughly **capital-neutral for a normal book**, and would therefore bite only where the tails are actually fat, which is where the regulator wanted it to bite. The measure is not a blunt increase in capital, it is a reallocation of capital toward books with tail risk, and saying that is a much better answer than "Basel moved to ES".

REAL INTERVIEW QUESTION (Credit Suisse · Market Risk)

What is VaR and what is ES (Expected Shortfall)?

WHAT THEY TEST / ANSWER

Define both in one sentence each, in a way that makes the relationship obvious, then give the picture.

VaR at confidence α over a horizon is the loss that is not exceeded with probability α. At 99% over one day, it is the loss that a bad day exceeds one time in a hundred. It is a *quantile* of the loss distribution.

ES at the same confidence is the *average* loss given that you are beyond the VaR. It is the mean of the tail, not its edge, and it is therefore always at least as large as VaR.

The picture that makes it stick. Draw the loss distribution. VaR is a vertical line at the percentile; ES is the centre of mass of everything to the right of that line. VaR tells you where the bad region starts, ES tells you how bad it is inside it.

Then the one-line consequence, which is what turns a definition into an answer: two books can have the same VaR and completely different ES, because VaR is blind to the shape of the tail beyond the cut-off. A book that loses just over the VaR on a bad day and a book that loses ten times the VaR on a bad day are indistinguishable to VaR. That is the entire argument for the second measure, and it is worth adding that both are computed from the same loss distribution, so ES is not extra information, it is a different summary of the information you already had.

REAL INTERVIEW QUESTION (Credit Suisse · Market Risk)

Between VaR and ES, which is best and why?

WHAT THEY TEST / ANSWER

Refuse the framing, briefly, and then answer it properly, because "ES is better" is the memorised answer and is only two thirds right.

Where ES wins.

It is coherent, in particular subadditive, so it cannot report that diversification increased risk. VaR can, and the two-bond example is the standard demonstration.

It sees the tail. VaR is a threshold, so it is insensitive to everything beyond it, and that insensitivity is directly exploitable: a book can be constructed to have a small VaR and an enormous loss just past it, by selling deep out-of-the-money options. ES prices that construction correctly.

It varies smoothly with the confidence level, since it averages the tail rather than reading off one order statistic, so it does not jump the way a sample VaR does when the observation sitting at the quantile is replaced by the next one. Do not overstate this into robustness: ES is *more* sensitive than VaR to the *size* of the worst observation, since doubling the largest loss moves ES and leaves the quantile untouched. Smooth in the confidence level, fragile in the tail, which is the same trade-off that reappears as the estimation problem below.

Where VaR wins, and this is the half that gets missed.

It is backtestable, cleanly. VaR makes a falsifiable statement you can count: exceptions should occur $(1 - \alpha)$ of the time, and comparing the count against a binomial is a genuine hypothesis test. ES is not *elicitable*, so there is no equivalently simple statistic to compare it against, and the backtests that exist are weaker and need more data. This is not academic: Basel adopted ES for capital while keeping VaR-based backtesting for model approval, which is a regulator saying in public that each measure does something the other cannot.

It is simpler and more robust to estimate. ES is an average over the few worst observations, so at 97.5% on 250 days it is the mean of about six data points and its estimate has a wide error bar. VaR at least has the whole sample either side of it.

It is universally understood. VaR is the number a board, a regulator and a client have all seen before.

The resolution. They answer different questions and a desk runs both, with VaR as the daily control that gets backtested and ES as the capital and tail measure. And then say the thing that outranks the whole comparison: both are estimated from a historical sample, so both assume the future resembles the past. The measure that covers what neither can is a stress test, which asks what happens in a scenario that is not in the data at all. A candidate who ends there has answered a better question than the one asked.

REAL INTERVIEW QUESTION (ING · Rates)

What is a CVA / CVaR, and what is the difference between the two?

WHAT THEY TEST / ANSWER

The question is built on an acronym collision and the answer is to separate them immediately, because a candidate who blends them has revealed that they know neither.

CVaR is conditional value at risk, which is this section's subject: expected shortfall under a third name. It is a *risk measure*, computed on a portfolio's loss distribution, and it answers how bad the tail is.

CVA is credit valuation adjustment, which is a *price*. It is the market value of the counterparty credit risk embedded in a derivative: the expected loss from your counterparty defaulting before the trade matures, computed as the loss given default applied to the expected exposure profile, weighted by the probability of default at each future date. It sits in the fair value of the trade and is charged to the desk, usually by a dedicated XVA desk that then hedges it.

So they differ on every axis worth naming. One is a risk measure and the other is a valuation adjustment. One is a quantile average of a loss distribution and the other is an expectation of a credit loss. One is measured in units of loss on a portfolio and the other is a currency amount attached to a specific trade. One is owned by market risk and the other by the XVA desk. They share three letters and nothing else, which is precisely why the question exists.

Then the connection that earns the follow-up, because there is one. CVA is itself volatile, since it moves with the counterparty's credit spread and with the exposure, so a bank carries market risk *on its CVA* and regulators require capital against that volatility. So there is such a thing as a VaR of CVA, which is a third object again, and being able to name it shows you understand that CVA is a position rather than a number.

INTERVIEW TRAP

1. **Saying ES is simply better than VaR.** It is coherent and tail-sensitive, and it is harder to estimate and harder to backtest. The regulator uses both, which is the clue.
2. **Confusing ES with the loss at the tail's edge.** ES is the *average* beyond VaR, not the VaR at a deeper confidence level, though the two are numerically close for well-behaved distributions.
3. **Confusing CVaR with CVA.** A risk measure against a valuation adjustment, as the ING answer above sets out. The names are an accident and interviewers use the collision deliberately.
4. **Believing coherence solves the estimation problem.** A coherent measure computed from a short sample of a fat-tailed distribution is still a bad number. Coherence is a property of the measure, not of the estimate.
5. **Assuming subadditivity always fails for VaR.** It holds for elliptical distributions, which includes the normal, so a variance-covariance VaR is subadditive by construction. The failures appear with skewed, discrete or fat-tailed losses, which is exactly credit and options.

VISUAL / CODE TO BUILD: Interactive

A single loss-distribution panel with a shape control. The user chooses a distribution, from normal through progressively fatter-tailed, and sets the confidence level with a slider. A vertical line marks VaR and a shaded region marks the tail beyond it, with the centre of mass of that shaded region marked as ES. Watching the line stay still while the shaded mass redistributes as the tail fattens is the entire distinction between the two measures.

A second panel makes the subadditivity failure concrete. Two independent defaultable positions are shown with sliders for the default probability and the confidence level, and the panel prints $\mathrm{VaR}(A)+\mathrm{VaR}(B)$ against $\mathrm{VaR}(A+B)$, and the same pair for ES, colouring the VaR row red whenever the sum is smaller than the combined figure. Sweeping the default probability through the confidence level and watching the VaR row flip into violation, while the ES row never does, is a proof the reader performs rather than reads.

21.8 Stress Testing

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VaR and expected shortfall are both estimated from a sample of history, which means both answer the question "how bad can it get, given that the future looks like the recent past". Stress testing exists because that qualification is where the money is lost. A stress test asks a different question: not what happens with some probability, but what happens in a *specified* scenario, whether or not that scenario appears in the data.

KEY IDEA

The distinction to state in an interview is **probabilistic versus deterministic**. VaR is a statement about a distribution: a loss level and the probability of exceeding it. A stress test is a statement about a scenario: a set of market moves and the loss they produce, with *no probability attached at all*. That absence is the feature, not a defect. It is what lets a stress test cover events that have never happened, and it is why the two measures are complements and not substitutes.

The four kinds

- **Sensitivity tests.** Move one factor and see what breaks: equities down 20%, the curve up 100 basis points, volatility up 10 points. Simple, fast and mainly diagnostic, because real crises do not move one thing.
- **Historical scenarios.** Replay an actual episode with its actual moves across every factor at once: October 1987, autumn 1998, September 2008, March 2020. The advantage is internal consistency, since the moves genuinely happened together. The limitation is that you are always fighting the last war.
- **Hypothetical scenarios.** Construct an episode that has not happened but could: a sovereign default, a currency peg breaking, a large counterparty failing. Harder to build defensibly, and the only way to cover a risk that has no precedent.
- **Reverse stress tests.** Invert the exercise. Instead of asking what a scenario costs, start from a loss large enough to threaten the firm and work backwards to the scenarios that

would produce it. This is the most useful of the four and the least often done, because it finds the exposures nobody thought to test.

Two design rules that decide whether the test is worth running

Shock the whole vector, consistently. A scenario in which equities fall 30% and nothing else moves is not a scenario, it is a sensitivity. In a real equity selloff, implied volatility rises, the skew steepens, correlations move toward one, credit spreads widen, dividends are cut, funding tightens and liquidity evaporates. A stress test that omits the co-movements will systematically understate the loss on a hedged book, because the whole point of a hedge is that it survives one factor moving and not all of them.

Revalue, do not extrapolate. This is the rule most often broken, and it is worth carrying an example, because the size of the error surprises people.

WORKED EXAMPLE

Greeks against a full revaluation, on a 30% shock. A desk is short 10,000 one-year at-the-money calls. Spot is 100, volatility 20%, rates zero. The option is worth **7.966**, with a delta of **0.540**, a gamma of **0.0198** and a vega of **39.70** per unit of volatility.

The scenario is spot down 30% to 70 and implied volatility up 15 points to 35%, which is a mild version of a real equity shock.

Estimated from the Greeks, per option short:

Term	**Effect**
Delta, $-0.540 \times (-30)$	$+16.195$
Gamma, $-\frac{1}{2}(0.0198)(30)^2$	-8.931
Vega, -39.70×0.15	-5.954
Total	$\mathbf{+1.309}$

From a full revaluation: at spot 70 and volatility 35% the option is worth **2.330**, so the short position has made $7.966 - 2.330 = +\mathbf{5.636}$.

The Greek estimate captures **23%** of the actual result. Across 10,000 options that is a difference of about 43,000 currency units on a position the risk system would have reported as making 13,000. And note which way the error runs: here it *understates* a gain, but for a long option position or a barrier product the same second-order failure understates a loss, which is the version that ends careers. Greeks are a local approximation, and a stress scenario is by construction not local, so a stress test that shocks the Greeks is not a stress test. Revalue the book.

The contrast with VaR is worth holding on four axes, and value at risk sets it out from its own side. *Question*: VaR asks what loss is exceeded with a given probability, a stress test what loss a given scenario produces. *Probability*: VaR carries one by construction, a stress test carries none and should not, since a scenario with a probability attached can be argued down as unlikely in a committee. *Data*: VaR cannot see outside its sample, a stress scenario is specified and can include what has never happened. *Regime*: VaR calibrated on a calm period reports low risk because the period was calm, which is procyclical and self-confirming, while a fixed scenario does not shrink just because the market has been quiet. Expected shortfall sits between them, looking into the tail but still estimated from the same sample and so inheriting the same blind spot.

REAL INTERVIEW QUESTION (Goldman Sachs · EQD)

Stress testing: as an equity structurer, how would you stress test a new structured product to evaluate its resilience under adverse market conditions? What scenarios would you consider, and how would you interpret the results?

WHAT THEY TEST / ANSWER

Answer as a process, in four steps, because the question is asking how you work rather than what you know.

Step one, identify what the product is actually short. Decompose it before shocking anything. A typical autocallable leaves the desk short a down-and-in put, long a stream of conditional coupons and short a call feature that terminates the trade. So the exposures that matter are downside spot, the volatility of the downside strikes, the skew, dividends, rates, and, in a worst-of structure, correlation. Naming those first means your scenarios test the product rather than the market in general.

Step two, build scenarios that move everything at once. Spot down 10, 20, 30, 40 and 50%, and at each level move the co-movers in the direction they actually move: implied volatility up, skew steeper, correlation toward one, dividends cut, funding wider. Add at least one historical replay, 2008 and March 2020 being the standard pair for an equity book, and one path-dependent scenario, since a product with a barrier and an observation schedule cannot be valued from a terminal shock. A slow grind to the barrier and a fast gap to the same level are completely different trades for the hedger, and only the second one leaves you unable to rehedge.

Step three, revalue rather than extrapolate. Full repricing on the shocked surface, not a Greek expansion, for the reason given above: barrier products have discontinuous deltas near the barrier and the local approximation is worst exactly where the risk is.

Step four, interpret, which is what the question really asks. Three readings.

Is the loss survivable? Compare it against the desk's limits and against the fee earned on the trade. A product that pays 2% and loses 40% of notional in a plausible scenario is not a good trade at any hedge ratio.

Where does the risk concentrate? If the loss is dominated by one scenario or one factor, that is a concentration, and the answer is usually to change the product rather than the hedge: move the barrier, cap the notional, or add a feature that truncates the exposure.

Does the hedge survive? Check that the hedging strategy is still executable in the scenario. Delta hedging through a barrier assumes liquidity that a stressed market does not supply, and a hedge that requires selling into a limit-down market is not a hedge.

Close on the structurer's version of the point. Stress testing at the design stage is cheaper than stress testing after issuance, because before you print it you can still change the terms, and afterwards you can only hedge. That is the answer a structuring desk wants to hear.

INTERVIEW TRAP

1. **Shocking one factor at a time and calling it a stress test.** Single-factor sensitivities miss exactly the co-movement that hurts a hedged book.
2. **Shocking the Greeks instead of revaluing.** A stress scenario is not local, and the worked example above shows a second-order expansion capturing under a quarter of the true P&L.
3. **Attaching probabilities to stress scenarios.** As soon as a scenario has a probability, it can be argued down as unlikely. The absence of a probability is what protects it.
4. **Only replaying history.** Historical scenarios are internally consistent and always describe the previous crisis. Hypothetical and reverse tests are what cover the next one.
5. **Ignoring the path.** A shock applied to the terminal spot cannot capture a barrier being touched, a knock-out triggering or a hedger being unable to trade. Path matters for any product with a schedule.
6. **Running the test and not acting on it.** A stress framework with no linked action, a limit, a reserve, a change in the product, is documentation rather than risk management, and limits frameworks is where the link is made.

VISUAL / CODE TO BUILD: Interactive

A two-axis stress grid. Spot shock runs along the horizontal axis and volatility shock up the vertical, and the cell colour is the book's P&L under a full revaluation, green through red. The reader picks a position from a short list, a short call, a long straddle, a call spread, an autocallable, and the shape of the coloured surface is the position's risk made visible in two dimensions rather than as a row of Greeks.

The teaching moment is a toggle labelled "Greeks only", which redraws the same grid from a delta-gamma-vega expansion. Near the centre the two surfaces agree; toward the corners they diverge sharply, and for the autocallable they disagree in *sign* in places. Flipping the toggle back and forth is the argument for full revaluation, delivered as a picture.

A second control adds correlated co-movement: a single "crisis intensity" slider that moves volatility, skew, correlation, dividends and funding together along a calibrated path, so the reader can compare a pure spot shock with a realistic one and see how much of the loss the naive scenario was missing.

21.9 Scenario analysis

sales · trading · structuring | core | cross-asset

Greeks tell you what happens for a small move. Value at Risk tells you the size of a bad day in probability terms. Neither tells you what happens *in a particular world*: equities down 25%, volatility up 15 points, dividends cancelled, and the correlation you were relying on gone to one. Constructing those worlds, valuing the book inside them, and using the answer to make a decision is scenario analysis. It is the most management-friendly risk tool there is, because its output is not a statistic but a sentence: if this happens, we lose that, and here is what we would do about it.

It is also the tool that structuring and exotics interviews test hardest, usually in disguise: "walk me through the scenarios for this product" is a scenario-analysis question wearing a term sheet.

Scenario analysis and stress testing are not the same thing

The two are constantly conflated, and separating them cleanly is worth marks.

- **Stress testing** asks whether the firm *survives* a severe adverse world. It is deliberately extreme, often regulator-specified, and the output is measured against capital and liquidity. Severity calibration, the regulatory programmes and the survival question are developed in stress testing.
- **Scenario analysis** is the broader discipline of building coherent what-if worlds, severe or not, and using them to *decide*: which hedge to put on, which strike to pick, whether to sell the product at all, where the book's non-linearity bites. Most scenarios a desk looks at every day are not stresses. They are plausible worlds a week or a month out.

> **KEY IDEA**
>
> The distinguishing feature of a scenario, against a sensitivity, is **coherence across factors**. A Greek moves one input and holds everything else fixed. A scenario moves *everything at once, consistently*: if equities fall 25%, then implied volatility rises, the skew steepens, dividends get cut, credit spreads widen, rates fall, correlations rise towards one and liquidity thins. A scenario that shocks spot by -25% while leaving volatility unchanged is not conservative, it is incoherent, and it will systematically misstate the P&L of any option book.

Four ways to build a scenario

1. **Historical replay.** Take the actual factor moves of a real episode, October 1987, autumn 2008, the March 2020 crash, the 2022 rate shock, and apply them to today's book. Advantage: internally consistent by construction, since the world really did behave that way, and immune to the accusation of being made up. Drawback: today's portfolio may have no exposure to the factors that drove the old crisis, and the next crisis will be driven by something else.
2. **Hypothetical shocks.** Specify the moves directly: rates +200bp parallel, equities -30%, EUR/USD -10%, all correlations to 0.9. Advantage: you can target the book's actual vulnerabilities and reach beyond anything in the data. Drawback: you now own the coherence problem, and an incoherent combination produces a number nobody believes.
3. **Macro-narrative scenarios.** Start from a story (a hard landing, an energy shock, a sovereign event, an inflation re-acceleration) and derive the factor moves from it. Advantage: it is the only form that senior management and clients genuinely engage with, and it forces the second-round effects into the open. Drawback: the translation from narrative to numbers is judgemental and easy to argue with.
4. **Reverse stress testing**, owned by stress testing and listed here only as one of the four construction routes. Invert the question. Instead of asking what a scenario costs, ask what scenario would cost a specified amount: what has to happen for this book to lose €100m, or to breach the capital floor. Advantage: it finds the vulnerabilities you did not think to test, which is precisely the ones that hurt. Drawback: the answer is a set rather than a point, so it takes judgement to report usefully.

The workhorse: the spot and volatility grid

For any book with optionality, the standard daily artefact is a two-dimensional matrix of revalued P&L: spot shocks along one axis, implied volatility shocks along the other, with the book fully repriced in every cell rather than approximated by Greeks. Traders live in this grid, and being able to read one is a genuinely testable skill.

Note what the grid is and is not. Shocking spot without moving volatility is exactly the incoherence warned about above, and most cells of the grid do it. That is deliberate: the grid is a *map of the book's shape*, not a set of scenarios. A scenario is a single coherent point, and on this map it is a diagonal, spot down and volatility up together. You read the grid to find where the book is convex, where it is short, and where the worst cell hides; then you pick the coherent diagonal to quote as the loss.

WORKED EXAMPLE

Reading a grid. A long-gamma, long-vega book, P&L in €thousands, spot shock across the top and a parallel implied volatility shock down the side.

	−20%	−10%	0%	+10%	+20%
vol −5 pts	+1,100	+150	−450	+50	+900
vol ±0	+1,900	+700	0	+600	+1,700
vol +5 pts	+2,700	+1,250	+450	+1,150	+2,500

Three readings, in the order a trader takes them. The **middle row** is convex in spot: losses near zero, gains in both directions, which is long gamma. The **middle column** rises as you go down the page, which is long vega. And the **worst cell of the whole grid** is not in a corner, it is spot unchanged with volatility five points lower, at −€450k. That is the signature of a long-option book, and it makes the practical point that a grid exists to make: the worst outcome for a non-linear book is frequently a *small* move, not a large one, so a scenario set containing only extreme shocks will report that everything is fine.

Two refinements matter in practice. Real scenario sets shock the volatility *surface* rather than a single number, because skew and term structure move differently from the level, and a book can be long vega and short skew at the same time. And the grid should be run with time moved forward as well as prices, since for a book with short-dated barriers the interaction between decay and spot is the entire risk.

Scenario analysis on a payoff: the structuring version

For a structured product, scenario analysis is not a grid but an **enumeration**: a complete, mutually exclusive list of the paths the underlying can take, with the client's cash flows in each. Interviewers ask for this constantly because it proves you understand a product rather than having memorised its name.

The method, which works on any term sheet:

1. **List the observation dates** and, at each, the levels that matter (call barrier, coupon barrier, protection barrier, strike).
2. **Sort the outcomes by the earliest date at which something happens**, because early redemption truncates every later scenario.
3. **For each terminal case, give the cash flow** in the client's terms, capital plus coupons or capital minus a participation in the fall.

4. **Check the extremes and the boundaries.** What happens exactly at a barrier, and what happens in the flat case, which is usually the most probable and the least discussed.
5. **Say who is long and short what**, so the payoff description turns into a risk description: the issuer's hedge is the mirror of the client's payoff.

The payoffs themselves are defined in autocallables and Phoenix notes, which also carries the canonical worked enumeration (a Phoenix with a coupon barrier at 70%, a call barrier at 110% and a down-and-in put at 50%), and the hedging of them is in hedging structured products. Read that enumeration with the five steps above in hand: what belongs here is not the product but the method, which is the only part of it a risk audience needs.

REAL INTERVIEW QUESTION (Société Générale · Structured Products)

From a term sheet, explain the product's mechanism and all the possible scenarios.

WHAT THEY TEST / ANSWER

This is a process question, so show the process rather than jumping to an answer. Read the term sheet in a fixed order and say it out loud as you go: underlying and its initial fixing, notional and currency, maturity, then every observation schedule, then every level expressed as a percentage of the initial fixing, then the coupon definition, then the redemption rules, then the small print that changes everything, which is the barrier observation convention (continuous or on dates only), the memory feature, whether it is a single name, a basket or a worst-of, and the issuer's call rights. Then decompose the product into its option components, since that is what makes the scenarios derivable instead of memorised: a zero-coupon bond plus a strip of digitals plus a short down-and-in put is the anatomy of most yield-enhancement notes. Then enumerate the scenarios by the earliest date at which something happens, cover the flat case explicitly because it is usually the most likely, check the boundary values at each barrier, and give the client's cash flow in each branch. Close on the two things a desk cares about that a purely descriptive answer misses: what the client is really short, which is volatility, skew and correlation, and where the issuer's hedging risk concentrates, which is around the barrier and near the observation dates. Adding one sentence on what the product needs in order to be a good idea for this client shows commercial judgement rather than mechanical knowledge.

REAL INTERVIEW QUESTION (Société Générale · Structured Products)

Explain how an autocall works according to the scenarios of the underlying's evolution.

WHAT THEY TEST / ANSWER

Structure the answer as the three regimes of the underlying, because that is what the question asks for. Underlying rises or stays firm: on the first observation date at or above the call level the note redeems early at par plus coupon, so the client's return is capped and short-dated, and the trade ends. Underlying drifts sideways below the call level: no early redemption, coupons paid whenever the level is above the coupon barrier, so the client earns an above-market yield for as long as nothing bad happens, which is the scenario the product is designed and sold for. Underlying falls sharply: coupons stop, and if the protection barrier is breached the client ends up long the underlying's loss at maturity. Then add the two dynamic observations that separate a good answer from a definition. The client is systematically **short the tails and long the middle**, so they are short volatility, short skew and, on a basket, short correlation, and their upside is capped while their downside is not. And the issuer's risk is concentrated near the barriers, where the hedge flips: as spot approaches an autocall trigger the effective maturity of the whole structure shortens abruptly, which is why hedging these products is a gamma and vega problem around specific levels rather than a smooth delta, developed in hedging structured products.

The 2020 dividend shock is the episode that makes all of this concrete. Exotics desks that had sold autocalls and other yield-enhancement notes were structurally long future dividends, and in the spring of 2020 dividends were suspended or cancelled across European banks and other sectors, with cuts of a size no historical window contained. The exposure itself, why the hedge carries the dividend stream and which way the sign runs, belongs to dividends; what belongs here is what the episode says about scenario design. First, a scenario must be *multi-factor and coherent*, because a spot-and-volatility grid would have shown none of it. Second, some risk factors are policy-driven and therefore not usefully modelled from history at all, which is exactly the case for reverse stress testing rather than historical replay. Third, the loss came from a term the desk treated as an assumption rather than as a traded risk, which is the recurring structure of exotic-book accidents: the danger sits in the parameter nobody put a limit on.

INTERVIEW TRAP

Four traps. First, **incoherent scenarios**: shocking spot down and leaving implied volatility, skew, dividends and correlation unchanged, which understates the loss on almost any option book. Second, **only testing extremes**, when a non-linear book's worst cell is often a small move with a volatility drop, as in the grid above. Third, **using Greeks instead of full revaluation**: a delta-gamma approximation breaks exactly where scenarios are interesting, at barriers, digitals and large moves. Fourth, **attaching a probability to a scenario and then treating it as a forecast**: the point of a scenario is conditional, what happens *if*, and its usefulness does not depend on knowing how likely it is.

VISUAL / CODE TO BUILD: Interactive

A scenario workbench. Left: a portfolio builder (spot positions, vanillas, a barrier option, an autocall). Centre: a live spot-and-volatility grid with editable axis steps, fully revalued per cell, colour-scaled, with the worst cell highlighted so the reader discovers that it is not always a corner. Right: three tabs. **Coherent shocks**, where dragging a single "equity crash severity" slider moves spot, implied volatility, skew, dividends and correlation together along an editable transmission map, with a toggle to break the coherence and see the reported loss shrink misleadingly. **Historical replay**, which applies the actual factor moves of 1987, 2008, March 2020 and 2022 to the current book. **Reverse stress**, where the reader types a loss number and the tool searches for the mildest coherent scenario that produces it. A separate payoff-enumeration panel takes a term sheet's parameters (call, coupon and protection barriers, memory flag, barrier convention) and prints the exhaustive scenario tree with cash flows, so the reader can check their own enumeration against it.

Where this goes next

The severe end of the discipline, including regulatory programmes and severity calibration, is stress testing, and the probabilistic counterpart is Value at Risk. Scenario grids are built on the sensitivities aggregated in portfolio-level Greeks, and the after-the-fact version of the same arithmetic, explaining a realised move by factor, is PnL explain. The products used as examples here are defined in autocallables and Phoenix notes.

21.10 Aggregation of Greeks

trading · structuring | advanced | cross-asset

A trader adds up Greeks to decide what to hedge in the next ten minutes. A risk function adds up the same Greeks to decide whether the firm can survive next month. Those are different questions, they call for different arithmetic, and they routinely produce different numbers for the same book.

The mechanics of combining Greeks correctly, cash delta, beta adjustment, square-root-of-time vega weighting, the gamma matrix and measuring convexity by revaluation, are in portfolio-level Greeks aggregation and are not repeated here. This section is about aggregation as a risk-management problem: what is lost when you add things up, what regulators allow you to claim from it, and why the number on the risk screen differs from the one on the desk.

Three different aggregation problems

The word covers three tasks that get harder in order.

- **Across products, within a desk.** An option, a future and a swap on the same underlying can be reduced to a common exposure, and this is the case portfolio-level aggregation handles.
- **Across desks.** Two desks may both be long delta on the same name while using different models, different marks and different conventions, so their Greeks are not strictly comparable even when they carry the same label.
- **Across risk types.** Market, credit and operational risk are measured on different horizons at different confidence levels with different distributions. There is no arithmetic that legitimately adds them, and the fact that firms produce a single capital number anyway is a convention rather than a calculation.

KEY IDEA

Aggregation destroys information, and *which* information it destroys is a design choice rather than a technical detail. A single firm-wide delta of zero is compatible with being enormously long one sector and enormously short another. The purpose of a risk aggregation framework is not to produce one number; it is to choose the small set of numbers that cannot all be zero while the firm is in danger.

Aggregate factors, not Greeks

The technically correct route is not to sum sensitivities but to decompose everything into a common set of **risk factors** first, and then aggregate exposures factor by factor.

The reason is that a Greek is a derivative with respect to *something*, and two Greeks with the same name can be derivatives with respect to different things. A vega computed against a parallel shift of a flat volatility and a vega computed against the five-year point of a surface are both called vega and cannot be added. Once every position is expressed as a sensitivity to the same named factors, the addition is meaningful, and everything downstream, value at risk, stress testing, limits, is computed on the factor exposures rather than on the labels.

That is also why the factor set is a governance object rather than a modelling convenience: adding a factor makes a risk visible and removing one makes it disappear from every report at once.

Diversification benefit, and why it is contested

Here is the arithmetic that drives most of the argument between a trading floor and its regulator.

Risk measured as a quantile is **subadditive** under reasonable assumptions: the risk of a combined portfolio is no more than the sum of the risks of its parts, because the parts do not all lose money at the same time. The gap is called the **diversification benefit**.

WORKED EXAMPLE

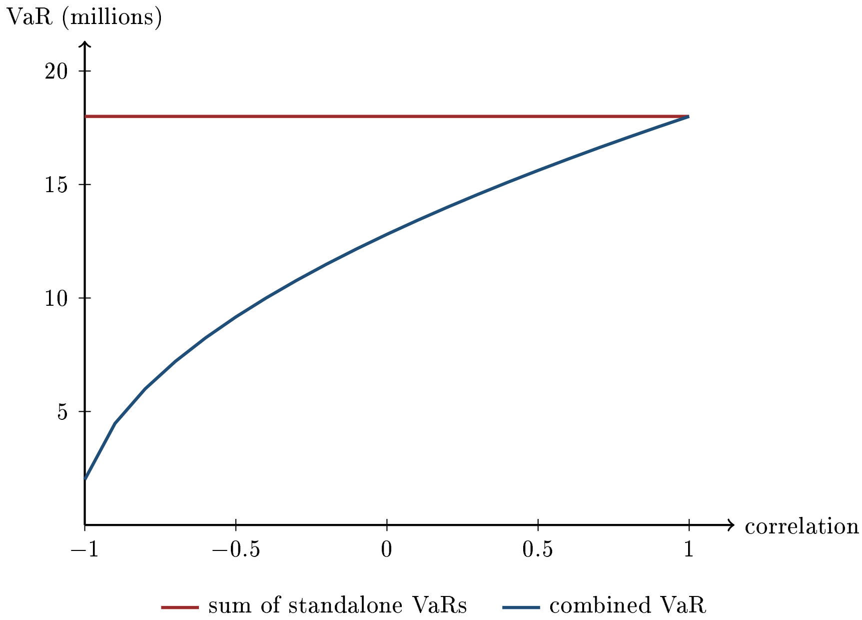
Two desks, with standalone value at risk of 10 million and 8 million. Under a normal approximation the combined figure is

$$\sqrt{10^2 + 8^2 + 2\rho \times 10 \times 8}\,.$$

Correlation	Combined VaR	Diversification benefit	As % of the sum
−0.5	9.17	8.83	49.1%
0	12.81	5.19	28.9%
0.3	14.56	3.44	19.1%
0.5	15.62	2.38	13.2%
1	18.00	0.00	0.0%

The sum of the parts is 18 in every row. At a correlation of zero the firm can claim it really only risks 12.8, which is 5.2 million of capital it does not have to hold. At a correlation of one it can claim nothing.

So the entire benefit rests on a correlation estimate, and the estimate has to be right in the scenario the capital is for.



The vertical gap between the two lines is the diversification benefit, and the picture makes the regulatory objection obvious: the whole claim lives in the left half of the chart, and it vanishes at the right-hand edge. Correlations estimated over a calm sample sit in the middle of the range and move toward the right-hand edge in a crisis, so the benefit is largest in the estimate exactly when it is smallest in reality.

That is why post-crisis capital rules are sceptical of it. Under the Fundamental Review of the Trading Book, diversification is recognised within a risk class through prescribed correlations rather than estimated ones, and **across** risk classes the charges are largely added rather than combined. Firms hold more capital than their own models suggest, on purpose, because a model that recognises diversification is making a claim about the tail from data that does not contain it. See capital requirements.

Three numbers for the same book

A large bank produces at least three views of the same position, and a risk manager spends much of the week on the differences.

- **The front office view**: the desk's own system, marked to the desk's own model, updated continuously. Fast, granular, and calibrated to how the desk hedges.
- **The risk view**: an independent system with an independent factor set, typically run overnight. Slower, coarser, and the one that limits are measured against.
- **The finance view**: the official profit and loss, produced by product control with independent price verification.

They disagree, always and legitimately, because they are computed at different times against different marks with different granularity. What matters is not that they agree but that the differences are explained and stable. An unexplained gap that grows is one of the earliest signals of a mismarked or mis-booked position, which is why P&L explain is a control and not an accounting exercise.

INTERVIEW TRAP

Trusting an aggregate that nobody can decompose. If a firm-wide vega number cannot be broken back down into the desks, maturities and underlyings that produced it, then nobody can act on it: the trader cannot hedge a number they cannot attribute, and the risk manager cannot challenge one they cannot trace. An aggregate is only useful if it comes with the path back to the positions.

Why there is never one limit

Because aggregation destroys information, a limit framework does not set one limit on one aggregate. It sets several deliberately overlapping ones, each catching what the others miss: notional and cash delta, which catch size; value at risk, which catches normal-market volatility; stress and scenario losses, which catch the tail that value at risk was never built to see; and concentration limits by name, sector and maturity bucket, which catch the position that nets to zero at the top level and is enormous underneath it.

The framework itself is risk limits frameworks, and the scenario machinery is stress testing.

Interview angle

REAL INTERVIEW QUESTION (Allianz Global Investors · Market Risk)

Once you have measured value at risk or expected shortfall for a portfolio, what do you do with it? Do you act before they are breached, and how?

WHAT THEY TEST / ANSWER

The second half is the question. Anybody can compute a risk measure; the interviewer wants to know whether you understand that a limit is a management tool rather than a threshold you watch approach.

Yes, you act before. A limit hit is already a failure, because unwinding under a breach means trading in whatever conditions caused it, usually a market that has just moved against you and become less liquid. The framework therefore has **soft triggers** well below the hard limit: a utilisation level that generates a warning, one that requires the desk to explain and produce a plan, and only then the limit itself.

What you actually do with the number, in order. Attribute it: which desks, which factors, which positions, because an aggregate you cannot decompose is not actionable. **Compare it**: against yesterday, against the limit, and against the desk's own view, and investigate the difference rather than the level. **Interrogate it**: value at risk is a quantile of a modelled distribution, so run the stress and scenario numbers alongside it, because the two disagree in exactly the situations that matter. **Act on it**: reduce, hedge, or ask for a temporary increase with a rationale, which is a legitimate outcome rather than a failure if the risk is understood and wanted.

Then the honest qualifications, which is what separates a risk answer from a recited one. Value at risk says nothing about how bad the tail is beyond the quantile, which is why expected shortfall replaced it as the regulatory measure. It is backward-looking, so a quiet sample produces a small number precisely when a regime is about to change. And it is model-dependent, so two defensible implementations give different answers on the same book. Being able to say what your risk measure does not see is the most useful thing a risk manager does.

REAL INTERVIEW QUESTION (Allianz Global Investors · Market Risk)

Tell me about hedging strategies using derivatives and about modifying a portfolio's duration: which instruments, and what is the impact of leverage?

WHAT THEY TEST / ANSWER

Three parts asked, so answer three parts, and keep the framing at portfolio level throughout because that is what the question is about.

Instruments, chosen by what you want to change. To change *direction* without selling the assets: index futures for equity, or bond futures and interest rate swaps for rates, because they are liquid, cheap and reversible. To change the *shape* of the outcome rather than its direction: options, which cost premium and buy convexity. To change a *specific* exposure that a broad hedge would not touch: single-name or sector instruments, which is where basis risk enters.

Duration, and this is where a portfolio manager is tested. Duration is additive in value terms, so the target is reached with notional × duration arithmetic rather than by trading a percentage of the portfolio. Receiving fixed on a swap adds duration and paying fixed removes it; long bond futures add it. The instruments to name are interest rate swaps for precision and bond futures for liquidity, and the practical caveat is that a futures hedge is measured in whole contracts against a cheapest-to-deliver bond, so it is approximate in a way a swap is not.

Leverage, which is the part the question is really aimed at. A derivative hedge requires almost no capital, so it does not reduce the portfolio's size, only its sensitivity, and that has three consequences worth naming. **Margin**: a futures or cleared swap hedge is margined daily, so a correct hedge still generates cash calls that must be funded, and a hedge that is economically right can fail for liquidity reasons. **Basis**: the hedge and the exposure are not the same instrument, so leverage magnifies the residual as well as the intended exposure. **Gap risk**: a hedge sized on normal sensitivities does not protect against a jump, and a levered portfolio has less room to survive one.

Close with the general point, since it is the one worth carrying: derivatives change a portfolio's sensitivities without changing what it owns, which is why they are the right tool and why the risk that remains after hedging is a different risk rather than a smaller version of the same one.

INTERVIEW TRAP

Four aggregation errors. (1) Adding Greeks that are derivatives with respect to different things, most often vegas against different points of a surface. (2) Reporting a net figure without the gross, so a book that is long one thing and short another looks flat. (3) Claiming a diversification benefit from a correlation estimated in calm conditions, which the table above shows is worth 29% of capital at zero correlation and nothing at one. (4) Reconciling the front-office and risk views by adjusting one to match the other rather than by explaining the gap, which removes the signal the gap was carrying.

Where this goes next

The arithmetic of combining Greeks correctly is portfolio-level Greeks aggregation. The measures being aggregated are value at risk and expected shortfall, what they miss is stress testing, the structure the numbers feed is risk limits frameworks, and the control that lives on the difference between two views of the same book is P&L explain.

21.11 Risk Limits Frameworks

trading · structuring · sales core cross-asset

VaR, expected shortfall and stress testing all measure risk. A limits framework is what turns a measurement into a decision, and without it the measurement is documentation. This section is about the framework itself: how limits are set, what each type catches and misses, and what happens when one is breached. Living with limits day to day on a desk belongs to position limits; the design and governance belong here.

> **KEY IDEA**
>
> A limit is not a forecast and not a prediction that a loss will not happen. It is the **terms on which the firm lends you its balance sheet**. That framing matters because it settles the argument a trader eventually has with risk: the question is never whether the trade is good, it is how much of the firm's capacity that trader is entitled to consume. A limit that is never close to being hit is set too wide to be doing anything.

The hierarchy

Limits cascade, and each level exists to answer a different question.

- The **board** sets a **risk appetite**: the total amount of risk the firm is willing to run, usually expressed against capital and earnings volatility.
- That is allocated to **business lines**, which is a strategic decision about which businesses the firm wants to be in.
- Each business allocates to **desks**, which is where limits start to be expressed in tradeable quantities.
- Desks allocate to **traders** or to books.

The sum of the lower-level limits is normally *larger* than the level above, deliberately, because not every desk is at its limit at once. That overallocation is efficient and it is also the mechanism by which a firm discovers, in a crisis, that its aggregate limit was never really binding.

The types, and what each one misses

No single limit is sufficient, and the reason to know all of them is that each is blind exactly where another sees.

- **Notional and position limits.** Crude, unambiguous, hard to argue with. They miss everything about risk per unit of notional: a €100 million two-year bond and the same notional at thirty years are not comparable.
- **Sensitivity limits**, on delta, gamma, vega, DV01 and CS01. These are the working limits a desk lives against, aggregated as in Greek aggregation. They miss large moves, because a sensitivity is a local quantity.

- **VaR and ES limits.** Aggregate across factors and account for diversification. They miss anything outside the estimation sample.
- **Stress limits.** A ceiling on the loss in specified scenarios, which is the only limit type that constrains what a statistical measure cannot see.
- **Concentration limits**, by single name, sector, country or maturity bucket. These exist because a diversified measure will happily permit an undiversified book.
- **Stop-loss and drawdown limits.** The only backward-looking type: they act on realised P&L rather than on exposure, and they are the firm's admission that a position can be within every forward limit and still be wrong.
- **Holding-period and liquidity limits**, sizing a position against what can actually be exited, which liquidity risk shows is often the binding constraint.

Hard, soft and the levels in between

A **hard limit** cannot be exceeded; breaching it requires the position to be reduced. A **soft limit** triggers escalation and a conversation rather than a forced trade. In practice most frameworks add **trigger levels** below the limit itself, commonly at 75% and 90% of utilisation, so that the conversation happens before anyone is forced to trade in a market that may already be moving against them.

INTUITION

The distinction that decides how a breach is handled is **passive versus active**. An *active* breach means the trader traded into it, and it is a discipline problem. A *passive* breach means the market moved and the position's measured risk changed without anyone doing anything, and it is a risk problem. Both must be reported, and only the first is misconduct. Knowing the difference is what a risk interviewer is checking when they ask what you would do about a breach.

WORKED EXAMPLE

A passive breach, with no trade at all. A desk is short 10,000 one-year calls struck at 120. Spot is 100 and implied volatility 20%, so each option carries a vega of $S\,\varphi(d_1)\sqrt{T} = 100(0.28699) = 28.70$ per unit of volatility, or **0.287** per volatility point. The book's vega is **2,870** per point, against a desk limit of 3,500. Utilisation is **82%**: inside the limit, above the 75% trigger, nothing to do beyond a note.

Spot now rallies to 120. The options are at the money, where vega is largest:

$$120 \times \varphi(0.1) \;=\; 120(0.39695) \;=\; 47.63 \quad \text{per unit, or } 0.476 \text{ per point,}$$

so the book's vega is **4,763** per point. Utilisation is **136%** and the desk is in breach, having placed no order.

The vega rose 66% purely because spot moved into the strikes, which is what vega exposure means by vega being a function of moneyness. Two lessons come out of this and both are worth saying aloud. First, a limit framework built only on today's sensitivities is procyclical: it grants the most capacity exactly where the position is least stable. Second, this is why utilisation is monitored against triggers rather than only against the limit, and why a desk running at 82% of a vega limit with spot 17% below its strikes is not, in any useful sense, inside its limit.

REAL INTERVIEW QUESTION (Allianz Global Investors · Market Risk)

Once you have measured VaR and expected shortfall on a portfolio, what do you do with them? Do you act before they are breached, and how do you adjust the portfolio?

WHAT THEY TEST / ANSWER

The question has three parts and the middle one is the whole test.

What you do with them. The number is only useful against a reference. Compare it to the limit, to its own recent history, and to its decomposition: which positions and which factors are producing it. A VaR that has risen 40% in a week is information regardless of whether it is near the limit, and the first action is always to explain the change, since a move you cannot explain is a model or a booking problem before it is a risk problem.

Do you act before the breach. *Yes*, and say why rather than just yes. Acting only at the limit means acting exactly when everyone else subject to the same measure is also acting, and when the market that moved you there is least able to absorb the trade. That is the procyclicality problem, and it is why frameworks put triggers at 75% and 90%. The correct formulation is that the limit is a boundary you should never approach passively, and utilisation trending toward it is itself the signal, not the number crossing it.

How you adjust. In order of preference, because the ranking is the answer.

Hedge the factor that is producing the risk. The decomposition tells you which one; hedging that factor is cheaper and less disruptive than cutting the book.

Reduce the concentrated position. If the risk comes from one name or one strike, reducing that is more efficient per unit of risk removed than trimming everything.

Diversify. Only genuinely, and with the caveat that diversification measured on a covariance matrix is largest in calm markets and smallest when you need it.

Cut across the board. Blunt, expensive and what you are left with if you waited for the breach.

Then the sentence that finishes it. A risk measure with no action attached is reporting, not risk management. The framework is the link between the two, and the test of whether it works is not whether the number is computed but whether anything happens when it moves.

REAL INTERVIEW QUESTION (BNP · Market Risk)

A trader calls you about his counterparty limits while you are alone on the desk. What do you do?

WHAT THEY TEST / ANSWER

This is a judgement and integrity question dressed as a process question. Answer the process, and let the process demonstrate the judgement.

First, establish the facts. What is the limit, what is the current utilisation, and is the trade about to breach it or has it already breached? Do not act on the trader's characterisation; look it up. If the system says something different from the trader, the system is the record.

Second, and this is the point of the question, do not authorise it. You are alone on the desk, which changes nothing about who has authority to raise or waive a limit, and that person is not you. Saying this plainly and without hedging is the answer they are listening for. The pressure of being the only person available is exactly the condition under which limits get informally waived, and every well-known trading loss has a version of this moment in it.

Third, escalate rather than refuse and hang up. Call the person who does have the authority, and if they are unreachable, follow the documented escalation path to whoever is next. Being unable to reach an approver is not an approval. Meanwhile tell the trader what you are doing so they can make their own decision about the trade with correct information.

Fourth, record it. Time, what was asked, what you said, who you escalated to. If the trade happens and the limit is breached, the record is what makes it a manageable incident rather than a discovery weeks later.

Then the balancing sentence, so you do not sound obstructive. Counterparty limits exist to stop the firm concentrating exposure on one name, and a limit that blocks genuine business should be reviewed and resized through the proper process, quickly, rather than worked around. The right posture is to make the escalation fast, not to make the limit negotiable.

INTERVIEW TRAP

1. **Setting limits only on sensitivities.** A book can be inside every Greek limit and lose catastrophically in a stress scenario, because sensitivities are local. A framework needs at least one scenario-based limit.
2. **Treating a passive breach as misconduct.** The market moved, nobody traded. It still gets reported, but the response is to resize or hedge, not to discipline.
3. **Limits that are never near being hit.** They are not constraints, they are decoration, and they make a risk report look reassuring for no reason.
4. **Waiving a limit informally.** The waiver is either documented and approved by someone with the authority, or it did not happen. The interview version of this is the one-person-on-the-desk question above.
5. **Forgetting that the lower limits sum to more than the upper one.** That is intentional, and it means the firm is implicitly assuming its desks will not all be at their limits simultaneously, which is exactly what a crisis makes false.
6. **Confusing a limit with an appetite.** A limit is a ceiling, an appetite is a target. Running permanently at 95% of every limit is not the same as running the business the board intended, and risk taking makes the same point from the trader's side.

VISUAL / CODE TO BUILD: Interactive

A limit-utilisation dashboard. Several limit types are shown as horizontal bars, notional, delta, vega, VaR, stress loss and concentration, each filling toward its limit with the 75% and 90% triggers marked. A single control moves the market, spot and volatility, and every bar responds to a full revaluation of a fixed book.

The point of the panel is that the bars do not move together. Dragging spot toward the book's strikes fills the vega and gamma bars while the notional bar does not move at all, which is the passive-breach example made continuous, and it shows why a framework needs several limit types rather than a better single one.

A second panel shows the hierarchy as nested boxes, trader inside desk inside business inside firm, with each box's fill being its utilisation. The instructive setting is one where every desk sits at 70% and the sum overflows the business box, which makes the deliberate overallocation visible and shows what happens when the assumption behind it fails.

Chapter 22

Trading - Desk Reality

22.1 Market making

sales · trading · structuring core cross-asset

Most junior front-office seats are on a desk that quotes prices to clients. The role of a market maker inside the market's plumbing is described in market makers; what this section is about is the job: where the quote comes from, why it is not centred on your fair value, what happens the second and third time the same client hits the same side, and how a desk that seems to make money on every trade can lose a year's revenue in a week.

What you are actually selling

A market maker sells **immediacy**. The client could wait, work an order patiently and probably transact closer to mid, but they want the trade now, in size, with certainty. You provide that, and the bid-ask spread is your fee.

INTUITION

You run a **bureau de change** in an airport. You buy dollars at 0.90 and sell them at 0.94, and you have no view whatsoever on the euro. If a steady mix of travellers arrives, buyers and sellers in roughly equal numbers, you never hold much inventory and you collect four cents a dollar all day for doing nothing but being there. Two things can ruin this. Everyone might arrive wanting the same side, leaving you with a pile of dollars and a real directional risk you never wanted. Or the people arriving might know something you do not, which is the fatal case: if the only customers who trade with you are the ones who are right, your spread is not a fee, it is a small discount on a losing position.

KEY IDEA

The market maker's P&L has two parts pulling against each other:

$$\text{P\&L} = \underbrace{\text{spread captured} \times \text{volume}}_{\text{the business}} - \underbrace{\text{inventory risk} + \text{adverse selection}}_{\text{what kills it}}.$$

Everything a market maker does, the width of the quote, its skew, the size shown, how fast it is hedged, is an attempt to keep the first term large while holding the second small.

Where the quote comes from

A quote is built in layers, and being able to itemise them is the answer to “how do you price for a client”.

1. **Fair value.** Your best estimate of the mid, from the model and from observable markets. This is where junior candidates stop, and it is only the first step.
2. **Hedging cost.** You will not keep the risk, so the price must include the cost of laying it off: crossing the spread in the underlying or in listed options, market impact in size, borrow or repo cost on a short, and the funding of any margin. On an illiquid underlying this can exceed everything else.
3. **Expected adverse selection.** A premium for the fact that you tend to be filled precisely when your price is wrong. It grows with how informed you believe the flow is.
4. **Inventory and risk premium.** What you charge for the risk you will be holding between the trade and the hedge, and for the residual risk you can never hedge, see inventory management.
5. **Capital and xVA.** Counterparty and funding charges for the trade, which for a long-dated uncollateralised client are material.
6. **Competition.** What everyone else is showing, which sets the ceiling on all of the above. In a competitive name the spread is whatever the second-best axe allows.

Two structural features change the calculus. In a **request-for-quote** market the client tells you the size before you price, so you can price the size, but you also know that you are in competition and only win when you are the most aggressive, which is itself adverse selection. In a **streaming** or order-book market you show a price without knowing who will take it, so the defence is size limits and speed of withdrawal rather than width.

Skew, not width: the move that separates professionals

The instinct of an inexperienced trader who has just been hit is to *widen*. The instinct of a professional is to *move*.

If you have just sold, you are short, and your problem is not that your spread was too narrow, it is that your inventory is now on one side. Widening symmetrically leaves you just as likely to be sold to again. Shifting the whole quote up makes your bid more attractive, so you get taken out of your position, and makes your offer less attractive, so you stop accumulating. Width is the price of uncertainty; **skew is the price of inventory**.

WORKED EXAMPLE

Volatility market making, with the numbers. A client asks for a two-way price on one-year at-the-money volatility on a stock at €100, for 100,000 shares of notional. There is no listed market and realised volatility has been running at 30.

Using the desk approximation that an at-the-money option is worth about $0.4\,\sigma S\sqrt{T}$, developed in options, at 30 vol the option is worth $0.4 \times 0.30 \times 100 =$ €12.00 a share (the exact Black-Scholes value at zero rates is €11.92, so the rule is good to under a percent). Differentiating the same rule, each volatility point is worth about $0.4 \times 100 \times 0.01 =$ €0.40 a share, so on 100,000 shares your **vega is roughly €40,000 per volatility point** (the exact figure is €39,400).

You show 29.5 at 30.5, a one-point wide market, which is €40,000 of gross spread across the full width and €20,000 of half-spread if you trade at your own price.

The client buys at 30.5. You are now short €40,000 of vega per point. If realised and implied volatility settle at 32, the position is marked $(32 - 30.5) \times 40{,}000 =$ €60,000 against you, and that figure is *already* net of the edge you sold: measured from the fair value of 30 it is €20,000 of captured half-spread against an €80,000 move in fair value. **Being wrong by two volatility points, against half a point of edge, costs three times the half-spread you earned for taking the risk.** That ratio, and not the width of the quote, is the real business, and it is why market makers care far more about the quality of their fair value than about the size of their spread.

Adverse selection and the hit ratio

If half your quotes get filled, your prices are probably fine. If nearly all of them do, you are the cheapest in the market for a reason you have not worked out yet. The **hit ratio** is therefore a risk metric as much as a commercial one, and the danger sign is not a low one.

KEY IDEA

Repeated one-way flow is information, not luck. A single fill is a trade. The same client coming back for the same side, the same size, in a name that does not really trade, is a message: they know something, or they have a large order to work, and either way your price is behind the market. The correct response is to move your market until the flow stops, and to reduce the size you show. Widening while staying centred on the same fair value is the error that turns a small loss into a large one.

Client flow also has positive value, which is the other half of the business. Flow tells you where the market's demand is before it is in the price, and a desk that sees enough of it can often **internalise**: hold a client's risk briefly because the offsetting client trade is likely to arrive, capturing both spreads and never crossing the market at all. This is the core economics of a flow franchise, developed in flow trading.

REAL INTERVIEW QUESTION (Optiver · desk not recorded)

What is market making?

WHAT THEY TEST / ANSWER

Give the function, the revenue and the risk in three sentences, and do not stop at the definition. A market maker continuously quotes a two-way price, a bid and an offer, standing ready to buy and to sell, so they supply immediacy to whoever wants to trade now. They are paid the spread, and they are in principle indifferent to direction, since the business is turnover and not a view. The risks are the two things that stop it being free money: **inventory risk**, since the flow arrives one-sided and leaves you holding a position you did not choose, and **adverse selection**, since you are most likely to be filled when your price is the one that is wrong. Then add the sentence that shows you know it is a craft: the tools for managing those risks are the width of the quote, its skew relative to your fair value, the size you show and the speed at which you hedge or internalise. If they want more, the market-structure view of the same role, obligations on an exchange and the difference from a broker, is in market makers.

REAL INTERVIEW QUESTION (Optiver · desk not recorded)

How do you make money in market making? Why?

WHAT THEY TEST / ANSWER

The "why" is the whole question, so answer the mechanism and then justify the economics. Mechanically, you buy at your bid and sell at your offer, and if the flow is balanced you earn the spread on the round trip while holding little risk, so revenue scales with volume and with spread rather than with any prediction. The reason anyone pays it is that you are selling a genuine service: immediacy and certainty of execution in size. The client's alternative is to wait, and waiting has a cost, in price uncertainty and in opportunity, so both sides gain. Then earn the follow-up by explaining why it is not free: you are compensated for bearing inventory risk between the trade and the hedge, for the adverse selection that means informed traders systematically pick you off, and for the capital and technology you commit. Finish with what actually differentiates good market makers, since that is what a firm like this is listening for: a better estimate of fair value, faster and smarter hedging, disciplined skewing of the quote against your own inventory, sizing so that no single trade matters, and the willingness to move your market rather than defend a price when the flow tells you that you are wrong.

REAL INTERVIEW QUESTION (Société Générale · EQD)

A client asks me for a price on 1000 calls on an underlying. I give them a 20-22 spread. Then they ask me for the same price again, for another 1000 calls. How do I react? Same question, but this time with 15000 calls.

WHAT THEY TEST / ANSWER

The examiner is testing whether you skew or merely widen, and whether size changes your answer. First fill: the client bought 1000 at 22, so you are short 1000 calls and short gamma and vega. **Do not requote 20-22.** Move the market up, to something like 21-23: your new bid is more attractive so you can buy your position back, and your new offer is less attractive so you stop accumulating a position you did not choose. Say explicitly that this is a shift and not a widening, and that width should only increase to the extent your uncertainty about fair value has genuinely increased. Second identical request: now treat it as **information**. A client returning for the same side and the same size is either informed or working a larger order, so your fair value is probably behind the market, and you should move further, ask what they are trying to achieve, and start hedging what you already have. On 15000, fifteen times the size, everything changes at once: the position would be material against your limits, the hedge has market impact and cannot be executed at screen prices, and the risk is no longer a rounding error. So you widen substantially and skew, you price the hedging cost and impact into the level rather than hoping, you check limits and possibly your risk manager before quoting at all, and you consider showing a price for a partial size or a price contingent on how it is executed. Close on the professional point: quoting the same width for 1000 and for 15000 is the mistake, because a market maker prices *size*, and the whole skill is that the quote is a function of your inventory, your uncertainty and the size in front of you.

REAL INTERVIEW QUESTION (Société Générale · EQD)

You are asked for a bid/ask on the volatility of a tech name that does not trade, with historical volatility at 30. Your ask gets taken. What do you do with your spread? Same again a second time. And a third. You have only sold, at an average of 34 vol. Two days later you learn the accounts were falsified, the stock loses 25%. Did you make or lose money? Why? What could you have done so that this was not the case?

WHAT THEY TEST / ANSWER

The sting is in the last part, so answer the sequence honestly and then be unsparing about the mistake. On each fill you should move your market up, not just widen, because you are accumulating a short vega position, and by the third fill in an illiquid name where all the flow is one way you should be shifting aggressively and cutting the size you show: three one-way fills in a name that does not trade is as loud a signal as a market gives. Did you make or lose money? You **lost, and badly**. Selling at an average of 34 against a historical 30 looks like a four-point edge, but you are short volatility and short gamma, and a 25% gap is the realised move that short-gamma positions cannot hedge: your delta hedge is set for a small move, the stock jumps through it, and you lose far more on the crash than the premium ever paid you. Implied volatility on the name also explodes after the news, so you are marked down on vega at the same time, and if what you sold was downside strikes you are short skew as well, which repriced violently. What you could have done, in order of importance: read the one-way flow as informed rather than as free premium and stopped selling after the first or second fill; sized so that no single name could do this, since the failure here is position size rather than price; asked why a client wants this much volatility in a name that does not trade, which is a question with an answer; hedged the tail rather than only the delta, by buying cheap far out-of-the-money puts or a proxy such as index or sector volatility, which converts an unbounded short into a spread; and marked and monitored the position against a realistic gap scenario rather than a normal one, which is precisely what scenario analysis is for. The honest summary, and the one interviewers want to hear: the trade was not a mispricing of volatility, it was a failure to recognise that the client was the informed side.

INTERVIEW TRAP

Four traps. First, **quoting around your fair value regardless of inventory**: the mid you show should be skewed away from the position you are carrying, not centred on your model. Second, **widening instead of moving**, which keeps you exposed to exactly the same one-sided flow. Third, **quoting the same width irrespective of size**, when a market maker's job is to price size, hedging cost and impact. Fourth, **treating a high hit ratio as success**: if everyone trades with you, you are the cheapest liquidity in the market and you should find out why before the reason finds you.

VISUAL / CODE TO BUILD: Interactive

A market-making simulator, played rather than read. A price series runs, orders arrive from a mix of uninformed traders (random) and informed ones (they trade only when the price is about to move in their favour), and the reader sets bid, ask and size before each tick. The scoreboard shows spread captured, inventory, mark-to-market and the hit ratio, split by counterparty type so the reader watches the informed flow eat their spread. Three difficulty settings make the section's lessons unavoidable: **balanced flow**, where staying centred works and the spread accumulates steadily; **one-sided flow**, where only skewing survives and widening alone does not; and **informed flow with a jump**, where a short-gamma book is destroyed unless the player cuts size and buys tail protection, reproducing the falsified-accounts question above.

Where this goes next

The market-structure view of the same role, including exchange obligations and the contrast with brokers, is market makers, and what determines the width of a spread in the first place is the bid-ask spread and liquidity. Managing what you have accumulated is inventory management, the franchise economics of seeing client flow is flow trading, and the hedging machinery underneath all of it is delta neutrality and dynamic hedging. The estimation-and-quote games used to test this skill in interviews, pricing the number of screens on a floor, live in brain teasers.

22.2 Flow trading

trading · sales · structuring | core | cross-asset

A flow trader does not decide what to own. Clients decide, by trading, and the trader is left holding whatever the day produced. That single fact separates this seat from every picture of trading a candidate arrives with, and it determines the revenue model, the skills, the risks and the way the desk is judged.

KEY IDEA

A flow desk's risk is **inherited, not chosen**. The business is intermediation: you show a price, the client takes the side that suits them, and your position is the residue. So the job is not to predict the market, it is to **price the risk correctly on the way in** and to **recycle it efficiently once you have it**. A proprietary desk chooses its positions and is judged on them, which is proprietary trading and a different business entirely.

Where the money comes from

Four sources, in roughly descending order of reliability.

- **The bid-offer**, captured over volume. This is the core, and it is earned when flow is two-way, since buying at the bid and selling at the offer earns the spread with no residual position.
- **Structuring and packaging margin** on trades that are not simply a vanilla, which is larger per trade and smaller in volume.

- **The residual**, the profit or loss on risk held between the client trade and the offsetting one. It is genuine revenue when the desk positions well and the reason a bad month happens when it does not.
- **Financing and inventory**, the carry on positions and collateral, which is the largest source on some desks and invisible on others.

WORKED EXAMPLE

An equity derivatives desk quotes a one-year at-the-money option at 19.5 bid, 20.5 offered, in volatility terms, on a notional of 10 million.

An at-the-money option's vega is about 0.4% of spot per volatility point, so the notional carries $10{,}000{,}000 \times 0.004 = 40{,}000$ of value per volatility point. The full one-point spread is therefore worth **40,000**, and on a single one-way trade the desk earns half of it against mid, **20,000**, with the other half earned only if the offsetting flow arrives.

That last clause is the whole business. If the desk sells at 20.5 and buys back from another client at 19.5, it books 40,000 with no position. If it has to buy back in the market at 20.5 because no client came, it books nothing and has paid the street's spread as well. Flow revenue is therefore a function of **how much of the flow can be internalised**, which is why franchise and market share matter far more here than any view on direction.

REAL INTERVIEW QUESTION (BNP · Exo)

Where does the margin on a structured product come from?

WHAT THEY TEST / ANSWER

Decompose it, because "the bank takes a fee" is not an answer and the components are individually defensible or not.

The hedging spread. The desk buys the components of the payoff at its own execution levels and sells the package at a level built off slightly worse assumptions: a volatility a point higher on what it is short, a dividend forecast a touch conservative, a borrow cost on the wide side. Each is small and there are several.

The unhedgeable risks, which is the honest core. Some of what the desk takes on cannot be laid off at any price: correlation on a worst-of, the gap through a barrier, the skew between two strikes it cannot trade separately. The margin includes a **reserve** for those, and the reserve is not profit until the risk has run off.

Funding. The note raises money for the issuer, so the desk is credited with the difference between the bank's funding cost and the risk-free rate, which on a long-dated note is often the largest single component and has nothing to do with the option at all.

Distribution. The fee paid to whoever sold it, which is a cost to the issuer and a margin to the distributor, and on retail products it is frequently larger than the manufacturing margin.

Close on the two points that make it a desk answer. The margin is **not booked at once**: reserves are released as risk runs off, which is why a structured-products desk can look profitable on day one and lose the profit over three years. And the size of the margin tracks **how contestable the trade is**: a listed vanilla is priced to a basis point because five banks are quoting, while a bespoke multi-asset note is priced to a percentage because nobody else can price it in the time available. That is competition, not complexity, doing the work.

REAL INTERVIEW QUESTION (Exane · Structured Products)

What is the margin on a tracker or a certificate, and more generally on structured products?

WHAT THEY TEST / ANSWER

Answer with the ordering and the reason, rather than with numbers you cannot source.

A **tracker or delta-one certificate** carries the thinnest margin of anything on the shelf: it is a linear payoff on a liquid index that any competitor can manufacture, so the margin is a small number of basis points a year plus whatever the desk earns on financing and on lending the hedge. It is a volume business, and it competes directly with ETFs, which caps it externally.

A **capital-protected note** sits higher, because the client is paying for a guarantee, an issuer, and a term. A **retail autocall on a worst-of basket** sits higher again, typically by an order of magnitude over a tracker, and the reason is not that it costs more to build.

That is the point to make: the margin scales with **contestability and disclosure**, not with the desk's cost. A payoff that three banks can quote in an hour is priced tightly. A payoff whose value depends on an unobservable correlation and an issuer's own funding is priced by one bank against a client who cannot easily compare, which is exactly why regulators focused on cost disclosure in retail structured products rather than on the products themselves.

Finish with the professional framing rather than a cynical one: the wide margin is also paying for a genuinely warehoused risk that has no market, and a desk that prices it thin and is wrong does not have a business either. The honest statement is that the margin covers the hedging spread, the reserve for what cannot be hedged, and a rent for being one of few who can price it, and that a good structurer can say which of those three dominates on any given trade.

What you inherit, and what you do with it

The moment a client trades, the desk holds a position it did not want. What happens next is the job.

Hedge the first-order risk immediately. Delta is hedged in minutes with futures or the underlying; it is cheap, liquid and nobody is paid to carry it.

Warehouse the second-order risk deliberately. Vega, gamma and the rest are partly hedged and partly held, because hedging every unit of them costs the spread each time and the desk is paid to carry some. That is a decision, taken against limits, not a failure to hedge.

Recycle. The best outcome is an offsetting client trade, since it earns a second spread and cancels the risk. Failing that, the street. Failing that, the desk holds it and the risk turns into the residual, in inventory management.

Live with the residue. Some exposures have no market at all: correlation, the long end of the dividend curve, skew in a name with three listed strikes. These accumulate in one direction across the whole industry, because every bank is on the same side of the same client flow, and that concentration is what turns a manageable position into a crowded one.

The catalogue of risks a trader carries, and how the Greeks, liquidity, counterparty, model and conduct risk are grouped, belongs to market risk and is not repeated here. What is specific to this seat is the *joint* behaviour. On a flow book these risks are **not independent**, and the correlation between them is the real exposure. One market shock widens spreads, so liquidity risk arrives at the same moment as market risk; the same shock pushes counterparties towards default and pushes the model into a region it was never calibrated for; and the client flow that would normally let the desk recycle its position dries up in exactly that moment. Sizing each risk separately and adding them up is the mistake, and it is why a flow desk is run against stress scenarios rather than against a list of individual limits, in position limits.

The reality of the seat

The classic interview version of this, the client who insists on trading when nobody else is on the desk, is a conduct question and is answered in operational risk. The flow-desk half of it is about pricing rather than about authority, and it is worth stating on its own.

If you can trade it but cannot hedge it, because the underlying market is closed or because the hedge is illiquid, the price has to carry the risk of not being able to lay it off. Quote wide enough to cover that, and tell the client why the price is wide. A wide honest price is a service, and it is also the only price you can defend afterwards; a tight price you cannot hedge is a gift you are making with the bank's money, and the desk inherits it either way. This is the same reasoning as the bid-offer above, applied to a moment when the recycling leg is missing: the spread is compensation for warehousing, so when the warehousing gets harder the spread widens.

> **INTERVIEW TRAP**
>
> Four misconceptions. (1) That a flow trader makes money by being right about direction: the delta is hedged within minutes and the revenue is the spread. (2) That every risk should be hedged: hedging costs the spread each time, and the desk is paid to carry a budgeted amount. (3) That structuring margin is profit on day one: much of it is a reserve against risk that has not run off. (4) That the desk's risks are independent: on a flow book they arrive together, which is why the stress scenario matters more than any single limit.

Where this goes next

The quoting mechanics are market making and market makers, the business that chooses its risk instead of inheriting it is proprietary trading, and what happens to the position between the client trade and the offset is inventory management. How the resulting profit and loss is decomposed is PnL explain, the constraints the desk runs against are position limits, and the products that generate most of this flow are structured notes.

22.3 Proprietary trading

trading · sales · structuring | core | cross-asset

A trading desk can make money in three ways. It can earn the bid-offer spread by standing between buyers and sellers. It can use what it learns from client flow to position slightly ahead of it. Or it can simply take a view and be right.

The first is market making, the second is flow trading, and the third is proprietary trading. This section is about the third, and it has an unusual feature for a section in this book: the activity it describes has largely been legislated out of the banks a reader is interviewing with. Understanding why is more useful than the definition.

The definition, and the line that will not stay drawn

Proprietary trading is taking positions with the firm's own capital, for the firm's own account, with no client on the other side and no client purpose being served.

That sounds crisp and it is not, which is the entire problem.

INTUITION

A market maker buys 500,000 shares from a client because the client wanted to sell. So far this is market making. Now the desk decides to hold the shares overnight rather than work out of them, because it thinks the price is going up. Nothing about the trade has changed. The inventory is the same, the client is the same, the booking is the same. The only thing that changed is why the position is still there in the morning, and *intent is not observable.*

That is why the regulatory question is never "was this a proprietary trade" but "was this inventory reasonable for the client business the desk actually does". The rules attach to the size and turnover of the inventory rather than to the trader's state of mind, which is the only observable thing available.

Why the economics are so different

Market making and proprietary trading do not just carry different risk. They earn money from different quantities, and the arithmetic makes the contrast plain.

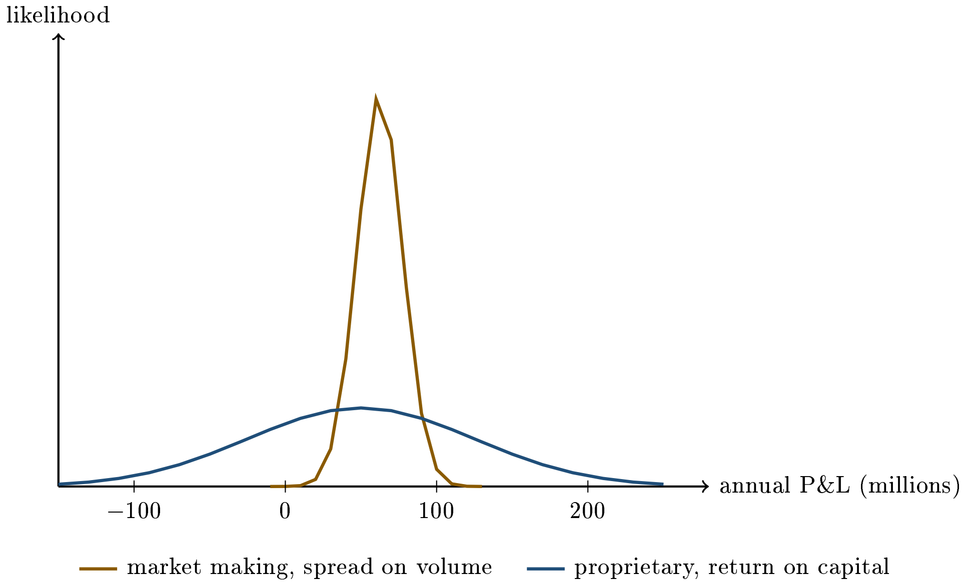
WORKED EXAMPLE

A market maker quotes a bid-offer of 5 basis points and turns over 500 million of notional a day. If it captures the full spread it earns

$$500{,}000{,}000 \times 0.0005 = 250{,}000 \text{ a day,}$$

so around 62.5 million over 250 trading days. Income is proportional to **volume**, and it barely depends on whether the market went up or down.

A proprietary desk runs 500 million of capital and targets a 10% return, so 50 million a year, with a return volatility of perhaps 15%, which is 75 million. A one standard deviation year therefore lands anywhere between a loss of 25 million and a gain of 125 million. Income is proportional to **capital and risk taken**, and it depends on nothing else.



Both businesses have a similar expected outcome and nothing else in common. One of them cannot plausibly lose money in a year; the other can lose a great deal, and its whole left tail sits in a region the first one never visits.

> **KEY IDEA**
>
> That picture is the policy argument in one image. A deposit-taking bank with a government safety net behind it can run the amber business without troubling anybody. The blue one puts the safety net behind a distribution with a large left tail, and the taxpayer is short that tail without being paid for it. Whatever one thinks of the rules, this is the reasoning they came from.

What the rules actually say

The Volcker Rule, part of the post-crisis reforms in the United States, prohibits banking entities from engaging in proprietary trading, and permits several things that look like it: market making, underwriting, risk-mitigating hedging, and trading in government securities. The European and British approaches differ in mechanism, generally through ring-fencing retail activity from trading rather than by banning an activity, but the direction is the same.

The interesting part is the market-making exemption, because it is where the unobservable-intent problem gets solved in practice. To qualify, a desk's inventory must be sized to **reasonably expected near-term customer demand**, which is a testable statement about turnover and holding periods rather than about motive. Around it sits a compliance apparatus of quantitative metrics, desk-level mandates and senior attestation, and it was simplified in amendments around 2019 and 2020 without changing the principle.

The rule itself is in the Volcker Rule.

Where it went

Proprietary risk-taking did not disappear from the financial system; it moved to firms without deposits and without a safety net. Standalone proprietary trading firms, electronic market makers that are not banks, and hedge funds now house most of it, and several bank prop desks became independent funds directly in the years after the rules landed.

That is the answer to the question a candidate should be asking: the activity still exists as a career, and it is mostly not at a bank.

What remains inside a bank

Several things that carry genuine principal risk, and the distinction that makes them lawful is that each is *incidental* to a client business rather than a view taken for its own sake.

- **Residual market-making inventory.** A desk that has bought a large block cannot hedge it perfectly or instantly, so it holds real risk while it works out of it. That is inventory management, and it is the largest source of principal risk on most floors.
- **Treasury and asset-liability management.** A bank's balance sheet has interest rate and currency mismatches, and somebody positions them deliberately.
- **Valuation adjustment desks.** An XVA desk hedging counterparty exposure runs substantial market risk arising from a client business it did not choose. See CVA, DVA and FVA.
- **Strategic and principal investments**, held for business reasons rather than as trading positions.

> INTERVIEW TRAP
>
> Telling an interviewer you want to do proprietary trading at a bank. It signals that you have not looked at the industry since about 2010, and it is one of the fastest ways to lose a front-office interview. If a directional, view-driven role is what you want, say so and name the firms that do it, which are prop shops and hedge funds. If you want a bank, understand that the risk you will run is inventory and flow risk, and that this is genuinely interesting rather than a consolation prize.

Interview angle

REAL INTERVIEW QUESTION (Natixis · Credit)

You buy one million dollars at 1.13, paying in euros. Do you make or lose money, and what is your P&L, if you sell at 1.12 in a year?

WHAT THEY TEST / ANSWER

There is a trap in the direction and most people fall into it, so slow down and fix the convention before computing anything.

EUR/USD is quoted as dollars per euro. A move from 1.13 to 1.12 means each euro buys fewer dollars, so **the dollar has strengthened**. You were long dollars. So you **make** money, even though the number went down, and anybody who answers "you lose, the rate fell" has misread which currency is which.

The arithmetic. Buying one million dollars at 1.13 costs 1,000,000/1.13 = €884,956. Selling them at 1.12 returns 1,000,000/1.12 = €892,857. The profit is

$$892{,}857 - 884{,}956 = €7{,}901,$$

which is 0.89% on the euros invested. Sanity-check it without the division: the gain is $1.13/1.12 - 1 = 0.893\%$, and getting the same answer two ways in front of an interviewer is worth doing out loud.

Then the part that makes it a real answer. Over a whole year this is not the complete P&L, because holding dollars earns the dollar interest rate while giving up the euro one. The honest comparison is against the one-year forward rather than against spot, since the forward is what you could have locked in at the start with no risk at all. If the dollar rate was above the euro rate then the forward was above 1.13 in dollar terms, you were paid to hold dollars, and your total return is better than the 0.89% spot move suggests. If it was below, the carry works against you and the spot gain may not have been enough. That relationship is interest rate parity.

Close with the observation that connects it to this section: a position like this, held for a year with a view and no client on the other side, is a proprietary position by any definition. Which is why, at a bank, it would sit in treasury or in a client-facing book, and would need somebody to explain what it was doing there.

INTERVIEW TRAP

Three conceptual errors. (1) Believing the prop-versus-market-making line can be drawn from a single trade, when the same trade is either depending on why it is still open the next morning. (2) Assuming banks take no principal risk now, when inventory, treasury and XVA all carry substantial amounts. (3) Comparing a market-making business and a proprietary one on expected return alone, when the figure above shows two distributions that share a mean and nothing else.

Where this goes next

The two businesses this one is defined against are market making and flow trading, and the risk a desk genuinely does carry is risk taking and inventory management. The rule that reshaped all of it is the Volcker Rule, and the structure that decides how much principal risk anybody may hold is risk limits frameworks.

22.4 Risk taking

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Everything in market risk is about measuring and containing risk, which can leave the impression that the goal is to have as little of it as possible. On a trading floor the opposite is true. A desk that runs no risk earns no return and is closed, and a trader who is permanently at 10% of their limits is not being prudent, they are failing to do the job they were hired for. The skill is not whether to take risk but *which* risks and *how much*, and those are two separate decisions that people conflate constantly.

> KEY IDEA
>
> **Edge and size are different questions.** Edge decides whether to do the trade at all: is the expected value positive after costs. Size decides how much, and size is not determined by how confident you are, it is determined by what you can survive being wrong about. Traders lose careers by getting the first question right and the second one wrong, and it is worth saying that sentence in an interview because most candidates only discuss the first.

The arithmetic of size

There is a formal answer to how much to bet, and knowing it is useful even though nobody follows it literally.

For a strategy with expected excess return μ and variance σ^2 per period, the leverage that maximises the long-run growth rate of capital is the **Kelly** fraction

$$f^* = \frac{\mu}{\sigma^2},$$

and the growth rate achieved at leverage f is $g(f) = f\mu - \frac{1}{2}f^2\sigma^2$.

WORKED EXAMPLE

Why nobody runs full Kelly. Take a strategy with a Sharpe ratio of 1.0 at 15% volatility, so $\mu = 0.15$ and $\sigma = 0.15$. Then

$$f^* = \frac{0.15}{0.15^2} = \mathbf{6.67} \times \text{leverage},$$

which is growth-optimal and completely unrunnable: at that size the strategy runs at 100% annual volatility and expects drawdowns of half the capital.

Now halve it. At $f = f^*/2$ the growth rate is

$$\frac{g(f^*/2)}{g(f^*)} = \frac{3\mu^2/8\sigma^2}{4\mu^2/8\sigma^2} = \mathbf{75}\%,$$

so **half the leverage buys three quarters of the growth**. Quarter Kelly keeps 43.75% of the growth at a quarter of the risk. The curve is flat near its peak and steep beyond it, which is the whole argument for running well inside the optimum: you give up very little expected growth and you buy a great deal of survivability. It is also why the penalty for oversizing is severe and asymmetric, since $g(f)$ turns negative at $f = 2f^*$: at twice the optimal leverage the same strategy has an expected growth rate of *zero*.

Why the downside dominates

The reason size matters more than edge is that losses and gains are not symmetric in their effect on capital.

Drawdown	Gain needed to recover
10%	11.1%
20%	25.0%
30%	42.9%
40%	66.7%
50%	100.0%
80%	400.0%

A 50% loss requires doubling to get back. That convexity is why **risk of ruin** is the binding constraint rather than expected return: a trader who breaches enough to be stopped out is finished regardless of whether the position was right, and a strategy with genuine edge run at a size that can lose 80% will not survive to collect it.

INTUITION

The trader's version of the constraint is simply that you have to be there tomorrow. This is not a psychological point, it is the reason limits exist and the reason a desk sizes to survive its worst plausible scenario rather than to maximise its expected one. It is also, uncomfortably, an argument for cutting a position you still believe in, because conviction does not pay margin calls.

The hardest question: adding to a loser

Averaging down is either the most disciplined thing a trader does or the way they blow up, and the difference is not the P&L.

The test is whether the *thesis* has changed or only the price. If you were long because a spread was wide against fair value and it has widened further with nothing else changed, the trade is now better and adding is rational. If it widened because something you did not know has come out, then the market is telling you your fair value was wrong, and adding is paying to be more wrong.

The practical rules that separate the two are unglamorous and worth stating as rules. Decide the maximum size before you put the trade on, so adding is a plan rather than a reaction. Write down what would make you wrong at the time you enter, because you will not be able to think clearly about it later. And notice the asymmetry of information: a market moving against you is evidence, weak but real, and treating it as noise every single time is not conviction, it is a policy of ignoring data.

REAL INTERVIEW QUESTION (UBS · desk not recorded)

What is a bad trader?

WHAT THEY TEST / ANSWER

Do not answer with personality, because everyone does. Answer with behaviours that are visible in a P&L or a risk report, which is how a desk head would actually identify one.

Sizing that does not reflect conviction. The same size on every trade means either the good ideas are too small or the bad ones too large. A bad trader's best trade and worst trade are the same size.

Asymmetric handling of winners and losers. Cutting winners quickly to book a gain and holding losers hoping to get back to flat. It is the most common failure and it is measurable, since the average winning trade is smaller than the average losing one, which is a P&L profile that cannot survive.

No process for being wrong. Not defining, in advance, what would change their mind. Without that, every adverse move is reinterpreted rather than acted on, and the position management becomes narrative.

Confusing P&L with skill. Judging a decision by its outcome. A good trade can lose and a bad trade can win, and with a Sharpe of 1.0 you need nearly four years of data before the P&L is statistically distinguishable from luck, so a trader who reasons from last quarter's result is reasoning from noise.

Hiding. Not marking a position honestly, not flagging a loss early, arguing with risk instead of informing them. Every large trading loss has this in it, and it is the only item on the list that is a firing offence rather than a coaching problem.

Then close by inverting it, which is what makes this an answer rather than a list. A good trader is not the one with the best ideas, it is the one whose losses are small and whose process is repeatable, because over enough trades the edge per trade is small and everything else is survivability. And add the uncomfortable half: on any short sample you cannot distinguish the two from the P&L, which is exactly why firms judge traders on process and not only on the number.

INTERVIEW TRAP

1. **Sizing by conviction rather than by survivability.** Certainty is not a risk measure, and the trades people are most certain about are frequently the crowded ones.
2. **Judging a decision by its outcome.** Given how long it takes to establish edge statistically, a single quarter's P&L carries almost no information about the quality of the decisions in it.
3. **Running near full Kelly.** It is growth-optimal and unsurvivable. At twice the optimal leverage the expected growth rate is zero, which is the same as not trading, with all of the risk.
4. **Treating a limit breach as a technicality.** It is the firm's capacity you are consuming, and the person who must find out first is risk, not your manager afterwards.
5. **Averaging down without a pre-set maximum size.** The rule is set before the trade, not during it, and a plan made while losing money is not a plan.
6. **Believing risk aversion is prudence.** A desk that never approaches its limits is misallocating the firm's capital just as surely as one that breaches them, and it will be closed for it.

VISUAL / CODE TO BUILD: Interactive

A Kelly curve with a survival panel underneath. The upper panel plots the long-run growth rate $g(f) = f\mu - \frac{1}{2}f^2\sigma^2$ against leverage, with sliders for the strategy's Sharpe and volatility, marking f^*, half Kelly and the zero-growth point at $2f^*$. The shape is the lesson: the peak is broad and the right-hand side falls off a cliff.

The lower panel runs simulated capital paths at the leverage the reader has selected, several hundred of them from a fixed seed, and reports the median outcome, the fifth percentile and the fraction of paths that fall below a ruin threshold. Setting the slider to full Kelly and watching a majority of paths survive while a substantial minority are destroyed, then halving it and watching the median barely move while the tail disappears, is the argument of the whole section performed rather than asserted.

A third control shows the drawdown-recovery curve as a simple plot of required gain against loss taken, with the reader's current worst simulated path marked on it.

22.5 Inventory management

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A flow desk's positions are mostly not decisions. They are residue: what is left on the book after servicing clients who all wanted the same side. Managing that residue well is the difference between a franchise that earns its spread and one that gives it back, and it is the part of trading that interviews probe when they stop asking about pricing and start asking "so now what do you do". The distinction to hold on to throughout is between a position you *chose*, which belongs to risk taking, and inventory you *acquired*, which is what this section is about.

Four things you can do with inventory

> **INTUITION**
>
> A **wholesaler** takes delivery of a thousand crates because a supplier needed to sell them today. They now own crates they never wanted, and the crates are costing them warehouse rent every day. They have exactly four options: sell the crates on the open market at whatever the going rate is, find another customer who wants crates and sell to them at a margin, discount their own price to move stock faster while raising the price at which they will buy more, or decide the crates will be worth more next month and deliberately keep them. A trading desk has the same four options and the same rent.

- **Hedge it.** Neutralise the risk in the liquid market. Fast and certain, but you pay the hedge's spread and impact, and a hedge is almost never perfect, so you convert the position into basis risk rather than removing it.
- **Recycle it.** Find the client on the other side. Costs nothing in spread and earns a second one, and it is the whole economic point of a franchise, see flow trading. It requires that you actually see the offsetting flow, and that you can wait.
- **Skew your quotes.** Move both sides of your market so the flow itself unwinds you, as set out in market making. Free, but slow, and it only works if you keep seeing flow.
- **Warehouse it.** Deliberately hold the risk because you are being paid enough for it. This is legitimate and often the correct answer, but it must be a decision with a size, a time limit and a reason, not the default that happens when nobody acts.

> **KEY IDEA**
>
> The choice among the four is an **economic comparison**, not a matter of temperament: the cost of hedging now, against the cost of carrying the position for the time you expect to need to recycle it, against the risk of what could happen in that time. That is why the first thing a good trader computes is the *daily carry* on the inventory, since it turns "how long can I wait" from an instinct into a number.

What holding inventory costs

Six lines, and a junior candidate typically names only the first.

- **Funding.** Cash tied up in a long position, at the desk's internal funding rate rather than at some theoretical risk-free rate.
- **Borrow or repo.** A short position must be borrowed and the fee paid; a long position can be lent out and the fee earned. On hard-to-borrow single names this dominates everything, see repo markets.
- **Capital and leverage.** Balance sheet, risk-weighted assets and leverage ratio usage, all of which are charged to the desk and none of which are free.
- **Carry and decay.** For an option book the daily theta, and for a futures position the roll, both of which run whether or not the market moves.
- **Hedge slippage.** Every hedge crosses a spread and, in size, moves the market. Rehedging a large position repeatedly is a real recurring cost.
- **The residual risk.** What the hedge does not cover: basis, skew, correlation, dividend and the pieces that no instrument trades.

WORKED EXAMPLE

How long can you afford to wait? You have bought 1,000,000 shares from a client at €100, so you hold €100m of inventory. Your desk funds at 4% and you can lend the stock out at 0.5%.

- Funding cost: 100m × 0.04/365 = €10,959 a day.
- Stock lending income: 100m × 0.005/365 = €1,370 a day.
- **Net carry: €9,589 a day.**

Now suppose recycling the position to another client would earn you two cents a share, that is 1,000,000 × €0.02 = €20,000. Dividing, 20,000/9,589 = 2.1 days. **You can afford roughly two days of patience before the carry has eaten the entire expected margin**, and that is before any price risk. Compute that number first and the decision usually makes itself: if the offsetting client is plausibly a week away, hedge now rather than hoping.

The risk you cannot hedge, and the cost of pretending you can

Hedging is easy where a liquid market exists. Inventory management is hard precisely where one does not, and every desk carries some risk it cannot lay off: vega on a single name with no listed options, correlation on a bespoke basket, dividend exposure on a long-dated structure, skew on a strike nobody quotes.

The honest approach has three parts, and stating them is what a good answer to any "how do you hedge this" question looks like. **Hedge the part you can**, usually delta and the liquid slice of the vega, with a proxy. **Price the part you cannot** into the quote at inception, since an unhedgeable risk must be paid for up front because there will be no second chance. **Limit and age it**, so the unhedgeable book stays small and does not quietly accumulate, which is how it always accumulates.

Proxy hedging deserves its own warning. Hedging a single name's volatility with index volatility works until the day the name moves for its own reasons, which is exactly the day you needed the hedge. The proxy converts an unhedgeable exposure into a basis exposure that is smaller in normal times and correlated with disaster in bad ones.

Ageing, limits and the behavioural traps

Two positions of the same size are not the same risk if one arrived this morning and the other has been there for three months. Desks therefore run **ageing reports**, and old inventory is treated as a warning rather than as a neutral fact: it usually means the position cannot be sold at the marked price, that the mark is optimistic, or that nobody has owned the decision. Alongside ageing sit the limits that cap the whole exercise, gross and net position, greek limits, concentration by name and by tenor, and the ageing triggers that force a review, developed in position limits and risk limits frameworks.

INTERVIEW TRAP

The traps here are behavioural rather than technical, which is why they persist. The **disposition effect**: cutting winners quickly and holding losers, because closing a losing position makes the loss real. **Marking your own book kindly**, where an illiquid position is valued at the price you wish existed, which turns a trading problem into a valuation problem and then into a control problem. **Anchoring on your entry price**, which is irrelevant to what you should do next. And **letting warehousing happen by default**: keeping risk because no one decided to remove it is not a decision to warehouse, and it is the most common way an inventory problem becomes a P&L event.

REAL INTERVIEW QUESTION (Société Générale · EQD)

You have a bid ask at 900, 700. If you get a buyer, how do you modify your bid ask?

WHAT THEY TEST / ANSWER

Read the quote back before answering, since the pair is presumably 700 bid and 900 offered, and saying so shows you handle a sloppy price without guessing. A buyer trades on your offer, so you have **sold at 900 and are now short**. Move the whole market *up*, for example to 800 bid at 1000 offered: your new bid is more attractive, so someone is more likely to sell to you and flatten your position, and your new offer is less attractive, so you stop accumulating a short you never chose. Say the principle out loud, because it is the answer they are grading: you shift the mid against your inventory rather than simply widening the spread, since width prices your uncertainty about fair value while skew prices the position you are carrying. Then add the refinements. How far you move depends on the size traded against your limits and on the liquidity of the underlying, and on a wide market like this one the move can be large. If the same buyer returns, treat it as information that your fair value is behind the market rather than as more spread to collect. And in parallel with skewing you should be hedging or recycling, since skewing alone only helps if more flow arrives.

REAL INTERVIEW QUESTION (Commerzbank · EQD)

You are short a call and delta-hedged. You priced the call with a repo of 1% but you realise the true repo is 10%: how will your P&L evolve and how do you adjust your hedge?

WHAT THEY TEST / ANSWER

Work through the forward, because everything follows from it. With continuous rates and no dividends, the forward is $F = Se^{(r-\text{repo})T}$, so a higher repo means a lower forward. Take a €100 stock, one year, 30 volatility and zero rates: at a 1% repo the forward is €99.01 and the strike-100 call is worth €11.37, while at a 10% repo the forward is €90.48 and the same call is worth €7.22. You **sold** the option, so you sold it about €4.15 too expensive relative to its true value, which is a gain, and the economic reason is concrete rather than abstract: your delta hedge is a *long* stock position, and you can lend that stock out at 10% rather than the 1% you assumed, so you earn nine points a year of carry you never priced. The P&L therefore accrues over the life of the trade as lending income, not as an instant mark-up, provided the borrow market stays that rich, and that proviso is the risk you have swapped into.

For the hedge, the spot delta is approximately $e^{-\text{repo}\,T}\Phi(d_1)$, which falls from 0.541 at a 1% repo to 0.387 at 10%, so you are **over-hedged by about 0.15 shares per option and should sell stock**. Then say the three things that make the answer complete. Re-mark the book on the correct repo curve, since the mispricing is in your valuation as well as your hedge. Recognise that you are now running **repo risk**, a short position in the borrow cost, since if the name becomes easy to borrow again the income disappears, and that risk can be reduced by locking a term borrow instead of relying on the overnight rate. And note the direction for the opposite position, because they often ask straight afterwards: a *long* call delta-hedged with a short stock position has to pay the borrow, so an underestimated repo is a loss, which is precisely how expensive-to-borrow names hurt desks that priced them as if they were general collateral.

VISUAL / CODE TO BUILD: Interactive

An inventory cockpit driven by a replayable flow of client trades. The top strip shows the position through time with the four actions available as buttons on each tick: hedge, recycle (available only when an offsetting client actually appears in the simulated flow), skew, or hold. A live panel decomposes the running cost into funding, borrow, capital, theta and cumulative hedge slippage, and displays the "days of patience" number from the worked example, recomputed as the position changes, so the reader sees it shrink as the inventory grows. A second tab makes the unhedgeable case concrete: a single-name vega position with a proxy-hedge slider, plotting hedged P&L in a quiet regime and then in a single-name shock where the proxy correlation breaks, so the basis risk appears as a loss rather than as a caveat. A third tab overlays an ageing report and lets the reader set concentration and ageing limits, then shows which historical decisions the limits would have forced.

Where this goes next

The quoting behaviour that creates and unwinds inventory is market making, and the franchise economics of seeing enough flow to recycle it is flow trading. Position taken deliberately rather than acquired belongs to risk taking, and the constraints that bound all of it are position limits and risk limits frameworks. The borrow and funding mechanics that set the carry are repo markets, the vega side of the residual is vega exposure and volatility regimes, and the attribution of what the inventory actually earned or cost is PnL explain.

22.6 PnL explain

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Every morning a derivatives desk is told what it made or lost yesterday, and then it has to say *why*. Not approximately: line by line, attributed to a cause, with a residual small enough to be ignored. That report is the PnL explain, and it is simultaneously the best test of whether the desk understands its own book and the fastest way anyone discovers that something is wrong.

The expansion

The value of a book depends on the spot, the volatility surface, rates and time. Expand the change in value in those variables and each term is a Greek multiplied by a market move:

$$\Delta V \approx \underbrace{\Delta \cdot \Delta S}_{\text{delta}} + \underbrace{\tfrac{1}{2}\Gamma(\Delta S)^2}_{\text{gamma}} + \underbrace{\Theta\,\Delta t}_{\text{theta}} + \underbrace{\mathcal{V}\,\Delta\sigma}_{\text{vega}} + \underbrace{\rho\,\Delta r}_{\text{rho}} + \text{cross terms} + \text{new trades} + \varepsilon.$$

The whole discipline is contained in the last symbol. ε is the **unexplained**, and it is what everyone actually looks at: if the explained terms reproduce the day's profit and loss, the desk's risk representation is right, and if they do not, either the risk is not what the system says or the marks are wrong.

> **KEY IDEA**
>
> For a **delta-hedged** option position the delta term vanishes and the first two survivors fight each other:
>
> $$\text{daily PnL} \approx \tfrac{1}{2}\Gamma S^2\big(r^2 - \sigma^2_{\text{daily}}\big),$$
>
> where r is the realized return of the day. Theta is not an independent leak, it is exactly the amount of gamma profit the market has to deliver for you to break even. So a long gamma position makes money on any day the market moves *more* than the implied volatility said it would, and loses on any quiet day, and a short gamma position is the mirror. That single equation is why a delta-hedged trade is a bet on realized against implied volatility and on nothing else, as in implied versus realized volatility.

Desk conventions, before any arithmetic

Greeks are quoted in currency on a desk, not in the units of a textbook, and the conventions are what make the interview questions below computable.

- **Delta** in currency: the notional exposure, so a delta of 5 million means a 1% market move changes the book by 50,000.
- **Gamma** in currency *per 1% move*: the change in that delta notional for a 1% move. Long 5 million of gamma means a 1% rally leaves you 5 million longer.
- **Vega** in currency *per volatility point*.
- **Theta** in currency *per calendar day*.

With gamma in that convention, a move of n percent produces a gamma profit of

$$\tfrac{1}{2}\,\Gamma_{\$}\,n^2 \times 1\%,$$

which is the formula every question below uses.

REAL INTERVIEW QUESTION (Société Générale · EQD)

How does an option trader's profit and loss decompose between delta, gamma, theta and vega?

WHAT THEY TEST / ANSWER

Give the expansion, then say what each term *means* on a desk, because the second half is what is being marked.

Delta is the directional term, and on a hedged book it should be close to zero by construction. If it is not, either the hedge was not done or the book moved between hedges, which is the subject of intraday hedging.

Gamma is the convexity term, $\frac{1}{2}\Gamma(\Delta S)^2$. It is always positive for a long option position, whichever way the market went, because it depends on the square of the move. This is the term that pays you for re-hedging: long gamma means you buy low and sell high mechanically.

Theta is what you pay for that privilege, and the two are locked together: for a delta-hedged position, theta is exactly the gamma profit that a move of one daily standard deviation would generate. Quoting that relation rather than treating theta as a separate decay is the single best thing you can say in this answer.

Vega is the mark-to-market term: the position is revalued at a new implied volatility. Note the distinction that separates candidates, since it is the crux of the whole subject. **Gamma and theta together are realized volatility**, they are cash generated by the market actually moving. **Vega is implied volatility**, a revaluation that has not yet produced any cash. A delta-hedged book can make money on gamma and lose more on vega on the same day, and that is not a contradiction, it is the difference between what happened and what the market now thinks will happen.

Close on the rest: rho and the cross terms, dividends and borrow for an equity book, the funding cost of the position, the profit on trades done during the day, and finally the unexplained residual, which is the number a risk manager reads first.

Working the numbers

REAL INTERVIEW QUESTION (BNP · Trading)

16% volatility, a 2% move in the underlying, and 5 million of gamma: what is your profit and loss? And your break-even?

WHAT THEY TEST / ANSWER

Two numbers, and they are connected.

The gamma profit. With gamma quoted as the delta change per 1% move, a move of n percent gives $\frac{1}{2}\Gamma_{\$}n^2 \times 1\%$. Here

$$\tfrac{1}{2} \times 5{,}000{,}000 \times 2^2 \times 1\% = 100{,}000.$$

The break-even. Use the rule of 16: a 16% annual volatility is $16/\sqrt{252} \approx 1\%$ a day, so the break-even daily move is **1%**. Any day the market moves more than that, a long gamma book makes money; any day it moves less, it loses.

Now show that the two are consistent, which is what turns a calculation into an answer. At the break-even move of 1% the gamma profit is $\frac{1}{2} \times 5{,}000{,}000 \times 1 \times 1\% = 25{,}000$, so the theta on this book must be about 25,000 a day. Check it directly, ignoring rates and dividends: $\Theta = -\frac{1}{2}\sigma^2 S^2 \Gamma$ per year, and $S^2\Gamma = 100\,\Gamma_{\$}$, giving $-\frac{1}{2} \times 0.16^2 \times 500{,}000{,}000 = -6.4$ million a year, which over 252 days is $-25{,}400$ a day. The two agree.

So the day's answer is $+100{,}000$ of gamma against $-25{,}400$ of theta, roughly $+75{,}000$ **net**, and the general statement is that the profit scales with the square of the ratio of the realized move to the break-even move: at twice the break-even you earn four days of theta and pay one, netting three times the theta, and at half of it you earn a quarter and pay one, losing three quarters of it.

REAL INTERVIEW QUESTION (Société Générale · EQD)

We sell a call on a stock with 16% volatility and delta hedge it. The theta is 1,000 a day. The spot moves 2% in one day. What is the realized profit and loss?

WHAT THEY TEST / ANSWER

A loss of about **3,000**, and the route matters more than the number.

Find the break-even. 16% annual volatility is about 1% a day by the rule of 16. That is the move at which the gamma loss exactly offsets the theta you collect.

Scale by the square. The market moved 2%, twice the break-even, and gamma profit and loss scales with the square of the move, so the gamma loss is $1{,}000 \times 2^2 = 4{,}000$.

Net. You are short the option, so you collect the theta and pay the gamma: $+1{,}000 - 4{,}000 = -3{,}000$.

Then give the general form, since it is the whole of this section in one line:

$$\mathrm{PnL} \;=\; \Theta_{\mathrm{day}} \left[1 - \left(\frac{r}{\sigma_{\mathrm{daily}}}\right)^2\right]$$

for a short-gamma position, so the profit and loss depends only on the ratio of the realized move to the implied daily move. Add the two qualifications a trader would. This is *one* day and the option is short-dated enough for gamma to be roughly constant across a 2% move; over a larger move you would integrate rather than use the square. And it assumes a single rebalance, since hedging more often through a choppy day changes the realized variance that the book actually captures, which is the point of intraday hedging.

REAL INTERVIEW QUESTION (JP Morgan · Delta One)

You are long 100k of vega and 1 million of gamma. The market moves +3% and the underlying's volatility rises by 1 point. What is your profit and loss?

WHAT THEY TEST / ANSWER

Two terms, computed separately and then added, with the conventions stated first because the numbers are meaningless without them: vega in currency per volatility point, gamma in currency per 1% move.

Gamma: $\frac{1}{2} \times 1{,}000{,}000 \times 3^2 \times 1\% = 45{,}000$. Positive, and note that it would be the same for a -3% move, since only the square enters.

Vega: $100{,}000 \times 1 = 100{,}000$. Also positive, since you are long vega and volatility rose.

Total: about $+145{,}000$, before theta, which was not given and which would be a negative offset of the order of the gamma profit on a one-standard-deviation day.

Then say the three things that make this a real answer rather than an arithmetic exercise. **Theta is missing from the question and should be named**, because on a long gamma and long vega book it is the price of both. **The two moves are not independent**: equity volatility usually falls when the market rallies 3%, so being long vega and seeing volatility rise on a rally is an unusual day and worth remarking on. And **the gamma figure assumes gamma is constant over the move**, which for a 3% move in a short-dated book is optimistic; the third-order term, sometimes called speed, is what the desk's own system would capture and your estimate would not.

The residual is the point

A clean explain has a small unexplained term. When it is large, the causes are worth knowing because each points somewhere different.

- **Higher-order terms**: large moves make the second-order expansion insufficient, and the fix is to revalue rather than to add another Greek.
- **Cross terms**: the spot and the volatility moved together, so the vanna and volga contributions are real and are missing from a first-pass explain.
- **Marking**: the surface was moved, a dividend forecast was revised, a correlation was remarked. This is genuine profit and loss and it belongs in its own line, not in the residual.
- **New trades**: business done during the day, which should be attributed to the spread earned rather than left to pollute the market attribution.
- **Something wrong**: a booking error, a stale market data point, a model change nobody flagged. This is why the residual is monitored rather than merely computed, and a persistent residual is investigated even when it is *favourable*.

> INTERVIEW TRAP
>
> Four errors. (1) Treating theta as an independent cost rather than as the rent on gamma: they are two views of the same number, and quoting one without the other misses the trade. (2) Confusing gamma profit with vega profit: gamma is cash from realized moves, vega is a revaluation at a new implied level, and a book can gain on one and lose on the other on the same day. (3) Using Greeks quoted in one convention with a formula written in another, which is where factors of 100 come from. (4) Ignoring a small but persistent residual, which is exactly the pattern that reveals a mismarked position.

Where this goes next

The sensitivities being multiplied are the Greeks, and the identity that each Greek multiplies its own market move, stated from the Greeks side rather than from the desk's, is Greeks and PnL explain. The full attribution, the conventions it depends on and the residual live here. The hedging decisions that determine how much of the realized move the book actually captures are delta neutrality and dynamic hedging and intraday hedging, the trade this decomposition is really measuring is implied versus realized volatility, and the positions being explained come from inventory management.

22.7 Book management

trading · structuring · sales | core | cross-asset

A trader does not manage trades. Trades are done and gone within seconds. What a trader manages is a **book**: a standing collection of positions with a mandate, a set of limits, a profit and loss, and one person whose name is against it.

Almost everything that distinguishes a good trader from a good pricer happens at that level. Anyone can decide whether a single trade is attractive. Deciding what a hundred overlapping positions add up to, what they are costing to hold, and which of them you would be relieved to be rid of is a different skill and it is what the job actually consists of.

What a book is

A book is the unit of accountability in a bank. It has four things:

- a **mandate**: what it is allowed to trade, in what products and currencies, and for what purpose;
- a **limit set**: the boundaries within which it may run risk, covered in position limits;
- a **profit and loss**, produced independently and reconciled daily; and
- an **owner**, which is why books rather than desks are the level at which risk reports are cut.

Everything is booked somewhere, and a position that is in the wrong book is invisible to the person responsible for it. That is not administrative pedantry: several of the losses in operational risk were positions in books whose owners did not know they were there.

The day

The rhythm is the same on every floor, and it is worth knowing because an interviewer will ask what you would actually be doing.

1. **Before the open.** Read the overnight: where did markets go, what did the book make or lose while nobody was watching, and does the P&L match what the risk report said it should. Check what expires or fixes today. Check the limits with the new marks.
2. **The open.** The first prints are the least reliable and the most consequential, because overnight risk gets adjusted into them.
3. **Through the day.** Quote, hedge, and manage inventory. The hedging cadence is intraday hedging, and the inventory question is inventory management.
4. **The close.** Mark the book, which is a decision and not an observation.
5. **After the close.** P&L is produced, explained and signed off, and the explanation is a control rather than a report: see P&L explain. Then the handover, if the book follows the sun.

> **KEY IDEA**
>
> The end of the day is when the book is most exposed to its own errors. Marks are set, the P&L is struck, limits are measured, and everything that was going to be spotted has to be spotted now. A trader who is only interested in the part of the day when the market is open is doing half the job.

Three questions about every position

The discipline that separates a managed book from an accumulated one is asking three things about everything in it.

Why is it here? Every position was put on for a reason. Some were client trades that have not been worked out, some were hedges for something that has since gone, and some are views nobody has revisited. A position whose reason has expired is a position nobody is managing.

What is it costing? Not just theta. Funding on the balance sheet it consumes, the capital charge it attracts, the borrow on any short, and the operational attention it takes. These are invisible in the P&L line and very visible in the return on the book.

What gets me out of it? The one nobody teaches. Before a position is put on, there should be an answer to how it is closed, at what cost, and in what market conditions. A position with no exit is not a trade, it is a commitment.

Ageing, and why old inventory is bad inventory

Every desk runs an ageing report showing how long each position has been held, and the reason is a selection effect rather than an accounting convention.

Inventory that has been on the book for a long time is, by construction, inventory that could not be sold. It is therefore more likely to be illiquid, more likely to be mismarked, because the mark has not been tested against a real trade for months, and more likely to be worth less than the screen says. Old positions are not neutral, they are adversely selected, and a book that never turns over is not a stable book; it is a book quietly accumulating the things nobody wanted.

The number that is not P&L

Here is the arithmetic that changes how a book is judged, and it surprises people who think trading is measured by profit.

WORKED EXAMPLE

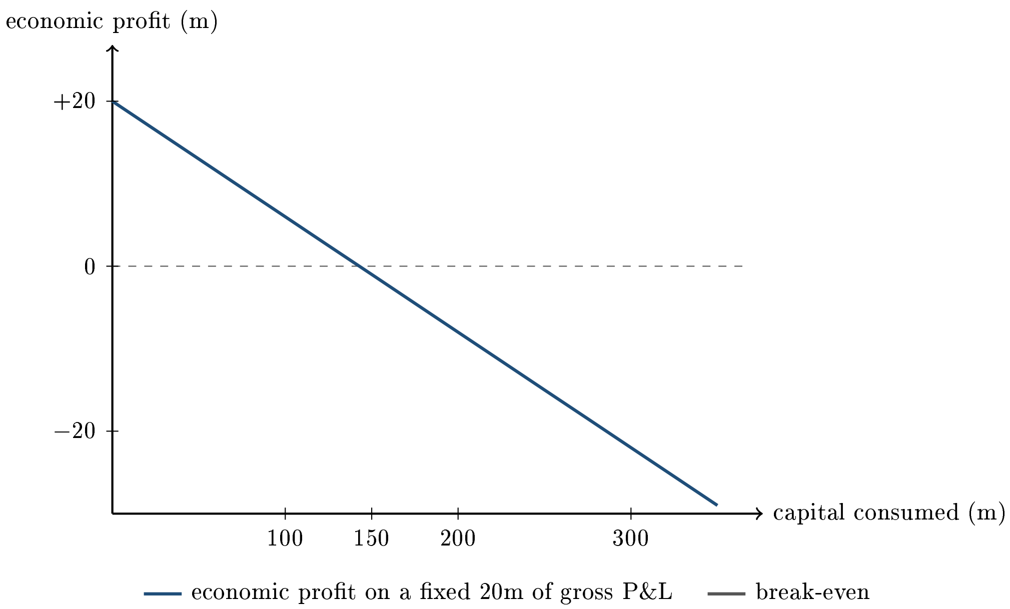
Two books, both making **20 million** of gross trading P&L a year.

Book A consumes 100 million of regulatory capital and 400 million of balance sheet. Book B consumes 300 million of capital and 1,200 million of balance sheet.

Charge funding at 0.5% of balance sheet and equity at a 12% cost of capital:

	Book A	Book B
Gross P&L	20.0	20.0
Funding charge	−2.0	−6.0
Capital charge	−12.0	−36.0
Economic profit	**+6.0**	**−22.0**
Return on capital	18.0%	4.7%

Identical headline profit. One book creates six million of value and the other destroys twenty-two. Nothing in the P&L line distinguishes them, and everything about whether the business should exist does.



On a fixed 20 million of gross P&L the line crosses zero at about **143 million** of capital. Past that point the book is losing money for the firm however good its trading looks, and the trader will still be reporting a profit every month. This is why balance sheet and capital are rationed on a trading floor, and why "can I have more limit" is a conversation about return rather than about risk appetite. The rules that set the capital number are in capital requirements.

Marking is a decision

The single most consequential thing a book manager does each day is decide where the book is marked, because that decision creates or destroys reported P&L immediately and reverses only when the position trades.

The pressure is obvious and it always points one way. Marking a difficult position optimistically turns a bad month into a good one and postpones the problem. That is why independent price verification exists, why illiquid positions carry valuation adjustments, and why an unexplained gap between the desk's marks and an independent source is treated as an incident rather than a disagreement.

> **INTERVIEW TRAP**
>
> Three failures of book management. (1) Judging a book by P&L alone, when the example above shows two identical P&Ls where one destroys value. (2) Letting positions age without a challenge, since old inventory is adversely selected rather than merely old. (3) Running a book nobody else can describe. If it cannot be handed over in ten minutes, it cannot be covered when you are ill, and a position that only one person understands is a single point of failure with a P&L attached.

Interview angle

REAL INTERVIEW QUESTION (CACIB · Cross Asset)

Rate moves and mark to market.

WHAT THEY TEST / ANSWER

The prompt is deliberately open, so impose a structure: what marking to market means, what a rate move does to a book through it, and where the difficulty actually lies.

What it means. Marking to market is restating every position at what it is currently worth rather than at what was paid for it, so unrealised gains and losses are recognised as they occur rather than at maturity. For a rates book that is the whole P&L: nothing has been sold, and the day's profit is entirely a revaluation.

What a rate move does. Discount every future cashflow at the new curve. To first order the change is minus duration times the rate move times the value, with convexity as the second-order correction, so a book with a positive duration loses when rates rise. Two refinements worth adding: the move is rarely parallel, so what matters is the exposure to each part of the curve rather than one duration number; and a derivative book reprices through its discount curve as well as through its forwards, which is why the choice of discounting curve is itself a P&L event.

Where the difficulty is, which is the part that makes it a real answer. Marking is easy where there are prices and hard where there are not. The liquid points on the curve are observable and everything between and beyond them is interpolated or extrapolated from a model, so a large part of a book's mark is a modelling choice rather than an observation. That is why banks run independent price verification, apply valuation adjustments to positions with no observable price, and treat a persistent gap between a desk's marks and an independent source as a control incident. Mismarking is the cheapest way to hide a loss and it is the reason the control exists.

If they push for a number, give the sensitivity rather than the level: a one basis point move on a ten-year position of a given size, which is the DV01, is the unit a rates trader actually thinks in.

Where this goes next

The daily explanation that validates everything above is P&L explain; the hedging that happens between the marks is intraday hedging; and the boundaries the book runs inside are position limits. The inventory question that generates most of the ageing problem is inventory management, and the firm-level view of all the books together is aggregation of Greeks.

22.8 Intraday hedging

trading · structuring · sales | advanced | cross-asset

Black-Scholes assumes you rehedge continuously and for free. You cannot, and you do not, which means the theory tells you *what* to hedge but not *when*. Deciding when is a real trading skill with real money attached, and it is what a desk actually argues about during the day. The definitions of delta, gamma and theta belong to delta neutrality and dynamic hedging, gamma and convexity and theta; this section is about the timing decision that sits on top of them.

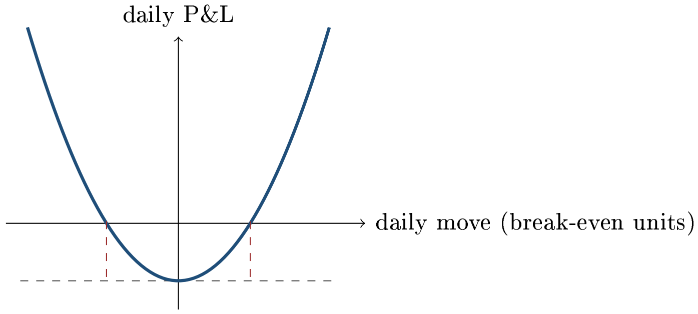
The one identity you need

For a delta-hedged option position over a short interval, with the move measured as a return $r = \Delta S/S$ and $\Gamma_\$ = \Gamma S^2$ the cash gamma,

$$\text{P\&L} \;\approx\; \tfrac{1}{2}\,\Gamma_\$\, r^2 \;+\; \Theta\,\Delta t \;+\; \text{vega} \times \Delta\sigma_{\text{implied}}.$$

Hold implied volatility fixed and set the first two terms against each other. There is a **break-even move**, the daily return at which the gamma term exactly pays the theta bill, and it is not a coincidence that it equals the daily volatility implied by the price you paid. The convenient desk approximation is that the daily volatility is the annual figure divided by 16, since $\sqrt{256} = 16$: a 16 volatility means a 1% break-even daily move, a 32 volatility means 2%.

> **KEY IDEA**
>
> The gamma term is **quadratic** in the move while the theta term is linear in time. So a move of twice the break-even does not cost twice the theta, it costs *four* times. That single asymmetry is why hedging frequency matters, why short-gamma books die in gaps rather than in drifts, and why almost every numerical question on this topic is answered by squaring a ratio.



long-gamma delta-hedged P&L — break-even moves, at −1 and +1 on the axis — theta paid on a flat day (short gamma is this curve reflected)

Frequency is a variance decision, not a profit decision

In the frictionless limit, hedging more often does not change your expected P&L. What it changes is the *dispersion* around it: the discrete-hedging error has a standard deviation that falls roughly like $1/\sqrt{n}$ in the number of rehedges, so quadrupling the hedge frequency halves the noise. Introduce transaction costs and the picture becomes a trade-off, since each rehedge crosses a spread and, in size, moves the market.

That gives the two competing pressures behind every rehedging rule:

- **Hedge more often** to cut the variance of the outcome and to avoid carrying an unintended directional position.
- **Hedge less often** to avoid paying the spread repeatedly on moves that are just noise.

The standard resolution is the **no-hedge band**, constructed in delta neutrality and dynamic hedging along with its cube-root dependence on the transaction cost. What belongs to an intraday book is that the band is never the only trigger. Desks combine three, which is more robust than any one of them: a move trigger (delta outside the band), a time trigger (a scheduled rehedge, at least into the close), and an event trigger (a data release, an earnings print, an auction).

Short gamma and long gamma are different jobs

- **Short gamma.** Every move costs you, quadratically, and you rehedge *into* the move: buying as the market rises, selling as it falls. You are a trend-follower, and your worst case is a single large gap you never got to hedge inside. So the discipline is tight bands and frequent hedging, accepting the transaction cost as the price of not being destroyed, plus explicit gap protection through cheap wings rather than through more frequent hedging, which cannot help against a gap.
- **Long gamma.** Every move pays you, and you rehedge *against* the move: selling as the market rises, buying as it falls. You are a mean-reverter, and the risk is not ruin but leakage: paying the spread on noise, and paying theta while nothing happens. So you can afford wider bands and more patience, hedging at levels you choose rather than mechanically, and monetising the move rather than immediately flattening it.

> KEY IDEA
>
> The asymmetry is not about who is more careful, it is about the shape of the loss. A long-gamma trader who hedges badly *underperforms*. A short-gamma trader who hedges badly can be *wiped out*, because the loss grows with the square of a move that has no upper bound. When in doubt, the side of the book you are on decides which mistake you can afford.

What intraday hedging cannot fix

Two things, and naming them is what stops the section being naive.

Gaps. Overnight, weekends, earnings and scheduled news move the price when you cannot trade. No hedging frequency helps, because the interval is imposed on you. The only real defences are position size, option wings, and reducing exposure before the event.

The weekend and the calendar. Theta accrues on calendar days but you can only hedge on trading days, so a Friday close carries three days of decay into one hedging decision. This is why short-gamma books like quiet Fridays and long-gamma books do not, and why some desks mark theta on a business-day basis while risk systems accrue it on a calendar basis, which is a recurring source of P&L explain noise, see PnL explain.

Two further practical points. Hedge the *big* deltas first and do not chase small ones, since transaction costs are certain while noise is not. And near a barrier or a digital, delta is unstable and can flip sign, so mechanical rehedging is exactly wrong there: those positions are managed by anticipating the level rather than by reacting to it. The single-day version of this question, where a 2% move against €1,000 of theta on a 16 volatility book costs three times the theta collected, is an attribution question rather than a cadence one, and it is worked in PnL explain along with the general form, theta times $1 - (\text{move/break-even})^2$. What that identity means *for how often you hedge* is the subject here, and the next question is where it bites.

REAL INTERVIEW QUESTION (Société Générale · EQD)

I short a delta-hedged option. The stock loses 3%. Assume that on the following days the stock stays flat. How many days do I need to recover my losses?

WHAT THEY TEST / ANSWER

Use the same squared ratio and answer in days of theta. If the break-even daily move is 1%, consistent with a 16 volatility, then a 3% move costs $3^2 = 9$ times one day's theta. You collect that same day's theta as well, so the position ends the day of the move down $9 - 1 = 8$ thetas, and with the stock flat from then on you collect exactly one theta per day, so you need **8 more flat days**. That is (move/break-even)$^2 = 9$ days in total, counting the day of the move, which is the general answer they are really after, so say the formula and then substitute. Then give the three second-order effects, in the order they matter, because a flat answer of nine is only half the marks. The option is now 3% away from the strike, so it is no longer at the money: gamma and theta both fall, so the daily theta you collect from here is *smaller* than the one used in the calculation and the true recovery takes longer than nine days. Time also passes, which raises theta for a still-near-the-money short-dated option and works the other way. And if the stock really does sit still, realised volatility collapses, so implied volatility would typically fall too, which helps a short-vega position and could recover the loss faster than the theta arithmetic alone suggests. Finish with the honest framing: this question is testing whether you know that hedged option P&L is realised volatility against implied volatility, and the answer in days is just that comparison expressed on a clock.

INTERVIEW TRAP

Four traps. First, **forgetting that gamma P&L is quadratic**: a double-sized move is a quadruple-sized number, and candidates who scale linearly get every one of these questions wrong. Second, **believing more frequent hedging protects against gaps**: it reduces diffusion error, not jump risk. Third, **hedging mechanically near barriers and digitals**, where delta is unstable and reacting to it can double the loss. Fourth, **ignoring transaction costs when arguing for continuous hedging**: in the real world the optimal frequency is finite precisely because the spread is not zero, and a long-gamma book's edge is most often lost to over-hedging rather than to being wrong.

VISUAL / CODE TO BUILD: Interactive

A hedging simulator on a replayable intraday price path. The reader chooses a position (long or short a straddle), a rehedging rule (continuous, fixed time interval, delta band with adjustable width, or manual clicks) and a transaction cost in basis points. The panel plots cumulative P&L decomposed into gamma, theta and transaction costs, so the reader can watch a long-gamma book's edge disappear into over-hedging and a short-gamma book's theta vanish in a trend. Two switches carry the section's lessons. **Distribution mode** runs the same rule over a thousand simulated paths and shows the P&L histogram, making visible that frequency narrows the distribution without moving its centre until costs are switched on. **Gap mode** inserts an overnight jump and lets the reader try to fix it with frequency, which fails, and then with wings, which works.

Where this goes next

The Greeks used here are defined in delta neutrality and dynamic hedging, gamma and convexity and theta as the cost of convexity, and their aggregation across a book in portfolio-level Greeks. The flow that creates the position in the first place is market making, what to do with the position you did not want is inventory management, and the attribution of what the hedging actually earned or cost is PnL explain.

22.9 Position limits

trading · structuring · sales core cross-asset

A trader does not decide how much risk to take. The firm decides, and a limit is how that decision is written down in a form a system can check overnight. This section is the desk view: what the limits are, why there are so many of them, and how a trader actually behaves as one approaches. How the framework is designed and governed belongs to risk limits frameworks.

The layers

No single number captures a derivatives book's risk, so limits are stacked, and each layer exists because it catches a failure the others miss.

- **Mandate.** What you are allowed to trade at all: products, currencies, maturities, counterparties. The most important limit and the least discussed, because breaching it is a conduct matter rather than a risk matter.
- **Notional and size.** Crude, easy to check, and the only one that catches leverage directly.
- **Greek limits**, bucketed: delta, gamma, vega by maturity and often by strike region, plus the asset-specific ones, dividends and borrow in equities. These are the day-to-day binding constraints.
- **VaR or expected shortfall**, a statistical summary across all risk factors at once, which is the only layer that accounts for how the factors move *together*, in value at risk.
- **Stress loss**, the loss under named scenarios. This is the backstop for everything the statistics miss, in stress testing.
- **Concentration**, capping single names, sectors, maturity buckets and counterparties, because a book can be inside every aggregate limit and still be one position.
- **Stop-loss and drawdown**, which act on the profit and loss rather than on the position, and which force a decision rather than describe a risk.

> KEY IDEA
>
> The layers are not redundancy, they are coverage of different failure modes. Greeks are a local approximation and miss the gap. VaR is a quantile and says nothing about what happens beyond it, and it assumes yesterday's correlations. Stress catches a scenario someone wrote down and misses the one nobody imagined. Notional catches leverage and nothing else. **A book that is comfortably inside every limit can still be one scenario from a large loss**, which is precisely why the set is layered and why the binding constraint moves between layers as the market changes.

WORKED EXAMPLE

A desk with a delta limit of 50 million and a gamma limit of 3 million per 1% move is running 5 million of delta and 2.5 million of gamma. It is at 10% of its delta limit and 83% of its gamma limit, so gamma is the binding constraint and delta is nowhere near.

On an ordinary 1% day that gamma is worth $\frac{1}{2} \times 2{,}500{,}000 \times 1 \times 1\% = 12{,}500$, which is unremarkable in either direction. On a 5% gap it is worth $\frac{1}{2} \times 2{,}500{,}000 \times 25 \times 1\% = 312{,}500$, twenty-five times as much, because gamma profit and loss scales with the square of the move.

A one-day 99% VaR will barely register the second case, since a 5% gap sits outside the quantile it measures. The gamma limit and the stress limit are what stand between the desk and that number, and they are doing a job the statistical limit structurally cannot. That is the argument for layers in one example.

Living with them

REAL INTERVIEW QUESTION (Allianz Global Investors · Market Risk)

Once you have measured VaR or expected shortfall on a portfolio, what do you do with it? Do you act before they are breached, and how do you adjust the portfolio?

WHAT THEY TEST / ANSWER

Three parts, and the middle one is the real question.

What you do with the number. On its own it is a monitoring output. It becomes useful when it is decomposed: which positions contribute the most, which risk factors drive it, and how it has moved since yesterday and why. A VaR that rose 20% overnight with no new trades is telling you the market's volatility or correlations moved, and that is more actionable than the level itself.

Do you act before a breach? Yes, always, and the reason is the answer they want. A limit is not a target to run against. Utilisation is managed with headroom, typically with soft triggers well below the hard limit, for two reasons. First, **the measure moves without you**: VaR rises when volatility rises, so a static book can breach a limit purely because the market changed, and if you were at 98% you are now over. Second, and more seriously, **limits bind when liquidity is worst**. A limit that forces you to reduce during a stress event forces you to sell into the same market that caused the breach, at the widest spreads of the year, alongside everyone else in the same position. Waiting until the breach converts a risk limit into a mechanism for realising the loss.

How you adjust. In order of preference: stop adding, which is free; hedge the dominant factor rather than reducing everything, since the decomposition tells you which one is driving the number; reduce the least liquid positions first, because they are the ones you will not be able to exit later; and only then cut broadly. Note that hedging can be more efficient than selling: a macro hedge on an index can cut the measure quickly while the underlying positions are unwound in an orderly way over days.

Close on the discipline: any breach is reported and escalated regardless of how quickly it is cured, and a breach that is fixed before anyone notices is a worse problem than one that is reported, because it means the control did not work.

Where limits fail

Four failure modes are worth knowing, because interviewers ask about them and because they are how real losses happen.

The unmeasured axis. Every limit that exists is monitored; the risk that grows is the one nobody set a limit on. Correlation on a multi-asset book, the far end of the dividend curve, borrow on a crowded name: these accumulate precisely because there is no number turning red. The question to ask of any book is not "where are we against limits" but "what are we long that has no limit".

Netting that is not offsetting. An aggregate Greek can be small while its components are large and opposite, which is fine if they genuinely offset and dangerous if they are different maturities, different strikes or different underlyings, in portfolio-level Greeks aggregation.

Measurement timing. Limits checked at the close say nothing about intraday exposure, and a book that is flat every evening can carry significant risk during the day.

The limit as an incentive. A trader rewarded for revenue and constrained by limits will find the axis with the most headroom, which is rational and is exactly why the set of limits has to cover the space rather than the few risks that are easy to compute.

Stress results are one of the inputs a limit is checked against, and the comparison is the step that matters on this desk: **the worst stress outcome is measured against the stress limit, and separately against what the desk could actually hedge in that scenario**. A structure whose worst case sits inside the limit but cannot be hedged at any price in that state is not acceptable merely because the number fits, and the right output is then a reserve, a size cap, or a decision not to write the trade. Designing the scenarios themselves is stress testing.

> INTERVIEW TRAP
>
> Four errors. (1) Treating a limit as a target and running at 95% utilisation, when the measure moves on its own and leaves no room. (2) Assuming a book inside every limit is safe, when the layers exist precisely because each one is blind to something. (3) Netting Greeks across buckets that do not genuinely offset. (4) Curing a breach quietly instead of reporting it, which is a conduct failure regardless of the outcome, as in case studies.

Where this goes next

The framework these limits belong to is risk limits frameworks, the statistical measures they are set on are value at risk and expected shortfall, and the backstop layer is stress testing. The positions being constrained come from flow trading and inventory management, and the aggregation problem is portfolio-level Greeks aggregation.

Chapter 23

Sales - Desk Reality

23.1 Client coverage

sales · structuring · trading core cross-asset

A client does not call “the bank”. They call a person, whose mobile number they have, who answers at seven in the morning, and who they will stop calling the moment somebody else is more useful. Client coverage is the business of being that person.

It is the least mathematical seat on a trading floor and one of the hardest to be good at, because the skill is not knowledge of products, which can be looked up, but judgement about which of the many things you could say to a client is worth their attention today.

What coverage actually means

Coverage is a named person made responsible for a named set of clients across a defined product range. Everything else follows from those three nouns.

Named person: the relationship is personal, and it moves with the individual. When a salesperson changes firm, a measurable share of their clients’ business follows within a year, which is the reason coverage seats are expensive and the reason handovers are done carefully.

Named clients: coverage is exclusive. Two people from the same bank calling the same client with different views is a failure, and the internal machinery for deciding who owns what is more contested than any outsider expects.

Defined products: which is where the tension sits. Cover a narrow product deeply and you are useful on that product and irrelevant on everything else. Cover a client broadly and you are the first call for everything and expert in nothing, so you rely on specialists behind you.

The triangle

Three roles touch every client trade, and knowing who owns what is the single most useful structural fact for a candidate.

	Owns	Is measured on
Sales	the client relationship and the client price	client revenue, wallet share
Trading	the risk and the price shown to sales	P&L on the book
Structuring	the product that does not exist yet	products sold, reusability

The mechanics that follow from this split are the subject of the rest of the chapter: how a client request becomes a price is RFQs, how the client price differs from the trader’s price is client pricing, and what sales sends to clients unprompted is axes and trade ideas and market colour.

KEY IDEA

Sales does not take market risk and trading does not talk to clients. That division is not a convention, it is the design: the person who knows what the client will pay must not be the person deciding what the risk is worth, or the price stops being a price. Every conversation between the two seats is a negotiation, and understanding that it is meant to be one is what stops a junior salesperson taking a trader's first number as an answer.

The day

Roughly, and it starts earlier than the trading day.

1. **Before the open**: read the overnight, read the research, and work out which two or three of your clients have a reason to care about what happened. Not all of them. The judgement is in the filtering.
2. **The morning call**: the desk's view is presented internally, and what a salesperson takes from it is the part their clients can act on.
3. **Calls and messages**: the substance of the job. Colour, axes, ideas, and answering the things clients ask.
4. **Requests**: a client asks for a price, sales asks the trader, sales shows the client, the client deals or does not. Minutes, sometimes seconds.
5. **After the close**: bookings checked, revenue attributed, and the next day's list of who to call prepared.

Who gets a human, and why

Coverage is expensive, so it is rationed, and the arithmetic that rations it is worth working because it explains the entire structure of the sales business.

WORKED EXAMPLE

A client buys 50 million of a bond. The desk's mid is 100.00 and the client is shown 100.05, five basis points. Gross revenue on the trade is

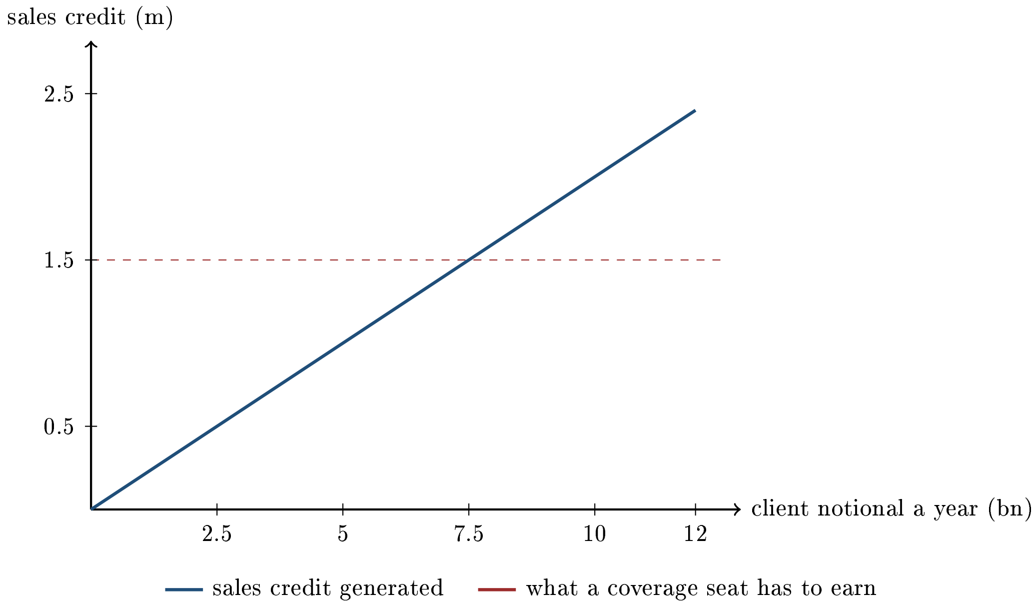
$$0.0005 \times 50{,}000{,}000 = 25{,}000,$$

of which the **sales credit**, the share attributed to coverage rather than to the risk the trader took, might be 40%, so 10,000.

Now size a coverage seat. Fully loaded, a salesperson costs the bank on the order of 500,000 a year once compensation, support, technology and premises are counted, and a business typically wants revenue of some multiple of that, say three times. So the portfolio must generate about **1.5 million of sales credit**.

At 10,000 a trade that is 150 trades a year, roughly three a week. Working back, it implies client gross revenue of $1.5\,\text{m}/0.40 = 3.75$ million, and at five basis points that means

$$\frac{3{,}750{,}000}{0.0005} = 7.5 \text{ billion of notional a year.}$$



The crossing point is the whole sales business model. A client trading less than about seven and a half billion a year, on these assumptions, cannot support a dedicated person. That is why smaller accounts are covered electronically or by a shared desk, why platforms exist, and why every salesperson is under continuous pressure to move up the client list rather than across it.

> **INTERVIEW TRAP**
>
> Reading the chart as saying small clients do not matter. It says they cannot support a *dedicated* seat at those margins, which is a different statement, and the two ways round it are the whole story of how the business has changed: serve them at lower cost through electronic channels, or earn more per trade by selling something more complex than a five-basis-point bond. Structured products exist partly because of the second route.

The conflict at the centre of the job

A salesperson is paid by the bank and trusted by the client. Those two facts point in opposite directions on every trade: a wider price is better for the bank today and worse for the relationship tomorrow.

There is no clever resolution, only an honest one. The business is repeated, the client can compare prices, and a client who works out they have been badly treated stops calling and tells others. So the incentive to be fair is real and comes from repetition rather than from virtue, which is exactly why the trades that go wrong are the one-off ones with the least sophisticated clients.

Around that sits the regulation: best execution obligations, suitability and appropriateness requirements, disclosure of what is being charged, and the rules on what may be said to whom. Those are covered in best execution, and the practical point is that they are floors rather than standards. A trade can be perfectly compliant and still be the wrong trade for the client.

Interview angle

REAL INTERVIEW QUESTION (Pictet AM · Sales)

What makes a good salesperson?

WHAT THEY TEST / ANSWER

Resist the temptation to answer "good communication skills", which is true of everybody and tells an interviewer nothing. Give three qualities that are specific to this job, and justify each with what it prevents.

Judgement about relevance. A salesperson has far more information than any client wants, so the skill is deciding what *not* to send. The measure of a good one is that their clients open the message, which is earned by not sending the other four. Anyone can forward research.

Being trusted on the bad days. Anyone can call a client when a trade has worked. The relationship is built when it has not: calling first, before the client calls you, explaining what happened and what the options are. Clients do not remember who gave them the best price; they remember who was there when the position went wrong.

Knowing enough to be dangerous, and knowing where that stops. You need enough product knowledge to have the conversation and enough honesty to say "I do not know, I will find out and call you back in ten minutes", and then actually to call back in ten minutes. The failure mode is guessing, which is discovered eventually and ends the relationship at once.

Then add the two that a thoughtful interviewer is waiting for. **Internal credibility**: a salesperson who cannot get a competitive price out of their own trading desk is not useful to anybody, and that is earned by bringing business the desk wants and by being right about what the client will actually do. **Willingness to say no**: to a client, when the trade is wrong for them, and to a trader, when the price is not good enough. The salespeople with the longest careers are the ones who declined business.

If asked what makes a bad one, the answer is short: somebody who optimises the current trade instead of the relationship, because that is the one mistake that cannot be recovered from.

REAL INTERVIEW QUESTION (Morgan Stanley · Sales)

Which asset class is traded the most? How much of the market does that represent?

WHAT THEY TEST / ANSWER

This is a check on whether you have a sense of scale for the market you are asking to work in, so give the answer with numbers and then say what the numbers mean.

By turnover, foreign exchange is the largest by a wide margin. The Bank for International Settlements triennial survey put global FX turnover at roughly 7.5 trillion dollars a day in 2022, and interest rate derivatives at a broadly comparable order of magnitude. Global equity trading is smaller by around an order of magnitude, in the hundreds of billions a day.

Then explain the gap, because the raw ranking is misleading and saying so is the answer. FX volumes are enormous partly because most of it is not investment: a large share is very short-dated swaps rolling positions, and every cross-border transaction in any other asset class generates an FX trade as a by-product. So FX turnover counts the plumbing of the whole system, not just views being expressed.

And the point that matters for a sales seat: notional traded and revenue earned are not the same ranking. FX is vast and extremely tight, often fractions of a basis point in the major pairs, so enormous volume converts into modest revenue per unit. Equity derivatives and structured products trade far less notional at far wider margins. A salesperson's revenue follows the margin, not the turnover, which is why the biggest market is not the biggest business.

If you do not know a figure, give the order of magnitude and say it is approximate rather than inventing precision. Trillions a day for FX, hundreds of billions for equities, and the relationship between them is the part being tested.

INTERVIEW TRAP

Three coverage errors. (1) Treating coverage as order-taking, when the value is in the filtering that happens before the client calls. (2) Optimising one trade's margin, which is the only mistake in this job that compounds. (3) Assuming the client is less sophisticated than you are: on the other side of the phone is frequently somebody who used to have your seat.

Where this goes next

Who the clients are and how differently they behave is flow versus real money clients. The mechanics of a request are RFQs and client pricing, what you send unprompted is axes and trade ideas and market colour, and the long game is relationship management.

23.2 Flow vs real money clients

sales · trading · structuring | core | cross-asset

"The client" is not one thing. A pension fund hedging a liability twenty years out and a hedge fund lifting your offer because it thinks you are slow are both clients, and almost nothing about serving them is the same: not the horizon, not the products, not the margin, not the risk you inherit from trading with them. A salesperson who cannot tell them apart is pricing blind.

The four archetypes

Real money. Pension funds, insurers, long-only asset managers, sovereign wealth funds. They invest money that belongs to someone with a long horizon, they are usually constrained by a mandate, by a benchmark, by ratings rules or by a capital regime, and they are hedging liabilities or reaching for yield rather than expressing a short-term view. They trade infrequently and in size. They are unlevered or lightly levered.

Fast money. Hedge funds, proprietary trading firms, some multi-strategy desks. Absolute return, leveraged, short horizon, and price-sensitive to the last basis point. They trade often, they compare quotes across dealers, and they will move for a fraction of a spread.

Corporates. A treasurer hedging a real exposure: a receivable in a foreign currency, a floating-rate loan, a commodity input. They are not trying to make money on the trade, they are trying to remove a risk from the business. They trade rarely, care about documentation and accounting treatment as much as about price, and value being able to call the same person again.

Distributors. Private banks, retail networks, wealth platforms, who buy structured products to sell on to end investors. The buying decision is made by the distributor rather than by the end client, and the products carry the widest margins on the shelf, which is the subject of retail versus institutional structuring.

> KEY IDEA
>
> The distinction that matters to a trading desk is not size, it is the **information content of the flow**. A corporate hedging an export receivable is trading for a reason that has nothing to do with your price being wrong, so their flow is benign and you can quote it tightly. A hedge fund that lifts your offer three times in an illiquid name is telling you something, and the correct response is to move the price, as worked through in case studies. Same notional, same product, completely different risk, and pricing them identically is how a desk loses money politely.

What each one wants from you

- **Real money** wants size without moving the market, certainty of execution, research and structuring input, and a counterparty that will still be there in five years. They will pay a wider spread for a large clean print than for a tight quote they cannot fill.
- **Fast money** wants the price, and then wants it faster. Value is added through liquidity, axes, and being right about where risk can actually be moved, in axes and trade ideas.
- **Corporates** want to be understood: the right hedge for the actual exposure, documentation that works, an accounting treatment their auditor will accept, and no surprises. Price matters less than being wrong about the structure.
- **Distributors** want a payoff that can be explained to an end client, a headline number that competes with what the network next door is showing, and support materials, secondary prices and after-sales service.

REAL INTERVIEW QUESTION (Société Générale · EQD)

Pitch me an equity swap, and tell me who your clients are.

WHAT THEY TEST / ANSWER

Pitch it in one breath, then spend the answer on the client map, because that is the half being tested.

The pitch. An equity swap gives you the total return of a stock, a basket or an index, price performance plus dividends, against paying a floating funding rate plus a spread. You get the exposure without buying the shares: no settlement, no custody, no register, and the position is financed by us rather than by your balance sheet.

Who buys it, and why each one does. A **hedge fund** uses it for leverage and for the ability to go short as easily as long, since the swap handles the borrow, and because it is often the cheapest access to a market where they are not set up to trade directly. An **asset manager** uses it to gain or hedge exposure quickly around a flow, to access a market with ownership restrictions or punitive settlement, or to run a position without appearing on the share register. A **corporate or a holding company** uses it around its own or a strategic stake, for hedging or monetisation, which is a specialised and heavily documented business. An **insurer or pension fund** is the least natural buyer, because the funding leg is an expense they can often avoid by just owning the shares, unless there is a regulatory or tax reason not to.

Then say what makes it a real answer: the reason a client chooses a swap over the cash shares is almost always **funding, access or reporting**, not a market view, so the first question to a client is which of the three it is. That determines whether they care about the spread, the tenor, the termination rights or the dividend treatment, and quoting before you know is how you win the trade you did not want. The instrument itself, its two legs and its risks, belongs to swaps; what is developed here is the client map on top of it.

Knowing what you are talking to

The distributor archetype is the one with the most rules attached to it, and the question *how does a private bank work* is answered in retail versus institutional structuring, which owns the retail channel. The one refinement the client map needs on top of the archetype above is who inside the distributor actually decides: a product committee and the relationship managers who have to sell the result, not a portfolio manager taking a view.

Why the same trade is priced differently

A desk quoting two clients on an identical trade will often show different prices, and a salesperson has to be able to say why without embarrassment. The legitimate reasons are concrete: the **credit** of the counterparty and whether the trade is collateralised; the **size** and how it compares to what can be hedged; the **information content** of the flow, since a price shown to a counterparty likely to be right must be wider; the **cost of the hedge** at that moment; and the **overall relationship**, since a client who gives the desk two-way business is genuinely cheaper to serve than one who only ever takes the same side.

What is not legitimate is charging a client more because they cannot tell. That line is where commercial judgement becomes a conduct question, and it is codified in best execution and suitability rules that distinguish between professional and retail counterparties precisely because the second cannot check. Pricing on risk is defensible in front of anyone; pricing on ignorance is not, and it is the distinction on which a sales answer to this question stands or falls, developed in client pricing.

INTERVIEW TRAP

Four errors. (1) Treating notional as the measure of a client's importance, when a smaller two-way client can be worth more than a larger one-way one. (2) Quoting a hedge fund and a corporate identically, which ignores the information in the flow. (3) Assuming a distributor's decision is the end client's decision: they are different people with different incentives. (4) Explaining a price difference by "the relationship" when the real reasons are credit, size, hedging cost and flow, which are all defensible and should be said out loud.

Where this goes next

The organisation of the coverage model is client coverage, the mechanics of being asked for a price are RFQs, and how the price is arrived at is client pricing. What the desk does with the resulting risk is flow trading, the ideas that start the conversation are axes and trade ideas, and the distribution channel with the most stringent obligations is retail versus institutional structuring.

23.3 RFQs

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Most of what a sales person does all day is handle requests for quotes. A client wants a price on something that does not trade on a screen, or does not trade on a screen in their size, so they ask a handful of dealers and pick the best answer. Everything about the job lives inside that loop: what you ask the client before you ask the trader, how fast you come back, how wide you go, and what you do when you lose. It is also the single most realistic thing an interviewer can simulate, which is why sales and structuring interviews are full of "a client calls you and wants . . . ".

Why the market works this way

An exchange order book gives continuous two-way prices in standardised instruments. A request for quote exists for everything else: an interest rate swap to a specific date, a block of stock too large for the book, a bespoke basket, a five-year note with a payoff nobody else will ever want. There is no resting liquidity for those, so the client *creates* liquidity by asking dealers to price on demand.

INTUITION

Buying a standard car is an order-book trade: the model exists, the price is published, you take it or leave it. Having a kitchen fitted is an RFQ. You describe what you want, three firms come and measure, each comes back with a number, and you pick one. Notice what follows. The firms need to ask you questions before they can quote at all. Their numbers are not comparable unless you specified the job identically to each. Whoever quotes cheapest wins, which means the winner is disproportionately the one who underestimated the job. And the mere act of asking three firms tells all three that a kitchen is about to be bought.

The loop, end to end

1. **Request.** The client calls, messages, or fires an electronic RFQ into a multi-dealer platform, usually to several dealers at once.
2. **Intake.** Sales pins down the exact trade. This is the step that separates a good sales person from a switchboard, and the checklist is below.
3. **Pricing.** The trader prices it: fair value, hedging cost, risk, capital, margin. How the margin itself is set belongs to client pricing, and how a quote is built out of its components to market making.
4. **Quote.** Sales shows the price with its terms: indicative or firm, the size it is good for, and how long it lives.
5. **Outcome.** "Done", and you have a trade, or "off" and you lost. Either way you ask where it traded, because that is how you calibrate.
6. **Post-trade.** Booking, hedging, confirmation, settlement. The risk exists from the moment the client says done, not from the moment it is booked, and the gap between those two moments is where operational accidents happen.

The words, used precisely

Term	What it means
Indicative	A guide price, not tradeable. Also "for information".
Firm / live	Tradeable now, for a stated size and a stated time.
Level check	The client wants a number, is not asking to trade.
On the wire	The client is live and deciding right now: do not walk away.
Your risk	The quote has expired, the market has moved, ask again.
Choice price	Bid equals offer, no spread, usually a favour or a fill-or-kill.
In comp	In competition, against a stated or unstated number of dealers.
Done / off	Traded, or lost.
Axe	A position the desk wants to do, so the price improves on that side.

Two of these carry real money. **Indicative versus firm** is the difference between a conversation and an obligation, and a sales person who blurs it will eventually have a trader holding a position they never agreed to. And **axe** is the cheapest source of edge in the business, since a price that helps the desk out of existing risk can be better for the client and better for the bank at the same time, which is the subject of axes and trade ideas.

Intake: what you must know before you ask the trader

Going to a trader with half a trade wastes the one thing an RFQ has no spare of, which is time. The checklist:

- **The instrument, exactly**: underlying or reference, strike or level, start and end dates, day count and calendar, currency, cash or physical settlement.
- **Size**, in the client's units, and whether it is all-or-nothing.
- **Direction**, if the client will say. Many will not, and asking bluntly can be the wrong move, so you ask for a two-way price instead.
- **Competition**: are you in comp, and against how many. It changes how wide you can be.
- **Urgency and validity**: do they need it now, and how long must the price live.
- **Documentation and credit**: is there a master agreement and a CSA, is there limit headroom, and does the trade need approval. Never quote a client you cannot face, and see counterparty risk.
- **The objective.** What are they actually trying to achieve? This is the question junior sales forget, and it is the one that turns an RFQ into a relationship, because the trade they asked

for is often not the best trade for the problem they have.

Competition, win rates and the winner's curse

In a multi-dealer RFQ you win only when you are the most aggressive, which means winning is itself evidence that your price was the outlier. So the RFQ business has adverse selection built into its structure, and the right way to think about a quote is as an expected value across all the requests you see, not as a margin on the ones you win.

WORKED EXAMPLE

Tight is not the same as profitable. Suppose that on a given flow you win about 20% of requests when you quote a margin of 0.30, and about 50% when you cut to 0.10.

- Wide: $0.30 \times 0.20 = 0.06$ of expected margin per request.
- Tight: $0.10 \times 0.50 = 0.05$ per request.

The wider quote earns more per request despite winning far less often. Now add the winner's curse. If, when you win at the tight price, the market on average moves 0.04 against you before you can hedge, because you were the dealer who was furthest from where the market really was, then the tight quote is worth $(0.10 - 0.04) \times 0.50 = 0.03$ per request, half the wide one.

This is the arithmetic behind an unintuitive rule: **a high win rate on a competitive flow is a warning sign**. It usually means your prices are the cheapest liquidity in the market, and the reason will find you eventually. The counterweight is that flow has value beyond its margin, in market colour and in the offsetting risk it brings, so the target win rate is a commercial decision rather than a maximisation, see flow vs real money clients.

A firm quote is an option you gave away

This is the piece of rigour that most candidates miss. When you show a firm price good for thirty seconds, you have handed the client a free option: they trade if the market moves in their favour and walk away if it does not. The value of that option grows with the volatility of the underlying and with the square root of the time you leave it live.

Three consequences a desk actually acts on. Quote validity is stated, and short: "good for the next thirty seconds" is not rudeness, it is the option expiring. Around a data release or an auction, quotes are withheld or widened, because volatility has spiked and the giveaway option is suddenly expensive. And on electronic platforms this is exactly what the contested practice of "last look" is trying to address, giving the dealer a brief window to reject a trade against a stale price, which shifts the option back towards the dealer and is why it is regulated and disclosed.

KEY IDEA

Information leakage runs both ways. An RFQ tells every dealer asked that the client is about to trade, and in size that information alone can move the market before the trade is done, which is why sophisticated clients limit how many dealers they ask and split large trades. On the dealer side, the flow of RFQs you see, including the ones you lose, is genuine information about who wants what, and it feeds the desk's view, see market color.

REAL INTERVIEW QUESTION (Société Générale · Rates)

A client asks you for a price for 100 calls, you quote them a vol of 20. Ten minutes later they ask you for a price for 200 calls, what vol do you quote? More than 20 or less than 20?

WHAT THEY TEST / ANSWER

Say **higher than 20** and then justify it on three separate grounds, because the question is testing whether you have one reason or several. First, **size**: 200 is twice the risk, the hedge costs more and has more market impact, and the position is a bigger fraction of your limits, so the price must move against the client as size grows. Second, **information**: the same client returning ten minutes later for more of the same side tells you that your first price was attractive, so your volatility mark is probably behind the market, and repeated one-way demand from an informed client is precisely the flow that loses money at unchanged prices. Third, **inventory**: if you traded the first 100 you are now short volatility, so you skew the mark up to be paid more for adding to a position you did not choose, exactly as in market making.

Then handle the sharpening they usually add. If you did *not* trade the first 100, the size and information arguments still stand but the inventory one does not, so you move less. If the client is in competition against several dealers, the market caps how far you can go, so you may quote higher and lose, which is an acceptable outcome and better than winning at a price that is wrong. And if they had asked for 200 up front rather than in two clips, you would have quoted the 200 price straight away, so mention that splitting a request in two is a known way of working an order and you should recognise it.

The most-asked worked example of this intake discipline is the equity financing simulation, where a client wants a total return swap on a portfolio and the questions you ask determine the price you can quote. It is not repeated here: the product and its spread build-up belong to equity financing, and the five-move case method that the simulation is usually used to examine belongs to case studies. What this section owns is the point the example illustrates, which the checklist above already states: on a bespoke request the intake *is* the deliverable, because a trader cannot price what sales has not pinned down, and every question left unasked becomes either a wider quote or a risk nobody priced.

REAL INTERVIEW QUESTION (BNP · Structured Products)

If a client calls you, nobody is on the desk, and they absolutely want to trade, what do you do?

WHAT THEY TEST / ANSWER

A judgement question, and there is a right answer: you do not invent a price and you do not leave the client stranded. Work in that order. First, **stay with the client and gather the trade**: instrument, size, direction if they will give it, timing, and how long they can wait, so that whoever prices it can do so in one pass. Second, **be honest about what you can and cannot show**: you can give levels that are genuinely observable, or an indicative based on a published screen, clearly labelled indicative, and you cannot show a firm price on risk you have no authority to commit. Third, **escalate immediately and in parallel**, to the trader's mobile, to the cover trader on another desk, to the regional desk if a different time zone is open, or to your manager. Say explicitly that quoting outside your mandate is not a commercial risk but a control breach, and that the honest version, "I can have a price for you in five minutes, let me get the trader", keeps far more business long term than a made-up number that has to be repriced.

Fourth, **if it is genuinely time-critical**, work the alternatives with the client: a smaller size within a pre-approved framework, a listed equivalent they can execute themselves now, or a firm price at the next liquidity point. Then close with the follow-up, because interviewers listen for it: log the request and the outcome, tell the desk what happened when they are back, and if this gap recurs, raise it as a coverage problem to be fixed with a documented backup rather than absorbed by improvisation.

INTERVIEW TRAP

Four traps. First, **blurring indicative and firm**, which is how a desk ends up owning risk nobody agreed to take. Second, **going to the trader with an incomplete trade**, which burns the only scarce resource in an RFQ. Third, **quoting the same width regardless of size and competition**: both change the answer, and a price that ignores them is either uncompetitive or a gift. Fourth, **treating a lost RFQ as nothing happened**: the level where it traded is free information about the market and about your own pricing, and a desk that never asks where it went never finds out that it is systematically wide, or systematically the cheapest.

VISUAL / CODE TO BUILD: Interactive

An RFQ simulator played from the sales seat. Requests arrive with partial information and the reader must choose which intake questions to ask, each costing simulated seconds, before sending the trade to a modelled trader who returns a fair value. The reader then sets a margin and a quote validity, and the engine resolves the outcome against simulated competitors and a live price path: win or lose, and if you win, the realised P&L after the market has moved during your validity window. The scoreboard tracks win rate, average margin quoted, average margin realised and P&L per request, so the reader discovers the two central results experimentally, that the profit-maximising win rate is well below 100%, and that a longer validity window is a real cost that grows with volatility. A stress toggle drops a data release into the middle of a live quote.

Where this goes next

Who the client is and how the relationship is run are client coverage and relationship management, and the difference between flow and real money counterparties is flow vs real money clients. How the margin inside the quote is decided is client pricing, how the trader builds the price underneath it is market making, and what happens to the risk once the client says done is inventory management. The regulatory frame around quoting and execution, including record-keeping obligations, is best execution and MiFID.

23.4 Client pricing

sales · structuring · trading | core | cross-asset

A client price is not a model output with something added. It is a decision, taken in seconds, about how much risk the desk wants, what it costs to carry, who is asking and what they will do next. This section takes it apart, because a salesperson who cannot name the components cannot defend the number, and defending the number is most of the job.

The anatomy of a price

> **KEY IDEA**
>
> Every client price decomposes the same way:
>
> $$\begin{aligned}\text{client price} &= \text{mid} + \text{hedging cost} + \text{risk charge}\\ &\qquad + \text{credit and funding} + \text{capital} + \text{sales margin}.\end{aligned}$$
>
> **Mid** is where the risk is genuinely valued. **Hedging cost** is what it takes to get flat, which is not the mid: it is the spread the desk must cross, and it grows with size and shrinks with liquidity. **Risk charge** covers what cannot be hedged and is released over time as a reserve. **Credit and funding** are the adjustments of CVA, DVA and FVA for an uncollateralised counterparty. **Capital** is the return required on what the trade consumes for its life. **Sales margin** is the last item and the only one that is a commercial choice rather than a cost.

Two people set those pieces. The **trader** owns everything up to and including capital: that is the risk price, the level at which the desk is indifferent. The **salesperson** adds the margin, and the internal negotiation over how much is a daily feature of the seat. Presenting a client price as though it were the model's answer is the fastest way to lose an argument with a client who knows better.

Ask before you quote

REAL INTERVIEW QUESTION (Natixis · EQD)

I offer you an investment: if Total's share price rises I give you a million, and if it falls you lose your investment. Rates are 2% and the maturity is one year. Can you give a range of prices you would put on such an investment?

WHAT THEY TEST / ANSWER

Recognise the payoff, notice what is missing, ask for it, and only then compute.

The payoff is a **digital call** struck at today's spot, paying 1,000,000 if the stock is higher in a year and nothing otherwise. Its value is $e^{-rT}\,N(d_2)$ per unit of payout, with the strike equal to spot so the log term vanishes and

$$d_2 = \frac{\left(r - \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}.$$

What is missing is the volatility, and saying so is the answer. Without it the question is not computable, and a candidate who produces a number has assumed one silently.

Then compute across plausible volatilities, which shows the sensitivity rather than pretending to a precision you do not have. At 20% volatility, d_2 is exactly zero, because $r - \sigma^2/2 = 0$ when $\sigma = \sqrt{2r} = 20\%$, so the probability is exactly one half and the value is $e^{-0.02} \times 0.5 \times 1{,}000{,}000 = 490{,}000$. At 25% it falls to 473,000 and at 15% it rises to 513,000. Note the elegance worth pointing out: for this particular pairing of rate and volatility the answer is exactly half the discounted payout.

Then say the two things that turn a valuation into a price. First, the probability of a rise under the risk-neutral measure is a bit *below* one half at realistic equity volatilities, because the drift is $r - \sigma^2/2$ in log terms, and that surprises people who expect fifty-fifty. Second, and this is the point of the question on a sales desk, **this is the fair value, not the price**. If you are buying it you would pay less: the seller has to hedge a digital, which means a call spread and a gap risk they cannot eliminate, and that cost belongs to you. Quoting the model value as the price is exactly the mistake this section is about.

Where the margin actually sits

REAL INTERVIEW QUESTION (Société Générale · EQD)

How do you price a digital? How does the price depend on the call spread's gap, and how does the trader's margin depend on it?

WHAT THEY TEST / ANSWER

Three linked answers, and the third is the one being tested.

Pricing. A digital paying 1 above K is worth $e^{-rT}N(d_2)$ in Black-Scholes, but that is not how it is priced on a desk, because the model number assumes it can be hedged perfectly and it cannot. It is priced as the **call spread that replicates it**: $1/\varepsilon$ spreads struck $(K-\varepsilon, K)$, which dominates the digital and is therefore the safe hedge for a seller.

Dependence on the gap. The tighter the gap, the closer the replicating spread's value falls towards the theoretical digital value, and the further above it the spread sits when the gap is wide. So the quoted price is a decreasing function of tightness, and at any usable gap it is strictly above the model number.

Where the margin is. The margin *is* the gap, or rather the difference between the spread the desk trades and the digital it sold. A wide gap means a large cushion and a large margin; a tight gap means a competitive price and very little protection. So the choice of ε is simultaneously a hedging decision and a pricing decision, and that is the insight: on this product the risk management parameter and the profit margin are the same number.

Then give the constraints, so it does not sound like a free choice. The gap is bounded below by the **listed strike grid and the liquidity** of those strikes, by the **size** relative to what trades, and by the gamma the desk is willing to carry near the barrier at expiry, which grows like $1/\varepsilon$. And it is bounded above by competition, since another house will quote tighter. Close on the disclosure point, which is what a good sales interviewer wants: the client should be told what gap has been assumed, because a digital quoted without it is not a quoted price. The replication itself is in common structuring mistakes.

Transparent, embedded, and the difference that matters

The same margin can be charged in three ways, and they are economically similar and regulatorily different.

- **Explicit commission**: a stated fee per trade. Fully visible, easily compared, and therefore competed down.
- **Spread**: embedded in the level shown. Visible to anyone who can see a mid, invisible to anyone who cannot.
- **Structural margin**: built into a payoff's parameters, a coupon, a participation, a barrier. Invisible without the ability to reprice the product, which most end clients do not have.

That ordering is exactly the ordering of margins in the market, and it is not a coincidence: what can be compared gets competed away. It is also why disclosure rules target the third category specifically, in retail versus institutional structuring.

REAL INTERVIEW QUESTION (Amundi · Delta One)

What are the average fees on an ETF?

WHAT THEY TEST / ANSWER

Give the ordering and the reason, then correct the question, because the fee is not the cost.

Ordering. Broad developed-market equity index trackers are the cheapest thing on the shelf, competing at single-digit basis points a year. Broad fixed income sits a little higher. Emerging markets, small caps, sector and thematic products are far higher, running into tens of basis points, and actively managed or leveraged products higher again. The driver is not the cost of running the fund, it is **how substitutable the product is**: five providers track the same large-cap index, and nobody else tracks a niche theme.

The correction. The management fee is the advertised number and the wrong one. What a holder actually loses to the product is the **tracking difference** against the index, which nets the fee, the taxes, the rebalancing costs and the securities lending revenue *received*, and a fund with a higher fee and a good lending programme can beat a cheaper one. On top of that a buyer pays the **bid-offer** on the exchange and any premium or discount to net asset value at the moment they trade, which for a short-horizon holder dominates the annual fee entirely.

Close with the general lesson, since this is a pricing section rather than a product one: the ETF is the clearest case in the market of **transparent pricing driving margins to the floor**, and the contrast with a structured note carrying a percentage margin on the same underlying exposure is a statement about comparability rather than about complexity. The instrument itself is in ETFs and index products.

Defending the number

Clients push back on price. The professional response has three moves and none of them is an apology.

Decompose. “The level is wide because the size is three times what trades in a day, and I have to cross the spread on the hedge” is a defensible sentence. “That is the price” is not, and it invites the client to go elsewhere and find out.

Offer the alternative. Most price objections are really structure objections: smaller size, worked over a day rather than in one print, a different strike where the market is liquid, a different maturity where the desk has an axe. Moving the trade to where the desk is naturally positioned is worth more than any concession on margin, and it is in axes and trade ideas.

Be willing to lose it. A trade lost at the right price is a good outcome, and a trade won at the wrong one is a loss that has not been booked yet. Say this explicitly in an interview, because the instinct to win every ticket is what the question is testing.

INTERVIEW TRAP

Four errors. (1) Quoting the model value as the price, when the price includes hedging cost, reserve, credit and capital. (2) Treating the margin as the only thing you can move, when size, strike, tenor and structure are all negotiable and cheaper to give. (3) Explaining a wide price by “the market”, when the client wants the component that made it wide. (4) Winning a trade by dropping to a level the desk cannot hedge, which is not a sale, it is a transfer.

Where this goes next

The mechanics of being asked are RFQs, who is asking is flow versus real money clients, and what the desk does with the risk afterwards is flow trading. The market-wide version of the spread is the bid-ask spread, the ideas that let you move a trade rather than a margin are axes and trade ideas, and the disclosure obligations that shape all of this are in retail versus institutional structuring.

23.5 Axes and trade ideas

sales · structuring · trading | core | cross-asset

Most of what a salesperson sends a client was not asked for. Two things dominate that outbound traffic, they look similar on a screen, and confusing them is the fastest way to lose a client's trust.

An **axe** is the bank's own interest: the desk has something it wants to get rid of, or something it wants to buy, and it will show a better price than usual to trade in that direction. A **trade idea** is a view about the market that the client might want to express. The first is about the bank's book. The second is supposed to be about the client's.

What an axe is, and why it is legitimate

"We are axed to sell 50 million of that bond" means exactly what it says: the desk owns it, would like not to, and is prepared to be competitive to that end. Axes are circulated internally every morning and passed to clients as a matter of course.

The reason this is a service rather than a conflict is that it is *disclosed* and the interests genuinely align. The desk wants its risk down and will pay for that in price; the client wanted to buy something anyway and gets it cheaper. Both sides are better off, and neither has been misled about why.

> WORKED EXAMPLE
>
> A desk is long 50 million of a bond it wants to be rid of. Normally it would offer at 100.05, five basis points over its mid. Axed, it shows 100.01.
>
> The client saves $0.0004 \times 50{,}000{,}000 = \mathbf{20{,}000}$ against the normal offer. Why would the desk give that away?
>
> Because holding the position is not free. Funding the balance sheet, the capital it attracts and the market risk it carries might cost the desk on the order of three basis points a month, which on 50 million is 15,000. If the desk expects to be stuck with the bond for another three months, the expected cost of holding is
>
> $$3 \times 15{,}000 = \mathbf{45{,}000},$$
>
> against 20,000 to be rid of it today. The axe is not generosity; it is the cheaper of two costs, and it stops being offered the moment the position is flat.

KEY IDEA

An axe is a price concession bought with the desk's own carrying cost. That is why axes are real, why they expire, and why "we were axed there yesterday" is not an argument. It also explains the one thing clients care about: an axe is genuinely better than the standard price, and it is available only in one direction.

What a trade idea is

An idea is a view plus a way to express it. Both halves are required, and most of what gets sent to clients has only the first.

A complete idea has five parts.

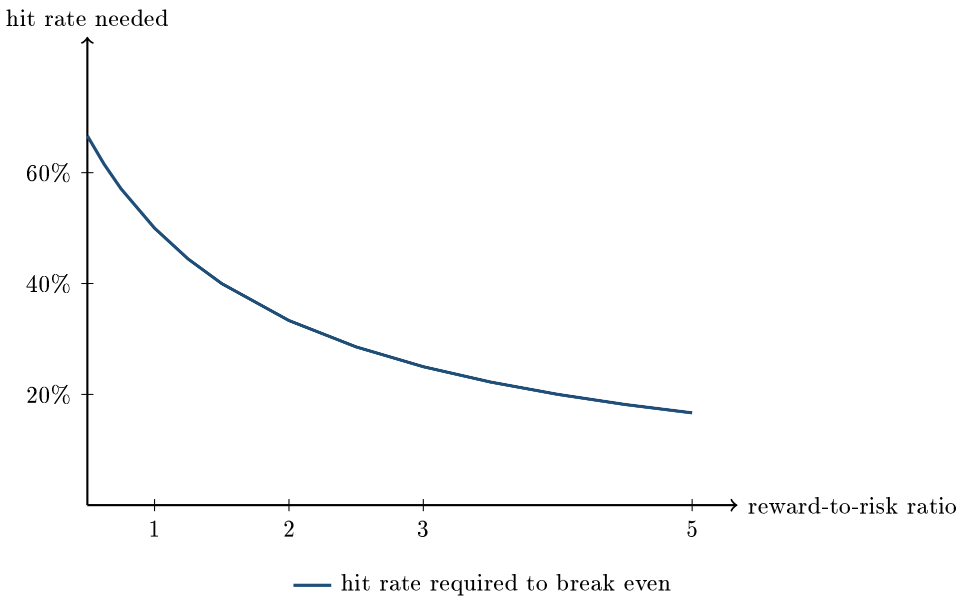
1. **The view**: what you think happens and why, in one sentence.
2. **Why now**: what has changed, because a view that was equally true last month is not an idea, it is an opinion.
3. **The instrument**: how to express it, and specifically why this instrument rather than the obvious one. This is where a derivatives salesperson adds value, since the obvious expression is usually the underlying and it is usually not the best one.
4. **The reward and the risk**: entry, target, and the level or event at which you are wrong.
5. **What invalidates it**: which is the same as the last point but stated as a decision rather than a number. An idea with no way of being wrong is not an idea.

The arithmetic nobody sends with the idea

A reward-to-risk ratio implies a hit rate, and stating it turns a story into a proposition.

If an idea makes R times what it loses, and it works with probability p, then it breaks even when $pR = 1 - p$, that is when

$$p^* = \frac{1}{1+R}.$$



At one-to-one you must be right more than half the time, which is a strong claim. At two-to-one you need only **33%**, and at three-to-one **25%**. That is the entire case for expressing a view with options rather than with the underlying: an option idea is usually a low-hit-rate, high-ratio proposition, and it is a perfectly good one provided everybody understands that most of them will not work.

> **KEY IDEA**
>
> Quote the hit rate the structure needs, not the confidence you have. "This needs to work one time in three" is a checkable statement a client can hold you to. "I am confident about this" is not, and after four losing ideas the difference is the whole conversation.

Where ideas come from, and why most are bad

Good ideas come from three places: the desk's own flow, which is information nobody else has and which must be used in aggregate rather than by naming clients; the research function; and the salesperson noticing that a client's stated problem has a solution nobody has offered them.

The bad ones come from three others, and they are recognisable.

- **An axe in a costume.** The desk wants to move a position and it is dressed as a view. This is the one that destroys trust, because clients work it out, and because the client's alternative was to receive it as an axe and get a better price.
- **The consensus idea.** If four banks sent it this morning it is in the price, and sending it says only that you read the same research as everybody else.
- **The idea with no exit.** A view with no invalidation level, which means the client cannot size it and cannot know when to stop.

> **INTERVIEW TRAP**
>
> Sending an axe as an idea. It is the single most damaging thing in this part of the job, and the reason is arithmetic rather than ethics: the client would have received a *better price* had you called it an axe, so they lose twice, once on the price and once on discovering that you presented the bank's interest as theirs. The regulatory framing is disclosure and conflicts, in best execution, but the commercial reason to avoid it is stronger than the regulatory one.

Reverse enquiry, which is where the good business is

Sometimes the client arrives with the idea: they want a specific payoff on a specific underlying, and they are asking whether you can build it. That is a **reverse enquiry**, and it is the best business a structuring desk gets.

The reasons are worth naming because they explain the whole shape of the structured products market. The client has already decided, so there is no persuasion cost and a much higher conversion rate. The payoff is unusual, so it is harder to price-compare and the margin is wider. And the request tells the desk something about what that class of client currently wants, which becomes the next idea sent to everybody else.

The discipline that goes with it is to say no when the answer should be no. A client asking for a structure that is wrong for them is still asking, and the fact that they asked does not make it suitable.

Interview angle

REAL INTERVIEW QUESTION (CACIB · FX)

Give me a trade idea.

WHAT THEY TEST / ANSWER

The answer is not a trade, it is a structure with a trade in it, and the structure is what is being marked. Give five things in order and it does not much matter whether the interviewer agrees with the view.

The view, in one sentence, with a reason. **Why now**: what has changed, because a view that has been true for six months is not an idea. **The instrument**, and crucially why this one rather than the obvious one: if you are on an FX desk, say why an option or a risk reversal expresses the view better than spot, whether that is because the carry is against you, because you want a defined loss, or because the skew is paying you to take the position in a particular shape. **The numbers**: entry level, target, and the level at which you are wrong. **The invalidation**, stated as a decision rather than a level.

Then add the two things that most candidates leave out and that a desk will notice. **The reward-to-risk ratio and the hit rate it implies**: at two-to-one you need to be right one time in three, and saying so converts a story into a proposition the interviewer can test. And **what the trade costs to hold**, since in FX that is carry, and a view that is right over six months can still lose money if you are paying to wait.

Two things not to do. Do not give a view without an instrument, which is a newspaper column rather than an idea. And do not claim high conviction on something you cannot size, because the follow-up is always "how much would you do?", and an idea you cannot size is one you have not thought about.

REAL INTERVIEW QUESTION (Morgan Stanley · Structured Products)

Can you give me an investment idea you would have at the moment? And why would you choose that idea?

WHAT THEY TEST / ANSWER

Note the second question, because it is the real one. The first asks for an idea and the second asks for your *selection process*, which is a different and harder thing.

Give the idea properly: view, why now, instrument, entry and target, and what would make you wrong. Keep it to a structure you can actually explain, because on a structured products desk you will be asked to decompose it into its components and price the pieces.

Then answer why this one, and answer it as a filter rather than as enthusiasm. The questions worth naming out loud: is the view differentiated, or is it what everybody sent this morning and therefore already in the price? Is the instrument the efficient expression, or am I paying for optionality the view does not need? Is it *suitable* for the client I would send it to, which means matching their mandate, their horizon and what they are allowed to hold, not just their view? And what is the risk-reward, stated as a ratio and a hit rate rather than as conviction?

Then the part specific to this seat. A structured product idea has to work for two parties at once. It must be attractive to the client, and it must leave the issuing desk with risk it can actually hedge, because a payoff that is wonderful for the client and unhedgeable for the desk does not get priced. Saying that shows you understand which side of the trade you would be sitting on.

Close by naming the client type you would send it to and why it fits them specifically. "I would show this to an insurer because they need yield and can hold ten-year paper" is a structuring answer. "I think the market goes up" is not.

INTERVIEW TRAP

Four errors. (1) Blurring an axe and an idea, which costs the client price and costs you trust. (2) Sending everything you have, which spends a client's attention until they stop reading. (3) Quoting conviction instead of a hit rate, which cannot be checked and therefore cannot be credited when you are right. (4) Pitching an idea you cannot size, since the immediate question is always how much.

Where this goes next

What the desk says about the market generally, as opposed to what it wants you to do, is market colour. When the client responds and asks for a price, the process is RFQs and the number is client pricing. The inventory position that generates the axe in the first place is inventory management, and the long-run account of whether any of this was any good is relationship management.

23.6 Market color

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Every client has the same screens you do. What they do not have is the view from inside a dealer, and that view is the product a sales person gives away all day in order to be the first call when the client wants to trade. "Colour" is that view: what is actually happening in the market, who is doing what in aggregate, where things traded rather than where they are quoted, and what held or broke. Interviews test it bluntly and continuously, because it is the one thing you cannot fake in a five-minute conversation.

What colour is, and what it is not

KEY IDEA

Colour is **observation**, not prediction. Research forecasts, strategists opine, and a sales person reports what the desk can see and the client cannot: the size and direction of flow in aggregate, where positioning is crowded, which levels attracted real buyers, how liquidity is behaving, and what did not happen that people expected. A forecast dressed up as colour is worth nothing, because the client already has ten of those in their inbox.

INTUITION

Colour is the difference between a **weather forecast** and **standing outside**. Everyone can read the forecast. You are the person on the street corner who can say: it started raining twenty minutes ago, it is heavier than forecast, the traffic has already stopped on the main road, and the three people who walked past me were all carrying umbrellas, so this was expected by some and not by others. None of that is a prediction, all of it is useful, and it is only available to someone who is out there.

The useful kinds, in rough order of value:

- **Flow colour.** What the desk has been trading, aggregated and anonymised: two-way in the front end, better buyers of downside in the index, real money selling into strength. This is the only kind clients cannot obtain anywhere else.
- **Positioning colour.** Where the market is crowded, inferred from what the desk has absorbed, from futures and options open interest, and from how the market reacts to news (a market that cannot rally on good news is long).
- **Level and technical colour.** Where size traded, which levels held, where the large strikes and barriers sit, which is genuinely predictive of hedging flows.
- **Liquidity colour.** How wide markets are, in what size, and how quickly they refill. In stress this is the most valuable thing you can tell anyone.
- **Structural colour.** Index rebalances, dividend and expiry calendars, issuance pipelines, and regulatory or tax changes with a date attached.

Good colour is specific, timely, and falsifiable. "The market feels heavy" is not colour; "we have absorbed a hundred million of index futures on the offer since the open and it has not lifted" is.

The boundary you must not cross

This is the part that gets people fired, so learn it as precisely as you learn a pricing formula. The rules differ by jurisdiction and by house, and the desk's own policy always wins, but the shape is consistent.

- **Never identify a client**, or make them identifiable. Naming, or describing a counterparty so narrowly that it amounts to naming ("a large Nordic pension fund"), breaches confidentiality even when the trade is done.

- **Never disclose a live order**, its size, or its limit. Telling anyone that a client has a large order to work, before it is done, is the textbook abuse, and the wave of benchmark and FX enforcement in the 2010s is precisely why sharing order information in chat rooms is now treated as a career-ending offence rather than as banter.
- **Aggregate and anonymise.** Colour is legitimate when it describes the market in aggregate and after the fact. The general standard is that the information must not be attributable to a client and must not give the recipient an advantage over the client whose trade generated it.
- **Do not pass inside information.** If you are wall-crossed on a deal you are restricted, and "colour" is not a route around a restricted list.
- **Do not front-run.** Hedging a client's trade is legitimate and expected; trading ahead of it for the desk's benefit is not, and the line between pre-hedging and front-running is drawn by disclosure and by the client's consent.

The detailed regime, information barriers, personal account dealing, communications surveillance and record-keeping, is in trading floor compliance and MiFID. In an interview, volunteering the boundary unprompted is a strong signal: it shows you understand that the value of colour comes precisely from the fact that it is constrained.

The daily rhythm

Colour is a habit before it is a skill, and the habit has a shape.

- **Before the open.** What happened overnight in Asia and in US futures, what moved and why, the day's data calendar and central bank speakers, corporate results, auctions and expiries. This is where the morning comment to clients comes from.
- **The open.** Whether the gap was bought or sold, and in what size, which tells you more about positioning than the level does.
- **Through the day.** What the desk is seeing, what is unusual, and what the desk wants to do, which becomes an axe, see axes and trade ideas.
- **The close and the auction.** Where size printed, and any index or rebalance flow. Marks are struck here, so this is where a lot of real risk changes hands.

Underneath all of it sits a discipline that interviewers test directly: **know your levels**. Three or four numbers per asset class you follow, from memory, with the direction of the last move and the reason for it. Not because a level is intrinsically interesting, but because a person who does not know where things are cannot possibly have a view about them.

WORKED EXAMPLE

A quick correlation estimate, from signs alone. A client asks how correlated rates and equities have been lately, and you have no terminal in front of you. Count instead of guessing. Suppose that over the last twelve months, yields and the equity index moved in the *same* direction in eight months and in opposite directions in four.

The statistic you have counted is $P(\text{same}) - P(\text{opposite}) = \frac{8}{12} - \frac{4}{12} = \frac{1}{3}$. For two jointly normal variables with zero mean this equals Kendall's tau, and tau maps to the linear correlation by $\tau = \frac{2}{\pi} \arcsin \rho$, so

$$\rho \;=\; \sin\!\left(\frac{\pi\tau}{2}\right) \;=\; \sin\!\left(\frac{\pi}{6}\right) \;=\; 0.5.$$

So a two-thirds sign-agreement rate corresponds to a correlation of about 0.5. Useful reference points on the same map: 60% agreement is about 0.31, 75% is about 0.71, and 80% is about 0.81.

Say the caveats out loud, because they are the difference between a trick and a tool. Twelve observations is a small sample, so the estimate is noisy. The mapping assumes joint normality, which fat tails violate. It also assumes neither series drifted materially over the window, since a trending market raises sign agreement on its own and the count would then read that trend as correlation. And correlation is regime-dependent here in a way that matters: a positive rates-equity correlation is the signature of an inflation-driven market, where higher yields hurt equities, while the negative correlation of the previous two decades was the signature of a growth-driven market where bonds hedged equities. The relationship between these measures is developed in independence and correlation.

REAL INTERVIEW QUESTION (Barclays · Credit)

What do you think of the markets at the moment?

WHAT THEY TEST / ANSWER

An open question with a hidden structure, and the mistake is to ramble. Answer in four beats and keep it to ninety seconds. **One view**, stated as a sentence, not a hedge: say what you think and about what. **Three observations with numbers** that support it, drawn from different places so it does not sound like one headline: a level and its recent move, a spread or a curve shape, and a flow or positioning observation. **What would change your mind**, which is the beat almost everyone omits and the one that signals you hold views as probabilities rather than as identity. **The trade**, because on a desk a view that cannot be expressed is not a view: name the instrument, the direction, and roughly what it costs to be wrong. Since this was asked on a credit desk, pitch it in credit terms, spreads against their range, dispersion between ratings buckets, primary issuance and how it is being absorbed, and where the basis between cash and CDS sits. Two practical rules: always state your as-of date, because "as of yesterday's close" is discipline rather than ignorance, and never claim a level you are not sure of, since being caught inventing a number ends the interview more surely than not knowing it. The interview-specific drilling of these questions is in market questions.

REAL INTERVIEW QUESTION (Barclays · EQD)

Where are the CAC 40, the Euro Stoxx 50 and 3-month Euribor today?

WHAT THEY TEST / ANSWER

A pure preparation check, and there is no clever way through it: either you have the numbers or you do not, and everybody on that desk knows them. So the real advice is about how to carry them. Quote the three levels with your as-of date, then add one sentence of movement and one of cause for each, since a level with no context is a recital while a level with a reason is a view: an index level with the move on the week and what drove it, and the three-month rate with where it sits relative to the policy rate and what the forward market implies next. Because this book cannot supply live numbers, build the habit instead: keep a levels sheet of three or four numbers per asset class you claim to follow, update it daily, and rehearse it out loud. Roughly the right set for a European equity-derivatives interview is the two index levels, three-month Euribor and the ECB deposit rate, the 10-year Bund and OAT yields, the index dividend yield and the front implied volatility (V2X), EUR/USD, and Brent. If you are genuinely caught out on one, say so in four words and give the direction and the recent range rather than bluffing a figure, then move immediately to something you do know precisely. The recovery is judged as heavily as the answer.

REAL INTERVIEW QUESTION (Bank of America · desk not recorded)

What happened in the markets this weekend?

WHAT THEY TEST / ANSWER

The question is checking whether you actually follow markets or only read about them on weekdays, and there is a hidden trap in it: most markets are closed at the weekend. Say that first, cleanly, because it shows you understand the mechanics rather than the vocabulary. Cash equity and bond markets do not trade, so "what happened" means one of three things, and you should cover all three. **News that will be priced at the open**: politics and elections, central bank commentary and weekend interviews, geopolitical events, weekend policy announcements, and results from Asian data published Sunday night. **Markets that do trade**: crypto, which is continuous and is therefore the only live read on weekend risk appetite, and futures once Asia opens Sunday evening in European time. **Scheduled weekend business**: OPEC meetings, European summits, and the fact that banking-sector interventions are deliberately timed for a weekend so that a resolution can be announced before the open. Then answer the actual question with what you followed, with your as-of date, and finish with the desk-relevant part: what it implies for the open, whether the gap is likely to be bought or sold, and which of your own positions would care. That last step is what makes it colour rather than a news summary.

REAL INTERVIEW QUESTION (Bank of America · Exo)

What is the correlation of rates and indices at the moment? Could you estimate for me the correlation between rates and indices over the past year?

WHAT THEY TEST / ANSWER

Two questions: the current level, which needs a number and an as-of date, and an estimate, which is testing method rather than memory. On the **sign and regime**, which is the part you must get right: the equity-rates correlation is not a constant of nature but a signature of what is driving the market. When inflation is the driver, higher yields hurt equity valuations and the correlation between yields and equity prices is *positive*, which was the regime of the post-2021 period. When growth is the driver, weak data pushes yields down and equities down together, so bonds hedge equities and the correlation is *negative*, which was the regime of the two prior decades. Say which regime you think you are in and why, and be careful to state whether you mean correlation with yields or with bond prices, since the sign flips. On the **estimate**, show a method that works without a terminal: count sign agreement over the last twelve monthly moves, then map it, since a sign-agreement rate of two thirds corresponds to a Kendall tau of one third and, under joint normality, $\rho = \sin(\pi\tau/2) = 0.5$. Give the reference points (60% agreement is about 0.3, 75% about 0.7) and then the caveats: twelve points is noisy, normality is an assumption, the answer depends on frequency since daily and monthly correlations differ materially, and the estimate is unstable across regimes. Close with why an exotics desk cares: this correlation is a direct input to hybrid and quanto payoffs and to the equity-rates term of an autocall's hedge, so a sign error is a hedging error rather than an academic one.

INTERVIEW TRAP

Four traps. First, **dressing a forecast as colour**: clients have plenty of views and want observation. Second, **crossing the confidentiality line**, whether by naming a client, describing them identifiably, or mentioning a live order, which is the fastest way off a trading floor. Third, **quoting levels you are not sure of**, since one invented number destroys the credibility of everything else you say. Fourth, **colour with no as-of date**, which is unusable, and which in an interview reads as though you are repeating something you read rather than something you follow.

VISUAL / CODE TO BUILD: Interactive

A colour dashboard and a drill in one. The dashboard shows a compact levels sheet across asset classes with each number's move on the day, week and month and its position in its one-year range, so the reader learns the shape of the numbers rather than only the values. The drill hides the values and asks the reader to type them, scoring accuracy and tracking improvement over days, which is exactly how the levels question is prepared. A third panel is a colour generator: it takes a simulated day of desk flow and produces two write-ups side by side, one that breaches confidentiality and one that does not, with the offending phrases highlighted, so the boundary is learned by contrast. A fourth panel plots rolling equity-rates correlation against realised inflation and lets the reader flip between daily and monthly frequency, making the regime dependence visible.

Where this goes next

Who receives the colour and how the relationship is built are client coverage and relationship management. Colour that turns into a tradeable suggestion becomes axes and trade ideas, and the request it generates is handled in RFQs. The rules that bound all of it are trading floor compliance and MiFID, the interview drill for market questions is market questions, and the statistics behind the estimate above are in independence and correlation.

23.7 Relationship Management

sales · structuring · trading | core | cross-asset

A bank's franchise is the sum of its client relationships, and a relationship is not a feeling. It is an asset with a present value, and almost every question about how to behave with a client resolves once you treat it as one.

Client coverage is about who covers whom and how a desk is organised around its accounts. This section is about the thing the coverage model exists to protect, and specifically about the one decision that separates a good salesperson from a merely competent one: knowing when *not* to maximise the current trade.

Every trade is a repeated game

> **KEY IDEA**
>
> A salesperson is paid on flow that arrives one trade at a time, but the relationship producing that flow is a **multi-period asset**. The temptation is therefore structural, not moral: the gain from widening a price is immediate and attributable, while the cost is diffuse, delayed and lands on someone else's P&L. Every rule of thumb in this section is a way of correcting for that asymmetry.

The correction is arithmetic, and it is worth carrying because it settles the argument faster than any appeal to principle.

WORKED EXAMPLE

What a squeeze actually costs. A client trades \$600 million of notional a year at an average margin of 3 basis points, so the relationship produces

$$600{,}000{,}000 \times 0.0003 \;=\; \$180{,}000 \text{ a year.}$$

Assume it lasts five years and discount at 5%: the annuity factor is 4.3295, so the relationship is worth

$$180{,}000 \times 4.3295 \;=\; \$\mathbf{779{,}306}.$$

Now you widen a single \$50 million trade by an extra 2 basis points. That earns

$$50{,}000{,}000 \times 0.0002 \;=\; \$10{,}000, \text{ once.}$$

The break-even is the number to remember. Squeezing is worth it only if it raises the probability of losing the account by less than

$$\frac{10{,}000}{779{,}306} \;=\; \mathbf{1.28}\%.$$

If it raises that probability by a mere 5%, the expected cost is \$38,965, nearly four times the gain. And a client who notices a bad price does not usually leave, which is worse: they downgrade you from first call to third, and you lose the trades you never see rather than the ones you priced badly.

The asymmetry is the point. The gain is certain, small and yours; the loss is probabilistic, large and the franchise's.

INTUITION

This is also why a desk will show a genuinely good price on a trade it does not want. Quoting wide to discourage a trade and quoting wide to profit from one look identical to the client, and they will assume the second. If you do not want the risk, say so and offer an alternative, because "I would rather not" costs nothing and a bad price costs a place in the rotation.

Trust is built on the bad days

Nothing about a relationship is tested when the trade works. What is remembered is what happened when it did not, and the behaviours that build credibility are specific and mostly uncomfortable.

- **Call before they call you.** A client who learns from their own screens that a position has moved against them has learned something else as well.
- **Show the mark you actually have.** Marking a client's losing position optimistically buys a week and costs the relationship, because the truth arrives anyway and arrives with your name on the delay.
- **Stand behind a price in size.** A quote that is only good for a fraction of what was asked is not a quote, and doing this once is remembered longer than a year of good levels.

- **Say when you do not know.** The fastest way to lose a sophisticated client is to answer a technical question you did not understand. They will check.
- **Own the error.** Mispriced, misbooked, missed a call: report it yourself, immediately, with what you propose to do about it.

Not every relationship is worth the same

Segmentation by revenue is the obvious approach and it is incomplete. Three dimensions matter and desks weight them differently.

Revenue, the direct measure, adjusted for the capital and balance sheet the relationship consumes. A large flow account that ties up balance sheet in financing may be worth less than a smaller one that does not.

Information, which is real and rarely stated. A large real-money account's flow tells you something about positioning that you cannot see elsewhere, and market colour is only as good as the flow behind it. That value accrues to the desk even on trades that make no margin.

Franchise, the value of being on a client's panel at all. Being one of three banks a large institution deals with is worth more than the current revenue, because it is an option on future business, on being shown the difficult trades, and on being called first when something large needs doing.

The practical consequence is that ranking accounts by last year's revenue will systematically underrate the account that gives you information and the account that has just started, and desks that reallocate coverage purely on that ranking tend to discover the error a year later.

REAL INTERVIEW QUESTION (Goldman Sachs · desk not recorded)

Wealth Management is client-centric: how do you continually strengthen relationships with your existing client base?

WHAT THEY TEST / ANSWER

Resist the temptation to answer with adjectives. The question wants a process, and the strongest answer is built on the distinction between *contact* and *value*.

Start by naming the failure mode. Most relationship management degrades into contact for its own sake: calls with nothing in them, mailings nobody reads, a lunch a year. Contact without content trains the client to ignore you, so the first principle is that every interaction should leave them with something they did not have.

Then give the four things that actually do it.

Know the portfolio better than they expect. Being able to say what a move in rates does to their specific book, unprompted, is the single most effective thing, because it demonstrates that the work happened before the call rather than during it.

Bring ideas that fit their constraints, not your axe. An idea a client cannot execute, because of their mandate, their accounting treatment or their governance, proves you have not understood them. Ideas that respect the constraints are rarer and land much harder, and axes and trade ideas is about the difference.

Be present when it goes wrong. Recommendations lose money. The relationship survives if you call first, explain what changed and say what you would do now; it does not survive going quiet.

Broaden the footprint deliberately. Relationships held by one person on each side are fragile. Introducing the research team, a structurer, a specialist for a product they are looking at, makes the relationship institutional rather than personal, which is what protects it when either person moves.

Close with the measurement, because it shows you would manage this rather than merely perform it. Track share of wallet and the trend in it, not just revenue; revenue can rise while your share falls, and the falling share is the number that predicts the loss. And say plainly that the goal is to be the first call, since first call is where the good business is and everything else in the answer is instrumental to it.

REAL INTERVIEW QUESTION (Credit Suisse · FX)

What qualities are needed to be a good salesperson?

WHAT THEY TEST / ANSWER

The trap is a list of adjectives, which every candidate gives. Answer with functions, and attach each one to something the job actually requires.

Technical credibility, first and non-negotiable. You cannot sell a structure you cannot explain, and a client will find the edge of your understanding within two questions. This is the quality candidates most often leave out of the list because it does not sound like a sales quality, and naming it first distinguishes you.

Judgement about the client's problem rather than your product. The skill is diagnosing what the client is actually trying to do, which is often not what they asked for, and sometimes the right answer is that they should not do the trade. A salesperson who has occasionally told a client not to trade is believed on the occasions they say the opposite.

The ability to be the bridge. A sales role sits between a client who wants certainty and a trader who wants risk they can hedge, and both sides are under pressure. Translating faithfully in both directions, including telling a client something they do not want to hear, is most of the daily work.

Discipline about the long game. Concretely: not maximising the current trade. The arithmetic is that a couple of basis points on one ticket is worth a fraction of a percent of a relationship's present value, so squeezing is only rational if it costs you essentially no probability of losing the account, which it never does.

Resilience, said precisely. Not "I handle pressure well", but the specific version: you will recommend trades that lose money, and the job is to make the next call anyway, with the same honesty.

Then, if the desk is named, tailor it. On an FX desk the flow is fast and the technical content per trade is lower than in structuring, so speed, accuracy under time pressure and the quality of your market colour matter more than depth on any one product. Saying that shows you know the desks differ, which is the half-mark most candidates leave on the table.

INTERVIEW TRAP

1. **Optimising the current ticket.** The arithmetic above is the answer, and it is worth being able to produce it rather than appealing to principle.
2. **Confusing contact with value.** Frequency of calls is not a relationship metric. Share of wallet and being the first call are.
3. **Hiding a bad mark or a mistake.** It buys days and costs years, and it is the one failure that is a firing offence rather than a coaching point.
4. **Pretending there is no conflict.** You are on the other side of the trade, and the client knows it. Acknowledging the conflict is what makes the rest of what you say credible; pretending to be a pure adviser makes all of it suspect.
5. **Letting the relationship rest on one person.** If it lives entirely in one pair of contacts, it leaves when either of them does.
6. **Ranking accounts on last year's revenue alone.** It underrates information value and underrates any account that has just started, which is exactly the account that most repays coverage.

VISUAL / CODE TO BUILD: Interactive

A relationship-value calculator with a squeeze decision built into it. Inputs are annual notional, average margin in basis points, expected relationship length and a discount rate; the output is the present value of the relationship, drawn as a bar.

Underneath, a second control sets the size and extra margin of a single trade and a slider sets how much that squeeze raises the annual probability of losing the account. The panel prints the one-off gain against the expected loss and colours the decision green or red, with the break-even probability shown explicitly. The intended moment is discovering how small the break-even is: at the default inputs it is 1.3%, which is far below any honest estimate of the effect of showing a client a bad price.

A second panel plots share of wallet and revenue as two lines over time from a simulated account, arranged so that revenue rises while share falls. It makes visible the failure the section warns about: the account looks healthy on the number the desk reports and is being lost on the number it does not.

Chapter 24

Regulation & Banking Environment

24.1 Basel III and IV

trading · structuring · sales | core | cross-asset

Regulation looks like someone else's problem until you understand that it prices your trades. A bank's willingness to do business is not set by whether a transaction is profitable but by whether it is profitable *per unit of capital it consumes*, and the Basel framework is what decides how much capital that is. This is why a sales or trading candidate is expected to know it: not the paragraph numbers, but the fact that two economically similar trades can carry completely different prices because one is cheap in capital and the other is not.

Who writes it and what the rounds did

The **Basel Committee on Banking Supervision**, hosted at the Bank for International Settlements, is not a legislature. It publishes standards that member jurisdictions then implement in their own law, which is why timing and detail diverge between the EU, the UK, the US and Asia, and why "Basel III is in force" is a statement that needs a country attached to it.

- **Basel I**, 1988. A single crude idea: hold capital equal to 8% of assets, with assets weighted into a handful of risk buckets. Simple and easy to arbitrage, because anything inside a bucket had the same weight regardless of its actual risk.
- **Basel II**, 2004. Introduced the **three pillars** that still structure everything: Pillar 1, the minimum capital calculation, now including operational risk; Pillar 2, supervisory review of the risks Pillar 1 misses; Pillar 3, disclosure so the market can discipline what supervisors do not. It also let banks use internal models to compute their own risk weights, which is the decision the next two rounds spent their time walking back.
- **Basel III**, from 2010. The post-crisis response, and the subject of the next subsection.
- **"Basel IV"**, the 2017 finalisation. The Committee insists this is still Basel III; the industry calls it Basel IV because the changes to risk weighting are large enough to feel like a new regime.

What Basel III actually diagnosed

The finding after 2008 was not that banks held no capital. Many met their ratios on the way into the crisis. The findings were that the capital was of the wrong *kind*, that risk weights understated the risk, and that no rule addressed liquidity at all. Basel III answers each.

- **Quality of capital.** Instruments that had counted as capital turned out not to absorb losses in a going concern, so the framework was rebuilt around common equity.

- **Quantity, through buffers.** Minimum ratios were raised and topped with a conservation buffer, a countercyclical buffer that supervisors can switch on in booms, and a surcharge for globally systemic banks. Breaching a buffer does not close the bank, it restricts dividends and bonuses, which is a deliberately calibrated pressure.
- **A leverage ratio**, capital against total exposure with **no risk weighting at all**, as a backstop against models that produce implausibly low weights.
- **Liquidity standards**, which had never existed. The **LCR** requires enough high-quality liquid assets to survive thirty days of stressed outflows; the **NSFR** requires stable funding against illiquid assets over a one-year horizon. Together they attack the maturity mismatch that actually killed institutions in 2008, most of which were solvent on paper when they failed. Both ratios are developed in liquidity risk, which owns them.

KEY IDEA

The two ratios are meant to bind in different places, and knowing which one bites where is the most useful thing in this section.

The **risk-weighted** ratio bites on risky assets: a corporate loan book, an unsecured derivatives portfolio. The **leverage ratio** bites on large, low-risk balance sheets: a repo book, a government bond inventory, a clearing business. A portfolio of zero risk-weighted sovereigns consumes no risk-weighted capital and still consumes leverage capital, so for those businesses the backstop is the real constraint.

That is why post-crisis regulation shrank repo and market-making inventory even though neither was blamed for the crisis. **A rule aimed at models ended up pricing balance sheet itself.**

WORKED EXAMPLE

The mechanic in one calculation. A bank makes a €100m corporate loan with a 100% risk weight, so it creates €100m of risk-weighted assets.

At the 4.5% CET1 minimum, that requires €4.5m of common equity. A large bank carrying the conservation buffer, a countercyclical buffer and a systemic surcharge is realistically managing to something above 10%, so call it **€10.5m of common equity** for one €100m loan.

Now the comparison that matters. The same €100m invested in a zero risk-weighted government bond creates €0 of risk-weighted assets and requires **no** risk-weighted capital. Under a 3% leverage ratio it still requires €3m of Tier 1, because the leverage measure does not care what the asset is.

So the two positions differ by a factor of several in capital consumed, and that difference, not the credit view, is frequently what decides whether the trade is done.

WORKED EXAMPLE

Turn it into the number a desk head actually uses. A trade earning €1m a year against €10m of CET1 returns 10% on capital. The same €1m earned against €20m of CET1 returns 5%.

If the bank's hurdle is 10%, **the first trade gets done and the second does not**, though they are identical in revenue. This is why client pricing includes a capital charge, why a collateralised trade is quoted more tightly than an uncollateralised one, and why a desk will push a client towards a cleared or margined structure. The pricing mechanics are in client pricing.

REAL INTERVIEW QUESTION (Natixis · Credit)

What is CET1?

WHAT THEY TEST / ANSWER

Expand it, say what qualifies, then say why the category exists, which is the part that shows you understand the regulation rather than the acronym.

Common Equity Tier 1. The highest quality layer of a bank's regulatory capital: ordinary shares and retained earnings, less deductions such as goodwill and other intangibles. It is reported as a ratio, CET1 divided by risk-weighted assets, and that ratio is the single number analysts and supervisors look at first.

Why the definition is narrow. Capital is regulatory only if it absorbs losses *while the bank is still operating*. Common equity does: losses reduce it directly, and dividends can be cut without any event of default. Instruments further down the stack absorb losses later or only in resolution, so they count towards Tier 1 and total capital but not towards CET1. Basel III narrowed the definition precisely because the 2008 experience showed that several things previously counted as capital did not absorb anything until the institution had already failed.

The two moving parts, and the one candidates forget. The ratio can rise because the numerator grows, by retaining earnings, cutting the dividend or issuing shares, or because the denominator falls, by shrinking or de-risking the balance sheet. **Managing the denominator is usually faster**, which is why a capital shortfall shows up on a trading floor as pressure to reduce risk-weighted assets rather than as a rights issue. If you have seen a desk asked to cut RWA before a reporting date, this is the reason.

Then the levels, carefully. The Pillar 1 minimum is 4.5%, but no large bank runs there: the conservation buffer, any countercyclical buffer, the systemic surcharge and a Pillar 2 requirement stack on top, and above all of that banks hold a management buffer. So a European G-SIB typically reports low-to-mid teens against a requirement somewhat below that. Say that you would check the current figure for the specific bank rather than quoting one, since it varies by institution and moves each quarter. The full stack is in capital requirements.

Close on what it means commercially, since this is a credit desk asking: the distance between a bank's reported CET1 and its requirement is its capacity to absorb losses and to grow the balance sheet, so it is a direct read on both its resilience and its appetite to take your trade.

Why the industry calls the 2017 package Basel IV

The finalisation left the pillars alone and attacked the thing Basel II had introduced: internal models producing risk weights that varied enormously between banks holding similar assets.

- **The output floor**, the headline. Risk-weighted assets computed with internal models may not fall below **72.5%** of what the standardised approach would give.
- **Revised standardised approaches** for credit and operational risk, which matter far more once they set the floor for everyone.
- **FRTB** for market risk: a hard boundary between the trading book and the banking book to stop assets being moved to whichever side is cheaper, **expected shortfall in place of value at risk**, model approval granted and withdrawn desk by desk, and punitive treatment of risk factors without enough observable data. The measures themselves are value at risk and expected shortfall; what is regulatory here is the *choice* between them.
- **SA-CCR** for counterparty exposure, replacing an older method that had become indefensible for portfolios of derivatives, and which is the subject of counterparty risk.

WORKED EXAMPLE

The output floor is easiest to see in one arithmetic step. A bank's internal model produces €60m of risk-weighted assets for a portfolio where the standardised approach would produce €100m.

The floor sets the minimum at $0.725 \times 100 = \mathbf{72.5}$ million euro, so the bank's risk-weighted assets rise from €60m to €72.5m, an increase of $72.5/60 - 1 = \mathbf{20.8}\%$, with no change whatsoever in the portfolio.

That is the whole of "Basel IV" in one line: **the capital cost of a business can rise by a fifth because of a change in how it is measured**, and the businesses affected are exactly those where internal models had produced the lowest weights, low-default corporate and mortgage lending above all. Implementation dates have moved repeatedly and differ by jurisdiction, so check the current status rather than quoting one.

INTERVIEW TRAP

Three errors. (1) Saying banks failed in 2008 because they had no capital. Many met their ratios; the capital was the wrong kind, the weights were too low, and there was no liquidity requirement at all. Naming the diagnosis is what the question tests. (2) Treating the leverage ratio as a lesser version of the capital ratio. It is a different constraint that binds on different businesses, and for repo and sovereign inventory it is the binding one. (3) Quoting a CET1 minimum of 4.5% as though it were what a bank must hold. Buffers, surcharges and Pillar 2 sit on top, and the management buffer sits on top of those.

Where this goes next

The capital stack and the ratio arithmetic in full are capital requirements, and the post-crisis rule that moved derivatives onto clearing houses is mandatory clearing. The risks being measured are market risk and counterparty risk, the measure FRTB moved to is expected shortfall, the episode that produced all of it is the 2008 financial crisis, and the place a capital charge reaches the client is client pricing.

24.2 Capital Requirements

trading · structuring · sales | core | cross-asset

Capital is the reason a desk can or cannot do a trade, and it is the least understood constraint on a trading floor because it is presented as compliance and behaves as economics. This section is about what capital is for, how much of it is required, and how that requirement converts into a limit on business.

> **KEY IDEA**
>
> Capital exists to absorb **unexpected** losses. The two quantities are defined and decomposed in credit risk; what matters here is the consequence. Expected losses are a cost of doing business: they are provisioned for and priced into the spread, exactly as a shop prices in shoplifting. What capital covers is the *variance* around that expectation, the year that turns out worse than the average.
>
> That distinction answers the question people ask first, which is why a bank cannot simply hold more provisions instead. Provisions are a liability against a loss you expect; capital is equity that can be consumed by a loss you did not. They sit on opposite sides of the balance sheet and do different jobs.

The stack, and why it has layers

Basel III and IV answers "what is CET1" as a question and hands the stack to this section, so the stack is what follows.

Common Equity Tier 1 is the highest-quality capital a bank has: ordinary shares and retained earnings, less regulatory deductions such as goodwill and certain deferred tax assets. Its defining property is that it absorbs losses *immediately and without any trigger*, because it is simply equity. Nothing has to be declared, converted or negotiated.

Additional Tier 1 sits above it: perpetual instruments, typically contingent convertibles, that convert to equity or are written down when the CET1 ratio falls through a trigger. **Tier 2** is subordinated debt of at least five years. The distinction that organises all three is **going concern** against **gone concern**: CET1 and AT1 absorb losses while the bank is still operating, Tier 2 absorbs them in resolution. That is the whole architecture in one sentence, and it is why the layers are priced so differently.

The ratio is CET1 divided by risk-weighted assets. The Basel III minimum is 4.5%, plus a 2.5% capital conservation buffer, plus a countercyclical buffer and a systemic surcharge for larger institutions, so a large bank's actual requirement typically lands somewhere around 10% to 12%.

Then the part that matters on a desk, and it is the reason this section exists separately from the regulation that defines the ratio. Breaching the buffers does not close the bank, it triggers the **maximum distributable amount** restriction: dividends, share buybacks and *AT1 coupons* are limited. That is why banks manage to a self-imposed buffer above the regulatory requirement, and it is why a CET1 ratio approaching the MDA threshold moves the bank's AT1 bonds long before it moves anything else. A credit desk that understands the MDA mechanism can read a bank's capital ratio as a signal about its subordinated paper.

Risk-weighted assets, and the arithmetic of headroom

The denominator is not the balance sheet, it is the balance sheet weighted by risk: each exposure multiplied by a factor reflecting its quality. A government bond in its own currency may carry zero, a corporate loan one hundred percent.

Two ways to get those weights, and the difference between them is most of what "Basel IV" means. The **standardised approach** uses the regulator's table. **Internal models** let a bank compute its own, which historically produced lower numbers and a suspicion that the models were being optimised rather than estimated. The Basel III finalisation answers that with an **output floor**: internal-model RWA cannot fall below 72.5% of what the standardised approach would give, which caps the benefit of modelling and is the single change with the largest effect on European banks.

WORKED EXAMPLE

How much business a capital ratio permits. A bank has €50 billion of risk-weighted assets and €6 billion of CET1, so its ratio is **12.0%**. Its requirement including buffers is 10.5%.

The maximum RWA it can carry on that capital is

$$\frac{6}{0.105} = \text{€}57.14 \text{ billion,}$$

so the headroom is **€7.14 billion of RWA**. Converting that into business depends entirely on the risk weight of what it writes:

Risk weight	Notional the headroom supports
20%	€35.7 billion
50%	€14.3 billion
100%	€7.1 billion

The same capital supports five times as much business at a 20% weight as at 100%. That is why capital is not a compliance topic: it is the mechanism by which a regulator decides which businesses a bank finds profitable, and a desk whose product carries a high risk weight is competing for a scarce resource against desks whose product does not.

Note also the direction of the constraint. A bank near its buffer does not stop trading, it reallocates: it prices high-weight business wider, or exits it, long before it raises equity. Those exits are visible in the market as a withdrawal of liquidity from particular products, and they are usually attributed to risk appetite when the cause is arithmetic.

REAL INTERVIEW QUESTION (Bank not recorded · Credit)

What is Basel 2 capital? Give the formula and its impact on a loan book compared with Basel 1.

WHAT THEY TEST / ANSWER

Answer in three moves: what changed conceptually, the formula, then the effect on a book. The conceptual change is the one to lead with, because the formula only makes sense once it is said: **Basel 1 weighted assets by category, Basel 2 weights them by measured credit quality**.

Under **Basel 1** the risk weight came from a short table, and every corporate loan carried 100% whether the borrower was AAA or nearly in default. Capital was 8% of risk-weighted assets, so capital $= 0.08 \times w \times \mathrm{EAD}$ with w drawn from $\{0, 20\%, 50\%, 100\%\}$. A bank could therefore reduce its capital only by shrinking, not by improving, which is precisely what made regulatory arbitrage worthwhile.

Under **Basel 2** the internal-ratings-based approach computes the capital requirement per exposure from its own risk parameters. Ignoring the maturity adjustment, the requirement per unit of exposure is

$$K \;=\; \mathrm{LGD}\left[\Phi\!\left(\frac{\Phi^{-1}(\mathrm{PD}) + \sqrt{\rho}\,\Phi^{-1}(0.999)}{\sqrt{1-\rho}}\right) - \mathrm{PD}\right], \qquad \mathrm{RWA} \;=\; 12.5 \times K \times \mathrm{EAD},$$

where PD is the one-year default probability, LGD the loss given default, ρ a prescribed asset correlation and $12.5 = 1/0.08$, which is what re-expresses a capital number as a risk-weighted asset number so the 8% ratio still applies.

Two features of that formula are worth pointing at, because they are what an interviewer is checking you noticed. The 0.999 is a **one-year, 99.9% confidence** loss, so the requirement is calibrated to a one-in-a-thousand-year outcome. And the $-\,\mathrm{PD}$ subtracts the expected loss, so K **covers unexpected loss only**, exactly the split this section opened with: expected loss is provisioned and priced into the spread, capital carries the variance around it.

The impact on a loan book is a redistribution before it is a reduction. Capital against high-quality borrowers falls sharply, since a low PD produces a weight far below 100%; capital against weak borrowers rises, and for genuinely distressed exposures the implied weight can exceed 100%. So the same book, unchanged, requires a different amount of capital depending on what is in it, and lending becomes priced against the capital it consumes rather than against a flat charge. For the investment-grade-heavy books most large banks ran, the aggregate requirement fell, which is the honest criticism to volunteer: banks were estimating the inputs to their own capital, and that is what the output floor described above was introduced to cap.

Close on the defect that 2008 exposed, which turns this from a definition into a judgement: the formula is **procyclical**. In a downturn realised PDs and LGDs rise, so the capital requirement rises precisely when capital is expensive and lending is most needed, which forces deleveraging into a recession. The countercyclical buffer above is the direct answer to that.

Two backstops and one trading-book regime

The leverage ratio is Tier 1 capital over total exposure with *no* risk weighting, at a minimum of 3%. It exists because risk-weighted measures can be gamed and because a portfolio of assets that all carry near-zero weights is still a large balance sheet. For a bank whose business is government bonds and repo, the leverage ratio rather than the risk-based ratio is frequently the binding constraint, which is why repo capacity contracts at quarter ends when balance sheets are reported.

Liquidity requirements sit alongside capital rather than inside it: the LCR and NSFR from liquidity risk are separate tests that a well-capitalised bank can still fail, and a bank fails for want of liquidity long before it fails for want of capital.

FRTB is the market-risk regime for the trading book, and three of its features matter to a desk. Capital is computed on expected shortfall at 97.5% rather than VaR at 99%. Liquidity horizons are differentiated by risk factor, so an illiquid exposure attracts capital as though it took longer to exit, which it does. And internal-model approval is granted *per desk* and must be maintained by passing ongoing tests, including a P&L attribution test comparing the risk model's predicted P&L against the actual. A desk that fails it moves to the standardised approach and its capital charge rises, so P&L explain stops being a control report and becomes a capital issue.

INTERVIEW TRAP

1. **Confusing capital with provisions or with liquidity.** Provisions cover expected loss, capital covers unexpected loss, liquidity is a separate test entirely, and a bank can be well-capitalised and still fail.
2. **Reading the requirement as the minimum.** The binding number is the MDA threshold, and banks run a management buffer above it because breaching restricts dividends and AT1 coupons.
3. **Treating RWA as the balance sheet.** It is the balance sheet weighted, and the weights are what decide which businesses survive.
4. **Ignoring the leverage ratio.** For low-risk-weight businesses it binds first, which is why repo capacity is scarce at reporting dates.
5. **Thinking capital rules stop trades.** They price them. A high-weight trade is not forbidden, it has to earn more, and the capital charge worked through in international financial institutions is what that means in a quote.
6. **Treating the P&L attribution test as reporting.** Under FRTB it determines whether a desk keeps model approval, and failing it raises the desk's capital charge.

VISUAL / CODE TO BUILD: Interactive

A capital-stack panel drawn as a column: CET1, AT1 and Tier 2 stacked against a requirement line with the conservation buffer, countercyclical buffer and systemic surcharge shown as separate bands. A loss slider eats into the stack from the bottom and the panel labels what happens at each level crossed, first the management buffer, then the MDA threshold with dividends and AT1 coupons flagged as restricted, then the AT1 trigger, then the minimum.

Watching the AT1 conversion fire before the regulatory minimum is reached is the mechanic that explains why AT1 bonds trade the way they do, and it is hard to convey any other way.

A second panel is a headroom calculator: CET1, RWA and the requirement as inputs, with the output shown as RWA headroom and then as notional at a chosen risk weight. Dragging the risk weight from 20% to 100% shrinks the business the same capital supports by a factor of five, which is the section's central point delivered in one gesture, and adding a leverage ratio line shows the point at which the unweighted constraint takes over from the weighted one.

24.3 Mandatory clearing

trading · structuring · sales | core | rates · credit · cross-asset

In September 2009 the G20 met in Pittsburgh and committed to pushing standardised over-the-counter derivatives through central counterparties. That sentence is the origin of a large fraction of the operational apparatus a modern derivatives desk lives inside, and it is worth understanding as a piece of policy design rather than as a rule to comply with, because the design has a cost that shows up directly in the margin a desk posts.

INTUITION

The problem 2008 exposed was not that banks traded derivatives with each other. It was that nobody, including the regulators and often including the banks, could say who was exposed to whom and by how much. The bilateral web was opaque, and opacity is what turns one failure into a system-wide freeze. Central clearing replaces the web with a hub.

What is actually mandated, and on whom

A clearing obligation has two dimensions, and both matter.

By product. Not everything can be cleared. A CCP has to be able to margin a contract daily and close it out on a member default, which requires the contract to be standardised, to have a liquid market, and to have a pricing source that survives the default of the largest participant. That is why the mandates landed where they did: vanilla interest rate swaps in the major currencies and the main credit index products, and not bespoke structures. A contract that cannot be reliably priced under stress cannot be safely cleared, and mandating it would move risk into the CCP rather than out of the system.

By counterparty. Regimes distinguish financial counterparties, which are generally caught in full, from non-financial ones, which are caught only above a threshold and are generally allowed to exclude positions that hedge commercial activity. That carve-out is deliberate: a manufacturer hedging its currency exposure is not what the mandate was written for, and forcing it to post daily variation margin would convert a hedge into a liquidity risk. This is the same conversion described in clearing houses, and for a corporate treasury it is not a small one.

The second lever, which does most of the work

A prohibition is only half of the design. The other half is that the products which are *not* mandated were made considerably more expensive to trade bilaterally.

Uncleared margin rules require both parties to exchange variation margin and, above thresholds, to post segregated initial margin calculated on a stressed ten-day horizon rather than the shorter horizon a CCP uses. The result is that an uncleared trade often consumes more collateral than the same trade cleared, on top of higher capital charges.

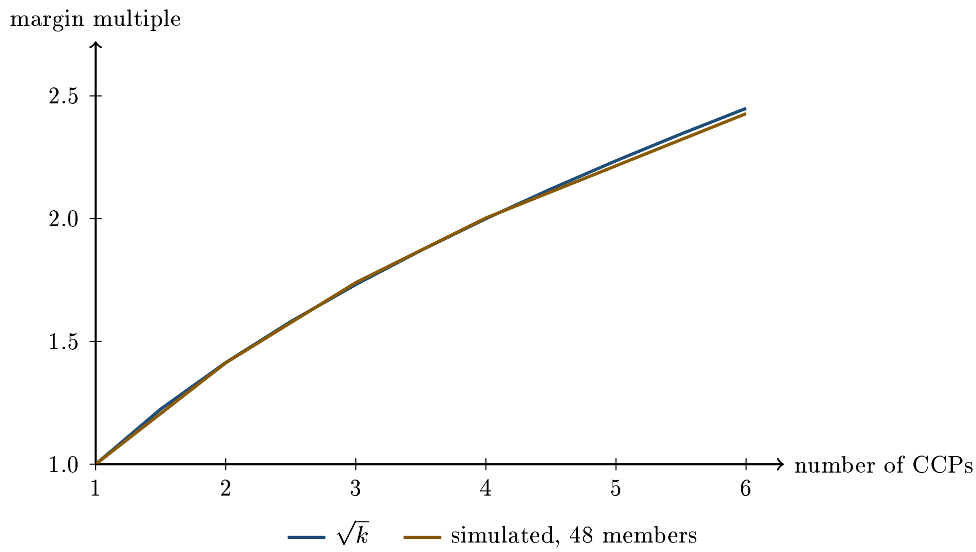
KEY IDEA

The clearing mandate works through price at least as much as through prohibition. Very little had to be banned, because once bilateral trading became the expensive option, standardised flow migrated to clearing on its own. That is worth recognising as a general pattern: the effective part of a regulation is often the relative cost it creates, not the activity it forbids, and that part is the one a desk actually feels.

The cost the design imposed

The benefit of clearing is multilateral netting, which is derived in clearing houses. That benefit has a property the mandates did not fully respect: **it only exists inside a single netting set**.

Different product classes were mandated onto different CCPs, and different jurisdictions authorised different ones. A dealer with an interest rate book at one CCP and a credit book at another nets neither against the other, so it posts margin twice on positions that would partly offset if they sat together.



same members, same trades, split across k CCPs instead of one

WORKED EXAMPLE

The fragmentation multiple, simulated. Take 48 dealers trading k independent product classes with each other. Under a single CCP netting everything, each dealer posts on its total net position. Under k CCPs it posts on each product's net separately, and the offsets between products are lost.

CCPs	Margin, one CCP	Margin, k CCPs	Multiple	$\sqrt{k}$
1	130.7	130.7	1.000	1.000
2	186.0	262.8	1.413	1.414
3	226.8	394.7	1.740	1.732
4	261.4	523.6	2.003	2.000
6	324.2	787.3	2.428	2.449

The simulated multiple tracks $\sqrt{k}$ to within about one percent throughout, which is the result you would expect: independent net positions add in quadrature when combined and add linearly when they are not.

So splitting the same book across four CCPs doubles the margin required to support it. Nothing about the risk changed. The trades are identical, the counterparties are identical, and the netting that would have offset them is simply not permitted to happen because the offsetting positions sit in different legal netting sets.

KEY IDEA

This is why cross-margining agreements between CCPs, and the competition between them, are a live regulatory issue rather than a commercial detail. A mandate that fragments clearing is partly working against its own purpose, and the industry cost of that is measured in the collateral that has to be found and funded to sit idle.

What it fixed and what it moved

Genuinely fixed. Bilateral counterparty exposure between dealers in the mandated products largely went away, replaced by exposure to a CCP that margins daily and has a defined default procedure. Trade repositories mean the aggregate positions are now visible to supervisors, which was the specific failure of 2008. Standardisation improved because clearing eligibility rewarded it.

Moved rather than removed. Credit risk became liquidity risk: a cleared position makes no credit demand but makes a daily cash demand, and that demand is largest exactly when markets move. Several episodes since have been margin-call events rather than default events, which is a different failure mode rather than no failure mode.

Concentrated. A handful of CCPs are now nodes whose failure would be catastrophic, which is why their recovery and resolution regimes are among the most argued-over pieces of financial regulation. The mandate did not eliminate too-big-to-fail from derivatives; it relocated it into institutions designed for the job and supervised accordingly, which is a real improvement and not the same as a solution.

Interview angle

The question database contains no row on clearing mandates, EMIR or Dodd-Frank, which is consistent with the pattern elsewhere: candidates are tested on instruments, not on the rules around them. The topic reaches an interview indirectly, usually through a question about what a clearing house does or why a trade is expensive.

The observation worth having ready is the one in the figure. Anyone can say that clearing reduces risk through netting. What shows you have thought about it is that netting only works inside one netting set, so a mandate that pushes products onto different CCPs gives back part of the benefit it was created to capture, and the give-back scales as the square root of the number of CCPs involved. Pair that with the observation that credit risk was converted into liquidity risk rather than removed, and you have said the two things about this regime that practitioners actually argue about.

> **INTERVIEW TRAP**
>
> Four errors. (1) Saying central clearing removes counterparty risk, when it replaces it with exposure to the CCP and a daily cash obligation. (2) Assuming everything must be cleared; bespoke products cannot be, and non-financial hedgers are largely exempt by design. (3) Treating the uncleared margin rules as a side issue, when the relative cost they create is what moved the flow. (4) Adding up netting benefits across CCPs as though they combined, when the entire point is that they do not.

Where this goes next

How a CCP actually works, what novation does and how the default waterfall is structured are in clearing houses. The two regimes that implemented the mandate are the Dodd-Frank Act and EMIR. The collateral that all of this consumes, and how it is valued, is collateral and haircuts; the exposure it is designed to contain is counterparty risk; and the crisis that produced the commitment is the 2008 financial crisis.

24.4 Dodd-Frank Act

sales · trading · structuring | core | cross-asset

Dodd-Frank is the United States legislative answer to 2008, and if you work on anything that touches a US counterparty its consequences are in your daily workflow whether or not you can name them. The useful way to hold it is not as a list of titles but as a set of answers to specific failures: the crisis showed that swap exposures were invisible, that a large dealer could not be wound down, that ratings were relied on by rule, and that originators had no stake in what they sold. Each of those became a programme.

KEY IDEA

Dodd-Frank is a rulemaking mandate, not a rulebook. The 2010 statute set objectives and delegated the detail to the agencies, chiefly the CFTC, the SEC, the Federal Reserve, the FDIC and the OCC. That single design fact explains most of what puzzles people about it: why implementation ran for the better part of a decade, why the operative requirements live in agency rules rather than in the Act, why the substance shifted with successive administrations, and why parts were later amended without the statute being repealed. If someone asks what Dodd-Frank requires, the honest answer starts by naming the rule, not the title.

The parts that reach a trading floor

Title VII, derivatives. The one that changed market structure, and the reason it exists is that in 2008 nobody, including the regulators, could see who was exposed to whom. It created, through agency rules:

- **Mandatory clearing** of standardised swaps, so a bilateral exposure becomes an exposure to a clearing house, developed in mandatory clearing.
- **Mandatory execution** of those swaps on regulated venues, swap execution facilities, which pushed price formation from the telephone to a screen with quote and trade transparency.
- **Reporting** of every swap to a swap data repository, in near real time for public price dissemination and in full detail to the regulator. This is the direct answer to the invisibility problem.
- **Registration** of swap dealers and major swap participants, which brings capital, margin, documentation, recordkeeping and business-conduct obligations, including duties when dealing with less sophisticated counterparties.
- **Margin for uncleared swaps**, initial and variation, which made bespoke trades materially more expensive to carry and is the single largest economic push towards standardisation.
- **Split jurisdiction**, with the CFTC taking swaps and the SEC security-based swaps, which is an ongoing source of definitional work rather than a tidy boundary.

Title VI, the Volcker Rule. Section 619 restricts proprietary trading and certain fund activities at banking entities, and it is substantial enough to have its own section: the Volcker rule.

Titles I and II, systemic risk and failure. Title I created the Financial Stability Oversight Council and the Office of Financial Research, imposed enhanced prudential standards on large institutions, and made supervisory **stress testing** and **resolution plans**, the living wills, recurring obligations, see stress testing. Title II created the **Orderly Liquidation Authority**, which is the direct answer to the Lehman problem described in the 2008 financial crisis: a way to wind down a large financial firm that is neither a disorderly bankruptcy nor a public rescue.

Title IX, investor protection and market integrity. Three provisions matter beyond their size. **Risk retention** requires securitisation sponsors to keep economic exposure, generally five percent, so that the originate-to-distribute incentive that degraded mortgage underwriting is at least partly reversed. **Ratings reform** removed statutory and regulatory references that compelled reliance on credit ratings, and put the rating agencies under a supervisory regime, which answers the manufactured-AAA problem. And the **whistleblower programme** pays awards for information leading to enforcement, which changed the internal calculus around misconduct more than its statutory length suggests.

Title X created the Consumer Financial Protection Bureau, which matters little to a derivatives desk and a great deal to the politics of the Act.

What it actually changed on a desk

1. A standardised swap is now executed on a venue, cleared, margined daily and reported within minutes. The workflow that used to be a phone call and a confirmation is a regulated pipeline.
2. **Pre-trade anonymity shrank.** Real-time public reporting means a large trade is visible quickly, which changes how a block is worked and is a genuine cost to a client executing size.
3. **Bespoke became expensive.** Uncleared margin requirements and capital charges put a price on non-standard risk, which is why structuring gravitated towards payoffs that can be hedged with cleared building blocks, a pattern traced in the evolution of derivatives.
4. **Cross-border complexity became a permanent feature.** A trade between a US person and a European counterparty can fall under both Dodd-Frank and the EU regime, resolved through substituted compliance and equivalence determinations rather than through a single rulebook, which is why the same economic trade can be booked differently depending on entity and location.
5. **Capital planning became a constraint on risk-taking.** Annual supervisory stress tests turned capital from an accounting outcome into a forward-looking limit that a desk's proposed positions must fit inside.

Two comparisons worth being able to make

Against Basel. Basel is an international standard on *capital and liquidity*, agreed by a committee and then implemented in national law, covered in Basel III and IV and capital requirements. Dodd-Frank is a national *statute* that also reaches market structure, resolution, conduct and consumer protection. They overlap, since Dodd-Frank is one vehicle through which the United States implements Basel standards for large banks, but they are not the same kind of object and confusing them is a common slip.

Against the European regime. Dodd-Frank Title VII and the European EMIR plus MiFID package implement the same post-crisis G20 commitments, clearing, reporting, venue trading and margin, through different instruments and on different timetables. The practical consequence is duplicative reporting and definitional mismatches rather than a level playing field, which is exactly the friction a cross-border desk manages.

It has not stood still

Two amendments are worth knowing so you do not describe a decade-old snapshot as current. In 2018 a regulatory relief act raised the asset threshold at which enhanced prudential standards and supervisory stress testing apply, tailoring the regime so that the heaviest requirements fall on the largest institutions. And the Volcker Rule was revised at the end of the decade to simplify its compliance and metrics regime. The direction of travel has been calibration rather than repeal, so the correct framing in an interview is that the architecture is intact and the thresholds and details have moved.

INTERVIEW TRAP

Four traps. First, **treating Dodd-Frank as a rulebook**: the operative requirements are agency rules, so a precise answer names the rule and the agency. Second, **confusing it with Basel**, which is an international capital and liquidity standard rather than a national statute covering market structure and resolution. Third, **saying it eliminated counterparty risk in swaps**: it moved much of that risk to clearing houses and margin, which concentrates and mutualises it rather than removing it, as counterparty risk sets out. Fourth, **describing it as unchanged since 2010**, when the thresholds and the Volcker compliance regime have both been recalibrated.

VISUAL / CODE TO BUILD: Interactive

A failure-to-remedy tracer, which is the only presentation of this material that makes it memorable. The left column lists the specific failures of 2008 (nobody could see swap exposures, a large dealer could not be resolved, ratings were relied on by rule, originators kept no stake, banks took proprietary risk against insured deposits, capital was inadequate in stress). The right column lists the Dodd-Frank programmes. The reader draws the connections and the tool grades them, then shows the implementing rule, the agency and the year for each, so the statute becomes a map from problems to responses rather than a list of titles. A second panel is a trade-workflow comparison: the same interest rate swap booked in 2006 and today, side by side, with each added step labelled by the rule that requires it, so the cost of the regime is visible as steps rather than described as a burden. A third panel overlays the US and EU timelines for the same G20 commitments, making the cross-border mismatch a picture.

Where this goes next

The clearing mandate is developed in mandatory clearing and the proprietary trading restriction in the Volcker rule. The capital and liquidity standards that run alongside it are Basel III and IV and capital requirements, the European counterparts are EMIR and MiFID, and the conduct layer that sits on the floor itself is trading floor compliance. The failures it responds to are catalogued in the 2008 financial crisis, and the institution that absorbs the risk it redirects is clearing houses.

24.5 EMIR

sales · trading · structuring core rates · credit · cross-asset

EMIR is not a compliance footnote. It is the reason the swap you price this morning is cleared, the reason you post initial margin on the one that is not, the reason your discount curve is built from overnight rates rather than from LIBOR, and part of the reason the same trade is quoted at different prices to two different clients. A rates, credit or structuring interviewer will not ask you to recite the regulation, but they will absolutely ask what changed in the OTC market after 2008, and EMIR is the European half of that answer.

What it is, in one paragraph

The **European Market Infrastructure Regulation**, Regulation (EU) No 648/2012, entered into force in August 2012. It is the EU's implementation of the commitments that the G20 made in Pittsburgh in 2009, after the crisis exposed how little anyone knew about the bilateral OTC derivatives network. Those commitments were: push standardised OTC derivatives into central clearing, collateralise what cannot be cleared, and report everything. The United States implemented the same agenda through Title VII of the Dodd-Frank Act, covered in the Dodd-Frank Act. The two regimes are cousins, which is why an EU and a US bank end up with broadly the same plumbing and endless arguments about equivalence. EMIR has since been revised, notably by the 2019 REFIT package, which simplified the regime for smaller counterparties and reformed reporting, and by a further revision adopted in 2024 that introduces a requirement for EU counterparties to keep active accounts at EU-authorised CCPs.

INTUITION

Before 2008 the OTC derivatives market was a dense web of bilateral promises. Every bank owed every other bank, mostly on trades nobody outside the two counterparties could see, and much of it was uncollateralised. When one node in that web looked like failing, nobody, including the regulators, could work out who was exposed to whom or by how much, so everybody assumed the worst and stopped dealing with everybody. EMIR attacks that in three ways: it replaces the web with a hub for the standardised products (clearing), it forces the remaining bilateral strands to be backed by cash or securities (margin), and it makes everyone photograph the web every day (reporting).

The three pillars

This is the structure to answer with, because “what is EMIR” is a three-pillar question and a candidate who gives the three in order sounds like they have worked next to the thing.

Pillar 1: the clearing obligation. Classes of OTC derivative that ESMA declares sufficiently standardised and liquid must be cleared through an authorised central counterparty. In practice this captures the plain interest rate swap complex in the major currencies and the main credit index products. Clearing works by **novation**: the original bilateral contract is torn up and replaced by two contracts facing the CCP, which becomes buyer to every seller and seller to every buyer.

before novation (left) after novation (right)

The point is not that risk disappears. It is that credit exposure is standardised, margined daily, netted multilaterally, and mutualised through a **default waterfall** whose order every markets person should be able to recite: the defaulting member’s own initial margin first, then that member’s contribution to the default fund, then a slice of the CCP’s own capital (its “skin in the game”), then the mutualised default fund contributed by surviving members, and only then the recovery tools such as further assessments on members. See clearing houses for the mechanics and mandatory clearing for the obligation itself.

Pillar 2: risk mitigation for uncleared trades. What cannot be cleared, which includes most genuinely bespoke structured trades, must instead be managed: timely confirmation, portfolio reconciliation, portfolio compression, an agreed dispute resolution process, daily valuation, and, since the phase-in that began in 2016, the exchange of **variation margin** and **initial margin** under the bilateral margin rules. Variation margin settles the daily change in mark to market. Initial margin is a buffer held against the move that could occur between a counterparty's default and the close-out of the position, it is posted by both sides, and it must be segregated so it is not available to the surviving party as a general asset. Collateral eligibility and haircuts are in collateral and haircuts.

Pillar 3: reporting. Both counterparties report every derivative contract, listed as well as OTC, to a registered trade repository, by the working day after the trade. This is the pillar that touches the most people and produces the least market comment, and it exists so that a supervisor can answer the question nobody could answer in September 2008: who is exposed to whom.

Who has to do what

EMIR sorts counterparties, and the classification is what determines whether the client on the other side of your request for quote has to clear.

- **Financial counterparties**: banks, insurers, funds and pension schemes. In scope for the margin rules, and for clearing unless they are small enough to stay under the same thresholds, a category REFIT created and labelled FC-.
- **Non-financial counterparties**: corporates. They are only pulled into the clearing obligation if their speculative positions exceed a clearing threshold, which is set separately for each asset class in the technical standards and is periodically revised. Above it they are known as NFC+, below it NFC-.

The detail that matters commercially, and that a sales candidate should know, is that positions objectively reducing risks directly related to the corporate's commercial activity, that is genuine hedges, are excluded from the threshold calculation. A manufacturer hedging its floating-rate debt or its input costs is not dragged into clearing for doing so. That carve-out is the reason a corporate sales desk can still offer a bespoke uncleared structure to a hedging client, and it is a good thing to be able to explain in a client-facing interview. Client pricing is in client pricing.

Interviewers often approach all of this through the listed-versus-OTC contrast, which is covered in exchanges. The EMIR-specific point worth adding when it comes up is that the line has genuinely blurred: a cleared interest rate swap now carries most of the credit and margin characteristics of a listed future while remaining, legally, an OTC derivative. What still separates them is customisation, and that is exactly what the clearing obligation is drawn around, since only classes standardised enough to be cleared are caught by it.

Why a front-office desk actually cares

It changed the discount curve. This is the most direct front-office consequence and the one most often tested. If a trade is collateralised under a credit support annex that pays the overnight rate on cash collateral, then the funding cost of that position is the overnight rate, and the correct discount curve is built from overnight index swaps rather than from LIBOR-style term rates. Cleared trades are collateralised by construction, so the shift to central clearing was also the shift to OIS discounting, and later to €STR and SOFR. See OIS, LIBOR, SOFR and €STR and interest rate swaps.

WORKED EXAMPLE

A single payment of €10m due in five years. Discount it at a collateralised rate of 2.00% and at an uncollateralised rate of 2.50%, annually compounded:

$$\frac{10{,}000{,}000}{1.02^5} = \frac{10{,}000{,}000}{1.104081} = 9{,}057{,}308, \qquad \frac{10{,}000{,}000}{1.025^5} = \frac{10{,}000{,}000}{1.131408} = 8{,}838{,}543.$$

The difference is €218,765, roughly 2.2% of the payment, from a 50 basis point difference in the discount rate on one cash flow. Multiply that across the schedule of a large swap book and you see why the industry rebuilt its curve infrastructure rather than treating this as a rounding error. The same arithmetic is why the price you quote a client depends on whether the trade will be cleared, collateralised under a CSA, or left uncollateralised.

REAL INTERVIEW QUESTION (ING · Rates)

How do you distinguish collateralised discount curves (OIS) from uncollateralised ones?

WHAT THEY TEST / ANSWER

The principle is that you discount at the rate you actually fund the position at, and the collateral agreement tells you what that rate is. Under a cash CSA paying the overnight rate, posting or receiving collateral finances the mark to market at that overnight rate, so the risk-free discounting curve is the OIS curve. With no CSA, there is no collateral to fund you, so the position is funded on the bank's own unsecured curve and the valuation picks up a funding adjustment on top, which is where FVA comes from. Then give the two refinements that show real desk exposure. First, this is why the market moved to dual-curve bootstrapping: you project forward rates off the relevant index curve and discount off the collateral curve, and the two are no longer the same object. Second, the discount rate follows the *cheapest to deliver* collateral permitted by the CSA, so a multi-currency CSA embeds an option on which currency to post, and that option has to be valued. Close by naming the family of adjustments the uncleared case attracts, CVA, FVA and MVA, covered in CVA, DVA and FVA.

It changed what a trade costs. The same economic swap now has a different all-in price depending on its regulatory route. Cleared: initial margin at the CCP plus a default fund contribution, but small counterparty credit charges. Uncleared: bilateral initial margin, funded for the life of the trade, which is the funding cost known as MVA, plus CVA and the capital charges described in Basel III and IV and capital requirements. Structurers therefore design toward clearable components where they can, and price the rest for what it truly consumes. Counterparty risk itself is in counterparty risk.

It concentrated risk rather than removing it. Central clearing converts a diffuse network of bilateral exposures into a small number of systemically critical hubs. That is a real improvement in transparency and netting, and it is also a concentration that regulators now spend considerable effort stress testing. Being able to say both halves of that sentence, rather than presenting clearing as an unambiguous good, is what a thoughtful answer sounds like.

The collateral machinery that makes all of this work, counterparty eligibility, eligible asset lists, haircuts calibrated to the margin period of risk, and wrong-way risk, is not re-derived here: it is in collateral and haircuts, with the secured funding market behind it in repo markets.

INTERVIEW TRAP

Four traps. First, saying central clearing "removes counterparty risk"; it transforms and mutualises it, and it creates concentration risk at the CCP, so say that instead. Second, confusing variation margin with initial margin: VM settles today's move and is a transfer of value, IM is a segregated buffer against the future move during close-out and is posted by both parties. Third, treating EMIR and MiFID as the same thing; EMIR governs clearing, margin and reporting of derivatives, whereas MiFID governs trading venues, transparency, investor protection and best execution. Fourth, assuming every derivative must be cleared; only classes ESMA has declared subject to the obligation must be, and genuine corporate hedging is carved out of the threshold test.

VISUAL / CODE TO BUILD: Interactive

Two linked views. The first is a network animation: a dense bilateral graph of twenty counterparties that the reader can rewire into a hub-and-spoke CCP structure, with live counters for total gross exposure and net exposure after multilateral netting, so the netting benefit is measured rather than asserted. Toggling a node to "default" then walks the waterfall layer by layer, showing which resource absorbs the loss and at what point surviving members start paying. The second is a pricing comparator: the same swap priced cleared, CSA-collateralised and uncollateralised, with the discount curve, initial margin, MVA and CVA broken out so the reader can see the same economics carry three different prices.

Where this goes next

The obligation at the centre of EMIR has its own section in mandatory clearing, and its US mirror is the Dodd-Frank Act. The trading-venue and transparency half of the post-crisis rulebook is MiFID, and the capital half is Basel III and IV. The valuation adjustments that the uncleared regime generates are in CVA, DVA and FVA, and the crisis that produced the whole agenda is in the 2008 financial crisis.

24.6 MiFID

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MiFID is the European rulebook for how investment services are provided and where financial instruments are traded. Where the Basel framework asks how much capital a bank must hold, MiFID asks a different question entirely: **what must a firm tell its client, and where must a trade happen**. It is therefore the regulation a salesperson meets daily and a capital rule is not.

The first directive applied from 2007. **MiFID II**, together with the accompanying regulation MiFIR, applied from January 2018 and is what people mean when they say MiFID without qualification. It is enormous, and the way to hold it is not as a list of articles but as four intentions.

Four things MiFID II was trying to do

- **Make prices visible.** Extend pre-trade and post-trade transparency beyond equities to bonds, derivatives, structured products and exchange-traded funds, where before it had been possible to trade without anyone outside the transaction knowing the price.

- **Move trading onto regulated venues.** Define the venue types precisely, create a new category for the trading that did not fit, and require certain instruments to trade on a venue rather than bilaterally.
- **Separate research from execution.** Stop investment research being paid for through dealing commissions, on the grounds that a bundled payment hides both the cost of the research and the cost of the trading.
- **Make costs and suitability explicit.** Require firms to define who a product is for, to check it is suitable, and to disclose every cost in cash terms before and after the fact.

> KEY IDEA
>
> The unifying idea is that **a client cannot judge value they cannot see**. Almost every requirement in MiFID II converts something previously embedded, in a spread, in a commission, in an undisclosed margin, into a number the client is shown. That reframing is the answer to give when someone asks what MiFID II was for, because it explains rules that otherwise look unconnected: transparency, unbundling and costs disclosure are the same instinct applied to three different places money was hidden.

The venue taxonomy

Worth knowing precisely, because the four acronyms come up and the distinctions are real.

Type	What it is	Discretion
RM	Regulated market, a traditional exchange	None, non-discretionary rules
MTF	Multilateral trading facility	None, non-discretionary rules
OTF	Organised trading facility	Discretionary, non-equities only
SI	Systematic internaliser	A firm dealing on own account

Two distinctions carry the meaning. **An OTF may exercise discretion** over how orders are matched, which an RM or MTF may not, and it exists only for non-equity instruments; it was created to capture broker crossing networks and voice broking that were previously outside the framework. **A systematic internaliser is not a venue at all**: it is an investment firm that deals on its own account against client orders often enough to be caught, which means it takes on quoting and transparency obligations of its own. That is the regime a bank's own market making usually falls into, so it is the one most likely to matter to the desk you are interviewing with. The market structure these sit inside is market microstructure.

REAL INTERVIEW QUESTION (Deutsche Bank · desk not recorded)

What is MiFID 2?

WHAT THEY TEST / ANSWER

Do not recite articles. Give the one-line definition, the four intentions, then one consequence you can actually discuss, and finish by saying which parts have since moved.

The definition. The European framework governing how investment services are provided and where instruments trade, applying from January 2018, and a substantial expansion of the 2007 original.

The four intentions, which is the structure worth having ready. *Transparency*, extending pre- and post-trade price publication from equities to bonds, derivatives and structured products. *Venues*, defining regulated markets, multilateral trading facilities, the new organised trading facilities and the systematic internaliser regime, and pushing trading onto them. *Unbundling*, forcing research to be paid for separately from execution. *Investor protection*, through product governance and target markets, suitability, and full disclosure of costs and charges in cash terms.

Then pick one and go deeper, because breadth alone sounds memorised. The best choice depends on the desk. For equities, unbundling: research budgets became explicit, asset managers had to decide whether to absorb the cost or charge it on, and sell-side research coverage contracted sharply as a result, particularly for smaller companies. For structuring or sales, costs and charges: the embedded margin in a structured product has to be shown to the client as a number, which changed how products are designed as much as how they are sold. For rates or credit, transparency: instruments that had traded entirely on the phone acquired a reporting obligation, with deferrals for large or illiquid trades so that a market maker is not immediately exposed after taking down a block.

Then the honest closing, which is what separates a good answer. Say that parts of the regime have been revisited. Unbundling in particular has been relaxed in both the UK and the EU after concerns that it reduced research coverage of smaller issuers, and the transparency regime has been reviewed. So state the framework confidently and add that you would check the current position on unbundling and on the transparency calibrations, because they have moved and are still moving. That is a better answer than a confident recital of the 2018 position.

If asked what it means for you personally: every client conversation about a product is now also a conversation about its target market and its total cost, and that is a change in the job rather than in the paperwork.

WORKED EXAMPLE

Costs and charges disclosure, in numbers, because this is where the rule bites commercially.

Take a five-year structured note issued at 100 carrying a headline coupon of 6%. Suppose the total embedded cost is 3% of notional: a distribution fee plus the structuring margin.

Before MiFID II, that 3% lived inside the pricing and the client saw a note at par. After, it must be shown in cash terms. Amortised over the life it is $3\%/5 = $ **0.6% per year**, which is $0.6/6 = $ **10% of the headline coupon**.

Nothing about the product changed. What changed is that the conversation now starts from a disclosed number, and a client who sees that a tenth of the advertised coupon is fee asks different questions. That is why the rule affected product design and not only documentation, and how a margin is arrived at in the first place is client pricing.

WORKED EXAMPLE

Unbundling, also in numbers, because the scale is not obvious.

A fund with €5bn of assets turning over half of the portfolio in a year trades €2.5bn. Under a bundled arrangement of 10 bp all-in, of which 4 bp was notionally research, it paid €2.5bn $\times$ 0.0010 = €2.5m in commissions, with €2.5bn $\times$ 0.0004 = €1m of that implicitly buying research.

Unbundled, the execution commission falls to 6 bp and the €1m of research becomes a line item that someone has to authorise, either the manager out of its own profits or the client through an explicit charge.

The money did not change; the visibility did. And once a manager had to sign off €1m a year as a cost rather than have it arrive inside a commission, research budgets fell and coverage of smaller companies fell with them. That outcome is the reason the rule has since been revisited, and it is a good illustration of a general point: making a cost visible does not leave the quantity unchanged.

What it did not do

INTUITION

It is worth being clear about MiFID's boundaries, because candidates routinely attribute the whole post-crisis rulebook to it. MiFID governs **conduct and market structure**: who you may sell what to, what you must disclose, and where trades may happen. It does not set capital requirements, which are the Basel framework, and it does not impose the clearing and reporting obligations on derivatives, which are EMIR in Europe and the corresponding parts of Dodd-Frank in the United States. A trade can be perfectly MiFID compliant and still fail an EMIR obligation, because the two ask different questions about the same transaction. Those sit in EMIR and mandatory clearing.

MiFID II also tightened **best execution**, moving the standard from taking reasonable steps to taking sufficient steps and adding reporting on execution quality and venue selection. That is a large enough subject to have its own treatment in best execution, and the record-keeping obligations, including the recording of client communications, belong with trading floor compliance.

INTERVIEW TRAP

Three errors. (1) Using MiFID as a synonym for European financial regulation. It is conduct and market structure; capital is Basel and derivatives clearing is EMIR. (2) Calling a systematic internaliser a trading venue. It is a firm dealing on its own account with venue-like obligations attached, and the distinction is exactly what the regime was created to handle. (3) Describing the 2018 rules as the current state without qualification. Unbundling and the transparency calibrations have both been revisited, so say what the framework does and add that you would check the current position.

Where this goes next

The obligation MiFID II raised and that deserves its own treatment is best execution, and the record-keeping and conduct rules a desk lives with day to day are trading floor compliance. The derivatives half of the European rulebook is EMIR, with the clearing obligation itself in mandatory clearing. The venues being classified are exchanges, the structure they sit in is market microstructure, and the margin that costs disclosure makes visible is set in client pricing.

24.7 The Volcker Rule

trading · sales core cross-asset

The Volcker rule, a provision of the Dodd-Frank Act named after Paul Volcker, prohibits US banking entities from **proprietary trading** and from owning or sponsoring hedge funds and private equity funds. It is worth a section not because the prohibition is complicated but because it is almost impossible to enforce, and the machinery built to enforce it changed how trading desks are run far more than the prohibition itself did.

The rationale is about the subsidy, not the risk

KEY IDEA

The argument is not that proprietary trading is too risky in general. It is that a *banking entity* has deposit insurance and access to central bank facilities, so its funding is cheaper than its risk would otherwise command, and using that subsidised funding to take directional positions transfers the downside to the public while the upside stays private.

Read it that way and the shape of the rule follows. It restricts an *activity within a particular kind of institution*, not an activity. The same trade at a hedge fund is untouched, which is not an oversight but the entire logic.

The problem: the prohibited and the permitted look identical

Here is the difficulty that makes the rule what it is. A market maker buys from a client who wants to sell, and is now long. That inventory is a position, exposed to price moves, and indistinguishable from a proprietary position by inspection. A hedger sells futures against a book, and that short is a position too.

So the rule cannot be written in terms of what is held. It has to be written in terms of **why** it is held, and intent is not observable. Every complication in Volcker compliance follows from trying to make an unobservable thing auditable.

The exemptions, and the phrase that carries the rule

Market making is permitted, subject to a condition that is the heart of the whole regime: inventory must not exceed the **reasonably expected near-term demand** of clients, universally abbreviated **RENTD**. A desk must be able to show that what it holds is sized to what its clients are likely to want, based on historical demand, and not to what the desk thinks the market will do.

Risk-mitigating hedging is permitted, and this exemption was tightened for a specific reason. A hedge must be tied to a *specific, identifiable* risk, its effectiveness must be documented and monitored on an ongoing basis, and it must not itself create significant new exposure. The motivating case is the 2012 London Whale loss, where a position described as a hedge grew until it was the dominant position in its market and its own risk had nothing to do with what it was said to be hedging. Naming that case, and naming what it demonstrates, is what a good answer on this topic contains: the word "hedge" is a claim about intent, and a claim about intent is exactly what cannot be verified without measurement.

Also permitted: underwriting, trading in US government obligations, transactions on behalf of clients as agent, and liquidity management.

How intent gets measured

Since intent cannot be observed, the rule proxies it with quantitative metrics computed per trading desk, and the metrics are where the rule actually lives. Inventory turnover and inventory ageing, risk factor sensitivities and their limits, VaR usage, customer-facing trade ratios, and a comprehensive profit and loss attribution separating what came from customer spread from what came from position value.

WORKED EXAMPLE

Turnover as a proxy for intent. A market maker holds inventory briefly and repeatedly; a proprietary trader holds a view. So the ratio of traded volume to average inventory separates them without anyone having to read a mind.

Annual turnover	Trading days of inventory held
200×	1.3
50×	5.0
20×	12.6
5×	50.4

A desk turning its book two hundred times a year holds a day of inventory and is obviously intermediating. A desk turning it five times a year holds fifty trading days of it and is, whatever it says, carrying a position. The question RENTD asks is whether the level is consistent with client demand: if clients trade €50 million a day net and the desk holds €500 million, that is ten days of expected demand, and the desk has to explain why it needs ten.

The metric proves nothing on its own, which is the honest thing to say about it. A genuinely illiquid product turns slowly for legitimate reasons. What the regime actually does is force the desk to have an answer, in advance, in writing, and that is a different mechanism from prohibition.

The apparatus around those metrics is the part traders experience: **desk mandates** that state in writing what each desk is permitted to trade and why, position and ageing limits, escalation when a metric moves outside its normal range, and senior attestation that the compliance programme is effective. Volcker's practical legacy is that a trading desk must be able to articulate its business in a document, which trading floor compliance covers as a daily reality.

INTUITION

The most interesting consequence is not on banks but on markets. Dealers carry less inventory than they did, so in ordinary conditions bid-ask spreads are fine and in stressed conditions there is less balance sheet standing between a seller and the next price. Corporate bond markets are the usual illustration.

Meanwhile the risk did not disappear, it **moved**: to non-bank market makers, to hedge funds, to principal trading firms, none of which are inside the rule. Whether that is safer is a genuine and unresolved argument. It is safer in the sense that losses now land outside the insured banking system. It is more dangerous in the sense that the entities holding the risk have no central bank facility and no obligation to keep quoting, so they withdraw exactly when the market most needs them, which is the mechanism liquidity risk describes.

A candidate who gives both halves of that argument has answered better than one who picks a side.

Volcker 2.0

The 2019 and 2020 revisions kept the prohibition and reduced the compliance burden: tiering obligations by trading assets so smaller banks face far less, replacing the presumption that a short-term position was proprietary with a presumption based on desk-level risk limits, and narrowing the fund restrictions. The direction of travel is toward relying on a desk's own documented risk limits rather than on a regulator's inference from holding periods, which is a reasonable adaptation and also an admission that inferring intent from data did not work as well as hoped.

INTERVIEW TRAP

1. **Saying the rule bans risk-taking.** It bans a specific activity in a specific kind of institution. Market making, underwriting and hedging all involve positions and all remain permitted.
2. **Thinking a position can be classified by looking at it.** Inventory and a prop position are the same object. Only intent differs, which is why the regime is built on metrics and mandates.
3. **Treating "it is a hedge" as a defence.** The London Whale was described as a hedge. The exemption requires a specific identified risk and ongoing documented effectiveness.
4. **Assuming the risk was eliminated.** It migrated outside the banking perimeter, to entities with no central bank facility and no obligation to quote.
5. **Ignoring turnover as a diagnostic.** It does not prove intent, and it is the number a compliance function will ask a desk to explain, so a desk should know its own.
6. **Forgetting the rule is jurisdictional.** It applies to US banking entities and their affiliates worldwide, so a desk in London inside a US group is covered and a competitor next door may not be.

VISUAL / CODE TO BUILD: Interactive

An inventory-path panel. A desk's position is drawn over time against its client flow, with a slider that shifts the desk's behaviour continuously from pure intermediation, where inventory oscillates around zero and mean-reverts within days, to position-taking, where it trends and persists. The panel computes turnover, average holding period and the share of P&L attributable to spread rather than to price movement, and shades the RENTD band implied by the client flow.

The point of the panel is that there is no line. Dragging the slider produces a continuum, and the reader has to decide where intermediation stops, which is precisely the problem the regulator has and cannot solve with a definition. Watching the metrics move smoothly rather than jumping at a threshold is a better explanation of why the regime looks the way it does than any description of the rule.

A second panel shows dealer inventory and bid-ask spreads through a simulated stress, with a control for how much inventory dealers are willing to carry. Reducing it leaves spreads unchanged in calm conditions and widens them sharply in the stress, which is the trade-off the rule made, drawn rather than argued.

24.8 Trading floor compliance

sales · trading · structuring core cross-asset

Trading losses end positions. Conduct failures end careers, and they do it faster and with less warning. That asymmetry is why the rules in this section are worth more of your attention than their intellectual content deserves, and why interviewers test them with situational questions rather than with definitions: they are checking whether you will do the safe thing when someone senior is pressing you to do the fast thing.

> **KEY IDEA**
>
> Every rule below is an instance of one principle: **a trading floor operates on the assumption that everything is recorded, reviewable and attributable.** Calls are taped, chats are archived, orders are timestamped, and surveillance runs across all of it looking for patterns. So the practical test for any borderline action is not whether it is defensible but whether it is *explicable in writing, later, to someone unsympathetic.* If it is not, do not do it, and if you have already done it, escalate it yourself before it is found.

The approved-channel rule

Business must happen on approved, recorded, archived channels. Not on a personal phone, a personal email address, or a consumer messaging app, and not on a channel that deletes messages.

This used to be treated as hygiene and is now enforced ferociously: regulators have imposed very substantial penalties across many major firms specifically for off-channel communications, including against senior people who were otherwise doing nothing wrong on the substance. The reason the enforcement is so hard-edged is that unrecorded communication defeats the entire supervisory model, since surveillance, market-abuse detection and best-execution evidence all assume the record is complete. A gap in the record is therefore a breach in itself, independent of what was said.

The practical rules: move any business conversation onto a recorded channel immediately, keep personal devices out of business use, never use a disappearing-message setting for anything work-related, and if a counterparty or client contacts you off-channel, redirect them and tell your supervisor and compliance rather than quietly complying.

Information: MNPI, walls and restricted lists

Material non-public information is information a reasonable investor would consider important that has not been made public. Trading on it, or passing it to someone who does, is insider dealing. The floor manages this structurally rather than by trusting judgement:

- **Information barriers** separate the parts of the bank that routinely hold MNPI, advisory and origination, from the parts that trade.
- **Wall-crossing** is the controlled, logged process by which someone is brought over the barrier for a specific purpose, which usually means they become restricted from trading the name.
- **Restricted and watch lists** record names in which activity is prohibited or monitored, and they are maintained centrally precisely so that individual judgement is not the control.

The procedure when you receive something you should not have is worth memorising because it happens and there is no time to think: **stop reading, do not forward it, do not act on it, do not discuss it, and tell compliance immediately.** Self-reporting a received leak is a non-event; sitting on it is the career-ending version of the same fact.

Adjacent to this is **client confidentiality**, which is why market colour must be aggregated and anonymised and why a live order is never mentioned, as set out in market color.

Market abuse, concretely

Naming the specific behaviours is more useful than the legal definition, because the definitions are broad and the behaviours are recognisable.

- **Spoofing and layering**: entering orders you intend to cancel in order to create a false impression of supply or demand. Intent is the element, and surveillance infers it from cancellation patterns.
- **Wash trades and matched orders**: trades with no change in beneficial ownership, creating volume that is not real.
- **Marking the close, or banging the fix**: trading into a reference window to move the print that a mark, a coupon or a settlement depends on. This is what much of the benchmark enforcement of the 2010s was about, and it is why fixing windows are now designed and monitored the way they are.
- **Front-running**: trading ahead of a client order for your own or the firm's benefit.
- **Improper disclosure**: telling anyone about a client's order or intention.

> KEY IDEA
>
> **Pre-hedging is not front-running, and the difference is disclosure and consent.** A dealer asked to price a large risk may legitimately begin hedging in anticipation of winning it, because otherwise it could not quote at all, and the client benefits from the tighter price. It becomes abuse when it is undisclosed, when it is done on information the client did not agree to have used that way, or when it is sized or timed to profit from the client's order rather than to manage the risk of it. If you cannot explain your hedging in those terms to the client afterwards, it was the other thing.

The rest of the perimeter, briefly

Personal account dealing is pre-cleared, subject to holding periods and blackout windows, and reported. **Gifts and entertainment** are capped and logged, because the concern is inducement rather than extravagance. **Outside business interests** and **conflicts** are declared. **Mandatory leave** exists so that a book cannot be run indefinitely by one person, which is a control against concealment as much as a wellbeing measure. And **attestations and training** are the mechanism by which the firm can demonstrate you knew the rule, which is why treating them as an annoyance is a mistake.

> KEY IDEA
>
> **The escalation rule, which is the single most useful thing in this section.** Escalate early, escalate in writing, and never resolve a control question alone under pressure. A control function saying no costs the firm a trade; a junior person saying yes to avoid friction costs the firm a regulatory finding and costs that person their job. Interviewers ask situational questions precisely to see whether you know that the correct answer is almost always "I would get the right person involved now, and record what I did", and that saying so is a sign of judgement rather than of timidity.

REAL INTERVIEW QUESTION (BNP · desk not recorded)

What do you do if a counterparty sends you an instant message on your personal messaging?

WHAT THEY TEST / ANSWER

There is a right answer here and it should come out without hesitation. **Do not conduct business on it.** Reply only to redirect: something to the effect that you cannot discuss business on this channel and will pick it up on the recorded firm channel or by phone, then actually move it there. **Preserve rather than delete**, because a record you have already received is potentially evidence and destroying it is far worse than receiving it. **Report it**, promptly, to your supervisor and to compliance, in writing, whether or not anything sensitive was said, since the reporting is what makes it a non-event.

Then give the reasoning, because they are testing understanding rather than obedience: unrecorded communication defeats supervision, market-abuse surveillance and best-execution evidence all at once, so the gap in the record is itself the breach independent of the content, which is exactly why regulators have penalised firms and individuals heavily for off-channel messaging even where the substance was innocuous. Add the two refinements that show you have thought about the practical side. Prevention beats cure, so business contacts belong on business channels and personal devices stay out of business use, and if a client habitually reaches for the convenient channel then it is a relationship conversation to have early rather than a rule to bend repeatedly. And if you realise you have already responded substantively, self-report immediately, because a promptly disclosed mistake is a training matter while a discovered one is a disciplinary matter.

REAL INTERVIEW QUESTION (Optiver · desk not recorded)

On the floor, your manager shouts at you to sell but you want to buy: what do you do and why?

WHAT THEY TEST / ANSWER

Separate two questions that sound like one, because the whole test is whether you can tell them apart. **Whose risk is it, and is this an instruction or an opinion?** If it is the manager's book, or if they hold the mandate and are giving an instruction within it, you execute it: that is what a hierarchy on a trading floor is for, speed matters, and they may hold information or a risk picture you do not. You do not silently substitute your own view for a legitimate instruction on someone else's risk. **Then, immediately afterwards and not during**, say what you think, with your reasoning and briefly: "done, and for what it is worth I liked it the other way for these two reasons". A good desk wants that, and the ability to voice disagreement without theatre is precisely the trait a firm like this is probing.

If instead it is *your* risk and your mandate, and it is an opinion rather than an instruction, then you own the decision and should say so respectfully and act on your own view, because being overridden into a position you do not believe in is how people lose money they cannot explain. Then the boundary that matters more than either case: if the instruction would breach a limit, a restriction, a client obligation or any control, it stops being a hierarchy question and you do not do it, you escalate, however loudly it is delivered, since "I was told to" is not a defence that has ever worked. Close with the tone point, since this is as much a temperament question as a technical one: on a busy floor you comply or you escalate now and you discuss later, and you do not argue at volume in the middle of a market.

INTERVIEW TRAP

Four traps. First, **treating off-channel contact as harmless when nothing sensitive was said**: the gap in the record is the breach. Second, **deleting an inconvenient message**, which converts a reportable minor issue into destruction of evidence. Third, **trying to resolve a control question yourself under pressure** rather than escalating, which is the exact scenario the framework exists to survive. Fourth, **assuming pre-hedging is always fine**: it is legitimate only with disclosure and consent, and sized to manage the risk rather than to profit from the client's order.

VISUAL / CODE TO BUILD: Interactive

A conduct simulator, since this material is learned by deciding rather than by reading. Short scenarios arrive in sequence, each on a timer to reproduce the pressure that is the actual variable: a client messages on a consumer app, a colleague forwards a deck containing something that looks like MNPI, a manager instructs a trade that would breach a limit, a portfolio manager asks what another client has been doing, a large order arrives and you must decide how much to pre-hedge. The reader picks an action and the tool shows the consequence twice, once as the immediate commercial outcome and once as how it reads in a surveillance review six months later, which is the gap the section is teaching. A second panel is a reference card organised by situation rather than by rule, with the escalation path and the exact wording of a redirect message, since knowing the sentence in advance is what makes people actually say it.

Where this goes next

The regulatory framework these obligations come from is MiFID, with the execution-quality strand in best execution and the statutory US programme in the Dodd-Frank Act. The client-facing judgement calls appear again in market color and RFQs, the desk behaviour that pre-hedging sits inside is market making, and conduct as a risk category with capital attached is operational risk. How these situational questions are graded in interviews is covered in behavioral signals desks look for.

24.9 Best execution

sales · trading · structuring | core | cross-asset

Best execution is the obligation to take all sufficient steps to obtain the best possible result for a client when executing an order. Two things about that sentence are routinely misread, and getting both right is most of what an interviewer is checking.

KEY IDEA

Best execution is not best price, and it is not a promise about the outcome.

It is not best price because the obligation is multi-factor: price, costs, speed, likelihood of execution and of settlement, size, nature of the order, and anything else relevant. A fill at a slightly worse price that actually completes can be the better result than a better price that leaves half the order unexecuted.

It is not about outcome because no firm can guarantee it obtained the best available price on any given trade. The obligation is one of **process**: have a policy, follow it, monitor whether it is working, and be able to demonstrate all three. This is why the regime consists of policies, monitoring and evidence rather than of price guarantees, and why a firm that got a poor price on one trade has not necessarily breached anything while a firm with no monitoring has, even if every one of its fills happened to be excellent.

The factors are not weighted equally for everyone. For a retail client the rule is explicit that **total consideration**, the price plus all costs, is the dominant factor, because a retail client is not in a position to trade off speed against price knowingly. For a professional client the weighting can legitimately differ: an asset manager unwinding a large position may care far more about market impact and certainty of completion than about the touch price, and a firm's policy is allowed to say so provided it does say so.

The question a bank markets desk actually faces

On an agency order the obligation is straightforward: you are handling someone else's order and you route it as well as you can. The harder case, and the one that applies to most of a derivatives or fixed income desk's business, is a client asking for a price that the desk will fill from its own book.

INTUITION

When a bank quotes a price on its own account, the client is not giving it an order to work, they are negotiating a transaction. So does best execution apply at all? The regulatory answer is that it depends on whether the client **legitimately relies** on the firm, and the test looks at four things: who initiated the transaction, whether market convention is for clients to shop around, how transparent prices are in that market, and what the firm told the client.

The practical reading is intuitive once stated. A client who calls three banks for a quote on a liquid instrument is shopping around and relying on competition, not on any one dealer. A client who accepts a quote on a bespoke structure they cannot price independently, from the only dealer they asked, is relying on the firm entirely. **The obligation tracks the client's ability to check you**, which is a sensible principle rather than an arbitrary one, and it is why the RFQ process matters commercially as well as legally. That process is RFQs, and the margin inside the quote is client pricing.

Measuring it, and why the benchmark decides the verdict

Execution quality is measured against a benchmark, and the choice of benchmark does more work than the execution does. The two that matter are the **arrival price**, the market price when the order was received, and **VWAP**, the volume-weighted average price over the execution window.

WORKED EXAMPLE

An order to buy 500,000 shares arrives when the market is at 100.00. The order is worked through the day and fills at an average of 100.12. The day's VWAP turns out to be 100.15.

Against arrival price, the execution cost $(100.12 - 100.00)/100.00 = \mathbf{12}$ bp, or $500{,}000 \times 0.12 = \$60{,}000$. That is the implementation shortfall: the money lost between deciding to trade and being done.

Against VWAP, the execution beat the benchmark by $(100.15 - 100.12)/100.00 = \mathbf{3}$ bp, or \$15,000 better than average.

Same trade. One report says you cost the client \$60,000, the other says you saved them \$15,000. Neither is wrong; they answer different questions. Arrival price measures the full cost of the decision to trade, including the market impact you caused. VWAP measures whether you traded better or worse than the day's average participant, and deliberately excludes the drift you may have created.

INTERVIEW TRAP

VWAP has a specific defect worth knowing: **a large order moves the VWAP towards itself**. If your order is a substantial share of the day's volume, your own fills dominate the average you are being measured against. Write your share as f: your measured slippage against VWAP works out to $(1 - f)$ times the gap between your price and everyone else's, so as f grows the number is squeezed toward zero whatever you actually did. You are being marked against yourself, and the comparison stops carrying information. That is precisely the case, a large order in a thin name, where execution quality matters most. Arrival price does not have this problem, which is why implementation shortfall is the benchmark a serious buy-side desk uses, and VWAP is the one most often quoted to clients.

MiFID II raised the standard from taking *reasonable* steps to taking *sufficient* steps and added public reporting: venues publishing execution quality statistics and firms publishing their top five execution venues by class. Those reports were widely judged to be unread and have since been scaled back in both the EU and the UK, so state that the reporting existed and that you would check its current status rather than describing it as live. The framework it sits in is MiFID.

REAL INTERVIEW QUESTION (BNP · Structured Products)

A client calls you, nobody else is on the desk, and they absolutely want to trade. What do you do?

WHAT THEY TEST / ANSWER

This is a judgement question, and the wrong answers are equally at both extremes: refusing to engage, and doing the trade because the client insisted. Structure the answer as what you can do now, what you cannot, and how you close the gap.

Start with the client, because the obligation runs to them. Take the enquiry properly: what they want, size, direction, urgency, and whether there is a hard deadline or whether absolutely wanting to trade means within the hour. A great deal of apparent urgency turns out to be a preference once you ask, and knowing which one it is determines everything that follows.

Then be clear about the boundary, and do not blur it. If pricing or committing risk is outside your mandate, it is outside your mandate, and the absence of colleagues does not extend it. Trading beyond your authority is not helpfulness, it is the error that ends careers, and an interviewer asking this question is checking that you know that without needing to be told. The mandate itself is trading floor compliance.

Then do the things that are genuinely yours to do. Escalate immediately, by whatever route the desk has for exactly this, since every desk has one: a cover trader, another desk, the head of desk on a mobile. Tell the client precisely what you are doing and when you will come back to them, then come back when you said you would even if the answer is not ready. Nothing damages a relationship like silence, and being told "I do not have a price yet, I will call you in twenty minutes" is not a bad outcome for a client.

Then the point that lifts the answer. Say that acting in the client's best interest and staying inside your authority are not in conflict here, and that treating them as a conflict is the trap. Executing a trade you cannot properly price is not a service to the client: an unhedgeable or mispriced fill is worse for them than a short wait, because they carry the consequence. **The client's interest is a good execution, not a fast one**, and where the two genuinely diverge, that is a conversation to have with them explicitly rather than a decision to make on their behalf.

Finish by saying you would write up what happened afterwards, so that the gap in cover becomes a known problem rather than a story. That is the answer of someone who has thought about the second time it happens.

INTERVIEW TRAP

Three errors. (1) Defining best execution as best price. It is multi-factor, and for a professional client speed or certainty of completion can dominate. (2) Treating it as a guarantee about a single trade. It is an obligation of process, evidenced by policy, monitoring and records. (3) Assuming it applies identically to a quote on own account and to an agency order. Whether it applies at all turns on legitimate reliance, and the four-fold test is the part of this subject a markets desk actually uses.

Where this goes next

The framework this obligation sits in is MiFID, and the desk-level conduct rules around mandates and escalation are trading floor compliance. The process by which a client puts dealers in competition is RFQs, the margin inside the price quoted is client pricing, and the cost the client pays before any of this is the bid-ask spread. The structure that determines what a good execution even looks like is market microstructure, and the technology used to achieve it is low-latency trading.

Chapter 25

Programming

25.1 Python for finance

trading · structuring · sales core cross-asset

Nobody on a front-office desk is hired to be a programmer, and almost everybody is expected to write code. The distinction matters for what you should learn and for what an interview will actually ask: not software engineering, but the ability to turn a quantitative idea into a short, correct program, and to say how long it will take to run.

What Python is used for, and what it is not

- **Prototyping and analysis.** Testing a pricing idea, checking a calibration, plotting a surface, pulling a time series and looking at it. This is the bulk of it.
- **Automation.** Reports, reconciliations, term-sheet generation, anything done by hand more than twice.
- **Model validation and research**, where the point is clarity rather than speed.

What it is usually *not* is the production pricing library, which is C++ for latency and for the discipline of a compiled language, in intro to C++ for finance, nor the tool the whole desk actually lives in, which is still Excel, in advanced Excel and VBA. A candidate who claims Python has replaced the spreadsheet has not sat on a desk.

KEY IDEA

An interview is not testing whether you can code, it is testing three things in order: whether your program is **correct**, whether you can state its **complexity**, and whether you know what would **break it**. Write the simplest thing that works, say its complexity unprompted, then name the edge cases. Candidates lose this question by optimising before they are correct, and by going silent while they think.

The idioms worth having

Python's value on a desk comes from writing very little. Three habits cover most of it.

Comprehensions instead of loops that build lists:

```
evens = [x for x in numbers if x % 2 == 0]
```

The right container. A list for ordered data, a dict for lookups, a set for membership tests. Checking membership in a list is a linear scan and in a set is constant time, which is the difference between a script that finishes and one that does not on a large book.

Vectorisation rather than element-by-element arithmetic, which is the single largest performance idea in numerical Python and belongs to NumPy and Pandas.

REAL INTERVIEW QUESTION (Natixis · QIS)

In Python, I give you a list of integers: return the list of even numbers.

WHAT THEY TEST / ANSWER

One line, and then the discussion, because the line is not what is being tested.

```
[x for x in numbers if x % 2 == 0]
```

Say why a comprehension rather than a loop: it is shorter, it is faster than appending in a Python loop, and it reads as the specification. Then volunteer the variants, since the follow-up is always one of them. If the input is large and you only need to iterate once, use a **generator**, `(x for x in numbers if x % 2 == 0)`, which does not build the list in memory. If the data is numerical and already in an array, use a **boolean mask**, `arr[arr % 2 == 0]`, which is far faster and is the answer they want if the seat is quantitative.

Then name the edge cases, which is what separates a careful candidate: **negative numbers** are fine here, since `-4 % 2` is 0, and the related trap worth naming is that Python's modulo takes the sign of the *divisor*, so `-3 % 2` is 1 where C-like languages give -1, which is why testing oddness with `x % 2 == 1` is safe in Python and broken elsewhere; **non-integer inputs** will either raise or give a surprising answer and should be validated or excluded; and an **empty list** returns an empty list, which is correct and worth saying out loud rather than discovering.

Complexity, out loud

REAL INTERVIEW QUESTION (Société Générale · EQD)

Give me the code to compute the Fibonacci sequence in Python, and its complexity.

WHAT THEY TEST / ANSWER

Give both versions, because the comparison is the question.

The **naive recursion** is the one people write first:

```
def fib(n):  return n if n < 2 else fib(n-1) + fib(n-2)
```

It is correct and it is a trap: each call spawns two more, so the number of calls grows like φ^n with φ the golden ratio, bounded above by $O(2^n)$. At $n = 40$ it is already slow, and at $n = 60$ it will not finish. Space is $O(n)$ for the call stack.

The **iterative** version is what you should actually write:

```
def fib(n):
    a, b = 0, 1
for _ in range(n):
    a, b = b, a + b
    return a
```

$O(n)$ time and $O(1)$ space, and in Python the tuple assignment does the swap without a temporary variable, which is the idiomatic touch worth showing.

Then give the two refinements that earn the follow-up. **Memoisation** turns the recursion into $O(n)$ time at the cost of $O(n)$ space, one decorator line, and it is the general lesson rather than a trick about this sequence: exponential recursions are almost always recomputing the same subproblems. And there is a **closed form**, Binet's formula, which is $O(1)$ but loses exactness to floating point for large n, so it is the right answer to "can you do better" and the wrong one to "give me the value".

Close by naming the trap in the question itself: `fib(0)` and `fib(1)` must be handled, and the interviewer's convention for where the sequence starts should be confirmed before you write anything.

Numerical building blocks

Four small programs cover most of what a desk actually asks a junior to write: a pricing formula, an integral, a root finder and a simulation.

```
from math import log, sqrt, exp, erf

def norm_cdf(x):
    return 0.5 * (1.0 + erf(x / sqrt(2.0)))

def bs_call(S, K, T, r, sigma):
    d1 = (log(S / K) + (r + 0.5 * sigma ** 2) * T) / (sigma * sqrt(T))
    d2 = d1 - sigma * sqrt(T)
    return S * norm_cdf(d1) - K * exp(-r * T) * norm_cdf(d2)
```

Note what is not there: no library beyond the standard one, and `erf` rather than an approximation, because the standard library already has it. Test it against a value you know, such as the at-the-money rule of thumb from the Black-Scholes model, before trusting it.

REAL INTERVIEW QUESTION (Société Générale · Delta One)

Code a left Riemann sum in Python.

WHAT THEY TEST / ANSWER

Write it, state the error, then say what you would use instead.

```
def left_riemann(f, a, b, n):
    h = (b - a) / n
    return h * sum(f(a + i * h) for i in range(n))
```

Point out the choices as you write. The generator inside `sum` avoids building a list of n values. The loop runs from 0 to $n-1$, which is what makes it the *left* rule; ending at n instead would be the right rule and is the classic off-by-one.

Complexity and accuracy. $O(n)$ time, $O(1)$ space. The error of the left rule is $O(h)$, first order in the step size, so halving h only halves the error. Say the alternatives and their orders, because that is the substance: the **midpoint** and **trapezoid** rules are $O(h^2)$ and cost the same number of evaluations, and **Simpson's** rule is $O(h^4)$. There is no reason to use the left rule in production, which is why the question is really about whether you know that.

Finish with what a desk would actually do: for a smooth one-dimensional integral, call `scipy.integrate.quad`, which is adaptive and returns an error estimate; for a high-dimensional one, integration by quadrature is hopeless and you simulate instead, as in convergence concepts. Knowing which regime you are in is the actual skill.

Talking about your own code

REAL INTERVIEW QUESTION (BNP · Trading)

Explain a Python project you have worked on.

WHAT THEY TEST / ANSWER

Structure it as problem, approach, result, limitation, and keep the code detail proportionate to how technical the interviewer is.

Problem: what was being done by hand or not at all, and why it mattered. A project without a business reason sounds like a course exercise. **Approach**: the data, the method and the two or three real decisions you made, which is where the interviewer will probe. Say why you chose an approach over the obvious alternative. **Result**: something concrete. Minutes saved, an error caught, a calibration that converged where the previous one did not, a number a desk actually used. **Limitation**: what it did not handle and what you would do differently. This is the part that makes the rest credible.

Three rules. Claim only what you **personally wrote**, since the second question is always about a specific choice and the answer collapses if the work was somebody else's, as in deal-breakers. Be ready to **go one level deeper** on any component you name, so do not mention a library you have only imported. And prefer a **small project you fully understand** to a large one you contributed to vaguely, because the interviewer is measuring depth rather than scope.

Habits that make code trustworthy

- **Test against something known** before anything else: a closed-form value, a limiting case, a symmetry, put-call parity. A pricer that has never reproduced a known number is not a pricer.
- **Check the limits.** Zero volatility, zero time, very deep in and out of the money. Most numerical bugs live at the boundary, and `sigma * sqrt(T)` in a denominator is the classic division by zero.
- **Seed the randomness** in any simulation you want to reproduce or compare.
- **Watch the floating point.** Never test two floats for equality, and be aware that summing many small numbers loses precision.
- **Keep it readable.** The reader is a colleague who has to trust the number, and code nobody can check does not get used.

INTERVIEW TRAP

Four interview errors. (1) Optimising before being correct: write the simple version, then say how you would speed it up. (2) Not stating the complexity unprompted, which is half the mark on any coding question. (3) Writing a Python loop over a large array when the answer is vectorisation. (4) Claiming a level of proficiency you cannot defend, since the follow-up is always a specific question and this is the cheapest deal-breaker to avoid.

Where this goes next

The array and dataframe layer that most desk work actually uses is NumPy and Pandas, the ready-made pricing tools are pricing libraries, and the data plumbing is financial data pipelines.

The languages a desk uses alongside it are VBA, SQL and C++, and the simulation methods these programs implement are in convergence concepts.

25.2 NumPy, Pandas

trading · structuring · sales | core | cross-asset

These two libraries are the reason Python displaced spreadsheets and specialist numerical languages on trading floors. NumPy provides a fast typed array; pandas provides that array with labels and a time index attached. Almost every piece of analysis a junior does, and a large share of what runs in production, is one of those two objects being operated on.

The Python language itself is Python for finance. This section is about the two libraries, why they are fast, and the specific ways they go wrong.

NumPy: one type, one block of memory

A Python list holds pointers to boxed objects scattered across memory, each carrying its own type information. A NumPy `ndarray` holds raw values of one type in one contiguous block. That single difference produces everything else.

- **Memory**: a million doubles in an ndarray occupy exactly 8 million bytes. The equivalent Python list of floats takes roughly four times that once the pointers and the object headers are counted.
- **Speed**: an operation on an array runs a single compiled loop over that block, so the per-element interpreter overhead disappears. That is what "vectorised" means. It is not that the loop is gone, it is that the loop is no longer in Python.
- **Broadcasting**: operations between arrays of compatible shapes work without writing the loop or copying data to match dimensions.

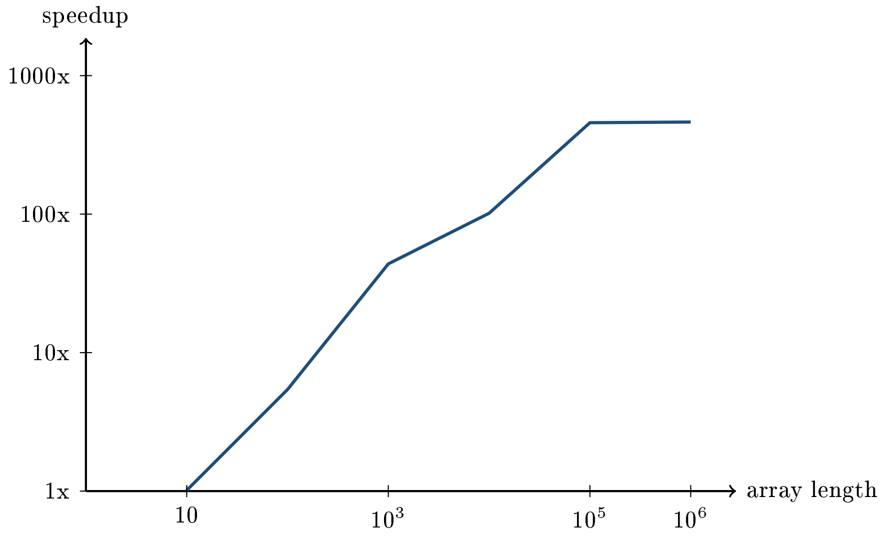
WORKED EXAMPLE

Summing the squares of a million floats, with `a = np.arange(1_000_000) / 2`, best of several runs on the machine this section was written on:

Method	Time	Speedup
Python generator with `sum()`	23.0 ms	1x
NumPy `(a*a).sum()`	1.03 ms	22x
NumPy `a.dot(a)`	0.05 ms	440x

Two things to take from it. The gap between the two NumPy forms is as instructive as the gap to Python: `(a*a)` allocates a second million-element array before summing it, while `dot` does the multiply and the accumulate in one pass with no allocation. Knowing that intermediate arrays cost memory bandwidth is most of practical NumPy performance.

And the vectorised version is not only faster, it is **more accurate**. The exact answer here is 83,333,208,333,375,000, whose nearest double is 83,333,208,333,375,008. The Python loop, accumulating strictly left to right, has a relative error of about 1.1×10^{-12}; `a.dot(a)` has 1.9×10^{-16} and `(a*a).sum()` is exact to double precision. NumPy sums in blocks rather than sequentially, so the running total never becomes large relative to the terms being added to it. Faster and more accurate is an unusual trade to be offered, and it is worth knowing it is on the table.



Both axes are logarithmic. The useful part is the left-hand end: at ten elements NumPy is no faster at all, because the fixed cost of calling into it swamps the work. Vectorising is worth doing on arrays, not on scalars, and a loop that calls a NumPy function on three numbers at a time is slower than the plain Python it replaced.

Pandas: the array plus its labels

A pandas `Series` is a NumPy array with an index. A `DataFrame` is a set of Series sharing one index. Everything distinctive about pandas comes from that index, and so does everything dangerous.

KEY IDEA

Pandas aligns on the index automatically, on every operation. Adding two Series adds the values that share a label, not the values that share a position. This is the single most useful thing about the library and the single most common source of silent errors, and the two are the same feature.

WORKED EXAMPLE

Two Series with overlapping but different indices:

`a`	Mon 1.0, Tue 2.0, Wed 3.0
`b`	Tue 10.0, Wed 20.0, Thu 30.0

`a + b` gives `Mon NaN, Tue 12.0, Wed 23.0, Thu NaN`. Only the shared labels combine, and the union of the indices is returned with missing entries marked.

Now the dangerous version. `a.values + b.values` strips the labels and adds by position, giving `[11, 22, 33]`. Every one of those numbers is wrong, because Monday has been added to Tuesday, and nothing anywhere reports an error. If two price series have different holiday calendars, this is how a correlation gets computed on data shifted by a day.

The operations that matter in finance

A small set of pandas verbs covers most of what a desk does with data.

- `pct_change` and `diff`: prices into returns, which is the first step in nearly everything, for the reasons in time series analysis.
- `shift`: move a series in time. The most important and most misused function in the library, discussed below.
- `rolling` and `ewm`: moving windows and exponentially weighted moving averages, which is where realised volatility and correlation estimates come from.
- `resample`: change frequency, daily to weekly to monthly, which needs a real datetime index to work.
- `groupby`: split, apply, combine. Aggregation by name, by sector, by maturity bucket.
- `merge` and `join` against `concat`: the first two combine on keys and the third stacks on an axis, and using the wrong one is a classic way to silently duplicate rows.

Two traps worth naming

The shift direction. A signal computed from today's close cannot be traded at today's close. Whether the code says `signal.shift(1) * returns` or `signal * returns` is the difference between a backtest and a fiction, and the second version will show excellent results. Any backtest whose Sharpe ratio looks implausible should be checked here first, before anything else. This is the look-ahead problem in backtesting frameworks, arriving as one character of code.

Missing data is a modelling decision. `dropna`, `fillna(0)` and `ffill` encode three different assumptions about what the gap means, and they are not interchangeable. Forward-filling a *price* across a holiday is defensible, since the instrument did have a value that day and nobody traded it. Forward-filling a *return* invents a zero return that never happened and quietly lowers the measured volatility. Dropping rows is safe for the statistics and dangerous for alignment, because it changes the index and the next operation will align against the shortened version.

Interview angle

REAL INTERVIEW QUESTION (Société Générale · Delta One)

What is NumPy?

WHAT THEY TEST / ANSWER

Answer in one sentence, then justify it with the reason it is fast, because everyone gives the sentence and few give the reason.

The sentence: NumPy is Python's numerical array library, providing a fixed-type, fixed-size, contiguous n-dimensional array and a large set of operations that act on the whole array at once.

Why it is fast, which is the actual question. A Python list is a sequence of pointers to individually boxed objects, so iterating it means chasing pointers and re-checking types a million times in the interpreter. A NumPy array is one block of raw doubles, so an operation runs a single compiled loop over contiguous memory. Vectorisation does not remove the loop; it moves it out of Python. Say the numbers if you have them: on a million elements, summing squares takes about 23 milliseconds in a Python loop and about 0.05 milliseconds with `dot`, so a factor of several hundred, and the array uses about a quarter of the memory.

Two additions that separate a real user from a reader. First, vectorising is worth it on arrays and counterproductive on scalars: at ten elements NumPy is no faster, because the call overhead dominates. Second, intermediate arrays are not free, so `a.dot(a)` beats `(a*a).sum()` by an order of magnitude simply because it never allocates the product.

And one thing almost nobody says: vectorised summation is usually more *accurate* than a Python loop, not less, because NumPy accumulates in blocks rather than strictly left to right, so the running total never dwarfs the terms being added. On the example above the loop carries a relative error around 10^{-12} and `dot` around 10^{-16}.

If they push on what you use it for: array arithmetic on prices and returns, linear algebra for covariance matrices and portfolio weights, and random number generation for simulation, which is Monte Carlo methods.

REAL INTERVIEW QUESTION (Société Générale · Delta One)

What is pandas?

WHAT THEY TEST / ANSWER

Define it against NumPy, because that is the relationship and it makes the answer precise rather than a list of features.

The definition: pandas is a data manipulation library built on NumPy. A Series is a NumPy array with an index; a DataFrame is a set of Series sharing that index. What pandas adds is labels, heterogeneous column types, and a large set of operations built around a time index.

What it is for: everything between the data and the analysis. Reading files and feeds, joining sources, changing frequency, computing rolling statistics, grouping and aggregating. On a desk it is what turns a price history into returns, a set of returns into a covariance matrix, and a trade file into a position report.

The feature to name, because it is what an interviewer is listening for: **automatic alignment on the index**. Operations combine values by label, not by position. That is why joining two price series with different holiday calendars works correctly and returns NaN where one is missing, and it is also the biggest trap in the library, because dropping to `.values` silently reverts to positional arithmetic and pairs Monday with Tuesday without any error.

The finance-specific verbs worth naming: `pct_change`, `shift`, `rolling`, `resample` and `groupby`. If you name only one, name `shift`, and say why: a signal from today's close cannot be traded at today's close, and the difference between shifting and not shifting is the difference between a backtest and a fiction.

If asked about limits, be honest: pandas is single-threaded, holds everything in memory, and copies more than people expect. Past a few gigabytes the answer is a database or a columnar engine rather than a bigger machine, which is financial data pipelines and SQL.

REAL INTERVIEW QUESTION (Société Générale · Index)

How do you remove NaN values from a pandas dataset?

WHAT THEY TEST / ANSWER

Give the mechanics in one line and then refuse to stop there, because the question looks like syntax and is actually about judgement.

The mechanics: `dropna` removes them, with `axis` and `how` controlling whether rows or columns go and whether any or all missing values trigger it; `fillna` replaces them with a constant, a computed value, or the previous observation via `ffill`; `interpolate` fills between known points.

Then the real answer: which one is correct depends entirely on *why the value is missing*, and there are three different reasons that look identical in the data.

A **market holiday**: the instrument existed and had a value, nobody traded. Forward-fill is defensible for a price. It is not defensible for a return, because it invents a zero return that never happened and lowers your measured volatility.

A **missing observation** in an otherwise live series, from a feed failure or a bad tick. Interpolation may be reasonable if the gap is short, and if it is long the honest answer is to exclude the period rather than manufacture it.

A **genuine absence**: the instrument did not exist yet, or the company had not listed. Filling this is always wrong, and it is the one that causes real damage, because filling pre-listing data with anything creates a backtest that traded something unavailable.

Two operational points to add. Dropping rows changes the index, so the next operation aligns against the shortened version, which is how a dropna in one place produces a silent misalignment three lines later. And always count what you removed and report it: a cleaning step that deletes 30% of the sample without anyone noticing is a bigger problem than the NaNs were.

INTERVIEW TRAP

Four errors. (1) Using `.values` or `.to_numpy()` to combine two Series, which discards the alignment that was protecting you. (2) Looping over a DataFrame with `iterrows` when a vectorised expression exists, which is typically two orders of magnitude slower and usually less readable. (3) Getting the `shift` direction wrong, which produces a backtest that works and a strategy that does not. (4) Choosing a NaN treatment by convenience rather than by what the gap means.

Where this goes next

Getting the data in and keeping it clean at scale is financial data pipelines, and the query language for when it no longer fits in memory is SQL. The place where the `shift` trap does the most damage is backtesting frameworks, the statistical reasoning behind the transformations is time series analysis, and the heaviest NumPy workload on a desk is Monte Carlo methods.

25.3 Pricing Libraries

trading · structuring | advanced | cross-asset

A pricing library is the piece of software that turns a trade and a set of market data into a price and a set of Greeks. Every bank has one, it is usually decades old, and it is the single most consequential piece of code on the floor, because the trading desk, the risk function and the finance function all price the same book with it and are required to agree.

This section is about the *engineering*, which is a different subject from the mathematics. The numerical methods themselves belong to Monte Carlo methods and PDE methods; what is here is how a library is put together and how it decides which of them to use.

The separation that makes a library possible

KEY IDEA

The design that every serious library converges on is a three-way separation: the **instrument** says what the payoff is, the **model** says what the dynamics are, and the **method** says how the expectation is computed. The three are orthogonal, so an autocallable prices under Black-Scholes or under Heston, by Monte Carlo or by PDE, without any of those four things knowing about the others.

The reason this matters is combinatorial. With N instruments, M models and K methods, a library that keeps them separate needs $N+M+K$ pieces of code; one that does not needs $N \times M \times K$. Every legacy library that has become unmaintainable got there by writing a pricer per product.

Two consequences fall out immediately. Adding a new payoff should not require touching a model, which is what lets a structuring desk quote something new the same week. And a new model should price the whole existing book on the day it is added, which is what makes a model change reviewable rather than terrifying.

Choosing the engine

REAL INTERVIEW QUESTION (Axa IM · Credit)

Pricing by finite differences: compare it with Monte Carlo pricing.

WHAT THEY TEST / ANSWER

Answer on the axes that decide which one a library actually calls, in order of how often they decide it.

Dimensionality is the first and usually the last question. A finite difference scheme discretises the state space, so its cost grows exponentially with the number of factors: one or two factors is routine, three is painful, four is not done. Monte Carlo's cost grows roughly *linearly* in the number of factors, because an extra factor is one more random number per step. So a single-asset barrier goes to the PDE and a worst-of on five underlyings goes to Monte Carlo, and that is not a preference, it is the only thing that runs.

Early exercise is the second. A PDE steps backwards in time, so at every node you already know the continuation value and comparing it against immediate exercise is free. That makes American and Bermudan options natural on a grid. Monte Carlo runs forwards, so the continuation value is not known, and you need a regression method such as Longstaff-Schwartz to estimate it, which works but adds a bias you have to control.

Path dependence is the third, and it runs the other way. Asians, lookbacks and anything with an observation schedule are trivial in Monte Carlo, since you simply record what you need along the path, and awkward on a grid because the payoff depends on history the grid does not carry, so you must add a state variable and pay the dimensional cost.

Accuracy and speed. A PDE converges smoothly in the grid spacing, so you can extrapolate and you get a deterministic answer: run it twice and get the same number. Monte Carlo converges as $1/\sqrt{N}$, so halving the error needs **four times** the paths, and the answer has a standard error attached to it.

Greeks, which is the practical difference nobody mentions. A PDE gives delta and gamma almost free, by finite differencing the solution grid you already computed. Monte Carlo Greeks by bumping are noisy unless you reuse the random numbers across the bumped and unbumped runs, and that detail, using the same seed, is the thing an interviewer is checking for.

So, in one line: PDE for low dimension and early exercise, Monte Carlo for high dimension and path dependence, closed form whenever it exists. A production library implements all three and routes each trade to the cheapest one that is correct for it.

WORKED EXAMPLE

What $1/\sqrt{N}$ costs in practice. Suppose a payoff has a standard deviation across paths of 15 on a price near 8, which is typical for an option with real optionality. The standard error of a Monte Carlo estimate is $s/\sqrt{N}$, so:

Target standard error	Paths required
0.05	90,000
0.01	2,250,000
0.001	225,000,000

Each extra decimal place costs a hundred times the compute. This is the reason variance reduction is not an optimisation but a necessity: a control variate that cuts the payoff's variance by a factor of ten cuts the required paths by ten, which is worth more than any amount of faster hardware. It is also why a desk quoting a price to a client to the nearest basis point is making a statement about its Monte Carlo configuration, not only about its model.

Greeks are the expensive part

A price is computed once. A risk report needs a sensitivity to every market input, and this is where the compute budget actually goes.

The naive approach is **bump and revalue**: perturb each input, reprice, difference. With n risk factors this costs $n + 1$ valuations, and a cross-asset book has hundreds of factors once you count every curve tenor and every volatility surface point.

Adjoint algorithmic differentiation computes all n sensitivities in a fixed multiple of the cost of a single valuation, typically around four times, *independently of n*. For a book with 500 risk factors that is 501 valuations against roughly 4, a speed-up of about **125 times**, and it is the reason AAD went from a research topic to a production requirement once regulators started asking for daily sensitivities on entire books.

The catch is worth knowing because it is the follow-up: AAD requires the whole valuation to be differentiable code, so it is invasive. A library retro-fitted with it is a rewrite, not a feature, which is why banks with old libraries still bump.

Compiled core, scripted skin

What a compiled language is, how it differs from an interpreted one and why the line is blurrier than the textbook version are treated in introduction to C++ for finance, which owns that distinction. What belongs here is the consequence, because a trading floor does not choose between the two, it *layers* them.

The pricing library's numerical core is C++, because a Monte Carlo running a hundred million paths in an interpreted loop is not a product. The interface is Python, because the people using it are changing what they ask for several times an hour. Between them sits a binding layer, and the rule that makes the arrangement work is that *nothing numerical is written above the boundary*: Python assembles trades, market data and requests and reads results back, while every loop over paths or grid nodes lives underneath.

That rule comes with its own diagnostic, which is the practical form of the whole idea: if you find yourself writing a `for` loop over paths in Python, you have put the code on the wrong side of the boundary, and the fix is to express it as an array operation or to push it into the compiled layer.

What actually goes wrong

The failures in production libraries are almost never the mathematics.

Two numbers for one trade. The front office prices with one configuration and risk with another, and the difference is real money in the P&L explain. Reconciling them is a large fraction of what a quant development team does, and preventing it is why the same library, not merely the same model, must serve both.

Irreproducibility. A price is a function of the trade, the market data, the model, the model's calibration and the library version. If any of those is not stamped alongside the price, the number cannot be reproduced next week, which matters enormously the first time a valuation is disputed. Calibration is the input people most often forget to version.

Silent model fallback. A library that quietly falls back to a simpler model when calibration fails will produce a plausible number for a trade nobody realises is unpriced. Failing loudly is a feature.

Performance cliffs. A change that is harmless on a vanilla book makes the overnight risk run miss its deadline, and a risk report that arrives after the desk starts trading is not a risk report.

INTERVIEW TRAP

1. **Writing a pricer per product.** It is faster for the first three products and unmaintainable by the thirtieth. The separation of instrument, model and method is the whole design.
2. **Bumping Monte Carlo Greeks without fixing the seed.** The difference between two noisy estimates is dominated by the noise. Reuse the random numbers.
3. **Treating a Monte Carlo price as exact.** It has a standard error, and quoting more digits than that error supports is a fiction. Report the error alongside the price.
4. **Loops over paths in an interpreted language.** Two orders of magnitude of performance are on the wrong side of a boundary.
5. **Not versioning the market data and the calibration.** A price you cannot reproduce is a price you cannot defend, and disputes arrive months later.
6. **Choosing the engine by preference rather than by the trade.** Dimensionality, early exercise and path dependence decide it, and a library that routes automatically on those three is doing what a good one should.

VISUAL / CODE TO BUILD: Interactive

A convergence panel that runs both engines on the same contract. The reader picks a payoff, a vanilla call, an American put, a barrier, an Asian, a worst-of on a chosen number of underlyings, and the panel plots price against compute budget for Monte Carlo and for the PDE on the same axes, with the Monte Carlo line drawn as a band of plus and minus two standard errors that narrows as $1/\sqrt{N}$.

The intended sequence is: on a vanilla, watch the PDE converge to a clean line while the Monte Carlo band is still visibly wide; switch to an Asian and watch the PDE line disappear because the state variable has been added; then raise the number of underlyings and watch the PDE cost explode while the Monte Carlo cost grows gently. Three drags, and the routing rule is derived rather than asserted.

A second panel contrasts Greeks by bumping against adjoint mode: a slider sets the number of risk factors and two bars show total valuations required, one growing linearly and one flat at about four. The crossover is immediate, which is the point.

25.4 Backtesting frameworks

trading · structuring · sales | advanced | cross-asset

A backtest is a simulation of a decision rule against historical data. It is the cheapest way to reject a bad idea and the easiest way to convince yourself of a false one, which is why the engineering of a backtesting framework is mostly the engineering of *defences against yourself*. Interviewers on systematic, QIS and index desks ask about it because the distinction between someone who has run one and someone who has read about one is immediately audible: the first thing an experienced person mentions is a bias, and the first thing an inexperienced one mentions is a Sharpe ratio.

KEY IDEA

One word, two meanings, and confusing them is an easy own goal. **Strategy backtesting** asks whether a trading rule would have made money, which is what this section is about. **Model backtesting** asks whether a risk model's predictions were honest, for example counting how many days a loss exceeded the reported VaR, which is covered in Value at Risk. If an interviewer says "backtesting" on a risk desk, they usually mean the second.

Anatomy of a framework

Any serious framework has the same six components, and being able to name them is a good answer to "how would you build one".

1. **Data layer.** Prices, corporate actions, index membership, reference and fundamental data, all **point-in-time**: stored as it was known on each date, not as it was later revised. This layer is where most backtests are silently ruined.
2. **Signal.** The rule that turns data into a desired exposure. Must be a pure function of information available strictly before the decision time.
3. **Portfolio construction.** Turning signals into positions: sizing, risk targeting, constraints, neutralisations (sector, beta, currency), and rebalancing schedule.

4. **Execution and cost model.** When and at what price you are assumed to trade, including spread, fees, borrow, slippage and market impact. The most under-specified part of most backtests and, as shown below, often the only part that matters.
5. **Accounting.** Positions, cash, financing, dividends, P&L, and exposure, with the same conventions your production books use. Unit-test this, because a bug here produces a plausible and completely fictitious equity curve.
6. **Metrics and reporting.** Returns, risk, drawdowns, turnover, capacity, and attribution, plus the diagnostics that let someone else attack the result.

Two architectures. A **vectorised** backtest computes signals and returns as whole arrays, which is fast and ideal for research sweeps, but it makes look-ahead bias easy to introduce and cannot represent order-level logic. An **event-driven** backtest walks forward through time delivering one event at a time to the strategy, which is slower but structurally prevents peeking and can model queues, partial fills and path-dependent rules. Serious desks use both: vectorised to screen ideas, event-driven to validate the survivors with the same code path that will run in production.

The biases, in the order they cost money

- **Look-ahead bias.** Using information that was not available at the decision time. The classic instances are trading at a close you used to compute the signal, using a restated fundamental figure at its original date, and joining a table on a report date rather than a publication date. The defence is structural rather than diligent: build the data layer so that a query for date t physically cannot return anything stamped after t.
- **Survivorship bias.** Testing on the names that still exist. Universes must include delistings, defaults and mergers, with their terminal values, or every equity strategy looks brilliant and every credit strategy looks riskless.
- **Point-in-time membership.** Index constituents change. Backtesting today's index members through history is survivorship bias with a different hat, and it is the standard flaw in an index-strategy backtest.
- **Transaction costs and capacity.** A cost assumption is a forecast, and at high turnover the entire result is a bet on it. Capacity is the same issue in size: impact grows with the fraction of volume you take, so a strategy that works at €10m may not exist at €500m.
- **Data snooping and overfitting.** The deepest problem, because it survives every other fix. If you try enough rules, one of them will look excellent by chance.

WORKED EXAMPLE

How impressive is a Sharpe ratio, really? Two calculations make the snooping problem concrete.

First, the significance of a single Sharpe ratio. Over T years the t statistic of an estimated Sharpe ratio is approximately

$$t \approx SR \times \sqrt{T},$$

so a Sharpe of 1.0 over four years gives $t = 2.0$, which is only marginally significant. Anyone quoting a Sharpe without a sample length is quoting half a number.

Second, and far more important, the effect of *searching*. The expected maximum of N independent standard normal draws grows like $\sqrt{2 \ln N}$. So if you test N strategies that all have **genuinely zero edge**, the best one you find will have a t statistic of about

$$\sqrt{2 \ln 100} = 3.03 \quad (N = 100), \qquad \sqrt{2 \ln 1000} = 3.72 \quad (N = 1000).$$

A t statistic of 3 is conventionally "highly significant", and here it is the expected result of a hundred coin flips with no skill whatsoever. This is why practitioners deflate a Sharpe ratio by the number of trials attempted, and why an honest research process records how many variants were tried, including the ones abandoned.

WORKED EXAMPLE

Transaction costs, and why turnover decides everything. A strategy shows a gross annual return of 12% with 8% volatility, so a gross Sharpe of 1.5. It turns its book over 200 times a year, and you assume 5bp of one-way cost.

- Annual cost: $200 \times 5\text{bp} = 200 \times 0.0005 = 10\%$.
- Net return: $12\% - 10\% = 2\%$, so the net Sharpe is $2/8 = 0.25$.

The gross Sharpe of 1.5 was almost entirely a cost assumption. Two follow-ups make the point unavoidable. The **break-even cost** is $12\%/200 = 6\text{bp}$, so if execution is one basis point worse than assumed the strategy is dead. And the **sensitivity** is $200 \times 1\text{bp} = 2\%$ of annual return per basis point of cost error, which is a quarter of a Sharpe point per basis point. At this turnover, the research question is not "does the signal predict" but "can we execute at 5bp", and those are answered by different people with different tools.

Validation that does not lie to you

- **Hold out, and mean it.** Fit on one period, evaluate once on another, and do not return to the holdout after seeing the result. A holdout looked at repeatedly is training data.
- **Walk forward.** Refit on a rolling window and trade the next window, repeatedly, which mimics how the strategy would actually have been run and exposes parameter instability.
- **Purge and embargo.** Standard k-fold cross-validation is invalid on time series with overlapping labels, because a training fold adjacent to a test fold shares information with it. The fix is to purge observations whose label windows overlap the test fold and to embargo a gap after it.
- **Prefer fewer parameters.** A rule with two parameters that works across many markets is more believable than one with nine tuned to a single market, and this is a judgement about the shape of the evidence rather than a statistical test.

- **Perturb everything.** Shift the rebalance date by a day, add a lag to the signal, double the costs, and drop the best month. A result that survives all four is worth discussing; one that does not is arithmetic about the past.

Engineering that keeps you honest

Reproducibility first: pin the data snapshot, the code version, the configuration and the random seed, so any result can be regenerated exactly. Use the **same code path** for research and production wherever possible, because backtest-to-live divergence is nearly always a difference in the plumbing rather than in the alpha. Unit-test the accounting against hand-computed cases, including a corporate action and a dividend, since those are where fictitious P&L is born. Log the number of variants tried, so the deflation above can actually be applied. And insert a paper-trading stage before real capital, which is the only test that catches data-latency and operational problems at all.

REAL INTERVIEW QUESTION (Société Générale · EQD)

What parameters should be taken into account for a statistical study to determine arbitrage strategies?

WHAT THEY TEST / ANSWER

Answer as a research protocol in four blocks, since a list of statistics would miss the point. **Data**: universe and how it was chosen, history length against the frequency of the effect you are looking for, point-in-time integrity, corporate actions, survivorship including delistings, and the exact timestamps of every input. **Signal and statistics**: the economic reason the mispricing should exist and who is on the other side, the estimator and its assumptions, stationarity of the relationship, and the significance of the result adjusted for how many variants you tried. **Frictions**: bid-offer, fees, borrow availability and cost for the short leg, financing, taxes and dividend withholding, and market impact at the size you intend, since an equity arbitrage that ignores borrow is not a strategy. **Risk and implementation**: turnover, capacity, drawdown and time to recovery, leverage and margin, the correlation of the strategy with the rest of the book, and what happens to the relationship in a crisis, which is usually when a convergence trade widens rather than converges. Then finish with the sentence that shows judgement: a genuine arbitrage should have an identifiable mechanism and an identifiable counterparty who is paying you for something, so if you cannot say what you are being paid for, the safe assumption is that you have found a statistical artefact and not an arbitrage.

REAL INTERVIEW QUESTION (Macquarie · QIS)

How do you construct a systematic index: selection rules, weighting and rebalancing?

WHAT THEY TEST / ANSWER

A systematic index is a backtested strategy that has been made contractual, so the answer is a strategy design plus the constraints that come from being investable and published. Take it in the order asked. **Selection**: define the eligible universe with objective filters (listing, sector, size, liquidity thresholds such as minimum average daily volume and free float) and then the selection rule itself, ranked on the target characteristic, with buffers so that a name near the boundary does not enter and exit repeatedly. **Weighting**: market cap, equal, inverse-volatility, risk parity or optimised, each with caps to control concentration, and note that the weighting scheme usually contributes more of the index's behaviour than the selection rule does. **Rebalancing**: frequency, announcement and effective dates, the tolerance band that triggers an off-cycle rebalance, and the treatment of corporate actions and cash. Then the parts a QIS interviewer is really listening for. Replicability and cost: the index must be tradeable by the desk that hedges products on it, so turnover, liquidity and the predictability of rebalance flow matter more than backtested performance, and a published rebalance is front-runnable, which is a real cost to the index holder. Rules must be complete and unambiguous, including what happens on a suspension or a missing price, because the index has to be computable every day without discretion. The backtest must be point-in-time on membership and must include the frictions, since an index whose live performance diverges from its backtest destroys the product built on it. And say that the decrement or fee mechanics, if any, belong in the index definition rather than in the marketing.

INTERVIEW TRAP

Four traps. First, **trading on the bar you used to decide**: signal from today's close, execution at today's close, which is look-ahead in its most common form. Second, **optimising and then reporting the optimum** as if it were an out-of-sample result, which is the single most frequent piece of dishonesty in strategy research, usually unintentional. Third, **quoting a Sharpe with no sample length, no turnover and no cost assumption**, which makes the number uninterpretable. Fourth, **a backtest that cannot be reproduced**, because unpinned data and code mean that when the result changes next month nobody will be able to say why.

VISUAL / CODE TO BUILD: Interactive

A backtest sandbox built to be broken. The reader configures a simple rule on a supplied price history and gets the equity curve, drawdown and metrics. Then four switches let them create each bias and watch its effect, which is far more instructive than reading about it: **look-ahead** (shift execution from the next open to the signal bar and watch the Sharpe jump), **survivorship** (include or exclude delisted names), **costs** (a slider from 0 to 20bp with the break-even cost marked, reproducing the turnover example), and **snooping** (a button that silently tries N parameter combinations and reports both the best Sharpe found and the $\sqrt{2\ln N}$ benchmark for zero true edge, so the reader sees their own best result sitting exactly on the noise line). A final panel runs walk-forward against a single in-sample fit so the parameter instability is visible as two diverging curves.

Where this goes next

The tooling this sits on is Python for finance and NumPy and pandas, with the data plumbing in financial data pipelines. The statistics behind the significance discussion are hypothesis testing and linear and multiple regression, and the annualisation caveat is autocorrelation. The other meaning of backtesting, validating a risk model rather than a strategy, is in Value at Risk.

25.5 Financial data pipelines

trading · structuring · sales core cross-asset

Every analysis a desk runs begins with getting the data right, and that is not a preliminary to the work, it is most of the work. The modelling in this book assumes a clean series of prices arriving on the right dates with the right adjustments. Producing that series is harder than anything the model does with it, and almost every wrong answer a junior produces comes from here rather than from the mathematics.

The stages

A pipeline is six steps, and each one is a place where a number can go quietly wrong.

- **Source**: a vendor, an exchange feed, an internal risk system, a file somebody emails. Each has its own identifiers, conventions and failure modes.
- **Ingest**: read it, parse it, and get it into a consistent shape.
- **Validate**: check it is what you expected before anything downstream uses it.
- **Align**: put several series on a common calendar, a common time zone and a common definition of a day.
- **Store**: keep it in a form that can be queried, and keep the raw input as well as the processed output.
- **Serve**: hand it to the model, the report or the dashboard.

> KEY IDEA
>
> The single most important property of a financial dataset is whether it is **point in time**: does it tell you what was known *on* the date, or what is known *about* the date today? Fundamentals are restated, economic releases are revised, ratings are changed retroactively in some databases, and index membership is usually stored as it is now rather than as it was. A backtest run on a non-point-in-time dataset uses information nobody had, and the result is not merely optimistic, it is meaningless. Every serious data question in an interview is a version of this one.

The hazards, specific to this data

Corporate actions. A two-for-one split halves the price overnight, and a raw series shows a 50% crash. Splits, dividends, rights issues, spin-offs and mergers all require adjustment, and the adjustment differs by purpose: a *price* series adjusts for splits only, a *total return* series also reinvests dividends, and the two answer different questions, as in corporate actions.

Survivorship bias. A universe built from today's index members contains only the companies that survived, so any strategy backtested on it inherits a performance boost it could not have had. The fix is a point-in-time membership history, and its absence is the most common silent flaw in a student project.

Calendars and time zones. Two exchanges do not share holidays, and aligning them naively produces either spurious zero returns or dropped days. A "daily close" is not one thing across regions, and comparing a European close to a US close embeds a several-hour lag that will look like predictive power.

Stale and missing values. A price that has not changed for three days may be a quiet market or a broken feed, and those need different treatment.

Bad ticks and outliers. A single mistyped price produces a return that dominates any volatility estimate computed from the series, which is why a validation step that flags implausible moves is worth more than any cleverness downstream.

REAL INTERVIEW QUESTION (TP ICAP · Structuring)

How do you read a CSV file in Python?

WHAT THEY TEST / ANSWER

Give the one-liner immediately, then show that you know it is never that simple, because the second half is what distinguishes someone who has actually loaded market data.

`import pandas as pd` then `df = pd.read_csv("prices.csv")`, or the standard library's `csv` module if the file is small and you want no dependency.

Then the arguments you will actually need, each tied to a real failure. `parse_dates` and `dayfirst`, because a European file writes 03/04 as the third of April and the default parser reads it as the fourth of March, which silently reorders your series. `dtype`, because an identifier with leading zeros or a ticker like `NA` will be inferred as a number or as a missing value. `sep` and `decimal`, because a French export uses semicolons and commas. `encoding`, because a file with accented names in `latin-1` will raise or mangle. `na_values`, because vendors write missing data as blanks, as `#N/A`, as `-999` or as `NULL`. And `chunksize` if the file does not fit in memory.

Close on what you do *after* reading, which is the answer a desk wants: check the row count against what you expected, check the date range and that it is sorted and unique, check for duplicates, and look at the first and last rows with your own eyes. A file that loads without error is not a file that loaded correctly.

Missing data is a decision, not a cleanup

REAL INTERVIEW QUESTION (Société Générale · Index)

How do you remove the "N/A" values from a Pandas dataset?

WHAT THEY TEST / ANSWER

Give the mechanics in one sentence, then spend the answer on the choice, because dropping missing values is rarely the right thing and the interviewer knows it.

The mechanics: `df.dropna()` removes rows, with `subset` to restrict which columns matter and `how` to choose between any and all; `df.fillna(value)` substitutes; `df.ffill()` carries the last observation forward.

The choice is the substance, and it depends entirely on *why* the value is missing. A **market holiday** in one country while another trades is not missing data, it is a calendar mismatch, and forward-filling is the correct treatment because the price genuinely did not change. A **failed feed** is missing data, and forward-filling it fabricates a zero return, which understates volatility and can hide a real move. A **non-existent** value, a company before it listed, must be left missing rather than filled, or you will backtest a position in a stock that did not exist.

Then name the trap that makes this a finance question rather than a Pandas one: **forward-filling introduces stale prices**, so any risk number computed from the result understates risk, and **filling with a mean or an interpolation uses future information**, which is a look-ahead bias and invalidates any backtest. Dropping rows is safest for a cross-sectional analysis and dangerous for a time series, because it silently changes the spacing between observations and therefore the annualisation.

Close on the discipline: whatever you do, **count and log** how many values were affected. A pipeline that quietly fills ten thousand missing prices and reports a clean result is worse than one that fails.

Validate before you trust

A pipeline should fail loudly rather than pass a plausible but wrong number downstream. The checks worth having are cheap:

```
assert df.index.is_monotonic_increasing
assert not df.index.duplicated().any()
assert df["price"].gt(0).all()
moves = df["price"].pct_change().abs()
suspects = moves[moves > 0.25]          # inspect, do not delete
```

Note the comment. A large move is a *flag*, not an error: it might be a bad tick, or it might be the day the stock actually fell 30%, and the pipeline's job is to surface it for a human rather than to decide. Deleting outliers automatically is how a genuine crash disappears from a volatility estimate.

Two more habits complete it. **Reconcile across sources** where you can, since two vendors disagreeing is the fastest way to find an error in either. And keep the pipeline **idempotent and reproducible**: re-running it on the same inputs must give the same outputs, the raw input must be kept alongside the processed result, and the data version should be recorded next to the code version, or an analysis cannot be reproduced six months later.

REAL INTERVIEW QUESTION (Macquarie · QIS)

Build a trading algorithm from a fictitious CSV file.

WHAT THEY TEST / ANSWER

Treat it as a pipeline exercise with a strategy at the end, because that is what it is, and narrate the order.

Load and inspect first, with the parsing care above, then check the date range, the frequency, missing days and duplicates, and plot the series before writing any logic. Say out loud that you would look at it, because candidates who go straight to the signal miss the broken data that invalidates everything after.

Define the signal explicitly, for example a moving-average crossover or a simple momentum rule, and state its parameters. Keep it simple: the exercise is not testing the strategy.

Compute positions from the signal with an explicit lag. This is the point of the whole exercise. A signal computed from today's close cannot be traded at today's close, so the position must be shifted by at least one period, and forgetting that is the single most common error in these tests. It manufactures spectacular returns from nothing.

Compute returns and costs. Strategy return is the position times the next period's return. Then subtract transaction costs on every position change, because a crossover strategy trades often and a costless backtest of it is fiction.

Evaluate honestly: cumulative return, annualised volatility, Sharpe ratio, maximum drawdown, turnover, and the number of trades. Report the strategy against holding the asset, since a strategy that underperforms buy-and-hold has not earned its complexity.

Then name the biases you have not eliminated, which is what a quantitative interviewer is listening for: look-ahead if any parameter was chosen after seeing the whole sample, overfitting if several rules were tried and the best reported, survivorship if the universe is today's, and the absence of slippage and market impact. Saying which of these your five-minute answer is still exposed to is worth more than a higher Sharpe ratio. The proper framework for this is backtesting frameworks.

Where the data comes from

REAL INTERVIEW QUESTION (BNP · Fixed Income)

Are you proficient in Excel and Bloomberg?

WHAT THEY TEST / ANSWER

Answer honestly and specifically, and then show you know what each tool is *for*, because the question is a proxy for whether you have worked with real market data.

For **Excel**, name what you actually do rather than claiming a level: lookups and index-match, pivot tables, array formulas, and the fact that a spreadsheet used by more than one person needs its inputs separated from its calculations. Say where you would stop using it, which is usually where the dataset stops fitting or where the calculation needs to be repeated and audited, in advanced Excel.

For **Bloomberg**, the useful answer names functions and what they give you rather than claiming familiarity: a description page for an instrument, a historical price series, a yield or option calculator, a curve, a news screen, and the spreadsheet add-in that pulls all of it into Excel, which is how most desks actually consume it. If you have used Refinitiv or another vendor instead, say so and say what the equivalents are.

Then the point that makes this a data answer: a terminal is a **source**, and everything in this section still applies to what comes out of it. The identifiers have to be mapped to your own, the history has to be checked for corporate action adjustment, and a pull that silently returns fewer rows than requested is the standard way an analysis goes wrong. Ending on the discipline rather than on the tool is what a desk wants to hear, and if you have not used a terminal, saying so plainly and describing what you have used is a far better answer than a vague claim, as in deal-breakers.

INTERVIEW TRAP

Four errors. (1) Using a non-point-in-time dataset for anything backward looking, which imports information nobody had. (2) Forward-filling prices without asking why they are missing, which fabricates zero returns and understates risk. (3) Deleting outliers automatically, which removes exactly the days that matter. (4) Trusting a file because it loaded, when the parser silently reordered your dates or turned an identifier into a number.

Where this goes next

The language is Python for finance and the array and dataframe tools are NumPy and Pandas. Querying stored data is SQL, pulling it from a service is APIs, and testing a strategy on it properly is backtesting frameworks. The adjustments the series needs are corporate actions, and the statistical treatment of the result is time series analysis.

25.6 APIs

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Nothing on a trading floor is entered by hand twice. Prices arrive from a vendor, orders leave for an exchange, trades land in a booking system, risk is pulled from the pricing library, and every one of those handoffs is an API. Knowing how they behave is not a software-engineering nicety; the failure modes in this section are the ones that produce duplicate orders and stale prices, which are expensive in a way that a slow script is not.

The three you will meet

Market data. Vendor APIs such as Bloomberg's and Refinitiv's, and direct exchange feeds. These come in two shapes that matter enormously, covered below, and they are also where most of the cost sits: market data is licensed per user and per use, and the licence terms constrain what you may redistribute or store.

Execution. Overwhelmingly **FIX**, the Financial Information eXchange protocol, which has been the industry standard for order flow since the 1990s. It is a tag-value message format carried over a session that guarantees ordering and recovery: each message is a sequence of `tag=value` pairs, a new order is a `NewOrderSingle`, and the venue answers with `ExecutionReport` messages as the order is acknowledged, partially filled, filled or cancelled. What matters for an interview is not the tag numbers but the structure: FIX separates a *session* layer, which handles sequence numbers, heartbeats and resending anything missed, from an *application* layer that carries the orders. That separation is the reason a dropped connection does not lose orders.

Internal. Trade capture, risk, reference data, the pricing library's own binding. These are the ones a junior actually writes against, and the ones with the least documentation.

Request-response against streaming

KEY IDEA

A **request-response** API, typically REST over HTTP, gives you a snapshot when you ask for one. A **streaming** API, typically a websocket or a vendor subscription, pushes you an update when something changes. The distinction is not a style preference: polling a request-response endpoint to simulate a stream is the most common and most expensive beginner mistake in market data.

WORKED EXAMPLE

Why polling does not work, in numbers. You want live prices on 500 instruments, and each updates on average five times a second, so the true event rate is $500 \times 5 = 2{,}500$ updates a second.

Subscribe and you receive 2,500 messages a second, every update, in order, with latency limited by the network.

Poll once a second and you make 500 requests a second, receive 500 snapshots a second, and therefore see one update in five: you miss **80%** of them, and the price you hold is on average **half a second** stale.

Poll five times a second to catch up and you are making $500 \times 5 = 2{,}500$ requests a second against a rate limit that is almost certainly lower, to obtain exactly the information the subscription would have pushed you for free, and you still cannot tell whether two consecutive snapshots represent one move or three.

The rule that follows: **poll for things that change slowly and you need on demand**, such as reference data, holiday calendars and end-of-day marks; **subscribe to things that change fast and you need continuously**, which is every live price. Getting this backwards is visible in a system's latency profile immediately and in its vendor bill a month later.

The failure mode that costs money

Everything above is ordinary software engineering. This part is specific to finance and is the thing worth carrying out of the section.

INTUITION

A timed-out order request leaves you in an unknown state. You sent a new order, the connection dropped before the acknowledgement came back, and you do not know whether the venue received it. There are exactly two possibilities and they require opposite actions, so guessing is not available.

The wrong answer is to retry. If the first request did arrive, you now have *two* orders in the market, and on a large order that is a real position nobody intended and a loss nobody budgeted.

The right answer has two parts. Send a **client order id** that you generate, unique per order, on every request, so the venue can recognise a duplicate and reject it rather than fill it: that is **idempotency**, and it turns a dangerous retry into a safe one. And when a request times out, *query* rather than resend: ask the venue for the status of that client order id and act on the answer.

The same discipline generalises. Any API call that changes state, submitting an order, booking a trade, cancelling, needs an idempotency key and a query path. Any call that only reads state can be retried freely.

The rest of the discipline, briefly

Rate limits. Every commercial API has them, and hitting one usually returns an error that looks transient. Retry with **exponential backoff** and jitter rather than immediately, because a fleet of clients all retrying on the same schedule after an outage is how the outage is extended.

Authentication. API keys, OAuth tokens or mutual TLS. The rule that matters is that credentials never go into source control, never into a notebook, and are rotated when someone leaves. This is checked in real onboarding and is worth saying out loud.

Pagination. A query that returns a large result set returns it in pages. Code that reads the first page and stops is a silent data-loss bug, which is worse than a loud one because the numbers still look plausible.

Versioning. Providers change APIs. A client pinned to a version keeps working; one that follows the latest breaks on a morning you did not choose. Pin, and upgrade deliberately.

Time. Everything is timestamped, and the timestamps come from different clocks in different zones. Store UTC, keep the venue's own timestamp alongside your receipt timestamp, and never compare the two as though they measured the same thing. Reconstructing what happened during an incident depends entirely on having done this beforehand.

INTERVIEW TRAP

1. **Polling a REST endpoint for live prices.** It is slower, more expensive and lossy compared with subscribing, and the arithmetic above shows by how much.
2. **Blind retries on order submission.** Without an idempotency key this doubles positions. Query the client order id instead.
3. **Retrying immediately on a rate-limit error.** Back off exponentially, add jitter, and do not synchronise a fleet of clients onto one retry schedule.
4. **Ignoring pagination.** Silent truncation produces answers that look right, which is the worst kind of wrong.
5. **Credentials in code.** Environment or a secrets manager, never the repository, and rotate them.
6. **Mixing timestamps.** Venue time and receipt time are different measurements from different clocks, and conflating them makes an incident unreconstructable.
7. **Assuming the vendor's data licence permits what you are doing.** Redistribution and storage are contractual questions, not technical ones, and the compliance consequences are the expensive part.

VISUAL / CODE TO BUILD: Interactive

A polling-against-streaming simulator. A price series is generated at a chosen event rate and drawn as the true path. Over it, the panel draws what a subscriber sees, which is the same path, and what a poller sees at a chosen frequency, drawn as a step function, with the gap between them shaded as staleness. A counter reports requests per second, updates missed and mean staleness for the poller.

The intended sequence is to start at one poll per second and watch the step function lag visibly through a fast move, then drag the poll rate up and watch the request counter climb into rate-limit territory while the step function still only approximates the path. That the subscriber line is exact at zero request cost is the argument.

A second panel animates the timeout problem: an order is sent, the acknowledgement is dropped, and the reader chooses “retry” or “query”. Retry shows two fills and a doubled position; query shows one fill and a correct state. Running it once is more persuasive than the paragraph, which is why the paragraph exists as well.

25.7 Advanced Excel

sales · trading · structuring core cross-asset

Every prediction that Excel is finished has been wrong, and it will be wrong again. A trading floor runs on spreadsheets because they are instantaneous, universally readable, and visible: anyone can open the file and see the number and the formula that produced it. That transparency is a genuine engineering property and it is why the tool survives against better ones. It is also why spreadsheet errors are a recognised category of operational loss.

Looking things up, and the three ways it breaks

REAL INTERVIEW QUESTION (Société Générale · Volatility)

What is a VLOOKUP in Excel?

WHAT THEY TEST / ANSWER

Define it, then immediately give its failure modes, because anyone can define it and the failures are what the question is for.

`VLOOKUP` searches for a value in the first column of a range and returns a value from a column to its right, specified by number: `=VLOOKUP(key, table, col_index, FALSE)`.

Three things break it, and each has cost somebody a number. **It cannot look left**: the key must be in the first column of the range, so a lookup that returns something to the left of the key requires rearranging the data. **The column index is a number**, so inserting a column into the table silently changes what is returned, with no error and no warning. And **the fourth argument defaults to approximate match** if omitted, which on unsorted data returns a plausible wrong answer rather than an error, which is the worst possible failure mode.

Then give the alternatives, since that is what shows current practice. `INDEX` with `MATCH` is the robust classic: it looks in any direction, it refers to the return column rather than counting to it, and it survives column insertions. `XLOOKUP` does all of that in one function with an exact match by default and a built-in "if not found" argument, and it is what you should use where it is available.

Close on the habit rather than the syntax: always force exact match, always wrap a lookup in `IFERROR` or `XLOOKUP`'s not-found argument so a missing key shows as a label rather than as `#N/A` propagating through a model, and never look up on a key you have not checked for duplicates.

The toolkit worth having

REAL INTERVIEW QUESTION (ECB · Market Risk)

Which Excel formulas do you know?

WHAT THEY TEST / ANSWER

Group them by what they do rather than reciting names, and name the ones a desk actually uses.

Lookup and reference: `XLOOKUP`, `INDEX` and `MATCH`, `VLOOKUP` with its caveats, `OFFSET` and `INDIRECT` with the warning that both are volatile and slow.

Conditional aggregation: `SUMIFS`, `COUNTIFS`, `AVERAGEIFS`, and `SUMPRODUCT` as the general-purpose tool that does weighted sums and multi-condition counts in one expression.

Logic and error handling: `IF`, `IFS`, `AND`, `OR`, `IFERROR`. Wrapping lookups in error handling is a discipline rather than a formula.

Dates: `EDATE`, `EOMONTH`, `WORKDAY` and `NETWORKDAYS`, which matter more in finance than anywhere else because every schedule is built from business days and holiday calendars.

Financial and numerical: `NPV`, `IRR`, `XIRR` for uneven dates, `RATE`, `NORM.S.DIST` and `NORM.S.INV`, which are what you need to write Black-Scholes in a cell, plus `Goal Seek` and `Solver` for implied volatility and calibration.

Dynamic arrays, if you want to sound current: `FILTER`, `SORT`, `UNIQUE`, `SEQUENCE`, and `LET` and `LAMBDA` for naming intermediate results and writing reusable functions without VBA. These are the biggest change to the product in twenty years and most candidates have not noticed.

Finish with a concrete example rather than the list, since the interviewer will ask for one: counting negative values in a column is `=COUNTIF(A:A,"<0")`, and the same question with two conditions is `COUNTIFS`. Being able to produce a working formula on demand is what the question is checking.

Making a model somebody else can trust

A spreadsheet on a desk is read and used by people who did not build it, so the discipline matters more than the formulas.

- **Separate inputs, calculations and outputs**, ideally onto different sheets or clearly marked blocks. The widespread convention of colouring hardcoded inputs blue and formulas black exists so that a reader can see instantly what is an assumption.
- **Never bury a constant inside a formula.** A discount rate typed into twenty cells is twenty places to update and nineteen chances to miss one. Put it in a labelled cell and refer to it.
- **One formula per row, copied across.** A row where three cells differ from their neighbours is the single most common source of spreadsheet error, and it is invisible unless you look for it.
- **Name what matters**, so that a formula reads `=Notional * Spread` rather than `=$B$4 * $D$12`.
- **Make errors visible.** A model that shows `#N/A` where a lookup failed is safer than one that shows a zero, because a zero flows silently into a total.
- **Document the units and the conventions** in the sheet: basis points or percent, annual or

daily, actual or 360. Most spreadsheet disputes are unit disputes.

> **KEY IDEA**
>
> Spreadsheet errors are a recognised operational risk, not a joke about typos. The two most publicised categories are both mechanical: a **range that does not cover all the rows**, so an average or a sum silently omits data, and **manual copying between files**, where a step done by hand every day is eventually done wrong. Both have produced consequential public errors, in academic research and in a bank's own risk reporting. The defence is not care, it is **structure**: fewer manual steps, ranges that extend automatically, checks that fail loudly, and a reconciliation total that must agree.

Two performance notes are worth knowing because they turn a slow file into a fast one. **Volatile functions**, `OFFSET`, `INDIRECT`, `TODAY`, `RAND`, force a recalculation of everything that depends on them on every edit, so a model built on them crawls. And **full-column references** such as `A:A` in an array formula ask Excel to consider a million rows; they are harmless in a `COUNTIF` and expensive inside a large array calculation.

Knowing when to stop

> **REAL INTERVIEW QUESTION** (BNP · EQD)
>
> *What is your level in Excel, what is your favourite function, and what function have you discovered recently that you find useful?*
>
> **WHAT THEY TEST / ANSWER**
>
> Three parts, and the third is the one that reveals whether you actually use the tool.
>
> **Level**: describe what you do rather than choosing a number out of ten. Something like: comfortable with lookups, conditional aggregation, pivot tables and dynamic arrays, have built models that other people use, and know when a task has outgrown a spreadsheet. A self-assessed score invites a test; a description invites a conversation.
>
> **Favourite function**: pick one and justify it in terms of what it prevents. `SUMPRODUCT` is a good answer because it does weighted sums and multi-condition logic in one readable expression. `XLOOKUP` is a good answer because it removes three failure modes of `VLOOKUP` at once. `LET` is a good answer because it lets a long formula name its intermediate steps and become readable. Any of these is better than `VLOOKUP`, which signals that you stopped learning some years ago.
>
> **Recently discovered**: this is the real question, because it tests whether you are still learning. Name something genuine and say what it replaced: perhaps `FILTER`, which returns a whole matching subset in one formula and replaces a helper column plus a sort; or `LAMBDA`, which lets you define a reusable function in the workbook without VBA; or `Power Query`, which replaces the weekly manual import described in financial data pipelines. What you must not do is invent one, since the follow-up is always "show me how you would use it".
>
> Then close on judgement, which is what an EQD interviewer is really listening for: the most useful thing to know about Excel is **when to stop using it**. When the dataset stops fitting comfortably, when the same manipulation is repeated daily, when the calculation needs to be tested or version-controlled, or when several people need to run it, the answer is a script rather than a sheet, in Python for finance. Saying that unprompted is worth more than any function name.

INTERVIEW TRAP

Four errors. (1) Using VLOOKUP without the exact-match argument, which returns a plausible wrong answer on unsorted data. (2) Hardcoding a constant inside formulas instead of referring to a labelled input cell. (3) Suppressing errors with IFERROR returning zero, which hides a broken lookup inside a total. (4) Claiming Excel proficiency without being able to write a working formula on request, which is the cheapest failure in this section.

Where this goes next

Automating a spreadsheet is VBA, and the point at which a spreadsheet should become a program is Python for finance. Getting data into it reliably is financial data pipelines, and querying data that has outgrown a sheet entirely is SQL.

25.8 VBA

sales · trading · structuring core cross-asset

VBA is old, unfashionable, and still asked about in a startling proportion of front-office interviews, for one reason: Excel is the universal interface of a trading floor, and VBA is what makes Excel do things. Term sheet generators, RFQ logs, ad hoc pricers, risk reports and reconciliation tools all run on it. Interviews test it concretely, often on a whiteboard, and they test the same handful of things: declaring variables, walking a range, handling errors, and writing a short loop without syntax mistakes. This section covers exactly that, plus the judgement about when to stop using it.

The minimum you must be able to write

Code lives in **modules** inside a workbook's VBA project. A **Sub** performs actions and returns nothing; a **Function** returns a value and, if it is in a standard module, can be called from a worksheet cell as a user-defined function. Every module should begin with `Option Explicit`, which forces you to declare every variable and turns a silent typo into a compile error.

```
Option Explicit

Function AtmApprox(spot As Double, vol As Double, t As Double) As Double
    ' desk rule of thumb: an ATM option is worth about 0.4 * vol * S * sqrt(T)
    AtmApprox = 0.4 * vol * spot * Sqr(t)
End Function
```

Note `Sqr` rather than `Sqrt`, which is the kind of detail a whiteboard question is checking. Declarations use `Dim name As Type`, and the types worth knowing, with their documented sizes, are:

Type	Size	Use
`Boolean`	2 bytes	flags
`Integer`	2 bytes	avoid: overflows above 32,767
`Long`	4 bytes	the default choice for counters and row numbers
`Single`	4 bytes	rarely worth it
`Double`	8 bytes	all prices, rates and volatilities
`Currency`	8 bytes	fixed-point, 4 decimals, for money where rounding matters
`Date`	8 bytes	dates and times
`String`	10 bytes + length	text
`Variant`	16 bytes +	anything, including whole ranges as arrays

Two practical rules. Use `Long` rather than `Integer` for anything that counts rows, since a worksheet has more than a million of them and `Integer` overflows at 32,767. And use `Double` for every market number, because `Single` loses precision where it matters and integers silently truncate.

Talking to the sheet, and the one performance rule

`Range` is the object you spend your life with. `Cells(row, col)` addresses one cell numerically, `Offset(dr, dc)` moves relative to a range, and `End(xlUp)` finds the last used cell, which is how you size a table without hardcoding.

```
Dim lastRow As Long
lastRow = Cells(Rows.Count, 1).End(xlUp).Row     ' last used row in column A
```

Then the single rule that separates code that runs in a second from code that takes a minute: **do not touch cells one at a time**. Each read or write of a cell crosses from VBA into Excel, and that crossing is what costs the time. Read the whole block into a Variant array, compute in memory, and write it back once.

```
Sub FastMultiply()
    Dim v As Variant, i As Long
    Application.ScreenUpdating = False
    Application.Calculation = xlCalculationManual

    v = Range("A1:C1000").Value      ' one call: a 1-based 2-D array
    For i = 1 To UBound(v, 1)
        v(i, 3) = v(i, 1) * v(i, 2)
    Next i
    Range("A1:C1000").Value = v      ' one call back

    Application.Calculation = xlCalculationAutomatic
    Application.ScreenUpdating = True
End Sub
```

The same loop written as `Cells(i, 3) = Cells(i, 1) * Cells(i, 2)` is typically one to two orders of magnitude slower, and on a large report that is the difference between usable and abandoned.

Error handling, properly

VBA has no exceptions in the modern sense. You get `On Error` and the `Err` object, and the professional pattern is a single handler with a guaranteed cleanup path:

```
Sub BuildReport()
    On Error GoTo Fail
    Application.ScreenUpdating = False

    If Range("Notional").Value <= 0 Then
        Err.Raise 5, "BuildReport", "Notional must be positive"
    End If
    ' ... work ...

Done:
    Application.ScreenUpdating = True     ' always restored
    Exit Sub

Fail:
    MsgBox "BuildReport failed: " & Err.Number & " " & Err.Description
    Resume Done
End Sub
```

Three things to note. `Err.Raise` is how you signal your own errors, and error 5 is "invalid procedure call"; for custom codes use `vbObjectError + n`. The `Resume Done` sends control back into the normal path so the cleanup runs exactly once. And `On Error GoTo 0` cancels an active handler, which you need when you want the next error to surface rather than be swallowed.

That last point is the whole story of `On Error Resume Next`, the answer to one of the most frequently asked VBA interview questions, and the most abused statement in the language: it tells VBA to ignore a failing line and continue with the next one. It has exactly one legitimate use, which is wrapping a single statement whose failure is expected and immediately checked.

```
On Error Resume Next
Set ws = ThisWorkbook.Worksheets("Prices")    ' may not exist
On Error GoTo 0                               ' stop ignoring immediately
If ws Is Nothing Then Set ws = ThisWorkbook.Worksheets.Add
```

The two functions you will be asked to write

Whiteboard VBA questions are drawn from a very short list. These two cover most of it, and both are worth being able to type without hesitating.

First, a hand-written VLOOKUP: find a key in the first column of a table and return the value from the third.

```
Function LookupCol3(key As Variant, rng As Range) As Variant
    Dim c As Range
    For Each c In rng.Columns(1).Cells
        If c.Value = key Then
            LookupCol3 = c.Offset(0, 2).Value   ' 0-based offset: 3rd column
            Exit Function
        End If
    Next c
    LookupCol3 = CVErr(xlErrNA)                 ' behave like #N/A
End Function
```

The offset is `(0, 2)` and not `(0, 3)` because `Offset` is relative and zero-based. Returning

`CVErr(xlErrNA)` rather than an empty string makes the cell show `#N/A`, so downstream formulas behave the way users expect.

Second, a sort. Bubble sort is the right answer under pressure: short, easy to get right, and easy to reason about.

```
Sub SortAscending(a() As Double)
    Dim i As Long, j As Long, tmp As Double
    For i = LBound(a) To UBound(a) - 1
        For j = LBound(a) To UBound(a) - 1 - (i - LBound(a))
            If a(j) > a(j + 1) Then
                tmp = a(j)
                a(j) = a(j + 1)
                a(j + 1) = tmp
            End If
        Next j
    Next i
End Sub
```

Arrays are passed by reference in VBA, so this sorts in place and returns nothing. Using `LBound` and `UBound` instead of 0 and $n - 1$ makes it correct whatever the array's base, which matters because a range read into a Variant is 1-based. The inner bound shrinks on each pass because the largest remaining element has already reached the end. It is $O(n^2)$, which is fine for hundreds of elements and wrong for hundreds of thousands.

Debugging, and when to leave VBA behind

The debugger is the reason VBA is pleasant to learn. **F9** sets a breakpoint on a line, so execution pauses there; **F8** then steps one line at a time; hovering over a variable shows its value; the **Locals** window lists every variable in scope with its current value; the **Watch** window tracks a specific expression and can break when it changes; and the **Immediate** window evaluates anything on the spot, with `?x` printing a value and `Debug.Print` logging from code without interrupting the run.

Know when to stop, though. VBA is single-threaded, has no package manager, no real testing framework, and no practical way to version-control code that lives inside a binary workbook. So anything data-heavy, statistical, or intended to be shared and maintained belongs in Python, with the spreadsheet reduced to a front end. This is not only an engineering preference: uncontrolled spreadsheet logic is a recognised **operational risk** category, since a hidden hardcode or a broken reference in a widely-used workbook has caused real losses, which is why banks maintain model inventories and end-user-computing controls, see operational risk and trading floor compliance.

REAL INTERVIEW QUESTION (HSBC · Structured Products)

Name different variable types in VBA.

WHAT THEY TEST / ANSWER

List them with what each is for, since a bare list sounds memorised. `Boolean` for flags, `Integer` and `Long` for whole numbers, `Single` and `Double` for decimals, `Currency` for fixed-point money, `Date` for dates and times, `String` for text, `Object` for references, and `Variant` for anything, which is what you get if you fail to declare a type. Then add the two judgements that make it a real answer. Use `Long` rather than `Integer` for row counters, because `Integer` is a 2-byte type that overflows at 32,767 while a worksheet has over a million rows, and use `Double` for every price, rate and volatility. And say why you always write `Option Explicit` at the top of a module: without it an undeclared variable is silently created as an empty `Variant`, so a typo becomes a wrong number rather than an error. If they push on sizes, `Integer` is 2 bytes and `Double` is 8, with `Variant` the most expensive at 16 bytes or more, though on a modern machine correctness matters far more than the memory.

REAL INTERVIEW QUESTION (BNP · Market Risk)

To force the declaration of all variables in a VBA project, what do you put in a module: `Option Explicit` or `Option Variables`?

WHAT THEY TEST / ANSWER

`Option Explicit`. `Option Variables` does not exist. Then use the remaining ten seconds to show you know *why* it matters, because that is the real question. Without it, referring to an undeclared name silently creates a new empty `Variant`, so `notionl = notional * 2` does not fail, it quietly computes zero, and in a risk report that is a wrong number delivered with confidence rather than an error you can see. Two practical additions: it goes at the top of *every* module, and you can have the editor insert it automatically by ticking "Require Variable Declaration" in the VBA editor options, which is the first thing to set on a new machine. Mentioning that this is exactly the class of silent spreadsheet error that end-user-computing controls exist to prevent is a good touch on a risk desk.

REAL INTERVIEW QUESTION (BNP · Market Risk)

What VBA code do you use to continue execution even if a line causes an error?

WHAT THEY TEST / ANSWER

`On Error Resume Next`. Say it immediately, then spend the rest of the answer on the caveat, because on a risk desk the caveat is what is being tested. It suppresses the error and carries on with the next statement, which means every subsequent failure is silent too, so used carelessly it converts a crash into a wrong answer, which is strictly worse. The disciplined pattern is to scope it to a single expected failure and cancel it immediately with `On Error GoTo 0`, then check the outcome explicitly, either `If Err.Number <> 0` or by testing whether the object you tried to obtain is `Nothing`. For anything else, use a real handler: `On Error GoTo Fail` with a labelled block that reports `Err.Number` and `Err.Description`, restores application state such as `ScreenUpdating` and `Calculation`, and exits cleanly via `Resume` into a shared cleanup label. Closing with "and I would never leave `On Error Resume Next` active across a whole procedure" is the sentence they are waiting for.

REAL INTERVIEW QUESTION (BNP · Market Risk)

In VBA, write the syntax that returns a value from the 3rd column of a table based on an entry matching the value in the 1st column.

WHAT THEY TEST / ANSWER

This is a VLOOKUP written by hand, and they want to see `Offset` used correctly. Offer the one-liner first, since the fastest correct answer is often the built-in: `result = Application.WorksheetFunction.VLookup(key, rng, 3, False)`, noting that `False` forces an exact match and that it raises an error when the key is absent, which is why `Application.VLookup` returning an error value is often preferable. Then write the explicit loop, which is what a whiteboard question is really asking for: iterate over `rng.Columns(1).Cells`, compare each cell's value to the key, and on a match return `c.Offset(0, 2).Value` and exit, falling through to `CVErr(xlErrNA)` if nothing matched. The full function is given earlier in this section. Then volunteer the three details that show experience: `Offset(0, 2)` moves two columns across because the offset is relative and zero-based, returning `CVErr(xlErrNA)` rather than an empty string makes the cell show `#N/A` so downstream formulas behave correctly, and on a large table you would read the range into a Variant array first and loop in memory, or build a `Dictionary`, since a cell-by-cell scan is slow.

REAL INTERVIEW QUESTION (HSBC · Structured Products)

Write a function to sort an array in VBA.

WHAT THEY TEST / ANSWER

They are testing whether you can write a correct nested loop under pressure, not whether you know quicksort. Write a bubble sort, as given earlier in this section: two nested `For` loops from `LBound` to `UBound`, swapping neighbours through a temporary variable when they are out of order, with the inner bound shrinking on each pass.

Then the commentary that earns the marks, on top of the by-reference, `LBound`/`UBound` and shrinking-bound points already made above. Say where $O(n^2)$ stops being acceptable and what you would do instead: at a hundred thousand elements you would use quicksort, or simply sort on the sheet, or hand the job to Python. And if asked to make it generic, change the signature to `Variant` and compare with a helper, remembering that comparing mixed types in VBA is a source of surprises. Declaring the loop counters as `Long` rather than `Integer` is a small detail an interviewer on a risk desk will notice.

REAL INTERVIEW QUESTION (BNP · Market Risk)

What is a breakpoint and how do you see the value of a variable in your code?

WHAT THEY TEST / ANSWER

Answer operationally, as though you were showing them. A breakpoint is a marker on a line that pauses execution just before that line runs, so you can inspect the program's state at that moment; you set one by clicking the margin or pressing **F9**, and the paused state is called break mode. From there **F8** steps one line at a time and **Shift+F8** steps over a called procedure. To see a value you have four tools and should name more than one: hover over the variable for a tooltip, open the **Locals** window to see every variable in scope at once with its type and value, add a **Watch** on an expression to follow it and optionally break when it changes, or use the **Immediate** window with `?myVar` to evaluate anything on the spot. Then add the two things practitioners actually rely on: `Debug.Print` writes to the Immediate window without stopping the run, which is how you inspect a long loop, and `Debug.Assert` breaks only when a condition is false, which is a cheap invariant check. Finishing with a conditional-breakpoint technique, such as `If i = 5000 Then Debug.Assert False`, shows real debugging experience rather than familiarity with the menus.

INTERVIEW TRAP

Four traps. First, **omitting `Option Explicit`**, which turns typos into silent wrong numbers, the single most common cause of spreadsheet errors that reach a client. Second, **`Integer` for row counters**, which overflows at 32,767 on a sheet with a million rows. Third, **cell-by-cell loops**, which are one to two orders of magnitude slower than reading a range into an array. Fourth, **leaving `On Error Resume Next` active**, or leaving `ScreenUpdating` or manual calculation switched off after an error, which is why every handler needs a guaranteed cleanup path.

VISUAL / CODE TO BUILD: Interactive

A browser VBA drill, since the skill being tested is typing correct syntax under pressure. Short tasks appear (walk a range and total a column, look up a value in the third column, sort an array, wrap a risky call in an error handler) and the reader types the code into an editor that checks it against a reference implementation and flags the classic mistakes specifically: missing `Option Explicit`, `Integer` where `Long` is needed, `Offset` off by one, a cell-by-cell loop where an array read would do, and an uncancelled `On Error Resume Next`. A second panel visualises the performance rule by animating the number of VBA-to-Excel crossings for a cell-by-cell loop against a single array read, with a timer. A third panel simulates the debugger: a short program runs line by line while the reader sets breakpoints and inspects a Locals pane, which is far easier to learn by doing than to read about.

Where this goes next

The spreadsheet skills VBA automates are in advanced Excel, and the language to graduate to for anything data-heavy or statistical is Python for finance, with querying handled in SQL. The reason uncontrolled spreadsheets are a risk category rather than a style complaint is operational risk, with the surrounding control environment in trading floor compliance.

25.9 SQL

trading · structuring · sales core cross-asset

Everything a desk relies on lives in a database: trades, positions, prices, static data, risk runs, client history. SQL is how you ask it questions, and it is the most portable technical skill in this book, because the language is essentially the same at every bank and has been for decades.

It is also the skill where a small amount of precision goes a long way. Most people can write a query that returns rows. Rather fewer can say with confidence that the number it returned is right, and the difference is almost always one of the three things below.

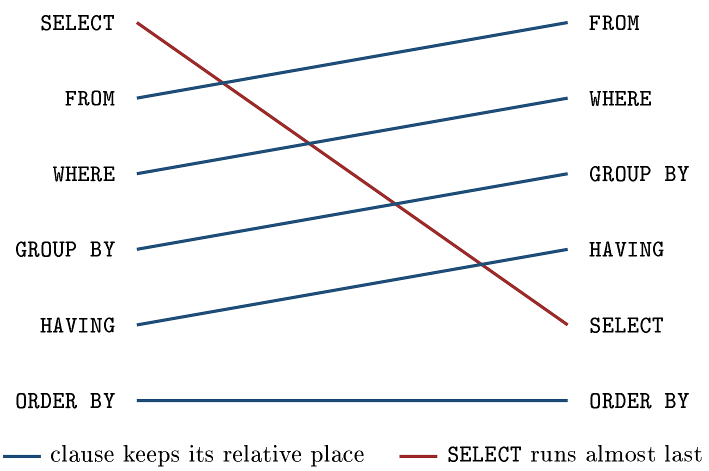
Declarative, which is the whole idea

SQL does not describe a procedure. You state the properties of the result you want and the engine decides how to obtain it: which table to read first, which index to use, how to perform the join. Two queries that look completely different can produce identical plans, and two that look similar can differ by orders of magnitude.

The consequence for a beginner is a habit rather than a technique. Describe the result, not the steps, and resist writing a loop in a language that does not have one.

The order clauses run in is not the order you write them

This is the single most useful fact about SQL and it answers a large share of the questions people actually have.



Written order is on the left, execution order on the right. Three consequences follow immediately and they are what interviewers are testing:

- **`WHERE` filters rows, `HAVING` filters groups.** `WHERE` runs before `GROUP BY` and so cannot see an aggregate; `HAVING` runs after and exists precisely to filter on one. "Trades over 10 million" is a `WHERE`; "clients whose total is over 10 million" is a `HAVING`.
- **You cannot use a `SELECT` alias in `WHERE`**, because `SELECT` has not run yet. You can use it in `ORDER BY`, because that runs after. People discover this rule by trial and error and never learn why.
- **Filter as early as possible.** A condition in `WHERE` reduces the rows before the grouping work happens; the same condition in `HAVING` does the work first and throws it away.

Joins, and the error that does not announce itself

An inner join keeps rows that match on both sides; a left join keeps every row from the left table and fills the right with nulls where there is no match. That much is in every tutorial.

What the tutorials underplay is that a join can *increase* the row count, and that this is the most common way a query returns a plausible and wrong number.

WORKED EXAMPLE

A trades table and a ratings table:

`trades`

id	instrument	notional
1	BOND_A	10
2	BOND_A	20
3	BOND_B	30

`ratings`

instrument	agency	rating
BOND_A	SP	A
BOND_A	Moodys	A2
BOND_B	SP	BBB

The total notional is plainly $10 + 20 + 30 = \mathbf{60}$. Now join to the ratings table to report the total by rating:

```
SELECT SUM(t.notional) FROM trades t
JOIN ratings r ON t.instrument = r.instrument
```

The answer comes back as **90**.

Nothing is broken. BOND_A has two ratings, so each of its two trades matches two rating rows, turning three trade rows into five joined rows and counting BOND_A's 30 twice. The query is valid, the engine is correct, and the number is 50% too high with no warning anywhere.

The fix is to make the right-hand side unique before joining, either by filtering (`AND r.agency = 'SP'`, which restores 60) or by aggregating the ratings table down to one row per instrument first.

KEY IDEA

Before joining, ask what the key is on each side. One-to-one and many-to-one are safe; many-to-many multiplies rows and silently inflates every sum downstream. The habit worth building is to check the row count before and after every join you write, because a join that changed it unexpectedly has changed your answer too.

Window functions, which is where finance lives

An aggregate collapses rows. A **window function** computes across a set of rows and *keeps* them, which is exactly what time series work needs.

- `LAG` and `LEAD`: the previous or next value. A return becomes `price / LAG(price) OVER (PARTITION BY instrument ORDER BY date)` minus one.
- `ROW_NUMBER` and `RANK`: numbering within a group, which is how you get the latest record per instrument, the standard shape being to number by date descending and keep number one.
- `SUM` and `AVG` with `OVER`: running totals and moving averages without a self-join.

Anyone who reaches for a correlated subquery to answer "the most recent price for each instrument" is writing something a window function does more clearly and usually far faster.

The query that matters most: as of a date

Financial data is **bitemporal**. Every fact has two dates: the date it refers to, and the date you learned it. A closing price for last Tuesday may have been revised on Thursday, and a credit rating has both an effective date and an announcement date.

A query that ignores the distinction produces a backtest that used information it could not have had, which is the same look-ahead problem that appears as a one-character error in pandas and as a design flaw in backtesting frameworks. In SQL it appears as a missing predicate: the correct historical query filters on *both* the effective date and the knowledge date, taking the latest version of each fact that was known as of the simulation date rather than the latest version that exists now.

> INTERVIEW TRAP
>
> Selecting the current value of a static attribute for a historical analysis. A query that joins today's sector classification onto five years of returns has quietly given every company the sector it ended up in, which is survivorship and look-ahead bias arriving through a join rather than through a model.

Enough performance to not be a nuisance

Four things account for most of the difference between a query that runs and one that finishes.

- **Indexes** exist on the columns you filter and join on. If a query is slow, the first question is whether it is using one.
- **Do not wrap an indexed column in a function.** A predicate written as `YEAR(trade_date) = 2024` usually cannot use the index on `trade_date`, while the equivalent range condition `trade_date >= '2024-01-01'` together with `trade_date < '2025-01-01'` can.
- **Select the columns you need**, not `*`, particularly on wide tables and columnar stores where the unread columns are genuinely free.
- **Read the plan.** Every database will show you what it intends to do, and the difference between an index seek and a full scan is visible there before you run anything.

Interview angle

REAL INTERVIEW QUESTION (ECB · Market Risk)

How do the SQL `ORDER BY` and `GROUP BY` clauses work? Give examples.

WHAT THEY TEST / ANSWER

Answer each one, then give the fact that ties them together, because the third part is what distinguishes someone who uses SQL from someone who has read about it.

`GROUP BY` collapses. It partitions the rows into groups sharing the listed values and returns one row per group, so every selected column must either appear in the `GROUP BY` or be wrapped in an aggregate. Example: total notional per instrument is `SELECT instrument, SUM(notional) FROM trades GROUP BY instrument`.

`ORDER BY` sorts. It does not change which rows come back, only their sequence, and it is the last thing that happens. Example: add `ORDER BY SUM(notional) DESC` to rank instruments by size, and note that you may sort by a `SELECT` alias here even though you could not have used it in `WHERE`.

Then the tying fact: clauses do not execute in the order they are written. The engine runs `FROM`, then `WHERE`, then `GROUP BY`, then `HAVING`, then `SELECT`, and `ORDER BY` last. Three things fall straight out of that and are worth saying explicitly, because they are what the question is really checking. `WHERE` cannot filter on an aggregate because it runs before the grouping, which is what `HAVING` is for. A `SELECT` alias is unavailable in `WHERE` and available in `ORDER BY`, for the same reason. And filtering in `WHERE` rather than `HAVING` is faster, because it removes rows before the grouping work is done rather than after.

A concrete contrast finishes it: “trades larger than 10 million” is a `WHERE`, and “clients whose *total* is larger than 10 million” is a `HAVING`. Candidates who can produce that pair on demand rarely get the rest wrong.

INTERVIEW TRAP

Four errors. (1) Joining on a non-unique key and trusting the resulting sum, which the example above showed inflating 60 into 90 without any error. (2) Filtering a left join’s right-hand table in `WHERE` rather than in the `ON` clause, which turns it back into an inner join and silently drops the unmatched rows you kept the left join for. (3) Forgetting that `COUNT(column)` ignores nulls while `COUNT(*)` does not. (4) Querying today’s version of a historical fact, which is look-ahead bias arriving through a join.

Where this goes next

Once the data is out of the database it usually lands in NumPy and pandas, and the join and alignment traps there are the same ideas in a different syntax. How the data gets into the database and stays correct is financial data pipelines, the place where a point-in-time error does the most damage is backtesting frameworks, and the statistical framing of why any of it matters is time series analysis.

25.10 Intro to C++ for finance

sales · trading · structuring core cross-asset

Walk onto any derivatives floor and you will find the same split. The screens, the notebooks, the ad hoc analysis a trader asks for at 9am and wants by 10am: that is Python. The library that actually prices the book, the engine that revalues fifty thousand trades overnight, the process that answers a market-data tick in microseconds: that is almost always C++. Python is the cockpit, C++ is the engine.

You are not expected to arrive as a C++ expert. You are expected to know what the language is for, to be able to read it, and to answer a handful of questions that come up again and again because they separate people who have compiled something from people who have only heard of it. This section covers exactly that: enough C++ to hold the conversation and to understand what the pricing library on your desk is doing.

REAL INTERVIEW QUESTION (Société Générale · desk not recorded)

What is the difference between a compiled language and an interpreted language?

WHAT THEY TEST / ANSWER

A compiled language (C++, C, Rust) is translated ahead of time by a compiler into machine code for a specific CPU. You ship an executable that the hardware runs directly, type errors are caught before the program ever starts, and the compiler gets one long opportunity to optimise (inlining, loop unrolling, vectorisation). An interpreted language (classically Python) is executed by a runtime that walks the program and performs each operation as it meets it, so each operation carries interpreter overhead, but you get dynamic typing and a much faster edit-run cycle.

What they are really testing is whether you understand the trade-off well enough to choose. The sharp answer adds that the boundary is blurry: CPython compiles to bytecode and then interprets it, Java and C# use a just-in-time compiler, and NumPy is a thin Python layer sitting on top of compiled C. So the desk answer is not "C++ is fast, Python is slow", it is: use Python for research, glue and anything a human iterates on, and push into C++ the inner loop that runs a hundred million times.

INTUITION

Think of two ways to use a foreign cookbook. **Compiling** is paying a translator once to produce a full English edition: the translation is slow and must be redone whenever the recipe changes, but afterwards you cook straight from the page at full speed. **Interpreting** is cooking with a translator standing next to you reading each line aloud: you can change the recipe mid-way and start instantly, but every single line costs you the translator's time. If you cook the dish once, take the translator. If you cook it a million times, pay for the edition.

The shape of a C++ program

C++ is *statically typed*: every variable has a type fixed at compile time, and the compiler refuses to build the program if the types do not line up. That feels heavy after Python, and it is exactly the point. On a risk engine, a type error you discover at compile time costs a minute, and the same error discovered at 2am during the overnight batch costs a lot more.

```
#include <cmath>
#include <vector>

// Present value of a fixed set of cashflows under continuous compounding,
// ignoring credit and any day-count subtlety.
double presentValue(const std::vector<double>& cashflows,
                    const std::vector<double>& times,
                    double r)
{
    double pv = 0.0;
    for (std::size_t i = 0; i < cashflows.size(); ++i)
        pv += cashflows[i] * std::exp(-r * times[i]);
    return pv;
}
```

Three things in that snippet are worth naming, because each is a question in disguise. `double` is a 64-bit floating point number, the default numeric type of every pricing library. The standard dynamic array, `std::vector<double>`, is contiguous in memory, which is why it is fast to walk. And `const std::vector<double>&` is a *reference to a constant*: we pass the vector without copying it, and promise not to modify it. Passing a large object by value instead, dropping that `&`, would silently copy every element on every call. That is the single most common performance bug in beginner C++ on a desk.

Memory: the part Python hides from you

Python never asks you where an object lives. C++ does, and this is where most interview questions land.

A local variable lives on the **stack**: it is created when execution enters the block, destroyed automatically when it leaves, and the whole thing costs essentially nothing. An object created with `new` lives on the **heap**: it survives until someone destroys it, it can be as large as you like, and allocating it is comparatively expensive. A **pointer** is how you refer to something that lives somewhere else.

REAL INTERVIEW QUESTION (Credit Suisse · desk not recorded)

What is a pointer?

WHAT THEY TEST / ANSWER

A pointer is a variable whose value is a memory address: it does not hold the object, it holds where the object is. You declare it `double* p`, you take the address of an existing object with `&x`, and you read through it with `*p` (dereferencing). A pointer can be null, meaning it points at nothing, and it can be *dangling*, meaning it points at memory that has already been freed, which is undefined behaviour and the classic source of crashes that only appear in production.

The follow-up you should pre-empt is the difference with a **reference** (`double& r`). A reference must be bound when it is created, can never be rebound to something else, and cannot be null: it is an alias for an existing object. Rule of thumb on a modern codebase: use references for “I need to look at this object without copying it”, use `std::unique_ptr` or `std::shared_ptr` when someone must *own* a heap object, and keep raw pointers only as non-owning observers. Saying that last sentence is what marks you as having written C++ this decade rather than in 1998.

The mechanism behind those smart pointers is the idea C++ is genuinely built on, and it has an ugly name: **RAII**, resource acquisition is initialisation. A C++ object's destructor runs automatically and deterministically when the object goes out of scope, so if you acquire a resource in the constructor and release it in the destructor, the resource can never leak, even if an exception is thrown halfway through. That is why `std::vector` frees its own memory, why `std::unique_ptr` deletes what it owns, and why well-written C++ contains almost no explicit `delete`.

Objects: how a pricing library is organised

REAL INTERVIEW QUESTION (CACIB · Exo)

Can you code? Which languages? What is object-oriented programming?

WHAT THEY TEST / ANSWER

Answer the first two honestly and specifically (name the language, the project, what it produced), then give object-oriented programming as four words with a reason attached, not as a recited list. **Encapsulation**: an object hides its internal state behind an interface, so a caller cannot corrupt it. **Abstraction**: the caller programs against what an object does, not how it does it. **Inheritance**: a derived class reuses and specialises a base class. **Polymorphism**: the same call, through the same base-class handle, dispatches to the right derived implementation at run time.

What they are testing is whether you can connect this to a pricing library. The connection is: an exotic desk adds new payoffs every month. If pricing is one enormous function full of `if` branches, every new product means editing and re-testing code that already works, and every trader edit risks breaking last month's product. If instead a payoff is an *object* implementing a common interface, a new product is a new class, and nothing that already works is touched. Interviewers on an exotics desk are listening for that sentence.

Here is the pattern in its smallest honest form. A payoff is an object that turns a terminal spot level into a cashflow, so it exposes exactly one operation.

```
#include <algorithm>   // std::max

class Payoff {
public:
    virtual double operator()(double S) const = 0;  // pure virtual
    virtual ~Payoff() = default;                    // virtual destructor
};

class CallPayoff : public Payoff {
public:
    explicit CallPayoff(double strike) : K_(strike) {}
    double operator()(double S) const override {
        return std::max(S - K_, 0.0);
    }
private:
    double K_;
};

class DigitalPayoff : public Payoff {
```

```
public:
    DigitalPayoff(double strike, double coupon)
        : K_(strike), c_(coupon) {}
    double operator()(double S) const override {
        return S > K_ ? c_ : 0.0;
    }
private:
    double K_, c_;
};
```

The pricing engine never mentions calls or digitals. It takes a `const Payoff&`, generates paths, and evaluates the payoff at maturity. Adding a new product means adding a class, and the engine is not recompiled in anger. The payoffs themselves are the ones you already know from the options section, and are not re-derived here.

REAL INTERVIEW QUESTION (JP Morgan · Exo)

In C++, what is an abstract class?

WHAT THEY TEST / ANSWER

A class containing at least one **pure virtual** function, that is a virtual function declared `= 0` with no required implementation, as with `operator()` in `Payoff` above. You cannot instantiate an abstract class: `Payoff p;` does not compile. You can only derive from it, and a derived class becomes concrete once it overrides every pure virtual function it inherited. Its purpose is to define an *interface*: a contract that every payoff, every model, every calibration routine must satisfy. Note that an abstract class may still hold data members and fully implemented methods, and a pure virtual function may even have a body that derived classes call explicitly.

There are two follow-up traps, and both are in the snippet above. First, the **virtual destructor**: if you hold derived objects through a `Payoff*` and delete one, and the base destructor is not virtual, the behaviour is undefined and in practice the derived destructor never runs, so resources leak. Any abstract base class needs a virtual destructor. Second, `override`: writing it makes the compiler verify that you are genuinely overriding a base method rather than accidentally declaring a new one with a slightly different signature, which is a silent bug that would otherwise show up as "my new payoff prices exactly like the vanilla".

Templates, the STL, and the cost of a container

The other half of the language is *generic* programming. A **template** is a class or function written once with the type left as a parameter, and instantiated by the compiler for each concrete type actually used. `std::vector<double>` and `std::vector<Trade>` are two different classes generated from one piece of source. Crucially this costs nothing at run time: the specialisation happens during compilation, so a template is not the same thing as run-time polymorphism through a virtual function. Pricing libraries use templates heavily to write one Monte Carlo loop that works for `double` and for an automatic-differentiation type that carries derivatives alongside the value.

The standard library ships the containers you will actually use, and knowing their cost is a fair interview question:

Container	Lookup	Use it when
`std::vector`	$O(1)$ by index	you walk it in order (almost always)
`std::map`	$O(\log n)$	you need the keys kept sorted
`std::unordered_map`	$O(1)$ average, $O(n)$ worst	you look things up by key
`std::deque`	$O(1)$ by index	you push and pop at both ends

The practical point behind the table is that `std::vector` is contiguous in memory, so walking it streams cleanly through the CPU cache, while a `std::map` scatters its nodes across the heap and every step is a potential cache miss. This is why the honest default on a desk is: use a vector, and justify anything else. That reasoning is developed properly in low-latency trading.

REAL INTERVIEW QUESTION (Natixis · Delta One)

I have the digits 1 to 9 in an array. I replace one of them with 0, without telling you which. Write code that finds the missing digit, optimising both time and memory complexity. Now I also want the index of the entry that was changed, while still knowing the digit. How do you do it?

WHAT THEY TEST / ANSWER

The digits 1 through 9 sum to 45. Replacing one digit x by 0 removes x and adds nothing, so the array now sums to $45 - x$, and therefore $x = 45 - \sum a_i$. The zero is still sitting in the array, so its index is found by the same scan: one pass, one running total, one saved index. That is $O(n)$ time and $O(1)$ extra memory, and both parts of the question fall out of a single loop.

```
std::pair<int,int> findReplaced(const std::vector<int>& a) {
    int sum = 0, idx = -1;
    for (std::size_t i = 0; i < a.size(); ++i) {
        sum += a[i];
        if (a[i] == 0) idx = static_cast<int>(i);
    }
    return {45 - sum, idx};   // {missing digit, its index}
}
```

What they are testing is whether you reach for arithmetic or for a data structure. The tempting answer, build a `std::unordered_set` of what you saw and check 1..9 against it, is also $O(n)$ in time but $O(n)$ in memory and much slower in practice. Say the sum trick, then say why it beats the set. The generalisation they may push you to: if the digit had been replaced by an arbitrary unknown value b rather than by 0, one sum gives you only $b - x$, so you need a second equation, and the standard choice is the sum of squares ($1^2 + \cdots + 9^2 = 285$), which gives $b^2 - x^2$ and determines both.

Worked example: why memory layout is a pricing decision

C++ makes you choose how your numbers are laid out, and that choice is sometimes the whole performance story. Take a recombining binomial tree with N steps (the model itself is not our subject here, only the storage).

WORKED EXAMPLE

A recombining tree has $k+1$ nodes at step k, so storing the whole tree from step 0 to step N takes

$$\sum_{k=0}^{N}(k+1) \;=\; \frac{(N+1)(N+2)}{2} \text{ nodes.}$$

With $N = 1000$ steps that is $\frac{1001\times1002}{2} = 501{,}501$ nodes. At 8 bytes per `double`, the full tree occupies $501{,}501 \times 8 = 4{,}012{,}008$ bytes, roughly 4 MB.

But a backward induction only ever needs the column it is writing and the column it is reading, and those can be the same array if you sweep it in the right direction. So one `std::vector<double>` of $N+1 = 1001$ entries is enough: $1001 \times 8 = 8{,}008$ bytes, roughly 8 kB. The ratio is $501{,}501/1{,}001 = 501$ exactly, so the rolling version uses 501 times less memory, and, more importantly, its whole working set fits comfortably in CPU cache while the 4 MB version does not.

The arithmetic is unremarkable. The habit is not: in C++ you are expected to ask "what does this actually allocate?" before you ask "is my loop tight?"

The same instinct applies to Monte Carlo, which is where most C++ time on a derivatives desk is really spent: the method is described in Monte Carlo methods, and the reason it lives in C++ rather than in Python is simply that the inner loop runs hundreds of millions of times and is trivially parallel across paths.

VISUAL / CODE TO BUILD: Interactive

A side-by-side "same algorithm, two languages" panel. The reader picks an algorithm (binomial backward induction, or a Monte Carlo European call), sets the size parameter (N steps, or number of paths), and the panel shows: the Python and C++ source next to each other, the measured run time of each on the server, and a memory bar for the full-tree versus rolling-vector layouts. The intended realisation is twofold: the gap widens as the size parameter grows, and switching the C++ version from the full tree to the rolling vector changes the run time even though the operation count is identical. Ship the measured numbers as a table too, so the point survives without the widget.

INTERVIEW TRAP

Four ways candidates lose points here. (1) Claiming "C++ is fast and Python is slow" with nothing behind it: the interviewer will ask why, and the answer is compilation, static types and memory layout, not folklore. (2) Passing large objects by value because the code compiles anyway, then wondering why the revaluation is slow. (3) Writing an abstract base class without a virtual destructor, which is the most reliable way to fail a C++ screen on an exotics desk. (4) Reciting "encapsulation, inheritance, polymorphism, abstraction" without being able to say what any of them buys a pricing library. Every one of these is fixed by having actually built something small and being ready to talk about it.

Where this goes next

C++ is one leg of a pair. The other is Python for finance, which is what you will write daily and what usually calls into the C++ library rather than replacing it. The libraries that this section's object model actually describes are covered in pricing libraries, and the point where every microsecond of the language starts to matter is low-latency trading. The finance those engines implement sits elsewhere in the book: the Black-Scholes model for what is being computed, and the Greeks for what the engine is asked to produce alongside the price.

25.11 Low-latency trading

trading advanced cross-asset

At the fast end of the market, the winner of a trade is decided by who reacts first, and the units are microseconds. This is a different engineering problem from everything else in this book, and it is worth understanding even though almost no bank front-office role requires you to do it: it explains a large part of how modern markets behave, and an interviewer will use it to see whether you know where your own seat sits.

The physics comes first

Before any code, the speed of light sets the floor. Light travels about 300,000 kilometres per second in vacuum and roughly two thirds of that in optical fibre, so **one microsecond is about 200 metres of fibre**.

WORKED EXAMPLE

New York to Chicago is roughly 1,150 kilometres. A round trip in fibre at 200,000 km/s takes at least

$$\frac{2 \times 1{,}150}{200{,}000} = 11.5 \text{ ms},$$

and real fibre routes are longer than the straight line, which is why the historical figure was around 13 milliseconds. Microwave through air travels at close to the vacuum speed, giving a theoretical round trip of 7.7 ms and a real one under 9.

That gap, several milliseconds, was worth building a chain of microwave towers for, and it is the clearest illustration of the economics: at this end of the market **latency is bought with capital**, in colocation, in hardware and in physical routes, not with cleverness.

Where the time actually goes

The latency budget runs from a market data packet arriving to an order leaving, and it has four parts.

- **The wire**: propagation, plus every switch the packet passes through. Solved by **colocation**, putting the machine in the same building as the matching engine, which removes propagation almost entirely and makes the rest matter.
- **The network stack**: the kernel's handling of the packet, which is avoided by **kernel bypass**, letting the application read from the network card directly.
- **The application**: decoding the message, updating the book, deciding, and encoding an order. This is the part software can control and where most of this section applies.

- **The exchange**: gateway and matching engine time, which is outside your control and is the reason the queue position you achieve matters as much as your own speed.

The extreme answer to the middle two is to stop using a general-purpose computer at all and implement the logic in an **FPGA**, where the decision is made in hardware as the packet arrives, at the cost of a development cycle measured in weeks rather than minutes.

> KEY IDEA
>
> Optimise the **tail**, not the average. A strategy's profit is destroyed by the occasional slow response, because that is exactly when the market has moved and you are the one being picked off. So the metric is the 99th and 99.9th percentile of response time, and the enemy is anything that is fast *usually*: garbage collection, memory allocation, page faults, lock contention, an operating system deciding to schedule something else. This is why low-latency code is written in C++ rather than a managed language, and it is the single most important idea in the section.

What that does to the code

The constraints follow from the key idea, and every one of them is about avoiding an occasional expensive event.

- **No allocation on the hot path.** Memory is allocated up front and reused, because a call to the allocator is fast on average and occasionally not.
- **No locks**, since contention is unbounded. Communication between threads uses lock-free queues, and often the hot path is pinned to a single core with nothing else on it.
- **Cache locality above everything.** A main-memory access costs on the order of a hundred times a cache hit, so data is laid out contiguously and processed in order. This is why arrays of structures are rearranged into structures of arrays.
- **Predictable branches**, since a mispredicted branch costs tens of cycles, and the rare-case code is moved out of the common path.
- **Busy-wait rather than sleep**, because waking a thread costs more than spinning.

REAL INTERVIEW QUESTION (Société Générale · desk not recorded)

How do you optimise a piece of code, for instance loops that run too long or unnecessary initialisations?

WHAT THEY TEST / ANSWER

Resist the urge to list tricks. The answer that scores is an *order of operations*, because the tricks are worthless applied in the wrong order.

Measure first. Profile before changing anything, and find where the time actually goes. Intuition about hot spots is wrong often enough that guessing is the single most expensive habit in performance work. On this kind of system you profile the *distribution*, not the mean, for the reason given above: the 99.9th percentile is what costs money.

Then remove work, in decreasing order of size. A better algorithm or data structure first, since no amount of tuning rescues the wrong one. Then work done repeatedly that need not be: hoist invariant computation out of the loop, avoid recomputing what can be cached, and do not initialise or allocate anything the loop does not use. Only then reach for the micro level, and on a hot path that means the constraints above, no allocation, no locks, contiguous data, predictable branches.

Then let the compiler help. Optimisation flags, and code written so the compiler can vectorise it: simple loops over contiguous arrays, no aliasing it cannot rule out, no call it cannot inline. A great deal of hand tuning is the compiler's job done badly.

Then measure again, because roughly a third of plausible optimisations make things worse, and on a hot path you will not notice without a number.

Close on judgement, since it is what separates the answer. Optimise the code that is on the hot path and leave the rest legible: the setup, the configuration and the recovery paths run once and should be written for clarity. Knowing which part of the program you are in is the actual skill. The language mechanics underneath this, virtual dispatch, templates and the rest, are in intro to C++ for finance.

Complexity is not the whole story

REAL INTERVIEW QUESTION (Société Générale · QIS)

Does algorithmic complexity mean anything to you?

WHAT THEY TEST / ANSWER

Give the definition, then the caveat that matters in this context, because the caveat is what shows you have measured something rather than read about it.

The definition. Big-O describes how the running time or memory of an algorithm grows with the size of the input, ignoring constant factors: $O(1)$ constant, $O(\log n)$ for a binary search, $O(n)$ for a scan, $O(n \log n)$ for a comparison sort, $O(n^2)$ for a naive double loop, $O(2^n)$ for the naive Fibonacci recursion. It is the right tool for choosing between algorithms when n can grow.

The caveat. It deliberately discards the constants, and on a hot path the constants are the whole problem. A hash lookup is $O(1)$ and can be slower than a linear scan of a contiguous array of twenty elements that sits in cache, because the scan has perfect locality and the hash has a pointer chase and a probable cache miss. So for small n, **memory layout beats asymptotics**, and the only way to know is to measure.

Then give the finance-specific version, which is the real answer here: in a low-latency system the input size is usually small and fixed, an order book of a few levels, so complexity rarely decides anything and cache behaviour, allocation and branch prediction decide everything. In a research or risk context the input is large and complexity decides everything, which is why an $O(n^2)$ correlation matrix build or an exponential recursion is a real problem there and not here. Knowing which regime you are in, and saying so, is the answer.

Risk at speed

Speed does not remove the obligation to control risk, and the controls have to live in the hot path.

Pre-trade checks on price, size, position and rate of ordering are required by regulation and by common sense, and they must run before every order, which makes them a latency cost that cannot be optimised away. **Kill switches** allow a strategy or a whole session to be stopped instantly, from outside the application. And **deployment discipline** is the control that matters most: the best-known catastrophic loss in this business came from a software deployment that left old code running on one server, which then traded uncontrollably for a matter of minutes. That was not a strategy failure or a latency failure, it was a release process failure, and it is why this business treats deployment as a risk control rather than as an engineering detail.

INTERVIEW TRAP

Four errors. (1) Optimising the average when the tail is what costs money. (2) Assuming faster is always better, when queue position, fee structure and being right about the signal often matter more. (3) Applying these techniques to code that is not on the hot path, which trades clarity for nothing. (4) Claiming low-latency experience in a bank interview when the desk in question trades in seconds, which signals you do not know what the seat does.

Where this goes next

The language this is written in is intro to C++ for finance, the market mechanics being raced against are market microstructure and exchanges, and the business these techniques serve is market making and market makers. Everything a desk does that is not on a hot path is written in Python.

Chapter 26

Interviews & Front-Office Mindset

26.1 How Trading Interviews Work

sales · trading · structuring   core   cross-asset

This chapter opens with the mechanics, because most of the preparation people do is misdirected rather than insufficient. They study the wrong material for the wrong stage and are surprised by a format nobody described to them.

What is owned here is the *process*: the stages, who runs each one, what the question database says about who asks what, and the shape a technical round actually takes. How to construct a technical answer is technical interview logic, what the interviewer is inferring from it is what interviewers really test, and the behavioural side is behavioural signals.

You are not hired by a bank

> **KEY IDEA**
>
> You are hired by a **desk**, usually by the two or three people you will sit with. The bank's brand runs the screening and the human resources process; the desk decides. This single fact explains the format of everything that follows: why the technical round is narrow and deep rather than broad, why "why this desk" is a harder question than "why this bank", and why a strong candidate can be rejected by one desk and hired by another in the same building the same week.

The stages are conventional, and the weight is not evenly spread across them.

- **Screening.** Curriculum vitae, sometimes an online test. Filters on school, grades and prior internships. Nothing you say matters yet.
- **Human resources round.** Motivation, mobility, languages, availability. Rarely decisive in your favour, frequently decisive against you, which is the asymmetry to plan around.
- **Desk technical rounds.** One to four conversations with traders, structurers or salespeople. This is the interview. Everything in this book that is not this chapter is preparation for these conversations.
- **Final round or superday.** Several back-to-back meetings, often including a senior manager and a desk head, sometimes a case study or a pitch.

In the French market there is an extra stage that is easy to miss because it is not called an interview: the **internship**. A six-month stage or a VIE on the desk is the real assessment, and the graduate offer is very often decided by people who have already watched you work for months. Treat the internship as the interview it is, and the eventual "interview" is a formality by comparison.

What the question database actually contains

The 3,394 questions in `questionsentretients.csv` are a record of what was really asked rather than of what anyone thinks should be asked, and its shape is worth reading before you plan any preparation.

Questions recorded	3,394
Distinct banks	60
Distinct desks	18
Questions with a desk recorded	2,519 (74%)
Share from the five most-represented banks	1,799 (53%)
Société Générale alone	885 (26%)
EQD and Structured Products, of those with a desk	1,035 (41%)

Three conclusions follow, and they are the practical content of this section.

One, preparation should be desk-weighted, not evenly spread. Two desks account for two fifths of everything asked where the desk is known. If you are interviewing for equity derivatives or structured products, the chapters on options, Greeks, strategies and structuring are not one topic among many, they are the syllabus. If you are interviewing for rates or credit, a large part of the equity material is background rather than examinable.

Two, the distribution is concentrated by bank as well. Five institutions produced half the questions, which means the observed style is mostly the style of the large French houses: technical, fast, and comfortable asking you to compute something on the spot. That is a real feature of the market and not an artefact, since those are also the desks that hire the most graduates in this market.

Three, a quarter of the entries carry no desk at all, and those are the general rounds: motivation, brain teasers, market questions. Preparing only desk technicals leaves a quarter of the exposure uncovered, and it is the quarter that comes first chronologically.

INTUITION

Read the database as a frequency table, not as a syllabus to memorise. Nobody can prepare 3,394 answers, and nobody needs to: the questions repeat, and they repeat because they test a small number of underlying ideas from different angles. The same put-call parity relation is behind a dozen apparently different questions across five desks. Learning the idea answers all of them; learning the answers answers one.

The format of a technical round

A technical round is not a quiz with independent items. It is a small number of opening questions, each of which is followed until you stop being able to answer. The interviewer is measuring *where* you stop, not whether you start.

This has a concrete consequence for how to speak. Answer the question that was asked, completely and briefly, and then volunteer the next layer yourself rather than waiting to be pushed onto it. Doing so moves the conversation to ground you have chosen, and it demonstrates the depth the follow-up was going to test anyway. At least 293 of the recorded questions contain more than one question mark, which is a lower bound on how many were asked as explicit chains, since most interviewers chain without punctuation.

The second consequence is that "I do not know" is a usable move, and using it well is a skill. The right form is to say what you do not know, say what you would need to work it out, and then work out as much of it as you can out loud. Technical interview logic develops the technique.

REAL INTERVIEW QUESTION (Société Générale · Credit)

Tell me about yourself.

WHAT THEY TEST / ANSWER

This is the opening move of nearly every process and it is not small talk: it sets the agenda for the next forty minutes, because the interviewer will take their follow-ups from whatever you raise.

Structure it in three parts and keep it to about ninety seconds. **Where you come from**, in one sentence, education and the one thing about it that is relevant. **What you have done**, meaning the internships, with the emphasis on what you actually did rather than on the team's mandate. **Why you are here**, connecting the two to this desk specifically.

The tactical point, which is the reason to prepare it rather than improvise: whatever you mention, you will be asked about. So mention the things you want to be asked about. If you name a product you priced, expect to be asked how it is priced, and only name it if you can. A candidate who says "I built a pricer for autocalls" has just chosen the topic of the next twenty minutes, and that is either the best or the worst decision they make in the interview.

Do not recite the curriculum vitae chronologically. They have it, they have read it, and reading it aloud is the commonest way to lose the first two minutes.

REAL INTERVIEW QUESTION (CACIB · Structured Products)

Why the structured products desk rather than another desk?

WHAT THEY TEST / ANSWER

The comparative is the whole question. "Why structured products" can be answered with enthusiasm; "rather than another desk" can only be answered by someone who knows what the other desks do, and that is what is being checked.

So answer it as a comparison, in the vocabulary of the work.

- Against **flow trading**: structuring is a slower cycle, days rather than seconds, and the intellectual content is in designing the product rather than in pricing it fast.
- Against **sales**: you own the construction and the risk of the payoff, not the relationship.
- Against **exotics trading**: you are closer to the client problem and further from the hedging book, though you have to understand both to price honestly.

Then give one concrete reason drawn from something you actually did, since a preference with no evidence behind it reads as a preference for the job title. And close on the part of the work you find hard rather than the part you find attractive, because naming a real difficulty is the fastest way to show you know what the job is.

The reason this question carries so much weight is the one at the top of this section: the desk is doing the hiring, and it is deciding whether you want *this seat* or a front-office seat in general. The second answer loses.

REAL INTERVIEW QUESTION (CACIB · FX)

Why trading, when you have done sales?

WHAT THEY TEST / ANSWER

This is the mismatch question, asked whenever your record points at a different seat from the one you are applying for, and it is asked to find out whether you are moving toward something or away from something.

Answer in that order. **What the experience gave you** that transfers: sales work means you have talked to clients about risk, watched flow, and seen which products actually trade, all of which a trading desk values and cannot get from a graduate who has only studied. **What you found missing**, stated as a preference rather than as a complaint: you want to own the position rather than intermediate it. **Evidence that the preference is informed**, meaning something specific you did on the trading side, a book you followed, a hedge you understood, a pricing you checked.

Two traps. Do not criticise the previous desk or the people on it, because the interviewer's colleague may have trained you and the market is small. And do not claim the switch is trivial: sales and trading are different jobs, and saying so, then explaining why you want the harder-for-you one, is more convincing than pretending the distance is not there.

REAL INTERVIEW QUESTION (BNP · Fixed Income)

What do you know about BNP, and why BNP?

WHAT THEY TEST / ANSWER

This is a screening-round question rather than a desk question, and answering it at the wrong level of detail is the usual failure. They are not testing whether you memorised the annual report.

Give three things, each specific and each checkable. **Something structural**: the business the bank is actually strong in, the desks that lead their market, the regions it is built around. **Something current**: a recent development you can discuss, an acquisition, a change of strategy, a market it has entered or left, drawn from the last few months rather than from the corporate history page. **Something personal**: who you have spoken to there and what they told you, which is the only part of the answer another candidate cannot also give.

Then pivot to the desk within two sentences, because that is where the decision is made, and because a candidate whose answer is entirely about the institution has told the interviewer they applied to a brand.

The preparation that makes this easy is not reading, it is talking to people who work there. One conversation gives you a better answer to this question than a week of research, and it also tells you whether you want the job.

INTERVIEW TRAP

1. **Preparing evenly across topics.** The database is concentrated: two desks account for two fifths of the desk-tagged questions. Prepare in proportion to the seat you are interviewing for.

2. **Preparing only technicals.** A quarter of the recorded questions carry no desk, and those rounds come first. Failing the human resources round means the desk never meets you.

3. **Memorising answers rather than ideas.** The questions repeat because a small number of ideas are being tested from many angles. The answers do not transfer; the ideas do.

4. **Volunteering material you cannot defend.** Everything you raise becomes the next question. Naming a product, a model or a project is choosing the topic.

5. **Treating the internship as a rehearsal.** On a French desk it is the assessment, and the people deciding your offer have already seen you work.

6. **Answering "why this bank" when you were asked "why this desk".** They are different questions with different answers, and only one of them is asked by the person who decides.

VISUAL / CODE TO BUILD: Interactive

A preparation planner driven by the real database. The reader picks a target desk and a target bank, and the page filters the 3,394 recorded questions to that combination, reporting how many there are, which topics they cluster into, and which chapters of this book those topics map to. A coverage bar tracks how much of that filtered set the reader has worked through, so the plan is measured against what was actually asked of people applying to the same seat rather than against a generic syllabus. A second mode drops the desk filter and serves the general-round questions, which is the quarter of the database most readers under-prepare, and a timer mode asks them in a chain, following each answer with the next question from the same row so the format matches the round it is preparing for.

26.2 Technical interview logic

sales · trading · structuring core cross-asset

Candidates prepare for technical interviews by collecting answers. That is the wrong unit of preparation. There are thousands of questions and only about half a dozen question *shapes*, and an interviewer choosing between two candidates who both know the fact is deciding on the shape of the answer, not the fact.

This section is about that. What the shapes are, what each one is really testing, and the structure that turns a correct answer into a convincing one. The material is drawn from the same question database as the rest of the book, used here as specimens of form rather than as content.

The answer template

Almost every technical answer in this book follows the same five moves, in the same order. Use it until it is automatic, because under pressure structure is what survives.

KEY IDEA

1. **State the conventions.** "Ignoring rates and dividends", "European exercise", "annual compounding". One clause. It costs two seconds and it removes the interviewer's first objection before they make it.
2. **Give the headline answer.** One sentence, committed. Not "it depends", which is where weak answers live.
3. **Justify in one step.** The single mechanism, not the full derivation.
4. **Volunteer the nuance.** The exception, the second-order correction, or the reason the naive answer is slightly wrong. This is where the marks are, because it is the part the next candidate will not say.
5. **Stop.** Silence after a complete answer is a strong signal. Filling it is how prepared candidates talk themselves into an error.

Shape one: the ladder

The commonest shape. A trivially easy opening question, then the same question applied to a harder object, then a version where the obvious answer is wrong. The interviewer is not testing the first rung; they are finding the rung where you stop.

REAL INTERVIEW QUESTION (BNP · EQD)

What is the delta of an at-the-money call? Of a put? Of a call spread? Of an at-the-money straddle? And how does the delta of a deep in-the-money call evolve when volatility rises?

WHAT THEY TEST / ANSWER

Five rungs, and the last one is the whole question. Answer the first four crisply: about $+0.5$ for the call, about -0.5 for the put, a hump that peaks well below 1 for the call spread, and about zero for the straddle. Then earn the rung they are climbing to. A deep in-the-money call has a delta near 1; raising volatility widens the distribution, which makes it more likely the option ends up out of the money, so the delta *falls*. The general statement, which is the one they want, is that volatility **lowers in-the-money deltas and lifts out-of-the-money ones**, because more uncertainty makes the option look more at-the-money.

Do not promise a limit of 0.5, because there is not one. At spot 150 against a strike of 100 over one year, the delta falls from 0.98 at 20% volatility only as far as about **0.82**, reached at a volatility of $\sigma^* = \sqrt{2\ln(S/K)/T} = 90\%$, and then turns back *up*: once $\frac{1}{2}\sigma\sqrt{T}$ dominates $\ln(S/K)/(\sigma\sqrt{T})$ in d_1, every call delta rises towards 1 as volatility grows without bound. The out-of-the-money side does not stop at the middle either, it passes straight through it: at spot 100 against a strike of 130 the delta runs 0.11 at 20% volatility, 0.32 at 40%, 0.53 at 80% and 0.81 at 200%. If you want one sentence that is safe at every volatility, say that higher volatility compresses the *difference* between deltas rather than driving them to any particular level.

The tactical lesson is about pace. On a ladder, answer the easy rungs in a few words each and save your time and your caveats for the last one, because the interviewer is only marking the last one. Candidates who deliver a careful lecture on rung one run out of clock. Deltas themselves are in the Greeks and the composite case in Greeks evolution within strategies.

Shape two: the underspecified question

Some questions are missing a parameter, and the omission is deliberate. Answering anyway, with a silently assumed value, is the failure. Asking for it is the answer.

REAL INTERVIEW QUESTION (Société Générale · EQD)

Stock X has a spot of 1000, volatility 10%, rates at 6%, no dividends. Estimate the price of a digital paying 1,000,000 if the spot ends above 1000.

WHAT THEY TEST / ANSWER

The first move is a question, not a calculation: *what maturity?* Nothing here is computable without it, and noticing that is most of the mark. Say so, then show you could do it for any maturity by setting up the general form.

A cash-or-nothing digital struck at K is worth $e^{-rT}N(d_2)$ per unit of payout, with

$$d_2 = \frac{\ln(S/K) + (r - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}} = \frac{(0.06 - 0.005)}{0.10}\sqrt{T} = 0.55\sqrt{T},$$

since the option is struck at the spot so the log term vanishes. Now put in a maturity. At one year, $d_2 = 0.55$, $N(d_2) \approx 0.709$, and discounting at $e^{-0.06} = 0.942$ gives 0.668, so the digital is worth about 668,000. At three months it is about 599,000, and at two years about 693,000: the answer moves a lot, so guessing the maturity would have been guessing the answer.

Two things to add. The probability is above 50% even though the option is struck at the spot, because the drift under the risk-neutral measure carries the forward above the spot, and the $-\sigma^2/2$ term partly offsets it: at these parameters the drift dominates because the volatility is low. And this is a risk-neutral probability, not a forecast. Digitals themselves are in exotic options.

Shape three: the approximation, then its origin

Desks want numbers in seconds, so the approximations are asked constantly. The follow-up is always the same, and it is the real question: where does the constant come from?

REAL INTERVIEW QUESTION (Exane · Exo)

Price a one-year at-the-money call in your head at 20% volatility: give the price and the vega, and explain where the simplified formula $S_0 \times \sigma\sqrt{T} \times 0.4$ comes from, in particular the 0.4.

WHAT THEY TEST / ANSWER

Number first: $0.4 \times 0.20 = 8\%$ of spot, and the vega is about 0.4% of spot per volatility point, so 0.4 on a spot of 100. The exact Black-Scholes values are 7.97 and 0.397, so the mental estimate is right to within a few basis points.

Now the derivation, which is short enough to give in full and is the point of the question. With zero rates and the strike at the spot, $d_1 = \frac{1}{2}\sigma\sqrt{T}$ and $d_2 = -\frac{1}{2}\sigma\sqrt{T}$, so

$$C = S\big[N(d_1) - N(d_2)\big] \;\approx\; S \times 2\,n(0) \times \tfrac{1}{2}\sigma\sqrt{T} \;=\; \frac{S\,\sigma\sqrt{T}}{\sqrt{2\pi}},$$

using a first-order expansion of N around zero, and $1/\sqrt{2\pi} = 0.3989$, which is the 0.4. Say the two consequences that show you understand rather than remember. The price is linear in volatility at the money, so the price divided by volatility *is* the vega, which is why the same 0.4 answers both halves of the question. And the approximation degrades once the option is far from the money or once rates and dividends move the forward away from the spot, at which point you strike the approximation at the forward instead. The model itself is in the Black-Scholes model.

Shape four: list, then critique

A question that asks for a list is never marked on the list. It is marked on what you say after it.

REAL INTERVIEW QUESTION (Société Générale · Exo)

What are the assumptions of the Black-Scholes model, and what are its limits?

WHAT THEY TEST / ANSWER

Give the list quickly, because everyone can: a lognormal diffusion for the underlying with constant volatility, constant and known interest rates, no dividends in the basic form, continuous frictionless trading with no transaction costs and unlimited divisibility, no arbitrage, and European exercise.

Then spend your time on the second half, matched assumption by assumption to what the market actually does, because that is the part being marked. **Constant volatility** is the big one, and its fingerprint is the smile: if the assumption held, every strike would trade at the same implied volatility, and none of them do. **Lognormal returns** understate the tails, so crashes are far more frequent than the model allows, which is the same fact seen from a different angle. **Continuous costless hedging** is impossible, so a delta hedge is discrete and leaves a gamma-driven error, and transaction costs make the replication strictly more expensive than the theoretical price. **Constant rates** matters little for short-dated equity options and matters a great deal for long-dated and for rates products.

Close on the sentence that separates a good candidate from a well-read one: the model is wrong and is used anyway, because it is a *quoting convention.* Nobody believes the volatility is constant; the market quotes prices as implied volatilities so that one number per strike carries the price, and Black-Scholes is the invertible function that does the translation. Attacking the assumptions without saying that is a half-answer, since it leaves unexplained why an industry uses a model it knows to be false.

Shape five: explain it simply

Common on sales and structuring desks and increasingly everywhere. It tests whether you own the idea or only its vocabulary, and it is the one candidates most reliably over-engineer.

REAL INTERVIEW QUESTION (Société Générale · EQD)

How would you explain what a call is to an eight-year-old?

WHAT THEY TEST / ANSWER

Use one concrete story, no jargon, and no numbers beyond the ones a child can hold. Something like: you really want a bike that costs 100 euros, and you have saved up the 100, but you are not sure yet. You pay the shopkeeper 10 euros today so that they promise to sell you the bike for 100 euros at the end of the summer, whatever happens. If the bike becomes very popular and the price goes to 150, you still pay 100, so you have done well. If nobody wants that bike any more and it drops to 70, you simply do not buy it, and all you have lost is the 10 euros. That is a call.

The interviewer is checking three things and none of them is finance. Did you choose a *single* analogy rather than three competing ones. Did you avoid every technical word, including strike, premium and expiry, or introduce them only after the story. And did you get the asymmetry right, which is the only part that actually matters: a small known loss against a large unknown gain. Say it in under a minute, then stop and offer to add the technical vocabulary on top, which shows you can move between registers rather than only downwards.

The rules that apply to every shape

- **Think out loud, but only once you have a direction.** Five seconds of silence to choose an approach reads as composure. Five minutes of narrated wandering reads as panic.
- **Do arithmetic aloud and sanity-check the magnitude.** A call worth more than the spot, a probability above one, a negative variance: catching your own impossible number is worth more than not making the slip.
- **Check the sign before you speak.** Most technical errors in interviews are sign errors, and most sign errors are avoidable by naming who is long what.
- **Never bluff.** Interviewers on these desks bluff-test deliberately, and the cost of being caught inventing is not the question, it is the interview.
- **Answer the question that was asked.** Payoff or profit, price or value, at issuance or on an existing position: these pairs are where prepared candidates lose points by answering the neighbouring question.

REAL INTERVIEW QUESTION (Edmond de Rothschild · Sales)

What do you say when you do not have the answer to a client's question?

WHAT THEY TEST / ANSWER

Asked as a sales question, it is really the interview's own rule stated out loud. Give the three-part move. **Say you do not know**, plainly and without decorating it. **Say what you do know** that bounds the answer: the direction, the order of magnitude, the mechanism, the person or desk who has the exact number. **Commit to a specific follow-up**, with a time, and then do it. The reason this is the right answer for a client is the reason it is the right answer in the room: a wrong number delivered confidently destroys trust permanently, while an admitted gap closed within the hour builds it.

Apply it to yourself during the interview. "I have not worked with that product, but I would reason about it as follows" is a strong answer and often gets you most of the credit, because the interviewer is testing reasoning under uncertainty, which is the job. What loses the interview is the plausible-sounding invention, since the person opposite prices this for a living and will know within one follow-up.

INTERVIEW TRAP

Five habits that cost marks. (1) Starting with "it depends" instead of committing and then qualifying. (2) Giving the full derivation when one mechanism was asked for, which burns the clock the ladder needs. (3) Assuming a missing parameter silently rather than asking for it. (4) Reciting a list and stopping, when the list was the setup and the critique was the question. (5) Continuing to talk after a complete answer, which is how most candidates introduce the error that sinks them.

VISUAL / CODE TO BUILD: Interactive

A drill trainer built on the question database. The reader picks a track and a topic and gets one real question at a time, tagged by shape, with a timer calibrated to how long a desk would actually give: fifteen seconds for an approximation, ninety for a ladder. The reader types or records an answer, then sees the model answer split into the five template moves, with each move shown as present or missing in what they wrote, so the feedback is about structure rather than about the fact. A shape filter lets someone drill only underspecified questions, or only list-then-critique ones, and a progress view shows which shape they are weakest on. Underspecified questions are scored on whether the reader asked for the missing parameter before computing anything, which is the single highest-value habit in this section.

Where this goes next

The format and sequencing of the process itself is how trading interviews work, the probability and market-making puzzles are brain teasers, opinions on markets are market questions, and turning a view into a recommendation is trade pitching. What the whole exercise is actually measuring is in what interviewers really test.

26.3 Brain teasers

sales · trading · structuring core cross-asset

Brain teasers are the part of a trading interview candidates most fear and most misunderstand. They are not an IQ test and the interviewer is rarely interested in whether you have seen the puzzle before. They are a cheap way to watch you do, in two minutes and out loud, the thing a trader does all day: take an under-specified problem, impose structure on it, produce a number, and then check whether that number is sane. A candidate who reasons audibly and arrives at a slightly wrong answer usually scores better than one who goes silent and produces the right one, because only the first has shown the interviewer anything.

Every teaser in this section is a real question from the database, and each one is chosen because it teaches a different technique.

The method

Use the same five steps every time, and say the first four aloud.

1. **Restate the problem** in your own words and confirm the rules. Half of all teaser failures are a misread, and asking costs nothing.
2. **Name the random variable** and what you are computing: an expectation, a probability, an optimal decision.
3. **Look for the shortcut before computing.** Symmetry, conditioning, and linearity of expectation between them collapse most teasers to one line. Enumeration is the fallback, not the plan.
4. **Compute**, showing the steps.
5. **Sanity check**: bounds, a degenerate case, an extreme value. “It has to be between 3.5 and 6, and 4.25 sits comfortably inside” is worth as much as the arithmetic.

KEY IDEA

Three techniques solve most teasers. **Conditioning**: split on the first thing that happens and use the value of the remaining game. **Symmetry**: if the setup treats several objects identically, their probabilities are equal and you never have to compute one. **Linearity of expectation**: the expectation of a sum is the sum of the expectations *even when the terms are dependent*, which is what makes otherwise brutal problems trivial. If none of the three applies, only then start enumerating.

Conditioning: the value of a game you can replay

REAL INTERVIEW QUESTION (JP Morgan · EQD)

What is this game worth: you roll a die, a value appears, and you can either take that value or reroll once and take the new value? In your head.

WHAT THEY TEST / ANSWER

4.25. Do it by conditioning, out loud, in three sentences. First, the value of the game you can fall back on: a single roll is worth 3.5. Second, the decision rule: after the first roll you keep the number if it beats what a reroll is worth, so you keep 4, 5 or 6 and reroll on 1, 2 or 3. Third, the value: with probability 1/2 you keep a number that is uniform on $\{4, 5, 6\}$ and worth 5 on average, and with probability 1/2 you reroll and get 3.5, so

$$\tfrac{1}{2} \times 5 + \tfrac{1}{2} \times 3.5 = 4.25.$$

Then sanity check it in one line, because that is the part being graded: the answer must lie between 3.5, the value with no option, and 6, the best possible outcome, and it is comfortably inside. The interviewer's likely follow-up is "and with two rerolls?", so have the recursion ready rather than the number: the same logic applies with 4.25 replacing 3.5 as the fallback, so you now keep 5 or 6 and reroll otherwise. Finish with the connection they are fishing for on a trading desk: this is an American option. The reroll is the right to abandon a payoff you already hold in exchange for an unknown one, the value 3.5 is the continuation value, and "keep if the number beats the continuation value" is the exercise boundary.

Symmetry: when you should refuse to compute

REAL INTERVIEW QUESTION (Société Générale · EQD)

An urn holds $N-1$ red balls and one blue ball. N candidates for an internship each draw one ball in turn, without replacement, and whoever draws the blue ball gets the internship. Which position, from 1 to N, has the best chance?

WHAT THEY TEST / ANSWER

None of them. Every position has probability exactly $1/N$, and the interviewer is testing whether you will resist the urge to compute. Give the symmetry argument, which is the whole answer: shuffle the balls into a random order first and then hand them out in sequence, which is the same experiment. The blue ball is equally likely to be in any of the N places in that shuffled order, so each candidate's chance is $1/N$. If they push, confirm it the direct way for the second drawer to show the algebra agrees: $\frac{N-1}{N} \times \frac{1}{N-1} = \frac{1}{N}$, where the first factor is the first candidate missing and the second is drawing the blue ball from what is left. Then add the point that makes it a trading answer rather than a maths answer: the later candidates have more *information* as they go, since they can see the draw shrinking, but information you receive after your decision is made changes nothing about the unconditional probability. That distinction, between what you know now and what you will know later, is the same one that separates a conditional from an unconditional expectation on a desk.

Linearity of expectation: the technique that feels like cheating

REAL INTERVIEW QUESTION (Morgan Stanley · QIS)

N coats and N people. What is the expected number who get their own coat back?

WHAT THEY TEST / ANSWER

1, for every N, and the reason is linearity of expectation. Define $X_i = 1$ if person i gets their own coat and 0 otherwise, so the total is $X = \sum_i X_i$. Each person is equally likely to receive any of the N coats, so $\mathbb{E}[X_i] = 1/N$, and therefore

$$\mathbb{E}[X] \;=\; \sum_{i=1}^{N} \mathbb{E}[X_i] \;=\; N \times \tfrac{1}{N} \;=\; 1.$$

Say explicitly why this is legitimate, because it is the point of the question: the X_i are *not* independent, since one person getting their coat changes the odds for everyone else, and linearity of expectation does not care. That is exactly the property that makes it powerful. The distribution of X is genuinely hard, and its expectation is one line. Two things to have ready. The answer does not depend on N, which is worth flagging as surprising rather than letting them find it. And the same trick is how you compute the expected number of anything on a desk: sum the indicator expectations for each name, each day or each observation date, without ever modelling the dependence between them.

Optimising over a decision

REAL INTERVIEW QUESTION (BNP · EQD)

There are two baskets, 50 white balls and 50 black balls. How do you arrange the balls in the two baskets so that, when a basket is chosen at random and then a ball drawn at random from it, the chance of a white ball is maximised?

WHAT THEY TEST / ANSWER

Put **one white ball alone** in the first basket and everything else, 49 white and 50 black, in the second. Then

$$\tfrac{1}{2} \times 1 \;+\; \tfrac{1}{2} \times \tfrac{49}{99} \;=\; \mathbf{0.747},$$

against 0.5 for any even split. Explain the intuition rather than presenting it as a trick: you get a free half of the probability mass, so you should make one basket a certainty and accept that the other is barely worse than a coin flip, because 49 out of 99 is almost one half. Then bound it, which is the step that shows discipline: you cannot do better, since the best possible first basket is a certain white and the best possible second basket given that is the remaining 49 out of 99, so 0.747 is the maximum. Do not claim you can reach 1. And if they ask what happens with more baskets, the same principle extends, one certainty per basket until the remaining pool runs out, which is the general shape of the answer.

The one that is actually about finance

Most teasers test reasoning. One family tests whether you understand what a price is, and it is the most revealing question in this section.

REAL INTERVIEW QUESTION (Société Générale · EQD)

A small game. You have a roulette with no zero. You are told it is rigged: 70% chance of black and 30% of red, but it pays normally. Next to it someone offers you a bet: if the roulette comes up black you give them 100 euros, and nothing otherwise. What do you price the bet at, and why?

WHAT THEY TEST / ANSWER

50, not 70, and the whole point of the question is the gap between those two numbers. The expected value of the liability is $0.7 \times 100 = 70$, and that is the answer most candidates give. But the roulette itself is a tradable instrument that pays even money, so you can *replicate*. Stake S on black: if black comes you win S and owe 100, for $S - 100$; if red comes you lose S, for $-S$. Set them equal, $S - 100 = -S$, so $S = 50$ and both outcomes are -50. A stake of 50 on black turns the bet into a certain loss of 50 whatever happens, so 50 is the replication cost and 50 is the price. State the lesson plainly, because it is the reason this question is asked on a derivatives desk: once a hedging instrument exists, the *real-world probability is irrelevant to the price.* What matters is the cost of the replicating portfolio, which is what risk-neutral valuation means and why nobody prices an option using their forecast of the stock. Two additions finish it. Anyone paying you more than 50 is handing you free money, so as the seller you charge as much above 50 as the counterparty will bear, and 70 is a fine outcome, but 50 is your floor. And note that the rigged wheel is a standing positive-expectation bet, though *not* an arbitrage: betting on black at even money with a 70% chance wins $0.7 - 0.3 = 0.40$ per euro staked in expectation, but it still loses outright 30% of the time. The real answer to the whole situation is to bet on black repeatedly and size it so you cannot be ruined, which is an edge to be sized rather than a riskless profit to be maximised. The riskless part of this problem is the hedge, and that distinction is what the question is built on.

Continuous distributions, and making a market

REAL INTERVIEW QUESTION (BNP · Volatility)

I have a pen that I cut at a random point, and I take the smaller piece to give me the price of an asset. The pen has length one. Can you give me a price?

WHAT THEY TEST / ANSWER

Two answers, and giving both is what the question is for. **The expectation**: if the cut point U is uniform on $[0, 1]$, the smaller piece is $\min(U, 1 - U)$, which is uniform on $[0, 1/2]$ by symmetry, so its expected value is $\mathbf{0.25}$. Sanity check it: the two pieces sum to 1 and the smaller is by definition below 0.5, so a quarter sits in the right place. **The price**: they said "give me a price", not "give me the expectation", so quote a two-way market around it, something like 0.22 at 0.28, and be ready to trade either side. That is the actual test, and it has two parts. Your *mid* shows whether you can compute; your *width* shows whether you understand that you are about to be picked off by someone who knows the answer, so the width should reflect your uncertainty and your size, and it should widen if they ask for a bigger clip. Expect them to hit or lift you immediately and then ask you to quote again, which is a test of whether you move your market after being traded on. You should: you have just learned something about which side they want.

Market-making games

The pen question is the entry point to a genre that dominates trading interviews at some firms: you are asked to quote a two-way price on something uncertain, the interviewer trades against you, and the game repeats. What is being tested is not arithmetic.

- **Quote quickly and quote something.** A slow perfect quote loses to a fast reasonable one, because in the game being simulated the flow goes elsewhere.
- **Width is information, not timidity.** A wide market says you are uncertain; a tight one says you are confident and invites size. Both are legitimate, and being unable to justify your width is not.
- **Update after being traded on.** If they lift your offer, they think it is cheap. Move your market up. A candidate who requotes the same price after three lifts is showing that they will not learn from flow, which is disqualifying for a market maker.
- **Track your position.** They will ask what you are long at the end. Knowing your average price and your size, without being reminded, is most of the score.

The desk version of all of this is in market making, and it is worth reading before an interview at a firm that plays these games, because the concepts are identical and only the stakes differ.

The teaser that is not a teaser

REAL INTERVIEW QUESTION (BNP · EQD)

If I offered to decide your internship on a coin flip, would you accept?

WHAT THEY TEST / ANSWER

There is no correct answer and the interviewer knows it, so what is being tested is how you handle a question designed to unbalance you. Do not treat it as probability. Treat it as a risk decision and reason about it out loud: what am I giving up, what am I getting, and are the two comparable? The strong response says no and gives a reason with structure, for example that you are being asked to swap a process in which your own effort and preparation affect the outcome for one in which they do not, and that you would take a fair gamble on something you cannot influence but not on something you can. That framing, that you accept risk you are paid for and decline risk that merely destroys information about you, is a genuinely good trading instinct. What matters more than the direction is that you commit to an answer, justify it in one breath, and do not fold when the interviewer pushes back, since they will push back regardless of what you say. Changing your mind under pressure with no new information is the failure mode this question is built to detect.

INTERVIEW TRAP

Four ways to lose a teaser you could have solved. (1) Going silent: the interviewer is grading the reasoning, so a candidate who thinks for ninety seconds and then says a number has shown nothing. (2) Reaching for enumeration before checking for symmetry, conditioning or linearity, which turns a one-line problem into a page of arithmetic and a mistake. (3) Answering with an expectation when you were asked for a *price*, which on a trading desk is a two-way market with a width. (4) Failing to sanity check: a stated bound, a degenerate case or an extreme value costs one sentence and catches most errors before the interviewer does.

VISUAL / CODE TO BUILD: Interactive

A teaser trainer with two modes. **Practice** presents a real teaser from the database, starts a timer, and asks the reader to type the technique they will use before revealing any working, so the habit being trained is technique-selection rather than recall; the solution then unfolds in the five steps of the method. **Market-making game** simulates the second half of the pen question: the reader quotes a two-way market on a random quantity, the engine trades against them with a bias they cannot see, and the game repeats for twenty rounds, tracking their position, their average price and their final profit and loss against the true value. The scoreboard reports the two things that actually matter, whether they moved their market after being traded on and whether their width tracked their uncertainty, rather than whether they made money on any single round.

Where this goes next

How the rest of a technical interview is structured, and how to handle questions that build on each other, is technical interview logic. Questions about markets and levels rather than puzzles are market questions. What the whole process is trying to measure is what interviewers really test. The probability machinery these teasers use is built in conditional probability and expectation and variance.

26.4 Market Questions

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Market questions are the cheapest points available in an interview and the ones candidates most often drop. "What is the level of the CAC?" takes no preparation beyond the habit of looking, and failing it says something a technical answer cannot repair: that you want to work in markets without watching them.

This section owns the practice. The macro content itself lives elsewhere, in central banks, monetary policy and inflation; forming and defending a trade idea is trade pitching. What is here is what to carry in your head, how to answer when you do not know, and what an interviewer is listening for underneath the number.

KEY IDEA

A market question has three parts and only the first is the one asked. **The level**, which is table stakes. **The move**, meaning where it was a month ago and how it got here. **The reason**, meaning the one or two things driving it. Answering only the level is a pass; answering all three, unprompted, in fifteen seconds, is the whole point of the question.

This book deliberately prints no levels. Anything numerical here would be wrong by the time you read it, and a candidate reciting a stale number from a textbook is worse off than one who says they have not checked today. What follows is the discipline instead.

The market card

Carry a short list, and refresh it before every interview and ideally every morning. It fits on one page.

- **Equity**: the local index, one US index, one Asian index, and the equity volatility index. Levels, and the move over a month.
- **Rates**: the policy rates of the central banks that matter to your desk, the 10-year government yield in your currency and in dollars, and the shape of the curve, meaning whether it is steepening or flattening and where the inversion is if there is one.
- **Credit**: the main investment-grade and high-yield indices, at least directionally.
- **FX**: the majors against your currency, and whether the move is being driven by rate differentials or by risk sentiment.
- **Commodities**: crude, and gold. On a commodity desk, far more, including the Brent-WTI spread and the shape of the curve.
- **The calendar**: what is released this week and what the central bank meetings are, from macro calendars.

Weight the card toward your desk. A rates interview will ask about policy rates and the 10-year and may never mention equities; an FX interview will ask what is driving the currency pair rather than where it is. The composition of the question database in how trading interviews work makes the same point about technicals, and it applies here with more force, because market questions are more desk-specific than technical ones.

REAL INTERVIEW QUESTION (Société Générale · Delta One)

What is the level of the Dow, the CAC, the Hang Seng and the Shanghai?

WHAT THEY TEST / ANSWER

Four indices across three regions, which tells you what is being tested: not one number but whether you look at the world every day. Give all four, in the order asked, quickly, and then do not stop there.

Add the move and the reason for at least one of them, unprompted. "The CAC is around X, up about two percent on the month, mostly on the back of Y" turns a recall question into a conversation and makes the interviewer's follow-up your choice rather than theirs.

If you are genuinely unsure of one, the recovery is to bracket it rather than to guess a precise figure: "on the Hang Seng I would give you a range rather than a point, I have not looked since yesterday's close." A range with an honest caveat costs almost nothing. A confidently wrong precise number costs a great deal, because the interviewer now has to wonder which of your other confident numbers were invented, and that doubt follows you into the technical questions.

Two things worth volunteering if the conversation allows. Say what the indices are, since the Dow is price-weighted and unusual and the CAC is a free-float market-cap index, and the Delta One desk cares about that distinction professionally. And say which ones you actually watch, and why, because a candidate for a Delta One seat who follows Asian index futures and their basis is showing the right instinct.

When you do not know

INTUITION

The interviewer knows you cannot know everything. What they are measuring is what you do at the edge of your knowledge, and that is the same thing they will need from you on the desk, where the honest answer to a client's question is often "I do not have that in front of me, here is what I do know". Say what you know, say what you would need, give a range, and never invent a number. Inventing is the one response that is disqualifying, and it is disqualifying for a reason that has nothing to do with the question: a desk cannot use someone whose confidence is uncorrelated with their knowledge.

Having a view

Some market questions do not want a fact, they want an opinion. The answer has to be a real one, and a real opinion has three properties: a direction, a reason, and a level at which you would admit you were wrong.

REAL INTERVIEW QUESTION (Société Générale · Rates)

What are the central banks' monetary policies (Fed, ECB), and what is your view on how they will evolve?

WHAT THEY TEST / ANSWER

Two parts, and give them separately, because the first is fact and the second is judgement and confusing them is the failure mode.

The facts. Where each policy rate stands, when it last moved and in which direction, what the balance sheet is doing, and what guidance has been given about the path. Give the two central banks in parallel rather than in sequence, since the interesting content is the divergence: they face different inflation, different labour markets and different energy exposure, and the policy gap between them is what drives the currency.

The judgement. Take a position. "It depends" is not an answer, and neither is repeating consensus without saying it is consensus. State what you think happens next, give the one indicator you are watching to confirm it, and give the observation that would change your mind. Something of the form "I expect the easing path to be slower than the market prices, because services inflation is proving stickier than goods inflation; I would change my view if the next two wage prints came in below expectations" is a complete answer, and it is complete precisely because it can be wrong.

Then the piece that makes it a rates answer rather than a macro essay: say what is already priced. The market has a path embedded in the forward curve, so a view is only tradeable if it differs from that path. A candidate who says "I am hawkish relative to what the curve prices" has said something with content; one who says "I am hawkish" has not. That distinction is the same one made in trade pitching, and on a rates desk it is the whole job.

REAL INTERVIEW QUESTION (Morgan Stanley · EQD)

Can you tell me about a current topic that interested you recently, outside finance if possible, and give me your view?

WHAT THEY TEST / ANSWER

The "outside finance" clause is the question. They have already established that you can talk about markets; this is a check on whether you are a person who thinks about things, and on whether you can explain something unfamiliar clearly to someone who has not read what you read, which is most of what sales and structuring work actually is.

So pick something you genuinely followed, not something you think sounds impressive, and avoid the two categories that go wrong: party politics, which risks an argument nobody wanted, and anything on which you only have a headline. A technology shift, a scientific result, a legal decision, a sporting or cultural story with a structural angle all work.

Then structure it as you would a market comment. What happened, in two sentences. Why it is more interesting than it looks. What you think, and what would change your mind. Ninety seconds, then stop and let them ask.

The mistake to avoid is forcing a link back to finance. "And this shows why volatility will rise" is transparent and it wastes the one part of the conversation where you were being assessed as a colleague rather than as a candidate. If a genuine market implication exists, mention it in one clause and move on. If it does not, say what you found interesting and leave it there.

REAL INTERVIEW QUESTION (Société Générale · Delta One)

Do you read the financial news daily? Why, and what are your sources?

WHAT THEY TEST / ANSWER

The only wrong answer is a vague yes. Everyone says yes, so the content is entirely in the specifics, and the specifics are checkable.

Name actual sources and say what you use each for: a wire or terminal for prices and headlines, one or two publications for analysis, one or two newsletters or research pieces you genuinely read, and, if you have it, access to sell-side research through your school or a prior internship. Say which podcasts or feeds you follow if that is true. The list should look like a routine rather than a bibliography.

Then describe the routine itself, because that is what is being tested: what you look at before the open, roughly how long it takes, and what you are actually looking for. Something like checking overnight moves in Asia, the day's calendar, and anything that moved more than expected, is a real answer that a person on a desk recognises.

Then close by making it specific to this desk. For a Delta One seat that means index levels, futures basis, dividend futures and flows around index rebalancing dates. Saying what you watch *for this seat* converts a generic question into evidence that you understand what the desk does all day, and that is a much better outcome than the question was offering.

INTERVIEW TRAP

1. **Inventing a precise number.** A bracket with a caveat costs nothing; a confident wrong figure contaminates every other number you give.
2. **Quoting a stale level from a book or a course.** Levels have a date. Check them the morning of the interview.
3. **Giving the level and stopping.** The level is the question asked; the move and the reason are the question meant.
4. **Having no view.** "It depends" and "the consensus is" are both ways of declining to answer, and both are noticed.
5. **Having a view that cannot be wrong.** An opinion without a falsifying observation is a slogan. Say what would change your mind.
6. **Ignoring what is priced.** A view only has content relative to the forward curve or the implied level, not relative to today's spot.
7. **Preparing the same card for every desk.** An FX desk and a credit desk want different pages, and the one you bring says which job you thought you were interviewing for.

VISUAL / CODE TO BUILD: Interactive

A live market card. The page pulls current levels for the instruments listed above and presents them as flashcards, asking for the level, then the one-month move, then the driver, and scoring the answer against a tolerance band rather than exactly, since that is how the question is really marked. A desk selector reweights the deck, so a rates candidate is drilled on policy rates, the 10-year and the curve shape while an FX candidate is drilled on the majors and their rate differentials. A second mode presents the week's calendar and asks what each release is expected to be and what would count as a surprise, which is the preparation that turns a level into a view. Because the data is live, this is the one part of the book that is never out of date, which is precisely why the printed section contains no numbers.

26.5 Case studies

sales · trading · structuring core cross-asset

A case is not a longer technical question. A technical question has an answer that is right or wrong; a case has a situation, missing information, several defensible outcomes and a decision you have to own. Desks use them because they are the closest thing an interview can get to the job, and they are marked almost entirely on process.

The good news is that the process is the same every time.

KEY IDEA

Five moves, in order, out loud.

1. **Clarify.** Restate the situation in one sentence, then ask the two or three questions whose answers would change what you do. Not ten questions: the ones that move the decision.
2. **Structure.** Name the parts of the problem before touching any of them. "There are three things to price here: funding, borrow, and the client's credit."
3. **Quantify.** Put numbers on each part, stating your conventions and saying which numbers you are assuming rather than knowing.
4. **Decide.** Give a specific answer. A price, a level, an action. Cases are lost by candidates who analyse well and never commit.
5. **Risks.** Say what would make you wrong, what you would monitor, and what you would do if it moved against you.

Every case below is that skeleton with different flesh.

The client case: what would you ask, and what would you quote

REAL INTERVIEW QUESTION (Société Générale · Delta One)

Simulation: a client wants to finance a 100 million dollar equity portfolio. What questions do you ask them, and what price do you quote on the total return swap?

WHAT THEY TEST / ANSWER

Clarify. The client wants exposure to a portfolio they do not want to fund on their own balance sheet, so you will pay them the total return of the basket and they will pay you a floating rate plus a spread. The questions that change the price, in order of how much they change it: *what is in the basket*, since large liquid index names and small hard-to-borrow names are completely different trades; *what maturity and can either side terminate early*; *what collateral, how much initial margin and daily variation*; *how are dividends passed through*, gross, net or at a fixed percentage; *what currency, and who carries the FX*; and *who is the client*, since their credit and their regulatory status set the capital charge.

Structure. The spread you quote is built from four things: your own funding cost for 100 million of balance sheet, the cost of actually holding the hedge including any borrow if you need to short, the capital and balance-sheet charge for carrying the position, and the client's credit net of collateral. Add the desk's margin last, not first.

Quantify. Show the arithmetic even with illustrative inputs, and label them as illustrative: if internal funding is the reference rate plus 25 basis points, the balance-sheet and capital charge is 15, and the client's credit after collateral is 10, then your cost is reference plus 50 and you quote reference plus 60 to 65. On 100 million, each basis point is 10,000 dollars a year, which is the number that tells the client why the questions above mattered.

Decide and state the risks. Quote the number, then name what you are exposed to: the client's credit between margin calls, gap risk on an illiquid name in the basket, dividend risk if the pass-through is not exactly what you can hedge, and the funding itself repricing if the trade is long-dated and your own cost of balance sheet moves. Ending on the risks is what makes it a trader's answer rather than a salesperson's.

The quoting case: what repeated flow is telling you

This is the single most instructive case in the database, and it is really a lesson about information rather than about volatility.

REAL INTERVIEW QUESTION (Société Générale · EQD)

You are asked for a bid-ask on the volatility of a technology name. It does not trade, and the historical volatility is 30. They lift your offer. What do you do with your quote? Again a second time. A third time. You have only ever sold, at an average volatility of 34. Two days later you learn the accounts were faked, the stock falls 25%. Did you make or lose money? Why? And what could you have done so that this did not happen?

WHAT THEY TEST / ANSWER

Did you lose? Yes, and heavily. You sold volatility three times, so you are short options and short gamma. A 25% gap is a realised move far beyond anything a 34 volatility prices, and because it is a gap you cannot delta hedge through it: the loss on a short gamma position grows with the square of the move, and the move here is many standard deviations. On top of that, implied volatility on the name explodes after the news, so your short vega is marked against you as well. You lost on realised and on implied at the same time, which is what happens when a short volatility position meets an event.

Why it happened is the real answer, and it is not about the 25%. You were being picked off. Being lifted once is flow; being lifted three times, one way, in a name that does not trade, is information, and the information was that the counterparty knew something you did not. The average of 34 against a historical 30 looked like a comfortable margin, and the comfort was the trap: historical volatility is a measure of a past that the counterparty knew was fabricated.

What you should have done. Move the price after the first fill, not after the third: raise the offer and widen the spread each time you are lifted, because in an illiquid name your quote is the only defence you have. Cut the size you show. Ask why they want it, and look for the reason, an earnings date, an auditor change, unusual option volume, a borrow that has become expensive. Cap your total position in a single name rather than treating each trade as independent. And if you still want to be short volatility, be short it with a defined tail: buy the low-strike puts back, so you are short a **put spread**, or short a strangle *with the wings bought*, rather than short an open wing. Be precise about this, because a bare short strangle is not a defined tail at all: it is a sold put and a sold call, so both wings are still open and the downside is the same exposure that just lost the money, merely struck further away. The distinction is the whole lesson of the case. The loss came from the tail, and a position whose tail is bounded could not have produced it. The general habit is in market making and the position side of it in position limits; the realised-versus-implied distinction is in implied versus realized volatility.

The sensitivity case: one input moves, what happens

REAL INTERVIEW QUESTION (BNP · Hybrid)

Case: an Athena on the SX5E, ten years, put down-and-in at 50%, coupon 5%, priced at 100. The next day, everything else equal, rates rise by 10 basis points. What is the impact on the price, and how does the trader hedge this product?

WHAT THEY TEST / ANSWER

Structure the answer as two competing effects, because the mark is for knowing that they compete.

Discounting. The note is a stream of future cashflows, so higher rates lower their present value. This is the dominant effect and it is negative for the holder. The size depends on the note's *expected* life, not its ten-year maturity: an Athena is usually called long before maturity, so the effective duration is a few years rather than ten. If you assume an expected life of about three years, a 10 basis point move costs roughly $3 \times 10 = 30$ basis points of price, so the note goes from 100 to about 99.7. Say the assumption out loud, because the number is only as good as that assumed life.

Forward. Higher rates raise the equity forward, which makes an early call more likely and the down-and-in put less likely to knock in, and both of those help the holder. This effect is positive and smaller.

Decide. Net, the price falls slightly: an autocall is short rates. Roughly 99.7 to 99.8 on these assumptions, and you should say explicitly that the exact number comes out of the Monte Carlo, not out of a mental estimate, because the call-date distribution is what sets the duration.

The hedge. The trader is on the other side of everything the client owns. They are long the down-and-in put, which on its own is short equity delta, hedged with futures or the underlying and rebalanced as the delta moves. Say in the same breath that the *net* book delta is not simply short: the autocall trigger and the coupon digitals pull the other way, and the total flips sign as spot passes the barrier and the trigger, which is exactly the follow-up an interviewer asks next. They are short the conditional coupons, hedged with the digital and vanilla legs; they are long volatility from the barrier option, so they sell vanilla volatility against it, with the notorious problem that the vega is concentrated at the barrier and cannot be matched by a single vanilla; they are **long** rates through the funding leg, since being short the note means being short its discounted cashflows and that liability shrinks when rates rise, and the exposure is swapped out with interest-rate swaps; and on a worst-of they carry a correlation position that has no clean hedge and is warehoused. The decomposition is in embedded optionality and the running of it in hedging structured products.

The arithmetic case: a level, in one line

Some cases are short. They exist to check that you can convert a client's sentence into a number without getting lost.

REAL INTERVIEW QUESTION (Société Générale · FX)

Situation: the floating rate is negative and I do not want to pay more than 1% of interest. Where do you set the strike of your cap, given that the credit margin is 0.5%?

WHAT THEY TEST / ANSWER

One line of arithmetic and then the caveats. The client pays the floating reference plus the 0.5% margin. They want the total capped at 1%, so the reference must be capped at $1\% - 0.5\% = 0.5\%$: buy a cap struck at **0.5%** on the reference rate. The margin is contractual and is not covered by the cap, which is the point being tested, since a candidate who answers 1% has capped the wrong leg.

Then the caveats, which is where the negative rate in the question comes in. Ask whether the loan contract **floors the reference at zero**, as most do: if it does, the client is already effectively paying the margin alone while rates are negative, they are short a floor at zero that they may not know about, and the cap should be priced against a floored reference rather than the raw index. Confirm the conventions before quoting: which index and tenor, which day-count, whether the cap settles on the same dates as the loan, and whether they want it on the full notional or on an amortising schedule matching the loan. And offer the alternative, since a client focused on a maximum cost is often better served by a collar, giving up the benefit of rates below some level to reduce or remove the premium. That last move, proposing the structure the client actually needs rather than the one they named, is what a sales interviewer is listening for.

The judgement case: no numbers at all

Some cases contain no mathematics and are the easiest to fail, because candidates answer them as if being helpful were the objective.

The worked example is the counterparty-limit call, which belongs to risk limits frameworks: that section owns the breach-escalation process and builds the answer on the limit framework it has just set out. What matters here is the shape of the case rather than the escalation route. Say the principle before the procedure, because a judgement case is testing whether you know which of the two governs. Then name the two failure modes out loud, since they are what is actually being tested: granting a temporary exception to be helpful, and stonewalling with no escalation. The first turns a limit into a negotiation; the second leaves a legitimate trade unexecuted and teaches the desk that risk is an obstacle rather than a function. The answer they want is that you can hold the line and remain useful at the same time, and it is the same answer for any question about pressure from a revenue-generating colleague.

INTERVIEW TRAP

Five ways to lose a case you understood. (1) Diving into arithmetic before clarifying, then discovering the missing assumption after five minutes of work. (2) Asking ten clarifying questions, which reads as stalling; ask the two that move the answer. (3) Presenting alternatives without choosing one. (4) Quoting a market level you cannot justify instead of labelling your inputs as assumptions and showing the build-up. (5) Finishing on the answer instead of on the risks, which is the half that shows you would survive the position rather than merely price it.

VISUAL / CODE TO BUILD: Interactive

A case simulator. The reader is given a situation one screen at a time and must act before the next screen arrives, so the case unfolds rather than being read whole. The quoting case is the model: the reader sets a bid-ask on volatility, the counterparty lifts the offer, and the reader must requote before the next request, with position and running profit shown; after three rounds the news breaks and the profit and loss is decomposed into realised gamma, vega mark and the cost of not having moved the price. Other cases follow the same pattern with different clocks: a client on the phone wanting a swap price, a risk limit call, a rates move overnight. Each ends with a scorecard against the five moves, clarify, structure, quantify, decide, risks, so the reader sees which move they skip under pressure, which is almost always the first or the last.

Where this goes next

The shorter technical questions that cases are built out of are in technical interview logic, opinions about the market in market questions, and the case that ends in a recommendation is trade pitching. The desk reality these cases imitate is developed in market making and inventory management, and what the exercise is really measuring is in what interviewers really test.

26.6 Trade Pitching

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"Pitch me a stock" is asked at almost every bank in the database, in a dozen forms, from "give me a trade idea" to "pitch me a structured product as if I were twelve". It is the closest an interview gets to the job, which is why it is asked so often and why it is worth preparing properly rather than hoping for a good day.

What is owned here is the pitch as an act of communication: its structure, how far it compresses, how it changes with the audience, and the follow-ups it must survive. Forming the market view is market questions, turning a view into a payoff is payoff design, and choosing between structures given the market you are in is strategy selection by regime. This section assumes the idea and works on the delivery.

KEY IDEA

A pitch is not an opinion. It is a **proposal with a trade attached**, and it has six parts: the thesis in one sentence, two or three reasons, what the market already prices, the expression, the size and risk, and the observation that would make you exit. Candidates almost always deliver the first two and stop. The last four are where the marks are, because they are the parts that require you to have thought about being wrong.

The part everyone omits: what is already priced

An idea is only worth acting on where it differs from consensus. "This company is growing fast" is not a pitch, because the price already contains that; "the market is pricing this growth to fade within two years and I think it persists for four" is a pitch, because it says what you believe that the price does not.

That gap has a name worth using, the **variant perception**, and articulating it does three things at once: it proves you know what the price implies, it makes the idea falsifiable, and it tells the interviewer what would have to happen for you to make money. If you cannot state it, you do not have a trade, you have an observation.

The same discipline applies away from equities. On rates the consensus is in the forward curve, on credit it is in the spread, on volatility it is in the implied level. In every case the question is the same: what do I think that this number does not say?

Compression

Prepare each idea at three lengths and know which is being asked for.

- **One sentence.** Thesis and expression. "Long the euro against the dollar through a six-month call spread, because I think the policy gap narrows faster than the curve prices."
- **One minute.** Add the two or three reasons, the variant perception and the risk.
- **Ten minutes.** Add the numbers, the alternatives you rejected and why, the sizing, and the exit.

Being asked for one minute and delivering ten is a failure of listening, and it is read as one.

REAL INTERVIEW QUESTION (Société Générale · EQD)

Imagine I do not have time to listen to you, but I still have one minute a day for you to give me a good idea. Go ahead, give me an arbitrage or investment idea.

WHAT THEY TEST / ANSWER

The constraint is the question. They are testing whether you can compress, because that is what the job is: a portfolio manager gives a salesperson thirty seconds and decides in that time whether to give them more.

So respect the minute, out loud. Open with the trade in one sentence, so that if you are cut off after five seconds the listener already has the idea. Then three supporting clauses, then stop deliberately rather than trailing off. Something of the shape: here is the trade; here is what the market prices; here is why I disagree; here is what kills it.

Two things to get right in the choice of idea. Pick something **you can defend at ten minutes** even though you are giving one, because the follow-up is the real test and a pitch you cannot deepen collapses on the first question. And pick something with a **clean expression**: an idea that requires four legs and a footnote is not a one-minute idea, whatever its merits.

The word "arbitrage" in the question deserves one clause of honesty. A true arbitrage, a riskless profit, is essentially never available to a graduate with a screen, so say what you mean: a relative value trade, long one thing against another, where the risk is the relationship rather than the direction. Using the word precisely is itself a signal, and misusing it is the sort of thing an interviewer remembers.

Then close the loop the question opened: offer to send a one-line version each morning. It answers the framing they gave you and it describes the habit the job actually requires.

REAL INTERVIEW QUESTION (Citi · EQD)

Pitch me a stock.

WHAT THEY TEST / ANSWER

Have two ready, in different sectors, and have followed them for months rather than assembled them the night before, because the follow-ups will find out.

Structure, in about ninety seconds:

- **What the company does**, in one sentence. If you cannot, you do not understand it well enough to pitch it.
- **The thesis**, in one sentence, with a direction.
- **Two or three reasons**, at most one of which is a number. Reasons should be about the business, not about the chart.
- **What is priced**, meaning the multiple or the growth the market is assuming, and where you differ.
- **The risk**, meaning the two things that would break the thesis, named specifically.
- **The exit**, meaning the level or the event at which you would close it.

Then expect these four follow-ups, because they always come. *Who is on the other side, and why are they wrong? What would change your mind? Why now rather than six months ago? How would you express it, and why not just buy the shares?* The last one is the derivatives desk's question and it is your opening: a view with a horizon and a magnitude may be better expressed with a call spread than with stock, and saying so on an equity derivatives desk is the answer they were fishing for.

Two things to avoid. Do not pitch a short unless you can explain the borrow and the crowding, and do not pitch the largest company in the index, since everything about it is known and the pitch has nowhere to go.

Pitching to the person in front of you

The same trade is described differently to a trader, to an institutional client and to a private client, and getting that wrong is a sales failure rather than a knowledge failure. A trader wants the risk and the expression first. An institutional client wants the thesis and the mandate fit. A private client wants to know what can go wrong, in words.

REAL INTERVIEW QUESTION (Société Générale · Structured Products)

Pitch me a structured product as if I were 12 years old (I took a reverse convertible).

WHAT THEY TEST / ANSWER

The instruction is the assessment: they want to see whether you understand the product well enough to strip the vocabulary off it. Every technical word you use is a mark lost, and the ones to avoid are the ones you most want to say: volatility, barrier, put, coupon, underlying.

For a reverse convertible, something like this works.

"I will pay you a big rent for lending me your money for a year. Much bigger than a bank would. In exchange, there is one condition. We pick a company, say a big carmaker. At the end of the year, if its share price has not fallen a lot, you get all your money back plus the rent. But if it has fallen a long way, I do not give you your money back. I give you the shares instead, which are now worth less than what you lent me. So you can lose a lot. The rent is big precisely because you are taking that risk. You are being paid to promise you will buy those shares if they get cheap."

Then, unprompted, add the sentence that shows you know what you are selling: "So the worst case is that you lose most of your money, and the best case is that you get the rent. Your upside is capped at the rent, and your downside follows the share price most of the way down, and that trade-off is what the rent is buying." Volunteering the asymmetry is the whole test. A candidate who describes the coupon and stops has just demonstrated the behaviour that gets structured products mis-sold, and interviewers on this desk are alert to it precisely because they have seen it done.

If they push into the mechanics afterwards, you can switch registers and decompose it properly, bond plus short put. Be ready for one precision there: "fallen a long way" describes the barrier version, which is what retail actually buys, whereas a vanilla reverse convertible converts on any close below the strike. The lay wording is right for the audience you were given, but say which one you were describing when asked. Do not lead with any of that: the question asked for the twelve-year-old version and answering the question asked is the point.

REAL INTERVIEW QUESTION (Leonteq · Structured Products)

Pitch a stock for a classic Athena that you would sell.

WHAT THEY TEST / ANSWER

This is two questions welded together, and doing them in the right order is most of the answer: first what makes a stock *suitable* for an Athena, then which one and why anyone would buy it.

What the product needs. The coupon is funded by the down-and-in put the investor sells, as autocallables sets out, so a good candidate underlying is one where that put is expensive: decent implied volatility and a healthy dividend yield, both of which raise the put and therefore the coupon. It also needs to be a name the client is willing to own at the barrier level, because that is the outcome they are being paid to accept, and it needs liquid listed options so the desk can hedge it.

What to avoid, said out loud, because it is where the judgement shows. A name whose volatility is high for a company-specific reason, a pending legal case or a refinancing, is not a good coupon, it is a warning: you are being paid a lot because the barrier is genuinely likely to break. The right underlying is a solid, boring, high-dividend large cap whose volatility is elevated by the market rather than by its own accounts.

Then the pitch itself, aimed at a client who already owns the sector: you have a name you are comfortable holding for years, you do not expect it to rise much this year, and you would happily buy more of it thirty percent lower. This turns that opinion into an income: you are paid a coupon while the view holds, redeemed early if it rises, and you end up owning the stock at a level you already said you liked if it falls a long way.

Close on the honest sentence, which on this desk is the one being marked: the risk is not the barrier being touched, it is the client discovering they did not really want the stock at that level. If they would not buy it there in cash, they should not sell the put.

REAL INTERVIEW QUESTION (Natixis · Fixed Income)

If you had 1,000,000 euros right now, what would you invest it in? Why?

WHAT THEY TEST / ANSWER

The trap is answering with an asset. It is an allocation question, so answer with a framework and then populate it, which takes fifteen seconds longer and is a completely different answer.

Start by asking, or by stating, the two things that determine everything: the **horizon** and the **constraint**. A million to be spent in two years and a million to be left for twenty are not the same problem, and saying so first shows you know that allocation is driven by the liability, not by the view.

Then give a split, with a reason for each line rather than a percentage for each line. Something like a core in government bonds at the maturity that matches the horizon, since a bond held to maturity in the currency you spend is the only part with no nominal risk left in it (inflation still bites, which is the honest caveat to add); a growth allocation in broad equity, index rather than single names, because you have no edge in stock selection and paying for the attempt is a certain cost against an uncertain benefit; a small allocation to whatever your genuine view is, sized as if it might be wrong; and cash for the part you might need.

Then say what you are *not* doing and why: no leverage, because you cannot fund it cheaply; nothing illiquid unless the horizon truly allows it; no product whose fees you cannot see.

For a fixed income desk, finish on their ground. Say where you would sit on the curve and why, in terms of what the curve currently prices about the policy path, and whether you would take that duration or stay short of it. That converts a generic allocation answer into a rates answer, which is what the seat is for.

INTERVIEW TRAP

1. **Pitching a view instead of a trade.** No expression, no size, no exit means no pitch.
2. **Omitting what is priced.** An idea has content only relative to the consensus embedded in the price.
3. **Pitching something you followed for a day.** The follow-ups are specific and they find this immediately.
4. **Ignoring the requested length.** Being given a minute and taking ten is read as not listening, which is the trait the question was designed to expose.
5. **Using jargon with the wrong audience.** Especially when the question has explicitly told you the audience.
6. **Having no way to be wrong.** An idea with no falsifier and no exit cannot be sized, and anything that cannot be sized cannot be traded.
7. **Selling only the upside.** On a structuring or sales desk, volunteering the downside is not a weakness in the pitch, it is the thing being tested.

VISUAL / CODE TO BUILD: Interactive

A pitch builder with a timer. The reader picks an asset and fills six fields, thesis, reasons, what is priced, expression, size and exit, and the page refuses to accept the pitch until all six are present, which is the entire lesson enforced mechanically. It then renders the pitch at three lengths, one sentence, one minute and ten minutes, from the same six inputs, so the reader can see that compression is selection rather than rewriting. A practice mode runs a countdown and cuts the reader off at the stated length, then serves the four standard follow-ups (who is on the other side, what would change your mind, why now, why this expression) one at a time. An audience toggle re-renders the same pitch for a trader, an institutional client and a twelve-year-old, flagging every term that would not survive the switch.

26.7 Behavioral signals desks look for

sales · trading · structuring | core | cross-asset

Technical questions have answers, so candidates prepare for them. Behavioural questions look like small talk, so candidates improvise, and that is where most offers are lost. The interviewer is not making conversation. They are sampling for five specific traits, using whatever question is nearest to hand, and they are doing it because they will sit two metres from you for a year and because the desk will hand you real risk within months.

This section is about what those five traits are and how each is probed, which is more useful than a list of questions, since the questions vary and the traits do not.

KEY IDEA

Every behavioural question is a proxy for the same underlying question: *what will this person do when they are wrong, under pressure, and nobody is watching?* A desk cannot test that directly, so it tests the nearest observable things: how you describe a failure, whether you change your mind when pushed, whether you know what the job actually involves, and whether your interest in the market survives one follow-up question.

The five signals

1. Risk instinct. Do you think in terms of what can go wrong and how much, or only in terms of what you expect? This is probed in technical questions as often as in behavioural ones, and the tell is whether you volunteer the downside without being asked.

2. Intellectual honesty. Will you say "I do not know"? A desk can teach almost anything except the willingness to admit an error early, and the cost of someone who hides a mistake for a day is measured in money. The version of this that interviewers watch most closely is what happens when they tell you that you are wrong: a candidate who folds immediately, and one who never folds, are both failing.

3. Coachability. Can you take a correction, apply it, and carry it forward to the next question? Interviews are often deliberately structured so that a hint given in question three is needed in question five. Whether you use it is the measurement.

4. Temperament and stamina. The job involves early starts, long days and being wrong in public. The signal is not enthusiasm, which everyone performs, but whether your account of your own past effort is specific and believable.

5. Genuine, specific interest. Not "I am passionate about markets", which is free to say, but a view you hold, a number you know, and a reason you care. This one collapses under a single follow-up if it is not real, which is exactly why it is asked.

How to structure an answer

Use four beats, and keep the fourth. Situation, decision, outcome, and what you would do differently. The first three are what most candidates give; the fourth is where the signal is, because it demonstrates the reflex the desk is actually hiring for, which is reviewing your own decisions after the fact rather than only your results.

Two constraints matter as much as the structure. Keep it to about ninety seconds, because a long answer is itself a negative signal about how you will brief a trader mid-morning. And make it specific: names, numbers, dates. A generic answer is indistinguishable from a fabricated one, and interviewers treat them the same way.

REAL INTERVIEW QUESTION (Barclays · Rates)

What was your last failure and how did you overcome it?

WHAT THEY TEST / ANSWER

Pick a real failure with a real cost, and resist the two standard evasions: the disguised strength ("I work too hard") and the failure that was somebody else's fault. Both are recognised instantly and both answer the question badly, because the interviewer is not measuring the failure, they are measuring your relationship with it. Use the four beats. **Situation**, briefly, with enough specifics to be credible. **Decision**: what you chose, and why it seemed right at the time, which is important because a failure with no defensible reasoning behind it reads as carelessness rather than as a judgement call. **Outcome**: what it cost, stated plainly and without minimising. **What you would do differently**, which should be a change of *process*, not a resolution to try harder: I now check X before committing, I now escalate at this threshold. Two things distinguish a strong answer. Own it in the first sentence rather than building to it, since the interviewer is listening for how long you take to say the word. And choose a failure from something you cared about, because the ones candidates pick from low-stakes situations reveal that they are managing the interview rather than answering it.

REAL INTERVIEW QUESTION (Société Générale · EQD)

Why you, and how are you going to make my life easier?

WHAT THEY TEST / ANSWER

This is the whole interview compressed into one question, and the second clause is the real one. Do not answer with adjectives. Answer with the job. Work out what this specific desk spends its time on, then name the parts of it a junior can absorb in the first six months, and give evidence that you can do them: building and checking a pricer, owning a daily risk report and noticing when a number looks wrong, turning a client request into a term sheet without three rounds of corrections, being the person who has already looked up the answer before the meeting. "Making my life easier" means removing work from a senior person, and a candidate who understands that is describing the actual job rather than a career aspiration. Then close on the honest version of "why you": not that you are better than the other candidates, which you cannot know, but that you have done the specific things this desk needs and you are easy to work with under pressure, with one concrete example of each. If you do not know what the desk does in enough detail to answer this, that is the answer the question was designed to produce, and it is why preparation on the desk's actual products beats preparation on your own biography.

Compression tests

Some questions exist purely to see whether you can be concise under pressure, which is a genuine job skill: a trader asking you something at 8:55 wants a sentence, not a paragraph.

REAL INTERVIEW QUESTION (Société Générale · EQD)

Describe yourself in one word.

WHAT THEY TEST / ANSWER

Give one word, then one sentence of evidence, then stop. The failure mode here is not the word you pick, it is answering with three words, or with a paragraph, or with a question about whether they really mean one. The question is a compression test and the compression is the answer. Pick a word that is both true and relevant to the desk, and be ready for the follow-up, which is always some version of "give me an example" or "when has that been a weakness". That second one is the real test, so choose a word whose downside you can discuss honestly: "persistent" has an obvious failure mode, "curious" has an obvious failure mode, and being able to name it immediately shows self-awareness rather than damaging you. Avoid words that are claims about outcomes rather than about you, such as "successful", and avoid words nobody would ever disclaim, such as "hardworking", since a word that no candidate would refuse carries no information and the interviewer knows it.

Explaining a pivot

Career changes and unusual paths are not a problem, but an unexplained one is. The signal being read is whether you make decisions deliberately.

Do you know what the job is?

REAL INTERVIEW QUESTION (Société Générale · Delta One)

Do you know what the team's expectations of your work will be?

WHAT THEY TEST / ANSWER

Answer at three time horizons and be concrete at each, because a vague answer here tells the desk you have not thought about the actual role. **First weeks**: absorb the systems, the booking flow and the daily reports, ask questions early rather than guessing, and make no unsupervised changes to anything that touches risk. **First months**: own a recurring piece of work end to end, typically a report, a reconciliation or a pricer, be the person who notices when a number is wrong before anyone else does, and be reliable enough that nobody checks your output twice. **First year**: understand the book well enough to answer a client or a control function without escalating everything, and start contributing on pricing or hedging under supervision. Then add the part that shows you know the culture rather than the tasks: on a desk, being wrong quickly and loudly is far better than being wrong quietly, and the expectation is that you escalate immediately, take the correction and do not repeat it. Ending on that is worth more than any list, because it is the one expectation a desk genuinely cannot compromise on, and it is where a junior most often fails. See the different roles within the bank for what the job actually consists of.

What differs by desk

The five signals are common; their weights are not.

- **Trading** weights risk instinct and the response to being wrong most heavily, and tests them by pushing back on your answers regardless of whether they are correct.
- **Sales** weights clarity, listening and the ability to hold a room, and tests them by asking you to explain something complicated simply, or to pitch. See trade pitching.
- **Structuring** weights precision and follow-through, since an error in a term sheet is permanent, and tests them by checking whether your answers stay consistent across a long conversation.

Knowing which one you are sitting in front of, and adjusting the emphasis rather than the content, is a legitimate and expected piece of preparation.

INTERVIEW TRAP

Four behavioural errors that cost offers. (1) Rehearsed answers delivered verbatim: they are audible, and they suggest that any unscripted question will find nothing behind them. (2) The disguised-strength failure story, which answers a different question and signals that you will not admit a real error. (3) Folding the moment an interviewer disagrees, or refusing to move at all; both are failures of the same test, which is whether you update on evidence rather than on pressure. (4) Claiming interest in markets without a single specific view, level or story to support it, which collapses on the first follow-up and taints everything you said before it.

VISUAL / CODE TO BUILD: Interactive

A behavioural rehearsal tool. The reader records or types a ninety-second answer to a real question from the database, and the tool scores it on the four beats, flagging a missing "what I would do differently" and highlighting generic phrases that could belong to any candidate. A second mode runs the pushback drill that trading interviews actually use: the reader answers, the tool disagrees with them regardless of whether they were right, and asks again; the score depends on whether they changed their answer for a reason or for the pressure, which is measured by whether their second answer contains new evidence. A third panel maps each question in the database's behavioural set to which of the five signals it probes, so the reader prepares traits rather than memorising responses.

Where this goes next

The shape of the whole process is how trading interviews work, and how technical questions are chained together within it is technical interview logic. The behaviours that end a process outright, rather than merely costing points, are deal-breakers, and the summary of what the whole exercise is measuring is what interviewers really test.

26.8 Deal-breakers

sales · trading · structuring | core | cross-asset

Most of this chapter is about accumulating points. This section is about the things that stop the count. A deal-breaker is not a weak answer, it is an answer after which the rest of the interview no longer matters, and the distinguishing feature is that you almost never find out it happened: the conversation stays polite, the remaining questions get asked, and the decision was made twenty minutes earlier.

They fall into four families, and only one of them is about knowledge.

Family one: integrity

The shortest way to end a front-office process is to say something the interviewer can check and find false. Three versions account for nearly all of it.

Inventing a technical answer. Covered as a rule in technical interview logic and repeated here because of what it costs. The person opposite prices these instruments for a living. A confident, wrong, invented answer is not scored as a wrong answer; it is scored as evidence about what you would do on the desk when you did not know something, and that is not a technical question.

Inflating the CV. Claiming you built a pricer you used, or ran a book you watched, is a bet that nobody will ask a second question. They always ask a second question, and the failure mode is specific: the candidate who cannot explain a choice they say they made. If you supported a project, say you supported it, and then say precisely what you did. A small, honest contribution described exactly is worth more than a large one described vaguely, because the second is indistinguishable from a fabrication.

Misrepresenting a situation you were in. Interviewers ask about failures and conflicts partly to see whether you will tell an unflattering truth. The candidate who has never failed, never disagreed with anyone and was never at fault is not describing a career, they are describing a script.

Conduct questions belong in this family too. Asked what you would do about a limit breach, a mispriced trade in your favour, a colleague cutting a corner, or a client given the wrong information, the answer is always the same shape: name it, escalate it through the documented route, and do not resolve it privately to be helpful. The worked version of that is the limits case in case studies. An answer that treats the rule as negotiable when the trade is profitable ends the process regardless of everything else.

Family two: risk instinct

Some answers reveal how you would behave with money at stake, and a desk reads them literally. What these signals *are*, and how a desk reads them positively, is behavioural signals desks look for; that section owns risk instinct and coachability as signals. What this section adds is only the **threshold**: the point at which each stops being a weak answer that costs you marks and becomes one that ends the process regardless of the rest.

- **Refusing to give a number.** On a market-making or estimation question, "I would need more information" as a final answer reads as an inability to act under uncertainty, which is the job. Ask for the parameter that matters, then commit to a number with your assumptions stated.
- **Doubling down.** Asked what you would do after a losing position moves further against you, an instinctive "add to it, it will come back" is a deal-breaker on a trading desk. So is refusing ever to add: the answer is a process, a level at which you were wrong, and a size you decided in advance.
- **Hiding the loss.** Any answer implying you would wait before reporting a problem ends it. Desks have lost more to concealed positions than to bad ones.
- **Not being correctable.** If an interviewer corrects you and you argue past the point where you can see they are right, they have learned what supervising you would be like. Take the correction, restate it in your own words, move on. Genuine disagreement is fine and often welcome, but only with an argument, and only once.

Family three: motivation and preparation

The largest family by volume, and the most avoidable. These questions look soft and are scored hard, because the answers are the cheapest possible signal of how much the candidate actually wants this particular job.

REAL INTERVIEW QUESTION (Société Générale · Delta One)

Why finance?

WHAT THEY TEST / ANSWER

The deal-breaker here is the generic answer: "I like markets, they are dynamic and fast moving." Everyone says it, it describes no decision, and it suggests the candidate is running a process rather than choosing a career.

A working answer has three parts. **A specific origin**: something concrete you did, an internship task, a paper, a portfolio you ran, a moment where you noticed you enjoyed the actual work rather than the idea of it. **What that told you about the work itself**: the feedback loop, being priced by the market rather than by an appraisal, the fact that a position tells you whether you were right. **Why this part of it**: not "finance" but this asset class and this seat, which is where the answer stops being interchangeable.

The test to run on your own answer is whether it could be pasted into an application for a different industry. If it could, it is not an answer. Say something that would be false of an unfriendly alternative career, and it will automatically be specific.

REAL INTERVIEW QUESTION (Credit Suisse · FX)

Why sales and not trading?

WHAT THEY TEST / ANSWER

This question is a trap in two directions and candidates fall into both. The first is answering it as a ranking, implying sales is the fallback for someone who could not do trading, which insults the desk you are sitting in front of. The second is answering it so diplomatically that it says nothing.

Answer it as a genuine preference between two different jobs, and show you know how they differ. Trading is a position and a screen; the feedback is immediate, impersonal and quantitative, and the skill is managing risk you already own. Sales is a franchise; the feedback is a client who calls you first next time, the horizon is longer, and the skill is understanding a client's problem well enough to bring them something before they ask. Then say which of those two descriptions is what you want your day to consist of, and support it with something you have actually done.

The deal-breaker is any answer suggesting you do not know that these are different jobs, which usually surfaces as a candidate praising their own quantitative skills in answer to a question about sales. Naming what you are giving up, which is the immediacy and the direct scoreboard, makes the choice sound like a choice.

Family four: the self-inflicted answers

Two questions cause more avoidable damage than the rest of the behavioural set combined, because candidates prepare answers to them that are transparently constructed.

REAL INTERVIEW QUESTION (BNP · Fixed Income)

What are your main character traits? Give three strengths and one weakness, with examples.

WHAT THEY TEST / ANSWER

The strengths are the easy half: pick three that are relevant to this seat, not three that are flattering, and attach a specific example to each. Unexamplified strengths are noise.

The weakness is where the interview is lost. Do not give a disguised strength, and "perfectionist", "I work too hard" and "impatient with slow people" are all disguised strengths; every interviewer has heard them and they are read as evasion, which is a worse signal than any real weakness. Give a genuine one, bounded in three moves: what it is, a concrete instance where it cost something, and what you now do about it that has visibly worked. "I used to go too deep into detail before sharing progress, so on my last project my manager did not see the model until it was late; I now send an intermediate version at the halfway point and it has stopped happening."

Note the shape: the weakness is real, the damage is admitted, the fix is specific and already in place. That answer is scored as self-awareness. The disguised version is scored as either self-deception or a lack of candour, and on a trading floor both are serious.

REAL INTERVIEW QUESTION (Pictet AM · Sales)

Describe what you particularly liked about your last experience, and what you disliked or found frustrating.

WHAT THEY TEST / ANSWER

The first half is safe and almost nobody fails it. The deal-breaker is hidden in the second half, and there are two ways to walk into it.

Disparaging the people. A candidate who answers the frustration half by criticising former colleagues, a manager or the firm has told the desk exactly how they will speak about this desk in a year. It reads as a disposition rather than as a description of one job, and it is the single most common way this question ends a process. The interviewer is not weighing whether your complaint was justified; they are noting that you made it to a stranger.

Claiming nothing frustrated you. The mirror failure. It is not modesty, it reads as either no engagement with the work or an unwillingness to be candid, and the follow-up question will find out which.

The answer that works keeps the frustration **structural rather than personal**: name something about the process, the pace, the visibility of the output, the distance between your work and a decision, then say what you did within your control about it, and what it told you about the environment you want next. That last clause is the one being marked, because it turns a complaint into a reason for applying here.

Do not manufacture a frustration that is a compliment in disguise, since it fails on the same ground as the disguised weakness: the interviewer has heard it, and pretending costs more than the honest answer would have.

REAL INTERVIEW QUESTION (Pictet AM · Sales)

What qualities would your former employers attribute to you, and what areas for improvement might they have criticised?

WHAT THEY TEST / ANSWER

This is the weakness question with the escape route removed, because it is asked in someone else's voice, and it is often a reference check in disguise. Two rules follow.

First, say only what those people would actually say, because a reference call can contradict you and sometimes does. If you are unsure what they would say, that is worth finding out before the interview, and asking a former manager directly is a normal and well-received thing to do.

Second, treat the criticism half as the substance rather than as the toll. A specific, attributable, past-tense criticism with a visible correction is one of the strongest answers available in a behavioural interview: "my manager on the structuring desk thought I asked for validation too often before moving on a task; by the end of the internship I was sending decisions with my reasoning rather than questions, and that was in my review." It is verifiable, it is unflattering, and it demonstrates that you can absorb feedback from a senior person, which is the entire question. The deal-breaker version is the candidate who cannot name a single criticism, which reads either as never having been given real feedback or as not having listened to it.

INTERVIEW TRAP

The six that end processes, in rough order of frequency. (1) The generic motivation answer that would fit any bank and any industry. (2) The disguised weakness. (3) The failure that is somebody else's fault. (4) The invented technical answer, given confidently. (5) The CV claim that collapses on the second question. (6) Any answer implying a rule is negotiable when the trade is profitable. None of these is a knowledge problem, and all of them are fixable the week before.

VISUAL / CODE TO BUILD: Interactive

A red-flag rehearsal tool. The reader records or types an answer to one of the behavioural questions in this section, and the tool marks it against the specific failure modes rather than giving a score: for a motivation answer it highlights every sentence that would remain true with a competitor's name substituted; for a weakness answer it flags the phrases on the disguised-strength list and checks whether a concrete cost and a specific fix are both present; for a failure answer it detects attribution of the cause to another person or to circumstances. A swap-the-name button literally rewrites the answer with a different bank and a different industry and shows it back to the reader, which makes the generic-answer problem impossible to miss. Each question also shows two anonymised real answers, one that worked and one that did not, with the difference annotated.

Where this goes next

The positive side of the same coin, what desks are actively looking for rather than screening out, is behavioural signals desks look for. The technical version of the integrity rule is in technical interview logic, the conduct scenario is worked in case studies, and the summary of what the whole process is measuring is what interviewers really test.

26.9 What Interviewers Really Test

sales · trading · structuring | core | cross-asset

The question database holds 3,394 entries and nobody is testing 3,394 things. Behind them sit a handful of properties an interviewer is trying to establish, and every question is a probe for one of them. This section is the mapping: what is being measured, which questions measure it, and what that implies for how to answer.

This closes the chapter, so it depends on the rest of it. How interviews work covers the process, technical interview logic how to build an answer, behavioural signals what your conduct communicates, and deal-breakers what ends a process. What is owned here is the inference the interviewer is making.

> **KEY IDEA**
>
> An interviewer is not scoring your answers, they are **locating your boundary** and then watching what you do at it. Every follow-up moves toward the edge of what you know, and the interview effectively ends when they find it. Which means two things. Where the boundary sits matters less than most candidates think, because everyone has one and they have interviewed people with theirs in every possible place. How you behave once you reach it matters far more, because that is the only part of the interview that resembles the job.

The six things

- **Do you understand it, or have you memorised it?** Probed by asking "why" until either a reason appears or a phrase repeats. The reason a single topic appears in the database in eight different costumes is that the eighth costume is where memorisation fails.
- **Can you compute without a calculator?** Probed by mental arithmetic and orders of magnitude. Being roughly right in ten seconds beats being exactly right in two minutes, because the desk needs the first and has software for the second.
- **Do you know the sign?** Long or short, up or down, who pays whom. Sign errors are the ones that lose money, and they are the ones interviewers weight most heavily relative to how easy they look.
- **Do you think in risk?** Probed by "and what could go wrong", which follows every idea you offer. A candidate who volunteers the risk before being asked has demonstrated something no correct answer can.
- **Do you know where your knowledge stops?** Probed by pushing past it and seeing whether you say so or start improvising.
- **Would they want to sit next to you?** Probed by everything, all the time, and developed in behavioural signals.

Questions with a false premise

A specific and common device deserves naming, because it catches well-prepared candidates more often than unprepared ones: the question that contains an error and invites you to agree with it. Accepting the premise is the failure, and the better you have memorised the conventional statement, the more likely you are to accept it.

REAL INTERVIEW QUESTION (Commerzbank · EQD)

Why is the delta of an ATM call 0.5?

WHAT THEY TEST / ANSWER

It is not, and the question is checking whether you will say so.

Answer in three moves. **Correct the premise politely**: an at-the-money call's delta is close to one half but above it, and the gap grows with maturity and with volatility. **Give the reason**: the delta is $N(d_1)$ and at the money with zero rates $d_1 = \frac{1}{2}\sigma\sqrt{T}$, which is positive, so $N(d_1) > 0.5$. **Give the number**, since a correction with no figure is just contradiction; the levels at each maturity are in Greek definitions.

Then explain why the approximation persists, because that is the generous version of the answer and it shows you understand rather than merely disagree: for short maturities the error is small enough to ignore on a trading floor, so the shorthand is useful and survives, and it only becomes wrong when someone applies it to a two-year option.

The transferable skill is the point of this box. When a question asserts something, check it before building on it. Say "that is nearly right, and here is the exact version" rather than either agreeing or announcing that the interviewer is wrong. Both extremes cost you; the middle is what a desk sounds like when someone misquotes a level.

Questions that look like arithmetic

REAL INTERVIEW QUESTION (JP Morgan · Rates)

Calculate 12,374 plus 10% minus 10% in your head, without a pen.

WHAT THEY TEST / ANSWER

The trap is that people answer 12,374. Adding ten percent and then subtracting ten percent does not return you to where you started, because the second percentage is taken from a larger number.

Do it multiplicatively, which is both the fast way and the way that makes the trap visible:

$$1.10 \times 0.90 = 1 - 0.10^2 = 0.99,$$

so the answer is 99% of the original, i.e. one percent lower. The number itself is $12{,}374 \times 0.99 = 12{,}250.26$.

One clarification is worth making out loud before you compute, and it earns more than the arithmetic does: in French notation the comma is a decimal separator, so "12,374" may mean twelve point three seven four rather than twelve thousand. Ask, or answer both, and note that it does not matter here because the operation is a multiplication by 0.99 either way, giving 12.25026 or 12,250.26 with the same digits. Checking what a number means before working with it is a habit desks care about considerably more than mental arithmetic speed.

Then generalise it, because that is what the question is for: successive percentage moves multiply rather than add, so a fall of 50% needs a rise of 100% to recover, and $(1+x)(1-x) = 1 - x^2$ means any round trip in percentage terms loses x^2. That is the same convexity that makes volatility drag real, and connecting the two is the answer that ends the follow-ups.

Questions that look like pricing

REAL INTERVIEW QUESTION (Optiver · EQD)

We are talking about options on the CAC 40, and the CAC 40 future trades at 5650. Which option is more expensive, the 5500 put or the 5800 call? Why?

WHAT THEY TEST / ANSWER

The strikes are deliberately equidistant from the future, 150 points either way, so the question has removed the easy answer and what remains is whichever effect you reach for first. There are two, they point in opposite directions, and naming both is the complete answer.

Lognormality favours the call, slightly. Equal distances in points are not equal distances in returns: $\ln(5650/5500) = 0.0269$ against $\ln(5800/5650) = 0.0262$, so the put is marginally further out of the money in the units the distribution actually uses. Most of the gap, though, is more basic than that: under a lognormal the right tail is unbounded and the left one is not, so even at perfectly *log*-symmetric strikes $Fe^{\pm m}$ the put is worth exactly e^{-m} times the call, about 2.7% less here. Priced flat at 20% volatility for one year, the put is worth 373.08 and the call 384.92, so on this effect alone **the call is dearer, by about 12**.

Skew favours the put, decisively. In equity indices the downside strike carries a higher implied volatility, and each of these options is worth about 22 per volatility point. So the lognormal advantage of 12 is erased by a skew of roughly a quarter of a volatility point each way, and real CAC skew at these strikes is several points. Price the put at 22% and the call at 18% and the put is worth 416.95 against 339.89, dearer by 77.

So: the put, on any real surface. The reason is skew, and the reason it is a good question is that the flat-volatility answer is the opposite one. A candidate who answers "they are equidistant so about the same" has priced with a model they were not asked about; one who says "the put, because of skew" has the right answer for the right reason. The mechanism is the volatility smile.

If the interviewer pushes, volunteer the market-structure version: index puts are bid because everyone hedging equity risk buys downside and few people want to own upside, so the demand imbalance is what the skew prices. That is a market maker's answer and this question was asked by a market maker.

Questions that look like they have no answer

REAL INTERVIEW QUESTION (Credit Suisse · FX)

I have an order to place, a colleague asks me for help, and an internal client wants a presentation in an hour: how do you proceed?

WHAT THEY TEST / ANSWER

There is no correct ordering, and looking for one is the mistake. The question tests whether you triage on the right axis, and the right axis is not urgency, it is **which of these is irreversible**.

So say the principle first, then apply it. The order goes to market, because an unexecuted order is exposure that is live right now and every minute of delay has a cost that cannot be recovered. The colleague gets thirty seconds immediately, not to solve their problem but to establish whether it is bigger than yours, since a colleague asking for help on a desk is sometimes asking you to check a trade before it goes wrong. The presentation is the only one of the three with a deadline rather than a cost of delay, so it is last, and if the hour is genuinely not enough you say so now rather than at minute fifty-five.

Then say the thing that shows you have worked somewhere: you communicate while you triage. "Give me two minutes" to the colleague and a message to the internal client saying the deck will be with them at a stated time costs you nothing and removes both of them from your attention. Silence is what turns a queue into three escalations.

The interviewer is checking three things at once: that you know execution risk is the live one, that you do not treat a colleague as an interruption, and that you manage expectations rather than absorb pressure quietly. The last is the one people fail, usually by heroically trying to do all three and delivering the deck late without warning anyone.

INTERVIEW TRAP

1. **Improvising past your boundary.** The interview was going fine until the invented answer. Everything you said before it is now suspect, which is a far larger cost than the question was worth.
2. **Accepting a false premise.** If the question asserts something, check it. The better prepared you are, the more fluently you will agree with a familiar error.
3. **Answering the surface question.** An arithmetic question is rarely about arithmetic and a pricing question is often about a convention. Ask what the question is for before answering it.
4. **Being exactly right, slowly.** Orders of magnitude fast beat decimals slow, and saying "roughly X, I can be precise if you want" gets both.
5. **Never volunteering a risk.** Every idea you offer should arrive with the thing that would break it, before you are asked.
6. **Treating "I do not know" as a loss.** It is a normal move, and the version that scores is "I do not know, here is what I would need to work it out, and here is how far I can get without it".

VISUAL / CODE TO BUILD: Interactive

A probe classifier. Each question drawn from the database is presented with the six properties as buttons, and the reader guesses what it is testing before seeing the answer and a worked response. Guessing the probe correctly is scored separately from answering the question, because the two skills are different and the first is the one this section teaches. A second mode runs an adaptive chain: the page keeps asking “why” on whatever the reader answers, following a real follow-up tree until the reader types “I do not know”, and then scores the exit rather than the depth, showing how a good and a bad exit read to an interviewer from the same boundary. A third mode seeds a fraction of the questions with false premises, so the reader practises checking assertions rather than building on them.